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Third-Party Information

Chapter 1
Simulation Goals and Work Flows

You can use HyperLynx SI/PI to evaluate your design with a variety of simulation types. For a post-layout design, you can run simulation to validate its performance. For a pre-layout design, you can run simulation to plan key design elements (such as a SERDES channel) and run sweep simulation to define routing constraints.

Work Flow Category	Description
Pre-Layout Design Work Flows	Deciding on the types of simulation to run on your pre-layout design depends on the design behaviors that you want to evaluate. Simulation can help you design nets with good signal integrity and design power-distribution networks (PDNs) with good power integrity. Simulating “what if” design variations can help you optimize the design. Running signal-integrity and power-integrity simulation prior to board layout can help you define post-layout constraints and avoid performance problems later in the design process.
Find Post-Layout Design Problems	Deciding on the types of simulation to run on your post-layout design depends on the design behaviors that you want to evaluate. Simulation can help you find nets with poor signal integrity and design power-distribution networks (PDNs) with poor power integrity. Simulating “what if” design variations can help you determine design changes that improve SI or PI.

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Pre-Layout Design Work Flows

Deciding on the types of simulation to run on your pre-layout design depends on the design behaviors that you want to evaluate. Simulation can help you design nets with good signal integrity and design power-distribution networks (PDNs) with good power integrity. Simulating "what if" design variations can help you optimize the design. Running signal-integrity and power-integrity simulation prior to board layout can help you define post-layout constraints and avoid performance problems later in the design process.

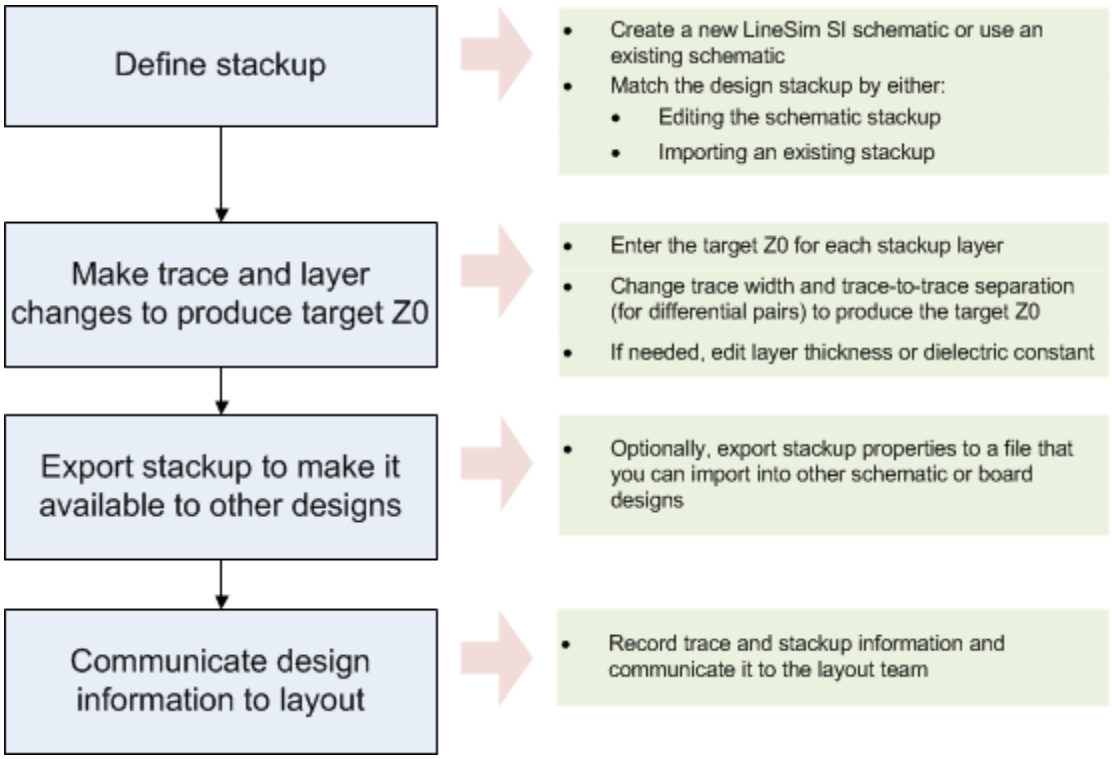
Goal	Description
Designing Trace and Stackup Geometries to Meet Target Impedance	Determine the signal trace width, trace-to-trace separation (for differential pairs), and stackup properties that produce the target characteristic impedance (Z0).
Designing Vias to Meet Impedance and Bypassing Requirements	Determine the signal via, pad, and antipad properties that produce the target characteristic impedance (Z0). This work flow uses the default signal via model. For single-ended signals, you can determine the location and number of stitching vias needed to provide low-impedance return current paths.
Designing Vias to Meet Loss Requirements	Determine the signal via properties that produce acceptable loss. You can determine the location and number of stitching vias needed to provide low-impedance return current paths for signal vias. This work flow uses 3-D electromagnetic simulation.
Designing Net Topologies to Meet Crosstalk Requirements	Determine net properties that meet crosstalk requirements. You can optimize design tradeoffs and see the effects of manufacturing tolerances that affect crosstalk. You can run sweeps to generate constraints for the post-layout design.
Designing Net Topologies to Meet Signal Quality Requirements	Determine net properties and terminations that meet signal quality requirements. You can optimize design tradeoffs and see the effects of manufacturing tolerances that affect signal quality. You can run sweeps to generate constraints for the post-layout design.
Designing Net Topologies to Meet Timing Requirements	Determine net properties that meet timing requirements. You can optimize design tradeoffs and see the effects of manufacturing tolerances that affect signal timing. You can run SI simulation with sweeps to generate constraints for the post-layout design.

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Goal	Description
Designing PDNs to Meet DC Power Loss and Current Density Requirements	Determine power-supply net geometries and stitching via quantities and locations to design power distribution networks (PDNs) with acceptable DC power loss and current density.
Designing PDNs to Meet Low Impedance Requirements Across a Range of Frequencies	Determine power-supply net geometries and decoupling capacitor quantities and locations to design power distribution networks (PDNs) with acceptable impedance across a frequency range. Evaluate decoupling capacitor mounting technologies, such as via-in-pad, microvias, and X2Y capacitors. Evaluate dielectric properties for embedded capacitors, such as a C-ply material or ultra-thin thickness.

Designing Trace and Stackup Geometries to Meet Target Impedance

Determine the signal trace width, trace-to-trace separation (for differential pairs), and stackup properties that produce the target characteristic impedance (Z0).



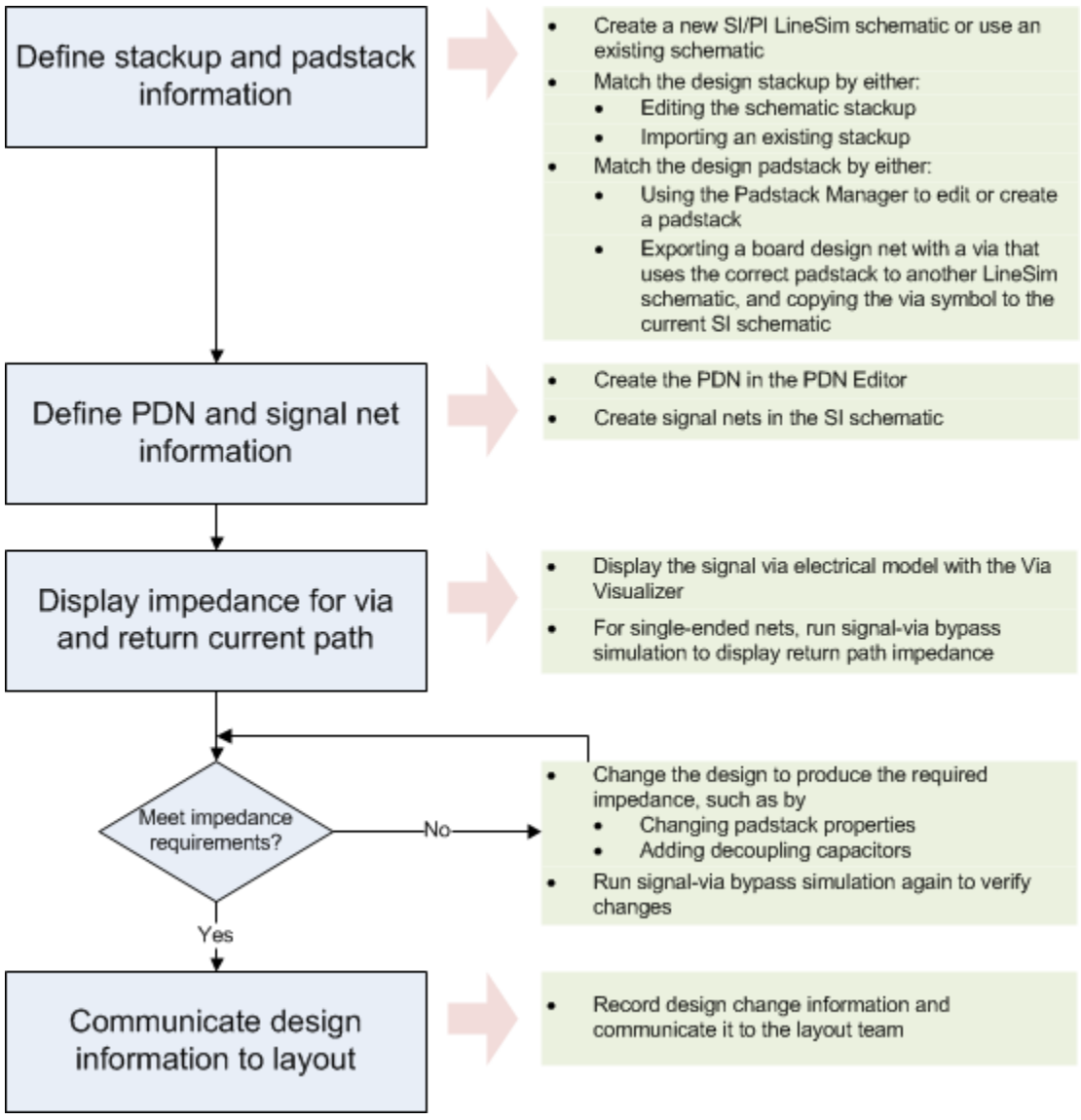
Related Topics

Exporting and Importing a Stackup

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Designing Vias to Meet Impedance and Bypassing Requirements

Determine the signal via, pad, and antipad properties that produce the target characteristic impedance (Z0). This work flow uses the default signal via model. For single-ended signals, you can determine the location and number of stitching vias needed to provide low-impedance return current paths.



Related Topics

Creating a Schematic-PDN Design

Exporting and Importing a Stackup

Editing a Padstack Definition

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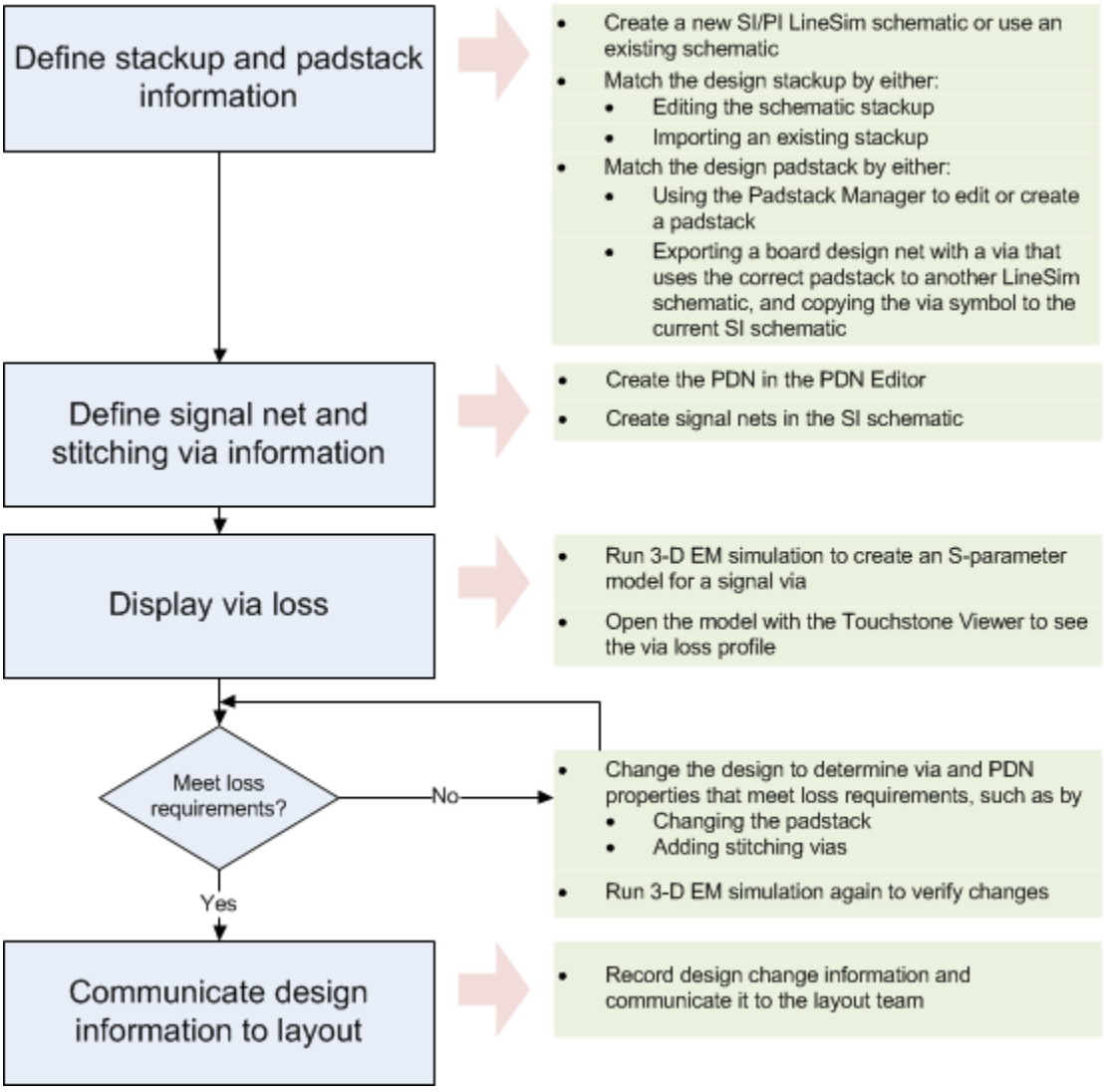
Exporting a Net from BoardSim to LineSim

Via Visualizer Dialog Box

Running Signal-Via Bypass Simulation

Designing Vias to Meet Loss Requirements

Determine the signal via properties that produce acceptable loss. You can determine the location and number of stitching vias needed to provide low-impedance return current paths for signal vias. This work flow uses 3-D electromagnetic simulation.



Related Topics

Creating a Schematic-PDN Design

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Exporting and Importing a Stackup

Editing a Padstack Definition

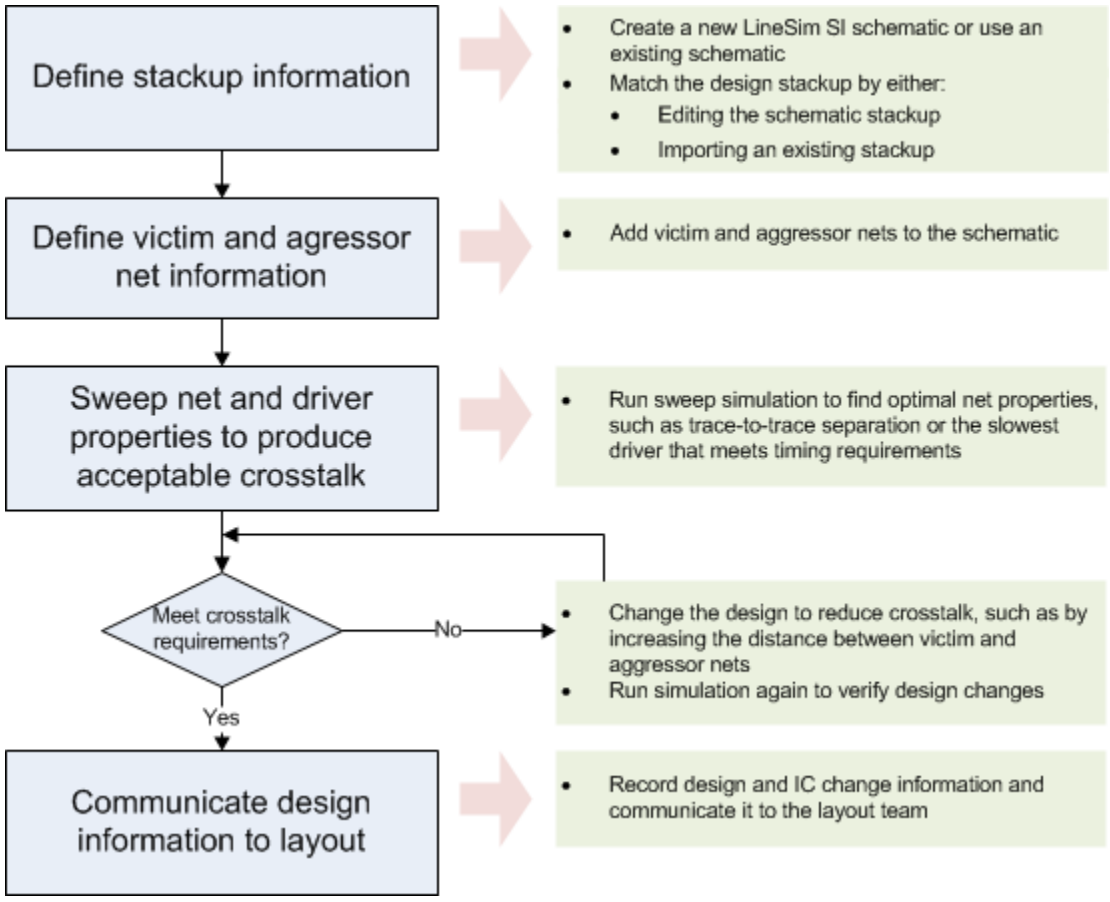
Exporting a Net from BoardSim to LineSim

Modeling a Via with a 3D EM Model in a Schematic

Viewing and Converting Touchstone and Fitted-Poles Models

Designing Net Topologies to Meet Crosstalk Requirements

Determine net properties that meet crosstalk requirements. You can optimize design tradeoffs and see the effects of manufacturing tolerances that affect crosstalk. You can run sweeps to generate constraints for the post-layout design.



Related Topics

Creating a Schematic-PDN Design

Exporting and Importing a Stackup

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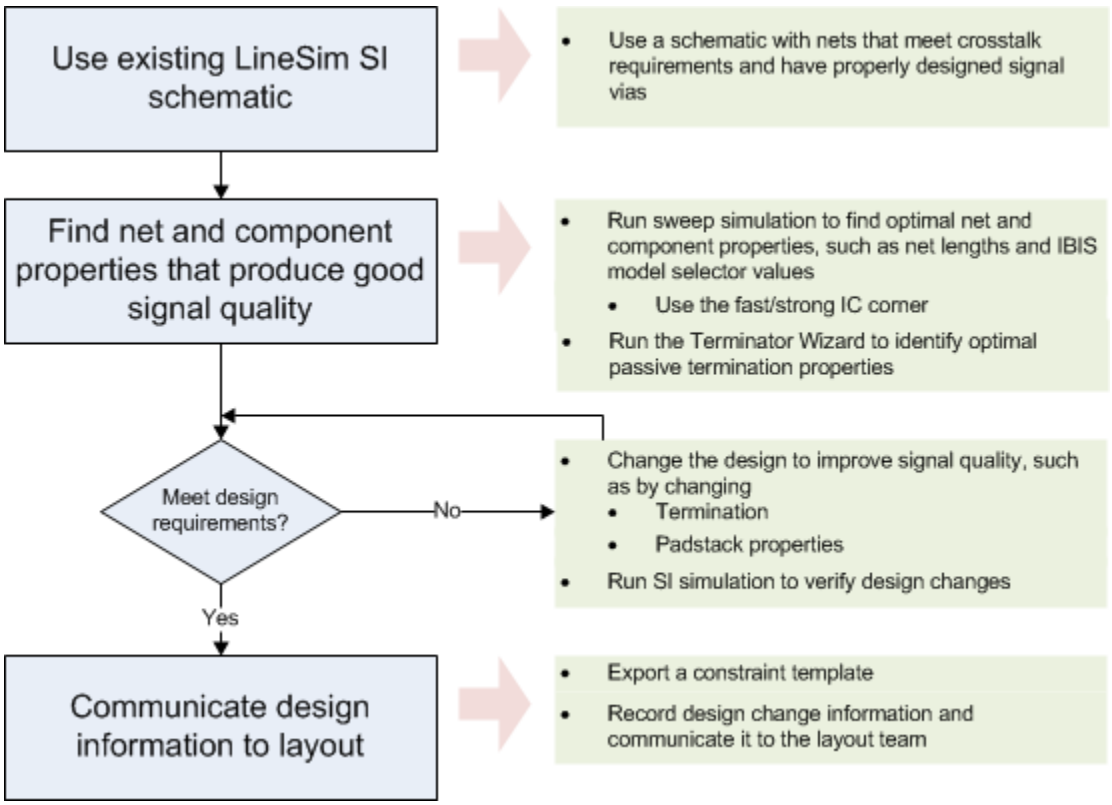
Running Signal Integrity Simulation

Designing Net Topologies to Meet Signal Quality Requirements

Determine net properties and terminations that meet signal quality requirements. You can optimize design tradeoffs and see the effects of manufacturing tolerances that affect signal quality. You can run sweeps to generate constraints for the post-layout design.

Note

Before you change a design to meet signal-quality requirements, net topologies must meet crosstalk requirements and net padstacks must meet impedance requirements.



Related Topics

Opening a Design

Running Signal Integrity Simulation

Running the Terminator Wizard

Creating a Schematic-PDN Design

Padstack Editor Dialog Box

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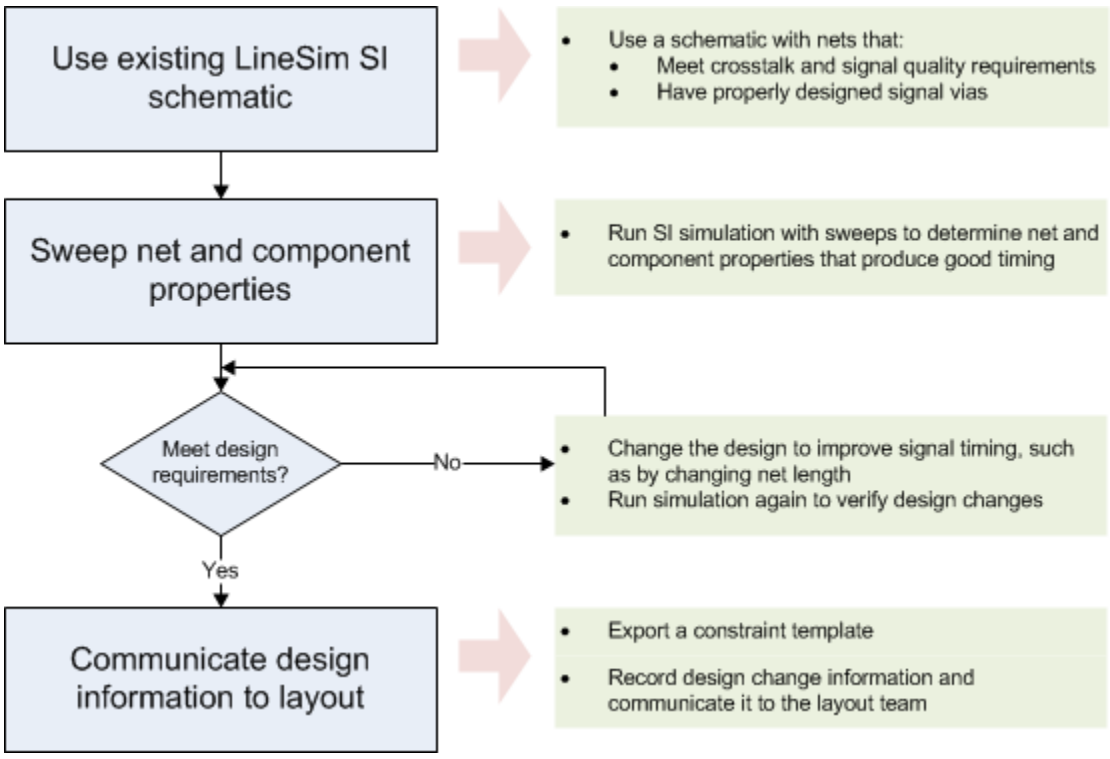
Exporting a Constraint Template From LineSim

Designing Net Topologies to Meet Timing Requirements

Determine net properties that meet timing requirements. You can optimize design tradeoffs and see the effects of manufacturing tolerances that affect signal timing. You can run SI simulation with sweeps to generate constraints for the post-layout design.

Note

Before you change a design to meet timing requirements, net topologies must meet crosstalk and signal-quality requirements.



Related Topics

Opening a Design

Running Signal Integrity Simulation

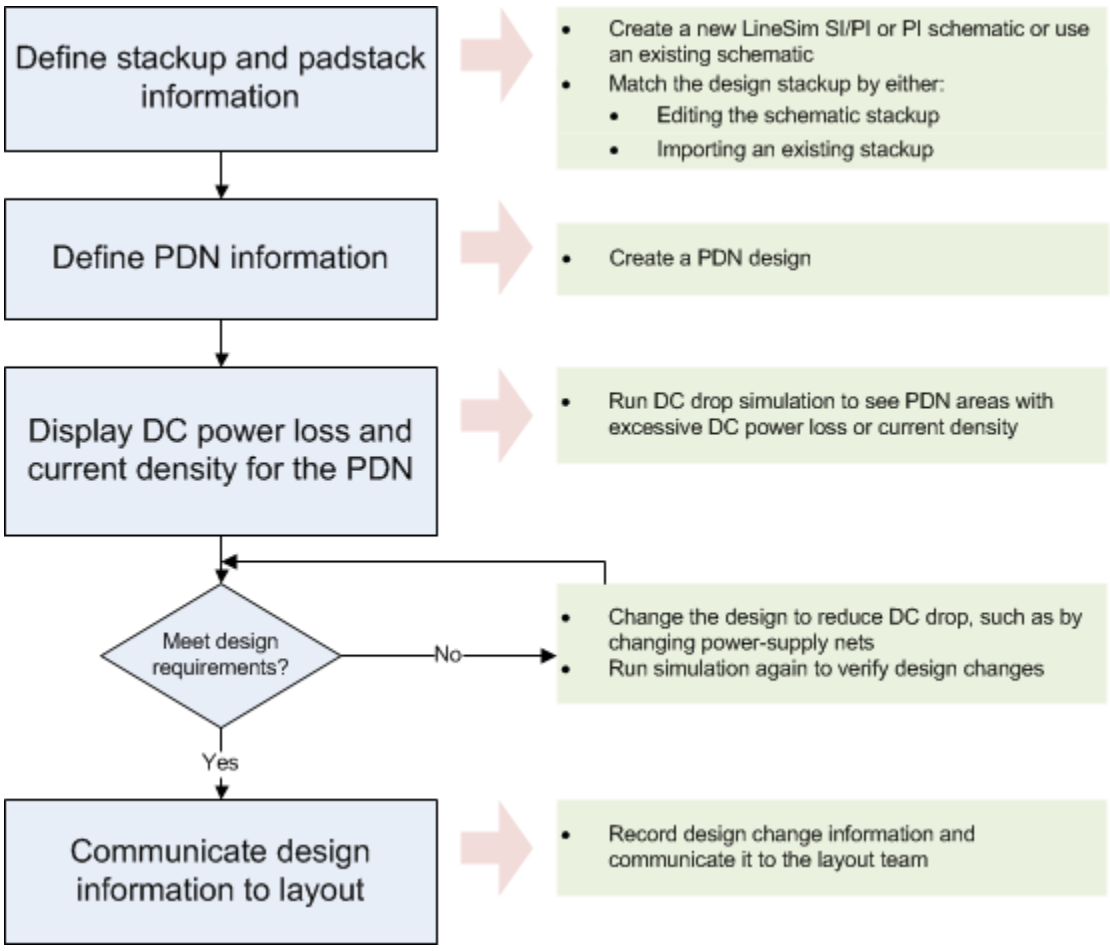
Creating a Schematic-PDN Design

Exporting a Constraint Template From LineSim

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Designing PDNs to Meet DC Power Loss and Current Density Requirements

Determine power-supply net geometries and stitching via quantities and locations to design power distribution networks (PDNs) with acceptable DC power loss and current density.



Related Topics

Creating a Schematic-PDN Design

Exporting and Importing a Stackup

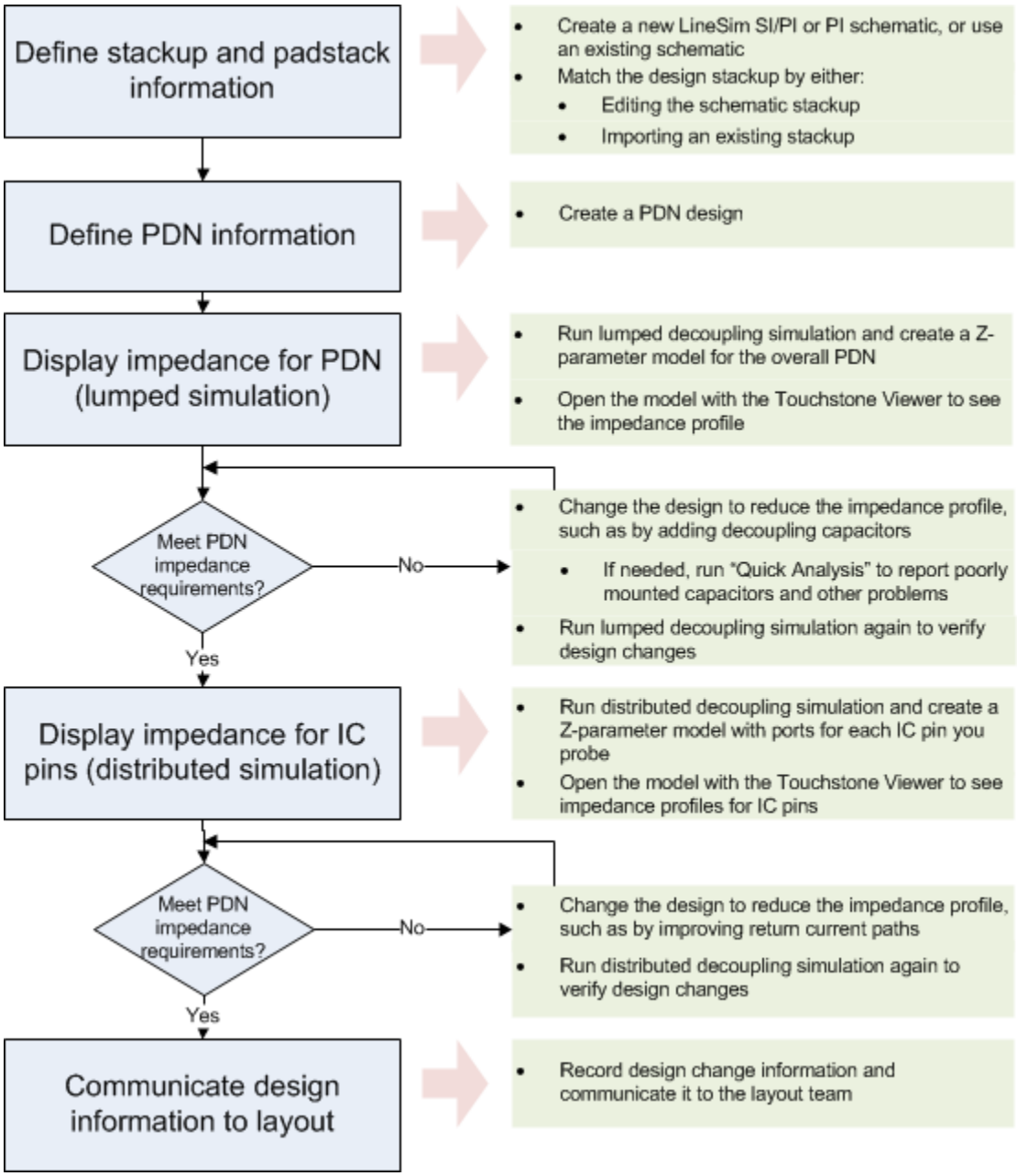
Running DC Drop Simulation

Designing PDNs to Meet Low Impedance Requirements Across a Range of Frequencies

Determine power-supply net geometries and decoupling capacitor quantities and locations to design power distribution networks (PDNs) with acceptable impedance across a frequency

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range. Evaluate decoupling capacitor mounting technologies, such as via-in-pad, microvias, and X2Y capacitors. Evaluate dielectric properties for embedded capacitors, such as a C-ply material or ultra-thin thickness.



Related Topics

Creating a Schematic-PDN Design

Exporting and Importing a Stackup

Decoupling Simulation

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Viewing and Converting Touchstone and Fitted-Poles Models

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Find Post-Layout Design Problems

Deciding on the types of simulation to run on your post-layout design depends on the design behaviors that you want to evaluate. Simulation can help you find nets with poor signal integrity and design power-distribution networks (PDNs) with poor power integrity. Simulating "what if" design variations can help you determine design changes that improve SI or PI.

Goal	Description
Finding Nets With Excessive Crosstalk	Find victim nets with excessive coupling to aggressor nets. You can screen an entire design for problems, simulate a group of critical nets, or verify that design revisions have not introduced problems on critical nets.
Finding Nets With Poor Signal Quality	Find nets with excessive overshoot, ringback, non-monotonicity, and so on. For DDRx interfaces, you can screen the entire interface for signal quality problems. For other interfaces, you can screen an entire design for problems, simulate a group of critical nets, or verify that design revisions have not introduced problems on critical nets.
Finding Nets With Incorrect Timing	For DDRx interfaces, you can screen the entire interface for timing problems. For other interfaces, find nets with wrong flight times.
Measuring Crosstalk Between Signal Nets	Measure the crosstalk on a victim net caused by coupling from switching aggressor nets.
Measuring Signal Quality Characteristics for Nets	Measure signal quality for one or more selected nets.
Measuring Timing for Signal Nets	Measure the flight time for one or more selected nets. Identify nets with unsatisfactory timing and use simulation results to fill out a timing budget spreadsheet.
Measuring Impedance for Trace Segments and Signal Vias	Measure impedance for trace segments and signal vias. This work flow uses default via models and not S-parameter models created by 3-D electromagnetic simulation.
Evaluating the Electrical Behavior of Interconnect and Signal Vias in the Frequency Domain	Measure return loss (reflections) and insertion loss (transmission) for signal nets. You can evaluate interconnect behaviors that are represented by S-parameters.
Measuring DC Power Loss and Current Density	Find metal areas with high DC power loss. You can find metal areas and stitching vias with excessive current density.

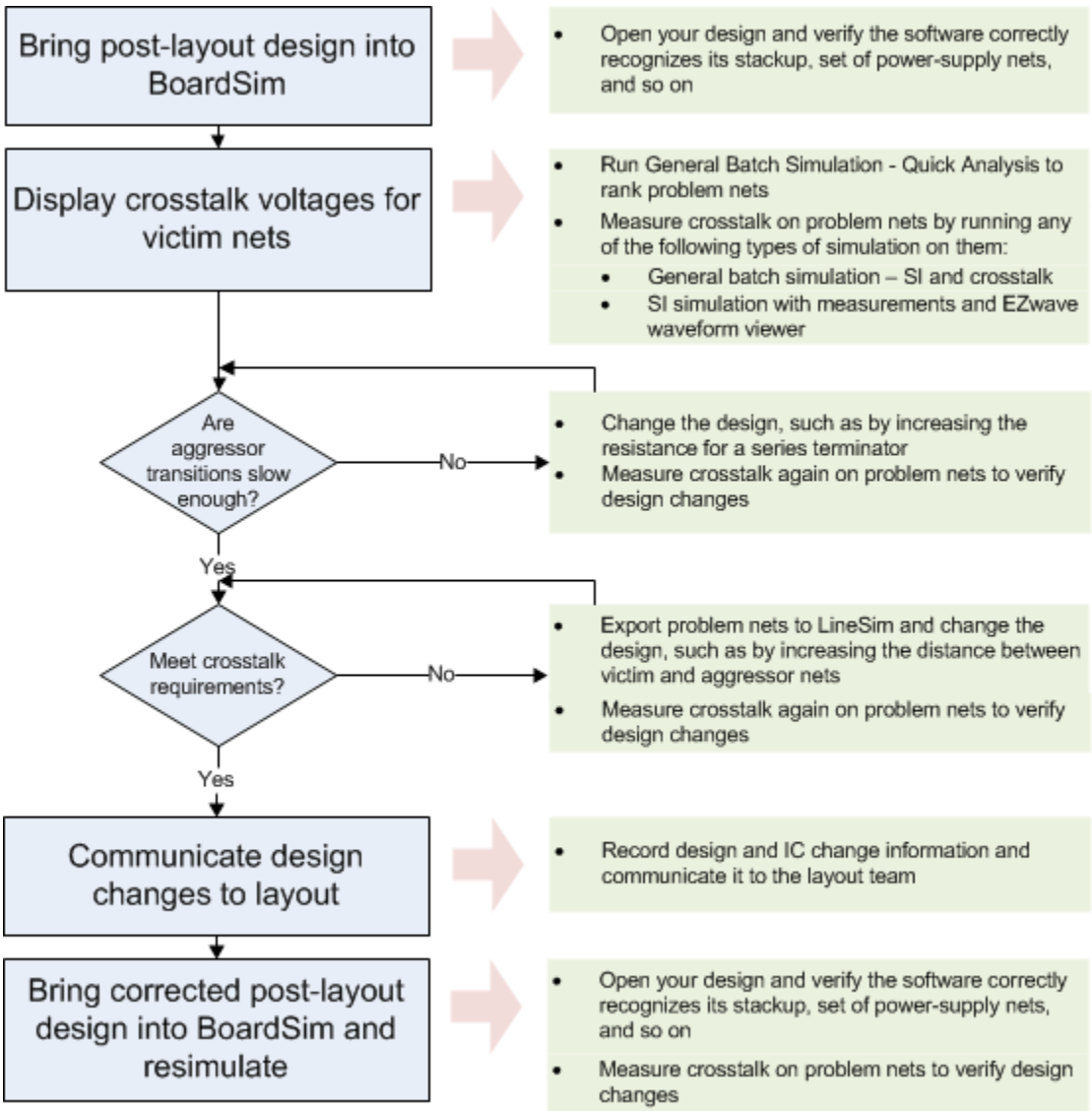
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Goal	Description
Measuring PDN Impedance at Key Locations on the Board	Measure power-distribution network (PDN) impedance over a frequency range. Find the minimum number of capacitors needed to meet the target PDN impedance. Find capacitors that connect to the PDN with highly-inductive mounting. Evaluate decoupling capacitor mounting technologies, such as via-in-pad, microvias, and X2Y capacitors. Evaluate dielectric properties for embedded capacitors, such as a C-ply material or ultra-thin thickness.
Measuring PDN Noise From IC Power Draw (Plane Noise)	Measure plane noise transmitted to the PDN by IC power-supply pins drawing large amounts of transient current. You can find PDN locations that need better decoupling.
Measuring Interaction Between Single-Ended Signal Vias and the PDN	Measure signal integrity with enhanced accuracy by modeling the energy transferred from a signal via to the PDN as a signal propagates through the signal via and return current flows through the PDN. Simulation takes into account via impedance and via-to-via noise coupling.
Measuring Bypass Quality for Single-Ended Signal Vias	Find single-ended signal vias with high-impedance return current paths.
Measuring Board Temperature From Component Heating	Measure the ability of the board and components to dissipate heat. This work flow does not take into account the metal heating from current flowing between VRM and DC sink component pins.
Measuring Board Temperature From Component and PDN Heating	Measure the ability of the board and components to dissipate heat. This work flow takes into account the metal heating from current flowing between VRM and DC sink component pins.

Finding Nets With Excessive Crosstalk

Find victim nets with excessive coupling to aggressor nets. You can screen an entire design for problems, simulate a group of critical nets, or verify that design revisions have not introduced problems on critical nets.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.



Related Topics

Opening and Verifying a Design

Running General Batch Simulation

Exporting a Net from BoardSim to LineSim

Finding Nets With Poor Signal Quality

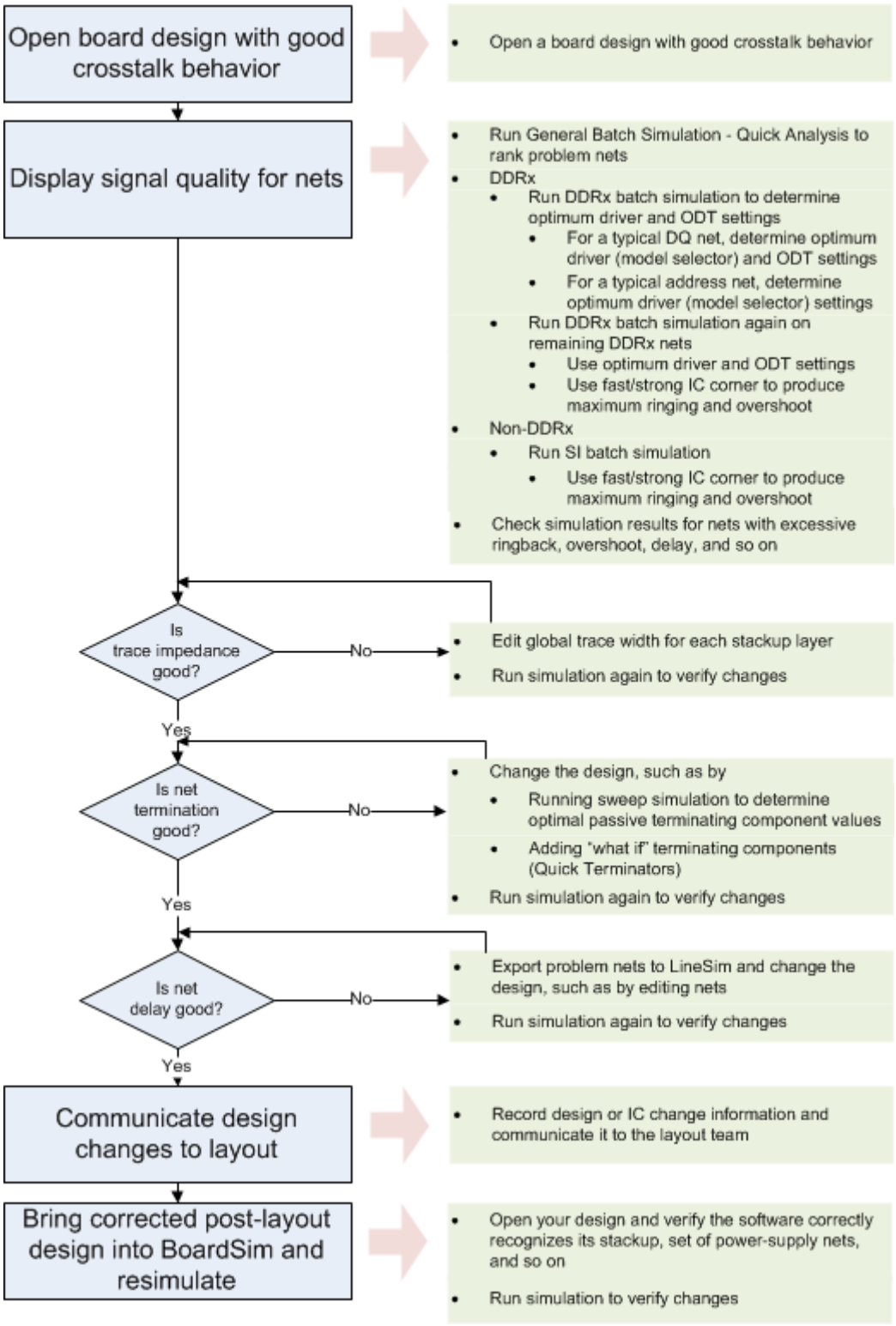
Find nets with excessive overshoot, ringback, non-monotonicity, and so on. For DDRx interfaces, you can screen the entire interface for signal quality problems. For other interfaces, you can screen an entire design for problems, simulate a group of critical nets, or verify that design revisions have not introduced problems on critical nets.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

Before you find nets with poor signal quality, net topologies must meet crosstalk requirements.

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Related Topics

Opening and Verifying a Design

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Running General Batch Simulation

Running a DDRx Memory Interface Simulation

Editing Trace Widths

Exporting a Net from BoardSim to LineSim

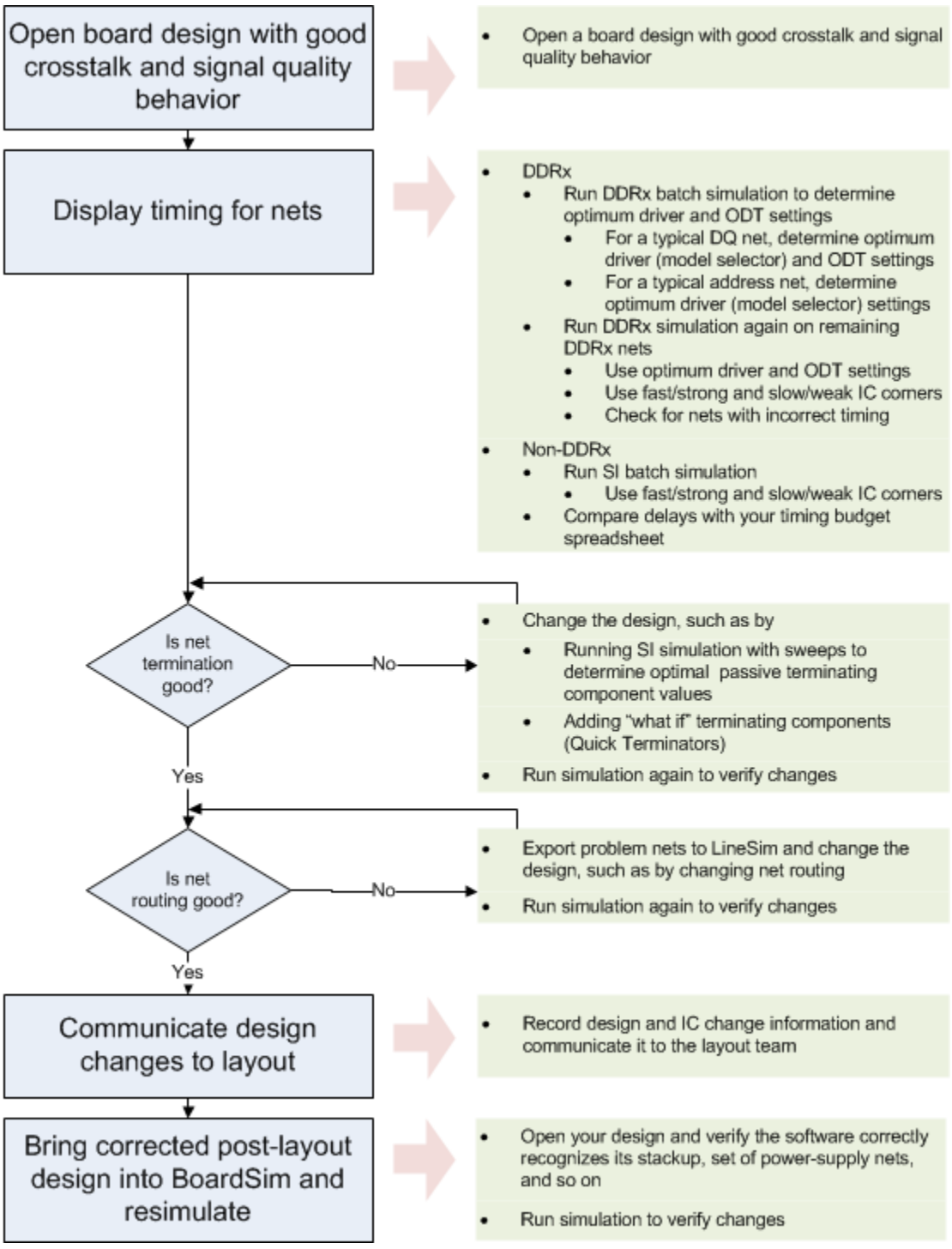
Finding Nets With Incorrect Timing

For DDRx interfaces, you can screen the entire interface for timing problems. For other interfaces, find nets with wrong flight times.

Note

Before you find nets with incorrect timing, net topologies must meet crosstalk and signal-quality requirements.

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Related Topics

Opening and Verifying a Design

Running a DDRx Memory Interface Simulation

Running Signal Integrity Simulation

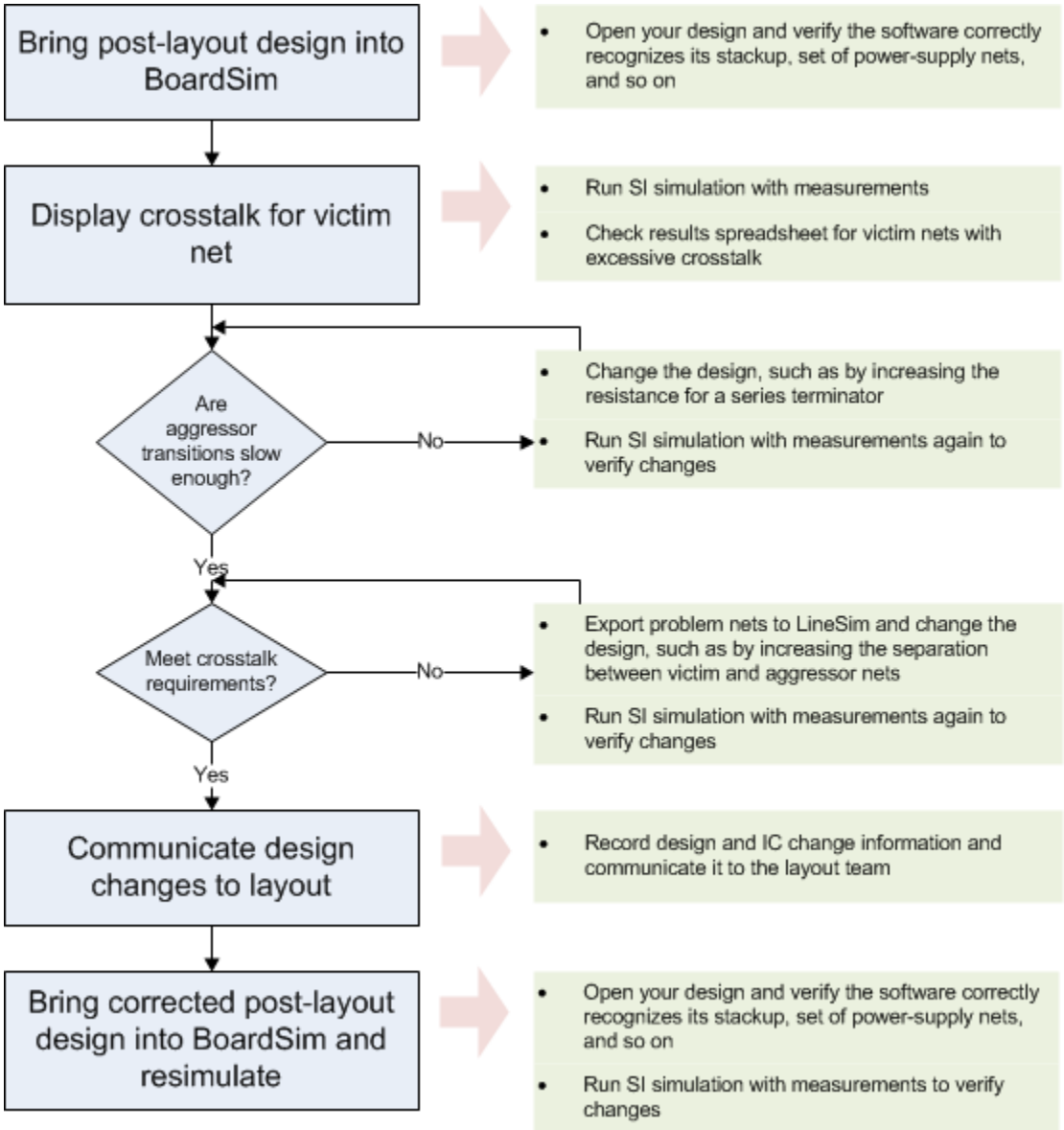
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Adding a Quick Terminator

Exporting a Net from BoardSim to LineSim

Measuring Crosstalk Between Signal Nets

Measure the crosstalk on a victim net caused by coupling from switching aggressor nets.



Related Topics

Opening and Verifying a Design

Running Signal Integrity Simulation with the EZwave Waveform Viewer

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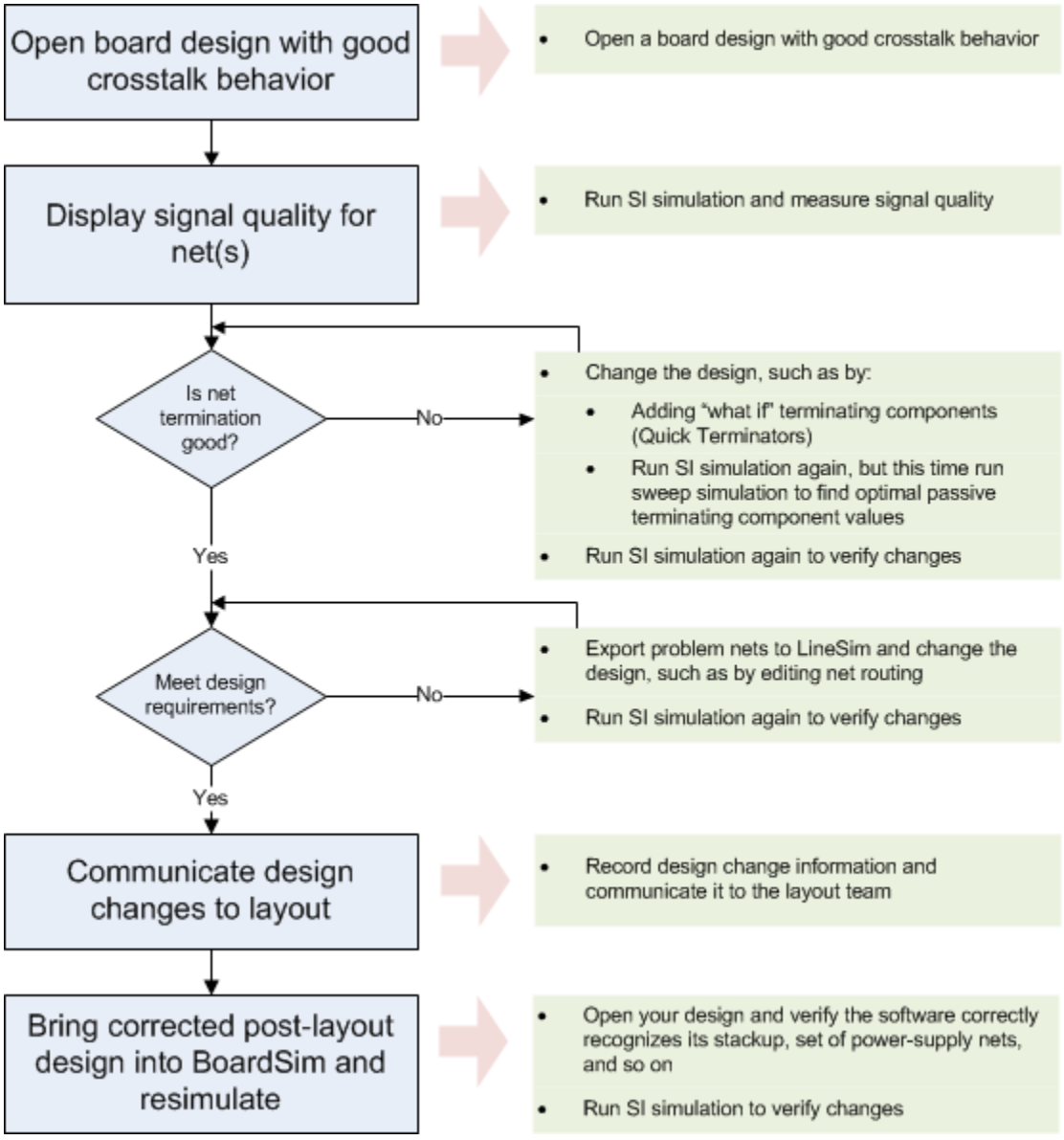
Exporting a Net from BoardSim to LineSim

Measuring Signal Quality Characteristics for Nets

Measure signal quality for one or more selected nets.

Note

Before you measure signal quality, net topologies must meet crosstalk requirements.



Related Topics

Opening and Verifying a Design

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Running Signal Integrity Simulation

Quick Terminators

Exporting a Net from BoardSim to LineSim

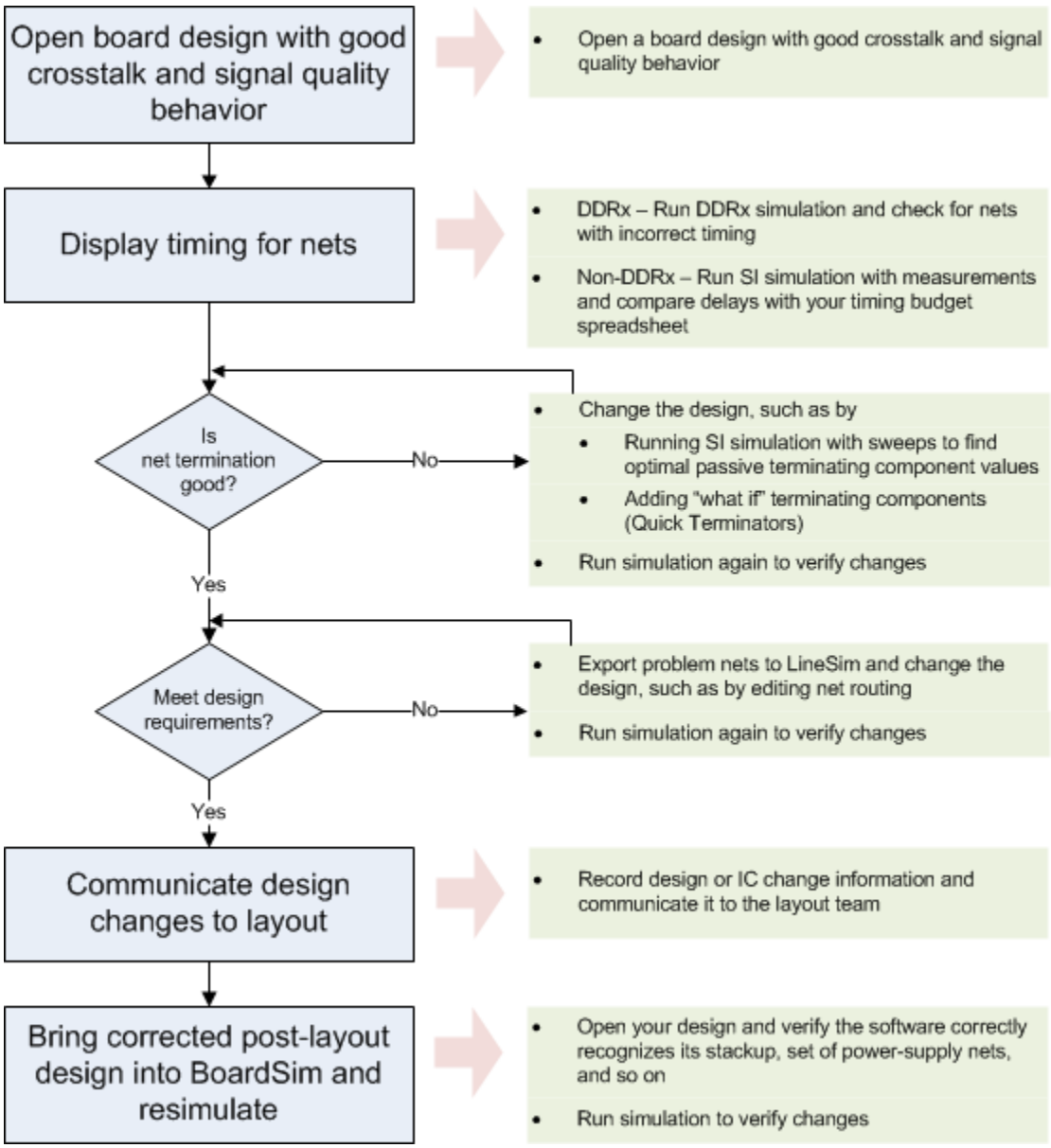
Measuring Timing for Signal Nets

Measure the flight time for one or more selected nets. Identify nets with unsatisfactory timing and use simulation results to fill out a timing budget spreadsheet.

Note

Before you measure timing, net topologies must meet crosstalk and signal-quality requirements.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.



Related Topics

Opening and Verifying a Design

Running a DDRx Memory Interface Simulation

Running Signal Integrity Simulation

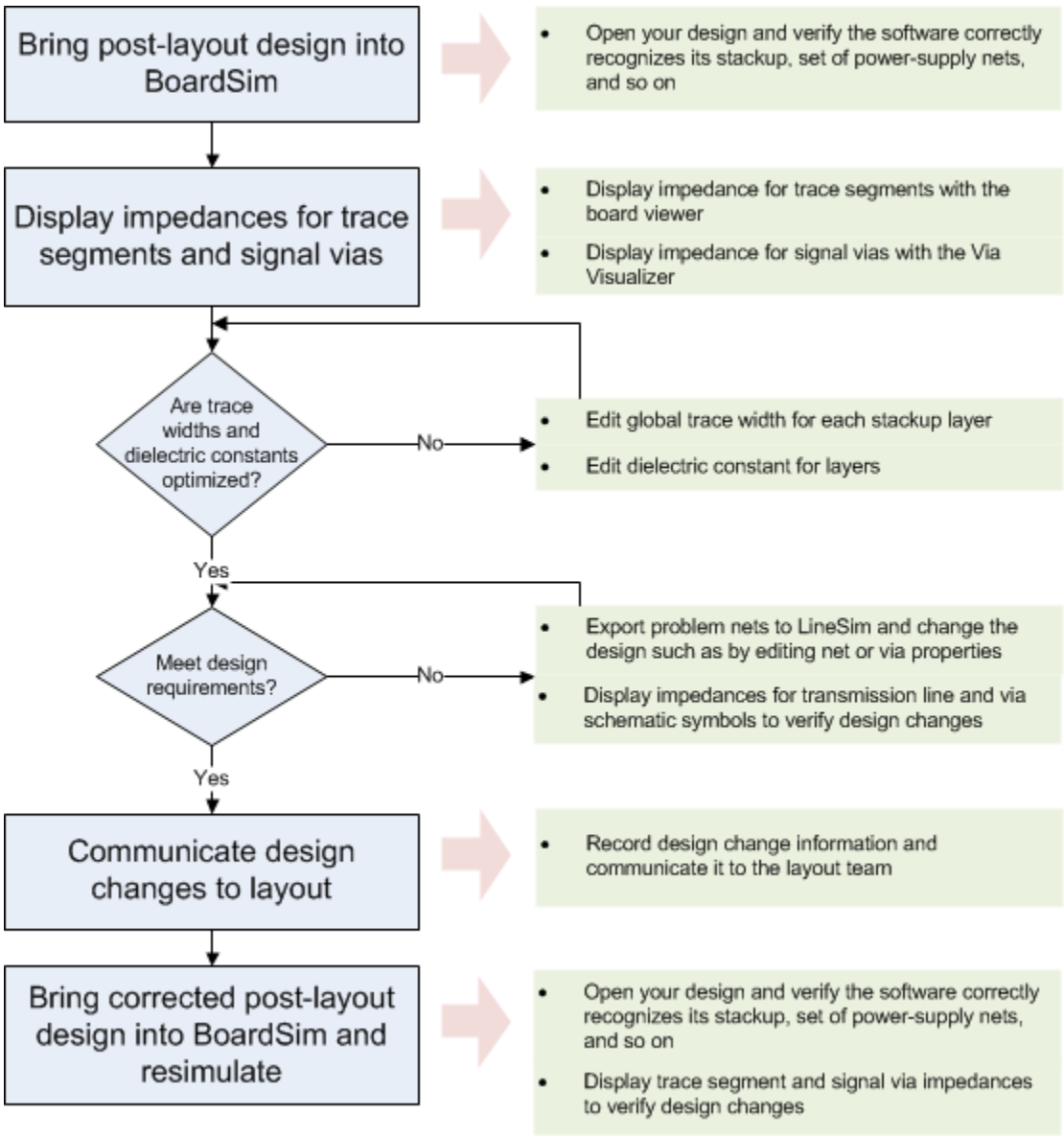
Quick Terminators

Exporting a Net from BoardSim to LineSim

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Measuring Impedance for Trace Segments and Signal Vias

Measure impedance for trace segments and signal vias. This work flow uses default via models and not S-parameter models created by 3-D electromagnetic simulation.



Related Topics

Opening and Verifying a Design

Viewing Net Segment Properties

Via Visualizer Dialog Box

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

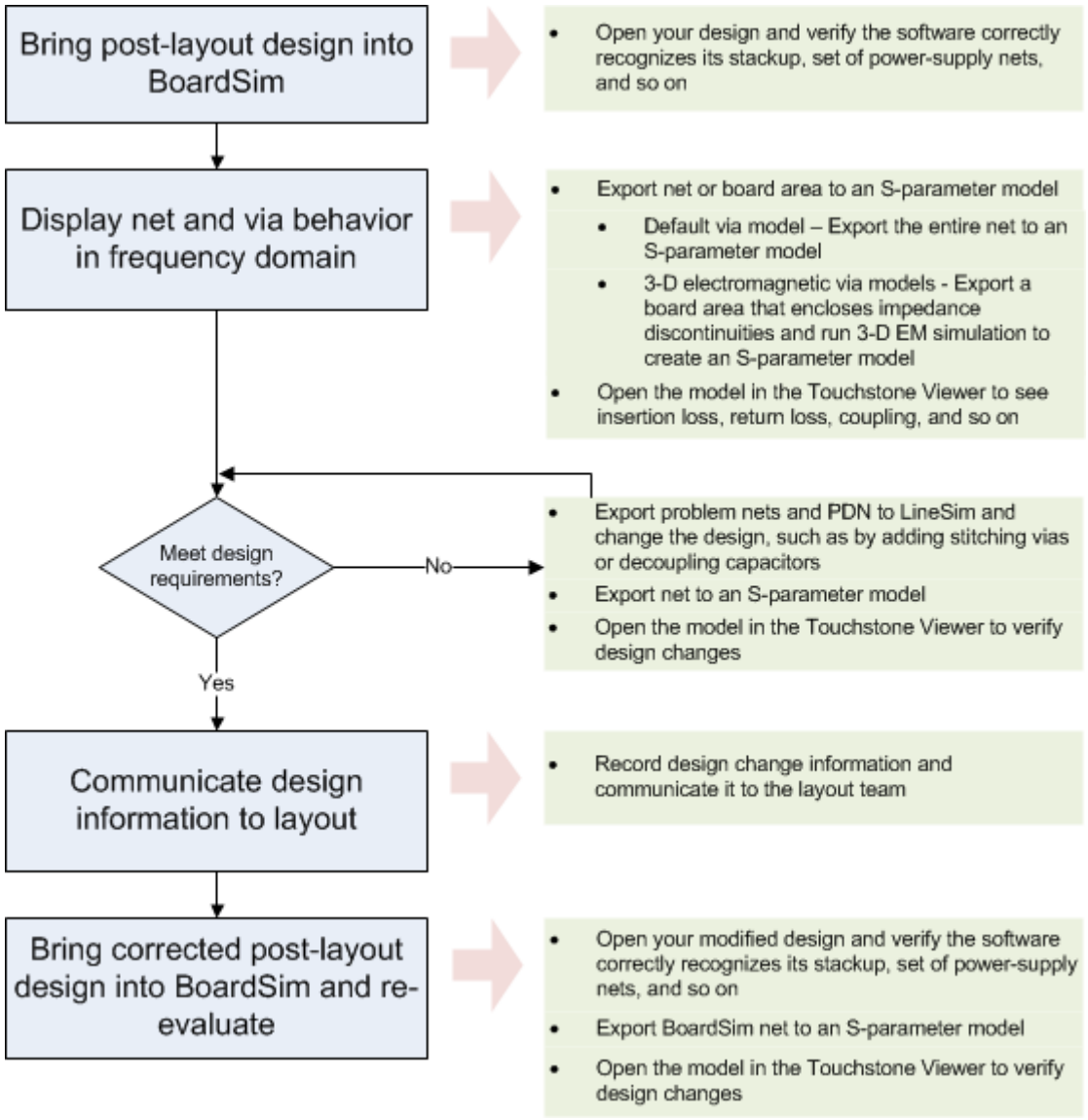
Editing Trace Widths

Defining the Basic Stackup

Exporting a Net from BoardSim to LineSim

Evaluating the Electrical Behavior of Interconnect and Signal Vias in the Frequency Domain

Measure return loss (reflections) and insertion loss (transmission) for signal nets. You can evaluate interconnect behaviors that are represented by S-parameters.



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Opening and Verifying a Design

Exporting a Net to an S-Parameter Model

Exporting Part of a Board Design for Analysis in HyperLynx Full-Wave Solver

Viewing and Measuring Model Curves

Exporting a Net from BoardSim to LineSim

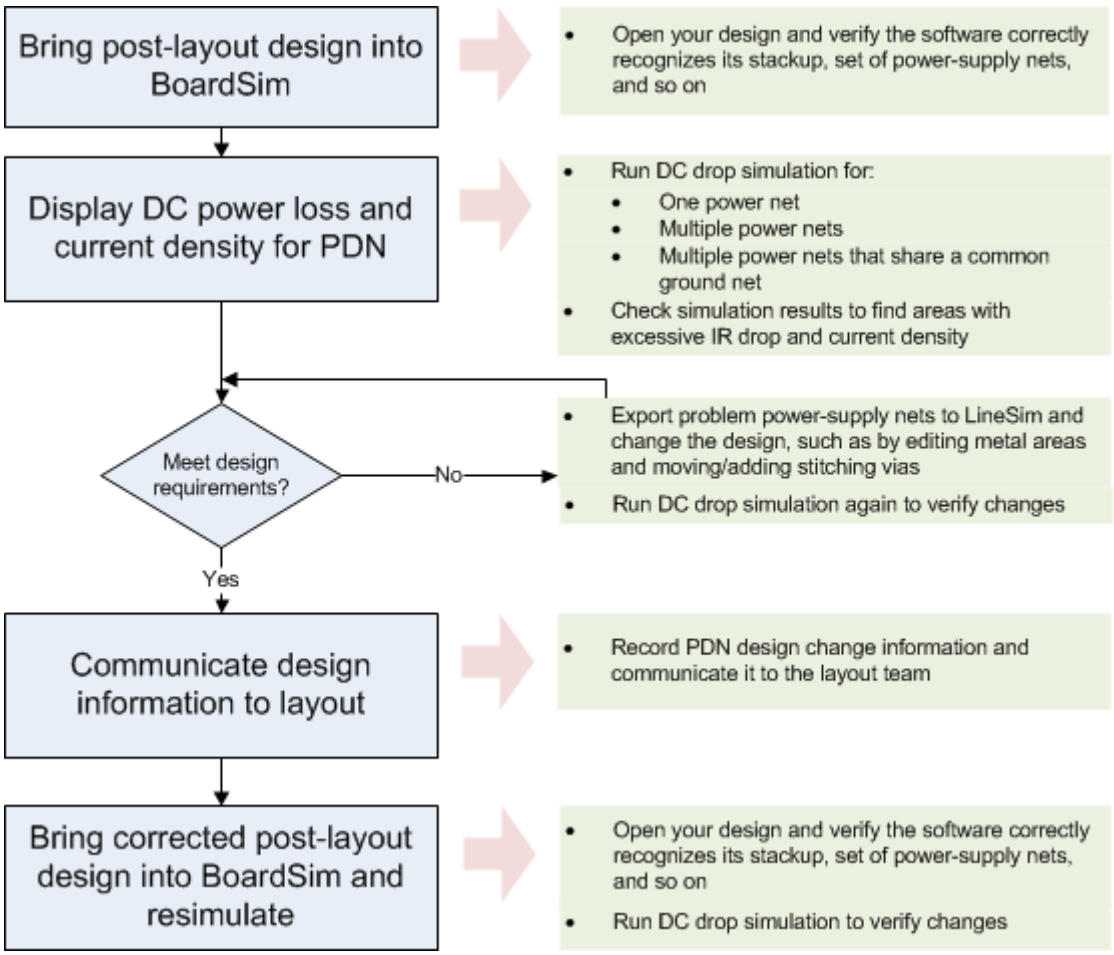
Measuring DC Power Loss and Current Density

Find metal areas with high DC power loss. You can find metal areas and stitching vias with excessive current density.

Note

You can also simulate DC drop for a design loaded in Xpedition® Layout or another Siemens EDA layout tool with an Analysis Control capability that supports the DC Drop client. See “Running DC Drop Simulation From Layout Software” on page 211.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.



Related Topics

Opening and Verifying a Design

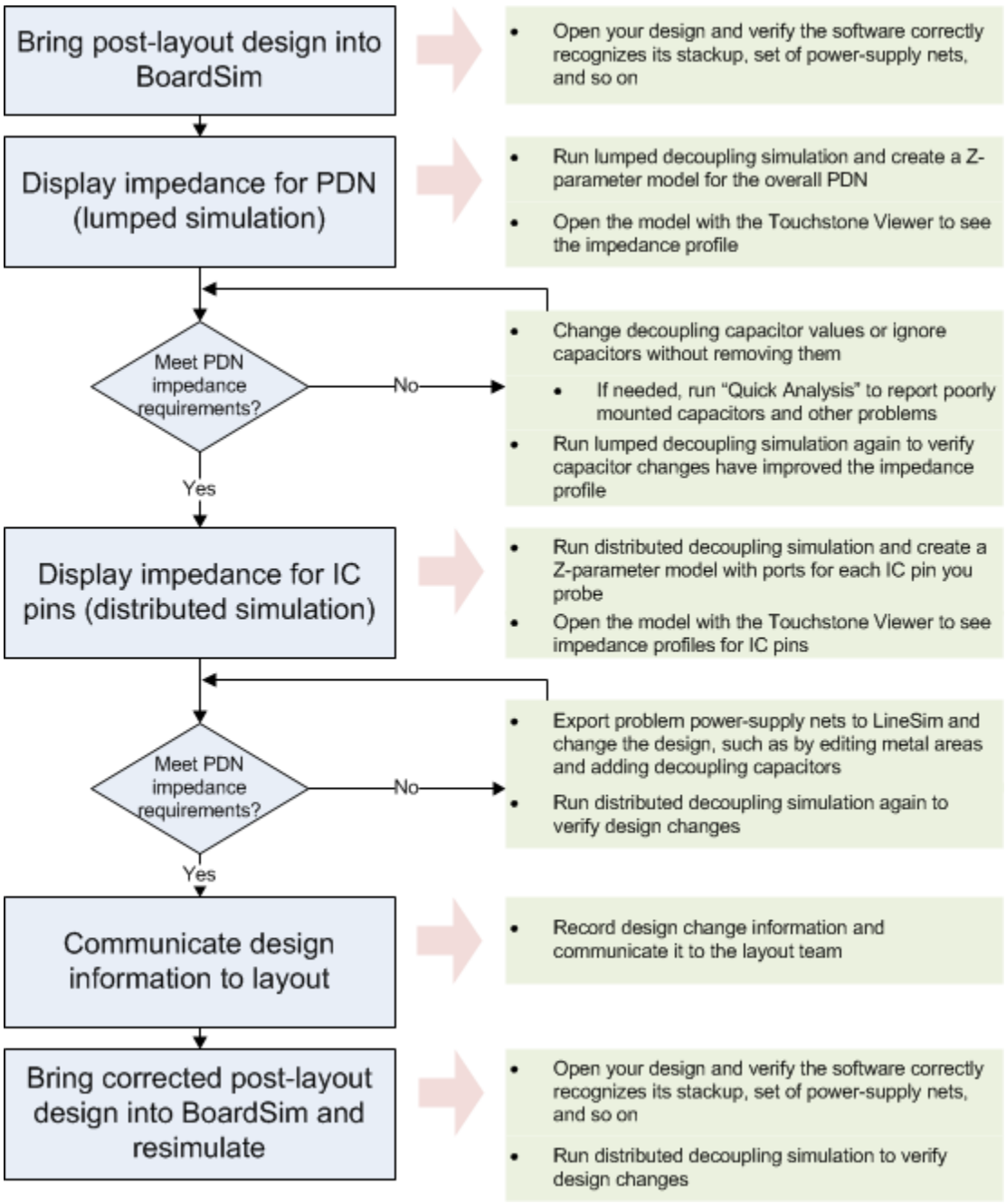
DC Drop Simulation

Exporting a Net from BoardSim to LineSim

Measuring PDN Impedance at Key Locations on the Board

Measure power-distribution network (PDN) impedance over a frequency range. Find the minimum number of capacitors needed to meet the target PDN impedance. Find capacitors that connect to the PDN with highly-inductive mounting. Evaluate decoupling capacitor mounting technologies, such as via-in-pad, microvias, and X2Y capacitors. Evaluate dielectric properties for embedded capacitors, such as a C-ply material or ultra-thin thickness.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.



Related Topics

Opening and Verifying a Design

Decoupling Simulation

Viewing and Measuring Model Curves

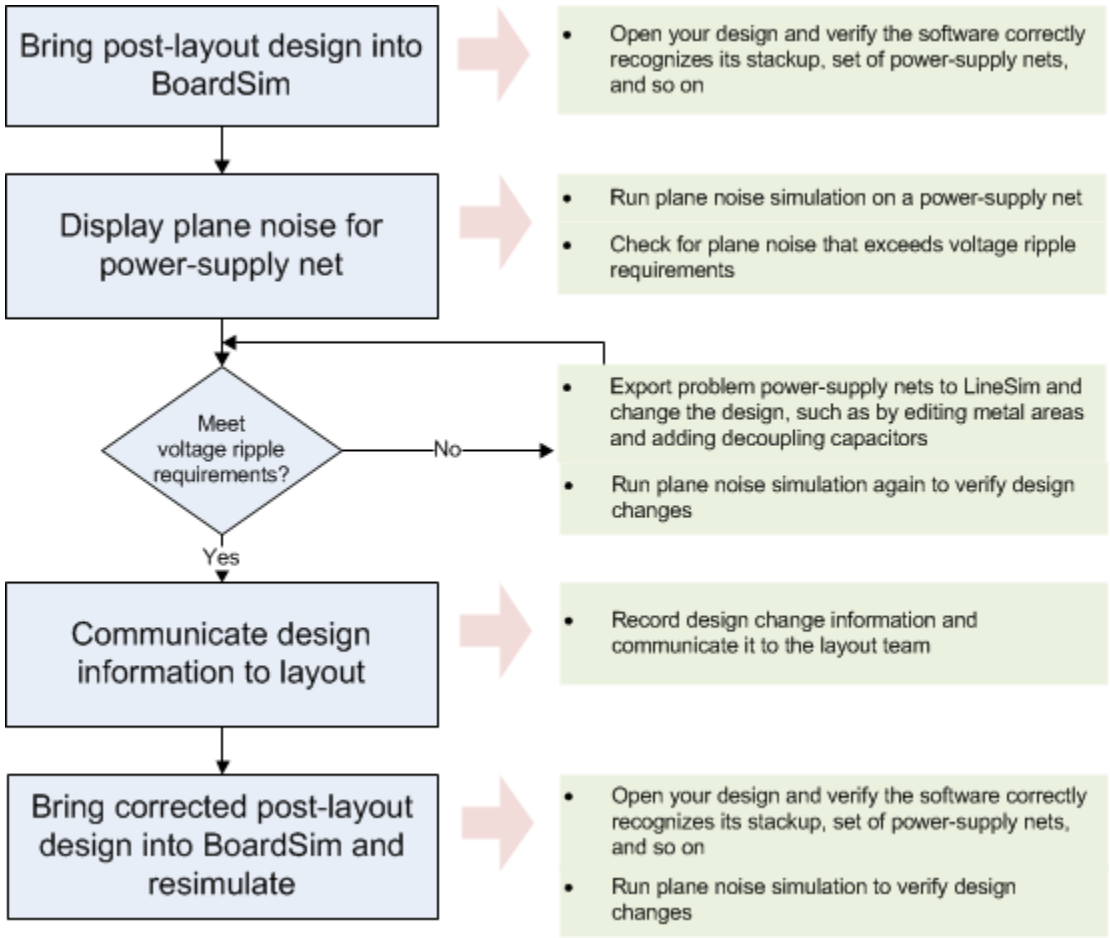
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Measuring PDN Noise From IC Power Draw (Plane Noise)

Measure plane noise transmitted to the PDN by IC power-supply pins drawing large amounts of transient current. You can find PDN locations that need better decoupling.

Prerequisites

- PDN has an optimum impedance profile. If the PDN impedance is too high, the simulated plane-noise voltages may also be too high and exceed the voltage ripple requirements.



Related Topics

Opening and Verifying a Design

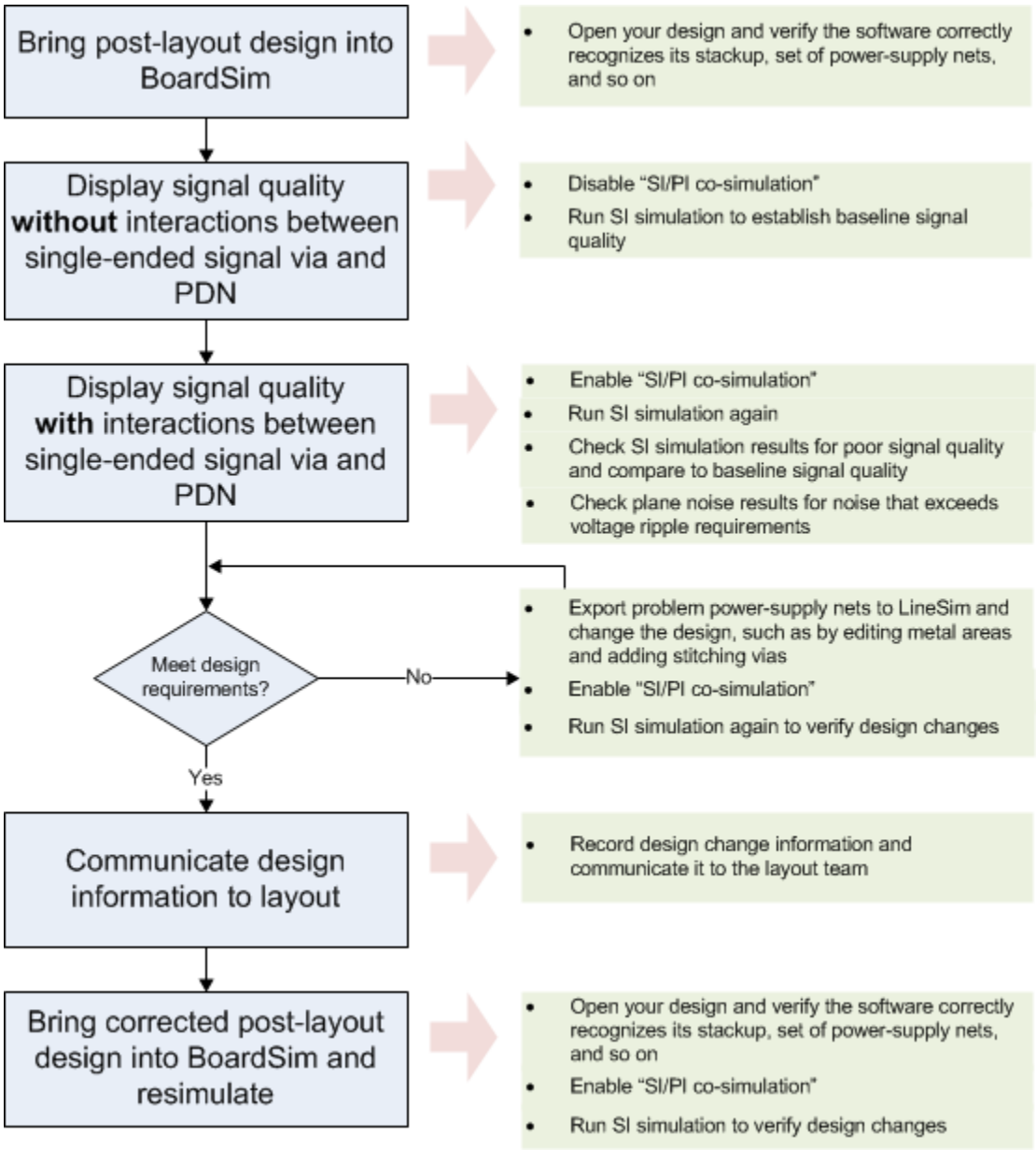
Running Plane Noise Simulation

Exporting a Net from BoardSim to LineSim

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Measuring Interaction Between Single-Ended Signal Vias and the PDN

Measure signal integrity with enhanced accuracy by modeling the energy transferred from a signal via to the PDN as a signal propagates through the signal via and return current flows through the PDN. Simulation takes into account via impedance and via-to-via noise coupling.



Related Topics

Opening and Verifying a Design

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Accounting for Noise Between Single-Ended Signal Via and Power Planes in SI Simulation (Co-Simulation)

Running Signal Integrity Simulation

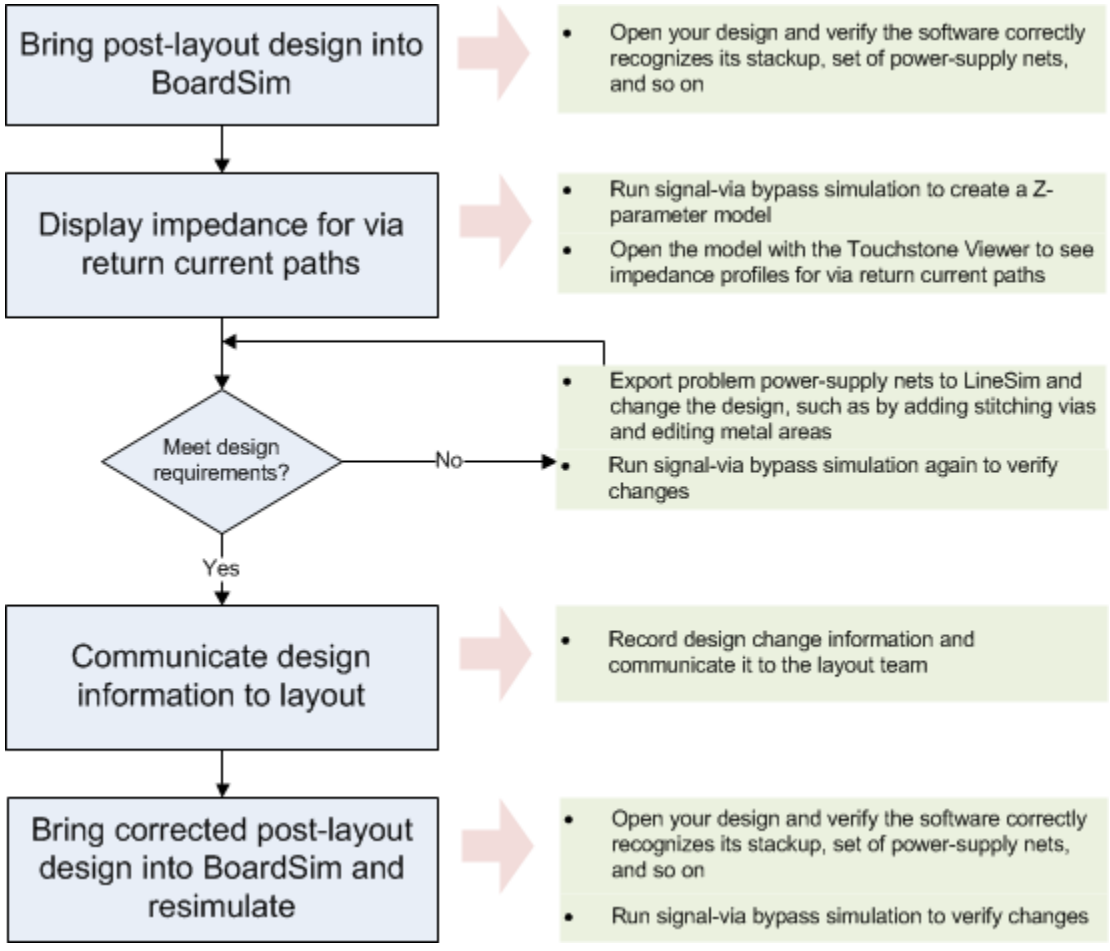
Plane-Noise Simulation Results

Measuring Bypass Quality for Single-Ended Signal Vias

Find single-ended signal vias with high-impedance return current paths.

Note

Bypass simulation measures one signal via at a time.



Related Topics

Opening and Verifying a Design

Running Signal-Via Bypass Simulation

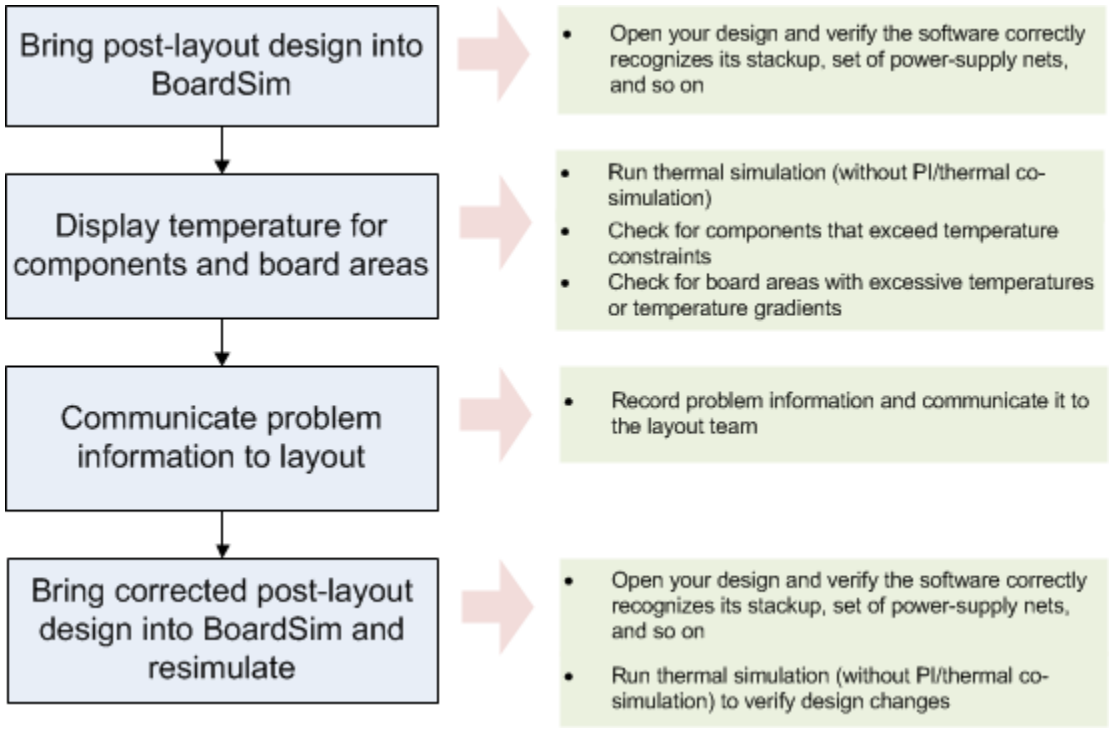
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Viewing and Measuring Model Curves

Exporting a Net from BoardSim to LineSim

Measuring Board Temperature From Component Heating

Measure the ability of the board and components to dissipate heat. This work flow does not take into account the metal heating from current flowing between VRM and DC sink component pins.



Related Topics

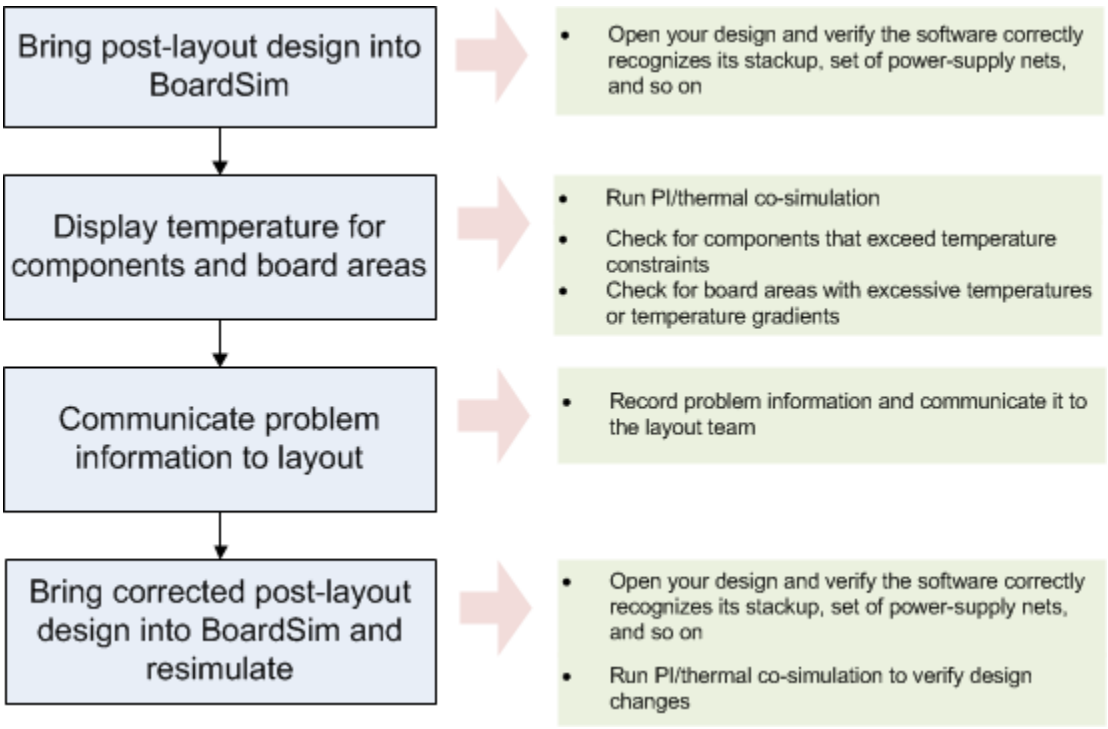
Opening and Verifying a Design

Running a Thermal Simulation [HyperLynx Thermal User Guide]

Measuring Board Temperature From Component and PDN Heating

Measure the ability of the board and components to dissipate heat. This work flow takes into account the metal heating from current flowing between VRM and DC sink component pins.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.



Related Topics

Opening and Verifying a Design

Running Batch DC Drop Simulation or Thermal Co-Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Chapter 2
Opening and Verifying a Design

You can export your design from your layout or schematic design software, or translate the design to a format that HyperLynx can read, if needed. Open your design and verify that the software recognizes it correctly before running simulation.

Topic	Description
Opening a Design	Open a board design file in *.cce*, ODB++, IPC-2581A, or *.hyp* format, or a schematic design in *.ffs* format. Open a multiple-board design from a *.pjh* file.
Setting Up a Multiple Board Design	If your design includes multiple boards, you can create a MultiBoard project to define connections between the boards, model electrical characteristics of the connectors, and simulate nets that span more than one board.
Verifying That the Software Recognizes Your Design Correctly	After you open a board, verify that the software recognizes your design correctly.
Defining Net Classes	You can categorize a group of nets as a net class. This enables you to select a group of nets at once (by net class) or edit settings for a group of nets when the software provides this option.
Creating a Schematic-PDN Design	You can simulate your design before determining the final layout by creating a schematic-PDN. To run SI simulation, create a schematic in the Free-Form Schematic Editor. To run PI simulation, use the PDN Editor to create the power-distribution network (PDN) layout.

Opening a Design

Open a board design file in *.cce*, ODB++, IPC-2581A, or *.hyp* format, or a schematic design in *.ffs* format. Open a multiple-board design from a *.pjh* file.

In HyperLynx, the board design environment is called BoardSim®. The schematic design environment is called LineSim®.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Siemens EDA layout software can export board designs that HyperLynx can read. To load a board design created in another tool, export your design to text files, and use HyperLynx to translate the design to *.hyp* or ODB++ format.

These Siemens EDA software tools can create a *.cce* or *.hyp* board design file:

- Siemens EDA Xpedition® Layout, Board Station® XE, PADS® Layout.

These software tools can create a *.ffs* schematic design file:

- Siemens EDA Xpedition Designer, Constraint Manager.

HyperLynx can translate board designs exported from these software tools:

- Accel EDA (Not supported for power integrity simulation.)
- Cadence Allegro
- Siemens EDA Board Station or Board Station RE
- Spectra DSN (Not supported for power integrity simulation.)
- Zuken Visula/CADStar for Windows (Not supported for power integrity simulation.)
- Zuken CR-3000 (Not supported for power integrity simulation.)
- Zuken CR-5000

Prerequisites

- You have configured software preferences, license check in/out options, and specified folder locations for design, model, and stimulus files. See Setting Up the Software.

Procedure

1. To load a board or schematic, choose **File**, then:
   - **Open Board**
   - **Open ODB++** (ODB++ design folder)
   - **Open Multiboard Project**
   - **Open Schematic**
2. Depending on your design, the following dialog boxes can appear:

Dialog box	Reason
Restore Session Edits	Allows you to select and load the most recent session edit information or session edits from a previous backup.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Dialog box	Reason
Illegal Single-Pin Components Found	The design contains single-pin components that are not recognized as ICs or test points. Specify if you want the software to convert them to test points.
Connect Nets With Manhattan Routing	The design contains fully unrouted nets. Specify if you want the software to route them with Manhattan routing.
Stackup Verifier	The design is missing stackup information or specifies a metal layer type that does not match the recommended type. Use the dialog box to fix the problem.
Select the Instance	The board you are loading is one of several instances in a MultiBoard project. Choose which instance to load.

Related Topics

Translating a Board Design

Viewing a Board in 2-D

Setting Up a Multiple Board Design

Board Designs With Multiple Stackups

Saving Session Edits

Restore Session Edits Dialog Box

Connect Nets with Manhattan Routing Dialog Box

Setting Up a Multiple Board Design

If your design includes multiple boards, you can create a MultiBoard project to define connections between the boards, model electrical characteristics of the connectors, and simulate nets that span more than one board.

Use the MultiBoard Wizard to set up a MultiBoard project file (PJH) that contains board names, board-to-board interconnection mapping, and electrical characteristics for the interconnections.

The following video contains additional information. Note that the video shows an old Connection Editor button behavior. Please refer to the procedure for current behavior.



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Restrictions and Limitations

- The sum of metal stackup layers used by all boards in a MultiBoard project cannot exceed 252. The software does not limit the number of dielectric stackup layers.

  If your MultiBoard project includes a specific board file multiple times, the software counts the metal stackup layers for only one instance of that board. For example, if your MultiBoard project contains four instances of a memory module file (such as *DIMM.cce*) that has five metal stackup layers, the software counts five metal stackup layers, not 20.

Prerequisites

- You have a MultiBoard license.
- If you intend to use SPICE models to model interconnects, you have an Advanced MultiBoard license.
- If you intend to use Touchstone models to model interconnects, you have the GHz license bundle.
- If you want to insert ODB++ data that is uncompressed, you must first convert the data into *.TGZ* format.

Procedure

1. Open the MultiBoard Project Wizard in one of the following ways:

If you want to...	Do the following...
Create a new MultiBoard project	**File > New MultiBoard Project** ( ). Enter a new name and click **Browse** to specify the location where the project file is created. **Caution:** Create a unique name for the exported board file that is not used for an existing schematic design (*.FFS*) or MultiBoard project (*.PJH*) file. When you save a *.HYP* file, an associated *.PJH* file is automatically created that can overwrite existing *.PJH* files or those associated with a *.FFS* file. For example, if you have an existing file named *myboard.ffs*, do not export a *.HYP* file named *myboard.hyp* to the same folder.
Edit an existing MultiBoard project	1. **File > Open MultiBoard Project** ( ). Browse to an existing MultiBoard project file (*.PJH*). 2. **Edit > MultiBoard Project**.

2. Click **Insert** to select a file for each board in your design. The file can be any of the following types: *.HYP*, *.CCE*, or ODB++ data in *.TGZ* format. Edit the **Comment** field if you want to describe your board.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

You can create a MultiBoard project that uses multiple instances of a single file type. For example, you can create a MultiBoard project that includes a main board called *mainboard.hyp*, and four identical DIMMs that are represented by a file called *DIMM.hyp*. You can also mix file types within a single MultiBoard project.

If your MultiBoard project includes a board file multiple times, the software counts the metal stackup layers in only file instance.

3. Click **Next**.
4. Define the connections between your boards by selecting the reference designators for each board, and click **Insert**.
5. If the connector on one board has a different number of pins than the connector on the other board, or if connector pins do not map one-to-one, edit the pin mapping between connectors.
   a. Insert a connection row for each pin you want to map. For example. a pin on “B00:J1 MAINBOARD” connects to“B01:J1 SDRAM”.
   b. Select a row in the **Left** column and add a period and pin number to the end of the reference designator (before the board ID comment). For example. “B00:J1 MAINBOARD” becomes “B00:J1.1 MAINBOARD”.
   c. In the same row, add a period and the pin number of the connected pin to the end of the reference designator in the **Right** column. For example. “B01:J1.1 SDRAM”.
6. Click **Next**.
7. In the Connector Model area, define connections as a short, or assign a simple or advanced (SPICE or S-parameter) model. Click **Connection Editor** to define, edit, or manage multiple connections using the Visual Connection Editor and the Table Connection Editor.
8. To define a connection, select a model type in the list, and assign or edit models for each interconnection as explained on the wizard page:

If you want to...	Do the following...
Model the interconnection as an electrical short	Select **Short**. This model effectively removes the connector from the circuit. Use this model to see how simulation results are affected by excluding the effects of the interconnect. **Note:** Click **Connection Editor** to view a graphical representation of the connection.

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If you want to...	Do the following...
Model the interconnection by defining simple electrical characteristics (R,L,C, Z, Delay)	Select **Simple** and specify the values for either: • **Resistance**, **Inductance**, **Capacitance** (RLC); or • **Resistance**, **Impedance**, **Delay** (R, Z, Delay). **Note:** Click **Connection Editor** to view a graphical representation of the connection.
Model the interconnection by assigning an S-Parameter or SPICE model	Select **Advanced** then do one of the following: • Click **Assign** to assign a model in the Assign S-Parameter/SPICE Model dialog box. • Click **Edit** to assign a different model. **Note:** Click **Connection Editor** to edit the pin-to-pin mapping for the model in a drag-and-drop graphic interface.

9. To define connections in the Connector Editor, click **Connection Editor**.

   The Table Connection Editor and Visual Connection Editor open. The Table Connection Editor enables you to edit connections in table format. The Visual Connection Editor provides a visual connection reference, and is only editable when you select the advanced model type.

10. If you are working on a new version of a design that you have previously set up in the software and saved a connection mapping file, in the Table Connection Editor, click **Import** to load that file now.

    After you set up connection mapping information for a connector, you can export the mapping information as a CSV file to reuse it again later. See step 15.

11. If you selected Advanced as the model type, click **Add model**.

12. In the Add Connector Model dialog box, select a model type, and select a model from the list.

    To specify the locations of model files, click **Library Path** if needed.

    The software can automatically determine connections and create a *.connect* file that describes port connections. The software creates the *.connect* file in the same location as the model file and names it with a *.connect* extension.

13. Next to Auto-Place Ports, select the port connection numbering pattern that corresponds with your design.

    All left side ports appear in the Left side ports list, and all right side ports appear in the Right side ports list.

14. If you have a model with a non-standard port numbering pattern, or the software does not recognize the port numbering pattern correctly for your design, you can manually

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move ports, or, for an S-parameter model, you can allow the software to automatically detect connections between ports.

If you want to...	Do the following...
To manually move ports	Ctrl-click or Shift-click and to select multiple rows in the list and click an arrow button to move them to the left or right side
To enable the software to automatically detect connections between ports in (S-parameter model only)	1. Click **Model Connections**. 2. In the Model Connections dialog box, click **Auto-detect connections**. 3. Review the list of ports in the Preview connections list. 4. If the connections are correct, click **Save connections**, then click **Close**. If the connections are not correct, adjust the coupling threshold from **Weak** to **Medium** or **Strong** and allow the software to determine connections again.

15. If you want to save the connection information for the connector, click **Export**.
16. Click **Finish**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Verifying That the Software Recognizes Your Design Correctly

After you open a board, verify that the software recognizes your design correctly.

Topic	Description
Verifying That Power Supply and Signal Nets are Recognized Correctly	Ensure that the software recognizes the power supply nets in your design correctly. If it does not, specify them manually. Nets that are not recognized as power supply nets are recognized as signal nets.
Verifying That Component Types are Recognized Correctly	To ensure that the software recognizes components correctly, and your components appear in model assignment lists correctly, edit reference designator mappings.
Verifying That Differential Pairs are Recognized Correctly	You must ensure that the software recognizes differentially paired nets correctly, so it can simulate them correctly. Create net name pairing rules to specify how the software automatically identifies paired nets when a board is loaded. You can also specify differential pairs manually for a design.
Verifying the Stackup Definition	Verify that the stackup definition accurately represents your design.
Verifying the Stackup Definition for a Board Design With Multiple Stackups	Verify that the stackup layer and stackup area definitions accurately represent your design.

Verifying That Power Supply and Signal Nets are Recognized Correctly

Ensure that the software recognizes the power supply nets in your design correctly. If it does not, specify them manually. Nets that are not recognized as power supply nets are recognized as signal nets.

The software automatically attempts to recognize power supply nets. The automatic identification algorithm can miss power supply nets with arbitrary names or few capacitor connections, and short power supply nets that are connected to a voltage regulator module.

The software automatically recognizes a net as a power supply net if three or more capacitor are connected to it, or if it is connected to 100 or more vias. You can change these values in the Net handling area of the Preferences dialog box (**Setup** > **Options** > **General** > **BoardSim** tab).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Look for these indications that power supply nets are not recognized correctly:

- Undetected power supply nets can lead to some nets looking complicated and huge in the board viewer. This occurs because the software displays not only the chosen net, but also all non-power supply nets connected to the chosen net through passive components (e.g., resistors and capacitors). The connected nets are called associated nets.
- Simulation runs very slow, as the software attempts to simultaneously analyze huge sets of nets that are erroneously tied together.

The following video contains additional information. Note that the video does not describe the Power Network button. See the procedure.



Procedure

1. Choose **Setup > Power Supplies**. The Edit Power Supply Nets dialog box opens.
2. Ensure that all power supply (power and ground) nets in your design are checked in the list. If your design includes many obscurely-named power supply nets, use the Power Network assistant to find other power supply nets that connect to the main power supply nets by components. To use the assistant:
   a. Select the main power supply nets in your design, then click **Power Network**.
   b. In the Power Network dialog box, select a power supply net. Connected components appear in a list to the right.
   c. Select any non-terminating components from the Connected components list that are likely to connect to other power supply nets. For example, ferrite beads, inductors, bypass capacitors, and some resistors.
   d. Select any connected power supply nets from the Connected nets list and click **Add to power-supplies**.
3. Click **OK**.
4. In the Edit Power-Supply Nets dialog box, ensure that the correct voltage is assigned to power supply nets in the “Edit supply voltages” list. To change the power supply voltage for specific nets, type a value in the Voltage column.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

5. Ensure that any power supply nets connected to a plane layer (defined as type “plane” without any copper shapes) are listed in the “Assign supply nets to plane layers” list. To associate a plane layer with a supply net, next to each plane layer, click the supply net name and select the correct net from the list. This step is not required for most fully routed designs.
6. Click **OK**.

Verifying That Component Types are Recognized Correctly

To ensure that the software recognizes components correctly, and your components appear in model assignment lists correctly, edit reference designator mappings.

The software recognizes a component as a resistor, inductor, IC, or other component type, by reading the prefix of the reference designator associated with the component.

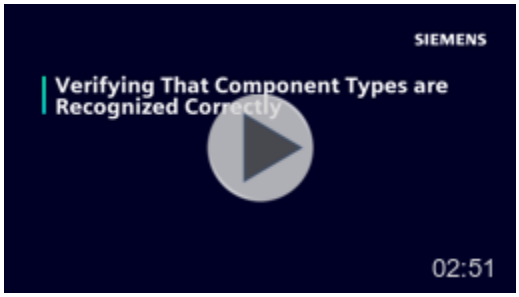
Tip

Review these tips before proceeding:

- Unrecognized components are mapped as ICs by default.
- The software does not recognize diodes, transistors, relays, or crystals as specific component types. Map these components as ICs.
- To describe a discrete clamp diode or diode-terminating network, map diodes as ICs and assign an IBIS model that contains information describing the diode.
- If you choose the “Design independent” option, reference-designator mappings are global and apply to all designs. If you choose the “Design-specific” option, reference-designator mappings apply only to the current design.
- If you make “Design-specific” changes to reference-designator mappings, you must re-load the design for the changes to take effect.
- Mappings at the top of the list have highest priority. Mappings are sorted into groups based on the number of characters in the reference prefix and then alphabetically sorted within the groups.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The following video contains additional information:



Procedure

1. Choose **Setup > Options > Reference Designator Mappings**.
2. To review current mappings, scroll through the Mappings list.
3. If you have loaded a design, choose either to have your mapping apply to all designs (“Design independent”) or apply only to the current design (“Design-specific”). If you haven’t loaded a design, your mapping applies to all designs.
4. You can either create a new component type or change the mapping of an existing component type:
   - To create a new mapping, type the new reference designator (Ref.) prefix and choose the component type from the radio buttons.
   - To change an existing mapping, select a component type from the Mappings list and then choose a new component type from the radio buttons.
5. Click **Add/Apply**.
6. Choose **Models** > **Assign Models/Values by Reference Designator** to open the Ref File Editor. Scroll through the list of parts to ensure that all components are recognized correctly.

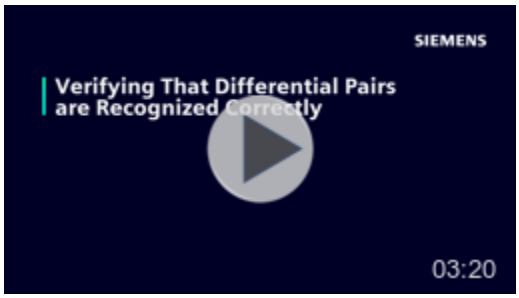
Verifying That Differential Pairs are Recognized Correctly

You must ensure that the software recognizes differentially paired nets correctly, so it can simulate them correctly. Create net name pairing rules to specify how the software automatically identifies paired nets when a board is loaded. You can also specify differential pairs manually for a design.

When you open a board design, the software reads differential pair identifying information from any assigned IBIS models, identifies nets connected by parallel terminators, and then reads the names of the remaining nets and attempts to recognize differentially paired nets based on the names of the nets.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The following video contains additional information. Note that the video does not describe manually defining differential pin pairs. See the procedure.



Restrictions and Limitations

- Net naming rules do not apply to net pairs that are already identified by information in assigned IBIS models, or net pairs connected by parallel terminators. The net names in IBIS models must match the net names in your design to be recognized correctly.

Note

Net name pairing rules are not design-specific, and apply to all boards that you open.

Prerequisites

- You have opened your board design.

Procedure

1. Choose the **Setup > Differential Pairs** menu item. In the Differential Pairs dialog box, review the list of differential pairs and verify that any differentially-paired nets that you plan to simulate are recognized.

   If the software does not recognize a differential pair, it simulates its members as single-ended nets.

2. Set up differential pairing rules to enable the software to recognize differential pairs automatically when you load a board, or manually specify paired nets for your design.

If you want to...	Do this...
Create a pairing rule that applies to all designs	1. In the “Differential pair net names” fields, type characters and wildcards, then click **Add**. 2. Click **OK**.
Manually specify paired nets for your design	1. In the Nets list, select two nets and click the arrow button to add them to the list of paired nets on the right. 2. Click **OK**. **Note:** The Status column displays “Invalid” if the software does not support the pairing.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to...	Do this...
Manually specify a differential pin pair for your design **Note:** You typically do this to override an incorrect automatically-defined differential pin pair.	1. Click **Setup diff twin pins**. The "Setup diff twin pins" dialog box opens. 2. Click **Add**, type the reference designator and pin number for both members of a differential pin pair, then click **OK**. Text in this dialog box is case-sensitive. For example: `J25.1 J26.1`

3. If you want to save manually specified pairings when you save the session, uncheck "Rebuild differential pairs when loading".
4. Click **OK**.

Related Topics

Differential Pairs Dialog Box

Verifying the Stackup Definition

Verify that the stackup definition accurately represents your design.

The following video contains additional information:



Note

If you have a board design with multiple stackups, see Verifying the Stackup Definition for a Board Design With Multiple Stackups.

Prerequisites

- Disable the **Enable Multiple Stackups** menu item. Choose **Setup > Stackup > Enable Multiple Stackups** to deactivate its check mark.

Procedure

1. Choose **Setup > Stackup > Edit**. The Stackup Editor opens.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

2. Verify the stackup definition, and make changes as needed. To change the stackup definition, see “Defining the Basic Stackup” on page 1233.

Verifying the Stackup Definition for a Board Design With Multiple Stackups

Verify that the stackup layer and stackup area definitions accurately represent your design.

The following video contains additional information:



Prerequisites

- You understand Board Designs With Multiple Stackups.
- The master stackup definition contains all the layers used by stackup areas in your design.
- Enable the **Enable Multiple Stackups** menu item. Choose **Setup > Stackup > Enable Multiple Stackups** to activate its check mark.

Procedure

1. Choose **Setup > Stackup > Stackup Manager**. The Stackup Manager dialog box opens.
2. Verify stackup definitions:
   a. Select a stackup and click **Edit**. The Stackup Editor opens.
   b. Verify that the stackup represents your design, and make changes as needed. See Defining the Basic Stackup.
   c. In the Stackup Manager dialog box, verify that the spreadsheet contains a stackup definition for each stackup area in your design, and add stackups as needed. See Defining a Local Stackup for a Board Design With Multiple Stackups.
   d. Repeat steps a - c for other stackups and boards (for a multiple-board design).

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3. Verify that each area of your board design has the correct stackup assigned to it:

   a. In the Stackup Manager dialog box, select a stackup and click **Select Area**. The board viewer highlights the area(s) assigned to the stackup. To remove highlighting, click **Select Area** again.

   b. Make stackup area changes or additions as needed. See Defining a Stackup Area for a Board Design With Multiple Stackups.

   Note

   The software automatically assigns the master stackup to areas of your board design without a stackup area assigned to them.

Related Topics

Modeling a Board Design With Multiple Stackups

Defining Net Classes

Context: BoardSim

You can categorize a group of nets as a net class. This enables you to select a group of nets at once (by net class) or edit settings for a group of nets when the software provides this option.

For example, you can create a net class for a set of data nets on the top layer in a DDRx design and name the class “DDR_data_top”.

Procedure

1. Choose the **Setup > Net Classes and Constraints** menu item.
2. Click **Set Up Net Classes**.
3. Click **New Net Class**, type a name for the class, then click **OK**.
4. Select nets from the list on the left, select the class from the list on the right, then click the arrow button to assign the nets to the class.

   Note

   When you assign a net to a class, any other nets associated with the net (including the other net of a selected differential pair) are also added to the net class. For example, if you select a net that is connected to another net by a passive component, both nets are added to the net class.

   Note

   You cannot add a net to more than one net class.

5. Click **OK**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Creating a Schematic-PDN Design

You can simulate your design before determining the final layout by creating a schematic-PDN. To run SI simulation, create a schematic in the Free-Form Schematic Editor. To run PI simulation, use the PDN Editor to create the power-distribution network (PDN) layout.

You can also edit and simulate parts of a board design, by exporting to a schematic-PDN. See Exporting a Net from BoardSim to LineSim.

Topic	Description
Creating a Schematic Design	Create a schematic and run SI simulation in the Free-Form Schematic Editor. You can add IC buffers (drivers and receivers), IC components, transmission lines, R, C, and L components, ferrite beads, vias, series bus switches, and power and ground connections. You can also add S-parameter or SPICE models to represent interconnections, connectors, and so on.
Creating a PDN Design	To run PI simulation on your design, use the PDN Editor to create the power-distribution network (PDN) layout. You can add or remove metal shapes from plane layers, add signal or stitching vias, decoupling capacitors, IC power pins, and VRMs.

Creating a Schematic Design

Create a schematic and run SI simulation in the Free-Form Schematic Editor. You can add IC buffers (drivers and receivers), IC components, transmission lines, R, C, and L components, ferrite beads, vias, series bus switches, and power and ground connections. You can also add S-parameter or SPICE models to represent interconnections, connectors, and so on.

Prerequisites

- If your design includes vias:
  - You have acquired the Via Modeling license.
  - You have defined padstacks that describe the vias in your design. See “Editing a Padstack Definition” on page 143.
- If your design has multiple stackups, you understand “Board Designs With Multiple Stackups” on page 157.
- You have verified that your stackup is correct:
  - If your design has one stackup, see “Defining the Basic Stackup” on page 1233.
  - If your design has multiple stackups, see “Defining a Local Stackup for a Board Design With Multiple Stackups” on page 154.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

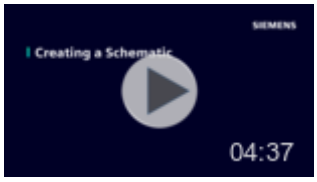
- You have defined power supply nets and voltages (**Setup > Power Supplies**). See "Verifying That Power Supply and Signal Nets are Recognized Correctly" on page 62.
- If you plan to use HyperLynx 3D Explorer to create an S-parameter model, you have acquired the nbgeomcore3d license and you understand "Exploring Design Variations" in the *HyperLynx Advanced Solvers User Guide*.

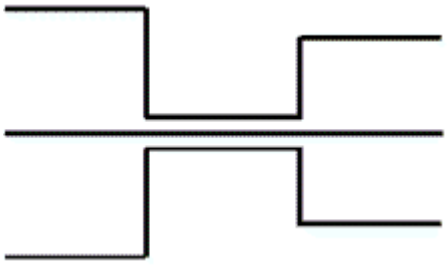
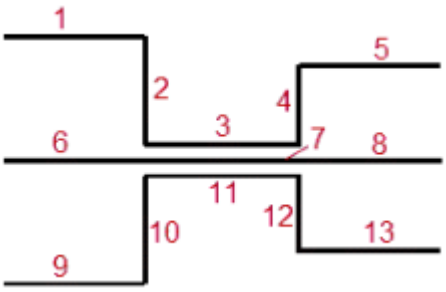
Procedure

1. Choose the **File > New Free-Form Schematic > SI**, or **SI/PI** menu item.
2. Click an icon in the schematic editor toolbar and click the location in the schematic where you want to place the symbol.

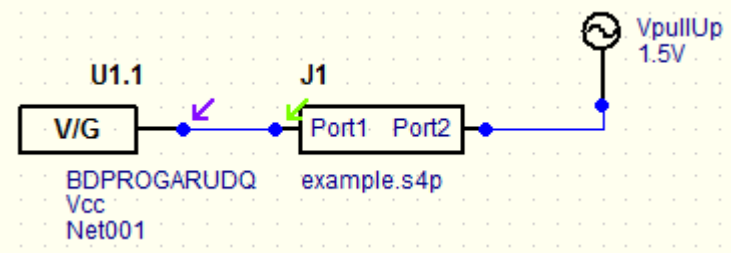
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The following video contains additional information:



For this symbol...	Do this...
Transmission line	Add a separate transmission line model to represent each part of a trace that has a unique topology or cross section. For example, the following image shows a top view of three traces.  You can use 13 transmission line models to model the trace segments.  Trace segments 3, 7, and 11 are likely to be affected by crosstalk. Assign a coupled transmission line model to each of the three transmission lines that represent segments 3, 7, and 11, and define a coupling region that describes the physical relationship between the three trace segments. See step 3.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

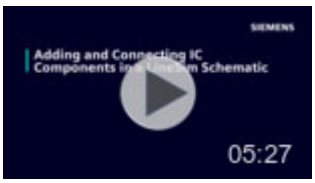
For this symbol...	Do this...
IC buffer (driver/receiver)	Add a single-ended or differential buffer symbol, and drag the blue connection point to connect it to the connection point of a transmission line symbol. To account for voltage variation at the power supply terminals of switching buffers in SI simulation: 1. Verify that an IBIS model with the keywords [ISSO_PD], [ISSO_PU], and [Composite Current] is assigned to IC symbols with the same reference designator. 2. Right-click an IC symbol and click **Add power/ground Pins**. Power and ground symbols marked V/G appear in the schematic. Do this once per group of IC symbols with the same reference designator. 3. Add a pullup symbol to the schematic and connect it to the V/G power symbol. Double-click the pullup symbol to select the power supply net. Optionally, account for the effects of the PDN connecting the VRM to the IC power pin by inserting a model that represents the PDN between the pullup and V/G power symbols. The figure below shows an S-parameter symbol, but you can also use passive components to represent PDN behavior.  4. Add a ground symbol to the schematic and connect it to the V/G ground symbol. Do this once per group of IC symbols with the same reference designator. 5. Enable power-bus coupling. See “Accounting for Non-Ideal Power Supplies in SI Simulation” on page 113.
Via	Add vias in the schematic if you intend to run SI simulation only. You can also add a via in the PDN Editor, so that it appears in the Schematic Editor, allowing you to connect it to other symbols in the schematic and run SI and PI simulation together. See “Accounting for Noise Between Single-Ended Signal Via and Power Planes in SI Simulation (Co-Simulation)” on page 471.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

3. Double-click a symbol to specify modeling parameters.

For this symbol...	Do this...
S-parameter/ SPICE model	To assign an existing S-parameter or SPICE model to the symbol, select a model from the list, and specify the port orientation. To use HyperLynx 3D Explorer to create an S-parameter model and assign it to the symbol: 1. Select Touchstone. 2. Click **Create in 3D Explorer**. HyperLynx 3D Explorer opens. 3. Define a set of "what-if" design variations, run a solver, and see which variations have acceptable performance. For help, see "Exploring Design Variations" in the *HyperLynx Advanced Solvers User Guide*. 4. Assign an S-parameter model to the schematic symbol: a. In HyperLynx 3D Explorer, on the **Results** tab, use an X in the Used column for design variations that you want to use for simulation in HyperLynx SI/PI. You can select multiple cells and press X to toggle the cell contents. If the Used column has more than one X, the software assigns the model for the bottom row with an X to the symbol. b. Near the bottom of the **Results** tab, click **Choose 'Used' Results**. c. The software assigns a model to the symbol. The software copies S-parameter models for the "Used" design variations to the design folder and assigns them names ending with *<experiment_name>-S_PARAMS.sNp*. These files are available for sweep simulations. 5. Close HyperLynx 3D Explorer and HyperLynx Full-Wave Solver.
Transmission line	In the Edit Transmission Line dialog box, select a model type (uncoupled or coupled), then specify modeling parameters in the Values tab. For coupled transmission lines, define a coupling region that describes the physical relationship between the transmission line and any other transmission lines that are coupled to it. 1. In the Add to Coupling Region tab, double-click the coupling region to add the transmission line to it. 2. Click the **Edit Coupling Region** tab to define the physical relationship between the coupled transmission lines (trace layer, length, width, separation, and so on). **Tip:** To quickly couple two or more transmission lines, press Ctrl-click over each symbol, right-click and choose **Couple**.

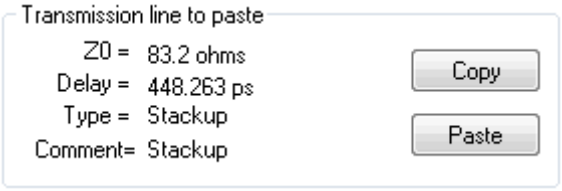
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

For this symbol...	Do this...
IC buffer (driver/ receiver), passive component (R,L, C, or ferrite bead)	In the Assign Models dialog box, select or double click a pin and select an appropriate model or value (R,L,C) for the component. See "Assigning a Model to an IC Pin" on page 86 and "Assigning a Model to a Passive Component Using the Assign Models Dialog Box" on page 91.
IC component	1. In the Assign IC Component Model dialog box, select an appropriate model for the component. 2. To make it easier to route connections to IC components, you can hide pins that you do not plan to include in simulation, or move pins to different locations. See "Hiding or Moving an IC Component Pin" on page 544. 3. Click **OK** to close the Assign IC Component Model dialog box. 4. In the Manage Assigned Models dialog box (**Models > Manage Assigned Models**), specify pin properties, such as I/O buffer direction, model selector, and stimulus. If your schematic also contains an IC buffer, you can use the Assign Models dialog box to set I/O buffer direction. Double-click an IC buffer to open the Assign Models dialog box. The following video contains additional information: 
Series bus switch	In the Bus Switch dialog box, select the pins that you want to be visible in the schematic. **Note:** Assign a bus switch IBIS model using a *.REF* file, and edit the IBIS file to enable the correct bus switch model for your design. See "Enabling Series Bus Switches for Simulation" on page 99.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

For this symbol...	Do this...
Via	In the Via Properties dialog box, select a via modeling method. If the via transmits a signal above 3 GHz, or is surrounded by complex design/stackup geometry, consider modeling the via with an S-parameter file (created by a 3D field solver) that represents the via behavior, and includes the effects of adjacent stitching vias and short sections of connected signal traces. Select the 3D modeling method and browse to the file, or run 3D EM simulation to create a model. See "Modeling a Via with a 3D EM Model in a Schematic" on page 127 and "Via Properties Dialog Box" on page 1062. Otherwise, select **None** for the 3D method and specify the layer span and connections, select a stackup (for a design with multiple stackups), and select a padstack that describes the via. To specify backdrilling information, enable Backdrilling (**Setup > Backdrilling/Plating/Constraint Settings**).

You can copy transmission line, IC, or passive component model parameters from one symbol to another. Click **Copy** (or **Paste**) in the appropriate dialog box, as shown below from the Edit Transmission Line dialog box, Transmission Type tab.

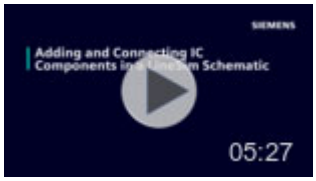


If you have a design with multiple stackups, you can assign a different stackup definition to selected transmission lines and via symbols (select symbols, right-click and choose **Change Stackup**). If you change the stackup for a via without also changing the stackup for a transmission line connected to it, the software disconnects the symbols.

4. Define connections.

If you want to...	Do this...
Connect one pin to another pin.	Click a symbol end point and drag a line to another symbol end point. If you want to assign a different name to a net, double-click a connection line and type the name.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to...	Do this...
Connect multiple pins at once between IC, S-parameter, and SPICE model components.	1. Select multiple pins on the first component by dragging a box around a block of pins or pressing Ctrl-click over a pin. 2. If you want to connect pins in top to bottom order, drag a connection line from a selected pin to a pin on a second component. **Note:** The software attaches connection lines to consecutive pins on a second component, even if you selected non-adjacent pins on the first component. 3. If you want to connect pins in another order: a. Choose the **Edit > Connect Symbols** menu item. The Confirm Connections dialog box opens and displays the selected pins in the left Pin column. b. Select multiple pins on a second component by dragging a box around a block of pins or pressing Ctrl-click over a pin. c. To reverse the order of connections, check “Reverse order” for the appropriate component. d. Click **Connect**. The following video contains additional information: 

Note

Via symbols can only connect to transmission line symbols that are associated with an uncoupled or coupled stackup type model.

5. Choose the **File > Save** menu item to save the design as an *.ffs* file.

Related Topics

Hiding or Moving an IC Component Pin

Creating a PDN Design

To run PI simulation on your design, use the PDN Editor to create the power-distribution network (PDN) layout. You can add or remove metal shapes from plane layers, add signal or stitching vias, decoupling capacitors, IC power pins, and VRMs.

Prerequisites

- Acquire a power integrity license.
- Edit the stackup definition as needed. See Defining the Basic Stackup.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- If your design includes signal vias, edit the padstack definitions as needed. See Editing a Padstack Definition.
- Define power supply nets and voltages. Click **Net Manager** () in the PDN Editor toolbar. See Verifying That Power Supply and Signal Nets are Recognized Correctly.

Procedure

1. Select **File > New Free-Form Schematic > PI**, or **SI/PI**.
2. Draw the board outline:
   a. Click **Draw Board Outline** ().
   b. Select the type of shape you want to draw by clicking **Rectangle** (), **Ellipse** (), or **Polygon** ().
   c. Draw or type coordinates for the board outline.

   When creating a polygon, you can force the line to be perfectly horizontal or vertical by pressing the Shift key while dragging the mouse.
3. Add voids and copper area fills.
   a. Click **Select Active Layers** () to select the layers on which the void or copper area fill is located.
   b. Click **Add Void Area** () or **Add Copper Area** ().
   c. Select the type of shape you want to draw by clicking **Rectangle** (), **Ellipse** (), or **Polygon** () and draw or type coordinates for the shape.
4. Add signal vias ( or ).

   In the Add Signal Via dialog box, type a name for the via and select a padstack that describes the via. Specify a coordinate location for the via, or click the location in the PDN Editor.

   To specify backdrilling information, enable Backdrilling (**Setup > Backdrilling Settings**).

   When you add a signal via in the PDN Editor, it also appears in the Schematic Editor, allowing you to connect it to other symbols in the schematic and run SI and PI simulation together. See Accounting for Noise Between Single-Ended Signal Via and Power Planes in SI Simulation (Co-Simulation).
5. Add stitching vias ().

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

a. In the Edit Stitching Via dialog box, provide coordinates for the via, or click the location in the PDN editor. To add a group of stitching vias, next to Place, select **Array**, set the array area or pitch, and specify a location for the array.

b. Select the layers that the via is connected to.

c. Specify a padstack that describes the via.

6. Add IC power pins ( ).

a. In the Add IC Power Pins dialog box dialog box, provide coordinates for the power pin, or click the location in the PDN editor. To add a group of pins, next to Place, select **Array**, set the array area or pitch, and specify a location for the array.

b. Specify a coordinate location for the IC pin, or click the location in the PDN Editor.

c. Specify the distance between the power supply and reference pins.

d. In the Connected/Reference Layers area, select the layers that the power supply pin is connected to in the Conn column. Select the layers that the return current pin is connected to in the Ref column.

Note

The Ref column is unavailable when you check the “Automatically Assign Reference Layers” option on the Preferences Dialog Box - Power Integrity Tab.

e. Select the power and reference nets that the pins connect to. Select <auto> when the pin connects to a layer that contains only a single power supply net.

f. Select the outer layer where the VRM is mounted (Top or Bottom).

g. Select a padstack that describes the power pin via.

h. Define AC or DC electrical models that describe the IC. See Power Integrity Models.

7. Add VRM ( ).

Note

You must add at least one IC power pin before you can add a VRM.

a. In the VRM Manager dialog box, click **New**. The Add/Edit VRM dialog box opens.

b. In the Add/Edit VRM dialog box, type a name for the VRM and select a model type.

Select Simple to model a VRM that includes a buck switching regulator. Select Advanced to model a VRM that includes a linear regulator. See “Add/Edit VRM Dialog Box” on page 723.

c. Type the VRM supply voltage.

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d. If the VRM is a multi-phase VRM, select the number of phases.

e. In the Phases area, for each phase, click in the I% column and type the percentage of total current that flows through the phase.

f. Click in the Power Net column to select the power net connected to the VRM.

g. Click in the Power Pin(s) column to select the power pin that connects to the PDN.

Note

If the VRM is not located on the same board, find a power supply pin located closest to the connection to the off-board VRM and assign a VRM model to it. Because VRMs work only at very low frequencies, the location of the VRM has little effect on the impedance profile.

h. Click in the Ref Net column to assign a reference net for each VRM.

i. Click in the Ref Pin(s) column to assign a specific reference pin. Click in the cell and select **Browse** to search for a pin in the list.

j. As needed, click the R and L columns to specify values for the VRM model type that you selected.

For R, (or R_0 for the advanced model) type the value of the equivalent resistance of the entire VRM. If the VRM is a multi-phase VRM, set the value of each phase to the value of the equivalent resistance of the entire VRM. The software calculates the voltage at the output pin based on the full current load through this resistance. If you do not know the VRM resistance, set the value to 0, and set the voltage output to an approximate value that corresponds with the expected current load on the VRM. For example, for a VRM that provides a nominal supply voltage of 1.5 V, type a voltage value of 1.47.

k. If the VRM includes a sense line, choose <select> in the sense line drop-down menu, and select power and reference pins in the VRM Sense Line dialog box.

l. If the VRM is capable of boosting VRM voltage to compensate for IR drop, in the Max Voltage Boost field, type the maximum additional voltage that the VRM can supply.

m. Click **OK**.

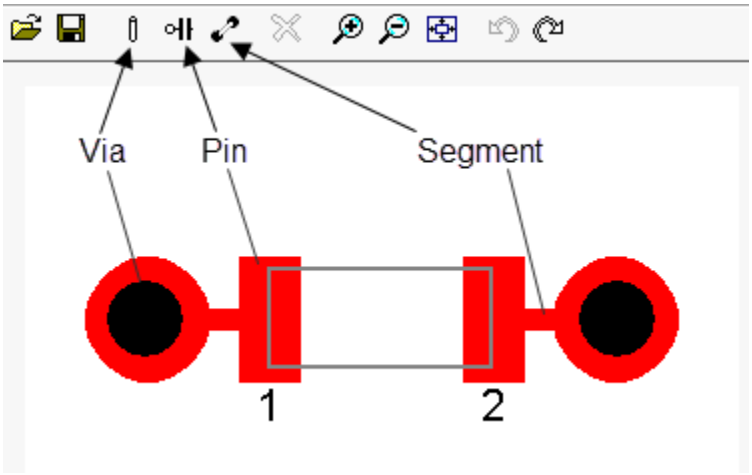
8. Add decoupling capacitors ().

a. In the Add Decoupling Capacitors dialog box, provide coordinates for the decoupling capacitor, or click the location in the PDN editor. To add a group of decoupling capacitors, next to Place, select **Array**, set the array area or pitch, and specify a location for the array.

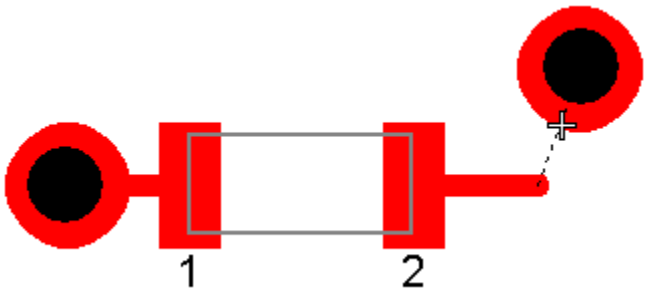
b. To edit the mounting scheme, Click **Edit Mounting Scheme**. You can create a custom scheme, save it as a *.DMS* file, and apply it to other decoupling capacitors.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

In the Decoupling Capacitor Mounting Scheme Editor, double-click a via, pin, or trace segment to edit properties. You can move vias or pads, or draw trace segments.



To draw a segment, click and drag. To connect segments, draw one segment, then click on the endpoint to draw the second connected endpoint.



c. Specify layer connections and power supply and reference net connections.

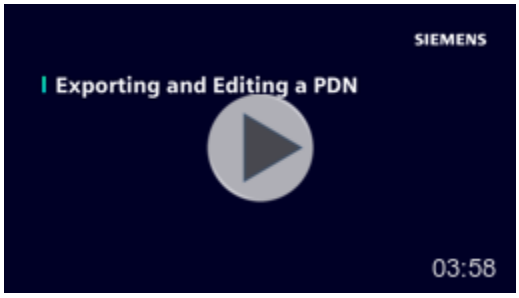
d. Click **Assign Model** to describe the capacitor in the Assign / Edit Capacitor Model Dialog Box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Results

The following videos contain additional information:





Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Chapter 3
Preparing for Simulation

Set up simulation options and assign models to components in your design before you run simulation.

Topic	Description
Assigning Models to Components and Pins	HyperLynx provides different methods to assign a model to a pin or an entire component. If you plan to simulate just a few nets, assign models to individual pins. If you plan to validate many nets, such as during batch simulation of the entire board, assign a model to an entire component by creating a *.REF* or *.QPL* file.
Assigning Models for PI Simulation	Before you can run power-integrity simulation, assign models to IC power supply pins, voltage-regulator module (VRM) pins, and decoupling capacitors.
Assigning Models for Thermal Analysis	Before you can run thermal simulation or power integrity with thermal co-simulation, assign thermal models in ThermalSim.
Accounting for Board and Package PDN Effects in SI Simulation	You can increase the accuracy of signal-integrity simulation by modeling board and package PDN effects.
Modeling a Board Area With an S-Parameter Model	You can run 3-D electromagnetic (EM) simulation to create an S-parameter model that accurately represents the behavior of interconnect structures with poor return paths or other non-TEM (transverse electromagnetic mode) behaviors, for a specific area of your board design.
Modeling a Via with a 3D EM Model in a Schematic	You can run 3-D electromagnetic (EM) simulation to create an S-parameter model that accurately represents the behavior of a via or differential via symbol in your schematic design.
Setting Simulation Options	Before you run simulation, set options to account for coupling, loss, signal, and noise between single-ended via and power planes.
Selecting Nets for SI Simulation	Some non-batch signal integrity simulation types require you to select the nets that you want to include in simulation by picking nets in the viewer or from a list in a dialog box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
Editing a Padstack Definition	Use the Padstack Manager dialog box to define padstacks for a schematic. Padstack information is stored in the schematic file (*.FFS*).
Preparing for DDRx Simulation	Find and fix common setup problems, gather and assign model files, gather reference designators for controller and memory components, and so on.
Modeling a Board Design With Multiple Stackups	If you have a board design with multiple stackups, you can define all the stackups and their locations in the board design.
Adding or Updating SERDES Protocols	You can make new or updated protocols available to SerDes wizards. Typically a Siemens EDA representative will send you a *.zip* file that contains configuration and supporting files that describe one or more protocols. You import a *.zip* file into a library file to make new or updated protocols available to a SerDes wizard.
Troubleshooting Simulation Setup	This topic provides solutions to common problems that can occur when you set up your design for simulation.
Saving Session Edits	When you save a board design, the software saves modeling and setup changes (session edits) to the *.BUD* file. The software saves schematic and PDN editor changes to the *.FFS* file.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Assigning Models to Components and Pins

HyperLynx provides different methods to assign a model to a pin or an entire component. If you plan to simulate just a few nets, assign models to individual pins. If you plan to validate many nets, such as during batch simulation of the entire board, assign a model to an entire component by creating a *.REF* or *.QPL* file.

A *.REF* file assigns models to reference designators whereas a *.QPL* file assigns models to part names. You can also use a combination of the methods, initially using a *.REF* and *.QPL* file, and then assign a model to a specific pin using the Assign Models dialog box. For information about how the software determines which model to choose, refer to "Precedence Among Model and Value Selection Methods" on page 630.

When assigning a model pin-by-pin, the model assignment to a pin on the same IC does not need to belong to the same model family. This gives you the freedom to experiment with different driver/receiver types more easily. The *.BUD* file stores pin model assignments.

You can use multiple model assignment methods (Assign Models dialog box, a *.REF* file, and *.QPL* files) together. When multiple model assignments conflict, the software uses only the model specified by the method with the highest priority (precedence). The software first assigns models from your *.QPL* files, then your design *.REF* file, and then any assignments made using the Assign Models dialog box (session edit file *.BUD*). The software overrides conflicting assignments in this order. Model assignments have precedence over value assignments.

If you do not have an exact model for a pin or component you are trying to simulate, you can create a new IBIS model with the Easy IBIS Wizard. See Creating IBIS Models with the Easy IBIS Wizard.

Topic	Description
Assigning a Model to an IC Pin	Use the Assign Models dialog box to assign an IC model to a component pin. You can assign an IC model to any pin that the software recognizes as an IC in the Assign Models pins list.
Assigning a Model to a Passive Component Using the Assign Models Dialog Box	Assign models to describe the resistors, capacitors, inductors, ferrite beads, or bus switches in your design. To account for package parasitics, include parasitic R, L, and C values in IBIS models.
Assigning a Model or Value to an Entire Component Using a .REF File	Use the REF-File Editor to create or edit a *.REF* automapping file, which assigns models and values to components with specific reference designators.
Assigning a Model or Value to an Entire Component Using a .QPL File	Use the QPL-File Editor to create or edit a *.QPL* automapping file, which BoardSim uses to assign models and values to components with specific part names.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
Enabling Series Bus Switches for Simulation	The [Series Switch Groups] keyword determines which series bus switch is enabled in simulation. The Assign Models dialog box does not provide a way to enable different series bus switches.
Disabling a REF or QPL File	Temporarily disable *.REF* and *.QPL* model and value assignments when you want to assign different models to perform a "what if" experiment.
Overriding a Model From an Automapping Model Assignment	You cannot always use the Assign Models dialog box to change model assignments made using an automapping file.

Assigning a Model to an IC Pin

Use the Assign Models dialog box to assign an IC model to a component pin. You can assign an IC model to any pin that the software recognizes as an IC in the Assign Models pins list.

If you assign an *.EBD/EMD* model to a component using a *.REF* or *.QPL* file, or created a MultiBoard project that uses an *.EBD/EMD* model to represent a board, you can use the Assign Models dialog box to change buffer settings and edit model selections mapped from the *.EBD/EMD* model.

Note

You can also assign a model using a *.REF* or *.QPL* file. When a model is assigned from a *.REF* or *.QPL* file, changes made using the Assign Models dialog box override *.REF* and *.QPL* mappings. You can save model assignments made using the Assign Models dialog box as session edits in the *.BUD* file for your design.

Caution

If you assign multiple Touchstone models, make sure the filename (not just the extension) of each model is different, so the simulator does not use the wrong model. The ADMS simulator converts a Touchstone model to a fitted-poles model before running simulation. Because a fitted-poles model has the *.PLS* extension, a model named *connector.s2p* or *connector.s4p* for example, are both converted to *connector.pls*. When a fitted-poles file is available, ADMS uses it instead of creating a new fitted-poles model for *connector.s4p*, producing inaccurate simulation results.

Restrictions and Limitations

- If you assign an IBIS differential model to a pin, be sure to apply the correct model to the pin. For example, if you have a model that pairs pins 1-2 and 7-8, and you choose the model for pin 1 and assign it to pin 7 on your board, the software will not recognize the net as part of a differential pair.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- To assign a SPICE or S-parameter model to more than one symbol in a schematic, use the same reference designator for these symbols.
- *.EBD/EMD* models can only be assigned using a *.REF* or *.QPL* file.

Prerequisites

- Ensure that the models you want to assign are in the correct location. See Setting Up the Software.
- Ensure that the models you want to use are in the correct format and contain the information you need for simulation. See Modeling.
- If you intend to assign a SPICE model, ensure that you set circuit simulator options to support your SPICE model type. See Simulation Controls Dialog Box.
- Ensure that the software recognizes component types correctly. See Verifying That Component Types are Recognized Correctly.

Procedure

1. Open the Assign Models dialog box.

Design Type	Do one of the following:
Board design	• Select a net and choose **Models > Assign models/Values by Net** (), and click the **IC** tab. See Selecting Nets for SI Simulation. • Select nets and open the Manage Assigned Models dialog box (**Models > Manage Assigned Models**) and double-click a pin row. • Right-click a component pin and select **Assign Model**, and click the **IC** tab.
Schematic design	• Double-click a symbol. • Open the Manage Assigned Models dialog box (**Models > Manage Assigned Models**) and double-click a pin row.

2. If the selected net spans multiple boards, from the Design File list, select the board that contains the component you want to model. To edit buffer settings and model assignments inside an *.EBD/EMD* model, select the *.EBD/EMD* model from this list.
3. If your design contains multiple instances of the same board, check “Apply To All Similar Boards” to model the selected pin the same way for all other instances of the currently selected board. Models are assigned to pins on similar boards when you perform step 4.
4. Double-click an IC driver or receiver pin from the Pins list. The Select IC Model dialog box opens.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The Pins list only displays pins on components recognized as ICs in the set of nets included in simulation.

-  indicates that a model is not assigned.
-  indicates that a model is assigned.
-  indicates that a driver model is assigned.
-  indicates that a receiver model is assigned.

For more information about the pins list icons, see the Assign Models Dialog Box.

To assign an...	Do the following...
IBIS model	1. Select the model you want to use from the Libraries, Devices, and Signal/Pin lists and click **OK**, or click **Find Model** to search all libraries. 2. Select **Model Selector** (if available) to select a model, and then click **OK**. This option is often available for ICs with programmable buffers, if the IBIS model includes the [Model Selector] keyword. 3. For schematic designs only, select **Assign model's pin name** to automatically assign the IBIS pin name to the IC symbol in the schematic.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

To assign an...	Do the following...
SPICE or S-Parameter (Touchstone) model	1. Select the model you want to use from the Files and Models lists and click **OK**, or click **Find Model** to search. 2. Use the spreadsheet to map the model ports to pins and power supply nets. For each port, click in the Circuit Connection cell and select the appropriate connection from the list. **Note:** For schematics, the Circuit Connection list displays only pins for one reference designator. To assign a model to more than one symbol in a schematic, use the same reference designator for these symbols. See Port-Mapping Examples. 3. Select Input or Output to specify if the buffer is a driver or receiver. 4. For a SPICE driver, you can describe the stimulus by selecting a stimulus circuit connection type for Vin, and editing the following values as needed: • Approximate Output Switching Time • Stimulus V high • Stimulus V low • Stimulus Delay time (for delayed stimulus circuit connection types) • Delayed Stimulus V high (for delayed stimulus circuit connection types) • Delayed Stimulus V low (for delayed stimulus circuit connection types) 5. If you use a complex model that has different modes of operation, you can pass parameters to the simulator to enable those modes. Click **Edit Parameters** and specify the parameter-value pairs as needed. See the examples located immediately below this table.

If you use a complex SPICE or S-parameter model and want to pass parameters to the simulator (see step 5 in the table above), some of the following parameter-value pairs may apply:

- If you set HyperLynx to use the ADMS simulator (Eldo native), and have assigned a strictly passive Touchstone model to a port, set the FORCE_PASSIVITY parameter to 2. When enforcing passivity, the ADMS simulator tries to detect and eliminate modeling discrepancies that can lead to instability or non-physical behavior.
- If you know a model is not symmetric, we recommend that you set the SYMMETRY parameter to 0 to avoid potentially wrong simulation results. Symmetrical models typically contain resistance, inductance, capacitance, and inductive capacitance models, but do not contain controlled sources.
- If the Touchstone model does not provide data at zero frequency, we recommend that you set the EXTRAP_TO_DC parameter to 1. This setting extrapolates to DC the curve from low frequency points given in the Touchstone model. This setting has no effect if the Touchstone model provides data at zero frequency.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

5. For IBIS model driver pins, set the buffer direction and state.

If you assigned an...	Do the following...
Output or bidirectional model	Select its direction or state in the Buffer Settings area.
Differential model	1. Select Output for the positive (+) pin, or select Output Inverted for the negative (-) pin. 2. In the Model to Paste area, click **Copy**. 3. In the Pins list, select the other differential pair pin. 4. In the Model to Paste area, click **Paste**. 5. Repeat for the other differential pair pin.

6. For IBIS models, select the power supply net that is connected to the Vcc and Vss pins in the Vcc Pin and Vss Pin lists, or select "use model's internal value".

Note

To override information in an IBIS model, select "When assigning a model to an IC pin, use a power supply net connected to the IC" (**Setup > Options > General, General** tab).

7. To apply the model assignment to other pins on the selected net or associated nets, click **Copy**, and then do one of the following:

If you want to...	Do the following...
Apply the value to one other component.	1. Select the other component in the Pins list. 2. Click **Paste**.
Apply the value to all other components.	Click **Paste All**.

- For multiple-board designs that include multiple instances of a board, when you check "Apply to all similar boards" and click **Paste** or **Paste All**, models are pasted to all similar boards.
- When you copy and paste models, the contents of the Pins list may change. This indicates that information in the model has changed the current set of nets. For example, when crosstalk simulation is enabled, replacing a driver IC model with a slower model might cause a net to be removed from the set of nets because it no longer couples with the selected nets. See Selecting Nets for SI Simulation.

8. To remove a model assignment, click **Remove**. If you used the Assign Models dialog box to override model assignments made in *.REF* or *.QPL* files, those assignments now take precedence again.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

9. To save model assignments, select **File > Save BoardSim Session File (.BUD)**. For a schematic, select **File > Save**.

Results

Models are now assigned to the IC components on your board. You can continue setting up your design or run a simulation.

Related Topics

Precedence Among Model and Value Selection Methods

Assign Models Dialog Box

Assigning a Model to a Passive Component Using the Assign Models Dialog Box

Assign models to describe the resistors, capacitors, inductors, ferrite beads, or bus switches in your design. To account for package parasitics, include parasitic R, L, and C values in IBIS models.

Note

You can also assign a model or value using a *.REF* or *.QPL* file. When a model is assigned from a *.REF* or *.QPL* file, changes made using the Assign Models dialog box override *.REF* and *.QPL* mappings. You can save model assignments made using the Assign Models dialog box as session edits in the *.BUD* file for your design.

Restrictions and Limitations

You cannot assign a model to a passive component that is connected to a power supply net.

Prerequisites

- Ensure that the models you want to assign are in the correct location. See Setting Up the Software.
- If using an IBIS model, ensure that the model exists and contains package parasitic information under the [PACKAGE] keyword. See Modeling.
- Ensure that components are recognized correctly. See Verifying That Component Types are Recognized Correctly.

Procedure

1. Open the Assign Models dialog box.
   - For a board design, select a net and click select **Models > Assign Models/Values by Net** (), or right-click a component pin in the board viewer and select **Assign Model**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- For a schematic design, double-click a symbol in the schematic.

2. If you have loaded a MultiBoard project, select the appropriate board from the Design File list.
3. In the Pins section, select a component pin and do one of the following:

If you want to...	Do the following...
Model an L, R, or C component by specifying a single value	1. Select **Value** and enter a number. 2. If the passive component is networked, click **Select** to specify the appropriate package description. If an appropriate package description is not available, you can create a custom package description. See Creating a Custom Package Description. **Note:** You can enter values using scientific notation. For example, type 1e6 for 1,000,000.
Model an L, R, C, bus switch, or ferrite bead component by assigning an IBIS model	1. Select **Model**, then click **Select** to open the Select IC Model dialog box. 2. From the Libraries, Devices, and Signal or Pin lists, select the appropriate model for the passive component. 3. Click **OK**. **Note:** The software recognizes L, R, or C IBIS models that include package parasitic information. Ensure that bus switch models contain appropriately structured connectivity information. See Enabling Series Bus Switches for Simulation.

A checkmark next to a passive component indicates that a model is assigned to that component, and an R or Q indicates that a *.REF* or *.QPL* file made the model assignment.

4. To apply the value or model assignment to other components of the same type on the selected net and associated nets, click **Copy**, and do one of the following:

If you want to...	Do the following...
Apply the value to one other component	1. Select the other component in the Pins list. 2. Click **Paste**.
Apply the value to all other components	Click **Paste All**.

5. To save changes to resistor, capacitor, and inductor values and model assignments made to a board, select **File > Save BoardSim Session File (.BUD)**.

   To save model assignment changes for a schematic (*.FFS*), select **File > Save**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Results

Models are now assigned to the passive components on your board. You can continue setting up your design or run a simulation.

Related Topics

Precedence Among Model and Value Selection Methods

Assign Models Dialog Box

Assigning a Model or Value to an Entire Component Using a .REF File

Assigning a Model or Value to an Entire Component Using a .QPL File

Assigning a Model or Value to an Entire Component Using a .REF File

Use the REF-File Editor to create or edit a *.REF* automapping file, which assigns models and values to components with specific reference designators.

Note

If you load a MultiBoard project, BoardSim automatically reads in the *.REF* file for the individual board and then creates an all-new composite *.REF* file that applies to the entire MultiBoard project. When you close the MultiBoard project, the composite *.REF* file is split into individual *.REF* files for the individual boards.

If you edit the composite *.REF* file for the MultiBoard project, the board ID suffix (such as _B00) in the RefDes column of the lower spreadsheet indicates on which board the reference designator is located. Edit the *.REF* file for an individual board by closing the MultiBoard project, loading the board, and then editing the *.REF* file.

Restrictions and Limitations

- You must use the Assign Models dialog box to assign SPICE, Touchstone, ferrite-bead, or series bus switch models.
- When you assign a model using an automapping file, you cannot remove the model from a pin using the Assign Models dialog box.
- The REF-File Editor does not support decoupling capacitor models. Use a *.QPL* file instead.

Procedure

1. Load the board or schematic.

   You must open the board or schematic so the REF-File Editor can identify which reference designators are present in the design.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

2. Select **Models > Assign Models/Values by Reference Designator (.REF file)**.

   The REF-File Editor opens.

3. In the Design's Part List area, select spreadsheet rows to identify the reference designators to receive the model or value assignment.

   Large designs contain many reference designators. To make finding and selecting individual reference designators easier, you can display a subset of all the reference designators in the design. To filter the *.REF* file reference designator spreadsheet, do any of the following:

   - Enter the filter value and click **Apply**.
   - Check "Show only parts without models" to hide reference designators with model assignments (ICs) or values (passive components).
   - Check "Show ICs" to display reference designators for ICs.
   - Check "Show passive components" to display reference designators for resistors and capacitors.

   The filter box supports the wildcards * (substitute any number of characters) and ? (substitute one character).

4. Assign a model.

   Press **Ctrl+Z** to undo an edit.

If you want to...	Do the following...
Assign a model to an IC reference designator	1. In the Model/Value To Insert area, select a library. 2. Select a component in the library. 3. Click **Assign Model**.
Assign a value to a discrete (two pin) resistor, inductor, or capacitor reference designator	1. Click By Value. 2. In the Model/Value To Insert area, click Discrete. 3. Enter a value. 4. Specify the units in exponent of form exxx or Exxx, where xxx is any integer value, positive or negative. For example, 1e3, or 10u (u=micro). 5. Click **Assign Model**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to...	Do the following...
Assign a value to a networked resistor or capacitor reference designator	1. Click **By Value**. 2. In the Model/Value To Insert area, click **Package / network**. 3. Select the package from the list. 4. Enter a value. 5. Click **Assign Model**.
Assign a model to a resistor or capacitor reference designator	1. Click **By Model**. 2. In the Model/Value To Insert area, click **Select Model**. The Select IC models dialog box opens. 3. Select a library, select a component in the library, and click **OK**. 4. In the Model/Value To Insert area, click **Assign Model**. **Restriction:** LineSim does not support resistor or capacitor packages. You cannot use the *.REF* File Editor to assign models to passive components in LineSim. If you want to model package parasitics, include parasitic R, L, and C values in the IBIS model that you assign to the component.

5. To remove assignments, select one or more rows in the assignment spreadsheet and click **Remove** or press the **Delete** key.
6. To use a series bus switch model, see Enabling Series Bus Switches for Simulation.
7. Select **File > Save**.

Caution

While you can save a *.REF* file to a different file name (using **File > Save As**), only *<design_file_name>.ref* works with a board or schematic named *<design_file_name>.[hyp, ffs]*.

Save your *.REF* file in the same folder as your design file. Name your *.REF* file in form <design_file_name>.*REF*, where <design_file_name> is the name of the HYP/FFS file. For example, the *.REF* file for demo.hyp is named demo.ref and located in the same folder as *demo.hyp*.

8. Select**File > Exit**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Results

Your *.REF* file is now set up and its model assignments will be included in your next simulation.

Related Topics

Disabling a REF or QPL File

REF and QPL File Syntax

Troubleshooting Automapping Model Assignment Errors

Automapping Files

Precedence Among Model and Value Selection Methods

Assigning a Model or Value to an Entire Component Using a .QPL File

Use the QPL-File Editor to create or edit a *.QPL* automapping file, which BoardSim uses to assign models and values to components with specific part names.

Restrictions and Limitations

- You must use the Assign Models dialog box to assign SPICE, Touchstone, ferrite-bead, or series bus switch models.
- When you assign a model using an automapping file, you cannot remove the model from a pin using the Assign Models dialog box.

Prerequisites

- Note the Qualified-Parts-List File(s) (QPL) field in the Set Directories dialog box (**Setup > Options > Directories**). The behavior upon opening the editor changes depending on this setting.

Procedure

1. Select **Models > Assign Models/Values by Part Name.**

If Set Directory QPL field...	Do this...
Is empty	The editor creates a new file. After you add rows to the file and click **Save**, the editor prompts you to specify the name of the file.
Contains only one file	The editor automatically opens the specified file.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If Set Directory QPL field...	Do this...
Contains multiple files	The Select QPL-file dialog box opens, which enables you to select a file to edit or create a new file. 1. To edit a different file or a new file, select **File > Open** or **File > New**.

2. Select the part type for model assignment.
3. Enter the part name for the selected part type.
4. In the Description field, enter a brief description of the component or a general comment. Comments can contain only printable characters, except for commas, which are reserved for field delimiters. Comments can be up to 80 characters long.
5. Specify a model or value for the part. The contents of this area depends on the Part Type you selected in the Part Type Info area.

For Part Type...	Do this...
IC	1. Select a library. 2. Select a component in the library. 3. Click **Assign Model**. **Note:** To search for a model, click **Find Model**.
Resistor, Capacitor, Inductor or Ferrite Bead- By Value	1. Click **By Value**. 2. Do one of the following: • Assign a value to a discrete (two pin) passive component. a. Click **Discrete.** b. Enter a value. c. Click **Assign Model**. • Assign a value to a networked resistor or capacitor. a. Click **Package / network.** b. Select the package from the list. c. Enter component values. d. Click **Assign Model**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

For Part Type...	Do this...
Resistor, Capacitor, Inductor or Ferrite Bead- By Model	1. Click **By Model**. 2. Click **Select Model**. The Select IC models dialog box opens. 3. Select a library. 4. Select a component in the library. 5. Click **OK**. 6. Click **Assign Model**. To model package parasitics, include parasitic R, L, and C values in the IBIS model that you assign to the component.
Decoupling Capacitor	1. Select the type of model to assign: • Simple C-L-R — Enter values for capacitance, ESR (equivalent series resistance), and ESL (equivalent series inductance). Select **ESL by capacitor size** and **<Auto-estimate>** to take capacitor package size and mounting via lengths (BoardSim only) into account when calculating ESL. Use these settings unless you have a specific reason to manually enter the ESL value or package information. • Library — Select a library and a model. Use the Assign / Edit Capacitor Model Dialog Box to create decoupling-capacitor libraries. • SPICE — Select a library and a device, and then assign capacitor pin names (numbers) to model node names. • Touchstone — Select a library (but not a device) and assign capacitor pin names (numbers) to model port index numbers. 2. Check “Includes mounting inductance” only if the decoupling capacitor model includes the effects of the via and its connectivity to the capacitor package. This situation happens when the vendor does not separate the capacitor model from how the capacitor package is mounted in the test fixture. 3. Click **Assign Model**.

6. Verify your assignments in the QPL-file model assignment spreadsheet. To edit an assignment, double-click a row in the spreadsheet. Press Ctrl+Z to undo an edit.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

If you assigned a model by entering the C-L-R values, the values do not display in the Value fields of the QPL-file model assignment spreadsheet. They are saved to the *.QPL* file in the order you entered them: Capacitance, ESR, ESL. To view the values, select a Part Name and look in the "Model/value to insert" section of the QPL file Editor.

7. Choose **File > Save** or **File > Save As**.

   Store your *.QPL* file anywhere on the computer or network. Name your *.QPL* file using the form *<alpha_numeric>.QPL*.

   For example, the *.QPL* file for demo.hyp can be named foo.qpl and located on the network.

8. Add your *.QPL* file to the Qualified-parts-list field in the Set Directories dialog box (**Setup > Options > Directories**).

   Note

   You can use multiple *.QPL* files for each design. Files are loaded in order, from left to right. Be sure to list your files in the intended order.

9. Ensure that "Use QPL file(s) to assign models" is checked.
10. Click **OK** to close the Set Directories dialog box.

Results

Your *.QPL* files are now set up and its model assignments will be included in your next simulation.

Related Topics

Disabling a REF or QPL File

REF and QPL File Syntax

Troubleshooting Automapping Model Assignment Errors

Automapping Files

Precedence Among Model and Value Selection Methods

Assigning a Model or Value to an Entire Component Using a .REF File

Enabling Series Bus Switches for Simulation

The [Series Switch Groups] keyword determines which series bus switch is enabled in simulation. The Assign Models dialog box does not provide a way to enable different series bus switches.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

A series bus switch pin is permitted by the IBIS specification to loop back out to another external pin. To ensure a valid signal-integrity simulation in this case, the load at the second pin and its reflection effects must be taken into account.

To ensure proper signal-integrity simulation in the general case, it is only safe to assign a series bus switch models on a component-wide basis.

Procedure

1. Select **Models > Edit IBIS IC Models (.IBS)**.
2. Select **File > Open** and select the IBIS file you want to edit.
3. Select **Edit > Find** and search for the keyword [Series Switch Groups].
4. Identify the active series pin pairs in the [Series Switch Groups] keyword and edit the On/Off values. For example:

Enable group 1	[Series Switch Groups] On 1 / Off 2 /
Enable group 2	[Series Switch Groups] On 2 / Off 1 /

Note

If the [Series Switch Groups] keyword contains multiple groups with the same entries, simulation uses the first definition. For example, this declaration enables groups 1, 2, 3, 4: [Series Switch Groups] On 1 2 3 4 / Off 1 2 3 4 /

This declaration disables groups 1, 2, 3, 4: [Series Switch Groups] Off 1 2 3 4 / On 1 2 3 4 /

5. Save your edits and close the IBIS editor.

Disabling a REF or QPL File

Temporarily disable *.REF* and *.QPL* model and value assignments when you want to assign different models to perform a “what if” experiment.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

Disable an automapping file.

To disable a...	Do this...
.REF file	1. Do one of the following: • Rename the *.REF* file name or extension so that it does not match <design_file_name>.ref. For example, if the design is named demo.hyp, rename demo.ref to demox.ref. • Move the *.REF* file to another folder. 2. Reload the design to activate the changes.
.QPL file	1. Select **Setup > Options > Directories**. 2. Uncheck Use QPL File To Assign Models. 3. Click **OK**.

Related Topics

Assigning a Model or Value to an Entire Component Using a .REF File

Assigning a Model or Value to an Entire Component Using a .QPL File

Overriding a Model From an Automapping Model Assignment

You cannot always use the Assign Models dialog box to change model assignments made using an automapping file.

Procedure

1. Change an automapped model assignment.

If you want to...	Do this...
Change an automapped *.EBD/EMD* model	Do one of the following: • Edit the automapping file to either change the model assignment, or delete the model assignment. If you delete the assignment, reassign a model using the Assign Models dialog box. • Delete or rename the automapping file, reload the schematic, and then assign an IC model using the Assign Models dialog box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to...	Do this...
Change a non-*.EBD/EMD* automapped model	Overwrite the automapping assignment using the Assign Models dialog box to assign a different IC model for the pin.

2. For *.QPL* files only, in the Set Directories dialog box, uncheck Use QPL file(s) to assign models or delete the *.QPL* file from the Qualified Part File(s) edit box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Assigning Models for PI Simulation

Before you can run power-integrity simulation, assign models to IC power supply pins, voltage-regulator module (VRM) pins, and decoupling capacitors.

Topic	Description
Assigning VRM Source, DC Sink, and AC Models	Assign power-integrity models to IC power supply pins to describe their current sink and voltage source behaviors for power integrity simulation.
Assigning Models to Decoupling Capacitors	Assign models to decoupling capacitors in your board design to describe their behavior for decoupling, plane noise, and signal-via bypassing simulation.
Creating Power Supply Pin Groups	You can create pin groups to probe multiple power supply pins in parallel during decoupling simulation and see results (PDN impedance) for each group. Power supply pins included in a pin group connect to the same component and pair of primary and reference nets.

Assigning VRM Source, DC Sink, and AC Models

Assign power-integrity models to IC power supply pins to describe their current sink and voltage source behaviors for power integrity simulation.

- VRM source models and DC sink models are required for DC drop simulation.
- AC models are required for Plane Noise analysis.

Restrictions and Limitations

- The software does not support SPICE sink and source models.
- Model topology is read only. Figures in the model-assignment dialog boxes show the supported model topology.

Prerequisites

- You have obtained current consumption properties and on/off switching times for the ICs from the manufacturer’s data sheets. For FPGAs, use the power calculator provided by the FPGA development system. For ASICs, ask the in-house IC designers at your company. If you cannot determine how the total current is distributed among the set of individual component pins, you can specify a total current value for all pins to assign an average value to each component pin.
- You have obtained voltage-regulator module (VRM) voltage, resistance, and inductance properties. Note the VRM design type (buck converter or linear regulator).

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- If you are using BoardSim, ensure that the software correctly recognizes power supply nets and connected components in your design. See Verifying That the Software Recognizes Your Design Correctly.

Procedure

1. Open the Assign Power Integrity Models dialog box (**Models > Assign Power Integrity Models** or **Xpedition Layout > Analysis Control > HyperLynx > Assign PI Models**), and click the **IC** tab.
2. For a DC Sink or AC model, in the Assign Power Integrity Models dialog box:
   a. Select the pins that you want to assign a model to by selecting the appropriate rows in the spreadsheet. Drag over the row headers or press Ctrl-click to select multiple pins.
   b. In the AC Model area or DC Sink Model area, click **Assign**.
   c. Provide the IC properties that you obtained from manufacturer data in the appropriate dialog box, and click **OK**.

   Tip

   For DC Sink models, if you only have a total current value for all pins, use the **Whole Group** option.

   See "Edit AC Power Pin Model Dialog Box" on page 852, and "Edit DC Power Pin Model Dialog Box" on page 854.
3. For a VRM model, in the Assign Power Integrity Models dialog box:
   a. Select the power supply pins that you want to assign a model to by selecting the appropriate rows in the spreadsheet. Drag over the row headers or press Ctrl-click to select multiple pins.

      If the VRM is not located on the same board, find a power supply pin located closest to the connector to the off-board VRM and assign a VRM model to it. Because VRMs work only at very low frequencies, the location of the VRM has little effect on an impedance profile.
   b. In the VRM Model area, click **Assign**. The Add/Edit VRM dialog box opens.
   c. Type a model name (the VRM Manager dialog box displays this name), select a model type, then provide information that represents the VRM as needed.

      Select Simple to model a VRM that includes a buck switching regulator. Select Advanced to model a VRM that includes a linear regulator. See "Add/Edit VRM Dialog Box" on page 723.
   d. Type the VRM supply voltage.
   e. For a multi-phase VRM, type the number of phases.

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f. In the Phases area, for each phase, click in the I% column and type the percentage of total current that flows through the phase.

g. Click in the Power Net column to select the power net connected to the VRM.

h. Click in the Power Pin(s) column to select the power pin(s) that connect to the VRM.

   If your design includes many pins, select <Browse…> in the drop-down menu to filter the list of pins, or select multiple pins an once using Ctrl-click or Shift-click.

i. Click in the Ref Net column to select the reference net connected to the VRM.

j. Click in the Ref Pin(s) column to select the reference pin(s) that connect to the VRM.

   If your design includes many pins, select <Browse…> in the drop-down menu to filter the list of pins, or select multiple pins an once using Ctrl-click or Shift-click.

k. As needed, click the R and L columns to specify values for the VRM model type that you selected.

   For R, (or R_0 for the advanced model) type the value of the equivalent resistance of the entire VRM. If the VRM is a multi-phase VRM, set the value of each phase to the value of the equivalent resistance of the entire VRM. The software calculates the voltage at the output pin based on the full current load through this resistance. If you do not know the VRM resistance, set the value to 0, and set the voltage output to an approximate value that corresponds with the expected current load on the VRM. For example, for a VRM that provides a nominal supply voltage of 1.5 V, type a voltage value of 1.47.

l. If the VRM includes a sense line, choose <Select…> in the sense line drop-down menu and select power and (optionally) reference pins in the VRM Sense Line dialog box.

m. If the VRM is capable of boosting VRM voltage to compensate for IR drop, in the Max Voltage Boost field, type the maximum additional voltage that the VRM can supply.

n. Click **OK**.

4. To save a DC Sink model for use in other simulations, do the following:

   a. Click in the Pin or Net column of a component and click **Save Model**.

   b. In the Save DC Models dialog box, edit the model name and save location as needed.

   c. Select how you want to assign the model for the current design:

      - **Assign the Model by REF file —** Creates a REF file containing the model assignment.

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- **Assign the Model by QPL file —** Appends the model assignment to the QPL file you specify.
- **Save the model and keep current assignment —** Saves the model in an HLPIMODEL file without making a model assignment.

This option is not available if a “Q” appears in the column next to the component, indicating the a QPL file has already assigned a model to the component.

d. Click **OK**.

An “R” appears in the column to the right of the component if you selected to assign your model using a REF file. A “Q” appears if you used a QPL file to make the assignment.

5. To simulate the entire current loop, including power supply nets and reference nets, specify the reference net that is connected to the IC. This enables the software to assign a power integrity model to the IC pin that is connected to the reference net. In the Reference Net area, click **Reference Net**. In the Set Reference Nets Dialog Box, select the reference net that is connected to the IC.
6. Assign models to series components (resistors or inductors):
   a. In the Assign Power Integrity Models dialog box, click the **Supply-Net Resistors** or **Supply-Net Inductors** tab.
   b. Click a component in the list and type values as needed. For networked components, select a package description.
   c. Click **Assign**.
7. Assign models to other series components, such as high-current power FETs.
   a. In the Assign Power Integrity Models dialog box, click the **Other Supply-Net Components** tab.
   b. Select an IC from the list.
   c. Select the spreadsheet row header for the first and second pins of the component, and click **Connect**.
8. Click **OK**.

Results

Models are now assigned to IC and VRM components on your board. You can continue setting up your design or run a simulation.

Related Topics

Power Integrity Models

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Assigning Models to Decoupling Capacitors

Scope: BoardSim

Assign models to decoupling capacitors in your board design to describe their behavior for decoupling, plane noise, and signal-via bypassing simulation.

Note

For schematic designs, you can assign models to decoupling capacitors in the PDN Editor. See Creating a PDN Design.

You can also assign decoupling capacitor models from the Check Capacitor Models page for the Decoupling Wizard, Via Model Extractor Wizard, and so on. The behavior of this wizard page is very similar to the behavior of the Assign Decoupling-Capacitor Models dialog box that is described below.

The following video contains additional information:



Prerequisites

- Gather decoupling capacitor model information in any of the following forms:
  - Values from datasheets — Obtain capacitance and equivalent series resistance (ESR) values from a decoupling capacitor datasheet. Siemens EDA recommends that you have the software automatically calculate the equivalent series inductance (ESL) value, but you can manually specify ESL using a value from a datasheet. You can enter datasheet values in a dialog box to create a simple C-L-R model.
  - Simulation models—Obtain SPICE or Touchstone models that represent decoupling capacitor behavior.
- If your design has many decoupling capacitors, you may save model-assignment time by creating a *.QPL* file that the software uses to automatically assign models. Create the file and enable the software to use it before loading your design. See Assigning a Model or Value to an Entire Component Using a .QPL File.
- If you have a decoupling capacitor library, copy its *.DECAP* file to the *HypFiles* or *Libs* folder under *\MentorGraphics\<release>\SDD_HOME\hyperlynx<# bits>*.

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Procedure

1. Select **Models > Edit Decoupling-Capacitor Models**.

   The Assign Decoupling-Capacitor Models dialog box opens.

2. To show capacitors for a specific pair of power supply nets, check Show these nets only, and select power and ground nets.

3. Verify any existing model assignments (made by a *.QPL* file, for example) by double-clicking a spreadsheet row and reviewing information displayed by the Assign / Edit Capacitor Model dialog box.

4. To assign a model to many capacitors of the same type, create a capacitor group that receives the same model assignment. If you choose to automatically create groups, the software groups decoupling capacitors with the same capacitance and maximum pin-to-pin dimensions.

If you want to...	Do this...
Automatically create groups	1. Click **Edit Groups**. The Assign Decoupling-Capacitor Groups dialog box opens. 2. Click **Auto-Grouping**. The Auto-Grouping dialog box opens. 3. Select **All available capacitors (preserve existing groups)**, unless you want to delete all existing groups and create new groups, and click **OK**. 4. Expand each group and verify that it contains the correct set of capacitors. If needed, remove a capacitor from the group by selecting its row and clicking <<. 5. When you have finished creating groups, click **OK**.
Manually create groups	1. Click **Edit Groups**. The Assign Decoupling-Capacitor Groups dialog box opens. 2. In the Capacitor groups area, select **<new>**. 3. In the Decoupling capacitors area, select the capacitors that you want to use the same model and click >>. 4. In the Capacitor groups area, specify the name of the new group. 5. When you have finished creating groups, click **OK**.

5. Assign a decoupling capacitor model.

   a. Select one or more decoupling capacitors or groups.

   b. Click **Assign Model**.

      The Assign / Edit Capacitor Model dialog box opens.

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c. If a model or an ESL parameter that you specify includes the inductance effects of the via and its connection to the capacitor package, select “Capacitor model includes mounting inductance (no additional inductance will be calculated/added during analysis)”.

d. Assign a CLR, SPICE, or Touchstone model.

If you want to...	Do this...
Assign a model from a decoupling capacitor library	From the Library Models area, select a library and model, and click **OK**.
Create and assign a simple CLR model	1. Select **Simple C-L-R**. 2. Specify capacitance and ESR values. 3. Select **Auto-calculate** (recommended, because ESL is strongly dependent on mounting and stackup) unless you have a reason to manually specify ESL.
Assign a SPICE model	1. Select **SPICE**. 2. Select a model file from the SPICE Files list. 3. To hide sub-circuits with more or fewer ports than the number of capacitor pins, check **Show only compatible subcircuits**. 4. Select a sub-circuit from the Sub-circuits list.
Assign a Touchstone model	1. Select **Touchstone**. 2. To hide model files with more or fewer ports than the number of capacitor pins, check **Show only compatible models** and select a number of ports option. 3. Select a model file from the Files list. 4. If the software cannot automatically identify a two-port model as a series or shunt type, select a type from the “Two-port model type” area. See “Two-port model type” in the table in Assign / Edit Capacitor Model Dialog Box.

e. If you assigned a CLR, SPICE, or Touchstone model that you want to use again, click  , specify the library name, library location, and model name. and then click **OK**.

f. Click **OK**.

Results

Models are now assigned to decoupling capacitors on your board. You can continue setting up your design or run a simulation.

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To see the effect that a capacitor or capacitor group has on the impedance profile of a PDN, you can run decoupling simulations with and without its model enabled and compare the results. Uncheck **Enabled** to disable a model.

Related Topics

Assigning a Model or Value to an Entire Component Using a .QPL File

Creating Power Supply Pin Groups

Scope: BoardSim

You can create pin groups to probe multiple power supply pins in parallel during decoupling simulation and see results (PDN impedance) for each group. Power supply pins included in a pin group connect to the same component and pair of primary and reference nets.

Automatic pin grouping works best if you assign IBIS models that contain pin mapping information ([Pin Mapping] keyword). The software can also group pins using information in the [Pin] keyword, and information in your design, such as the connected power supply net name. If any information conflicts, the software attempts to determine the most likely pin grouping. If needed, you can edit the way automatic pin grouping works. See Auto-Create Groups Options Dialog Box.

Restrictions and Limitations

- The software does not simulate pin groups with N/A in the reference net column of the Pin Group Manager dialog box.
- The software does not use information from *.EBD/EMD* models to automatically create pin groups.

Prerequisites

- Enable “Automatically assign reference layers” on the **Power Integrity** tab of the Preferences dialog box.
- To enable the software to automatically create pin groups based on IBIS model information, use a *.REF* or *.QPL* file to assign an IBIS model to each IC component you plan to probe during decoupling simulation.
- To model the path between the package pin and IC power bus as an RLC circuit instead of a short, obtain the RLC values from the component vendor and enter the values into the Edit Pin Group dialog box.

Procedure

1. Select **Models > Assign Power Integrity Pin Groups**.

   The Pin Group Manager dialog box opens.

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2. Specify how you want the software to group pins, or group them yourself.

If you want to...	Do this...
Automatically create pin groups for all components, using information from IBIS models (if assigned) and power supply nets.	Click **Create Groups**.
Automatically create pin groups for selected components, using information from IBIS models (if assigned) and power supply nets.	1. Select one or more components. 2. Right-click over the first column for a component and click **Auto-create Groups**.
Automatically create pin groups for selected components, using information only from assigned IBIS models.	1. Select one or more components. 2. Right-click over the first column for a component and click **Create IBIS Groups**.
Manually create a pin group for one component.	1. Right-click over the first column for a component and click **Add New Group**. 2. In the Edit Pin Group dialog box, add pins to the group, assign the model between the package pin and IC power bus, and assign the reference net.

3. If the software automatically created pin groups, verify that each group contains the correct set of pins. You can edit a pin group to add or remove pins, or change the way the software models the pin group as a circuit. Right-click in the first column, and click **Edit Group**. Make changes in the Edit Pin Group dialog box. If needed, you can also edit the advanced automatic pin group creation options in the Auto-Create Groups Options dialog box.
4. If N/A appears in the reference net column for a pin group that you plan to probe, click **N/A** and select a reference net.

Results

The Select IC Pin Group Probes page in the Decoupling Wizard displays power supply pin groups for the pair of power supply nets that you selected on the Select Nets for Analysis page.

Related Topics

Auto-Create Groups Options Dialog Box

Preferences Dialog Box - Power Integrity Tab

Pin Group Manager Dialog Box

Assigning a Model or Value to an Entire Component Using a .QPL File

Decoupling Wizard - Select IC Pin Group Probes Page

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Assigning Models for Thermal Analysis

Before you can run thermal simulation or power integrity with thermal co-simulation, assign thermal models in ThermalSim.

Select **Models > Assign Thermal Models**. See *ThermalSim Help*.

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Accounting for Board and Package PDN Effects in SI Simulation

You can increase the accuracy of signal-integrity simulation by modeling board and package PDN effects.

If you want to...	Do the following:
Analyze a high-speed memory interface (such as DDRx), where the switching of multiple, single-ended signals can generate SSN and other PDN-related effects that affect signal timing	Model PDN effects with power-aware SI models. See "Modeling a Non-Ideal PDN or Via-to-Via Coupling With an S-Parameter Model" on page 115.
Model package PDN effects such as non-ideal power supplies and simultaneous switching	Model critical ICs with power-aware IBIS models. See "Accounting for Non-Ideal Power Supplies in SI Simulation" on page 113.
Analyze a SERDES channel (where nets are differential and multiple buffers do not switch at the same time)	Model critical areas of your board design with a 3-D full-wave-solved area (S-parameter model) that captures detailed high-frequency effects at signal-path discontinuities. See "Modeling a Board Area With an S-Parameter Model" on page 122.

You can also account for PDN effects in SI simulation by enabling "Co-Simulation" to see time-domain plots that show how noise propagates in PDN power/ground cavities. Co-Simulation runs considerably slower than simulation using power-aware-SI models or 3D areas, and it uses a FDTD solver rather than using frequency-domain PDN-model extraction.

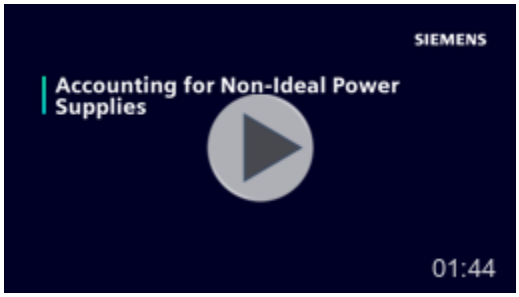
Accounting for Non-Ideal Power Supplies in SI Simulation

You can increase the accuracy of signal integrity simulation by accounting for voltage noise on power and ground buses caused by the current draw of switching drivers.

Accounting for non-ideal power supplies improves simulation accuracy by including the voltage noise that switching drivers add to the power and ground terminals of buffers on the same IC as well as power/ground buses.

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The following video contains additional information:



Prerequisites

- Assign IBIS models that contain the keywords [Composite Current], [ISSO_PU], [ISSO_PD], and [Pin Mapping] to ICs on the nets to simulate.

  For information about these keywords and how the software uses them, see KB article MG580201 on Support Center.

- For a schematic design, add IC power/ground pin symbols. You can also add symbols to represent how a power-distribution network (PDN) connects a VRM to an IC power pin. For details, see step 2 and the "IC buffer (driver/receiver)" table row in "Creating a Schematic Design" on page 70.

- You have acquired the DDRx license.

Procedure

1. For a board design only, you can include other nets with buffers that connect to the same power supply net as the selected net in simulation. Enable the "Include power bus coupling" PI effect.

   a. Choose the **Setup > PI Effects Settings** menu item.

      The PI Effects Settings dialog box opens.

   b. Enable "Include power bus coupling".

2. Enable the **Setup > Enable PI Effects in SI Simulations** menu item  .

Results

When you enable the "Include power bus coupling" PI effect and select a net for simulation, the software also includes nets with buffers that connect to the same power and ground buses used by the buffer on the selected net. The [Pin Mapping] keyword in an assigned IBIS model specifies power and ground bus names for a buffer pin. Simulation run time increases when the software includes these additional nets.

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Modeling a Non-Ideal PDN or Via-to-Via Coupling With an S-Parameter Model

You can increase the accuracy of signal integrity simulation by modeling the PDN or signal via-to-via coupling through the PDN with a power-aware SI model. Including power-aware SI models in simulation also helps you isolate noise coupling issues to better understand the sources of noise in your design.

A power-aware SI model is an S-parameter model that the software creates using HyperLynx Hybrid Solver or the HyperLynx PI simulation engine. The model can include ports at IC signal and power pins, and/or signal vias. The software enables you to connect the model by mapping its ports to the IC pins or vias that they represent in the board design.

You can include more than one model in SI simulation. For example, you can include one that describes the PDN and one that describes via-to-via coupling.

Restrictions and Limitations

- You cannot export a PDN model from a MultiBoard project.
- You cannot export a PDN model from a board design that includes multiple stackups.

Prerequisites

- If you plan to create a model that describes the PDN, obtain and assign power-aware IBIS models to the components connected to the nets of interest. Power-aware IBIS models must include accurate die capacitance information to produce accurate simulation results.
- Evaluate your design to optimize PDN performance. Run Lumped or Quick Analysis decoupling simulation if needed. See “Decoupling Simulation” on page 238.
- Ensure that you have assigned PI models that accurately describe the PDN, including a VRM model and associated reference nets. VRMs (with their low resistance DC paths) can significantly lower PDN impedance at low frequencies. See “Assigning Models for PI Simulation” on page 103.
- To include a power supply net in simulation that is connected by a series passive component to a power supply net you have probed, check “Allow AC supply models for non-capacitor components” in the Preferences dialog box. See Preferences Dialog Box - Power Integrity Tab. If you check this option, the wizard displays the Supply Component Models page instead of the Check Capacitor Models page.
- To enable the software to measure PDN impedance through a group of power supply pins in parallel, create power supply pin groups. See “Creating Power Supply Pin Groups” on page 110.
- If you plan to refine project and simulation settings in HyperLynx Hybrid Solver, you have a license for Hybrid Solver.

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Procedure

1. Choose the **Analyze 3D EM > Manage Power Aware SI Models** menu item.

   The Manage Power Aware SI Models dialog box opens.

2. Create a model or import one (created in HyperLynx Hybrid Solver, for example). Do one of the following:

If you want to...	Do this...
Create a model	Click **Create**. The PDN and Channel Model Extractor wizard opens.
Import a model	Click **Import**, select the S-parameter file, then click **Open**.

3. For an imported model, check or change port mappings for the model, if needed. Select the model row and click **View/Edit Ports**.

   Note

   S-parameter files created by the Advanced Decoupling Wizard, or in HyperLynx Hybrid Solver contain port mapping information in the header that enables the software to automatically map ports.

4. For a new model, in the PDN and Channel Model Extractor wizard, on the Start Analysis page, select **New** to create a new configuration setup file (*.dao*) that contains all the settings you make in the wizard.

   Select “Use last configuration” to load the most recently used setup file, or select “Load saved configuration” to select a specific configuration setup file.

   To save wizard settings for future analysis or channel characterization click **Save** and specify a name and path for the file.

5. On the Choose Export Type page, choose the type of model you want to create:

If you want to create a model that describes...	Do this...
Only the PDN	Select “Power-distribution network (PDN)”.
Via-via coupling within the PDN	Select “Signal vias for crosstalk through PDN”.
The PDN and signal nets	Select “Signal nets and PDN including non-ideal return paths”.
Only signal nets	Select “Signal nets only including non-ideal return paths”.

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Note

The option you choose on this page determines the configuration pages that appear in the wizard. Some of the following steps in this procedure only apply to specific model types.

6. On the Choose Probe Type page, select one of the following:
   - Automatically — Displays the Auto-assign Ports page, where you select components that enables the software to automatically assign probes for them.
   - Manually — Displays probe-related pages, where you manually assign probes.
7. (If available) On the Select Power Nets page, select power and reference nets from the "Available nets" list on the left, then click the appropriate arrow to add each net to the "Power nets to probe" or "Reference nets" list on the right.
8. (If available) On the Select Signal Nets page:
   a. Select an option to choose nets by name, net class, or from the nets currently selected in the board viewer.
   b. Select the signal nets that you want to include in the model from the "Available nets" list on the left, then click the arrow to add nets to the "Signal nets to export" list on the right.
   c. (Optional) To ensure that the Select Power Nets page includes power-supply nets associated with signal nets that you select on this page, do the following:

Note

The software associates a signal net to a power-supply net in two ways. They can associate with a passive component that connects them, such as a two-pin resistor or a resistor package. If you have assigned an IBIS model to a component with a pin that connects to a selected signal net, a [Pin Mapping] keyword can associate a signal net to power and reference nets/busses.

      i. Click **Add accompanying power nets**.
      ii. If you selected "Signal nets and PDN" on the Choose Export Type page:
         a. The software adds all associated power and ground nets to the "Power nets to probe" list on the Select Power Nets page.
         b. Go to the Select Power Nets page, move reference nets to the "Power nets to probe" list. For example, if the software added GND to the "Power nets to probe" list, move it to the "Reference nets" list.
9. On the Check Capacitor Models or Supply Component Models page:
   a. Verify model assignments and values. If needed, assign or remove models, and edit groups.

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Note

Assign a model to a resistor or inductor that connects a power supply net that you probe to a power supply net that you do not probe but want to include in simulation.

b. To perform a "what if" experiment by excluding a specific model from simulation, uncheck Enabled for the model you want to exclude.

10. (If available) On the Specify Via Ports page, select the vias that you want to include in the model.

11. On the Check VRM Models page, verify model assignments, then choose whether to probe VRM pins:

    a. To display VRMs assigned only to nets that you selected on the Select Power Nets page, check "Show VRM(s) for selected nets only".

    b. To add or edit a VRM model or reference net, click **VRM Manager**. The Assign Power Integrity Models dialog box opens. For information, see step 3 in "Assigning VRM Source, DC Sink, and AC Models" on page 103.

    c. To probe pins for a VRM, ensure that its right-most spreadsheet column has a check mark. An empty blue box in the right-most column indicates missing model information, such as a reference net.

12. (If available) On the Auto-assign Ports page, select the components that you want to probe:

    a. To enable the software to automatically create pin groups, select "For components with no user-defined pin groups, automatically create groups before creating probes".

    b. To enable the software to minimize the distance between port terminals by adding pins to neighboring reference area fills, select "For signal pins, create virtual reference pins on nearby metal".

    When this option is checked, the software creates a new pin on neighboring reference area fills that surround each signal pin. The software then creates the port between the signal pin and the newly-created pin.

    When unchecked, the software creates the port across the signal pin and an existing neighboring reference pin.

    c. If you need to manually specify pin groups, click **Manage Pin Groups**. See "Pin Group Manager Dialog Box" on page 976.

13. If you chose to specify probes manually (on the Choose Probe Type page) specify the pin pair or pin pair groups that you want to include in simulation.

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Note

You cannot select a pin pair that contains a pin that belongs to a pin group that you have selected on the Select Group Pair Probes page.

Point to an empty blue box to see the pin group name.

a. To include a pin pair in simulation, click the Select IC Pin-Pair Probes page. Check all the IC power supply pin pairs that you want to include in simulation.

   The software represents each pin pair that you select as a port in the output model. The more pin pairs you select, the longer simulation takes and the larger the model file becomes.

   If a pin pair that you want to simulate is missing from the list:

   i. Click **Edit Pin Pair Probes**. The Create IC power pin pair dialog box opens.

   ii. Click a "+" symbol for a reference designator to see a list of pins. Click a row in the supply net list and the reference net list to specify a supply and return pin pair.

   The board viewer or PDN Editor marks pin locations with blue and red arrows.

   iii. Click **Add Probe** then **OK**.

   You can also select a supply pin in the list and click **Find Reference** to automatically find the reference pin and add the probe.

b. To include a pin pair group in simulation, click the Select Group Pair Probes page. Check all the IC power supply pin pair groups that you want to include in simulation.

   The software represents each pin pair group that you select as a port in the output model. The more pin pair groups you select, the longer simulation takes and the larger the model file becomes.

   If a pin pair group that you want to simulate is missing from the list:

   i. Click **Manage Pin Groups**.

   The Pin Group Manager dialog box opens. If pin groups do not appear in the list, create them now and click **OK**. See Creating Power Supply Pin Groups.

   ii. On the Select Group Pair Probes page, specify a pin pair group that you want to include in simulation. Click **Edit Pin-Group Probes**. The Create IC pin group pair dialog box opens.

   iii. Click a "+" symbol for a reference designator to see a list of pins. Click a row to specify a supply and return pin group pair. The board viewer or PDN Editor marks pin locations with blue and red arrows.

   iv. Click **Add Probe** then **OK**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

14. On the Normalization Impedance page, use the default normalization impedance unless you intend to compare the exported S-parameter model to a model that was created with a different normalization impedance.
15. On the Choose Easy/Custom page, select Custom only if you need to edit any of the less-commonly used options on the Customize Settings page.
16. On the Customize Settings page:

If you want to reduce simulation run time and memory consumption by…	Do this…
Merging small details in complex metal shapes	Check “Defeature metal areas” and specify the minimum dimension of a feature to include in simulation.
Excluding metal located outside of a perimeter that encloses all the pin pairs and pin pair groups.	Check “Crop layout data”.
Excluding metal located on a power supply net with no probes.	Check “Include only metal for nets with probes”.

If you want to…	Do this…
Include additional power supply nets in simulation that are connected by a resistor or inductor to power supply nets you have probed.	Uncheck “Include only metal for nets with probes”.
Load the design and project into HyperLynx Hybrid Solver, where you can make design changes and run simulation.	Check “Show HyperLynx Hybrid Solver GUI”.

For a via-to-via coupling model only:

If you want to…	Do this…
Account for energy coming from transmission planes formed by other pairs of power supply nets and transmitted through stitching vias that run vertically throughout the stackup.	Check “Include all power / ground nets in the analysis”.
Account for capacitor mounting inductance	Check “Include capacitor mounting inductance”.

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If you want to…	Do this…
Reduce simulation run time and memory consumption by automatically finding stitching vias that are located close together, and then merging their individual models into an equivalent model.	Check “Enable stitching-via optimization”. See “Stitching-Via Optimization” on page 529.

17. On the Control Frequency Sweep page, type the minimum and maximum extraction frequency. Set the maximum frequency to include the largest significant frequency in your design signals.

    a. If you plan to use the model in time domain simulation, check “Include DC Point” (if the option is available for the selected model type), and enter the number of samples for the entire frequency range.

       The software uses mixed linear-log sampling.

       Note

       Time-domain simulation requires a voltage value at zero Hz in the S-parameter model. Without it, the model may simulate with incorrect DC levels (DC offset).

    b. If you plan to use the model in frequency domain simulation only, you can select one of the other sampling types:

If you want to…	Do this…
Vary the sampling step size depending on model characteristics (via-via coupling model only)	1. Select “Adaptive sampling”. Adaptive balances simulation run time and accuracy by increasing the sampling rate near frequencies with resonances. 2. Adjust the “accuracy at resonances” to specify the relative accuracy at resonant frequencies. If the design has a large number of power supply nets, hundreds of decoupling capacitors, and hundreds or thousands of stitching vias, enabling the Low option provides preliminary results with decreased analysis run time. After evaluating the preliminary results, you can identify which frequency ranges interest you the most and run analysis with higher accuracy on each range of interest. 3. Specify the number of samples the software applies to flat, non-resonant regions of a model.

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If you want to…	Do this…
Distribute sampling points across the frequency range at logarithmic intervals	1. Select “Logarithmic sampling”. The intervals between sampling points are smaller at lower frequencies and larger at higher frequencies. With logarithmic sampling, every next frequency point is equal to the previous value times a factor K > 1. 2. Enter the number of samples for the entire frequency range.
Distribute sampling points across the frequency range at equal intervals	1. Select “Linear sampling”. 2. Enter the number of samples for the entire frequency range.

18. On the Run Analysis page, rename the folder where the software saves the S-parameter file, if needed, then click **Run Analysis**.

    Note

    The **Check Data** button changes to **Run Analysis** when you have provided the minimum required information on all wizard pages.

    When simulation completes, the software displays the S-parameter model in the Touchstone and Fitted-Poles Viewer, and adds it to the list of models in the Manage Power Aware SI Models dialog box.

19. Click **Close** to exit the wizard and return to the Manage Power Aware SI Models dialog box.

20. In the Manage Power Aware SI Models dialog box, ensure that all of the models that you want to include in simulation are checked in the On column of the list.

21. Click **Close**.

Results

When simulation completes, the Touchstone Viewer displays an S-parameter model. The software saves results and wizard settings file (*.dao*) in a timestamped folder in the design folder.

To help you map Touchstone model ports to component pins, see the *<design>_<iteration>.z<number_of_ports>p.ports* file.

Modeling a Board Area With an S-Parameter Model

You can run 3-D electromagnetic (EM) simulation to create an S-parameter model that accurately represents the behavior of interconnect structures with poor return paths or other

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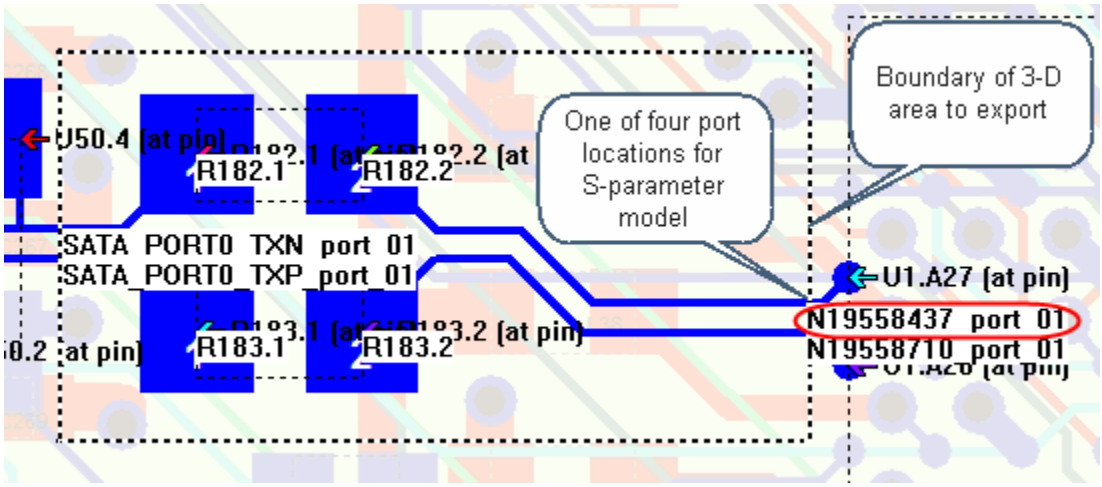
non-TEM (transverse electromagnetic mode) behaviors, for a specific area of your board design.

The software uses a “3-D area” to identify a specific subset of your board design to represent with an S-parameter model. A 3-D area encloses a three-dimensional board area and its contents, such as nets, stackup layers, signal vias, and so on.

Consider creating an S-parameter model to represent a board area (a “3-D area”) that contains:

- Differential vias that transfer signals over 6 Gb/s
- Differential vias that are near each other and likely to couple (barrel-to-barrel)
- Single-ended vias
- Traces crossing gaps or holes
- BGA breakout routing
- Series blocking capacitors located over voids
- A connector
- Traces with tight serpentine routing

Dashes in the figure below shows a 3-D area that includes series blocking capacitors.



The software can run HyperLynx DRC to automatically define 3-D areas for selected nets that include:

- Differential vias that are near each other and likely to couple (barrel-to-barrel)
- Single-ended vias
- BGA breakout routing
- Series blocking capacitors (including capacitors placed over planes)

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If your design has many similar areas that you want to model with a 3-D area, you can define a 3-D area for one area (a differential via, for example), the software can then run HyperLynx DRC to find matching areas and model them with the same S-parameter model.

Note

For a schematic design, you can create an S-parameter model that represents the behavior of signal vias and nearby stitching vias, then assign the model to a via or differential via symbol. See "Modeling a Via with a 3D EM Model in a Schematic" on page 127.

Restrictions and Limitations

- The software does not model via *backdrill setback* in 3-D areas. If you specify via backdrill information with the Backdrilling/Plating/Constraints Settings dialog box, the software ignores Setback option values when defining a 3-D area.
- The software models vias as filled. If you specify a hollow via with the Backdrilling/Plating/Constraints Settings dialog box, the software ignores "Conductively filled" option values when exporting a 3-D area.
- If you have a board design with multiple stackups, you can define a 3-D area that contains only one stackup definition.

Prerequisites

- You have enabled the option "Show anti-objects in board viewer and use in analysis" (choose the **Setup > Anti-Objects** menu item).
- You have enabled the option "Replace 3D Area with corresponding S-parameter model during SI analysis" (choose the **Setup > Options > General > BoardSim** tab).
- You have acquired the 3D Area Model Export license.
- You have acquired a license for either HyperLynx Full-Wave Solver HPC or HyperLynx Full-Wave Solver.
- If you want to use HyperLynx DRC to automatically define 3-D areas or find matching 3-D areas in your design, you have deleted any old SDD_HLDRC_HOME environment variables that may remain on your computer from older installations of HyperLynx DRC, to enable the software to use HyperLynx DRC.
- If your design has flooded plane layers, you have assigned power-supply nets to them (choose the **Setup > Power Supplies > Edit Power-Supply Nets** dialog box).
- If you plan to solve 3-D areas with multiple computers in parallel, you have set up job distribution. See "Job Distribution" in the *HyperLynx Advanced Solvers User Guide*.

Procedure

1. Choose the **Analyze 3D EM > Manage Areas for 3D EM Simulation** menu item.

   The 3D Area Manager dialog box opens.

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2. (Optional) If you are an experienced HyperLynx Advanced Solver customer who wants detailed control over the geometric model and solver settings used to create the S-parameter model representing a 3-D area, specify a folder location in the HLAS Options Folder field.

   This folder contains options files (*.opt*) for Full-Wave Solver HPC or another HyperLynx Advanced Solvers tool. See "HyperLynx Advanced Solvers Option Files for 3-D Areas" on page 551.

3. Define 3-D areas for your design:

If you want to...	Do this...
Automatically define 3-D areas for selected net(s)	1. To review or edit 3-D area creation options, click **Options**, specify option values in the 3D-Area Creation Options dialog box, then click **OK**. Pad-to-pad distance thresholds and other settings determine the nets and objects that the software includes in a 3-D area. If needed, see "3-D Areas That Include Multiple Nets" on page 467. 2. If you have opened a multiple-board project, enable only one board in the "Filter nets and areas by boards" area. 3. If you have not already selected a net or nets, click **Select Nets**, then select nets that you want to include in a 3-D area. 4. Click **Create Areas for Selected Nets**. The software adds new 3-D area and model names to the table. 3-D area name format is *3D_<victim_net name>_x_and_y_coordinates_for_3-D_area_center>_<primary_net_quantity>_PNT*. Model name format is *3D_PV_<signal_pin_and_via_quantity>_P_<port_quantity>_N_<signal_net_quantity>_<x_and_y_coordinates_for_3-D_area_center>.sNp*. 5. (Optional) Review the nets and objects included by a 3-D area: a. Check "Zoom to selection". b. Use the left (unlabeled) table column to select a 3-D area. c. Verify that the 3-D area includes the vias or other design objects that you want. d. If needed, repeat steps 1 - 4.

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If you want to...	Do this...
Manually define a 3-D area	Perform the steps described in "Exporting Part of a Board Design for Analysis in HyperLynx Full-Wave Solver" on page 417. **Note:** If you define a 3-D area that overlaps another, you can enable the S-parameter model assignment for one overlapping area at a time. For example, if you define a large 3-D area with multiple differential vias, define small 3-D areas with one differential via (that is located within the large 3-D area), you can enable either the large 3-D area or the small 3-D areas.

4. Match a 3-D area to similar areas of your design:

   a. In the 3D areas table, use the left (unlabeled) table column to select one or more rows for 3-D areas that you want to match to other places in your design.

   b. Choose a search scope:

      - To search selected nets for matching 3-D areas, select nets, then click **Find Matches for Highlighted Areas**.
      - To search all nets for matching 3-D areas, de-select all nets (you can Ctrl-Click a selected net in the Select Net by Name dialog box), then click **Find Matches for Highlighted Areas**.

      The software adds any matching 3-D areas to the table and assigns them names ending with *<existing 3-D area name>_M<x coordinate>_<y_coordinate>*.

5. To create S-parameter models and assign them to 3-D areas, use the left (unlabeled) table column to select one or more rows for 3-D areas, then do one of the following:

If you want to...	Do this...
Solve using one computer	Click **Solve Highlighted Areas**.
Solving using multiple computers in parallel	1. Click **Solve on Multiple Machines**. The HyperLynx Job Distribution dialog box opens and loads the HyperLynx Advanced Solvers project file (PHYS) for each 3-D area you selected. You can use this dialog box to solve a 3-D area with multiple computers in parallel. See "Job Distribution" in the *HyperLynx Advanced Solvers User Guide.* 2. Specify job distribution options and solve.

The software writes S-parameter models to the folder containing the design file.

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Results

You are ready to run simulation. When you run simulation that includes any nets that are part of a 3-D area, the software uses the assigned S-parameter model to represent that part of the net. You can temporarily disable the S-parameter model assignment and use standard board design information to represent a 3-D area. From the 3D Area Manager dialog box, uncheck the “On” column for a 3-D area.

You can evaluate a generated S-parameter model with the Touchstone Viewer.

Note

When you include a passive component in a 3-D area and choose to add ports for it, the software includes two ports in the S-parameter model - one for the component and one at the component pads. If you export a net that includes the component to a schematic, the software represents the passive component correctly, but the schematic may appear unintuitive. See KB article MG594821 on Support Center.

Related Topics

Setup Anti-Pads and Anti-Segments Dialog Box

Preferences Dialog Box - BoardSim Tab

Add or Edit 3D Area Dialog Box

Viewing Touchstone and Fitted-Poles Model Curves

Modeling a Via with a 3D EM Model in a Schematic

You can run 3-D electromagnetic (EM) simulation to create an S-parameter model that accurately represents the behavior of a via or differential via symbol in your schematic design.

Consider generating an S-parameter model for differential vias that transfer signals over 6 Gb/s, or vias that are near each other and likely to couple (barrel-to-barrel).

Prerequisites

- You have acquired a 3D Area Model Export license.
- You have acquired a Full-Wave Solver license.
- If you plan to model the cross section for feeding traces as a trapezoidal shape, you have acquired the Lossy Lines license.
- You are familiar with the basics of creating a schematic design. See “Creating a Schematic Design” on page 70.

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Procedure

1. Double-click a via or differential via symbol in a schematic design.
2. In the Via Properties dialog box, next to 3D EM Modeling, select HyperLynx Full-Wave Solver.
3. Next to 3D EM Project File, click **New**.
4. In the New Full-Wave Solver Project dialog box, type a name for the file, select Default, and click **OK**.

   Return to the Via Properties dialog box.

   Note

   To visually confirm that you have correctly specified via geometric properties in any of the steps that follow, click **View 3D**. It may take a few moments for HyperLynx Full-Wave Solver to display via geometries.

5. On the **Via Properties** tab:
   a. For a design with multiple stackups, select the stackup that describes the via from the Stackup list.
   b. For a multiple-board design, select the board that contains the via from the Board list.
   c. Select the padstack that describes the via. If you want to select an SMD padstack, check “Show SMD padstacks”.

      For a differential via pair, specify whether to model a common anti-pad and specify via barrel separation distance. If needed, enable backdrill modeling and define backdrilling properties. See “Accounting for Backdrilling” on page 135.
   d. (Optional, to perform “what if” analysis.) Uncheck “Use Padstack” and either edit the distance from the via to a metal area on the same layer or specify “No metal”.
6. On the **Backdrilling Settings** tab, enable backdrilling and specify the backdrilling layer and diameter settings that you want to include in the model. To enable backdrilling, see “Accounting for Backdrilling” on page 135.

   Note

   The software does not model via *backdrill setback*. The software ignores Setback option values.

7. On the **Stitching Vias** tab, describe any stitching vias that you want to include in the model:
   a. To add stitching via(s), click **Add** or **Multiple Add**.

      The Add/Edit Stitching Via dialog box opens.

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b. Select the padstack that describes the via, then specify its layer connections.

c. Do either of the following:

- To add multiple stitching vias, click **Multiple Add** and specify the number of stitching vias and their distance from the signal via.

  The software evenly distributes the stitching vias around a signal via. You can later refine their locations when you return to the Via Properties dialog box.

- To add one stitching via, click **Add** and specify its X/Y coordinates (relative to the signal via).

d. If needed, enable backdrill modeling and define stitching via backdrilling properties. See "Accounting for Backdrilling" on page 135.

e. Click **OK**.

Return to the Via Properties dialog box.

8. On the **Feeding Traces** tab, add a check mark for each layer connected to the signal via, then select a connected layer and describe the feeding traces.

   To attach a feeding trace to the left side of the signal via, set Direction to Input. To attach a feeding trace to the right side of the signal via, set Direction to Output.

Note

You can accidentally assign stitching vias and feeding traces to an overlapping location. If this happens, the software reports an error when you click **OK** or **View 3D**. Manually adjust stitching via or feeding trace locations to fix the problem.

9. On the **Global Parameters** tab:

Note

For detailed definitions of options on this tab, see "Via Properties Dialog Box - Global Parameters Tab" on page 1068.

a. Specify the model type and boundary size.

b. Check Use Absorbing Boundaries to eliminate artificial resonances in a model by representing the area boundary edges as an absorbing material (PCB dielectric, not air).

c. Specify a sampling frequency range by doing either of the following:

- Check Minimum, then enter the meshing frequency and the number of frequencies written to the S-parameter model.

- Uncheck Minimum to enable the software to calculate these values for you.

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d. To omit via pads on stackup layers that do not connect to feeding traces, check Remove Non-Functional Pads.

Note

Simulation does not omit pads on the top and bottom stackup layers because they are required for the plating process during PCB manufacturing.

e. Specify the default anti-pad clearance and via plating material.

f. To specify conductor surface roughness properties, check Enable Surface Roughness, then specify properties.

g. If the via has a feeding trace routed on a plane stackup layer, specify the default distance from each side of a feeding trace to plane metal.

h. To model the cross section for feeding traces as a trapezoidal shape, check "Use Trapezoidal trace shapes".

10. Click **Simulate** to run simulation and create the S-parameter model. Refer to the Output window to see simulation (3-D EM solving) progress.

Results

When simulation completes, the software saves the generated model with the 3-D via project file (*.v3d*). For example, *...\MentorGraphics\<release>\SDD_HOME\hyperlynx64\HypFiles*.

To assign the generated model to another via symbol, perform steps 1 and 2, then browse to the newly-generated *.v3d* file. For additional information, see "Via Project File and S-Parameter File Reuse" on page 1079.

Related Topics

New HyperLynx Full-Wave Project Dialog Box

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Setting Simulation Options

Before you run simulation, set options to account for coupling, loss, signal, and noise between single-ended via and power planes.

Topic	Description
Accounting for Coupling	Account for coupling in your design by enabling trace coupling and specifying electrical or geometrical criteria that identifies coupled nets to include in simulation.
Accounting for Loss	To account for dielectric and metal losses in simulation, enable lossy transmission-line modeling. You can also account for loss due to metal surface roughness.
Accounting for Backdrilling	You can increase simulation accuracy by accounting for the removal of via stubs (backdrilling).
Accounting for Per-Padstack Plating and Constraints	You can increase the simulation accuracy by accounting for padstack-specific plating properties, such as plating thickness, filled/not filled, and metal type. You can also specify a maximum via current constraint for each padstack.
Assigning a Stimulus	Assign a stimulus to drivers in your design before you run SI simulation with the Digital Oscilloscope or EZwave, or General Batch Simulation. You can assign a global stimulus that applies to all driver pins, or define multiple driver waveforms and assign them to specific driver nets or pins.

Accounting for Coupling

Account for coupling in your design by enabling trace coupling and specifying electrical or geometrical criteria that identifies coupled nets to include in simulation.

When you select a net for simulation, that net is the victim net, and any nets that exceed the coupling threshold settings are included in simulation as aggressor nets.

To account for coupling in schematic design simulations, you create coupling regions that define spatial relationships between coupled nets or net segments. See “Creating a Schematic Design” on page 70.

The following simulation types use independent crosstalk threshold settings. These settings are available from the dialog boxes associated with each simulation type:

- General Batch simulation
- SERDES Batch simulation
- DDRx simulation

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- IBIS-AMI and FastEye channel analysis - for SERDES designs

The following video contains additional information. Note that the video does not describe the current set of Coupling Settings dialog box options. See “Coupling Settings Dialog Box” on page 756.



Restrictions and Limitations

- The software always recognizes nets that are identified as differential pairs as coupled nets. For help specifying differential pairs, see “Verifying That Differential Pairs are Recognized Correctly” on page 65.
- The software always recognizes nets that are selected together as coupled nets. For help selecting multiple nets at a time, see “Selecting Nets for SI Simulation” on page 141.
- The software does not account for coupling between nets routed by *Manhattan routing*. Manhattan routing does not provide the detailed trace routing information required to account for coupling.

Prerequisites

You have the Crosstalk license.

Procedure

1. Enable crosstalk simulation. Select the **Setup > Enable Crosstalk Simulation** menu item ( ).

   Note

   Turn crosstalk simulation on or off to see the effect of crosstalk on your design.

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2. Select a method for defining the nets that are included in simulation. Select the **Setup > Coupling Thresholds** menu item. The Set Coupling Thresholds dialog box opens.

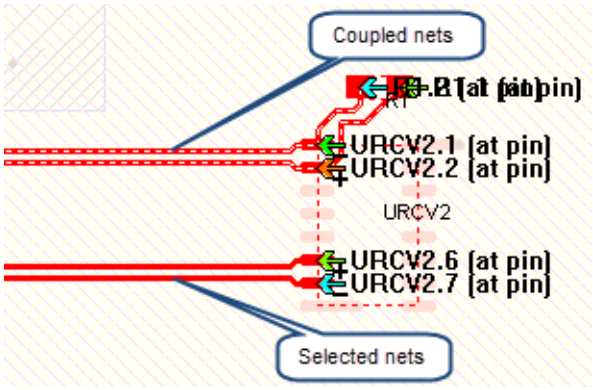
If you want to ...	Do this ...
Include nets by specifying the maximum acceptable combined voltage contribution to victim nets from aggressor nets.	1. Select **Use electrical thresholds**. 2. Specify the threshold voltage. 3. Specify the default model values that the software assigns to any components that do not already have models assigned to them. Simulation includes aggressor nets that exceed this threshold.
• Include a specific number of nearest aggressor nets in simulation, regardless of the voltage contribution from these nets. • Account for coupling to area fills and hatched planes. • Run sweep simulations with the oscilloscope or EZwave™ waveform viewer.	1. Select **Use geometric thresholds**. 2. Define a spatial area around a victim net to find aggressor nets to include in simulation: • Maximum distance (center-to-center) between the victim and aggressor nets. • Minimum coupled length, which represents the sum of one or more trace segment lengths on the selected victim net. • Maximum number of nearby nets and layers. **Note:** For information about how the software applies geometric thresholds, see “Horizontal and Vertical Geometric Search Range for Coupled Nets” on page 546.

3. To set trace-to-trace coupling settings, S-parameter/package coupling settings, or more detailed geometric threshold options such as trace-to-area-fill coupling settings, click **Coupling Settings**. The Coupling Settings dialog box opens.

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Results

You are now ready to select nets for simulation. When you select nets for simulation, the board viewer displays aggressor nets with dashed highlighting. See “Selecting Nets for SI Simulation” on page 141.



To examine coupled nets in more detail, select a net and choose the **View > Coupling Regions** menu item, or right-click a net in the board viewer and choose the **Walk Coupling Region** menu item. See “Viewing Coupling Region Details” on page 612.

Related Topics

Set Coupling Thresholds Dialog Box

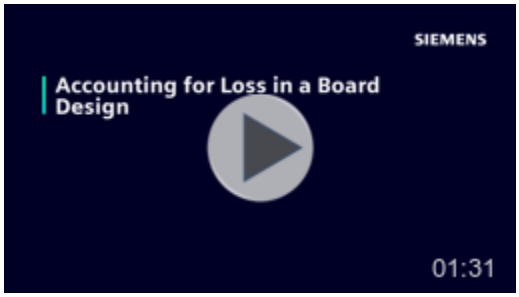
Coupling Settings Dialog Box

Accounting for Loss

To account for dielectric and metal losses in simulation, enable lossy transmission-line modeling. You can also account for loss due to metal surface roughness.

As signal frequency increases, diminishing skin depth approaches the surface roughness amplitude, and surface roughness contributes more significantly to signal attenuation.

The following video contains additional information:



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Prerequisites

- You have the Lossy Lines license.
- If you want to model surface roughness, you have the Surface Roughness license.

Procedure

1. Enable lossy transmission line modeling. Select **Setup > Enable Lossy Simulation** ().
2. If needed, select **Setup > Enable Surface Roughness**.

   Note

   You can turn off surface roughness modeling to decrease simulation run time if your board design does not include a dielectric material with extremely low loss.

3. Edit material properties that affect loss, if needed.

If you want to...	Do this...
Specify the dielectric loss tangent and bulk resistivity of a trace/plane and its coating.	Edit the stackup definition in the Stackup Editor (**Setup > Stackup > Edit** ). Enter specific values for bulk resistivity by creating a Custom metal on the Metal tab.
Specify the bulk resistivity and loss tangent for a specific transmission line in a schematic.	Edit material properties on the Values tab of the Edit Transmission Line dialog box (double-click a transmission line in a schematic).
Specify surface roughness properties.	Edit the stackup definition in the Stackup Editor (**Setup > Stackup > Edit** ). Enter specific values for surface roughness on the Manufacturing tab.

Related Topics

Stackup Editor - Metal Tab

Stackup Editor - Manufacturing Tab

Edit Transmission Line Dialog Box - Values Tab

Accounting for Backdrilling

You can increase simulation accuracy by accounting for the removal of via stubs (backdrilling).

Backdrilling improves signal integrity by removing a via stub’s impedance discontinuity. Backdrilling has a degrading effect on power integrity if a via is drilled to a diameter that is larger than the antipad, or a buried via or blind via is drilled through an outer plane layer

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because this reduces the metal in the power-distribution network (PDN) available for current (supply and return) flow.

Note

To account for backdrilling in schematic designs, you can use either the Via Properties dialog box or the **Backdrilling** tab of the Backdrilling/Plating/Constraints Settings dialog box to specify backdrilling information.

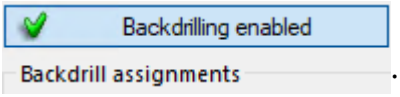
Restrictions and Limitations

- For designs formatted in ODB++, the software supports backdrill data in ODB++ 8.0 and newer.
- The software does not support backdrilling for board designs with multiple stackups.

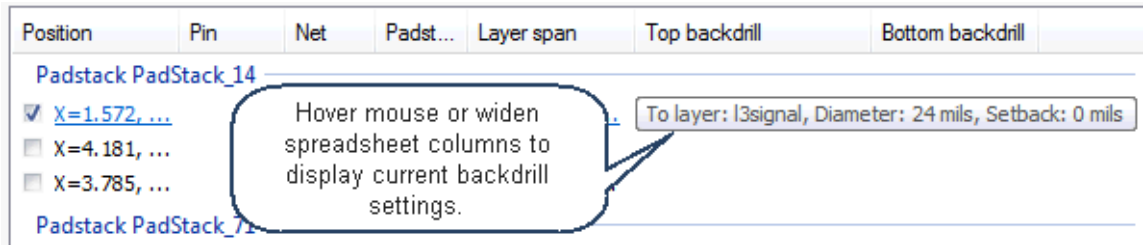
Prerequisites

- You have acquired the Via Models license.
- Although Xpedition Layout and Board Station XE automatically create backdrill information when you run backdrill analysis before exporting a design, other layout software does not create backdrill information in an exported design.
- You have obtained drill diameter(s) and backdrill setback values from your PCB manufacturer.

Procedure

1. Choose the **Setup > Backdrilling/Plating/Constraints Settings** menu item. The Backdrilling/Plating/Constraint Settings dialog box opens.
2. On the **Backdrilling** tab, ensure that the button near the top of the dialog box displays "Backdrilling enabled" 

. If it does not, click the "Backdrilling disabled" button.
3. If you have loaded a MultiBoard project, select a board.
4. Review the current settings. Filter the set of vias by net or padstack as needed to display detailed information in the spreadsheet. Type an asterisk (*) to match any number of characters and a question mark (?) to match any one character.



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

5. To add or edit settings, in the “Backdrill assignments” area, select vias by adding check marks, specify information, then click **Assign**.

   The default behavior for “Backdrill from Top/Bottom to layer” <Auto> is to remove all layers with pads that do not connect to traces or planes. The default value for Diameter <Auto> is the pad diameter. The default value for Setback <Auto> is 0.

   (BoardSim only) If you want to manually specify a backdrill layer, but do not already know the layer span for a via, you can click its spreadsheet row to highlight its location in the board viewer, temporarily close this dialog box, right-click the via to open the Via Visualizer and display the layer span, close the Via Visualizer, then re-open this dialog box.

Related Topics

Accounting for Per-Padstack Plating and Constraints

Accounting for Per-Padstack Plating and Constraints

You can increase the simulation accuracy by accounting for padstack-specific plating properties, such as plating thickness, filled/not filled, and metal type. You can also specify a maximum via current constraint for each padstack.

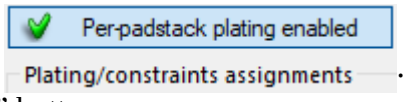
Restrictions and Limitations

- The software does not support per-padstack plating and constraints for board designs that include multiple stackups.

Prerequisites

- You have obtained plating and constraint values provided from your PCB manufacturer.
- You have acquired the Via Models license.

Procedure

1. Choose the **Setup > Backdrilling/Plating/Constraints Settings** menu item. The Backdrilling/Plating/Constraint Settings dialog box opens.
2. On the **Plating/Constraints** tab, ensure that the button near the top of the dialog box displays “Per-padstack plating enabled” 

. If it does not, click the “Per-padstack plating disabled” button.
3. If you have loaded a MultiBoard project, select a board.
4. Filter the set of vias by net or padstack as needed to display detailed information in the spreadsheet. Type an asterisk (*) to match any number of characters and a question mark (?) to match any one character.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

5. To add or edit settings, in the “Plating/constraints assignments” area, select via(s) by adding check mark(s), specify plating and constraint information, then click **Assign**.

   The Metal option value determines the Resistivity value. To specify a custom Resistivity value, set the Metal option to Custom, then type a value into the Resistivity field.

   The **Default Padstack** tab of the Preferences dialog box provides default plating values. See the table section “Via Barrel Plating” in “Preferences Dialog Box - Default Padstack Tab” on page 988.

   The default “Max via current” value is no constraint (unlimited via current).

Related Topics

Accounting for Backdrilling

Assigning a Stimulus

Assign a stimulus to drivers in your design before you run SI simulation with the Digital Oscilloscope or EZwave, or General Batch Simulation. You can assign a global stimulus that applies to all driver pins, or define multiple driver waveforms and assign them to specific driver nets or pins.

The software provides several types of stimulus, including edge, pulse, PRBS, 8B/10B, and 128B/132B. You can also define a custom stimulus.

You can assign a stimulus to individual nets or pins to simulate timing relationships between nets/pins when you want to:

- Analyze crosstalk with different waveforms on aggressor nets, to help examine the pattern dependency of crosstalk
- Simulate a design that uses source-synchronous signaling, such as DDRx and similar technologies, where one IC transmits both the clock and data signals (as opposed to a master system clock, which is transmitted by a different IC) with slightly different timing

When you save a board design, the software saves stimulus assignments to the PJH file (not the BUD file) to ensure that a net spanning multiple boards in a multiple-board project has only one stimulus assignment. When you save a schematic design, the software saves stimulus assignments to the FFS file.

Prerequisites

- Verify your design. See Verifying That the Software Recognizes Your Design Correctly.
- If you want to assign a stimulus to a specific pin in a board design, select the net that connects to the pin. See Selecting Nets for SI Simulation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

1. To assign a stimulus to a specific net or pin, choose the **Setup > Stimulus** menu item.

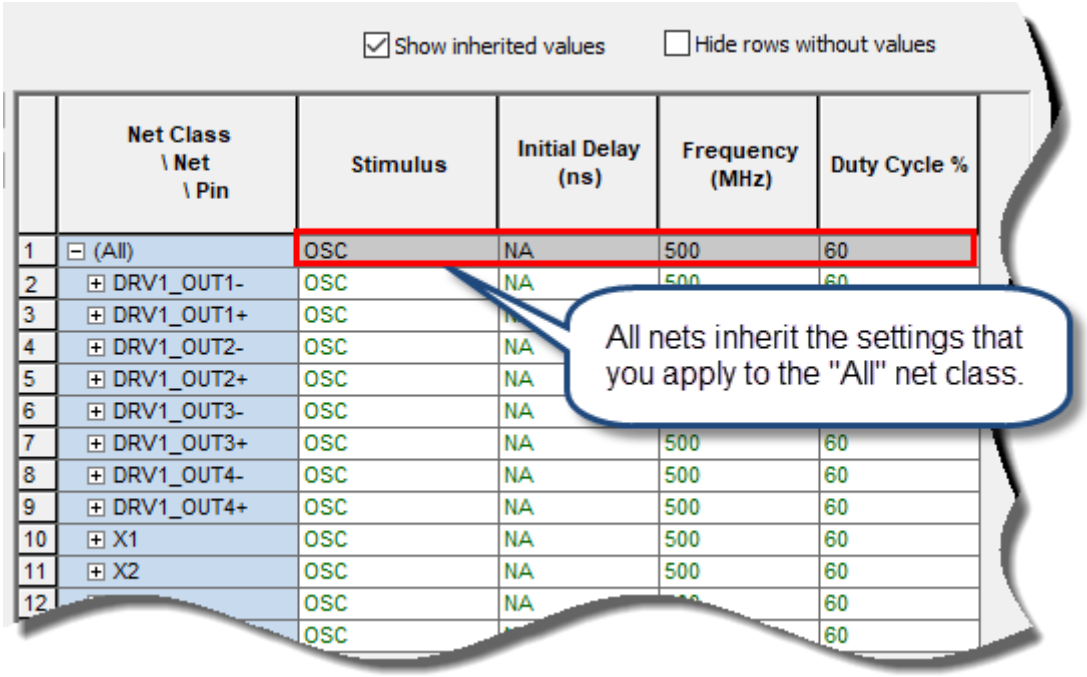
   The Assign Stimulus dialog box opens.

   Note

   If you have assigned models, you can edit stimulus settings by double-clicking a stimulus setting in the Manage Assigned Models dialog box (**Models > Manage Assigned Models**).

2. In the Assign Stimulus dialog box:

   a. In the "All" net class row, click in the Stimulus column, select a stimulus type from the list, and type default initial delay, frequency and duty cycle values that apply to all nets.

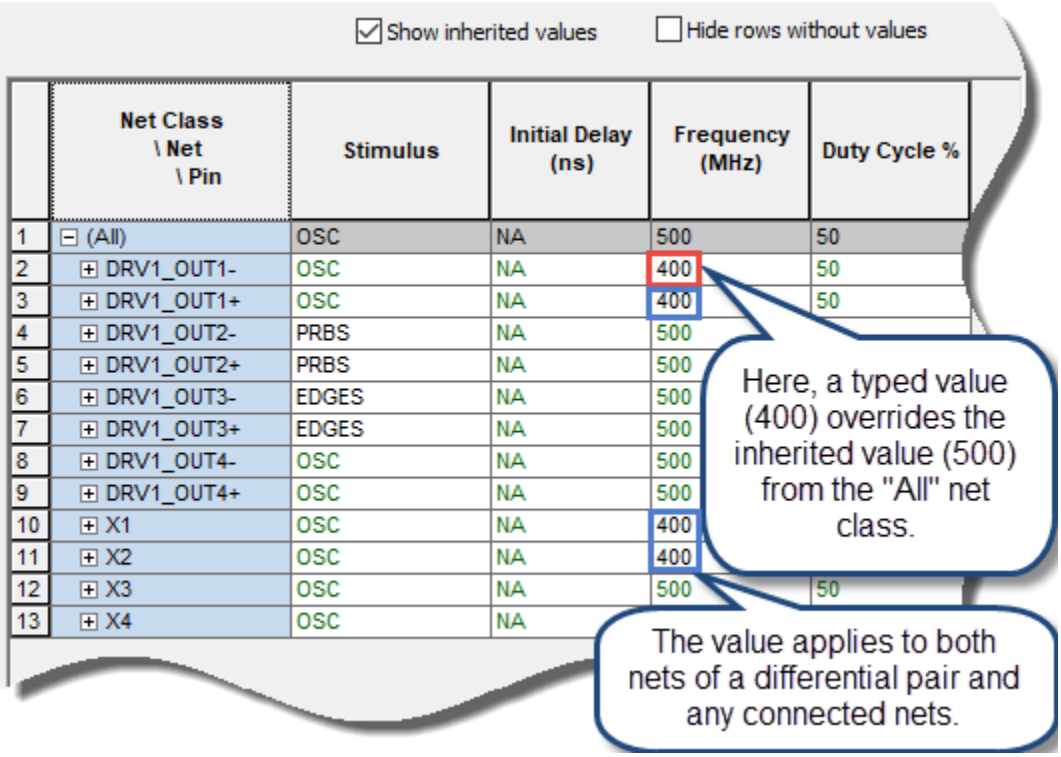


   b. If your design contains definitions for net classes, you can assign a stimulus to groups of nets by editing the stimulus setting for a net class row. Nets within the class inherit the stimulus setting. To define net classes, see "Defining Net Classes" on page 69.

   c. Click and edit a setting for any specific net or pin rows that require a different stimulus setting.

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When you assign a value to a net row, pins connected to the net inherit the settings. Any other connected nets also inherit the settings, along with the other net of a differential pair.



	Net Class \ Net \ Pin	Stimulus	Initial Delay (ns)	Frequency (MHz)	Duty Cycle %
1	⊟ (All)	OSC	NA	500	50
2	⊞ DRV1_OUT1-	OSC	NA	400	50
3	⊞ DRV1_OUT1+	OSC	NA	400	50
4	⊞ DRV1_OUT2-	PRBS	NA	500	
5	⊞ DRV1_OUT2+	PRBS	NA	500	
6	⊞ DRV1_OUT3-	EDGES	NA	500	
7	⊞ DRV1_OUT3+	EDGES	NA	500	
8	⊞ DRV1_OUT4-	OSC	NA	500	
9	⊞ DRV1_OUT4+	OSC	NA	500	
10	⊞ X1	OSC	NA	400	
11	⊞ X2	OSC	NA	400	
12	⊞ X3	OSC	NA	500	50
13	⊞ X4	OSC	NA		

d. If the stimulus you need is not available, click **Edit Stimulus** to create a stimulus. See Creating a Custom Stimulus.

e. To import stimulus settings that you have already assigned to nets in a separate DDRx batch simulation, click **Import DDRx Stimulus**, select the DDRx stimuli list *.txt* file, and click **Open**.

3. If you run Digital Oscilloscope or EZwave (Interactive Simulation) simulation, you can assign a global stimulus that applies to all nets. In the Stimulus area of the Digital Oscilloscope or EZwave (Interactive Simulation) dialog boxes, select **Global** and select edge or oscillator options as needed.

   For a board design, global stimulus is applied to all enabled driver pins on the selected net and its associated nets. For a schematic design, global stimulus is applied to all enabled driver pins in the schematic.

Note

A global stimulus overrides any stimulus settings for specific nets or pins in the Assign Stimulus or Manage Assigned Model dialog boxes.

Related Topics

Assign Stimulus Dialog Box

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Selecting Nets for SI Simulation

Some non-batch signal integrity simulation types require you to select the nets that you want to include in simulation by picking nets in the viewer or from a list in a dialog box.

When you select nets for simulation, nets that are connected or coupled to the selected net are also included in simulation, including nets that span multiple boards. Nets can be connected directly, by passive components, or by association in assigned IBIS models (differential pairs).

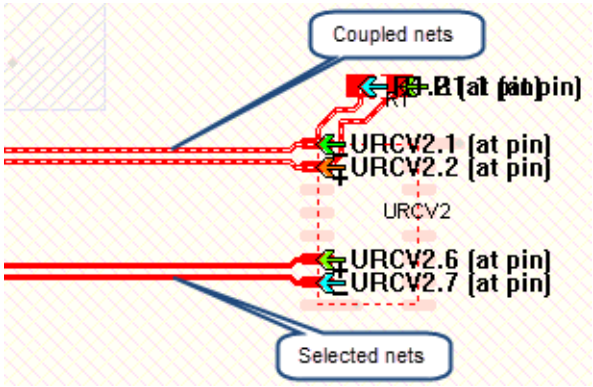
Note

Power supply nets are not included in signal integrity simulation, even if they are connected to signal nets.

Note

You can use the procedure in this topic to select nets for non-SI simulation tasks, such as exporting a SPICE netlist, modeling a board area with an S-parameter model, and exporting a power-distribution network (PDN) to an S-parameter model. For information, see the Related Topics section below.

When you select nets for simulation, the nets that are coupled to selected nets are displayed in the viewer with dashed highlighting.



Nets that are not selected display as dimmed.

The following video contains additional information about selecting nets:



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Restrictions and Limitations

- Batch signal integrity simulation types provide a separate spreadsheet or wizard to help you select nets.
- Acquire the Crosstalk license to select multiple nets. Crosstalk simulation automatically enables when you select multiple nets. If you do not have the Crosstalk license, only the first net you select is recognized when you select more than one net.

Prerequisites

- You have verified that power supply and signal nets are recognized correctly. See Verifying That Power Supply and Signal Nets are Recognized Correctly.
- You have set coupling thresholds to specify the set of coupled nets to include in the set of selected nets. Accounting for Coupling.

Procedure

1. Select nets.

If you want to...	Do this...
Select from a list of nets, sorted by net name, length or width.	1. Choose **Select > Net by Name for SI Analysis** (). 2. Select a net, or hold **Ctrl** or **Shift** to select multiple nets from the list and click **OK**. **Note:** For board designs only, net lengths that appear in this dialog box depend on the "Remove redundant metal from a board's nets as the board is loaded" setting (**Setup > Options > BoardSim** tab), and do not include the length of associated nets.
Select nets connected to a pin from a list of reference designators.	Choose **Select > Net by Reference Designator for SI Analysis**.
Click a net in the board viewer (board designs only).	1. Right-click the net and click **Select Net <net name>**. 2. To add or remove a net from the set of selected nets, right-click the net and click **Include in selection** or **Exclude from selection**.
Click a net in the schematic viewer (schematic designs only)	Right-click the net and click **Select Net <net name>**.

2. Verify your net selection is correct.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Results

Now you can assign models. See Assigning Models to Components and Pins.

To see more information about the models and stimulus assignments for components connected to the set of selected nets, select **Models > Manage Assigned Models**.

Related Topics

Assigning Models to Components and Pins

Exporting a Net to a SPICE Netlist

Exporting a PDN to an S-Parameter Model

Modeling a Board Area With an S-Parameter Model

Editing a Padstack Definition

Context: LineSim

Use the Padstack Manager dialog box to define padstacks for a schematic. Padstack information is stored in the schematic file (*.FFS*).

Restrictions and Limitations

- The padstack definitions shown in the Padstack Manager are unique to each schematic. You cannot create a padstack library and share padstacks with other schematics.
- If you have a schematic design with multiple stackups, such as a schematic with a signal net that spans a rigid area and a flexible area, a new padstack uses the master stackup. You can use the Via Properties dialog box to assign a local stackup to a via symbol. To display the padstack for a via with a local stackup, open the Padstack Editor dialog box from the Via Properties dialog box.

Prerequisites

- You have acquired the Via Models license.

Procedure

1. Open a schematic and select **Setup > Padstacks**. The Padstack Manager dialog box opens.
2. Double-click the default padstack to edit it, or click **New** to create a new padstack.

   Note

   If a padstack consists of pads of different sizes, the Pad-Size column shows the largest size. Round shapes are indicated by D (circular) or O (oval or oblong).

3. Describe the padstack in the PadStack Editor.

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You can describe padstack properties for each layer specifically, or edit the definition for the default layer. The software applies the default layer description to all layers for which you have not created a specific description. When describing properties for a layer, define both pad and anti-pad properties, so that you can use the same padstack definition for both plane and signal layers when needed.

a. In the Layer Span area, specify the layers that define the ends of the via barrel. <top> and <bottom> layers represent the outer layers of the board, no matter what the actual layer names are.

   To model a surface mount device (SMD) pad, specify the same surface layer in both the From and To fields, and select **SMD**.

b. Type the inside diameter of the via barrel in the Finished Diameter box.

a. To edit layer properties, click the cell and select or type the property value.

   For the Pad Shape cell, <None> specifies that no pad exists on this layer and Oblong specifies a rectangle with routed corners. Also, <None> specifies a round pad with the same diameter as the drill size.

   For the Anti-Pad Shape cell, <Auto> specifies an anti-pad with the same shape as the pad and using the value from the Clearance box in the Default Padstack tab of the Preferences dialog box (**Setup menu > Options > General**).

4. Click **OK**.

Related Topics

Padstack Manager Dialog Box

Padstack Editor Dialog Box

Preferences Dialog Box - Default Padstack Tab

Board Designs With Multiple Stackups

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Preparing for DDRx Simulation

Find and fix common setup problems, gather and assign model files, gather reference designators for controller and memory components, and so on.

Decide whether to use S-parameter models to represent portions of critical nets that contain vias, DC blocking capacitors, BGA breakout fields, or other structures with poor return current paths or non-TEM (transverse electromagnetic mode) behaviors. For a board design, see "Modeling a Board Area With an S-Parameter Model" on page 122. For a schematic design, see "Modeling a Via with a 3D EM Model in a Schematic" on page 127.

Decide whether to account for board and package power-distribution network (PDN) effects. See "Accounting for Board and Package PDN Effects in SI Simulation" on page 113.

If a memory interface includes signal nets with series capacitors, the software can assign a settled voltage to the capacitors when you enable the "Settle series capacitors before every simulation" option on the Preferences Dialog Box - Advanced Tab. Enabling this option means that simulation can obtain a settled voltage for a series capacitor on a signal net without having you run a very long time-domain simulation or assign a special ADMS or SPICE model to the differential pair.

Topic	Description
Strategy to Quickly Find and Fix DDRx Simulation Setup Problems	Find and fix common setup problems on a subset of nets in a DDRx memory interface before simulating all nets.
Required Information for DDRx Simulation	Gather and create files, and gather information required to run DDRx simulation.
DDRx AMI Simulation Flows	If your DDRx memory interface has single-ended nets with IBIS-AMI models that represent equalization capabilities, the software provides multiple simulation flows to choose from.
Specifying Locations for Stacked-Die DRAMs	DDRx simulation supports stacked-die DRAMs with IBIS models when you have an EBD/EMD model for your module that points to IBIS models which define the DRAMs.

Strategy to Quickly Find and Fix DDRx Simulation Setup Problems

Find and fix common setup problems on a subset of nets in a DDRx memory interface before simulating all nets.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The following steps can help you to find and fix setup problems more quickly than if you immediately simulated the full interface:

1. Use the oscilloscope to simulate a net from each interface signal category (such as data, address, command, and control), then fix any setup problems. For nets with multiple drivers, you may want to run a simulation for each enabled driver.
2. If you have a SPICE/S-parameter connector model in a multiple-board project, ensure that DDRx nets correctly pass through a board-to-board connector model by selecting a net and inspecting the connectivity.
3. Run DDRx batch simulation on:
   a. A few nets from each interface signal category to find additional setup problems, such as driver and ODT settings.
   b. All nets for an interface.

Required Information for DDRx Simulation

Gather and create files, and gather information required to run DDRx simulation.

Design Files

Design Type	Description
Board design	A design with all DRAM components on a main board requires only a single design file. A design with DIMMs requires you to create a multiple-board project that contains: • A main board with slots for one or more DIMMs. • A plug-in board for each DIMM. If your design uses a DIMM plug-in board multiple times, use the same board file for each instance. If your design uses unique DIMM plug-in boards, such as a 1-rank DIMM and a 2-rank DIMM, use a separate board file for each instance. See “Setting Up a Multiple Board Design” on page 57.
Schematic design	Create a schematic that represents a section of your DDRx interface. Assign the same reference designator to all pins that belong to the same DRAM or controller component. This enables the software to identify the IC symbols that collectively belong to the controller and DRAM components, based on the reference designator assignment. See “Preparing a Schematic for DDRx Batch Simulation” on page 1225.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Model Files

Model Type	When to Assign	Description
IBIS	Before opening a DDRx wizard	The [Model Spec] and [Receiver Thresholds] keywords provide voltage threshold and other measurement information. The [Model Selector] keyword identifies the on-die termination (ODT) component model to use for simulation. If a model you receive from the DRAM or controller component vendor does not contain the [Model Selector] keyword, manually add it. See "Adding Model Selector Keywords to IBIS Models" on page 625. **Note:** If an IBIS model for a controller has a single [Model Selector] keyword, you can use the IBIS Model Selector Editor (**Models > Update Model Selectors in IBIS** menu item) to create a new IBIS model with a separate [Model Selector] keyword for strobe and data nets. Use a *.ref* or *.qpl* file to assign IBIS or EBD/EMD models to entire controller and DRAM components, and PLL and register components for registered DIMMs (RDIMMs). See "Assigning a Model or Value to an Entire Component Using a .REF File" on page 93 and "Assigning a Model or Value to an Entire Component Using a .QPL File" on page 96.
IBIS-AMI (for DDRx types that implement equalization)	Before opening a DDRx wizard	An IBIS-AMI model represents signal equalization behaviors for drivers and receivers in controller and DRAM components. Assign model files for drivers and receivers in your design, including IBIS-AMI files and their associated executable files, and have copied them to the correct location (**Models > Edit Model Library Paths** menu item). Store *.dll* or *.so* files in the same folder as the IBIS model or in another folder displayed in the "Model-library file path(s)" list in the Set Directories dialog box. If you want to account for driver and receiver filtering, you have assigned IBIS-AMI models with GetWave functions (reserved parameter GetwaveExists = True). If you plan to edit or sweep equalization parameters, you have obtained detailed parameter descriptions from the IBIS-AMI model vendor.
"Power aware" IBIS	Before opening a DDRx wizard	If you plan to account for simultaneous switching noise (SSN), you have assigned power-aware IBIS models that include the keywords [ISSO_PD], [ISSO_PU] and [Composite Current]. See "Accounting for Non-Ideal Power Supplies in SI Simulation" on page 113.

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Model Type	When to Assign	Description
EBD/EMD	Before opening a DDRx wizard	In a multiple-board design, an EBD/EMD model usually represents a board, such as a DIMM. See "MultiBoard Simulation with EBD/EMD Models" on page 1215. In LineSim, you must use a *.ref* file to assign an EBD/EMD model.
Timing	When you run a wizard	Timing models represent timing information provided by DRAM and controller component datasheets. Timing models have a *.v* filename extension and must be formatted in the HyperLynx timing model format. See "Creating Controller and DRAM Timing Models" on page 655. In most cases, you use a standard DRAM timing model because a JEDEC standard specifies the timing requirements at the DRAM components. For controllers, timing requirements can vary widely. Create your own timing model if the parameters of your controller differ from an example controller timing model. The HyperLynx installation includes example timing models. See the *\MentorGraphics\<release>\SDD_HOME\hyperlynx64\Libs* folder.
SPICE	Before opening a DDRx wizard	If you have a controller or DRAM component model in SPICE format, you must reference them from IBIS models. See "Referencing a SPICE Model with the External Model Keyword" on page 623.
Resistor package	Before opening a DDRx wizard	If a DDRx net passes through a resistor package, assign a *.pak* model to it. The model provides pin-to-pin connectivity information needed by the software to map DDRx nets to components.

Design Information

Note

DDRx batch-mode simulation uses information in this section. DDRx interactive simulation does not use information in this section.

Required Information	Description
Controller reference designator	Specify a reference designator for a controller component used by a specific DDRx interface. If you use a multiple-board project with DIMMs on different boards, you also specify a board name.
Number of ranks	Specify the number of ranks for a specific DDRx interface. A chip select signal typically represents a rank in a design.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Required Information	Description
DRAM slot, rank, and reference designator	For each DRAM, specify the number of slot and ranks, and its reference designator. If you use a multiple-board project with DIMMs on different boards, you also specify a board name. In BoardSim, you can gather this information by doing the following: 1. Open a DIMM board. 2. Select a DDRx address net and open the Assign Models dialog box (**Models > Assign Models/Values by Net** menu item). 3. Write down the reference designator for each DRAM.
DIMM combinations	If needed, specify different combinations of DIMM types. Examples for a two slot DDRx interface: • A dual-rank DIMM in both slots. • A dual-rank DIMM in slit 1 and a single-rank DIMM in slot 2.
1T or 2T (for DDRx types that support this timing)	Specify whether the address and command timing is one or two clock cycles.

DDRx AMI Simulation Flows

If your DDRx memory interface has single-ended nets with IBIS-AMI models that represent equalization capabilities, the software provides multiple simulation flows to choose from.

Note

"Equalization" can generally help to compensate for signal degradation by manipulating a waveform at either a driver or receiver. IBIS-AMI models can represent equalization capabilities for drivers and receivers. For an example benefit, equalization can help to open the eye in an eye diagram. For some IBIS-AMI models, you can edit or sweep parameter values to change equalization behaviors to improve signal quality.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Overview

Simulation Flow	Description	Example Applications or Benefits
Statistical AMI	Uses a statistical-based AMI simulation methodology that assumes linear time-invariant (LTI) behavior, so all non-LTI effects (including DFE — decision-feedback equalization) are either approximated or omitted.	• Quickly estimate quality for a net or wide net bus. • Quickly run many simulation iterations, such as sweeping AMI parameter values to optimize equalization for a DQ net.
Time-domain AMI	Uses a time-based AMI simulation methodology that was initially intended to analyze SERDES channels, which does not account for asymmetric driver behavior.	• More accurate than “Statistical AMI”. • Accounts for DFE. • Faster than “Advanced DDRx AMI”.
Advanced DDRx AMI	Uses an advanced simulation methodology to accurately account for all sources of non-linearity, including drivers, receivers, and power-supply voltage variation due to simultaneous switching drivers.	Verify net timing and quality with best accuracy.

Detailed Capabilities and Properties

Capability or Property	Statistical AMI	Time-Domain AMI	Advanced DDRx AMI
Simulation run time	Shortest	Medium	Longest **Note:** An advanced simulation with 100 K stimulus bits and a 16 bit DQ bus can potentially consume 100 GB of disk space.
Account for analog buffer rise/fall asymmetry and high/low impedance asymmetry	No	No	Yes
Account for non-zero volt DC offset	Yes	Yes	Yes

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Capability or Property	Statistical AMI	Time-Domain AMI	Advanced DDRx AMI
Account for driver/receiver equalization provided by filtering capabilities (such as DFE, FFE, CTLE) of an IBIS-AMI model.	Yes (DFE support is approximate and DFE quality depends on AMI model quality)	Yes	Yes
Support power-aware IBIS models, which help represent simultaneous switching noise (SSN)	DDRx Batch-Mode Wizard		
	No	No	Yes
	DDR5 Wizard		
	No	No	No
An eye diagram for a DQ signal net accounts for jitter contributed by its related DQS signal net.	DDRx Batch-Mode Wizard		
	No	Yes	Yes
	DDR5 Wizard		
	No	No	No
	Yes — The software aligns each UI in an eye diagram for a DQ signal net by measuring DQS signal net offset (jitter) on a cycle-by-cycle basis. No — The software aligns each UI in an eye diagram for a DQ signal net with an ideal clock and no jitter.		
Create measurement spreadsheets	No (produces only eye diagrams)	Yes	Yes
Support stimulus pattern that you select or specify	N/A	Yes	Yes

Related Topics

DDRx Memory Interface Simulation

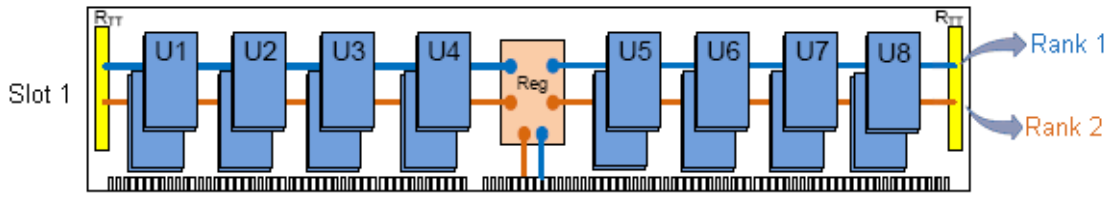
Statistical and Time Domain Simulation Comparison

Specifying Locations for Stacked-Die DRAMs

DDRx simulation supports stacked-die DRAMs with IBIS models when you have an EBD/EMD model for your module that points to IBIS models which define the DRAMs.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 3-1. Slot and Rank Landmarks for DDRx - Stacked Dual-Die DRAM



Restrictions and Limitations

- The software does not support the use of EBD/EMD models that point to other EBD/EMD models.

Prerequisites

- You have opened a design and a DDRx wizard. See "DDRx Memory Interface Simulation" on page 185.

Procedure

1. On the DRAMs page, specify the number of slots.

   Tip

   If your design does not contain DIMMs, set the number of slots to 1.

2. Specify the number of ranks per slot.

   Designs that contain stacked-die DRAMs have one rank for each set of dies. If you have four stacked dies, you have four ranks.

   For example, specify two ranks per slot for a dual-die DRAM and specify four ranks per slot for a quad-die DRAM.

3. For a board design, from the Board list, select the EBD/EMD model for the first DRAM.
4. In the DRAM to Rank Assignments area, click the row header to select the Slot 1, Rank 1 row.
5. In the DRAM Reference Designators area, click the row header to select the Ref Des for each DRAM in rank 1 (U0), then click →.

   Note

   If your design includes ECC DRAM, ensure that you assign both DRAM and ECC DRAM reference designators to the same rank.

6. In the DRAM to Rank Assignments area, click the row header to select the Slot 1, Rank 2 row.
7. In the DRAM Reference Designators area, select the IBIS model for the DRAM at rank 2 (U2), then click →.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

8. For a board design, select the next EBD/EMD model from the Board list and repeat steps 3-7 until you assign models to all DRAMs.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Modeling a Board Design With Multiple Stackups

If you have a board design with multiple stackups, you can define all the stackups and their locations in the board design.

Topic	Description
Defining a Local Stackup for a Board Design With Multiple Stackups	You can create a stackup definition that uses a subset of layers defined by the master stackup. You can assign this new “local” stackup definition to an area of the board design.
Defining a Stackup Area for a Board Design With Multiple Stackups	You can assign a stackup to an area of your board design.
Board Designs With Multiple Stackups	A board design can be divided into areas with different stackup definitions. For example, an area that implements a flexible cable has a different stackup definition than a rigid area with components mounted to it. To represent this kind of design for simulation, you can define a set of stackup definitions and assign them to areas where the stackup is different.

Defining a Local Stackup for a Board Design With Multiple Stackups

You can create a stackup definition that uses a subset of layers defined by the master stackup. You can assign this new “local” stackup definition to an area of the board design.

Prerequisites

- You understand “Board Designs With Multiple Stackups” on page 157.
- Acquire the Multiple Stackup license.
- Enable the **Enable Multiple Stackups** menu item (choose the menu item to activate its check mark).

Procedure

1. Choose **Setup** > **Stackup** > **Stackup Manager**. The Stackup Manager dialog box opens.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

2. Create a new local stackup definition:

If you want to...	Do this...
Base a new stackup on the master stackup	1. Click **New**. The Create Stackup dialog box opens. 2. Add a checkmark for each layer you want to include in the new stackup. 3. If needed, view or edit the new stackup. Click **View Stackup** to open the Stackup Editor. When done, click **Close**. For information about editing stackups, see "Defining the Basic Stackup" on page 1233. 4. Click **OK**.
Base a new stackup on an existing local stackup	1. Select an existing stackup that provides a good basis for the stackup you want to create. 2. Click **Copy** and type the name of the new stackup. 3. If needed, edit the new stackup. For information about editing stackups, see "Defining the Basic Stackup" on page 1233.

Results

You can now assign a local stackup definition to a specific area of the board. See "Defining a Stackup Area for a Board Design With Multiple Stackups" on page 155.

Defining a Stackup Area for a Board Design With Multiple Stackups

You can assign a stackup to an area of your board design.

Restrictions and Limitations

- The software does not support overlapping stackup areas. However, if you use Xpedition Layout VX.2 or newer and your design has a Stiffener object, exporting the design to HyperLynx SI/PI creates a stackup area for the Stiffener object that can overlap another stackup area. Where these two stackup areas overlap, the software uses the stackup area representing the Stiffener object.

Prerequisites

- Acquire the Multiple Stackup license.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- You understand Board Designs With Multiple Stackups.
- All stackup definitions that you want to assign to areas of your design have been defined. See Verifying the Stackup Definition for a Board Design With Multiple Stackups.
- Enable the **Enable Multiple Stackups** menu item (choose the menu item to activate its check mark).

Procedure

1. To create a stackup area:
   a. Choose **Setup > Stackup > Add Stackup Area**. The Add or Edit Stackup Area dialog box opens.
   b. Type a name, assign a stackup, and select a shape.
   c. Specify snap options:

If you want to...	Do this...
Snap a corner or center point to a grid.	1. Check **Snap To The Grid**. 2. Type the distance between grid elements.
Snap a corner or center point to a nearby stackup area.	Check **Snap to Neighbor Stackup Area**.

Note
When you drag a line segment, the software does not use the snap options.

   d. Specify the location of a stackup area:
      i. In the board viewer, click to add the first point.
      ii. For a rectangular shape, click again to close the outline.
      iii. For a polygon shape, click to add points, and close the outline by clicking either the first point or **Close** (located below the Points spreadsheet).

Note
Press Backspace to undo a change and press Ctrl-Backspace to redo a change.

You can right-click an area outline to display editing options.

Note
Press Shift when dragging a polygon corner to draw a diagonal line segment.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

2. To change a stackup area:

   a. In the board viewer, right-click the outline for the stackup area and click **Edit Stackup Area**.

   b. Edit a stackup area:

If you want to...	Do this...
Move a corner.	Drag a corner point.
Move a line segment.	Avoiding the center and corner points, drag a line segment.
Add a corner.	Drag a center point and release the mouse button.
Delete a corner.	Hover the pointer over a corner point and press Delete.

Note

Press Backspace to undo a change and press Ctrl-Backspace to redo a change.

You can right-click an area outline to display editing options.

Board Designs With Multiple Stackups

A board design can be divided into areas with different stackup definitions. For example, an area that implements a flexible cable has a different stackup definition than a rigid area with components mounted to it. To represent this kind of design for simulation, you can define a set of stackup definitions and assign them to areas where the stackup is different.

Figure 3-2 shows an example board design loaded into Xpedition Layout. The design has flexible cables that permanently attach to rigid areas with components or connectors.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 3-2. Example Board Design With Multiple Stackup Areas

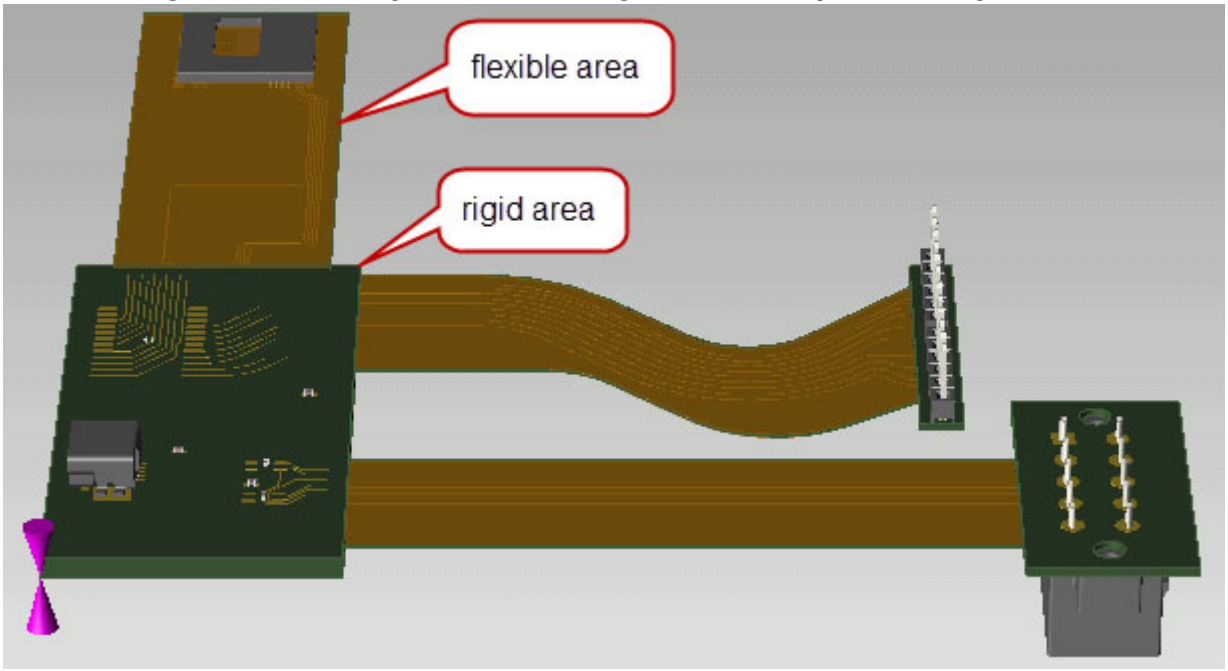
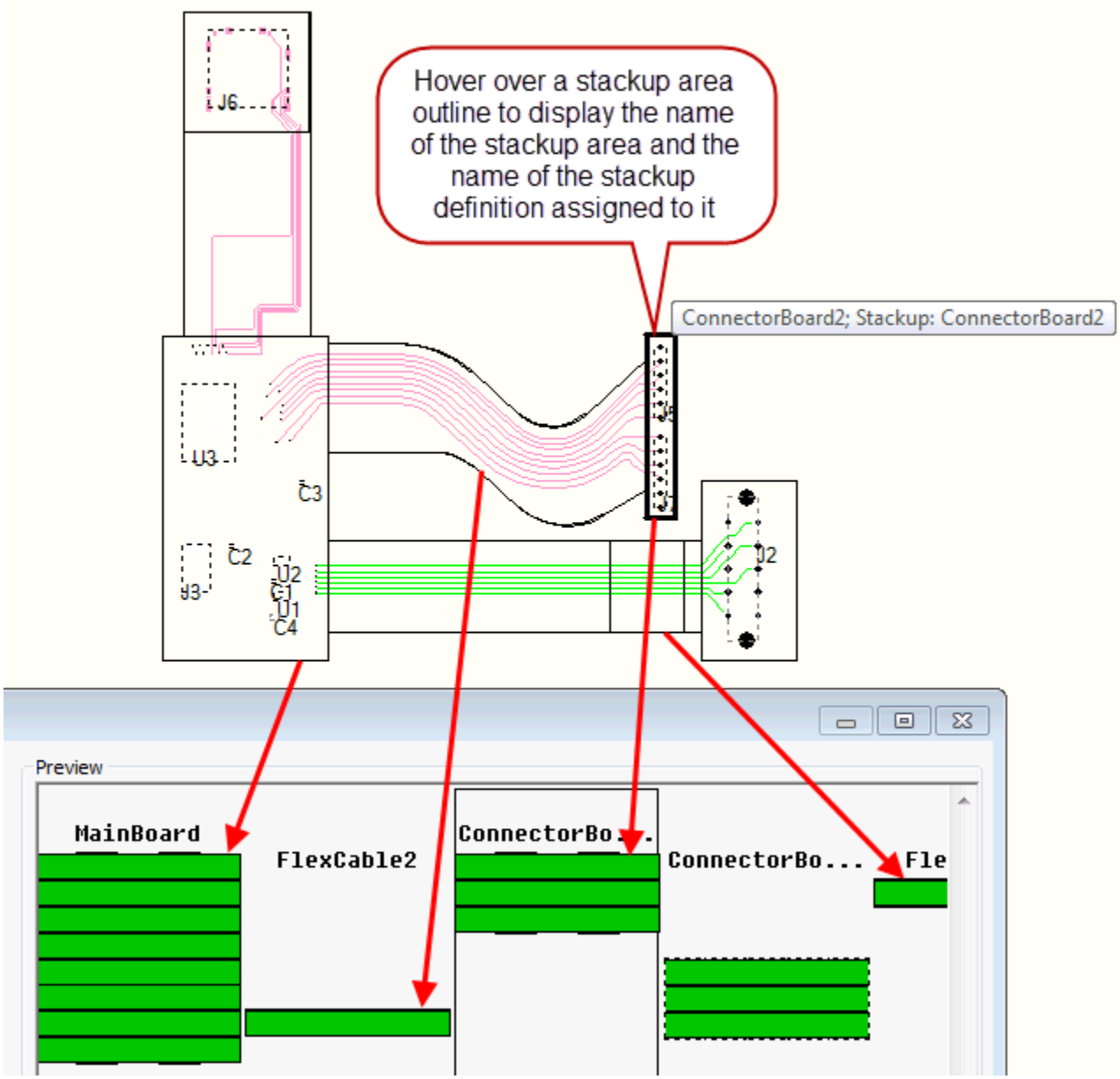


Figure 3-3 shows the same example board design loaded into BoardSim. Arrows show the stackup layers for some of the stackup areas.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 3-3. Stackup Areas Displayed in the Board Viewer



When you export a board design that contains multiple stackups from Xpedition Layout VX.2 and newer, HyperLynx SI/PI automatically creates a set of stackup definitions and assigns them to the appropriate stackup areas in the BoardSim board. Exporting creates a stackup area for each Board Outline and Stiffener object in your Xpedition Layout design.

Note

You can specify folded and unfolded representations of a design in Xpedition Layout. Simulation runs on an unfolded representation of a board design.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

For other layout tools, translated board designs that contain multiple stackups open in HyperLynx SI/PI with only a master stackup definition that describes all layers for all areas of the design. The software automatically assigns the master stackup definition to all design areas by default. For each design area (such as a flexible cable), you manually create a new stackup area and assign a stackup definition to it. You can make a copy of the master stackup definition, edit the copy, and assign it to one or more stackup areas.

In LineSim, you can also create a set of stackups and assign a specific stackup definition to a transmission line or signal via.

Adding or Updating SERDES Protocols

You can make new or updated protocols available to SerDes wizards. Typically a Siemens EDA representative will send you a *.zip* file that contains configuration and supporting files that describe one or more protocols. You import a *.zip* file into a library file to make new or updated protocols available to a SerDes wizard.

Prerequisites

- You have a *.zip* file containing configuration and supporting files that describe one or more SerDes protocols.

Procedure

1. Choose the **Simulate SERDES > Add SerDes Protocols** menu item. The Add/Update SerDes Protocol Definitions dialog box opens.
2. Specify the path to a *.zip* file that contains protocol files.
3. Choose a protocol library folder and file name:

If you want to...	Do this...
Add a protocol to the library file named *SerDesWizard.ini* file in the HyperLynx installation folder.	Select “HyperLynx install: SDD_HOME\<path>\SerDesWizard.ini”.
Add a protocol to a custom library file and location that you specify. **Note:** You can define only one custom library file. You can use this capability when your company does not give you permission to modify the folder containing a HyperLynx installation.	1. Select “My local SerDes library”. 2. Type the path to a library file, then click **Create**. A library file must have an *.ini* file name extension. When you have previously defined a library file, the dialog box displays its path and makes the Create button unavailable.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

You can import a protocol to either *SerDesWizard.ini* or a custom library file, but not both. For example, if *SerDesWizard.ini* contains protocol XYZ, you cannot import protocol XYZ into a custom library file.

4. If a *.zip* file and selected library file contain a SerDes protocol with the same name, choose whether or not to update the library file by selecting one of the following options:
   - Replace it with the new version
   - Keep the old version
5. Click **OK**.

Related Topics

Analyzing Multiple SERDES Channels Using the SERDES Wizard - Compliance Mode

Evaluating SERDES Channel Compliance From a Measured S-Parameter Model in the SERDES Wizard

Troubleshooting Simulation Setup

This topic provides solutions to common problems that can occur when you set up your design for simulation.

Problem	Solution
The pins list of the Assign Models dialog box does not show pins that you expect to see.	The software does not recognize a component type correctly, or the pin is not included in the set of selected nets. See Verifying That Component Types are Recognized Correctly, or Selecting Nets for SI Simulation.

Saving Session Edits

When you save a board design, the software saves modeling and setup changes (session edits) to the *.BUD* file. The software saves schematic and PDN editor changes to the *.FFS* file.

To save board design session edits, select **File > Save BoardSim Session File (.BUD)** . For a schematic, select **File > Save**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Design Type	Information Saved
Board or Schematic	• Changes to the stackup definition • Model assignments made using the Assign Models dialog box • IC and ferrite-bead models • Passive-component values and package descriptions • For IC models, the Vcc-pin and Vss-pin settings • New components (Quick Terminators) • Simulation settings • Temperature • Net-by-net batch simulation settings
Board	• Identification of design (specification of power supply and signal nets, and differential pairs). • Changes to a multiple board design (multiple instances of a single board) • Manhattan routing **Note:** If you unroute a net without re-routing it, the unrouting changes will be absent when you reload your board.
Schematic	Schematic or PDN design changes made in the schematic editor or PDN editor.

For a board design, the software saves the last two versions of the session file. The *.BUD* file contains the most recent set of changes. The *.BBD* file (backup *.BUD* file) contains the second most recent set of changes. These files are stored in the same folder as the board file.

Restriction

For designs that contain an *.EBD/EMD* model, the software does not save session edits inside the *.EBD/EMD* models, such as buffer direction or model assignments made using the Assign Models dialog box.

Caution

The component information in a *.BUD* session file (that is, all the information other than the stackup information) is based on reference designators. If you renumber the reference designators on your board, you will invalidate most or all of the information in your session file. This may force you to re-enter much of your component data.

If the session file has saved some IC-model assignments that reference models that are not available, the affected edits are ignored and no warning messages appear. This prevents the session file from becoming a barrier to loading a board.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Tip

If the software cannot find IC models (that you assigned and saved to a session file) when you re-load a board, check your Model Library File Path setting (**Setup** > **Options** > **Directories**).

If your multiple-board design uses multiple copies (instances) of a board design file, when you save session edits, the software allows you to save separate files for each instance or share settings for all instances of the board. For example, you might save separate files if you have a data bus that connects multiple memory module instances, and you want to set data bus pins on one instance to the output direction and set data bus pins on the other instances to the input direction.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Chapter 4
Simulating a Design

HyperLynx provides several signal integrity and power integrity simulation types that you can run to produce results that help you validate or plan your design.

Topic	Description
Running Signal Integrity Simulation	Signal integrity simulation provides options that run simulation with parametric sweeps, and allows you to see results as a waveform, eye diagram, or spreadsheet.
Running General Batch Simulation	General Batch simulation provides detailed signal quality, delay, crosstalk, and radiated emissions measurement results, and compares measurement results to limits that you specify. The software provides measurement results in a text or spreadsheet format, and optionally saves simulation waveforms and measurements to files.
DDRx Memory Interface Simulation	DDRx batch-mode simulation helps you verify timing among nets for a memory interface. You can also evaluate a single-ended DDR5/LPDDR5 net with IBIS-AMI models.
DC Drop Simulation	DC drop simulation helps you identify areas in your design with excessive IR drop or current density.
Analyzing a SERDES Design	You can use the SERDES Wizard to analyze multiple SERDES channels and run separate simulations for varying transmitter or receiver settings and channel properties.
Decoupling Simulation	Depending on the accuracy and information you seek, the software provides different methods for running PDN decoupling simulation.
Running Plane Noise Simulation	Use plane noise simulation to observe how noise propagates across plane regions of the power-distribution network (PDN) when power supply pins draw large amounts of transient current.
Running Signal-Via Bypass Simulation	Signal-via bypass simulation helps you evaluate the ability of the power-distribution network (PDN) to provide low-impedance return current paths for signals transmitted through a single-ended via. Signal-via bypassing simulation creates a Z-parameter model showing the return current impedance across a frequency range. Values greater than several ohms indicate insufficient bypassing. The Z-parameter model accounts for the effects of nearby stitching vias, bypass capacitors, and interplane capacitance.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
Pulse Response Simulation	Pulse response simulation is a quick method for pre-screening nets to find nets that require more detailed simulation.

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Running Signal Integrity Simulation

Signal integrity simulation provides options that run simulation with parametric sweeps, and allows you to see results as a waveform, eye diagram, or spreadsheet.

You can simulate a single net, or multiple nets together to see their timing relationships and crosstalk effects. If you have a large number of nets, or if you want to simulate multiple nets independently, use General batch simulation.

The software runs simulation and shows results in either the Digital Oscilloscope or EZwave waveform viewer. Simulate using the oscilloscope unless you need to see measurement results in spreadsheet form provided with EZwave.

Topic	Description
Running Signal Integrity Simulation with the Oscilloscope Waveform Viewer	Signal integrity simulation with the Digital Oscilloscope provides options that run simulation with parametric sweeps, and allows you to see results as a waveform or eye diagram.
Running Signal Integrity Simulation with the EZwave Waveform Viewer	Signal integrity simulation with EZwave provides options that run simulation with parametric sweeps, and allows you to see results as a waveform or eye diagram, or measurements in spreadsheet form.

Running Signal Integrity Simulation with the Oscilloscope Waveform Viewer

Signal integrity simulation with the Digital Oscilloscope provides options that run simulation with parametric sweeps, and allows you to see results as a waveform or eye diagram.

Prerequisites

- On the Welcome Screen, use the waveform viewer options to make the Digital Oscilloscope waveform viewer available in toolbars and menus.
- Acquire the following licenses as needed:

If...	Acquire this license...
You want to see results as eye diagrams.	Advanced Scope
Your design includes an IBIS model that references an encrypted HSPICE model, the software requires the HSPICE simulation engine (that you install and license separately).	Advanced Scope and SPICE Output

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If...	Acquire this license...
Your design includes an IBIS model that references an Eldo or VDHL model (with an [External Model] or [External Circuit] keyword), or a Verilog-A model that is embedded in a SPICE netlist using HSPICE or ADMS syntax, the software requires the ADMS simulation engine.	GHZ license bundle
You want to account for noise between single-ended signal via and power planes (SI/PI co-simulation).	Co-simulation

- Copy model files for drivers and receivers in your design to the correct location (**Models > Edit Model Library Paths)**.
- Verify that the software recognizes your design correctly. See Verifying That the Software Recognizes Your Design Correctly.
- Assign models and set bi-directional driver and receiver buffer states as needed. See Assigning Models to Components and Pins. For bi-directional IC buffers, use the Assign Models dialog box to set the driver pin to the output direction and set the receiver pin to the input direction.
- Configure simulation options to account for coupling, loss, and noise between single-ended signal vias and power planes. See Setting Simulation Options.
- Select the nets that you want to simulate. See Selecting Nets for SI Simulation.
- If your simulation includes signal nets with series capacitors, the software can assign a settled voltage to the capacitors when you enable the “Settle series capacitors before every simulation” option on the Preferences Dialog Box - Advanced Tab (see the topic for technical details). Enabling this option means that simulation can obtain a settled voltage for a series capacitor on a signal net without having you run a very long time-domain simulation or assign a special ADMS or SPICE model to the differential pair.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

1. Open the dialog box for the simulation method you need:

If you want to...	Do the following...
Run sweeps	1. Choose the **Simulate SI > Run Interactive Sweeps** () menu item. 2. In the Sweep Manager dialog box, select a parameter that you want to sweep. 3. Click **Add Range** to specify range and increment values to define the set of simulations. To review the set of simulations, click the **Simulation Cases** tab. 4. Click **Run Sweeps**.
Not run sweeps	Choose the **Simulate SI > Run Interactive Simulation** () menu item.

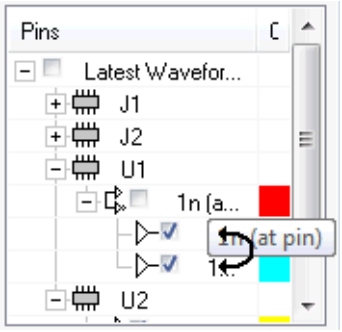
2. In the Digital Oscilloscope dialog box, click **Standard** or **Eye Diagram** to see results as waveforms or an eye diagram.
3. If needed, edit simulation engine options:
   - In the Simulator area, set the simulation engine to Auto unless you need to use a specific simulator. See Supported SI Models and Simulators.
   - If you have assigned SPICE models that require SPICE simulation parameters, options, or files, click **SPICE Options**.
4. Set the simulation resolution to Auto unless you need to specify a specific value. See "Automatic Time Step Calculation for Time-Domain SI Simulation" in Digital Oscilloscope Dialog Box.
5. To include the effect of noise between single-ended signal vias and power planes, check SI/PI Co-Sim. This option runs PI simulation with SI simulation. See Setting Simulation Options.
6. Adjust the Horizontal Scale to set the simulation run time.
7. If needed, define the stimulus and eye mask.

For waveforms...	For an eye diagram...
To specify simple stimulus options, select **Global** and click **Edge** or **Oscillator** and specify values as needed.	To load or define a specific stimulus, click **Global** and click **Configure**. To define an eye mask, click the **Eye Mask** tab. To show the eye mask, check **Eye Mask**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

For waveforms...	For an eye diagram...
To specify a specific stimulus for each pin click **Per-net/pin** and click **Assign** to open the Assign Stimulus dialog box.	To specify a specific stimulus for each pin click **Per-net/pin** and click **Assign** to open the Assign Stimulus dialog box. To define an eye mask, click the **Eye Mask** tab. To show the eye mask, check **Eye Mask**.

8. Select IC modeling conditions, Slow-Weak, Typical, or Fast-Strong.
9. Select the pin locations (probe points) in the Pins list, for the points where you want to see simulation results. For waveforms, all probe points in the list are selected by default. For eye diagrams, no probe points are selected by default. If the software does not automatically recognize differential pins and attach differential probes to them, such as for SPICE models, you can manually specify them. If the software displays the waveform of a differential probe with reversed polarity, meaning the inverted pin is located to the left of the non-inverted pin in the Pins spreadsheet, you can swap the polarity of the differential probes.

If you want to...	Do the following...
Manually specify a differential probe	1. In the bottom row of the Pins spreadsheet, double-click **<Insert diff probe>**. 2. Select pins in the Positive pin (+) and Negative pin (-) lists. 3. Click **OK**.
Swap differential probe polarity	Point to either pin number (to the right of the check box) of the differential pair, so that pointer shape changes to a U-shaped arrow, and then click. 

10. Click **Start Simulation** or **Start Sweeps**.

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Results

Show, hide, or save Oscilloscope waveform or eye diagram data, take measurements, and adjust the view:

If you want to...	Do the following...
Hide or show specific results for a probe location in your design	Check the probe (or sweep simulation case for that location) in the Pins list.
To adjust the size and scale of the waveform or eye diagram	Adjust the vertical position and scale, and horizontal delay and scale as needed.
See waveform results from multiple simulations	Check **Previous Results**, **Latest Results**, or **Loaded Results**. You can save and load one set of simulation results to see a maximum of three sets of results in the oscilloscope at one time. You can only see one eye diagram at a time.
Take measurements from a waveform or eye diagram	See "Taking Measurements From an Oscilloscope Waveform or Eye Diagram" on page 382.
Save simulation results	Click **Save/Load** and select the file in the Load/ Save Waveforms dialog box.

Related Topics

Accounting for Noise Between Single-Ended Signal Via and Power Planes in SI Simulation (Co-Simulation)

Sweep Manager Dialog Box - Setup Tab

Running Signal Integrity Simulation with the EZwave Waveform Viewer

Signal integrity simulation with EZwave provides options that run simulation with parametric sweeps, and allows you to see results as a waveform or eye diagram, or measurements in spreadsheet form.

Restrictions and Limitations

- Signal integrity simulation with EZwave does not support measurements for SPICE drivers and receivers because they do not provide threshold information. Assigned SPICE models are ignored unless they are contained within an IBIS wrapper. See Referencing an External Model From an IBIS Model.

Prerequisites

- On the Welcome Screen, use the waveform viewer options to make the EZwave waveform viewer available in toolbars and menus.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- Acquire the following licenses as needed:

If...	Acquire this license...
You want to see results as eye diagrams.	Advanced Scope
Your design includes an IBIS model that references an encrypted HSPICE model, the software requires the HSPICE simulation engine (that you install and license separately).	Advanced Scope and SPICE Output
Your design includes an IBIS model that references an Eldo or VDHL model (with an [External Model] or [External Circuit] keyword), or a Verilog-A model that is embedded in a SPICE netlist using HSPICE or ADMS syntax, the software requires the ADMS simulation engine.	GHz license bundle
You want to account for noise between single-ended signal via and power planes.	Co-Simulation

- Copy model files for drivers and receivers in your design to the correct location (**Models > Edit Model Library Paths)**.
- Verify that the software recognizes your design correctly. See Verifying That the Software Recognizes Your Design Correctly.
- Assign models and set bi-directional driver and receiver buffer states as needed. See Assigning Models to Components and Pins. For bi-directional IC buffers, use the Assign Models dialog box to set the driver pin to the output direction and set the receiver pin to the input direction.
- Configure simulation options to account for coupling, loss, and noise between single-ended signal vias and power planes. See Setting Simulation Options.
- Select the nets that you want to simulate. See Selecting Nets for SI Simulation.
- If your simulation includes signal nets with series capacitors, the software can assign a settled voltage to the capacitors when you enable the “Settle Series Capacitors Before Every Simulation” option on the Preferences Dialog Box - Advanced Tab (see the topic for technical details). Enabling this option means that simulation can obtain a settled voltage for a series capacitor on a signal net without having you run a very long time-domain simulation or assign a special ADMS or SPICE model to the differential pair.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

1. Open the dialog box for the simulation method you need:

If you want to...	Do the following...
Run sweeps	1. Choose a simulation method that produces results in the form that you want. • Waveform or eye diagram: **Simulate SI** > **Run Interactive Sweeps** > **And Show Waveforms** () menu item. • Spreadsheet and waveform/eye diagram: **Simulate SI** > **Run Interactive Sweeps** > **And Show Waveforms and Measurements** () menu item. 2. In the Sweep Manager dialog box, select a parameter that you want to sweep. 3. Click **Add Range** to specify range and increment values to define the set of simulations. To review the set of simulations, click the **Simulation Cases** tab. 4. Click **Run Sweeps**.
Not run sweeps	Choose a simulation method that produces results in the form that you need. • Waveform or eye diagram: **Simulate SI > Run Interactive Simulation > And Show Waveforms** () menu item. • Spreadsheet and waveform/eye diagram: **Simulate SI > Run Interactive Simulation > And Show Waveforms and Measurements** () item.

2. In the Interactive Simulation dialog box, click **Standard** or **Eye Diagram** to see results as waveforms or an eye diagram. Eye diagram results are not available for simulation with sweeps or measurements.

3. If needed, click **Simulation Controls** to edit simulation engine options:

   - Set the simulation engine to Auto unless you need to use a specific simulator. See Supported SI Models and Simulators.
   - Set the simulation resolution to Auto unless you need to specify a specific value. See “Automatic Time Step Calculation for Time-Domain SI Simulation” in Digital Oscilloscope Dialog Box.
   - To include the effect of noise between single-ended signal vias and power planes, check **SI/PI Co-Simulation**. This option runs PI simulation with SI simulation. See “Setting Simulation Options” on page 131.
   - If you have assigned SPICE models that require SPICE simulation parameters, options or files, click **SPICE Options**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

4. Define the stimulus:

For waveforms...	For an eye diagram...
To specify simple stimulus options, click **Global** and specify a frequency and duty cycle.	To load or define a specific stimulus, click **Global** and click **Configure**.
To specify a specific stimulus for each pin click **Per-net/pin** and click **Assign** to open the Assign Stimulus dialog box.	To specify a specific stimulus for each pin click **Per-net/pin** and click **Assign** to open the Assign Stimulus dialog box.

5. Choose IC modeling conditions, **Slow-Weak**, **Typical**, or **Fast-Strong**.
6. For simulation with measurements:
   a. In the Analysis type area, select the measurement types that you want to include in spreadsheet results, and click **Measurements** to specify measurement details.
   b. Select what parts of your design are included in simulation:
      - For a board design, click **Set Constraints** to edit the default constraints for the selected net in your design. The software uses these constraint values to flag a net that fails a measurement requirement in the results spreadsheet.
      - For a schematic design, click either **Entire Schematic** to see measurement results for all nets, or click **Net** and click a schematic symbol that connects to the net to simulate.
   c. To enable manually saving simulation results from EZwave to disk, check **Store analysis waveforms**.

Note

Disable this option to save runtime when running long simulations.

7. For simulation without measurements:
   a. Set the simulation run time. In the Simulation duration area, click **Auto** to let the software determine the run time based on the stimulus sequence you specify in the Stimulus area, or click **Manual** to specify a specific time.
   b. For an eye diagram, select display options in the Eye diagrams area as needed. You can configure an eye mask in EZwave after simulation completes (**Tools** > **Eye Diagram**).

Caution

If you disable “Plot eye waveforms”, simulation data is not automatically saved. Before you close HyperLynx, you must open EZwave manually (**Simulate SI** > **Open EZwave**) and save simulation results.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

8. Click **Start Sweeps** or **Start Simulation**.

Results

When simulation completes, EZwave opens and the Simulation Results dialog box displays results in spreadsheet form.

For information on measuring waveform data in EZwave, see "Using the Measurement Tool" and "Signal Integrity Measurements" in the *EZwave SI Measurement Guide*. You can also see EZwave Help for information about viewing and measuring waveform data.

Related Topics

Accounting for Noise Between Single-Ended Signal Via and Power Planes in SI Simulation (Co-Simulation)

Interactive Sweeps Dialog Box

Interactive Sweeps with Measurements Dialog Box

Interactive Simulation Dialog Box

Interactive Simulation with Measurements Dialog Box

Sweep Manager Dialog Box - Setup Tab

Measurements Dialog Box

Simulation Results Dialog Box

Running General Batch Simulation

Scope: BoardSim

General Batch simulation provides detailed signal quality, delay, crosstalk, and radiated emissions measurement results, and compares measurement results to limits that you specify. The software provides measurement results in a text or spreadsheet format, and optionally saves simulation waveforms and measurements to files.

General Batch simulation contains three types of simulation:

Simulation Type	Description
Quick Analysis	• Does not require you to assign driver and receiver models. Models all drivers using a basic default model that you can customize. • Provides basic results, including crosstalk estimations and termination suggestions.
Signal-integrity and crosstalk	Provides detailed signal quality, delay, or crosstalk measurement results.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Simulation Type	Description
EMC	Provides radiated emissions measurement results and compares them to standard regulatory limits or other limits that you specify.

You can enable or disable each of these simulation types when you run General Batch simulation. For example, you can run Quick Analysis alone (disable Signal-integrity and crosstalk and EMC in general batch simulation) to scan an entire board for signal integrity problems, and then run Signal-integrity and crosstalk simulation for only the smaller set of critical nets.

You can also run Quick Analysis for each important subset of IC buffer types. For example, when you have an important set of nets with ICs that switch 3.3 volts in three nanoseconds and another important set of nets with ICs that switch 1.5 volts in 500 picoseconds, run Quick Analysis twice, once with each IC buffer type.

Restrictions and Limitations

- EMC simulation is not available for multiple-board designs.
- Quick Analysis has unique coupling/crosstalk threshold settings. It ignores the crosstalk enable/disable setting (**Setup > Enable Crosstalk Simulation**) and does not use settings in the Set Coupling Thresholds and Coupling Settings dialog boxes.
- You cannot disable lossy transmission line modeling or via modeling. General Batch simulation accounts for these effects.
- The software ignores any SPICE models assigned to drivers and receivers because these models do not provide threshold information that the software requires. However, you can wrap a SPICE model in an IBIS model that the software will recognize. See Referencing an External Model From an IBIS Model.

Prerequisites

- Verify that the software recognizes your design correctly. See Verifying That the Software Recognizes Your Design Correctly.
- Unless you only plan to run Quick Analysis, assign models to drivers and receivers on the nets that you plan to include in simulation, including associated nets and coupled nets. See Assigning Models to Components and Pins.
- For nets that span more than one board in a multiple-board project, assign models to drivers and receivers on the nets on the other boards. (If an IC pin has no model assignment, the software assumes that the pin is open.)

  To model a connector pin as electrically open, assign the OPEN-CIRCUIT model to the pin from the *Open_mod.ibs* library.

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- Acquire the following licenses as needed:

If you want to...	Acquire this license...
Account for crosstalk	BoardSim Crosstalk
Account for loss and advanced (2D) via models	Advanced Scope and SPICE Output
Account for noise between single-ended signal via and power planes (SI/PI co-simulation)	Co-Simulation
Run EMC simulation	BoardSim EMC

Procedure

1. Choose the **Simulate SI > Run General Batch Simulation** ( ) menu item.
2. On the Overview page, select the simulation types that you want to run (signal-integrity and crosstalk, Quick Analysis, EMC).

   You can also load settings from a previous simulation. When you run a simulation, the software creates a *Batch.gbs* settings file in the design folder (…<*design folder*>\ *Batch_Results_*<*date_time*>).

   To include the effects of crosstalk in signal quality and delay measurements, select "Include crosstalk aggressors when simulating quality/delay".

   Note

   This option can significantly increase simulation run time. If you plan to include many nets in simulation, consider running two separate batch simulations. Run one simulation with this option enabled, that includes only differential pairs and other nets strongly affected by coupling, and run another simulation with this option disabled, that includes single-ended nets and other nets that are not strongly affected by coupling.

3. On the Nets page, select the nets that you want to include in simulation. In the "Choose nets for analysis" area, select how you want the list of available nets to appear (sorted by name or by net class) or select "nets selected in board viewer" to display nets that you selected in the board viewer.
4. Select nets or net classes from the "Available nets/net classes" list on the left, then click the arrows to add nets to the "Nets for analysis" list on the right. If you have run a Quick Analysis simulation or a set of Hyperlynx DRC rules as part of a pre-screening strategy, you can click **Import** to import a list of nets from an RPT file (Quick Analysis) or an REP file (HyperLynx DRC).
5. Click **Edit Net Classes, Constraints and Stimuli** to specify limits that the software uses for pass/fail evaluation of measurement results.

   The **Constraints** tab of the "Edit Net Classes, Constraints and Stimuli" dialog box opens.

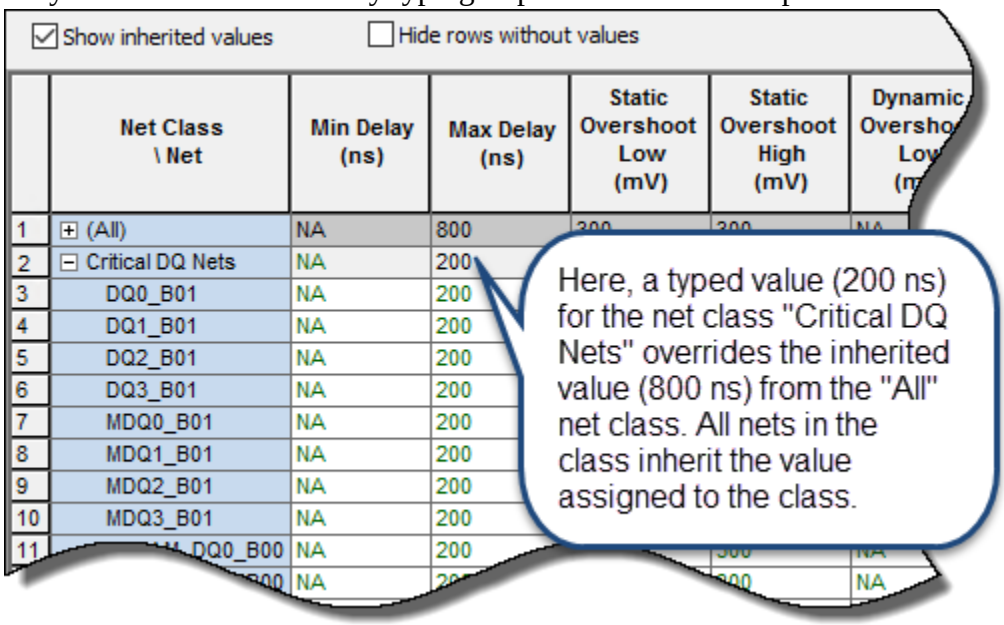
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6. Click **Set Up Net Classes** to categorize a group of nets as a net class. This enables you to select a group of nets at once (by net class) or edit settings for a group of nets when the software provides this option. See "Defining Net Classes" on page 69.

7. Specify constraints in the spreadsheet, or click **Import/Export Net Classes and Constraints** to import constraints from Constraint Manager or a CSV file. See "Importing and Exporting Constraints, Net Classes, or Stimulus" on page 431.

   a. Check "Show inherited values".

      This option displays each constraint limit in the spreadsheet as either a green inherited value or a black specific value. By default, all net classes and the nets within them inherit the values assigned to the "All" net class. If you assign a value to a net class, the software automatically assigns this value to all nets in the class unless you override that value by typing a specific value in the spreadsheet.



      To only show the net classes and nets to which you assigned a specific value, check "Hide rows without values".

   b. You can sort the list of nets using the filter options in the filter area. To see only the nets on one board of a multiple-board design, select the board from the Board list at the top right of the dialog box.

   To see the location in the board viewer of specific nets that you select in the spreadsheet, in the "Push net selection to" area at the top of the page, click **Board Viewer**. Click **Spreadsheet** to select spreadsheet rows that correspond with nets that you have selected in the board viewer.

8. To edit stimulus settings, click the **Stimulus** tab.

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Similar to inherited constraint values, nets inherit the stimulus setting of a net class, and pins inherit the stimulus setting of a net. Inherited settings display in green text.

If you want to…	Do this…
Show model and edge rate information from assigned models in the Models and Edge Rate columns	Click **Load Model Info**.
Edit the stimulus setting for a net class, net, or pin	Click the stimulus setting in the Stimulus column and select the stimulus from the list.
Define a custom stimulus	Click **Edit Stimulus**. See "Creating a Custom Stimulus" on page 492. After you define a custom stimulus, you can select it from the Stimulus column in the spreadsheet.
Import stimulus settings that you have already assigned to nets in a separate DDRx batch simulation	Click **Import DDRx Stimulus**, select the DDRx stimuli list *.txt* file, and click **Open**.

9. Click **Apply** to save changes and click **Close** to return to the wizard.
10. On the Quick Analysis page, if you only want to run crosstalk simulation, click **Run only crosstalk simulation**. This option quickly checks dialog box options to include crosstalk information and exclude other information. Otherwise, select the information

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that you want to include in the simulation report. The options you select determines the pages and fields available in the wizard.

If you want to include...	Do the following...
Crosstalk simulation	1. Check “Show crosstalk strength estimates, sorted by largest crosstalk value”, then enter a coupling threshold. The software identifies nets that produce crosstalk greater than the threshold as aggressor nets. **Tip:** Increase the coupling threshold value if you want to list only the strongest aggressors or you have a large number of nets that are flagged with violations or warnings and want to focus on the most severe problems. Decrease the coupling threshold value if your design has a tight crosstalk noise budget, your driver IC voltage swing is very low, or you want to see more data in the report. 2. To exclude nets that have no aggressors that exceed the coupling threshold from the simulation report, check “Do not show nets without aggressors”.
Termination suggestions	Check one or both of the following: • “Show signal-integrity problems caused by line lengths“ • “Suggest termination changes and optimal values“
Edits made to passive components or terminators	Check “Show component changes“.
Nets that were unrouted or rerouted with Manhattan routing in BoardSim	Check “Show net changes“.
Quick terminators and their values	Check “Show new components“.
Stackup information	Check “Show stackup“.
Detailed electrical information	Check “Show interconnect statistics“.
Numeric statistics for each net	Check “Show counts“.

11. Click **Default IC Model** to review and edit basic model parameters that the software assigns to all drivers when running Quick Analysis.

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Tip

Use the 0%-100% rise/fall time (not 10%-90% or 20%-80%) of the worst-case driver IC that is most commonly used on the board. For example, when the board has many ICs with rise/fall times of 2-3 ns, a few ICs with rise/fall times of 5-10 ns, and a few ICs with rise/fall times of 0.5-1 ns, then specify a value of 2 ns.

When the faster-switching ICs on the board have asymmetric rise/fall times, such as when the falling edge is consistently faster than the rising edge, use the time representing the faster edge. The faster edge nearly always constrains the signal-integrity problems for the board.

12. If you chose to run Quick Analysis and see termination suggestions, on the Terminator Wizard page, check the termination options to include in the report. See "General Batch Wizard - Quick Analysis Terminator Wizard Page" on page 729.
13. If you chose to run Quick Analysis and see interconnect statistics, on the Interconnect Statistics page, select the trace property information to include in the report. See "General Batch Wizard - Quick-Analysis Interconnect Statistics Page" on page 730.
14. If you chose to run EMC simulation, on the EMC page, select simulation options and the regulatory constraints that you want the software to use when evaluating radiated emission results for compliance. See "General Batch Wizard - EMC Page" on page 731.
15. On the Simulation Settings page:
    a. Select the IBIS IC model corners to use in simulation. Check both Fast-strong and Slow-weak to see best-case and worst-case corner simulations since it is unlikely that the typical IC results will exceed corner results. Check Typical to save time when you need only approximate results. For more information on min/typ/max data in an IBIS model, see IC Operating Settings.
    b. To specify how the software identifies coupled nets for crosstalk analysis, click **Coupling Settings** and **Coupling Thresholds**. To specify the simulation engine, simulation resolution, SPICE simulation options, or waveform generation options, click **Simulation Controls**. If you have already specified coupling and simulation control settings for the design (outside the wizard) that you want to use in simulation, click **Copy Interactive Settings to Wizard**.

    Note

    You can also copy wizard settings to Interactive Mode (outside of the wizard) if needed. This option also enables lossy simulation and via simulation outside of the wizard.

    c. In the Measurement interval area, specify the number of cycles in which you want the software to take a measurement, and specify an edge, or choose Both to take measurements for both edges.

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A cycle is not a bit interval. It is the time between two opposite edges that form a "pulse." Check "Auto for required number of cycles" to let the software determine the simulation run time based on the number of cycles, including sufficient time for the net to settle to a reasonable steady state, taking into account the total flight time. If additional delay causes this simulation run time to exclude a measurement for the final cycle of interest, specify a slightly longer simulation run time in the Fixed field.

d. Specify a monotonicity threshold. The threshold value defines the maximum amplitude of non-monotonicities to ignore. Use this option to filter out glitches that are insignificant or so small they would not have any effect inside the receiver IC package. Any non-monotonicities that exceed this value are reported in the Rise/Fall Monotonic [Pass/Fail] column of the simulation results.

e. To assign a specific IBIS model to all unassigned IC pins, in the Default IC Model area, check "Use default IBIS model for unassigned IC pins" and click **Select Model** to choose a model.

16. On the Driver/Receiver Cases page, specify a case for each driver and receiver combination to include in simulation results. If you do not specify a case on this page, the software runs a simulation for each possible combination of driver and receiver on each net.

    a. Select a net class in the "Applies to Net Class" column.

    b. Click a cell in the Driver or Receiver columns to select the driver or receiver component.

    c. Click **Add** to specify another case.

17. On the Crosstalk page, specify options related to crosstalk analysis.

    a. In the Analysis type area, select the algorithm that the software uses to enable and switch drivers on aggressor nets to determine crosstalk results. When you enable "Total crosstalk", the software runs one set of simulations where all aggressors drive simultaneously. When you enable "Total + additional simulations…", the software runs one set of simulations where all aggressors drive simultaneously, and an additional simulation for each aggressor, where that aggressor drives and all other aggressors do not. See Crosstalk Simulation Algorithm for Total Contributions and Crosstalk Simulation Algorithm for Individual Contributions.

    b. In the "Iterations for each aggressor's bidirectional pins" area, select "Single driver per aggressor" if you want the software to automatically set one of the bi-directional buffer pins on each aggressor net as the driver.

       The software determines the driver that is likely to contribute the most crosstalk from the set of allowed driver/receiver cases (limited by the settings you make on the Driver/Receiver page). In the case where more than one driver may contribute the most crosstalk, the software considers the driving direction and selects the one that is

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closest to the victim driver, for example, an aggressor driver pin on the same IC as the victim driver pin, driving in the same direction.

Select “Exhaustive round robin” to run one simulation for each possible combination of bidirectional pins set as drivers, up to the specified maximum number of cases (Max combination of aggressors). When you limit the number of cases, the software selects the cases that are likely to produce the most crosstalk.

Note

The software determines aggressor nets based on the coupling settings you specify on the Simulation Settings page (Coupling Settings and Coupling Thresholds).

c. In the “Victim logic states” area, select each constant state of the victim net (high, low, high-impedance) for which you want to run a simulation.

Note

The total number of simulations that the software runs depends on the number of victim nets and driver/receiver cases (selected nets and valid driver/receiver cases), the number of victim logic states, the number of aggressor nets (Crosstalk iterations), and the number of aggressor driver/receiver cases (Analysis type).

d. In the “Stimulus for aggressor nets” area, select the stimulus that the software applies to aggressor nets. Check “Same as victim” or “Opposite to victim”, or check “Per spreadsheet” to use the stimulus assignments you make in the Stimulus tab of the Net Grouping and Constraints dialog box.

18. On the Reporting Options / Audit page, specify the name and location of the folder where the software creates report files, and select the report file formats that you want to open after simulation completes.

    To save waveforms to disk, so you can display them later in EZwave, check “Save waveforms”.

    To automatically open EZwave and display waveforms when simulation completes, check “Show waveforms”.

    The root folder location depends on options you set in the “Reports and log files directory” section of the Set Directories Dialog Box.

19. Click **Run Audit**. The software checks that each net included in simulation has an assigned driver model and at least one assigned receiver model. The software reports unassigned models in a *Batch-Audit.csv* report file saved in the results folder.

20. Click **Run** to start simulation.

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Results

When simulation completes, the results that you selected on the Reporting Options / Audit page now display.

Note

The software saves results and *.sdf* files, even if you choose to not automatically display them.

Simulation Result	Description
Summary report file	A summary report contains Quick Analysis results.
HTML report	A report that includes the ability to search, sort, and filter results data. You can also export results to Microsoft Excel, comma-separated values, and other formats. You can access HTML reports from previous simulations in the HTML Report Manager (**View > HTML Report Manager**).
Excel file	A detailed report spreadsheet contains simulation results. For definitions of spreadsheet headers, see General Batch Simulation Results Spreadsheet. On Linux, the software writes simulation results to a CSV file.
Waveforms	A waveform file shows signal behavior at times near and during a measurement. You can use EZwave to open a waveform file.
Standard delay files (*.sdf*)	A standard delay file contains interconnect delays between driver and receiver pins. You can use this file to transfer batch simulation results to other applications, such as a timing analysis program. For basic syntax information, see Standard Delay Format.

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DDRx Memory Interface Simulation

DDRx batch-mode simulation helps you verify timing among nets for a memory interface. You can also evaluate a single-ended DDR5/LPDDR5 net with IBIS-AMI models.

DDRx batch simulation performs timing measurements and slew-rate derating adjustments between pairs of signals for every cycle in the simulation. This cycle-by-cycle approach takes into account the effects of noise or intersymbol interference (ISI) on individual waveform transitions. This cycle-by-cycle approach requires thousands of simulations to account for all nets in a memory interface and can take a significant amount of time.

Topic	Description
Running a DDRx Memory Interface Simulation	DDRx batch simulation reports timing and signal-integrity properties for signals between a memory controller and its memory components. You can use this information to verify timing requirements and find nets with unsatisfactory performance.
Running Simulation on a DDRx Single-Ended Net With IBIS-AMI Models	You can use the DDR5 Wizard to simulate a single-ended net with IBIS-AMI models. The wizard can account for driver/receiver equalization behaviors represented by IBIS-AMI models, sweep AMI parameter values, and report results as eye diagrams. If you have finished layout, you may want to use the wizard to quickly check a DQ net before running batch simulation. If you have run DDRx batch-mode simulation and want to investigate ways to improve DQ net performance, you may want to use the wizard to sweep its receiver/driver AMI parameters.

Running a DDRx Memory Interface Simulation

DDRx batch simulation reports timing and signal-integrity properties for signals between a memory controller and its memory components. You can use this information to verify timing requirements and find nets with unsatisfactory performance.

When simulating a memory interface for the first time, you may save time by finding and fixing common setup problems on a subset of nets in the interface before simulating all nets. See “Strategy to Quickly Find and Fix DDRx Simulation Setup Problems” on page 145.

The steps below provide the main ideas to set up and run simulation. To see detailed definitions for wizard page options, click the Help button on a page.

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Restrictions and Limitations

- You can simulate one clock for a memory interface at a time. For example, if your design has a DRAM package with dual die, it has two clocks and requires you to simulate the interface with one clock, then simulate the interface with the other clock.
- Your memory interface conforms to JEDEC specifications. Specifications are available at www.jedec.org.
- You cannot disable lossy transmission line modeling or via modeling. Simulation accounts for these effects.
- Sweeps for design properties, such as trace length, are available only for schematic designs.

Prerequisites

- You have reviewed "Preparing for DDRx Simulation" on page 145.
- Verify that the software correctly recognizes your design. See "Verifying That the Software Recognizes Your Design Correctly" on page 62.
- If your controller can calibrate skew among memory interface signals, you understand its calibration capabilities to complete step 13.
- If you plan to edit or sweep equalization parameters, you have obtained detailed parameter descriptions from the IBIS-AMI model vendor.
- You have the DDRx Wizard license.
- For a multiple-board design, you have:
  - The MultiBoard license.
  - If a multiple-board project has a Touchstone file assigned to an advanced connector model, you have the GHz license bundle.

Procedure

1. Click **Run DDRx Batch Simulation**  , or choose the **Simulate SI > Run DDRx Batch Simulation** menu item. The DDRx Batch-Mode Wizard opens.
2. On the Initialization page:
   a. When you run the wizard to completion on a design, the software prompts you to save a setup file. If you want to import an existing setup file, do the following:

If you want to...	Do this...
Use an existing setup file.	Click **Import** and select a setup file (*.ddr*).

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If you want to...	Do this...
Verify a setup file that you manually created.	1. Click **Import** and select a setup file (*.ddr*). 2. Click **Verify**. 3. Correct any syntax errors and re-import the file.

b. Select a DDRx interface type and data rate. If the Unbuffered option is available, specify whether the interface is unbuffered or registered.

c. (If available) Select a WCK:CK ratio to define the frequency relationship between a write clock and a command/address clock.

3. On the Controller page, find the table row with the reference designator for the controller, click the small square next to the Ref Des column, then click  .

4. On the DRAMs page:

Note

If your design uses stacked-die DRAMs, perform the steps in "Specifying Locations for Stacked-Die DRAMs" on page 151.

a. Specify the number of slots and ranks.

b. On the left side of the page, find the table row with the reference designator for a DRAM in the specific memory interface you want to simulate, then click the small square to the left of the Ref Des column.

You can select multiple reference designators by using Shift+click or by clicking and dragging in the RefDes column. This capability is especially helpful for a schematic design.

c. On the right side of the page, find the table row for the slot or rank that uses the DRAM you selected in step 4.b, then click the cell in the DRAMs column.

d. Click  .

5. On the IBIS Models page, verify model assignments. If errors exist, click **Assign Component Models** to open the REF-File Editor and fix them.

6. On the Nets to Simulate page:

a. Enable timing measurements for signal categories (such as data, clock, address/command) to include in simulation.

b. Select a type of receiver threshold to use for timing measurements.

Most DRAM specifications measure timing using AC thresholds above and below the Vref for setup, and use DC thresholds above and below Vref for the hold requirements. Most controllers measure timing using a single threshold above and

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below Vref for both setup and hold times. For both of these cases, select “Use all four receiver AC/DC thresholds”.

In the unusual situation where the reference voltage is used to measure both setup and hold timing, select “Use only the Vtt threshold”.

c. Review advanced options and enable as needed.

7. On the DRAM Signals page:

   a. Click **Perform Automatic Net Mapping**. The software maps memory interface nets to each DRAM.

   b. If any mappings are incorrect, make corrections in step 8. If you want to see paths for controller-to-DRAM nets, click **View Controller Net Topology**.

8. On the next several wizard pages, verify that the software has correctly mapped nets to signal categories such as data, clock, and address/command.

9. On the Disable Nets page, uncheck any nets that you want to exclude from simulation.

Note

Simulation will include an unchecked net when it is needed to perform a timing measurement that you specify in step 6.a.

10. On the ODT Models page, click in each DQ, DQS, or DM net cell to assign the IBIS model you want to use for each ODT disabled or enabled state.

    If you plan to sweep ODT models for a schematic design, you can save multiple model configurations for signal nets by doing the following:

    a. Select an ODT setting from the list at the bottom of the page.

    b. Click in each ODT enabled/disabled cell and select a model.

    c. Repeat until you have defined all of your model configurations.

11. On the ODT Behavior page, verify the ODT states for write and read operations.

12. On the IBIS Model Selectors page, verify model selections for non-ODT signals. Click in a cell to change a model.

13. (If available) On the Leveling&Calibration page, specify the write leveling delays to include for simulation:

    a. Enable the skew calibration options supported by your controller.

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b. Specify a source for skew values:

If you want to specify skews...	Do this...
Automatically, where the software runs one simulation to measure skews and automatically runs a second simulation to apply skews.	Check “Automatically run simulation 2 times”. **Note:** If you have assigned power-aware IBIS models and want to save overall run time, you can exclude PI effects before creating a delays file. To exclude PI effects, on the Stimulus and Crosstalk page, uncheck “Include PI effects in SI simulations”. For the next simulation, check “Include PI effects in SI simulations”, then perform the steps in the next row in this table.
By importing a file that specifies skews.	1. Uncheck “Automatically run simulation 2 times”. 2. Select “Use values from the DDRx Delays external file”, then click **Select** to specify a file location. For example, *DDR*Delays_autogenerated.txt* located in the design folder. 3. Specify whether the table represents skew data for a specific IC model corner or for all corners.
By manually specifying skews.	1. Uncheck “Automatically run simulation 2 times”. 2. Select “Use values from the table on this page”, then type values into the table. 3. Specify whether the table represents skew data for a specific IC model corner or for all corners.

For information about skew file contents and locations, see “Creating and Applying a DDRx Delays File” on page 490.

14. (If available) On the Equalization page, review AMI parameter values and edit them as needed:

    a. Specify whether you want to see parameters for a pin, lane, or device.

    Your selection determines the scope of your equalization settings, such as for only one pin, for all signals on a given lane, or for all pins on a given device.

    b. Select controller or DRAM pins, then a read or write cycle.

    c. Select a pin, lane, or device. The tree view on the left side of the page displays model parameters.

    d. Select a parameter and edit its value if needed.

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Note

The "Ignore initial DQ bits" and "Ignore initial ACC bits" options on the Simulation Options page override a value specified by an AMI Ignore_bits parameter displayed on this page.

e. If you want to copy the value for the current parameter to all other pins on a device, click either of the following:

- **Use setting for all Controller pins**
- **Use setting for all DRAM pins**. This button copies a value to all DRAM devices and pins.

f. If you want to save values for the current pin to a file, click **Save As**, then specify a location. You can load values from this file to other interfaces in your design, or to other designs.

15. On the Timing Models page, review data rate and model assignments, then change if needed.

    If you change a model assignment, click **Verify All** to verify the timing model syntax.

16. (If available) On the Vref Training page, specify training options.

17. (If available) On the Round Trip Time page, either type limits into the table or import a file that specifies limits.

18. On the Stimulus and Crosstalk page, specify stimulus for each net. You can enable options to include crosstalk or PI effects in simulation.

Restriction

If your interface supports equalization, you have assigned power-aware IBIS models, and you have enabled either Statistical AMI or Time-domain AMI in step 20.b, the software does not include PI effects (including, simultaneous switching noise).

Also, if you enable Statistical AMI in step 20.b, simulation ignores stimulus settings on this page.

19. (If available) On the Sweep Manager page, specify sweep ranges:

    a. Select a design property ODT model setting to sweep and click **Add Range**. The Sweeping dialog box opens.

    b. Specify parameter value ranges, then click **OK**.

20. On the Simulation Options page:

    a. Enable the IBIS IC model corners to use in simulation, and whether to apply the IC model voltage or power-supply net voltage to passive termination components.

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Check both Fast-strong and Slow-weak to see best-case and worst-case corner simulations since it is unlikely that the typical IC results will exceed corner results. Check only Typical to save time when you need approximate results.

b. If your DDRx interface has nets with drivers or receivers with IBIS-AMI models that represent equalization behaviors, enable a simulation flow.

For more information, see "DDRx AMI Simulation Flows" on page 149.

c. If needed, edit other options.

21. (Optional) On the Quality Checks page, choose whether to override JEDEC measurement thresholds with values that you specify.

22. On the Report Options page, select whether to perform an audit, then specify measurements and report options:

If you want to...	Do this...
Create an HTML report that contains worst-case measurements. You can click a measurement link to display related waveforms.	Select "HTML report". **Note:** This option additionally creates CSV spreadsheets.
Create a spreadsheet that contains detailed measurements.	Select either of the following: • XLS spreadsheets. Restrictions: • The software automatically switches to CSV format for spreadsheets with over 5000 rows. • Unavailable for computers running Linux. • CSV spreadsheets — Recommended for reports with more than 5000 rows.
Compare measurement values in spreadsheets to waveforms.	1. Check "Save all <DDRx> simulation waveforms to disk. This option saves waveform data to CSV files. 2. If you want to reduce disk space consumption by saving only one in ten waveform data points, uncheck "Save all simulated points in waveforms".
Avoid overwriting previous results.	Check "add time tags to the folder name". The root results folder location depends on options you set in the "Reports and log files directory" section of the "Set Directories Dialog Box" on page 1016.

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23. On the Simulate page, review any reported errors. After correcting any errors, click **Run Batch Simulation**.

    Save a wizard setup file when prompted. Specify a unique file name to preserve existing setup files.

    The “DDRx Batch Mode - Run Simulation” dialog box opens.

24. Click **Run**.

Results

If you enabled “HTML report” on the Report Options page, an HTML report opens. Click a measurement to display related waveforms in EZwave. The report provides many search, filter, sort, navigation, format, and export capabilities. For numerical columns, you can use < (less than) and > (greater than) operators to filter the data. After you close an HTML report, you can re-open it. See “Re-Opening a DDRx HTML Report” on page 296.

DDRx simulation can produce spreadsheets, report files, and waveform and stimulus files. Spreadsheets contain timing and signal-integrity measurements. The Oscilloscope and EZwave can display waveform files. See “DDRx Batch Simulation Results” on page 277.

If a single-ended net with a receiver or driver that uses equalization has simulation results that you want to investigate, you can simulate that net by itself and sweep its receiver/driver AMI parameters. See “Running Simulation on a DDRx Single-Ended Net With IBIS-AMI Models” on page 192.

If a routed net (in BoardSim) has poor results, you can export it to LineSim to debug and optimize. “Exporting a Net from BoardSim to LineSim” on page 411

The wizard automatically saves stimulus from the Stimulus and Crosstalk page. For information, see “Usage Notes” on page 809.

If you enabled XLS or CSV spreadsheets on the Report Options page, the software writes each combination of swept parameters to its own folder.

Related Topics

Preparing for DDRx Simulation

Running Simulation on a DDRx Single-Ended Net With IBIS-AMI Models

You can use the DDR5 Wizard to simulate a single-ended net with IBIS-AMI models. The wizard can account for driver/receiver equalization behaviors represented by IBIS-AMI models, sweep AMI parameter values, and report results as eye diagrams. If you have finished layout, you may want to use the wizard to quickly check a DQ net before running batch simulation. If you have run DDRx batch-mode simulation and want to investigate ways to improve DQ net performance, you may want to use the wizard to sweep its receiver/driver AMI parameters.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Restrictions and Limitations

- You can simulate one single-ended net at a time.
- The software can only use bit interval settings that you make in step 5.b. It does not use settings from the Assign Stimulus dialog box. If you want to specify jitter and your AMI model supports it, edit AMI parameters in step 4.
- You cannot disable lossy transmission line modeling or via modeling. Simulation accounts for these effects.

Prerequisites

- You have reviewed "Preparing for DDRx Simulation" on page 145.
- You have:
  - Assigned IBIS-AMI models to drivers and receivers on the net that you plan to simulate.
  - Assigned IBIS or IBIS-AMI models to aggressor nets that you plan to include in simulation.
  - Copied IBIS-AMI files (*.ami*) and their associated library files (*.dll* or *.so*) to the correct location (**Models > Edit Model Library Paths** menu item).
- If you plan to edit or sweep equalization parameters, you have obtained detailed parameter descriptions from the IBIS-AMI model vendor.
- You have the DDRx Wizard license.
- For a multiple-board design, you have:
  - The MultiBoard license.
  - If a multiple-board project has a Touchstone file assigned to an advanced connector model, you have the GHz license bundle.

Procedure

1. If you have opened a board design in BoardSim, select a single-ended net with equalization.
2. Choose the **Simulate SI > Run DDR5 Interactive Simulation** menu item. The DDR5 Wizard opens.
3. On the Channel page:
   a. Select driver (TX) and receiver (RX) pins, then enable IC modeling and crosstalk options.

      If you have saved DDR5 Wizard settings from a previous simulation and want to reuse them, click **Import settings**, then specify a *.d5s* file.

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If you want to restore default option values for all wizard pages, click **Reset all settings**.

b. Specify a simulation flow. For information, see "DDRx AMI Simulation Flows" on page 149.

c. If you want to review or edit model assignments or model selector values, click **Set Model Selectors and Check Models**.

4. On the "TX settings" and "RX settings" pages, review IBIS-AMI parameter values and edit as needed.

Tip

If the Settings field is blank, verify that you have assigned an IBIS-AMI model to the pin. See step 3.c.

If you want to assign parameter values for the selected pin to other pins, you can save them to a settings file (*.ami*). When you simulate another pin, you can load settings from this file.

5. On the Stimulus page:

a. Click **Assign Stimulus**. The Assign Stimulus dialog box opens.

b. Assign stimulus to pins:

If you want to...	Do this...
Use stimulus previously created for this net by the DDRx Batch-Mode Wizard.	Click **Import DDRx Stimulus** and browse to the stimulus file. The stimulus file location is \<*design folder*>\<*DDR_Results folder*>*StimulusForNets*\<*stimulus.txt*>.
Manually assign stimulus.	Edit table values.

Note

If you enable "AMI Statistical" in step 3.b, the software ignores a bit pattern you specify, but applies a stimulus bit rate that you specify.

c. Click **Close**.

d. If needed, edit "Number of bits to simulate" and advanced AMI options. For information, click **Help**.

6. (Optional) On the Sweep Parameters page:

a. Expand a branch, select a parameter, then click **Add Range** or **Edit Range**. The Sweeping dialog box opens.

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b. Specify a sweep range, then click **OK**.

7. (If available and optional) On the Sweep Cases page, review the model and design properties for each simulation.

8. On the Output Options page:

If you want to...	Do this...
Avoid overwriting previous results.	Check “Add time tags to the folder name”.
Set the vertical center of an eye diagram to DC level ((V_dc_high - V_dc_low)/2) from an RX waveform, instead of 0 V.	Check “Restore DC offset in eye diagram”.
In the Eye Density Viewer, delete eye diagrams created by previous simulations.	Check “Clear previous eyes”.
Display an eye diagram from a previous simulation.	Click **Open Eye Density Viewer**.

Advanced options:

If you want to...	Enable this option...
Investigate the effects of equalization by opening EZwave and showing waveforms. The simulation flow you enabled in step 3.b determines the available kinds of waveforms.	Show waveforms.
Save EZwave waveform database files (*.wdb*) to the results sub-folder. If you enabled “Advanced DDR AMI” in step 3.b, the software also creates a <*design_name*>.*waves* folder that contains additional waveforms.	Save waveforms
Save debug files to *DebugFolder* in the results sub-folder.	Create AMI model debug info.

9. Click **Run**.

Results

The HyperLynx Eye Density/BER Viewer opens and displays an eye diagram.

If you enabled the “Show waveforms” option in step 8, EZwave opens and displays waveforms. If you additionally enabled “Advanced DDR AMI” in step 3.b, step response waveforms for victim and aggressor nets are available.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Running a DDRx Memory Interface Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DC Drop Simulation

DC drop simulation helps you identify areas in your design with excessive IR drop or current density.

DC drop simulation reports IR drop (voltage drop) and current density across power and ground nets. These reports help you see the effects of IC and connector pins that draw large amounts of current through power supply nets at DC operating conditions.

DC drop simulation helps you find these areas in your design that have these problems:

- Excessive voltage drop, sometimes known as rail collapse, can cause the voltage supplied to an IC power pin to fall below the recommended minimum operating voltage.
- Excessive current density in voltage island neckdowns can generate excessive heat in the power supply net, which can cause board failures such as PCB delamination and fusing.
- Excessive current density in stitching vias can lead to via failures, such as an opened connection. DC drop simulation does not translate current density to temperature because it does not model how the heat spreads away from the regions with high current density. However it does show regions in the design with concentrated current flow that, depending on design details, can lead to excessive heat.

Topic	Description
Running DC Drop Simulation	Use DC drop simulation to report IR drop, current density, and resistance between pairs of VRM and IC pins for a power-supply net in a board or schematic design.
Running DC Drop Simulation for Multiple Power Nets That Share a Common Ground Net	Use DC drop simulation to analyze IR drop and current density for multiple power nets that share a common ground net a in a board design. The software measures IR drop and current density for the power nets and the ground net (while current flows on all of its power nets at once).
Running Batch DC Drop Simulation or Thermal Co-Simulation	Use batch DC drop simulation on a board design to measure IR (voltage) drop and current density, and measure resistance between pairs of VRM and IC pins, for multiple power-supply nets at a time. You can also run thermal and batch DC drop simulation together, to account for metal resistivity changes caused by heating from components and current flowing between VRM and IC power-supply pins.
Running DC Drop Simulation From Layout Software	Use DC drop simulation to analyze IR drop and current density for multiple power-supply nets in a design loaded into a Siemens EDA layout tool with an Analysis Control capability that supports the DC Drop client.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Running DC Drop Simulation

Use DC drop simulation to report IR drop, current density, and resistance between pairs of VRM and IC pins for a power-supply net in a board or schematic design.

The following video contains additional information. Note that the video does not describe assigning models to a multiple-phase VRM (see "Assigning Models for PI Simulation" on page 103), specifying design constraints for specific padstacks (see "Accounting for Per-Padstack Plating and Constraints" on page 137), and measuring resistance between pairs of VRM and IC pins (see the procedure).



Restrictions and Limitations

- For a multiple-board design:
  - If you select a net that connects to a net on another board with a connector you have modeled with an S-parameter or SPICE model (Advanced connector model), the software excludes the net on the other board from simulation.
  - HTML reports do not contain graphical results for voltage drop and current density.
- DC drop simulation does not account for metal resistivity changes caused by heating from components and current flowing between VRM and IC power-supply pins. If you want to account for the effects of heating on IR drop, you can run thermal and batch DC drop simulation together. See "Running Batch DC Drop Simulation or Thermal Co-Simulation" on page 206.

Prerequisites

- You have the DC Drop license.
- The software correctly recognizes your design. See "Verifying That the Software Recognizes Your Design Correctly" on page 62.
- You have assigned power-integrity (PI) models to VRMs, IC sinks, and passive components (resistors, inductors, and ferrite beads) that connect power nets. See "Assigning Models for PI Simulation" on page 103.

  To include reference nets in simulation, ensure that you have specified a reference net for each VRM and IC pin when you assigned PI models.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- You have defined constraints for your design, including maximum voltage drop (in percentage), current density, and via current.
- If needed, you have specified per-padstack plating and maximum via current constraints. See "Accounting for Per-Padstack Plating and Constraints" on page 137.

Procedure

1. Choose the **Simulate PI > Run DC Drop Simulation (PowerScope)** menu item or click  . The DC Drop Analysis dialog box opens.
2. (Optional for a multiple-board design) Use the Board list to display information for all boards or a selected board.
3. In the "Power/Ground Net to Analyze" area, select a power-supply net.
4. To include reference nets in simulation, check "Include Reference net(s)".

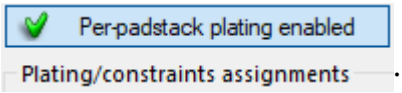
   Note
   This option increases simulation run time.

5. Verify the model assignments displayed in the Assigned Models area:
   a. (Optional) To display pins without model assignments, check "Show all Pins".
      - "<none>" means that a DC sink or VRM model is not assigned to the pin.
      - "(disconnected)" applies only to IC pins and means that the pin does not connect to the power-supply net through a via or trace segments. These pins could represent board-geometry problems. Click **Pre-Process Geometry** to display the "(disconnected)" label.

   b. To add or edit model assignments, click **Assign**. Use the Assign Power Integrity Models dialog box to make model assignments.
6. (Optional) Verify the accuracy of the geometries in the display pane:
   - Right-click in the display pane to zoom and pan.
   - Click a stackup layer name in the Display Areas section to highlight geometries located on that layer.
   - Click **Pre-Process Geometry** to display anti-pads and anti-segments in the display pane.
7. Specify design constraints. The software highlights constraint violations in simulation text results.

If you want to specify design constraints for...	Do this...
A board	Type values into the fields.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to specify design constraints for...	Do this...
Specific padstacks **Note:** This option overrides a value in the "Max Via Current" field.	1. Click **Via Constraints**. The Backdrilling/Plating/Constraints Settings dialog box opens. 2. On the **Plating/Constraints** tab, ensure that the top button displays "Per-padstack plating enabled"  3. Specify constraints. See "Accounting for Per-Padstack Plating and Constraints" on page 137.

8. To create a report, check Create Report and select a format:

If you want...	Do this...
An easy-to-navigate report that includes screen captures of measurement locations in the board design.	Select HTML and specify via reporting options. **Restriction:** HTML reports do not include voltage drop and current density graphs for a multiple-board design.
A spreadsheet.	Select "Microsoft Excel (.XLS)".
A comma-separated values file.	Select "Comma-separated (.CSV)".

9. To save power dissipation results to files that you can import into another Siemens EDA tool, check "Write power-map files for Simcenter Flotherm/XT/FLOEFD". You can import the saved files into any of the following:
   - Simcenter™ Flotherm™
   - Simcenter Flotherm XT
   - Simcenter FLOEFD
   - HyperLynx ThermalSim
10. Specify the type of information you want to include in a report:

If you want a report to include...	Select this option...
DC drop and current density.	DC Drop
Resistance between pairs of VRM and IC pins.	Resistances

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want a report to include...	Select this option...
DC drop, current density, and resistance between pairs of VRM and IC pins.	DC + Resistances

11. Click **Simulate**.

Results

The software can create the following results:

Simulation Result	Description
HTML Report	The DC Drop Simulation Report dialog box automatically opens and can display these types of results: • Text representing voltage drop, current density, and via current. The report highlights results that violate design constraints. The report provides many search, filter, sort, navigation, format, and export capabilities. • 2-D graphical results for voltage drop and current density measurements. • Resistance between pairs of VRM and IC pins. You can access HTML reports from previous simulations in the HTML Report Manager (**View > HTML Report Manager**).
Graphs	The HyperLynx PI PowerScope dialog box automatically opens and displays simulation results in 2-D and 3-D graphs that represent areas of your design. It displays DC drop voltage using color coding to indicate areas of higher and lower voltage drop, DC current distribution, and DC current density for metal and vias. You can save graphs. For example results, see “DC Drop Simulation Results” on page 351. When you include reference nets in simulation, the Layer list additionally sorts the geometries into Power Layers and Reference Layers groups. Clearing one of the groups makes it easier to see DC drop values with a magnitude of a few millivolts. **Note:** A graph shows the voltage drop across the metal for a plane layer and does not include the effect of VRM resistance. By contrast text results in the Reporter dialog box represent absolute voltage. For example, if you run simulation, change the VRM model resistance, re-run simulation, then the Reporter will show different voltage drop values and the graphs will show the same voltage drop value.
Text	The Reporter dialog box automatically opens and displays voltage and current information at each power source, load, and via. Click a link for a pin or via to highlight its location in the board viewer.
Spreadsheet	Results stored in a file with an XLS or CSV format. Spreadsheets do not contain current-density measurements.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Simulation Result	Description
Power dissipation file	A file that contains power dissipation results. You can import this file into the Siemens EDA tools listed in step 9. The result file name is *Thermal_<net_name>.txt*. For a multiple-board design, the file name is *Thermal_<net_name>_<board_ID>.txt*.
DC resistance matrix file	For each power net that you simulated, a file that contains resistance between pairs of VRM and IC pins. The file name is *DCResMatrix_<power_supply_net>.txt*.

Note

If your design includes a multi-phase VRM with a sense line and boost capability, the reported maximum voltage boost value for the VRM is the maximum boost among all phases, not the average maximum boost.

The software saves results to a time-stamped folder in the design folder by default. You can change the report folder location in the "Reports and log files directory" section of the "Set Directories Dialog Box" on page 1016.

To share an HTML report with someone, you can send them the results folder. Double-click *index.html* in the *report_data* sub-folder to open an HTML report. For example, *DCDROP_<date>-<time>\<design_name>\report_data\index.html*. Note that an HTML report opened in this way cannot launch an external viewer, such as the HyperLynx PI PowerScope.

To identify possible ways to change a board design to fix problems, select the problem power-support nets and export them to LineSim. See "Exporting a Net from BoardSim to LineSim" on page 411.

Related Topics

DC Drop Analysis Dialog Box

HyperLynx PI PowerScope Dialog Box

Reporter Dialog Box

Running DC Drop Simulation for Multiple Power Nets That Share a Common Ground Net

Use DC drop simulation to analyze IR drop and current density for multiple power nets that share a common ground net a in a board design. The software measures IR drop and current density for the power nets and the ground net (while current flows on all of its power nets at once).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Restrictions and Limitations

- DC drop simulation for a common ground net is not available for multiple-board designs.
- HTML reports do not contain graphical results for voltage drop and current density.
- DC drop simulation does not:
  - Account for metal resistivity changes caused by heating from components and current flowing between VRM and IC power-supply pins. If you want to account for the effects of heating on IR drop, you can run thermal and batch DC drop simulation together. See "Running Batch DC Drop Simulation or Thermal Co-Simulation" on page 206.
  - Create graphical results for the PowerScope.
  - Report resistance between pairs of IC sink and VRM pins. You can measure this pin-to-pin resistance for a power or ground net with other product capabilities. See "Running DC Drop Simulation" on page 198 and "Running Batch DC Drop Simulation or Thermal Co-Simulation" on page 206.

Prerequisites

- You have the DC Drop license.
- The software correctly recognizes your design. See "Verifying That the Software Recognizes Your Design Correctly" on page 62.
- You have assigned power-integrity (PI) models to VRMs, IC sinks, and passive components (resistors, inductors, and ferrite beads) that connect power nets. Ensure that you have specified a reference net for each VRM and IC pin when you assigned PI models. See "Assigning Models for PI Simulation" on page 103.
- You have defined constraints for your design, including maximum voltage drop (in percentage), current density, and via current.

Procedure

1. Choose the **Simulate PI > Run DC Drop for Multiple Nets with Combined Ground Currents** menu item. The DC Drop Simulation for Multiple Nets with Combined Ground Currents dialog box opens.
2. In the "Select Ground Net to analyze" area, select a ground net.

   Note

   If the table is missing a ground net, ensure that your PI model assignments include reference nets. To review or edit PI model assignments, click **Assign Models**.

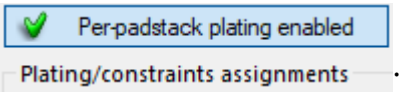
3. In the Include Power Nets area, select power nets and specify their voltage drop constraints. The software highlights constraint violations in simulation text results.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

If the table is missing a power net, ensure that your PI model assignments include reference nets. To review or edit PI model assignments, click **Assign Models**.

4. In the Constraints area, specify design constraints for current density and via current:

If you want to specify design constraints for...	Do this...
All nets	Type values into the fields.
Specific padstacks **Note:** This option overrides a value in the "Max Via Current" field.	1. Click **Via Constraints**. The Backdrilling/ Plating/Constraints Settings dialog box opens. 2. On the **Plating/Constraints** tab, ensure that the top button displays "Per-padstack plating enabled"  . 3. Specify constraints. See "Accounting for Per-Padstack Plating and Constraints" on page 137.

5. To create 2-D and 3-D graphical results that show the locations of high voltage drop, current density, and current distribution that you can display in the HyperLynx PI PowerScope, check Create PowerScope Data.
6. To create a report, check Create Reports and select a format. All report formats include voltage and current measurements for pins and vias, and the maximum current density measurement for the power-supply net.

If you want...	Do this...
An easy-to-navigate report	Select HTML and specify via reporting options.
A simple spreadsheet	Select "Microsoft Excel (.XLS)".
A comma-separated values file	Select "Comma-separated (.CSV)".

7. To save power dissipation results to files that you can import into another Siemens EDA tool, check "Write power-map files for Simcenter Flotherm/XT/FLOEFD". You can import the saved files into any of the following:
   - Simcenter Flotherm
   - Simcenter Flotherm XT
   - Simcenter FLOEFD
   - HyperLynx ThermalSim

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

8. Click **Simulate**.

Results

The software can create the following results:

Simulation Result	Description
Text	The Reporter dialog box automatically opens and displays voltage and current information at each power source, load, and via. Click a link for a pin or via to highlight its location in the board viewer.
HTML Report	The DC Drop Simulation Report automatically opens and displays voltage drop, current density, and via current in a tabular format. The report highlights results that violate design constraints. The report provides many search, filter, sort, navigation, format, and export capabilities. You can access HTML reports from previous simulations in the HTML Report Manager (**View > HTML Report Manager**).
Spreadsheet	Results stored in a file with an XLS or CSV format. Spreadsheets do not contain current-density measurements.
Power dissipation file	A file that contains power dissipation results. You can import this file into the Siemens EDA tools listed in step 7. The result file name is *Thermal_<net_name>.txt*. For a multiple-board design, the file name is *Thermal_<net_name>_<board_ID>.txt*.

Note

If your design includes a multi-phase VRM with a sense line and boost capability, the reported maximum voltage boost value for the VRM is the maximum boost among all phases, not the average maximum boost.

The software saves simulation results to the reports folder. The root report location depends on options you set in the "Reports and log files directory" section of the "Set Directories Dialog Box" on page 1016.

To share an HTML report with someone, you can send them the results folder. Double-click *index.html* in the *report_data* sub-folder to open an HTML report. For example, *DCDROP_<date>-<time>\<design_name>\report_data\index.html*. Note that an HTML report opened in this way cannot launch an external viewer, such as the HyperLynx PI PowerScope.

To identify possible ways to change a board design to fix problems, select the problem power-support nets and export them to LineSim. See "Exporting a Net from BoardSim to LineSim" on page 411.

Related Topics

HyperLynx PI PowerScope Dialog Box

Reporter Dialog Box

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Running Batch DC Drop Simulation or Thermal Co-Simulation

Use batch DC drop simulation on a board design to measure IR (voltage) drop and current density, and measure resistance between pairs of VRM and IC pins, for multiple power-supply nets at a time. You can also run thermal and batch DC drop simulation together, to account for metal resistivity changes caused by heating from components and current flowing between VRM and IC power-supply pins.

Note

Simulating several power-supply nets with complex topologies can take hours to run and consume a lot of memory.

The following video contains additional information. Note that the video does not describe assigning models to a multiple-phase VRM (see "Assigning Models for PI Simulation" on page 103), specifying design constraints for specific padstacks (see "Accounting for Per-Padstack Plating and Constraints" on page 137), measuring resistance between pairs of VRM and IC pins (see the procedure), and creating an HTML report (see the procedure).



Restrictions and Limitations

- For a multiple-board design:
  - If you select a net that uses an Advanced model to connect to a net on another board, DC drop simulation excludes the net on the other board.
  - Thermal simulation is not available.
  - HTML reports do not contain graphical results for voltage drop and current density.
- Thermal and batch DC drop co-simulation is not available for board designs with multiple stackups, such as a rigid/flex design.

Prerequisites

- You have the DC Drop license.
- The software correctly recognizes your design. See "Verifying That the Software Recognizes Your Design Correctly" on page 62.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- You have assigned power-integrity (PI) models to VRMs, IC sinks, and passive components (resistors, inductors, and ferrite beads) that connect power nets. See "Assigning Models for PI Simulation" on page 103.

  To include reference nets in simulation, ensure that you have specified a reference net for each VRM and IC pin when you assigned PI models.

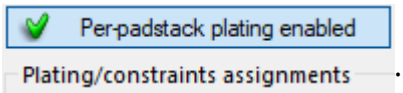
- You have defined constraints for your design, including maximum voltage drop (in percentage), current density, and via current.

- To run thermal co-simulation, you have assigned thermal component models, thermal conductivity values for metal and dielectric layers, and environment properties. See Running a Thermal Simulation in the *HyperLynx Thermal User Guide*.

Procedure

1. Choose the **Simulate PI > Run DC Drop Batch Simulation** menu item or **Simulate Thermal > Run PI/Thermal Co-Simulation** menu item. The Batch DC Drop Simulation dialog box opens.

2. Select each power-supply net that you want to simulate, and specify its constraints if needed. The software highlights constraint violations in simulation text results.

If you want to specify...	Do this...
Per net design constraints	Type a value in the table.
Per padstack: • Maximum via current constraints **Note:** This option overrides a value in the "Max Via Current" column for a net. • Plating properties	1. Click **Via Constraints**. The Backdrilling/Plating/Constraints Settings dialog box opens. 2. On the **Plating/Constraints** tab, ensure that the top button displays "Per-padstack plating enabled"  3. Specify constraint or plating values. See "Accounting for Per-Padstack Plating and Constraints" on page 137.

   For a multiple-board design, if you select a net that connects (with a Simple or Short interconnect model) to a net on another board, simulation includes the net on the other board.

3. To include reference nets in simulation, check Include Reference Nets.

   Note

   This option increases simulation run time.

4. To verify or edit model assignments, click **Assign Models**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

5. To create graphical information to help you see the locations of maximum measurements in your board design, check Create PowerScope Data.

Note

Creating PowerScope data requires significantly more simulation run time than textual results. If you have a large board with many power-supply nets, you may want to first run batch simulation without creating PowerScope data, and then use the textual results to determine which power-supply nets to graphically examine.

6. To create a report for DC drop simulation without thermal co-simulation, check Create Reports and select a format. All report formats include voltage and current measurements for pins and vias, and the maximum current density measurement for the power-supply net.

If you want...	Do this...
An easy-to-navigate report that optionally includes screen captures of measurement locations in the board design.	1. Select HTML and specify via reporting options. 2. To include screen captures of measurement locations and links to measurement locations in a board design, check Create PowerScope Data.
A simple spreadsheet.	Select “Microsoft Excel (.XLS)”.
A comma-separated values file.	Select “Comma-separated (.CSV)”.

7. To save power dissipation results to files that you can import into any of the following Siemens EDA tools, check “Write power-map files for Simcenter Flotherm/XT/ FLOEFD”:

- Simcenter Flotherm
- Simcenter Flotherm XT
- Simcenter FLOEFD
- HyperLynx ThermalSim

Restriction

This option is automatically enabled when you run thermal co-simulation.

8. Specify the type of information you want to include in a report:

If you want a report to include...	Select this option...
DC drop and current density.	DC Drop

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want a report to include...	Select this option...
Resistance between pairs of VRM and IC pins.	Resistances
DC drop, current density, and resistance between pairs of VRM and IC pins.	DC + Resistances

9. If you intend to run simulation again, save DC drop simulation settings in a session file (DCS). Click **Save** (near the top of the dialog box).
10. Click **Run**.

Results

The software can create the following results:

Simulation Result	Description
Text	The Reporter dialog box automatically opens and displays a summary of simulation results. Click the "Detailed report" link to display voltage and current measurements at each power source pin, load pin, and via. From the detailed report, you can click a link for a pin or via to highlight its location in the board viewer.
Graphs	The HyperLynx PI PowerScope dialog box displays simulation results in 2-D and 3-D graphs that represent areas of your design. To open the PowerScope, you can either: • In the Reporter dialog box, click a link to the PowerScope data. • Choose the **View > Simulation Reports > DC Drop Simulation** menu item. Graphs show DC drop voltage using color coding to indicate areas of higher and lower voltage drop, DC current distribution, and DC current density for metal and vias. You can save graphs. For example results, see "DC Drop Simulation Results" on page 351. When you include reference nets in simulation, the Layer list additionally sorts geometries into Power Layers and Reference Layers groups. Clearing one of the groups makes it easier to see DC drop values with a magnitude of a few millivolts. **Note:** A graph shows the voltage drop across the metal for a plane layer and does not include the effect of VRM resistance. By contrast text results in the Reporter dialog box represent absolute voltage. For example, if you run simulation, change the VRM model resistance, re-run simulation, then the Reporter will show different voltage drop values and the graphs will show the same voltage drop value.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Simulation Result	Description
HTML Report	The DC Drop Simulation Report automatically opens and displays: • Text representing voltage drop, current density, and via current. The report highlights results that violate design constraints. The report provides many search, filter, sort, navigation, format, and export capabilities. • 2-D graphical results for voltage drop and current density measurements. • Resistance between pairs of VRM and IC pins. You can access HTML reports from previous simulations in the HTML Report Manager (**View > HTML Report Manager**).
Spreadsheet	Results stored in a file with an XLS or CSV format. Spreadsheets do not contain current-density measurements.
Power dissipation file	For each net you select in the spreadsheet, simulation writes a file containing power dissipation results. You can import this file into Siemens EDA tools listed in step 7. The file name is *Thermal_<net_name>.txt*.
DC resistance matrix file	For each power net that you simulated, a file that contains resistance between pairs of VRM and IC pins. The file name is *DCResMatrix_<power_supply_net>.txt*.

Note

If your design includes a multi-phase VRM with a sense line and boost capability, the reported maximum voltage boost value for the VRM is the maximum boost among all phases, not the average maximum boost.

The software saves DC drop simulation results to the reports folder and a sub-folder named *DCDROP<date>-<time>*. The software saves power dissipation results to the reports folder. The root report location depends on options you set in the "Reports and log files directory" section of the "Set Directories Dialog Box" on page 1016.

To share an HTML report with someone, you can send them the results folder. Double-click *index.html* in the *report_data* sub-folder to open an HTML report. For example, *DCDROP_<date>-<time>\<design_name>\report_data\index.html*. Note that an HTML report opened in this way cannot launch an external viewer, such as the HyperLynx PI PowerScope.

To identify possible ways to change a board design to fix problems, select the problem power-supply nets and export them to LineSim. See "Exporting a Net from BoardSim to LineSim" on page 411.

Related Topics

Batch DC Drop Simulation Dialog Box

HyperLynx PI PowerScope Dialog Box

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Statistical Contour Chart Dialog Box

Running DC Drop Simulation From Layout Software

Use DC drop simulation to analyze IR drop and current density for multiple power-supply nets in a design loaded into a Siemens EDA layout tool with an Analysis Control capability that supports the DC Drop client.

Use Hazard Explorer to review design objects that violate constraints (hazards), by viewing them in a Siemens EDA layout tool. If you want to view hazards and simulation results together, and in 3-D, you can display them in a layout tool **3D View** tab or the HyperLynx SI PowerScope dialog box.

Note

Simulating several power-supply nets with complex topologies can take hours to run and consume a lot of memory.

Restrictions and Limitations

- DC drop simulation does not report temperatures due to current density. You can display current density for metal regions (as opposed to trace routing) and see where current is concentrated.
- When displaying results in 3-D, the **3D View** tab and HyperLynx PI PowerScope do not display Via Current hazards.
- The software reads design constraints from Constraint Manager only once: the first time you open a design.
- DC drop simulation does not support designs with multiple stackups or negative planes.
- Analysis Control does not report resistances between pairs of IC sink and VRM pins.
- You cannot run DC drop simulation from layout software on Linux.

Prerequisites

- You have the DC Drop license.
- You have installed:
  - Compatible versions of HyperLynx and a Siemens EDA layout tool that supports the Analysis Control capability. For details, see *HyperLynx Release Highlights*.
  - Xpedition Layout or another Siemens EDA layout tool with an Analysis Control capability that supports the DC Drop client. By contrast, an Xpedition Layout Team Client does not support the DC Drop client.
  - To view hazards in 3-D with a Siemens EDA layout tool, you have installed the 3D Plug-In product from the layout tool installation media.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- You understand the basic capabilities of DC drop simulation. See "DC Drop Simulation" on page 197.
- You have defined constraints for your design, including maximum voltage drop, current density, and via current. You can optionally specify them in Constraint Manager.
- You have assigned power-integrity (PI) models to VRMs, IC sinks, and passive components (resistors, inductors, and ferrite beads) that connect power nets. See "Assigning Models for PI Simulation" on page 103.

  To include reference nets in simulation, ensure that you have specified a reference net for each VRM and IC pin when you assigned PI models.

Procedure

1. Choose the **Analysis > Analysis Control** menu item.
2. From the Analysis Control window, from the **HyperLynx** menu 

, choose **Start Client**. The HyperLynx SI/PI client starts as a background process and loads your design.
3. Review constraints and change them as needed:

If you want to change...	Do this...
An individual constraint	Type a value, with or without units, and press Enter.
Multiple constraints at once	1. Click a cell to start a selection and press Shift-Click or Ctrl-Click to continue a selection. **Note:** You cannot click a cell in column 1 to start or continue a selection. 2. Type a value, with or without units, and press Enter.

4. To include measurements for reference nets for nets you select in step 5:

If you want to include...	Do this...
Reference net(s) for an individual net	In the Include Reference Net(s) column, choose Yes.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to include...	Do this...
Reference net(s) for multiple nets at once	1. Click a cell to start a selection and press Shift-Click or Ctrl-Click to continue a selection. **Note:** You cannot click a cell in column 1 to start or continue a selection. 2. In the Include Reference Net(s) column, choose Yes.

Ensure that you specified a reference net for each VRM and IC pin when you assigned PI models.

5. Select nets that you want to include in simulation:

If you want to select nets with…	Do this...
A Siemens EDA layout tool	In the Siemens EDA layout tool, select one or more nets.
Analysis Control	Check one or more nets. **Note:** If you want to check/uncheck multiple nets, click a cell to start the selection and press Shift-Click or Ctrl-Click to continue the selection. You cannot click a cell in column 1 to start or continue a selection.

6. Run a simulation:

If you want to simulate nets you selected in…	Do this...
A Siemens EDA layout tool	Click **Analyze Selection**.
Analysis Control	Click **Analyze Checked**.

Note

If simulation immediately finishes, it is possible that a model assignment problem prevented simulation from running. To investigate with Hazard Explorer, see step 7. In step 7.c, click a Setup spreadsheet row to display potential setup problems.

7. View hazards:
   a. Click **Hazards View**.
   b. In Hazard Explorer, in the left pane, click the **HyperLynx** tab and select a hazard branch (such as Voltage Drop).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

The All Hazards branch contains hazards for nets that you checked in Analysis Control.

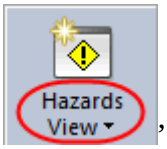
The Selection Hazards branch contains hazards for nets that you selected in the layout tool workspace.

c. In the right pane of Hazard Explorer, click a spreadsheet row to highlight a hazard in the layout tool.

d. If it is hard to see hazard objects, you can dim other objects by enabling the **Color By Hazard** and **Shadow Mode** buttons on the Hazard Explorer toolbar.

For information about Hazard Explorer, see “Displaying Hazards in the Design” in the *PCB Verification Guide*.

8. (Optional) View hazards in 3-D:

If you want to view hazards using the…	Do this...
3D View tab in a Siemens EDA layout tool	1. In the Siemens EDA layout tool, choose the **Window > Add 3D View** menu item. 2. Below the layout tool workspace, click the **3D View** tab. 3. In the left pane of Hazard Explorer, click the **HyperLynx** tab and select a hazard branch. 4. In the right pane of Hazard Explorer, click a spreadsheet row to highlight a hazard in the **3D View** tab. 5. If it is hard to see the results graphics, you can move and scale them in the Z-axis: a. From Analysis Control, from the **Hazards View** menu  , choose **3D Positioning**. b. In the 3D Positioning dialog box, change Span to scale the results graphics and change Origin to move away from the surface of the board.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to view hazards using the…	Do this...
HyperLynx PI PowerScope dialog box (which can additionally display current density and distribution)	1. In Analysis Control, from the **Hazards View** menu  , choose **Use PowerScope**. The HyperLynx PI PowerScope opens. 2. In the left pane of Hazard Explorer, click the **HyperLynx** tab and select a hazard branch. 3. In the right pane of Hazard Explorer, click a spreadsheet row to highlight a hazard in the HyperLynx PI PowerScope dialog box.

Results

The HyperLynx PI PowerScope dialog box can display graphical simulation results in both 2-D and 3-D. It displays DC drop voltage using color coding to indicate areas of higher and lower voltage drop, DC current distribution, and DC current density. For example results, see “Voltage Drop Graphs” on page 353 and “Current Density Graphs” on page 355.

Related Topics

HyperLynx PI PowerScope Dialog Box

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Analyzing a SERDES Design

You can use the SERDES Wizard to analyze multiple SERDES channels and run separate simulations for varying transmitter or receiver settings and channel properties.

The software can extract S-parameter models from the nets you include in simulation, and run S-parameter metrics and protocol compliance checks to produce results that include eye density plots, optimized transmitter and receiver settings, and other results in an interactive HTML spreadsheet format.

The wizard has several modes:

- Compliance Mode - The software creates an S-parameter model that the software uses to determine if channel performance meets the minimum standards for a specific protocol.
- Measured Mode - The software runs metric and compliance checks from an S-parameter model that you provide — measured in a test lab for example.
- IBIS-AMI Mode - If you have IBIS-AMI models for transmitters and receivers in your design, the software runs simulation and optimization to determine optimal transmitter and receiver settings.

If you...	Do the following...
Want the software to extract an S-parameter model of one or more SERDES channels that you want to analyze	Run SERDES batch simulation or SERDES batch simulation in Compliance Mode. See “Analyzing Multiple SERDES Channels Using the SERDES Wizard - Compliance Mode” on page 217.
Have a measured or extracted S-parameter model that represents one or more SERDES channels that you want to analyze	Run SERDES batch simulation in Measured Mode. See “Evaluating SERDES Channel Compliance From a Measured S-Parameter Model in the SERDES Wizard” on page 222.
Have IBIS-AMI models that describe transmitters and receivers in SERDES channels that you want to analyze, and want to determine optimal transmitter and receiver settings for your design	Run SERDES IBIS-AMI batch simulation See “Analyzing Multiple SERDES Channels Using the SERDES Wizard - IBIS AMI” on page 225.

You can also simulate one SERDES channel at a time using the FastEye Channel Analyzer Wizard. This wizard requires transmitter and receiver models and does not extract S-parameter models or run metrics or compliance checks. See “Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard” on page 232.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Analyzing Multiple SERDES Channels Using the SERDES Wizard - Compliance Mode

You can use the SERDES Batch Wizard to simulate multiple SERDES channels in a schematic or board design to investigate how channel topology, Rx/Tx parameters, jitter, and crosstalk affect channel performance.

The wizard also enables you to set up channel sweeps to run separate simulations for varying channel properties, such as stackup layer properties, passive component values and others.

SERDES batch simulation creates S-parameter models from the nets you include in simulation, and runs S-parameter metrics and protocol compliance checks to produce results that include eye density plots, optimized Tx and Rx settings, and other results in an interactive HTML spreadsheet format.

The wizard has several modes. Each mode is available from a different menu item. Compliance Mode only contains the wizard pages related to protocol compliance checking. In Compliance Mode, the software creates an S-parameter model that the software uses to determine if channel performance meets the minimum standards for a specific protocol.

Restrictions and Limitations

- Channel analysis is based on analytical modeling and simulation methods that are valid only for channels that behave linearly or nearly linear and time-invariant.
- SERDES batch simulation has unique coupling/crosstalk threshold settings. It ignores the crosstalk enable/disable setting (**Setup > Enable Crosstalk Simulation**) and does not use settings in the Set Coupling Thresholds and Coupling Settings dialog boxes.
- You can evaluate only one SERDES bus at a time, even if your design contains multiple SERDES busses.
- You cannot disable lossy transmission line modeling. Simulation accounts for this effect.

Prerequisites

- You have acquired the SERDES / FastEye / AMI license.
- For crosstalk analysis, you have acquired the Crosstalk license.
- The software recognizes your design correctly. The software must recognize differential pairs correctly to include them in simulation. See “Verifying That the Software Recognizes Your Design Correctly” on page 62.
- Decide whether to use S-parameter models to represent portions of critical nets that contain vias, DC blocking capacitors, BGA breakout fields, or other structures with poor return current paths or non-TEM (transverse electromagnetic mode) behaviors. For a board design, see “Modeling a Board Area With an S-Parameter Model” on page 122.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

For a schematic design, see "Modeling a Via with a 3D EM Model in a Schematic" on page 127.

- If you plan to run simulation using a protocol that is not part of the installation and have received from a Siemens EDA representative a *.zip* file that contains configuration and supporting files that describe a protocol, you have completed the steps in "Adding or Updating SERDES Protocols" on page 160.
- If you plan to include the crosstalk effects from additional aggressor nets in simulation, you have assigned models to drivers and receivers on those nets. Additional aggressor nets are differential nets that you do not want to analyze, but do want include in simulation, so you can see the crosstalk effect that these nets have on victim nets.
- If you have IBIS models that contain pin mapping information or package modeling information that you want to include in simulation, you have assigned these models to the Tx and Rx components that you want to include in simulation. See "Assigning Models to Components and Pins" on page 85.

Procedure

1. Choose the **Simulate SERDES > Run SERDES Compliance Check Wizard** () menu item to open the wizard in Compliance Mode.
2. On the Analysis Options page:
   a. On the left, select nets from the All Differential Pairs list on the left, and click the double right arrow () to move them to the Selected Differential Pairs list on the right and include them in simulation.

      Note

      Only differential pairs that the software recognizes appear in the list. See "Verifying That Differential Pairs are Recognized Correctly" on page 65.

   b. Select the channel operating protocol and type, and specify the symbol interval and symbol rate if needed.

      The list of protocols and types that appear in the list depend on the set of supporting .INI and .CNFG files in the *C:MentorGraphics\<release>\SDD_HOME\hyperlynx64* and *C:MentorGraphics\<release>\SDD_HOME\hyperlynx64\SerDes* folders. You can customize the list, or update the list when new protocols become available by acquiring the set of required files from a Siemens representative.

   c. When you run simulation, the software automatically saves a setup file (*HL_last.sws*) in the design folder that contains all the settings you made in the wizard. To save settings for later use, click **Save**. To load settings from a previous simulation, click "**…**", select the appropriate setup file, then click **Import**.

   d. Specify the results folder location. The default location depends on options you set in the "Reports and log files directory" section of the Set Directories Dialog Box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

e. Select “Generate PDF Report” if you want the software to create simulation results in PDF format.

3. On the Transmitters/Receivers page:

   a. If you have assigned IBIS models that contain information that identifies Tx and Rx buffers, the software automatically configures this page based on that information. If needed, click the reference designator in the table to change the endpoint, so that each endpoint is labeled correctly as a TX or RX, reflecting the correct driving direction.

   b. Repeat for “Endpoint #2” on the right. “Endpoint #2” can include multiple reference designators to describe, for example, a USB bus with multiple end points.

4. On the Compliance Verification page:

Note

In the Extract channel S-parameter area, the option to “Include packages from IC device models assigned to Tx/Rx pins’ is unavailable for compliance checking because the software includes protocol-specific parameters that specify parasitics for standard-reference packages, or reference Touchstone files that describe package parasitics.

   a. In the “Modeling parameters” area, you can change the default protocol-specific parameters that the software uses to create the S-parameter models if needed. Type the frequency range and reference impedance of the S-parameter model. Select a sampling (sweeping) type:

Sweeping Type	Instructions
Linear	Enter the number of simulation points. Increase the number if you simulate with the exported S-parameter model and run out of simulation points.
Logarithmic	Enter the number of points per decade.
Adaptive	Setting “Tolerance” to Low produces the most-accurate S-parameter model. Adaptive sweeping varies the sampling step size depending on model characteristics. It increases the sampling rate at resonant frequencies and other high-activity response events.

   b. In the Crosstalk Options area, define the conditions that the software uses to determine aggressor nets from the set of nets selected for simulation, and select any additional nets that the software should consider as potential aggressor nets.

   These additional nets (additional aggressors) are differential nets that you did not select to include in simulation on the Analysis Options page. The software does not

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analyze additional aggressor nets, but includes the effect from them on victim nets in analysis results.

Coupling thresholds and settings determine the set of aggressor nets from within the set of nets that you selected to include in simulation on the Analysis Options page. See "Accounting for Coupling" on page 131.

To account for the crosstalk effect from other nearby additional aggressor nets that meet the coupling threshold settings:

c. Select **Include additional pairs**, and click **Setup**. The Additional Aggressors dialog box opens.

d. Select a sorting method for the Differential Pairs list on the left (**by net name** or **by net class**).

e. Click pairs that you want to include in simulation in the Differential Pairs list, and click >> to add them to the Selected Aggressor Pairs list on the right.

f. Click **OK**.

Note

Ensure that you assign models to additional aggressor nets to define signal directionality, which enables the software to determine correct NEXT and FEXT results.

Note

SERDES batch simulation has unique coupling/crosstalk threshold settings. It ignores the crosstalk enable/disable setting (**Setup > Enable Crosstalk Simulation**) and settings you make in the Set Coupling Thresholds and Coupling Settings dialog boxes outside of the wizard (**Setup > Coupling Settings**).

Note

For a schematic design, coupling settings are not available on this page. Associate coupled nets by defining coupling regions in the schematic. See "Creating a Schematic Design" on page 70.

g. In the "Advanced options" area (for some protocols) you can edit the default compliance parameters if needed. Click **Edit Parameters**. Click and edit values in the Compliance Parameter Editor and click **Save**, or click **Load** and specify a previously-saved customized parameters file (*.cfng*). Customized parameter values appear in the Customized parameters list.

To understand the impact of editing a parameter value, refer to the latest version of the applicable protocol-specific specification. Specifications are available from the following groups online, often to registered users only:

- Ethernet - IEEE Standards Association

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- OIF CEI - Optical Internetworking Forum
- PCIe - PCI-SIG
- Fibre Channel - INCITS/T11
- USB - USB Implementers Forum (USB-IF)
- JESD - JEDEC

h. Select options to disable transmitter FFE or jitter, and receiver CTLE or DFE if you want to evaluate channel compliance without these effects.

i. If you want the software to run simulation without accounting for the transmitter or receiver package, select “Disable the assumed package model”. The assumed package model is a protocol-specific worst-case package model.

j. Check “Disable assumed PCB model” if you have already modeled the inter-cable PCB - with an S-parameter model in LineSim for example.

Note

This option is only available for some protocols, such as Ethernet CR.

5. On the Channel Sweep page, set a range for each set of channel parameter values that you want to run a separate simulation for. Expand the property tree and double-click a parameter to set a range in the Sweeping dialog box. See “Sweeping Dialog Box” on page 1045.
6. Click **Run**.

Results

When simulation completes, results open in an interactive HTML format. The software saves results files with your design file in a folder named *SerDesResults_<date-time>*. The HTML report contains links that open results files. The result types that appear in the report depend on the protocol.

Simulation result	Description
S-parameter files	S-parameter files that the software creates or uses to create results. Includes S-parameter files with and without crosstalk and mixed-mode conversions.
Pulse and step response waveform files	In *.lis* format, open in EZwave waveform viewer.
Eye density or bit error rate (BER) plots	Results in *.sds* format open in HyperLynx Eye Density/BER Viewer. BER plots help identify valid data sampling locations by reporting BER as a function of the sampling location across the unit interval (UI, same as bit interval) and voltage. The color of the contour indicates its BER.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

You can access HTML reports from previous simulations in the HTML Report Manager (**View > HTML Report Manager**).

Related Topics

Set Coupling Thresholds Dialog Box

Coupling Settings Dialog Box

Defining Net Classes

General Batch Wizard

General Batch Simulation Results Spreadsheet

Evaluating SERDES Channel Compliance From a Measured S-Parameter Model in the SERDES Wizard

You can use the SERDES wizard to analyze a measured S-parameter model that you provide, and run protocol compliance checks to produce results that include eye density plots, optimized Tx and Rx settings, and other results in an interactive HTML spreadsheet format.

The wizard has several modes. Each mode is available from a different menu item. Measured Mode enables you to run metric and compliance checks from an S-parameter model that you provide.

Note

The software does not require Tx or Rx models to produce simulation results.

Restrictions and Limitations

- In Measured Mode, the wizard only analyzes a single channel at a time.

Prerequisites

- You have acquired the SERDES / FastEye / AMI license.
- You have a measured 8-port (or higher) S-parameter model that includes a single victim and one or more aggressor channels. You can also use a measured 4-port S-parameter model that represent the SERDES channel that you want to analyze and additional 4-port S-parameter models that represent NEXT or FEXT from each aggressor channel (if you want to include the effects of crosstalk).

  Note

  If you need to de-embed the test fixture from a measured S-parameter model, see “Removing the Effect of Measurement Fixtures From an S-Parameter Model (De-embedding)” on page 1361 .

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- If you plan to use a protocol that is not part of the installation and have received a *.zip* file that contains configuration and supporting files that describe a protocol from a Siemens EDA representative, you have completed the steps in "Adding or Updating SERDES Protocols" on page 160.

Procedure

1. Choose the **Simulate SERDES > Use Measured S-parameters in > SERDES Batch Wizard (Compliance Checks Only)** menu item to open the SERDES Batch Wizard.
2. On the Analysis Options page:
   a. In the first row, click ... and select the S-parameter model file that includes the channel that you want to analyze.
   b. In the Port Conn column, select the correct port connectivity numbering pattern. See "S-Parameter Port Numbering" on page 1346.
   c. If you want to include the effects of crosstalk in simulation, in the following rows, click ... and select the S-parameter model file that represents an aggressor channel that you want to include in simulation. Repeat this step for any additional aggressor channels. If you are using an 8-port or higher model that includes both victim and aggressor nets, the software considers the first 4 ports to be the victim channel and highlights both victim and aggressor channels in red.
   d. In the Type column, for each aggressor channel, select NEXT or FEXT to specify the crosstalk relationship to the victim channel S-parameter model.
   e. Select the channel operating protocol and type, and specify the symbol interval and symbol rate if needed.

      The list of protocols and types that appear in the list depend on the set of supporting .INI and .CNFG files in the *C:MentorGraphics\<release>\SDD_HOME\hyperlynx64* and *C:MentorGraphics\<release>\SDD_HOME\hyperlynx64\SerDes* folders. You can customize the list, or update the list when new protocols become available by acquiring the set of required files from a Siemens representative.
   f. When you run simulation, the software automatically saves a setup file (*HL_last.sws*) in the design folder that contains all the settings you made in the wizard. To save settings for later use, click **Save**. To load all settings from a previous simulation, click "**…**", select the appropriate setup file, and click **Import**.
   g. Specify the results folder location. The default location depends on options you set in the "Reports and log files directory" section of the Set Directories Dialog Box.
   h. Select "Generate PDF Report" if you want the software to create simulation results in PDF format.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

3. On the Compliance Verification page, set options for protocol compliance checking:

   Note

   Options in the Extract channel S-parameter area, Modeling parameters area, and crosstalk options area are unavailable in Measured Mode, since the software uses models you provide, and does not extract S-parameter models.

   a. In the "Advanced options" area (for some protocols) you can edit the default compliance parameters if needed. Click **Edit Parameters**. Click and edit values in the Compliance Parameter Editor and click **Save**, or click **Load** and specify a previously-saved customized parameters file (.cfng). Customized parameter values appear in the Customized parameters list.

      To understand the impact of editing a parameter value, refer to the latest version of the applicable protocol-specific specification. Specifications are available from the following groups online, often to registered users only:

      - Ethernet - IEEE Standards Association
      - OIF CEI - Optical Internetworking Forum
      - PCIe - PCI-SIG
      - Fibre Channel - INCITS/T11
      - USB - USB Implementers Forum (USB-IF)
      - JESD - JEDEC

   b. Select options to disable transmitter FFE or jitter, and receiver CTLE or DFE if you want to evaluate channel compliance without these effects.

   c. If you want the software to run simulation without accounting for the transmitter or receiver package, select "Disable the assumed package model". The assumed package model is a protocol-specific worst-case package model.

   d. Check "Disable assumed PCB model" if you have already modeled the inter-cable PCB - with an S-parameter model in LineSim for example.

      Note

      This option is only available for some protocols, such as Ethernet CR.

4. On the Channel Sweep page, set a range for each set of channel parameter values that you want to run a separate simulation for. Expand the property tree and double-click a parameter to set a range in the Sweeping dialog box. See "Sweeping Dialog Box" on page 1045.

5. Click **Run**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Results

When simulation completes, results open in an interactive HTML format. The software saves results files with your design file in a folder named *SerDesResults_<date-time>*. The HTML report contains links that open results files. The result types that appear in the report depend on the protocol.

Simulation result	Description
S-parameter files	S-parameter files that the software creates or uses to create results. Includes S-parameter files with and without crosstalk and mixed-mode conversions.
Pulse and step response waveform files	In *.lis* format, open in EZwave waveform viewer.
Eye density or bit error rate (BER) plots	Results in *.sds* format, open in HyperLynx Eye Density/BER Viewer. BER plots help identify valid data sampling locations by reporting BER as a function of the sampling location across the unit interval (UI, same as bit interval) and voltage. The color of the contour indicates its BER.

You can access HTML reports from previous simulations in the HTML Report Manager (**View > HTML Report Manager**).

Analyzing Multiple SERDES Channels Using the SERDES Wizard - IBIS AMI

If you have IBIS-AMI models that describe transmitters and receivers in your design, you can use the SERDES Wizard to simulate multiple SERDES channels in a schematic or board design to investigate how channel topology, Rx/Tx parameters, jitter, and crosstalk affect channel performance. The wizard can also run optimization to determine optimal transmitter and receiver settings for your design.

The wizard also enables you to set up channel sweeps to run separate simulations for varying channel properties, such as stackup layer properties, passive component values and others.

SERDES batch simulation runs time domain or statistical simulation using IBIS AMI models to create S-parameter models from the nets you include in simulation, and runs S-parameter metrics and protocol compliance checks to produce results that include eye density plots, bit error rate (BER) plots, and bathtub curves, optimized Tx and Rx settings, and other results in an interactive HTML spreadsheet format.

Restrictions and Limitations

- Channel analysis is based on analytical modeling and simulation methods that are valid only for channels that behave linearly or nearly linear and time-invariant.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- SERDES batch simulation has unique coupling/crosstalk threshold settings. It ignores the crosstalk enable/disable setting (**Setup > Enable Crosstalk Simulation**) and does not use settings in the Set Coupling Thresholds and Coupling Settings dialog boxes.
- You can evaluate only one SERDES bus at a time, even if your design contains multiple SERDES busses.
- You cannot disable lossy transmission line modeling. Simulation accounts for this effect.

Prerequisites

- You have acquired the SERDES / FastEye / AMI license.
- For crosstalk analysis, you have acquired the Crosstalk license.
- The software recognizes your design correctly. The software must recognize differential pairs correctly to include them in simulation. See “Verifying That the Software Recognizes Your Design Correctly” on page 62.
- You have model files for transmitters and receivers in your design, including IBIS-AMI files and their associated executable files, and have copied them to the correct location (**Models > Edit Model Library Paths)**.

  Store *.dll* or *.so* files in the same folder as IBIS models that reference them.
- Decide whether to use S-parameter models to represent portions of critical nets that contain vias, DC blocking capacitors, BGA breakout fields, or other structures with poor return current paths or non-TEM (transverse electromagnetic mode) behaviors. For a board design, see “Modeling a Board Area With an S-Parameter Model” on page 122. For a schematic design, see “Modeling a Via with a 3D EM Model in a Schematic” on page 127.
- If you plan to run simulation using a protocol that is not part of the installation and have received from a Siemens EDA representative a *.zip* file that contains configuration and supporting files that describe a protocol, you have completed the steps in “Adding or Updating SERDES Protocols” on page 160.
- If you plan to include the crosstalk effects from additional aggressor nets in simulation, you have assigned models to drivers and receivers on those nets. Additional aggressor nets are differential nets that you do not want to analyze, but do want include in simulation, so you can see the crosstalk effect that these nets have on victim nets.
- If you have IBIS models that contain pin mapping information or package modeling information that you want to include in simulation, you have assigned these models to the Tx and Rx components that you want to include in simulation. See “Assigning Models to Components and Pins” on page 85.

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Procedure

1. Choose the **Simulate SERDES > Run SERDES IBIS-AMI Batch Wizard** () menu item.
2. On the Analysis Options page:
   a. On the left, select nets from the All Differential Pairs list on the left, and click the double right arrow () to move them to the Selected Differential Pairs list on the right and include them in simulation.

   Note

   Only differential pairs that the software recognizes appear in the list. See "Verifying That Differential Pairs are Recognized Correctly" on page 65.

   b. Select the channel operating protocol and type, and specify the symbol interval and symbol rate if needed.

   The list of protocols and types that appear in the list depend on the set of supporting .INI and .CNFG files in the *C:MentorGraphics\<release>\SDD_HOME\hyperlynx64* and *C:MentorGraphics\<release>\SDD_HOME\hyperlynx64\SerDes* folders. You can customize the list, or update the list when new protocols become available by acquiring the set of required files from a Siemens representative.

   c. When you run simulation, the software automatically saves a setup file (*HL_last.sws*) in the design folder that contains all the settings you made in the wizard. To save settings for later use, click **Save**. To load settings from a previous simulation, click "**…**", select the appropriate setup file, then click **Import**.

   d. Specify the results folder location. The default location depends on options you set in the "Reports and log files directory" section of the Set Directories Dialog Box.

   e. Select "Generate PDF Report" if you want the software to create simulation results in PDF format in addition to an HTML report.
3. On the Transmitters/Receivers page:
   a. If you have assigned IBIS models that contain information that identifies Tx and Rx buffers, the software automatically configures this page based on that information. Otherwise, select the reference designator that corresponds with the driving device from the "Endpoint #1" list.

   b. Repeat for "Endpoint #2" on the right. "Endpoint #2" can include multiple reference designators to describe, for example, a USB bus with multiple end points.

   c. If needed, to simplify the set of nets that appear in the list, type filter text in the Pair Filter or Component fields, click **Edit Net Classes** to configure or edit groups of nets, or click **Edit Coupling** to configure settings that determine the set of additional aggressors nets that the software includes in simulation.

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4. On the Advanced Eye Diagram page:

   a. In the Analysis options area, select an analysis type and select the IC modeling corners for which you want to see simulation results. See “Statistical and Time Domain Simulation Comparison” on page 600.

   b. In the Stimulus area, for time-domain simulation, select a Sequence type, then specify the duration of simulation by typing the total number of symbols to simulate. The software repeats the specified symbol pattern until it reaches the total number of symbols.

   Note

   If the protocol you select on the Analysis Options page specifies a modulation type, this page displays that modulation type. If the protocol does not specify a modulation type, this page obtains a modulation type from the Device 1 Settings and Device 2 Settings pages.

   c. If you want to set advanced AMI-related options, such as the number of samples per symbol interval, number of symbols per GetWave call to an AMI DLL, number of starting symbol to ignore, or advanced troubleshooting options, click **Advanced AMI Options**. See “Advanced AMI Options Dialog Box” on page 694.

   This option is available when you enable “Time-domain” in the “Analysis options” area.

   d. In the “Eye mask” area, select the eye mask that you want to use when viewing simulation results, or click the ellipsis button to create and save an eye mask. To use a simple BER threshold instead of a mask, select **<No Mask>** and type the threshold value.

   e. In the “Channel characterization” area, check “Automatic channel characterization”. See “Bit Sequence for Automatic Channel Characterization” on page 478.

   If you think you need to set the characterization type (step/pulse response or PRBS), change channel response settings, or enable channel linearity checking, uncheck “Automatic channel characterization”, then click **Advanced**. See “Advanced Channel Characterization Dialog Box” on page 695.

   f. If you intend to troubleshoot AMI models, in the Model Testing area, click **Setup** to set options. See “AMI Model Testing Dialog Box” on page 699. Click **Preview channel characterization** to view the response waveform(s) at the receiver.

   Note

   Model testing options are unavailable if you have selected more than one differential pair for analysis on the Analysis Options page.

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Note

When assigned AMI models contain [Ts4file] keywords that reference S-parameter models to describe Tx and Rx on-die interconnect behavior, the software does not create or use time domain waveforms in channel characterization, and channel characterization preview is unavailable.

g. In the Crosstalk Options area, define the conditions that the software uses to determine aggressor nets from the set of nets selected for simulation, and select any additional nets that the software should consider as potential aggressor nets.

   These additional nets (additional aggressors) are differential nets that you did not select to include in simulation on the Analysis Options page. The software does not analyze additional aggressor nets, but includes the effect from them on victim nets in analysis results.

   Coupling thresholds and settings determine the set of aggressor nets from within the set of nets that you selected to include in simulation on the Analysis Options page. See “Accounting for Coupling” on page 131.

Note

For Advanced Eye Analysis, the software determines aggressor driver behavior from victim Rx voltage step responses, unless AMI models provide S-parameters for the victim Tx and Rx. In this case, the AMI analysis uses S-parameters for the aggressor Tx-to-victim Rx coupling, along with the assumption that the aggressor Tx has the same edge rate, voltage swing, and impedance as the victim Tx.

   To account for the crosstalk effect from other nearby additional aggressor nets that meet the coupling threshold settings:

h. Select **Include additional pairs**, and click **Setup**. The Additional Aggressors dialog box opens.

i. Select a sorting method for the Differential Pairs list on the left (**by net name** or **by net class**).

j. Click pairs that you want to include in simulation in the Differential Pairs list, and click >> to add them to the Selected Aggressor Pairs list on the right.

k. Click **OK**.

Note

Ensure that you assign models to additional aggressor nets to define signal directionality, which enables the software to determine correct NEXT and FEXT results.

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Note

SERDES batch simulation has unique coupling/crosstalk threshold settings. It ignores the crosstalk enable/disable setting (**Setup > Enable Crosstalk Simulation**) and settings you make in the Set Coupling Thresholds and Coupling Settings dialog boxes outside of the wizard (**Setup > Coupling Settings**).

Note

For a schematic design, coupling settings are not available on this page. Associate coupled nets by defining coupling regions in the schematic. See "Creating a Schematic Design" on page 70.

5. On the Endpoint 1 and Endpoint 2 Settings pages, specify the transmitter and receiver IBIS AMI parameters (equalization for example) and jitter settings that you want to use in simulation. Click a setting in the Transmitter or Receiver list and type a value. The software uses these settings to create a temporary AMI file for simulation, preserving your original files.

   Note

   You cannot type a value for a parameter in the Reserved_Parameters tree.

   The software saves independent settings for each endpoint (device). Click **Copy Settings From Endpoint 1** or **Copy Settings From Endpoint 2** if needed.

   If you do not want all channels to use the same endpoint settings (for example, a channel with longer nets may require different equalization settings) click **Per-channel overrides**. In the "Per-channel overrides for <reference designator>" dialog box, click values to edit them as needed for each channel. Click **Customize Tx** or **Customize Rx** if you only need to edit a few settings for each channel and want only those settings to appear in the "Per-channel overrides" dialog box.

6. On the Parameter Sweep or Parameter Optimization page, you can set the software to run multiple simulations and vary a specific range of AMI parameters, or set the software to automatically run the minimum required number of simulations while varying parameters to find the optimal settings. Select **Sweep Parameters** or **Optimize Parameters**.

   Note

   Consider the run time required by a set of sweep simulations when selecting parameters and parameter values. A simulation runs for each combination of parameter and range of values (number of parameters multiplied by values in the sweep range). The wizard displays the number of simulations near the bottom of the page.

   If you need to run sweep simulations with many parameter combinations, consider running more than one sweep simulation to cover the parametric range. You can also synchronize, or lock the range of values (Paste Range as a Lock), for one model

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parameter to another model parameter, so that the software runs a single simulation for all locked parameters.

a. Double-click a parameter and select values as needed to set a range of varying values.

b. If you selected Optimize Parameters, click Optimization Options change the target figure of merit and other advanced options. The value used for stimulus PRBS order should be low, is used only during optimization, and is separate from the stimulus used to produce simulation results with AMI models. Do not change these options unless you have a specific reason to do so, or if you are working with a Siemens EDA representative.

7. On the Channel Sweep page, set a range for each set of channel parameter values that you want to run a separate simulation for. Expand the property tree and double-click a parameter to set a range in the Sweeping dialog box. See "Sweeping Dialog Box" on page 1045.

8. Click **Run**.

Results

When simulation completes, results open in an interactive HTML format. The software saves results files with your design file in a folder named *SerDesResults_<date-time>*. The HTML report contains links that open results files. The result types that appear in the report depend on the protocol.

Simulation result	Description
S-parameter files	S-parameter files that the software creates or uses to create results. Includes S-parameter files with and without crosstalk and mixed-mode conversions.
Pulse and step response waveform files	In *.lis* format, open in EZwave waveform viewer.
Eye density or bit error rate (BER) plots	Results in *.sds* format open in HyperLynx Eye Density/BER Viewer. BER plots help identify valid data sampling locations by reporting BER as a function of the sampling location across the unit interval (UI, same as bit interval) and voltage. The color of the contour indicates its BER.

You can access HTML reports from previous simulations in the HTML Report Manager (**View > HTML Report Manager**).

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Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

If you do not have IBIS-AMI models that describe transmitters or receivers in your SERDES design, you can use the FastEye Channel Analyzer to simulate a SERDES channel to investigate how channel topology, Rx/Tx equalization and pre-emphasis parameters, jitter, and crosstalk affect channel performance. FastEye channel analysis runs much faster than standard eye-diagram simulations (using the Oscilloscope or EZwave) when simulating many bits.

FastEye channel analysis produces eye diagrams, bit error rate (BER) plots, and bathtub curves to help you see how channel topology and jitter affect channel performance.

FastEye channel analysis can also account for crosstalk from aggressor nets on the selected victim channel. The FastEye wizard can either automatically create the crosstalk files or use crosstalk files that you created with hardware measurements or saved from a previous FastEye channel analysis.

Restrictions and Limitations

- You can only analyze one single-ended or differential channel at a time.
- Crosstalk analysis does not run *round robin* simulations for primary (victim) or aggressor channels with more than one transmitter. To run FastEye channel analysis with different victim/aggressor pins driving the channel, manually enable/disable the appropriate model pins and run separate analyses.
- Channel analysis is based on analytical modeling and simulation methods that are valid only for channels that behave linearly or nearly linear. For non-LTI channels, you can still use the FastEye Wizard to generate a worst-case bit sequence that is likely to close the eye more than even a very long PRBS stimulus. This allows you to estimate eye closure using the oscilloscope or EZwave.

Prerequisites

- You have acquired the SERDES / FastEye / AMI license.
- For crosstalk analysis, you have the Crosstalk license.
- You have model files for transmitters and receivers in your design and have copied them to the correct location (**Setup > Options > Directories**).
- You have copies of channel characterization files or crosstalk characterization files (*.pls* or *.sp*) if you do not want the FastEye Channel Analysis wizard to create them.
- Verify that the software recognizes your design correctly. See “Verifying That the Software Recognizes Your Design Correctly” on page 62.
- Decide whether to use S-parameter models to represent portions of critical nets that contain vias, DC blocking capacitors, BGA breakout fields, or other structures with poor return current paths or non-TEM (transverse electromagnetic mode) behaviors. For a

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board design, see "Modeling a Board Area With an S-Parameter Model" on page 122. For a schematic design, see "Modeling a Via with a 3D EM Model in a Schematic" on page 127.

- Configure simulation options to account for coupling, loss, and noise between single-ended signal vias and power planes. See "Setting Simulation Options" on page 131.
- Assign models and set transmitter and receiver buffer states as needed. See "Assigning Models to Components and Pins" on page 85. For bidirectional IC buffers, use the Assign Models dialog box to set the transmitter pin to the output direction and set the receiver pin to the input direction. If you include crosstalk in channel analysis, also do this for aggressor nets.
- For a board design, select the nets of the channel that you want to simulate. See "Selecting Nets for SI Simulation" on page 141.
- If your simulation includes signal nets with series capacitors, the software can assign a settled voltage to the capacitors when you enable the "Settle Series Capacitors Before Every Simulation" option on the Preferences Dialog Box - Advanced Tab. Enabling this option means that automatic channel characterization can obtain a settled voltage for a series capacitor on a signal net without having you assign a special ADMS or SPICE model to it.

Procedure

1. Choose **Simulate SERDES > Run FastEye Channel Analysis** or click  to open the FastEye Channel Analyzer wizard. Select simulation options on each page and click **Next**.
2. On the Choose New/Saved Analysis page, select **New** to create a new configuration setup file (*.few*) that contains all the settings you make in the wizard. Select **Use last configuration** to load the most recently used setup file, or select **Load saved configuration** to select a specific configuration setup file. To save wizard settings for future analysis or channel characterization select **Save settings to file** and specify a path for the saved file.
3. On the Choose Analysis Type page, select an option:
   - Choose "Perform FastEye analysis and optionally generate worst-case sequence" for a time domain simulation.
   - Choose "Only generate worst-case sequence" to only save the worst-case stimulus file (*.bit*).

     Use this option if you want to create a stimulus for lab-based measurements of eye density, or to generate an eye diagram in the oscilloscope, EZwave, or a third-party simulator. You can generate a worst-case sequence even when the channel does not exhibit linear and time-invariant (LTI) behavior. For such channels, the worst-case pattern may not produce maximal closure, but it is likely to close the eye more than

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even a very long random (PRBS) stimulus. This allows you to estimate eye closure using the oscilloscope or EZwave.

- Choose “Perform statistical analysis” for a statistical simulation.

  See “Statistical and Time Domain Simulation Comparison” on page 600, “Checking Channels for Linear and Time-Invariant Behavior” on page 484, and “Worst-Case Bit Patterns - FastEye” on page 891.

4. On the Set Up Channel Characterizations page, specify the transmitter and receiver pins in your design, and set the probe location to “Always at the die”.

   a. Select **New** to allow the wizard to create a channel characterization file. If needed, click **New/View** to configure channel characterization options. To use an existing characterization file, click **Load**.

   b. To include the effects of crosstalk from nearby aggressor channels or nets, select **Include crosstalk effects from aggressor channels** and select the aggressor channels that you want to include in the simulation. The channels that appear in the list depend on crosstalk threshold settings.

      If needed, you can also add the crosstalk effects from other “external” aggressor channels to simulation by specifying a separate characterization file created outside the wizard. Select “Allow external aggressor channels” and click **+ Channel**. Double click the cell in the Path column to specify the location of the external crosstalk characterization files.

      Select the Low inactive stuck state unless you have a specific reason not to. For most driver ICs, the impedance of the low stage is lower than or equal to the impedance of the high stage and so the worst-case reflections of crosstalk signals come from the low stage.

   Channel characterization files are created automatically when you start analysis, or you can select a channel and click **Characterize Selected** to create them now. For more information on external characterization files, see “External Characterization Files” on page 539.

5. On the Define Stimulus page, define the stimulus by specifying the simulation length (total bits), bit interval/rate, and bit pattern type. You can also simulate eye stress by specifying a period (in bits) within which the software inserts worst-case patterns into the stimulus. To generate a worst-case stimulus file, select **Worst-case PRBS** or **Worst-case 8b/10b** as the bit pattern.

   Check **Convert to PAM-4** to convert the stimulus to PAM-4 encoding.

6. On the Add Jitter page, select deterministic or random types of jitter to add to the channel input stimulus to model effects such as crosstalk or supply noise. Specify a jitter distribution that represents both driver and receiver jitter. Drivers and receivers are active devices that contribute random jitter due to thermal and transistor device noise, PLL (that is, CDR circuitry) behavior, and so on. If you are unsure about which type of

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jitter to use, select only Gaussian, and use the standard deviation (sigma) value from your driver IC's data sheet.

Click the arrow button to specify more details.

Note

Do not specify the jitter produced by the following effects, unless you have a specific reason to do so:

- PCB layout effects—Such as impedance mismatches and signal dispersion
- Data-dependent effects—Such as ISI, duty-cycle distortion, pseudo-random bit sequence periodicity

7. On the Add Pre-Emphasis/DFE/CTLE page:

If you want to...	Do the following...
Have the software determine optimal pre-emphasis, DFE or CTLE settings	1. Click **Synthesize optimal values**. 2. Click **Add pre-emphasis**, **Add DFE** or **Add CTLE**. 3. For pre-emphasis or DFE, click **Taps/weights** to specify the total number of taps.
Specify pre-emphasis, DFE or CTLE settings	1. Click **Specify taps/parameters**. 2. Click **Add pre-emphasis**, **Add DFE** or **Add CTLE**. 3. For pre-emphasis or DFE, click **Taps/weights** to specify taps and weights. 4. For CTLE, click **Specify parameters** to define the CTLE filter behavior.

Note

If the channel characterization simulation uses a driver model with pre-emphasis enabled or a receiver model with CTLE enabled, these filters are already accounted for and you do not need to re-specify them.

If you chose worst-case sequence generation only on the Choose Analysis Type page, DFE is disabled. However, you can still specify pre-emphasis and CTLE filter settings.

8. On the Choose Fitting/Convolution page, select **Complex-pole fitting**. Select **Convolution** for more accurate results if your channel has a short pulse/step response (ISI effects attenuate quickly) and a wide, complex spectrum (both high- and low-frequency resonances).

   See "Model Channel Frequency Response with Complex-Pole Models" on page 564.

9. On the Set Up Crosstalk Analysis page, configure crosstalk analysis settings.

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a. Select synchronous or asynchronous crosstalk timing, to describe the phase relationship between the victim channel and aggressor channels.

b. Select time domain or statistical analysis. See "Statistical and Time Domain Simulation Comparison" on page 600.

c. Set the stimulus type, or load a custom bit sequence file. To set a different stimulus for a channel, click the Stimulus cell in the spreadsheet for the channel, and select from the list.

Select **Convert to PAM-4** to convert the stimulus to PAM-4 encoding. The software applies a PAM-4 stimulus to both aggressor and victim nets.

10. On the View Analysis Results page, select the results that you want to view.
11. Click **Run** to start simulation.

Results

When simulation completes, the results that you selected on the View Analysis Results page appear.

Simulation result	Description
Eye diagrams	The software displays an eye diagram in the FastEye Viewer.
Eye density or bit error rate (BER) plots	BER plots help identify valid data sampling locations by reporting BER as a function of the sampling location across the unit interval (UI, same as bit interval) and voltage. The color of the contour indicates its BER. The software displays eye density or BER plots in the HyperLynx Eye Density/BER Viewer. Eye density plot (*.tps*) files are stored in the design folder.
Bathtub curves	Bathtub curves help identify valid data sampling locations by reporting the BER as a function of the sampling location across the unit interval at several voltage offsets. The software displays bathtub curves in the Bathtub Chart dialog box. **Note:** Bathtub curves are not available for simulations that include parametric sweeps.
Statistical Contour Chart	A statistical contour chart shows a nested series of eye opening contours and their BER. The color of the contour indicates its BER. Like bathtub curves, statistical contours indicate the quality of sampling locations across the unit interval. An advantage of statistical contours over bathtub curves is that the inner-eye contours display both sampling time and voltage information. The software displays statistical contours in the Statistical Contour Chart dialog box.

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You can use the View buttons on the View Analysis Results page to re-open dialog boxes that display results. The results are available until you close the wizard or run a new analysis.

Related Topics

Assign Models Dialog Box

FastEye Viewer

Bathtub Chart Dialog Box

Statistical Contour Chart Dialog Box

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Decoupling Simulation

Depending on the accuracy and information you seek, the software provides different methods for running PDN decoupling simulation.

To achieve the best decoupling simulation results and efficiently use your time, use the PDN Decoupling Optimizer. If you do not have a license that includes the PDN Decoupling Optimizer, you should follow this process:

- First run lumped decoupling simulation. It quickly produces an approximate impedance profile for the overall PDN. This simulation type requires decoupling capacitor models.
- If you do not get a sufficiently low PDN impedance profile from lumped simulation, run Quick Analysis to help you find ineffective decoupling capacitors by listing their mounting inductance and resonant frequency.
- Run distributed decoupling simulation after you get good results from lumped decoupling simulation. It can have a long simulation run time and produce accurate impedance profiles for individual power supply pins or groups of power supply pins. This simulation type requires decoupling capacitor models and can use optional VRM models.

For detailed diagrams of the above process, see Measuring PDN Impedance at Key Locations on the Board, Designing PDNs to Meet Low Impedance Requirements Across a Range of Frequencies.

Topic	Description
Simulating PDN Decoupling - Lumped	Evaluate the ability of an overall PDN to provide low-impedance paths for IC current loads. Lumped decoupling simulation runs quickly and provides approximate results. It accounts for the mounting parasitics and loss for decoupling capacitors, but ignores the location of capacitors and the board outline.
Simulating PDN Decoupling - Quick Analysis	Find ineffective decoupling capacitors by listing information about them, such as mounting inductance and resonant frequency.
Distributed Decoupling Simulation Comparison	Depending on the simulation capabilities you seek, the software provides different methods for running distributed decoupling simulation.
Simulating PDN Decoupling - Distributed	Evaluate the ability of a PDN to provide low-impedance paths for specific power-supply pins or pin groups. Distributed PDN decoupling simulation provides more accurate results than the lumped modeling method, but takes longer to complete. It accounts for the location of each decoupling capacitor, inter-plane capacitance and inductance, the board outline, and the mounting parasitics and loss for decoupling capacitors.

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Topic	Description
Simulating PDN Decoupling - Advanced Distributed	Evaluate the ability of a PDN to provide low-impedance paths for specific power supply pin pairs or pin group pairs. Advanced distributed PDN decoupling simulation accounts for the location of each decoupling capacitor, inter-plane capacitance and inductance, the board outline, and the mounting parasitics and loss for decoupling capacitors.
Optimizing PDN Decoupling Capacitors - HyperLynx PI	The PDN Decoupling Optimizer enables you to analyze existing decoupling capacitors and their locations in a board or schematic (PDN Editor) design to determine the optimum decoupling capacitor for each location. An optimized design includes the minimum number of capacitors, the minimum number of different capacitor types, and has the lowest cost.

Simulating PDN Decoupling - Lumped

Evaluate the ability of an overall PDN to provide low-impedance paths for IC current loads. Lumped decoupling simulation runs quickly and provides approximate results. It accounts for the mounting parasitics and loss for decoupling capacitors, but ignores the location of capacitors and the board outline.

Restrictions and Limitations

- Decoupling simulation runs on one pair of power supply nets at a time, where one net provides power and the other provides return current. For ICs with multiple pairs of power supply nets, run decoupling simulation for each pair of power supply nets that you want to simulate.
- You cannot run decoupling simulation on power supply nets formed entirely by trace segments. You can simulate power supply nets that contain metal areas that form at least one transmission plane.
- If you have a multiple-board design, you can run decoupling simulation on one board at a time.
- Decoupling simulation is unavailable for board designs with multiple stackups.

Prerequisites

- Assign decoupling capacitor models. See Assigning Models to Decoupling Capacitors.
- Acquire the information needed to define the target impedance for the PDN. See Information Needed to Calculate Target PDN Impedance.
- Acquire the Decoupling license.

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Procedure

1. Click **Analyze Decoupling**  , or choose the **Simulate PI > Analyze Decoupling**.
2. On the Start Analysis page:
   a. Do one of the following.

If you want to...	Do this...
Run the wizard for the first time on your design.	Select New.
Run the wizard another time on your design.	Select “Use last configuration”. **Note:** This option is unavailable after you close the design.
Load wizard settings from a previously-saved setup file, and run the wizard another time on your design.	1. Select “Load saved configuration”. 2. Click **Load**, specify the location of the *.dao* wizard settings file, and click <*Open*>.

   b. To save the wizard settings file, check “Save settings to file”, click **Browse**, and specify the location of the *.dao* file.
3. On the Choose a Type of Analysis page, select Lumped Analysis.
4. On the Select Nets for Analysis page, add a pair of power supply nets to the Nets to analyze area. You can double-click net names to move them between the Available nets area and Nets to analyze area.
5. On the Check Capacitor Models page:
   a. Verify decoupling capacitor model assignments and values. If needed, you can assign or remove models, and edit groups.
   b. To perform a “what if” experiment by excluding a model from simulation, uncheck Enabled.
6. On the Check VRM Models page, verify model assignments, then choose whether to probe VRM pins:
   a. To display VRMs assigned only to nets that you selected on the Select Power Nets page, check “Show VRM(s) for selected nets only”.
   b. To add or edit a VRM model or reference net, click **VRM Manager**. The Assign Power Integrity Models dialog box opens. For information, see step 3 in “Assigning VRM Source, DC Sink, and AC Models” on page 103.

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c. To probe pins for a VRM, ensure that its right-most spreadsheet column has a check mark. An empty blue box in the right-most column indicates missing model information, such as a reference net.

7. On the Set the Target Impedance page, either enter the target impedance or click **Calculator** to open a wizard that can help you define the target impedance.
8. On the Choose Easy / Custom page, choose between default and custom decoupling simulation options.

   If you select Easy, some options on the following wizard pages become unavailable.
9. On the Customize Settings page, do any of the following:

If you want to...	Do this...
Account for energy coming from transmission planes formed by other pairs of power supply nets and transmitted through stitching vias that run vertically throughout the stackup. **Note:** Checking this option can increase simulation run time.	Check “Include all power / ground nets in the analysis”.
Account for decoupling capacitor mounting inductance.	Check “Include capacitor mounting inductance”.
Account for capacitance between plane layers.	Check “Include inter-plane capacitance”.
Account for capacitors that may be useful only at low frequencies because they have poor or non-ideal mounting properties.	Check “Include poorly connected capacitors”.

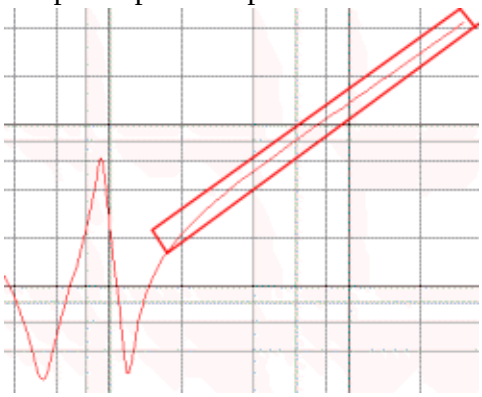
10. On the Control Frequency Sweep page:

    a. Enter the frequency range in MHz.

       Above a certain frequency, such as 150 MHz, many ICs have in-package decoupling that provide most of the decoupling. Decoupling capacitors and buried capacitance in the PCB contribute little or no decoupling above this design-dependent frequency.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

b. Do one of the following:

If you want to distribute sampling points across the frequency range...	Do this...
In a variable way, by increasing the sampling rate near frequencies with resonances. Use this sampling method unless you have a reason to not use it.	1. Select “Adaptive sampling”. 2. Specify an “Accuracy at resonances” value. Selecting High produces the longest simulation run time, but it may still be reasonably short for lumped simulation. 3. Specify a value for the “Minimum number of samples in flat, non-resonant regions”. The enclosed region in the figure below shows a flat and non-resonant region of an example impedance profile. 
At logarithmic intervals.	1. Select “Logarithmic sampling”. 2. Specify a value for the “Number of samples” for the entire frequency range.
At equal intervals.	1. Select “Linear sampling”. 2. Specify a value for the “Number of samples” for the entire frequency range.

11. On the Run Analysis page, rename the results folder if needed, and click **Run Analysis**.

Results

When simulation completes, the software displays results and saves them with wizard settings (*.dao* file) in a timestamped folder within the design folder.

Simulation Result	Description
Output file (Z-parameter file)	The Touchstone Viewer automatically opens and displays a Z-parameter model created by decoupling simulation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Simulation Result	Description
HyperLynx Lumped Decoupling Report (HTML)	A report that shows the PDN impedance profile as a Z-parameter plot. You can access HTML reports from previous simulations in the HTML Report Manager (**View > HTML Report Manager**). To share an HTML report with someone, you can send them the results folder. Double-click *index.html* in the *report_data* sub-folder to open an HTML report. For example, *Decoupling_Results_<date>-<time>\<design_name>_<any>_Decoupling_Report>\report_data\index.html*. Note that an HTML report opened in this way cannot launch an external viewer, such as the Touchstone Viewer.

Check the Z-parameter model to see if the impedance profile is below the target impedance across the frequency range.

If the impedance profile is good, the next step is to run distributed decoupling simulation.

If the impedance profile is not good:

- Run Quick Analysis to find ineffective decoupling capacitors.
- Investigate analysis failures or unexpected results, by seeing information created by the decoupling simulation engine. Choose the **View > Simulation Reports > Decoupling Analysis** menu item.
- Determine the contribution of specific capacitors (enabled on the Check Capacitor Models page) or PDN modeling details (enabled on the Customize Settings page), you can run simulation with one or more options checked, run it again with those options unchecked, and then compare the impedance profiles of the two Z-parameter models.
- If the design is in BoardSim, export the pair of power supply nets to LineSim, change the design to improve PDN decoupling, and run decoupling simulation in LineSim to verify the fixes.

Related Topics

Viewing and Converting Touchstone and Fitted-Poles Models

Reporter Dialog Box

Exporting a Net from BoardSim to LineSim

Simulating PDN Decoupling - Quick Analysis

Find ineffective decoupling capacitors by listing information about them, such as mounting inductance and resonant frequency.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Restrictions and Limitations

- If you have a multiple-board design, you can only run decoupling simulation on one board at a time.
- Decoupling simulation is unavailable for board designs with multiple stackups.

Prerequisites

- Assign decoupling capacitor models. See Assigning Models to Decoupling Capacitors.
- Acquire the Decoupling license.

Procedure

1. Click **Analyze Decoupling**  , or choose the **Simulate PI > Analyze Decoupling** menu item.
2. On the Start Analysis page:
   a. Do one of the following.

If you want to...	Do this...
Run the wizard for the first time on your design.	Select **New**.
Rerun the wizard on your design.	Select “Use last configuration”. **Note:** This option is unavailable after you close the design.
Load wizard settings from a previously-saved setup file, and rerun the wizard on your design.	1. Select “Load saved configuration”. 2. Click **Load**, specify the location of the *.dao* wizard settings file, and click **Open**.

   b. To save the wizard settings file, check “Save settings to file”, click **Browse**, and specify the location of the *.dao* file.
3. On the Choose a Type of Analysis page, select Quick Analysis.
4. On the Check Capacitor Models page:
   a. Verify decoupling capacitor model assignments and values. If needed, assign or remove models, and edit groups.
   b. To perform a “what if” experiment by excluding a model from simulation, uncheck Enabled.
5. On the Run Analysis page, rename the results folder if needed, and click **Run Analysis**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Results

When simulation completes, HTML results display, and the Reporter dialog box opens. Check the results to find ineffective decoupling capacitors and other potential problems. The software saves spreadsheet results and wizard settings (*.dao* file) in a timestamped folder within the design folder.

You can access HTML reports from previous simulations in the HTML Report Manager (**View > HTML Report Manager**).

To share an HTML report with someone, you can send them the results folder. Double-click *index.html* in the *report_data* sub-folder to open an HTML report. For example, *Decoupling_Results_<date>-<time>\<design_name>_<any>_Decoupling_Report>\report_data\index.html*. Note that an HTML report opened in this way cannot launch an external viewer, such as the Touchstone Viewer.

Related Topics

Reporter Dialog Box

Distributed Decoupling Simulation Comparison

Depending on the simulation capabilities you seek, the software provides different methods for running distributed decoupling simulation.

If you want ...	Run standard distributed decoupling simulation	Run advanced distributed decoupling simulation
Fast simulation run times (compared to other decoupling technology in the market).	X	
To simulate power-supply nets (power or ground) with trace routing in some places, rather than only metal areas. **Note:** Standard distributed decoupling simulation includes short traces that connect decoupling capacitor or IC pins to metal areas.		X
To include in simulation a power-supply net that connects with a series passive component to a power-supply net you have probed. Use this capability to include in simulation a power-supply that has no probe and is “associated” by a resistor or inductor to a power-supply net with a probe.		X

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want ...	Run standard distributed decoupling simulation	Run advanced distributed decoupling simulation
To account for strong coupling between *transmission plane*s formed by a power-supply net you have selected for simulation and transmission planes formed by other power-supply nets.		X

Simulating PDN Decoupling - Distributed

Evaluate the ability of a PDN to provide low-impedance paths for specific power-supply pins or pin groups. Distributed PDN decoupling simulation provides more accurate results than the lumped modeling method, but takes longer to complete. It accounts for the location of each decoupling capacitor, inter-plane capacitance and inductance, the board outline, and the mounting parasitics and loss for decoupling capacitors.

Restrictions and Limitations

- Decoupling simulation runs on one pair of power-supply nets at a time, where one net provides power and the other provides return current. For ICs with multiple pairs of power-supply nets, run decoupling simulation for each pair of power-supply nets that you want to simulate.
- You cannot run decoupling simulation on power-supply nets formed entirely by trace segments. You can simulate power-supply nets that contain metal areas that form at least one *transmission plane*.
- If you have a multiple-board design, you can run decoupling simulation on one board at a time.
- Decoupling simulation is unavailable for board designs with multiple stackups.

Prerequisites

- You understand the capabilities of standard and advanced distributed decoupling simulation. See Distributed Decoupling Simulation Comparison.
- Run lumped decoupling simulation and modify your design until results show good performance. See Simulating PDN Decoupling - Lumped.
- Optionally assign VRM models. VRMs (with their low resistance DC paths) can significantly lower PDN impedance at low frequencies. See Assigning VRM Source, DC Sink, and AC Models.
- Acquire the information needed to define the target impedance for the PDN. See Information Needed to Calculate Target PDN Impedance.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- To enable the software to measure PDN impedance through a group of power-supply pins in parallel:
  - Check “Automatically assign reference layers” on the **Power Integrity** tab of the Preferences dialog box.
  - Create power-supply pin groups. See Creating Power Supply Pin Groups.
- Acquire the Decoupling license.

Procedure

1. Click **Analyze Decoupling**  , or choose the **Simulate PI > Analyze Decoupling** menu item.
2. On the Start Analysis page:
   a. Do one of the following.

If you want to...	Do this...
Run the wizard for the first time on your design.	Select New.
Run the wizard another time on your design.	Select “Use last configuration”. **Note:** This option is unavailable after you close the design.
Load wizard settings from a previously-saved setup file, and run the wizard another time on your design.	1. Select “Load saved configuration”. 2. Click **Load**, specify the location of the *.dao* wizard settings file, and click **Open**.

   b. To save a wizard settings file, click **Save**, then specify the location of the *.dao* file.
3. On the Choose a Type of Analysis page, select Distributed Analysis.
4. On the Select Nets for Analysis page, add a pair of power-supply nets to the Nets to analyze area. You can double-click net names to move them between the Available nets and Nets to analyze areas.
5. On the Check Capacitor Models page:
   a. Verify decoupling capacitor model assignments and values. If needed, you can assign or remove models, and edit groups.
   b. To perform a “what if” experiment by excluding a model from simulation, uncheck Enabled.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

6. On the Check VRM Models page, verify model assignments, then choose whether to observe PDN behavior at VRM pins:

   a. To display VRMs assigned only to nets that you selected on the Select Power Nets page, check “Show VRM(s) for selected nets only”.

   b. To add or edit a VRM model or reference net, click **VRM Manager**. The Assign Power Integrity Models dialog box opens. For information, see step 3 in “Assigning VRM Source, DC Sink, and AC Models” on page 103.

   c. To observe PDN behavior at pins for a VRM, ensure that its right-most spreadsheet column has a check mark. An empty blue box in the right-most column indicates missing model information, such as a reference net.

7. On the Set the Target Impedance page, either enter the target impedance or click **Calculator** to open a wizard that can help you define the target impedance.

8. On the Select IC Power Pins page:

   a. Verify the spreadsheet contains all the IC power-supply pins that you want to simulate.

      i. If a board design pin that you want to simulate is missing, click **Add IC Power Pin** and specify the pin and its reference net.

      ii. If a schematic design pin that you want to simulate is missing, close the wizard, add to the schematic an IC sink or VRM symbol, and assign a reference net and model (for a VRM symbol) to it.

   If a transmission plane does not enclose the pin with sufficient overlap, the software does not add the pin to the spreadsheet. See Power-Supply Pins That Can Be Selected for Distributed Decoupling Simulation and Exporting a PDN.

   b. Check an IC power-supply pin to include it in simulation and as a port in the output Z-parameter model. The more pins you select, the longer simulation takes and the larger the model file becomes.

   Restriction

   You cannot check a pin that also belongs to a pin group that you have checked on the Select IC Pin Group Probes page. Point to an empty blue box for a pin to see the name of its pin group.

9. On the Select IC Pin Group Probes page:

   a. Verify the spreadsheet contains all the IC power-supply pin groups that you want to simulate. If needed, click **Manage Supply Pin Groups** to create add or edit pin groups.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

b. Check a pin group to simulate its member pins in parallel and add the group as a port in the output Z-parameter model. The more pin groups you select, the longer simulation takes and the larger the model file becomes.

10. On the Choose Easy / Custom page, choose between default and custom decoupling simulation options.

    If you select Easy, some options on the following wizard pages become unavailable.

11. On the Customize Settings page, do any of the following:

If you want to...	Do this...
Account for energy coming from transmission planes formed by other pairs of power-supply nets and transmitted through stitching vias that run vertically throughout the stackup. **Note:** Checking this option can increase simulation run time.	Check “Include all power / ground nets in the analysis”.
Account for decoupling capacitor mounting inductance.	Check “Include capacitor mounting inductance”.
Ignore series mounting inductance for each power-supply pin. For example, you might do this to more easily isolate the effects of changing decoupling capacitor values or dielectric thicknesses on PDN impedance.	Check “Remove series inductance unique to each power pin, to see plane decoupling more clearly”.
Reduce simulation run time and memory consumption by automatically finding stitching vias that are located close together and merging their individual models into an equivalent model. See Stitching-Via Optimization.	1. Check “Enable stitching-via optimization”. 2. Select the Tolerance value.

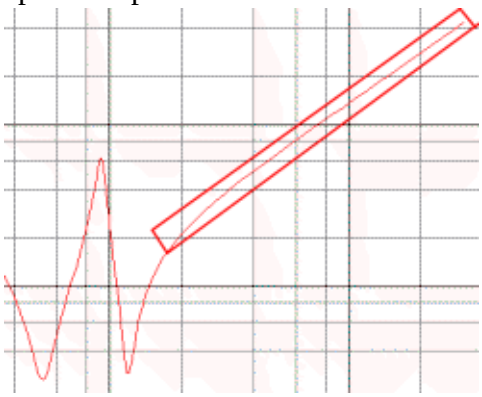
12. On the Control Frequency Sweep page, set the sampling method and interval:

    a. Enter the frequency range in MHz.

    Above a certain frequency, such as 150 MHz, many ICs have in-package decoupling that provide most of the decoupling. Decoupling capacitors and buried capacitance in the PCB contribute little or no decoupling above this design-dependent frequency.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

b. Do one of the following:

If you want to distribute sampling points across the frequency range...	Do this...
In a variable way, by increasing the sampling rate near frequencies with resonances. Use this sampling method unless you have a reason to use another sampling method.	1. Select “Adaptive sampling”. 2. Select an “Accuracy at resonances” value. Take the complexity of the design into account. If the design has large numbers of power-supply nets, hundreds of decoupling capacitors, and hundreds or thousands of stitching vias, selecting Low provides preliminary results with decreased simulation run time. After evaluating the preliminary results, you can identify which frequency ranges interest you the most and try running simulation with higher accuracy on each range of interest. 3. Specify a value for “Minimum number of samples in flat, non-resonant regions”. The enclosed region in the figure below shows a flat and non-resonant region of an example impedance profile. 
At logarithmic intervals.	1. Select “Logarithmic sampling”. 2. Specify a value for the “Number of samples” for the entire frequency range.
At equal intervals.	1. Select “Linear sampling”. 2. Specify a value for the “Number of samples” for the entire frequency range.

13. On the Run Analysis page, rename the results folder if needed, and click **Run Analysis**.

Results

When simulation completes, the software displays results and saves them with wizard settings (*.dao* file) in a timestamped folder within the design folder.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Simulation Result	Description
Output file (Z-parameter file)	The Touchstone Viewer automatically opens and displays a Z-parameter model created by decoupling simulation.
HyperLynx Distributed Decoupling Report (HTML)	A report that shows the PDN impedance profile as a Z-parameter plot. You can access HTML reports from previous simulations in the HTML Report Manager (**View > HTML Report Manager**). To share an HTML report with someone, you can send them the results folder. Double-click *index.html* in the *report_data* sub-folder to open an HTML report. For example, *Decoupling_Results_<date>-<time>\<design_name>_<any>_Decoupling_Report>\report_data\index.html*. Note that an HTML report opened in this way cannot launch an external viewer, such as the Touchstone Viewer.

Check the Z-parameter model to see if the impedance profile is below the target impedance across the frequency range. To help you map Touchstone model ports to component pins, see the *<design>_<iteration>.z<number_of_ports>p.ports* file.

If the impedance profile is not good, you can:

- Investigate analysis failures or unexpected results, by seeing information created by the decoupling simulation engine. Choose the **View > Simulation Reports > Decoupling Analysis**.
- Determine the contribution of specific capacitors (enabled on the Check Capacitor Models page) or PDN modeling details (enabled on the Customize Settings page), you can run simulation with one or more options checked, run it again with those options unchecked, and then compare the impedance profiles of the two Z-parameter models.
- Change your design to improve PDN decoupling performance. For a board design, you can export nets to a schematic design, change the design, and run simulation again.

Related Topics

Preferences Dialog Box - Power Integrity Tab

Viewing and Converting Touchstone and Fitted-Poles Models

Reporter Dialog Box

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Simulating PDN Decoupling - Advanced Distributed

Evaluate the ability of a PDN to provide low-impedance paths for specific power supply pin pairs or pin group pairs. Advanced distributed PDN decoupling simulation accounts for the location of each decoupling capacitor, inter-plane capacitance and inductance, the board outline, and the mounting parasitics and loss for decoupling capacitors.

Restrictions and Limitations

- You cannot disable the effects of dielectric loss. Advanced Decoupling Analysis always accounts for dielectric loss, even when you disable lossy transmission line modeling from the main HyperLynx toolbar.
- If you have a multiple-board design, you can run decoupling simulation on one board at a time.
- Decoupling simulation is unavailable for board designs with multiple stackups.

Prerequisites

- Acquire the Decoupling license.
- Run lumped decoupling simulation and modify your design until results show good performance. See Simulating PDN Decoupling - Lumped.
- You understand the capabilities of standard and advanced distributed decoupling simulation. See Distributed Decoupling Simulation Comparison.
- Optionally assign VRM models. VRMs (with their low resistance DC paths) can significantly lower PDN impedance at low frequencies. See Assigning VRM Source, DC Sink, and AC Models.
- To include in simulation a power supply net that connects with a series passive component to a power supply net you have probed, check “Allow AC supply models for non-capacitor components” in the Preferences dialog box. See Preferences Dialog Box - Power Integrity Tab. If you enable this option, the wizard displays the Supply Component Models page instead of the Check Capacitor Models page.
- To enable the software to measure PDN impedance through a group of power supply pins in parallel, create power supply pin groups. See Creating Power Supply Pin Groups.
- Acquire the information needed to define the target impedance for the PDN. See Information Needed to Calculate Target PDN Impedance.
- If you want to refine project and simulation settings in HyperLynx Hybrid Solver:
  - Install and license the full version of the HyperLynx Hybrid Solver software.
  - Specify the location of the HyperLynx Hybrid Solver software. See the “3D HyperLynx Advanced Solvers” area in Preferences Dialog Box - Simulators Tab.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

1. Choose the **Simulate PI > Advanced Decoupling Analysis** menu item.
2. On the Start Analysis page:
   a. Do one of the following:

If you want to...	Do this...
Run the wizard for the first time on your design.	Select New.
Run the wizard another time on your design.	Select “Use last configuration”. **Note:** This option is unavailable after you close the design.
Load wizard settings from a previously-saved setup file, and run the wizard another time on your design.	1. Select “Load saved configuration”. 2. Click **Load**, specify the location of the wizard settings file (*.dao*), and click **Open**.

   b. To save the wizard settings file, click **Save**, then specify a name and location.
3. On the Choose a Type of Analysis page, select Decoupling Analysis.

   Do not select PDN Decoupling Optimizer unless you intend to run the PDN Optimizer to enable the software to determine optimum decoupling capacitor values. See “Optimizing PDN Decoupling Capacitors - HyperLynx PI” on page 258.
4. On the Select Nets for Analysis page, specify power supply nets that:
   - You want to make available for probing on pages that appear later in the wizard.
   - You do not probe, want to include in simulation, and connect with a resistor or inductor to a power supply net that you probe in step 8.

   Double-click net names to move them between the “Available nets” and “Probable nets” areas.
5. On the Supply Component Models or Check Capacitor Models page:
   a. Verify model assignments and values. If needed, you can assign or remove models, and edit groups.

Note

Assign a model to a resistor or inductor that connects a power supply net that you probe in step 8 to a power supply net that you do not probe and want to include in simulation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

For decoupling capacitor ESL, the software uses a manually-specified ESL value, instead of calculating it based on capacitor body size. If you define a decoupling capacitor model with the:

- Assign / Edit Capacitor Model Dialog Box, the software uses the “Specify value” field value, even if you select the “Auto-calculate” option.
- QPL-File Editor, the software uses the “ESL by value” field value, even if you select the “ESL by capacitor size” option.

b. To perform a “what if” experiment by excluding a model from simulation, uncheck Enabled.

6. On the Check VRM Models page, verify model assignments, then choose whether to observe PDN behavior at VRM pins:

   a. To display VRMs assigned only to nets that you selected on the Select Power Nets page, check “Show VRM(s) for selected nets only”.

   b. To add or edit a VRM model or reference net, click **VRM Manager**. The Assign Power Integrity Models dialog box opens. For information, see step 3 in “Assigning VRM Source, DC Sink, and AC Models” on page 103.

   c. To observe PDN behavior at pins for a VRM, ensure that its right-most spreadsheet column has a check mark. An empty blue box in the right-most column indicates missing model information, such as a reference net.

7. On the Set the Target Impedance page, either enter the target impedance or click **Calculator** to open a wizard that can help you define the target impedance.

8. On the next pages, specify the pin pair or pin pair group that you want to include in simulation. Select a pin pair group if the IC manufacturer provides a target impedance based on a group of pins because it produces more accurate results. When you specify pin pair groups, the software determines PDN impedance through a group of power-supply pins in parallel. Otherwise, select a pin pair.

   Note

   You cannot check a pin pair with a pin that also belongs to a pin group that you have checked on the Select Group Pair Probes page. Point to an empty blue box to see the pin group name.

   a. To include a pin pair in simulation, click the “Select IC Pin-Pair Probes” page. Check all the IC power supply pin pairs that you want to include in simulation. The more pin pairs you select, the longer simulation takes and the larger the model file becomes. The software represents each pin pair that you select as a port in the output Z-parameter model.

      If a pin pair that you want to simulate is missing from the list:

      i. Click **Edit Pin-Pair Probes**. The Create IC power pin pair dialog box opens.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

ii. Click a “+” symbol for a reference designator to see a list of pins. Click a row in the supply net list and the reference net list to specify a supply and return pin pair. The board viewer or PDN Editor marks pin locations with blue and red arrows.

iii. Click **Add Probe** and **OK**. You can also select a supply pin in the list and click **Find Reference** to automatically find the reference pin and add the probe.

b. To include a pin pair group in simulation, click the “Select Group Pair Probes” page. Check all the IC power supply pin pair groups that you want to include in simulation. The more pin pair groups you select, the longer simulation takes and the larger the model file becomes. The software represents each pin pair group that you select as a port in the output Z-parameter model.

If a pin pair group that you want to simulate is missing from the list (indicated by an empty blue box):

i. Click **Manage Pin Groups**. The Pin Group Manager dialog box opens. If pin groups do not appear in the list, click **Create Groups** to create them now, then click **OK**. See Creating Power Supply Pin Groups.

ii. On the Select Group Pair Probes page, specify a pin group pair that you want to include in simulation. Click **Edit Pin-Group Probes**. The Create IC pin group pair dialog box opens.

iii. Click a “+” symbol for a reference designator to see a list of pins. Click a row to specify a supply and return pin group pair.

iv. Click **Add Probe** and **OK**.

9. On the Choose Easy / Custom page, choose between default and custom decoupling simulation options.

If you select Easy, all options on the Customize Settings page become unavailable. You can still edit options on the Control Frequency Sweep wizard page.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

10. On the Customize Settings page:

If you want to reduce simulation run time and memory consumption by...	Do this…
Merging small details in complex metal shapes	Check “Defeature metal areas” and specify the minimum dimension of a feature to include in simulation.
Excluding metal located outside of a perimeter that encloses all the pin pairs and group pairs.	Check “Crop layout data”.

If you want to include in simulation a power supply net that you have not probed and…	Do this…
Connects with a resistor or inductor to power supply net you have probed in step 8.	Uncheck “Include only metal for nets with probes”.

If you want to…	Do this…
Load the design and project into HyperLynx Hybrid Solver, where you can make design changes and run simulation.	Check “Show HyperLynx Hybrid Solver GUI”.

11. On the Control Frequency Sweep page, set the sampling method and interval:
    a. Enter the frequency range in MHz.

       Above a certain frequency, such as 150 MHz, many ICs have in-package decoupling that provide most of the decoupling. Decoupling capacitors and buried capacitance in the PCB contribute little or no decoupling above this design-dependent frequency.

    b. Do one of the following:

If you want to distribute sampling points across the frequency range…	Do this…
At logarithmic intervals.	1. Select Logarithmic. 2. Enter the number of samples for the entire frequency range.
At equal intervals.	1. Select Linear. 2. Enter the number of samples for the entire frequency range.

12. On the Run Analysis page, rename the results folder if needed, and click **Run Analysis**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Results

When simulation completes, the software displays results and saves them with wizard settings (*.dao* file) in a timestamped folder within the design folder.

Note

If you selected "Show HyperLynx Hybrid Solver GUI" on the Customize Settings, page, HyperLynx Hybrid Solver opens. Edit your design if needed, solve, and view results impedance profile results in HyperLynx Hybrid Solver.

Simulation Result	Description
Output file (Z-parameter file)	The Touchstone Viewer automatically opens and displays a Z-parameter model created by decoupling simulation.
Simulation Report (HTML)	A report that shows the PDN impedance profile as a Z-parameter plot and provides a shorted loop inductance value for each capacitor. The report also provides the ability to search, sort, and filter results data. You can also export results to Microsoft Excel, comma-separated values, and other formats. You can access HTML reports from previous simulations in the HTML Report Manager (**View > HTML Report Manager**). To share an HTML report with someone, you can send them the results folder. Double-click *index.html* in the *report_data* sub-folder to open an HTML report. For example, *Decoupling_Results_<date>-<time>\ <design_name>_<any>_Decoupling_Report>\ report_data\index.html*. Note that an HTML report opened in this way cannot launch an external viewer, such as the Touchstone Viewer.

Check the Z-parameter model to see if the impedance profile is below the target impedance across the frequency range. To help you map Touchstone model ports to component pins, see the *<design>_<iteration>.z<number_of_ports>p.ports* file.

If the impedance profile is not acceptable:

- Investigate analysis failures or unexpected results by viewing information created by the decoupling simulation engine. Choose the**View > Simulation Reports > Decoupling Analysis** menu item.
- Determine the contribution of specific capacitors (enabled on the Check Capacitor Models or Supply Component Models page) or PDN modeling details (enabled on the

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Customize Settings page), you can run simulation with one or more options checked, run it again with those options unchecked, and then compare the impedance profiles of the two Z-parameter models.

- Change your design to improve PDN decoupling performance. For a board design, you can export nets to a schematic design, change the design, and run simulation again.

Related Topics

Viewing and Converting Touchstone and Fitted-Poles Models

Reporter Dialog Box

Optimizing PDN Decoupling Capacitors - HyperLynx PI

The PDN Decoupling Optimizer enables you to analyze existing decoupling capacitors and their locations in a board or schematic (PDN Editor) design to determine the optimum decoupling capacitor for each location. An optimized design includes the minimum number of capacitors, the minimum number of different capacitor types, and has the lowest cost.

The software can use synthesis or optimization to determine the best design. Synthesis is a relatively quick-running iterative process, in which the software follows one of several methods to improve the design through each analysis iteration. Optimization is more complex, and is based on the genetic algorithm, in which the software begins with a design solution that consists of a selection of potential capacitors, and evolves the design in multiple iterations (generations) using a set of potential design variants (populations) that improve through each generation. The design evolves in a manner similar to natural selection in nature.

Optimization also creates results that include the impedance profile at each DUT probe location (for an IC power-supply pin or VRM pin), and loop inductance results from each capacitor location to each DUT probe.

Restrictions and Limitations

- You cannot disable the effects of dielectric loss. Decoupling optimization always accounts for dielectric loss, even when you disable lossy transmission line modeling.
- If you have a multiple-board design, run decoupling optimization on one board at a time.
- Decoupling capacitor optimization is unavailable for board designs with multiple stackups.

Prerequisites

- Review the list of capacitors included in the default *.decap* capacitor libraries, located in *C:\MentorGraphics\<release>\SDD_HOME\hyperlynx64\Libs*. If needed, create additional decoupling capacitor libraries in *.caplib* format to define all potential

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capacitors that the software considers when determining the optimum capacitor for each location. See "CAPLIB File Format" on page 482.

- To enable the software to measure PDN impedance through a group of power supply pins in parallel, create power supply pin groups. See "Creating Power Supply Pin Groups" on page 110.
- Optionally assign VRM models. VRMs (with their low resistance DC paths) can significantly lower PDN impedance at low frequencies. See Assigning VRM Source, DC Sink, and AC Models.
- Acquire the information needed to define the target impedance for the PDN. The software enables you to specify target impedance requirements at multiple frequencies, so that, for example, the requirement at higher frequencies can be relaxed. See "Information Needed to Calculate Target PDN Impedance" on page 522.
- To include in simulation a power supply net that connects with a series passive component to a power supply net you have probed, check "Allow AC supply models for non-capacitor components" in the Preferences dialog box. See "Preferences Dialog Box - Power Integrity Tab" on page 995. If you check this option, the wizard displays the Supply Component Models page instead of the Check Capacitor Models page.
- You have the PDN Decoupling Optimizer license.

Procedure

1. Choose the **Simulate PI > Advanced Decoupling Analysis** menu item.
2. On the Start Analysis page:
   a. Do one of the following:

If you want to...	Do this...
Run the wizard for the first time on your design.	Select New.
Run the wizard another time on your design.	Select "Use last configuration". **Note:** This option is unavailable after you close the design.
Load wizard settings from a previously-saved setup file, and run the wizard another time on your design.	1. Select "Load saved configuration". 2. Click **Load**, specify the location of the *.dao* wizard settings file, and click **Open**.

   b. To save the wizard settings file, check "Save settings to file", click **Browse**, and specify the location of the *.dao* file.
3. On the Choose a Type of Analysis page, select PDN Decoupling Optimizer.

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4. On the Select Nets for Analysis page, specify power supply nets that:

   - You want to make available for probing on pages that appear later in the wizard.
   - You do not probe, want to include in simulation, and connect with a resistor or inductor to a power supply net that you probe in Step 6.

   Double-click net names to move them between the Available nets and Probable nets areas.

5. On the Supply Component Models or Check Capacitor Models page, Double-click a capacitor or capacitor group row to assign a model to each capacitor that you want to include in the baseline analysis. The set of models that you assign defines the baseline design that the software compares to the best-case design (all capacitors modeled as shorts).

   Note

   Assign a model to a resistor or inductor that connects a power supply net that you probe in Step 6 to a power supply net that you do not probe and want to include in simulation.

6. On the Check VRM Models page, verify model assignments, then choose whether to probe VRM pins:

   a. To display VRMs assigned only to nets that you selected on the Select Power Nets page, check “Show VRM(s) for selected nets only”.

   b. To add or edit a VRM model or reference net, click **VRM Manager**. The Assign Power Integrity Models dialog box opens. For information, see step 3 in “Assigning VRM Source, DC Sink, and AC Models” on page 103.

   c. To probe pins for a VRM, ensure that its right-most spreadsheet column has a check mark. An empty blue box in the right-most column indicates missing model information, such as a reference net.

7. Specify the pin pair or pin pair group that you want to include in simulation. Select a pin pair group if the IC manufacturer provides a target impedance based on a group of pins because it produces more accurate results. When you specify pin pair groups, the software determines PDN impedance through a group of power-supply pins in parallel. Otherwise, select a pin pair.

   Note

   You cannot check a pin pair with a pin that also belongs to a pin group that you have already checked on the Select Group Pair Probes page. Point to an empty blue box for a pin to see the pin group name.

   a. To include a pin pair in simulation, click the Select IC Pin-Pair Probes page. Check all the IC power supply pin pairs that you want to include in simulation. The more pin pairs you select, the longer simulation takes and the larger the model file

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becomes. The software represents each pin pair that you select as a port in the output Z-parameter model. If a pin pair that you want to simulate is missing from the list:

i. Click **Edit Pin Pair Probes**. The Create IC power pin pair dialog box opens.

ii. Click a "+" symbol for a reference designator to see a list of pins. Click a row in the supply net list and the reference net list to specify a supply and return pin pair. The board viewer or PDN Editor marks pin locations with blue and red arrows.

iii. Click **Add Probe** and **OK**. You can also select a supply pin in the list and click **Find Reference** to automatically find the reference pin and add the probe.

b. To include a pin pair group in simulation, click the Select Group Pair Probes page. Check all the IC power supply pin pair groups that you want to include in simulation. The more pin pairs you select, the longer simulation takes and the larger the model file becomes. The software represents each pin pair that you select as a port in the output Z-parameter model. If a pin group that you want to simulate is missing from the list (indicated by an empty blue box):

i. Click **Manage Pin Groups**. The Pin Group Manager dialog box opens. If pin groups do not appear in the list, create them now and click **OK**. See "Creating Power Supply Pin Groups" on page 110.

ii. On the Select Group Pair Probes page, specify a pin group pair that you want to include in simulation. Click **Edit Pin-Group Probes**. The Create IC pin group pair dialog box opens.

iii. Click a "+" symbol for a reference designator to see a list of pins. Click a row to specify a supply and return pin group pair.

iv. Click **Add Probe** and **OK**.

8. On the Choose Easy / Custom page, choose between default and custom decoupling simulation options.

   If you select Easy, all options on the Customize Settings page become unavailable. You can still edit options on the Control Frequency Sweep wizard page.

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9. On the Customize Settings page:

If you want to reduce simulation run time and memory consumption by…	Do this…
Merging small details in complex metal shapes	Check “Defeature metal areas” and specify the minimum dimension of a feature to include in simulation.
Excluding metal located outside of a perimeter that encloses all the pin pairs and group pairs.	Check “Crop layout data”.
Excluding metal located on a power supply net with no probes.	Check “Include only metal for nets with probes”.

If you want to include in simulation a power supply net that you have not probed and…	Do this…
Connects with a resistor or inductor to power supply net you have probed in Step 6.	Uncheck “Include only metal for nets with probes”.

If you want to...	Do this…
Load the design and project into HyperLynx Hybrid Solver, where you can make design changes and run simulation.	Check “Show HyperLynx Hybrid Solver GUI”.

10. On the Control Frequency Sweep page, set the sampling method and interval:
    a. Enter the frequency range in MHz.

       Above a certain frequency, such as 150 MHz, many ICs have in-package decoupling that provide most of the decoupling. Decoupling capacitors and buried capacitance in the PCB contribute little or no decoupling above this design-dependent frequency.

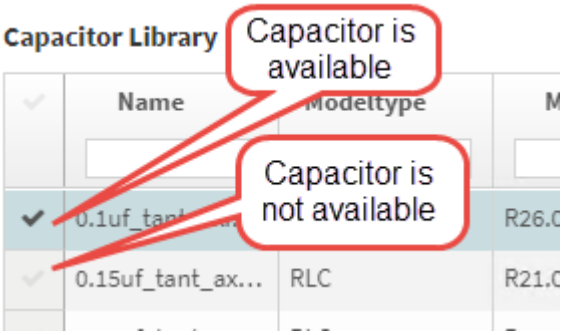
    b. Do one of the following:

If you want to distribute sampling points across the frequency range…	Do this…
At logarithmic intervals.	1. Select Logarithmic. 2. Enter the total number of samples for the entire frequency range.
At equal intervals.	1. Select Linear. 2. Enter the total number of samples for the entire frequency range.

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11. On the Run Analysis page, rename the results folder if needed, and click **Run Analysis**. The software creates the folder in the design folder.

12. The PDN Decoupling Optimizer opens. On the **Library Setup** tab, ensure that the list includes the capacitors that you want to include during optimization:

If you want to...	Do this...
Reduce run time, by making only specific capacitors in a library available to optimization **Note:** If you do not enable specific capacitors, the software can include all capacitors with suitable footprints during optimization.	Click the first column for a capacitor that you want to be available during optimization, so that it is available: 
Add capacitors to an existing capacitor list	Click **Append** and select a capacitor library file.
Erase an existing capacitor list, then add capacitors	Do either of the following: • To add all capacitor library files in a folder, click **Import Folder**, then select the folder. • To add a specific capacitor library file, click **Import** and select a capacitor library file.

The table initially lists capacitors available from *.decap* and *.caplib* library files in the folder *C:\MentorGraphics\<release>\SDD_HOME\hyperlynx64\Libs*.

13. On the **Model Setup** tab, map each library capacitor footprint to locations on the board where the capacitor will physically fit. Also specify impedance requirements for each probe location, then run a baseline analysis:

Caution

Do not click **CREATE DEFAULT**. It erases port configuration data provided by the Advanced Decoupling Wizard.

a. Click **FOOTPRINT REMAP**. The Footprint Remap dialog box opens.

b. Select a row in the Sizes table, select the corresponding footprint on the right, and click **ASSIGN**.

c. Repeat for all sizes, then click **SAVE** and **CLOSE**.

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d. Specify impedance requirements for each probe location by double-clicking a DUT type row in the Termination column of the Port Configuration table. The Impedance Requirement dialog box opens.

e. Edit the frequency and impedance values as needed. Click **ADD** to specify additional values.

f. Repeat for all DUT type rows.

g. To see if the baseline design and best-case design (capacitors modeled as shorts) meet the impedance requirement, click **RUN ANALYSIS**.

   In the Baseline Analysis graph, click legend labels to hide or show impedance profiles and impedance requirements as needed.

   If the best-case design does not meet the impedance requirement, stop and change your design. Consider changing the number of decoupling capacitors in your design, or the sampling frequency range, and run the baseline analysis again to ensure that the best-case design meets the impedance requirement with some margin.

14. On the **Run** tab, you can run synthesis, optimization, or both:

    a. In the Process list, select Recommended Flow.

       This option runs all six synthesizer strategies and enables you to compare results to find the best design solution. Synthesis runs faster than optimization, so run synthesis alone before running optimization to ensure that the design meets (or nearly meets) the impedance requirement.

    b. If you want to run synthesis using only one strategy, in the Process list, select Expert System instead of Recommended Flow. Select a strategy in the Strategy list, and edit the Similar Tolerance value if needed.

       By default, the software attempts to use a single capacitor in place of three different capacitors with similar performance and value when possible. For example, a higher tolerance value produces a design solution that contains fewer different capacitor types, reducing the potential cost of the design, and also decreases the number of different capacitor types that the synthesizer considers.

    c. Click **RUN PROCESS**.

       As simulation runs, the Results table and Plot update to show progress. When simulation completes, results appear on the **Report** tab.

    d. On the **Run** tab, in the Plot area, evaluate the impedance profile. If the design meets the impedance requirement or nearly meets the requirement, consider running optimization to further improve the design and reduce the number and different types of capacitors.

    e. To run optimization, in the Process list, select Genetic Algorithm (Optimizer).

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f. Use the default values set in the Population and Generations list unless you need to change them.

   The population size is the number of potential design solutions in each analysis iteration (generation). Increasing the number for population and generations increases run time. A population value between 20 (for a low number of capacitors) and 50 (for a high number of capacitors) produces good results. If you specify a generations value, it should be approximately three times the number of capacitor locations to ensure convergence.

g. Click **RUN PROCESS**.

Results

When simulation completes, select the **Report** tab to see loop inductance results at each capacitor location and result details. A capacitor marked as Open in the New Termination column indicates a capacitor location that you can remove from the design.

The report contains five sections:

Section	Description
Version	Lists the software version for documentation purposes.
Summary	An overview of the analysis setup.
Loop Inductance	A plot and table of the loop inductance of each capacitor for a given device with its power and ground pins shorted. This provides a quick way to identify capacitors with poor layout by using the quantitative score of each capacitor layout/mounting as well as a comparative metric among the capacitors on the PCB.
Model Setup	The Touchstone file name and location, pass/fail criteria, and margin, as well as plots/configuration data of the baseline analysis.
Results	Lists of passing optimized PDNs with margin, number of capacitors, number of capacitor parts, and cost. Select the PDN to display detailed information.

The software saves simulation files to a timestamped folder in the design folder. These files include a *.dao* wizard settings file, the *.pdn* port/footprint mapping file, S and Z parameter files for the PDN, and a log file. To help you map Touchstone model ports to component pins, see the *<design>_<iteration>.z<number_of_ports>p.ports* file.

Running Plane Noise Simulation

Use plane noise simulation to observe how noise propagates across plane regions of the power-distribution network (PDN) when power supply pins draw large amounts of transient current.

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Excessive plane noise can offset the localized voltage on a IC power supply pin so much that a receiver pin connected to it can switch logic state, even when the driver voltage is held constant.

Plane-noise simulation applies a current pulse to one or more IC power supply pins to imitate the large currents required for I/O or core logic switching, and then reports the layer-to-layer voltage difference at all X/Y locations across the transmission plane. It also reports surface and capacitor currents across the transmission plane.

The HyperLynx PI PowerScope dialog box displays plane-noise results in a 3-D graphical form. As simulation runs, you can see how noise voltage varies in time across the PDN. When simulation completes, the HyperLynx PI PowerScope displays the maximum plane-noise voltages across the PDN. The 3-D plot can help identify PDN locations that need more decoupling capacitors (or possibly less inductive mounting for existing capacitors) by making it easy to see “hot spots” on the board and to visualize the effectiveness of specific decoupling capacitors. You can save the 3-D plot to a file for later viewing.

You can simulate plane noise on a board or schematic design. Perform a “what if” experiment on a board design by exporting the power-distribution network from BoardSim (you do not have to select a signal net) to a LineSim schematic and edit it in the PDN Editor.

Note

Before running plane-noise simulation, run decoupling analysis to verify that the PDN impedance satisfies the target impedance requirements. If the PDN impedance is too high, it is possible that simulated plane-noise voltages will be too high and exceed the voltage ripple requirements.

If you observe high plane-noise voltage, you may have to modify the PDN design to lower its impedance, run decoupling analysis to verify the PDN impedance profiles meet the target impedance requirements, and rerun plane-noise simulation again.

Restrictions and Limitations

- For a board design, plane noise simulation does not include power supply nets formed entirely by trace segments.
- Plane noise simulation is unavailable for a multiple-board design and for board designs with multiple stackups.
- Power supply nets routed with trace segments are simulated, but the HyperLynx PI PowerScope does not display 3-D plots for them.
- If your schematic design includes a stackup layer with more than one power supply net, you must create a separate design for each net that you plan to simulate (one power supply net on the layer).

Prerequisites

- Acquire the Plane Noise license.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- Verify that the software recognizes your design correctly. See Verifying That the Software Recognizes Your Design Correctly.
- Configure simulation options to account for coupling, loss, and noise between single-ended signal vias and power planes. See Setting Simulation Options.
- Assign models to decoupling capacitors and run decoupling analysis to verify that the PDN impedance satisfies the target impedance requirements. See Decoupling Simulation.
- Assign VRM models, AC models, and specify reference nets. See Assigning VRM Source, DC Sink, and AC Models.
- For a schematic design, edit the PDN in the LineSim PDN Editor to create the exact layout geometries, if needed.

Procedure

1. Choose the **Simulate PI > Run Plane-Noise Simulation** menu item.
2. For a board design:
   a. In the Plane Noise Analysis dialog box, select a power supply net to simulate.
   b. Assigned PI models (AC models) appear in the Assigned Models list. Click **Assign** to edit AC model and reference net assignments.
   c. In the Simulation Time field, type the simulation run time.

      The default stop time is usually adequate for AC models with a rising edge or single pulse current waveform. However, if you assign a repeating stimulus, specify an initial delay or wide pulse, or specify double pulses, consider increasing the simulation time to ensure that simulation reports the maximum-amplitude plane noise.

      The simulation time value overrides the period length (for pulse signal types) in the AC model. For example, if the AC model contains a repeating current waveform that extends beyond the simulation time, the software truncates the current waveform.
3. For a schematic design, in the HyperLynx PI PowerScope dialog box, in the Stop field, type the simulation run time.
4. Click **Run Analysis** or **Start Simulation**.

Results

When simulation completes, the HyperLynx PI PowerScope dialog box opens to display graphical results. See Plane-Noise Simulation Results.

Click **Save** to save simulation results (for each transmission plane) to a (*.TPS*) file in the design folder.

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Related Topics

HyperLynx PI PowerScope Dialog Box

Exporting a Net from BoardSim to LineSim

Creating a PDN Design

Transmission Planes Overview

Running Signal-Via Bypass Simulation

Signal-via bypass simulation helps you evaluate the ability of the power-distribution network (PDN) to provide low-impedance return current paths for signals transmitted through a single-ended via. Signal-via bypassing simulation creates a Z-parameter model showing the return current impedance across a frequency range. Values greater than several ohms indicate insufficient bypassing. The Z-parameter model accounts for the effects of nearby stitching vias, bypass capacitors, and interplane capacitance.

Bypass simulation runs on one signal via at a time and one pair of connected stackup layers at a time. Run the Bypass wizard additional times to simulate other signal vias in your design. If the signal via connects to trace segments located on more than two stackup layers, select which pair of stackup layers to include in simulation. Run bypass simulation to evaluate via bypassing for other pairs of connected stackup layers.

The color of a non-highlighted page name indicates the following:

Color	Description
White	All required information is present.
Red	Some required information is missing.
Gray	Similar to red, some required information is missing. Gray appears when information on a previous page is missing. Gray also appears on the first wizard page if you click the Load Saved Configuration option and do not specify a file.

Restrictions and Limitations

- For a board design, signal-via bypass simulation does not include power supply nets formed entirely by trace segments.
- Signal-via bypass modeling does not support differential signal-via pairs, although you can create bypass models for individual vias in the via pair.
- In a schematic, you can export models only for vias connected to stackup type (coupled or uncoupled) transmission lines.
- You cannot export an S-parameter model for a signal via from a net in a schematic that contains a MOSFET (series bus switch) component.

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- Signal-via bypass simulation is not available for a multiple-board design or a board design with multiple stackups.

Prerequisites

- Acquire the Signal-Via Bypass Models license.
- Verify that the software recognizes your design correctly. See Verifying That the Software Recognizes Your Design Correctly.
- Configure simulation options to account for coupling, loss, and noise between single-ended signal vias and power planes. See Setting Simulation Options.
- If your design uses plane layers, ensure power supply nets are correctly assigned.
- Assign models to decoupling capacitors and run decoupling analysis to verify that the PDN impedance satisfies the target impedance requirements. See Decoupling Simulation.
- For a schematic design, edit the PDN in the LineSim PDN Editor to create the exact layout geometries, if needed.
- Verify the simulation via methods are set correctly (**Setup > Via Simulation Method**). Default settings are recommended.

Procedure

1. Choose the **Simulate PI > Analyze Signal-Via Bypassing** menu item.

   The Bypass Wizard opens.

2. On the Start Analysis page, start a new analysis or load the settings for a saved analysis. Editing the setting on this page also edits the same setting on the Run Analysis page.

   a. Select one of the following:

If you want to...	Do this...
Run simulation for the first time, or create a configuration file	Select **New**.
Reuse settings from the current session.	Select **Use last configuration**. If the option is not available, close and reopen the wizard in the current session.
Use a Bypass Wizard configuration file (*.DAO*) from a previous simulation.	1. Select **Load save configuration**. 2. Click **Load**. 3. Open a settings file by selecting **Load save configuration**, selecting **Load**, Browse to the *.DAO* file, and select **Open**.

3. On the Select Signal Via page, select the signal via that you want to analyze.

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If the signal via connects to trace segments located on more than two stackup layers, select which pair of stackup layers to analyze. You can run bypass analysis additional times to analyze other stackup layer pairs.

Note that the spreadsheet on this page does not display differential signal vias because signal-via bypass modeling does not support differential signal-via pairs.

a. Do one of the following:

If you are simulating a via...	Do this...
in a schematic	1. Check the signal via to include it in simulation. The schematic via name displays the reference designator for the signal via symbol. 2. Select the connected layers to include in simulation. The connected layers area displays the complete set of stackup layers that connect to the selected signal via. • Select the plus sign (+) to expand the spreadsheet row to display all connected stackup layers. • Select the minus sign (-) to collapse the spreadsheet row to display only the top-most connected stackup layer. 3. Click the Layer Pair cell to display the stackup layer pairs for the signal via. 4. Choose the portion of the via barrel to model by selecting a pair of stackup layers from the Layer cell.
on a board	1. Display the via in the board viewer, zooming and panning if needed. See Viewing a Board in 2-D. 2. Follow the wizard directions to select a signal via. The name and the X/Y coordinates of the selected signal via display on this page. This field is blank until you select a signal via in the board viewer. To replace an existing via selection, select a different via. 3. If necessary, click **Pan to** to pan to a selected via. The selected via displays in the center of the available board viewer area. 4. Choose the portion of the via barrel to model by selecting a pair of stackup layers from the “Pair of connected layers to analyze area” list. The Connected layers box displays the complete set of stackup layers that connects to the signal via. 5. To see the stackup layers with the Via Visualizer, right-click the via in the board viewer and select **View Via Properties**.

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4. On the Check Capacitor Models page, review and edit decoupling capacitor model assignments. Double-click a capacitor to assign or edit capacitor models.

   See Assigning Models to Decoupling Capacitors.

5. On the Set the Target Impedance page, specify the target impedance of the PDN return current paths for signals transmitted through a single-ended via. The value you specify displays as a green reference line in the Touchstone Viewer when you display the output Z-parameter file.

6. On the Choose Easy/Custom page, select **Easy**, unless you need to include capacitor mounting inductance, optimize stitching-via modeling, or specify a particular sampling frequency.

   Selecting Easy automatically selects typical simulation settings on some of the wizard pages. Many of the settings become read only.

7. On the Customize Settings page, select detailed simulation options. You can simulate your design with different combinations of options to help determine how individual types of design properties contribute to signal-via bypassing performance.

   Restriction

   This page is editable only when you select the **Custom** option on the Choose Easy / Custom Page.

   a. Select any of the following:

Option	Description
Include capacitor mounting inductance	Check to determine the contribution of capacitor mounting inductance to the overall signal-via bypassing performance. After simulating, uncheck this option, run simulation again, and compare the results.
Enable stitching-via optimization	1. Check to merge the individual models of stitching vias that are located close together into an equivalent model, for all clustered stitching vias across the transmission plane. Reducing the number of stitching-via models speeds up simulation and reduces memory consumption. 2. Set the Tolerance slider to the merging radius for optimization: • Low—1/30th of the minimum wavelength of the signal • Medium—1/20th of the minimum wavelength of the signal • High—1/10th of the minimum wavelength of the signal See Stitching-Via Optimization.

8. On the Control Frequency Sweep page, edit the frequency range and sampling options, both of which affect simulation run time and the resolution of the exported Z-parameter model.

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a. Follow the wizard instructions to set the Min and Max simulation frequencies.

b. Select one type of frequency sampling:

If you want to...	Do this...
Vary the sampling step size depending on model characteristics. The adaptive scale is better than logarithmic and linear because it increases the sampling rate near frequencies with resonances.	1. Select Adaptive sampling. 2. Set the Accuracy at resonances. For lumped analysis, enabling the High option can still yield reasonably fast simulation run times. For distributed analysis, take the complexity of the design into account. If the design has a large number of power supply nets, hundreds of decoupling capacitors, and hundreds or thousands of stitching vias, enable the Low option to obtain preliminary results with decreased simulation run time. After evaluating the preliminary results, identify which frequency ranges to include in a second simulation run using higher accuracy on each range of interest. 3. Set the minimum number of samples in flat, non-resonant regions. The number of samples you specify applies to flat, non-resonant regions of an impedance profile. See the enclosed curve region in Figure 10-9.
Use sampling points that are distributed at logarithmic intervals across the frequency range.	1. Select Logarithmic sampling. 2. Set the number of samples. The number of samples you specify applies to the entire frequency range. The intervals between sampling points are smaller at lower frequencies and larger for higher frequencies. With logarithmic sampling, each next frequency point is equal to the previous value times a factor K > 1. This produces a constant increase ratio. However, the absolute distance between sampling points grows.
Use sampling points that are distributed at equal intervals across the frequency range.	1. Select Linear sampling. 2. Set the number of samples. **Restriction:** This option is available only when you select **Custom** in the Choose Easy / Custom Page.

c. Optionally, select **Default** to restore the initial settings.

d. Click **Next**.

9. On the Run Analysis page, rename the results folder if needed, and click **Run Analysis**.

10. Repeat this procedure as needed, varying vias and/or stackup layer pairs.

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Results

When simulation completes, the software displays results and saves them with wizard settings (*.DAO* file) in a timestamped folder within the design folder.

Simulation Result	Description
Output file (Z-parameter file)	Shows the impedance of the signal-via bypassing over a frequency range. Use the Touchstone and Fitted-Poles Viewer to view Z-parameter files. See "Viewing Touchstone and Fitted-Poles Model Curves" on page 1295. The file name is in the form *<design>_<analysis_iteration>.z1p*, where: <design> is the name of the board or schematic <analysis_iteration> begins with "empty" and increments by one for each analysis. For example, *design_.z1p* and *design_1.z1p*.

Related Topics

Creating a PDN Design

Pulse Response Simulation

Pulse response simulation is a quick method for pre-screening nets to find nets that require more detailed simulation.

When pulse response simulation runs, the software uses a pulse response to determine the effective signal amount, ISI, and crosstalk, and evaluates the channel by comparing results to metrics based on a "pseudo eye".

To run Pulse Response Simulation, choose the **SI Simulation > Run Pulse Response Simulation** menu item.

The wizard is very similar to the General Batch Wizard. For more information on this type of analysis see this IEEE paper (requires login): *Board-Level Signal Integrity Methodology* by Yuan-Liang Li, Kai Xiao, Xiaoning Ye, Yanjie Zhu, Edward Hsiung, Thonas Su, Kai-bin Wu, Jimmy Hsu, and Karen Kang.

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Chapter 5
Simulation Results

Complex simulations yield detailed results that require additional explanation, which are included here.

Topic	Description
DDRx Batch Simulation Results	The DDRx Batch Mode Wizard stores simulation results in an HTML report or various spreadsheets. You can analyze the results to improve a design, validate constraints, or verify design performance.
General Batch Simulation Results Spreadsheet	A General Batch simulation spreadsheet contains measurement values and pass/fail status, and is in HTML format, or Microsoft Excel (XLS) format on Windows and CSV format on Linux.
DC Drop Simulation Results	DC Drop simulation provides three types of results: visual (HyperLynx PI PowerScope), text file (Reporter), and spreadsheet (*.XLS* or *.CSV*).
Decoupling Simulation Results - Decoupling Capacitor Spreadsheet	Decoupling simulation, in Quick Analysis mode, creates a spreadsheet that reports information about decoupling capacitors on your design.
Field Solver Results	The Field Solver provides capacitances, inductances, propagation velocities, and characteristic impedances for a cross section of a set of coupled transmission lines.
Taking Measurements From an Oscilloscope Waveform or Eye Diagram	You can take measurements from a Digital Oscilloscope waveform or eye diagram, and specify an eye mask that overlays the eye diagram.
Automatic Measurements in an Oscilloscope Waveform or Eye Diagram	Use automatic measurements to perform voltage and timing measurements on voltage waveforms or eye diagrams currently displayed in Oscilloscope.
Reading FastEye Diagram Automatic Measurements	The FastEye Channel Analyzer automatically reports FastEye diagram measurements in the FastEye Viewer.
Measuring FastEye Diagrams Manually	You can perform precise time, voltage, current, and slew-rate measurements from the FastEye Viewer screen using measurement crosshairs or simply by observing pointer position information.
Displaying Waveform Results in EZwave	You can open manually EZwave to see simulation results previously saved to a waveform database.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
Plane-Noise Simulation Results	When plane noise simulation completes, the HyperLynx PI PowerScope dialog box opens to display voltage and current graphical results.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDRx Batch Simulation Results

The DDRx Batch Mode Wizard stores simulation results in an HTML report or various spreadsheets. You can analyze the results to improve a design, validate constraints, or verify design performance.

Each DDRx batch simulation creates a subfolder to contain results files. The root folder location depends on options you set in the “Reports and log files directory” section of the Set Directories Dialog Box.

Subfolder names take the form:

DDR_Results_<month>-<day>-<year>_<hour>h-<minute>m

Example: *<design>\DDR_Results_Nov-30-2017_11h-33m*

The HTML results and report viewer executable file is located in a subfolder named *DDR_Report*.

You can access HTML reports from previous simulations that you performed with your computer in the HTML Report Manager (**View > HTML Report Manager**).

When you open a schematic and specify sweep simulations, DDRx batch simulation creates an additional subfolder for each sweep simulation case in the form:

SweepCase_<number>

Example: *<design>\DDR_Results_Nov-30-2017_11h-33m\SweepCase_0*

Topic	Description
DDRx Results HTML Report	The DDRx Batch-Mode Wizard can display a summary of worst-case simulation results in an easy-to-navigate HTML table format.
DDRx Results Spreadsheets	DDRx results spreadsheets contain measurement values and pass/fail status. Spreadsheet rows contain measurements for one write or read cycle.
DDRx Log File	View the *DDR_log{<DDRx_setup_file_name>}.txt* file to investigate results or simulation failures. This file contains simulation/measurement progress and messages for DDRx batch simulation, as well as audit results.
DDRx Waveform Files	Use waveform files to investigate measurements reported in spreadsheets. Many waveform files are created when you simulate all or many memory interface nets.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDRx Results HTML Report

The DDRx Batch-Mode Wizard can display a summary of worst-case simulation results in an easy-to-navigate HTML table format.

If you need more detailed information than is available in an HTML report, the software can write results to spreadsheets. See “DDRx Results Spreadsheets” on page 298.

Topic	Description
DDRx HTML Report Format	HyperLynx provides an HTML report viewer for displaying DDRx simulation results. Although you can use a web browser to display results, the report viewer has features that make it more useful for understanding simulation results.
Categories of Simulation Results in a DDRx HTML Report	An HTML report contains tabs that contain categories of DDRx simulation results plus supporting information.
Re-Opening a DDRx HTML Report	After you close a HTML report that automatically opens when DDRx batch-mode simulation completes, you can re-open it.

DDRx HTML Report Format

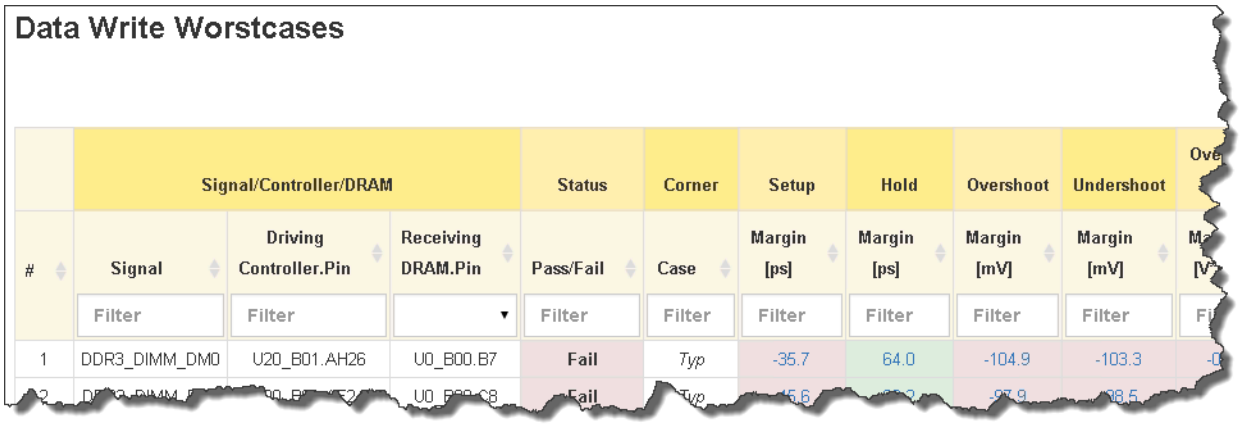
HyperLynx provides an HTML report viewer for displaying DDRx simulation results. Although you can use a web browser to display results, the report viewer has features that make it more useful for understanding simulation results.

The HTML report contains only worst-case simulation results, which are categorized by a group of tabs at the top of the page:



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Clicking the **Data Write** tab displays setup and hold margins plus all of the JEDEC-required SI measurements:



Data Write Worstcases

	Signal/Controller/DRAM			Status	Corner	Setup	Hold	Overshoot	Undershoot	Ove
#	Signal	Driving Controller.Pin	Receiving DRAM.Pin	Pass/Fail	Case	Margin [ps]	Margin [ps]	Margin [mV]	Margin [mV]	Ma [V]
	Filter	Filter		Filter	Filter	Filter	Filter	Filter	Filter	Fi
1	DDR3_DIMM_DM0	U20_B01.AH26	U0_B00.B7	Fail	*Typ*	-35.7	64.0	-104.9	-103.3	-0
2	DDR3_DIMM_[illegible]	[illegible]	U0_B00.C8	Fail	*Typ*	-15.6	[illegible]	-97.9	-98.5	

Each column enables you to filter results using alphanumeric strings, as well as the relational operators “>” and “<” for numeric columns. You can also sort any column based on its alphanumeric content by clicking the up/down arrows to the right of a column name.

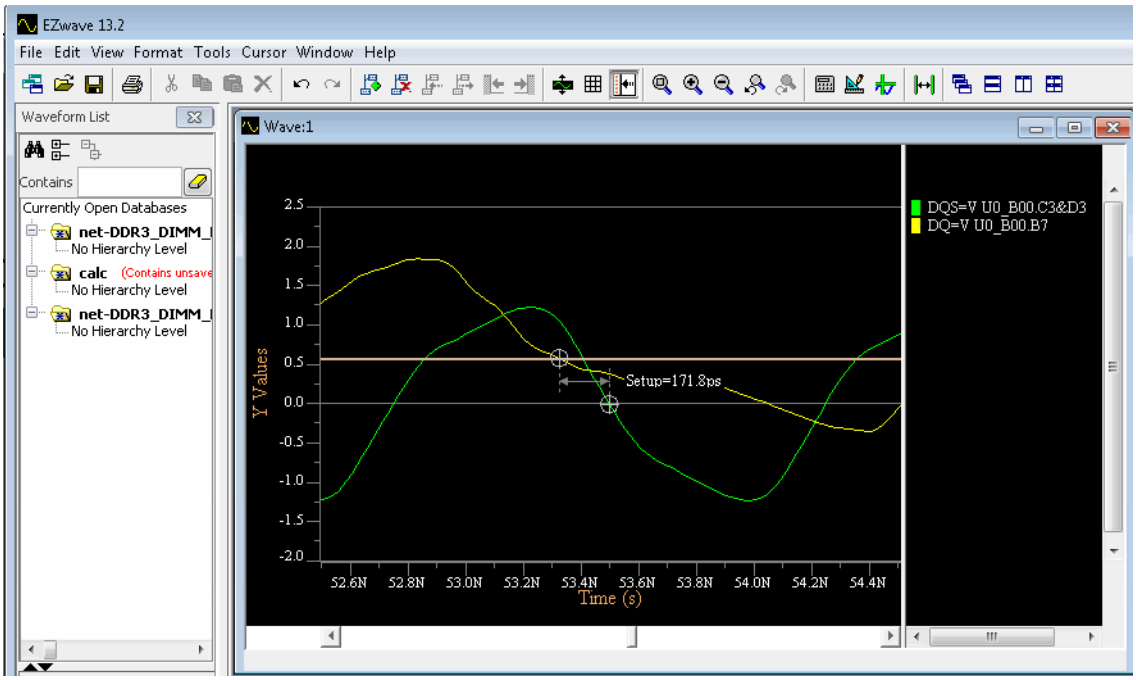
Some columns display additional columns when you click their header. For example, a column that initially displays the “Margin” name can expand into multiple columns that list all of the measurements used to calculate the final margin value. When you click the Setup column header, the software shows these additional columns:

Setup				
Measurement [ps]	Output Variation [ps]	Base Requirement [ps]	Derating [ps]	Margin [ps]
Filter	Filter	Filter	Filter	Filter
171.8	-100.0	-30.0	-77.5	-35.7

The Setup Margin is -35.7 ps, which is the sum of the values listed in the other four columns.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

To show measurement waveforms in EZwave (not available if you are using a web browser), click a measurement value link.



For additional information about these measurements, you can find the relevant XLS or CSV file and open it in a spreadsheet viewer, such as Excel. In the case of the signals listed under Data Write Worstcases that are Corner Case = Typ, you examine this file:

<design>\DDR_results_<date>\DDR_report_data_worstcases_Typ.xls

You will notice that the spreadsheet column names do not precisely match the HTML report, but you can often determine the mapping. Here are some example mappings:

HTML Report Column	Spreadsheet Column
Setup Measurement [ps]	Setup Time (From Sim), tDS, [ps]
Setup Output Variation [ps]	(calculated by subtracting Min Setup from Setup Time)
Setup Base Requirement [ps]	Base Setup Time, tDS(base), [ps]
Setup Derating [ps]	Setup Derate Time Delta, d(tDS), [ps]
Setup Margin [ps]	Setup Time Margin, tDS(margin) = tDS(min) - tDS(req), [ps]

For detailed information about mapping between the HTML report and spreadsheets, see “Categories of Simulation Results in a DDRx HTML Report” on page 281.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Categories of Simulation Results in a DDRx HTML Report

An HTML report contains tabs that contain categories of DDRx simulation results plus supporting information.

The following topics provide column definitions for tabs that include measurements. They do not describe the contents of the **Setup Info** tab.

Tab	Description
Data Write	Lists data write worstcase values from DDRx simulation.
Data Read	Lists data read worstcase values from DDRx simulation.
Address	Lists address worstcase values from DDRx simulation.
Differential Nets	Lists differential worstcase values from DDRx simulation.
CLK-DQS Skew	Lists CLK-DQS skew worstcase values from DDRx simulation.
Eye Density Links	Lists eye density plots and related measurements from DDRx simulation.

Data Write

Lists data write worstcase values from DDRx simulation.

Simulation Results for DDRx Interface Types

A set of simulation results varies by DDRx interface:

- "DDR4 / LPDDR4 / DDR5 / LPDDR5 Data Write Measurement Definitions" on page 281
- "DDR2 / DDR3 Data Write Measurement Definitions" on page 284

DDR4 / LPDDR4 / DDR5 / LPDDR5 Data Write Measurement Definitions

For information about measurements listed in the Definition column, see the following topics:

- "DDRx Address and Data Eye Aggregated Measurements Spreadsheets" on page 310
- "DDRx Address and Data Eye Per Bit Measurements Spreadsheets" on page 303
- "DDRx JEDEC Measurements Spreadsheets" on page 327

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
Signal/Controller/ DRAM	Lists the net name, driving pin, and receiving pin.
Status	Indicates whether the simulation results for the net meet the timing constraints. When Status = Pass, all margin measurements pass for that net.
Sweeps (You have specified sweep cases on the Sweep Manager page for a schematic design.)	
<blank header>	
Widest horizontal eye	Voltage at Widest Eye Opening, Vcent, [mV]
Reference voltage used	Average Center Voltage, Vcent(pin_mid), [mV]
Setup	
Measurement	Measured Setup Time, [ps]
Output Variation	-(Output Delay Uncertainty, [ps])
Base Requirement	-(Half Eye-Mask Width, TdIPW/2, [ps])
Margin	Setup Time Margin with Output Delay Uncertainty, [ps]
Hold	
Measurement	Measured Hold Time, [ps]
Output Variation	-(Output Delay Uncertainty, [ps])
Base Requirement	-(Half Eye-Mask Width, TdIPW/2, [ps])
Margin	Hold Time Margin with Output Delay Uncertainty, [ps]
Pulse Width	
Measurement	Vcent(pin_mid) to Vcent(pin_mid) Pulse Width, [ps]
Output Variation	-(((“Vcent(pin_mid) to Vcent(pin_mid) Pulse Width, [ps]”) - ((“Pulse Width Limit, tdIPW, [ps]”) + (“Pulse Width Margin, [ps]”)))
Base Requirement	-(Pulse Width Limit, tdIPW, [ps])
Margin	Pulse Width Margin, [ps]
tDIHL (LPDDR5) **tDIVW_1bit** (LPDDR4)	
Measurement	tDIHL [ps]

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
Output Variation	tDIHL Margin with Output Delay Uncertainty, [ps]
Base Requirement	tDIHL Limit, [ps]
Margin	tDIHL Margin, [ps]
Min Slew Rate	
Measurement	Min Slew Rate, [V/ns]
Limit	Min Slew Rate Limit, SRIN_VdIVW_min, [V/ns]
Margin	Min Slew Rate Margin, [V/ns]
Max Slew Rate	
Measurement	Max Slew Rate, [V/ns]
Limit	Max Slew Rate Limit, SRIN_VdIVW_max, [V/ns]
Margin	Max Slew Rate Margin, [V/ns]
Min Slew Rate 2 (DDR4)	
Measurement	Min Slew Rate (2), [V/ns]
Limit	Min Slew Rate (2) Limit, SRIN_VdIVW_min, [V/ns]
Margin	Min Slew Rate (2) Margin, [V/ns]
Max Slew Rate 2 (DDR4)	
Measurement	Max Slew Rate (2), [V/ns]
Limit	Max Slew Rate (2) Limit, SRIN_VdIVW_max, [V/ns]
Margin	Max Slew Rate (2) Margin, [V/ns]
Voltage Margin between Signal and Eye-mask	Min Voltage Above/Below Eye-Mask, [mV]
Peak Voltage	
Above/Below Reference Voltage	Peak Voltage Above/Below Vcent(pin_mid), [mV]
Above/Below Reference Voltage Required	-(VIHL_AC/2 Limit, [mV])
Above/Below Reference Voltage Margin	Peak Voltage Above/Below Vcent(pin_mid) Margin, [mV]
Overshoot	
Measurement	Rise Rail Overshoot, [mV]

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
Limit	Rise Rail Overshoot Limit, [mV]
Margin	Rise Rail Overshoot Margin, [mV]
Undershoot	
Measurement	Fall Rail Undershoot, [mV]
Limit	Fall Rail Undershoot Limit, [mV]
Margin	Fall Rail Undershoot Margin, [mV]
Overshoot Area	
Measurement	Rise Rail Overshoot Area, [V*ns]
Limit	Rise Rail Overshoot Area Limit, [V*ns]
Margin	Rise Rail Overshoot Area Margin, [V*ns]
Undershoot Area	
Measurement	Fall Rail Undershoot Area, [V*ns]
Limit	Fall Rail Undershoot Area Limit, [V*ns]
Margin	Fall Rail Undershoot Area Margin, [V*ns]
Overshoot Area 2 (DDR4)	
Measurement	Rise Rail Overshoot Area (2), [V*ns]
Limit	Rise Rail Overshoot Area Limit (2), [V*ns]
Margin	Rise Rail Overshoot Area Margin (2), [V*ns]
Undershoot Area 2 (DDR4)	
Measurement	Fall Rail Undershoot Area (2), [V*ns]
Limit	Fall Rail Undershoot Area Limit (2), [V*ns]
Margin	Fall Rail Undershoot Area Margin (2), [V*ns]
<blank header>	
Monotonic	Monotonic, [Pass/Fail]

DDR2 / DDR3 Data Write Measurement Definitions

For information about measurements listed in the Definition column, see the following topics:

- “DDRx Data Spreadsheets” on page 321
- “DDRx JEDEC Measurements Spreadsheets” on page 327

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
Signal/Controller/DRAM	Lists the net name, driving pin, and receiving pin.
Status	Indicates whether the simulation results for the net meet the timing constraints. When Status = Pass, you can assume all margins measurements pass for that net.
Setup	
Measurement	Setup Time (From Sim), tDS, [ps]
Output Variation	-((“Setup Time (From Sim), tDS, [ps]”) - (“Min Setup Time, tDS(min) = tDS - Initial Delay Delta, [ps]”)) = Initial Delay Delta
Base Requirement	Base Setup Time, tDS(base), [ps]
Derating	Setup Derate Time Delta, d(tDS), [ps]
Margin	Setup Time Margin, tDS(margin) = tDS(min) - tDS(req), [ps]
Hold	
Measurement	Hold Time (From Sim), tDH, [ps]
Output Variation	-((“Hold Time (From Sim), tDH, [ps]”) - (“Min Hold Time, tDH(min) = tDH - Initial Delay Delta, [ps]”) = Initial Delay Delta
Base Requirement	Base Hold Time, tDH(base), [ps]
Derating	Hold Derate Time Delta, d(tDH), [ps]
Margin	Hold Time Margin, tDH(margin) = tDH(min) - tDH(req), [ps]
Overshoot	
Measurement	Rise Rail Overshoot, [mV]
Limit	Rise Rail Overshoot Limit, [mV]
Margin	Rise Rail Overshoot Margin, [mV]
Undershoot	
Measurement	Fall Rail Undershoot, [mV]
Limit	Fall Rail Undershoot Limit, [mV]
Margin	Fall Rail Undershoot Margin, [mV]
Overshoot Area	
Measurement	Rise Rail Overshoot Area, [V*ns]
Limit	Rise Rail Overshoot Area Limit, [V*ns]
Margin	Rise Rail Overshoot Area Margin, [V*ns]
Undershoot Area	
Measurement	Fall Rail Undershoot Area, [V*ns]

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
Limit	Fall Rail Undershoot Area Limit, [V*ns]
Margin	Fall Rail Undershoot Area Margin, [V*ns]
tVAC	
Measurement	TVAC/TDVAC Min Time Over AC, ps
Requirement	TVAC/TDVAC Time Limit, ps
Margin	TVAC/TDVAC Margin, ps
VIH/L	
VIH/L(AC)	VIH/L(AC) Min Limit, [Pass/Fail]
VIH/L(DC)	VIH/L(DC) Monotonicity, [Pass/Fail]

Data Read

Lists data read worstcase values from DDRx simulation.

Simulation Results for DDRx Interface Types

A set of simulation results varies by DDRx interface:

- "DDR4 / LPDDR4 / DDR5 / LPDDR5 Data Read Measurement Definitions" on page 286
- "DDR2 / DDR3 Data Read Measurement Definitions" on page 287

DDR4 / LPDDR4 / DDR5 / LPDDR5 Data Read Measurement Definitions

For information about measurements listed in the Definition column, see the following topics:

- "DDRx Address and Data Eye Aggregated Measurements Spreadsheets" on page 310
- "DDRx Address and Data Eye Per Bit Measurements Spreadsheets" on page 303

HTML Report Column	Definition
Signal/DRAM/ Controller	Lists the net name, driving pin, and receiving pin.
Status	Indicates whether the simulation results for the net meet the timing constraints. When Status = Pass, you can assume all margin measurements pass for that net.
Sweeps (You have specified sweep cases on the Sweep Manager page for a schematic design.)	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
<blank header>	
Widest Horizontal Eye	Voltage at Widest Eye Opening, Vcent, [mV]
Reference Voltage Used	Average Center Voltage, Vcent(pin_mid), [mV]
Setup	
Measurement	Measured Setup Time, [ps]
Output Variation	-(Output Delay Uncertainty, [ps])
Base Requirement	-(Half Eye-Mask Width, TdIPW/2, [ps])
Margin	Setup Time Margin with Output Delay Uncertainty, [ps]
Hold	
Measurement	Measured Hold Time, [ps]
Output Variation	-(Output Delay Uncertainty, [ps])
Base Requirement	-(Half Eye-Mask Width, TdIPW/2, [ps])
Margin	Hold Time Margin, tDH(margin) = tDH(min) - tDH(req), [ps]
Pulse Width	
Measurement	Vcent(pin_mid) to Vcent(pin_mid) Pulse Width, [ps]
Output Variation	-((("Vcent(pin_mid) to Vcent(pin_mid) Pulse Width, [ps]") - (("Pulse Width Limit, tdIPW, [ps]") + ("Pulse Width Margin, [ps]")))
Base Requirement	-(Pulse Width Limit, tdIPW, [ps])
Margin	Pulse Width Margin, [ps]

DDR2 / DDR3 Data Read Measurement Definitions

For detailed information about the definitions in the following table, see "DDRx Data Spreadsheets" on page 321.

HTML Report Column	Definition
Signal/DRAM/Controller	Lists the net name, driving pin, and receiving pin.
Status	Indicates whether the simulation results for the net meet the timing constraints. When Status = Pass, you can assume all margins measurements pass for that net.
Setup	
Measurement	Setup Time (From Sim), tDS, [ps]
Output Variation	-(("Setup Time (From Sim), tDS, [ps]") - ("Min Setup Time, tDS(min) = tDS - Initial Delay Delta, [ps]")) = Initial Delay Delta

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
Base Requirement	Base Setup Time, tDS(base), [ps]
Derating	Setup Derate Time Delta, d(tDS), [ps]
Margin	Setup Time Margin, tDS(margin) = tDS(min) - tDS(req), [ps]
Hold	
Measurement	Hold Time (From Sim), tDH, [ps]
Output Variation	-((“Hold Time (From Sim), tDH, [ps]” - (“Min Hold Time, tDH(min) = tDH - Initial Delay Delta, [ps]”)) = Initial Delay Delta
Base Requirement	Base Hold Time, tDH(base), [ps]
Derating	Hold Derate Time Delta, d(tDH), [ps]
Margin	Hold Time Margin, tDH(margin) = tDH(min) - tDH(req), [ps]

Address

Lists address worstcase values from DDRx simulation.

For information about measurements listed in the Definition column, see the following topics:

- “DDRx Address Spreadsheets” on page 299
- “DDRx JEDEC Measurements Spreadsheets” on page 327

HTML Report Column	Definition
Signal/Dram	Lists the net name and accessed DRAM.
Status	Indicates whether the simulation results for the net meet the timing constraints. When Status = Pass, you can assume all margins measurements pass for that net.
Sweeps (You have specified sweep cases on the Sweep Manager page for a schematic design.)	
<blank header>	
Widest horizontal eye (DDR4, LPDDR4, DDR5, LPDDR5)	Voltage at Widest Eye Opening, Vcent, [mV]
Reference voltage (DDR4, LPDDR4, DDR5, LPDDR5)	Average Center Voltage, Vcent(pin_mid), [mV]
Setup	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
Measurement	Setup Time (From Sim), tIS, [ps]
Output Variation	-((“Setup Time (From Sim), tIS, [ps]”) - (“Min Setup Time, tIS(min) = tIS - Initial Delay Delta, [ps]”)) = Initial Delay Delta
Base Requirement	-(Base Setup Time, tIS(base), [ps])
Derating	-(Setup Derate Time Delta, d(tIS), [ps])
Margin	Setup Time Margin, tIS(margin) = tIS(min) - tIS(req), [ps]
Hold	
Measurement	Hold Time (From Sim), tIH, [ps]
Output Variation	-((“Hold Time (From Sim), tIH, [ps]”) - (“Min Hold Time, tIH(min) = tIH - Initial Delay Delta, [ps]”)) = Initial Delay Delta
Base Requirement	-(Base Hold Time, tIH(base), [ps])
Derating	-(Hold Derate Time Delta, d(tIH), [ps])
Margin	Hold Time Margin, tIH(margin) = tIH(min) - tIH(req), [ps]
Pulse Width (DDR5, LPDDR4, LPDDR5)	
Measurement	Min Pulse Width [ps]
Output Variation	Min Pulse Width Margin with Output Delay Uncertainty, [ps]
Base Requirement	Min Pulse Width Limit, TdIPW, [ps]
Margin	Min Pulse Width Margin, [ps]
Min Slew Rate	
Measurement	Measurement Min Slew Rate, [V/ns]
Limit	Min Slew Rate Limit, SRIN_VdIVW_min, [V/ns]
Margin	Min Slew Rate Margin, [V/ns]
Max Slew Rate	
Measurement	Max Slew Rate, [V/ns]
Limit	Max Slew Rate Limit, SRIN_VdIVW_max, [V/ns]
Margin	Max Slew Rate Margin, [V/ns]
Voltage Margin between Signal and Eye-Mask (DDR5, LPDDR5)	
Margin	Vertical gap between eye mask and worst-case signal.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
Peak Voltage (DDR5, LPDDR5)	
Above/Below Reference Voltage	Peak Voltage Above/Below Vcent(pin_mid), [mV]
Above/Below Reference Voltage Required	-(VIHL_AC/2 Limit, [mV])
Above/Below Reference Voltage Margin	Peak Voltage Above/Below Vcent(pin_mid) Margin, [mV]
Overshoot	
Measurement	Rise Rail Overshoot, [mV]
Limit	Rise Rail Overshoot Limit, [mV]
Margin	Rise Rail Overshoot Margin, [mV]
Undershoot	
Measurement	Fall Rail Undershoot, [mV]
Limit	Fall Rail Undershoot Limit, [mV]
Margin	Fall Rail Undershoot Margin, [mV]
Overshoot Area	
Measurement	Rise Rail Overshoot Area, [V*ns]
Limit	Rise Rail Overshoot Area Limit, [V*ns]
Margin	Rise Rail Overshoot Area Margin, [V*ns]
Undershoot Area	
Measurement	Fall Rail Undershoot Area, [V*ns]
Limit	Fall Rail Undershoot Area Limit, [V*ns]
Margin	Fall Rail Undershoot Area Margin, [V*ns]
Overshoot Area 2 (DDR4)	
Measurement	Rise Rail Overshoot Area (2), [V*ns]
Limit	Rise Rail Overshoot Area (2) Limit, [V*ns]
Margin	Rise Rail Overshoot Area (2) Margin, [V*ns]
Undershoot Area 2 (DDR4)	
Measurement	Fall Rail Undershoot Area (2), [V*ns]

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
Limit	Fall Rail Undershoot Area (2) Limit, [V*ns]
Margin	Fall Rail Undershoot Area (2) Margin, [V*ns]
tVAC	
Measurement	TVAC/TDVAC Min Time Over AC, ps
Requirement	TVAC/TDVAC Time Limit, ps
Margin	TVAC/TDVAC Margin, ps
VIH/L	
VIH/L(AC)	VIH/L(AC) Min Limit, [Pass/Fail]
VIH/L(DC)	VIH/L(DC) Monotonicity, [Pass/Fail]
<blank header>	
Monotonic (DDR5, LPDDR5)	Monotonic, [Pass/Fail]

Differential Nets

Lists differential worstcase values from DDRx simulation.

For information about measurements listed in the Definition column, see the following topics:

- "DDRx JEDEC Measurements Spreadsheets" on page 327
- "DDRx Address and Data Eye Aggregated Measurements Spreadsheets" on page 310

HTML Report Column	Definition
Signal/Driver/Receiver	Lists the net name, driving pin, receiving pin, and read/write operation.
Status	Indicates whether the simulation results for the net meet the timing constraints. When Status = Pass, you can assume all margins measurements pass for that net.
Sweeps (You have specified sweep cases on the Sweep Manager page for a schematic design.)	
Overshoot	
Measurement	Rise Rail Overshoot, [mV]
Limit	Rise Rail Overshoot Limit, [mV]
Margin	Rise Rail Overshoot Margin, [mV]
Undershoot	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
Measurement	Fall Rail Undershoot, [mV]
Limit	Fall Rail Undershoot Limit, [mV]
Margin	Fall Rail Undershoot Margin, [mV]
Overshoot Area 2 (DDR4)	
Measurement	Rise Rail Overshoot Area (2), [V*ns]
Limit	Rise Rail Overshoot Area (2) Limit, [V*ns]
Margin	Rise Rail Overshoot Area (2) Margin, [V*ns]
Undershoot Area 2 (DDR4)	
Measurement	Fall Rail Undershoot Area (2), [V*ns]
Limit	Fall Rail Undershoot Area (2) Limit, [V*ns]
Margin	Fall Rail Undershoot Area (2) Margin, [V*ns]
Overshoot Area	
Measurement	Rise Rail Overshoot Area, [V*ns]
Limit	Rise Rail Overshoot Area Limit, [V*ns]
Margin	Rise Rail Overshoot Area Margin, [V*ns]
Undershoot Area	
Measurement	Fall Rail Undershoot Area, [V*ns]
Limit	Fall Rail Undershoot Area Limit, [V*ns]
Margin	Fall Rail Undershoot Area Margin, [V*ns]
Min Slew (LPDDR5)	
Measurement	Min Slew Rate [V/ns]
Base Requirement	Min Slew Rate Limit, SRIN_dIVW_min, [V/ns]
Margin	Min Slew Rate Margin [V/ns]
Max Slew (LPDDR5)	
Measurement	Max Slew Rate [V/ns]
Base Requirement	Max Slew Rate Limit, SRIN_dIVW_max, [V/ns]
Margin	Max Slew Rate Margin [V/ns]

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
tDVAC	
Measurement	TVAC/TDVAC Min Time Over AC, ps
Requirement	TVAC/TDVAC Time Limit, ps
Margin	TVAC/TDVAC Margin, ps
Vix (DDR4, LPDDR4, DDR5, LPDDR5)	
Measurement	((VIX Limit Hi, mV - VIX Limit Low, mV) / 2) - VIX Margin, mV
Limit	(VIX Limit Hi, mV - VIX Limit Low, mV) / 2
Margin	VIX Margin, mV
VID(AC) [mV]	VID(AC) Margin, mV
VID(DC)	VID(DC), [Pass/Fail]
VRx/2 (DDR5)	
Measurement	Peak differential voltage ((positive signal voltage) - (negative signal voltage)) for CK and DQS.
Base Requirement	Vrx/2 Limit, mV
Margin	Vrx/2 Margin, mV
Vinse Peak-to-peak (LPDDR4, LPDDR5)	
Measurement	Vinse Peak To Peak Limit plus Margin, mV
Limit	Vinse Peak To Peak Limit, mV
Margin	Vinse Peak To Peak Margin, mV
Vinse High (LPDDR4, LPDDR5)	
Measurement	Vinse High Margin plus Vinse High Margin, mV
Limit	Vinse High Limit, mV
Margin	Vinse High Margin, mV
Vinse Low (LPDDR4, LPDDR5)	
Measurement	Vinse Low Margin plus Vinse Low Limit, mV
Limit	Vinse Low Limit, mV

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
Margin	Vinse Low Margin, mV
VSEH (Non-DDR4)	
Measurement	VSEH Limit Low, mV
Limit	VSEH Limit Hi, mV
Margin	VSEH Margin, mV
VSEL (Non-DDR4)	
Measurement	VSEL Limit Low, mV
Limit	VSEL Limit Hi, mV
Margin	VSEL Margin, mV
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VSEH/L (DDR4)	"Pass" when both of the following are "Pass": • VSEL, [Pass/Fail] • VSEH, [Pass/Fail]
Monotonic (Non-DDR4)	Monotonic, [Pass/Fail]

CLK-DQS Skew

Lists CLK-DQS skew worstcase values from DDRx simulation.

For information about measurements listed in the Definition column, see "DDRx Skew Spreadsheets" on page 338.

HTML Report Column	Definition
Signal/Dram	Lists the net name, accessed DRAM, CLK pins, and DQS pins.
Status	Indicates whether the simulation results for the net meet the timing constraints. When Status = Pass, you can assume all margins measurements pass for that net.
Sweeps (You have specified sweep cases on the Sweep Manager page for a schematic design.)	
DQSS - Earliest DQS	
Allowed Early DQS Skew	- (Negative tDQSS Limit, [ps])
DQS Arrival Time	Strobe Arrival Time (From Sim), [ps]

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
CLK Arrival Time	Clk Arrival Time (From Sim), [ps]
Max Early DQS Launch Skew	(Launch Skew Time (typ), [ps]) - (Launch Skew Time (min), [ps])
Margin	tDQSS (early) Time Margin = tDQSS(early) - tDQSS(negative limit), [ps]
DQSS - Latest DQS	
Allowed Late DQS Skew	Positive tDQSS Limit, [ps]
DQS Arrival Time	Strobe Arrival Time (From Sim), [ps]
CLK Arrival Time	Clk Arrival Time (From Sim), [ps]
Max Late DQS Launch Skew	(Launch Skew Time (max), [ps]) - (Launch Skew Time (typ), [ps])
Margin	tDQSS (late) Time Margin = tDQSS(positive limit) - tDQSS(late), [ps]
tDSS	
Measurement	(tDSS Setup Time (From Sim), [ps]) - “Earliest DQS Launch Skew”
Earliest DQS Launch Skew	-((“Launch Skew Time (typ), [ps]”) - (“Launch Skew Time (min), [ps]”))
Base Requirement	-(tDSS Limit, [ps])
Margin	tDSS Time Margin = tDSS - tDSS(limit), [ps]
tDSH	
Measurement	(tDSH Hold Time (From Sim), [ps]) - “Latest DQS Launch Skew”
Latest DQS Launch Skew	-((“Launch Skew Time (max), [ps]”) - (“Launch Skew Time (typ), [ps]”))
Base Requirement	-(tDSH Limit, [ps])
Margin	tDSH Time Margin = tDSH - tDSH(limit), [ps]

Eye Density Links

Lists eye density plots and related measurements from DDRx simulation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HTML Report Column	Definition
Eye Density Links	Lists the available eye density plots. Click a "Load Eye Density Plots" link to display an eye density plot and measurement data in a spreadsheet. For spreadsheet column definitions, see "DDRx Address and Data Eye Aggregated Measurements Spreadsheets" on page 310.
Type	Indicates whether an eye density plot shows data signals or address signals.
Sweeps (You have specified sweep cases on the Sweep Manager page for a schematic design.)	Indicates values for a specific sweep case.
Corner	Indicates whether the IC model corner is Slow, Typ, or Fast.

Re-Opening a DDRx HTML Report

After you close a HTML report that automatically opens when DDRx batch-mode simulation completes, you can re-open it.

Prerequisites

- You have enabled "HTML reports" on the Output page and run simulation. See "Running a DDRx Memory Interface Simulation" on page 185.

Procedure

View an HTML report in one of the following ways:

If you want to...	Do this...
View HTML reports from previous simulations	Choose the **View > HTML Report Manager** menu item.
View the HTML report plus the measurement waveforms	Open the report viewer by double-clicking this file: *<design>\DDR_Results_<date>\DDR_Report\report.bat*
View the HTML report without viewing the measurement waveforms	With a web browser, open this file: *<design>\DDR_Results_<date>\DDR_Report\report_data\index.html* **Note:** When you open an HTML report this way, clicking a link in the table does not open EZwave and display measurement waveforms.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Results

The HTML report opens. For information about the contents of a report, see “DDRx HTML Report Format” on page 278.

If you want to share results with someone, send the *DDR_Results* folder. The recipient can double-click the *report.bat* file to see the HTML report as long as HyperLynx is installed on the target machine. Otherwise, use web browser to open the *index.html* file and view the results table but no waveforms.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDRx Results Spreadsheets

DDRx results spreadsheets contain measurement values and pass/fail status. Spreadsheet rows contain measurements for one write or read cycle.

Note

The Windows version of the software can produce spreadsheets in XLS format with color coding. Spreadsheet cells with a red background indicate measurements that violate timing model requirements. Spreadsheet cells with a yellow background indicate a minimum setup or hold time margin.

Topic	Description
DDRx Address Spreadsheets	These spreadsheets report setup and hold measurements for address, control, and command signals in the DDRx interface. Address, command, and control nets are measured relative to clocks.
DDRx Address and Data Eye Per Bit Measurements Spreadsheets	These spreadsheets report the per bit measurements at each strobe/clock cycle for all nets in the DDRx interface.
DDRx Address and Data Eye Aggregated Measurements Spreadsheets	This spreadsheet reports the worst case measurements across all the bits seen on a given signal for all nets in the memory interface.
DDRx Clock Jitter Spreadsheets	These spreadsheets report clock jitter for all nets in the DDRx interface.
DDRx Clock Jitter Error Spreadsheets	These spreadsheets report clock jitter error for all nets in the DDRx interface.
DDRx Data Spreadsheets	These spreadsheets report setup and hold measurements for data and data mask signals in the DDRx interface. Data nets are measured relative to strobes.
DDRx JEDEC Measurements Spreadsheets	These spreadsheets report signal-integrity measurements for all nets in the DDRx interface.
DDRx Round Trip Time Spreadsheets	These spreadsheets report round trip time for all nets in the DDRx interface.
DDRx Skew Spreadsheets	These spreadsheets report skew measurements for clocks and strobes in the DDRx interface. Skew is produced by routing, crosstalk, and other factors. DDRx batch simulation does not introduce random jitter.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
DDRx Audit Spreadsheet	DDR_audit<DDRx_setup_file_name>.xls contains information about simulation setup problems for all pins in the DDRx interface. This is an optional file that the software writes when you enable the audit option.

DDRx Address Spreadsheets

These spreadsheets report setup and hold measurements for address, control, and command signals in the DDRx interface. Address, command, and control nets are measured relative to clocks.

Note

For some DDRx interfaces, the software can create "address eye" spreadsheets. See "DDRx Address and Data Eye Aggregated Measurements Spreadsheets" on page 310 and "DDRx Address and Data Eye Per Bit Measurements Spreadsheets" on page 303.

Report file name takes the form:

*DDR_report_address*_[violations | worstcases | allcases]_[Typ | Fast | Slow].[*csv* | *xls*]

where:

violations contains only nets that violate timing-model requirements.

worst cases contains all nets and their worst-case measurements.

allcases contains all nets and all measurements (for each cycle in the simulation).

Typ, Fast, Slow represent IC model corners.

Column Name	Definition
Net Name	Name of the measured net. For a multiple-board project, the net name does not include the board ID.
Pass/Fail	"Fail" if any measurements fail, based on timing model limits.
Operation	Read or Write, with rank definition as rank(slot number, rank within the slot).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Driver Comp Ref Des & Pin Name	Driver component reference designator and pin name for the measured net. For MultiBoard projects, the net name contains the board ID. Use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Receiver Comp Ref Des & Pin Name	Receiver component reference designator and pin name for the measured net. For a multiple-board project, the net name contains the board ID. Use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Associated Clk/Strobe Net Name	Name of the clock net paired with the measured net.
Associated Clk/Strobe Driver Comp Ref Des & Pin Name	Driver component reference designator and pin name for the clock net paired with the measured net.
Associated Clk/Strobe Receiver Comp Ref Des & Pin Name	Receiver component designator and pin name for the clock net paired with the measured net.
Clk/Strobe Crossing Threshold Time, [ns]	Time in the simulation when the clock waveforms cross zero. Setup/hold times are measured before/after this time. Investigate measurement results by using this time when viewing waveform files, if you enable saving waveform files.
Setup Time (From Sim), tIS, [ps]	The measured address and clock setup time, prior to derating. Measurements start when the address waveform last crosses the AC threshold and end when the corresponding clock waveform crosses zero. The measured setup time is based on the average driver delay. See DDRx Delay File Measurements.
Min Setup Time, tIS(min) = tIS - Initial Delay Delta, [ps]	Setup time obtained by shifting address waveforms to the right, within the specified range of initial delays. This represents the worst setup time possible when varying the initial delay. The calculation is (Setup Time (From Sim), tIS, [ps]) minus (Initial delay delta), where: Initial delay delta—(((Maximum initial delay for DQS and CK) minus (Minimum initial delay for DQS and CK))/2).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Base Setup Time, tIS(base), [ps]	The setup limit whose value is defined in the timing model.
Setup Derate Time Delta d(tIS), [ps]	The setup derating value obtained from derating tables built into DDRx batch simulation or specified in the timing model. Values can be positive or negative. The log file contains derating calculation details, such as slew rates for data and strobe nets. See DDRx Log File.
Required Setup Time, tIS(req) = tIS(base) + d(tIS), [ps]	The adjusted setup limit, where the value from the timing model is adjusted by the setup derating value. The calculation is (Base Setup Time, tIS(base), [ps]) plus (Setup Derate Time Delta d(tIS), [ps]).
Setup Time Margin, tIS(margin) = tIS(min) - tIS(req), [ps]	The margin (or slack) remaining after subtracting the slew-rate-adjusted setup limit from the measured setup time. Negative values produce errors. The calculation is (Min Setup Time, tIS(min) = tIS - Initial Delay Delta, [ps]) minus (Required Setup Time, tIS(req) = tIS(base) + d(tIS), [ps]). **Tip:** Spreadsheet cells with a yellow background indicate a minimum setup time margin.
Hold Time (From Sim), tIH, [ps]	The measured address and clock hold time, prior to derating. Measurements start when the address waveform crosses the DC threshold and end when the clock waveform crosses zero. If the IBIS model does not contain a DC threshold, the AC threshold is used instead. The measured hold time is based on the average driver delay. See DDRx Delay File Measurements.
Min Hold Time, tIH(min) = tIH - Initial Delay Delta, [ps]	Hold time obtained by shifting address waveforms to the left, within the specified range of initial delays. This represents the worst hold time possible when varying the initial delay. The calculation is (Hold Time (From Sim), tIH, [ps]) minus (Initial delay delta), where: Initial delay delta—(((Maximum initial delay for DQS and CK) minus (Minimum initial delay for DQS and CK))/2).
Base Hold Time, tIH(base), [ps]	The hold limit whose value is defined in the timing model.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Hold Derate Time Delta d(tIH), [ps]	The hold derating value obtained from derating tables built into DDRx batch simulation. Values can be positive or negative. The log file contains derating calculation details, such as slew rates for data and strobe nets. See DDRx Log File.
Required Hold Time, tIH(req) = tIH(base) + d(tIH), [ps]	The adjusted hold limit, where the value from the timing model is adjusted by the hold derating value. The calculation is (Base Hold Time, tIH(base), [ps]) plus (Hold Derate Time Delta d(tIH), [ps]).
Hold Time Margin, tIH(margin) = tIH(min) - tIH(req), [ps]	The margin (or slack) remaining after subtracting the slew-rate-adjusted hold limit from the measured hold time. Negative values produce errors. The calculation is (Min Hold Time, tIH(min) = tIH - Initial Delay Delta, [ps]) minus (Required Hold Time, tIH(req) = tIH(base) + d(tIH), [ps]). **Tip:** Spreadsheet cells with a yellow background indicate a minimum hold time margin.
Pulse Width, [ps] (DDR5, LPDDR4, LPDDR5)	Measured minimum high/low/high or low/high/low pulse width. Measurement begins and ends when a signal crosses Vref.
Pulse Width Limit, [ps] (DDR5, LPDDR4, LPDDR5)	TciPW(%) * UI, values from DRAM timing model.
Pulse Width Margin, [ps] (DDR5, LPDDR4, LPDDR5)	Pulse Width - Pulse Width Margin
Pulse Width Margin with Output Delay Uncertainty, [ps] (DDR5, LPDDR4, LPDDR5)	Pulse Width Margin - Output Delay Uncertainty
Stimulus Offset, [ns]	The timing offset between drivers on the address and clock nets, when the initial clock delay is average. See DDRx Delay File Measurements.
% of Cycles With Failures	Percentage of waveform measurements with either of the following properties: • The measured value fails the limit value • The measurement itself failed for some reason **Restriction:** Only the violations spreadsheets contain this column.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Comments	Displays error or other measurement problem information.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Address and Data Eye Per Bit Measurements Spreadsheets

These spreadsheets report the per bit measurements at each strobe/clock cycle for all nets in the DDRx interface.

File names are of the form:

*DDR_report_address_eye_perbit_*[Typ | Fast | Slow].[*csv* | *.xls*]

*DDR_report_data_eye_perbit_*[Typ | Fast | Slow].[*csv* | *.xls*]

where:

Typ, Fast, Slow represent IC model corners.

Column Name	Definition
Net Name	Name of the measured net.
Pass/Fail	Measurement status
Operation	Read or Write, with rank definition as rank(slot number, rank within the slot)
Driver Comp Ref Des & Pin Name	Driver pin identification information.
Receiver Comp Ref Des & Pin Name	Receiver pin identification information.
Associated Clk/Strobe Net Name	Name of the clock/strobe net associated with the measured net.
Associated Clk/Strobe Driver Comp Ref Des & Pin Name	Associated Clock/Strobe driver pin identification information.
Associated Clk/Strobe Receiver Comp Ref Des & Pin Name	Associated Clock/Strobe receiver pin identification information.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Clk/Strobe Crossing Threshold Time, [ns]	Where DQS crosses zero volts. This is where all DQ waveform measurements are taken.
Average Center Voltage, Vcent(pin_mid), [mV]	Write: average center voltage, Vcent(pin_mid) = (Min Vcent for Ux + Max Vcent for Ux)/2 Read: The same equation is used for read and write operations. The value(s) depends on the options you chose in the Vref Training page of the DDRx Wizard. • With the option “Enable separate Vref per rank” unchecked: Single Vref per DRAM - For multi-rank, Vref is calculated using one min and one max value per DRAM group. Single Vref per lane - Vref value is calculated using the min and max value for each lane. Single Vref for all the signals - Vref value is calculated using the min and max values across all signals. Therefore, all signals have the same Vref. • With the option “Enable separate Vref per rank “ checked: Single Vref per DRAM - The Vref value is calculated per rank, rather than per DRAM group. For example, (rank1max + rank1min)/2. Single Vref per lane - Each rank (specified in the Operation column) in each lane has a different Vcent value. For example, if you are simulating an interface with two lanes, each with two ranks, four Vcent values are calculated. Single Vref for all signals - Vcent value is calculated per rank using the value in the “Voltage at Widest Eye Opening, Vcent, [mV]” column. For additional information on how Vcent is calculated, see Vcent(pin_mid) Calculation Examples.
Upper Eye Mask, Vcent(pin_mid) + VdIVW_Total/2, [mV]	Upper voltage limit of the eye mask. VdIVW_Total is the Eye-Mask Height as reported in the DDRx Address and Data Eye Aggregated Measurements Spreadsheets.
Lower Eye Mask, Vcent(pin_mid) - VdIVW_Total/2, [mV]	Lower voltage limit of the eye mask. VdIVW_Total is the Eye-Mask Height as reported in the DDRx Address and Data Eye Aggregated Measurements Spreadsheets.
Half Eye-Mask Width, TdIPW/2, [ps]	Write: (TdIPW(%) * UI) / 2, values from DRAM timing model. Read: (EyeMaskWidthLimit (%) * UI) / 2, values from controller timing model.

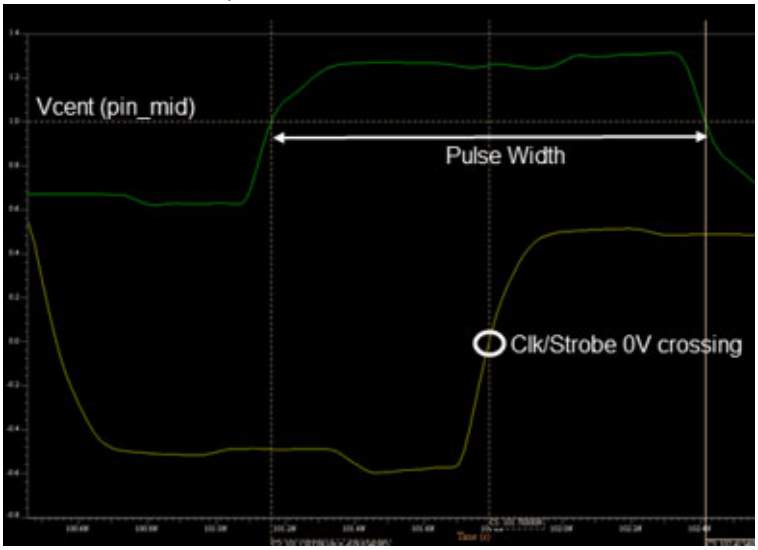
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Half Eye-Mask Width, TcIVW/2, [ps]	(TcIVW(%) * UI) / 2, values from the DRAM timing model. Measured from the center of a hexagonal eye mask.
Half Eye-Mask Width1, TcIVW1/2, [ps]	(TcIVW1(%) * UI) / 2, values from the DRAM timing model. Represents half the maximum or total width of a hexagonal eye mask, measured from the center of the eye mask.
Half Eye-Mask Width2, TcIVW2/2, [ps]	(TcIVW2(%) * UI) / 2, values from the DRAM timing model. Represents half the width of a hexagonal eye mask, within the rectangular region (for example, between the top-left and top-right corners of an eye mask), measured from the center of the eye mask.
Half Eye-Mask Width, RxEW/2, [ps]	(RxEW(%) * UI) / 2, values from DRAM timing model. Represents half the width of a kite-shaped eye mask.
Rx PeakOffset, [ps]	RxPeakOffset(%) * UI, values from DRAM timing model. Represents the horizontal offset of the maximum height of a kite-shaped eye mask, from the start of the UI.
Half Eye-Mask Width1, TdIVW1/2, [ps]	(TdIVW1(%) * UI) / 2, values from the DRAM timing model. Represents half the maximum or total width of a hexagonal eye mask, measured from the center of the eye mask.
Half Eye-Mask Width2, TdIVW2/2, [ps]	(TdIVW2(%) * UI) / 2, values from the DRAM timing model. Represents half the width of a hexagonal eye mask, within the rectangular region (for example, between the top-left and top-right corners of an eye mask), measured from the center of the eye mask.
Half Eye-Mask Width, TdIVW/2, [ps]	Time between a strobe signal and the side of an eye mask. The center of an eye mask is a strobe signal.
Output Delay Uncertainty, [ps]	Write: [tDQSDQ(max) – tDQSDQ(min)]/2, values from controller timing model. Read:[tDQSQ(max) – tDQSQ(min)]/2, values from the DRAM timing model.
Measured Setup Time, [ps]	DQ rising edge: Measured from zero volt DQS crossing to (Vcent(pin_mid) + VdIVW/2) DQ falling edge: Measured from zero volt DQS crossing to (Vcent(pin_mid) – VdIVW/2)
Setup Time Margin, [ps]	Measured Setup Time – (eye-mask width/2).
Setup Time Margin with Output Delay Uncertainty, [ps]	Setup Time Margin – Output Delay Uncertainty

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Column Name	Definition
Measured Hold Time, [ps]	Rising edge of DQ signal: Measured from DQS 0V crossing to (Vcent(pin_mid) – VdIVW/2) Falling edge of DQ signal: Measured from DQS 0V crossing to (Vcent(pin_mid) + VdIVW/2)
Hold Time Margin, [ps]	Equals Measured Hold Time - (TdIPW/2). TdIPW is the Eye-Mask Width.
Hold Time Margin with Output Delay Uncertainty, [ps]	Hold Time Margin – Output Delay Uncertainty
Min Voltage Above/Below Eye-Mask, [mV]	For high bit: Within the mask window time, the min value is the value from the upper mask voltage to closest data point. For low bit: Within the mask window time, the min value is the value from the lower mask voltage to closest data point.
Peak Voltage Above/Below Vcent(pin_mid), [mV]	For high bit: Within the mask window time, find the value from Vcent(pin_mid) to furthest data point above Vcent(pin_mid). For low bit: Within the mask window time, find the value from Vcent(pin_mid) to furthest data point below Vcent(pin_mid).
VIHL_AC/2 Limit, [mV]	Write: Equals VIHL_AC_DQ/2. Value from DRAM timing model. Read: Equals MaxEyeHeightLimit/2. Value from controller timing model.
Peak Voltage Above/Below Vcent(pin_mid) Margin, [mV]	Peak Voltage Above/Below Vcent(pin_mid) – VIHL_AC/2 Limit.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Vcent(pin_mid) to Vcent(pin_mid) Pulse Width, [ps]	Pulse width measured at the Vcent(pin_mid) voltage value of the pulse that corresponds to the point where the Clk/Strobe rising edge crosses zero volts (Clk/Strobe Crossing Threshold Time column value). 
Pulse Width Limit, TdIPW, [ps]	Write: TdIPW % * PW = 58% * 1/1600Mbps. Values from DRAM timing model. Read: VrefToVrefLimit% * PW. Values from controller timing model.
Pulse Width Margin, [ps]	Vcent(pin_mid) to Vcent(pin_mid) Pulse Width - Pulse Width Limit, TdIPW
Pulse Width Margin with Output Delay Uncertainty, [ps]	Pulse Width Margin – Output Delay Uncertainty
tDIHL, [ps]	Measured minimum high/low/high pulse width, from the falling edge crossing of VIH(AC) to the rising edge crossing VIH(AC).
tDIHL Limit, [ps]	tDIHL(%) * UI, values from the DRAM timing model.
tDIHL Margin, [ps]	tDIHL - tDIHL Limit.
tDIHL Margin with Output Delay Uncertainty, [ps]	tDIHL Margin - <variable>, where <variable> is: • tDQSQDDR4 — For a controller with per-bit DQ deskew capability. • tDQSQ or tQH — For a controller without per-bit DQ deskew capability.
tDIHL Signal Trace Point Left, [ns]	For a high/low/high pulse and its minimum width, the time when a falling edge crosses VIH(AC).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
tDIHL Signal Trace Point Right, [ns]	For a high/low/high pulse and its minimum width, the time when a rising edge crosses VIH(AC).
Min Slew Rate, [V/ns]	This measurement depends on the interface you use. For information, see: • "Note About Slew Rate - Non-DDR4" on page 310 • "Note About Slew Rate for DDR4" on page 309
Min Slew Rate Limit, SRIN_VdIVW_min, [V/ns]	Write: SRIN_dIVW_min, value from the DRAM timing model. Read: MinSlewRateLimit, value from the controller timing model.
Min Slew Rate Margin, [V/ns]	Min Slew Rate – Min Slew Rate Limit, SRIN_VdIVW_min
Max Slew Rate, [V/ns]	This measurement depends on the interface you use. For information, see: • "Note About Slew Rate - Non-DDR4" on page 310 • "Note About Slew Rate for DDR4" on page 309
Max Slew Rate Limit, SRIN_VdIVW_max, [V/ns]	Write: SRIN_dIVW_max, value from the DRAM timing model. Read: MaxSlewRateLimit, value from the controller timing model.
Max Slew Rate Margin, [V/ns]	Equals Max Slew Rate - Max Slew Rate Limit
Min Slew Rate (2), [V/ns] (DDR4)	See "Note About Slew Rate for DDR4" on page 309.
Min Slew Rate (2) Limit, SRIN_VdIVW_min, [V/ns] (DDR4)	Write: SRIN_dIVW_min, value from the DRAM timing model. Read: MinSlewRateLimit, value from the controller timing model.
Min Slew Rate (2) Margin, [V/ns] (DDR4)	Min Slew Rate – Min Slew Rate Limit, SRIN_VdIVW_min
Max Slew Rate (2), [V/ns] (DDR4)	See "Note About Slew Rate for DDR4" on page 309.
Max Slew Rate (2) Limit, SRIN_VdIVW_max, [V/ns] (DDR4)	Write: SRIN_dIVW_max, value from the DRAM timing model. Read: MaxSlewRateLimit, value from the controller timing model.

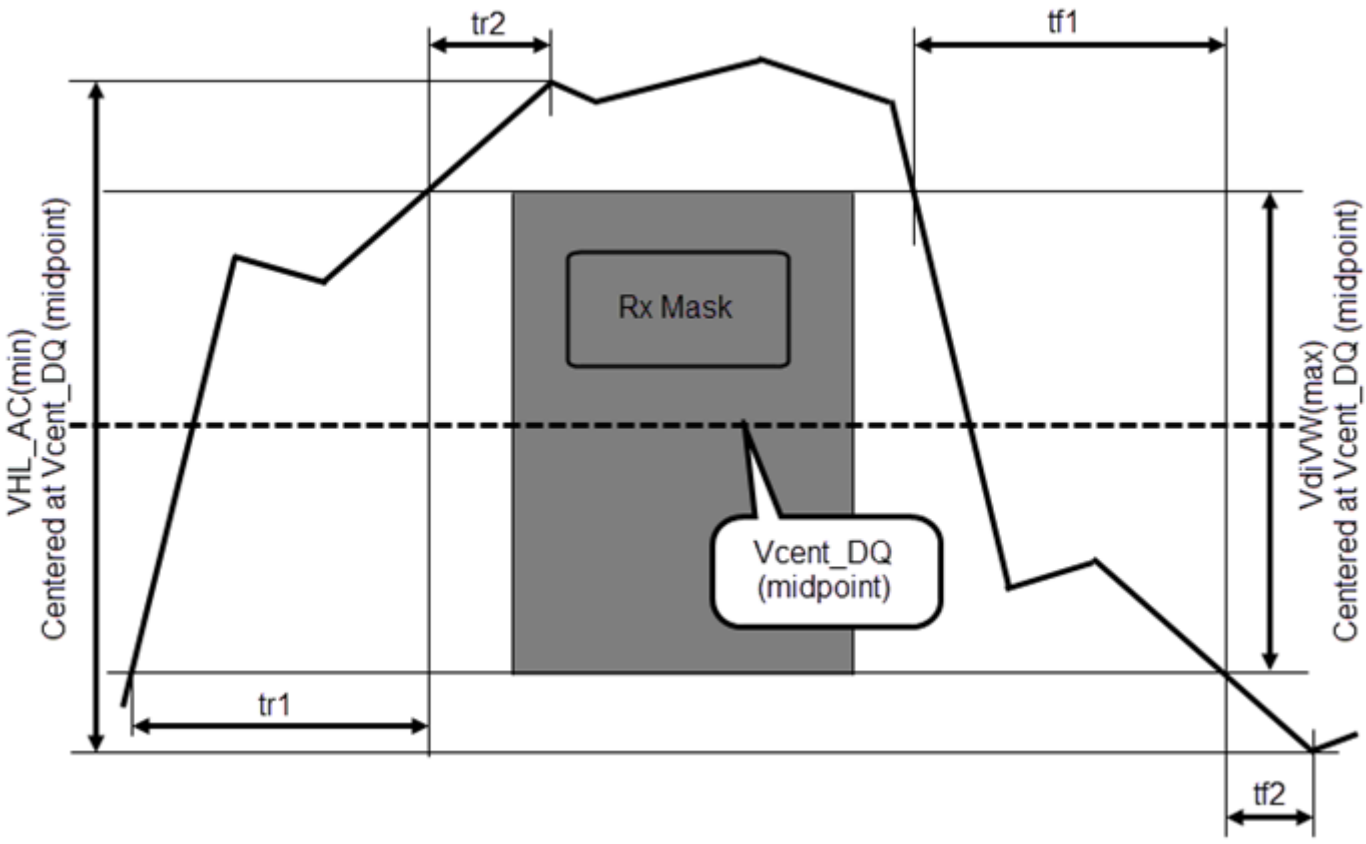
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Max Slew Rate (2) Margin, [V/ns] (DDR4)	Equals Max Slew Rate - Max Slew Rate Limit
Eye Mask Shape	One of the following: Eye-Rectangle, Eye-Kite, Eye-Hexagon.

Note About Slew Rate for DDR4

Slew rate has two components. The following table and figure provide measurement details:

Spreadsheet columns	Signal transition	Equation
Min Slew Rate Max Slew Rate	Rising edge	VdIVW(max) / tr1
	Falling edge	VdIVW(max) / tf1
Min Slew Rate (2) Max Slew Rate (2)	Rising edge	(VIHL_AC(min) - VDIVW(max) / (2*tr2)
	Falling edge	(VIHL_AC(min) - VDIVW(max) / (2*tf2)



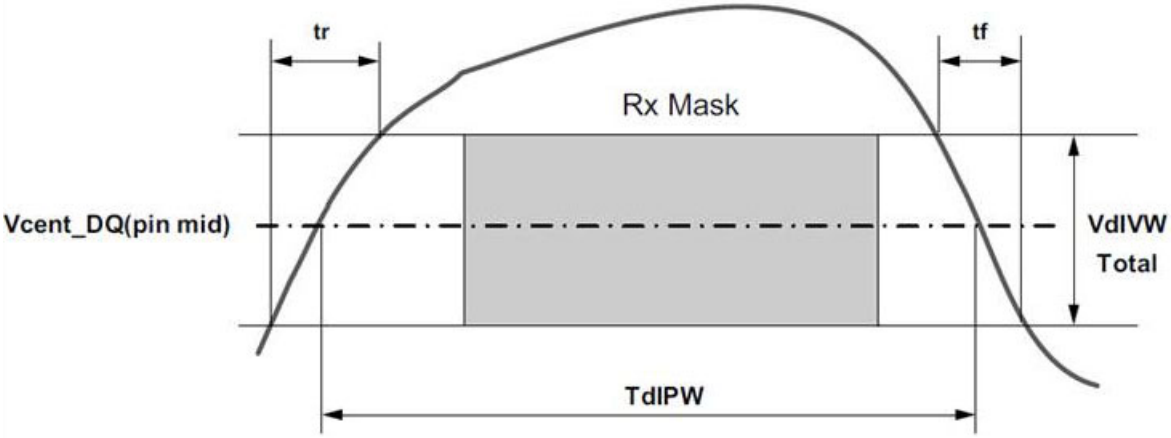
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note About Slew Rate - Non-DDR4

Slew rate is the rate of change of voltage of the waveform between defined points in time. The voltages used for the high and low points correspond to the high and low levels of the eye mask. The eye mask is in turn vertically centered around the reference voltage. Therefore, the slew rate is calculated as equal to (Eye Mask Height)/(time taken for signal to traverse between eye mask voltage levels).

If two edges exist at the zero volt crossing of DQS, the spreadsheet reports the maximum slew rate for both edges. If only one edge exists, both min and max values are reported as the same value.

In the following figure, the slew rate is VdIVW/tr or VdIVW/tf:



Related Topics

Running a DDRx Memory Interface Simulation

DDRx Address and Data Eye Aggregated Measurements Spreadsheets

This spreadsheet reports the worst case measurements across all the bits seen on a given signal for all nets in the memory interface.

File names are of the form:

*DDR_report_address_eye_aggregated_*[Typ | Fast | Slow].[*csv* | *.xls*]

*DDR_report_data_eye_aggregated_*[Typ | Fast | Slow].[*csv* | *.xls*]

where:

Typ, Fast, Slow represent IC model corners of the aggregate results.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Some of the column definitions refer to the DRAM and controller timing model parameters defined by a JEDEC specification. Visit www.jedec.org.

Note

To view results in the HyperLynx Eye Density/BER Viewer, double-click *ResAMI_data_aggregated.sds* in the *<Results>/RCV_Waveforms* folder.

Column Name	Definition
Net Name	Name of the measured net.
Pass/Fail	Measurement status for the net.
Operation	Read or Write, with rank definition as rank(slot number, rank within the slot).
Driver Comp Ref Des & Pin Name	Driver pin identification
Receiver Comp Ref Des & Pin Name	Receiver pin identification
Associated Strobe Net Name	Name of the strobe net associated with the measured net.
Eye-Mask Width, TcIVW_Total, [ps]	TcIVW(%) * UI, values from the DRAM timing model.
Eye-Mask Width1, TcIVW1, [ps]	TcIVW1(%) * UI, values from the DRAM timing model. Represents the maximum or total width of a hexagonal eye mask.
Eye-Mask Width2, TcIVW2, [ps]	TcIVW2(%) * UI, values from the DRAM timing model. Represents the width of a hexagonal eye mask, within the rectangular region (for example, between the top-left and top-right corners of an eye mask).
Eye-Mask Width, TdIVW_Total, [ps]	Write: TdIVW(%) * UI, values from the DRAM timing model. Read: EyeMaskWidthLimit (%) * UI, values from the controller timing model.
Eye-Mask Width1, TdIVW1, [ps]	TdIVW1(%) * UI, values from the DRAM timing model. Represents the maximum or total width of a hexagonal eye mask.
Eye-Mask Width2, TdIVW2, [ps]	TdIVW2(%) * UI, values from the DRAM timing model. Represents the width of a hexagonal eye mask, within the rectangular region (for example, between the top-left and top-right corners of an eye mask).
Eye-Mask Width, RxEW, [ps]	RxEW(%) * UI, values from DRAM timing model. Represents the width of a kite-shaped eye mask.

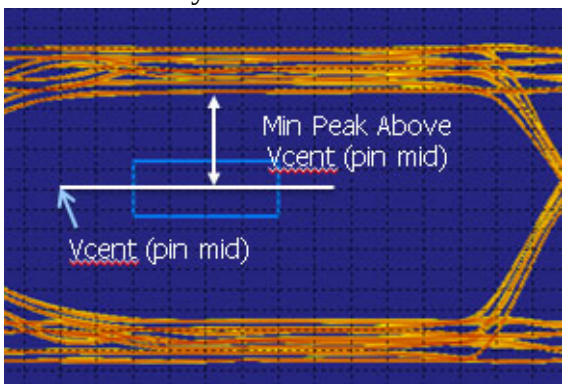
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Eye-Mask Rx PeakOffset, [ps]	RxPeakOffset(%) * UI, values from DRAM timing model. Represents the horizontal offset of the maximum height of a kite-shaped eye mask, from the start of the UI.
Eye-Mask Height, VdIVW_Total, [mV]	Write: VdIVW (Volt), values from the DRAM timing model. Read: EyeMaskHeightLimit (Volt), values from the controller timing model.
Voltage at Widest Eye Opening, Vcent, [mV]	The voltage measured where the eye is the widest for this particular pin.
Average Center Voltage, Vcent(pin_mid), [mV]	Voltage used as the reference voltage (Vref) of the receiving device for a given measurement. Write: Average center voltage, Vcent(pin_mid) = (Min Vcent for Ux + Max Vcent for Ux)/2 Read: The same equation is used for read and write operations. The value(s) depends on the options you chose in the Vref Training page of the DDRx Wizard. • With the option “Enable separate Vref per rank” unchecked: For multi-rank, Vref is calculated using one min and one max value per DRAM group. Single Vref per lane - Vref value is calculated using the min and max value for each lane, regardless of rank. Single Vref for all the signals - Vref value is calculated using the min and max values across all signals. Therefore, all signals from any rank have the same Vref value. • With the option “Enable separate Vref per rank” checked: Single Vref per lane - Each rank (specified in the Operation column) in each lane has a different Vcent(pin_mid) value. For example, if you are simulating an interface with two lanes, each with two ranks, four Vcent(pin_mid) values are calculated. Single Vref for all signals - Vcent(pin_mid) value is calculated for each rank using all the signals from that rank, per rank using the value in the “Voltage at Widest Eye Opening, Vcent, [mV]” column. For additional information on how Vcent is calculated, see Vcent(pin_mid) Calculation Examples.
Output Delay Uncertainty, [ps]	Write: [tDQSDQ(max) – tDQSDQ(min)]/2, values from the controller timing model. Read: [tDQSQ(max) – tDQSQ(min)]/2, values from the DRAM timing model.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Min Setup Time Measured, [ps]	DQ rising edge: Measured from zero volt DQS crossing to (Vcent(pin_mid) + VdIVW/2) DQ falling edge: Measured from zero volt DQS crossing to (Vcent(pin_mid) – VdIVW/2)
Min Setup Time Margin [ps]	Smallest value of all per bit setup time margins. The margin is measured from left side of the eye mask boundary to minimum value of eye diagram.
Min Setup Time Margin with Output Delay Uncertainty, [ps]	Min Setup Time Margin - Output Delay Uncertainty
Min Setup Time Margin Trace Point, [ns]	The time at the zero volt crossing of associated DQS where the minimum setup time was measured on the per bit waveform.
Min Hold Time Measured, [ps]	Rising edge of DQ signal: Measured from DQS 0V crossing to (Vcent(pin_mid) – VdIVW/2) Falling edge of DQ signal: Measured from DQS 0V crossing to (Vcent(pin_mid) + VdIVW/2)
Min Hold Time Margin [ps]	Smallest value of all per bit hold time margins. The margin is measurement from right side of the eye mask boundary to nearest value of eye diagram.
Min Hold Time Margin with Output Delay Uncertainty, [ps]	Min Hold Time Margin - Output Delay Uncertainty
Min Hold Time Margin Trace Point, [ns]	The time at the zero volt crossing of associated DQS where the minimum hold time was measured on the per bit waveform.
VIHL_AC/2 Limit, [mV]	Write: VIHL_AC/2 Limit = VIHL_AC_DQ/2, values from the DRAM timing model. Read: VIHL_AC/2 Limit = MaxEyeHeightLimit/2, values from the controller timing model.
Min Peak Above Vcent(pin mid) [mV]	Minimum voltage above the Vcent value across all cycles.
Min Peak Above Vcent(pin mid) Margin, [mV]	Min Peak Above Vcent(pin mid) – VIHL_AC/2 Limit
Min Peak Above Vcent(pin mid) Trace Point, [ns]	The time at the zero volt crossing of associated DQS where the minimum peak above Vcent was measured.
Min Peak Below Vcent(pin mid) [mV]	Minimum voltage below the Vcent value across all cycles.
Min Peak Below Vcent(pin mid) Margin, [mV]	Min Peak Below Vcent(pin mid) – VIHL_AC/2 Limit

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Min Peak Below Vcent(pin mid) Trace Point, [ns]	The time where DQS crosses zero volts and the minimum peak value of the eye diagram that is within the eye mask boundary.
Min Voltage Above Eye-Mask [mV]	Distance from the top of eye mask to the smallest eye diagram value located above the eye mask. 
Min Voltage Above Eye-Mask Trace Point, [ns]	The time when DQS crosses zero volts where the minimum voltage above the eye mask was measured.
Min Voltage Below Eye-Mask [mV]	Distance from the bottom of eye mask to the minimum eye diagram voltage directly below the eye mask.
Min Voltage Below Eye-Mask Trace Point, [ns]	The time where the minimum voltage below the eye-mask that is within the eye mask boundary and at the point where DQS crosses zero volts.
Min Pulse Width [ps]	Minimum measured pulse width at the trace point from Vcent(pin_mid).
Min Pulse Width Limit, TdIPW, [ps]	Write: TdIPW % * PW, values from the DRAM timing model. Read: VrefToVrefLimit% * PW, values from the controller timing model.
Min Pulse Width Margin, [ps]	Min Pulse Width – Pulse Width Limit
Min Pulse Width Margin with Output Delay Uncertainty, [ps]	Pulse Width Margin – OutputUncertaintyForPulseWidth **Note:** The controller timing file defines the OutputUncertaintyForPulseWidth variable.
Min Pulse Width Trace Point, [ns]	The time at the zero volt crossing of DQS where the minimum pulse width occurred.
tDIHL [ps]	Measured minimum pulse width, from the falling edge crossing of VIH(AC) to the rising edge crossing VIH(AC).
tDIHL Limit, TdIHL, [ps]	tDIHL(%) * UI, values from the DRAM timing model.
tDIHL Margin, [ps]	tDIHL - tDIHL Limit.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
tDIHL Margin with Output Delay Uncertainty, [ps]	tDIHL Margin - <variable>, where <variable> is: • tDQSQDDR4 — For a controller with per-bit DQ deskew capability. • tDQSQ or tQH — For a controller without per-bit DQ deskew capability.
tDIHL Trace Point, [ns]	The time of a strobe crossing Vref for a high/low/high pulse and its minimum width.
tDIHL Signal Trace Point Left, [ns]	For a high/low/high pulse and its minimum width, the time when a falling edge crosses VIH(AC).
tDIHL Signal Trace Point Right, [ns]	For a high/low/high pulse and its minimum width, the time when a rising edge crosses VIH(AC).
Min Slew Rate [V/ns]	This measurement depends on the interface you use. For information, see: • "Note About Slew Rate - Non-DDR4" on page 310 • "Note About Slew Rate for DDR4" on page 309
Min Slew Rate Limit, SRIN_dIVW_min, [V/ns]	Write: SRIN_dIVW_min, value from the DRAM timing model. Read: MinSlewRateLimit, value from the controller timing model.
Min Slew Rate Margin, [V/ns]	Min Slew Rate – Min Slew Rate Limit, SRIN_VdIVW_min
Min Slew Rate Trace Point, [ns]	The time where the minimum slew rate is measured, which is the point where DQS crosses zero volts.
Max Slew Rate [V/ns]	This measurement depends on the interface you use. For information, see: • "Note About Slew Rate - Non-DDR4" on page 310 • "Note About Slew Rate for DDR4" on page 309
Max Slew Rate Limit, SRIN_dIVW_max, [V/ns]	Write: SRIN_dIVW_max, value from the DRAM timing model. Read: MaxSlewRateLimit, value from the controller timing model.
Max Slew Rate Margin, [V/ns]	Max Slew Rate – Max Slew Rate Limit, SRIN_dIVW_max
Max Slew Rate Trace Point, [V/ns]	The time where the maximum slew rate measured from the point where DQS crosses zero volts.
Min Slew Rate (2), [V/ns] (DDR4)	See "Note About Slew Rate for DDR4" on page 309.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Min Slew Rate (2) Limit, SRIN_VdIVW_min, [V/ns] (DDR4)	Write: SRIN_dIVW_min, value from the DRAM timing model. Read: MinSlewRateLimit, value from the controller timing model.
Min Slew Rate (2) Margin, [V/ns] (DDR4)	Min Slew Rate – Min Slew Rate Limit, SRIN_VdIVW_min
Max Slew Rate (2), [V/ns] (DDR4)	See “Note About Slew Rate for DDR4” on page 309.
Max Slew Rate (2) Limit, SRIN_VdIVW_max, [V/ns] (DDR4)	Write: SRIN_dIVW_max, value from the DRAM timing model. Read: MaxSlewRateLimit, value from the controller timing model.
Max Slew Rate (2) Margin, [V/ns] (DDR4)	Equals Max Slew Rate - Max Slew Rate Limit
Eye Mask Shape	One of the following: Eye-Rectangle, Eye-Kite, Eye-Hexagon.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Clock Jitter Spreadsheets

These spreadsheets report clock jitter for all nets in the DDRx interface.

File names take the form:

*DDR_report_clock_jitter*_[Typ | Fast | Slow].*csv*

where:

Typ, Fast, Slow represent IC model corners

Spreadsheet Columns

Column Name	Definition
Clock/Strobe Net Name	Name of the measured clock or strobe net. For a multiple-board project, the net name does not include the board ID.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Clock/Strobe Driver Comp Ref Des & Pin Name	Clock or strobe driver component reference designator and pin name. For a multiple-board project, the net name contains the board ID. Use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Clock/Strobe Receiver Comp Ref Des & Pin Name	Clock or strobe receiver component reference designator and pin name. For MultiBoard projects, the net name contains the board ID. Use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Min tCK(avg), [ps]	Minimum average clock period, measured across any consecutive 200-cycle window.
Min tCK(avg) Trace Point, [ns]	Start time of a rolling window of 200 periods in a tCK waveform. In this window, the average tCK is minimum compared to other 200-cycle windows.
Max tCK(avg), [ps]	Maximum average clock period, measured across any consecutive 200-cycle window.
Max tCK(avg) Trace Point, [ns]	Start time of a rolling window of 200 periods in a tCK waveform. In this window, average tCK is maximum compared to other 200-cycle windows.
Min tCK(abs), [ps]	Minimum absolute clock period, as measured from one rising edge to the next consecutive rising edge.
Min tCK(abs) Trace Point, [ns]	Start time of a period in a waveform, where tCK is minimum comparing to other periods.
Max tCK(abs), [ps]	Maximum absolute clock period, as measured from one rising edge to the next consecutive rising edge.
Max tCK(abs) Trace Point, [ns]	Start time of a period in a waveform, where tCK is maximum comparing to other periods.
Min tCH(avg), [ps]	Minimum average clock high pulse width, measured across any consecutive 200-cycle window.
Min tCH(avg) Trace Point, [ns]	Start time of a rolling window of 200 periods in a tCH waveform. In this window, the average tCH is minimum compared to other 200-cycle windows.
Max tCH(avg), [ps]	Maximum average clock high pulse width, measured across any consecutive 200-cycle window.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Max tCH(avg) Trace Point, [ns]	Start time of a rolling window of 200 periods in a tCH waveform. In this window, the average tCH is maximum compared to other 200-cycle windows.
Min tCH(abs)/tDQSH, [ps]	Minimum absolute clock high pulse width or the minimum DQS high pulse width.
Min tCH(abs)/tDQSH Trace Point, [ns]	Start time of a pulse in a waveform, where either tCH or tDQSH is minimum compared to other pulses.
Max tCH(abs)/tDQSH, [ps]	Maximum absolute clock high pulse width or the maximum DQS high pulse width.
Max tCH(abs)/tDQSH Trace Point, [ns]	Start time of a pulse in a waveform, where either tCH or tDQSH is maximum compared to other pulses.
Min tCL(avg), [ps]	Minimum average clock low pulse width.
Min tCL(avg) Trace Point, [ns]	Start time of a rolling window of 200 periods in a tCL waveform. In this window, the average tCL is minimum compared to other 200-cycle windows.
Max tCL(avg), [ps]	Maximum average clock low pulse width.
Max tCL(avg) Trace Point, [ns]	Start time of a rolling window of 200 periods in a tCL waveform. In this window, the average tCL is maximum compared to other 200-cycle windows.
Min tCL(abs)/tDQSL, [ps]	Minimum absolute clock low pulse width or the minimum DQS low pulse width.
Min tCL(abs)/tDQSL Trace Point, [ns]	Start time of a pulse in a waveform, where either tCL or tDQSL is minimum compared to other pulses.
Max tCL(abs)/tDQSL, [ps]	Maximum absolute clock low pulse width or the maximum DQS low pulse width.
Max tCL(abs)/tDQSL Trace Point, [ns]	Start time of a pulse in a waveform, where either tCL or tDQSL is maximum compared to other pulses.
Min tJIT(per), [ps]	Minimum single period jitter, defined as the largest deviation of any signal tCK from tCK(avg).
Min tJIT(per) Trace Point, [ns]	Start time of a period in a waveform where tCK - tCK(avg) is minimum.
Max tJIT(per), [ps]	Maximum single period jitter, defined as the largest deviation of any signal tCK from tCK(avg).
Max tJIT(per) Trace Point, [ns]	Start time of a period in a waveform where tCK - tCK(avg) is maximum.
Max tJIT(cc), [ps]	Maximum clock jitter between two consecutive cycles.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Max tJIT(cc) Trace Point, [ns]	Start time of a period in a waveform where (tCK – tCK prev) is maximum.
Min tJIT(duty), [ps]	Minimum deviation of the tCH pulse from tCH(avg) and the tCL pulse from tCL(avg).
Min tJIT(duty) Trace Point, [ns]	Start time of a pulse in a waveform where the pulse size is minimum.
Max tJIT(duty), [ps]	Maximum deviation of the tCH pulse from tCH(avg) and the tCL pulse from tCL(avg).
Max tJIT(duty) Trace Point, [ns]	Start time of a pulse in a waveform where the pulse size is maximum.
PCB Jitter Abs High Pulse, (Min tCH(abs) - tCK/2), [ps]	Minimum absolute high pulse width - tCK/2
PCB Jitter Abs Low Pulse, (tCK/2 - Min tCL(abs)), [ps]	tCK/2 - minimum absolute low pulse width
CHIP Jitter Abs Positive, [ps]	The value of tCKPW_Skew_abs_p, from the timing model file.
CHIP Jitter Abs Negative, [ps]	The value of tCKPW_Skew_abs_n, from the timing model file.
Total Jitter High Pulse Abs, (PCB Jitter High + CHIP Jitter Abs Negative [ps]	PCB jitter high + CHIP jitter absolute negative
Total Jitter Low Pulse Abs, (PCB Jitter Low + CHIP Jitter Abs Positive [ps]	PCB jitter low + CHIP jitter absolute positive
Total Jitter Abs Limit, [ps]	The value of tCKPW_Limit_abs, from the timing model file.
Total Jitter High Pulse Abs Margin, [ps]	tCK/2 + Total High Pulse Jitter - Total Jitter Abs Limit
Total Jitter Low Pulse Abs Margin, [ps]	tCK/2 - Total Low Pulse Jitter - Total Jitter Abs Limit
PCB Jitter Avg High Pulse, (tCH(avg) - tCK/2), [ps]	[(High Pulse (1) + High Pulse(2) + High Pulse(3) … + High Pulse(n)) / n] - tCK/2
PCB Jitter Avg Low Pulse, (tCK/2 - tCL(avg)), [ps]	(Low Pulse (1) + Low Pulse(2) + Low Pulse(3) … + Low Pulse(n)) / n
CHIP Jitter Avg Positive, [ps]	The value of tCKPW_Skew_avg_p, from the timing model file.

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Column Name	Definition
CHIP Jitter Avg Negative, [ps]	The value of tCKPW_Skew_avg_n, from the timing model file.
Total Jitter High Pulse Avg, (PCB Jitter High + CHIP Jitter Avg Negative [ps]	PCB jitter high + CHIP jitter average negative
Total Jitter Low Pulse Avg, (PCB Jitter Low + CHIP Jitter Avg Positive [ps]	PCB jitter low + CHIP jitter average positive
Total Jitter Avg Limit, [ps]	The value of tCKPW_Limit_avg, from timing model file.
Total Jitter High Pulse Avg Margin, [ps]	tCK/2 + Total Jitter High Pulse Avg - Total Jitter Avg Limit
Total Jitter Low Pulse Avg Margin, [ps]	tCK/2 - Total Jitter Low Pulse Avg - Total Jitter Avg Limit
Total Jitter, [Pass/Fail]	“Pass” when all margins are positive.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Clock Jitter Error Spreadsheets

These spreadsheets report clock jitter error for all nets in the DDRx interface.

File names take the form:

*DDR_report_clock_jitter_err_*[Typ | Fast | Slow].*csv*

where:

Typ, Fast, Slow represent IC model corners

Spreadsheet Columns

Column Name	Definition
Clock Net Name	Name of the measured clock net. For a multiple-board project, the net name does not include the board ID.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Clock Driver Comp Ref Des & Pin Name	Driver component reference designator and pin name for the measured net. For a multiple-board project, the net name contains the board ID. Use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Clock Receiver Comp Ref Des & Pin Name	Receiver component reference designator and pin name for the measured net. For a multiple-board project, the net name contains the board ID. Use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Cumulative Error Across n Cycles	The number of cycles (n). The software determines the required number of cycles (between 2 and 50) based on the variation of clock jitter measurements.
tERR, [ps]	Cumulative error for n cycles. tERR(n) = (tCK(1) + tCK(2) … + tCK(n)) - tCK * n
tERR Limit, [ps]	Maximum allowable tERR time.
tERR Margin, [ps]	Time difference from the tERR Limit, when the worst violation occurs (tERR - tERR Limit).

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Data Spreadsheets

These spreadsheets report setup and hold measurements for data and data mask signals in the DDRx interface. Data nets are measured relative to strobes.

Report file name takes the form:

*DDR_report_data*_[violations | worstcases | allcases]_[Typ | Fast | Slow].*xls*

where:

violations contains only nets that violate timing-model requirements.

worst cases contains all nets and their worst-case measurements.

allcases contains all nets and all measurements (for each cycle in the simulation).

Typ, Fast, Slow represent IC model corners.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Tip

The data spreadsheets contain two rows for each signal. The first row contains setup measurements, while the second row contains hold measurements.

Table 5-1. DDRx Data Spreadsheet Column Definitions

Column Name	Definition
Net Name	Name of the measured net. For a multiple-board project, the net name does not include the board ID.
Pass/Fail	Fails are based on timing model limits. If either the setup margin in (O) or the hold margin in (U) is less than 0ps, this column displays “Fail,” is shaded red, and the negative margin is shaded red in its respective cell. A Fail can also occur when a measurement cannot be performed for an edge. In this case, all numerical cells are empty and the Comments column contains a description of the issue.
Operation	Read or Write, with rank definition as rank(slot number, rank within the slot).
Driver Comp Ref Des & Pin Name	Driver component reference designator and pin name for the measured net. For a multiple-board project, the net name contains the board ID. Use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Receiver Comp Ref Des & Pin Name	Receiver component reference designator and pin name for the measured net. For a multiple-board project, the net name contains the board ID. Use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Associated Clk/Strobe Net Name	Name of the strobe net paired with the measured net.
Associated Clk/Strobe Driver Comp Ref Des & Pin Name	Driver component reference designator and pin name for the strobe net paired with the measured net.
Associated Clk/Strobe Receiver Comp Ref Des & Pin Name	Receiver component designator and pin name for the strobe net paired with the measured net.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 5-1. DDRx Data Spreadsheet Column Definitions (cont.)

Column Name	Definition
Clk/Strobe Crossing Threshold Time, [ns]	Time in the simulation when the strobe waveforms cross zero (differential) or V_{REF} (single-ended). Setup/hold times are measured before/after this time. Investigate measurement results by using this time when viewing waveform files (if you enable saving waveform files).
Setup Time (From Sim), tDS, [ps]	The measured data and strobe setup time, prior to derating. Measurements start when the data waveform either: • First crosses the AC threshold after the last crossing of the DC threshold. • Last crosses the AC threshold (in the case where the model does not provide a DC threshold or the DC and AC thresholds are the same). Measurements end when the corresponding strobe waveform crosses zero (differential) or V_{REF} (single-ended). The measured setup time is based on the average driver delay. See DDRx Delay File Measurements.
Min Setup Time, tDS(min) = tDS - Initial Delay Delta, [ps]	Setup time obtained by shifting data waveforms to the right, within the specified range of initial delays. This represents the worst setup time possible when varying the initial delay. The calculation is (Setup Time (From Sim), tIS, [ps]) minus (Initial delay delta), where: Initial delay delta—([(Maximum initial delay for DQS and CK) minus (Minimum initial delay for DQS and CK)]/2). See “Stimulus Offset” in the table in DDRx Address Spreadsheets.
Base Setup Time, tDS(base), [ps]	The setup limit whose value is defined in the timing model.
Setup Derate Time Delta d(tDS), [ps]	The setup derating value obtained from derating tables built into DDRx batch simulation or contained in the timing model. Values can be positive or negative. The log file contains derating calculation details, such as slew rates for data and strobe nets. See DDRx Log File. For instructions on how derating values are determined, see JEDEC specifications JESD29-2* or JES72-3* at www.jedec.org.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 5-1. DDRx Data Spreadsheet Column Definitions (cont.)

Column Name	Definition
Required Setup Time, tDS(req) = tDS(base) + d(tDS), [ps]	The adjusted setup limit, where the value from the timing model is adjusted by the setup derating value. The calculation is (Base Setup Time, tIS(base), [ps]) plus (Setup Derate Time Delta d(tIS), [ps]).
Setup Time Margin, tDS(margin) = tDS(min) - tDS(req), [ps]	The margin (or slack) remaining after subtracting the slew-rate-adjusted setup limit from the measured setup time. Negative values produce errors. The calculation is (Min Setup Time, tIS(min)) minus (Required Setup Time, tIS(req)), where: tIS(min) = (tIS - Initial Delay Delta, [ps]; tIS(req) = tIS(base) + d(tIS), [ps]. **Tip:** Spreadsheet cells with a yellow background indicate a minimum setup time margin.
Hold Time (From Sim), tDH, [ps]	The measured data hold time, prior to derating. Measurements start when the strobe waveform crosses zero (differential) or V_{REF} (single-ended). Measurements start when the data waveform either: • First crosses the AC threshold after the last crossing of the DC threshold, or • Last crosses the AC threshold (in the case where the model does not provide a DC threshold or the DC and AC thresholds are the same). The measured hold time is based on the average driver delay. See DDRx Delay File Measurements.
Min Hold Time, tDH(min) = tDH - Initial Delay Delta, [ps]	Hold time obtained by shifting data waveforms to the left, within the specified range of initial delays. This represents the worst hold time that can be obtained by varying the initial delay. The calculation is (Hold Time (From Sim), tIH, [ps]) minus (Initial delay delta), where: Initial delay delta = ([(Maximum initial delay for DQS and CK) minus (Minimum initial delay for DQS and CK)]/2).
Base Hold Time, tDH(base), [ps]	The hold limit whose value is defined in the timing model.
Hold Derate Time Delta d(tDH), [ps]	The hold derating value obtained from derating tables built into DDRx batch simulation. Values can be positive or negative. The log file contains derating calculation details, such as slew rates for data and strobe nets. See DDRx Log File.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 5-1. DDRx Data Spreadsheet Column Definitions (cont.)

Column Name	Definition
Required Hold Time, tDH(req) = tDH(base) + d(tDH), [ps]	The adjusted hold limit, where the value from the timing model is adjusted by the hold derating value. The calculation is (Base Hold Time, tIH(base), [ps]) plus (Hold Derate Time Delta d(tIH), [ps]).
Hold Time Margin, tDH(margin) = tDH(min) - tDH(req), [ps]	The margin (or slack) remaining after subtracting the slew-rate-adjusted hold limit from the measured hold time. Negative values produce errors. The calculation is Min Hold Time, tIH(min) minus Required Hold Time, tIH(req), where tIH(min) = tIH - Initial Delay Delta, [ps]); tIH(req) = tIH(base) + d(tIH), [ps]). **Tip:** Spreadsheet cells with a yellow background indicate a minimum hold time margin.
Pulse Width, [ps]	The pulse width at the trace point from Vdd/2.
Pulse Width Limit, [ps]	Write: TdIPW % * PW, values from the DRAM timing model. Read: VrefToVrefLimit% * PW, values from the controller timing model.
Pulse Width Margin, [ps]	Pulse Width – Pulse Width Limit

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 5-1. DDRx Data Spreadsheet Column Definitions (cont.)

Column Name	Definition
Pulse Width Margin with Output Delay Uncertainty, [ps]	Case #1: The controller timing file completely specifies pulse width uncertainty. For DQ/DM nets: Pulse Width Margin – OutputUncertaintyForPulseWidth For Address, Control, and Command nets: Pulse Width Margin – OutputUncertaintyForPulseWidth_CA **Note:** OutputUncertaintyForPulseWidth and OutputUncertaintyForPulseWidth_CA are variables defined in the controller timing file.
	Case #2: The controller timing file specifies only the extra part of pulse width uncertainty. For DQ/DM nets: Pulse Width Margin - 2 * (Initial Delay Delta + ExtraOutputUncertaintyForPulseWidth) For Address, Control, and Command nets: Pulse Width Margin - 2 * (Initial Delay Delta + ExtraOutputUncertaintyForPulseWidth) **Note:** ExtraOutputUncertaintyForPulseWidth and ExtraOutputUncertaintyForPulseWidth_CA are variables defined in the controller timing file.
	Case #3: The controller timing file defines no special variables. Pulse Width Margin - 2 * Initial Delay Delta
Stimulus Offset, [ns]	The timing offset between drivers on the data and strobe nets, when the initial strobe delay is average. See DDRx Delay File Measurements.
% of Cycles With Failures	Percentage of waveform measurements with either of the following properties: • The measured value fails the limit value • The measurement itself failed for some reason **Restriction:** Only the violations spreadsheets contain this column.
Comments	Displays error or other measurement problem information.

Related Topics

Running a DDRx Memory Interface Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDRx JEDEC Measurements Spreadsheets

These spreadsheets report signal-integrity measurements for all nets in the DDRx interface.

File names take the form:

*DDR_report_Jedec_measurements*_[Typ | Fast | Slow].*csv*

where:

Typ, Fast, Slow represent IC model corners

Table 5-2. DDRx JEDEC Measurements Spreadsheet Column Definitions

Column Name	Definition
Net Name	Name of the measured net. For a multiple-board project, the net name does not include the board ID.
Pass/Fail	"Fail" if any measurements fail, based on IBIS model requirements or requirements built into DDRx batch simulation.
Operation	Read or Write, with rank definition as rank(slot number, rank within the slot).
Driver Comp Ref Des & Pin Name	Driver component reference designator and pin name for the measured net. For a multiple-board project, the net name contains the board ID. You can use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Receiver Comp Ref Des & Pin Name	Receiver component reference designator and pin name for the measured net. For a multiple-board project, the net name contains the board ID. You can use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Rise Rail Overshoot, [mV]	Maximum voltage above the power rail, for a rising-edge transition at the receiver.
Rise Rail Overshoot Limit, [mV]	Voltage limit above the power rail, for a rising-edge transition at the receiver. Change the value for this limit on the DDRx Batch-Mode Wizard - Quality Checks Page.
Rise Rail Overshoot Margin, [mV]	Voltage difference between the signal and the Rise Rail Overshoot Limit at the time of maximum undershoot.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 5-2. DDRx JEDEC Measurements Spreadsheet Column Definitions

Column Name	Definition
Rise Rail Overshoot, [Pass/Fail]	"Fail" if the waveform at the receiver passes more than a threshold beyond the power rail. You can edit the threshold on the DDRx Batch-Mode Wizard - Simulation Options Page.
Rise Rail Overshoot Trace Point, [ns]	The time when the signal reaches the maximum overshoot voltage level.
Fall Rail Undershoot, [mV]	Minimum voltage below the ground rail, for a falling-edge transition at the receiver.
Fall Rail Undershoot Limit, [mV]	Voltage limit below the power rail, for a falling-edge transition at the receiver. You can change the value for this limit on the DDRx Batch-Mode Wizard - Quality Checks Page.
Fall Rail Undershoot Margin, [mV]	Voltage difference between the signal and the Fall Rail Undershoot Limit at the time of the maximum undershoot.
Fall Rail Undershoot, [Pass/Fail]	"Fail" if the waveform at the receiver passes more than a threshold beyond the ground rail. You can edit the threshold on the DDRx Batch-Mode Wizard - Simulation Options Page.
Fall Rail Undershoot Trace Point, [ns]	The time when the signal reaches the peak undershoot voltage level.
Rise Rail Overshoot Area, [V*ns]	(Non-DDR4) Area under the signal above VDD, measured at the time of the maximum overshoot. (DDR4) Area under the signal between VDD and VDOS (VCOS for a clock), measured at the time of the maximum overshoot.
Rise Rail Overshoot Area Limit, [V*ns]	Maximum Rise Rail Overshoot Area. You can change the value for this limit on the DDRx Batch-Mode Wizard - Quality Checks Page.
Rise Rail Overshoot Area Margin, [V*ns]	Difference between the Rise Rail Overshoot Area, and the Rise Rail Overshoot Area Limit, at the time of the maximum overshoot.
Rise Rail Overshoot Area, [Pass/Fail]	"Fail" if the Rise Rail Overshoot Area exceeds the Rise Rail Overshoot Area Limit.
Rise Rail Overshoot Area Trace Point, [ns]	The time when the signal reaches the maximum overshoot area.
Fall Rail Undershoot Area, [V*ns]	(Non-DDR4) Area above the signal below VSS, measured at the time of the maximum undershoot. (DDR4) Area under the signal between VCC and VDUS (VCUS for a clock), measured at the time of the maximum undershoot.

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Table 5-2. DDRx JEDEC Measurements Spreadsheet Column Definitions

Column Name	Definition
Fall Rail Undershoot Area Limit, [V*ns]	Maximum Fall Rail Undershoot Area. You can change the value for this limit on the DDRx Batch-Mode Wizard - Quality Checks Page.
Fall Rail Undershoot Area Margin, [V*ns]	Difference between the Fall Rail Undershoot Area, and the Fall Rail Undershoot Area Limit, at the time of the maximum undershoot.
Fall Rail Undershoot Area, [Pass/Fail]	"Fail" if the Fall Rail Undershoot Area exceeds the Rise Rail Undershoot Area Limit.
Fall Rail Undershoot Area Trace Point, [ns]	The time when the signal reaches the maximum undershoot area.
Rise Rail Overshoot Area (2), [V*ns] (DDR4)	Area under the signal between VDOS (VCOS for a clock) and VDOSP (VCOSP for a clock), measured at the time of the maximum undershoot.
Rise Rail Overshoot Area (2) Limit, [V*ns] (DDR4)	Maximum Rise Rail Overshoot Area (2).
Rise Rail Overshoot Area (2) Margin, [V*ns] (DDR4)	Difference between the Rise Rail Overshoot Area (2), and the Rise Rail Overshoot Area (2) Limit, at the time of the maximum overshoot.
Rise Rail Overshoot Area (2), [Pass/Fail] (DDR4)	"Fail" if the Rise Rail Overshoot Area (2) exceeds the Rise Rail Overshoot Area (2) Limit.
Rise Rail Overshoot Area (2) Trace Point, [ns] (DDR4)	The time when the signal reaches the maximum overshoot area.
Fall Rail Undershoot Area (2), [V*ns] (DDR4)	Area above the signal between VDUS (VCUS for a clock) and VDUSP (VCUSP for a clock), measured at the time of the maximum undershoot.
Fall Rail Undershoot Area (2) Limit, [V*ns] (DDR4)	Maximum Fall Rail Undershoot Area (2).
Fall Rail Undershoot Area (2) Margin, [V*ns] (DDR4)	Difference between the Fall Rail Undershoot Area (2), and the Fall Rail Undershoot Area (2) Limit, at the time of the maximum undershoot.

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Table 5-2. DDRx JEDEC Measurements Spreadsheet Column Definitions

Column Name	Definition
Fall Rail Undershoot Area (2), [Pass/Fail] (DDR4)	"Fail" if the Fall Rail Undershoot Area (2) exceeds the Fall Rail Undershoot Area (2) Limit.
Fall Rail Undershoot Area (2) Trace Point, [ns] (DDR4)	The time when the signal reaches the maximum undershoot area.
Monotonic [Pass/Fail]	"Fail" if the waveform at the receiver reverses direction while between VIL_{ac} and VIH_{ac}. The monotonicity threshold is set on the DDRx Batch-Mode Wizard - Quality Checks Page.
Monotonic Trace Point [ns]	The time of the worst monotonic violation.
VIH/L(AC) Min Limit, [Pass/Fail]	"Fail" if VIH/L_{ac} is not crossed at least once in each cycle. "N/A" for differential nets.
VIH/L(AC) Min Limit, Trace Point, [ns]	The time when the signal is farthest from crossing VIH/L_{ac}. "N/A" when no violations occur.
VIH/L(DC) Monotonicity, [Pass/Fail]	"Fail" when VIH/L_{dc} is crossed more than once after exceeding $VIH/L_{ac.}$ "N/A" for differential nets. The monotonicity threshold is set on the Quality Checks page of the DDRx Batch-Mode wizard.
VIH/L(DC) Monotonicity Trace Point, [ns]	The time of the worst violation. "N/A" when no violations occur.
Vref Threshold Multi Cross [Pass/Fail]	"Fail" if the rising or falling waveform at the receiver passes through V_{REF} more than once.
Max Slew Time, [ps]	The maximum time the waveform at the receiver takes to pass between VIL_{ac} and VIH_{ac}. "N/A" if the IBIS model does not specify [Receiver Thresholds] with sub-parameters Tslew_ac for single-ended signals and Tdiffslew_ac for differential signals.
Max Slew Time [Pass/Fail]	"Fail" if the waveform at the receiver has excessive slew time. "N/A" if the IBIS model does not specify [Receiver Thresholds] keywords with sub-parameters Tslew_ac for single-ended signals and Tdiffslew_ac for differential signals.

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Table 5-2. DDRx JEDEC Measurements Spreadsheet Column Definitions

Column Name	Definition
VIX Limit Low, mV	Minimum acceptable voltage level for differential signal crossing: VDD/2 - Max VIX where: • VDD is the typical, maximum, or minimum value used during simulation. • Max VIX from the DDRx Batch-Mode Wizard - Quality Checks Page You can change the value for this limit on the DDRx Batch-Mode Wizard - Quality Checks Page.
VIX Limit Hi, mV	Maximum acceptable voltage level for differential signal crossing: VDD/2 + Max VIX where: • VDD is the typical, maximum, or minimum value used during simulation. • Max VIX comes from the DDRx Batch-Mode Wizard - Quality Checks Page. You can change the value for this limit on the DDRx Batch-Mode Wizard - Quality Checks Page.
VIX Margin, mV	Minimum of the following: • Differential signal crossover measurement – VIX Limit Low • Differential signal crossover measurement – VIX Limit High
VIX, [Pass/Fail]	“Fail” if the differential signals cross above VIX Limit Hi, or below VIX Limit Low.
VIX Trace Point, [ns]	Time when the worst violation occurs.
Vrx/2 Limit, mV	Minimum acceptable peak differential voltage ((positive signal voltage) - (negative signal voltage)) for CK and DQS.
Vrx/2 Margin, mV	Measured peak differential voltage - Vrx/2 Limit.
Vrx/2, [Pass/Fail]	“Fail” if the measured peak differential voltage is less than Vrx/2 Limit.
Vrx/2 Trace Point, [ns]	Time when the worst violation occurs.
VID(AC) Limit, mV	Differential voltage level that the signal must exceed at least once in each cycle. This value is VIHdiff/VILdiff(AC) for DDR3. You can change the value for this limit on the DDRx Batch-Mode Wizard - Quality Checks Page.
VID(AC) Margin, mV	Voltage difference from the VID(AC) Limit, when the worst violation occurs.

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Table 5-2. DDRx JEDEC Measurements Spreadsheet Column Definitions

Column Name	Definition
VID(AC), [Pass/Fail]	“Fail” if the differential signal does not cross the VID(AC) Limit at least once in each cycle.
VID(AC) Trace Point, [ns]	Time when the worst violation occurs.
VID(DC) Limit, mV	Differential voltage level that the signal must exceed and remain beyond for each entire cycle. This value is VIHdiff/VILdiff(DC) for DDR3. You can change the value for this limit on the DDRx Batch-Mode Wizard - Quality Checks Page.
VID(DC) Margin, mV	Voltage difference from the VID(DC) Limit, when the worst violation occurs.
VID(DC), [Pass/Fail]	“Fail” if the differential signal does not cross the VID(DC) Limit at least once in each cycle.
VID(DC) Trace Point, [ns]	Time when the worst violation occurs.
Vdd/2, mV (non-LPDDRx)	Voltage used as the reference voltage (Vref) of the receiving device for a given measurement.

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Table 5-2. DDRx JEDEC Measurements Spreadsheet Column Definitions

Column Name	Definition
Vref, mV (LPDDR4, LPDDR5)	Voltage used as the reference voltage (Vref) of the receiving device for a given measurement. Write: Average center voltage, Vcent(pin_mid) = (Min Vcent + Max Vcent) / 2 Read: The same equation is used for read and write operations. The value(s) depends on the options you chose in the Vref Training page of the DDRx Wizard. • With the option “Enable separate Vref per rank” unchecked: For multi-rank, Vref is calculated using one min and one max value per DRAM group. Single Vref per lane: Vref value is calculated using the min and max value for each lane, regardless of rank. Single Vref for all the signals: Vref value is calculated using the min and max values across all signals. Therefore, all signals from any rank have the same Vref value. • With the option “Enable separate Vref per rank” checked: Single Vref per lane: Each rank (specified in the Operation column) in each lane has a different Vcent(pin_mid) value. For example, if you are simulating an interface with two lanes, each with two ranks, four Vcent(pin_mid) values are calculated. Single Vref for all signals: Vcent(pin_mid) value is calculated for each rank using all the signals from that rank, per rank using the value in the “Voltage at Widest Eye Opening, Vcent, [mV]” column. For additional information on how to calculate Vcent, see “Vcent(pin_mid) Calculation Examples” on page 516.
Vref - Vinse Low Limit, mV (LPDDR4, LPDDR5)	Vref minus the minimum Vinse required value for the low voltage swing. This difference shows the minimum peak value that the signal needs to swing on the low side with respect to ground, similar to VSEL in other DDR technologies.
Vinse Peak To Peak Limit, mV (DDR5)	Minimum acceptable single-ended input voltage swing (measured from Vref) for CK and DQS.
Vinse Peak To Peak Margin, mV (DDR5)	Measured single-ended input voltage swing (measured from Vref) for CK and DQS - Vinse Peak To Peak Limit.

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Table 5-2. DDRx JEDEC Measurements Spreadsheet Column Definitions

Column Name	Definition
Vinse Peak To Peak, [Pass/Fail] (DDR5)	"Fail" if the measured single-ended input voltage swing (measured from Vref) for CK and DQS is less than Vinse Peak To Peak Limit.
Vinse Peak To Peak Trace Point, [ns] (DDR5)	The time with the minimum single-ended input voltage swing (measured from Vref) for CK and DQS.
Vinse Low Margin, mV	The amount of the low voltage swing (measured from Vref) beyond what Vinse requires.
Vinse Low, [Pass/Fail]	If the Vinse Low Margin is positive, then Pass.
Vinse Low Trace Point, [ns]	The time at which the differential signals cross immediately before the worst-case low state.
Vref + Vinse High Limit, mV	Vref plus the minimum Vinse required value for the high voltage swing. This difference shows the minimum peak value that the signal needs to swing on the high side with respect to ground, similar to VSEH in other DDR technologies.
Vinse High Margin, mV	The amount of the high voltage swing measured from Vref beyond what Vinse requires.
Vinse High, [Pass/Fail]	If the Vinse High Margin is positive, then Pass.
Vinse High Trace Point, [ns]	The time at which the differential signals cross immediately before the worst-case high state.
VSEL Limit, mV	Positive and negative differential signals must exceed this voltage level at least once in each cycle. You can change the value for this limit on the DDRx Batch-Mode Wizard - Quality Checks Page.
VSEL Margin, mV	Voltage difference from VSEL Limit when the limit is exceeded for the worst violation.
VSEL, [Pass/Fail]	"Fail" if VSEL Limit is not exceeded.
VSEL Trace Point, [ns]	Time when the worst violation occurs.
VSEH Limit, mV	Positive and negative differential signals must exceed this voltage level at least once in each cycle. You can change the value for this limit on the DDRx Batch-Mode Wizard - Quality Checks Page.
VSEH Margin, mV	Voltage difference from VSEH Limit when the limit is exceeded for the worst violation.
VSEH, [Pass/Fail]	"Fail" if VSEH Limit is not exceeded.

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Table 5-2. DDRx JEDEC Measurements Spreadsheet Column Definitions

Column Name	Definition
VSEH Trace Point, [ns]	Time when the worst violation occurs.
TVAC/TDVAC Min Time Over AC, ps	Measured time over AC Threshold for single-ended (tVAC) or differential (tDVAC) nets. You can enable or disable this limit on the DDRx Batch-Mode Wizard - Quality Checks Page.
TVAC/TDVAC Time Limit, ps	Minimum required time over AC threshold for single-ended (tVAC) or differential (tDVAC) nets. You can enable or disable this limit on the DDRx Batch-Mode Wizard - Quality Checks Page.
TVAC/TDVAC Margin, ps	Time difference from (tVAC/tDVAC Min Time Over AC) or (tVAC/tDVAC Time Limit) when either limit is exceeded.
TVAC/TDVAC, [Pass/Fail]	“Fail” if (tVAC/tDVAC Min Time Over AC) or (tVAC/tDVAC Time Limit) are exceeded.
TVAC/TDVAC Trace Point, [ns]	Time when the worst violation occurs.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Round Trip Time Spreadsheets

These spreadsheets report round trip time for all nets in the DDRx interface.

File names take the form:

*DDR_report_round_trip_time*_[Typ | Fast | Slow].*csv*

where:

Typ, Fast, Slow represent IC model corners

Spreadsheet Columns

Column Name	Definition
Strobe Net Name	Name of the measured strobe net. For a multiple-board project, the net name does not include the board ID.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Strobe Driver Comp Ref Des & Pin Name, Reading	Strobe driver component reference designator and pin name, when reading. For a multiple-board project, the net name contains the board ID. Use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Strobe Receiver Comp Ref Des & Pin Name, Reading	Strobe receiver component reference designator and pin name, when reading. For a multiple-board project, the net name contains the board ID. Use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Strobe Initial Edge,[Rise/Fall]	The *round trip time* is measured for an initial rising edge strobe signal, and an initial falling edge strobe signal.
Associated Clk Net Name	Name of the clock net associated with the strobe.
Associated Clk Driver Comp Ref Des & Pin Name, Writing	Associated clock driver component reference designator and pin name, when writing. For a multiple-board project, the net name contains the board ID. Use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Associated Clk Receiver Comp Ref Des & Pin Name, Writing	Associated clock receiver component reference designator and pin name, when writing. For a multiple-board project, the net name contains the board ID. Use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Strobe Time-To-Vmeas, [ps]	The measured reference-load delay for the strobe driver. This value is zero when the “Compensate signal launch skews…” option is disabled on the DDRx Batch-Mode Wizard - Nets to Simulate Page.
Strobe Arrival Time, [ps]	The time when the receiver switches.
Strobe Flight Time, [ps]	Strobe Arrival Time - Strobe Time-To-Vmeas
Clk Time-To-Vmeas, [ps]	The measured reference-load delay for the clock driver. This value is zero when the “Compensate signal launch skews…” option is disabled on the DDRx Batch-Mode Wizard - Nets to Simulate Page.
Clk Arrival Time, [ps]	The time when the receiver switches.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Clk Flight Time, [ps]	Clk Arrival Time - Clk Time-To-Vmeas
tDQSCK (min), [ps]	tDQSCK (min), from timing model file
tDQSCK (max), [ps]	tDQSCK (max), from timing model file
Strobe Read Shift, [ps]	Strobe time shift when reading. Value is a 90 degree phase shift unless a value for DQSReadShift is provided in a timing model.
Round Trip Time1 (min), (Clk Flight Time + Strobe Flight Time + tDQSCK(min)), [ps]	Clk Flight Time + Strobe Flight Time + tDQSCK(min)
Round Trip Time1, (Clk Flight Time + Strobe Flight Time), [ps]	Clk Flight Time + Strobe Flight Time
Round Trip Time1 (max), (Clk Flight Time + Strobe Flight Time + tDQSCK(max)), [ps]	Clk Flight Time + Strobe Flight Time + tDQSCK(max)
Round Trip Time1 Limit (min), [ps]	The value you provided on the DDRx Batch-Mode Wizard - Round Trip Time Page.
Round Trip Time1 Limit (max), [ps]	The value you provided on the DDRx Batch-Mode Wizard - Round Trip Time Page.
Round Trip Time1 Setup Margin, [ps]	Round Trip Time1 + tDQSCK (min) - Round Trip Time1 Limit (min)
Round Trip Time1 Hold Margin, [ps]	Round Trip Time1 Limit (max) - Round Trip Time1 - tDQSCK (max)
Round Trip Time1, [Pass/Fail]	“Pass” when both margins are not negative
Round Trip Time2 (min), (Clk Flight Time + Strobe Flight Time + tDQSCK(min) + Strobe Read Shift), [ps]	Clk Flight Time + Strobe Flight Time + tDQSCK(min) + Strobe Read Shift
Round Trip Time2, (Clk Flight Time + Strobe Flight Time + Strobe Read Shift), [ps]	Clk Flight Time + Strobe Flight Time + Strobe Read Shift
Round Trip Time2 (max), (Clk Flight Time + Strobe Flight Time + tDQSCK(max) + Strobe Read Shift), [ps]	Clk Flight Time + Strobe Flight Time + tDQSCK(max) + Strobe Read Shift
Round Trip Time2 Limit (min), [ps]	The value you provided on the DDRx Batch-Mode Wizard - Round Trip Time Page.

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Column Name	Definition
Round Trip Time2 Limit (max), [ps]	The value you provided on the DDRx Batch-Mode Wizard - Round Trip Time Page.
Round Trip Time2 Setup Margin, [ps]	Round Trip Time2 + tDQSCK (min) - Round Trip Time2 Limit (min) + Strobe Read Shift
Round Trip Time2 Hold Margin, [ps]	Round Trip Time2 Limit (max) - Round Trip Time2 - tDQSCK (max) + Strobe Read Shift
Round Trip Time2, [Pass/Fail]	“Pass” when both margins are not negative

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Skew Spreadsheets

These spreadsheets report skew measurements for clocks and strobes in the DDRx interface. Skew is produced by routing, crosstalk, and other factors. DDRx batch simulation does not introduce random jitter.

File names take the form:

*DDR_report_skew*_[violations | worstcases | allcases]_[Typ | Fast | Slow].*[csv | xls]*

where:

violations contains only nets that violate timing-model requirements.

worst cases contains all nets and their worst-case measurements.

allcases contains all nets and all measurements (for each cycle in the simulation).

Typ, Fast, Slow represent IC model corners.

The timing model provides limits. To graphically display tDQSS, tDSS, and other timing model parameters, see DDRx Batch-Mode Wizard - Timing Models Page. JEDEC specification JESD79* defines timing model parameters.

Note

Measurements in this spreadsheet are based the “average” driver delay, unless specified otherwise. See DDRx Delay File Measurements.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Spreadsheet Columns

Column Name	Definition
Net Name	Name of the measured net. For a multiple-board project, the net name does not include the board ID.
Pass/Fail	"Fail" if any measurements fail, based on timing model limits.
Operation	Read or Write, with rank definition as rank(slot number, rank within the slot).
Driver Comp Ref Des & Pin Name	Driver component reference designator and pin name for the measured net. For a multiple-board project, the net name contains the board ID. Use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Receiver Comp Ref Des & Pin Name	Receiver component reference designator and pin name for the measured net. For a multiple-board project, the net name contains the board ID. Use this information to find the related waveform file, if you enabled the save waveform files option on the DDRx Batch-Mode Wizard - Report Options Page.
Associated Clk Net Name Associated Strobe Net Name	Name of the clock or strobe net paired with the measured net.
Associated Clk Driver Comp Ref Des & Pin Name Associated Strobe Driver Comp Ref Des & Pin Name	Driver component reference designator and pin name for the clock or strobe net paired with the measured net.
Associated Clk Receiver Comp Ref Des & Pin Name Associated Strobe Receiver Comp Ref Des & Pin Name	Receiver component reference designator and pin name for the clock or strobe net paired with the measured net.
Clk Launch Time (From Sim), [ns]	Time at driver when the differential clock signals cross zero.
Clk Arrival Time (From Sim), [ns]	Time at receiver when the differential clock signals cross zero.
Clk Delay Time, [ps]	The calculation is (Clk Launch Time (From Sim), [ns]) minus (Clk Arrival Time (From Sim), [ns]).
Strobe Launch Time (From Sim), [ns]	Time at driver when the strobe crosses zero (differential) or VREF (single-ended).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Strobe Arrival Time (From Sim), [ns]	Time at receiver when the strobe crosses zero (differential) or VREF (single-ended).
Strobe Delay Time, [ps]	The calculation is (Strobe Launch Time (From Sim), [ns]) minus (Strobe Arrival Time (From Sim), [ns]).
Launch Skew Time (min), [ps]	Difference between Strobe Launch Time (From Sim), [ns] and Clk Launch Time (From Sim), [ns] when the initial strobe/clock delay for the strobe and clock signals is minimum. The calculation is (Launch Skew Time (typ), [ps]) minus (Initial delay delta), where: Initial delay delta = ([(Maximum initial strobe/clock delay) minus (Minimum initial strobe/clock delay)]/2).
Launch Skew Time (typ), [ps]	Difference between Strobe Launch Time (From Sim), [ns] and Strobe Launch Time (From Sim), [ns] when the initial strobe/clock delay for DQS and CK is average. The calculation is (Strobe Launch Time (From Sim), [ns]) minus (Clk Launch Time (From Sim), [ns]).
Launch Skew Time (max), [ps]	Difference between Strobe Launch Time (From Sim), [ns] and Clk Launch Time (From Sim), [ns] when the initial strobe/clock delay for the strobe and clock signals is maximum. The calculation is (Launch Skew Time (typ), [ps]) plus (Initial delay delta), where: Initial delay delta = ([(Maximum initial strobe/clock delay) minus (Minimum initial strobe/clock delay)]/2).
tDQSS Skew Time (early), [ps]	Difference between Strobe Arrival Time (From Sim), [ns] and Clk Arrival Time (From Sim), [ns] when the initial strobe/clock delay is minimum. The calculation is (tDQSS Skew Time (typ), [ps]) minus (tCKDQSave), where: tCKDQSave = [(tCKDQSmax) minus (tCKDQSmin)]/2.
tDQSS Skew Time (typ), [ps]	Difference between Strobe Arrival Time (From Sim), [ns] and Clk Arrival Time (From Sim), [ns] when the initial strobe/clock delay is average. The calculation is (Strobe Arrival Time (From Sim), [ns]) minus (Clk Arrival Time (From Sim), [ns]). The value is zero when the strobe and clock are aligned.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
tDQSS Skew Time (late), [ps]	Difference between Strobe Arrival Time (From Sim), [ns] and Clk Arrival Time (From Sim), [ns] when the initial strobe/clock delay is maximum. The calculation is (tDQSS Skew Time (typ), [ps]) plus (tCKDQSave), where: tCKDQSave = [(tCKDQSmax) - (tCKDQSmin)]/2.
tDSS Setup Time (From Sim), [ps]	Difference between the clock arrival time (current cycle) and the strobe arrive time (previous cycle) when the initial strobe/ clock delay is maximum (that is, a late strobe). The calculation is (Clk Arrival Time (From Sim), [ns]) - (previous measured DQS falling edge) - (tCKDQSave), where: tCKDQSave = (tCKDQSmax - tCKDQSmin)/2. When tDQSS Skew Time (late), [ps] is a negative value, tDSS Setup Time is not reported because the next preceding edge of DQS (late) is a rising edge and this measurement is taken against the next preceding falling edge of the DQS (late). This makes the value more than half of the CK period. The ideal value is one bit time.
tDSH Hold Time (From Sim), [ps]	Difference between the strobe arrival time (next cycle) and the clock arrival time (current cycle) when the initial strobe/clock delay is minimum (that is, an early strobe). The calculation is (next measured DQS falling edge) - (Clk Arrival Time (From Sim), [ns]) - (tCKDQSave), where: tCKDQSave = (tCKDQSmax - tCKDQSmin)/2. When tDQSS Skew Time (early), [ps] is a positive value, tDSH Hold Time is not reported because the next edge is a rising edge and this measurement is taken against the falling edge of the DQS (early). This makes the value more than half of the CK period. The ideal value is one bit time.
Negative tDQSS Limit, [ps]	Value from DRAM timing model. See DDRx Batch-Mode Wizard - Timing Models Page.
Positive tDQSS Limit, [ps]	
tDSS Limit, [ps]	
tDSH Limit, [ps]	
tDQSS (early) Time Margin = tDQSS(early) - tDQSS(negative limit), [ps]	The calculation is (tDQSS Skew Time (early), [ps]) - (Negative tDQSS Limit, [ps]). Value is negative when it exceeds the limit. **Tip:** Spreadsheet cells with a yellow background indicate a minimum tDQSS margin.

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Column Name	Definition
tDQSS (late) Time Margin = tDQSS(positive limit) - tDQSS(late), [ps]	The calculation is (Positive tDQSS Limit, [ps]) - (tDQSS Skew Time (late), [ps]). Value is negative when it exceeds the limit. **Tip:** Spreadsheet cells with a yellow background indicate a minimum tDQss margin.
tDSS Time Margin = tDSS - tDSS(limit), [ps]	The calculation is (tDSS Limit, [ps]) - (tDSS Setup Time (From Sim), [ps]). Value is negative when it exceeds the limit.
tDSH Time Margin = tDSH - tDSH(limit), [ps]	The calculation is (tDSH Limit, [ps]) - (tDSH Hold Time (From Sim), [ps]). Value is negative when it exceeds the limit.
Stimulus Offset, [ps]	The timing offset between the drivers on the clock and strobe nets, when the initial strobe/clock delay is average.
Comments	Displays error or other measurement problem information.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Audit Spreadsheet

DDR_audit<DDRx_setup_file_name>.xls contains information about simulation setup problems for all pins in the DDRx interface. This is an optional file that the software writes when you enable the audit option.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Log File

View the *DDR_log{<DDRx_setup_file_name>}.txt* file to investigate results or simulation failures. This file contains simulation/measurement progress and messages for DDRx batch simulation, as well as audit results.

The log file contains the following:

- Names of simulated nets and driver/receiver configuration
- For ODT, model selector values for specific simulations
- Initial delay applied to the driver

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- Errors and warnings from simulation and measurements
- Measurement details
- Names of nets coupled by crosstalk, if required
- General errors for each net
- Total run time

DDRx Waveform Files

Use waveform files to investigate measurements reported in spreadsheets. Many waveform files are created when you simulate all or many memory interface nets.

Each waveform is written to comma-separated values (CSV) format. You can display waveform files in the oscilloscope, EZwave, or with third-party software.

Folder Naming Format

The DDR_Results* folder contains additional sub-folders that contain optional waveform files. To help you manage the large number of waveform files, sub-folders sort the waveform files by drive/receive modes and IC model corner values. The waveform subfolder names are of form:

[DRV | RCV]_Waveforms_[Fast | Typ | Slow]

where:

Typ, Fast, Slow represent IC model corners.

Example: *<design>\DDR_Results_Nov-30-2008_11h-33m\DRV_Waveforms_Slow*

Note

Waveform measurements are described at the pin of the device by default. If the IBIS model includes the parameter “Timing_location = die”, the measurements for that device use the die waveforms.

File Naming Format

Driver waveform file names are of the form:

net-<net_name>_drv-<refdes>.<pin1>&<pin2>_<operation>_[after | before]_shift.csv

Receiver waveform file names are of the form:

net-<net_name>_drv-<refdes>.<pin1>&<pin2>_rcv-<refdes>.<pin1>&<pin2>_<operation>_[after | before]_shift.csv

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The following table explains the placeholders in the file naming format:

Placeholder	Description
<*net_name*>	The name of the net.
<*refdes*>	The reference designator and, for a multiple-board project only, the board name of the of the component.
<*pin1*>	The name of the pin. For a differential signal, it is the positive pin.
<*pin2*>	The name of the negative pin of a differential signal. It is absent for single-ended signals. “&” separates <pin1> and <pin2>.
<*operation*>	W for write, R for read, number for rank number (1 = slot1/rank1)
_[after | before]_shift	Present for strobe signals only. The “before_shift” waveform is what you would see at the controller pin with an oscilloscope at a test bench. The “after_shift” waveform is what you see inside the controller, after it is shifted by controller circuitry. Setup and hold measurements use this measurement.

For example:

net-RDSQ0_drv-U123_B00.DQS0+&DQS0-_drv-U321_B01.A0&A1_R1_after_shift.csv

Usage Notes

Waveform files contain either a sampling or all data created by simulation.

- Use the DDRx Batch-Mode Wizard - Report Options Page to choose whether waveforms contain sampled and all data points.
- You can save a sampling of all data points to conserve disk space. The sampling rate is 1 in 10, meaning that every tenth data point from simulation is written to the waveform file.

General Batch Simulation Results Spreadsheet

A General Batch simulation spreadsheet contains measurement values and pass/fail status, and is in HTML format, or Microsoft Excel (XLS) format on Windows and CSV format on Linux.

You can view and edit constraints in the Edit Net Classes Constraints and Stimuli dialog box.

When you uncheck “Report limits and margins” on the General Batch Wizard - Reporting Options / Audit page, the results spreadsheet does not contain columns for margins and limits.

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Spreadsheet Columns

Column Name	Definition
Overview [Pass/Fail]	“Fail” when any other column value is Fail.
Net(s)	Name of the simulated net, any electrically connected nets, and the other member of a differential pair.
Coupled Net(s)	Aggressor nets coupled to the victim net.
Driver(s) [RefDes.Pin]	Reference designator and pin for the driver.
Receiver [RefDes.Pin]	Reference designator and pin for the receiver.
Simulation Corner	IC model corner used for a measurement. See IC Operating Settings.
Min Rise Delay	delay = receiver time - driver time Where: • delay is for a rising-edge transition. • receiver time is the first crossing of Vil at the receiver. • driver time is the crossing of Vmeas at the driver test waveform. For measurement details, see Min Delay.
Max Rise Delay	delay = receiver time - driver time Where: • delay is for a rising-edge transition. • receiver time is the final crossing of Vih at the receiver. • driver time is the crossing of Vmeas at the driver test waveform. For measurement details, see Max Delay.
Min Fall Delay	delay = receiver time - driver time Where: • delay is for a falling-edge transition. • receiver time is the first crossing of Vih at the receiver. • driver time is the crossing of Vmeas at the driver. For measurement details, see Min Delay.

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Column Name	Definition
Max Fall Delay	delay = receiver time - driver time Where: • delay is for a falling-edge transition. • receiver time is the final crossing of Vil at the receiver. • driver time is the crossing of Vmeas at the driver. For measurement details, see Max Delay.
Static Overshoot Low	overshoot = power rail - measurement Where: • overshoot is for a falling-edge transition. • power rail is the low rail voltage. • measurement is the minimum voltage at the receiver. For measurement details, see Static Overshoot Low. **Note:** When the falling dynamic overshoot limit is defined, batch simulation tests the waveform for dynamic overshoot instead of static overshoot.
Static Overshoot Low Limit	This value can come from the following sources: • The IBIS model, from the [Model] keyword and S_overshoot_low subparameter (highest priority). • Static Overshoot Low constraint. **Note:** When you specify both static and dynamic overshoot limits, the software uses the dynamic limit to determine overshoot pass/fail status. For example, if the signal does not exceed the dynamic overshoot limits, but does exceed the static limit, the overall result is “pass”.
Static Overshoot High	overshoot = measurement - power rail Where: • overshoot is for a rising-edge transition. • measurement is the maximum voltage at the receiver. • power rail is the high rail voltage. For measurement details, see Static Overshoot High.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Static Overshoot High Limit	Value provided by either: • An IBIS model, from the [Model] keyword and S_overshoot_high subparameter (highest priority). • The Static Overshoot High constraint. **Note:** When you specify both static and dynamic overshoot limits, the software uses the dynamic limit to determine overshoot pass/ fail status. For example, if the signal does not exceed the dynamic overshoot limits, but does exceed the static limit, the overall result is “pass”.
Dynamic Overshoot Low	The voltage that a falling signal goes below the low rail voltage for the receiver dynamic overshoot low = maximum voltage - low rail voltage
Dynamic Overshoot Low Limit	The value can come from the following sources: • The IBIS model, from the [Model] keyword and D_overshoot_low subparameter (highest priority). • Dynamic Overshoot Low constraint.
Dynamic Overshoot Time Low	The amount of time the waveform spends below the minimum acceptable static voltage. For measurement information, see Dynamic Overshoot Time.
Dynamic Overshoot Time Low Limit	The value can come from the following sources: • The IBIS model, from the [Model] keyword and D_overshoot_time and D_overshoot_low subparameters (highest priority). • Dynamic Overshoot Time Low constraint.
Dynamic Overshoot High	The voltage that a rising signal goes above the high rail voltage for the receiver dynamic overshoot high = maximum voltage - high rail voltage
Dynamic Overshoot High Limit	Value provided by either: • An IBIS model, from the [Model] keyword and D_overshoot_high subparameter (highest priority). • The Dynamic Overshoot High constraint.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Dynamic Overshoot Time High	The amount of time the waveform spends above the maximum acceptable static voltage. For measurement information, see Dynamic Overshoot Time.
Dynamic Overshoot Time High Limit	Value can come from the following sources: • The IBIS model, from the [Model] keyword and D_overshoot_time and D_overshoot_high subparameters (highest priority). • The Dynamic Overshoot Time constraint.
Ringback Delay	The Ringback Delay constraint. For measurement information, see Ringback Delay.
Ringback Margin Low	margin = threshold - measurement Where: • measurement is the minimum ringback voltage measured at the receiver. • threshold is Vil from the receiver IC
Ringback Margin Low Limit	The Ringback Margin Low constraint.
Ringback Margin High	margin = measurement - threshold Where: • threshold is Vih from the receiver IC model • measurement is the closest ringback voltage.
Ringback Margin High Limit	The Ringback Margin High constraint.
Monotonic [Pass/Fail]	“Fail” when a rising or falling transition reverses direction while between the receiver thresholds and for a voltage amplitude greater than the “Maximum allowed nonmonotonic glitches” constraint, specified on the Simulation Settings page of the General Batch Wizard. Figure 8-53 shows a non-monotonicity that is not reported in this column because it does not occur between the receiver thresholds.
Multi Cross Threshold [Pass/Fail]	“Fail” when a rising or falling transition has crossed the receiver Vih or Vil threshold more than once.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Definition
Threshold Cross [Pass/Fail]	"Fail" when the waveform does not cross the far receiver threshold for either of following conditions: • The falling-edge transition fails to cross Vil for the receiver. • The rising-edge transition fails to cross Vih for the receiver. When the result is Fail, a bad DC bias may exist.
Crosstalk	crosstalk = measurement - steady state Where: • crosstalk is for both rising- and falling-edge transitions. • measurement is the absolute maximum of either the positive crosstalk voltage or the negative crosstalk voltage for rising- and falling-edge transitions. • steady state is the steady-state DC voltage at the victim receiver. For measurement information, see Crosstalk.
Crosstalk Limit	The Crosstalk constraint.
Driver Model File	Name of the driver model library file. The file is located in a folder that you specify with the Select Directories for IC-Model Files Dialog Box.
Driver Model Name	Name of the driver model. The value comes from the [Model] keyword in the IBIS model.
Receiver Model File	Name of the receiver model library file. The file is located in a folder that you specify with the Select Directories for IC-Model Files Dialog Box.
Receiver Model Name	Name of the receiver model. The value comes from the [Model] keyword in the IBIS model.
Total Net Length	Sum of the trace segment lengths for net(s) listed in the Net(s) column.
Comments	Status and warning messages.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Differential Measurements in General Batch Simulation

For differential pairs, the software uses differential signals to perform the following measurements:

- Min and Max Rise Delay
- Min and Max Fall Delay
- Ringback Margin High
- Ringback Margin Low
- Monotonic [Pass/Fail]
- Multi Cross Threshold [Pass/Fail]
- Threshold Cross [Pass/Fail]
- Crosstalk

For other measurements, the software uses single-ended signals in a differential pair.

Related Topics

Running General Batch Simulation

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DC Drop Simulation Results

DC Drop simulation provides three types of results: visual (HyperLynx PI PowerScope), text file (Reporter), and spreadsheet (*.XLS* or *.CSV*).

The figure below shows a simple example design in the PDN Editor with one current sink and one voltage source.

Figure 5-1. Measuring DC Drop - Design

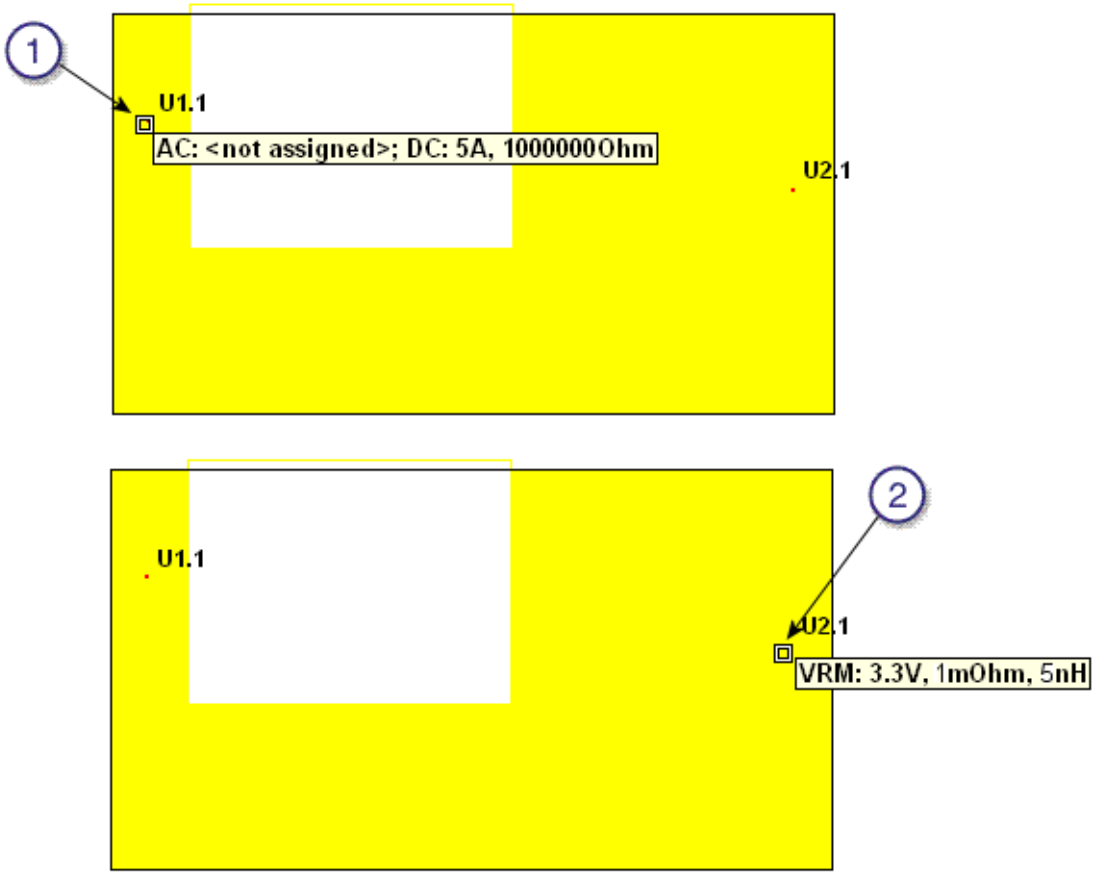


Table 5-3. Measuring DC Drop - Design

1	An IC power supply pin acting as a current sink. A DC model provides the electrical behavior. DC drop simulation ignores AC electrical models, if they are assigned.
2	A voltage-regulator module (VRM) pin acting as a voltage source. A VRM model provides the electrical behavior.

Topic	Description
Text Report	The following figure shows the textual results displayed in the Reporter Dialog Box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
Voltage Drop Graphs	A voltage drop graph uses color to visually display the voltage drop values found during simulation.
Current Density Graphs	A current density graph uses color to visually display the current changes in the design.

Text Report

The following figure shows the textual results displayed in the Reporter Dialog Box.

For information about mapping DC drop simulation circuit elements to terms used in this report, see DC Drop Conceptual Circuits on page 752.

Figure 5-2. Measuring DC Drop - Reporter

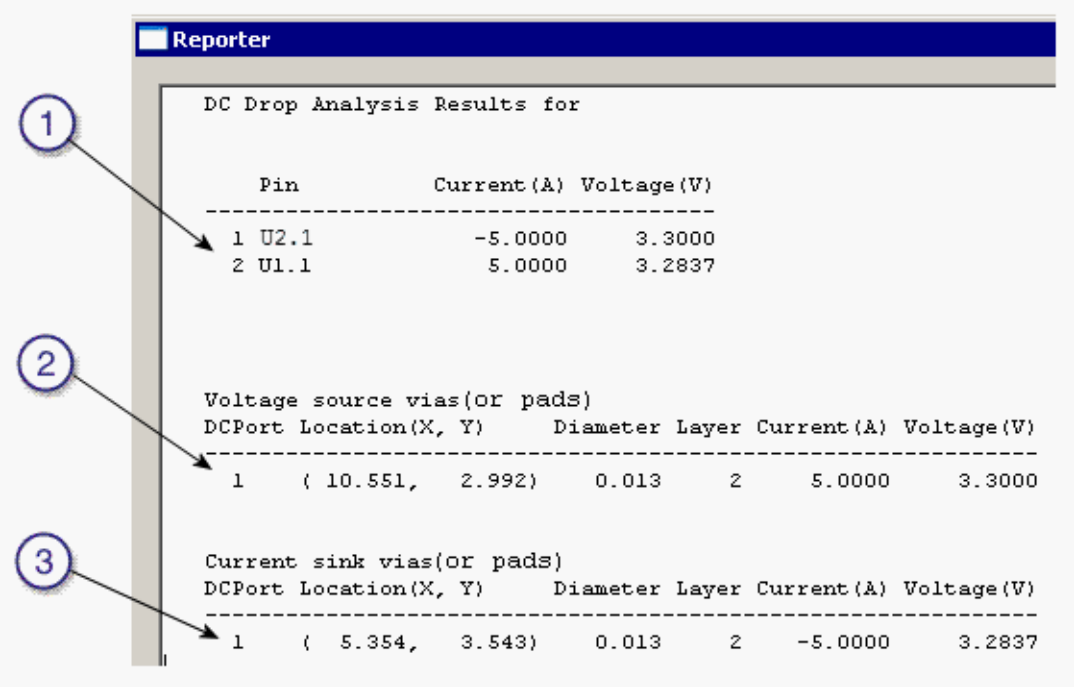


Table 5-4. Measuring DC Drop - Reporter

1	This section summarizes the model assignments. Current flowing into a pin has a positive sign. U2.1 has -5 A assigned to it, so this is the voltage source (VRM). U1.1 has +5 A assigned to it, so this is the current sink (IC power supply pin).
2	This section provides statistics for pins with voltage source models.
3	This section provides statistics for pins with current sink models.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Voltage Drop Graphs

A voltage drop graph uses color to visually display the voltage drop values found during simulation.

Note

HyperLynx PI PowerScope does not show results for some PDN elements that do not conduct current. Ensure that metal areas or traces connect to the power supply net that you want to simulate, and assign VRM and DC sink models to components in that net.

The following figure shows the optional graphical results for voltage drop displayed in the HyperLynx PI PowerScope Dialog Box. The HyperLynx PI PowerScope display is set to two dimensions, which produces a “top down and flat” display of the power-supply net geometries.

Figure 5-3. Measuring DC Drop - HyperLynx PI PowerScope 2-D

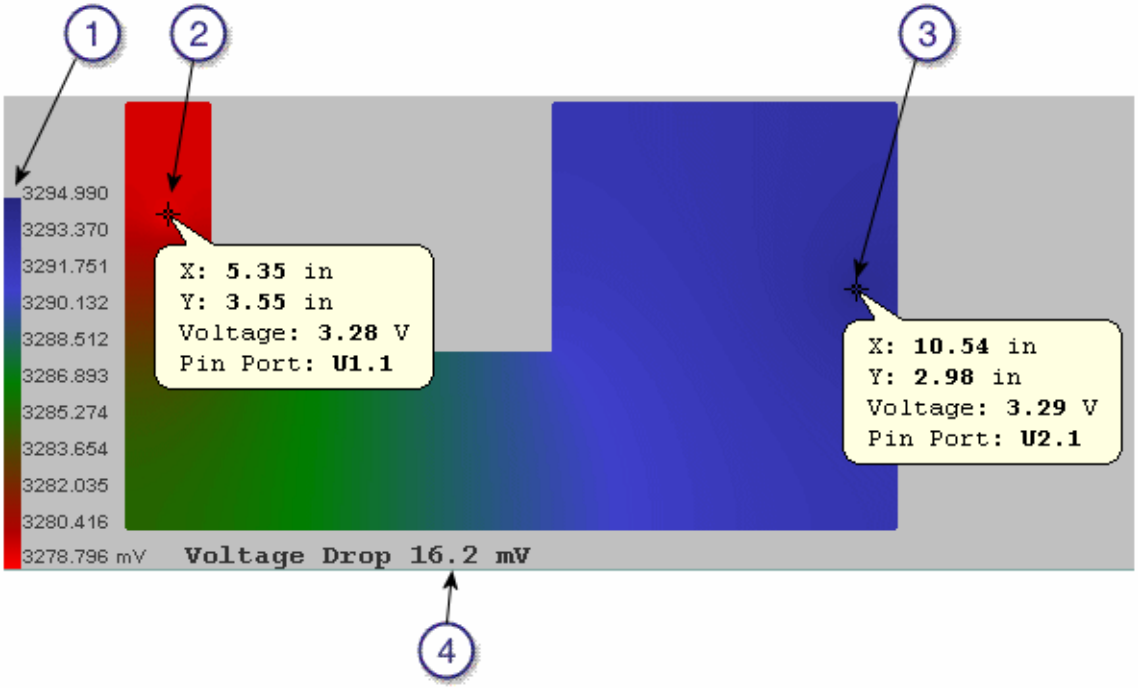


Table 5-5. Measuring DC Drop - HyperLynx PI PowerScope 2-D

1	Legend that maps the graph colors to the measured voltage. Voltage values are relative to the origin set in the HyperLynx PI PowerScope. The color shift from blue to red indicates an increase in voltage drop values.
2	Voltage source pin. To display the ToolTip containing model information, enable the HyperLynx PI PowerScope “Inspect” mode, and then point to the pin.
3	Current sink pin.

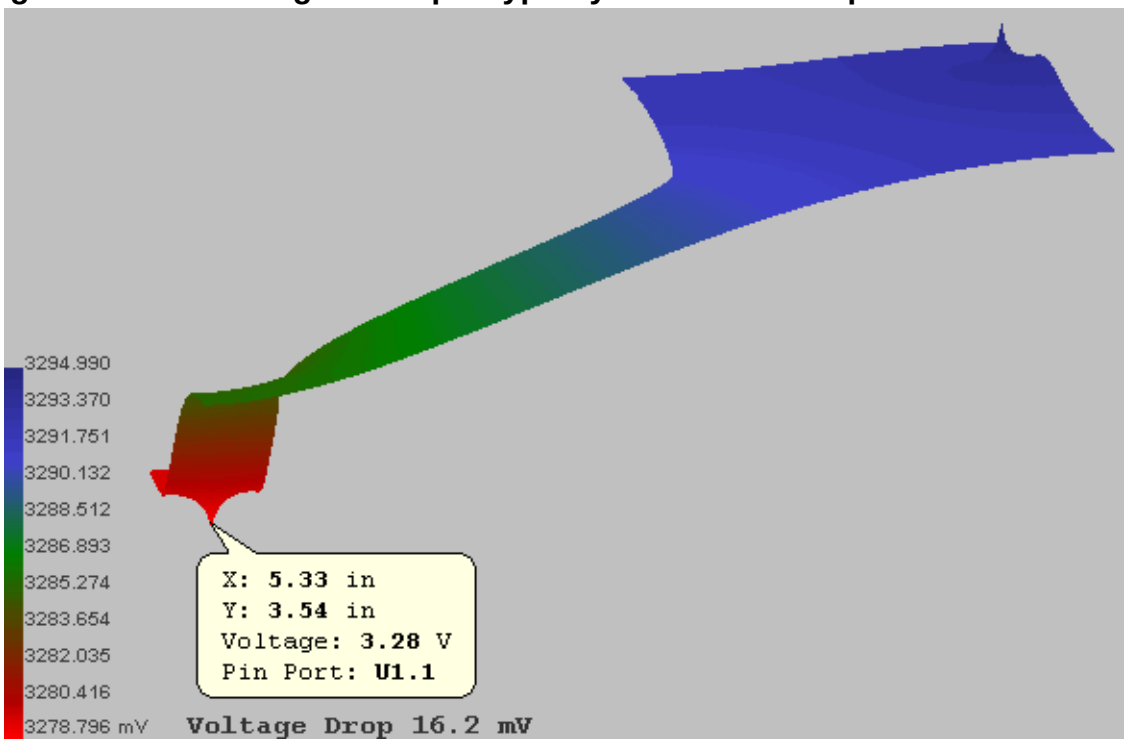
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 5-5. Measuring DC Drop - HyperLynx PI PowerScope 2-D (cont.)

4	Maximum voltage difference between the voltage source and current sink. Note the legend and this value have different number of significant digits, so rounding is likely. If there were multiple sources/sinks, The maximum voltage difference across all of them is reported.

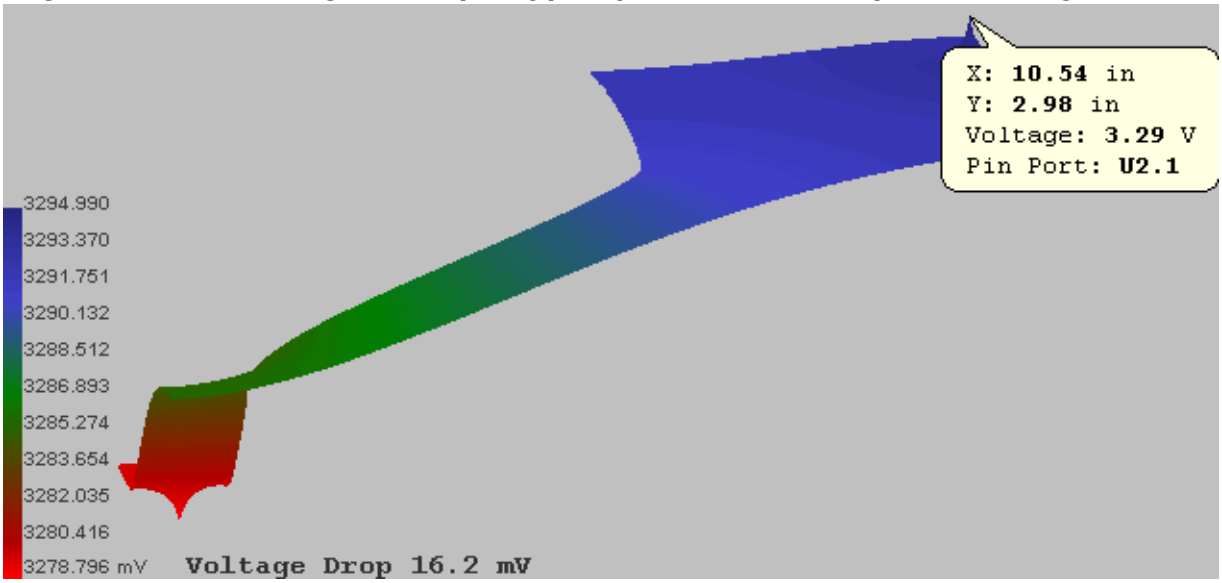
The following figures show the optional graphical results for voltage drop displayed in the HyperLynx PI PowerScope. The HyperLynx PI PowerScope display is set to three dimensions, which displays the voltage in the Z axis (or height). You can rotate the graph, and the orientation shown in the following figure emphasizes the location of the sink/source pins.

Figure 5-4. Measuring DC Drop - HyperLynx PI PowerScope 3-D Current Sink



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 5-5. Measuring DC Drop - HyperLynx PI PowerScope 3-D Voltage Source



Current Density Graphs

A current density graph uses color to visually display the current changes in the design.

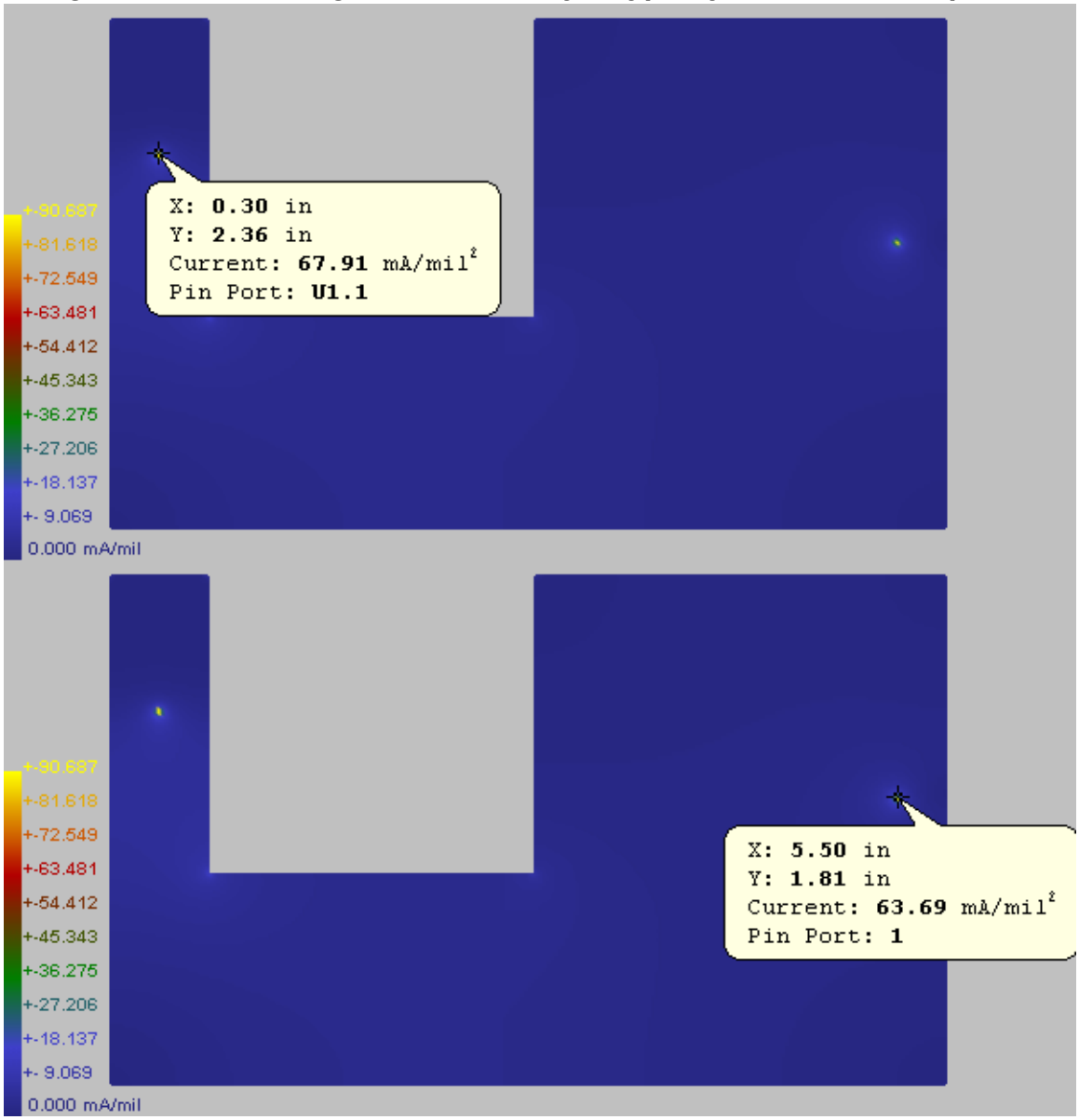
Note

HyperLynx PI PowerScope does not show results for some PDN elements that do not conduct current. Ensure that metal areas or traces connect to the power supply net that you want to simulate, and assign VRM and DC sink models components in that net.

The following figure shows the optional graphical results for current density displayed in the HyperLynx PI PowerScope Dialog Box. The HyperLynx PI PowerScope display is set to two dimensions, which produces a “top down and flat” display of the power supply net geometries.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

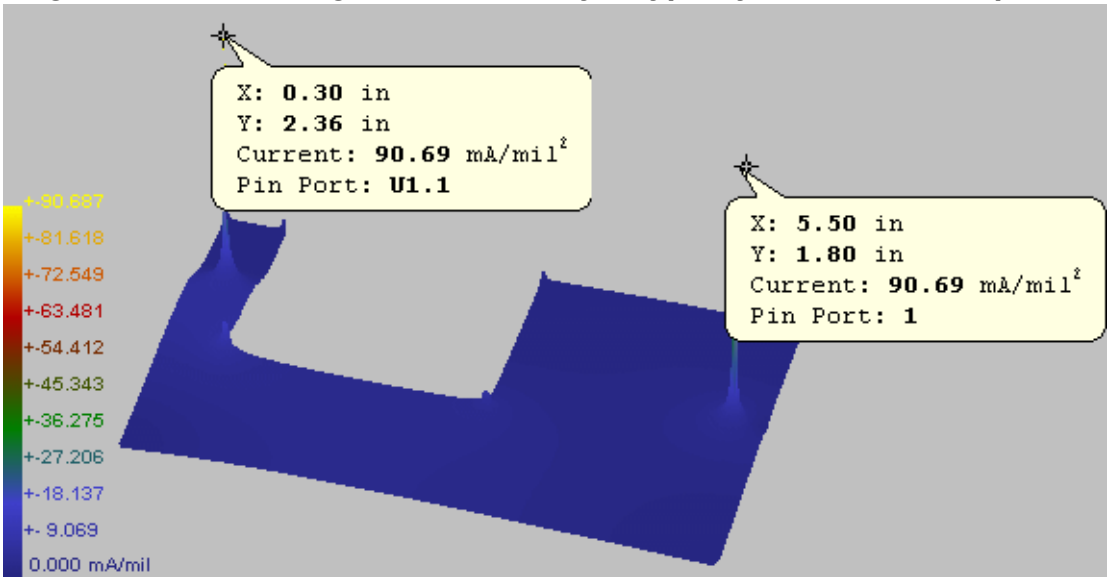
Figure 5-6. Measuring Current Density - HyperLynx PI PowerScope 2-D



The following figure shows the optional graphical results for current density displayed in the HyperLynx PI PowerScope. The HyperLynx PI PowerScope display is set to three dimensions, which displays the current density in the Z axis (or height). You can rotate the graph, and this particular orientation was chosen to emphasize the location of the sink/source pins.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 5-7. Measuring Current Density - HyperLynx PI PowerScope 3-D



Decoupling Simulation Results - Decoupling Capacitor Spreadsheet

Decoupling simulation, in Quick Analysis mode, creates a spreadsheet that reports information about decoupling capacitors on your design.

Columns

Column	Description
Capacitor	Reference designator for decoupling capacitor.
Model	C-L-R values or name of the assigned SPICE or Touchstone model.
Value, uF	The value comes from assignments you make in the Assign / Edit Capacitor Model Dialog Box. If you assigned a C-L-R model to the capacitor, the capacitance appears. If you assigned a Touchstone or SPICE model to the capacitor, “N/A” appears.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column	Description
Mounting Quality	Shows the quality rating of the connection between the decoupling capacitor and the power-distribution network (PDN) as one of the following values: • Good—Decoupling capacitor is properly connected and the total mounting inductance is less than 1.92 nH. • Marginal—Decoupling capacitor is properly connected and the total mounting inductance is more than 1.92 nH. Capacitors with low mounting inductance can be reported as marginal, because of ESL contributed by its model. • Virtual connection—Decoupling capacitor is connected by a "virtual connection" because it: • Contacts a metal shape that is of a negligible size or does not form part of a transmission plane. • Has mounting trace routing that is longer than the Pin-to-area connection search distance (BoardSim) value in the Preferences Dialog Box - Power Integrity Tab. For a description of how HyperLynx forms a virtual connection, see "Pin-to-area connection search distance (BoardSim)" in the Preferences Dialog Box - Power Integrity Tab. Capacitors with virtual connections have approximate mounting inductance and are available to both lumped and distributed decoupling analysis. • Poorly connected—Decoupling capacitor is poorly connected and decoupling analysis cannot find the set of coupled transmission planes needed to estimate its return path. Poorly connected capacitors have a roughly estimated mounting inductance, and are available only to lumped decoupling analysis when you enable Include poorly connected capacitors on the Decoupling Wizard - Customize Settings Page. • Rejected—HyperLynx cannot model the decoupling capacitor. The Comments column provides details.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column	Description
Total Mounting Inductance, nH	Shows the total mounting inductance from vias and traces, including equivalent series inductance (ESL) of the decoupling capacitor body. When you enable the “Simple C-L-R” model type in the Assign / Edit Capacitor Model Dialog Box, this column reports the inductance of the entire mounting loop, which includes the contributions from the following parts: • Capacitor body (ESL) • Inductance from mounting vias and traces When you enable the SPICE or Touchstone model type in the Assign / Edit Capacitor Model Dialog Box, the column shows the inductance of the entire mounting loop minus the estimated ESL of the capacitor. The SPICE or Touchstone model includes the capacitor ESL.
Estimated ESL, nH	Shows the part of the entire mounting loop formed by the decoupling capacitor itself. Restriction: This information is available only for two pin capacitors. The value comes from assignments you make in the Assign / Edit Capacitor Model Dialog Box. This column displays the equivalent series inductance when you assigned a C-L-R model to the capacitor. This column displays “N/A” when you assigned a Touchstone or SPICE model to the capacitor.
Actual Resonance Frequency, MHz	Shows the resonant frequency when you assigned a C-L-R model to the capacitor. See “Assign / Edit Capacitor Model Dialog Box” on page 700. The value comes from the following equation: $$F = \frac{1}{2 \times \pi \times \sqrt{LC}}$$ where L is the value in the Total Mounting Inductance, nH column. This column displays “N/A” when you assigned a Touchstone or SPICE model to the capacitor.
Resonant Frequency for User-Specified ESL (w/o Mounting), MHz	Same as the Actual Resonance Frequency, MHz column, except the equation uses only the L from the Estimated ESL, nH column. This column displays “N/A” when you enable Auto-calculate ESL in the Assign / Edit Capacitor Model Dialog Box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column	Description
Comments	Explains why the software excluded a decoupling capacitor from the simulation. Table 5-6 contains descriptions for some problems reported in this column.

Table 5-6. Decoupling Capacitor Spreadsheets - Comments Column Contents

Problem	Description
Failed to create decoupling capacitor model; decoupling capacitor omitted	The software cannot find a transmission plane connected to the capacitor or the capacitor has incorrect circuit connections. For example, the capacitor pin connects to a net with no metal areas.
Failed to initialize decoupling circuit	The circuit file is incorrect. This includes the circuit created by HypSim for a SPICE or Touchstone model.
File name is not specified for model	The filename is not specified for the SPICE or Touchstone model. This comment usually appears if you manually edit the schematic (*.FFS*) or BoardSim user session (*.BUD*) file and do not specify the model filename.
Model file isn't found	The file specified for the SPICE or Touchstone model is not located in the model-library file paths defined in the Set Directories Dialog Box.
Cannot find Decoupling Model	The file specified for the SPICE or Touchstone model is not located in the model-library file paths defined in the Set Directories Dialog Box.
Decoupling capacitor <name> is connected to only one net	All decoupling capacitor pins connect to the same power supply net.
Decoupling capacitor <name> is connected to more then two nets	The capacitor has three or more pins and connects to three or more power supply nets.
Cannot find pin <reference_designator>.<pin> in model	For SPICE and Touchstone models with three or more pins/ ports, the pin-to-port mapping does not contain the named pin. This may happen if the model file was changed after it was assigned.
Cannot determine model node-to-pin mapping	The software could not map a model node to a capacitor pin. This may happen if you manually edit the schematic (*.FFS*) or BoardSim user session (*.BUD*) file and you change a node name in the netlist.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 5-6. Decoupling Capacitor Spreadsheets - Comments Column Contents

Problem	Description
Cannot find connection of pin <ref des>.<pin> to metal area	The software could not find a connection from the capacitor pin to a metal area. Ensure the pin is connected to a metal area and the path from the pin to the metal area is shorter than the value you set for Pin-to-area connection search distance (2 cm by default) in the Preferences Dialog Box - Power Integrity Tab.
Pin <ref des>.<pin> is not connected to metal area of sufficient size	The capacitor pin connects to a metal area that forms a too-small transmission plane.

Related Topics

Simulating PDN Decoupling - Quick Analysis

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Field Solver Results

The Field Solver provides capacitances, inductances, propagation velocities, and characteristic impedances for a cross section of a set of coupled transmission lines.

Topic	Description
Viewing Field-Solver Results	Use this procedure to run the LineSim field solver, for any coupling region already defined.
How the Field Solver Runs in LineSim	In LineSim, the field solver performs calculations for every coupling region. This information is calculated from the purely geometric and material data you provided in specifying each coupling region's properties.
How Field Solver Results Display	In LineSim, uncoupled transmission-line impedance and delay values display explicitly in the schematic editor, inside each transmission line symbol. They are also shown in the Edit Transmission Line dialog box, in the Values tab.
Auto-Calculate Versus As-Needed Modes	LineSim Crosstalk's field solver is designed to run fast enough that it is interactive, i.e., it can afford to be run whenever its results are needed.
Viewing Detailed Field-Solver Results	For every coupling region, the field solver contains more information than is available in the summary information.
Viewing Electrical Field Lines	When a signal travels along a conductor, it transfers energy in the form of a wave that consists of time-changing electric and magnetic fields. The field solver predicts these fields, given a specific cross section containing conductive traces and various dielectrics.
How Field Lines are Displayed	In the graphical viewer, electric field lines are plotted in blue and electric equipotentials are plotted in red.
Choose a Propagation Mode to Plot	A propagation mode is the manner in which signals are arranged on a system of traces in order to propagate the signals. A basic set of propagation modes is a collection of modes that are combined to create any arbitrary set of real signals on the traces.
Generating a Report of the Field Solver's Numerical Results	Much of the field solver's output data is in the form of matrices or lists of numerical parameters (impedance, propagation speed, and so on). You can view all of this data in a report file.
Contents of the Results Report	The Detailed Field-Solver Results table gives an overview of the data contained in the field-solver numerical results file.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Viewing Field-Solver Results

Use this procedure to run the LineSim field solver, for any coupling region already defined.

Procedure

1. Double-click any transmission line belonging to the coupling region whose field properties you want to see.
2. Click the **Field Solver** tab.
3. To see field lines, in the Field Plotting area, click **Start**.

   OR

   To see numerical data (an, impedance matrix), in the Numerical Results area, click **View**.

Related Topics

How the Field Solver Runs in LineSim

How the Field Solver Runs in LineSim

In LineSim, the field solver performs calculations for every coupling region. This information is calculated from the purely geometric and material data you provided in specifying each coupling region's properties.

The calculations include:

- Capacitance matrix
- Inductance matrix
- Characteristic impedance matrix
- Propagation speed(s)
- If multiple propagation speeds, the percentage of energy in each trace traveling at each speed
- An optimal resistor termination array for the region's transmission lines

Therefore, the field solver can be thought of as a calculation engine that transforms geometric/material data into corresponding electromagnetic data.

Related Topics

How Field Solver Results Display

Viewing Field-Solver Results

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

How Field Solver Results Display

In LineSim, uncoupled transmission-line impedance and delay values display explicitly in the schematic editor, inside each transmission line symbol. They are also shown in the Edit Transmission Line dialog box, in the Values tab.

With coupled transmission lines, the software attempts to display electrical information in much the same way as with uncoupled. This is not entirely possible, because the information associated with a collection of coupled lines is more complex than the single-value parameters associated with uncoupled lines. For coupled lines, some information is displayed in the schematic editor; some is shown in the Edit Coupling Regions dialog box; and full details are available from the Field Solver dialog box. See Viewing Detailed Field-Solver Results.

Field-Solver Results in the Schematic Editor

For coupled transmission lines, there is no single value that describes each line's characteristic impedance. Similarly, if the lines are microstrips or buried microstrips (rather than striplines), so that the electromagnetic fields they generate lie in a mixture of dielectrics (such as FR-4 and air), then multiple propagation velocities exist per line and there is no single line-delay value.

However, for coupled lines, crosstalk does display a single value for impedance and delay in each transmission-line symbol. The values shown are as follows:

Table 5-7. Display of Impedance and Delay in Transmission-Line Symbols

Parameter	What is Displayed in a Transmission-Line Symbol in the Schematic Editor
characteristic impedance	line's diagonal value from the characteristic-impedance matrix
delay	if line is a stripline (a single dielectric): the single delay value if line is not a stripline (multiple dielectrics): weighted average of line's multiple delays; weighting based on percentage of energy traveling at each speed

Schematic Impedance

You can think of each transmission line's diagonal impedance as the impedance of the line to ground, accounting for the presence of the nearby, coupled lines. If the lines in the region are weakly coupled, the diagonal value is close to the calculation for the line in isolation (by ignoring neighboring traces); as the lines become more strongly coupled, the diagonal impedance deviates more from the isolated value.

Although it is not possible to completely terminate a coupled transmission line with a single resistor, if you are forced to use only one resistor and the signal on the line is not either purely differential or common-mode, then the diagonal impedance value is usually the best terminating value to use.

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Schematic Delay

For coupled striplines, whose electromagnetic fields exist entirely in dielectric of one type, there is only one signal propagation velocity and therefore a single delay value, which the schematic editor displays.

However, for coupled microstrips or buried microstrips, whose fields penetrate both PCB dielectric and air, there are multiple propagation velocities (specifically, as many velocities as there are traces in the coupling region). In this case, in order to display a single delay value in the schematic editor, crosstalk averages each line's multiple delays together. The calculation is a weighted average, with the weighting based on the percentage of signal energy that exists at each velocity. If only a small amount of energy travels at a given speed, then that speed's contribution to the average is small. The final result is displayed in the schematic editor.

Usually, unless a coupling region's geometry is very asymmetric, the difference between propagation velocities is small. (An example of an "asymmetric" geometry would be a microstrip of one width coupled to a buried microstrip of a different width, with the buried trace below and considerably off to the side of the outer-layer trace.) Therefore, the averaging effect described above is usually not major.

Field-Solver Results in the Edit Coupling Regions Dialog Box

Some field-solver results are displayed directly in the Edit Coupling Regions dialog box, so you can monitor the effects of coupling-region geometry changes as you make them. Specifically, in the dialog box's Impedance area, the list shows impedance data for the current cross section.

By default, the field solver recalculates impedances every time you make a change in the dialog box. You can optionally run with "auto-calculate" mode turned off, however; see Auto-Calculate Versus As-Needed Modes.

Exactly how impedances values are displayed varies depending on whether there are two or more than two traces in the coupling region.

Impedance Display Area in Field Solver

The list in the Impedance area always shows the following information.

Table 5-8. Contents of Impedance Display Area in Field Solver

Column	Description
Transmission Line	The name of the transmission line; includes schematic-cell coordinates plus any "comment" you enter for the line.
Impedance	An impedance value, in ohms; if multiple lines in the coupling region, taken from the diagonal values in the impedance matrix.
Notes	A description of what the impedance is.

For details on getting more-complete impedance information (e.g., the full impedance matrix), see "Viewing Detailed Field-Solver Results" on page 367.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Extra Impedance Information - Differential Z - When Only Two Traces

When there are only two traces in a coupling region, the software assumes that you are interested in the differential impedance between the traces. Therefore, with two traces, the impedance-area list shows, in addition to the "diagonal impedances" described above, the two-trace differential impedance. This value is labeled "differential" in the Transmission Line column.

The differential impedance is the correct terminating value to use, line-to-line, only if the two traces in the coupling region carry only differential signals.

Related Topics

Viewing Field-Solver Results

Viewing Detailed Field-Solver Results

Auto-Calculate Versus As-Needed Modes

LineSim Crosstalk's field solver is designed to run fast enough that it is interactive, i.e., it can afford to be run whenever its results are needed.

Table 5-9. Field Solver Modes

Mode	Description
Auto-Calculate Mode	If you are working on small coupling regions (regions with a small number of transmission lines) and if your computer is fast, you may want the field solver to run any time any change is made to a coupling region, even while you're in the middle of working in the Edit Coupling Regions dialog box. In this "auto-calculate" mode, to which LineSim Crosstalk defaults, the field solver runs each time you change a cross-section value anywhere in the dialog box. To place the field-solver in auto-calculate mode (if previously disabled): • In the Edit Coupling Regions dialog box, in the Impedance area, select the Auto Calc check box. The field solver runs and new results are placed in the Impedance list. If you are working on a large coupling region (one with many transmission lines) or if the field solver is taking several or more seconds to run each time it is run, then it is best to leave auto-calculate mode off.
As-Needed Mode	Optionally, the field solver can be set up to run whenever a coupling region's geometry or material data are changed, but not while you are working in the Edit Coupling Regions dialog box, in the middle of making changes. For details on opening the Edit Coupling Regions dialog box and changing coupling-region properties, see Edit Transmission Line Dialog Box - Edit Coupling Regions Tab. In this mode, the solver runs only when you close the dialog box or click the Transmission-Line Type tab.

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Table 5-9. Field Solver Modes (cont.)

Mode	Description
Disabling Auto-Calculate Mode	To disable auto-calculate mode: • In the Edit Coupling Regions dialog box, in the Impedance area, clear the Auto Calc check box. The field solver runs and new results are placed in the Impedance list. If you want see the field solver's output data before you are ready to close the Edit Coupling Regions dialog box, you can force the solver to run. Running it will refresh the data displayed in the Impedance list and also allow you to switch to the Field Solver tab and immediately click the Numerical Results View button.
Manually Forcing the Field Solver to Run	To manually force the field solver to run, without closing the Edit Coupling Regions dialog box (only if auto-calculate mode is disabled): • In the Edit Coupling Regions dialog box, in the Impedance area, click **Calculate**. The field solver runs and new results are placed in the Impedance list.

Viewing Detailed Field-Solver Results

For every coupling region, the field solver contains more information than is available in the summary information.

Table 5-10 describes the additional information.

Table 5-10. Detailed Field-Solver Results

Information	Description
capacitance matrix	Specifies the self and mutual capacitances of the traces in the coupling region.
inductance matrix	Specifies the self and mutual inductances of the traces.
characteristic-impedance matrix	Specifies the full matrix impedance for the system of coupled transmission lines; off-diagonal values are small for weak coupling and large for strong.
optimal terminator-resistor array	Specifies an array of resistors (line-to-ground and line-to-line) that perfectly terminates the system of coupled lines; in theory, this array can completely eliminate crosstalk amongst the lines.
list of propagation speeds	Specifies the velocities at which signals propagate on the traces; there multiple values if the traces' electromagnetic fields "see" more than one type of dielectric (microstrip or buried microstrip).

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Table 5-10. Detailed Field-Solver Results (cont.)

table of energy percentages in each propagation mode	Specifies the amount of each trace's energy that travels at each propagation velocity; for multi-speed coupling regions only.
recommended termination values	Specifies a list of impedance values, including the differential, common-mode, and line-to-ground values; for two-trace coupling regions only.
graphical field lines	Displays a picture showing the coupling region's electric field lines and electric equipotentials.

Related Topics

How Field Solver Results Display

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Viewing Electrical Field Lines

When a signal travels along a conductor, it transfers energy in the form of a wave that consists of time-changing electric and magnetic fields. The field solver predicts these fields, given a specific cross section containing conductive traces and various dielectrics.

It can be informative to view the field lines calculated by the field solver. Though the field plots provide no direct analytic value, they can give an intuitive feeling for how various traces are coupled to each other.

Restriction

The LineSim/BoardSim Crosstalk license is required to run crosstalk simulation.

Topic	Description
Viewing Electrical Field Lines in LineSim for Coupling Regions	View the electrical field lines for a coupling region
Viewing Electrical Field Lines in BoardSim for Trace Segments	View the electrical field lines for a trace segment.

Viewing Electrical Field Lines in LineSim for Coupling Regions

View the electrical field lines for a coupling region

Procedure

1. If needed, open the Edit Transmission Line dialog box by double-clicking a transmission line that is in the coupling region whose field lines you want to see.
2. Click the **Field Solver** tab. A large graphical view of the coupling region appears, see Field Solver Dialog Box.
3. In the Propagation Mode list, do one of the following:
   - Leave the mode at the default setting and go to the next step.
   - Select the propagation mode for which you want to see field lines.

   Restriction

   The list is unavailable when the selected transmission line is not coupled to another transmission line.

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4. Do one of the following:
   - If you changed the propagation mode in the previous step, plotting starts automatically.
   - If you did not change the mode, click **Start**. The software begins calculating and displaying electric field lines and electric equipotentials.

   See “How Field Solver Results Display” on page 364.
5. To view numerical results from the field solver, click **View**.

   See “Generating a Report of the Field Solver’s Numerical Results” on page 373.

Viewing Electrical Field Lines in BoardSim for Trace Segments

View the electrical field lines for a trace segment.

Procedure

1. In BoardSim, right-click over the trace segment and click **View Field-Solver Output**. See Field Solver Dialog Box.
2. If the trace segment is coupled to another trace segment, the Propagation Mode list becomes available. You can do any of the following:
   - Leave the mode at the default setting and go to step 3.
   - From the Propagation Mode list, select the propagation mode for which you want to see field lines.
3. Do one of the following:
   - If you changed the propagation mode in step 2, plotting starts automatically.
   - If you didn’t change the mode, click **Start**. The software begins calculating and displaying electric field lines and electric equipotentials.

   See “How Field Solver Results Display” on page 364
4. To view numerical results from the field solver, click **View**.

Related Topics

Generating a Report of the Field Solver’s Numerical Results

How Field Lines are Displayed

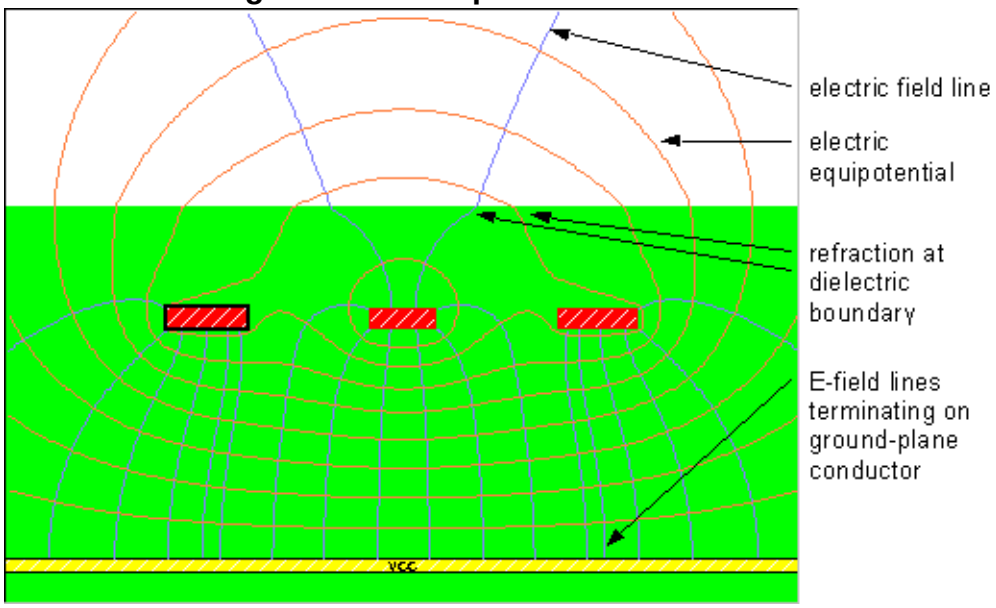
In the graphical viewer, electric field lines are plotted in blue and electric equipotentials are plotted in red.

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Electric field lines, described as the electric field's "lines of force," begin and end on conductor surfaces (where physical charges reside). They refract (change direction) at boundaries between different dielectrics. See Figure 5-8.

Electric equipotentials are curves along which the electric potential (voltage) is a constant. They form closed contours around one or more conductors, and refract at dissimilar-dielectric boundaries.

Figure 5-8. Example of a Field-Line Plot



The field lines are not plotted instantly, because the software calculates their positions on-the-fly, when you click the Start button (or change propagation mode). The plotting is fairly quick on most computers, but you can interrupt it before completion if you want.

Choose a Propagation Mode to Plot

A propagation mode is the manner in which signals are arranged on a system of traces in order to propagate the signals. A basic set of propagation modes is a collection of modes that are combined to create any arbitrary set of real signals on the traces.

Tip

The simulator automatically defines the propagation modes. You only need to think about propagation modes when plotting field lines or looking at signal propagation speeds.

For example, in the case of two traces coupled together, designers often think in terms of a set of modes consists of "differential mode" and "common mode." The differential propagation mode is one in which if one trace carries the voltage +V, the other trace carries -V (the two traces always carry opposite voltages). For the common mode, if one trace carries +V, the other also carries +V.

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Note that it is conceptually possible to describe any pair of real signals traveling on the two traces as some mixture of these two modes. For example, a mostly differential signal that had a small common-mode component to it could be constructed by mixing 80% differential mode with 20% common mode.

Note, too, that it is possible to conceive of other equally valid propagation-mode sets for the two traces. Another possibility, for example, is a set in which mode 1 consists of signal V on trace 1 and no signal on trace 2; and mode 2 consists of no signal on trace 1 and signal V on trace 2. This is a basis set just as valid as the set consisting of differential + common modes—you can conceptually use either set to construct any real set of signals on the traces.

Tip

The construction of a propagation-mode set is arbitrary and has nothing to do with the validity of the electromagnetic solutions generated by the field solver or the waveforms generated by the crosstalk simulator. For stripline configurations, any basis set is equally valid, and crosstalk only requires one so that it can display field lines in some reasonable manner. Thus, it makes sense to choose a set of modes that is conceptually simple.

Propagation Modes for Striplines - One Dielectric Only

For coupling regions that have only one propagation velocity, where all of the traces are striplines (whose fields exist only in one type of dielectric), the software crosstalk analysis constructs the conceptually simplest possible set of propagation modes: one in which mode 1 means trace 1 has a signal V and all other traces have no signal; mode 2 means trace 2 has a signal V and all other traces have no signal; and so forth.

Thus, when you choose mode 1 from the Propagation Mode combo box and click the Start button, you will see lines emanating from and surrounding trace 1. Mode 2 produces lines around trace 2—and so forth.

Propagation Modes for Microstrips and Buried Microstrips - Multiple Dielectrics

For coupling regions that have multiple propagation velocities, where the traces are microstrips or buried microstrips (whose fields penetrate multiple dielectrics), the concept of propagation modes takes on added physical significance. In this case, it is possible to construct a set of modes such that each mode represents the amount of signal on each trace that travels at one of the propagation speeds. The number of speeds (and therefore modes) equals the number of traces.

For example, for the coupling region shown in “How Field Solver Results Display” on page 364, there are three modes, one propagating energy with a speed of 51.6% of the speed of light; another propagating at 50.8% of light speed; and a third propagating at 49.1% of light speed. In general for the multi-speed case, each mode involves some amount of signal on each trace. Therefore, when you plot one of the modes (unlike with the single-velocity case; see “Propagation Modes for Striplines” above), you see lines emanating from and surrounding all of

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the traces. "How Field Solver Results Display" on page 364 shows the plot for this coupling region's propagation mode 1.

Following the theme of the discussion in the previous section, you could define other mode sets for the coupling region in Figure 5-8. Suppose, for example, that you defined mode 1 as [+V,0,0], mode 2 as [0,+V,0] and mode 3 [0,0,+V]. This is conceptually simple at first glance, but now each mode involves a mixture of three different propagating speeds. So the more physically significant mode set for these traces is the one in which each mode propagates signals at one "pure" speed.

For two-trace microstrip and buried-microstrip configurations in which the traces are symmetrically arranged (each trace is on the same layer, has the same width and thickness, and so on.), it turns out that the mode set that describes the two propagation speed and the differential/common mode set coincide (they are the same). Thus, for symmetric trace arrangements, driving purely differential signals means that only one mode is excited, and only one propagation speed results.

Generating a Report of the Field Solver's Numerical Results

Much of the field solver's output data is in the form of matrices or lists of numerical parameters (impedance, propagation speed, and so on). You can view all of this data in a report file.

Procedure

1. If you are using LineSim, open the Edit Transmission Line dialog box by double-clicking a transmission line that is in the coupling region whose field lines you want to see.

   Or

   If you are using BoardSim, right-click over a trace segment whose field lines you want to see, and then click **View Field-Solver Output**.

2. If needed, click the **Field Solver** tab. A large graphical view of the coupling region appears.

3. In the Numerical Results area, click **View**.

4. If the View button is unavailable (because the coupling region has changed and the field solver has not yet been run), click **Start**, and then click **View**.

Results

The software creates a report file and opens it in the HyperLynx File Editor. You can scroll up and down in the editor to see the report's data, and print it if desired. By default, the report is written into a file named <*Coupling_Region_Name*>.*TXT*, where <Coupling_Region_Name>

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is the name of the coupling region for which the data are being reported. The file is located in the same directory as the *.FFS* schematic file.

The results file is overwritten with each run. To preserve field-solver data, save the results file under a unique name by renaming the coupling region. For details on editing coupling region names, see "Edit Transmission Line Dialog Box - Edit Coupling Regions Tab" on page 862.

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Contents of the Results Report

The Detailed Field-Solver Results table gives an overview of the data contained in the field-solver numerical results file.

The following topics provide additional details. For additional information, see Impedance and Termination Summary for Two-Line Coupling Regions Only.

Topic	Description
Physical Input Data	This section of the report file shows for what geometric and material data the field-solver results were calculated. The data serves as a record of the input problem, for future reference. Also, for certain coupling regions whose characteristics are important in a key design decision, it may also be worth looking at the input data to verify that the problem on which the field solver ran was exactly as expected. The input data includes information on each trace in the coupling region as well as the region's PCB stackup.
Implementing Optimal Termination	Use this procedure to implement an optimal terminator-resistor array matrix termination.
Characteristic-Impedance Matrix	This matrix gives the characteristic impedance (in ohms) of the system of coupled transmission lines in the coupling region. Coupled lines do not have a single-value impedance, like uncoupled lines and a set of coupled lines share an impedance matrix.
Capacitance Matrix	This matrix gives the self and mutual capacitances (in pF/m) of the coupled transmission lines in the coupling region. More specifically, the diagonal values in the matrix give the capacitances to ground of the corresponding transmission lines, while the off-diagonal values give the capacitances between the corresponding pair of lines.
Inductance Matrix	This matrix gives the self and mutual inductances (in nH/m) of the coupled transmission lines in the coupling region. More specifically, the diagonal values in the matrix give the self inductances of the corresponding transmission lines, while the off-diagonal values give the mutual inductances of the corresponding pair of lines.
Propagation-Speeds List in LineSim	This list gives the speed(s) (in m/s) at which signals propagate along the transmission lines in the coupling region.

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Topic	Description
Percentage of Energy Matrix for Multiple-Speed Coupling Regions Only	If the coupling region supports multiple propagation speeds, this matrix gives, for each transmission line in the region, the percentage of signal energy that travels at each speed. In the matrix, each column represents a line (a trace); reading down the column shows how much of the signal energy in that line travels in each of the propagation modes listed in the propagation-speeds list. The percentages in each column add to approximately 100%, to fully account for the energy in each transmission line.
Impedance and Termination Summary for Two-Line Coupling Regions Only	For the special case of a two-line coupling region, the field-solver numerical results report gives additional information about specific termination options.

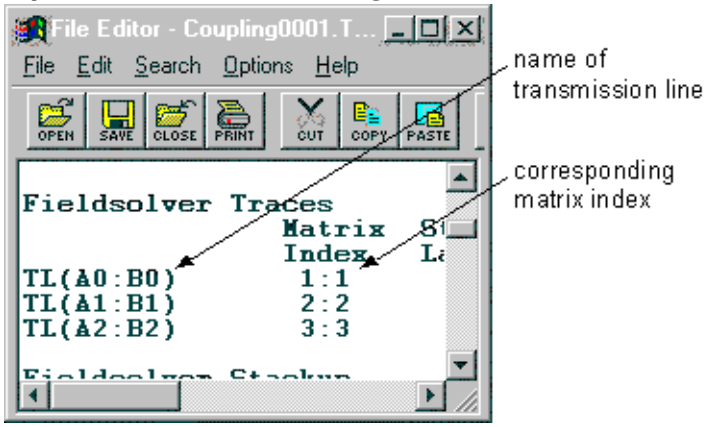
Physical Input Data

This section of the report file shows for what geometric and material data the field-solver results were calculated. The data serves as a record of the input problem, for future reference. Also, for certain coupling regions whose characteristics are important in a key design decision, it may also be worth looking at the input data to verify that the problem on which the field solver ran was exactly as expected. The input data includes information on each trace in the coupling region as well as the region's PCB stackup.

Correlating Transmission Lines and Matrix Indices

In the input data, the Field Solver Traces section lists by name each transmission line in the coupling region, and shows the corresponding trace index by which the line is referred to in the electrical matrix data elsewhere in the file. This data allows you to correlate transmission lines and trace indices. See Figure 5-9.

Figure 5-9. Example of Table Correlating Transmission Lines and Trace Indices



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Field Solver Output Data

This section of the report file shows the electrical data calculated by the field solver for the coupling region. As described earlier, much of this data are in matrix form, because the transmission lines in the region are coupled together.

Related Topics

Generating a Report of the Field Solver's Numerical Results

Implementing Optimal Termination

Use this procedure to implement an optimal terminator-resistor array matrix termination.

Many times, users of LineSim Crosstalk's crosstalk-analysis features are interested in how to terminate traces that are coupled to other traces. This matrix gives the theoretically optimal resistor termination array for the set of coupled lines in the coupling region.

A key fact about coupled lines is that they cannot be perfectly terminated individually. Instead, a matrix of resistors that prescribes both line-to-ground and line-to-line resistances is required.

This termination array has the remarkable property that it not only "kills" single-line reflections at the line ends, but also eliminates arriving crosstalk signals.

On the other hand, there are many situations in digital electronics where line-to-line resistors (in addition to adding undesirably to passive-component count) are simply not permissible for DC-bias reasons. For example, whereas two coupled data lines may require a 160-ohm resistor between them to eliminate line-to-line crosstalk, it is unlikely that the driver ICs on the lines would be "happy" with the resistor when one line was pulled high and the other low.

Still, in some critical situations, especially when the line-to-line coupling is relatively weak and therefore the line-to-line terminating resistances are fairly high, a matrix terminator may be workable.

There are some IC technologies which are specifically designed to work with line-to-line termination: differential drivers. For these devices, line-to-line termination serves not only to prevent line reflections and eliminate crosstalk, but is often also required to bias the ICs for correct operation.

Procedure

1. Place the resistors in the diagonal matrix positions between the corresponding trace to ground.

   Example: Resistor 2-2 should be placed from trace 2 to ground, at the trace end.

2. Place the resistors in the off-diagonal matrix positions line-to-line between the corresponding traces.

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Example: Resistor 2-1 should be placed between traces 1 and 2, at the trace ends.

3. Note that there are twice as many off-diagonal values as there are line-to-line resistors, since, for example, off-diagonal resistance 2-1 refers to the same resistor as resistance 1-2.

4. To correlate a specific transmission line in a coupling region to a matrix index, see the Fieldsolver Traces section of the Physical Input Data section of the field-solver report, as illustrated in Figure 5-9.

Related Topics

Generating a Report of the Field Solver's Numerical Results

Characteristic-Impedance Matrix

This matrix gives the characteristic impedance (in ohms) of the system of coupled transmission lines in the coupling region. Coupled lines do not have a single-value impedance, like uncoupled lines and a set of coupled lines share an impedance matrix.

The values in the diagonal matrix positions can be thought of as giving the impedances to ground of the corresponding transmission lines, accounting for the presence of the other nearby, coupled traces. When an IC drives into one of the lines, however, it "sees" not only the diagonal impedance for that line, but also some of the off-diagonal terms in the matrix.

For lines that are only weakly coupled, the diagonal impedance terms are dominant, and the diagonal values are close to what they would be if the lines were completely isolated from each other. As the coupling becomes stronger, the diagonal terms deviate more from their standalone values, and the off-diagonal terms increase. Note that small off-diagonal impedances mean weak coupling; large impedances mean strong coupling.

Barring special cases like two-line pairs in which the two signals are known to be either purely differential or purely common-mode, the diagonal impedances in the matrix are generally the best single-resistor terminators to use. Note, however, that coupled transmission lines cannot be perfectly terminated unless a full matrix termination (including both line-to-ground and line-to-line resistors) is employed. See Implementing Optimal Termination.

To correlate a specific transmission line in a coupling region to a matrix index, see the Fieldsolver Traces section of the Physical Input Data section of the field-solver report, as illustrated in Figure 5-9.

Related Topics

Generating a Report of the Field Solver's Numerical Results

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Capacitance Matrix

This matrix gives the self and mutual capacitances (in pF/m) of the coupled transmission lines in the coupling region. More specifically, the diagonal values in the matrix give the capacitances to ground of the corresponding transmission lines, while the off-diagonal values give the capacitances between the corresponding pair of lines.

Many users are surprised to see that the off-diagonal capacitance-matrix values are negative. The negative sign simply reflects the fact that if a positive charge is placed on a given trace, negative charge will accumulate on all others.

For purposes of judging how much capacitance exists between traces, you can ignore the negative signs. The off-diagonal values do represent real, physical capacitance.

However, in the mathematical formalism of coupled transmission lines, the negative signs are important. For example, if you transfer the capacitance matrix for a coupling region to another EDA tool (such as SPICE), the off-diagonal values must be negative.

Note that the values in the capacitance matrix have units of pF/m, rather than simply pF. This means that if you are trying to calculate, for example, the total capacitance-to-ground of a transmission line in the matrix, you must multiply the corresponding diagonal value in the matrix by the length (in meters) of the line.

To correlate a specific transmission line in a coupling region to a matrix index, see the Fieldsolver Traces section of the Physical Input Data section of the field-solver report, as illustrated in Figure 5-9.

Related Topics

Generating a Report of the Field Solver's Numerical Results

Inductance Matrix

This matrix gives the self and mutual inductances (in nH/m) of the coupled transmission lines in the coupling region. More specifically, the diagonal values in the matrix give the self inductances of the corresponding transmission lines, while the off-diagonal values give the mutual inductances of the corresponding pair of lines.

Note that the values in the inductance matrix have units of nH/m, rather than simply nH. This means that if you are trying to calculate, for example, the total self inductance of a transmission line in the matrix, you must multiply the corresponding diagonal value in the matrix by the length (in meters) of the line.

To correlate a specific transmission line in a coupling region to a matrix index, see the Field Solver Traces section of the Physical Input Data section of the field-solver report, as illustrated in Figure 5-9.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Generating a Report of the Field Solver's Numerical Results

Propagation-Speeds List in LineSim

This list gives the speed(s) (in m/s) at which signals propagate along the transmission lines in the coupling region.

Coupling regions in which there is only dielectric (such as for stripline traces) have only one propagation speed; the signals on all traces in the region propagate with this single velocity.

However, coupling regions in which there are boundaries between dissimilar dielectrics (such as microstrip or buried-microstrip traces) have multiple, discrete propagation speeds. Generally, each transmission line in the coupling region propagates some energy at each of the velocities prescribed by the region. There are as many speeds as there are transmission lines in the coupling region.

For most practical cross section geometries, the multiple speeds are all close to each other. However, it is possible to construct highly asymmetric cross sections in which the speeds are quite different. (An example of a "highly asymmetric" geometry would be a microstrip of one width coupled to a buried microstrip of a different width, with the buried trace below and considerably off to the side of the outer-layer trace.) This is an undesirable condition, however, because multiple, widely varying propagation speeds cause signal distortion, as one portion of the signal races ahead of the other(s).

For convenience, the propagation-speeds list displays velocities not only in m/s, but also as a fraction of the speed of light. For example, a value of "0.4822c" means 48.22% of the speed of light.

Tip

A misconception about propagation velocity on a transmission line is that electrons in the conductor are traveling along the line at the propagation velocity. This is absolutely not true! Electrons in a conductor spend almost all of their time randomly colliding with atoms in the conductor lattice; the mean time between collisions is on the order of 10 femtoseconds (1/100th of a ps).

As a result, conduction electrons have only a relatively tiny average forward velocity in the presence of a driving voltage. A typical electron "drift velocity" in a conductor is on the order of 1 foot/hour. Instead, what moves at the transmission line's propagation velocity is the electromagnetic wave that constitutes the actual signal on the line. Indeed, this wave is what you measure in the lab with an oscilloscope: a voltage waveform, which is really a measure of the electric field associated with the traveling electromagnetic wave.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Choose a Propagation Mode to Plot

Percentage of Energy Matrix for Multiple-Speed Coupling Regions Only

If the coupling region supports multiple propagation speeds, this matrix gives, for each transmission line in the region, the percentage of signal energy that travels at each speed. In the matrix, each column represents a line (a trace); reading down the column shows how much of the signal energy in that line travels in each of the propagation modes listed in the propagation-speeds list. The percentages in each column add to approximately 100%, to fully account for the energy in each transmission line.

The values in this matrix are usually only of limited interest, unless the matrix shows a very uneven breakdown in energy sharing between propagation modes. For example, for certain highly asymmetric (and unusual) geometries, it is possible to have certain transmission lines carrying most of their energy in one mode, while others carry a more even mixture of modes. If the velocities between modes differ significantly, this uneven distribution could lead to noticeable skew between signals on the lines.

Impedance and Termination Summary for Two-Line Coupling Regions Only

For the special case of a two-line coupling region, the field-solver numerical results report gives additional information about specific termination options.

Table 5-11. Impedance and Termination Summary (Two Transmission Lines Coupling Regions Only)

Termination Type	Description
differential	This is the proper line-to-line resistor to use if the two transmission lines are being driven differentially (with equal-but-opposite signals). Will not terminate common-mode signals at all.
common-mode	This is the proper line-to-ground resistor to use for each line if the two transmission lines are being driven identically, i.e., with equal signals of the same polarity. Not very useful for signals that sometimes switch together and sometimes oppositely, unless crosstalk is primarily of concern when they switch together.
line-to-ground	This is the best line-to-ground resistor to use for each line if the signals on the transmission lines are completely unrelated. Will not perfectly terminate the line, but is a good single-component “compromise” value.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 5-11. Impedance and Termination Summary (Two Transmission Lines Coupling Regions Only) (cont.)

optimal termination	Describes the theoretically optimal resistor-array termination; consists of a line-to-line resistor plus two line-to-ground resistors. Same as the values given in the Optimal Terminator-Resistor Array matrix. Successfully terminates differential, common-mode, or mixed signals, but may violate DC-bias conditions on the lines.

Taking Measurements From an Oscilloscope Waveform or Eye Diagram

You can take measurements from a Digital Oscilloscope waveform or eye diagram, and specify an eye mask that overlays the eye diagram.

Prerequisites

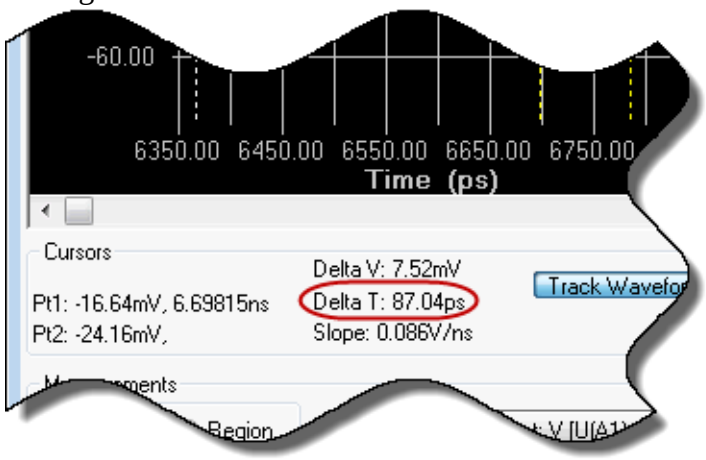
- Run simulation in the Digital Oscilloscope.

Procedure

Obtain measured values by using automatic measurements or by manually selecting points on waveforms or eye diagrams.

Type	Do the following...
Automatic	1. In the Measurements area, select Region and click-drag in the diagram to specify a time region where you want to take the measurement. This is helpful for multiple-period waveforms, when you want to exclude “warm up” bits that are transmitted before the channel, transmitter, or receiver arrives at the normal operating condition. 2. Select the waveform that you want to measure in the Waveform list. 3. From the Measurements toolbar, select a measurement button to display the measurement.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Type	Do the following...
Manual	1. To display threshold voltages from IBIS models as dashed horizontal lines on a diagram, select a pin from the “Thresholds for” list. 2. Pan and zoom to the part of the waveform that you want to measure. 3. Enable **Track Waveform**, select the waveform in the main screen, and select the specific location to place the first measurement marker. 4. Select the second point on the waveform. Voltage and time differences display in the Cursors area, as shown in the following image. To select a point on another waveform, disable and enable **Track Waveform** again and select the second waveform. 

Related Topics

Digital Oscilloscope Dialog Box

Automatic Measurements in an Oscilloscope Waveform or Eye Diagram

Use automatic measurements to perform voltage and timing measurements on voltage waveforms or eye diagrams currently displayed in Oscilloscope.

You can run automatic measurements on single-edge transitions, all transitions in the simulation including multiple-cycle waveforms, or on transitions located in a region of simulation time that you define. The oscilloscope finds all possible occurrences of the selected measurement and reports the worst and best-case results.

Measurements are made only on the waveforms or eye diagrams currently displayed in the Digital Oscilloscope, similar to a hardware oscilloscope. This means that additional information available from IC models, such as Vcc and Gnd voltage levels, are not used during automatic measurements. However, the flight time measurement runs a separate simulation to obtain time-to-Vmeas.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

Batch simulation performs a separate DC simulation to determine the high/low level voltages. In some cases, oscilloscope and batch simulation measurements may not match. By contrast, automatic measurements use only the waveforms displayed in the oscilloscope to establish high and low voltage levels.

Automatic measurements are only available for voltage waveforms (not current).

Determining High and Low Level Voltages - V_high and V_low

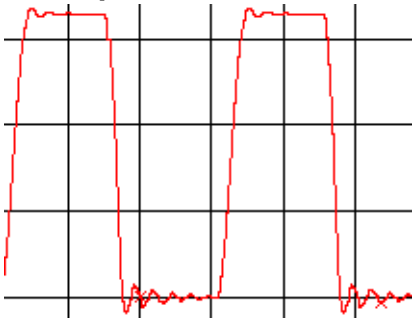
To measure overshoot, rise/fall time, and so on, the software identifies voltages representing the high level (V_high) and low level (V_low). The method it uses to identify high and low levels depends on the shape of the waveform and whether you have enabled eye diagram operation.

- V-high and V-low for a Waveform

  The oscilloscope evaluates the shape of the waveform and determines whether significant flat spots or plateaus exist between transitions.

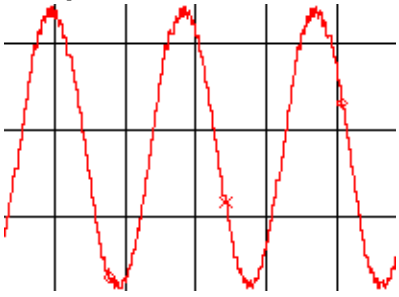
  When a waveform has significant flat spots or plateaus between signal transitions, the software considers the most-common high and low voltages to be the high and low levels.

Figure 5-10. Example of a Waveform With Plateaus



When a waveform does not have significant flat spots or plateaus between signal transitions, the software considers the minimum and maximum voltages in the waveform to be the high and low levels.

Figure 5-11. Example of a Waveform Without Plateaus



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

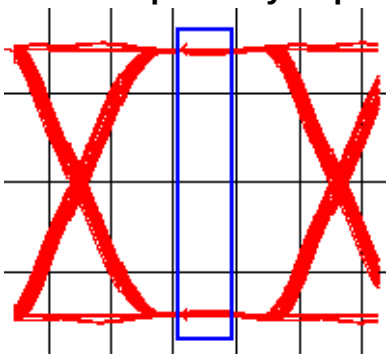
On a waveform without plateaus, measured overshoot is zero volts because the software cannot establish high or low level voltages, which overshoot measurements reference.

- V-high and V-low for an Eye Diagram

  The software calculates the average high and low voltages inside an eye aperture box. The aperture box width is 20% of the bit interval and it is positioned at the UI value you specify in the Eye Height Sampling dialog box.

  Figure 5-12 illustrates an eye diagram with an eye aperture box.

Figure 5-12. Example of Eye Aperture Box



Determining Measurement Threshold Voltages - V_high_ref and V_low_ref

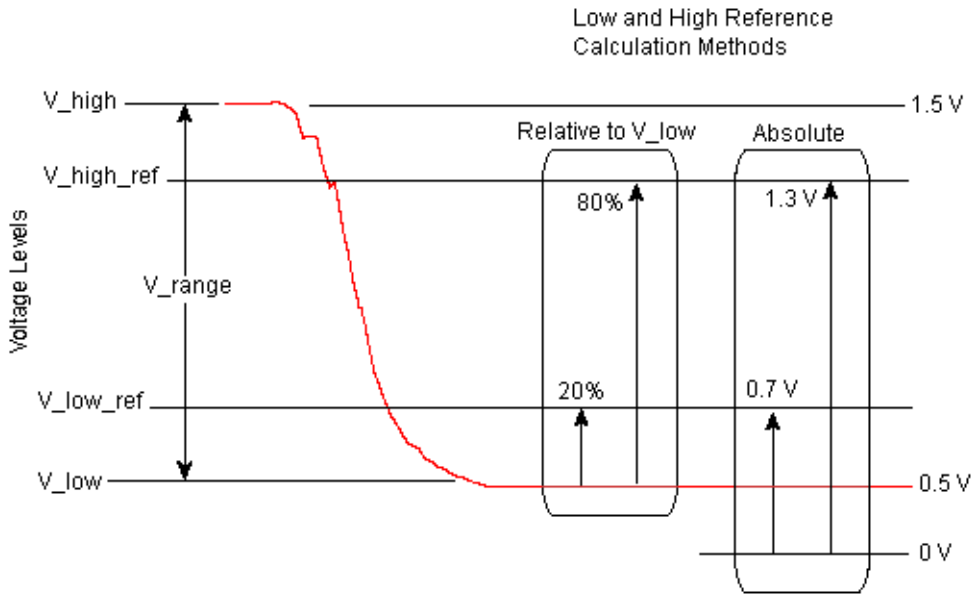
To perform timing-related measurements, such as rise/fall times, the software must know at which voltages to start and stop the measurements. These start/stop threshold voltages are high level reference voltage (V_high_ref) and low level reference voltage (V_low_ref).

The signaling technology used by IC models on the net determines threshold voltage values, and whether to specify values based on the high and low level voltages, or absolute voltages.

Figure 5-13 illustrates how the software calculates high/low level reference voltages.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 5-13. Calculating High/Low Level Reference Voltages



Where:

- V_high and V_low represent high and low level voltages
- V_range represents the absolute value of V_high - V_low
- V_high_ref represents either:
  - a relative voltage V_low + (V_range * % High Reference Offset)
  - an absolute high reference offset voltage
- V_low_ref represents either:
  - a relative voltage V_low + (V_range * % Low Reference Offset)
  - an absolute low reference offset voltage

Rising and Falling Overshoot

Overshoot voltage is measured separately for rising (positive) and falling (negative) transitions. The worst-case value is reported.

Rising overshoot = V_max – V_high

Falling overshoot = V_low - V_min

Where:

V_max and V_min are the maximum and minimum voltages for the waveform.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

On a waveform without plateaus, measured overshoot is zero volts because the oscilloscope cannot establish high or low level voltages, which overshoot measurements reference.

Peak-to-Peak Voltage

Peak-to-peak voltage = V_max - V_min

Where:

V_max and V_min are the maximum and minimum voltages for the entire waveform.

Rise and Fall Time

The minimum, average, and maximum rise/fall times are reported.

Rise time (in ns) = T_high_ref - T_low_ref

Fall time (in ns) = T_low_ref - T_high_ref

Where:

T_high_ref is the time at which the waveform crosses V_high_ref.

T_low_ref is the time at which the waveform crosses V_low_ref.

Only the first crossing within a bit interval is used. Any subsequent crossings, perhaps due to ringing, are ignored.

Non-monotonic waveform behavior between thresholds is ignored.

Rise and Fall Slew Rate

The minimum, average, and maximum rise/fall slew rates are reported.

Rising slew rate (in V/ns) = (V_high_ref - V_low_ref) / Rise time

Falling slew rate (in V/ns) = (V_high_ref - V_low_ref) / Fall time

Non-monotonic waveform behavior between thresholds is ignored.

Flight Time

The oscilloscope and batch simulation measure flight time in the same way. They both obtain time-to-Vmeas from a separate simulation. Non-monotonic waveform behavior between thresholds is ignored.

If the waveform contains multiple cycles, the oscilloscope reports best-case (minimum) and worst-case (maximum) flight times.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

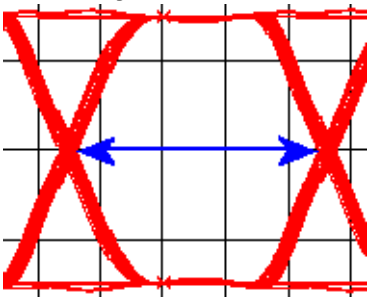
Restrictions:

- You can measure flight time only on waveforms for receivers.
- The receiver waveform can be associated with only one driver, that is, only one driver can be enabled on the net during flight time measurements.
- The oscilloscope cannot measure flight time when the receiver or driver has a SPICE model. This type of model does not provide Vmeasure, Vih, and Vil information required to perform flight time compensation.
- Flight time measurements are available only for the latest simulation. This is because previous and loaded results do not provide information about the simulation conditions that produced the waveform, such as IC models (which contain Vmeas, Vih, and Vil).

Eye Width

Eye width represents the distance in time between the right and left sides of the inner boundary of the eye diagram, as measured at the midpoint voltage, which is halfway between V_low and V_high. The following figure shows a symmetric eye diagram where the eye width is maximum at the midpoint voltage. This is not always the case, and the eye width may be narrower at the midpoint voltage than at another voltage.

Figure 5-14. Eye Width Measurement



Eye width measurements made by the oscilloscope do not include a guardband. By contrast, hardware digital oscilloscopes may apply a guardband such as three sigma.

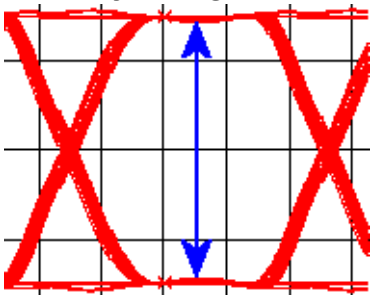
Eye Height

Eye height represents the distance in voltage between the top and bottom sides of the inner boundary of the eye diagram, as measured at a location you specify within the unit interval (UI).

Measure the eye height at the location you believe the receive circuitry actually samples the state of the eye. Figure 5-15 illustrates a measurement location halfway across the UI.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 5-15. Eye Height Measurement



Eye height measurements do not include a guardband. Digital oscilloscopes may apply a guardband, such as three sigma.

Related Topics

Digital Oscilloscope Dialog Box

Reading FastEye Diagram Automatic Measurements

The FastEye Channel Analyzer automatically reports FastEye diagram measurements in the FastEye Viewer.

Procedure

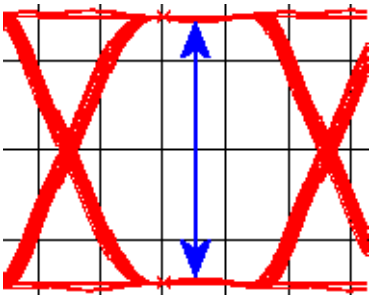
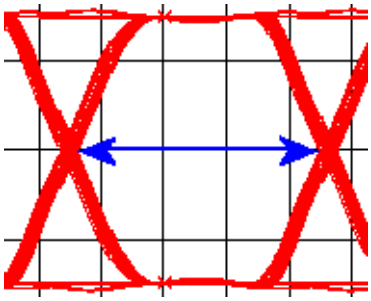
1. In the Show area, select the waveform to measure from the Probes spreadsheet.

   Waveform names correspond to the probe you specified in the FastEye Channel Analyzer - Set Up Channel Characterizations Page.

   If you re-run FastEye channel analysis on the same pin, a new row at the bottom of the Pins spreadsheet corresponds to the latest results.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

2. Measurement results are displayed near the lower-left corner of the dialog box.

Measurement	Description
Maximum eye opening	The maximum distance in voltage between the top and bottom sides of the inner boundary of the eye diagram.  Eye height measurements do not include a guardband. By contrast, test bench oscilloscopes may apply a guardband, such as three sigma.
Eye Width	The distance in time between the right and left sides of the inner boundary of the eye diagram, as measured at the midpoint voltage, which is halfway between V_low and V_high.  For information about V_low and V_high, see Determining Measurement Threshold Voltages - V_high_ref and V_low_ref. Eye width is reported in units of both time and unit interval (UI). You specify the UI length in the FastEye Channel Analyzer - Define Stimulus Page. Eye width measurements do not include a guardband. By contrast, test bench oscilloscopes may apply a guardband, such as three sigma.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Measurement	Description
Period that makes the smallest eye opening	Identifies which period, in the sequence of periods that make up the overall simulation, has the narrowest eye width. This information enables you to examine the waveforms and bit stimulus preceding the named period. You can display detailed FastEye waveforms by enabling the All traces option on the FastEye Channel Analyzer - View Analysis Results Page (prior to analysis) and enabling standard operation in the FastEye Channel Analyzer (when analysis completes). To calculate the offset in the simulation for the start of the period with the smallest eye opening, use the following expression: (period # - 1) * bit interval. The period number starts at 1. **Example:** If the period # is 10 and the bit interval is 3.3 ns, then (10 - 1) * 3.3 ns = 29.7 ns.

Related Topics

FastEye Viewer

Measuring FastEye Diagrams Manually

You can perform precise time, voltage, current, and slew-rate measurements from the FastEye Viewer screen using measurement crosshairs or simply by observing pointer position information.

You can place measurement crosshairs anywhere on the FastEye Viewer screen. When you place two measurement crosshairs, the viewer automatically displays delta time information.

Note

When you position the pointer over the FastEye Viewer screen, its voltage and time position is displayed in the Cursor field in the Cursors area.

You can use the pointer position as an alternative method to measurement crosshairs to perform quick measurements. Simply point to the waveform, hold the mouse steady, and then look at the Cursor field.

Procedure

1. Click on the screen where you want to make a measurement.

   The first measurement crosshairs appears and its voltage and time appear in the Cursors area next to Pt1.

2. To measure a delta voltage or delta time, click on the screen where you want to make a second measurement.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The second measurement crosshairs appears. Its voltage and time appear in the Cursors area next to Pt2. The time and voltage differences between the two measurement crosshairs appear next to Delta V, Delta T, and Slope.

3. To turn off the measurement crosshairs, either:
   - Click over the screen a third time.
   - Click **Erase**. Clicking Erase also erases the FastEye data.

Related Topics

FastEye Viewer

Displaying Waveform Results in EZwave

You can open manually EZwave to see simulation results previously saved to a waveform database.

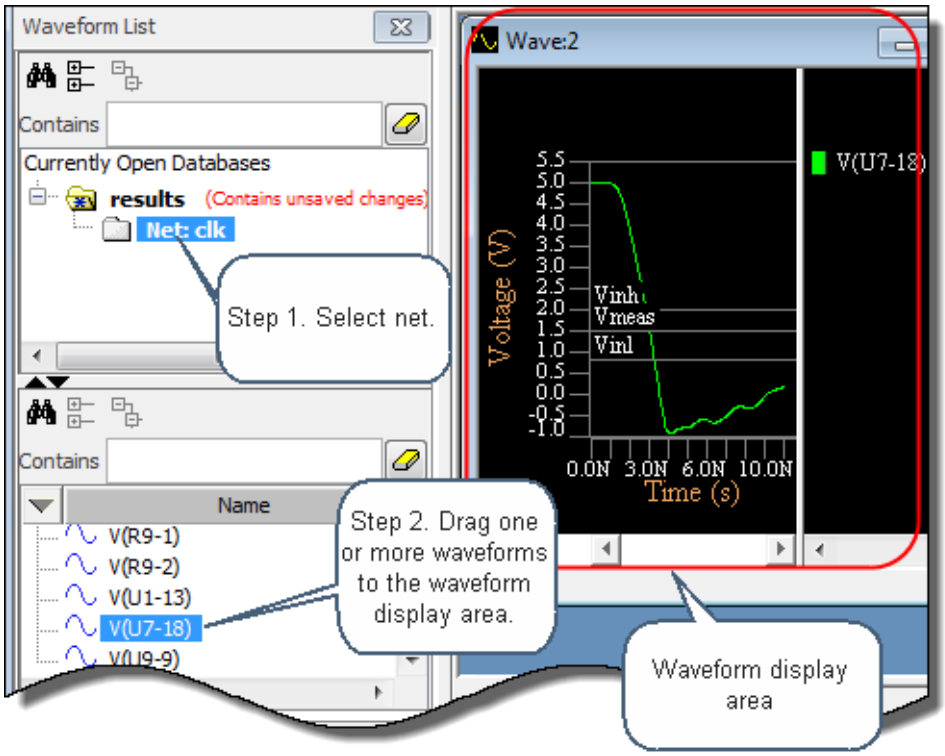
Procedure

1. Choose the **Simulate SI > Open EZwave** menu item. EZwave opens.
2. Choose the **File > Open** menu item, browse to a waveform database (*.wdb*), then click **Open**.
3. Select the net and drag one or more waveforms to the waveform display area.

   Dragging additional waveforms to the waveform display area overlays them. To display the waveform by itself in the waveform display area, double-click the pin. See the figure below.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

EZwave supports many other ways to plot waveforms, such as overlaying multiple waveforms by plotting them on the same row. For information, see "Add Waveforms" in the *EZwave User's and Reference Manual*.



Related Topics

Add Waveforms

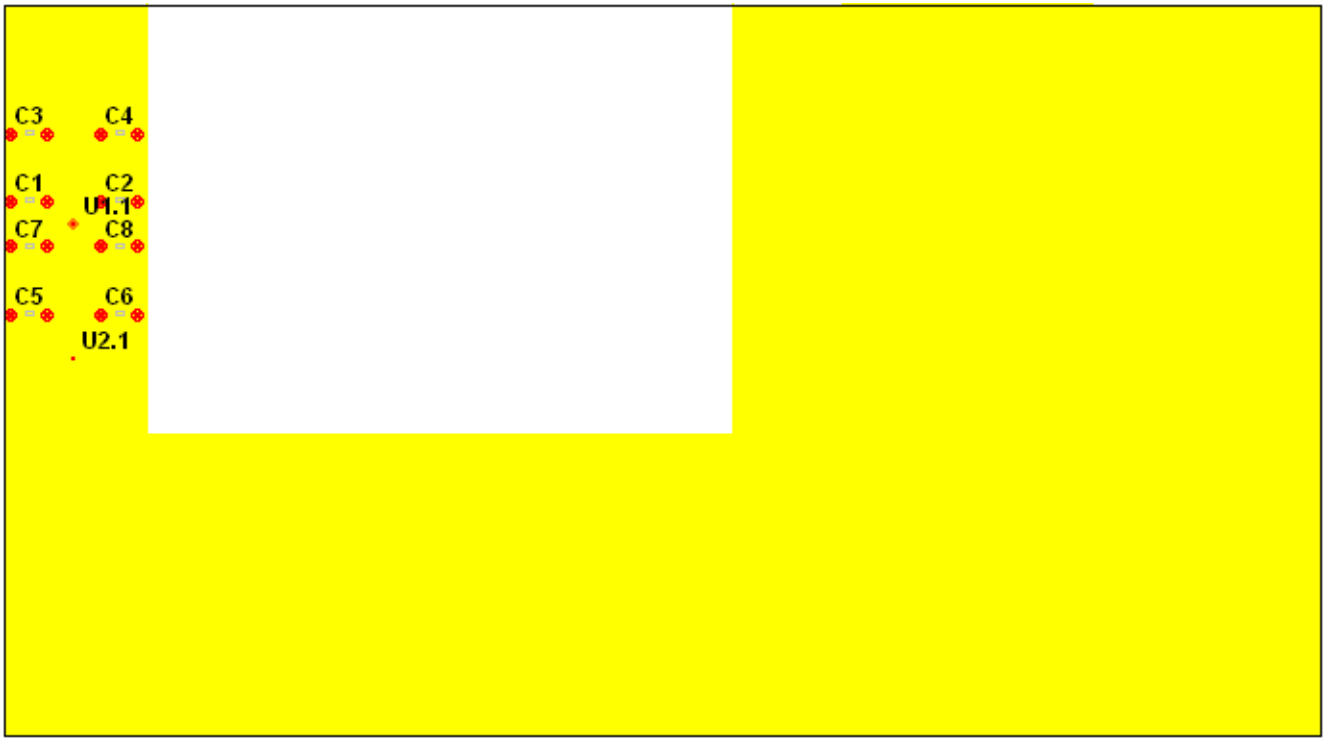
Plane-Noise Simulation Results

When plane noise simulation completes, the HyperLynx PI PowerScope dialog box opens to display voltage and current graphical results.

The figure below shows a simple example design in the PDN Editor with one IC pin and two arrays of decoupling capacitors. The design includes one transmission plane consisting of two layers that have the same geometry.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

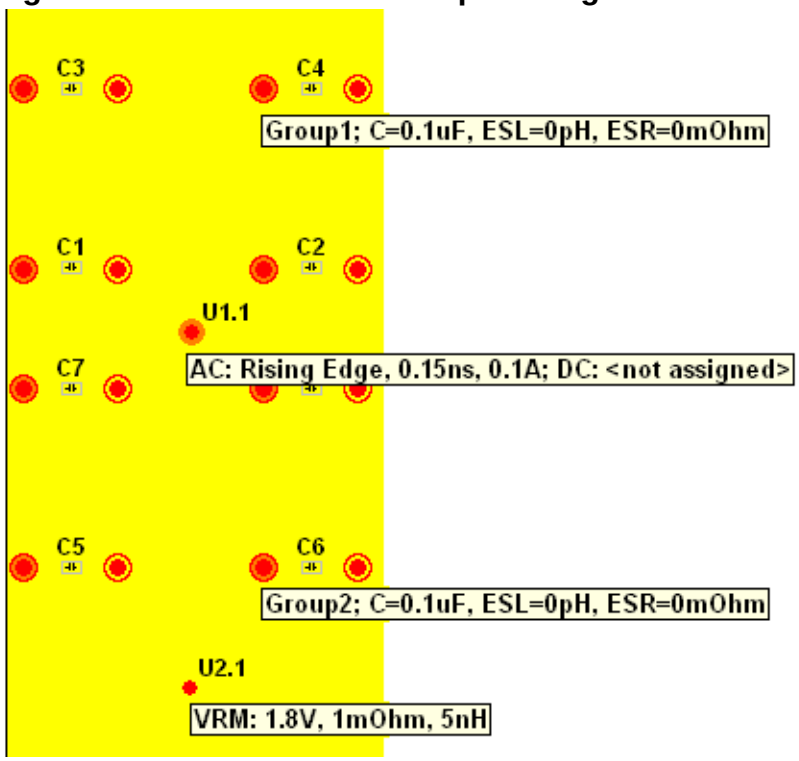
Figure 5-16. Plane Noise Example Design



The following figure shows a close up of the two capacitor arrays (C1-C4, C5-C8), an IC pin with an AC power-integrity model (U1.1), and an IC pin with a VRM model (U2.1). Hover over a component pin to see the assigned power-integrity model(s).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 5-17. Plane Noise Example Design - Zoomed In



Plane Noise Voltage Graphs

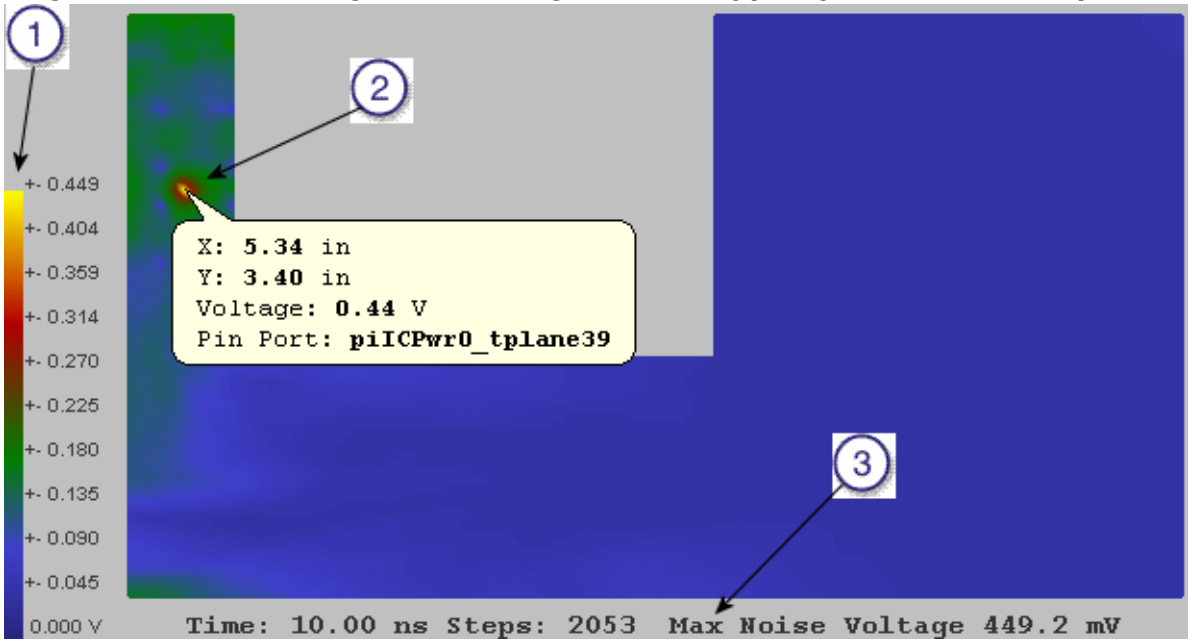
The figure below shows the graphical results for plane voltage noise displayed in the HyperLynx PI PowerScope dialog box.

The HyperLynx PI PowerScope display is set to 2-D, which produces a “top down and flat” display of the power supply net geometries.

To display the ToolTip containing X/Y coordinates, simulation results, and model port information, enable the HyperLynx PI PowerScope “Inspect” mode, and then point to the graph.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 5-18. Measuring Plane Voltage Noise - HyperLynx PI PowerScope 2-D

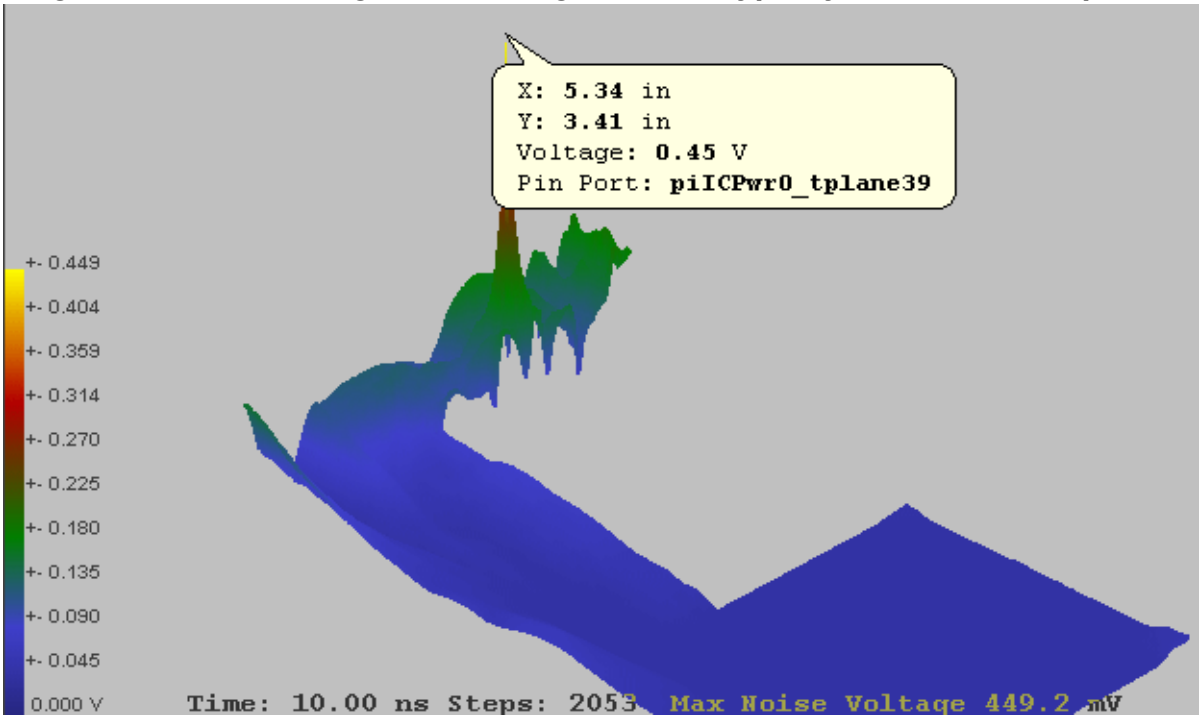


1	Legend that maps the graph colors to the voltage difference at the same X/Y coordinates between layers in the transmission plane. Voltage values increase as the color shifts from blue to yellow.
2	IC pin with AC model. To display the ToolTip containing X/Y coordinates, measured voltage, and model port name (when available), enable the HyperLynx PI PowerScope “Inspect” mode, and then point to the pin.
3	Voltage difference at the same X/Y coordinates between layers in the transmission plane. Voltage values are relative to the origin set in the HyperLynx PI PowerScope. Note the legend and this value have different number of significant digits, so rounding is likely.

The figure below shows graphical results for plane noise displayed in the HyperLynx PI PowerScope. The HyperLynx PI PowerScope display is set to 3-D, which displays the voltage difference between the layers in the transmission plane in the Z axis (or height). You can rotate the graph, and this particular orientation was chosen to emphasize the location of the IC pin. Also notice the low voltages at the decoupling-capacitor locations.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 5-19. Measuring Plane Voltage Noise - HyperLynx PI PowerScope 3-D

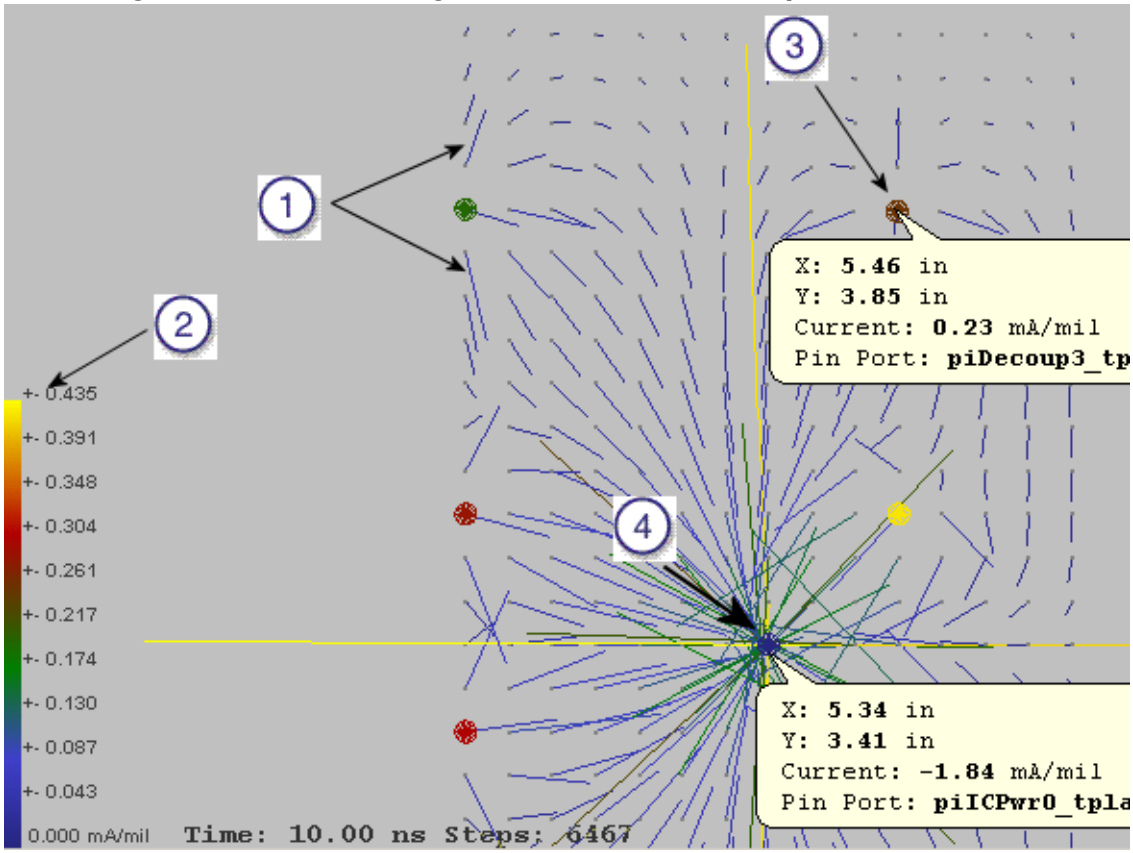


Plane Noise Current Graphs

The figure below shows the graphical results for surface and capacitor currents displayed in the HyperLynx PI PowerScope. The HyperLynx PI PowerScope display is set to 2-D, which produces a “top down and flat” display of the power supply net geometries.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 5-20. Measuring Plane Surface and Capacitor Currents



1	Current vectors that indicate current magnitude and direction.
2	Legend that maps the graph colors to the current magnitude on both layers in the transmission plane.
3	Decoupling capacitor pin. Short lines represent little current flow and may indicate the capacitor is not helping to decoupling the transmission planes.
4	IC pin with AC model.

Related Topics

Running Plane Noise Simulation

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Chapter 6
Exporting or Importing Design and Model Data

You can export nets, boards, power-distribution network (PDN) models, and so on to use them with other simulation software or to analyze their electrical behavior. You can also export constraint templates and design simulation file archives.

Topic	Description
Exporting a Net to an S-Parameter Model	You can create an S-parameter model by exporting selected nets. Use the Extract S-Parameter Model dialog box to export passive networks, such as BoardSim nets and LineSim schematics, to S-parameter models representing equivalent circuits.
Exporting a PDN to an S-Parameter Model	Use the PDN & Channel Model Extractor wizard to create a detailed S-parameter model of the entire power-distribution network (PDN), with ports at IC signal and power pins, and/or signal via locations. The exported S-parameter model accounts for the effects of the entire PDN, including stitching vias, decoupling capacitors, and buried capacitance.
Files Written by PDN Model Extraction	PDN model extraction writes S-parameter, log, and optional power-integrity wizard option files to the <design> folder.
Exporting a Net from BoardSim to LineSim	You can identify board design changes that solve specific SI or PI problems by selecting the problem signal or power nets and exporting them to LineSim.
Exporting a Net to a SPICE Netlist	You can extract the detailed physical information of a net, convert it to electrical data, and then write it to a SPICE netlist. Use the SPICE netlist to simulate the interconnect in SPICE or as a way to view the electrical characteristics of the interconnect.
Exporting Part of a Board Design for Analysis in HyperLynx Full-Wave Solver	You can export a 3-D region of a board design (such as a SERDES channel or signal via) to HyperLynx Full-Wave Solver to run 3-D electromagnetic (EM) simulation, create an S-parameter model, and observe the model to evaluate the interconnect behavior in the frequency domain.
Exporting a Board to IBIS EBD Models	Use the EBD model generator to create an IBIS EBD model from a board file. The EBD model generator is highly automated, so you need only a minimal knowledge of the syntax needed to create an EBD model.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
EBD Models Generated by BoardSim	When an EBD model is generated in BoardSim, the model contains only external nets and has certain limitations for simulation.
Exporting a Board to ICX	Create the files you need to simulate a board design in ICX.
Exporting a Schematic to BoardSim	Use this procedure to create a BoardSim board file that is electrically equivalent to a LineSim schematic. This allows you to define “what if” interconnects in LineSim for a board that has not been laid out yet.
Exporting a Constraint Template From LineSim	Use this procedure to export a constraint template file and to optionally update Constraint Manager, based on properties of the selected LineSim net. This capability enables you to define net topologies in LineSim that have good signal integrity and transfer that information to constraint-aware Siemens EDA software, such as Constraint Manager or Constraint Template Editor (CTE).
Exporting a Net from Constraint Manager to a Schematic	You can export one or more electrical nets from the Nets page of the Constraint Manager spreadsheet to a LineSim schematic. For information about this process, see the *Constraint Manager Users Manual*.
Importing and Exporting Constraints, Net Classes, or Stimulus	You can import or export constraints, net grouping (net classes), or stimulus assignments for nets that you want to include in interactive simulation with measurements, interactive sweeps with measurements, or General Batch simulation.
Exporting and Importing a Stackup	You can reuse a stackup among designs. Exporting and importing a proven stackup can save time when preparing a design for simulation. You can also create a backup copy of the stackup, which is helpful when performing multiple “what if” experiments.
Archiving Design Simulation Files	Use the Archive Design utility to automatically gather and compress design simulation files for your board or schematic. You can use the archive as a “snapshot” of your design simulation files at a specific moment in your PCB design cycle. If you have experienced problems with a board or schematic, you can send the design archive to technical support for investigation.

Exporting a Net to an S-Parameter Model

You can create an S-parameter model by exporting selected nets. Use the Extract S-Parameter Model dialog box to export passive networks, such as BoardSim nets and LineSim schematics, to S-parameter models representing equivalent circuits.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

You can also retain the SPICE netlist and simulation run file that are automatically created when exporting an S-parameter model by enabling “Do not delete generated simulation netlists” on the Preferences Dialog Box - Advanced Tab.

Prerequisites

- If you enabled HSPICE for interactive simulation on the Preferences Dialog Box - Simulators Tab, you must change simulators and enable ADMS before exporting S-parameter models.
- HyperLynx exports standard mode S-Parameter models. To generate a mixed mode s-parameter mode, you must translate the standard mode model after export, see Convert Mode Dialog Box.
- You have acquired the SPICE Output and Advance Scope licenses.

Procedure

1. If you have a board design:
   a. Select a net for simulation.
   b. The exported S-parameter model includes nets associated conductively or by coupling. If crosstalk is enabled, adjust the coupling threshold voltage so that the board viewer displays the aggressor nets you want to include in the S-parameter model.
   c. If you are exporting a differential pair from BoardSim, assign an IBIS model containing the [Diff Pin] keyword to the output pins. This ensures that all segments of the differential pair are considered coupled during the export.
2. If you have a schematic design:
   a. Draw the schematic you want to model. The S-parameter model includes the effects of the schematic elements, such as transmission lines and resistors.
   b. The hierarchical port symbol provides a convenient way to add a port when you do not plan to an assign IC model to that portion of the circuit.
3. To export the net:

If you have a...	Do this...
Board design	Choose the **Export > Net To > S-Parameter Model** menu item.
Schematic design	Choose the **Export > S-Parameter Model** menu item. The exported model can have ports for all IC pins in the schematic.

The Extract S-Parameter Model dialog box opens.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

4. Map IC pins to S-parameter model ports by doing either of the following:
   - Click in a Port cell type an integer port number to add the IC pin to the model, or select NC to omit the IC pin from the model.
   - Click **Map Auto** to automatically assign port numbers for all IC pins.
5. Type the frequency range of the S-parameter model in to the min and max frequency fields.
6. Select the type of sampling to perform from the Sweeping Type list and do the following:

Sampling Type	Instructions
Linear	Enter the number simulation points. Increase the number if you simulate with the exported S-parameter model and run out of simulation points.
Logarithmic	Enter the number of points per decade.
Adaptive	Setting “Accuracy at resonances” to High produces the most-accurate S-parameter model. Adaptive sweeping varies the sampling step size depending on model characteristics. It increases the sampling rate at resonant frequencies and other high-activity response events.

7. In the Reference Impedance area, specify the impedance to which the model parameters are normalized by selecting one of the following:
   - Typical for Channel/Connector — The value is 50 ohms.
   - Typical for Decoupling — The value is 0.01 ohms.
   - Custom — Type a value in the box.
8. To display the new S-parameter model in the HyperLynx Touchstone and Fitted-Poles Viewer, check “Automatically display results”.
9. Click **Create Model**.

   The Save S-Parameter Model dialog box opens.
10. Type or browse to the model folder and file name, and click **Save**.

    A standard mode S-parameter model is now saved.
11. (Optional) To generate a mixed mode S-parameter model, you must translate the standard mode model after exporting. See Convert Mode Dialog Box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Results

The software creates S-parameter models that are not purely linear or logarithmic. A sophisticated adaptive sweep technology increases the sampling rate at resonant frequencies and other high-activity response events to create a standard mode S-parameter model.

Related Topics

Viewing and Measuring Model Curves

Preferences Dialog Box - Simulators Tab

Exporting a PDN to an S-Parameter Model

Use the PDN & Channel Model Extractor wizard to create a detailed S-parameter model of the entire power-distribution network (PDN), with ports at IC signal and power pins, and/or signal via locations. The exported S-parameter model accounts for the effects of the entire PDN, including stitching vias, decoupling capacitors, and buried capacitance.

You can open the model in the Touchstone and Fitted-Poles Viewer to study insertion and return loss, or include the model in other simulations.

Restrictions and Limitations

- You cannot export a PDN model from a MultiBoard project.
- You cannot export a PDN model from a board design that includes multiple stackups.

Prerequisites

- You have set up and verified the design setup and model assignments. See Verifying That the Software Recognizes Your Design Correctly and Assigning Models for PI Simulation.
- You have evaluated your design to optimize PDN performance. Run Lumped or Quick Analysis decoupling simulation if needed. See “Decoupling Simulation” on page 238.
- You have ensured that assigned PI models accurately describe the PDN, including a VRM model and associated reference nets. VRMs (with their low resistance DC paths) can significantly lower PDN impedance at low frequencies. See “Assigning Models for PI Simulation” on page 103.
- To include a power supply net in simulation that is connected by a series passive component to a power supply net you have probed, you have checked “Allow AC supply models for non-capacitor components” in the Preferences dialog box. See Preferences Dialog Box - Power Integrity Tab. If you check this option, the wizard displays the Supply Component Models page instead of the Check Capacitor Models page.

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- To enable the software to measure PDN impedance through a group of power supply pins in parallel, you have created power supply pin groups. See “Creating Power Supply Pin Groups” on page 110.
- You have the PDN Model Export license. If you plan to refine project and simulation settings in HyperLynx Hybrid Solver, you have a license for Hybrid Solver.

Procedure

1. Choose the **Export > PDN/Channel Model** menu item. The PDN & Channel Model Extractor opens.
2. For a new model, on the Start Analysis page, select New to create a setup file (*.dao*) that contains all the settings you make in the wizard.

   Select “Use last configuration” to load the most recently used setup file, or select “Load saved configuration” to select a specific setup file.

   To save wizard settings for future analysis or channel characterization click **Save** and specify a file path and name.
3. On the Choose Export Type page, choose the type of model you want to create:

If you want to create a model that describes...	Do this...
Only the PDN	Select “Power-distribution network (PDN)”.
Via-via coupling within the PDN	Select “Signal vias for crosstalk through PDN”.
The PDN and signal nets	Select “Signal nets and PDN including non-ideal return paths”.
Only signal nets	Select “Signal nets only including non-ideal return paths”.

Note

The option you choose on this page determines the configuration pages that appear in the wizard. Some steps in this procedure apply to only specific model types.

4. On the Choose Probe Type page, select one of the following:
   - Automatically — Displays the Auto-assign Ports page, where you select components that enables the software to automatically assign probes for them.
   - Manually — Displays probe-related pages, where you manually assign probes.
5. (If available) On the Select Power Nets page, select power and reference nets from the “Available nets” list on the left, then click the appropriate arrow to add each net to the “Power nets to probe” or “Reference nets” list on the right.

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Note

To help navigate many nets, you can press a keyboard key for the first character of a net name to highlight a name that starts with that character. Press the key again to highlight the next net name.

6. (If available) On the Select Signal Nets page:
   a. Select an option to choose nets by name, net class, or from the nets currently selected in the board viewer.
   b. Select the signal nets that you want to include in the model from the “Available nets” list on the left, then click the arrow to add nets to the “Signal nets to export” list on the right.
   c. (Optional) To ensure that the Select Power Nets page includes power-supply nets associated with signal nets that you select on this page, do the following:

Note

The software associates a signal net to a power-supply net in two ways. They can associate with a passive component that connects them, such as a two-pin resistor or a resistor package. If you have assigned an IBIS model to a component with a pin that connects to a selected signal net, a [Pin Mapping] keyword can associate a signal net to power and reference nets/busses.

      i. Click **Add accompanying power nets**.
      ii. If you selected “Signal nets and PDN” on the Choose Export Type page:
         a. The software adds all associated power and ground nets to the “Power nets to probe” list on the Select Power Nets page.
         b. Go to the Select Power Nets page, move reference nets to the “Power nets to probe” list. For example, if the software added GND to the “Power nets to probe” list, move it to the “Reference nets” list.
7. On the Check Capacitor Models or Supply Component Models page:
   a. Verify model assignments and values. If needed, assign or remove models, and edit groups.

Note

Assign a model to a resistor or inductor that connects a power supply net that you probe to a power supply net that you do not probe, but want to include in simulation.

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For decoupling capacitor ESL, the software uses a manually-specified ESL value, instead of calculating it based on capacitor body size. If you define a decoupling capacitor model with the:

- Assign / Edit Capacitor Model Dialog Box, the software uses the “Specify value” field value, even if you select the “Auto-calculate” option.
- QPL-File Editor, the software uses the “ESL by value” field value, even if you select the “ESL by capacitor size” option.

b. To perform a “what if” experiment by excluding a specific model from simulation, uncheck Enabled for the model you want to exclude.

8. (If available) On the Specify Via Ports page, select the vias that you want to include in the model.
9. On the Check VRM Models page, verify model assignments, then choose whether to observe PDN behavior at VRM pins:
   a. To display VRMs assigned only to nets that you selected on the Select Power Nets page, check “Show VRM(s) for selected nets only”.
   b. To add or edit a VRM model or reference net, click **VRM Manager**. The Assign Power Integrity Models dialog box opens. For information, see step 3 in “Assigning VRM Source, DC Sink, and AC Models” on page 103.
   c. To observe PDN behavior at pins for a VRM, ensure that its right-most spreadsheet column has a check mark. An empty blue box in the right-most column indicates missing model information, such as a reference net.
10. (If available) On the Auto-assign Ports page, select component(s) to probe, then do any of the following:
    a. To enable the software to create pin groups, check “For components with no user-defined pin groups, automatically create groups before creating probes”.
    b. (If available) To enable the software to minimize the distance between port terminals for a signal net by adding a pin to a neighboring reference area fill, check “For signal pins, create virtual reference pins on nearby metal”.

       When checked, the software creates a port between the signal pin and the newly-created pin.

       When unchecked, the software creates the port across the signal pin and an existing neighboring reference pin.
    c. If you need to manually specify pin groups, click **Manage Pin Groups**. See “Creating Power Supply Pin Groups” on page 110.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

11. (If you chose to specify probes manually, on the Choose Probe Type page) Specify the pin pairs or pin group pairs that you want to include in simulation:

    Note

    You cannot select a pin pair that contains a pin that belongs to a pin group that you have selected on the Select Group Pair Probes page.

    Point to an empty blue box to see the pin group name.

    The software represents each pin pair (or pin group pair) that you select as a port in the output model. The more pin pairs (or pin group pairs) you select, the longer simulation takes and the larger the model file becomes.

    a. To include a pin pair in simulation, click the Select IC Pin-Pair Probes page. Check all the IC power supply pin pairs that you want to include in simulation.

       If a pin pair that you want to simulate is missing from the list:

       i. Click **Edit Pin-Pair Probes**. The "Create IC power pin pair" dialog box opens.

       ii. Click a "+" symbol for a reference designator to see a list of pins. You can specify a pin pair by doing either of the following:

           - Select a supply pin, then click **Find Reference**.
           - Click a row in the supply net list and the reference net list, then click **Add Probe**.

           The board viewer or PDN Editor marks pin locations with blue (return pin) and red (power pin) arrows.

       iii. Click **OK** to return to the wizard.

    b. To include a pin pair group in simulation, click the Select Group Pair Probes page. Check all the IC power supply pin pair groups that you want to include in simulation.

       If a pin pair group that you want to simulate is missing from the list:

       i. Click **Manage Pin Groups**.

          The Pin Group Manager dialog box opens. If pin groups do not appear in the list, click **Create Groups** to create them now, then click **OK**. See Creating Power Supply Pin Groups.

       ii. On the Select Group Pair Probes page, specify a pin pair group that you want to include in simulation. Click **Edit Pin-Group Probes**. The Create IC pin group pair dialog box opens.

       iii. Click a "+" symbol for a reference designator to see a list of pins. Click a row to specify a supply and return pin group pair. The board viewer or PDN Editor marks pin locations with blue and red arrows.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

iv. Click **Add Probe** then **OK**.

12. On the Normalization Impedance page, use the default normalization impedance unless you intend to compare the exported S-parameter model to a model that was created with a different normalization impedance.
13. On the Choose Easy/Custom page, select Custom only if you need to edit any of the less-commonly used options on these pages: Customize Settings, Control Frequency Sweep.
14. On the Customize Settings page:

If you want to reduce simulation run time and memory consumption by…	Do this…
Merging small details in complex metal shapes	Check “Defeature metal areas” and specify the minimum dimension of a feature to include in simulation.
Excluding metal located outside of a perimeter that encloses all the pin pairs and pin group pairs.	Check “Crop layout data”.
Excluding metal located on a power supply net with no probes.	Check “Include only metal for nets with probes”.

If you want to…	Do this…
Include additional power supply nets in simulation that are connected by a resistor or inductor to power supply nets you have probed.	Uncheck “Include only metal for nets with probes”.
Load the design and project into HyperLynx Hybrid Solver, where you can make design changes and run simulation.	Check “Show HyperLynx Hybrid Solver GUI”.

For a via-to-via coupling model only:

If you want to…	Do this…
Account for energy coming from transmission planes formed by other pairs of power supply nets and transmitted through stitching vias that run vertically throughout the stackup.	Check “Include all power / ground nets in the analysis”.
Account for capacitor mounting inductance. See on-screen text for additional information.	Check “Include capacitor mounting inductance”.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to…	Do this…
Reduce simulation run time and memory consumption by automatically finding stitching vias that are located close together, and then merging their individual models into an equivalent model.	Check “Enable stitching-via optimization”. See “Stitching-Via Optimization” on page 529.

15. On the Control Frequency Sweep page, type the minimum and maximum extraction frequency. Set the maximum frequency to include the largest significant frequency in your design signals.

    a. If you plan to use the model in time domain simulation, check “Include DC point” (if the option is available for the selected model type).

    Note

    Time-domain simulation requires a voltage value at zero Hz in the S-parameter model. Without it, the model may simulate with incorrect DC levels (DC offset).

    b. Select a sampling type:

If you want to…	Do this…
Vary the sampling step size with frequency. The software uses a mixture of linear and log sampling to solve a minimum number of samples at specific frequency ranges, then distributes the remaining number of samples by interpolating among the solved frequencies.	1. Select Auto. 2. Adjust the Accuracy slider: • Low — Your PDN has low Q resonance throughout the frequency range, or you want to get an initial idea of your PDN’s frequency response. This option has the shortest run time. • High — Your PDN has high Q resonance throughout the frequency range, or you want maximum accuracy for final verification. This option has the longest run time. 3. Enter the number of samples for the entire frequency range.
Distribute sampling points across the frequency range at logarithmic intervals	1. Select “Logarithmic sampling”. The intervals between sampling points are smaller at lower frequencies and larger at higher frequencies. With logarithmic sampling, every next frequency point is equal to the previous value times a factor K > 1. 2. Enter the number of samples for the entire frequency range.

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If you want to…	Do this…
Distribute sampling points across the frequency range at equal intervals	1. Select “Linear sampling”. 2. Enter the number of samples for the entire frequency range.

16. On the Run Analysis page, type a folder name to store simulation results, then click **Run Analysis**.

Note

The **Check Data** button changes to **Run Analysis** when you have provided the minimum required information on all wizard pages.

Results

When simulation completes, the Touchstone Viewer displays an S-parameter model. The software saves results and wizard settings file (*.dao*) in a timestamped folder in the design folder.

To help you map Touchstone model ports to component pins, see the *<design>_<iteration>.z<number_of_ports>p.ports* file.

Related Topics

PDN Model Extractor Wizard

Files Written by PDN Model Extraction

Files Written by PDN Model Extraction

PDN model extraction writes S-parameter, log, and optional power-integrity wizard option files to the <design> folder.

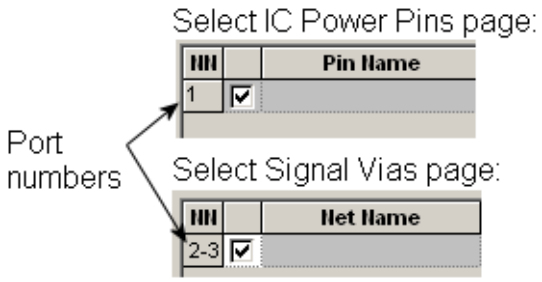
See Design Folder and HyperLynx Files.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 6-1. PDN Output Files - Exporting

File	Description
<design>_<M>.s<N>p	S-parameter file showing the signal scattering characteristics of the PDN over a frequency range. Use the Touchstone Viewer to view S-parameter files. See "Viewing and Converting Touchstone and Fitted-Poles Models" on page 1293. File name is of form <design>_<M>.s<N>p where: • design—name of the board or LineSim schematic • M—starts at "empty" and increments by one for each export. For example, *design_.s1p* and *design_1.s1p*. • N—<number_of_ports> There is one port for every IC pin. There is one port for every stackup layer connection to a single or differential via.
Port-to-component pin mapping file—<design>_<analysis_iteration>.ports	Maps Touchstone model ports to component pins. The port numbers correspond to wizard spreadsheet numbers. See Figure 6-1.
ME.log	Export PDN log file. Use the Reporter Dialog Box to view log files, such as when investigating analysis failures or unexpected results.
<design>.dao (optional)	Power-integrity wizard options file. Contains the PDN Model Extractor wizard settings.

Figure 6-1. Mapping Spreadsheet Ports to Exported PDN Model Ports



Related Topics

Exporting a PDN to an S-Parameter Model

Exporting a Net from BoardSim to LineSim

You can identify board design changes that solve specific SI or PI problems by selecting the problem signal or power nets and exporting them to LineSim.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

For example, you can use this capability to identify design changes to fix:

- Excessive delay, crosstalk, or incorrectly coupled differential signals by changing routing or component placement.
- Excessive power-distribution network (PDN) impedance or current density by changing decoupling capacitor locations or quantities, geometry of copper pours or voids, location and quantity of stitching vias, and so on.

Signal nets exported to the schematic design can include trace properties, associated nets, IC model assignments, via properties, termination components, coupled segments and area fills, and so on. When you have a multiple-board project open and select a net that connects to nets on other boards, the exported schematic contains those nets and their board-to-board connectors. Use the Schematic Editor to view and edit exported signal nets.

Power supply nets exported to the schematic design can include PDN elements, including IC power supply pins, capacitor pins, vias, board outlines, copper pours and voids, and so on. Use the PDN Editor to view and edit exported power supply nets.

The following video contains additional information:



Restrictions and Limitations

BoardSim does not export:

- Vias on unrouted nets.
- Power supply nets when a multiple-board project is loaded.
- Metal areas used to implement a portion of a signal net.

Prerequisites

- To export coupled segments, you have the BoardSim Crosstalk license.
- To automatically open LineSim after exporting the net, you have the LineSim license.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- If exporting coupled segments, you have enabled crosstalk simulation and have specified coupling thresholds. See “Set Coupling Thresholds Dialog Box” on page 1015.
- If exporting area fills coupled to the selected net, you have enabled crosstalk simulation and have specified geometric coupling thresholds. See “Set Coupling Thresholds Dialog Box” on page 1015.
- If needed, you have removed non-functional pads from signal vias. (**Setup > Remove > Non-Functional Pads** menu item)
- If exporting power supply nets to the PDN Editor, you have enabled “Automatically assign reference layers” in the Preferences Dialog Box - Power Integrity Tab. Enabling this option helps to ensure good power-integrity simulation correlation between BoardSim and LineSim.
- If exporting 3D areas, you have created S-parameter models representing the behavior of the 3D area. See “Add or Edit 3D Area Dialog Box” on page 682.

Procedure

1. Select the net(s) to export. See “Selecting Nets for SI Simulation” on page 141. You can also select nets from the Export to LineSim Free-Form Schematic dialog box.
2. Choose the **Export > Net To > Free-Form Schematic** menu item.
3. Specify the location of the exported schematic file.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

4. In the Export to LineSim Free-Form Schematic dialog box:

If you want to...	Do this...
Export one or more signal nets	1. Check “Export to Free-Form Schematic Editor”. 2. Click **Select** to select one or more signal nets. 3. To export coupled segments on other nets or to export coupled area fills, check “Export coupled segments”. **Restriction:** This option is unavailable when you have disabled crosstalk simulation or the net does not couple to another net or to an area fill. 4. To export signal vias as S-parameter models previously created by the Add or Edit 3D Area Dialog Box and HyperLynx Advanced Solvers, select Export 3D Areas. 5. Selected nets can include multiple pins on the same IC. To export an IC component symbol to represent these multiple pins, instead of multiple IC symbols: a. Check “Use IC component symbol if more than”. b. Type the minimum number of pins on the same IC required to represent an IC with an IC component symbol. 6. To reduce the number of transmission line symbols for an exported net, check “Merge Segments to Neighbors if shorter than”, then type a length.
Export a power supply net	1. Check Export to PDN Editor. 2. Select nets to export.

5. Specify how to represent vias in the exported schematic by selecting an item from the “Export Vias as” list.

Restriction

This option is unavailable when you have disabled crosstalk simulation or the net does not couple to another net or to an area fill.

6. To open LineSim and load the schematic for the exported net, check “Open exported file in LineSim”.

Restriction

This option is unavailable if no LineSim license is available.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

7. To include the electrical contents of EBD models assigned to pins on the net, check “Expand into EBD”.

   Restriction

   This option is available only when an EBD model is assigned to a pin on the net.

8. Click **Export**.

Results

If you enable cross probing, when you export to LineSim, you can select a transmission line symbol (or symbols using Ctrl-click) in LineSim and the BoardSim view will zoom to the net segment. You can also select a net segment (or segments using Ctrl-click) in BoardSim and the LineSim view will zoom to the transmission line symbol. To enable cross probing and “Zoom to Crossprobe” for both LineSim and BoardSim, choose **View > Cross Probing > Enable Cross Probing** and **View > Cross Probing > Zoom Cross Probing**.

LineSim represents area fills coupled to the selected net as reference conductors in coupling regions.

If you exported a PDN with decoupling capacitors, verify the accuracy of exported decoupling capacitor mounting. In some complex intersections of trace segments, vias, and pads, the exported mounting may include structures from adjacent nets. If needed, fix the mounting in LineSim using the Decoupling Mounting Scheme Editor dialog box.

Exporting a Net to a SPICE Netlist

You can extract the detailed physical information of a net, convert it to electrical data, and then write it to a SPICE netlist. Use the SPICE netlist to simulate the interconnect in SPICE or as a way to view the electrical characteristics of the interconnect.

Prerequisites

- The SPICE Output license is required to export SPICE netlists.

Procedure

1. If you want an exported SPICE netlist to include one or more include files that you specify, do the following:

   a. Specify SPICE include files with the SPICE Options Dialog Box.

   b. Do either of the following:

      - ADMS simulator — You have a full ADMS license and use the “Simulation engine” area of the Simulation Controls Dialog Box to manually set the simulation engine to ADMS.
      - HSPICE simulator — Use the “Simulation engine” area of the Simulation Controls Dialog Box to manually set the simulation engine to HSPICE.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

2. Set up the board or schematic design for exporting.

   a. Set up the circuit to export to a SPICE netlist, and optionally assign IC models and set simulation options.

If using...	Do this...
BoardSim	1. Select a net for simulation. See Selecting Nets for SI Simulation. The exported netlist includes the selected net, associated nets, and aggressor nets, when crosstalk is enabled. 2. If crosstalk is enabled, edit the electrical or geometric coupling threshold in the Set Coupling Thresholds Dialog Box so that the board viewer displays the set of aggressor nets to include in the SPICE netlist.
LineSim	Draw the schematic. The exported netlist includes all of the elements in the schematic, such as coupled transmission lines, resistors, and capacitors.

   b. (Optional) Enable lossy simulation by choosing the **Setup > Enable Lossy Simulation** menu item.

   c. (Optional) Assign IC models to the selected net or schematic. The presence or absence of a driver can influence the simulation time step.

   d. (Optional) Use the “Simulation resolution” area in the Simulation Controls Dialog Box to manually set the simulation time step. The simulation time step determines how to model short transmission lines:

Transmission Line Type	Model
Lumped RLC equivalent	The transmission line delay is shorter than the simulation time step.
Distributed	The transmission line delay is longer than the simulation time step.

3. To export the net to a SPICE netlist:

If using...	Do this...
BoardSim	Choose the **Export > Net To > SPICE Netlist** menu item.
LineSim	Choose the **Export > SPICE Netlist** menu item.

4. Enter the file location and click **Save**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Results

- SPICE netlists contain the following information:
  - The interconnect for the nets, which is modeled as SPICE transmission lines.
  - Passive components that are attached to the interconnect.
  - External nodes on the net that can connect to IC pins. When you enable the HyperSim or ADMS simulator from the Simulation engine area of the Simulation Controls Dialog Box, the netlist uses VHDL-AMS entities to specify IBIS I/O models. When you enable the HSPICE simulator, the netlist uses HSPICE B-elements to specify IBIS I/O models.
- Netlists with coupling and loss information use the HSPICE W element, which is compatible only with HSPICE and ADMS. Netlists with no coupling or loss information use the standard T element, which is compatible with most programs derived from Berkeley SPICE.

Related Topics

Creating a Schematic Design

Exporting Part of a Board Design for Analysis in HyperLynx Full-Wave Solver

You can export a 3-D region of a board design (such as a SERDES channel or signal via) to HyperLynx Full-Wave Solver to run 3-D electromagnetic (EM) simulation, create an S-parameter model, and observe the model to evaluate the interconnect behavior in the frequency domain.

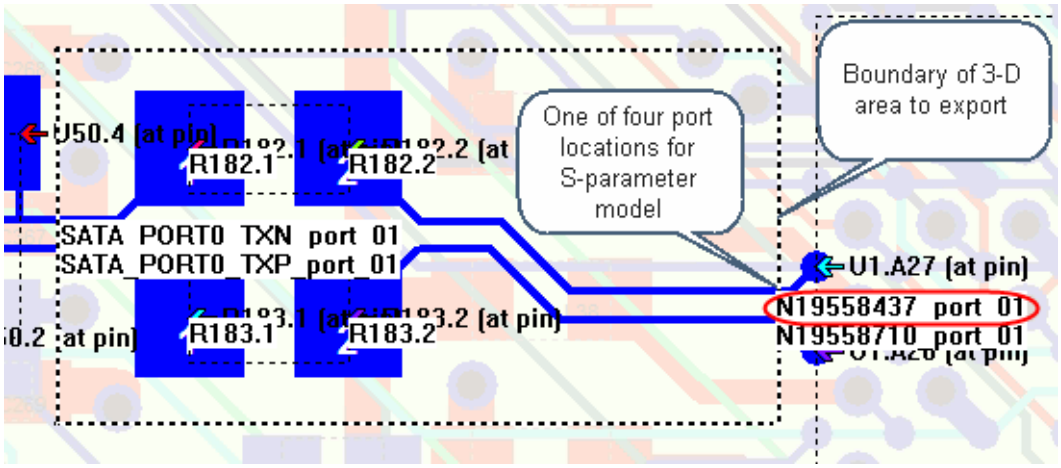
Note

This topic focuses on manually defining an individual 3-D area and creating an S-parameter model to represent it. The software can also automatically define 3-D areas for selected nets and create S-parameter models to represent them. See “Modeling a Board Area With an S-Parameter Model” on page 122.

The software can also use an S-parameter model to represent a 3-D area during simulation when you enable the option “Replace 3D Area with corresponding S-parameter model during SI analysis” (choose the **Setup > Options > General** menu item, click the **BoardSim** tab). See “Modeling a Board Area With an S-Parameter Model” on page 122.

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The dashed line in the figure below shows an area of a board design that is ready for export.



Note

For a schematic design, you can create a 3-D EM model that includes signal vias and nearby stitching vias, and assign the model to a via or differential via symbol. See “Modeling a Via with a 3D EM Model in a Schematic” on page 127.

Restrictions and Limitations

- If you create a 3-D area that overlaps another, you can enable the S-parameter model assignment for one overlapping area at a time. For example, if you create a large 3-D area with multiple differential vias, create small 3-D areas with one differential via (that is located within the large 3-D area), you can enable either the large 3-D area or the small 3-D areas.
- The software does not model via *backdrill setback*. If you specify via backdrill with the Backdrilling/Plating/Constraints Settings dialog box, the software ignores Setback option values when exporting a 3-D area.
- The software models vias as filled. If you specify a hollow via with the Backdrilling/Plating/Constraints Settings dialog box, the software ignores “Conductively filled” option values when exporting a 3-D area.
- If you have a board design with multiple stackups, you can export an area with only one stackup definition.

Prerequisites

- You have acquired the 3D Area Model Export license.
- You have acquired a license for either HyperLynx Full-Wave Solver HPC or HyperLynx Full-Wave Solver.
- If you want a 3-D area to include aggressor nets for a selected (victim) net, enable crosstalk (choose the **Setup > Enable Trace Coupling** menu item).

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- To evaluate the performance of a range of design properties for a 3-D area, you understand "Exploring Design Variations" in the *HyperLynx Advanced Solvers User Guide*, and have acquired the nbgeomcore3d license.
- You have enabled "Show anti-objects in board viewer and use in analysis" (choose the **Setup > Anti-Objects** menu item).
- If your design has flooded plane layers, you have assigned power-supply nets to them (choose the **Setup > Edit Power-Supply Nets** menu item).
- If you plan to solve a 3-D area with multiple computers in parallel, set up job distribution. See "Job Distribution" in the *HyperLynx Advanced Solvers User Guide*.

Procedure

1. Select net(s) to include in the 3-D area.
2. Choose the **Analyze 3D EM > Manage Areas for 3D EM Simulation** menu item.

   The 3D Area Manager dialog box opens.
3. (Optional) If you are an experienced HyperLynx Advanced Solver customer who wants detailed control over the geometric model and solver settings used to create the S-parameter model that represents a 3-D area, specify a folder location in the HLAS Options Folder field.

   This folder contains options files (*.opt*) for Full-Wave Solver HPC, Full-Wave Solver, Fast 3D Solver, or Hybrid Solver. See "HyperLynx Advanced Solvers Option Files for 3-D Areas" on page 551.
4. Click **New**.
5. In the Add or Edit 3D Area dialog box, type a name for the 3-D area. The software uses this name to create the corresponding S-parameter file name.
6. If you have loaded a multiple-board project, select the board that contains the 3-D area you want to export.
7. Specify the 3-D area location:

   Note

   To minimize 3-D EM solver run time, consider exporting only the minimum amount of data needed to simulate the area of interest. Ensure that signal traces are not running along an area boundary. When including signal vias, include all nearby stitching vias.

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Restriction

The software automatically identifies ports for signal nets that intersect the boundary of the export area.

If you want to...	Do this...
Export a rectangular area	1. Select Rectangular from the Shape list and, in the board viewer, drag a rectangle to enclose the area. 2. To edit the area location, you can: • Drag a corner or line to a new location. You can right-click an area outline to display editing options. • Enter new coordinates, then press Enter.
Export a polygonal area	Select Polygon from the Shape list and, in the board viewer, create a polygon to enclose the area: 1. Draw the first line by holding down the mouse button as you move the mouse to the end of the line. Press Shift while dragging a line to make it horizontal, vertical, or diagonal. 2. Draw additional lines by using the same mouse drag movement or by clicking where you want the next line to end (it automatically connects to the previous line). If needed, you can disable snap-to-grid behavior or specify a new grid pitch. You can right-click an area outline to display editing options. 3. Close the polygon by either clicking **Close** or dragging a line to the starting point of the first line. 4. To delete a corner, you can do either of the following: • From the board viewer, point to a corner (), then press Delete. • From the Points spreadsheet, select the first column for the row to delete, then click **Delete**.
Export the entire board	Select Rectangular from the Shape list and check Whole Board.

8. Verify that the area contains the correct set of signal nets, power-supply nets, ports, and stackup layers.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Caution

If the 3-D area includes a passive component that you want to model, ensure that you select the single port (formed across both pins) for that component in the Choose Ports list. The software represents an unchecked series passive component port as an open circuit.

9. To change port numbering, select a # value in the Port Map spreadsheet.
10. Specify a name, location, and frequency range for the S-parameter model.
11. Check Use Absorbing Boundaries to model the area boundary edges as an absorbing material (PCB) and eliminate artificial resonances from the model.

Note

The software slightly expands the exported area on all sides to provide room for an absorbing boundary.

12. Check Use External Reference when a board area under a BGA includes only a partial reference layer or inconsistent reference throughout the area under the BGA. When you select this option, the software adds a plane layer to the 3-D model that represents the reference layer of the BGA package to create a more accurate representation of the design. This ensures that the software creates ports for BGA pins in a consistent location (at the external reference).

Note

In step 3, in the HLAS Options Field, if you specified a folder that contains an *ExpertReferenceOptions.opt* file, consider displaying the added reference plane in HyperLynx Full-Wave Solver to review its geometries and connections to your board. For instructions, see the second table row in step 13.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

13. Create an S-parameter model and assign it to the 3-D area:

If you want to...	Do this...
Create an S-parameter model in HyperLynx Full-Wave Solver to represent the 3-D area and automatically assign it to the area in BoardSim	To use only your computer to create an S-parameter model for a 3-D area: 1. Close the Add or Edit 3D Area dialog box. 2. In the 3D-Area Manager dialog box, select one or more 3-D areas in the 3D areas table. 3. Click **Solve Highlighted Areas**. In the background, HyperLynx Full-Wave Solver creates an S-parameter model and assigns it to the 3-D area.
	To use multiple computers in parallel to create an S-parameter model for one or more 3-D areas: 1. Close the Add or Edit 3D Area dialog box. 2. In the 3D-Area Manager dialog box, select one or more 3-D areas in the 3D areas table. 3. Click **Solve on Multiple Machines**. The HyperLynx Job Distribution dialog box opens and loads the HyperLynx Advanced Solvers project file (PHYS) for each 3-D area you selected. See “Job Distribution” in the *HyperLynx Advanced Solvers User Guide*.
Open the 3-D area in HyperLynx Full-Wave Solver to review or edit it, run the solver to create an S-parameter model to represent the 3-D area, and then automatically assign it to the area in BoardSim	1. Choose an item from the Advanced Solver list: • Keep Existing Project — Opens an existing project with the name you specified in step 5, so you can edit it. • Create New Project — Creates a new project. If an existing project already has the name you specified in step 5, the software asks you whether to overwrite the existing project. HyperLynx Full-Wave Solver opens and loads the 3-D area into a Simulation named DUT. 2. Review Model properties and change them if needed. 3. Mesh and solve to create the S-parameter model and assign it to the 3-D area. For help, see “*HyperLynx Advanced Solvers User Guide*”. 4. Close HyperLynx Full-Wave Solver.

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If you want to...	Do this...
Open the 3-D area in HyperLynx 3D Explorer, create a set of S-parameter models to represent a range of design variations that you want to evaluate, and assign one of them to the area in BoardSim (part 1 of 2)	1. Choose an item from the 3D Explorer Variability list: • Open 3D Explorer — Supports post-layout design variations. • Create Pre-Layout Template — Supports pre-layout design variations (more flexible). HyperLynx 3D Explorer opens. 2. (Post-layout template only) In the New Study dialog box, specify the name and location of a customized template to represent design variations, then click **OK**. The software adds the template to the **Customized** branch in Template Browser. 3. (Pre-layout template only) In Template Browser, the **Customized** branch displays one or more new templates representing your 3-D area. Select a template and click **Customize**. **Note:** You can familiarize yourself with the parameters available for each template by reading its description in the **Help** tab. You can also evaluate the performance of different templates by repeating this procedure and selecting a different template. 4. Define a set of “what-if” design variations, run a solver, and see which variations have acceptable performance. For help, see “Exploring Design Variations” in the *HyperLynx Advanced Solvers User Guide*. **Note:** See instructions in the next table row to continue.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to...	Do this...
Open the 3-D area in HyperLynx 3D Explorer, create a set of S-parameter models to represent a range of design variations that you want to evaluate, and assign one of them to the area in BoardSim (part 2 of 2)	**Note:** These steps continue the instructions in the previous table row. 1. Assign an S-parameter model to a 3-D area: a. In HyperLynx 3D Explorer, on the **Results** tab, use an X in the Used column for design variations that you want to use for simulation in HyperLynx SI/PI. You can select multiple cells and press X to toggle the cell contents. If the Used column has more than one X, the software assigns the model for the top row with an X to the 3-D area. b. Near the bottom of the **Results** tab, click **Choose 'Used' Results**. 3D Explorer minimizes and the Add or Edit 3D Area dialog box opens in HyperLynx SI/PI and assigns an S-parameter model to the 3-D area. The software copies S-parameter models for the "Used" design variations to the location you specified in step 10, and assigns them names ending with *<experiment_name>-S_PARAMS.sNp*. **Note:** To assign an S-parameter model for a different design variation to the 3-D area, click **Browse** (next to the Model File field) and select a file. 2. Close HyperLynx 3D Explorer and HyperLynx Full-Wave Solver.

Note

You can temporarily disable an S-parameter model assignment and use standard board design information to represent a 3-D area. From the 3D Area Manager dialog box, uncheck the "On" column for a 3-D area.

14. To export a 3-D area to a CCE file that you can load into BoardSim or a HyperLynx Full-Wave Solver tool, click **Export**, then specify a location for the CCE file.

    The CCE file contains 3-D area information, such as geometries and ports.

    You can use this capability to help evaluate a subset of a very large design or multiple-board project that takes a long time to load.

Results

You can find additional matching 3-D areas in your design. See "Modeling a Board Area With an S-Parameter Model" on page 122.

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Related Topics

3D-Area Manager Dialog Box

Exporting a Board to IBIS EBD Models

Use the EBD model generator to create an IBIS EBD model from a board file. The EBD model generator is highly automated, so you need only a minimal knowledge of the syntax needed to create an EBD model.

The IBIS EBD format enables you to describe:

- The electrical properties of the interconnections on your board
- The components that plug into your board
- Your board as a single component
- The electrical properties of your board without revealing its physical properties

Prerequisites

- Reviewed "EBD Models Generated by BoardSim" on page 426.
- For each net you want to include in the generated EBD model, assign IBS or EBD models to all ICs.
- Acquire the BoardSim EBD Writer license.
- For a net with vias, L and C values for the via depend on the via simulation options you set in the "Select Method of Simulating Vias Dialog Box" on page 1012.

Procedure

1. Save session edits for the loaded board:

   Choose the **File > Save BoardSim Session File** menu item.

2. Choose the **Export > Board To > IBIS .EBD File** menu item.

   The Choose .EBD External Connector dialog box opens.

3. In the Reference Designator list, select the reference designator for the external connector for the EBD model you want to create.

4. To include vias in the EBD model, check "Include vias in EBD description".

5. Click **OK**.

   The Save EBD File dialog box opens.

6. Select or type the file name and click **Save**.

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7. If BoardSim cannot map an IC model to an IC reference designator in the EBD model, it writes a dummy File Name and Component Name parameter to the [Reference Designator Map] section of the EBD file it creates.

   Search for “Warning: Supply a valid mapping” in the EBD file to identify IC reference designators that were not properly mapped to an IC model and fix them.

   Note

   The IBIS specification allows an EBD model to point only to an IBS or EBD model. When ICs on the target net are assigned to SPICE or Touchstone models, BoardSim displays a warning to the screen and includes a warning in the [Reference Designator Map] section of the generated EBD file. To find this type of warning in the generated EBD file, search for “Warning: Supply a valid mapping.”

   Restriction

   BoardSim does not support EBD models that point to other EBD models. However, BoardSim does support EBD models that point to IBS models.

8. Check the syntax of the new EBD model. See “Checking IBIS File Syntax” on page 1262.

Results

Occasionally, you may have a board file in which two or more connectors together define the external interface of the EBD file. However, BoardSim supports only one connector to define the external interface of the EBD file.

EBD Models Generated by BoardSim

When an EBD model is generated in BoardSim, the model contains only external nets and has certain limitations for simulation.

Table 6-2. EBD Models Generated by BoardSim

Content/Limitation	Description
EBD Models Represent Only External Nets	Contains only nets that connect to external pins. The IBIS specification does not have a provision to include nets that do not touch an external connector. Therefore, the *.EBD* generator does not write any nets to the *.EBD* model that are purely internal to the board file. Nets that are conductively associated with an external net are included in the generated *.EBD* model.

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Table 6-2. EBD Models Generated by BoardSim (cont.)

Content/Limitation	Description
EBD Models Cannot Represent Coupling Information	The *.EBD* model cannot represent coupling between nets. The IBIS specification does not have a provision for including coupling information. Therefore, even if coupling is enabled, coupling data is ignored and not written to the *.EBD* model.
EBD Models Cannot Represent Ferrite Beads	The IBIS specification allows an *.EBD* model to point only to *.IBS* and other *.EBD* models. However, *.IBS* and *.EBD* models cannot represent ferrite bead components. An error displays if a component on the target net is assigned a ferrite bead model when BoardSim generates the *.EBD* model.
EBD Models Cannot Contain SPICE IC Models	The IBIS specification allows an *.EBD* model to point only to *.IBS* and other *.EBD* models. If a SPICE model is assigned to an IC on the target net, no model is assigned to that IC in the generated *.EBD* model. See Exporting a Board to IBIS EBD Models.
Power-Supply Net Names are Constants	Pins connected to power-supply nets and an external connector on your board are written out only as “POWER” or “GND” in the *.EBD* model generated by the software. If your design contains other names, you must edit the *.EBD* model generated by the software to restore the original power-supply pin names. Also, you can add nets to, or subtract nets from, the power supply list. See “Verifying That Power Supply and Signal Nets are Recognized Correctly” on page 62.

Exporting a Board to ICX

Create the files you need to simulate a board design in ICX.

Caution

We recommend that you do not select any nets before exporting the board to ICX. If you are unsure if nets are selected, close the board and re-open it before exporting the board to ICX. When you select a net, BoardSim “net cleaning” behavior automatically deletes segments that lie entirely within pads. ICX may not properly recognize connectivity if these segments are deleted.

The software exports the following board design properties:

- Component-wide IBIS model assignments made using the REF automapping file
- Interactive buffer direction assignments for IBIS bidirectional buffers
- Physical termination components, including values set interactively or by the REF file

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- Series passive components, which are automatically translated into IBIS files

Restrictions and Limitations

Export to ICX cannot export the following properties to the ICX design:

- Interactive IBIS model assignments
- MultiBoard projects
- EBD, SPICE, and Touchstone model assignments
- Quick Terminators (virtual terminators)

Procedure

1. (Optional) To set optional export settings in the *BSW.INI* file:
   a. If BoardSim is open, close it.
   b. Open *BSW.INI* with Notepad or another text editor. *BSW.INI* is stored in the same folder as the *BSW.EXE* application.
   c. In the [EXPORT_TO_ICX] section, add the parameter

      ```
      Extra_XFORM_options=<values>
      ```

      Set the parameter value to -n -w to add additional warnings and notes to the XFORM log file. See ICX documentation for other parameter values. If [EXPORT_TO_ICX] does not exist, add it to the last row in the file, and then add the new parameter below it.
   d. Save the edits and close the text editor. The changes take effect the next time you open BoardSim.
2. Open the design in BoardSim.
3. Choose the **Export > Board To > ICX NDD File**. The Export To ICX dialog box opens.
4. To change the output directory or the NDD (neutral design data) file name, click **Browse**, then specify the new location.
5. If HyperLynx and ICX are not installed on the same computer, skip to step 7. If HyperLynx and ICX are installed on the same computer, to translate the NDD file into an ICX file and other related files, check “Run XFORM utility to create .ICX file”, then click **OK**.
6. If you checked “Run XFORM utility to create *.ICX* file” in step 5, and XFORM fails to create the *.ICX* file, click **Show LOG File** to display XFORM messages.
7. If HyperLynx and ICX are not installed on the same computer, go to a computer with ICX and run the ICX XFORM utility to translate the exported NDD file into an ICX file

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and other related files. For information about running XFORM, see Appendix A in the *IS User's Guide* provided with ICX.

Exporting a Schematic to BoardSim

Use this procedure to create a BoardSim board file that is electrically equivalent to a LineSim schematic. This allows you to define "what if" interconnects in LineSim for a board that has not been laid out yet.

Note

The exported board may appear to have random component placement and routing. The schematic does not contain physical information, so the exported file contains components with arbitrary positions and virtual nets rather than routed nets.

Restrictions and Limitations

- PDN Editor geometries are not exported to BoardSim.
- Exporting a schematic to BoardSim is unavailable for schematic designs with multiple stackups.

Prerequisites

- Acquire the Via Models license.

Procedure

1. Open a schematic.
2. Choose the **Export > BoardSim Board** menu item.
3. Specify the file location and click **Save**.

Results

The exported board file contains all schematic components and IC model information. The board file has arbitrary placements for components and nets, but it should behave accurately from an electrical perspective.

Exporting a Constraint Template From LineSim

Use this procedure to export a constraint template file and to optionally update Constraint Manager, based on properties of the selected LineSim net. This capability enables you to define net topologies in LineSim that have good signal integrity and transfer that information to constraint-aware Siemens EDA software, such as Constraint Manager or Constraint Template Editor (CTE).

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Constraint templates are a reusable set of constraints that can be applied to similar nets in different designs. Constraint templates contain electrical constraints (such as maximum transmission-line lengths), physical constraints, FromTos (net scheduling), IC model assignments, and so on. You can collect constraint template libraries, which provides a way to apply proven design rules to nets in new designs.

Prerequisites

- If you plan to automatically update Constraint Manager with the contents of the exported template file, start Constraint Manager before performing step 7.
- During the export, the process encrypts the constraint data before sending it to CTE. EE 7.9.3 and newer can read the encrypted constraint data, but older EE releases cannot. If you use an older EE release, you can force the export to send unencrypted constraint data.

Procedure

1. Add part names to IC symbols on the net by double-clicking each symbol and typing a value in the Part field in the Assign Models dialog box.
2. Select the net by doing any of the following:
   - Right-click a driver IC symbol or an IC pin and click **Create Constraint Template**.
   - Click a driver IC symbol and choose the **Export > Constraint Template** menu item.
   - Choose the **Export > Constraint Template** menu item. If no driver IC symbol is selected and there is more than one driver in the schematic, the Select Driver dialog box opens. Select the driver pin and click **OK**.

   The Export Constraint Template dialog box opens.
3. To specify detailed constraint values, click **Edit Template**. Specify values in the Define Constraint Template dialog box.

   Note

   If the schematic contains virtual pins (where three or more transmission lines directly connect to each other) or custom topologies, specify the net scheduling in the Define Constraint Template Dialog Box - Net Scheduling Tab.

   Maximum delays are not exported for “simple” transmission lines because they do not contain physical transmission-line lengths.
4. Type the template name to display in the CTE or Constraint Manager application.
5. Type or browse to the template file name and folder. You can export template files in the following formats:
   - CTM — Used by Expedition 2007 and newer. Selected by default.

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- CMS — Used by pre-Expedition 2007.

6. To automatically open the exported template file, check “Open generated template in the Constraint Template Editor”.
7. To automatically update Constraint Manager with the contents of the exported template file, check “Update Constraint Manager with generated template”.

Note

This option is available only when Constraint Manager exported the original schematic.

Constraint Manager must be running to complete this operation.

8. Click **OK**.

Exporting a Net from Constraint Manager to a Schematic

You can export one or more electrical nets from the Nets page of the Constraint Manager spreadsheet to a LineSim schematic. For information about this process, see the *Constraint Manager Users Manual*.

Importing and Exporting Constraints, Net Classes, or Stimulus

Scope: BoardSim

You can import or export constraints, net grouping (net classes), or stimulus assignments for nets that you want to include in interactive simulation with measurements, interactive sweeps with measurements, or General Batch simulation.

You can import constraints directly from Constraint Manager, from a CSE file (exported with a CCE design file) from Xpedition Layout, or from a CSV file. You can also export constraints to a CSV file.

Prerequisites

- If you plan to import constraints directly from Constraint Manager:
  - Choose the **Setup > Options > Directories** menu item, then set the “PCB flow path” to the SDD_HOME folder of your Xpedition installation location.
  - Ensure that the same design is open in Xpedition Layout, Xpedition Designer, or Constraint Manager.

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- If you plan to import constraints from a CSE file, ensure that the CSE file and the design file (in CCE format) are in the same location, and open the CCE file in HyperLynx SI/PI.

Procedure

1. Choose the **Setup > Net Classes and Constraints** menu item. The Net Classes, Constraints and Stimuli dialog box opens.
2. Click **Import / Export Net Classes and Constraints**.
3. In the Import / Export Net Classes and Constraints dialog box, choose an import or export method.

If you want to...	Do the following...
Import directly from Constraint Manager	1. Click **Import from Constraint Manager**. 2. Click **OK**. The software overwrites all existing data with imported data.
Import from a CSE or CSV file	1. To remove any existing data that is not present in the imported data, select “Clear existing net classes, constraint and stimuli before import”. 2. Select the information you want to import. 3. Click **Import from CSV**. 4. In the dialog box that opens, select a CSE or CSV file that contains constraints. 5. Click **Open**. 6. Click **OK**. **Note:** Imported data always replaces any conflicting existing data.
Export to a CSV file	1. If you want to export constraints only for net or class rows that you selected in the spreadsheet in the Edit Net Classes, Constraints and Stimuli dialog box, check “Export selected rows only”. 2. If you want to export the specific inherited constraint values for each net, select “Export inherited values”. **Note:** This option saves inherited values as explicit values for each net. Do not use this option if you intend to re-import the CSV file and edit inherited values later. 3. Click **Export to CSV**. 4. Click **OK**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Exporting and Importing a Stackup

You can reuse a stackup among designs. Exporting and importing a proven stackup can save time when preparing a design for simulation. You can also create a backup copy of the stackup, which is helpful when performing multiple "what if" experiments.

Stackup (STK) files contain exported stackup properties. You can import a stackup directly from a LineSim schematic design file, so it is unnecessary to export the stackup from a schematic design before importing it into another design.

Restrictions and Limitations

- You cannot import a stackup that contains fewer layers than the current design.

Prerequisites

- If you use a HyperLynx SI/PI menu to import or export a stackup, disable the **Enable Multiple Stackups** menu item (choose the menu item to deactivate its check mark).

Procedure

1. Export a stackup:
   a. Choose the **File > Export** (Stackup Editor) or **Setup > Stackup > Export** (HyperLynx SI/PI, unavailable for a board design with multiple stackups) menu item.
   b. Specify the STK file location and click **Save**.
2. Import a stackup:
   a. Choose the **File > Import** (Stackup Editor) or **Setup > Stackup > Import** (HyperLynx SI/PI) menu item.

      The Open dialog box opens.
   b. From the Files of Type list, select whether to import from a stackup file (STK) or a LineSim schematic file (FFS).
   c. Select the file to import and click **Open**.
   d. If you import a stackup containing more layers than the current design, the Layer Mapping dialog box opens. The Used spreadsheet column identifies stackup layers that are used by nets in the current design. Assign a stackup layer from the imported source design to the current design:

For a selected Source Layer, if you want to...	Select this Destination Layer value...
Replace a Destination Layer with the Source Layer	<layer_name>
Not use the Source Layer	Not Imported

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

For a selected Source Layer, if you want to...	Select this Destination Layer value...
Insert the Source Layer	New Layer New layer names use the form new_layer_<number>.

e. Click **OK**.

Results

Now you can verify or edit the imported stackup. See Defining the Basic Stackup.

Archiving Design Simulation Files

Use the Archive Design utility to automatically gather and compress design simulation files for your board or schematic. You can use the archive as a "snapshot" of your design simulation files at a specific moment in your PCB design cycle. If you have experienced problems with a board or schematic, you can send the design archive to technical support for investigation.

Note

The Archive Design utility uses InfoZip technology to compress archive files. Copyright (c) 1990-2001 Info-ZIP. All rights reserved. Info-ZIP's software (Zip, UnZip and related utilities) is free and can be obtained as source code or executables from various anonymous-ftp sites, including ftp.uu.net:/pub/archiving/zip/*. There is no charge for this software.

Restrictions and Limitations

- Batch simulation report (*.txt*) and design change summary files (*.txt*) are not archived.

Procedure

1. Load your design.
2. If you make changes to the design setup, do one of the following:
   - Board design — Choose the**File > Save BoardSim Session File** menu item.
   - Schematic design — Choose the **File > Save** menu item.
3. Choose the **Export > Design Archive** menu item.

   The Archive Design Dialog Box opens.
4. Enable options, specify the archive folder location, then click **OK**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Chapter 7
Solving Problems Found in Simulation

After you identify an issue in your design, the software provides specific ways to improve your design.

Topic	Description
Editing Trace Widths	You can perform “what if” signal-integrity simulations by editing trace widths to vary the impedance of traces on the board.
Possible Bad Effects From Width Changes	Changing the widths of a board's traces usually does not affect the electrical validity of the traces, but this cannot always be guaranteed. Generally, reducing the trace width is safe. However, widening traces can cause electrical problems, especially if the board is densely routed.
Examples of Changing Trace Widths	These examples illustrate the effects of changing trace widths for various layers and geometries.
Evaluating Design Performance Changes by Varying Anti-Object Clearances	You can vary clearance values for all pads and segments in metal plane areas of your power distribution network and run simulation again to observe the effect on design performance.
Board and Net Property Reports	You can report design properties to help investigate problems found in simulation.
Net Terminations	The software provides different ways of quickly terminating nets in your schematic or board design.

Editing Trace Widths

You can perform “what if” signal-integrity simulations by editing trace widths to vary the impedance of traces on the board.

The widths of the traces on your board play a major role, along with the board stackup, in determining the trace impedance. Impedance, in turn, greatly affects signal quality and radiated-emissions behavior. For this reason, you may sometimes want to experimentally change the widths of certain traces on your board to see how signal integrity is affected.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Tip

A PCB trace normally consists of many individual “segments” which, taken together, make up the complete trace. When simulating, the software treats each of these segments individually as a separate transmission line. This means that if you have a trace that consists of a mixture segment widths, the software correctly accounts for the resulting impedance discontinuities and delay changes.

The software allows you to change trace widths directly, without having to go back to your PCB-layout tool. These changes are made to your current layout, and are experimental and temporary. When you exit the software or close your board, the changes are discarded. The next time you load your board into the software, the original layout is restored along with the original trace widths.

Procedure

1. Choose the **Edit > Trace Widths** menu item.
2. If you have a MultiBoard project loaded, select which board or boards you want to modify from the Traces on Boards list.
3. In the Select Trace Segments To Change area, select the net(s), stackup layer(s), and width(s) for which you want to change the widths.

If you want to...	Do this...
Select nets to modify	In the Traces On These NETS area, either make width changes on only a specific net, or on all of the nets on the board. If you are choosing a particular net, you can change the order in which the nets are listed in the Selected Net combo box by selecting the desired radio button in the Sort Nets By area.
Select a layer(s)	In the AND On These LAYERS area, select whether to change widths on only a specific stackup layer, or on all layers. The Selected Layer combo box lists all of the layers in the current board stackup.
Select a range of widths	In the AND With WIDTHS In This RANGE area, select to limit the changes to only trace segments in a selected range of original widths, or to all segments regardless of width. If you choose to enter a range, you enter the minimum and maximum widths in the range.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note that the three selection criteria (nets, layers, and width range) are ANDed together. To eliminate one of the criteria, click the All radio button in that criterion's area. For example, to eliminate the width range, click the All Widths radio button.

4. Type the new width in the Width box.
5. Click **Change Widths**. The widths are altered immediately, and are shown in the board viewer.
6. Click **Close** and resume analysis.

Results

The changed widths are in effect until you make additional changes that override them, or until you close and reload the board. You cannot restore your original widths except by re-loading your board.

To restore the original trace widths used in your PCB layout, you must re-load the board file. When you re-load, the width changes made in the previous session are discarded and the original widths from the board file are restored.

Trace-width changes you make are not reported in the Design Change Summary. Also, although changed widths are used when the Board Wizard analyzes your PCB, the changes are not summarized in the design-change sections of the board report.

Therefore, to keep a record of the traces and layers on which you've changed widths, you must do so manually.

Related Topics

Possible Bad Effects From Width Changes

Possible Bad Effects From Width Changes

Changing the widths of a board's traces usually does not affect the electrical validity of the traces, but this cannot always be guaranteed. Generally, reducing the trace width is safe. However, widening traces can cause electrical problems, especially if the board is densely routed.

In a given signal-integrity simulation, BoardSim only “looks” at the net you have chosen for simulation, plus any associated nets. Therefore, it is not as dangerous to widen traces as it might seem. Problems only arise if the widened trace touches another segment on the same net or a segment on an associated net.

In rare cases, narrowing a trace can cause electrical problems. For example, if a trace connects to a pad marginally, at the edge of the trace only, narrowing the trace can cause an open in the trace-to-pad connection.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Editing Trace Widths

Examples of Changing Trace Widths

These examples illustrate the effects of changing trace widths for various layers and geometries.

If you want to...	Do this...
Change the width of a single trace on all of the layers on which it is routed	1. Select **Edit > Trace Widths** 2. In the Traces On These NETS area, select **Selected Net Only**, and then select the net with the Selected Net list. 3. In the AND On These LAYERS area, click **All Layers**. 4. In the AND With WIDTHS In This RANGE area, select **All Widths**. 5. Click **Change Widths**. 6. Click **Close** and resume analysis.
Change the widths of all the traces on a single stackup layer	1. Select **Edit > Trace Widths** 2. In the Traces On These NETS area, select **All Nets**. 3. In the AND On These LAYERS area, select **Selected Layer Only**, and then select the desired layer in the Selected Layer list. 4. In the AND With WIDTHS In This RANGE area, select **All Widths**. 5. Click **Change Widths**. 6. Click **Close** and resume analysis.
Change all of the 10-mil and 8-mil traces on your board to 6 mils wide	1. Select **Edit > Trace Widths** 2. In the Traces On These NETS area, select **All Nets**. 3. In the AND On These LAYERS area, select **All Layers**. 4. In the AND With WIDTHS In This RANGE area, select **Selected Range**, and then type 8 into the Min Width box and 10 into the Max Width box. 5. In the Width To Change To area, type 6. 6. Click **Change Widths**. 7. Click **Close** and resume analysis.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to...	Do this...
Change the top-layer, 6-mil-wide segments on a net called “CLK” to 8 mils wide	1. Select **Edit > Trace Widths** 2. In the Traces On These NETS area, select **Selected Net Only**, and select **CLK** in the Selected Net list. 3. In the AND On These LAYERS area, select **Selected Layer Only**, and select layer “Top” on the Selected Layer list. 4. In the AND With WIDTHS In This Range area, select **Selected Range**, and type 6 into the Min Width box and 6 into the Max Width box. 5. In the Width To Change To area, type 8. 6. Click **Change Widths**. 7. Click **Close** and resume analysis.

Related Topics

Editing Trace Widths

Evaluating Design Performance Changes by Varying Anti-Object Clearances

Scope: BoardSim

You can vary clearance values for all pads and segments in metal plane areas of your power distribution network and run simulation again to observe the effect on design performance.

Note

This setting only affects anti-object clearances that are defined as properties of a padstack. Clearances that are defined as properties of a metal plane area are not affected.

Restrictions and Limitations

- Your design must contain plane areas, and you must identify them as power supply nets. See Verifying That Power Supply and Signal Nets are Recognized Correctly.
- All boards in a MultiBoard project must contain plane areas identified as power supply nets.

Procedure

1. Choose the **Setup > Anti-Objects** menu item.
2. Check “Show anti-objects in board viewer and use in analysis”.

   If you disable this option, any anti-objects that are described by explicit anti-pad geometry information in your design file are still displayed in the board viewer.

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Tip

If you have a very large board with many vias, disable this option to decrease simulation time and increase board viewer performance.

3. To specify specific clearances for all pads and segments, check “Force user-defined clearances” and type clearance values. The values you provide override clearance information defined in the board design file or schematic (padstack definition).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Board and Net Property Reports

You can report design properties to help investigate problems found in simulation.

Topic	Description
Reporting Board Properties	Create a report containing the total number of nets, segments, pins, and vias on the board.
Reporting Net Properties	Use the Statistics for Selected Net dialog box to report a set of statistics for the selected net.
Report File Format	The report file for boards and nets contains a number of common fields.
Creating a Design Change Report	Use the Design Changes dialog box to generate a concise report of all the component changes you have made on your board to improve signal quality or lower radiated emissions (EMC).
Viewing Net Segment Properties	Use the board viewer to view properties and field solver results for individual net segments that you select in the board viewer.

Reporting Board Properties

Scope: BoardSim

Create a report containing the total number of nets, segments, pins, and vias on the board.

Procedure

1. Choose the **Export > Reports > Board Statistics** menu item.

   The software reads the board design file to generate the report. Depending on how the software translates your design file, there may be small discrepancies from similar totals reported by your layout software.

2. A dialog box displays properties for your board.

Related Topics

Report File Format

Reporting Net Properties

Scope: BoardSim

Use the Statistics for Selected Net dialog box to report a set of statistics for the selected net.

Procedure

1. Select a net.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

2. Choose the **Export > Reports > Net Statistics** menu item.

   The Statistics for Selected Net dialog box displays several statistics for the selected net and its associated nets.

3. Click **Copy to Clip** to copy the report to the clipboard, so you can paste the information to a text editor.

4. Click **OK**.

Related Topics

Report File Format

Report File Format

The report file for boards and nets contains a number of common fields.

Note that in the report file definitions, a “net” represents both the selected net and associated nets.

Fields

Field	Description
Total delay	The summed propagation delay of every metal segment on the net.
Total length	The summed physical length of every segment.
Minimum characteristic impedance	Per-segment minimum characteristic impedance on the net, and provides a rough indication of how much impedance mismatch exists on the net.
Maximum characteristic impedance	Per-segment maximum characteristic impedance on the net, and provides a rough indication of how much impedance mismatch exists on the net.
Total receiver load capacitance	The summed value of all the receiver capacitances on the selected net. Large capacitance values may indicate increased signal delays.
Total resistance	The summed DC resistance of every segment on the net.
Effective net Z0	Attempts to show by how much the selected net’s actual characteristic impedance is effectively lowered by the presence of IC capacitance along the net. Use this value as a guide when choosing termination resistances, since the proper termination value is often lower than suggested by the net’s actual Z0 for nets that are significantly loaded by IC capacitance.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Estimated peak crosstalk	A rough estimate of the total amount of crosstalk that can occur on the net, based on the neighboring “aggressor” nets and the ICs driving them. **Restriction:** This value is displayed only if you are licensed for BoardSim Crosstalk, have crosstalk analysis enabled, and are using electrical (rather than geometric) thresholds. See Accounting for Coupling.
Associated nets	The list displays the nets associated with the selected net and — if you have a BoardSim Crosstalk license and have crosstalk analysis enabled—the aggressor nets coupled to it. Nets that are coupled are identified in the list with the tag “by coupling.”

Related Topics

Reporting Net Properties

Creating a Design Change Report

Scope: BoardSim

Use the Design Changes dialog box to generate a concise report of all the component changes you have made on your board to improve signal quality or lower radiated emissions (EMC).

Give this report to your layout designer or service bureau as a record of the changes you want made to your board in its next revision. Or you can use the list yourself to drive changes in schematics for the board.

The design change summary includes the following changes:

- Stackup — Such as thickness adjustments to affect impedances
- Changed components — Such as modified terminating component values to improve signal quality
- New components — Such as new terminators (Quick Terminators) added to improve signal quality

Restrictions and Limitations

- Trace-width changes you make are not reported in the Design Change Summary. To keep a record of the traces and layers on which you've changed widths, you must do so manually.

Procedure

1. Choose the **Export > Reports > Design Change Summary** menu item.
2. Click **Finish**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Results

The HyperLynx File Editor opens and displays the report. The report file is named *<board_file_name>.txt* and is located in the folder that contains the board file.

Viewing Net Segment Properties

Scope: BoardSim

Use the board viewer to view properties and field solver results for individual net segments that you select in the board viewer.

Procedure

1. Select a net. See Selecting Nets for SI Simulation.
2. View net properties:

If you want to...	Do this...
View properties for a net segment.	1. Right-click over a segment and click **View Segment Properties**. The Segment Properties dialog box displays various segment properties. 2. Click **Copy to Clip** to copy the segment properties to the clipboard, so you can paste the values into a text editor. 3. Click **Close**.
View field solver output for a net segment.	1. Right-click over a segment and click **View Field-Solver Output**. 2. Click the **Field Solver** tab. For Field Solver details, see Viewing Electrical Field Lines in BoardSim for Trace Segments. 3. To view numerical field-solver results, click **View**. The HyperLynx File Editor displays the results. 4. Click **Close**.
View attenuation over a frequency range for a net segment.	1. Right-click over the segment and click **View Field-Solver Output**. 2. Click the **Loss** tab. **Restriction:** The **Loss** tab is unavailable unless you enable the Lossy simulation option on the toolbar. 3. Click **Close**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Viewing a Board in 2-D

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Net Terminations

The software provides different ways of quickly terminating nets in your schematic or board design.

You can run the Terminator Wizard on a net in your schematic or board design. The wizard then makes suggestions for improving the performance of the net, which could include adding termination or changing the values of existing termination components. If you need to add termination:

- For a schematic design, you can add an RC terminator component to the net and then re-run the wizard to suggest and apply values.
- For a board design, the wizard can add a Quick Terminator to the net and apply suggested values. Alternatively, you can add a Quick Terminator to the net and then re-run the wizard to suggest and apply values. The software saves the Quick Terminator with the session file, and applies it each time you open your design.

You can also run sweep simulations to identify optimum passive terminating component values. See “Running Signal Integrity Simulation” on page 167. General batch simulation can provide termination suggestions for nets. See “Running General Batch Simulation” on page 175.

After you decide on which termination changes to make, you can use the Design Change Summary report to record your terminating-component changes for later back-annotation into your pre-layout design.

Topic	Description
Automatic Terminator Optimization	Use the Terminator Wizard to improve signal integrity and EMC properties for nets in your board or schematic design by finding optimal termination component values.
Quick Terminators	Quick Terminators are special components that allow you to experiment with terminations that are not currently on your board design by adding temporary terminating components (resistors and capacitors).

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Automatic Terminator Optimization

Use the Terminator Wizard to improve signal integrity and EMC properties for nets in your board or schematic design by finding optimal termination component values.

The Terminator Wizard automatically finds optimal values for the terminating components on your board or schematic. Terminating components include resistors and capacitors, both series and parallel; optimal values means resistances and capacitances that give the best waveforms from a signal-integrity and EMC standpoint.

For a board design, the Terminator Wizard uses quick terminator components in two ways. First, if one or more quick terminators are present on a net, the wizard treats them as “real” components and recommends values for them. Second, if a net is unterminated and the wizard is recommending a termination, it uses quick terminators to create the necessary components.

The wizard works in one of two ways depending on whether the selected net is terminated:

- If the net is terminated already, the wizard bases its analysis on the terminating components present on the net, suggesting, if possible, optimal component values. The terminating components can be actual components present in your schematic or board, or Quick Terminators added to your board design. The wizard works on any net with a single termination type (such as AC or series), and with a number of useful topologies involving multiple terminators.
- If the net is not terminated, and the software thinks the net is too long to be unterminated, it suggests a termination strategy—both a type of termination and optimal component values.

Topic	Description
Terminator Wizard Limitations	The models, topology, and termination type of a net determine Terminator Wizard limitations.
Running the Terminator Wizard	Use the Terminator Wizard dialog box display information about current and recommended termination for a net. You can also apply the recommend termination values from this dialog box.
How to Choose Between Multiple Terminators	When you run the Terminator Wizard on a net with multiple terminators, the wizard first examines the net to see if it can identify the multiple-component configuration.
Component Values and Recommendations	For a terminated net, the optimal component values are often not simple to determine, particularly since the loading effect of IC capacitances effectively alters the characteristic impedance of the net. The Terminator Wizard accounts for all of the IC models on the selected net and all associated nets, factoring the capacitances in the models into an effective characteristic-impedance calculation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
Signal-Integrity Checks and Warnings	When you run the Terminator Wizard, the wizard automatically runs various signal-integrity checks against the selected net. The wizard reports violations in the Terminator Analysis area.
Supported Termination Types and Net Topologies	If the Terminator Wizard finds terminating components already on the selected net (any mixture of “real” components and Quick Terminators), it attempts to identify the termination type and determine optimum values for the components.

Terminator Wizard Limitations

The models, topology, and termination type of a net determine Terminator Wizard limitations.

Limitation	Description
Terminator Wizard Unavailability	The Terminator Wizard is not available for a board design when a MultiBoard project is loaded, or when you select a net containing a pin with an .EBD model assigned to it. The Terminator Wizard is not available for a schematic whenever an .EBD model has been assigned to a net.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Limitation	Description
Differential Line-to-Line Termination	The Terminator Wizard recognizes differential termination when the circuit either meets all of the following conditions: • The selected net and the other net are connected to driver pins in an IBIS model. • A [DIFF PAIR] keyword in the IBIS model associates the driver pin on the selected net to the driver pin on the other net. • A resistor directly connects the selected net to the other net. The resistor can be native to the design or added to the design as a differential resistor Quick Terminator. OR • The selected net and the other net connect to driver pins with the same reference designator. For a schematic, you can use the Assign Models dialog box to assign a reference designator and pin name to the driver ICs. See "Assigning a Model or Value to an Entire Component Using a .REF File" on page 93. **Restriction:** The Crosstalk license is required to use a differential termination. Generally, the wizard does not support nets (or groups of nets) with multiple drivers present. However, since differential pairs require two drivers for proper circuit operation, an exception is made for them when the differential nets are connected in one of the circuit configurations listed above. To predict an optimal value for such a terminator, the wizard needs access to the Crosstalk field solver. Accordingly, recommendations for differential-terminator values are available only if you are licensed for Crosstalk analysis. The Terminator Wizard automatically identifies differential pairs that use differential IBIS IC models.
No Placement Checks for Differential Terminators	The Terminator Wizard does not check for proper positioning of a differential terminator (that is, whether a line-to-line terminator has excessive stub length or is otherwise misplaced). Use interactive simulation to gauge the effectiveness of a differential terminator's location.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Limitation	Description
Some Combinations of Multiple Terminators Not Supported	The wizard also does not support some complex termination schemes based on multiple terminators. See Supported Termination Types and Net Topologies. Any combinations not specifically described in the table are probably not supported. In situations where the wizard cannot recognize a complex termination type, use interactive simulation instead of the wizard to choose optimal component values.
Multiple Drivers Not Supported - Except for Differential IBIS Models	Except for the case of a trace pair driven by an IBIS differential IC model, the wizard will also not analyze any net that has more than one driver actively selected. In order to analyze such a net (provided it is not differential), change all but one of the drivers into a receiver (or remove the other driver models entirely).
Ferrite Beads Not Supported	The wizard does not support ferrite-bead terminators. Use interactive simulation to find an optimal ferrite bead.
Nets with Complex Routing Schemes	For nets with complex routing schemes (such as complicated, “non-obvious” branching), the wizard can sometimes not find an optimal termination scheme. Generally, the Terminator Wizard works best on nets that are single-receiver, or daisy-chained, or cleanly star-routed (that is, with clearly identifiable branches).
Terminator Wizard Requires Driver IC Model	In order for the Terminator Wizard to run a complete analysis and recommend component values, you must have a model selected for the driver IC on the current net. The presence of a driver model is critical because many of the driver’s properties have a profound effect on terminating-component values. The following are important driver properties: • Slew time • Output impedance • Physical position on the net If you run the wizard without a driver model selected, the software gives a warning in the Messages area; even if a termination is present on the net, the wizard lists the Termination Type as “unknown” (with a red question mark). Some of the statistics about the net are displayed, but no recommendation is made for terminating-component values.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Running the Terminator Wizard

Use the Terminator Wizard dialog box display information about current and recommended termination for a net. You can also apply the recommend termination values from this dialog box.

Prerequisites

- Review Automatic Terminator Optimization, Terminator Wizard Limitations and Supported Termination Types and Net Topologies.
- In order for the Terminator Wizard to run a complete analysis and recommend component values, you should have a model selected for the driver IC on the current net. The model must contain a slew time, output impedance, and the physical position on the net. The presence of a driver model is important because many of the driver's properties have a large effect on terminating-component values.

  However, if you plan to run the Terminator Wizard on a net without an IC driver model, ensure the default driver's rise/fall time is correct. See "Default Driver Characteristic" in "Preferences Dialog Box - General Tab" on page 990.

Procedure

1. Choose the **Simulate SI > Optimize Termination** menu item.
2. If you have not already selected a net, the Select Net By Name dialog box opens. Select a net, and click **OK**.

   If you have assigned an EBD/EMD model to a component to provide electrical information for the component package, the wizard can determine optimal termination for nets inside the model. Select a net with an EBD/EMD prefix.
3. If there are multiple nets in the schematic, the Select Net for Terminator Wizard dialog box opens. In the Select a Device Pin list, double-click on an IC pin attached to the net you want to analyze. The dialog box closes.
4. The Terminator Wizard dialog box opens and displays results in the Terminator Analysis area.

   If you are investigating a board design, the analysis is for the selected net and any nets attached directly to it through series components (or a set of differential pins on a driving IC). If you are investigating a schematic, the analysis is for the current schematic.
5. To specify component tolerance, select the tolerance from the Apply Tolerance list.

   By default, the Terminator Wizard calculates exact terminating-component values, without regard for the values you could actually purchase and install on a board.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If the wizard recommends resistor and/or capacitor values for a terminated net, you can easily apply the recommended values to the components on your board (or to a Quick Terminator), then re-simulate to see the resulting waveforms.

If the wizard recommends a terminator for an unterminated net, you can easily create the terminator and apply the recommended values to the components, then re-simulate to see the resulting waveforms. For board designs, the wizard can automatically create new termination components in the form of a Quick Terminator.

6. Add the recommended terminator:

To add a terminator to a...	Do this...
Board design	Click **Apply Values**. The wizard automatically creates the recommended terminator as a Quick Terminator, and applies the suggested component values. **Note:** You can also manually add a terminator to an unterminated net instead of automatically creating one based on the Terminator Wizard's recommendation. See "Adding a Quick Terminator" on page 458.
Schematic design	The Terminator Wizard cannot automatically modify a schematic to implement a terminator recommended for an unterminated net. Instead, you need to manually add the recommended components. 1. Note the terminator recommended by the wizard. Then click **OK** to close the wizard dialog box. 2. In the schematic, add the component(s) recommended by the wizard. 3. Re-open the Terminator Wizard. 4. Click **Apply Values**. The recommended value(s) is written into the component(s) you just added.

7. To re-simulate using the recommended values, click **OK** to close the Terminator Wizard.
8. Open the oscilloscope or Interactive Simulation dialog box, and re-simulate.

How to Choose Between Multiple Terminators

When you run the Terminator Wizard on a net with multiple terminators, the wizard first examines the net to see if it can identify the multiple-component configuration.

If the wizard successfully identifies the terminating scheme, it displays the combination of components in the Termination Type field (such as "series, AC, pull-up"). If not, the wizard marks the Termination Type with a red question mark, and cannot proceed with analysis.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

When the wizard correctly identifies the type, the wizard then displays a set of radio buttons that offer several choices for which terminator type to recommend values.

For example, if a net has three terminators in its layout, series, AC parallel, and DC pull-up, the wizard identifies the net as having terminator type “Series, AC, pull-up”. The wizard then presents these radio buttons in the Preferred Choice box:

- Best — Recommends the best termination type to use
- Series Termination — Forces analysis of the series terminator
- AC Termination — Forces analysis of the AC terminator
- DC Termination — Forces analysis of the DC pull-up terminator

Select the appropriate terminator in the Preferred Choice area.

After you make your choice, the wizard immediately shows its recommended value for that termination type. If you choose “Best,” the recommended terminator type is listed in the Terminator Analysis area as the “Suggested Termination.”

When you select a new net for analysis, the wizard does not save your choice of preferred terminator type for the previous net. If you return to the previous net to analyze it again, you must re-choose your preferred type.

Simulating With a Particular Terminator

To simulate with a particular terminator that you chose (or the wizard recommended) during analysis:

- In the Termination Suggestions area, click **Apply Values**.

The recommended termination values are automatically applied to your circuit. You can now close the wizard and open the oscilloscope, Interactive Simulation dialog box, or spectrum analyzer to simulate and see an actual waveform.

Unused Termination Components

When you set the component values of your preferred type of terminator using the Apply Values buttons, the wizard must also set the values for the “unused” terminating components in such a way that they do not interfere with the simulation of the preferred terminator. This is accomplished as follows:

- Unused series resistors are set to 0.0 ohms.
- In an unused AC terminator, the resistor value is set to 1 Mohm (the capacitor is unchanged).
- In an unused DC pull-up/down terminator, the resistor value is set to 1 Mohm.

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Component Values and Recommendations

For a terminated net, the optimal component values are often not simple to determine, particularly since the loading effect of IC capacitances effectively alters the characteristic impedance of the net. The Terminator Wizard accounts for all of the IC models on the selected net and all associated nets, factoring the capacitances in the models into an effective characteristic-impedance calculation.

Effective Z0 Value

The value of Effective Z0 attempts to show by how much the selected net's actual characteristic impedance is effectively lowered by the presence of IC capacitance along the net and associated nets. This value can be used as a guide when choosing termination resistances, since for nets that are significantly loaded by IC capacitance, the proper termination value is almost always lower than suggested by the net's actual Z0.

Results for Nets With Single Terminators

For nets with a single terminator already in place (composed either of "real" component(s) or a Quick Terminator), the wizard attempts to identify the termination type; if the identification succeeds, then the wizard calculates an optimal value for the terminating component(s).

Results for Nets With No Terminators

If you run the Terminator Wizard on a net that has no terminating components, the wizard first runs an analysis to determine if the net's signal integrity is likely to be acceptable without termination.

If the wizard believes that the net does not need termination, then in the Terminator Analysis area, the Termination Type is set to "No termination found"; no termination is suggested; and Apply Values button is unavailable.

If the wizard concludes that the net is too long to remain unterminated, it will attempt to recommend a termination type, and optimal values for the terminator's components. The algorithms used to determine the optimal terminator type are complex; they take into account the positions of driver and receiver ICs along the net, the topology of the net's routing (e.g., daisy-chained versus star-routed), comparison of driver versus net impedance, etc. Part of the process of recommending a terminator is to also recommend its position on the net, since location is often just as important as component values.

Recommended Terminating-Component Values

The Terminator Wizard displays its recommended resistor and/or capacitor values at the bottom of the Terminator Analysis area. If the currently selected net uses series or DC parallel termination, only one or more resistor values are recommended; if it uses AC parallel termination, resistor and capacitor values are suggested.

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If there are multiple resistors or capacitors on a net, then the wizard's recommended values are identified per-component in the following manner:

<component_type> <reference_designator.pin> suggested: <value>

where <*component_type*> is the type of component ("resistor" or "capacitor"); <*reference_designator.pin*> specifies the component's reference designator and a pin on the component; and <*value*> is the recommended value.

If you make changes to the net being analyzed (for example, change any of its IC models or alter the board's or schematic's stackup), re-run the wizard to see how the recommended termination values may have changed in response. The series-resistor value, for example, is strongly dependent on your current choice of driver IC.

Signal-Integrity Checks and Warnings

When you run the Terminator Wizard, the wizard automatically runs various signal-integrity checks against the selected net. The wizard reports violations in the Terminator Analysis area.

The wizard issues warnings that can help explain why the waveform you see in the oscilloscope or the Interactive Simulation dialog box is less than perfect, even if you are using the component values recommended by the Terminator Wizard. The wizard cannot compensate, for example, for improper component placement (such as a series resistor located too far from the driver IC) or poor routing topology.

The checks fall into two broad categories: searching for problematic component values (such as resistors that are too large or small), and searching for problematic component placement (such as a series resistor located too far from the driver it terminates).

Table 7-1 lists the signal-integrity checks run by the Terminator Wizard.

Table 7-1. Terminator Wizard - Types of Signal-Integrity Checks

Type of Check	Description
Unterminated-net length	For unterminated nets, checks if the net's length is too long to have no termination; suggests a solution, if possible
Component value non-optimal	If a terminating component's value is more than 25% different from the value the wizard thinks is optimal, the wizard issues a warning; this makes it easy to find components that probably need "fixing"
Driver-to-series-resistor length	For series-resistor terminators, checks if the distance from driver to resistor is too long for effective termination

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Table 7-1. Terminator Wizard - Types of Signal-Integrity Checks (cont.)

AC-terminator resistor-to-capacitor length	For AC parallel terminators, checks if the distance between resistor and capacitor is too long for effective termination
Pull-up/pull-down combo resistor-to-resistor length	For DC parallel pull-up/pull-down combo terminators, checks if the distance between the two resistors is too long for effective termination
Receiver-IC stub length	For each receiver IC, checks if its stub length (distance from the "main" trace routing) is too long
Resistor placement relative to receiver ICs	For any non-series type of terminating resistance, checks for improper placement relative to receiver ICs on the net; for example, will flag a DC parallel resistor that is located too far from a receiver IC
Driver impedance exceeding net's impedance	For each driver, issues a warning if the driver's impedance exceeds the net's impedance (that is, if the driver intrinsically over-terminates the net)
Driver impedance large enough to cause bad DC levels or excessive tolerance variation in the driver itself	For each driver, issues a warning if the driver impedance is large enough to cause any of the following problems: • Likelihood of invalid DC levels (when DC termination used) • Likelihood of an excessive portion of series termination residing in the driver itself and therefore subject to excessive tolerance variations

Supported Termination Types and Net Topologies

If the Terminator Wizard finds terminating components already on the selected net (any mixture of "real" components and Quick Terminators), it attempts to identify the termination type and determine optimum values for the components.

If there are terminating components present on the net, the analysis can succeed only if the wizard is able to automatically determine from the components what type of termination you are using (such as series resistor or AC parallel). The wizard examines the following to determine the termination type:

- What resistor and capacitor components are present

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- What nets are connected to each component (such as two signal nets or a signal net and an AC ground)
- Where the components are placed, especially relative to the driver IC
- Other topological details of the net's routing

If the wizard can identify the termination type, it displays the result in the Terminator Analysis area. If the wizard cannot identify the type, the Termination Type is left as "unknown" (red question mark). In this case, the wizard then cannot make a recommendation for component values.

The Terminator Wizard can recognize the following termination types and net topologies:

- Single DC parallel resistor (pull-up or pull-down), single DC parallel pull-up/pull-down combination, or single AC parallel resistor and capacitor.
- Multiple series resistors, each terminating one branch of a "star" route. See How the Wizard Recognizes Branched Topologies.
- Multiple parallel terminators in "any mixture of types."

  "Any mixture of types" can be any mix of DC parallel resistor, DC parallel pull-up/pull-down combo, and AC parallel resistor and capacitor.
- Single series resistor and multiple parallel terminators of the same type.

The wizard does not recognize the following:

- Series R + multiple parallel terminators of differing types (such as one pull-up R + one AC terminator).
- Differential trace-to-trace R, if the two traces are driven by an IBIS differential IC model or by the differential resistor Quick Terminator.

How the Wizard Recognizes Branched Topologies

The wizard supports termination schemes in which a "star-routed" net has each of its branches terminated by a separate series resistor.

However, in order to recognize such a topology and make useful component-value recommendations for it, the wizard must be able to automatically judge whether a given net is actually routed as a valid star.

To make a topological judgment about star routing, the wizard uses a path-tracing algorithm. If a net has multiple series resistors, it is considered to be a valid star route only if one end of each resistor traces back only to the driver IC, and the other end traces only to receiver ICs.

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Quick Terminators

Quick Terminators are special components that allow you to experiment with terminations that are not currently on your board design by adding temporary terminating components (resistors and capacitors).

You can use a Quick Terminator to see if adding termination can improve a signal-integrity or EMC problem on a net that is unterminated. You can also use a Quick Terminator to experiment with different termination types on nets that are already terminated with another type.

The software saves Quick Terminators with the session file, and applies it each time you open your design. After you decide on which termination changes to make, you can use the Design Change Summary report to record your terminating-component changes for later back-annotation into your design.

Topic	Description
Adding a Quick Terminator	You can add terminating components to a selected net and its associated nets in a board design, regardless of what other components are already present on the net.
Specifying a Differential Resistor Stub Value	Under certain circumstances, it is necessary for you to specify differential pins, a differential resistor, and the stub values.

Adding a Quick Terminator

Scope: BoardSim

You can add terminating components to a selected net and its associated nets in a board design, regardless of what other components are already present on the net.

You can also add a Quick Terminator to any IC pin on the net. This gives you the flexibility to place a resistor at a driver IC (series termination), a resistor and capacitor at the last receiver IC (AC parallel termination), and so forth.

Restrictions and Limitations

- You cannot place a Quick Terminator inside an EBD model.
- EMC simulation doesn't account for radiation emitted by a Quick Terminator because the software does not have any information about a Quick Terminator's physical package or its orientation on your board.

Prerequisites

- Review “Quick Terminators” on page 458.

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- If you plan to replace a real terminator (one physically present on your PCB) with a Quick Terminator, use the Assign Models dialog box to change the real terminator's component values such that the terminator no longer has any effect in the circuit. See “Assigning Models to Components and Pins” on page 85.
- Select the net to which you want to add a Quick Terminator.

Procedure

1. To add a Quick Terminator, do one of the following:
   - Choose the **Models > Assign Quick Terminator** menu item.
   - Click Add Quick Terminator  .
   - In the board viewer, right-click over a pin on the selected net, and click **Add Quick Terminator**.

   The **Quick Terminator** tab of the Assign Models dialog box opens.
2. In the Quick Terminator Location list, select the IC pin as the location to place the termination.
3. In the Terminator Style area, select the terminator type you want to add.

   A small resistor icon appears next to the selected pin in the Quick Terminator Location list.
4. In the Terminator Values area, type or select the component values.

   For parallel DC terminators, the values include selectable pull-up and pull-down voltages.
   - If you selected a Series resistor in the Terminator Style area, you can also specify what kind of interconnect separates the resistor from the driver or receiver IC.
   - If you selected R differential in the Terminator Style area, make sure the name of the second pin of the differential pair is displayed in the Opposite Pin box. If the pin name is incorrect or absent, do one of the following:
     - In the Opposite Pin box, type the opposite pin's <reference designator>.<pin> value; for example, U1.5.
     - Click **Browse**, select the other pin from a list.

   You can also specify a stub value for a differential resistor terminator. See Specifying a Differential Resistor Stub Value.
5. Click **Close**.

Results

The Assign Models dialog box displays a Quick Terminator resistor icon in the Pins list next to the pin with the terminator, as a reminder that a Quick Terminator exists on the pin. You can

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now use the Design Change Summary report to record your terminating-component changes for later back-annotation into your pre-layout design. See “Creating a Design Change Report” on page 443.

Note that you can use the **Quick Terminator** tab on the Assign Models dialog box to edit values of an existing Quick Terminator or remove it completely.

Specifying a Differential Resistor Stub Value

Scope: BoardSim

Under certain circumstances, it is necessary for you to specify differential pins, a differential resistor, and the stub values.

This is a two-step process: first you specify stub values for the selected pin and then you specify stub values for the opposite pin. The selected and opposite pins represent the differential pins terminated by the differential resistor.

The software simulates the stub by adding a transmission line between the IC and the existing routing on your board.

Prerequisites

- You have added a differential resistor Quick Terminator (see Adding a Quick Terminator).

Procedure

1. In the Terminator values area, on the Layer list, select the stackup layer for the stub.

   The upper half of the diagram shows stub values for the selected pin and the lower half of the diagram shows the stub values for the opposite pin, if the opposite pin has been defined.

2. Type values in the Length and Width boxes.

3. To identify the opposite pin, do one of the following:

   - In the Opposite Pin box, type the opposite pin's <reference designator>.<pin> value; for example, U1.5.

   - Click **Browse**, select the other pin using the Select Second Pin dialog box, and then click **OK**.

4. To specify the resistor stub values for the opposite pin, select the opposite pin from the “Quick-terminator location list” and repeat steps 1-2.

   Because the stub layer and width default to match the layer and width of the portion of your board’s actual routing that touches the IC, you usually do not need to change layer and width.

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Chapter 8
Support Information

Simulation involves technical concepts that sometimes require additional explanation. Refer to this information as needed.

Topic	Description
3-D Areas That Include Multiple Nets	The software can automatically define 3-D areas that include multiple nets. For example, your board design may have adjacent SERDES channels and you want a 3-D area to include a via array for all of them. A 3-D area that includes multiple nets accounts for coupling among them.
Accounting for Noise Between Single-Ended Signal Via and Power Planes in SI Simulation (Co-Simulation)	You can increase the accuracy of signal integrity simulation by accounting for the energy transferred to the power distribution network when signals propagate through single-ended signal vias.
Area Fill Edge Approximation Examples	The software represents an area fill or routed power-supply net with an approximate shape to model coupling to a selected signal net. The position of the selected net and the approximated area fill edge (or routed power-supply net) is a major part of how the software creates cross sections for the field solver.
Approximate Switching Time	The driver switching time reported in the Approx. Switching Time column of the SI Nets Spreadsheet is based on assigned IC model and net properties.
Automatic SI Simulator Selection	The software follows a specific selection process when you enable Auto in the “Simulation engine” of the Simulation Controls dialog box or the Simulator area of the oscilloscope.
Bit Sequence for Automatic Channel Characterization	If you enable the “Automatically characterize channel” and PRBS options on the Channel Characterization Dialog Box, FastEye channel extraction automatically defines the bit sequence in the Figure 12-4.
BoardSim Files	BoardSim reads and writes several kinds of files.
BoardSim Board File Contents	A BoardSim board file is an ASCII text file, with a *.HYP* extension, formatted in a HyperLynx-proprietary format. It contains all of the information about a PCB layout needed for signal-integrity simulation.

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Topic	Description
BoardSim Session Files	Session edits are any changes you make to the database while using BoardSim. BoardSim captures your edits during a session and saves them to a file when you exit or choose to save them in the middle of a session (select **File > Save BoardSim Session File**).
CAPLIB File Format	The CAPLIB decoupling capacitor library file format describes a custom set of capacitors that the software considers when running PDN decoupling optimization.
Checking Channels for Linear and Time-Invariant Behavior	SERDES channel analysis requires channels that exhibit linear and time-invariant (LTI) behavior, which consists of the following characteristics:
Contents of Waveform Files in CSV Format	The time data in the file is in seconds; voltages are in volts; currents are in amps.
Coupling Dots	Coupling dots indicate the orientation of coupled transmission lines. To correctly simulate the phase of crosstalk voltages and currents, ensure that the correct ends of the coupled transmission lines are marked with coupling dots.
Coupling Ratio for RLC Matrix Package Coupling	The coupling ratio (C_mutual/C_self) determines which package pins and nets SI simulation includes as aggressors when the package is modeled with RLC matrix information within the [Define Package Model] keyword. Use the Coupling Settings dialog box to specify the coupling threshold.
Creating and Applying a DDRx Delays File	If your memory interface supports leveling and calibration capabilities for read/write operations, the DDRx Wizard uses a delays file to minimize skew among the appropriate signals.
Creating a Custom Stimulus	If you need to assign a specific stimulus that the software does not provide, create a custom stimulus.
Current Flow For DC Drop	Current flowing between a voltage provider (source) and its current consumers (sinks) traverses the power supply net with some current density and voltage drop.
CURVE Subrecords with Invalid Coordinates	Translators can sometimes create *.HYP* files that contain CURVE subrecords (in a NET keyword) with coordinates that do not add up. The software checks the distance between the center point and each of the end points in the CURVE subrecord and if either of them mismatches the specified radius by more than 5%, the arc is invalid.
DC Drop Conceptual Circuits	Conceptual circuits demonstrate how current flows from a VRM model, through a power-supply net and to a current sink model. The information in this topic also helps you map circuit elements to terminology used in DC drop simulation results.

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Topic	Description
DDRx Batch Simulation	DDRx is a source-synchronous technology, meaning that instead of one master clock that applies to the entire memory interface, multiple strobes and clocks are used for sub-portions of the interface. To set up timing measurements on a specific data, address, command, or control signal, the DDRx Wizard automatically pairs it to its clock or strobe signal. Setup, hold, and other measurements on a data, address, command, or control signal are made relative to its clock or strobe signal. Signal pairing also exists between strobe and clock signals.
Design Factors Contributing to DC Drop	Performance constraints may produce PCB designs with numerous power-supply nets that are implemented as small isolated regions and trace segments.
Design Folder and HyperLynx Files	The design folder is the default location where the software creates and saves files related to your design. The default design folder also contains example schematic and board design files.
Decoupling Simulation Background	Understanding some concepts can help you understand simulation results and prepare for simulation.
Editing Power Aware SI Model Port Mappings	After you have assigned a power-aware SI model, you can edit port mappings if needed.
Evaluating SERDES Channel Compliance From an S-Parameter Model	If you have an S-parameter model of a SERDES channel that uses a supported protocol, the software can quickly calculate figures of merit (FOM) without using IBIS AMI transmitter and receiver models. This method provides a quick way to evaluate performance that is particularly useful when your design includes many channels and you do not have IBIS AMI models.
ExpertReferenceOptions.opt File	When BoardSim creates a 3-D area, you can have it add a plane layer to represent a return current path provided by a BGA (or similar) package and connect it to your board design. You can optionally provide detailed parameters for the added plane layer and its connections to your board design with an *ExpertReferenceOptions.opt* file. You manually create this file.
Exporting and Translating a Board Design	You can export a board design from Xpedition Layout, PADS Layout, and other Siemens EDA layout software. HyperLynx can translate a board design created in other layout software tools.
External Characterization Files	You can use external channel characterization files for FastEye or IBIS-AMI channel analysis that you create outside of the FastEye/IBIS AMI analysis wizard.
The Field Solver in LineSim	LineSim uses its field solver to solve for the capacitances, inductances, propagation velocities, and characteristic impedances of a coupling region’s cross section.

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Topic	Description
Flight-Time Compensation in General Batch Simulation	Flight-time compensation is available for IC pins with test fixture information, such as Vref and Cref.
Hiding or Moving an IC Component Pin	You can minimize the number of crossed connection wires in a schematic with IC components by hiding pins that you do not plan to include in simulation, or moving pins to different locations on an IC component symbol.
Horizontal and Vertical Geometric Search Range for Coupled Nets	Horizontal and vertical search ranges apply to each cross section that includes trace segments for the selected victim net and potential aggressor nets.
How BoardSim Recognizes Power Supply Nets	When you load a board, BoardSim attempts to identify power-supply nets and determine the voltage. If net names are not recognized correctly, you can manually specify power supply nets and voltages in the Edit Power-Supply Nets dialog box.
How BoardSim Recognizes Component Types	When BoardSim loads your board, it examines the list of devices in the design file and uses the reference-designator prefix to determine the component type of each device. Many commonly-used prefixes are automatically mapped. However, you may have to update the mapping for unrecognized prefixes.
How Duty Cycle Affects EMC Simulation	EMC simulation results are sensitive to the duty cycle of a signal.
HyperLynx Advanced Solvers Option Files for 3-D Areas	When creating a 3-D area, you can override default settings with values from HyperLynx Advanced Solvers tool options files (*.opt*). This capability supports experienced customers who want detailed control over the geometric model and solver settings used to create the S-parameter model that represents a 3-D area.
IBIS-AMI Model Requirements for Statistical Simulation	Accurate statistical channel analysis requires IBIS-AMI models that are designed for both statistical and time domain simulation. Avoid using a model when you cannot confirm that it was designed for statistical simulation.
Identifying Optimum Tap Weights	If you do not already know the best tap weights for the channel, FastEye analysis can automatically identify them for you.
Jitter Distribution Types	You can enable multiple jitter distributions at the same time. If you enable multiple jitter distributions, the total jitter probability distribution function (PDF) is a convolution of the individual jitter distributions.
Manipulating a 3-D View	You can rotate and view a board design from multiple angles in the 3D PCB Viewer to more easily see net topology and PDN implementation. 3-D viewing is especially useful for seeing how a signal via passes through metal areas and how a decoupling capacitor connects to a PDN.

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Topic	Description
Model Channel Frequency Response with Complex-Pole Models	FastEye channel analysis uses channel-response waveforms to create a model of the channel frequency-domain behavior.
Monotonicity Threshold	This General batch simulation setting defines the maximum amplitude of non-monotonicities to ignore.
MultiBoard Project Board IDs	Board IDs are unique identifiers the software assigns to each board in a multiple board design. Board IDs used by dialog boxes and reports make it easier to identify particular boards, or nets or components associated with particular boards.
Oscilloscope Probes	Oscilloscope probes enable you to see voltage waveforms (all simulators) or current waveforms (HyperSim simulator only) at points in the circuit during simulation.
Parametric Sweeps	Use sweeps to automatically vary and simulate design property values over a range that you specify.
Port-Mapping Examples	When you use the Assign Models dialog box to assign a SPICE or Touchstone model, use the spreadsheet to map ports to circuit connections.
Pre-Emphasis and DFE Structures	To use this optional feature, you must know the details of how pre-emphasis/DFE is implemented in the driver/receiver, such as the number of taps and their weights.
Preparing a Board Design File for Translation	In your layout software, prepare your design so that HyperLynx can convert it with minimal translation errors and less post-translation design cleanup.
Quick Analysis Crosstalk Estimation	Quick Analysis crosstalk estimation results provided by general batch simulation are conservative to ensure that the software does not overlook any nets in your design that might experience significant crosstalk.
Setting Up the Software	Edit design-independent options to specify the locations of IC models and design simulation files, license check in/out behaviors, measurement units, and simulation and appearance preferences.
Signal-Integrity Net Constraints	You can specify the following constraints before you run General Batch simulation or signal integrity simulation with measurements in EZwave.
Specifying Device Kits	Use this procedure to specify the location of a device kit and manually load a *<device_kit>.INI* file. In some cases, when you open the design that comes with a device kit, the software automatically loads a *<device_kit>.INI* file for the design.

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Topic	Description
Standard Delay Format	A standard delay format (SDF) file contains interconnect delays between driver and receiver pins. This is an industry-standard file format.
Statistical and Time Domain Simulation Comparison	IBIS-AMI and FastEye channel analyses can provide results from statistical or time domain simulation.
Tips for Running Simulation with Parametric Sweeps	It is easy to specify many sweep simulations, which can produce very long run times and memory overloads. This can happen for very large numbers of individual sweep simulations and for small numbers of individual sweep simulations with many probes.
Trace to Area Fill Coupling Examples	The software searches for area fills to couple to a selected net when you check “Include trace to area fill coupling” from the Coupling Settings dialog box.
Transferring HyperLynx Settings	You can transfer IC model, design folder name, and differential pair net name suffix settings (BoardSim only) that are stored in the HyperLynx initialization file (*BSW.INI*) from a previous HyperLynx installation to the latest HyperLynx installation.
Translating a Board Design	If your board layout software cannot save a board design in HYP, CCE, or ODB++ format, use HyperLynx to translate your board design file to HYP format.
Using the Field Solver	Run the field solver on your schematic for any coupling region that you already defined.
Viewing a Board in 2-D	Use the board viewer to display the topology and components for signal nets and power-distribution networks on a board.
Viewing Coupling Region Details	Use the Coupling Region dialog box to evaluate the crosstalk contribution from trace segments of nearby aggressor nets to the selected victim net, see a geometric relationship between coupled trace segments, or to see trace-to-trace impedance calculations for differential pairs.
Virtual Pins	A branch point exists where one transmission line connects to two or more other transmission lines. The software automatically creates a virtual pin at the branch point to provide a reference point when you create a constraint template.
Opening HyperLynx Tools on Linux	Open HyperLynx SI/PI/Thermal to observe signal integrity, power integrity, or thermal properties for your design. You also can directly open the Touchstone Viewer or Visual IBIS Editor.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

3-D Areas That Include Multiple Nets

The software can automatically define 3-D areas that include multiple nets. For example, your board design may have adjacent SERDES channels and you want a 3-D area to include a via array for all of them. A 3-D area that includes multiple nets accounts for coupling among them.

When including nets in an automatically-defined 3-D area, the software uses these key criteria:

- Set of selected nets:
  - Nets that you have manually selected.
  - Neighboring nets that act as aggressors to manually-selected nets. The software automatically selects neighboring nets that exceed the coupling threshold. See "Accounting for Coupling" on page 131.
- Proximity between neighboring selected nets within a board area:
  - If the pads on neighboring nets are too far apart, an automatically-defined 3-D area excludes neighboring nets.
  - If the pads on neighboring nets are close enough, an automatically-defined 3-D area includes neighboring nets.

  Use the 3D-Area Creation Options dialog box to specify pad-to-pad distance and whether to include neighboring nets only within a specific distance from a primary net. See step 1 in "Modeling a Board Area With an S-Parameter Model" on page 122.

When the software automatically defines a 3-D area with multiple nets, it:

- Makes a copy of the 3-D for each net. Each copy assigns one of the nets (or differential pairs) as a "primary net". When you simulate a net that is represented by a 3-D area with multiple nets, the software uses the copy that specifies the simulated net as a primary net.
- Creates one S-parameter model to represent all copies.

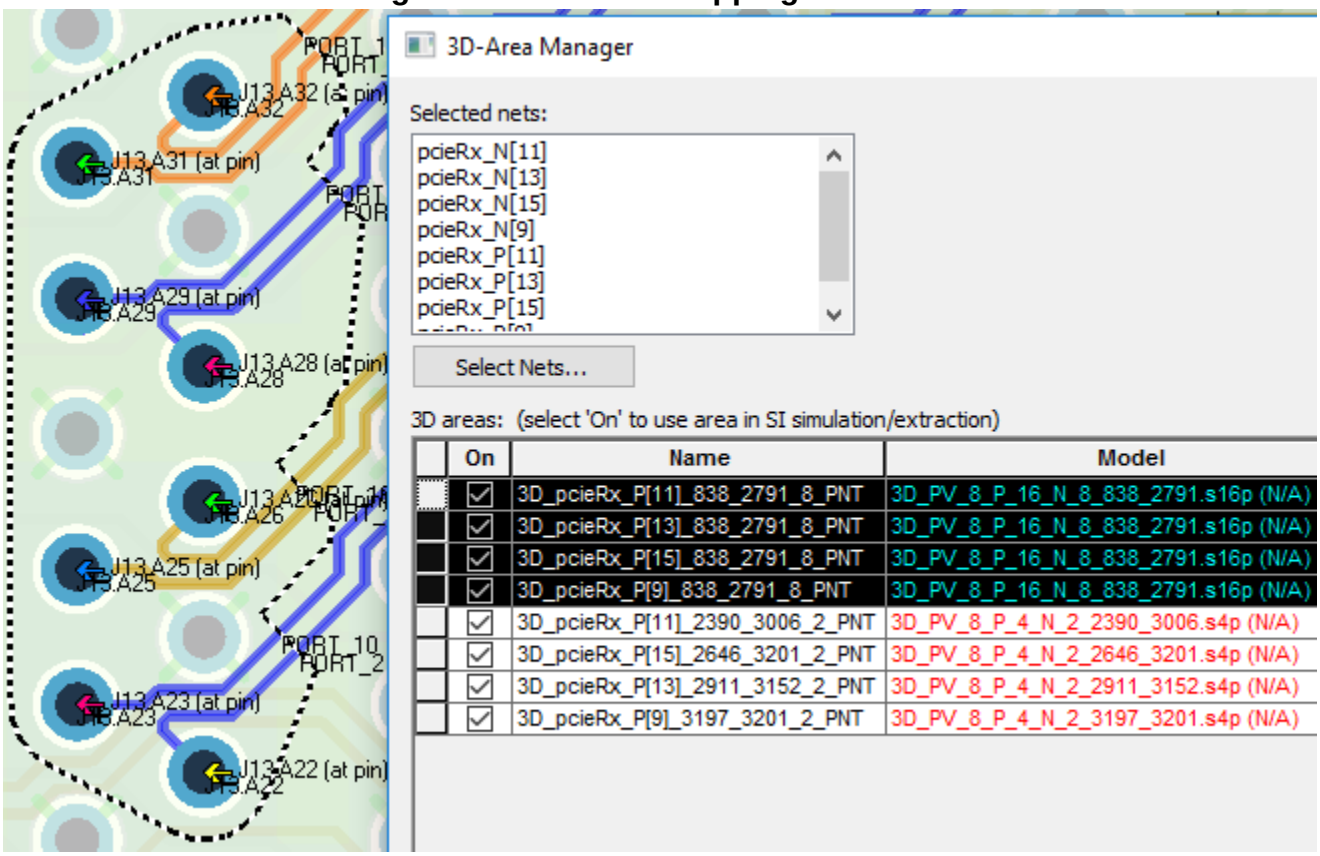
For example, a 16 port S-parameter model can represent an area that contains four differential pairs. The software creates four copies of the 3-D area, where each copy specifies the nets for one differential pair as primary nets. The primary nets are different for each copy.

Example of Automatically-Defined 3-D Areas That Contain Multiple Nets

Figure 8-1 on page 468 shows four differential pairs and four 3-D areas with boundaries that exactly overlap. In the 3D-Area Manager dialog box, the Name column shows a unique name for each 3-D area and the Model column shows the name of a single S-parameter model assigned to all four 3-D areas.

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Figure 8-1. Four Overlapping 3-D Areas



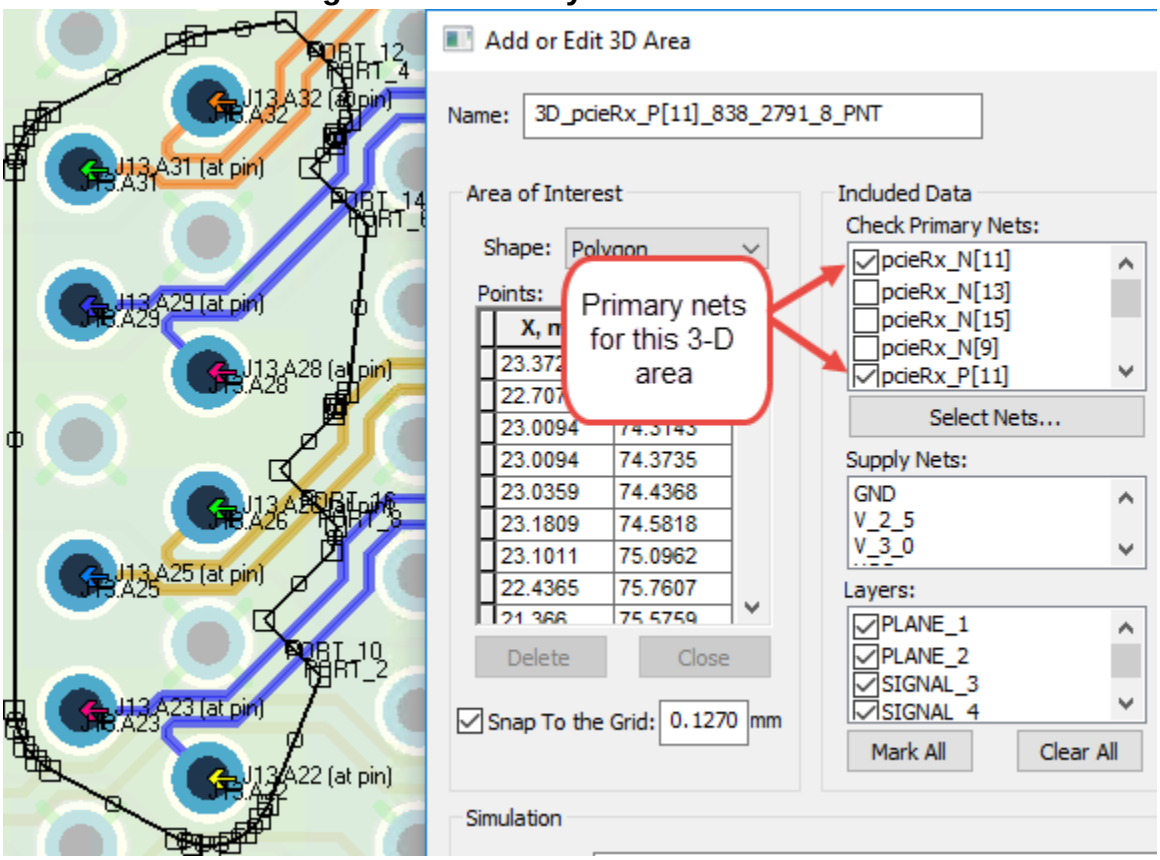
Note

For information about the naming formats for the 3-D areas and models in Figure 8-1 on page 468, see step 4 in "Modeling a Board Area With an S-Parameter Model" on page 122.

Figure 8-2 on page 469 shows the contents of the 3-D area listed in the top row in Figure 8-1 on page 468. Check marks identify the primary nets for a differential pair.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-2. Primary Nets in a 3-D Area

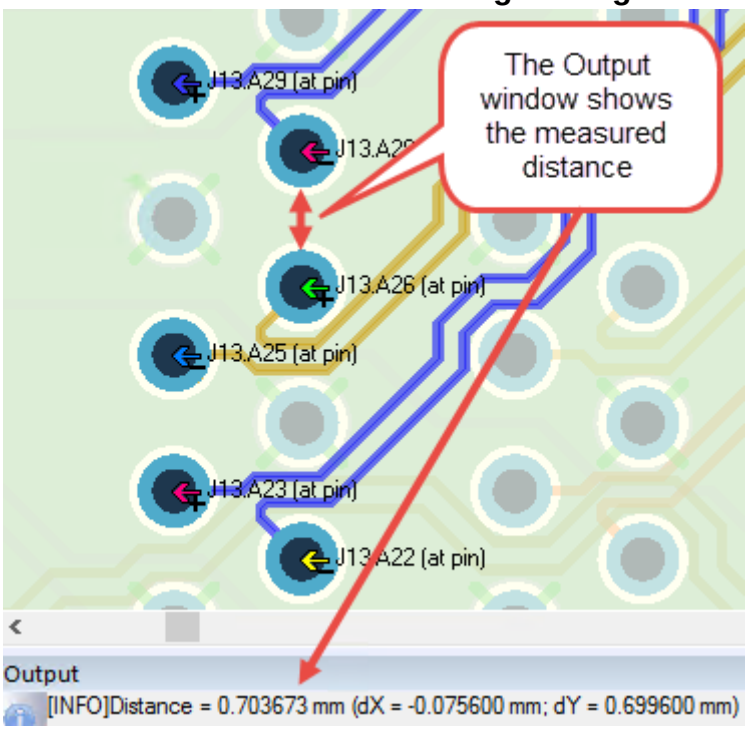


To automatically define 3-D areas that contain multiple nets, you select multiple nets and specify distance thresholds. A key distance threshold option is named “Coupling distance between victim & aggressor pins/pads” in the 3D-Area Creation Options dialog box.

Figure 8-3 on page 470 shows that the distance between pad edges among vias for neighboring differential pairs in this example design is around 0.7 millimeters.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-3. Distance Between Pads for Neighboring Differential Pairs

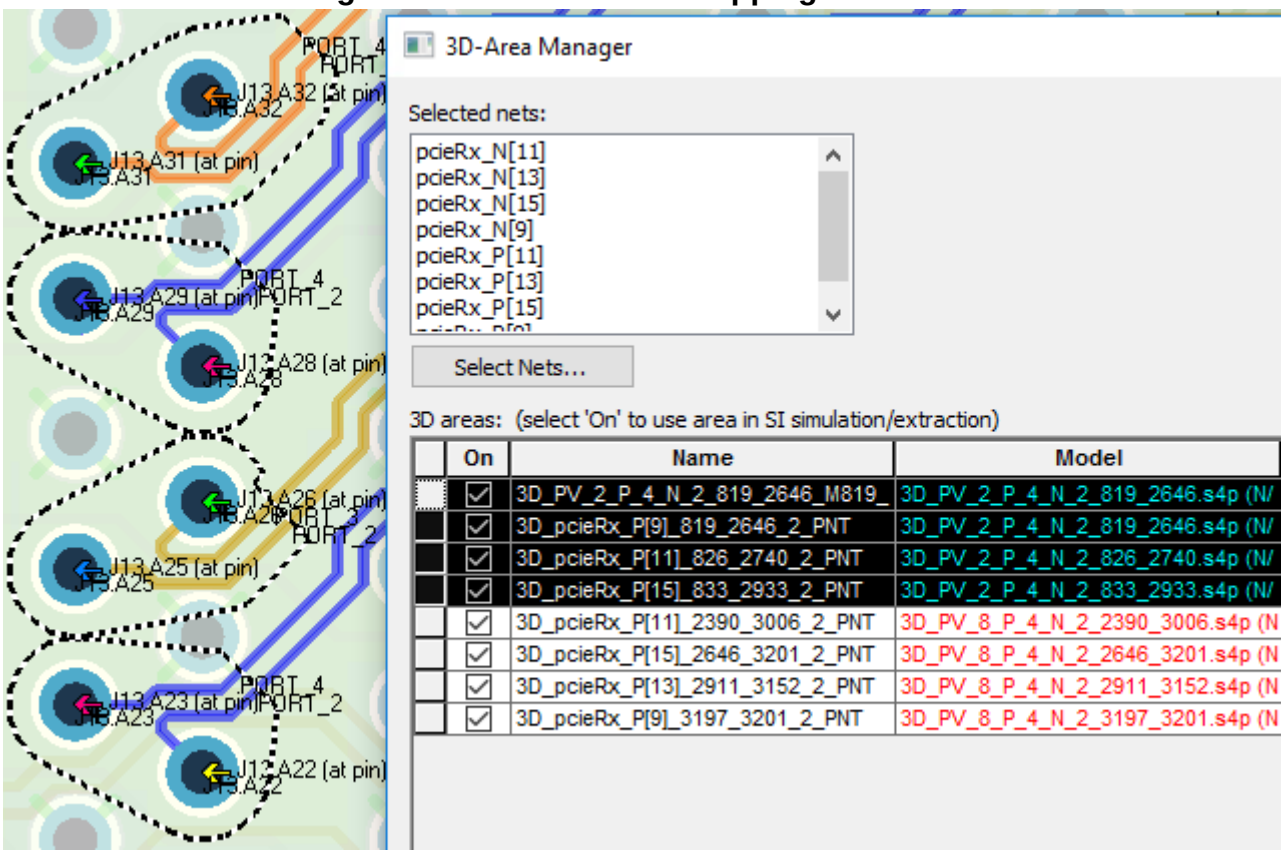


If you specify a value of 0.71 millimeters for “Coupling distance between victim & aggressor pins/pads”, the software includes neighboring differential pairs with pads that are 0.7 millimeters apart and defines the overlapping 3-D areas in Figure 8-1 on page 468.

By contrast, if you specify a value of 0.69 millimeters for “Coupling distance between victim & aggressor pins/pads”, the software excludes neighboring differential pairs with pads that are 0.7 millimeters apart and defines the non-overlapping 3-D areas in Figure 8-4 on page 471.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-4. Four Non-Overlapping 3-D Areas



Related Topics

Modeling a Board Area With an S-Parameter Model

Accounting for Noise Between Single-Ended Signal Via and Power Planes in SI Simulation (Co-Simulation)

You can increase the accuracy of signal integrity simulation by accounting for the energy transferred to the power distribution network when signals propagate through single-ended signal vias.

When you run signal integrity simulation with this option enabled, the software also models the PDN (transmission planes and the decoupling capacitors and stitching vias that connect them), runs power integrity analysis, and provides plane noise voltage (and surface and capacitor currents) in the HyperLynx PI PowerScope Dialog Box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

The energy radiated by the signal propagating through the via provides all plane-noise stimulus.

Note

This option is also called co-simulation because it uses both signal-integrity and power-integrity circuit modeling and simulation engines.

Restrictions and Limitations

- This option does not support SPICE models that contain active components, such as transistors and diodes.
- This option is not available for multiple-board designs.
- Do not set the simulation engine to Fixed ADMS or HSPICE. Set the simulation engine to Auto or Fixed > HyperSim.

Prerequisites

- You have the Co-Simulation license.
- You have assigned power integrity models. If you have information about a VRM in your design, assign a VRM power integrity model. AC power integrity models are ignored. The energy radiated by the signal propagating through the via provides all plane-noise stimulus.
- You have ensured good decoupling performance from the power-distribution network (PDN). PDNs with high impedance, especially at very low frequencies when there is no VRM model, can cause exaggerated plane-noise results.

Procedure

1. Choose the **Setup > Simulation Controls** menu item. The Simulation Controls dialog box opens.
2. Check “SI/PI Co-Simulation”.

   Note

   Some simulation types provide access to the SI/PI Co-Simulation option from their associated dialog boxes.

Results

Co-simulation includes transmission plane noise in signal-integrity results and shows noise from signal vias in power integrity simulation results (plane noise) in the HyperLynx PI PowerScope.

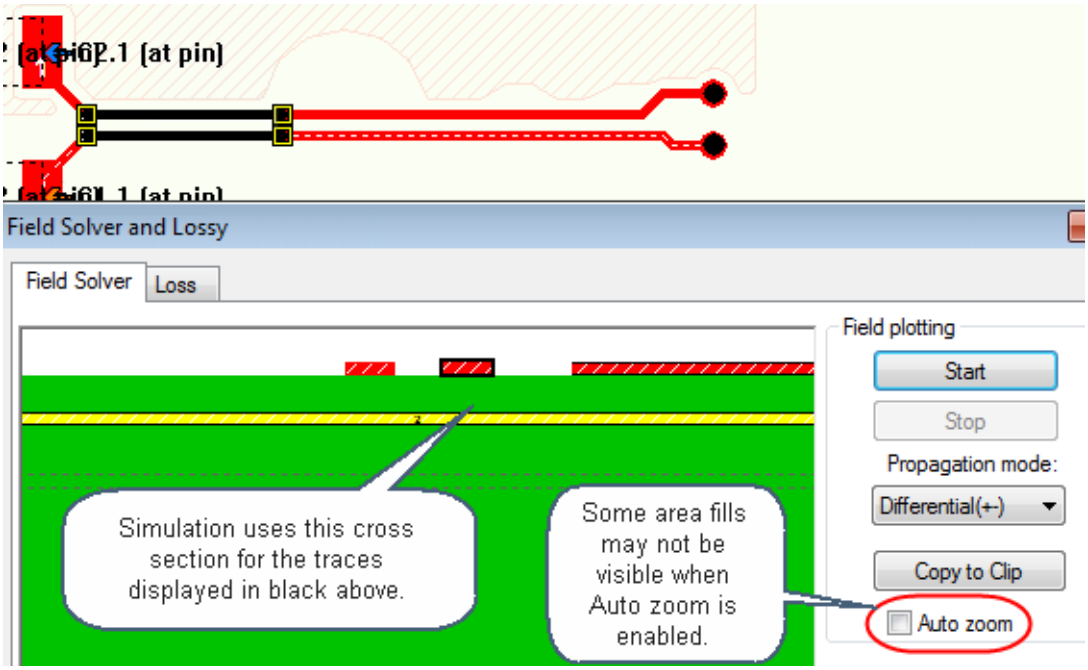
Consider adding stitching vias to your design in areas around single-ended signal vias that transmit significant noise to the PDN.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

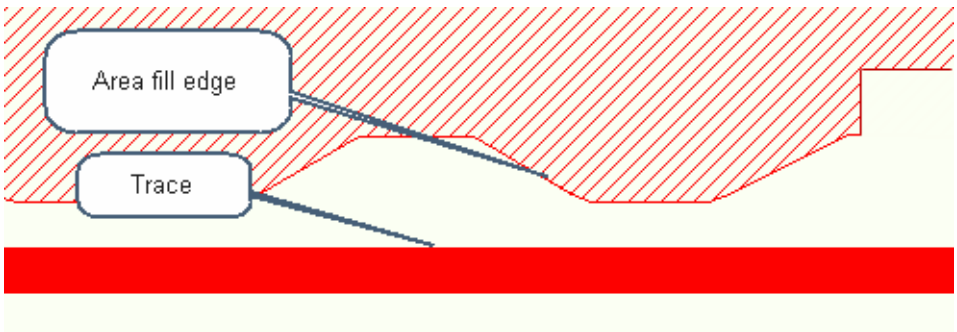
Area Fill Edge Approximation Examples

The software represents an area fill or routed power-supply net with an approximate shape to model coupling to a selected signal net. The position of the selected net and the approximated area fill edge (or routed power-supply net) is a major part of how the software creates cross sections for the field solver.

The figure below shows an example cross section displayed in the field solver.



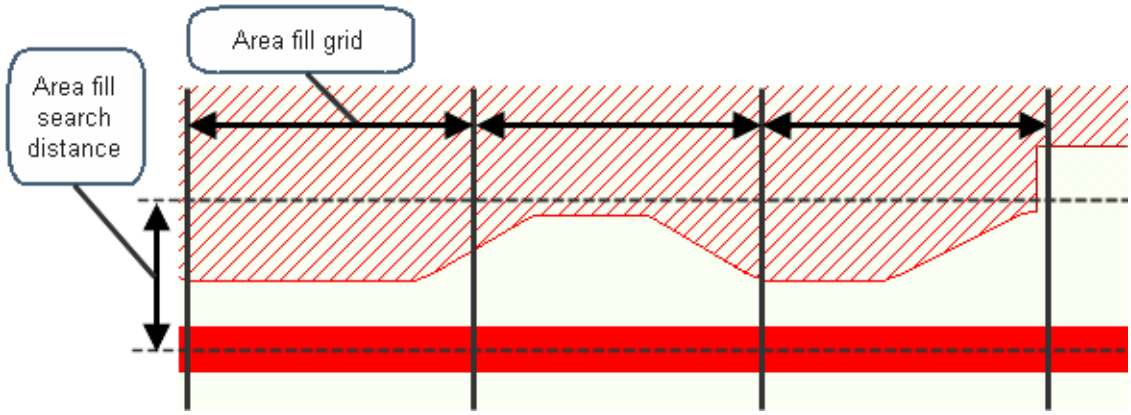
The figure below shows the area fill edge and trace displayed in the board viewer.



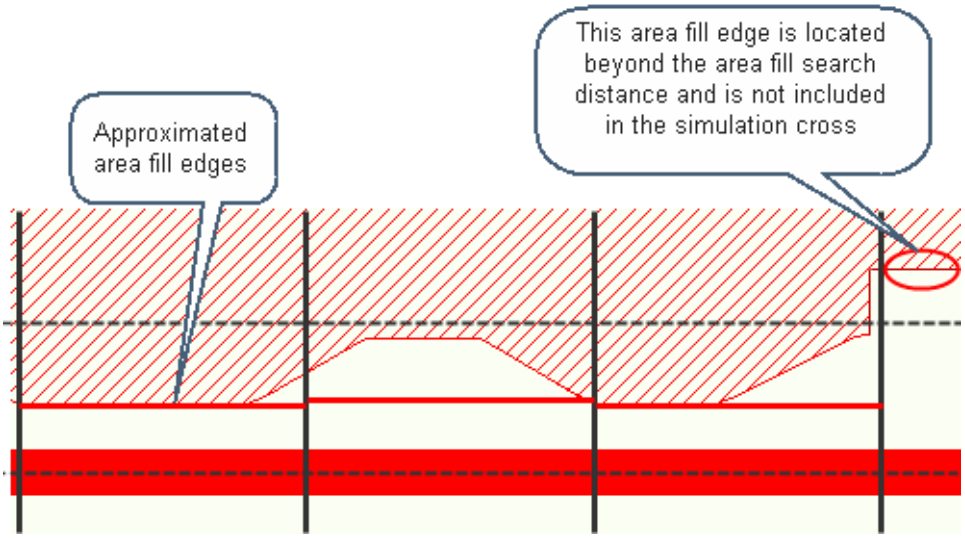
Options from the Coupling Settings Dialog Box define the size of the grid used to approximate the shape of area fills. The grid runs perpendicular to the trace centerline and rotates when the

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trace bends or curves. The figure below shows an example “large” grid and the names of the options that define the length and width of each grid cell.

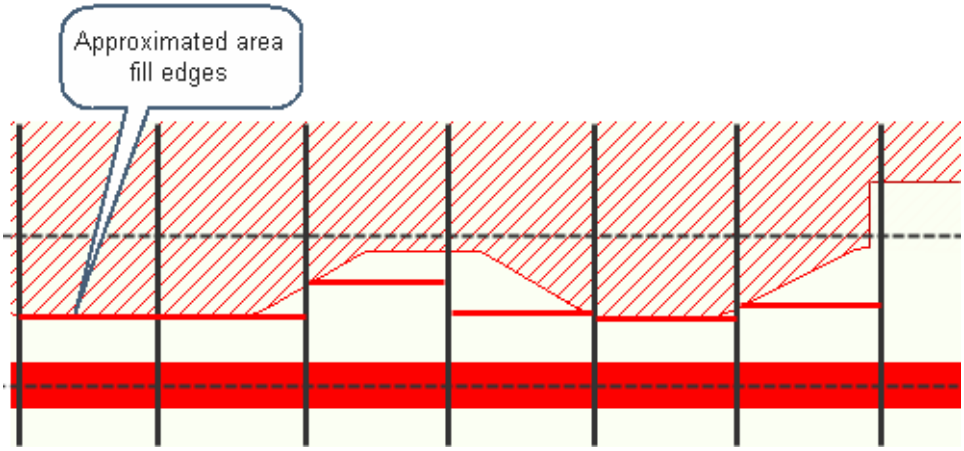


The figure below shows the approximate area fill edge resulting from the large grid.

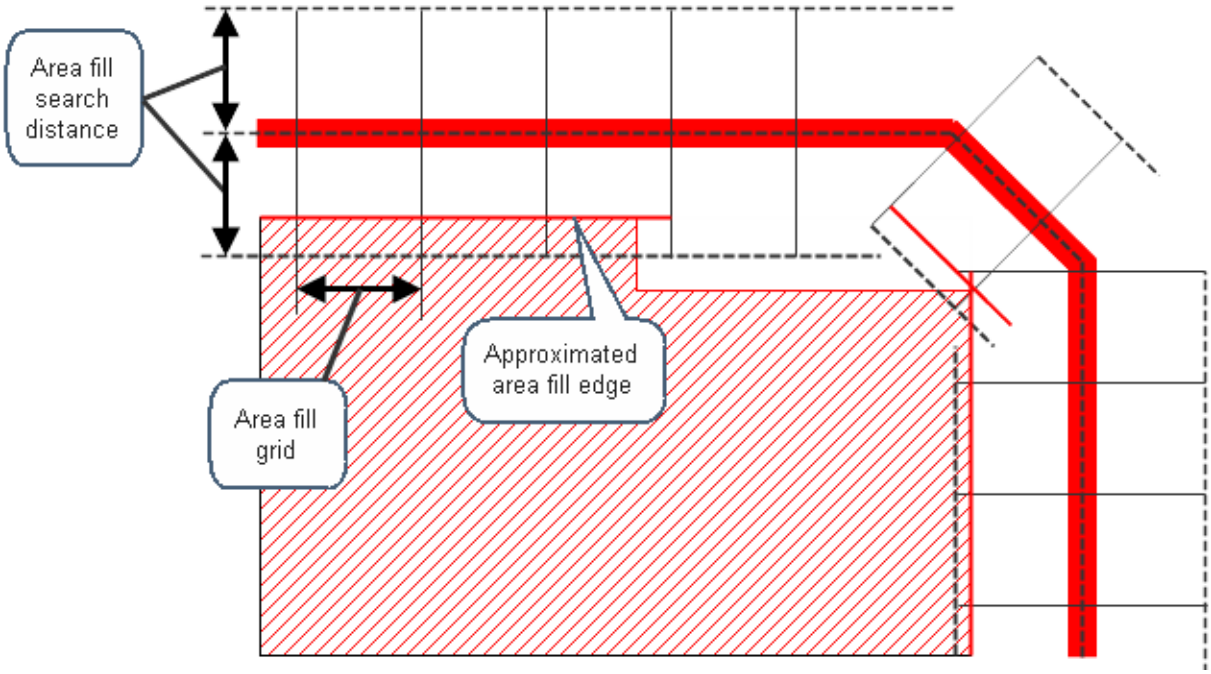


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The figure below shows how a smaller area fill grid can better model small variations in an area fill edge. Creating more cross sections with unique trace to area fill edge distances increases simulation run time.



When the trace bends, the grid rotates so that it continues to run perpendicular to the trace centerline. The figure below shows how the grid rotates as the trace bends around an area fill corner.



The grid also rotates for a curved trace. The curve is divided into many short and straight segments.

Related Topics

Field Solver Dialog Box

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Accounting for Coupling

Horizontal and Vertical Geometric Search Range for Coupled Nets

Coupling Ratio for RLC Matrix Package Coupling

Trace to Area Fill Coupling Examples

Approximate Switching Time

The driver switching time reported in the Approx. Switching Time column of the SI Nets Spreadsheet is based on assigned IC model and net properties.

When you assign an IBIS model to a driver, the software determines the approximate switching time as follows:

Condition	The software...
V-t table exists	Uses the switching time extracted from the table. If multiple tables exist per switching edge, the fastest time is used.
V-t table does not exist	Uses the Ramp keyword in the following way: switching time = typical power-supply range / Ramp value. The software assumes that the driver switches rail to rail because typically, older devices do not have V-t tables.
Devices with asymmetric rising and falling times	Uses the faster time
Devices with differential pair drivers	Uses the switching time from one of the drivers (not the differential switching time)
Nets with multiple drivers	Uses the fastest rising or falling time of all drivers
Nets with no driver	Uses the default driver characteristics rise/fall time value (**Setup** > **Options** > **Preferences** > **General** tab)
Nets with at least one missing IC model	Displays “no model”

Automatic SI Simulator Selection

The software follows a specific selection process when you enable Auto in the “Simulation engine” of the Simulation Controls dialog box or the Simulator area of the oscilloscope.

The software:

- Creates the simulation netlist, taking into account trace models, via models, and component models.
- Inspects the netlist to identify which simulator to use.

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- If the netlist contains neither SPICE nor VHDL-AMS models, uses HyperSim.
- If the netlist contains SPICE models:

Identify the model type(s).	When you assign the SPICE model: • Directly to the component, the model type comes from the "ADMS model syntax type" area of the Preferences Dialog Box - Simulators Tab. • Indirectly to the component, by referencing it from an IBIS model that uses the [External Circuit] or [External Model] keyword, the model type comes from the Language sub-parameter.
Identify the simulator to use, based on the model type.	When all models in the netlist are SPICE or HSPICE: • If HyperSim supports all the elements used by the SPICE models, use it. • Otherwise, use ADMS.
	When the netlist contains VHDL-AMS models, which only ADMS can simulate: • If the netlist also contains HSPICE-Encrypted models, which only HSPICE can simulate, HyperLynx stops and reports the conflict. • If the netlist also contains HSPICE models (that is, SPICE models with HSPICE-specific extensions), use ADMS in HSPICE compatible mode. • Otherwise, use ADMS.
	When the netlist contains HSPICE-Encrypted models: • If the netlist also contains Eldo-Encrypted models, HyperLynx stops and reports the conflict. • Otherwise, use HSPICE.

- Otherwise the software writes the netlist to the SPICE format and uses ADMS.

Related Topics

Supported SI Models and Simulators

Running Signal Integrity Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

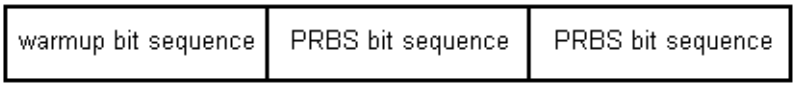
Bit Sequence for Automatic Channel Characterization

If you enable the “Automatically characterize channel” and PRBS options on the Channel Characterization Dialog Box, FastEye channel extraction automatically defines the bit sequence in the Figure 12-4.

This overall bit sequence consists of the following parts:

- Warmup bit sequence — The number of bits matches the value you specify in the “Number of warmup bits before the Tx/channel are stable” box. The bit sequence consists of the last n bits of the PRBS bit sequence plus zero or more full PRBS bit sequences. FastEye channel extraction automatically chooses a PRBS bit sequence with a bit order that is longer than the ISI for the channel. Zero full PRBS bit sequences are needed when the number of skipped bits is less than the full PRBS bit sequence length.
- PRBS bit sequence — FastEye channel extraction automatically chooses a PRBS bit sequence with a bit order that is longer than the channel ISI, and then applies it twice.

Figure 8-5. Bit Sequence for Automatic Channel Characterization



Related Topics

Channel Characterization Dialog Box

BoardSim Files

BoardSim reads and writes several kinds of files.

Table 8-1. BoardSim Associated Files

File Name	Description	Created	Location
<design_name>.HYP	PCB layout information	By the translator	Anywhere. Project directory recommended.
<design_name>.CCE	PCB layout information	By Xpedition Layout and other Siemens EDA software	Anywhere
<design_name>.TGZ	PCB layout information	By layout software that can create ODB++ files	Anywhere

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 8-1. BoardSim Associated Files (cont.)

File Name	Description	Created	Location
<design_name>.PJH	Contains design-specific preferences	When saved in BoardSim	Same location as PCB layout information file
<design_name>.REF	Contains reference designator to model mappings.	Using any text editor, if not automatically generated or through GUI	Same location as PCB layout information file
<any_name>.QPL	Contains part number to model mappings.	Using any text editor or GUI	Anywhere
<design_name>.BUD	Board User Data file. Contains session edits.	When saved in BoardSim	Same location as PCB layout information file
<design_name>.BBD	Backup copy of *.BUD*	When saved in BoardSim with existing *.BUD* file	Same location as PCB layout information file
BSW.INI	Contains non-design-specific preferences. Updates settings.	During installation (also applies to LineSim)	In the HyperLynx installation directory.
BSW.PAK	Contains package definitions of common variety of library.	Shipped with HyperLynx	

BoardSim Board File Contents

A BoardSim board file is an ASCII text file, with a *.HYP* extension, formatted in a HyperLynx-proprietary format. It contains all of the information about a PCB layout needed for signal-integrity simulation.

For each board you simulate, you run a translator on your PCB-layout data to produce a BoardSim Board file. Then, you load the BoardSim board into BoardSim and simulate.

You may never need to view the contents of a BoardSim board file, but it is helpful to have a basic understanding of what the BoardSim board contains.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

BoardSim Board Contents

Item	Description
Board Outline	The board outline data defines the shape of your board. An outline can include both linear and curved segments. The board-outline data are optional; not all PCB-layout tools provide it. If the data is missing, BoardSim creates a rectangular outline big enough to encompass all of the components on the board.
Stackup	The stackup data defines your board's layer stackup. A stackup includes information about signal, power-plane, and dielectric layers. The stackup data are optional; not all PCB-layout tools provide it. If the data are missing, BoardSim will attempt to create an electrically valid stackup, but will warn you to edit it.
Devices	The device data defines the components on your board. Device information includes reference designators, component names (for ICs), and component values (for passive components). The device data are required. BoardSim must have at least some information about the devices on a net to perform a simulation.
Pad Stacks	The pad-stack data defines the various pad stacks used on your board. Pad-stack definitions are optional. Some older .HYP-file translators do not use explicit pad-stack definitions; newer ones do.
Nets	The net data defines the nets on your board. Net information includes definitions for each metal segment, via, pad, and device pin on the board. The net information is required. BoardSim must have detailed information about trace metal to model and simulate the net.
Comment Lines	Comment lines in the .HYP file must have an asterisk (*) in the first column. On rare occasions, you may wish to remove an element from a .HYP file by commenting out the element's line. For example, if you wished to remove a resistor's pin from a certain net, you could precede the pin's record with an asterisk: *(PIN X=2.100 Y=2.350 R=Udrv1.1 P=PS4) This is now a comment line

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

BoardSim Session Files

Session edits are any changes you make to the database while using BoardSim. BoardSim captures your edits during a session and saves them to a file when you exit or choose to save them in the middle of a session (select **File > Save BoardSim Session File**).

When you first create a board file, there is no corresponding session file. The session file is created only after you load a design file for the first time, make some edits and either save session edits, or exit the software.

When you re-load the same board in another session, the software reads the edits from the file and automatically restores them.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

CAPLIB File Format

Used by: **Simulate PI > Advanced Decoupling Wizard > Choose a Type of Analysis page > PDN Decoupling Optimizer**

The CAPLIB decoupling capacitor library file format describes a custom set of capacitors that the software considers when running PDN decoupling optimization.

Format

A CAPLIB file is a text file in CSV format, with a *.caplib* extension.

You can add additional data to the file to help you filter a capacitor library. The additional data is not used by the software in analysis. For example, you can add PARTNUMBER, HEIGHT, ESR, ESL, ESC, or COMMENT data to the file.

The file must conform to the following formatting and syntax rules:

Parameters

Keyword	Example Value	Description
NAME	0.1UF_TANT_AXIAL_35V	Unique name
MODEL TYPE	RLC, SPICE, or TOUCHSTONE	Model type Value can be empty. **Note:** Provide a value if you want to use this data to filter the list of capacitors in the library when selecting capacitors to include in analysis. This data is not used by the software in analysis.
MODEL	R26.0;L2e-09;C1e-07 filename.sp filename.s2p	R, L, and C values or SPICE or Touchstone model filename **Note:** Model files must be in the same folder as the .caplib file.
FOOTPRINT	AXIAL RADIAL 0805 1210	Size of the capacitor
CAPACITANCE	0.000022 1e-8	Value of capacitance Value must be greater than zero.
PRICE	0.02	Price per capacitor Value must be greater than or equal to zero.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Keyword	Example Value	Description
MATERIAL	TANT C0GNP0 X5R	Material Value can be empty. **Note:** Provide a value if you want to use this data to filter the list of capacitors in the library when selecting capacitors to include in analysis. This data is not used by the software in analysis.
VOLTAGE	10V 16V	Voltage Value can be empty. **Note:** Provide a value if you want to use this data to filter the list of capacitors in the library when selecting capacitors to include in analysis. This data is not used by the software in analysis.
TOLERANCE	+10;-10	Capacitance Tolerance Value can be empty. **Note:** Provide a value if you want to use this data to filter the list of capacitors in the library when selecting capacitors to include in analysis. This data is not used by the software in analysis.

Examples

Example 1:

```
NAME,MODELTYPE,MODEL,FOOTPRINT,CAPACITANCE,PRICE,MATERIAL,VOLTAGE,TOLERANCE
0.1UF_TANT_AXIAL_35V,RLC,R26.0;L2e-09;C1e-07,AXIAL,1e-7,0.02,TANT,35V,+10;-10
0.15UF_TANT_AXIAL_35V,RLC,R21.0;L2e-09;C1.5e-07,AXIAL,1.5e-7,0.02,TANT,35V,+10;-10
0.22UF_TANT_AXIAL_35V,RLC,R17.0;L2e-09;C2.2e-07,AXIAL,2.2e-7,0.02,TANT,35V,+10;-10
```


Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Checking Channels for Linear and Time-Invariant Behavior

SERDES channel analysis requires channels that exhibit linear and time-invariant (LTI) behavior, which consists of the following characteristics:

- Driver output buffer can have a non-linear I-V characteristic, but not a large non-linear capacitance characteristic
- Driver output buffer must have a unique and unchanging response to any given stimulus sequence
- Driver output buffer can have asymmetric rise and fall transitions, as long as you do any of the following in the Channel Characterization Dialog Box:
  - If you characterize the channel automatically, enable PRBS in the Characterization type area.
  - If you characterize the channel manually and provide step/pulse waveforms, enable “Pulse and step waveforms (recommended)”.
  - If you characterize the channel manually, provide PRBS waveforms.
- Interconnect between driver and receiver must be linear, which is not a problem with passive interconnects (metal, dielectric)
- The receiver input stage must present a linear load (for example, no diode clamping)

If the channel has LTI behavior, FastEye channel analysis results match or nearly match standard (time domain) eye diagram results for the same channel.

FastEye channel analysis checks the linearity of the channel by comparing the energy in the actual pulse response to the energy in the calculated pulse response, where the calculated pulse response is the difference between the actual step response and the actual step response negated and delayed.

Non-linear channel behavior is usually produced by the attached ICs and not the interconnections, which are usually passive. The behavior of the driver usually affects channel linearity most. The I/V curve in the operation region must be linear or have exactly the same response to any given stimulus sequence.

Tip

Even if you use non-linear driver/receiver models, you might be able to use FastEye channel analysis to evaluate the intrinsic properties of the bare channel by temporarily assigning linear IC models. This capability enables you to see whether changing the net topology or interconnection structures, such as vias, opens or closes the eye.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

Contents of Waveform Files in CSV Format

The time data in the file is in seconds; voltages are in volts; currents are in amps.

The polarity for current measurements is positive for current flowing into the component and negative for current flowing out of the component. See the following examples:

- For a series resistor, you will see positive current on one end where it is flowing in, and negative current on the other end where it is flowing out.
- For a driver switching from low to high, you will mostly see negative currents which indicate that current is flowing out of the driver.
- For a driver switching from high to low, you will mostly see positive currents which indicate that current is flowing into the driver.

There can be a difference in how voltages and currents are measured in the CSV file. If you choose to probe at the pin, voltages are measured outside the package, but currents are measured inside. Thus, an oscilloscope probe that appears in the schematic to be outside of an IC is really located inside the IC package for current measurements, but may be located outside the IC package for voltage measurements.

Some European versions of Microsoft Excel cannot open the CSV file from the Open menu, possibly due to the use of commas as decimal indicators. A workaround is to open Windows Explorer and double-click the CSV file.

Related Topics

Digital Oscilloscope Dialog Box

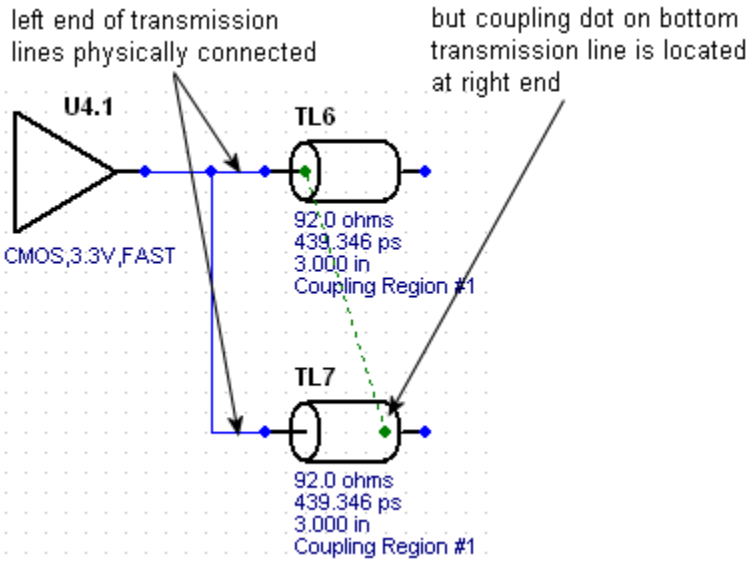
Coupling Dots

Coupling dots indicate the orientation of coupled transmission lines. To correctly simulate the phase of crosstalk voltages and currents, ensure that the correct ends of the coupled transmission lines are marked with coupling dots.

The following figure shows a schematic with both coupling dots displayed incorrectly.

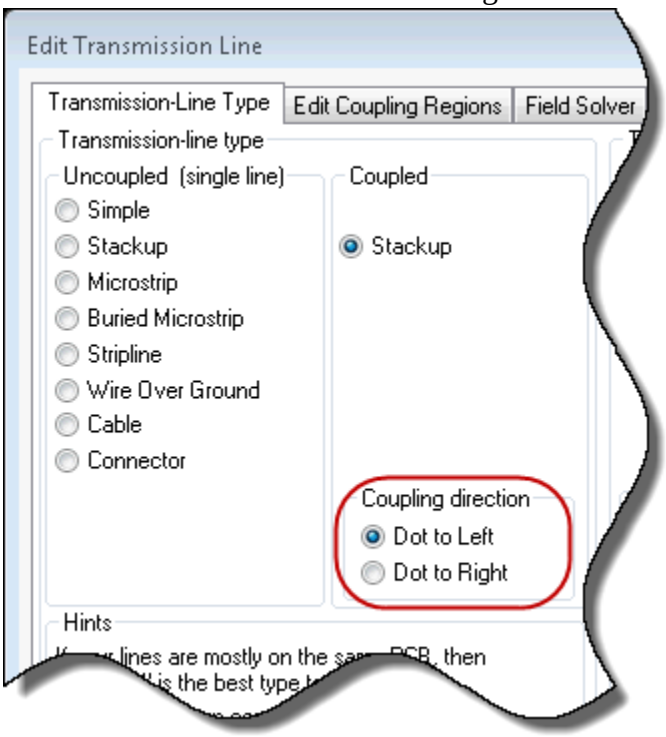
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Figure 8-6. Example of Incorrect Locations of Coupling Dots



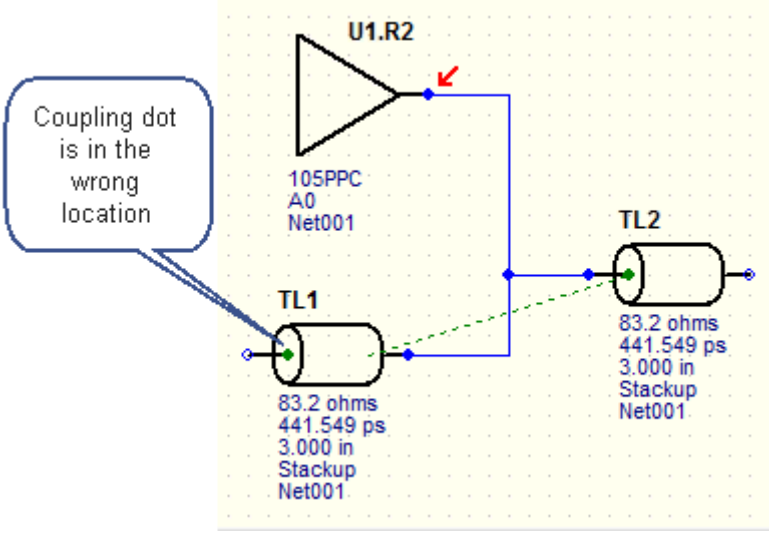
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

To switch a coupling dot to the opposite side, double click a transmission line and select the opposite dot direction in the Edit Transmission Line dialog box:



Examples

Figure 8-7. Example 1 - Incorrect Coupling Dot Location



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-8. Example 1 - Correct Coupling Dot Location

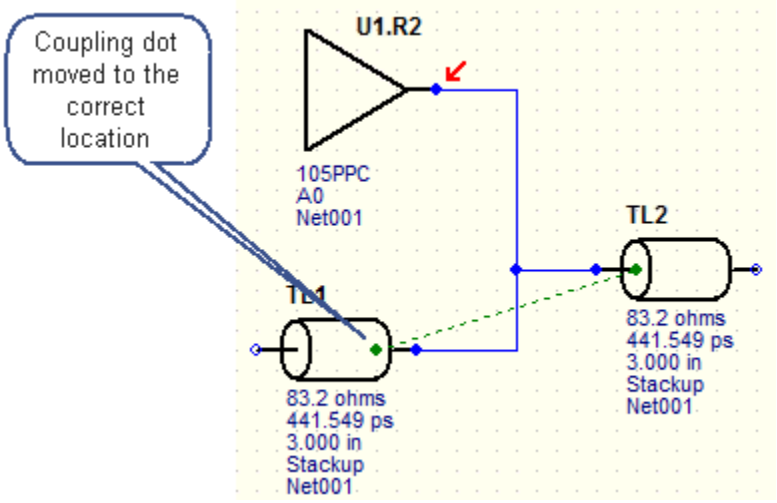
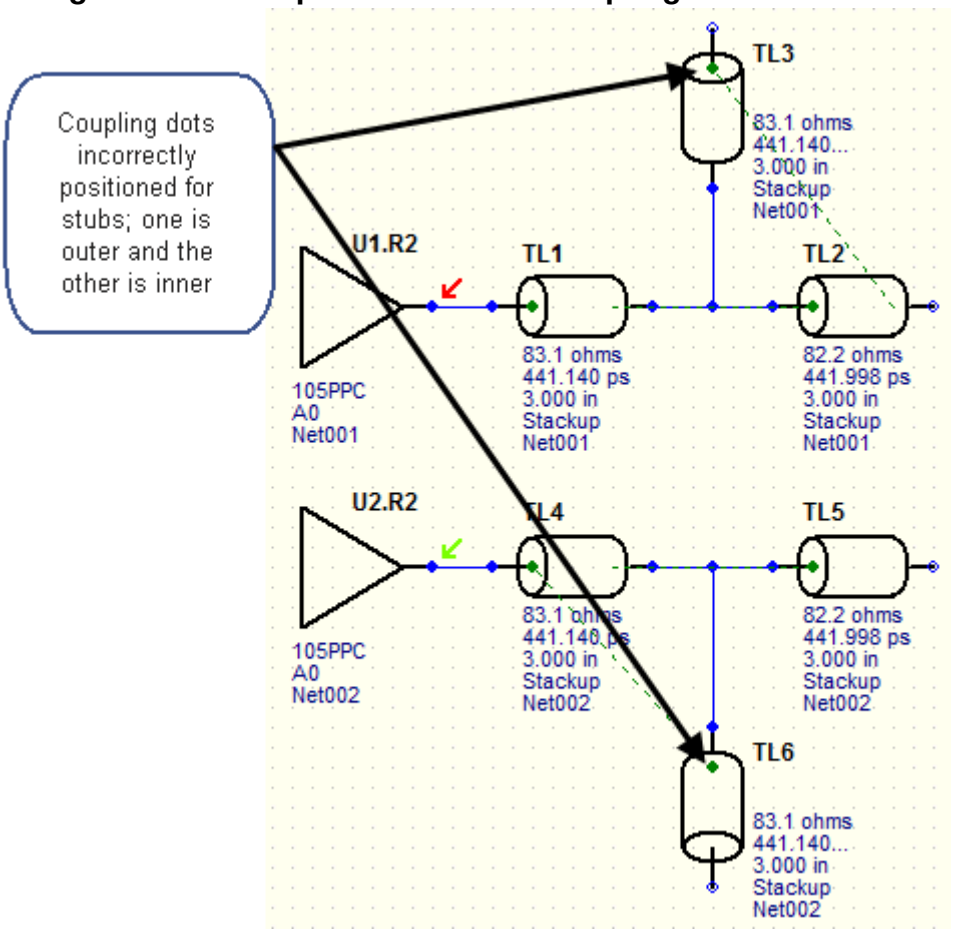
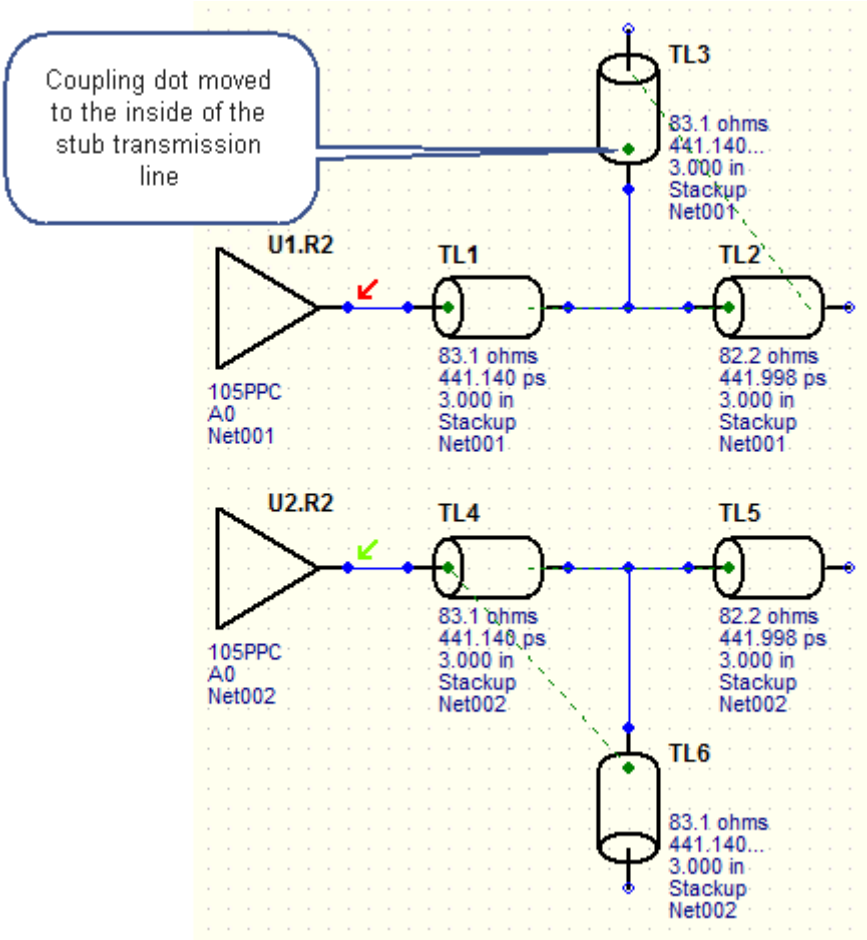


Figure 8-9. Example 2 - Incorrect Coupling Dot Location



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Figure 8-10. Example 2 - Correct Coupling Dot Location



Coupling Ratio for RLC Matrix Package Coupling

The coupling ratio (C_mutual/C_self) determines which package pins and nets SI simulation includes as aggressors when the package is modeled with RLC matrix information within the [Define Package Model] keyword. Use the Coupling Settings dialog box to specify the coupling threshold.

Simulation includes coupling from a neighboring pin when the following equation is true:

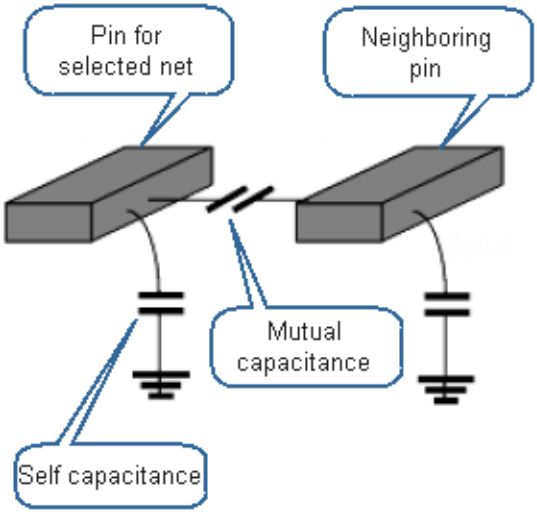
20*log(C_mutual/C_self) > the dB coupling threshold value

Where:

C_{mutual} is the mutual capacitance between the pin for the selected net and the pin for the neighboring net.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

C_{self} is the self capacitance of the pin for the selected net.



Related Topics

Accounting for Coupling

Horizontal and Vertical Geometric Search Range for Coupled Nets

Trace to Area Fill Coupling Examples

Area Fill Edge Approximation Examples

Creating and Applying a DDRx Delays File

If your memory interface supports leveling and calibration capabilities for read/write operations, the DDRx Wizard uses a delays file to minimize skew among the appropriate signals.

During a first simulation pass, the wizard measures the relative timing skews among the signals and saves the results to a delays file. This information specifies the amount of controller skew needed to optimally de-skew the timing among nets. During a second simulation pass, the software uses a delays file to create measurements that account for skew. For more information, see “Types of Calibration” on page 792 and step 3.

Procedure

1. Choose the **Simulate SI** > **Run DDRx Batch Simulation** menu item. The DDRx Batch-Mode Wizard opens.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

2. Ensure that you specify the following:

Initialization page	1. Select an interface that supports leveling and calibration capabilities for read/write operations. 2. Specify the data rate.
Nets to Simulate page	Uncheck all options except for the “Clock-to-strobe skew timing” option to minimize the simulation run time.

3. Choose one of the following methods:

If you want to...	**Do this...**
Automatically generate a delays file and apply to a second simulation	1. On the Leveling&Calibration page, check “Automatically run simulation 2 times”. The first simulation pass creates a *DDR_Delays_Pass1.txt* file, which the wizard applies to the second simulation pass, and saves it to a *DDR_Results** folder. 2. On the Simulate page, click **Run Batch Simulation**. Save your setup file and click **Run**.
Note: If you have assigned power-aware IBIS models and want to save overall run time, you can exclude PI effects before creating a delays file. To exclude PI effects, on the Stimulus and Crosstalk page, uncheck “Include PI effects in SI simulations”. For the next simulation, check “Include PI effects in SI simulations”, uncheck “Automatically run simulation 2 times”, then import the delays file.	
Generate a delays file and then import it for a second simulation	1. For a first simulation, on the Leveling&Calibration page, uncheck “Automatically run simulation 2 times”. 2. n the Simulate page, click **Run Batch Simulation**. Save your setup file and click **Run**. The wizard creates the *DDRxDelays_autogenerated.txt* file in the design directory (not in a *DDR_Results** folder). Rename the file so that it is not overwritten by the next simulation. 3. For a second simulation, on the Leveling & Calibration page, select “Use values from the DDRx Delays external file”, then import the delays file.

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Results

Using either method in step 3, the wizard creates and applies a delays file for more accurate simulation:

- The automatic method is more convenient and saves delays files in the results folder. The automatic second simulation pass creates a *DDR_Delays_Pass2.txt* file, which you can use for reference or troubleshooting. If you need to manually apply a delays file, you can later import the file on the Leveling&Calibration page and then run simulation.
- The manual method requires more steps, and saves a delays file (*DDRxDelays_autogenerated.txt*) in the design folder. If you want to preserve this file, you must rename it.

The wizard always creates new delays files for each DDRx simulation, and you must ensure that you apply the delays file if you make any of the following changes to your design:

- Board-to-board connector models in a multiple-board project
- IBIS or timing models
- Data rate
- Component speed grades

Related Topics

Preparing for DDRx Simulation

Write/Read Leveling for DDRx

Creating a Custom Stimulus

If you need to assign a specific stimulus that the software does not provide, create a custom stimulus.

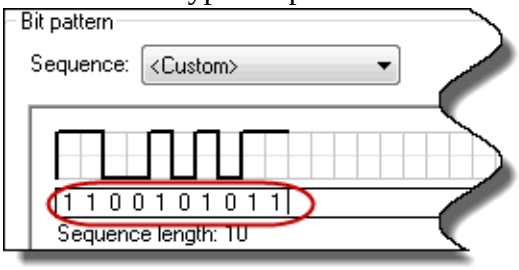
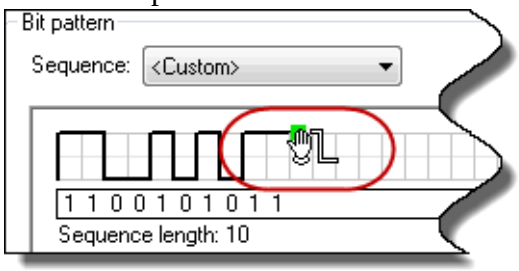
If you have a stimulus file (EDS) that you want to use, copy it to your stimulus file folder (the design folder by default). If necessary, choose the **Setup > Options > Directories** menu item to change this location.

Procedure

1. Open the Assign Stimulus dialog box (**Setup > Stimulus > Edit Stimulus**), then click **Edit Stimulus**.
2. In the Edit Stimulus dialog box, select a standard sequence or select <Custom> from the Sequence list.

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3. To create a custom bit sequence:

If you want to...	Do the following...
Type	Click in the field and type the pattern. 
Draw	Click and draw the pattern with the mouse. 
Load a bit pattern file (BIT)	1. Type the bit sequence in a text file, separating 0 and 1 values by a separator (space, comma, semicolon or return character), and save the file in ASCII format with the extension *.bit*. 2. Click **Load** and select the file. **Note:** When you save the bit pattern to a file, only the bit values are written. If you had previously loaded a bit pattern file with bit value separators, such as spaces or carriage returns into the Bit Pattern Editor, the separator characters are not saved.

4. Set options to describe the bit interval, bit rate and sequence repetitions, and add jitter as needed. See Edit Stimulus Dialog Box.

5. Click  to save the EDS file.

Results

The custom stimulus now appears as a selection when you assign a stimulus to a net or pin.

Related Topics

Assigning a Stimulus

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Current Flow For DC Drop

Current flowing between a voltage provider (source) and its current consumers (sinks) traverses the power supply net with some current density and voltage drop.

The following figure shows the current flow between a voltage source and current sink. This figure only shows the current that flows horizontally across a single metal stackup layer. For an example of current flowing vertically through vias, see Figure 8-12.

Figure 8-11. DC Drop Current Flow - Current Density

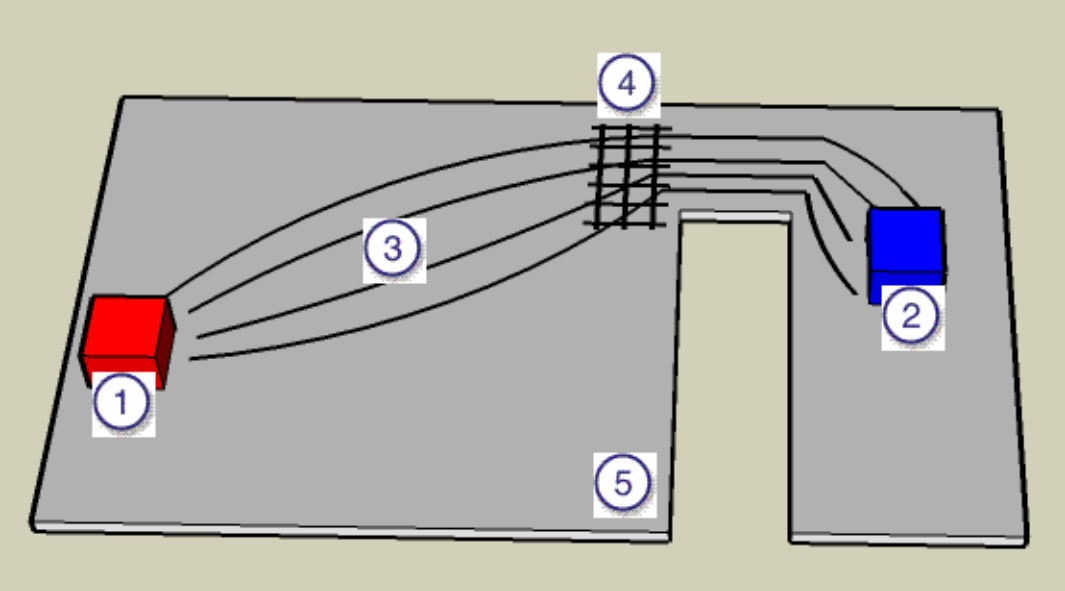


Table 8-2. DC Drop Current Flow - Current Density

1	Voltage source, such as a voltage-regulator module (VRM), which is a type of DC-to-DC converter.
2	Current sink, such as IC power supply pins.
3	Current flowing between the voltage source and current sink.
4	Current density—Everywhere on the metal, some current density J (mA/mil^2 or A/mm^2) and some associated voltage drop.
5	Metal stackup layer—A single layer of metal of a certain material and thickness.

Current always flows in a loop, so the total current flowing in a power net also flows in the ground net (but with a different distribution).

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Figure 8-12. DC Drop Current Flow - Current Loop

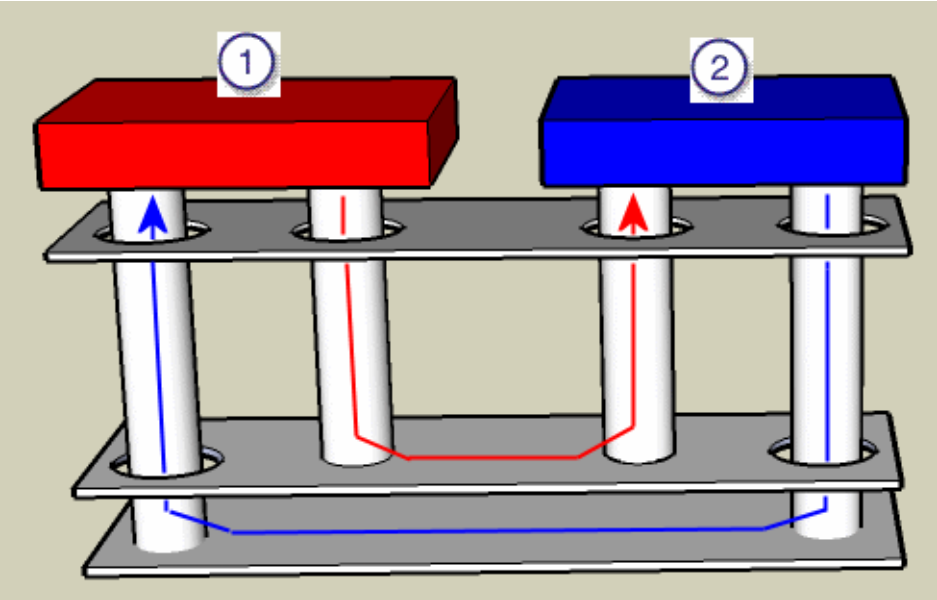


Table 8-3. DC Drop Current Flow - Current Loop

1	Voltage source, such as a voltage-regulator module (VRM), which is a type of DC-to-DC converter
2	Current sink, such as IC power supply pins

Related Topics

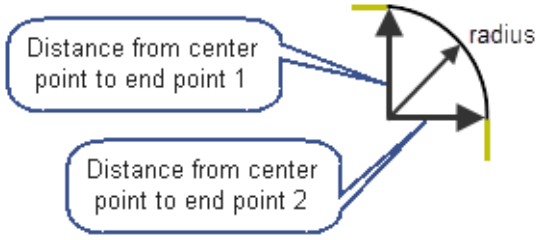
DC Drop Simulation

CURVE Subrecords with Invalid Coordinates

Translators can sometimes create *.HYP* files that contain CURVE subrecords (in a NET keyword) with coordinates that do not add up. The software checks the distance between the center point and each of the end points in the CURVE subrecord and if either of them mismatches the specified radius by more than 5%, the arc is invalid.

Figure 8-13 shows what the software measures to determine if an arc is valid.

Figure 8-13. CURVE Subrecord - Distance Between Center and End Points



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Even though this problem is more likely to happen for curves with very small radii, the problem is typically caused by bad data and not numerical rounding.

For example, in the CURVE subrecord below, the distance between the center point (XC/YC) and an end point (X1/Y1) is more than 5% different than the radius (R). The example does not provide units.

```
(CURVE X1=0.149555 Y1=-0.223520 X2=0.149631 Y2=-0.223698 XC=0.149631 YC=-
0.223596 R=0.000102)
```

Use the following Euclidian equation to calculate the distance between XC/YC and X1/Y1:

$$\text{Distance} = \sqrt{(XC - X1)^2 + (YC - Y1)^2}$$

Substituting values:

$$\text{Distance} = \sqrt{(0.149631 - 0.149555)^2 + ((-0.223596) - (-0.223520))^2}$$

Solution:

$$\text{Distance} = 1.0748e-4$$

The distance from XC/YC to X1/Y1 is 5.37% greater than the radius. The distance from XC/YC to X2/Y2 matches the radius.

DC Drop Conceptual Circuits

Conceptual circuits demonstrate how current flows from a VRM model, through a power-supply net and to a current sink model. The information in this topic also helps you map circuit elements to terminology used in DC drop simulation results.

DC Drop Simulation Circuit - Simulating a Single Power-Supply Net

The following figure shows the power-integrity models and main circuit elements used for DC drop simulation.

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Figure 8-14. DC Drop Circuit - Simulate One Power-Supply Net

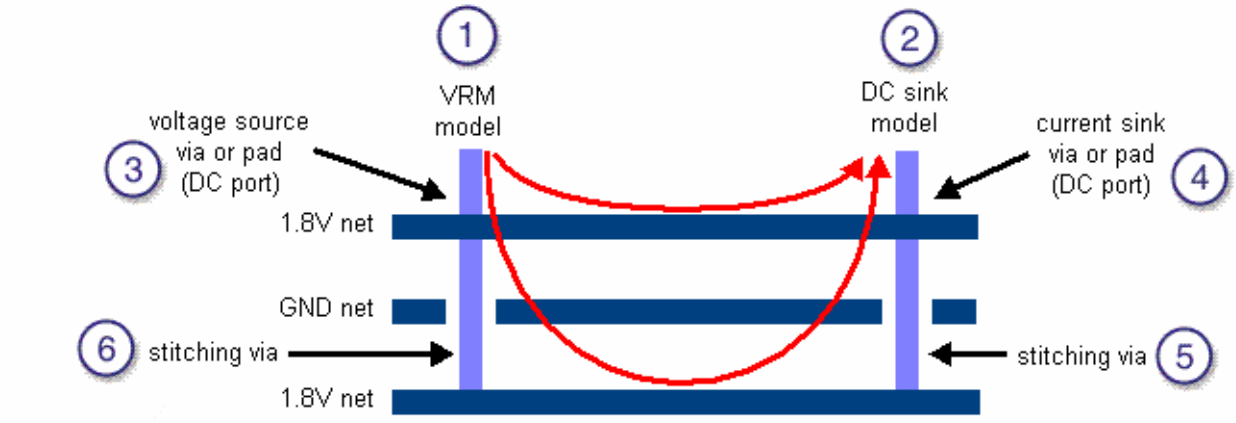
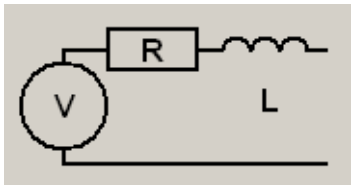
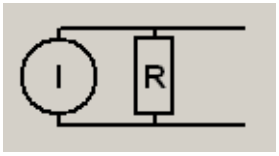


Table 8-4. DC Drop Circuit - Simulate One Power-Supply Net

Object	Description
1	VRM model—Voltage source implemented in the design by voltage-regulator modules (VRMs), which is a type of DC-to-DC converter. The figure below shows the simulation circuit for a VRM.  Negative polarity indicates current flow out of the pin. VRM pins often connect to the power-supply net through one or more of the following: surface-mount pads, component-pin vias, trace segments, and so on.
2	DC sink model—DC current sink implemented in the design as IC power-supply pins and perhaps other components. The figure below shows the simulation circuit for a DC sink.  Positive polarity indicates current flow into the pin. DC sink pins often connect to the power-supply net through one or more of the following: surface-mount pads, component-pin vias, trace segments, and so on.
3	Point where current enters a metal area, usually by a via or pad. **Note:** A point where current enters a routed trace is reported in the Other vias section of the numerical report. See HyperLynx PI PowerScope Dialog Box.

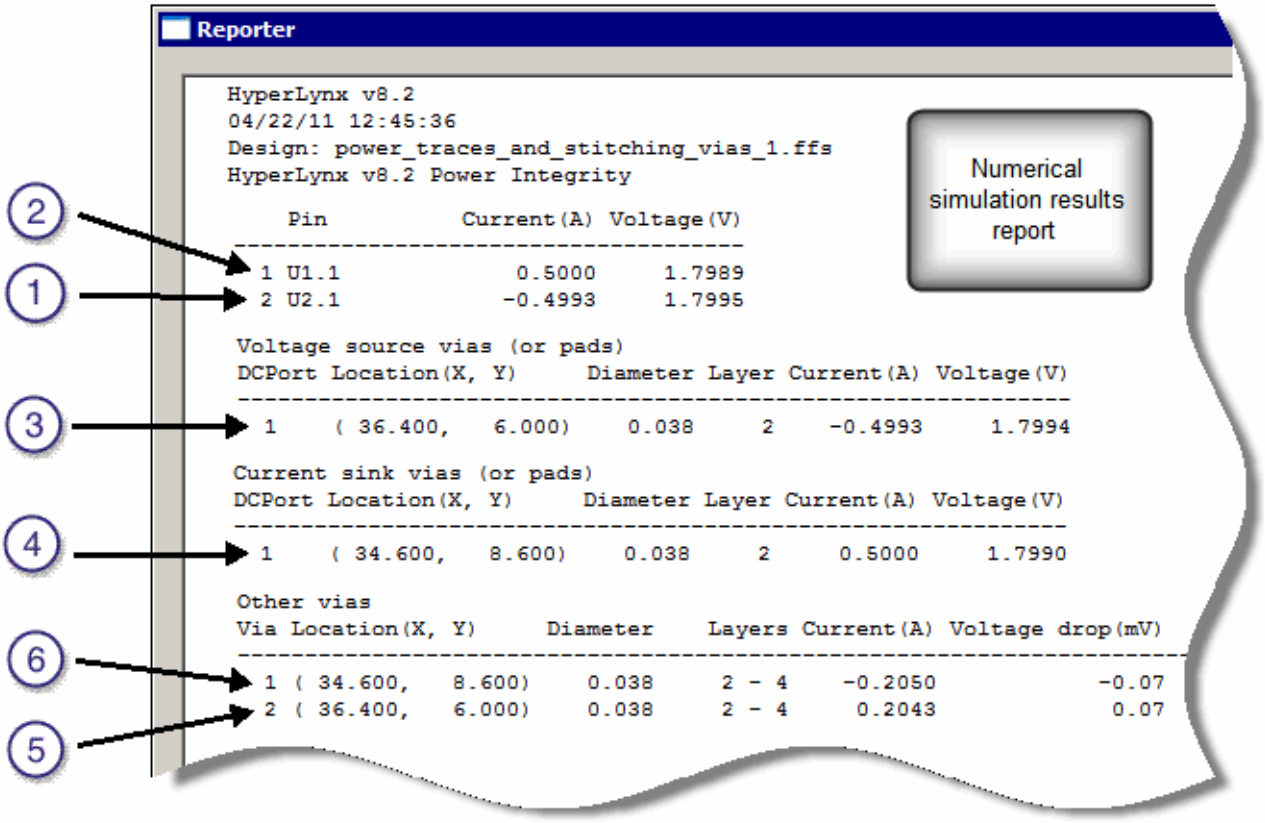
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 8-4. DC Drop Circuit - Simulate One Power-Supply Net (cont.)

Object	Description
4	Point where current exits a metal area, usually by a via or pad. **Note:** A point where current exits a routed trace is reported in the Other vias section of the numerical report. See HyperLynx PI PowerScope Dialog Box.
5	Stitching via that connects the DC sink to a metal area on another stackup layer.
6	Stitching via that connects the VRM to a metal area on another stackup layer.

The following figure maps the objects in Figure 8-14 to the numerical simulation results displayed in the Reporter Dialog Box.

Figure 8-15. DC Drop Numerical Simulation Results - Simulate One Power-Supply Net



DC Drop Simulation Circuit - Including Reference Nets

DC drop simulations can simulate the set of power and reference (ground) nets that form a DC current loop and report simulation results. To do this, select “Include reference net(s)” in either the Batch DC Drop Simulation dialog box or the DC Drop analysis dialog box. The software then automatically assigns VRM and DC sink models to pins on the reference net(s). This

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capability saves you setup time and ensures the selected power-supply net and reference nets use the same amount of overall DC current.

Note

If you manually assign VRM or DC sink models to pins on a reference net, and select the option to include reference nets in simulation, the software overrides manual assignments, and uses the automatic DC drop model assignments in simulation.

The figure below shows the VRM and DC sink models and main circuit elements used for running DC drop simulation on a pair of selected and reference power-supply nets.

Figure 8-16. DC Drop Simulation Circuit - Simulate Selected and Reference Power-Supply Nets

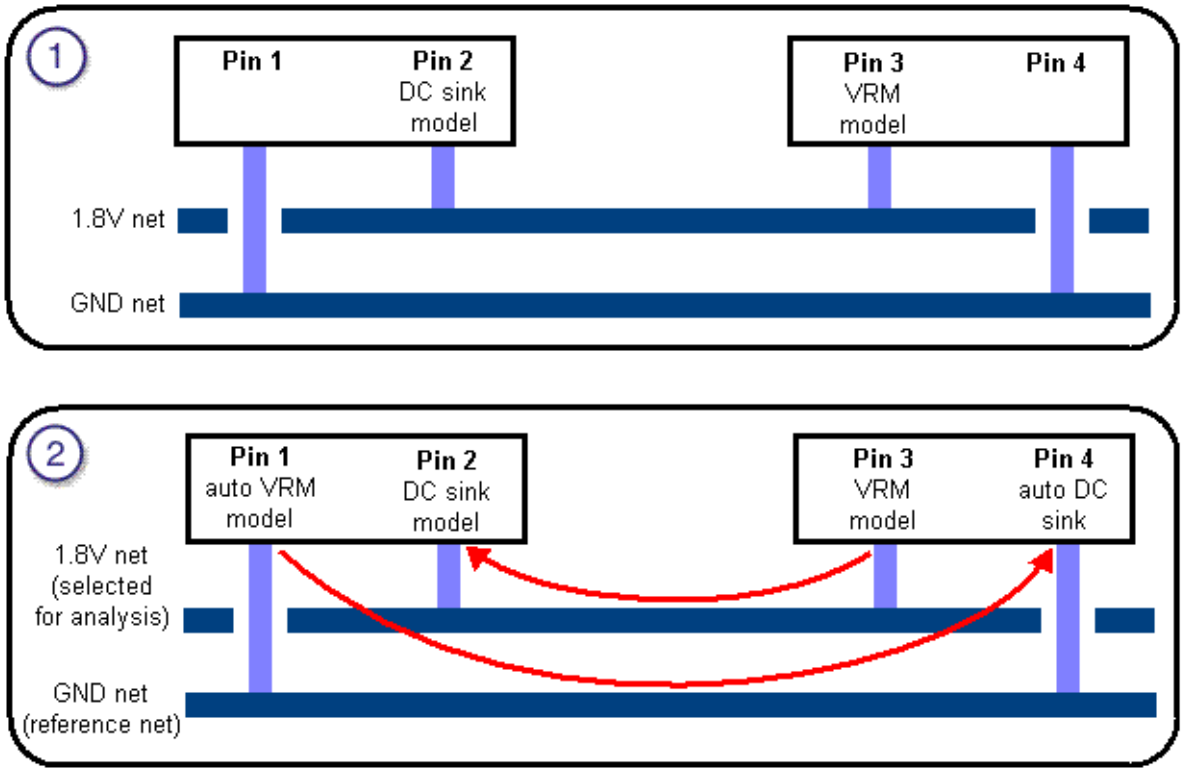


Table 8-5. DC Drop Simulation Circuit - Simulate Selected and Reference Power-Supply Nets

Circuit	Description
1	This circuit shows DC sink and VRM models assigned to pins 2 and 3. When assigning the VRM and DC sink models, 1.8V was identified as the “connected” net. When assigning the VRM model, GND was identified as the “reference” net. This circuit shows pins 1 and 4 with no assigned models.

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Table 8-5. DC Drop Simulation Circuit - Simulate Selected and Reference Power-Supply Nets (cont.)

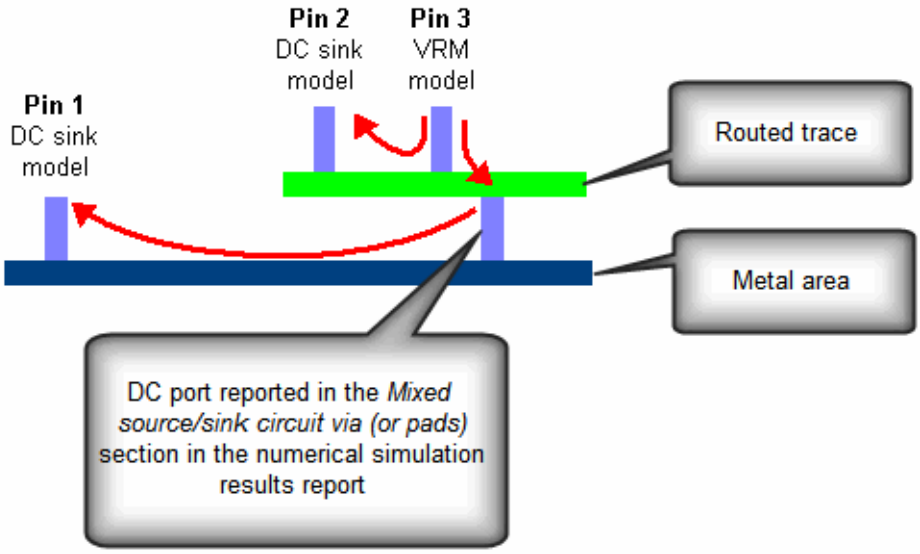
Circuit	Description
2	This circuit shows that you have used the DC Drop Analysis dialog box to do the following: 1. Select the 1.8V net for simulation. 2. Select "Include Reference Net(s)", which is the GND net in this example. When you start simulation, the software automatically assigns models to power-supply pins on the reference net and on the same components. See How BoardSim Recognizes Power Supply Nets. Pins 1 and 4 show this automatic and temporary model assignment.

Mixed Source/Sink Vias or Pads

The following figure shows a routed trace connected to both a VRM and DC sink. The via connecting the routed trace to the metal area is reported as a DC port in the "Mixed source/sink vias (or pads) area" of the numerical simulation results report.

Current from pin 3 flows partly through the routed trace to pin 2 and partly through the via and metal area to pin 1. If pins 1 and 2 sink 100 mA each, then pin 3 sources 200 mA. However, only 100 mA flows through the DC port and that is the value reported in the "Mixed source/sink vias (or pads) area" of the numerical simulation results report.

Figure 8-17. DC Drop Circuit - Mixed Source/Sink Vias

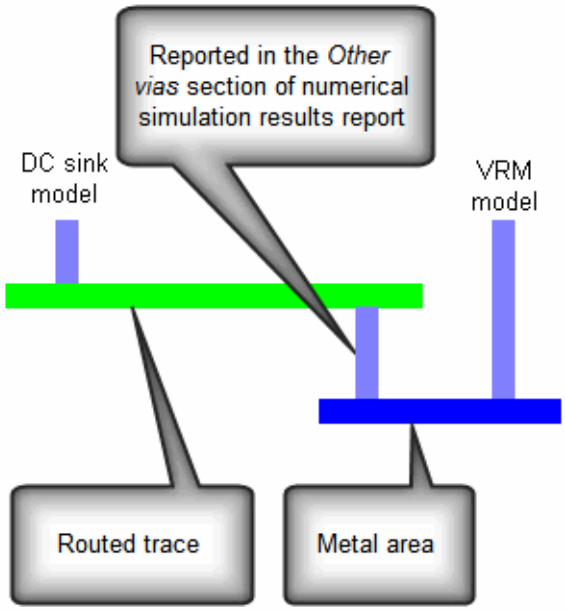


Other Vias

The figure below shows a routed trace that connects the DC sink to a via that connects to a metal area and VRM pin. The via connecting the routed trace to the metal area is reported in the "Other vias" section of the numerical simulation results report. The "Other vias" section also reports vias that connect two routed traces.

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Figure 8-18. DC Drop Circuit - Other Via



Related Topics

Design Factors Contributing to DC Drop

DC Drop Simulation

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DDRx Batch Simulation

DDRx is a source-synchronous technology, meaning that instead of one master clock that applies to the entire memory interface, multiple strobes and clocks are used for sub-portions of the interface. To set up timing measurements on a specific data, address, command, or control signal, the DDRx Wizard automatically pairs it to its clock or strobe signal. Setup, hold, and other measurements on a data, address, command, or control signal are made relative to its clock or strobe signal. Signal pairing also exists between strobe and clock signals.

Topic	Description
Algorithm to Map Nets to DDRx Interface Signal Functions	The DDRx Wizard automatically maps nets in the design to DDRx interface signal functions, such as data, clock, and strobe.
Data Flow for DDRx Batch Simulation	The following figure shows the main data inputs and outputs for DDRx batch simulation.
DDRxDelays_autogenerated.txt File Format	The *DDRxDelays_autogenerated.txt* is an external file containing write/read leveling delays for byte and bit signals.
DDRx Delay File Measurements	When you enable the “DDRx Delays external file” option on the DDRx Batch-Mode Wizard - Leveling&Calibration page, the initial simulation run generates a delay file that contains delays to compensate for timing delays that are inherent when your memory interface supports delay calibration.
DDRx Batch-Mode Simulation Results Folders and Files	Knowing the set of folders and files created by DDRx batch-mode simulation can help you find the information you need. The software writes this information to the design folder.
DDRx Setup File	The DDRx batch simulation wizard creates a batch simulation setup file (*.ddr*) that contains all of the information needed to simulate and measure DDRx memory interface signals. Advanced users can manually create or edit the setup file and load it into the wizard.
DDRx Write/Read Leveling Delay File	DDRxDelays_autogenerated.txt contains byte-lane specific delays for data signals during memory-write cycles. This file is located in the <design> folder.
Map Custom Data Rates to Standard JEDEC Derating Tables	If you specify a custom data rate in the Timing Models wizard page, the custom data rate maps to the standard JEDEC derating tables.
On-Die Termination - ODT	Data and data strobe signal circuits in some memory interfaces include ODT to improve signal integrity during read and write operations.
Pairing DDRx Interface Signals	To set up timing measurements on a specific data, address, command, or control signal, the DDRx Wizard automatically pairs it to its clock or strobe signal.

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Topic	Description
Round Robin for DDRx Batch Simulation	DDRx batch simulation creates round robin simulations based on the type of net being simulated.
RTT_Limits.txt File Format	An *RTT_Limits.txt* file contains the minimum and maximum round trip time limits for use during DDRx batch simulations.
Vcent(pin_mid) Calculation Examples	Vcent(pin_mid) is the average center voltage for a set of pins. The value differs based on the options you select on the Vref Training page of the DDRx Wizard. Understanding how a Vref value is selected is the key to understanding how Vcent(pin_mid) is calculated.
Write/Read Leveling for DDRx	Write and read leveling enables memory controllers for some types of memory interfaces to compensate for skew caused by different routing methods used by signal groups.

Algorithm to Map Nets to DDRx Interface Signal Functions

The DDRx Wizard automatically maps nets in the design to DDRx interface signal functions, such as data, clock, and strobe.

Note

Incomplete connectivity information can prevent the DDRx Wizard from mapping all the signals in the DDRx interface. For example, if the net passes through a resistor package, a *.PAK* model must be assigned to the reference designator for the DDRx Wizard to know the connectivity among resistor package pins. Note that this resistor package model assignment is also required to interactively simulate the net with the oscilloscope or Interactive Simulation Dialog Box, and is not unique to DDRx simulation. See “Assigning a Model to a Passive Component Using the Assign Models Dialog Box” on page 91.

The software uses the following algorithm to automatically map nets to the DDRx interface.

1. Map DRAM IBIS model pins to DDRx interface functions.

   For each unique IBIS model assigned to a DRAM, use signal names from the [Pin] keyword to map model pins to DDRx functions (Clock, Data Strobe, Data Mask, Data Bit, Address/Command, and Control).

   For differential signals, the first pin in the [Diff Pin] keyword is the positive signal.

   You identify DRAM reference designators on the DRAMs wizard page. See DDRx Batch-Mode Wizard - DRAMs Page.

2. Identify the signal nets for the controller.

   Create a list containing all the signal nets directly connected to the memory controller.

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You identify the memory controller reference designator on the DDRx Batch-Mode Wizard - Controller Page.

3. Identify the signal nets connecting the DRAMs to the controller.

   For each DRAM component pin, follow the signal path topology to create a list containing the signal nets directly connecting the DRAM to the controller. This is a subset of the list created in step 2.

4. Map DRAM component pins to controller nets with specific DDRx functions.

   Map the function groups created in step 1 to signal nets connecting the DRAM to the controller identified in step 3.

5. Build and refine a wizard database mapping signal nets to DDRx functions.

   In addition to containing information obtained in step 4, other signal net to DDRx function mapping occurs. Examples:

   - Map Data Mask and Data nets for a specific DRAM to the Data Strobe net.
   - If the DDR2 DRAM has eight data bits (x8), the Data Mask input pin can act as a redundant data strobe output pin (RDQS). This feature enables the memory-system designer to reduce loading on the DQS signals in cases where x4 DRAM devices are mixed with x8 DRAM devices, and is selected through an internal mode register. The automatic net-mapping algorithm always assumes these pins are used in a Data Mask function. You can reassign these pins as RDQS outputs by manually reassigning them on the Data Strobes and Data Nets pages.

6. Customer confirms automatic mapping.

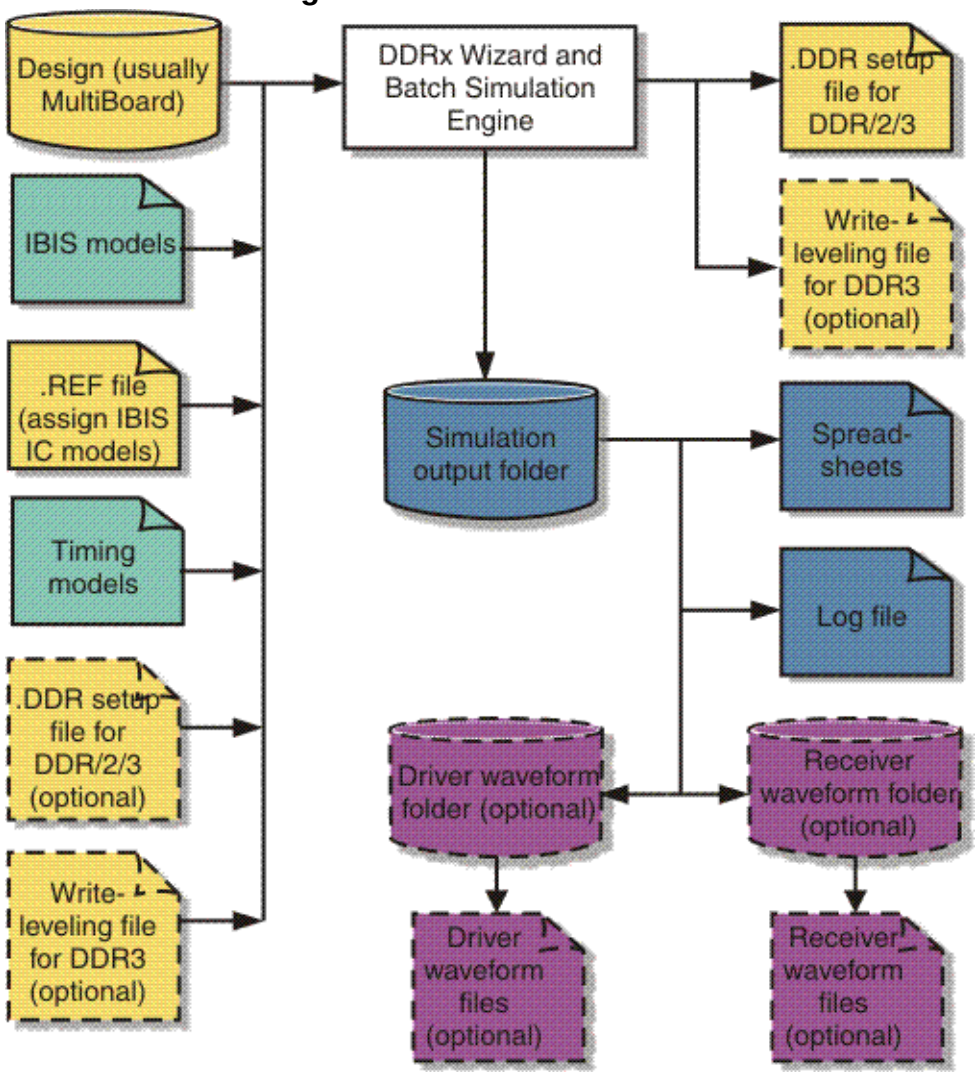
   You can edit automatic mapping assignments on the following wizard pages: Data Strobes, Data Nets, Clock Nets, Addr/Cmd Nets, and Control Nets.

Data Flow for DDRx Batch Simulation

The following figure shows the main data inputs and outputs for DDRx batch simulation.

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Figure 8-19. DDRx Data Flow



DDRx Folders Legend

Color	Description
	<design> folder. See Design Folder and HyperLynx Files.
	<model_library> folder. For example: C:\MentorGraphics\<release>\SDD_HOME\hyperlynx\Libs. Shipping timing models are located in the C:\MentorGraphics\<release>\SDD_HOME\hyperlynx\Libs folder, but you can open them from other locations. You can open IBIS models from other locations. See Select Directories for IC-Model Files Dialog Box.

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	<design>\DDR_Results folder. When you open a design in LineSim and specify sweep simulations, DDRx batch simulation creates an additional subfolder for each sweep simulation case. The subfolder location and name is <design>\DDR_Results\SweepCase_<number>.
	<design>\DDR_Results\<waveform_folders> folders. One sub-folder contains waveforms measured at driver pins and the other sub-folder contains waveforms measure at receiver pins.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDRxDelays_autogenerated.txt File Format

Used by: **DDRx Batch-Mode Wizard > Leveling & Calibration page**

The *DDRxDelays_autogenerated.txt* is an external file containing write/read leveling delays for byte and bit signals.

The *DDRxDelays_autogenerated.txt* file contains the absolute initial delays (for example, default initial delay + skew). The external file can also contain skew values. Theses skew values populate the table on the DDRx Batch-Mode Wizard - Leveling&Calibration Page and are included in the simulation. The wizard applies all of the read leveling or per bit deskew information contained in the external file during simulation.

The external file can contain any net from the DDR interface such as DQS, DQ, DM, CK, and ADDR. Absolute initial delays from the external file are not inherited by dependent nets. Therefore, when a delay is specified for net DQS0, delays must also be specified for nets DQ0-DQ7. Otherwise DQ0-DQ7 nets are assigned an initial delay of zero, which is the default value.

Format

A *DDRxDelays_autogenerated.txt* file must conform to the following syntax:

- Net: netName Delay: DelayValue Skew: SkewValue SimCase: operation

In the external file, absolute initial delays are required, (default initial delay + skew).

Parameters

Table 8-6. DDRx Write Leveling Delay File Contents

Keyword	Description
current delay	Unadjusted delay. For the write leveling, this is the stimulus offset of address with respect to the clock. For read leveling, this is the stimulus offset between data and strobe.
average skew	Average skew between the DQS and corresponding CK signal. In a multicycle strobe/clock waveform, the wizard measures the skew at each cycle and calculates average skew. For data nets, it also measures average setup/hold time differences at each cycle at the receiver and calculates the delay to produce worst-case values for setup and hold. In the DDRx Skew Spreadsheets, this value corresponds to the average of the tDQSS Skew Time (typ), [ps] values.
Net:	netName is the name of any physical net that is part of one electrical net. For differential nets, netName can be the positive or negative branch of the net.

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Table 8-6. DDRx Write Leveling Delay File Contents (cont.)

Keyword	Description
Delay:	DelayValue is the required absolute initial delay for this net, in ns. This is the recommended initial delay to apply to the net to minimize the difference between the average setup time and the average hold time. When you choose the "DDRx Delays external file" option on the Leveling & Calibration page of the DDRx batch-mode wizard, the wizard uses the delay value from this file as the initial delay value for each particular net. See Creating and Applying a DDRx Delays File.
Skew:	(Optional) SkewValue is the difference between the DelayValue and the default initial net delay, in ns. If specified, SkewValue is converted to ps before populating the GUI table on the Leveling & Calibration page of the DDRx Batch-Mode Wizard.
SimCase	SimCase – Specifies a write or read operation using the following syntax: • Write operation W<slotnumber> – Write to slot <slotnumber>. For example, W1 writes to slot1. or W(<slotnumber>,<banknumber>) – Write to bank <banknumber> of slot <slotnumber>. For example, W(1,2) writes to bank 2 of slot 1. • Read operations R<banknumber> – Read from bank<bank number>. For example, R1 reads from bank1 or R(<slotnumber>,<banknumber>) – Read from bank<banknumber> of slot <slotnumber>. For example, R(2,1) reads from bank 1 of slot 2.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Examples

```
*
* This is an automatically generated file with optimized delays for DDR3.
* It should be renamed to DDR3Delays.txt file to be used by the DDR batch
* mode runs.
*
***
*** current delay: 1.875, average skew: -0.309
Net: MDQS_P0  Delay: 2.184  Skew: -0.309  SimCase: W1
***
*** current delay: 0.937, average skew of its DQS: -0.309, average setup
margin: 0.196, average hold margin: 0.335
Net: MDM0  Delay: 1.176  Skew: -0.239  SimCase: W1
***
*** current delay: 1.875, average skew: -0.303
Net: MDQS_P1  Delay: 2.178  Skew: -0.303  SimCase: W1
***
*** current delay: 1.875, average skew: -0.340
Net: MDQS_P2  Delay: 2.215  Skew: -0.340  SimCase: W1
***
*** current delay: 1.875, average skew: -0.304
Net: MDQS_P3  Delay: 2.179  Skew: -0.304  SimCase: W1
***
*** current delay: 1.875, average skew: -0.334
Net: MDQS_P4  Delay: 2.209  Skew: -0.334  SimCase: W1
***
*** current delay: 1.875, average skew: -0.282
Net: MDQS_P5  Delay: 2.157  Skew: -0.282  SimCase: W1
***
*** current delay: 1.875, average skew: -0.283
Net: MDQS_P6  Delay: 2.158  Skew: -0.283  SimCase: W1
***
*** current delay: 1.875, average skew: -0.284
Net: MDQS_P7  Delay: 2.159  Skew: -0.284  SimCase: W1
```

DDRx Delay File Measurements

When you enable the “DDRx Delays external file” option on the DDRx Batch-Mode Wizard - Leveling&Calibration page, the initial simulation run generates a delay file that contains delays to compensate for timing delays that are inherent when your memory interface supports delay calibration.

The file is called *DDRxDelays_autogenerated.txt*. You should rename the initial file because the auto-generated file overwrites with each simulation run. When you select this option in subsequent simulation runs, the wizard uses the delay values from this file as the initial delay value for each particular net. The delay value is calculated as follows:

For DQ nets:

Skew = Average Skew of its Strobe Net + [(Average Hold Time – Average Setup Time)/2] in ns

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Delay = Current initial delay – Skew in ns

For DQS nets:

Skew = Average skew of this strobe net with associated clock net

Delay = Current initial delay – Skew

When the simulation is run for the first time, the required file does not exist and all delay corrections are set to zero. See Creating and Applying a DDRx Delays File.

The DQ write and read simulation results are different with and without write leveling delays when you use the delay file to specify separate delays for each bit (each DQ net).

The wizard generates the skew values in the *DDRxDelays_autogenerated.txt* file by measuring the time difference between the clock and DQS at the receiver. In multi-cycle strobe/clock waveforms, the wizard measures skew at each cycle and calculates the average skew. For data nets, it also measures average setup/hold time differences for each cycle at the receiver and calculates the worst-case delay for the setup and hold values.

The wizard calculates skew between the CK and DQS signals and the same skew is inherited by the appropriate DQ signals, for each 8 bit of the strobe. However, the additional delay correction is added to the data nets. Also, the DDRx Delays file delay correction for data nets may differ from the associated strobe nets. The DDR engine also uses calculated setup/hold times for each data net and attempts to balance the setup and hold times by adding an extra delay to each data net. The additional delay is calculates as:

[(average setup times) - (average hold times)]/2

For example, a net DQ0 has setup/hold times calculated for 3 cycles of one simulation: 30/70, 20/80 and 10/90 ps. To balance the setup/hold values, the DDR engine adds an additional delay to the DQ0 net, calculated as:

[(30+20+10)/3 - (70+80+90)/3]/2 = [20ps (average setup) – 80ps (average hold)]/2 = -30ps.

After applying this extra delay to the DQ0 net, the setup/hold values change to: 60/40, 50/50 and 40/60. For these numbers, the average setup and hold values are equal. Note that each data net has its own delay correction value.

DDRx Batch-Mode Simulation Results Folders and Files

Knowing the set of folders and files created by DDRx batch-mode simulation can help you find the information you need. The software writes this information to the design folder.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

The options you enable in the DDRx Batch-Mode Wizard and the type of DDRx interface you simulate determine which files and folders the software creates. This topic describes some of the files and folders.

Table 8-7. Folders With a Timestamp

Folder	Description
DDR_Report	Contains results formatted as an HTML table and displayed by a special viewer.
DRV_Waveforms_<various> *RCV_Waveforms_<various>*	Contains waveform files (*.csv*).
StimulusForNets	Contains stimulus files (*.eds*) for each net. You can import a stimulus file for another simulation.

Table 8-8. Files in Folders That Have a Timestamp

File	Description
DDR_log_<various>.txt	Contains detailed simulation messages.
DDR_report_<spreadsheets>	Contains measurements.
<various>.ddr	Contains settings for the DDRx Batch-Mode Wizard.

Table 8-9. Folders Without a Timestamp

Folder	Description
AMI_SETTINGS (DDRx interfaces that use equalization)	Contains AMI settings used by simulation.
DDR<design>.waves	Waveform files created by the DDR5 Wizard when you enable the “Advanced DDRx AMI” simulation flow. These files are intended for use by the simulator and not the end user.

DDRx Setup File

The DDRx batch simulation wizard creates a batch simulation setup file (*.ddr*) that contains all of the information needed to simulate and measure DDRx memory interface signals. Advanced users can manually create or edit the setup file and load it into the wizard.

The setup file maps reference designators to DDRx controller and DRAM ICs, sets [Model Selector] values for ODT, maps DRAM IC reference designators to slots and ranks, sets stimulus bit sequences, simulation options, and so on.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The setup file stores all wizard options and values, and saves to the results folder when you close the wizard. For more information about the results folder, see "DDRx Batch Simulation Results" on page 277.

You can import a saved setup file during a future wizard session and re-run simulation on the same design or on a variation of the design. To preserve the original simulation settings, copy and rename the setup file.

DDRx Write/Read Leveling Delay File

DDRxDelays_autogenerated.txt contains byte-lane specific delays for data signals during memory-write cycles. This file is located in the <design> folder.

Do not edit this file directly. See "Write/Read Leveling for DDRx" on page 517. To manually specify delays, use the spreadsheet in the "DDRx Batch-Mode Wizard - Leveling&Calibration Page" on page 790.

For information about the measurements used to create the write leveling delay file, see DDRx Delay File Measurements.

Map Custom Data Rates to Standard JEDEC Derating Tables

If you specify a custom data rate in the Timing Models wizard page, the custom data rate maps to the standard JEDEC derating tables.

For DDR2 and DDR3 designs, the data rate value on the Timing Models wizard page determines which built-in derating table to use. If you specify a standard data rate, the matching derating table is used from JEDEC specification JESD79*. If you specify a custom data rate, the custom data rate maps to the standard JEDEC derating tables as shown in Table 8-10.

Table 8-10. Mapping Custom Data Rates to Standard Derating Tables

Net Type	Rate Mapping
DDR2	
Clock nets and related nets; Differential strobe nets and related nets	• < 400 MHz, > 400 MHz to < 533 MHz maps to the 400/533 MHz table • > 533 MHz maps to the 667/800/1066 MHz table
Single-ended strobe nets and related nets	• < 400 MHz maps to the 400 MHz table • > 400 MHz to < 533 MHz maps to the 533 MHz table • > 533 MHz maps to the 667 MHz table
DDR3	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

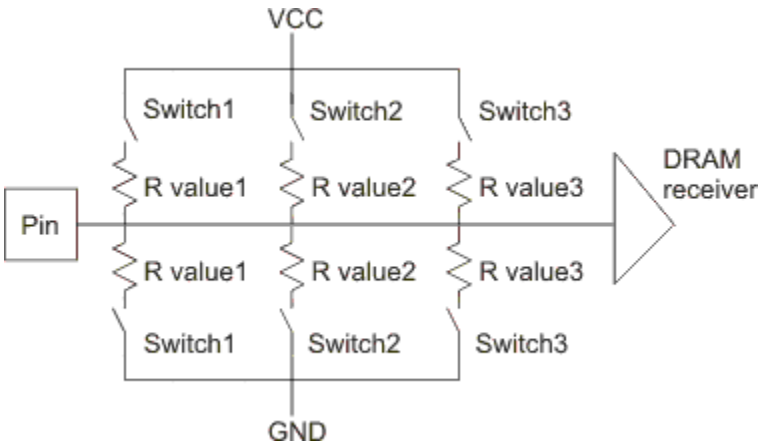
Table 8-10. Mapping Custom Data Rates to Standard Derating Tables (cont.)

Net Type	Rate Mapping
Clock nets and related nets	All custom frequencies map to the 800/1066/1333/1600 MHz table
Differential strobe nets and related nets	All custom frequencies map to the 800/1066 MHz table

On-Die Termination - ODT

Data and data strobe signal circuits in some memory interfaces include ODT to improve signal integrity during read and write operations.

The figure below shows an ODT circuit that consists of voltage-dividers that can be enabled independently to provide a range of resistance values, or be completely disabled.



ODT can change the receiver characteristics so much that you need separate models to represent the termination enabled/disabled behaviors. IBIS models use the [Model Selector] keyword to control which model within a component to use during simulation. The DDRx Wizard helps you specify models for enabled/disabled termination behaviors.

While ODT switches on and off dynamically in the actual design, ODT settings cannot change within a specific DDRx batch simulation. However, ODT settings can change between simulations.

Pairing DDRx Interface Signals

To set up timing measurements on a specific data, address, command, or control signal, the DDRx Wizard automatically pairs it to its clock or strobe signal.

Setup, hold, and other measurements on a data, address, command, or control signal are made relative to its clock or strobe signal. Signal pairing also exists between strobe and clock signals.

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DDRx is a source-synchronous technology, meaning that instead of one master clock that applies to the entire memory interface, multiple strobes and clocks are used for sub-portions of the interface.

Round Robin for DDRx Batch Simulation

DDRx batch simulation creates round robin simulations based on the type of net being simulated.

DDRx batch simulation uses information about the type of net to set up round robin simulations for it. Table 11-16 summarizes the basic driver-enabling rules for DDRx round robin. Note that Table 11-16 provides an incomplete description of driver-enabling rules, because it omits details about data/data strobe nets in DDR2/DDR3 interfaces that need multiple cases for some driver pins (to allow for varying ODT positions).

Table 8-11. Driver-Enabling Rules for DDRx Round Robin

Net Type	Enabled Pins on the Net
Data (DQ) Data strobe (DQS) Redundant data strobe (RDQS-DDR2 only) Check bits (CB)	All pins; the memory controller pin is enabled during a write and one DRAM pin at a time is enabled during a read.
Clocks	Controller pins Clocks are driven only by the controller; all DRAM pins are always receivers.
Data mask (DM)	Controller pins Data mask bits are unused during read operations.
Address (A) Bank address (BA) Command (RAS, CAS, WE) Chip select (CS) Clock enable (CKE) On-die termination (ODT)	Controller pins Address/command/control bits are driven only by the controller; all DRAM pins are always receivers.

RTT_Limits.txt File Format

Used by: DDRx Batch-Mode Wizard - Round Trip Time Page

An *RTT_Limits.txt* file contains the minimum and maximum round trip time limits for use during DDRx batch simulations.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Format

An *RTT_Limits.txt* file must conform to the following syntax:

- Net: netname_ROUND_TRIP_TIME1_MIN: val ROUND_TRIP_TIME1_MAX: val ROUND_TRIP_TIME2_MIN: val ROUND_TRIP_TIME2_MAX: val

Keywords

Keyword	Description
Net:	The DQS net name for the specified round trip time limits. Must be the names of the physical nets connected directly to the controller. Spaces are not allowed in the net name. Net name must be the positive part of the differential pair or use "/ "without a space to designate a differential pair. For example, DQS_P or DQS_P/DQS_N.
ROUND_TRIP_TIME1_MIN: ROUND_TRIP_TIME1_MAX:	Round Trip Time 1 (RTT-1) is calculated as CLK flight time plus tDQSCK (DQS from CLK delay uncertainty) plus DQS flight time (read). RTT-1 is specified at the controller where DQS is gated with DQSMASK. val is the minimum or maximum limit for the round trip time, in ps.
ROUND_TRIP_TIME2_MIN: ROUND_TRIP_TIME2_MAX:	Round Trip Time 2 (RTT-2) consists of RTT-1 and includes a 90 degree shift to DQS (read). RTT-2 is specified where read data switches from the DQS domain to the internal CLK domain. Enter the minimum and maximum limits for each round trip time in ps. val is the minimum or maximum limit for the round trip time, in ps.

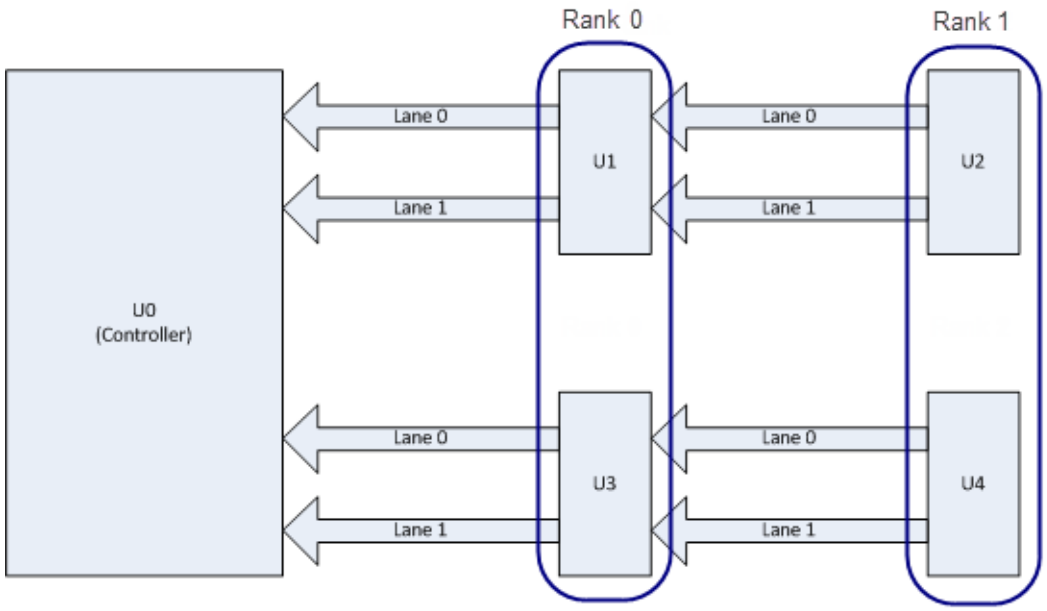
Examples

```
*
* File to import round trip time limits.
* Format:
*   Net: netname ROUND_TRIP_TIME1_MIN: val ROUND_TRIP_TIME1_MAX: val
ROUND_TRIP_TIME2_MIN: val ROUND_TRIP_TIME2_MAX: val
*
* netname - Can be only the positive part of the differential nets or in
the format: positive/negative;
*    Must be the names of the physical nets connected directly to the
controller.
*
* val - limit value in picoseconds
*
Net: SR_LDQSP/SR_LDQSN ROUND_TRIP_TIME1_MIN: -300 ROUND_TRIP_TIME1_MAX:
400 ROUND_TRIP_TIME2_MIN: -500 ROUND_TRIP_TIME2_MAX: 600
Net: SR_UDQSP/SR_UDQSN ROUND_TRIP_TIME1_MIN: -310 ROUND_TRIP_TIME1_MAX:
420 ROUND_TRIP_TIME2_MIN: -510 ROUND_TRIP_TIME2_MAX: 650
Net: VR_LDQSP ROUND_TRIP_TIME1_MIN: -340 ROUND_TRIP_TIME1_MAX: 460
ROUND_TRIP_TIME2_MIN: -530 ROUND_TRIP_TIME2_MAX: 620
```


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Vcent(pin_mid) Calculation Examples

Vcent(pin_mid) is the average center voltage for a set of pins. The value differs based on the options you select on the Vref Training page of the DDRx Wizard. Understanding how a Vref value is selected is the key to understanding how Vcent(pin_mid) is calculated.



Case 1: Enable separate Vref per rank, per lane

This case results in eight Vref values at the controller: U1-Lane0, U1-Lane1, U2-Lane0, U2-Lane1, U3-Lane0, U3-Lane1, U4-Lane0 and U4-Lane1. Each of these Vref voltages is calculated by taking the average of the widest and narrowest eye width center voltage (Vcent) driven by each of the eight lanes.

Case 2: Enable separate Vref per rank, per controller

This case results in two Vref values at the controller: one for Rank 0 (U1 and U3) and one for Rank 1 (U2 and U4). Each Vref value is calculated by taking the average of the widest and narrowest eye width center voltage (Vcent) of each of the ranks. The first Vref calculation uses the widest and narrowest values when U1 and U3 drive, and the second Vref calculation uses the widest and narrowest eye-width-centers of all DQ pins being driven by either U2 or U4. For example, if U1 has sixteen DQ nets and U3 has sixteen DQ nets, the first Vref is the average of the widest and narrowest eye-width-center voltage of all thirty-two DQ nets.

Case 3: Disable separate Vref per rank, per lane

This case results in four Vref values at the controller: one for the Lane 0 of U1/U2, one for Lane 1 of U1/U2, one for Lane 0 of U3/U4 and one for Lane 1 of U3/U4. If Lane 0 of U1/U2 has eight DQ signals, the average of the widest and narrowest eye-width-centers of all 16 options is

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calculated when the signals of Lane 0 are driven by U1; and then when the signals are driven by U2. For example, when U1 drives, the eye-width centers are 0.59, 0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 0.7, and when U2 drives, the eye-width centers are 0.4, 0.6, 0.6, 0.6, 0.6, 0.6, 0.6, 0.62. Therefore, the calculation finds the average of the largest of all values (0.7) and the smallest of all values (0.4) to find the Vref for Lane0: (0.7 + 0.4)/2.

Case 4: Disable separate Vref per rank, per controller

This case results in a single Vref for U0. This single Vref is the average of the widest and narrowest of any of the signals coming into U0, regardless of the DRAM.

Write/Read Leveling for DDRx

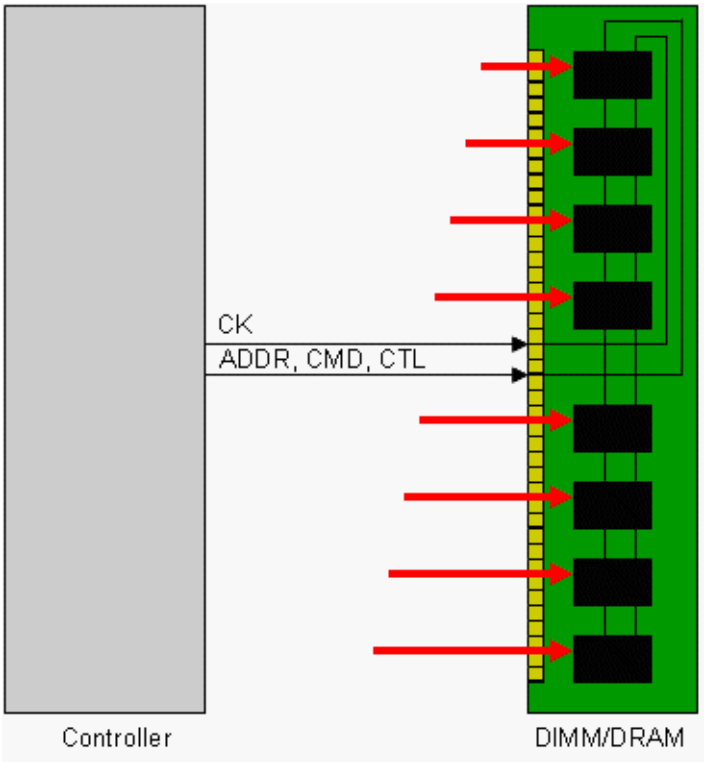
Write and read leveling enables memory controllers for some types of memory interfaces to compensate for skew caused by different routing methods used by signal groups.

To improve signal integrity at higher data transfer rates, DDR3 introduced “fly-by” routing topology for the [clock, address, command, control] signal group (see the black traces in Figure 8-20). By contrast, the [data, strobe, mask] signal group is typically routed (by byte lanes) directly to a specific DRAM. Fly-by topology reduces the quantity and length of stubs, and is easy to terminate. For a description of the fly-by routing topology, see the JESD79-3B, DDR3 SDRAM Specification.

Combining “fly-by” routing topology for [clock, address, command, control] and “direct” routing topology [data, strobe, mask] signal groups introduces skew into signal timing. When the controller drives the [data, strobe, mask] signals, the nominal arrival times at the memory pins are independent of DRAM placement on the DIMM. By contrast, when the controller drives the [clock, address, command, control] signals, the nominal arrival times at the memory pins are dependent on DRAM placement on the DIMM. See Figure 8-20. The black traces transmit [clock, address, command, control] signals. The length of the red arrows show the relative flight-time delays for these signals, where the top DRAM has the shortest flight-time delay and the bottom DRAM has the longest flight-time delay.

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Figure 8-20. Relative Flight Time Delays for Signals on Fly-by Routed Traces



To compensate for this intrinsic skew between [clock, address, command, control] and [data, strobe, mask] signal groups, DDRx memory controllers support write leveling delays, where the controller inserts a byte-lane-specific delay to individual [data, strobe, mask] signals during memory-write cycles. Specific byte-lane delay values are usually determined dynamically during hardware initialization within the controller, using a special “write leveling” mode in DRAM components where data (DQ) pins indicate the status of the alignment between strobe (DQS) and clock (CK) pins. More specifically, the controller sweeps the delay for the strobe (DQS) pin while monitoring the alignment status on the data (DQ) pin. Also, the controller can usually identify unique delays for strobe (DQS) pins located in different slots.

The software accepts write leveling delays for strobe signals, but not for data and data mask signals. The timing analysis process measures the skew between the strobe and the clock and setup/hold relationships at the DRAMs during write cycles. For information about the measurements used to create the write leveling delay file, see “DDRx Delay File Measurements” on page 509. Timing relationships between the data and mask signals relative to the strobes are not dependent on this strobe-to-clock relationship.

Tip

Assume the controller has sufficient capability to implement the write leveling delays that you identify by running simulations. Most memory controller vendors do not publish the range and resolution of write leveling delays supported by the controller.

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Note

DDRx batch simulation creates ideal write/read leveling delays, and does not take into account min/typ/max delay conditions.

Related Topics

Creating and Applying a DDRx Delays File

Design Factors Contributing to DC Drop

Performance constraints may produce PCB designs with numerous power-supply nets that are implemented as small isolated regions and trace segments.

This trend is driven by any of the following performance requirements:

- Reduce power consumption by doing any of the following:
  - Using the smallest VCC value at which the IC meets specifications and by running each major section of an IC at an optimized value. Independent power-supply nets can be biased to very similar voltages, such as 1.0V and 1.1V.
  - Powering-down idle circuity by shutting off voltage to isolated power-supply nets.
- Reduce the propagation of noise from the power-distribution network (PDN) for one part of an IC to other parts of the IC.

Cost constraints may produce PCB designs with few stackup layers, causing an individual stackup layer to contain a mixture of power-supply and signal nets, or multiple power-supply nets.

IC packaging and off-board connectors may produce PCB designs with highly-perforated metal regions. This condition is especially true for ball grid arrays (BGAs) where interior pins usually connect to the PCB through vias, whose antipads can perforate power-supply nets.

Depending on the design, ground nets or power nets may have more DC drop problems:

- Ground nets may have fewer cutouts/perforations because PCB designers often deliberately try to carry and preserve AC return currents on them.
- Areas under BGAs could be a problem, depending on the stackup, because the ground net could be perforated by antipads to make room for vias.

Related Topics

DC Drop Conceptual Circuits

DC Drop Simulation

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Design Folder and HyperLynx Files

The design folder is the default location where the software creates and saves files related to your design. The default design folder also contains example schematic and board design files.

The default location is:

- C:\MentorGraphics\<release>\SDD_HOME\hyperlynx64\HypFiles (64 bit software)
- C:\MentorGraphics\<release>\SDD_HOME\hyperlynx\HypFiles (32 bit software)

To change this default location, see Set Directories Dialog Box.

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Decoupling Simulation Background

Understanding some concepts can help you understand simulation results and prepare for simulation.

Topic	Description
Information Needed to Calculate Target PDN Impedance	Target PDN impedance is based on peak transient current consumed by ICs, nominal supply voltage, and maximum percentage ripple allowed by ICs or your power budget. Obtain this information before running decoupling simulation.
Transmission Planes Overview	Transmission planes propagate electric and magnetic fields along cavities formed between two metal areas on different stackup layers. A transmission plane helps store and propagate energy to IC power supply pins, and also accidentally carry other types of noise signals.
Circuit Topology for Lumped Decoupling Simulation	The software models the PDN as a set of directly-connected elements. It does not account for distributed current paths, such as decoupling capacitor and IC pin locations. Lumped simulation results are considered to be optimistic because they do not account for distributed current paths.
Circuit Topology for Distributed Decoupling Simulation	The software models the PDN as a set of transmission planes and decoupling capacitors. It accounts for distributed current paths, such as decoupling capacitor and IC pin locations. It does not account for signal-integrity structures and components, such as signal traces and vias, and IC driver/receiver pins.
Stitching-Via Optimization	You can speed up decoupling and signal-via bypassing simulation by having the software find stitching vias in the design that are located close together and merge their individual models into an equivalent model.
Power-Supply Pins That Can Be Selected for Distributed Decoupling Simulation and Exporting a PDN	The Select IC Power Pins page in the decoupling wizard and exporting a PDN to an S-parameter model wizard searches the design for transmission planes that either enclose IC power-supply pins or are very close (about 118 mils or 3 millimeters) to them. In LineSim, if at least one of the found transmission planes has a layer that connects to the IC power-supply pin and a layer that references the IC power-supply pin (that is, you identified it as a reference layer when you added the IC power-supply pin symbol to the PDN layout), the IC power-supply pin is available for probing. BoardSim has the same behavior as LineSim, except the reference layer for the IC power-supply pin does not have to be found.

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Information Needed to Calculate Target PDN Impedance

Target PDN impedance is based on peak transient current consumed by ICs, nominal supply voltage, and maximum percentage ripple allowed by ICs or your power budget. Obtain this information before running decoupling simulation.

Note

The decoupling wizard provides a Target-Z Wizard where you can enter this information and automatically calculate the target PDN impedance.

Peak Transient Current

You can obtain peak IC core and I/O transient current values from any of the following sources:

- Catalog IC— View the datasheet or ask the vendor. Datasheets may provide parameter values that vary by system operation mode.
- FPGA— Run the power calculator provided by the FPGA development system.
- ASIC— Ask the in-house IC designers at your company.

Nominal Supply Voltage

The nominal voltage provided by the voltage-regulator module (VRM).

Supply Voltage Ripple

The maximum offset from the nominal DC voltage allowed by an IC or your power budget.

Note

Ripple is not the peak-to-peak range of the nominal DC voltage.

For example, you may allocate 30% (or some other value) of a power budget for DC drop and the rest for AC. To translate this allocation to a ripple value, if you have a 5% ripple budget, then you would assign 1.5% (that is, 5% times 30%) to DC drop and 3.5% to AC impedance.

The 30% value for the DC drop share of a power budget may not apply to your design. If the design has very good AC impedance, you may allocate less to AC impedance and more to DC drop. Similarly, if the design has few DC drop problems, you may allocate more to AC impedance and less to DC drop.

Transmission Planes Overview

Transmission planes propagate electric and magnetic fields along cavities formed between two metal areas on different stackup layers. A transmission plane helps store and propagate energy to IC power supply pins, and also accidentally carry other types of noise signals.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-21 shows a basic transmission plane.

Figure 8-21. Transmission-Plane Model

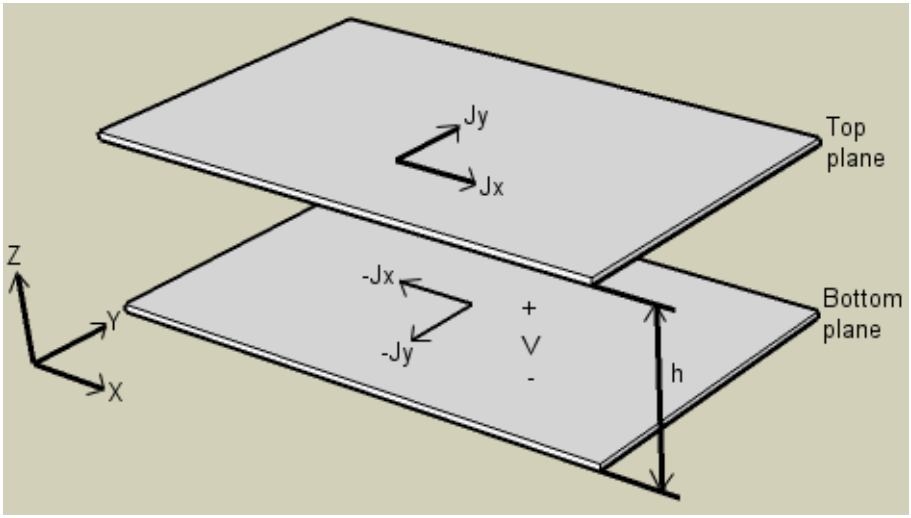
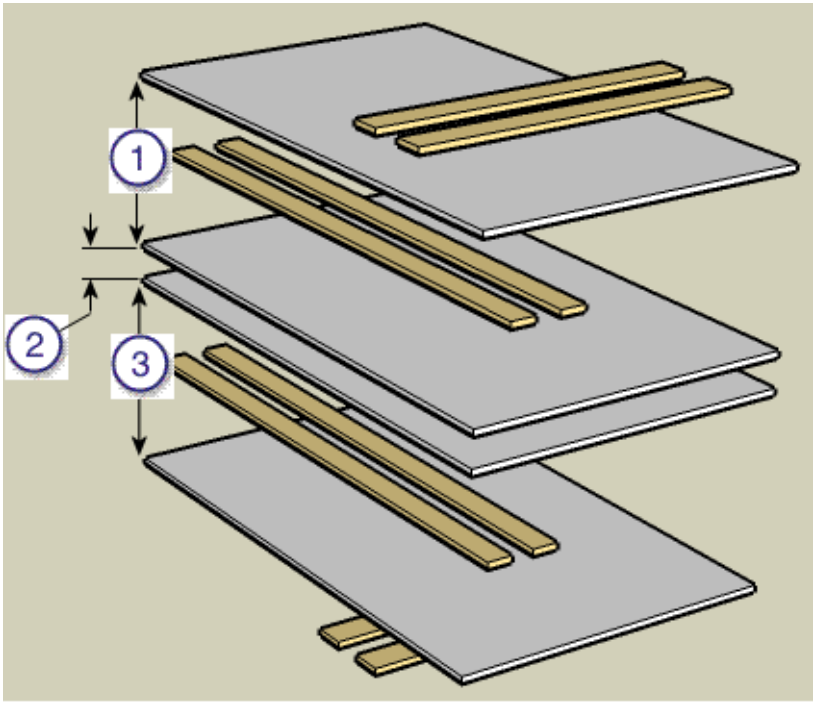


Figure 8-22 shows three transmission planes formed by metal areas located on four stackup layers.

Figure 8-22. Example Containing Three Transmission Planes



In simulation, a transmission plane model is analogous to a transmission line model. The transmission plane models how energy is transmitted on a conductor and received among power supply pins. The transmission line models how energy is transmitted on a conductor and received among signal pins. Table 8-12 provides further comparisons.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 8-12. Comparing Transmission Planes to Transmission Lines

Simulation Element	Transmission Plane	Transmission Line
Signal source	IC power supply pins that demand current and generate power flow	IC driver pins that switch to create signals
Conductor	Cavity formed by two metal shapes on different stackup layers	Trace segments
Signal loads (things that shape signals)	Decoupling capacitors, stitching vias, voltage-regulator modules (VRMs), power-distribution network (PDN) geometries (including stackup)	IC receiver pins, passive termination, signal trace geometries (including stackup)

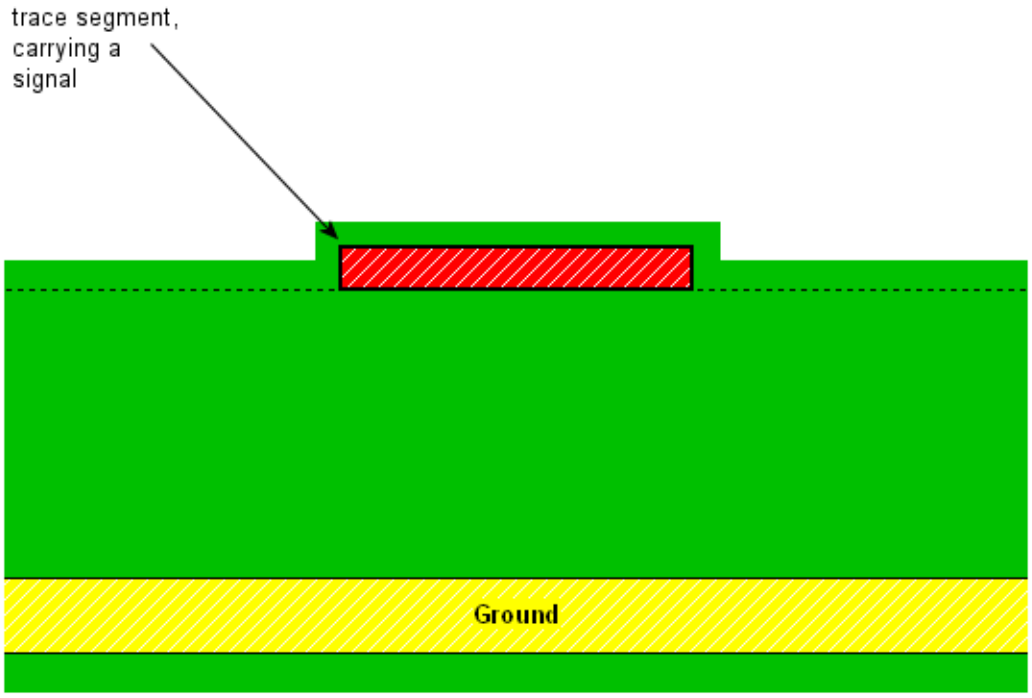
Transmission planes have inherent distributed capacitance that stores energy very near to IC power supply pins. Their inductance is low, especially if the dielectric layer separating the metal shapes is thin. Their inherent low impedance enables large amounts of energy to propagate, with little loss.

While many people think of a transmission line as a simulation circuit element, thinking about its electromagnetic behavior can help you understand how it is similar and different than a transmission plane.

Where does the signal energy reside in Figure 8-23?

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

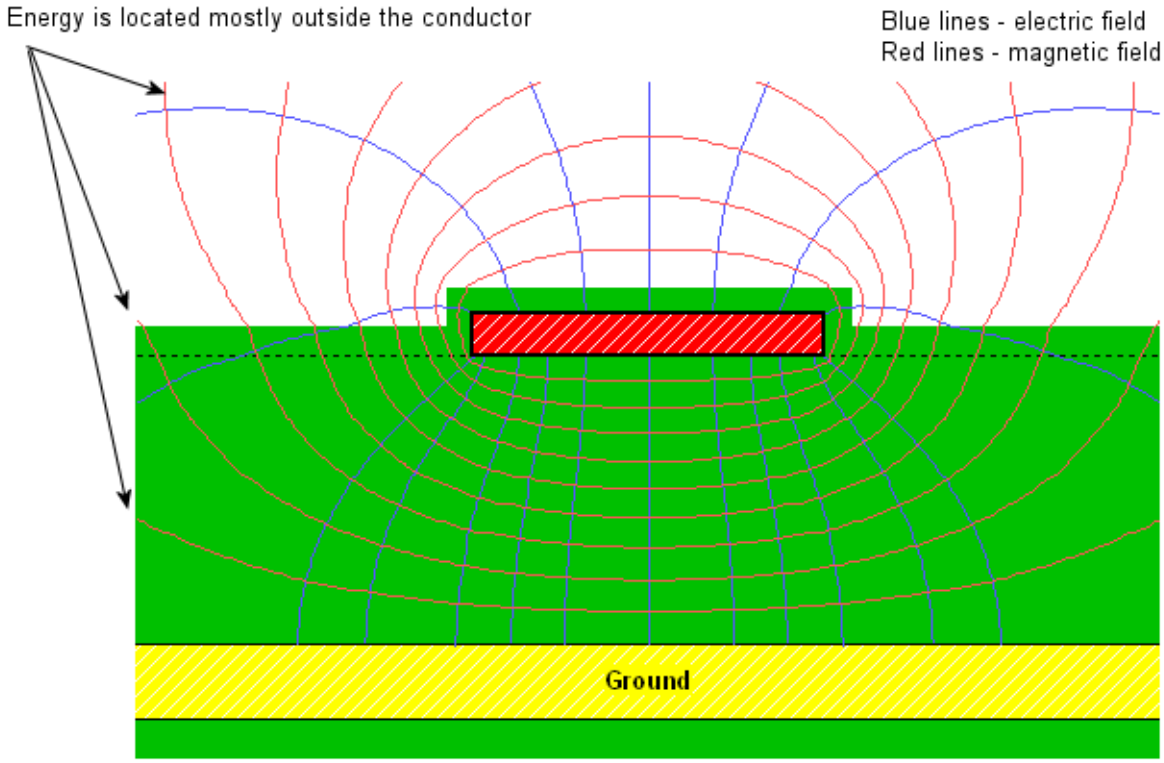
Figure 8-23. Transmission Line Structure



The electromagnetic fields contain the signal energy. Notice that in Figure 8-24, the signal energy is located almost entirely outside the conductors, in the air and dielectric materials. From a circuit point of view, by contrast, we usually think only of conductor currents and voltages.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

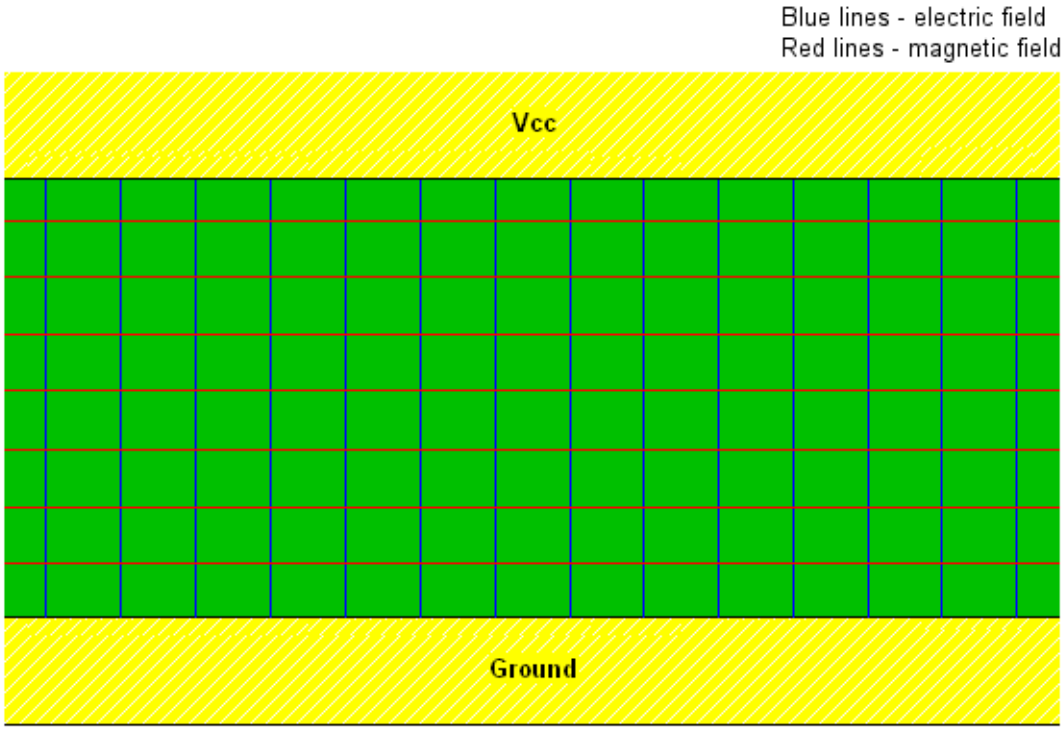
Figure 8-24. Transmission Line Electromagnetic Fields



The same effect is true in transmission planes, where the energy (in this case, power propagating to IC power supply pins) is carried in the dielectric layer, in the cavity between transmission plane layers. See Figure 8-25.

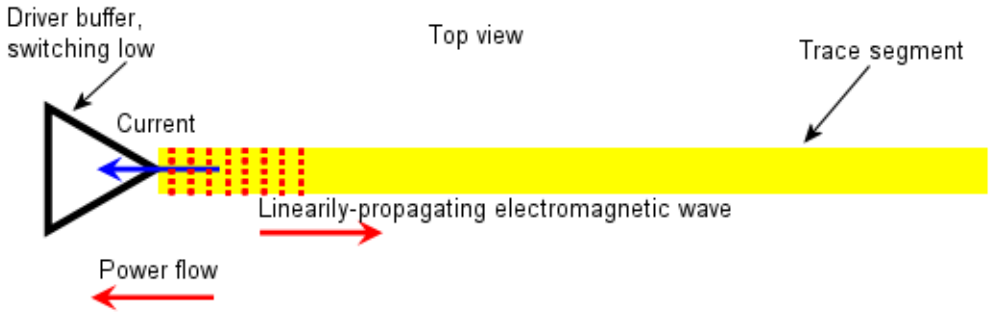
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-25. Transmission Plane Electromagnetic Fields



To help understand how transmission planes know when and where to propagate energy needed by IC power supply pins, consider a transmission line analogy. A transmission line begins to propagate energy when a driver IC pin switches state and causes a traveling “disturbance” (that is, electromagnetic wave) at one end of the transmission line. Figure 8-26 shows that the traveling wave pulls current from further and further along the transmission line into the IC pin.

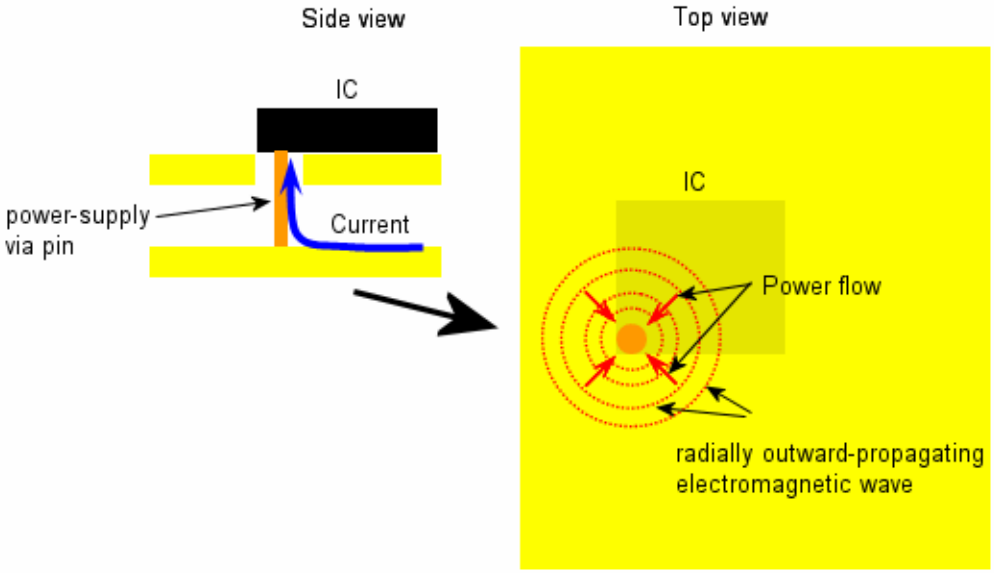
Figure 8-26. Electromagnetic Wave Propagating in a Transmission Line



A very similar thing happens in a transmission plane, except radially, when an IC power supply pin needs to pull in current. Figure 8-27 shows that the traveling electromagnetic wave pulls current further and further away in the transmission plane into the IC pin.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-27. Electromagnetic Wave Propagating in a Transmission Plane



Related Topics

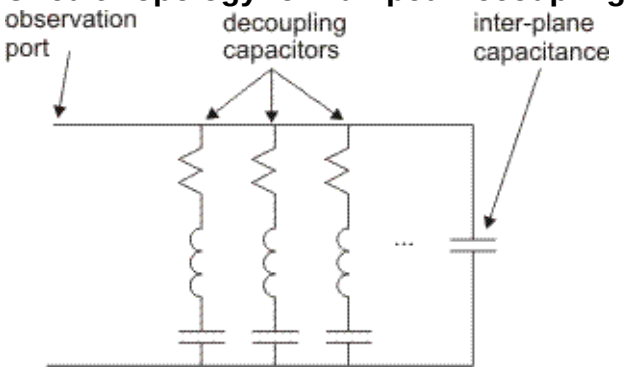
Simulating PDN Decoupling - Distributed

Circuit Topology for Lumped Decoupling Simulation

The software models the PDN as a set of directly-connected elements. It does not account for distributed current paths, such as decoupling capacitor and IC pin locations. Lumped simulation results are considered to be optimistic because they do not account for distributed current paths.

Figure 8-29 shows the circuit topology for lumped decoupling analysis. Although not shown, the topology can include VRM models.

Figure 8-28. Circuit Topology for Lumped Decoupling Simulation



You can exclude certain design elements, such as inter-plane capacitance, from the circuit topology. See Decoupling Wizard - Customize Settings Page (Standard Simulation).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Simulating PDN Decoupling - Lumped

Circuit Topology for Distributed Decoupling Simulation

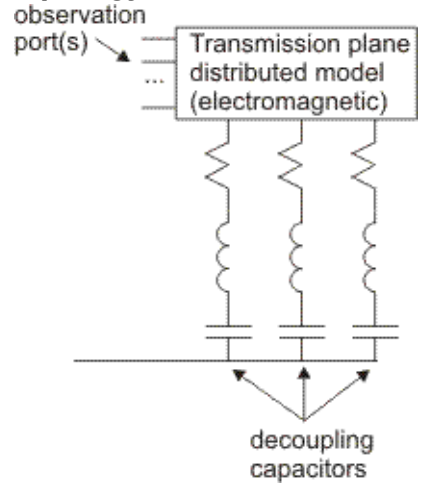
The software models the PDN as a set of transmission planes and decoupling capacitors. It accounts for distributed current paths, such as decoupling capacitor and IC pin locations. It does not account for signal-integrity structures and components, such as signal traces and vias, and IC driver/receiver pins.

Figure 8-29 shows the circuit topology for distributed decoupling analysis. Although not shown, the topology can include VRM models.

Note

This information does not apply to advanced distributed decoupling simulation.

Figure 8-29. Circuit Topology for Distributed Decoupling Simulation



Related Topics

Simulating PDN Decoupling - Distributed

Stitching-Via Optimization

You can speed up decoupling and signal-via bypassing simulation by having the software find stitching vias in the design that are located close together and merge their individual models into an equivalent model.

Stitching-via optimization takes advantage of the fact that when the size of an object (or a group of objects) is much smaller than the wavelength of a signal, the signal does not interact with

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them significantly. Therefore, approximate models can accurately represent these objects in simulation.

The Customize Settings page for the Decoupling Wizard and Bypass Wizard provides a Tolerance option that controls the merging radius for optimization:

- Low—1/30th of the minimum wavelength of the signal
- Medium—1/20th of the minimum wavelength of the signal
- High—1/10th of the minimum wavelength of the signal

For example, if you set the upper simulation frequency to 300 MHz, the wavelength of a 300 MHz signal in FR-4 is about 20 inches. In electromagnetic analysis, 1/10th wavelength is considered safely much smaller than the wavelength of the signal. Within a 2 inch radius, the software can use one equivalent via model to represent individual stitching vias.

Not all stitching vias are eligible for optimization. Most optimization takes place far away from IC and decoupling-capacitor pins. The optimization algorithm preserves individual models for caging vias and stitching vias that contribute significantly to transmission-plane or decoupling-capacitor inductance. Caging vias that are located very close to the IC or decoupling-capacitor pin are always modeled individually. If you run decoupling analysis to produce Z-parameters for an IC power-supply pin that uses a via with a stitching section, any very-nearby stitching vias are preserved as individual models to observe their full caging effect. The same is true if you run bypass simulation to produce Z-parameters for a signal net topology that uses a via with a stitching section.

As a result, this setting on the Customization Setting page may have little effect for designs where most of the stitching vias in the transmission plane contribute significantly to transmission-plane or decoupling-capacitor inductance.

Related Topics

Decoupling Wizard - Customize Settings Page (Standard Simulation)

PDN Model Extractor Wizard - Customize Settings Page

Via Model Extractor Wizard - Customize Settings Page

Running Signal-Via Bypass Simulation

Power-Supply Pins That Can Be Selected for Distributed Decoupling Simulation and Exporting a PDN

The Select IC Power Pins page in the decoupling wizard and exporting a PDN to an S-parameter model wizard searches the design for transmission planes that either enclose IC power-supply pins or are very close (about 118 mils or 3 millimeters) to them. In LineSim, if at least one of

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the found transmission planes has a layer that connects to the IC power-supply pin and a layer that references the IC power-supply pin (that is, you identified it as a reference layer when you added the IC power-supply pin symbol to the PDN layout), the IC power-supply pin is available for probing. BoardSim has the same behavior as LineSim, except the reference layer for the IC power-supply pin does not have to be found.

For example, in LineSim, let us assume an IC power-supply pin connects to a metal area on LAYER1 on net VCC and references metal areas on LAYER5 on net GND. If the wizard finds at least one transmission plane that contains a metal area on LAYER1 on net VCC and a metal area on LAYER5 on net GND, the IC power-supply pin is available for probing.

Note that even if an IC power-supply pin is available for probing, it can be rejected when you finish running the wizard, when a more-detailed analysis of its location is performed.

Related Topics

Simulating PDN Decoupling - Distributed

Exporting a PDN to an S-Parameter Model

Editing Power Aware SI Model Port Mappings

Scope: BoardSim

After you have assigned a power-aware SI model, you can edit port mappings if needed.

Prerequisites

- You have assigned a power aware SI model. See “Modeling a Non-Ideal PDN or Via-to-Via Coupling With an S-Parameter Model” on page 115.

Procedure

1. Open the Manage Power Aware SI Models dialog box if needed. Choose the **Analyze 3D EM > Manage Power Aware SI Models** menu item.
2. Select a model row and click **View/Edit Ports**.

   The Edit/View Power Aware SI Model dialog box appears. This dialog box shows the location of each of the port terminals and the nets that are connected to the pins or vias.
3. Click **Edit**. The “Edit Power Aware SI model port mappings” dialog box opens.
4. To delete a port mapping, select a port row in the Model Port Mapping list in the top half of the dialog box and click **Delete Mapping**.
5. To add a mapping, in the bottom half of the dialog box, select a port row, select a source pin/via, select a reference pin/via and click **Add Mapping** to add the mapping to the Model Port Mapping list above.
6. Add or delete any other mappings and click **OK** when finished.

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Evaluating SERDES Channel Compliance From an S-Parameter Model

If you have an S-parameter model of a SERDES channel that uses a supported protocol, the software can quickly calculate figures of merit (FOM) without using IBIS AMI transmitter and receiver models. This method provides a quick way to evaluate performance that is particularly useful when your design includes many channels and you do not have IBIS AMI models.

The software uses a default minimally compliant driver, receiver and package model in calculations to produce results that you can use to investigate how interconnect topology and crosstalk affect channel performance, and to identify optimal Tx and Rx equalization parameters.

For an Ethernet channel, the software uses algorithms from specification IEEE 802.3bj, annex 93A, to calculate channel operating metrics (COM) based on a signal-to-noise ratio FOM for all operating modes. The software can check several operating modes, such as 100GBASE-KR4 and CAUI-4, for COM compliance. For information about parameters, see the appropriate annex in the IEEE 802.3bj or IEEE 802.3bm specification.

Note

You can also check 10GBASE-KR and 40GBASE-KR4 operating modes for COM compliance, even though they pre-date the definition of COM. The software uses a set of COM channel parameters that were extracted from informative channel parameters as described in specification IEEE 802.3ap, annex 69B.

For a PCI Express channel, the software uses algorithms similar to COM and simulation guidelines from the PCI Express base specification to calculate metrics based on a maximum eye area FOM. For information about parameters, see section 4.3.6.4.2 Simulation Tool Requirements, section 4.3.6.4.3 Simulation Tool Chain Inputs, section 4.3.6.4.4 Processing Steps and section 4.3.6.4.5 Simulation Tool Outputs of the PCI Express base specification.

For more information, see Evaluating SERDES Channel Compliance From an S-Parameter Model in the Touchstone Viewer and Evaluating SERDES Channel Compliance From a Measured S-Parameter Model in the SERDES Wizard.

Related Topics

General Batch Simulation Results Spreadsheet

Analyzing Multiple SERDES Channels Using the SERDES Wizard - Compliance Mode

Evaluating SERDES Channel Compliance From a Measured S-Parameter Model in the SERDES Wizard

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

ExpertReferenceOptions.opt File

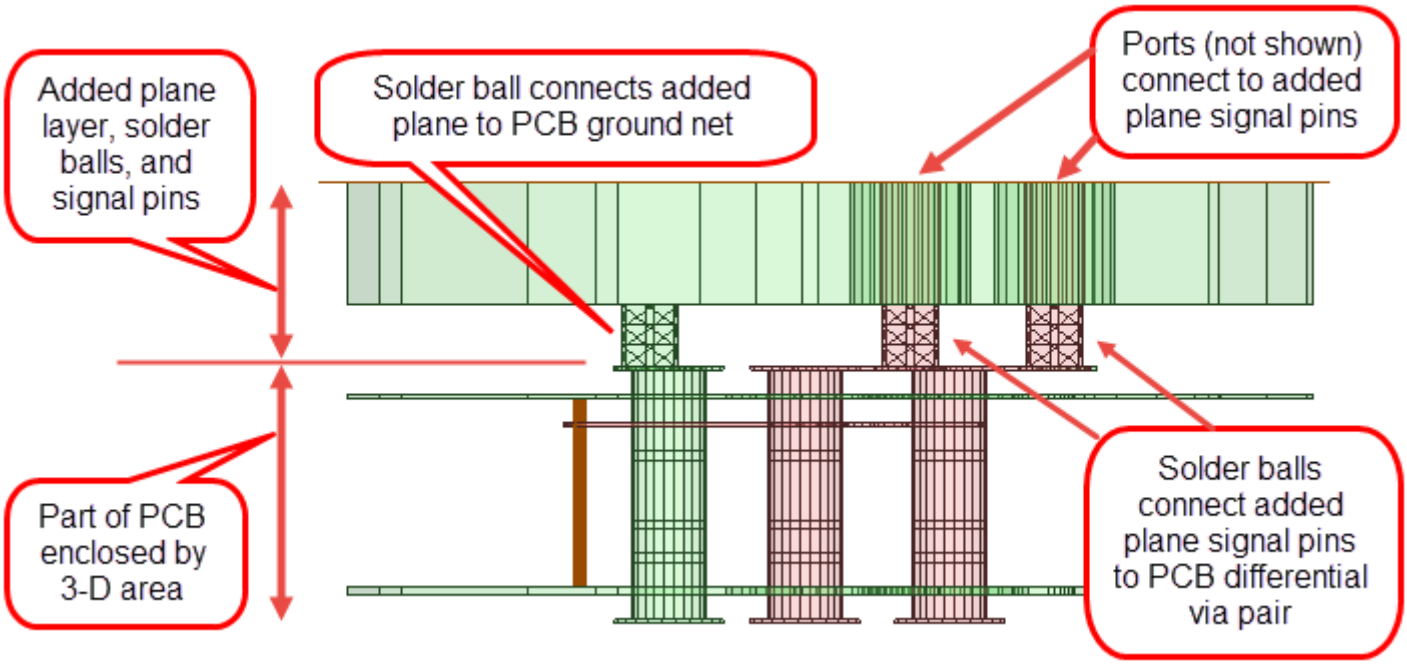
When BoardSim creates a 3-D area, you can have it add a plane layer to represent a return current path provided by a BGA (or similar) package and connect it to your board design. You can optionally provide detailed parameters for the added plane layer and its connections to your board design with an *ExpertReferenceOptions.opt* file. You manually create this file.

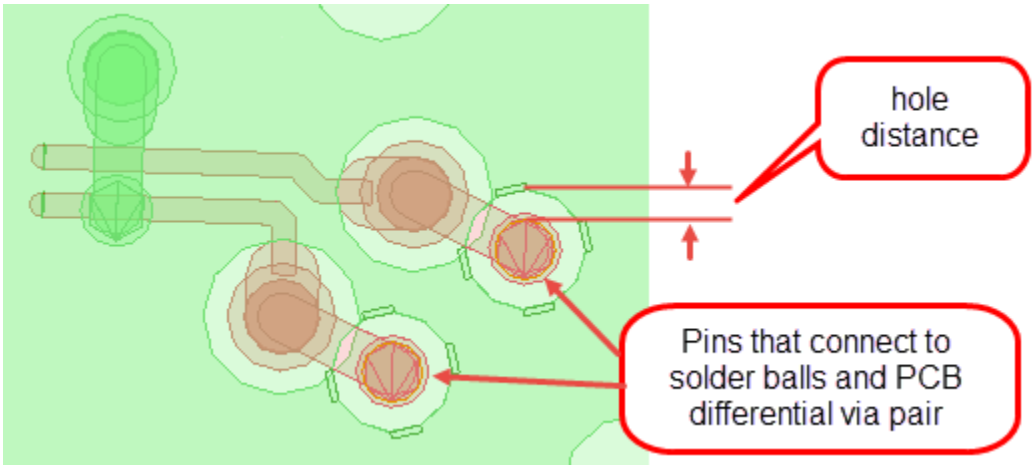
Example File and Added Plane Layer Geometries

This section contains an example file, and geometries for an example 3-D area and added plane layer.

```
<?xml version="1.0" ?>
<Options Type="ExpertReferenceOptions">
  <Parameters>
    <Parameter Display="Distance Units" Value="mil" />
    <Parameter Display="Solder Ball Diameter" Value="10.0" />
    <Parameter Display="Solder Ball Height" Value="10.0" />
    <Parameter Display="Hole Distance" Value="5.0" />
    <Parameter Display="Layer Thickness" Value="20.0" />
    <Parameter Display="Adjust Component Size" Value="True" />
    <Parameter Display="Pin Size" Value="10.0" />
  </Parameters>
  <Children>
    <Child Type="ComponentReferenceOptions">
      <Parameters>
        <Parameter Display="Component Name" Value="U9" />
        <Parameter Display="Solder Ball Diameter" Value="9.0" />
      </Parameters>
    </Child>
    <Child Type="ComponentReferenceOptions">
      <Parameters>
        <Parameter Display="Component Name" Value="U11" />
        <Parameter Display="Solder Ball Diameter" Value="1.10E+1" />
      </Parameters>
    </Child>
  </Children>
</Options>
```


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File Syntax

An *ExpertReferenceOptions.opt* file is a text file in XML format, with an *.opt* extension.

General information:

- Every parameter is optional.
- Parameter names and values are case-sensitive.
- Values must be enclosed by double or single quotes.

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- Numerical value format can be decimal or scientific notation. For example, 11.0 or 1.10E+1.
- This file supports two levels of hierarchy. Specify global parameters in the <Parameters> section. Specify component-specific parameters in the optional <Children> section.

  Component-specific parameters override global parameters. In the above example file "Example File and Added Plane Layer Geometries" on page 533), the solder ball diameter is 9 mils for U9, 11 mils for U11, and 10 mils for any other reference designator enclosed by a 3-D area.

Statement or Section	Description
<?xml version="1.0" ?>	XML declaration.
<Options Type="ExpertReferenceOptions"> </Options>	Root section.
<Parameters> </Parameters>	Contains global settings.
<Children> </Children>	Contains one or more Child sections.
<Child Type="ComponentReferenceOptions"> </Child>	Contains settings for a specific reference designator.

Parameter	Description
<Parameter Display="Adjust Component Size" Value="True" /> <Parameter Display="Adjust Component Size" Value="False" />	Specifies the extents of the added plane layer. "True" means the plane layer extends to the boundary of a Full-Wave Solver model layout. "False" means the plane layer extends to the boundary of an IC component.
<Parameter Display="Component Name" Value="*reference_designator*" />	Specifies the reference designator that other attributes in this <Child> section apply to. **Note:** Only the <Child> section supports this parameter.

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Parameter	Description
<Parameter Display=”Distance Units” Value=”um” /> <Parameter Display=”Distance Units” Value=”mil” />	Specifies units. **Note:** The software supports metric or English units. Specify “um” for metric units and “mil” for English units. **Note:** Only the <Parameters> section supports this parameter.
<Parameter Display=”Ground Net” Value=”*name*” />	Specifies the name of a power-supply net on the board design that you want to connect to the added plane layer.
<Parameter Display=”Hole Distance” Value=”*distance*” />	Specifies the radial distance from the surface of a pin to the surrounding hole in the added plane layer.
<Parameter Display=”Layer Thickness” Value=”*thickness*” />	Specifies the thickness of the added plane layer.
<Parameter Display=”Pin Size” Value=”*diameter*” />	Specifies the diameter of a pin that passes through the added plane layer.
<Parameter Display=”Solder Ball Diameter” Value=”*diameter*” />	Specifies the diameter of a solder ball that connects the added plane layer to a power-supply net on the PCB.
<Parameter Display=”Solder Ball Height” Value=”*height*” />	Specifies the height of a solder ball that connects the added plane layer to a power-supply net on the PCB.

Related Topics

HyperLynx Advanced Solvers Option Files for 3-D Areas

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Exporting and Translating a Board Design

You can export a board design from Xpedition Layout, PADS Layout, and other Siemens EDA layout software. HyperLynx can translate a board design created in other layout software tools.

Topic	Description
Exporting a Board File from Siemens EDA Layout Software	You can export a board design from Siemens EDA Xpedition® Layout or Board Station XE to a CCE file, which you can open in HyperLynx to analyze signal and power integrity.
Exporting a Board File From Siemens EDA PADS Layout	You can export a board design from Siemens EDA PADS® Layout to a HYP file, which you can load in HyperLynx to analyze signal and power integrity.

Exporting a Board File from Siemens EDA Layout Software

You can export a board design from Siemens EDA Xpedition® Layout or Board Station XE to a CCE file, which you can open in HyperLynx to analyze signal and power integrity.

Prerequisites

To include backdrill data with the exported design, create the *Backdrill.cfg* file. For syntax information, refer to Xpedition Layout documentation.

Procedure

1. If your board design contains negative planes, in Xpedition Layout or Board Station XE choose the **Planes > Generate Negative Planes** menu item.
2. If your design contains backdrill data, choose the **Output > Backdrill Analysis** menu item.
3. Choose the **Analysis > Export to HyperLynx SI/PI/Thermal** menu item.

   For detailed information, see “Exporting Design Data to HyperLynx” in the *Layout Verification Guide* that ships with Xpedition.

Results

You are now ready to open your board design in HyperLynx. See Opening a Design.

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Exporting a Board File From Siemens EDA PADS Layout

You can export a board design from Siemens EDA PADS® Layout to a HYP file, which you can load in HyperLynx to analyze signal and power integrity.

If HyperLynx is installed on your computer, you can open the file directly. Choose the **Tools > Analysis > Signal/Power Integrity** menu item.

Note

For the latest instructions about exporting a board from PADS Layout, see the PADS Layout documentation.

Procedure

1. In PADS Layout, ensure that any assigned resistor, inductor, or capacitor values use the format that HyperLynx recognizes. Specify the VALUE property for the component in the format:

   <numeric value><scaling suffix or scientific notation><units>

   HyperLynx can only read values if scaling suffix, scientific notation, and unit information are specified in the format:

Name	Scaling Suffix	Scientific Notation
mega	M	1,000,000x
kilo	K or k	1,000x
milli	m	0.001x
micro	u or U	1e-6x
nano	n or N	1e-9x
pico	p or P	1e-12x

Component	Units
R	O, ohm
L	H, henrys
C	F, Farads

2. Choose the **File > Export** menu item.
3. From the “Save as type” list, select HYP Files.
4. Click **Save**. The HYP Export dialog box opens.

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5. If needed, change export option values.
6. Click **OK**.

Results

You are now ready to open your board design in HyperLynx. See Opening a Design.

External Characterization Files

You can use external channel characterization files for FastEye or IBIS-AMI channel analysis that you create outside of the FastEye/IBIS AMI analysis wizard.

The channel characterization file must accurately represent the channel implementation and transmitter/receiver analog buffer behavior. The following types of changes can cause the file to not accurately represent the current channel behavior:

- Channel topology, which is the set of physical elements and geometries used to implement the channel and includes trace segments, stackup, signal vias, and so on
- Coupling thresholds (BoardSim) or coupling regions (LineSim)
- Transmitter/receiver analog buffer settings
- Transmitter output/input mode for channels with more than one transmitter

You may want to use external channel characterization files for any of the following reasons:

- Channel characterization simulation is slow (perhaps due to SPICE models) and you want to use the same channel topology and probe locations with different analysis settings.
- You prefer to analyze channels only in the frequency domain and want to use an S-parameter file to represent the analog channel characterization.

Analog Channel Characterization Files

Time Domain Files

You can manually save *.PLS* files from the Channel Characterization Dialog Box. These files are saved to the design folder unless you specify another location. The default design folder location is *C:\MentorGraphics\<release>\SDD_HOME\hyperlynx(or hyperlynx64)\HypFiles*. See Design Folder and HyperLynx Files.

Frequency Domain Files

You can provide S-parameter models that you create with third-party software. Provide one model to represent the channel behavior between the transmitter and receiver analog pins. If you have a chain of S-parameter models to represent individual channel elements, cascade them into a single S-parameter model. See Cascade Multiple S-Parameter Models in Series.

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Aggressor Channel Characterization Files

Provide characterization files that represent the victim receiver behavior when an aggressor switches from low to high or high to low. If the aggressor net has two drivers, provide files for signals being transferred in both directions.

Time Domain Files

These files contain the response of a single receiver to an aggressor making a single step from low to high or from high to low. You can load:

- .LIS file created by SPICE or the Digital Oscilloscope Dialog Box. The simulation length must be at least 165ns.
- .PLS files from other sources.

Frequency Domain Files

Restriction

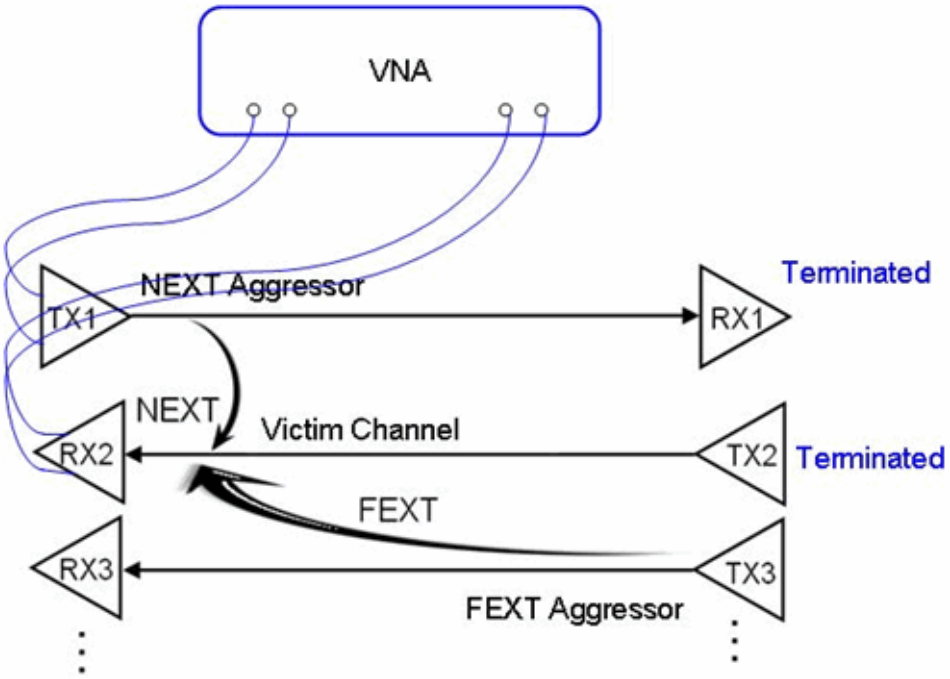
If the channel is non-linear, you cannot use S-parameter files as aggressor characterization files.

These files contain the response of a single receiver to an aggressor making a single step from low to high or from high to low. You can load *.S4P* (differential pair) or *.S2P* (single-ended) S-parameter files created by PCB hardware measurements.

Example: Figure 8-30 shows the channel buffer and VNA setup for a near-end crosstalk (NEXT) measurement for victim differential receiver RX2. Note that the VNA and channel buffers in this figure are not set up to measure far-end crosstalk (FEXT), but it is marked to provide the general idea of how to do it.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-30. PCB Measurement Set Up to Measure Crosstalk



Related Topics

Bit Sequence for Automatic Channel Characterization

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

The Field Solver in LineSim

LineSim uses its field solver to solve for the capacitances, inductances, propagation velocities, and characteristic impedances of a coupling region's cross section.

A field solver is a program that can solve for the electrical characteristics of a system of conductors and dielectrics, using one or more of the basic equations of electromagnetic theory ("Maxwell's equations").

Because coupling regions consist of two-dimensional cross sections that are assumed to be constant over some specified length, the field solver needs to work in only two dimensions. Taking advantage of this fact allows the software to calculate coupling parameters accurately, but also very quickly—in fact, interactively, as you work. For information about creating a coupling region, see Creating a Schematic Design.

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Tip

Three-dimensional electromagnetic solutions become important only if the frequencies of the signals traveling on a system of conductors is so high that the wavelengths of the signals' components are shorter than the various conductor structures in the system (e.g. vias, corner bends, etc.). This condition rarely occurs on PCBs carrying digital signals, so tools that analyze digital PCBs use two- rather than three-dimensional solvers. The big gain for users is speed: solvers run much faster in two dimensions than in three.

When more than one transmission line is present in a coupling region, the various electrical parameters of the system take on a matrix form. For example, for a two-trace coupling region, there is no longer a single value of capacitance that describes the region's cross section. Rather, there exists a 2x2 matrix which specifies both the capacitances of the individual traces to ground, and the capacitance between the traces.

It is worth noting that there is no need to understand any of the electromagnetic details in order to successfully perform crosstalk simulation. You can enter all of your geometric details, let the field solver take care of the electrical details automatically, and get results in the form of waveforms and report files. Even a parameter like differential impedance is calculated automatically to prevent you from having to know how to calculate it from a characteristic-impedance matrix.

How the Field Solver Works in LineSim

In order to completely determine the electromagnetic properties of a coupling region's cross section, the field solver must calculate the capacitance and inductance matrices for the cross section. These matrices give the conductor-to-ground and conductor-to-conductor capacitances and the self and mutual inductances of the traces in the coupling region.

To calculate capacitance values, the field solver finds the solution to Laplace's equation, a form of one of Maxwell's basic equations of electromagnetics:

$$\nabla^2 V = 0 \qquad (subject\ to\ all\ applicable\ boudary\ conditions)$$

In the solution, the solver seeks to find charge densities on the conductor surfaces and dielectric boundaries, rather than bothering to calculate the electric potential at all points between the conductors.

Note

This approach makes the field solver a "boundary-element" solver.

Several proprietary methods are used to speed calculations significantly while maintaining a high level of accuracy.

The solution to Laplace's equation occurs subject to all of the boundary conditions specified in the coupling region's cross section, i.e., it takes into account the exact shapes and locations of

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the conductors in the region and the locations and material properties of the dielectric boundaries. Special care is taken to calculate charge density accurately in regions in which it changes rapidly (e.g., at the corners of conductors).

Once the coupling region's capacitance values are found, then to calculate the inductance matrix, the field solver takes advantage of the following equation from transmission-line theory:

$$LC = \frac{\varepsilon}{c^2}, \quad \text{or} \quad L = \frac{\varepsilon}{Cc^2} = \frac{1}{C_0 c^2}$$

This allows a second solution to Laplace's equation — one in which all of the dielectrics are replaced by vacuum and the capacitance matrix C0 is found — to substitute for an explicit calculation of the coupling region's magnetic properties.

Once the capacitance and inductance matrices are both known, then the region's propagation speed(s) and characteristic impedances can be calculated. For the case of inhomogeneous dielectrics (i.e., a mixture of dielectric constants, as occurs with microstrip and buried-microstrip traces), multiple propagation speeds exist. These speeds are found from the eigenvalues of the matrix product LC.

How the Field Solver Runs in LineSim

For a schematic design, the field solver calculates the following information for every coupling region:

- Capacitance matrix
- Inductance matrix
- Characteristic impedance matrix
- Propagation speed(s)
- If multiple propagation speeds, the percentage of energy in each trace traveling at each speed
- An optimal resistor termination array for the region's transmission lines

This topic contains the following:

- How Field Solver Results Display
- Auto-Calculate Versus As-Needed Modes

Field Solver Messages

In preparation for performing crosstalk analysis, BoardSim briefly calls its field solver near the end of the board-loading process to characterize certain aspects of the PCB's stackup. Often, you will see a progress dialog box labeled "HyperLynx" and "Running field solver" while this analysis is running.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Tip

The field solver is called regardless of whether or not you are licensed for BoardSim's Crosstalk option. If you are not licensed for Crosstalk analysis, this (and when certain other changes, like stackup editing occur) is the only time the field solver runs; it is not available during simulation or any other kind of analysis unless you own the Crosstalk option.

Flight-Time Compensation in General Batch Simulation

Flight-time compensation is available for IC pins with test fixture information, such as Vref and Cref.

A portion of the pin-to-pin delay specified in an IC model, such as from clock to output, includes the duration of the signal transition between T=0 and Vmeasure as it switches into the test fixture load specified by the IC model. However the PCB interconnect load rarely matches the test fixture load, so the duration of the signal transition between the start of simulation and Vmeasure is different during PCB simulation and system operation than predicted by the IC model.

Batch simulation calculates delay times from driver IC pins to receiver IC pins (flight times), that compensate for the difference between the test fixture load and the PCB interconnect load. You can use flight times in spreadsheets used to manage timing budgets for system-level signals.

Batch simulation reports flight time for a differential pair as one of the following:

Both pins as a differential pair—The IBIS model contains the Rref_diff sub-keyword.

Each pin as a single-ended signal—The IBIS model either does not contain the Rref_diff sub-keyword or contains both Rref_diff and Cref_diff sub-keywords.

Hiding or Moving an IC Component Pin

Scope: LineSim

You can minimize the number of crossed connection wires in a schematic with IC components by hiding pins that you do not plan to include in simulation, or moving pins to different locations on an IC component symbol.

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The following video contains additional information:



Procedure

1. Double-click an IC component symbol in a schematic. The Assign IC Component Model dialog box opens.
2. Hide pins that you do not plan to include in simulation:

If you want to...	Do this...
Hide a pin.	In the Visible column, uncheck a pin row.
Hide multiple pins at once.	1. Click **Configure**. The Configure IC Component Symbol dialog box opens. 2. Select pin rows and click <<.

3. Edit pin locations:

If you want to...	Do this...
Move all pins automatically.	From the Auto-Place Ports list, select a pattern. **Note:** The software uses spreadsheet row numbers when moving pins. For example, the “Odd to Left, Even to Right” pattern moves pins on odd-numbered spreadsheet rows to the left side of a component.
Move a pin to the other side.	Click in the Side column and select Right or Left.
Move a block of pins to the other side.	1. In the Side column, click and drag to select a block of pins. A menu automatically appears. 2. Select Right or Left.
Move a pin up or down a side.	1. In the first column, click a row and release the mouse button. 2. In the first column, drag the selected row up or down.
Move multiple pins up or down.	1. Click **Configure**. The Configure IC Component Symbol dialog box opens. 2. Select pin rows and click **Up** or **Down**.

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Related Topics

Creating a Schematic Design

Horizontal and Vertical Geometric Search Range for Coupled Nets

Horizontal and vertical search ranges apply to each cross section that includes trace segments for the selected victim net and potential aggressor nets.

When the selected victim net consists of several trace segments, the software constructs several unique cross sections (or *coupling region*s) that contain trace segments for the victim and potential aggressor nets that are located within the search ranges.

The horizontal search range extends from both sides of the trace segment and ends at the first occurrence of any of the following limits or design objects:

- "Maximum distance from aggressor" option from the Set Coupling Thresholds dialog box.

  When the aggressor net is a member of a differential pair and its trace is located within the horizontal search range, the other member of the differential pair is always considered an aggressor, even when its traces are located outside the horizontal search range.

- "Horizontal Neighbor Limit" option from the Set Coupling Thresholds dialog box.

  The software does not count either of the following conditions:

  - Trace is located directly above or below the trace for the selected net.
  - Trace for a routed power-supply net is located between traces for the selected and coupled nets.

- An area fill or trace for a power-supply net.

The vertical search range extends above and below the trace segment, and depends on whether you check the "Include trace to area fill coupling" option from the Coupling Settings dialog box:

- Unchecked — The vertical search range ends at a plane layer.
- Checked — The vertical search range ends at the first occurrence of any of the following:
  - Number of stackup layers specified by "Vertical Layer Limit" from the Set Coupling Thresholds dialog box.
  - A plane layer with no area fills.

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- An area fill that extends beyond both sides of the horizontal search range.

The figure below shows how a selected net couple to other signal nets when “Horizontal Neighbor Limit” = 1 and “Vertical Layer Limit” = 1.

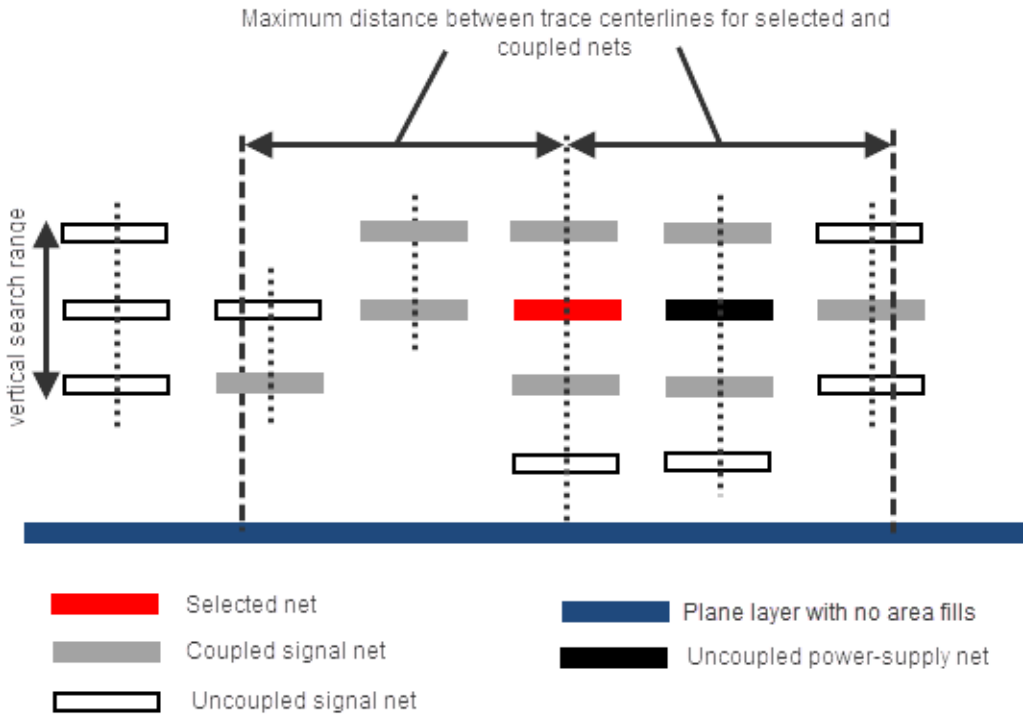


Figure Landmark	Description
Selected net	The origin of the horizontal and vertical search ranges.
Coupled signal net	Traces located within the horizontal and vertical search ranges, and coupled to traces on the selected net. The software does not count routed power-supply nets within the horizontal search range. The trace to the right of an “Uncoupled power-supply net” is coupled to the selected net.
Uncoupled trace	Traces for nets that are located outside the horizontal and vertical search ranges, and do not couple to traces for the selected net.
Uncoupled power-supply net	A power-supply net that is routed as a trace.
Plane layer with no area fills	A stackup layer that is identified as a plane layer and contains no area fills. A plane layer always ends the vertical search range.

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Related Topics

Accounting for Coupling

Coupling Ratio for RLC Matrix Package Coupling

How BoardSim Recognizes Power Supply Nets

When you load a board, BoardSim attempts to identify power-supply nets and determine the voltage. If net names are not recognized correctly, you can manually specify power supply nets and voltages in the Edit Power-Supply Nets dialog box.

BoardSim recognizes nets as power supply nets if:

- The net name indicates that it is a power supply net.
  - Nets named PWR, POWER, VCC, or VDD are recognized as 5V power supply nets.
  - Nets named GND, GRND, GROUND, or VSS are recognized as 0V power supply nets.
  - Net names that include a voltage number are recognized as power supply nets with the appropriate voltage. Net names must be in one of these formats:
    - +<number>V or +<number> or V+<number>
    - -<number>V or -<number> or V-<number>
    - <number>V or <number>

      For example, nets named +12V, v+12, -12, 12V, or 12, are recognized as 12V power supply nets.
- The net is connected to decoupling capacitors. BoardSim looks for nets that are connected to more than three capacitors by default. You can change the default number of capacitors in the “Assume net is a power-supply if” setting on the BoardSim tab of the Preferences dialog box.

Notes

- The name-matching is case-insensitive, e.g., “VCC”, “vcc”, and “Vcc” all match.
- Net names in this format are recognized as power supply nets if they are connected to at least one capacitor. This prevents an entire digital bus with supply-like net names from being mistaken as a collection of power supplies, since digital nets rarely have capacitors connected directly to them.
- BoardSim ignores unrouted nets when identifying power-supply nets.
- BoardSim may recognize any analog nets included in your design as power supply nets if they are connected to multiple capacitors.

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- To exclude analog nets from SI simulation, mark them as power supply nets in the Edit Power-Supply Nets dialog box dialog box.

Related Topics

Verifying That Power Supply and Signal Nets are Recognized Correctly

How BoardSim Recognizes Component Types

When BoardSim loads your board, it examines the list of devices in the design file and uses the reference-designator prefix to determine the component type of each device. Many commonly-used prefixes are automatically mapped. However, you may have to update the mapping for unrecognized prefixes.

A prefix is the first part of the reference designator and is the part that indicates the components type. For example, U for ICs (U1, U2) or R for resistors (R1, R2). Use the Edit Reference Designator Mappings Dialog Box to define the reference-designator mappings that the software uses to identify component types (IC, R, C, L, connector, and ferrite bead).

Supported Component Types

Table 8-13. Component Types Supported by BoardSim

Component Type	Default Prefix
IC (any driver or receiver device)	U
Resistor	R
Capacitor	C
Inductor	L
Ferrite bead	
Connector	J
Test point	TP

Although the software does not directly support other component types (such as transistors), this does not mean that you cannot simulate nets that include other types. See “Unsupported Components” below.

Note

The component type is unrelated to how a component is packaged. A discrete resistor and the resistors in an R network are both type resistor. Package types are handled separately from component types.

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Test Points

All one-pin components are treated as a test point, regardless of the prefix. This designation cannot be changed. BoardSim contains one default mapping for test points: the prefix TP. You can add additional mappings using the Edit Reference Designator Mappings Dialog Box.

By default, test points are filtered out as a board is loading. To include test points, enable Treat test points as IC pins in the Preferences Dialog box (**Setup > Options > General**, **Advanced** Tab). When you choose to ignore test points, you cannot attach oscilloscope probes to them. However, EZwave always probes test points. You also cannot assign device models to ignored test points.

You can choose to treat test points as IC pins. Treat test points as IC pins to simulate board performance in test fixture applications where signals are probed (loaded) at test points, or where test points inject signals and therefore need model assignments. Use the “Treat test Points as IC pins” option in the Preferences Dialog Box - Advanced Tab to treat them as IC pins.

Unsupported Components

The software supports the components IC, R, C, L, connector, and ferrite bead. Although the software does not directly support all component types, you can simulate a net that includes a non-supported component by substituting a component that is supported. For example, model a transistor, relay or crystal as an IC.

Although the software does not explicitly support diodes, IBIS models support clamp diodes, so you can use them to describe a discrete clamp diode or diode-terminating network. Note that the mappings for prefixes “CR” and “D” default to “IC”. For example, for a net that is clamped by pin A on a clamp diode CR3, choose a receiver-IC model for CR3.A.

Related Topics

Verifying That Component Types are Recognized Correctly

How Duty Cycle Affects EMC Simulation

EMC simulation results are sensitive to the duty cycle of a signal.

Even-harmonic radiation may be low or non-existent for a signal with an ideal 50% - 50% duty cycle and perfect signal integrity. Recall that a perfect square wave has only odd harmonics.

Since signals are rarely perfectly balanced and produce even-harmonic radiation, you may want to simulate with a less than perfect duty cycle, such as 49% - 51%.

Related Topics

Spectrum Analyzer Dialog Box

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HyperLynx Advanced Solvers Option Files for 3-D Areas

When creating a 3-D area, you can override default settings with values from HyperLynx Advanced Solvers tool options files (*.opt*). This capability supports experienced customers who want detailed control over the geometric model and solver settings used to create the S-parameter model that represents a 3-D area.

Use the HLAS Options Folder field in the 3D-Area Manager Dialog Box to specify a folder that contains any or all of these options files:

- *ExpertReferenceOptions.opt*
- *LayoutOptions.opt*
- *MeshOptions.opt*
- *SolverOptions.opt*

Note

Filenames are case insensitive for computers running Windows.

You manually create an *ExpertReferenceOptions.opt* file. See "ExpertReferenceOptions.opt File" on page 533.

Full-Wave Solver HPC, Full-Wave Solver, Fast 3D Solver, or Hybrid Solver create the remaining types of options files when you run a solve. You can find them in a project folder (for example, *\MentorGraphics\HLVX<release>\SDD_HOME\Nimbic\examples\diffpair*) or a simulation folder (for example, *\MentorGraphics\HLVX<release>\SDD_HOME\Nimbic\examples\diffpair\Simulation1*).

Related Topics

Exporting Part of a Board Design for Analysis in HyperLynx Full-Wave Solver

Modeling a Board Area With an S-Parameter Model

IBIS-AMI Model Requirements for Statistical Simulation

Accurate statistical channel analysis requires IBIS-AMI models that are designed for both statistical and time domain simulation. Avoid using a model when you cannot confirm that it was designed for statistical simulation.

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To ensure accurate statistical simulation, use models that at least describe the following behaviors in the AMI_Init function:

- Transmitter pre-emphasis
- Receiver equalization. Note that equalization is fixed and does not support clock and data recovery (CDR).
- Noise jitter for the transmitter and receiver. Reserved AMI parameters (defined by IBIS 5.1 and newer) can specify this behavior. Statistical simulation adds the transmitter and receiver jitter together and applies them at the receiver.

The most accurate models for statistical simulation also:

- Describe simulation frequency
- Describe simulation samples per bit
- Support the full range of AMI parameters available in the IBIS specification.

Note

To see similar results in statistical and time domain simulations, the AMI_Init and AMI_GetWave functions in the IBIS-AMI model must describe similar behavior. Statistical simulation uses the AMI_Init function, while time domain simulation uses the AMI_GetWave function.

Identifying Optimum Tap Weights

If you do not already know the best tap weights for the channel, FastEye analysis can automatically identify them for you.

Procedure

1. In the IC model, set all the tap weights to zero.
2. If you provide external step- and pulse-response waveforms, generate them prior to running FastEye channel analysis and specify them in the Channel Characterization Dialog Box.
3. On the FastEye Channel Analyzer - Add Pre-Emphasis/DFE/CTLE Page, enable the “Synthesize optimal values” option.
4. Set other wizard options as needed, run FastEye channel analysis, and see if the FastEye diagram, BER, and other measurements indicate acceptable results.
5. In the IC model, set the tap weights to the synthesized values.

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

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Jitter Distribution Types

You can enable multiple jitter distributions at the same time. If you enable multiple jitter distributions, the total jitter probability distribution function (PDF) is a convolution of the individual jitter distributions.

If you are not sure what jitter distribution to use, enable only Gaussian and obtain the combined driver and receiver jitter values (in sigma) from IC datasheets, design kit documentation, and so on. You may have your own reasoning on how to combine driver and receiver jitter values, by treating them statistically independent or not.

Although eye diagrams are based on the channel response measured at the receiver input pin, but the goal is to find the BER or eye diagram at the receiver decision point (which is beyond its amplifiers, DFE/CTLE, filters, and CDR circuitry). Because drivers and receivers are active devices, they both contribute random jitter due to thermal and transistor device noise, PLL (that is, CDR circuitry) behavior, and so on. This is why you should specify a jitter distribution representing both driver and receiver jitter.

Do not specify the jitter produced by the following effects, unless you have a specific reason to do so:

- PCB layout effects—Such as impedance mismatches and signal dispersion
- Data-dependent effects—Such as ISI, duty-cycle distortion, pseudo-random bit sequence periodicity

Table 8-14. Jitter Types Supported by HyperLynx Features

	FastEye Diagrams	Standard Eye Diagrams	IBIS-AMI Eye Diagrams
DjRj Jitter	X[1]		X
Dual-Dirac Jitter	X[2]		X
Duty Cycle Distortion Jitter	X		X[3]
Gaussian Jitter	X	X	X
Sinusoidal Deterministic Jitter	X	X	X[4]
Uniform Jitter	X	X	X[5]

1. Indirectly supported by using a combination of Gaussian and uniform jitter.
2. Indirectly supported by using a combination of sine and Gaussian jitter.
3. Supported by the Tx_DCD reserved parameter keyword.
4. Indirectly supported by the Dual-Dirac (mean1 mean2 sigma) reserved parameter keyword. Note that dual-Dirac jitter is a combination of sine and Gaussian jitter. By specifying a very small or zero sigma, you can define the magnitude of the sine jitter by setting mean1 and mean2 to an equal value and opposite signs (that is, the absolute value of mean1 equals the absolute value of mean2).
5. Indirectly supported by the DjRj(minDj MaxDj sigma) reserved parameter keyword. DjRj jitter is a combination of uniform and Gaussian jitter. The distribution for uniform jitter is between -minDj and maxDj.

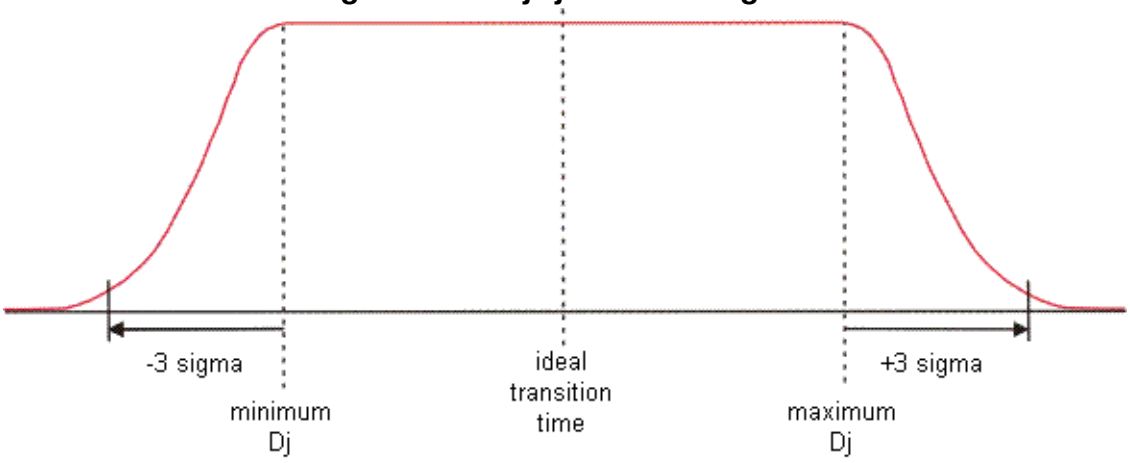
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DjRj Jitter

DjRj jitter represents a combination of Gaussian and uniform distributions.

A histogram consisting of a large number of DjRj-distributed jitter values resembles a uniform distribution with Gaussian distributions at its sides. See Figure 8-31. The Gaussian distribution portions represent the random and unbounded jitter contribution. The uniform distribution portion represents the deterministic and bounded jitter contribution.

Figure 8-31. DjRj Jitter Histogram



Dual-Dirac Jitter

Dual-Dirac jitter characterizes the cumulative effect of periodic (sine) and Gaussian jitter.

A histogram of the sine jitter has only two peaks that are often approximated by two Dirac functions. The distance between the Dirac peaks is two times the magnitude of the underlying periodic jitter. If the jitter has no DC or constant phase offset, the peaks of the sine jitter PDF are located at equal distances from the ideal transition time. If not, they could both be offset to the right or left.

The Gaussian component is characterized by its sigma.

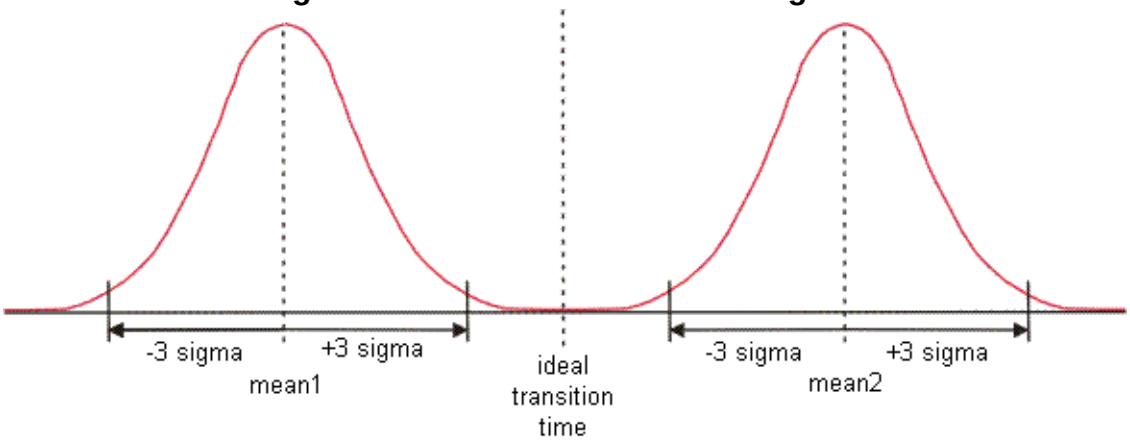
When both components are present, the total PDF becomes a convolution of partial PDFs and, since one of them consists of two Dirac functions, the result is the sum of the two Gaussian PDFs taken with a factor of 0.5. When the magnitude of the sine component is large compared to the sigma of the Gaussian component, the cumulative PDF resembles two bell curves, with their maximums located at “mean1” and “mean2”, where abs(mean1-mean2) is two times the magnitude of the sine jitter. See Figure 8-32.

With progressively smaller (magnitude of sine jitter) to (sigma of Gaussian jitter) ratios, the two bell curve peaks move closer together and the depression between them begins to lift until finally they form a single bell curve with twice the original magnitude. This happens if the

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magnitude of the sine component becomes negligible, and a single Gaussian distribution is formed.

Figure 8-32. Dual-Dirac Jitter Histogram



Duty Cycle Distortion Jitter

Duty cycle distortion (DCD) deterministic jitter represents a non-Gaussian and bimodal distribution of signals.

There are two forms of DCD:

- Clock duty cycle jitter or “clock DCD”—The transmitter produces even bits that are shorter or longer than odd bits, regardless of the bit pattern.

  Restriction

  IBIS-AMI channel analysis and the Tx_DCD keyword support this definition.

- Edge asymmetry or “data DCD”— The transmitter produces rising edge delays that are shorter or longer than falling edge delays. This jitter type makes all rising edges offset by the same amount regardless of how many 0 bits precede or follow them. Similarly, this jitter type makes all falling edges offset by the same amount regardless of how many 1 bits precede or follow them.

  Restriction

  FastEye channel analysis supports this definition.

The probability density function (PDF) for clock duty cycle distortion is:

$$PDF_{DCD}(t) = \frac{1}{2}\left[\delta\left(t - \frac{\delta_{DCD}}{2}\right) + \delta\left(t + \frac{\delta_{DCD}}{2}\right)\right]$$

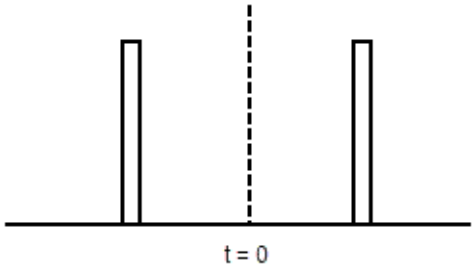
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Where:

δ_{DCD} is the peak-to-peak duty cycle distortion.

Figure 8-33 shows the bimodal distribution for DCD jitter.

Figure 8-33. Duty Cycle Distortion Jitter - Histogram



Gaussian Jitter

Gaussian jitter is added to the stimulus so that each transition is adjusted away from its ideal transition time by a random amount. A histogram built from increasingly long observations of Gaussian random jitter, with a sufficiently small size of subintervals and large number of such subintervals, resembles a bell curve.

See Figure 8-34. In the limit, the histogram approaches a smooth continuous function called a Gaussian probability density function (PDF), with one parameter:

$$p(x) = G(\text{sigma}, x) = \frac{1}{\text{sigma}\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\text{sigma}^2}\right)$$

Where:

sigma is called the standard or mean deviation.

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Figure 8-34. Gaussian Probability Density Function

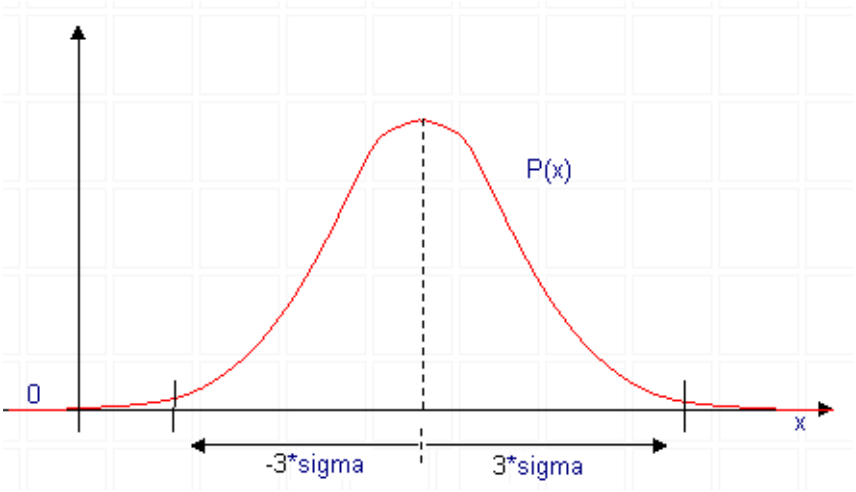


Figure 8-34 has the following properties:

- The Gaussian distribution is implemented as a symmetric PDF with a a mean of zero.
- The distribution has one maximum, which is located at the mean.
- The parameter sigma shows how widely (on average) the Gaussian random variable may deviate around its mean value. In most cases, with probability of 99.73%, the random variable gets into the +/-3 sigma interval.

  Specify the width (or magnitude) of the jitter at one sigma. HyperLynx automatically derives the width of other sigmas. The sigmas are equally spaced from one another.

  Specify sigma as an absolute value (for example, in nanoseconds) or as a relative value (for example, a fraction of the unit interval set for the simulation).

  Increasing the value of sigma increases (on average) the deviation of the timing of waveform transitions away from the ideal switching time.
- The function is non-negative, $p(x) > 0$.
- Although it starts from x = -infinity and goes up to x = +infinity (the function has no left or right bounds), the area enclosed by the curve and the X axis is finite and equals 1.

  If we compare two functions with different sigma, the one with smaller sigma will be narrower but higher, to preserve the area equal 1.

  While there is no theoretical limit on the Gaussian random jitter value, a practical limit exists for serialization/deserialization (SERDES) channels. For example, if adjacent transitions are each jittered toward each other by more than half the bit interval, we get a bit sampling error. HyperLynx prevents this from happening.
- Table 8-15 shows the relationship of the Gaussian distribution to the confidence interval. The first column shows the probability of any particular event falling inside the range of the sigma in the second column.

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To determine the likelihood of an event happening outside the range of sigma in the second column, subtract the value in the first column from 1. For example, the probability of an event falling outside 3 sigma is 1 - 0.9973 = 0.0027, or one in 370.

Table 8-15. Gaussian Distribution - Confidence Interval

Probability	Range (in sigma)
0.80	1.28155
0.90	1.64485
0.95	1.95996
0.98	2.32635
0.99	2.57583
0.995	2.80703
0.9973	3.000
0.998	3.09023
0.999	3.29052
0.9999	3.8906
0.99999	4.4172

See “Units for Gaussian and Uniform Jitter” on page 562.

Fast-Developing Jitter Sources

Starting with V8.0, HyperLynx uses “fast-developing jitter”. Fast-developing jitter means that jitter values can vary fast from bit to bit, as if they are almost uncorrelated. By contrast, slow-developing jitter means that jitter values vary slowly or continuously.

If you compared the distribution of fast-developing jitter and slow-developing jitter, while using the same sigma for both distributions, many more bits are required for slow-developing jitter to show its full variability. Because of this, the visible effect from the jitter was sometimes too small in eye-diagrams when the number of simulated bits was not sufficiently large. When using “fast-developing jitter”, V8.0 shows more jitter effect on the same bit length than previous releases.

Gaussian Random Jitter Conceptual Example

Specify jitter as 10% of the nominal bit interval of 3.8 ns.

Specify a stimulus length of more than 370 bits.

This indicates that 68.33% of the time (1 sigma), the jittered transition will occur between the times of (nominal - (3.8 ns * 10%)) and (nominal + (3.8 ns * 10%)).

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Similarly, 95.5% of the time (2 sigma) the jittered transition will occur between the times of (nominal - (3.8 ns * 20%)) and (nominal + (3.8 ns * 20%)).

Related Topics

Units for Gaussian and Uniform Jitter

Sinusoidal Deterministic Jitter

Sinusoidal deterministic jitter represents a non-Gaussian distribution with a finite peak-to-peak amplitude.

Figure 8-35 on page 559 shows how the timing offset varies over the sinusoidal jitter period, which is usually much greater than the bit interval. Zero timing offset is the ideal switching time for the signal.

Figure 8-35 shows a zero-degree initial phase, where the timing offset increases slowly to reach the maximum positive timing offset, decreases slowly to reach the maximum negative timing offset, and so on.

Figure 8-35. Timing Offset Over Sinusoidal Jitter Period - 0 Degrees Initial Phase

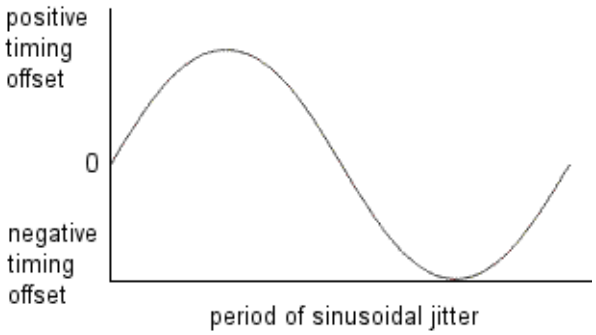
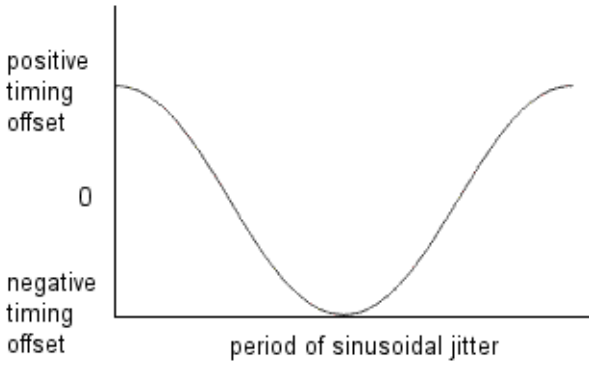


Figure 8-36 shows a ninety-degree initial phase, where the timing offset slowly decreases to reach the maximum negative timing offset, and then slowly increases to reach the maximum positive timing offset.

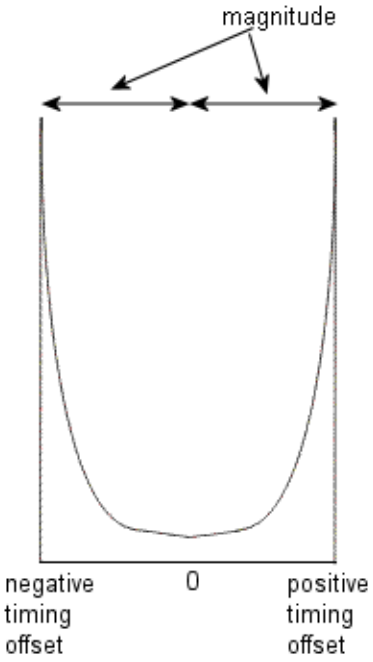
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-36. Timing Offset Over Sinusoidal Jitter Period - 90 Degrees Initial Phase



A histogram consisting of a large number of sinusoidally-distributed jitter values resembles Figure 8-37.

Figure 8-37. Sinusoidal Jitter - Histogram



Uniform Jitter

Uniform jitter represents a non-Gaussian distribution where each possible value has the same probability of happening.

A histogram consisting of a large number of uniformly-distributed jitter values with no offset (mean = 0) from ideal signal transition timing values resembles Figure 8-38. Figure 8-39 shows uniform jitter with a positive offset (mean > 0) from ideal signal transition timing.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-38. Uniform Jitter - Histogram - Mean = 0

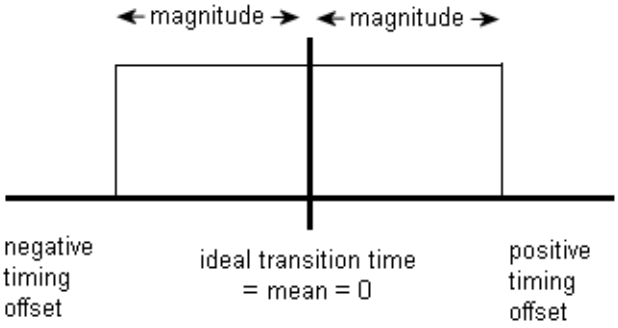
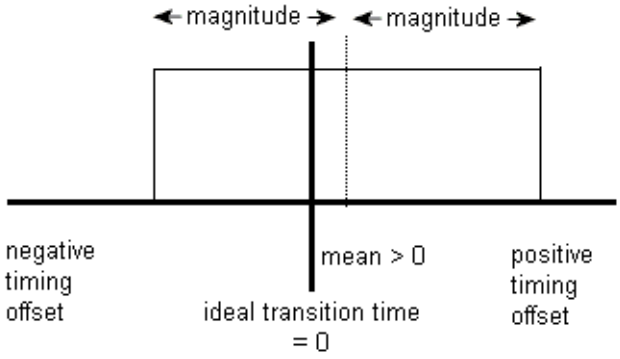


Figure 8-39. Uniform Jitter - Histogram - Mean > 0



Jitter Applications

This table shows some common uses for the various jitter distributions.

Table 8-16. Jitter Applications

Jitter Type	Can Represent
DjRj	Receiver clock-and-data recovery (CDR) jitter. There are indications that steady state phase distribution in receiver CDR circuitry alone has a flat “top”, that cannot be modeled with only Gaussian or sine jitter distributions.
Dual-Dirac	Characterize the cumulative effect of periodic (sine) and Gaussian jitter.
Duty Cycle Distortion—DCD	Clock duty cycle jitter or “clock DCD”: • Transmitter serialization clocks with duty cycle errors. Can result from transmitter or driver input threshold offset from the nominal voltage. Edge asymmetry or “data DCD”: • Transmitter or driver rise and fall delay asymmetry. Can result from non-linear loads, unbalanced transmitter source and sink currents, and common mode voltage in differential signals.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 8-16. Jitter Applications (cont.)

Jitter Type	Can Represent
Gaussian	• Random noise present in the system, such as thermal and transistor device noise, which is associated with conductor electron flow and increases with frequency and temperature. Common random noise sources include thermal noise, shot noise, and flick noise. • Power/substrate connection because of finite power-supply rejection ratio
Sine	• Deterministic noise sources • Power-supply voltage variations (that is, ripple) • Crosstalk. This includes electromagnetic effects that you cannot precisely predict and effects whose variations are much slower than the frequency of the data channel. **Restriction:** FastEye and IBIS-AMI channel analysis do not include crosstalk. Standard eye diagrams do include crosstalk.
Uniform	Receiver clock-and-data recovery (CDR) jitter. There are indications that steady state phase distribution in receiver CDR circuitry alone has a flat “top”, that cannot be modeled with only Gaussian or sine jitter distributions. You may need to combine uniform and Gaussian jitter distributions to fully represent this behavior. **Note:** You can also use uniform jitter to produce a worst-case distribution more quickly than Gaussian jitter, which can be helpful during “what if” experiments.

Units for Gaussian and Uniform Jitter

% and ns units represent the same thing. Some IC manufacturers specify maximum jitter in simple time units while others specify the maximum jitter relative to the bit interval or unit interval (UI) for which the part is designed to function. Signaling technology specifications might also specify the maximum allowable jitter for a driver in the circuit.

To convert between the two formats, use the equation:

UI * percent / 100 = time

Where:

UI is the unit interval or bit interval

percent is the specified maximum jitter in percentage of the UI

time is the maximum time in seconds for the specified jitter

For a Gaussian distribution, the percent or ns time is the one sigma value of the jitter distribution. For an illustration of three sigma, see Figure 8-34 on page 557.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

For a uniform distribution, the percent or ns time is the magnitude of the jitter distribution. For an illustration of magnitude, see Figure 8-38 on page 561.

Manipulating a 3-D View

Scope: BoardSim

You can rotate and view a board design from multiple angles in the 3D PCB Viewer to more easily see net topology and PDN implementation. 3-D viewing is especially useful for seeing how a signal via passes through metal areas and how a decoupling capacitor connects to a PDN.

Prerequisites

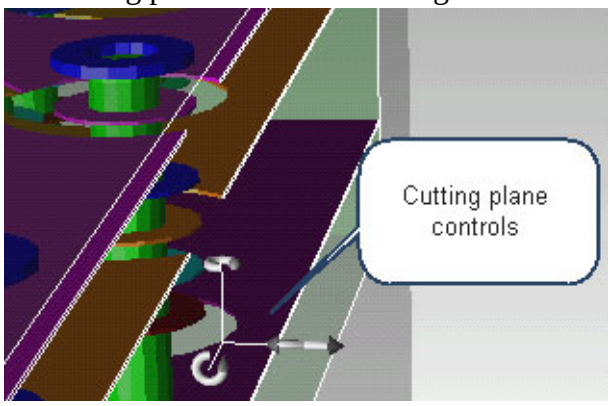
- You have installed the 3D PCB Viewer product, which is available from the HyperLynx software installation media.

Procedure

1. Choose the **View > Display in 3D > Board** menu item. The 3D PCB Viewer dialog box opens.
2. To manipulate the view, do any of the following:

If you want to...	Do this...
See a standard view (Top, Front, Left, and so on).	Select a standard view from the View menu, or click a standard view icon on the toolbar.
Zoom in or out.	Rotate the mouse wheel.
Rotate the board around the X- or Y-axis.	Press Shift+press and hold the middle mouse button as you drag the cursor up and down for X-axis rotation or sideways for Y-axis rotation.
Rotate the board around the Z-axis.	Press Ctrl+ and hold the middle mouse button as you drag the cursor in a circular motion.
Pan.	Press and hold the middle mouse button as you drag the cursor up and down or sideways.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to...	Do this...
See a cross section view.	1. Choose the **Visibility > Cutting Plane** menu item. 2. To change the location of the cross section, point to any of the cutting plane controls and drag the cursor. 
Hide a stackup layer.	1. Right-click an empty area of the (2D) board viewer and click **Show Viewing Filter**. 2. In the Viewing Filter dialog box, uncheck layers that you want to hide.

Related Topics

Display 3D Board Area Dialog Box

Model Channel Frequency Response with Complex-Pole Models

FastEye channel analysis uses channel-response waveforms to create a model of the channel frequency-domain behavior.

A proprietary complex-pole fitting (CPF) algorithm creates a complex-pole model, in the pole-residue form, to represent the frequency dependency of the channel.

FastEye channel analysis uses the complex-pole model to create the FastEye diagram and to display the results in the time domain. The complex-pole model exists in memory and is not stored as an external file.

CPF has the following advantages over alternative simulation technologies, such as convolution and equivalent circuits:

- Simulates much faster
- Is more accurate—Especially for long simulations, where errors accumulate due to truncation of unit step and pulse responses (also known as local truncation error)

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- Is more stable— Circuit passivity may be violated by truncation and (possibly) coarse time step of convolution and equivalent circuit methods
- Automatically enforces causality

FastEye channel analysis also supports convolution. For a comparison of the strengths of CPF and convolution, see Table 10-2 on page 885.

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

Monotonicity Threshold

This General batch simulation setting defines the maximum amplitude of non-monotonicities to ignore.

Use this option to filter out glitches that are insignificant or so small they would not have any effect inside the receiver IC package.

Any non-monotonicities larger than this amplitude are reported in the Rise/Fall Monotonic [Pass/Fail] column in simulation results.

For single-ended nets, the software performs this check while the waveform is between Vil(DC) and Vih(DC), and uses the following receiver threshold information from the IBIS model:

Vil(DC), highest priority first

[Receiver Thresholds] keyword, Vth - Vinl_dc subparameter

[Model Spec] or [Model] keyword, Vinl subparameter

Vih(DC)

[Receiver Thresholds] keyword, Vth + Vinh_dc subparameter

[Model Spec] or [Model] keyword, Vinh subparameter

For differential signals, the software performs this check while the differential waveforms are within the voltage range defined by the Vdiff(DC), and uses the following receiver information from the IBIS model (highest priority first):

[Receiver Thresholds] keyword, Vdiff_dc subparameter

[Diff Pin] keyword, Vdiff subparameter

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

MultiBoard Project Board IDs

Board IDs are unique identifiers the software assigns to each board in a multiple board design. Board IDs used by dialog boxes and reports make it easier to identify particular boards, or nets or components associated with particular boards.

The board ID values start with “B00” and increment by one (B00, B01, B02), without gaps in the sequence. If you delete (or add) boards from your MultiBoard project, the subsequent board ID values automatically decrement (or increment).

If you have multiple instances of a board, BoardSim assign different board IDs for each instance. For example, if your design uses a board file for a memory module four times, the BoardSim assigns each instance a different board ID.

To lookup the board ID to board file mapping, open the MultiBoard Wizard to the page that contains the board file names and their board IDs.

Use Board IDs to Display Data for a Particular Board

In some dialog boxes, the board ID displays data for a particular board (with data for other boards filtered out). Use the design file list to select a board ID in a dialog box.

Note

You must load a MultiBoard project to see the design file list.

You can then select nets or components by familiar the names and limit the quantity of nets from which to choose at one time. For example, the Select Net by Name dialog box displays only the nets for the board selected by the Design File list.

Other dialog boxes, and the board viewer, embed the board ID into the net or component name.

Board IDs Embedded into Net and Component Names

Depending on the dialog box, BoardSim may add the board ID to the net or component name as a suffix, or embed it into the name. For example, in the board viewer, the board ID is added to the end of every component name (for example, U3_B02). By contrast, in the Pins area of the Digital Oscilloscope Dialog Box and Waveform List area of EZwave embed the board ID into the probe name (for example, U3_B02.12).

Related Topics

Setting Up a Multiple Board Design

Oscilloscope Probes

Oscilloscope probes enable you to see voltage waveforms (all simulators) or current waveforms (HyperSim simulator only) at points in the circuit during simulation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

In the schematic or board viewer, probes look like arrows pointing to the pins to which they attach. The color of the probe indicates the color of the corresponding waveform in the oscilloscope.

Use the Pins spreadsheet in the oscilloscope to view and edit probes.

Differential Pairs

For differential pin pairs, the software attaches a differential probe in addition to the single-ended probe attached to the individual pins, resulting in a total of three probes for the two differential pins. If the software does not automatically recognize differential pins and attach differential probes to them, such as for SPICE models, you can manually specify them.

Automatic Probes on Unconnected SPICE Model Ports

For SPICE models only, the oscilloscope automatically attaches a single-ended probe to all the unconnected (NC) ports of a model, even if they are not connected to the net or schematic. This capability enables you to display waveforms for key signals that would otherwise be hidden from view, such as the following:

- On a SPICE SERDES (serializer/deserializer) receiver model with a special port to access the post-equalized signal, representing the waveform seen by the receiver.
- On a SPICE model that combines the silicon and package behaviors, but you want to probe between the silicon and package. You can modify the SPICE model so that the node(s) between silicon and package are brought out to external port(s).

Probing in Other Locations in a Schematic Design

For a schematic design, you can probe a location without a component, such as between two transmission lines, by adding an IC to the location but not assigning a model to it. An IC with no model has no effect on the circuit.

Note

Probing capability is not available for a board design. For a board design, probes can attach only to component pins, not to trace segments, pads, or vias. If you use the Interactive Simulation Dialog Box to run simulation, you can additionally probe signal vias by enabling Vias in the Simulation Controls Dialog Box.

An alternative method is to add a 10K pull-up or pull-down resistor to the net. Such a large resistor will have little or no effect on your circuit because it is a very small load, but it provides a component pin at which you can probe. Its package parasitics will be present, however, and for very-high-speed signals, you may want to reduce those to minimum values before simulating.

Related Topics

Digital Oscilloscope Dialog Box

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Parametric Sweeps

Use sweeps to automatically vary and simulate design property values over a range that you specify.

In a constrained design, sweeps can help identify a range of design property values that produce acceptable signal-integrity. You can also use sweeps to understand the effects of manufacturing tolerances on signal integrity.

IBIS Model Sweeping Uses Internal Supply Values

When sweeping [Model Selector] values or IBIS buffer models, the internal voltage for the buffer is used instead of external or on-board values.

When you enable the option “When assigning a model to an IC-pin, use a power-supply net connected to the IC” in the Preferences Dialog Box - General Tab, the option is temporarily overridden during sweeps and restored when sweeps finishes.

Sweeps Do Not Support Electrical Coupling Thresholds for a Board Design

For a board design, when you enable crosstalk, use geometric coupling thresholds to run sweep simulations.

When you start sweep simulations with electrical coupling thresholds enabled, the software displays a message describing this requirement.

Sweeps requires a constant set of nets to simulate and using geometric coupling thresholds supports this requirement. By contrast, if you enable electrical coupling thresholds, it is possible for the software to identify different sets of aggressor nets from one sweep simulation to another. For example, imagine sweeping (by decrementing) the dielectric thickness so much that the software finds a new aggressor net on a different metal layer.

For information about how the software automatically identifies the set of aggressor nets to include during crosstalk simulation based on the coupling thresholds you define, see “Finding Nets With Excessive Crosstalk” on page 35.

Unrouted Trace Segment Sweeps

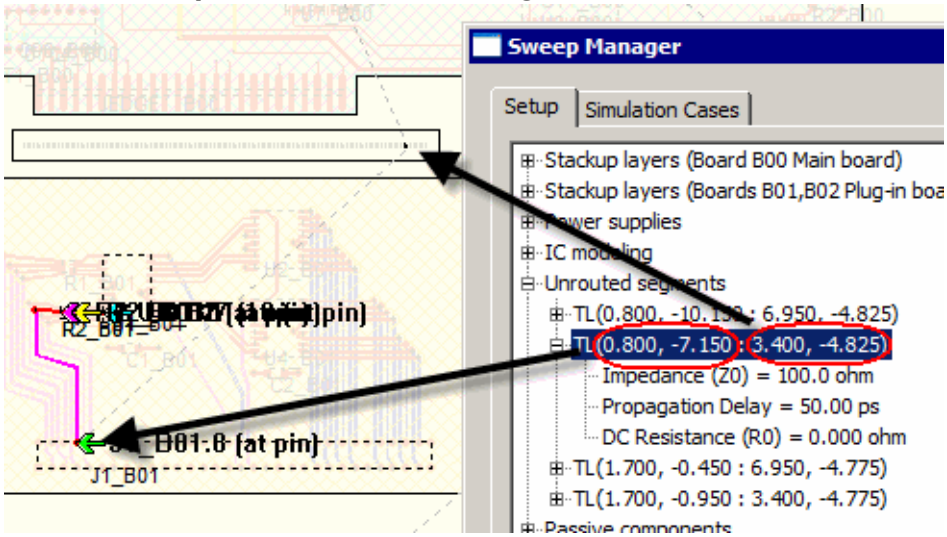
You can sweep unrouted trace segments that represent nets routed with Manhattan routing or “simple” board-to-board connector models in a MultiBoard project.

You cannot sweep routed trace segments unless you first reroute them with Manhattan routing.

The Sweep Manager Dialog Box - Setup Tab displays the coordinates of the end points of the unrouted trace segments in the board viewer. Figure 8-40 shows the end points of the dashed line for the unrouted trace segment representing the “simple” board-to-board connector model.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-40. Swept Unrouted Trace Segment Location in Board Viewer



Related Topics

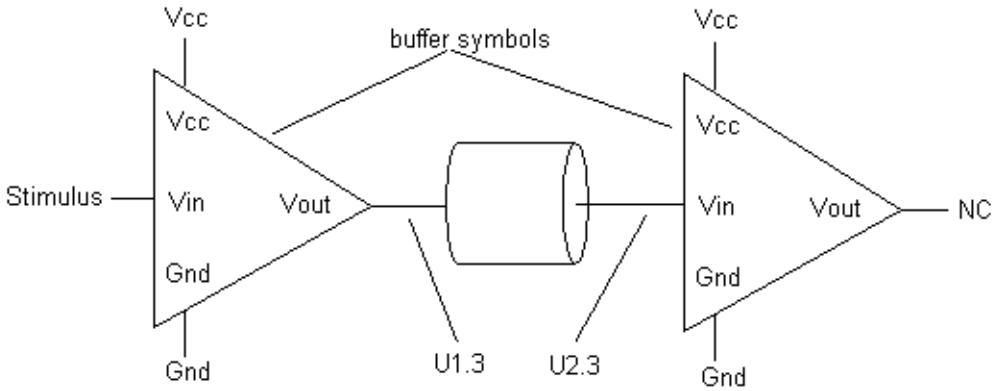
Tips for Running Simulation with Parametric Sweeps

Port-Mapping Examples

When you use the Assign Models dialog box to assign a SPICE or Touchstone model, use the spreadsheet to map ports to circuit connections.

For a Non-Differential SPICE Model

A SPICE model has the following ports: Vin, Vout, Vcc, and Gnd. The circuit consists of a transmission line connecting driver pin U1.3 to receiver pin U2.3. In the following image, Port Names are shown inside the buffer symbols and Circuit Connection names are outside the buffer symbols:



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Configure the spreadsheet for the driver pin to look like this:

Port	Circuit Connection
Vin	Stimulus
Vout	U1.3
Vcc	Vcc
Gnd	Gnd

Configure the spreadsheet for the receiver pin to look like this:

Port	Circuit Connection
Vin	U2.3
Vout	NC
Vcc	Vcc
Gnd	Gnd

For a Differential SPICE Model

A differential SPICE model has the following ports: Vin+, Vin-, Vout+, Vout-, Vcc, and Gnd. The circuit consists of a transmission line connecting driver pin U1.1 to receiver pin U2.1 and another transmission line connecting driver pin U1.2 to receiver pin U2.2.

Configure the spreadsheet for the driver pins to look like this:

Port	Circuit Connection
Vin+	Stimulus
Vin-	Inverted stimulus
Vout+	U1.1
Vout-	U1.2
Vcc	Vcc
Gnd	Gnd

Buffer States - Input Versus Output Hi-Z

Two buffer states—Input and Output Hi-Z— refer to a state in which an IC pin is high-impedance and not actively driving the net to which it is connected:

- Input is available for pins that can only receive or on I/O or bidirectional pins that can act as inputs when the output or driving circuitry is shut off.
- Output Hi-Z is available on pins that can only drive or be turned “off,” but have no receiver-input stage.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you have a data sheet that refers to an IC pin's high-impedance state, this means that the pin is I/O capable, and the driving circuitry is disabled and the pin is in a high-impedance receiving (i.e., "input") state. For these types of pins, select the Input buffer state, not the Output Hi-Z buffer state.

Viewing Series Bus Switch Pin Connectivity

Use the Bus Switch tab of the Assign Models dialog box to view the connections among series bus switch pins.

This information is available when a pin on the selected net is present in a [Series Pin Mapping] keyword in the IBIS model assigned to the reference designator.

See Figure 8-41 on page 571. In the Pins list, when you select a pin connected to a series bus switch, all the other pins in the series bus switch are also highlighted. If the series bus switch contains more than one series pin pair, such as (6, 7) and (6, 11) in Figure 8-41, the Connectivity area in the dialog box displays all of them.

Figure 8-41. Series Bus Switch Connectivity

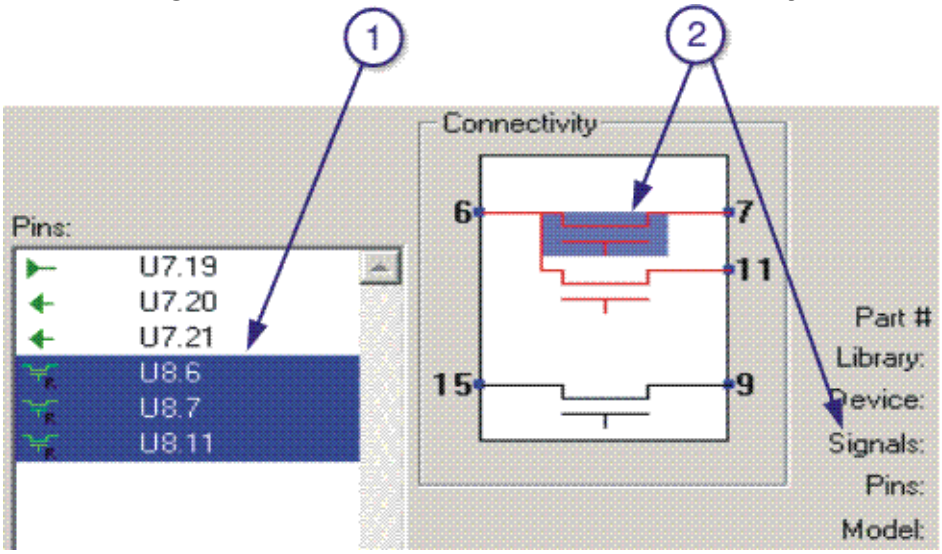


Table 8-17. Series Bus Switch Connectivity

1	Highlighted pins are connected to the series bus switch. Pins 15 and 9 of the example [Series Pin Mapping] keyword (below) do not appear in the Pins list because they are not part of the selected net. Example IBIS model syntax: [Series Pin Mapping] pin_2 model_name function_table_group 6 7 SWITCH1 1 6 11 SWITCH2 2 15 9 SWITCH1 1

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 8-17. Series Bus Switch Connectivity (cont.)

2	Highlighted MOSFET identifies the source of signals, pins, and model information. Click another MOSFET drawn with red lines to display its signals, pins and model information. You cannot display information for MOSFETs drawn with black lines because they do not connect to the selected net.

Related Topics

Assign Models Dialog Box

Pre-Emphasis and DFE Structures

To use this optional feature, you must know the details of how pre-emphasis/DFE is implemented in the driver/receiver, such as the number of taps and their weights.

Many PCB designers will not know this information and some IC vendors consider this information to be proprietary and do not distribute it. However, IC I/O designers will know this information and some IC vendors publish driver/receiver datasheets with this information.

FastEye channel analysis makes the following assumptions:

- Driver/receiver ICs implement pre-emphasis/DFE circuits commonly used by SERDES designs. Driver/receiver ICs implement linear pre-emphasis and non-linear DFE that contains clamping.
- Crosstalk is not taken into account.
- Noise is taken into account, if you specify jitter properties to represent it. See FastEye Channel Analyzer - Add Jitter Page.

Map Tap Weights to Pre-Emphasis and DFE Structures

FastEye channel analysis assumes that drivers/receivers implement circuitry for pre-emphasis/DFE that resemble the structures in the following figures.

FastEye channel analysis supports a large number of pre-taps (pre-emphasis only) and post-taps, not just the number of taps displayed in Figure 8-42 and Figure 8-43.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-42. FastEye Pre-Emphasis Filter - Feed-Forward

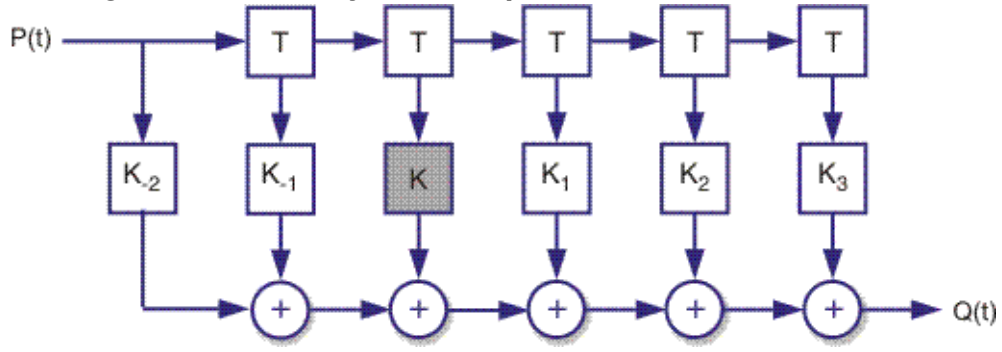


Table 8-18. FastEye Pre-Emphasis Filter - Feed-Forward

Symbol	Description
T	Delay element with duration of one bit interval
K_{-2}, K-1	Pre-taps
K	Main tap
K_1, K2, K_3	Post-taps
+	Addition operator

Figure 8-43. FastEye DFE Filter - Feed-Backward

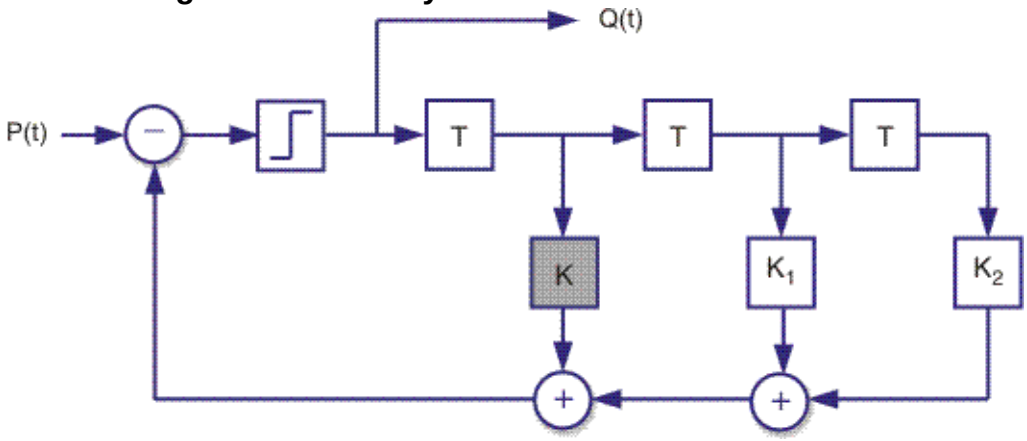


Table 8-19. FastEye DFE Filter - Feed-Backward Contents

Symbol	Description
—	Subtraction operator
⎍	Decision point or slicer. Converts an arbitrary waveform into a two-level waveform, for example, Y(t) = F(x(t)). If x > x0, then Y = 1, else Y = -1. FastEye analysis assumes the threshold is zero (centered waveforms) and the gain of the decision point is 1.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 8-19. FastEye DFE Filter - Feed-Backward Contents (cont.)

Symbol	Description
T	Delay element with duration of one bit interval
K	Main tap
K_1, K2	Post-taps. This filter does not use pre-taps.
+	Addition operator

Related Topics

FastEye Channel Analyzer - Add Pre-Emphasis/DFE/CTLE Page

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Preparing a Board Design File for Translation

In your layout software, prepare your design so that HyperLynx can convert it with minimal translation errors and less post-translation design cleanup.

Topic	Description
Preparing an Accel EDA Design for Translation	To prepare your Accel EDA (Sequoia) board design for translation, set component values and IC names and save your design as an ASCII text file.
Preparing Cadence Allegro Designs for Translation	To prepare your Cadence Allegro board design for translation, set metal shape options, and save your design as an ASCII text file. If HyperLynx and Cadence Allegro and the Extracta utility are installed on your computer, you can convert a native BRD file.
Preparing a Siemens EDA Board Station Layout or Board Station RE Design for Translation	To prepare your Siemens EDA Board Station Layout or Board Station RE design for translation, rename design files to the format HyperLynx requires for translation.
Creating a File Menu Item to Rename Board Station Files for Translation to HyperLynx	If you regularly prepare Siemens EDA Board Station Layout or Board Station RE designs for translation, you can set up a script that you access from the File menu in Layout to copy and rename the files needed for translation.
Preparing Zuken Visula/ CADStar for Windows Designs for Translation	To prepare your Zuken Visula or CADStar for Windows design for translation, create an alphanumeric pin name file if needed, and save your design as a PAF file.
Preparing Zuken CR-3000 Designs for Translation	To prepare your Zuken CR-3000 design for translation, run the ZUKXTRACT extraction script on a UNIX computer and move the files to a Windows computer that HyperLynx can access.
Preparing Zuken CR-5000 Board Designer Designs for Translation	To prepare your Zuken CR-5000 design for translation, create PCF, FTF, and RUF files.

Preparing an Accel EDA Design for Translation

To prepare your Accel EDA (Sequoia) board design for translation, set component values and IC names and save your design as an ASCII text file.

Restriction

You cannot translate designs created with older, pre-Sequoia versions of P-CAD or most old versions of Tango.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

1. In Accel EDA, set the Value attribute for component values and IC names so they are available for translation:
   a. Choose the **Edit > Components** menu item.
   b. In the Components list, select the component whose value you wish to set.
   c. Click **Properties**.
   d. Select the **Pattern** tab and, in the upper left corner, type the desired value in the Value field. For proper translation, type in a number with a scaling factor or scientific notation:

Scaling Factor or Scientific Notation	Definition
M or E6	mega = 1,000,000x
K or k, or E3	kilo = 1,000x
m or E-3	milli = 0.001x
u or U, or E-6	micro = 1e-6x
n or N, or E-9	nano=1e-9x
p or P, or E-12	pico=1e-12x

   e. Click **OK**.
2. Choose the **File > Save As** menu item.
3. In the file type list, select ASCII.
4. Click **OK**. This creates a text file of the design.

Results

Your design is now ready for translation. See Translating a Board Design.

Preparing Cadence Allegro Designs for Translation

To prepare your Cadence Allegro board design for translation, set metal shape options, and save your design as an ASCII text file. If HyperLynx and Cadence Allegro and the Extracta utility are installed on your computer, you can convert a native BRD file.

Prerequisites

- You have installed and licensed (licensing required for version 17.4 and newer) the Cadence Extracta ASCII-extraction utility on the same computer as Allegro.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

It is not sufficient to copy the Extracta executable file to the computer. Cadence Allegro and Allegro Physical Viewer (also known as Cadence Viewer Plus), and OrCAD Lite include Extracta. Allegro FREE Physical Viewer does not include Extracta.

Procedure

1. If you use Allegro 15.0 or newer, and your design contains dynamic metal shapes, which automatically create and adjust voids when the design changes, skip to step 3.
2. If your Allegro design contains static metal shapes, update the void data prior to creating an ASCII version of the design.
   a. Create a copy of the design to use for analysis purposes. (Optional)
   b. In Allegro, perform the void operation on each static metal shape. For example, if the void clearance settings in Allegro are correct, the void operation may be performed with the **Shape > Manual Void > Element > <select_a_static_metal_shape> > sequence** menu item.

   Proper clearances now exist around padstacks and between metal shapes. Maximum data fidelity is especially needed for accurate power-integrity simulations, such as decoupling capacitor analysis.
3. If HyperLynx and Allegro are installed on the same computer, you can translate the design using the BRD file. See Translating a Board Design.
4. If HyperLynx and Allegro are installed on different computers:
   a. Copy *control_hyp.txt* and *control_hyp2.txt* from the HyperLynx computer to the Allegro computer in the folder containing the design.

      These files are in the folder containing the HyperLynx application file *bsw.exe* (Windows) or *bsw* (Linux). For example *C:\MentorGraphics\<release>\SDD_HOME\hyperlynx64*.
   b. Open a command window on the Allegro computer, navigate to the design folder, and type the following:

      extracta <design>.brd control_hyp.txt <design>.a_b <design>_COMPONENT.txt <design>_COMPONENT_PIN.txt <design>_COMPOSITE_PAD.txt <design>_CONNECTIVITY.txt <design>_FULL_GEOMETRY.txt <design>_LAYER.txt

      Caution

      Be sure to name the output files exactly as described because the HyperLynx translator looks for only those names.

      The command window displays the extraction progress and completion message. The extractor creates log and error files in the current folder.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

c. In the same command window and folder, type the following:

extracta <design>.brd control_hyp2.txt <design>_NET.txt <design>_RAT_PIN.txt <design>_SYMBOL.txt

Note

This step uses *control_hyp2.txt* instead of *control_hyp.txt*.

The command window displays the extraction progress and completion message. The extractor creates log and error files in the current folder.

d. Copy the files listed in the table to the HyperLynx computer in a location that HyperLynx can access.

File Name <fname> is the basename of the Allegro board file	File Contents
<fname>.a_b	Stackup and outline information for the board
<fname>_COMPONENT.txt	Component information
<fname>_COMPONENT_PIN.txt	Component pin information
<fname>_COMPOSITE_PAD.txt	Pad data from symbol pins and vias in the design
<fname>_CONNECTIVITY.txt	Connectivity information
<fname>_FULL_GEOMETRY.txt	Geometry and pad data
<fname>_LAYER.txt	Physical layer data
<fname>_NET.txt	Net data
<fname>_RAT_PIN.txt	Component pin and ratsnesting data for a net
<fname>_SYMBOL.txt	Symbol data

Results

Your design is now ready for translation. See Translating a Board Design.

Preparing a Siemens EDA Board Station Layout or Board Station RE Design for Translation

To prepare your Siemens EDA Board Station Layout or Board Station RE design for translation, rename design files to the format HyperLynx requires for translation.

If you have assigned models to components by editing component properties in Board Station, when you translate your design, an automapping model assignment file (REF) is created with the board design file (HYP).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you translate Board Station files for HyperLynx often, you can create a file menu item in Board Station that copies and renames the files needed for translation. See Creating a File Menu Item to Rename Board Station Files for Translation to HyperLynx.

Procedure

Copy the latest version of these files from your Board Station design folders to a new folder, and rename them.

Copy this file...	From...	Rename to...
geoms_ascii	*<root design folder>*	*<boardname>.PRT*
traces.traces_<#>	*<root design folder>/PCB*	*<boardname>.WIR*
nets.nets_<#>	*<root design folder>/PCB*	*<boardname>.NET*
comps.comps_<#>	*<root design folder>/PCB*	*<boardname>.CMP*
tech.tech_<#>	*<root design folder>/PCB*	*<boardname>.TEC*

Results

Your design is now ready for translation. See Translating a Board Design.

Creating a File Menu Item to Rename Board Station Files for Translation to HyperLynx

If you regularly prepare Siemens EDA Board Station Layout or Board Station RE designs for translation, you can set up a script that you access from the File menu in Layout to copy and rename the files needed for translation.

Procedure

1. Create a startup file.

   a. Create a text file and paste the following text into the file. Be sure the text is on a single line in the file.

   ```
   $insert_menu_item($menu_text_item("Create _Hyperlynx Files",
   "create_hyperlynx_files()"), , "file_pulldown", @last);
   ```

   b. Save the file as *layout.startup* in ASCII format, in a folder named *startup*, if it does not already exist. Create the folder in one of the following locations.

If your installation is...	**Create the startup folder in this location...**
Site specific	*$HOME\shared\etc\cust\startup\layout.startup*
Workstation specific	*$HOME\etc\cust\startup\layout.startup*

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If your installation is...	Create the startup folder in this location...
User specific	*$HOME\mgc\startup\layout.startup*

Note

$HOME is *C:\MentorGraphics* by default.

2. Create the script file.

   a. Create a text file named *lay_area.ample* script file in the *$HOME\mgc\userware\layout* folder if it does not already exist.

   b. Set the environment variable AMPLE_PATH to point to *$HOME\mgc\userware*.

   c. Paste the following text into the *lay_area.ample* file and save it in ASCII format:

```
function create_hyperlynx_files()
{
local hyperfile_dir = $strcat($get_design_name(),"/
hyperlynx_files");
local board_name = $strcat("/", $get_board_name());
local hyperfile_name = $strcat(hyperfile_dir, board_name);
$writeln($strcat("writing hyperlynx files to ", hyperfile_dir));
$system($strcat("mkdir ", hyperfile_dir), @true);
$save_ascii_geometries($strcat(hyperfile_name, ".prt"), [""],
@replace, @all, @nodirectory, @version);
$save_traces($strcat(hyperfile_name, ".wir"), @replace,
@pathname, @noguide, @partial_routes_also);
$save_nets($strcat(hyperfile_name, ".net"), @replace,
@pathname);
$save_components($strcat(hyperfile_name, ".cmp"), @replace,
@pathname);
$save_technology($strcat(hyperfile_name, ".tec"), @replace,
@pathname);
}
```

3. To run the script, restart the software, load your design in Layout, and choose the **File > Create HyperLynx Files** menu item.

   The script copies and renames the files that HyperLynx requires for translation to the folder *$HOME\hyperlynx_files*.

Results

Your design is now ready for translation. See Translating a Board Design.

Preparing Zuken Visula/CADStar for Windows Designs for Translation

To prepare your Zuken Visula or CADStar for Windows design for translation, create an alphanumeric pin name file if needed, and save your design as a PAF file.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

1. If your Zuken CADStar for Windows design contains alphanumeric pin names, create a *<design>.cpa* (CADSTAR PCB Archive) file to allow HyperLynx to extract the pin names when translating the design.

   Note

   This capability is unavailable for Zuken Visula.

   a. In Zuken CADSTAR, choose the **File > File Export** menu item.

   b. In the Export to File dialog box, in the Format list, select PCB Archive.

   c. Save the CPA file to your design folder.

   d. Click **OK**.

2. Save your design as a PAF file.

   Note

   In Zuken Visula, generate the PAF file from a UNIX workstation, either directly or via a VPN session from your Windows computer.

Results

The design is converted with the following limitations.

- Padforms are derived from the PADASSIGN section, not from the padstack section. Therefore, PADFORMS are the same, regardless of the placement side of the shape.
- All pins are converted to the original defined pin numbers, not the optional pin names.
- Only 90-degree pad rotation is allowed.
- Diamond padforms are treated as round.
- Bullet padforms are treated as “finger.”
- Dimension entities are ignored.
- Taper line ends and pads are ignored.
- Copper areas and voids are not completely supported.
- Closed polylines containing arcs are not filled.

Your design is now ready for translation. See Translating a Board Design.

Preparing Zuken CR-3000 Designs for Translation

To prepare your Zuken CR-3000 design for translation, run the ZUKXTRACT extraction script on a UNIX computer and move the files to a Windows computer that HyperLynx can access.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The UNIX shell script ZUKXTRACT calls a series of CR-3000 utility programs to convert your board design to an ASCII text file.

Procedure

1. Copy the ZUKXTRACT UNIX shell script to a UNIX workstation that has access to your design files and the CR-3000 utility programs. The script is installed in the same directory as HyperLynx on your Windows computer.
2. The ZUKXTRACT script is designed to run in an English-language environment. If you are running in a non-English environment, add the following two lines to the beginning of the script file:

   ```
   setenv ZLANG english
   setenv ZNLSLANG english
   ```

3. On a UNIX workstation with access to the CR-3000 utility programs, run the shell script ZUKXTRACT.

   The script generates text files with the following extensions: BSF, UDF, MDF, WDF, WSF, CCF.

4. Move the files to a Windows computer that HyperLynx can access.

Results

Your design is now ready for translation. See Translating a Board Design.

Preparing Zuken CR-5000 Board Designer Designs for Translation

To prepare your Zuken CR-5000 design for translation, create PCF, FTF, and RUF files.

Restriction

The software cannot translate Zuken CR-5000 PWS designs.

Procedure

In CR-5000 Board Designer, generate these files:

File Name	File Contents
<board_name>.pcf	PCB data
<board_name>.ftf	Footprint data
<board_name>.ruf	Stackup layer data, including thickness, dielectric constant, and loss tangent

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Results

Your design is now ready for translation. See Translating a Board Design.

Quick Analysis Crosstalk Estimation

Quick Analysis crosstalk estimation results provided by general batch simulation are conservative to ensure that the software does not overlook any nets in your design that might experience significant crosstalk.

The algorithm for estimating crosstalk is based on the weak coupling theory of crosstalk, which yields a set of closed-form prediction equations that can run quickly. The resulting capability is sufficient to make reasonable guesses as to how much crosstalk each possible aggressor net can generate on a victim net.

For nets with clean linear routing, which run parallel to each other for a medium distance, the aggressor-finding algorithm is more accurate. As the routing topology becomes more complex, then the results become more approximate.

To obtain accurate crosstalk results, run signal integrity simulation with the Digital Oscilloscope or EZwave waveform viewer, or detailed signal integrity and crosstalk simulation in general batch simulation.

Setting Up the Software

Edit design-independent options to specify the locations of IC models and design simulation files, license check in/out behaviors, measurement units, and simulation and appearance preferences.

You can also transfer software settings from a previous HyperLynx installation. See Transferring HyperLynx Settings.

Procedure

Change the software setup according to your preferences:

If you want to...	Do this...
Change paths to design, models, stimulus, and other directories	1. Choose the **Setup > Options > Directories** menu item. The Set Directories dialog box opens. 2. Specify directory paths as needed. 3. Click **OK**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to...	Do this...
Change simulation and appearance preferences	1. Choose the **Setup > Options > General** menu item. The Preferences dialog box opens. 2. Select a tab and set options as needed. 3. Click **OK**.
Edit reference designator mappings to ensure that the software recognizes components correctly, and components appear in model assignment lists correctly.	1. Choose the **Setup > Options > Reference Designator Mappings** menu item. The Edit Reference Designator Mappings dialog box opens. 2. Select an existing mapping and edit, or create a new mapping. 3. Click **Apply/Add**.
Create differential pairing rules to enable the software to recognize differential pairs automatically when you load a board.	1. Choose the **Setup > Differential Pairs** menu item. The Differential Pair Net Names dialog box opens. 2. Create a pairing rule that matches paired net names by typing characters and wildcards, then click **Add**. 3. Click **OK**.
Set measurement units	1. Choose the **Setup > Options > Units** menu item. The Units dialog box opens. 2. Select measurement and metal thickness units. 3. Click **OK**.
Set the via modeling method to account for inductance. **Note:** The software represents vias as simple capacitors when this option is not enabled.	1. Acquire the Via Models license and open a board design. 2. Choose the **Setup > Via Simulation Method** menu item. The Select Method of Simulating Vias dialog box opens. 3. Check “Include via L and C”, then check additional options as needed. 4. In the “Via-modeling method” area, select a method for modeling via inductance and capacitance. 5. Click **OK**. **Note:** Via simulation options are ignored when you enable SI/PI co-simulation.
Omit signal via pads that do not connect to traces from simulation.	Choose the **Setup > Remove Non-Functional Pads** menu item. **Note:** Pads on the top and bottom stackup layers are not omitted from simulation, since they are required for the plating process during PCB manufacture.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to...	Do this...
Check license status	1. Close all design files. 2. Choose the **Setup > Options > License Status** menu item. The License Status dialog box opens and displays license information.
Select licenses and other licensing options	1. Close all design files. 2. Choose the **Setup > Options > License Checkout and Checkin** menu item. The HyperLynx Licensing dialog box opens. 3. Select licenses and other licensing options. 4. Click **OK**.

Results

You are now ready to open a design. See Opening a Design.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Signal-Integrity Net Constraints

You can specify the following constraints before you run General Batch simulation or signal integrity simulation with measurements in EZwave.

Topic	Description
Static Overshoot High	Specifies by how much voltage the rising signal transition can go above the high rail voltage for the receiver.
Static Overshoot Low	Specifies by how much voltage the falling signal transition can go below the low rail voltage for the receiver.
Dynamic Overshoot High	Specifies by how much voltage the rising signal transition can go above the high rail voltage for the receiver.
Dynamic Overshoot Low	Specifies by how much voltage the falling signal transition can go below the low rail voltage for the receiver.
Dynamic Overshoot Time	For a rising-edge transition, specifies the maximum amount of time that the waveform can be above the voltage specified by Static Overshoot High. For a falling-edge transition, specifies the maximum amount of time that the waveform can be below the voltage specified by Static Overshoot Low.
Max. Rise SI Overshoot	Specifies by how much voltage the rising signal transition can go above the final DC voltage for the receiver.
Max. Fall SI Overshoot	Specifies by how much voltage the falling signal transition can go below the final DC voltage for the receiver.
Ringback Margin High	Specifies how far the rising waveform is allowed to “fall back” or “rebound” after first passing through the receiver logic high timing threshold.
Ringback Margin Low	Specifies how far the falling waveform is allowed to “rise back” or “rebound” after first passing through the receiver logic low timing threshold.
Ringback Delay	Specifies how long to delay the ringback measurement, starting from the first time the waveform crosses the logic threshold for the receiver.
Max Delay	Specifies the maximum acceptable delay to any receiver on the net for both of the following conditions.
Min Delay	Specifies the minimum acceptable delay to any receiver on the net for both of the following conditions.
Crosstalk	Specifies the maximum acceptable amount of voltage, positive or negative, that can be induced on the victim net by signal switching on aggressor nets.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
Relationship Between Crosstalk Constraint and Interactive Coupling Threshold	When you interactively assign IC models, set the following options to the same value.

Static Overshoot High

Specifies by how much voltage the rising signal transition can go above the high rail voltage for the receiver.

The static overshoot high limit = maximum acceptable static overshoot voltage - high rail voltage

For example, the static overshoot high limit is 100 mV when the maximum acceptable static overshoot voltage is 1.6 V and the high rail voltage is 1.5 V:

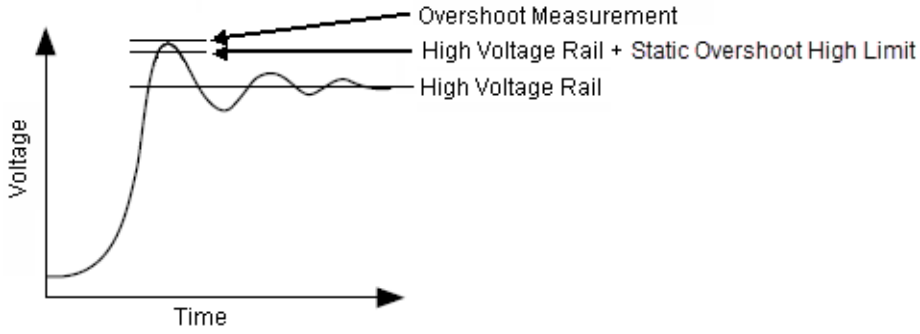
100 mV = 1.6 V - 1.5 V

Note

When you specify both static and dynamic overshoot limits, the software uses the dynamic limit to determine overshoot pass/fail status. For example, if the signal does not exceed the dynamic overshoot limits, but does exceed the static limit, the overall result is “pass”.

The following waveform exceeds the overshoot limit.

Figure 8-44. Static Overshoot High - Fail



This constraint applies to all receivers on the net.

For differential pairs, measurements are made on single-ended waveforms for the positive and negative pins.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Static Overshoot Low

Specifies by how much voltage the falling signal transition can go below the low rail voltage for the receiver.

static overshoot low limit = low rail voltage - minimum acceptable static overshoot voltage

For example, the overshoot limit is 100 mV when the low rail voltage is 0 V and the minimum acceptable static overshoot voltage is -100 mV:

100 mV = 0 V - (-100 mV)

Note

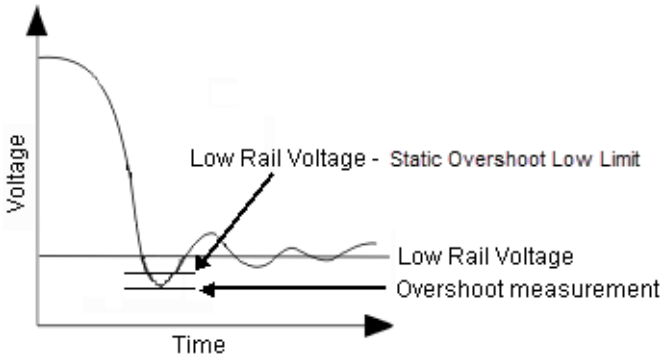
Falling overshoot is sometimes called undershoot.

Note

When you specify both static and dynamic overshoot limits, the software uses the dynamic limit to determine overshoot pass/fail status. For example, if the signal does not exceed the dynamic overshoot limits, but does exceed the static limit, the overall result is “pass”.

The following waveform fails the static overshoot low limit.

Figure 8-45. Static Overshoot Low - Fail



This constraint applies to all receivers on the net.

For differential pairs, measurements are made on single-ended waveforms for the positive and negative pins.

Dynamic Overshoot High

Specifies by how much voltage the rising signal transition can go above the high rail voltage for the receiver.

dynamic overshoot high limit = maximum acceptable dynamic voltage - high rail voltage

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Restriction

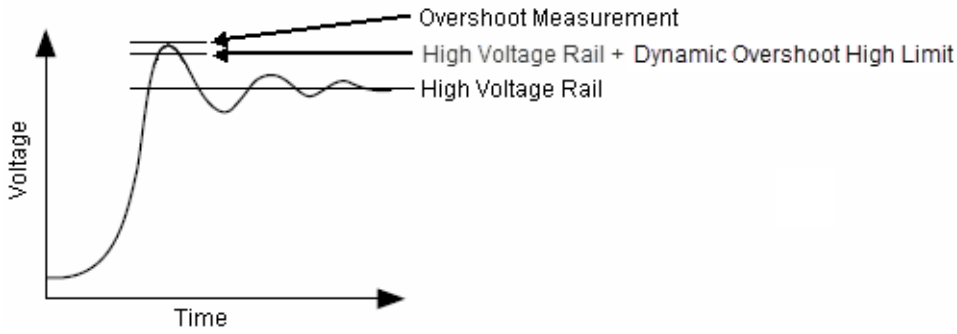
This limit cannot be less than Static Overshoot High.

For example, the dynamic overshoot high limit is 400 mV when the maximum dynamic overshoot voltage is 1.9 V and the high rail voltage is 1.5 V:

400 mV = 1.9 V - 1.5 V

The following shows a waveform that fails this constraint.

Figure 8-46. Dynamic Overshoot High - Fail



This constraint applies to all receivers on the net.

For differential pairs, measurements are made on single-ended waveforms for the positive and negative pins.

Dynamic Overshoot Low

Specifies by how much voltage the falling signal transition can go below the low rail voltage for the receiver.

dynamic overshoot low limit = low rail voltage - minimum acceptable dynamic overshoot voltage

For example, the dynamic overshoot low limit is 400 mV when the low rail voltage is 0 V and the minimum acceptable dynamic overshoot voltage is -400 mV:

400 mV = 0 V - (-400 mV)

Restriction

This limit cannot be less than Static Overshoot Low.

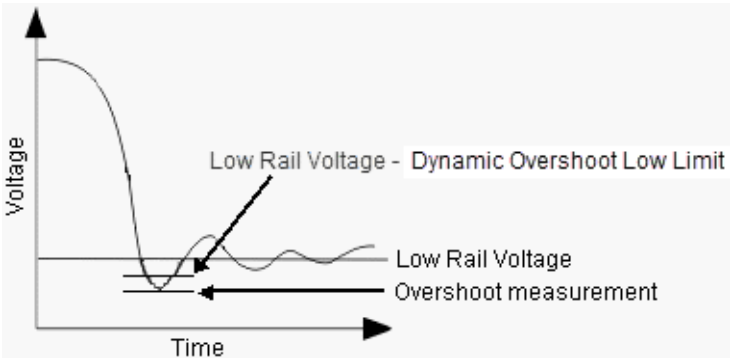
Note

Falling overshoot is sometimes called undershoot.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The following shows a waveform that fails this constraint.

Figure 8-47. Dynamic Overshoot Low - Fail



This constraint apples to all receivers on the net.

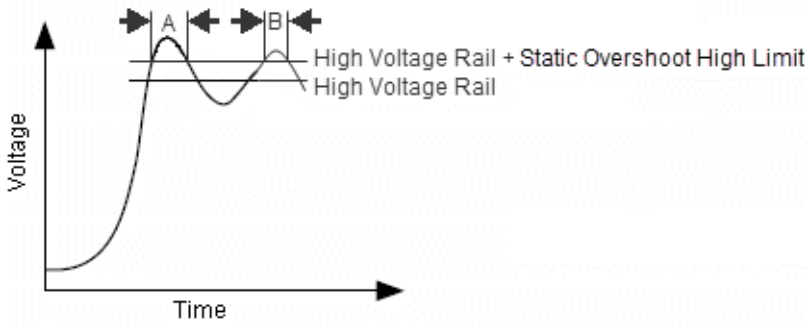
For differential pairs, measurements are made on single-ended waveforms for the positive and negative pins.

Dynamic Overshoot Time

For a rising-edge transition, specifies the maximum amount of time that the waveform can be above the voltage specified by Static Overshoot High. For a falling-edge transition, specifies the maximum amount of time that the waveform can be below the voltage specified by Static Overshoot Low.

The following shows a rising waveform that twice goes above the voltage specified by Static Overshoot High. The spreadsheet reports measurement A because it is larger than measurement B.

Figure 8-48. Dynamic Overshoot Time - Example Measurements



This constraint applies to all receivers on the net.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Max. Rise SI Overshoot

Specifies by how much voltage the rising signal transition can go above the final DC voltage for the receiver.

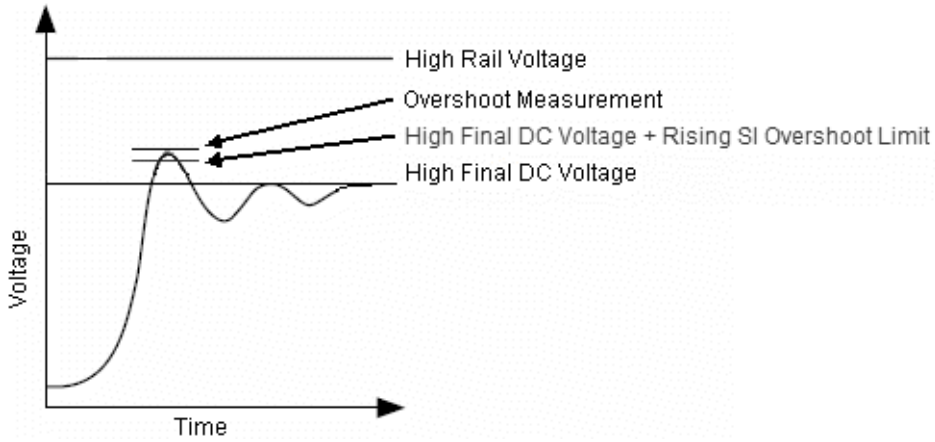
rising SI overshoot limit = maximum SI overshoot voltage - high final DC voltage

For example, the rising SI overshoot limit is 100 mV when the maximum SI overshoot voltage is 1 V and the high final DC voltage is 900 mV:

100 mV = 1 V - 900 mV

The following shows a waveform that fails this constraint.

Figure 8-49. Max. Rise SI Overshoot



For devices which run from a fairly wide-spaced set of power-supply voltages but swing between a smaller set of high/low voltages, this measurement can detect signal-quality problems, such as ringing, that are missed by Max. Rise Static Rail Overshoot.

This constraint applies to all receivers on the net.

For differential pairs, measurements are made on the differential waveform for the positive and negative pins.

Max. Fall SI Overshoot

Specifies by how much voltage the falling signal transition can go below the final DC voltage for the receiver.

falling SI overshoot limit = minimum SI overshoot voltage - low final DC voltage

Note

Falling overshoot is sometimes called undershoot.

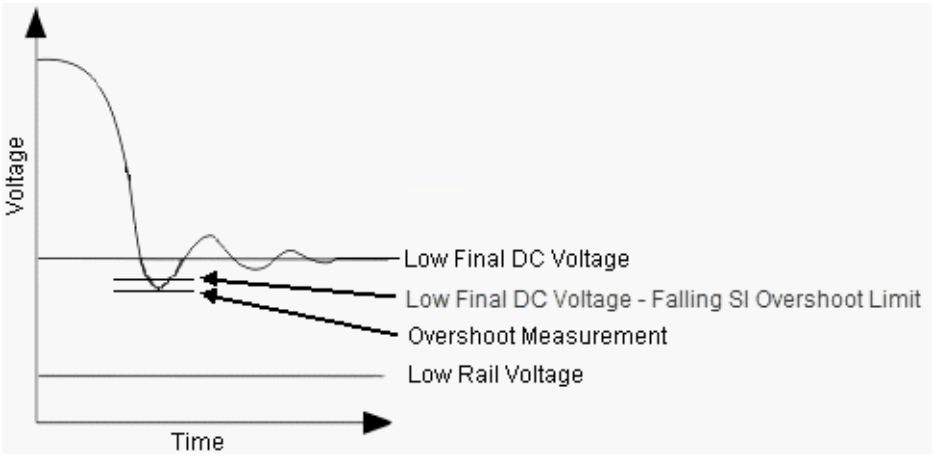
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

For example, the falling SI overshoot limit is 100 mV when the minimum SI overshoot voltage is 600 mV and the low final DC voltage is 500 mV:

100 mV = 600 mV - 500 mV

The following shows a waveform that fails this constraint.

Figure 8-50. Max. Fall SI Overshoot



For devices which run from a fairly wide-spaced set of power-supply voltages but swing between a smaller set of high/low voltages, this measurement can detect signal-quality problems, such as ringing, that are missed by Static Overshoot Low.

This constraint applies to all receivers on the net.

For differential pairs, measurements are made on the differential waveform for the positive and negative pins.

Ringback Margin High

Specifies how far the rising waveform is allowed to “fall back” or “rebound” after first passing through the receiver logic high timing threshold.

ringback margin high limit = minimum allowable rising ringback voltage - logic high timing threshold

For example, the ringback margin high limit is 100 mV when the rising waveform is allowed to fall back to 2.1 V and the logic high timing threshold is 2.0 V:

100 mV = 2.1 V - 2.0 V

The logic high timing threshold comes from the [Receiver Thresholds] keyword and the Vth and Vinh_dc subparameters. When this keyword is unavailable, the logic high timing threshold comes from the [Model] or [Model Spec] keyword and the Vinh subparameter.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-51. Ringback Margin High - Using [Receiver Thresholds] Keyword

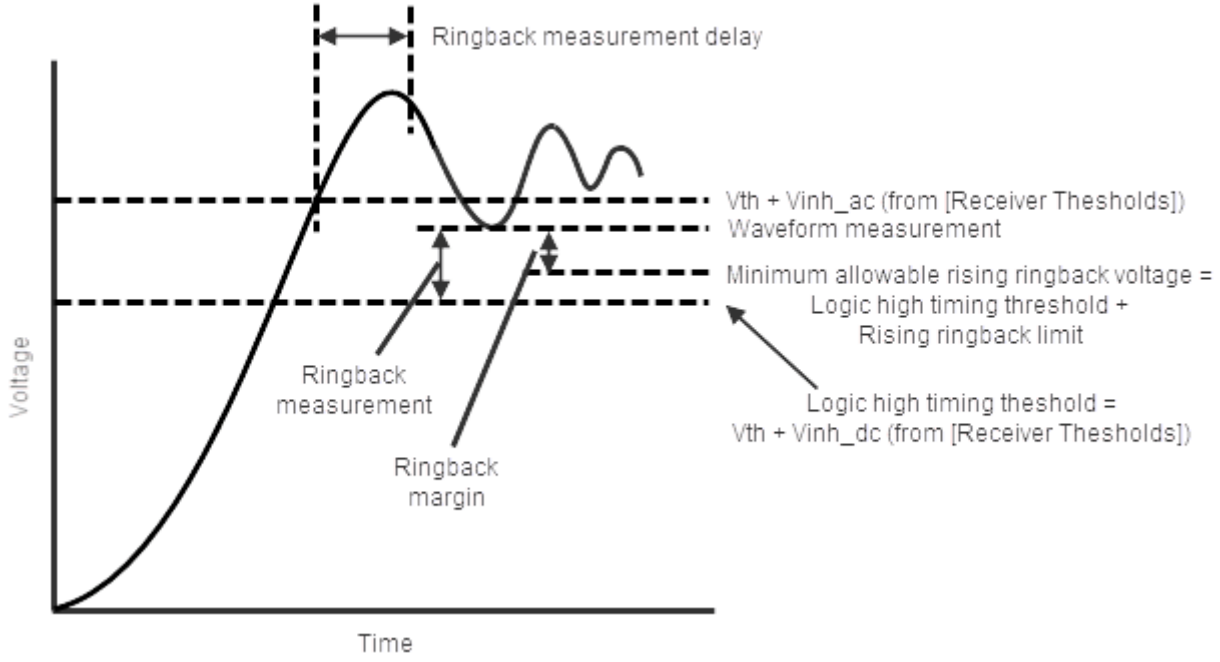
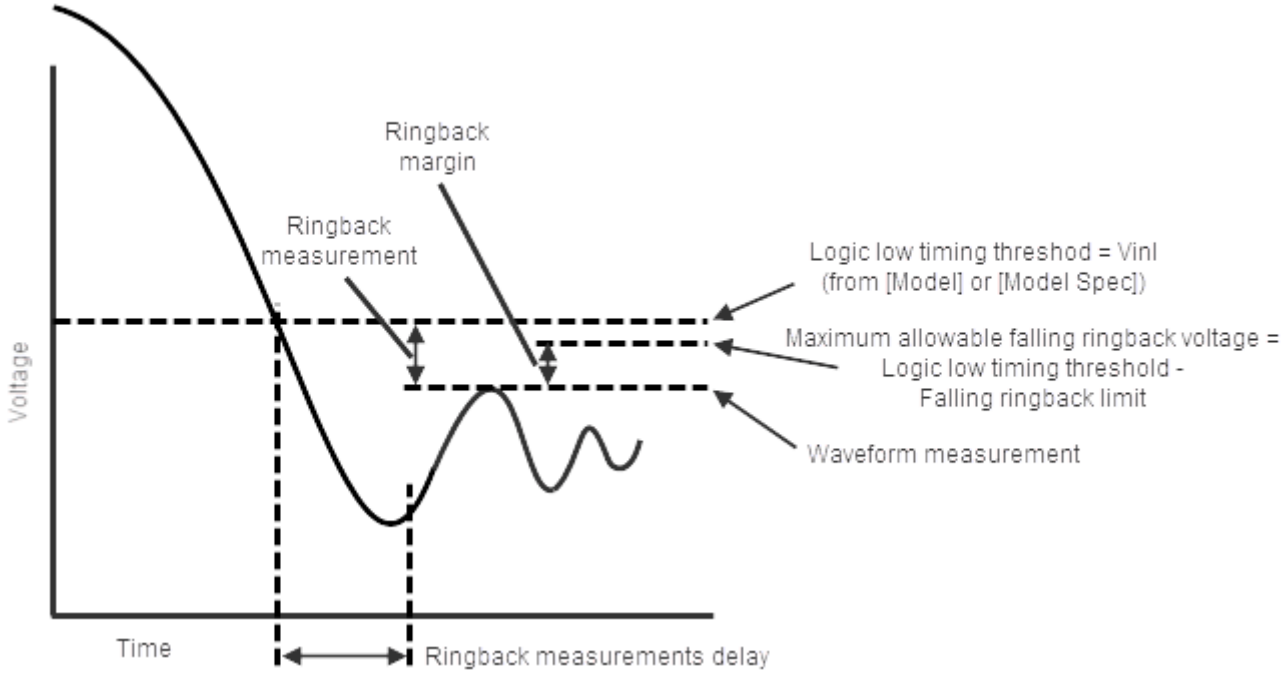


Figure 8-52. Ringback Margin High- Using [Model] or [Model Spec] Keyword

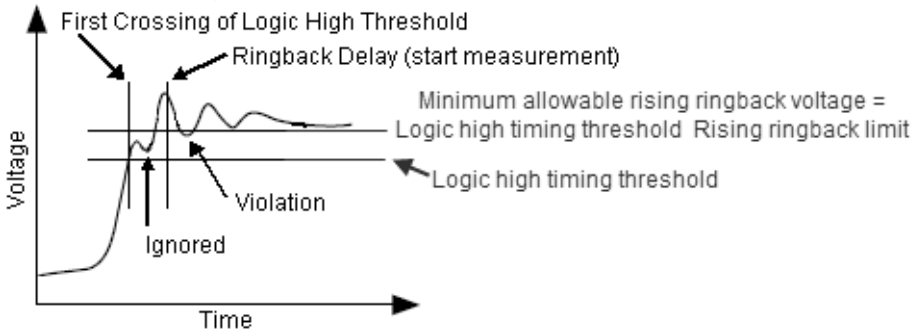


Tiny non-monotonicities located soon after the waveform first crosses the receiver logic high timing threshold can cause misleading measurements. Specify a Ringback Delay constraint value to delay the start of the measurement until some time after the first timing threshold crossing. The following shows the application of Ringback Delay. The non-monotonicity is not reported as a rising ringback failure because it is located between the first crossing of the logic

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

high threshold and the ringback delay. A constraint failure/violation occurs the second time the waveform falls below the minimum rising ringback voltage because it happens after the ringback delay has ended.

Figure 8-53. Ringback Margin High - Non-Monotonicity



Excessive ringback can cause unwanted switching at the receiver, because the waveform passes through the timing threshold more than once.

This constraint applies to all receivers on the net.

For differential pairs, measurements are made on the differential waveform for the positive and negative pins.

Ringback Margin Low

Specifies how far the falling waveform is allowed to “rise back” or “rebound” after first passing through the receiver logic low timing threshold.

> falling ringback limit = logic low timing threshold - maximum allowable falling ringback voltage

For example, the falling ringback limit is 100 mV when the logic low timing threshold is 0.8 V and the falling waveform is allowed to rise back to 0.7 V:

> 100 mV = 0.8 V - 0.7 V

The “logic low timing threshold” comes from the [Receiver Thresholds] keyword and the Vth and Vinl_dc subparameters. When this keyword is unavailable, the “logic low timing threshold” comes from the [Model] or [Model Spec] keyword and the Vinl subparameter.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-54. Ringback Margin Low- Using [Receiver Thresholds] Keyword

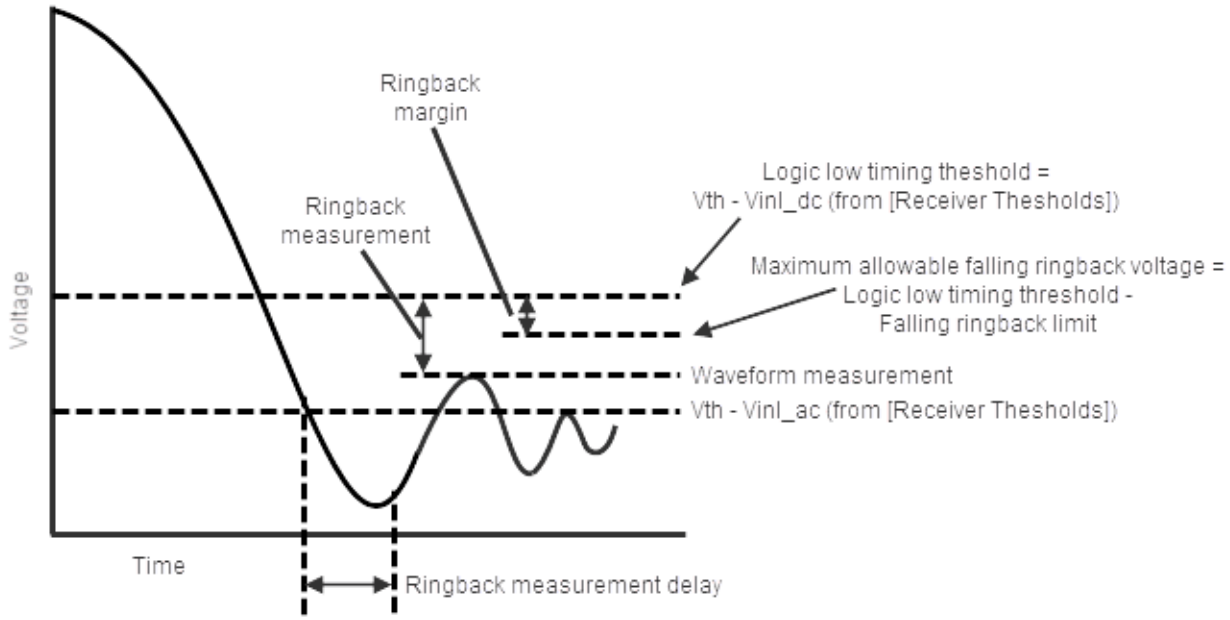
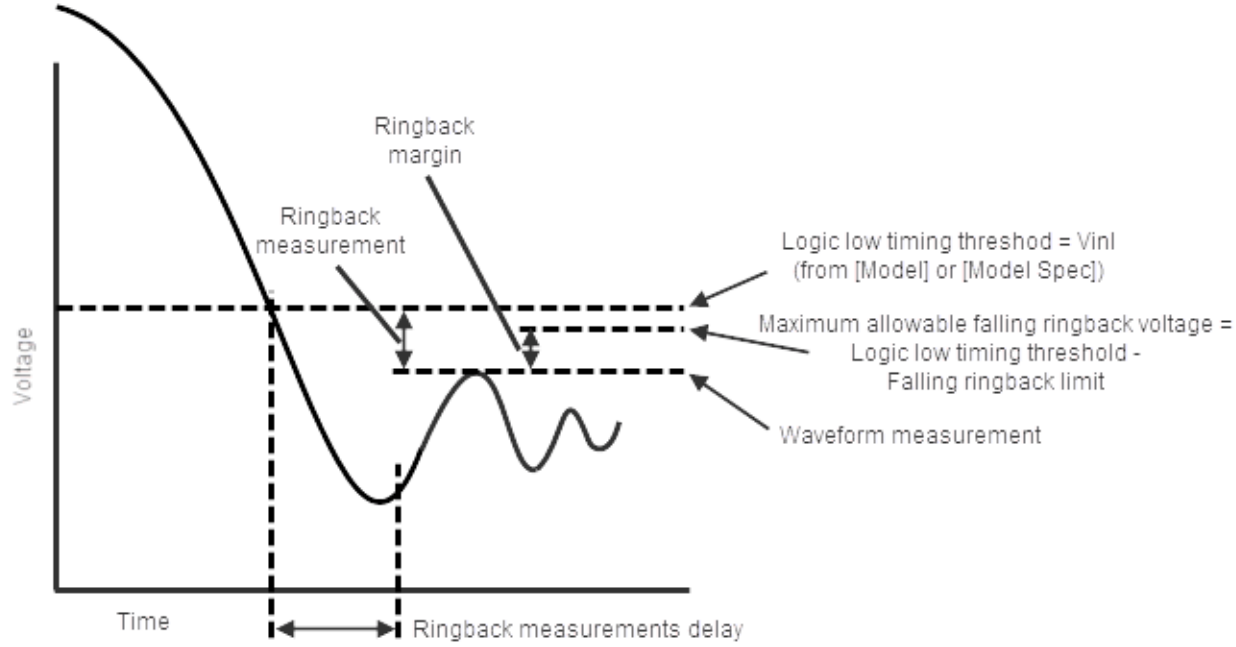


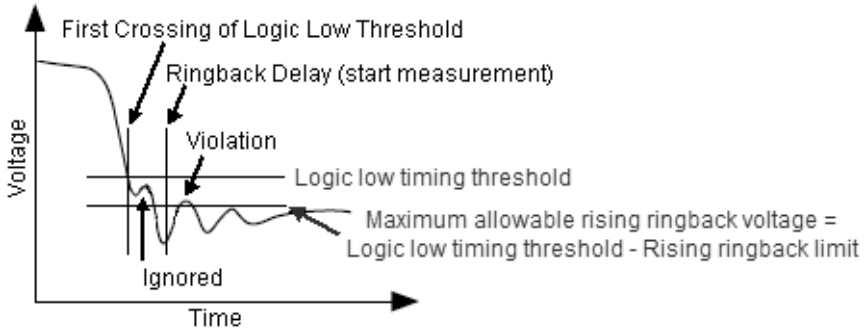
Figure 8-55. Ringback Margin Low- Using [Model] or [Model Spec] Keyword



Tiny non-monotonicities located soon after the waveform first crosses the receiver logic low timing threshold can cause misleading measurements. Use Ringback Delay to delay the start of the measurement until some time after the first timing threshold crossing. The following shows the application of Ringback Delay. The non-monotonicity is not reported as a falling ringback failure because it is located between the first crossing of the logic low threshold and the ringback delay. A constraint failure/violation occurs the second time the waveform rises above the minimum falling ringback voltage because it happens after the ringback delay has ended.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-56. Ringback Margin Low



Excessive ringback can cause unwanted switching at the receiver, because the waveform passes through the timing threshold more than once.

This constraint applies to all receivers on the net.

For differential pairs, measurements are made on the differential waveform for the positive and negative pins.

Ringback Delay

Specifies how long to delay the ringback measurement, starting from the first time the waveform crosses the logic threshold for the receiver.

See Figure 8-53 and Figure 8-56.

This constraint applies to all receivers on the net.

This constraint has no effect if you specify NA for both Ringback Margin High and Ringback Margin Low.

Max Delay

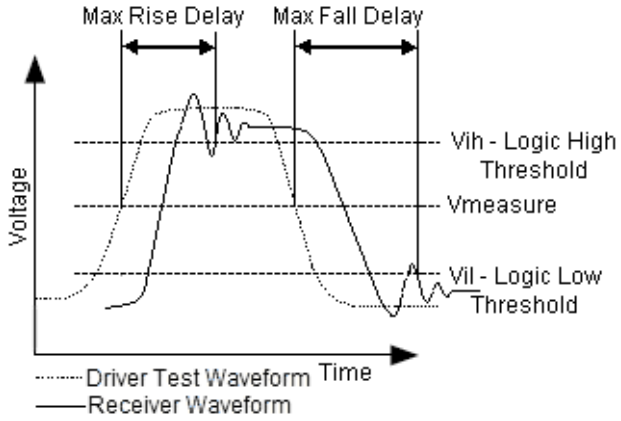
Specifies the maximum acceptable delay to any receiver on the net for both of the following conditions.

- Rising waveform — The measurement begins when the driver test waveform crosses Vmeasure and ends when the receiver waveform crosses the logic high threshold for the final time.
- Falling waveform — The measurement begins when the driver test waveform crosses Vmeasure and ends when the receiver waveform crosses logic low threshold for the final time.

The following shows the receiver waveform crossing each logic threshold twice. Measuring delays at the final crossing of the logic thresholds provides the most pessimistic delay values.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-57. Max Delay



For driver ICs, Vmeasure specifies the voltage at which the driver has switched to the other logic level.

For receiver ICs, logic high and logic low thresholds specify the lowest and highest voltage at which the receiver recognizes a state change.

If you run batch simulation with all three IC model corners, all delays are calculated from the smallest driver switching times. This provides the most conservative results.

For differential pairs, measurements are made on the differential waveform for the positive and negative pins.

Min Delay

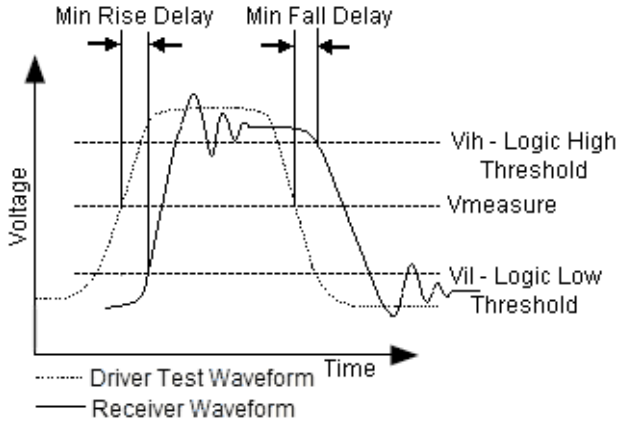
Specifies the minimum acceptable delay to any receiver on the net for both of the following conditions.

- Rising waveform—The measurement begins when the driver test waveform crosses Vmeasure and ends when the receiver waveform crosses the logic low threshold for the first time.
- Falling waveform—The measurement begins when the driver test waveform crosses Vmeasure and ends when the receiver waveform crosses logic high threshold for the first time.

The following shows the receiver waveform crossing each logic threshold once.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-58. Min. Delay



For driver ICs, Vmeasure specifies the voltage at which the driver is considered to be switched.

For receiver ICs, logic high and logic low thresholds specify the lowest and highest voltage at which the receiver recognizes a state change.

If you run batch simulation with all three IC model corners, all delays are calculated from the largest driver switching times. This provides the most conservative results.

For differential pairs, measurements are made on the differential waveform for the positive and negative pins.

Crosstalk

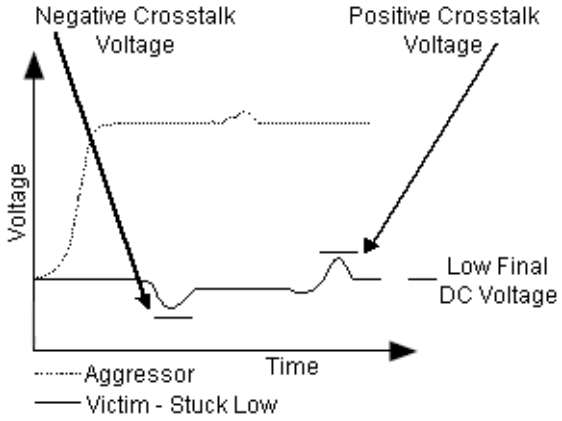
Specifies the maximum acceptable amount of voltage, positive or negative, that can be induced on the victim net by signal switching on aggressor nets.

For the selected victim net and its associated nets only, not its aggressor nets, find the maximum peak voltage excursion away from the DC voltage for the net, positive or negative, at any receiver IC. The magnitude, that is absolute value, of this excursion is the maximum crosstalk. When both stuck high and stuck low simulations are run, use the maximum crosstalk observed in either simulation.

The following shows a rising waveform on an aggressor net that causes both positive and negative crosstalk on the victim net. The crosstalk on the victim net does not fail the constraint. The spreadsheet reports only the larger of the positive and negative crosstalk values.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 8-59. Max. Rise Crosstalk



For differential pairs, measurements are made on the differential waveform for the positive and negative pins.

This constraint also specifies the electrical threshold for crosstalk simulations, which is used to find aggressor nets that are coupled to the selected net. See “Finding Nets With Excessive Crosstalk” on page 35.

Technically, every net on the design couples to the select net, but from a practical viewpoint only a small number of other nets couple strongly enough to generate any significant crosstalk. The noise budget for the design determines the minimum amount of crosstalk that simulation should report as a violation.

You can specify different threshold voltages for individual nets you enable for crosstalk simulation. However, for simplicity, you will probably use the same value for all nets.

Relationship Between Crosstalk Constraint and Interactive Coupling Threshold

When you interactively assign IC models, set the following options to the same value.

- Coupling threshold in the Set Coupling Thresholds dialog box
- Crosstalk constraint

The interactive coupling threshold from the Set Coupling Thresholds dialog box determines which aggressor nets to display in the Assign Models dialog box. When you set the interactive coupling threshold to a value larger than the Crosstalk value, some of the aggressor nets that you want to evaluate in batch simulation may not appear in the Assign Models dialog box. This means that you cannot interactively assign IC models to the aggressor nets in preparation for batch simulation and the contribution of aggressor nets to the victim net crosstalk is excluded from detailed batch simulation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Specifying Device Kits

Use this procedure to specify the location of a device kit and manually load a <*device_kit*>.*INI* file. In some cases, when you open the design that comes with a device kit, the software automatically loads a <*device_kit*>.*INI* file for the design.

Prerequisites

- Read the documentation that comes with the device kit to learn how to set it up.
- Open a schematic or board that comes with a device kit, or open a new schematic.

Procedure

1. Choose the **Setup > Device Kit** menu item.
2. In the Specify Device Kit for Current Design dialog box, browse to the <*device_kit*>.*INI* file for the device kit, set additional values according to the device kit documentation, and click **OK**.

Standard Delay Format

A standard delay format (SDF) file contains interconnect delays between driver and receiver pins. This is an industry-standard file format.

INTERCONNECT Entry Syntax

(INTERCONNECT <driver reference designator.pin> <receiver reference designator.pin> (min rise delay::max rise delay) (min fall delay:: max fall delay))

For example: (INTERCONNECT U1.G3 X2.34 (0.551:: 0.635) (0.508:: 0.590))

Delay values use the time units defined by the TIMESCALE entry.

Statistical and Time Domain Simulation Comparison

IBIS-AMI and FastEye channel analyses can provide results from statistical or time domain simulation.

When the assumption of a linear and/or time invariant (LTI) channel is valid, the greater speed of statistical simulation can help you screen channels on a board and quickly identify those that are clearly good. You can then use time domain simulation to examine the remaining channels in detail.

- Time domain simulation enables you to apply a stimulus pattern with a particular bit order or number of characters, apply a custom bit sequence, or have the wizard calculate

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

a worst-case stimulus to get the most-closed eye. The simulator applies the stimulus at the transmitter and observes the response at the receiver on a bit-by-bit basis. This enables the analysis to produce an eye diagram with all waveform details or a contour showing the eye diagram outline, in addition to bit error rate plots.

- Statistical simulation uses pulse and step responses extracted from the channel characterization to predict bit error rates more quickly than time domain simulation. You specify the general type of bit pattern, rather than a specific bit sequence. The analysis does not provide waveform details, but does produce the same bit error rate plots available in time domain simulation. These plots are accurate under the assumption that Tx, Rx, and channel interconnect are linear and time-invariant (LTI) in their behavior.

In statistical simulation, the simulator uses the channel response file and options that you set in the IBIS-AMI or FastEye wizard. In time domain simulation, the simulator applies the bit pattern with the transmitter and observes the channel response at the receiver on a bit-by-bit basis. Statistical simulation can capture some features of the bit pattern, such as 8B/10B, and the bit pattern that you enable in the wizard affects simulation results.

Note that for IBIS-AMI models, the validity and correlation between statistical and time domain results depends on whether the models have been designed to support each type of simulation. For information, see IBIS-AMI Model Requirements for Statistical Simulation.

The following table compares characteristics for statistical and time domain simulation, and provides details on how accurately statistical and time domain simulation handle specific aspects of transmitter and receiver modeling.

Table 8-20. Statistical and Time Domain Simulation Comparison for FastEye and IBIS-AMI Channel Analysis

Characteristic	Statistical Simulation	Time Domain Simulation
bit error rate (BER) and simulation run time	Predicts channel BER to 1e-20 and below in a few moments.	Predicts channel BER in the range of 1e-6 to 1e-8 in several minutes or several tens of minutes.
Non-linear and/or time invariant (non-LTI) transmitter and receiver modeling	Models the effect of clock and data recovery (CDR) statistically by including the jitter described by the Rx_Clock_PDF keyword, although this modeling is not absolutely accurate. Models linear and non-adaptable pre-emphasis. Approximately models equalization (DFE and CTLE), but excludes the effect of error propagation.	Models pre-emphasis, clock and data recovery (CDR), and equalization (DFE and CTLE) behaviors that are described by IBIS-AMI algorithmic files (*.DLL* or *.so*). FastEye channel analysis does not support IBIS-AMI models.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 8-20. Statistical and Time Domain Simulation Comparison for FastEye and IBIS-AMI Channel Analysis (cont.)

Characteristic	Statistical Simulation	Time Domain Simulation
Jitter and inter-symbol interference (ISI) modeling	Models receiver jitter when the Rx_Clock_PDF keyword is provided. Models transmitter jitter as if it were applied to the receiver. Models ISI, but cannot model interaction between transmitter jitter and ISI.	Models jitter at the transmitter and takes into account the effects of transmitter jitter on the channel, including jitter amplification caused by channel ISI.
Stimulus bit patterns	Emulates the effects of bit patterns, but does not apply them on a bit-by-bit basis.	Supports many bit pattern types and applies them on a bit-by-bit basis.
Simulation waveforms formatted as an eye diagram	Statistical simulation does not produce any waveforms.	Available
Channel behavior samples per bit	Uses a fixed resolution, on the order of hundreds of samples per bit, that you cannot change.	You can specify the quantity of samples per bit.

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

Tips for Running Simulation with Parametric Sweeps

It is easy to specify many sweep simulations, which can produce very long run times and memory overloads. This can happen for very large numbers of individual sweep simulations and for small numbers of individual sweep simulations with many probes.

Manage the Number of Sweep Simulations

Caution

It is possible to specify so many sweep simulations that all memory is consumed before sweep simulations complete. If this happens, all sweep simulation waveforms that were created prior to the memory overload are lost. Although sweeps attempts to allocate sufficient memory prior to starting simulation, it does not always accurately predict the memory needed by the simulator.

One workaround is to divide the sweeps you want to perform into two or more sessions.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

To help you manage the number of sweep simulations, the Sweep Manager Dialog Box - Setup Tab provides ways to disable specific sweep simulations.

Disable Specific Sweep Values Within a Range

When you specify sweep ranges by using the initial/final values or tolerance methods in the Sweeping Dialog Box, sweep values are automatically displayed in the By List box.

You can disable specific sweep values within a range by enabling the By List sweep range method and then removing values from the By List box.

Disable Specific Sweep Simulations

After you assign a sweep range to a design property, the Sweep Manager Dialog Box - Setup Tab adds simulation check boxes for the item and its branch in the tree view.

Clear a check box to disable simulation for an item or an entire branch. This capability may be useful if you want to rerun sweeps for only specific design properties.

Check boxes with a gray background indicate that some items lower in the branch are enabled while others are disabled.

Related Topics

Parametric Sweeps

Trace to Area Fill Coupling Examples

The software searches for area fills to couple to a selected net when you check “Include trace to area fill coupling” from the Coupling Settings dialog box.

The horizontal search range starts at the trace centerline for the selected net and ends at the “Area fills search distance” from the Coupling Settings Dialog Box.

The vertical search range extends up and down through all stackup layers, unless blocked by either of the following:

- A plane layer with no area fills.
- An area fill that extends beyond both sides of the horizontal search range defined by “Area fills search distance”. This area fill can be located on a plane or signal stackup layer.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The figure below shows how a signal trace couples to area fills and routed power-supply nets.

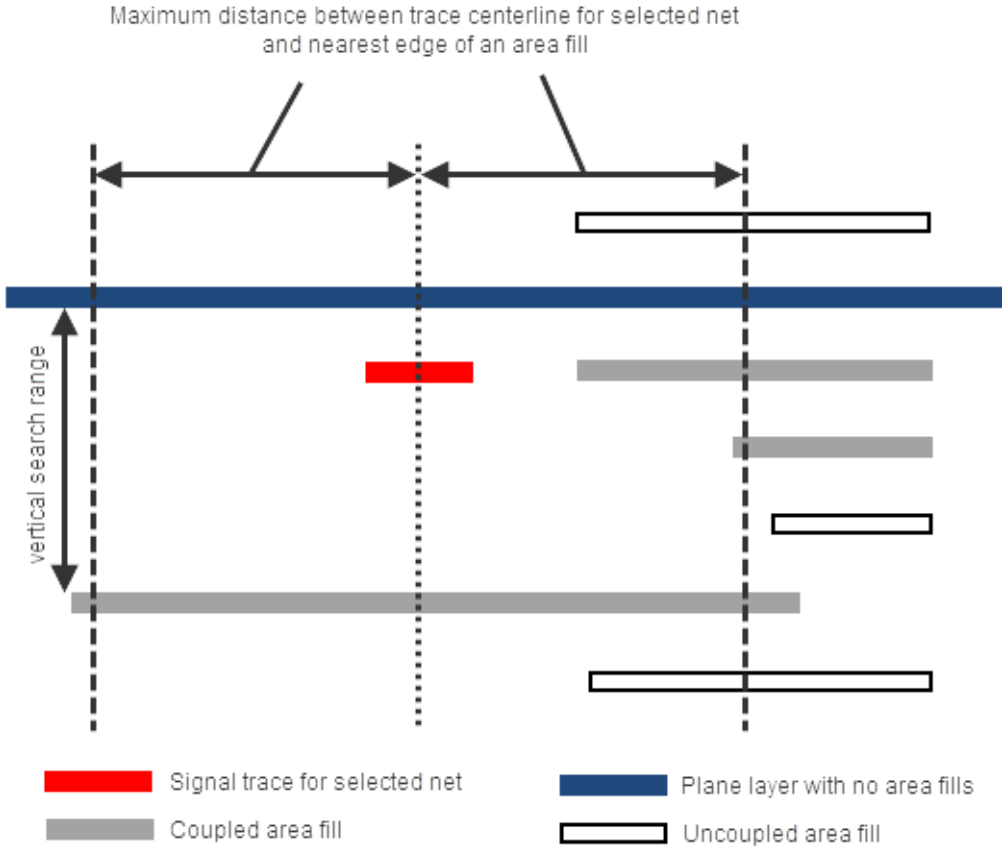
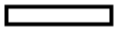
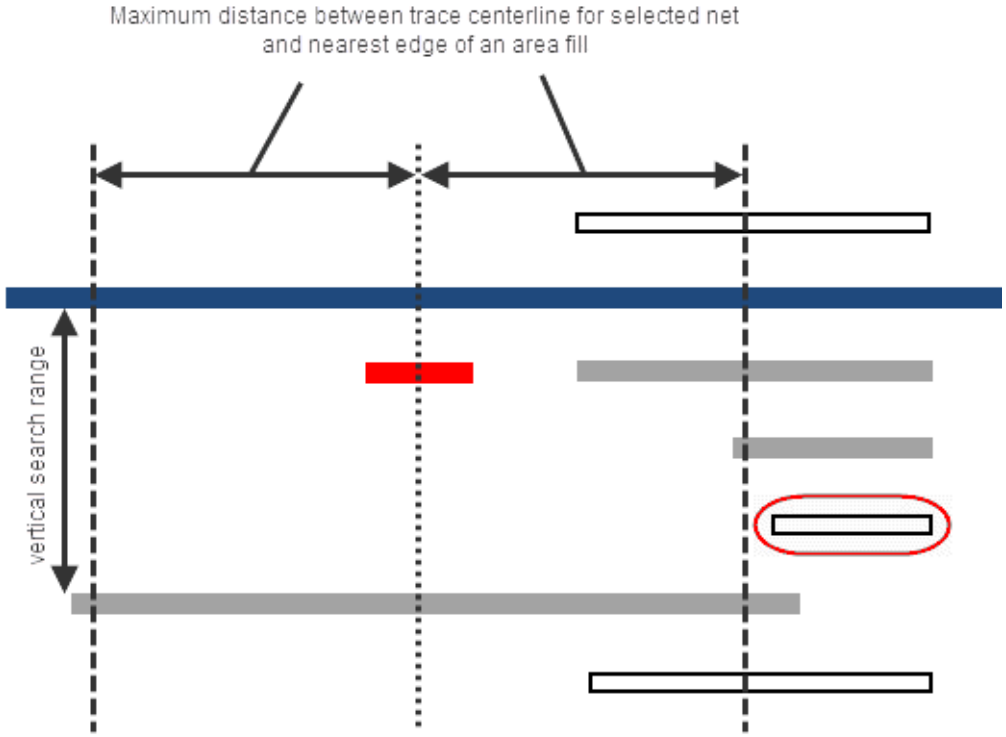


Figure Landmark	Description
Selected net	The origin of the horizontal and vertical search ranges.
Coupled area fill	Area fills that are located within the horizontal and vertical search ranges, and coupled to the selected net. The bottom area fill defines the bottom of the vertical search range because it extends beyond both sides of the horizontal search range.
Plane layer with no area fills	A stackup layer that is identified as a plane layer and contains no area fills, and is coupled to the selected net. This layer defines the top of the vertical search range.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure Landmark	Description
Uncoupled area fill 	Area fills that are located outside the horizontal and vertical search ranges. Specific examples of when the area fill is not coupled to the selected net: • Its nearest edge is located beyond the horizontal search range. • It is blocked by either a plane layer with no area fills or an area fill that extends beyond both sides of the horizontal search range.

The figure below shows an area fill that is too far away from the selected net to include its coupling during simulation.



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The figure below shows two area fills that are blocked by either a plane layer or an area fill that spans the horizontal search range. Simulation does not include coupling to the selected net for either of the area fills.



Related Topics

Accounting for Coupling

Horizontal and Vertical Geometric Search Range for Coupled Nets

Coupling Ratio for RLC Matrix Package Coupling

Area Fill Edge Approximation Examples

Transferring HyperLynx Settings

You can transfer IC model, design folder name, and differential pair net name suffix settings (BoardSim only) that are stored in the HyperLynx initialization file (*BSW.INI*) from a previous HyperLynx installation to the latest HyperLynx installation.

Caution

This procedure describes how to manually edit a file containing formatted information. Formatting errors can produce unexpected simulation results.

Procedure

1. If HyperLynx is running, close it.
2. Rename the *BSW.INI* file for the latest installation.

   For example, in the folder *\MentorGraphics\<latest_release>\SDD_HOME\hyperlynx64*, rename *BSW.INI* to *BSW.INI.save*.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

3. Copy the *BSW.INI* file from the previous installation to the latest installation.

   For example, copy *\MentorGraphics\<previous_release>\SDD_HOME\hyperlynx64\BSW.INI* to *\MentorGraphics\<latest_release>\SDD_HOME\hyperlynx64\BSW.INI*.

4. Edit *\MentorGraphics\<latest_release>\SDD_HOME\hyperlynx\BSW.INI* and change the keywords for transferring settings in the following sections:

Section	Keyword	Description
[BSW_LIBRARY]	ModLibPath##	Specify the path to the folder containing the IC model library. Specify up to 99 folders. For example, ModLibPath00=*C:\MentorGraphics\2009HL\SDD_HOME\hyperlynx\LIBS* becomes *ModLibPath00=C:\MentorGraphics\2009.1HL\SDD_HOME\hyperlynx\LIBS*.
[BSW_PREFERENCES]	HypPath	Specify the folder containing design files (HYP, CCE, FFS).
[DIFF_PAIR_SUFFIXES]	Pair#	Specify differential pair net name suffixes, such as _p and _n, to help the software automatically identify differential pairs. Examples: +=- _n=_p See Differential Pairs Dialog Box.

5. Save the edited file.
6. Open HyperLynx.

Tip

You can automatically create an all-new *BSW.INI* file by renaming the current *BSW.INI* file, opening HyperLynx, and then closing HyperLynx. If no *BSW.INI* file exists, the software automatically creates a new file when you close the program.

Translating a Board Design

If your board layout software cannot save a board design in HYP, CCE, or ODB++ format, use HyperLynx to translate your board design file to HYP format.

Caution

Some layout software tools create board design files that do not contain the detailed metal area and via-clearance information that power integrity simulation requires.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Board designs exported from these software tools require translation to HYP format:

- Accel EDA (Not supported for power integrity simulation.)
- Cadence Allegro
- Siemens EDA Board Station or Board Station RE
- Specctra DSN (Not supported for power integrity simulation.)
- Zuken Visula/CADStar for Windows (Not supported for power integrity simulation.)
- Zuken CR-3000 (Not supported for power integrity simulation.)
- Zuken CR-5000 (The software cannot translate Zuken CR-5000 PWS designs.)

Prerequisites

- You have prepared your board design and exported a file using your board layout software tool. See Preparing a Board Design File for Translation.
- The board design file has a unique name.
- If you have a Cadence Allegro design that has a complex board outline, enable the "Don't crop area fills ..." option on the **BoardSim** tab of the Preferences dialog box.

Procedure

1. Choose the **File > New Board** menu item. The Choose a File to Translate dialog box opens.
2. From the file type list, select the type of file to translate.
3. Browse to the board design file and click **Open**.

   Note

   Cadence Allegro BRD files only appear if the CDSROOT environment variable is defined. This environment variable is not defined when you install Cadence software. See KB article MG594641 on Support Center.

4. To specify translator options, do the following:
   a. Click **Options**.
   b. In the Translator Options dialog box, edit options in the Standard Options area as needed.
   c. If you have a Board Station Layout or Board Station RE design that contains variants, type -a <variant name> in the Non-Standard Command-Line Options field to specify the variant to translate. Otherwise, leave this field empty unless a Siemens representative provides instruction.
   d. Click **OK**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

5. Run the translation:
   - To run translation only, click **Translate**.
   - To run translation and automatically load the board, click **Translate & Open**.

Results

You are now ready to open your board design in HyperLynx. See Opening a Design.

Using the Field Solver

Scope: LineSim

Run the field solver on your schematic for any coupling region that you already defined.

Procedure

1. Double-click any transmission line belonging to the coupling region whose field properties you want to see.
2. Click the **Field Solver** tab.
3. To see field lines, in the Field Plotting area, click **Start**.

   Or

   To see numerical data (for example, impedance matrix), in the Numerical Results area, click **View**.

Viewing a Board in 2-D

Use the board viewer to display the topology and components for signal nets and power-distribution networks on a board.

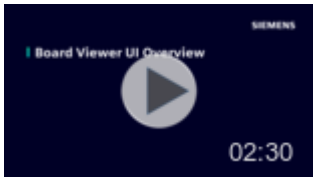
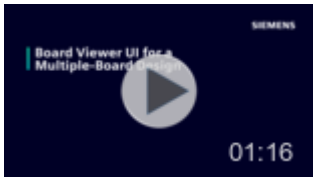
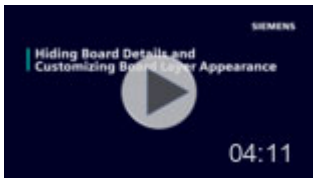
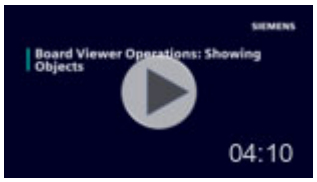
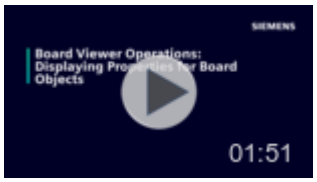
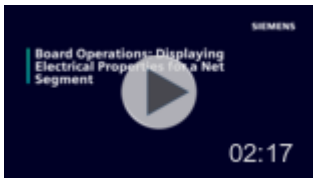
Note

Use xPCB/xDX View to display *.CCE* board files exported from Xpedition Layout or other Siemens EDA products. xPCB/xDX View displays custom pad shapes more accurately than the BoardSim board viewer. *.CCE* files provide additional layers to display manufacturing and other types of information.

Video Topics

View videos to learn how to use the board viewer.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Video Title	Description
	Learn how the board viewer displays layout and simulation-related information.
	Learn how the board viewer displays a multiple-board design.
	Learn how to hide design detail on a busy board: • Display specific types of reference designators • Choose which types of objects to highlight during mouse-over • Show or hide all pads, anti-pads and anti-segments • Zoom to an area of interest • Highlight an object/restore highlighting Learn how to customize your view of board layers: • Show and hide specific signal or plane layers • Identify stackup layers used to implement trace segments and metal shapes • Change the display pattern of copper pours
	Learn how to view: • Location of decoupling and bypass capacitors • Location of a specific component • Electrical connectivity of capacitor mounting • Location of specific power-supply nets
	Learn how to display properties for: • Nets • Trace segments • Vias • Pins • Passive components
	Learn how to view information about: • A net segment, such as characteristic impedance, delay, and loss versus frequency • Coupled aggressor net segments, such as net separation and impedance values • View coupling regions

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Board Viewer Operations

The board viewer provides several ways to help you navigate your board.

Table 8-21. Board Viewer Operations

If you want to...	Do this...
Fit the Board to the Window	Click Fit to Window  .
Pan in Board Viewer	Do any of the following: • Press the arrow keys for fine movement. • Press the Page Up and Page Down keys for coarse movement. • Choose**View > Zoom Pan**, move the pointer to the new center location of the board viewer, and click. • Press and hold down the mouse wheel while dragging the mouse.
Zoom By a Fixed Increment	Press and hold down the Shift key and rotate the mouse wheel.
Zoom In to an Area Defined by a Box	Draw a box around the area of the board you want to enlarge to fill the board viewer. The center of the box you draw becomes the new center of the board. 1. Click Zoom Area In  . 2. Drag a box around the area to enlarge.
Zoom to the Previous Zoom Level	Zoom to any previous zoom level in the current session by stepping through the previous zoom levels one at a time. 1. Click Zoom Previous  .
Measure Distance	Choose **View > Measure Distance**. Click and drag to display the distance between two points.
Flip the Board	Change the orientation. 1. Choose**View > Flip Board**. 2. Do any of the following: • To flip the board around the vertical axis, choose **Right**. • To flip the board around the horizontal axis, choose **Down**. Results: • Component outlines change sides on the board. Black component outlines become gray and gray outlines become black. • Vias change color because you are now viewing the via pads on the opposite side of the board.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Find Component Dialog Box

Highlight Net Dialog Box

Display 3D Board Area Dialog Box

Manipulating a 3-D View

Viewing Coupling Region Details

Scope: BoardSim

Use the Coupling Region dialog box to evaluate the crosstalk contribution from trace segments of nearby aggressor nets to the selected victim net, see a geometric relationship between coupled trace segments, or to see trace-to-trace impedance calculations for differential pairs.

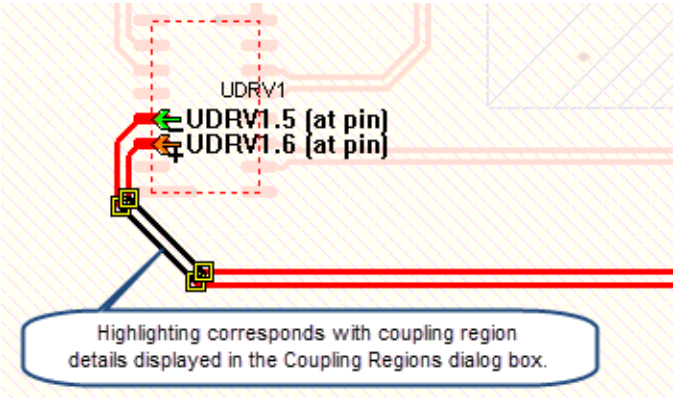
You must manually create coupling regions in schematic designs. See “Creating a Schematic Design” on page 70. View coupling region information for schematic designs in the Edit Transmission Line dialog box.

Prerequisites

- Enable Crosstalk Simulation and set coupling thresholds so that coupled nets are identified. See Accounting for Coupling.

Procedure

1. Select a net and choose the **View > Coupling Regions** menu item, or right-click a trace and choose the **Walk Coupling Region** menu item.
2. In the Coupling Region dialog box, click **Back** or **Next** to cycle through coupling regions. The board viewer highlights the trace segments in each coupling region.



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

To see the data for aggressor nets in each region, click:

- **Nets** to show a list of names of the nets in the coupling region and the geometric relationships among their trace segments, including the edge-edge distance and vertical separation in the stackup.
- **Impedance** to show electrical characteristics of the coupling region, (coupled trace impedance, signal-propagation delays, and capacitance and inductance matrices) as generated by the field solver.
- **Cross Section** to show a graphical representation of the geometric relationships of the nets in the coupling region.

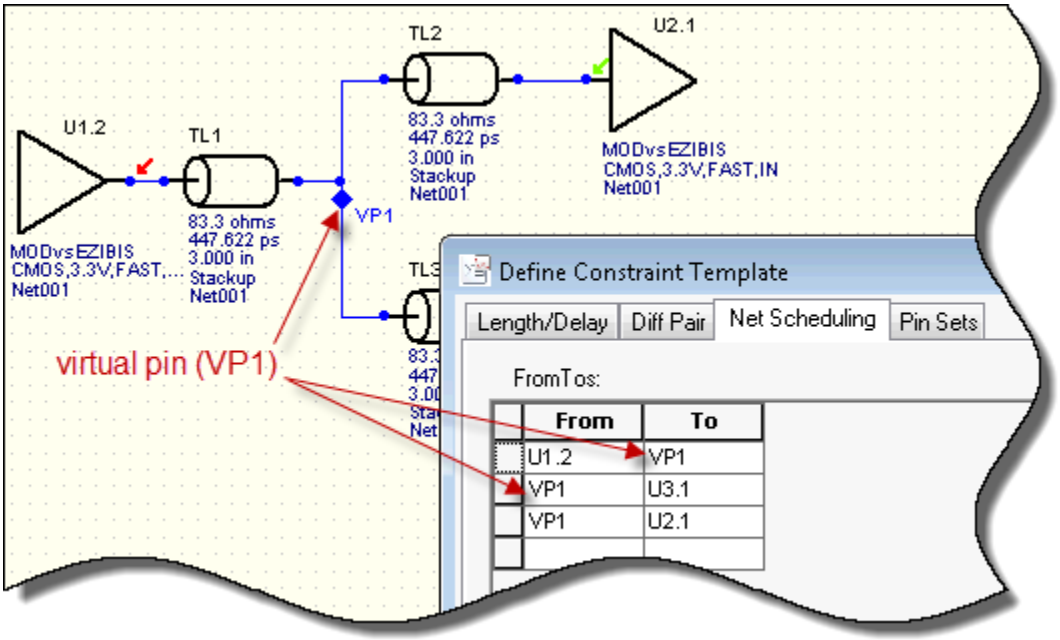
Related Topics

Horizontal and Vertical Geometric Search Range for Coupled Nets

Virtual Pins

A branch point exists where one transmission line connects to two or more other transmission lines. The software automatically creates a virtual pin at the branch point to provide a reference point when you create a constraint template.

Virtual pin symbols are diamond-shaped, and larger than intersection points. In the following figure, the smaller intersection point circle above the virtual pin named VP1 is not a virtual pin and cannot be named.



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Opening HyperLynx Tools on Linux

Open HyperLynx SI/PI/Thermal to observe signal integrity, power integrity, or thermal properties for your design. You also can directly open the Touchstone Viewer or Visual IBIS Editor.

Prerequisites

- Install and configure licensing. Refer to the *Installation and Administration Manual for HyperLynx*.

Procedure

Open the software:

If you want to open...	Run this Linux command...
HyperLynx SI/PI/ Thermal	`/MentorGraphics/<release>/SDD_HOME/hyperlynx/run_bsw`
Touchstone Viewer	`/MentorGraphics/<release>/SDD_HOME/hyperlynx/run_mwprog TouchstoneViewer`
Visual IBIS Editor	`/MentorGraphics/<release>/SDD_HOME/hyperlynx/run_mwprog VisIbis`

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Chapter 9
Modeling

Simulating a design requires the correct models for accurate results.

Topic	Description
Supported SI Models and Simulators	SI simulation supports several types of models and simulators.
IBIS Models	An IBIS model is a behavioral method of modeling input/output buffers based on V/I curve data that is derived from measurement or full circuit simulation.
S-Parameter Models	You can create an S-parameter model by exporting a net.
Z-Parameter Models	You can create a Z-parameter model by running decoupling simulation.
Model Mapping	Use *.REF* and *.QPL* files to assign models and values to components.
Package Modeling	You can create a custom package and model package parasitics.
Timing Models	DDRx simulation requires timing models for controller and DRAM components.
Power Integrity Models	You can describe the electrical behavior of IC and voltage-regulator module (VRM) components for PI simulation by assigning PI models.
Reference Nets	Reference nets provide return-current paths for current sinking into IC power-supply pins. Reference nets may be implemented across multiple stackup layers, where stitching vias connect metal areas on different stackup layers.
Series Components for Power-Supply Nets	Series components connect one power-supply net to another power-supply net. Simulation can include both nets that are connected to the component. These nets are called associated nets.

Supported SI Models and Simulators

SI simulation supports several types of models and simulators.

Restrictions:

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- Siemens EDA ADMS automatically installs with HyperLynx and requires the DDRx or GHz license bundle.
- Simulating S-parameters requires the GHz or DDRx license bundle.
- Verilog-AMS models are not supported.
- Synopsys HSPICE requires separate installation and licensing.

Note

For information about how the software automatically selects the simulator, see Automatic SI Simulator Selection.

Table 9-1. Supported SI Models and Simulators

Model Types	Simulators		
	HyperSim	ADMS	HSPICE
IBIS	Yes	Yes	Yes
EBD/EMD	Yes	Yes	Yes
IBIS-AMI[1] (model has [Algorithmic Model] keyword)	Yes	Yes	Yes
IBIS-ISS	Yes	Yes	Yes
S-parameter	Yes	Yes	Yes
Eldo (model has Eldo-specific SPICE syntax)[2]	--	Yes	--[3]
Encrypted Eldo[2]	--	Yes	--
HSPICE (model has HSPICE-specific syntax)[2]	--	Yes[4]	Yes
Encrypted HSPICE[2]	--	--	Yes
VHDL-AMS, VHDL-A[2]	--	Yes[5]	--
Verilog-A	--	Yes	--

1. Only IBIS-AMI channel analysissupports
IBIS-AMI models.
2. Batch and other types of simulations that automatically report signal-integrity measurements require Eldo, encrypted Eldo, VHDL-AMS, HSPICE, and HSPICE encrypted I/O buffer models to be referenced from an IBIS model by an [External Model] or [External Circuit] keyword. Note that HyperLynx supports the non-standard Language sub-parameter values of Eldo-Encrypted, HSPICE, and HSPICE-Encrypted.
3. The syntax for some object types is different for HSPICE and Eldo. Both ADMS and HSPICE support standard, Berkeley SPICE syntax.
4. HSPICE models assigned directly to components are supported when you enable the “HSPICE compatible (including Eldo encrypted)” option on the Preferences Dialog Box - Simulators Tab. ADMS may not support all HSPICE-specific syntax, such as recently-added syntax.

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5. Pre-compiled VHDL-AMS models are supported by ADMS in HyperLynx when they are embedded in a top-level SPICE netlist, which is typically provided by a Siemens EDA device kit. HyperLynx dynamically compiles only VHDL-AMS models that are referenced from an IBIS model by an [External Model] or [External Circuit] keyword.

Related Topics

Preferences Dialog Box - Simulators Tab

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IBIS Models

An IBIS model is a behavioral method of modeling input/output buffers based on V/I curve data that is derived from measurement or full circuit simulation.

Topic	Description
IBIS Editor	You can create, edit, verify, and maintain IBIS (I/O Buffer Information Specification) device models with the HyperLynx Visual IBIS Editor.
Referencing an External Model From an IBIS Model	Some simulation options require IBIS models. When you have a SPICE or VHDL-AMS model, you can create an IBIS model that references the SPICE or VHDL-AMS model as an external model.
Referencing a SPICE Model with the External Model Keyword	This procedure provides the general process for using the [External Model] keyword to reference an external SPICE model from an IBIS model.
IC Operating Settings	IC operating settings are combinations of the min and max data in an IBIS model.
Adding Model Selector Keywords to IBIS Models	Use this procedure to add [Model Selector] keywords to IBIS models. Use these keywords to specify buffer strength or on-demand termination (ODT) conditions you want to use during simulation.
Supported IBIS Model Spec and Receiver Threshold Keywords	HyperLynx supports the set of sub-keywords required to measure DDRx signals for the [Model Spec] and [Receiver Threshold] keywords in IBIS models.

IBIS Editor

You can create, edit, verify, and maintain IBIS (I/O Buffer Information Specification) device models with the HyperLynx Visual IBIS Editor.

See Creating and Editing IBIS Models.

Referencing an External Model From an IBIS Model

Some simulation options require IBIS models. When you have a SPICE or VHDL-AMS model, you can create an IBIS model that references the SPICE or VHDL-AMS model as an external model.

The external model provides IC component behaviors that are normally provided in IBIS models by I-V and V-T tables, the C_comp subparameter, and so on. The IBIS model provides

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test load and threshold information (such as Vinh, Vinl, and Vmeas) that is needed to perform signal-integrity measurements.

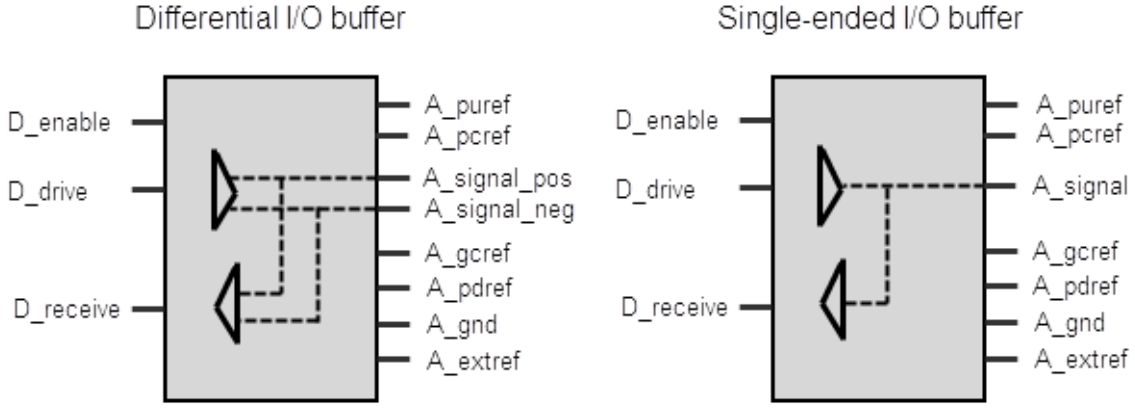
If the external model you want to reference can use pre-defined port names, use the [External Model] keyword. Otherwise, use the [External Circuit] keyword, which supports arbitrary port names, but requires you to manually map model ports to subcircuit nodes.

Predefined Ports for the [External Model] Keyword

Use the [External Model] keyword to map predefined ports to IC buffer model ports when you need to reference a SPICE or VHDL-AMS model from an IBIS model.

The figure below shows the full set of predefined port names for differential and singled-ended buffers.

Figure 9-1. Pre-Defined Port Names for [External Model] Keyword



The value of the Model_type subparameter for the [Model] keyword determines the set of required ports for the specific buffer model.

Table 9-2. Required [External Model] Ports for Single-Ended Buffers

	D_drive	D_enable	D_receive	A_signal
I/O I/O_open_drain I/O_open_sink I/O_open_source I/O_ECL	X	X	X	X
3-state 3-state_ECL	X	X		X

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Table 9-2. Required [External Model] Ports for Single-Ended Buffers (cont.)

	D_drive	D_enable	D_receive	A_signal
Output Output_ECL Open_drain Open_sink	X			X
Input Input_ECL			X	X

Table 9-3. Required [External Model] Ports for Differential Buffers

Model_type	D_drive	D_enable	D_receive	A_signal_pos	A_signal_neg
I/O_diff	X	X	X	X	X
3-state_diff	X	X		X	X
Output_diff	X			X	X
Input_diff			X	X	X

Example [External Model] Keyword for SPICE Model

This example contains an [External Model] and [End External Model] block that calls an external SPICE model for a differential buffer. Some of the key ports are also described.

The figure below is based on an example from the I/O Buffer Information Specification (IBIS), in the Multi-Lingual Model Extensions section. The specification also provides many more examples, including how to call external models in VHDL-AMS syntax and how to call external models that represent series switches.

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Figure 9-2. Example [External Model] Keyword for SPICE Model

```
[External Model]

Language SPICE

|Corner  corner_name  file_name        circuit_name (.subckt name)
Corner   typ          buffer_typ.sp    buffer_io_typ
Corner   min          buffer_min.sp    buffer_io_min
Corner   max          buffer_max.sp    buffer_io_max

|Ports   List of port names that are in the same order as in SPICE
Ports A_signal_pos A_signal_neg sp_receive sp_drive sp_enable
Ports A_puref A_pdref A_pcref A_gcref A_extref sp_ref A_gnd

| D_to_A  d_port    port1     port2   vlow  vhigh  trise  tfall  corner
D_to_A    D_drive   sp_drive  sp_ref  0.0   3.3    0.5n   0.3n   typ
D_to_A    D_drive   sp_drive  sp_ref  0.0   3.0    0.6n   0.3n   min
D_to_A    D_drive   sp_drive  sp_ref  0.0   3.6    0.4n   0.3n   max
D_to_A    D_enable  sp_enable sp_ref  0.0   3.3    0.5n   0.3n   typ
D_to_A    D_enable  sp_enable sp_ref  0.0   3.0    0.6n   0.3n   min
D_to_A    D_enable  sp_enable sp_ref  0.0   3.6    0.4n   0.3n   max

| A_to_D  d_port    port1        port2        vlow   vhigh  corner
A_to_D    D_receive A_signal_pos A_signal_neg -200m  200m   typ
A_to_D    D_receive A_signal_pos A_signal_neg -200m  200m   min
A_to_D    D_receive A_signal_pos A_signal_neg -200m  200m   max

[End External Model]
```

Note

The syntax of the subcircuit in the external SPICE model must be compatible with the simulator you enable in the Digital Oscilloscope Dialog Box or Simulation Controls Dialog Box, and in the Simulation Controls Dialog Box.

Table 9-4. Example Key Ports

Port	Description
A_signal_pos, A_signal_neg	Analog signals that connect to the pin model at the die. The external SPICE pin model replaces the IBIS model behavioral and tabular information. Although not shown in the example, a package model can be assigned to these ports.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 9-4. Example Key Ports (cont.)

Port	Description
(D_drive, sp_drive, sp_ref), (D_enable, sp_enable, sp_ref)	Because SPICE does not have a true digital signal, use the D_to_A subparameter to transform the D_drive and D_enable digital signals into analog signals that it can use. The transform behavior resembles a voltage source with a digital signal input that toggles the voltage across the two analog output signals. When D_drive is 1, the voltage across sp_drive and sp_ref is vhigh. When D_drive is 0, the voltage across sp_drive and sp_ref is vlow. When D_enable is 0, the voltage across sp_enable and sp_ref is vlow. When D_enable is 1, the voltage across sp_enable and sp_ref is vhigh. The voltage changes linearly from vlow to vhigh in trise seconds. The voltage changes linearly from vhigh to vlow in tfall seconds.
D_receive, A_signal_pos, A_signal_neg	Use the A_to_D subparameter to transform the analog signal from the buffer model to the digital D_receive signal. The transform behavior resembles a comparator with two analog inputs and a digital signal output. When the voltage across A_signal_pos and A_signal_neg is greater than vhigh, D_receive is 1. When the voltage across A_signal_pos and A_signal_neg is less than vlow, D_receive is 0.
A_puref	Connects to voltage specified in the [Pullup Reference] or [Voltage Range] keyword.
A_pdref	Connected to voltage specified in the [Pulldown Reference] keyword or 0 V.
A_pcref	Connected to voltage specified in the [POWER Clamp Reference] or [Voltage Range] keywords.
A_gcref	Connected to voltage specified in the [GND Clamp Reference] keyword or 0 V.

Notice that some port names, such as sp_drive, do not appear in the list of pre-defined ports. When using D_to_A or A_to_D converters, the Ports subparameter includes intermediate ports with non-predefined names to connect the converter to the IBIS and SPICE models.

Related Topics

Referencing a SPICE Model with the External Model Keyword

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Referencing a SPICE Model with the External Model Keyword

This procedure provides the general process for using the [External Model] keyword to reference an external SPICE model from an IBIS model.

Prerequisites

- Understanding of Referencing an External Model From an IBIS Model.
- The IBIS model already contains a [Model] keyword that provides test load and threshold information (such as Vinh, Vinl, and Vmeas) that is needed to perform signal-integrity measurements.

Procedure

1. Map the predefined port names in Table 9-2 or Table 9-3 to SPICE subcircuit node names.
2. Add the [External Model] and [End External Model] keywords under the appropriate [Model] keyword. Place all subparameters between these keywords.
3. Add the Language subparameter. For example: Language SPICE.
4. Add one or more Corner subparameters:

   Corner <corner> <file_name> <sub_circuit_name>
5. Add the Ports subparameter, using the predefined port names from step 1 and port names for the D_to_A and A_to_D converters. The list of port names must be in the same order as in the sub circuit definition.
6. Add the D_to_A and A_to_D subparameters. Map the digital signals to analog signals.

   Map the port names (D_drive, D_enable, D_receive) to analog signals.

   Set the vhigh argument equal to VDD [Voltage Range] and set the trise and tfall arguments to 0.1ns, unless the model vendor provides specific values.

Related Topics

Supported SI Models and Simulators

IC Operating Settings

IC operating settings are combinations of the min and max data in an IBIS model.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

The IBIS format supports min/typ/max data, but only requires typical. Changing IC operating parameters only affects the IBIS IC models in your circuit that contain min/max data. If you change the IC operating parameters but see no change in your simulation waveforms, it is because the IBIS model(s) you use do not have min/max data.

Table 9-5. IC Operating Setting Combinations for IBIS Models

Parameter	IBIS Keywords	Fast-Strong	Slow-Weak
driver current	[Model] > [Pullup] and [Pulldown]	max	min
slew rate	[Model] > [Rising Waveform] and [Falling Waveform] or [Ramp]	max	min
clamp-diode current	[Model] > [GND Clamp] and [Power Clamp]	max	min
component capacitance	[Model] > C_comp	min	max
package inductance	[Component] > [Package]	min	max
package capacitance	[Component] > [Package]	min	max
package resistance	[Component] > [Package]	min	max

In addition, you can set IBIS model pull-up and power clamp voltage to vary with the IC operating setting. For the oscilloscope, enable “When assigning a model to an IC-pin, use a power-supply net connected to the IC” in the Preferences Dialog Box - General Tab. For General batch simulation, enable corners on the Simulation settings page.

The following table shows how pull-up and power clamp voltages in IBIS models vary with IC operating settings.

Table 9-6. Pullup and Power Clamp Voltages - IBIS Models

Parameter	IBIS Keywords	Fast-Strong	Slow-Weak
pull-up voltage	[Model] > [Voltage Range] or [Pullup Reference]	max	min
power clamp voltage	[Model] > [Voltage Range] or [POWER Clamp Reference]	max	min

Related Topics

Digital Oscilloscope Dialog Box

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Adding Model Selector Keywords to IBIS Models

Use this procedure to add [Model Selector] keywords to IBIS models. Use these keywords to specify buffer strength or on-demand termination (ODT) conditions you want to use during simulation.

Prerequisites

- Review Creating and Editing IBIS Models.

Procedure

1. Open an IBIS model.
2. Create a [Model Selector] keyword containing all the [Model] names needed to simulate the pin for all buffer strength and ODT conditions you plan to simulate. See On-Die Termination - ODT.

   Syntax:

   ```
   [Model Selector] <model_selector_name>
   <programmable_buffer_model_name> <comment_to_end_of_line>
   ```

   Example:

   ```
   [Model Selector] DQ DQ_DRVFULL_ODTOFF Full-strength driver with
   disabled ODT DQ_DRVHALF_ODT50 Half-strength driver with 50 Ohm ODT
   DQ_DRVHALF_ODT100 Half-strength driver with 100 Ohm ODT
   ```

3. In the [Pin] keyword, find a data pin used by the memory interface and assign the appropriate <model_selector_name> defined in step 2.

   Syntax:

   ```
   [Pin] signal_name model_name R_pin L_pin C_pin <pin_name>
   <signal_name> <model_name>
   ```

   Example:

   ```
   [Pin] signal_name model_name R_pin L_pin C_pin DQ5 DQ5 DQ
   ```

   JEDEC specifications JESD79-2* and JESD79-3* refer to data pins as data (DQ), data mask (DM), check bit (CB), and data strobe (DQS). The pin names in the IBIS model can be different, making it harder for you to map the model-specific data pin names to the generic names in the specifications.

4. Repeat step 3 as needed to update the [Pin] keyword for all data pins in the memory interface.
5. Check the model syntax with the Visual IBIS Editor. See Checking IBIS File Syntax.

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Supported IBIS Model Spec and Receiver Threshold Keywords

HyperLynx supports the set of sub-keywords required to measure DDRx signals for the [Model Spec] and [Receiver Threshold] keywords in IBIS models.

Table 9-7. Supported IBIS Sub-keywords

Keyword	Supported sub-keyword
[Model Spec]	Vinl, Vinh—min/typ/max operating values Vmeas—min/typ/max operating values Vref—min/typ/max operating values
[Receiver Thresholds] - Single-ended buffers	Vth, Vth_min, Vth_max Vinh_ac, Vinh_dc Vinl_ac, Vinl_dc Threshold_sensitivity Reference_supply
[Receiver Thresholds] - Differential buffers	Vcross_low, Vcross_high Vdiff_ac, Vdiff_dc

S-Parameter Models

You can create an S-parameter model by exporting a net.

See Exporting a Net to an S-Parameter Model.

Defining Port Connections Within an S-Parameter File

You can add text to an S-parameter file that describes connections between ports. This is useful when using an S-parameter model in a schematic to represent branching net topologies, such as a connector in a multiple-board design.

Procedure

1. Open the S-parameter file in a text editor.
2. At the beginning of the file, type one connect statement for each set of connected ports as shown in the following example, and save the file.

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For example, if ports 1, 2, and 3 are connected, and ports 5, 8, and 9 are connected, type the following:

```
! CONNECT 1,2,3;
! CONNECT 5,8,9;
```

Results

You can now assign the file to an S-Parameter/SPICE Model symbol in the schematic design.

Z-Parameter Models

You can create a Z-parameter model by running decoupling simulation.

See Decoupling Simulation.

Touchstone Viewer

Use the Touchstone and Fitted-Poles Viewer to understand the contents and judge the quality of Touchstone and fitted-poles models. See Viewing and Converting Touchstone and Fitted-Poles Models.

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Model Mapping

Use *.REF* and *.QPL* files to assign models and values to components.

Topic	Description
Automapping Files	Use a *.REF* and *.QPL* automapping file to assign a model or value to pins on a component with a specific reference designator or part name. Automapping files provide an efficient method to assign models and values to many components on your board, which is useful when running detailed batch signal-integrity simulation on many or all nets on the board.
Precedence Among Model and Value Selection Methods	Model and value assignment precedence is a potential source of confusion when debugging a *.REF* or *.QPL* file. When you assign different models or values to a pin, the software uses the model or value assignment made by the method with the highest precedence. Model assignments have higher precedence than value assignments.
Searching for an IC Model in Model Directories	Use this dialog box to search for an IC model in your model directories and assign it to a pin.
Troubleshooting Unexpected Model Assignments	If the expected model name does not appear in the Assign Models dialog box for a pin on the selected net, you may need to investigate the origin of the error.
Troubleshooting Automapping Model Assignment Errors	When you create a large automapping file, you can accidentally introduce errors into the file. Use the REF-File Editor or QPL-File Editor to create and maintain automapping files to avoid these errors.
REF and QPL File Syntax	Use the REF- or QPL-File Editors to avoid syntax errors when editing or creating *.REF* and *.QPL* files. If your company already has qualified parts list files in a non-QPL format, knowledge of the syntax can help you convert them.

Automapping Files

Use a *.REF* and *.QPL* automapping file to assign a model or value to pins on a component with a specific reference designator or part name. Automapping files provide an efficient method to assign models and values to many components on your board, which is useful when running detailed batch signal-integrity simulation on many or all nets on the board.

A *.REF* file assigns a model or value to pins on a component with a specific reference designator while a *.QPL* file assigns a model or value to pins on all components with a specific part name, regardless of its reference designator.

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Your design has one *.REF* file, which is stored in the <design> directory. A design can use multiple *.QPL* files, which you can locate in any directory the software can access. When different models are assigned to a pin through multiple *.QPL* files, the software attaches the model from the *.QPL* file with the highest precedence. See Set Directories Dialog Box.

You create and edit *.REF* and *.QPL* files with the REF- and QPL-File Editors, which provide a fill-in-the-blank interface and create files without syntax errors. If there is no exact model for a pin you are trying to simulate, you can easily create one. See Creating IBIS Models with the Easy IBIS Wizard.

Automapping Component Assignment	Description
ICs	When you map an IC to an IC model with multiple pins, such as in an IBIS model, signal pins in the model are assigned to component pins with the same signal name.
IC Default Buffer Direction	When an automapping file assigns an IC model to a pin, as much information as possible about the model is set automatically. However you may need to manually set the buffer direction for the pin. The default buffer direction is based on the model format and model direction of the pin. An *.EBD/EMD* model pin takes on the characteristics of the IBIS model pin(s) that it references; if the IBIS model pin(s) creates a bi-directional signal, you may need to change the driver direction manually. When an automapping file sets a driver pin as a receiver, you must manually set the driver buffer direction by using the Assign Models dialog box.
Resistors and Capacitors	If the board file contains correct values for all of the resistors and capacitors in your design, you do not have to specify resistor and capacitor values in a *.REF* or *.QPL* file. **Restriction:** Decoupling capacitors do not use values set by the design. However, you can override resistor and capacitor values in the board file by specifying values for the components in a *.REF* or *.QPL* file, or by assigning values in the Assign Models dialog box. You can also use automapping files to assign a package model to resistors or capacitors contained in network packages. For schematics, using model automapping files to assign values to resistors and capacitors is not as efficient.

Related Topics

Assigning a Model or Value to an Entire Component Using a .REF File

Assigning a Model or Value to an Entire Component Using a .QPL File

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Disabling a REF or QPL File

REF-File Editor

REF and QPL File Syntax

Troubleshooting Automapping Model Assignment Errors

Precedence Among Model and Value Selection Methods

Precedence Among Model and Value Selection Methods

Model and value assignment precedence is a potential source of confusion when debugging a *.REF* or *.QPL* file. When you assign different models or values to a pin, the software uses the model or value assignment made by the method with the highest precedence. Model assignments have higher precedence than value assignments.

Table 9-8. Precedence Among Model/Value Assignment Methods

Precedence (priority)	Assignment Method	Comment
Highest	**Model Assignment Methods**	
	.REF file models	Models specified with full network paths have precedence over those with relative paths.
	EBD/EMD models in a *.REF* file	Unlike non-EBD/EMD models, an EBD/EMD model assignment made by the *.REF* file has the highest precedence.
	Assign Models dialog box	In memory and not yet saved to a *.BUD* file.
	.BUD file model	A saved interactive selection.
	non-EBD/EMD models in a *.REF* file	Unlike EBD/EMD models, a non-EBD/EMD model assignment made by the *.REF* file does not have the highest precedence.
	.QPL file models	Models specified with full network paths have precedence over those with relative paths. You set the precedence among multiple *.QPL* files with the Set Directories Dialog Box.
	Value Assignment Methods	
	Assign Models dialog box value	In memory and not yet saved to a *.BUD* file.
	.BUD file value	A saved interactive selection.
	.REF file value	--

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Table 9-8. Precedence Among Model/Value Assignment Methods (cont.)

Lowest	*.QPL* file value	You set the precedence among multiple *.QPL* files with the Set Directories Dialog Box.

Because models specified in an automapping file have lower precedence than models specified in the current session or a previously saved session (*.BUD* file), you can override, pin-by-pin, any models specified in a *.REF* or *.QPL* file. Any such overrides are stored in the session file (*.BUD*) when you close your board or the software, and take precedence the next time you load your design.

Searching for an IC Model in Model Directories

Use this dialog box to search for an IC model in your model directories and assign it to a pin.

Restrictions and Limitations

- You must manually update the model search index after adding or removing model library files from your model directories, or after adding or removing a model directory path. Search does not automatically recognize these changes.

Prerequisites

- Ensure that all model library files are in a folder that you specified as a IC model directory (Model-library path) (**Setup > Options > Directories**).
- Manually update the model search index (**Models > Generate Model Finder Index**).

Procedure

1. From the Assign Models dialog box, double-click a pin in the Pins list. The Select IC Model dialog box opens.
2. Click **Find Model**. The IC Model Finder dialog box opens.
3. Type a string into the “Search text” field. Search does not support wild-card characters.
4. Click **Search**.

Results

Search results display found models with a check mark in the Srch column of the spreadsheet. Click a column header to sort data. Double-click a model to assign it to a pin.

Troubleshooting Unexpected Model Assignments

If the expected model name does not appear in the Assign Models dialog box for a pin on the selected net, you may need to investigate the origin of the error.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 9-9. Possible Origins of a Model Assignment Error

Problem	Solution
Software recognizes a model assignment from an incorrect source (*.REF*, *.QPL*, or Assign Models dialog box)	When more than one source assigns a model to a pin, the software uses the model assignment from the source with the highest precedence. Assignments made from the Assign Models dialog box override *.REF* assignments, and *.REF* assignments override *.QPL* assignments. Models have higher precedence than values. The Assign Models dialog box identifies when IC model assignments are made by *.REF* and *.QPL* files. The Pins list displays an R next to the driver or receiver symbol when the *.REF* file assigns the model. Similarly, a Q is displayed when the *.QPL* file assigns the model. See Precedence Among Model and Value Selection Methods.
.REF or *.QPL* file syntax error.	The software ignores a line in a *.REF* or *.QPL* file that selects a model when it contains a syntax error. Additionally, for *.REF* files only, the software also ignores all lines that follow the syntax error. See Troubleshooting Automapping Model Assignment Errors.
Invalid *.REF* or *.QPL* model mapping.	While a line in a *.REF* or *.QPL* file may be syntactically correct, it can point to libraries or components/models that do not exist. See Troubleshooting Automapping Model Assignment Errors.
.QPL automapping is disabled.	If a *.QPL* file specifies the expected model selection, ensure that the ability to use a *.QPL* file for automapping is not disabled. See Disabling a REF or QPL File.
QPL File or Directory Name Error	If a *.QPL* file specifies the desired model selection, ensure the correct file or directory name is entered into the Set Directories dialog box. See Set Directories Dialog Box.

Troubleshooting Automapping Model Assignment Errors

When you create a large automapping file, you can accidentally introduce errors into the file. Use the REF-File Editor or QPL-File Editor to create and maintain automapping files to avoid these errors.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 9-10. Possible Errors in .REF and .QPL Files

Error	Description
Syntax errors in *.REF* files	Syntax errors in *.REF* files are reported immediately when you load the board or schematic. While you do not have to immediately correct syntax errors and reload the design, the erroneous line and all the lines following it are discarded. When a syntax error is reported for a *.REF* file, you can either ignore the error and use the Assign Models dialog box to load the model that failed to load automatically from the *.REF* file, or you can fix the *.REF* file to eliminate the errors, save the fixed file, and then continue working in BoardSim or LineSim. HyperLynx detects when the *.REF* file has changed, so if you choose to fix errors in the *.REF* file, the software automatically reloads it, and uses the information from the new version of the file.
Syntax errors in *.QPL* files	Syntax checks are not performed on *.QPL* file(s) when you load a design or when you load a *.QPL* file into the QPL-File Editor. If a *.QPL* file specifies the correct model selection, verify that the Set Directories dialog box contains the correct file information. See Set Directories Dialog Box. If a line in a *.QPL* file contains an error, the line is ignored, and the file continues to load. You have the following options when errors for a *.QPL* file are reported: • Ignore the errors, and interactively load the models that failed to load automatically from the *.QPL* file • Fix the *.QPL* file to eliminate the errors and save the file. When you edit a *.QPL* file, the software detects the new version and automatically reloads the file.
Model contents or model location error	Automapping files might specify a model or library that does not exist, or no longer exists at the location specified in the Model-Library File Path(s) box in the Set Directories dialog box. See Set Directories Dialog Box.
Errors reported during signal-integrity simulation	Model content or location errors are reported when the software loads models for the net under simulation. For batch signal-integrity simulation, the software suppresses real-time reporting of model-related errors because the simulation runs unattended. Therefore, model assignment errors are listed in the batch simulation report.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

REF and QPL File Syntax

Use the REF- or QPL-File Editors to avoid syntax errors when editing or creating *.REF* and *.QPL* files. If your company already has qualified parts list files in a non-QPL format, knowledge of the syntax can help you convert them.

Syntax Shared by REF and QPL Files

- Automapping files are ASCII format and cannot contain binary characters.
- Automapping files are case-insensitive.
- Comment lines are not supported. While the REF- and QPL-file editors insert comments at the top of the automapping files, any comments you manually add are not preserved.
- Each model record must be on a single line, with fields separated by commas.
- White space between fields is allowed.
- You can specify library or model files using a full network path if you want to be sure that the software uses a specific file.
- When you specify library or model files without a file path, the software assumes that the files are located in one of the directories listed in the Model-Library File Path(s) box in the Set Directories Dialog Box.
- Automapping files are comma delimited, which means the parameters on each line are separated from one another by a comma. Many other software packages can write data out in this format. This means that if you have bill-of-material data about the components on your board in another program, such as a database program, you may be able to create a template for the *.REF* or *.QPL* file by writing that data out in comma-delimited form.

Unique .REF File Formatting

Rules in this section apply only to *.REF* files. Square brackets [] enclose optional fields.

Table 9-11. .Ref File Formatting

Entry	Description
.IBS and .EBD/EMD models	Name of the component. The component name must match a component described in the specified library file. To specify an IBS IC model, use a line in the form: <reference_designator>, <library.IBS>, <component_name> To specify an EBD/EMD model, use a line in the form: <reference_designator>, <library.EBD>, <board_description> or <reference_designator>, <library.EMD>, <board_description>

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 9-11. .Ref File Formatting (cont.)

Entry	Description
Resistor, Capacitor, or Inductor models	The component name must match a component described in the specified library file. To specify an IBIS model, use a line in the form: <reference_designator>, MODEL, <library.IBS>, <component_name>
Resistor values	A resistor-value line can contain one or two values, depending on whether the resistor is a single component (discrete or a single-valued pull-up or pull-down network) or a pull-up/pull-down network. To specify a resistor value for a discrete resistor or for a network-package resistor with a single value, such as a pull-up resistor, use a line in the form: <reference_designator>, <value> To specify a resistor value for a network-package resistor with two values, such as for pullup and pulldown resistors, use a line in the form: <reference_designator>, <value1> [, [<value2>] [, <package_name>]] where <value1> maps to the common pin in the resistor package with the lower pin number. **Example:** If pin 5 represents a pullup common pin and pin 8 represents a pulldown common pin, <value1> represents the pullup common pin because pin 5 is a “lower” number than pin 8.
Capacitor values	To specify a capacitor value, use a line in the form: <reference_designator>, <value> To specify a capacitor value for a network-package capacitor with two values, use a line in the form: <reference_designator>, <value1> [, [<value2>] [, <package_name>]] where <value1> maps to the common pin in the capacitor package with the lower pin number. **Example:** If the package has two common pins 5 and 8, <value1> represents pin 5 because it is a “lower” number than pin 8.
Inductor values	To specify an inductor value, use a line of form: <reference_designator>, <value>

Unique .QPL File Formatting

Rules in this section apply only to *.QPL* files. Square brackets [] enclose optional fields.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 9-12. Unique .QPL Formatting

Entry	Description
.IBS IC model	Specify a component name. The component name must match a component described in the specified library file. To specify an .IBS IC model, use a line in the form: IC, <part_name>, “description”, <library.IBS>, <component_name>
EBD/EMD model	Specify a board description. The board description must match a board description described in the specified library file. To specify an .EBD/EMD IC model, use a line in the form: IC, <part_name>, “description”, <library.EBD>, <board_description> or IC, <part_name>, “description”, <library.EMD>, <board_description> The description field may be left blank by typing two double quotes followed by a comma or by typing just the comma. Examples: IC, <part_name>, “”, <library.EBD>, <component_name> IC, <part_name>, , <library.EBD>, <component_name>
Resistor, Capacitor, Inductor, or Ferrite bead models	Specify a component name. The component name must match a component described in the specified library file. To specify an .IBS model, use one of the following lines in the form: R, <part_name>, “description”, MODEL, <library.IBS>, <component_name> L, <part_name>, “description”, MODEL, <library.IBS>, <component_name> C, <part_name>, “description”, MODEL, <library.IBS>, <component_name> BD, <part_name>, “description”, <library.IBS>, <component_name>

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Table 9-12. Unique .QPL Formatting (cont.)

Entry	Description
Resistor values	A resistor-value line can contain one or two values, depending on whether the resistor is a single component (discrete or a single-valued pull-up or pull-down network) or a pull-up/pull-down network. To specify a resistor value for a discrete resistor or for a network-package resistor with a single value, such as a pull-up resistor, use a line in the form: R, <part_name>, “description”, <value> To specify a resistor value for a network-package resistor with two values, such as used for pullup and pulldown resistors, use a line in the form: R, <part_name>, “description”, <value1> [, [<value2>] [, <package_name>]] where <value1> maps to the common pin in the resistor package with the lower pin number. **Example:** If pin 5 represents a pullup common pin and pin 8 represents a pulldown common pin, then <value1> represents the pullup common pin because pin 5 is a “lower” number than pin 8.
Capacitor values	To specify a non-decoupling capacitor value, use a line in the form: C, <part_name>, “description”, <value> To specify a capacitor value for a network-package capacitor with two values, use a line of form: C, <part_name>, “description”, <value1> [, [<value2>] [, <package_name>]] where <value1> maps to the common pin in the capacitor package with the lower pin number. **Example:** If the package has two common pins 5 and 8, <value1> represents pin 5 because it is a “lower” number than pin 8.
Inductor values	To specify an inductor value, use a line in the form: L, <part_name>, “description”, <value>

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Table 9-12. Unique .QPL Formatting (cont.)

Entry	Description
Decoupling capacitor values	To specify a decoupling capacitor value and manually specify ESL, use a line in the form: DECAP, <part_name>, “description”, RLC, <C>, <ESR>, <ESL>, <capacitor_model_includes_mounting_inductance yes | no> To specify a decoupling capacitor value and automatically calculate ESL by automatically determining package dimensions (equivalent to using the QPL-File Editor ESL by capacitor size option) using the <Auto-estimate> list item), use a line in the form: DECAP, <part_name>, “description”, RLC, <C>, <ESR>, <capacitor_model_includes_mounting_inductance yes | no> To specify a decoupling capacitor value and automatically calculate ESL by manually specifying package dimensions, use a line in the form: DECAP, <part_name>, “description”, RLC, <C>, <ESR>, <package_width>, <package_length>, <package_height>, <capacitor_model_includes_mounting_inductance yes | no> Units for package dimensions can be: m, cm, mm, in, mil. To specify a decoupling capacitor model in a library, use a line in the form: DECAP, <part_name>, “description”, Library, <library_name>, <model_name> To specify a decoupling capacitor model in a SPICE file, use a line in the form: DECAP, <part_name>, “description”, SPICE, <file_name>, <device_name>, <capacitor_model_includes_mounting_inductance yes | no>, <model_node>=<capacitor_pin_name> To specify a decoupling capacitor model in a Touchstone file, use a line in the form: DECAP, <part_name>, “description”, Touchstone, <file_name>, <capacitor_model_includes_mounting_inductance yes | no>, <model_port>=<capacitor_pin_name>
Maximum length of the various fields	To specify the maximum length, use a line in the form: part_name - 40 description - 80 library_name - 100 component/model_name - 50

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Example REF File

```
U1, 701V.IBS, CGS701V
U2, MEMS.EBD, SDRAM512

R9, 69
RP1, 1000, , RES-SIP6-SERIES-1

C9, 81pF
C23, 33uF, , CAP-SIP14-PULLUP-1
```

Reference Designator	Assignment Description
U1	IC modeled as a CGS701V component from the .IBS library file 701V.IBS
U2	IC modeled as an SDRAM512 path description from the .EBD library file MEMS.EBD
R9	69 Ohm resistor
RP1	Resistor package modeled with the RES-SIP6-SERIES-1 component from the BSW.PAK package library, with a 1000 ohm value
C9	81pF discrete capacitor
C23	Capacitor package modeled with the CAP-SIP14-PULLUP-1 component from the BSW.PAK package library, with a 33uF value

Example QPL File

```
IC, CLOCK-701V, "National CGS701V", 701V.IBS, CGS701V
IC, RAM-512, "MemWell SDRAM512", MEMS.EBD, SDRAM512

R, R-1K, "", 1000
R, RPACK-1K, "", 1000, , RES-SIP6-SERIES-1

DECAP, C123, "", RLC, 10U, 5M, no
DECAP, C124, "", RLC, 10U, 5M, 3P, no
DECAP, C125, "", RLC, 10U, 5M, 0.032IN, 0.063IN, 0.035IN, no
DECAP, C126, "", LIB, DECAP.LIB, 10UF
```

Part Name	Description
CLOCK-701V	IC modeled as a CGS701V component from the IBIS library file 701V.IBS
RAM-512	IC modeled as a SDRAM512 path description from the EBD library file MEMS.EBD
R-1K	Resistor package modeled as RES-SIP6-SERIES-1 component from the BSW.PAK package library, with a 1000 ohm value
RPACK-1K	1000 ohm resistor

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Part Name	Description
C123	Decoupling capacitor with a nominal value of 10 uF, ESR of 5 milliohms, and an automatically-calculated ESL value based on a capacitor package size that is also automatically calculated. The ESL value does not account for mounting inductance.
C124	Decoupling capacitor with a nominal value of 10 uF, ESR of 5 milliohms, and ESL of 3 picoHenries. The ESL value does not account for mounting inductance.
C125	Decoupling capacitor with a nominal value of 10 uF, ESR of 5 milliohms, and package dimensions of width = 0.032 inch, length = 0.063 inch, and height = 0.35 inch. The ESL is automatically calculated from the capacitor package dimensions. The ESL value does not account for mounting inductance.
C126	Decoupling capacitor with values defined by the 10UF model in the DECAP.LIB library file. The ESL value comes from the model in the library.

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Package Modeling

You can create a custom package and model package parasitics.

Topic	Description
Modeling Package Parasitics	You can model package parasitics in an IBIS model file as RLC values, or with an S-parameter or SPICE model.
Modeling a Package in an IBIS File	At frequencies above 500 MHz, when an RLC package model may no longer be accurate, you can describe an IC package with one or more SPICE or S-parameter models from within an IBIS model by using the IBIS Package Wrapper.
Creating a Custom Package Description	If an appropriate package description is not available for a networked resistor or capacitor that you are trying to model, you can create a custom package description for the component.
USER.PAK File Format	The (*.PAK*) format describes the electrical connections in a resistor or capacitor network package. The software reads custom user package description definitions from *USER.PAK* and adds them to the default library in *BSW.PAK*.

Modeling Package Parasitics

You can model package parasitics in an IBIS model file as RLC values, or with an S-parameter or SPICE model.

Modeling a Package with RLC Values

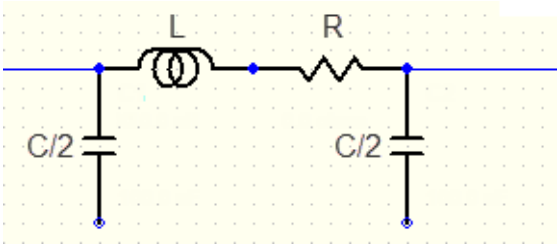
The software reads the following IBIS keywords and subparameters when you model package parasitics with R, L, and C values.

- [Package], R_pkg, L_pkg, C_pkg
- [Pin], R_pin, L_pin, C_pin
- [Define Package Model]
    - [Resistance Matrix], [Inductance Matrix], [Capacitance Matrix]

  When you enable “Use lumped representation of IBIS package parasitics” on the Preferences Dialog Box - Advanced Tab, the software models IBIS package parasitics as lumped elements. When the [Define Package Model] keyword contains R, L, or C matrices to define coupling between package pins, the simulation circuit may be more complex than illustrated by Figure 9-3. For example, it may include lumped mutual inductance elements.

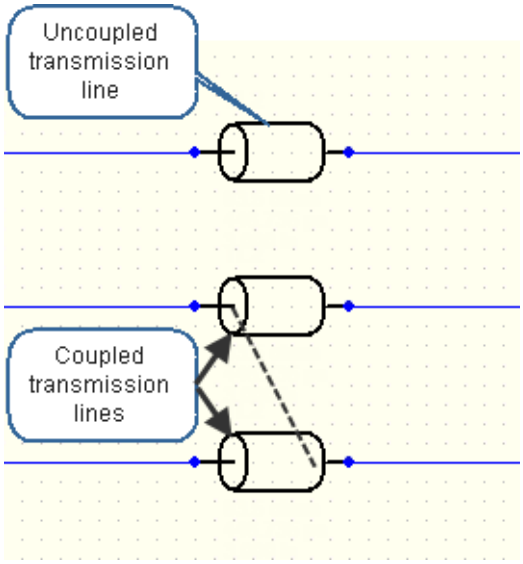
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Figure 9-3. IBIS Package Parasitics Modeled as Lumped Elements



When you disable, “Use lumped representation of IBIS package parasitics”, the software models IBIS package parasitics as equivalent transmission lines or coupled transmission lines. When the [Define Package Model] keyword contains R, L, or C matrices to define coupling between package pins, the simulation circuit may include coupled transmission lines, as illustrated by Figure 9-4.

Figure 9-4. IBIS Package Parasitics Modeled as Transmission Lines



Modeling a Package With an S-Parameter or SPICE Model

At frequencies above 500 MHz, when an RLC package model may no longer be accurate, you can describe an IC package with a SPICE or S-parameter model from within an IBIS model, by using the IBIS Package Wrapper. See “Modeling a Package in an IBIS File” on page 642.

Modeling a Package in an IBIS File

At frequencies above 500 MHz, when an RLC package model may no longer be accurate, you can describe an IC package with one or more SPICE or S-parameter models from within an IBIS model by using the IBIS Package Wrapper.

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You can describe package interconnects for both signal and power pins for power-aware SI simulation.

Restrictions and Limitations

- An IBIS model created by the IBIS Package Wrapper using the “old” [External Circuit]/[Circuit Call] interconnect modeling syntax can only be used in HyperLynx SI/PI.

Procedure

1. Choose the **Models > Wrap S-Parameter/SPICE Package Models in IBIS** menu item to open the IBIS Package Wrapper. The IBIS Package Wrapper dialog box opens.
2. In the IBIS File area, click **Select Master IBIS File** and select an *.ibs* model file. If needed, click **View File** to view the contents of the file in the Visual IBIS Editor.
3. Select the component to which you want to apply the package model from the Pick Component list.
4. In the Package Model area, click **Select Pkg File** and select an S-parameter or SPICE model file. If needed, click **View File** to view the contents of the file in the Touchstone and Fitted-Poles Viewer (S-parameter) or HyperLynx File Editor (SPICE).
5. If you have a SPICE model file that contains multiple models, select the correct one from the Pick Model list.
6. Next to Port Connectivity, select the pattern that correctly matches each die port to a pin port, or check Edit Manually and click the arrow buttons as needed.
7. If you need to map a single port combination (ports 1 and 2 for example) to multiple IC pin/die locations, click **Add Instance**.

   This makes an additional copy of ports available for assignment in the Map Ports/Pins list.
8. Map each die-pin port combination that you want to model (package ports) to the correct IC pin:

   a. Click a Package Port row from the left list, and click the corresponding IBIS Pin row from the right list.

      Hold the Ctrl or Shift keys to select multiple rows.

   b. Click the arrow button to assign the mapping.

   After you assign the last package port to a pin, the software creates another instance of all package ports.

   You can map a power package port to multiple reference pins if needed. For example, you can map one instance of a package port to one ground pin of an IC, and map a second instance of the package port to a pin on the same IC. The software highlights in yellow a pin that is mapped to multiple instances of a model.

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If you have a SPICE model, you can connect all negative port terminals to the simulator global reference node if needed. Select the reference node (SPICE node 0) in the Package Ports list and click **A_gnd**.

Note

For an S-parameter model, the software automatically connects negative port terminals to the simulator global reference.

Note

The software models unmapped package ports with terminators.

9. Click **Save Session**.

   The session file contains the port mapping information for the specified IBIS file. You can load the session file later, if you need to change the mapping or assign similar port mappings to another IBIS file.

10. Check “Use new interconnect modeling syntax” unless you want to continue to use the older HyperLynx-specific model syntax that includes [External Circuit] or [Circuit Call] references.

11. In the “Output IBIS file” field, edit the name of the new model file and click **Build Model**.

    The new model file opens in the HyperLynx Visual IBIS Editor.

Results

When “Use new interconnect modeling syntax” is checked, the software creates a new IBIS model file that includes an [Interconnect Model Group] keyword and the associated keywords.

When “Use new interconnect modeling syntax” is unchecked, the software does the following:

- Creates a new IBIS model file that includes [External Circuit] or [Circuit Call] references to the following files:
  - A *.sp* SPICE wrapper file (for an S-parameter-wrapped package model only) in the model folder.
  - A SPICE model that describes termination for any ports in the package model that you did not map in the IBIS Model Wrapper. S-parameter-ports are terminated with a resistor matching the reference impedance from the S-parameter model. SPICE model ports are terminated with a 50 ohm resistor.
- Sets any RLC package values (under the [Pin] keyword) for the mapped pins to zero in the original IBIS file.
- Marks package model descriptions ([Package Model] keywords) in the original IBIS file as comments.

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Creating a Custom Package Description

If an appropriate package description is not available for a networked resistor or capacitor that you are trying to model, you can create a custom package description for the component.

When you model a networked resistor or capacitor by assigning a value in the Assign Models dialog box or REF or QPL file editor, you must also select a package description that defines how the pins are connected. Assign a correct package description to ensure that the software recognizes net connections correctly.

When you load a board design, the software automatically attempts to determine the correct package description from this list. The software may not recognize a package description that is available in the default list. For example, if your design includes a 16-pin DIP series networked resistor, but pin 16 is unconnected, the software recognizes the component as a15-pin networked resistor, and cannot find a match.

The *BSW.PAK* file contains the list of default package descriptions that appear in the Assign Models dialog box and the REF or QPL file editor. To add additional package descriptions that appear at the end of this list, create a new file called *USER.PAK*.

Prerequisites

- Ensure that an appropriate package description for your networked resistor or capacitor is not already defined in the default list contained in the *BSW.PAK* file.
- Understand the PAK file format. See USER.PAK File Format.

Procedure

1. Create a text file named *USER.PAK* and save it as an ASCII file.
2. Using an ASCII-only text editor such as the HyperLynx File Editor (**Edit > Plane Text File**), edit the file:
   a. At the top of the file, place these two lines:

```
{PAK}
{VERSION=1.10}
```

   b. Add custom package descriptions as needed. Immediately following the two header lines, add the definition of the new package, followed by the {END} record. For

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example, to create a package description for a 9-pin networked resistor with pull-up style resistors, each with one end tied to a common pin, in a SIP package, type:

```
{PACK=9_PIN_SIP_PULLUP
(STYLE=R_PULLUP)
(SHAPE=SIP)
(TOTAL_PINS=9)
(PIN_PAIR=2,1)
(PIN_PAIR=3,1)
(PIN_PAIR=4,1)
(PIN_PAIR=5,1)
(PIN_PAIR=6,1)
(PIN_PAIR=7,1)
(PIN_PAIR=,1)
(PIN_PAIR=9,1)
(PIN_LOC=1,1)
(PIN_LOC=2,2)
(PIN_LOC=3,3)
(PIN_LOC=4,4)
(PIN_LOC=5,5)
(PIN_LOC=6,6)
(PIN_LOC=7,7)
(PIN_LOC=,)
(PIN_LOC=9,9)
}
{END}
```

To save time, you can copy and paste a similar package description from *BSW.PAK* as a starting point, then edit it as needed.

3. Save the file in ASCII format, and copy the file to your HyperLynx root directory (the same location as *BSW.PAK*). For example: *C:\MentorGraphics\<release>\SDD_HOME\hyperlynx*

Results

When you load a board design, the new package descriptions from *USER.PAK* appear at the end of the default package description list.

USER.PAK File Format

The (*.PAK*) format describes the electrical connections in a resistor or capacitor network package. The software reads custom user package description definitions from *USER.PAK* and adds them to the default library in *BSW.PAK*.

Format

A *USER.PAK* file must conform to the following formatting and syntax rules:

- An italicized field denotes a value that you must provide. Replace the italicized text with an appropriate value.

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- Square brackets [ ] denote optional parameters.
- All subrecords (lines beginning with “(“) must be on a single line.
- Curly braces { } can be separated from keywords and record ends by white space; the right brace } can be on the same line as the last subrecord or on the next line.
- Parentheses ( ) can be separated from keywords and record ends by white space; must be on the same line as the subrecord.
- If on the same line as other text, comments must be separated by at least one white space from the preceding text. If an entire line is a comment, it can begin in column 1, but must contain not contain the character '}'.
- The maximum allowed line length is 180 characters.
- Lines can be terminated by CR, LF, CR-LF, or LF-CR.
- White space is defined as space, horizontal tab, vertical tab, linefeed, form feed, or carriage return.
- Any characters are allowed in a name, except white-space characters.
- Numeric values can be followed by an exponent of the form exxx or Exxx, where xxx is any integer value, positive or negative.
- All numeric values can be followed by alphabetic scaling factors:

M	mega (1,000,000x)
K or k	kilo (1,000x)
m	milli (0.001x)
u or U	micro (1e-6x)
n or N	nano (1e-9x)
p or P	pico (1e-12x)

- Suffixes may be separated from their numeric values by white space.
- Scaling suffixes may be followed by other alphanumeric characters, e.g., uH or pF; the additional characters are terminated by white space

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- Template for a *.PAK* file:

```
Template
{PAK}
{VERSION=number [comment]}
{PACK=name of package description[comment]
(STYLE=type)
(SHAPE=type)
(TOTAL_PINS=number)
(PIN_PAIR=number,number,number)
(PIN_PAIR=number,number,number)
(PIN_LOC=1,number)
(PIN_LOC=2,number)
(PIN_LOC=3,number)
(PIN_LOC=4,number)
}
{END}
```

Parameters

Keyword/ Subrecord	Example Value	Description
PAK	{PAK}	• Required keyword that identifies the file format as *.PAK*. • Must be the first non-blank line in file. • Only one PAK keyword is allowed per file.
VERSION	{VERSION=*number* [comment]} {VERSION=1.02 new definition added by Dave S.}	• Required keyword that specifies the version of the file format. • Must be the second non-blank line in file. • Only one VERSION keyword is allowed per file.

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Keyword/ Subrecord	Example Value	Description
PACK	{PACK=*name* [comment] {PACK=RES-DIP4-PULLUPDN-1 (STYLE=R_PULLUP_PULLDOWN) (SHAPE=DIP) (TOTAL_PINS=4) (PIN_PAIR=1,2,4) (PIN_PAIR=3,2,4) (PIN_LOC=1,1) (PIN_LOC=2,2) (PIN_LOC=3,3) (PIN_LOC=4,4) }	• Identifies a package description as a resistor or capacitor package. • Must be at least one PACK record per file. • Name is the package's name; if it exceeds 20 characters, it will be truncated. • Three different subrecords must follow, in this order: STYLE, TOTAL_PINS, and PIN_PAIR
STYLE	(STYLE=*package_style*) [comment] (STYLE=R_PULLUP_PULLDOWN)	• Defines the package's component and connection style. • *package_style* specifies the package's style; valid values are: • R_SERIES • R_PULLUP • R_PULLUP_PULLDOWN • C_SERIES • C_PULLUP • R_xxx specifies a resistor package • C_xxx specifies a capacitor package • x_SERIES specifies that each element in the package has two independent pins, i.e., is independent of the other elements; • x_PULLUP specifies that each element in the package has one independent pin and one pin in common with the other elements; • x_PULLUP_PULLDOWN specifies that each element in the package has one independent pin and two pins in common with the other elements

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Keyword/ Subrecord	Example Value	Description
SHAPE	(SHAPE=*package_shape*) (SHAPE=DIP)	• Defines the package's physical shape • *package_shape* valid values are: DIP (dual-in-line package) SIP (single-in-line package)
TOTAL_PINS	(TOTAL_PINS=*number*) (TOTAL_PINS=4)	• Specifies the total number of pins on the package • *number* must be a positive integer
PIN_PAIR	(PIN_PAIR=*pin_name*, *pin_name*[,pin_name]) [comment] (PIN_PAIR=1,2,4) (PIN_PAIR=3,2,4)	• Specifies the pairing of two or three pins on a package. • *pin_name* can be any valid name string (typically an integer number, but can be alphabetic; if it exceeds 5 characters, it will be truncated. • All styles except R_PULLUP_PULLDOWN allow two pin_name fields; R_PULLUP_PULLDOWN requires three pin_name fields • For x_SERIES styles, both pin_name fields are for independent pins; • For x_PULLUP styles, the first pin_name field is for the independent pin and the second is for the common pin; • For the R_PULLUP_PULLDOWN style, the first pin_name field is for the independent pin, the second is for the common pull-up pin, and the third is for the common pull-down pin

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Keyword/ Subrecord	Example Value	Description
PIN_LOC (part 1 of 2)	(PIN_LOC=*pin_name*,*location_number*) (PIN_LOC=1,1) (PIN_LOC=2,2) (PIN_LOC=3,3) (PIN_LOC=4,4)	• Specifies the physical position of each pin on the package • *pin_name* is the name of the pin; must match a name specified in a PIN_PAIR record • *location_number* is an integer number that specifies the named pin's position on the package, according to the following rules: • if the package shape is SIP and the package has n pins, the pin names must be numbered from 1 to n with 1 defined as the top-most pin, n as the bottom-most pin; • if the package shape is DIP and the package has n pins, the pin names must be numbered from 1 to n with 1 in standard DIP style: pin 1 in the upper left-hand corner, pin n as the pin in upper right-hand corner
PIN_LOC (part 2 of 2)		• the PIN_LOC information is used for drawing the package's internal connections; if missing, the internal connections are not drawn when the package is displayed in BoardSim's user interface • the PIN_LOC records must come AFTER the PIN_PAIR records; if a PIN_LOC record for a pin comes before the PIN_PAIR record for that pin, the .PAK file is considered to have a syntax error • each STYLE, TOTAL_PINS, or PIN_PAIR record must be on a single line • every pin on a package must be mentioned at least once in a PIN_PAIR *pin_name* field; independent pins can be mentioned only once; common pins can be mentioned multiple times
END	{END}	Identifies the end of the file

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Examples

Example 1: (a 16-pin, series-style DIP resistor)

```
{PACK=R_16_SER_SIP 16-pin, series-style DIP resistor
(STYLE=R_SERIES) package
(SHAPE=DIP)
(TOTAL_PINS=16)
(PIN_PAIR=1,16)
(PIN_PAIR=2,15)
(PIN_PAIR=3,14)
(PIN_PAIR=4,13)
(PIN_PAIR=5,12)
(PIN_PAIR=6,11)
(PIN_PAIR=7,10)
(PIN_PAIR=8,9)
(PIN_LOC=1,1)
(PIN_LOC=2,2)
(PIN_LOC=3,3)
(PIN_LOC=4,4)
(PIN_LOC=5,5)
(PIN_LOC=6,6)
(PIN_LOC=7,7)
(PIN_LOC=8,8)
(PIN_LOC=9,9)
(PIN_LOC=10,10)
(PIN_LOC=11,11)
(PIN_LOC=12,12)
(PIN_LOC=13,13)
(PIN_LOC=14,14)
(PIN_LOC=15,15)
(PIN_LOC=16,16)
}
```

Example 2: (a 9-pin, pull-up-style SIP capacitor with alphanumeric pin names)

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```
{PACK=cap_9_comm_sip 9-pin, pull-up-style SIP capacitor
(STYLE=C_PULLUP) package, with alpha pin names
(SHAPE=SIP)
(TOTAL_PINS=9)
(PIN_PAIR=B,A) pin A is to the common pull-up voltage
(PIN_PAIR=C,A)
(PIN_PAIR=D,A)
(PIN_PAIR=E,A)
(PIN_PAIR=F,A)
(PIN_PAIR=G,A)
(PIN_PAIR=H,A)
(PIN_PAIR=I,A)
(PIN_LOC=A,1)
(PIN_LOC=B,2)
(PIN_LOC=C,3)
(PIN_LOC=D,4)
(PIN_LOC=E,5)
(PIN_LOC=F,6)
(PIN_LOC=G,7)
(PIN_LOC=H,8)
(PIN_LOC=I,9)
}
```

Example 3: (a 10-pin, SIP pull-up/pull-down resistor)

```
{PACK=R_PACK_9SIP 10-pin, SIP pullup/pull-down resistor
(STYLE=R_PULLUP_PULLDOWN)
(SHAPE=SIP)
(TOTAL_PINS=10)
(PIN_PAIR=2,1,10) pin 1 is to the common pull-up voltage;
(PIN_PAIR=3,1,10) pin 10 is to the common pull-down voltage
(PIN_PAIR=4,1,10)
(PIN_PAIR=5,1,10)
(PIN_PAIR=6,1,10)
(PIN_PAIR=7,1,10)
(PIN_PAIR=8,1,10)
(PIN_PAIR=9,1,10)
(PIN_LOC=1,1)
(PIN_LOC=2,2)
(PIN_LOC=3,3)
(PIN_LOC=4,4)
(PIN_LOC=5,5)
(PIN_LOC=6,6)
(PIN_LOC=7,7)
(PIN_LOC=8,8)
(PIN_LOC=9,9)
(PIN_LOC=10,10)
}
```


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Timing Models

DDRx simulation requires timing models for controller and DRAM components.

Topic	Description
Timing Model Editor	Use the HyperLynx Timing Model Editor (**Models** > **Edit DDRx Timing Models**) to edit timing models for controller and DRAM components.
Creating a Timing Model	Use this procedure to create your own controller timing model.
Creating Controller and DRAM Timing Models	Use the HyperLynx Timing Model Wizard to create a simple timing model for a DDRx memory controller. Use the HyperLynx Timing Model Editor to modify an existing timing model.
Required Controller Timing Model Parameters	The software requires timing models for controllers to contain a minimum set of parameters.

Timing Model Editor

Use the HyperLynx Timing Model Editor (**Models** > **Edit DDRx Timing Models**) to edit timing models for controller and DRAM components.

Creating a Timing Model

Use this procedure to create your own controller timing model.

A timing model contains the maximum or minimum setup and hold times for each type of receiver pin (such as data and address) relative to the associated strobe/clock, maximum skew between certain pin pairs, signal launch delay of one pin relative to another, and so on.

Some parameters exist in the DRAM timing model or in the controller timing model, but not in both timing models. Some parameters exist in both DRAM and controller timing models, which are likely to specify different values for the same parameter (depending on whether the measurement is performed on a read or write cycle).

Restrictions and Limitations

- All DRAMs must have the same timing requirements and use the same timing model. Differences are flagged in red.

Prerequisites

- You have acquired the DDRx Wizard license.
- You are familiar with the required parameters from the data sheet.

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- You understand how required parameters are defined.

Procedure

1. Open the DDRx Controller Timing Model Wizard (**Models > Run DDRx Controller Timing Model Wizard**).
2. Select the DDR technology to use in the timing model.
3. Follow the wizard pages to construct your timing model.
4. Click **Finish** to save your timing model.

Results

Your timing model is now ready to assign.

Creating Controller and DRAM Timing Models

Use the HyperLynx Timing Model Wizard to create a simple timing model for a DDRx memory controller. Use the HyperLynx Timing Model Editor to modify an existing timing model.

DRAM and Controller timing models are required by the DDRx Wizard to specify the timing requirements at both ends for the interfacing signals. The DDRx Wizard uses the parameters from the timing models to derive the final timing margins.

These timing parameters include skew, delay, and setup and hold time requirements on signals with respect to their associated strobe or clock signal. DRAM timing models are standardized since the timing specifications at the DRAMs are specified by the JEDEC standards. However, controllers can have timing requirements that differ from vendor to vendor.

Default timing models for memory and controller ICs ship with HyperLynx, and are located in the *Libs* folder. For example, *C:\MentorGraphics\<release>\SDD_HOME\hyperlynx\Libs*.

For more information on creating memory controller timing models, see MG575901 (DDR2 and DDR3) and MG592006 (DDR4 and LPDDR4).

Required Controller Timing Model Parameters

The software requires timing models for controllers to contain a minimum set of parameters.

The DDRx Batch-Mode Wizard and HyperLynx Timing Model Wizard require controller timing parameters as inputs. Include the timing relationships in Table 9-13 in the specific block of a Memory Controller timing model (signal edge qualifiers are ignored). Note that parameter tDQDQS and parameters tDS/tDH are complementary methods of describing the DQ-to-DQS timing during a read cycle; specify only one of the two methods in the timing model file.

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Table 9-13. Required Controller Timing Parameters

Model File Relationship	Active Cycle	Description
tCKAC(min) and tCKAC(max)		
$delay(ck, addr_cmd, min, max);	Read/ Write	The minimum and maximum delay between the address and command output signals transitioning to valid before the rising edge of the output clock (CK). See Figure 9-5. • Applicable to address (A) and command (BA, RAS, CAS, and WE) signals on both read and write cycles. • Supports both 1T and 2T timing on signals. 2T timing is when the delay between a valid transition of the Address and Command signals and the rising edge of the clock is more than one clock cycle.
tCKCTL(min) and tCKCTL(max)		
$delay(ck, ctl, min, max);	Read/ Write	The minimum and maximum delay between the control output signals transitioning to valid before the rising edge of the output clock. See Figure 9-6. • Applicable to control signals (CS, CKE, and ODT) for both read and write cycles. • Supports only 1T timing.
tCKDQS(min) and tCKDQS(max)		
$fullskew(dqs, dq, min, max);	Write	The minimum and maximum skew between the rising edge of the output data strobe (DQS) and the rising edge of the output clock (CK). See Figure 9-7. • Applicable to CK and DQS signals during a write cycle.
tDQSDQ(min) and tDQSDQ(max)		
$delay(dqs, dq, min, max);	Write	The minimum and maximum delay between the data and data mask output signals transitioning to valid and the associated output data strobe edge (either rising or falling). See Figure 9-8. • Applicable to data(DQ) and data mask (DM) signals and its associated DQS signal during a write cycle.
tDS and tDH		

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Table 9-13. Required Controller Timing Parameters (cont.)

Model File Relationship	Active Cycle	Description
$setup(dq, dqs, tDS); $hold(dqs, dq, tDH); Or $setuphold(dqs, dq, tDS, tDH);	Read	Data setup (tDS) and hold (tDH) window relative to the associated DQS. See Figure 9-9. • During a read cycle, the DRAMs output the transitions of the DQS signal approximately aligned with the transitions of the DQ signals. Typically, the controller captures the DQ signals by phase shifting the DQS signal internally by about 1/4 of a clock period. • Specify whether the setup and hold parameters are measured at the controller pins (phase shift = 0) or after an internal 90° phase shift. See Figure 9-9. • Setup values are positive if DQ must be valid before DQS (negative if after); Hold values are positive if DQ must remain valid until after DQS (negative if before). • Applicable to data(DQ) signals and its associated DQS signal during a read cycle.

Figure 9-5. tCKAC

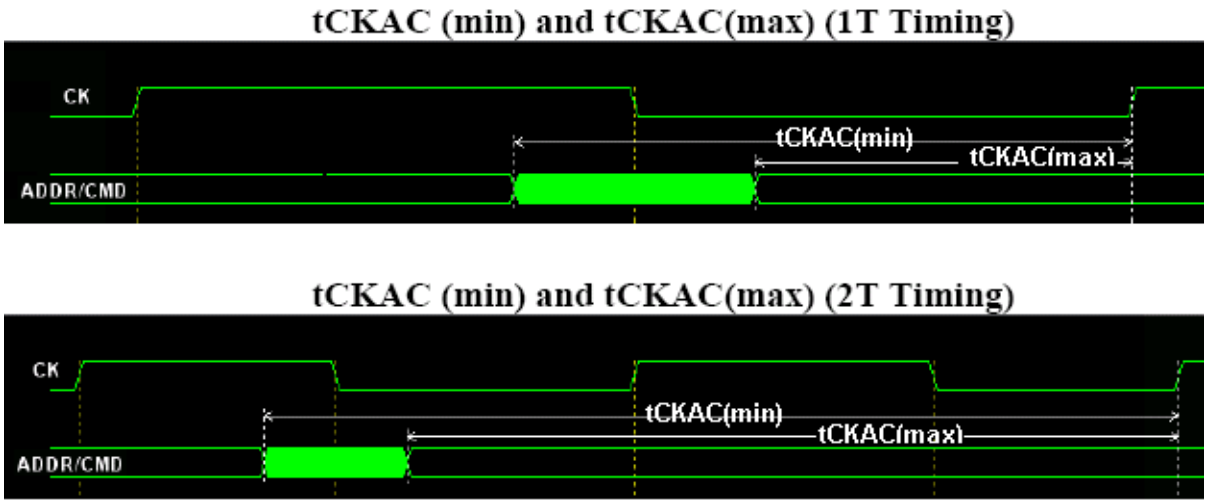
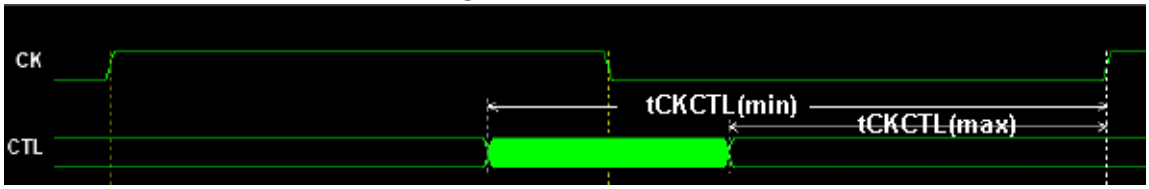


Figure 9-6. tCKCTL



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Figure 9-7. tCKDQS

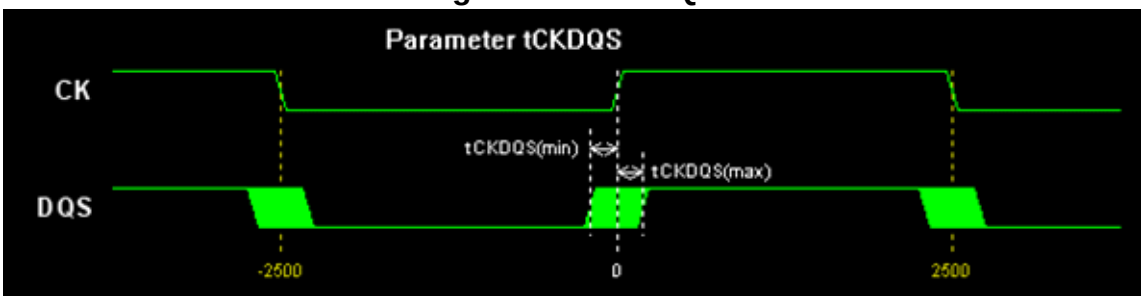


Figure 9-8. tDQSDQ

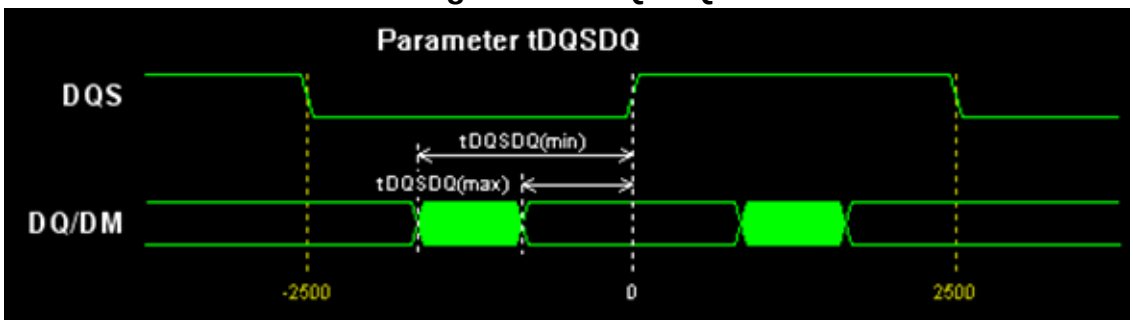
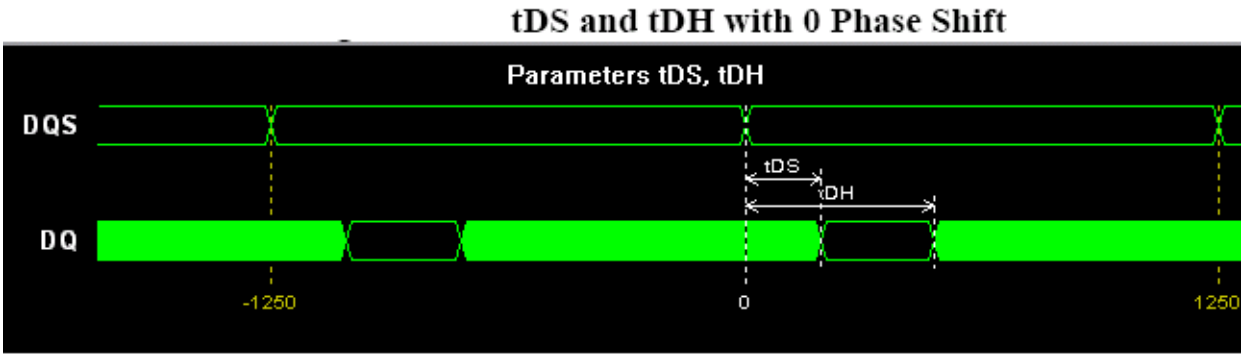
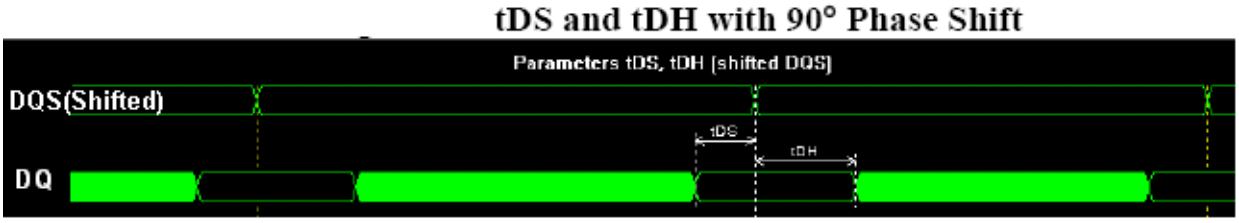


Figure 9-9. tDS and tDH with Phase Shift





Power Integrity Models

You can describe the electrical behavior of IC and voltage-regulator module (VRM) components for PI simulation by assigning PI models.

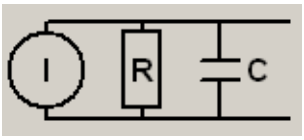
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You can specify current sink and voltage source information for IC and VRM components. If you want to account for return-current paths during PI simulation, assign a reference net when you assign a PI model.

AC Current Sink Model

An AC model typically represents I/O buffer switching and IC core-logic power on/off transitions. It represents a current sink waveform for a IC power supply pin.

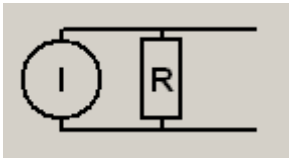
Figure 9-10. Electrical Schematic for an AC Current Sink Model



DC Current Sink Model

A DC model typically represents the steady state current consumption of an IC or other component in the design that draws significant current. It represents a constant current sink for a IC power supply pin.

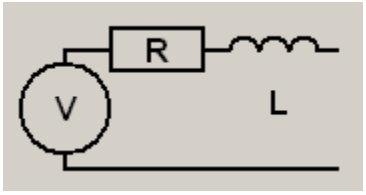
Figure 9-11. Electrical Schematic for a DC Current Sink Model



VRM Voltage Source Model

A VRM model typically represents the power supply voltage in the design. It represents a voltage source for an IC power supply pin.

Figure 9-12. Electrical Schematic for a VRM Voltage Source Model



For frequency-domain simulation, VRMs (with their low-R DC paths) can significantly lower PDN impedance at low frequencies. By contrast, a PDN without a VRM behaves like a simple capacitor at low frequencies. PDNs with high impedance, especially at very low frequencies when there is no VRM model, can cause exaggerated plane-noise results.

For time-domain simulation, VRMs work with the current-source stimulus to avoid a too-high voltage near time zero, due to high impedance at very low frequencies.

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Related Topics

Assigning Models for PI Simulation

Reference Nets

Reference nets provide return-current paths for current sinking into IC power-supply pins. Reference nets may be implemented across multiple stackup layers, where stitching vias connect metal areas on different stackup layers.

In LineSim, you specify the reference layer(s) when assigning AC or VRM models.

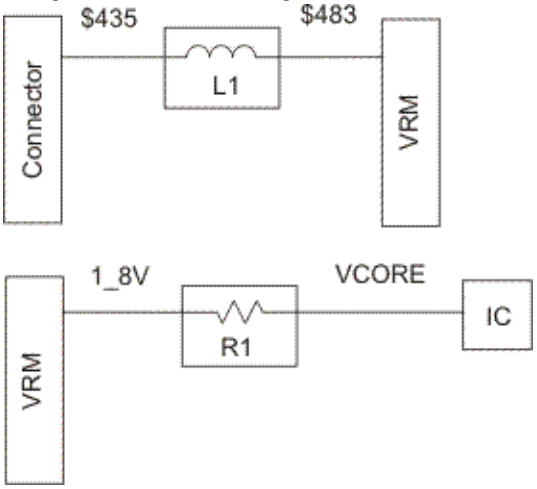
In BoardSim, you specify the reference net and layer(s) when assigning AC or VRM models. BoardSim automatically finds available reference layers when you specify the reference net. You can deselect an available reference layer if you know that it is not well connected by stitching vias to another reference layer.

Series Components for Power-Supply Nets

Series components connect one power-supply net to another power-supply net. Simulation can include both nets that are connected to the component. These nets are called associated nets.

Figure 9-13 shows how power supply nets are included in simulation when connected to resistors or inductors that connect to VRMs. Inductor L1 associates nets $435 and $483. Resistor R1 associates nets 1_8V and VCORE.

Figure 9-13. Example Series Components for Power-Supply Nets



Non-resistor/inductor components, such as high-current power FETs, can also associate power-supply nets.

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You assign values to series components in a BoardSim board. You can edit existing series connection resistance/inductance values, plus you can define series connections through a resistor package.

Restriction

You cannot model series components for power-supply nets in the PDN Editor.

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Chapter 10
Reference - Dialog Boxes

This chapter describes windows and dialog boxes for the software.

Topic	Description
3D-Area Creation Options	Use this dialog box to define options for creating new 3-D areas for selected nets, and for finding board areas that match existing 3-D areas.
3D-Area Manager Dialog Box	Use this dialog box to create or edit 3-D areas, and run HyperLynx DRC to find other matching 3-D areas in your design. The software can represent a 3-D area with an S-parameter model that you create with HyperLynx Full-Wave Solver.
Add Connector Model Dialog Box	Use this dialog box to assign a passive SPICE and S-Parameter (Touchstone) model to a connector.
Add or Edit 3D Area Dialog Box	Use this dialog box to define an area and topology on which to run a 3-D electromagnetic (EM) simulation to create a S-parameter model.
Add Signal Via Dialog Box	Use this dialog box to add single-ended and differential vias in the PDN.
Add/Edit Decoupling Capacitor(s) Dialog Box	Use this dialog box to add or edit decoupling capacitors to the PDN.
Add/Edit IC Power Pin(s) Dialog Box	Use this dialog box to add or edit IC power-supply pins in a power-distribution network (PDN).
Adding an Eye Mask to a FastEye Diagram	The FastEye Viewer enables you to overlay a FastEye diagram with a mask that displays the “keep out” regions that the eye must avoid.
Advanced AMI Options Dialog Box	Use this dialog box to set advanced AMI-related options, such as the number of samples per bit interval, number of bits per call to AMI DLL, number of starting bits to ignore, or advanced troubleshooting options.
Advanced Channel Characterization Dialog Box	Use this dialog box to set advanced channel characterization options, such as the characterization type, channel response, linearity check, and reporting options.
Advanced Jitter Settings Dialog Box	Use this dialog box to specify jitter settings, save jitter settings to a file, or load jitter settings from a file.

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Topic	Description
AMI File Assignment Dialog Box	Use this dialog box to assign *.AMI* and *.DLL (Windows)/.so (Linux)* files to the channel driver and receiver.
AMI Model Testing Dialog Box	Use this dialog box to set AMI model troubleshooting options.
Assign / Edit Capacitor Model Dialog Box	Use this dialog box to assign a model to a decoupling capacitor.
Assign Decoupling-Capacitor Groups Dialog Box	Use this dialog box to edit decoupling-capacitor groups.
Assign Decoupling-Capacitor Models Dialog Box	Use this dialog box to assign a model to a decoupling capacitor or a series passive component that connects two power supply nets.
Assign IC Component Model Dialog Box	Use this dialog box to assign an IBIS model to an IC component symbol, hide pins that you do not plan to include in simulation, and to edit pin location.
Assign Models Dialog Box	Use this dialog box to assign a model to a specific IC pin or ferrite bead. For board designs, this dialog box has additional tabs enabling you to edit passive component values, assign Quick Terminators, and to display series bus switch connectivity.
Assign Power Integrity Models Dialog Box	Use this dialog box to assign power-integrity models to power-supply pins and values to supply-net resistors, inductors and other components.
Assign S-Parameter/SPICE Model Dialog Box	Use this dialog box to assign passive SPICE and S-Parameter (Touchstone) models to a package/connector symbol.
Assign Stimulus Dialog Box	Use this dialog box to assign stimulus to specific pins or nets in the design. For a board design, you can assign stimulus to individual pins or to a net (and all its pins). For a schematic, you can assign stimulus to individual pins. You can also import a DDRx stimulus (*.txt*) file.
Add/Edit VRM Dialog Box	Use this dialog box to specify the electrical characteristics of the voltage-regulator model (VRM) assigned to the IC power supply pin.
Auto-Create Groups Options Dialog Box	Use this dialog box to choose the types of IBIS model information the software can use to automatically create power supply pin groups.
Auto-Grouping Dialog Box	Use this dialog box to automatically create decoupling-capacitor groups.
Batch DC Drop Simulation Dialog Box	Use this dialog box to simulate DC drop for multiple power supply nets at a time.
General Batch Wizard	Use this wizard to set up and run general batch SI and EMC simulations.

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Topic	Description
Bathtub Chart Dialog Box	Use this dialog box to display and document bathtub curves. Bathtub curves help identify valid data sampling locations by reporting the bit error rate (BER) as a function of the sampling location across the unit interval (UI, same as bit interval) at several voltage offsets.
Change Trace Widths Dialog Box	Use this dialog box to change the width of a trace on nets or stackup layers you specify. This capability helps you perform “what if” experiments to improve signal quality.
Channel Characterization Dialog Box	Use this dialog box to set up simulation properties for a new channel characterization. Optionally, you can view channel-response waveforms automatically created by channel analysis or manually created by this dialog box.
Configure Eye Diagram Dialog Box - Eye Mask Tab	Use this tab to edit eye mask properties for eye diagram analysis. You can load existing eye masks from a library or save new eye masks into a library.
Configure Eye Diagram Dialog Box - Stimulus Tab	Use this tab to define complex driver or transmitter stimulus.
Configure IC Component Symbol	Use this dialog box to hide pins that you do not plan to include in simulation, and to edit pin location.
Confirm Connections Dialog Box	Use this dialog box to reverse connections between two IC component symbols.
Connect Nets with Manhattan Routing Dialog Box	Use this dialog box to perform “what if” trace routing and component placement experiments on your board design by editing the length of a routed, unrouted, or partially-routed signal net.
Coupling Settings Dialog Box	Use this dialog box to specify trace, package, and area fill coupling settings.
Coupling Settings Dialog Box (LineSim)	Enables you to specify which coupled nets are included with those you select in the design area.
CTLE Settings Dialog Box	FastEye Analysis allows you to simulate the effect of a CTLE high-pass peaking filter at the receiver for your design.
DC Drop Analysis Dialog Box	Use this dialog box to interactively simulate DC drop for power-supply nets.
DDR2 Slew Rate Derating Dialog Box	Reports derated slew rates for DDR2 data, address, and control pins, and to report non-derated slew rates for DDR differential clock and strobe pins. As part of filling out a timing budget spreadsheet for the DDR2 signal, you can use the reported values to look up the following derating values in datasheet tables: delta tDS, delta tDH, delta tIS, and delta tIH.

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Topic	Description
DDR5 Wizard	Use this wizard to set up the parameters required to simulate a single-ended net (such as DQ) for a DDRx interface that supports equalization. If you have run DDRx batch simulation and want to investigate results for a single-ended net with equalization, you can simulate that net with the DDR5 Wizard. The wizard can account for equalization behaviors represented by IBIS-AMI models, sweep AMI parameters, and report results as eye diagrams.
DDRx Batch-Mode Wizard	Use this wizard to set up the parameters required to run a batch simulation on a DDRx interface.
Decoupling Wizard	Use this wizard to run Lumped Analysis, Quick Analysis, or Distributed Analysis decoupling simulation.
Define Constraint Template Dialog Box	Use this tab to specify length and delay constraints for the net and its pin pairs, constraints for differential pairs, FromTos that appear in the FromTos section of the exported template file, and view pin sets and modify the type property for pin sets.
Design Changes Dialog Box	Use the Design Changes dialog box to generate a concise report of all the component changes you have made on your board to improve signal quality or lower radiated emissions. A layout designer or service bureau uses this record of changes you want made to your board in its next revision. You could also use the list yourself to drive changes in schematics for the board.
Differential Pair Net Names Dialog Box	Use this dialog box to specify differential pair rules for all board designs and save them to *BSW.INI*.
Differential Pairs Dialog Box	Use this dialog box to create net name pairing rules that the software can use to automatically recognize differential pairs in your board design. You can also use this dialog box to manually specify differential net pairs and differential pin pairs.
Digital Oscilloscope Dialog Box	Use this dialog box to interactively simulate signal integrity and display the results as waveforms or eye diagrams. You can simulate the selected net and its associated nets in your board design, or simulate all the nets in the schematic that have an enabled driver.
Display 3D Board Area Dialog Box	Use this dialog box to rotate and view a board design from multiple angles to more easily see net topology and power-distribution network (PDN) implementation. 3-D viewing can be especially useful to see how a signal via passes through metal areas and how a decoupling capacitor connects to a PDN.
Edit AC Power Pin Model Dialog Box	Use this dialog box to specify the electrical characteristics and stimulus waveform of the current source model assigned to the IC power-supply pin. AC models typically represent I/O buffer switching and IC core-logic power on/off transitions.

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Topic	Description
Edit DC Power Pin Model Dialog Box	Use this dialog box to specify the electrical characteristics of the current sink model assigned to the IC power-supply pin.
Edit Power-Supply Nets Dialog Box	Use the Edit Power-Supply Nets dialog box to edit power supply net properties.
Edit Stimulus Dialog Box	Use this dialog box to create or edit a stimulus.
Edit Transmission Line Dialog Box	Use this dialog box to edit transmission line properties and define coupling regions.s
Export Constraint Template Dialog Box	Use this dialog box to generate a constraint template file based on a selected net, or modify an existing template file.
Export Nets to S-Parameters in Batch Mode Dialog Box	Use this dialog box to create S-parameter models from multiple single nets or differentially paired nets. The software creates an S-parameter model for each net or differential pair of nets that you select in the dialog box.
Export to LineSim Free-Form Schematic Dialog Box	Use this dialog box to create a LineSim schematic for selected signal and power supply nets.
Eye Height Sampling Dialog Box	Use this dialog box to specify the time within the unit interval (UI) you want to measure the height of the eye diagram. If you know when the receiver circuitry actually samples the logic state of the waveform, enter that time.
FastEye Channel Analyzer	Use the FastEye Channel Analyzer to simulate a SERDES channel to investigate how channel topology, Rx/Tx equalization and pre-emphasis parameters, jitter, and crosstalk, affect channel performance when you do not have IBIS-AMI models that describe transmitters or receivers in your SERDES design.
FastEye Viewer	Use the FastEye Viewer to display and measure FastEye diagrams created by the FastEye Channel Analyzer.
Field Solver Dialog Box	Use this dialog box to run the field solver and view electrical field lines.
Find Component Dialog Box	Use this dialog box to locate a specific component or pin on a board or power distribution network (PDN) layout. After you specify the component or pin, the view changes to show and highlight the component or pin. To remove the highlighting, click in the board viewer.
Free-Form Schematic Editor	Use this interface to draw or modify a signal-integrity schematic.

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Topic	Description
Generate Back-Annotation File/Data Dialog Box	Use this dialog box to save certain types of changes you made to your board design in a *.ECO* file so that you can pass those changes back to your board layout program or schematic editor. For example, if you changed values for several termination components, you can use this dialog box to automatically pass the new values back to your layout program.
Highlight Net Dialog Box	Use this dialog box to locate nets in the board viewer or PDN Editor. Nets are only highlighted and not selected for simulation.
HyperLynx Eye Density/BER Viewer	Use this dialog box to display eye density and BER simulation results.
HyperLynx PI PowerScope Dialog Box	Use this dialog box to display graphical simulation results for planes and metal areas in the design.
IBIS AMI Parameter Editor	Use this editor to display and change AMI parameter values.
Import Constraints from Constraint Manager Dialog Box	Use this dialog box to select the Constraint Manager project that contains the constraints or net groupings (net classes) that you want to import.
License Status Dialog Box	Use this dialog box to see the status of currently acquired licenses.
Interactive Simulation Dialog Box	Use this dialog box to run interactive SI simulation on selected nets in a board design or an entire schematic. When simulation completes, EZwave displays waveforms.
Interactive Simulation with Measurements Dialog Box	Use this dialog box to run interactive SI simulation and select the types of measurements to perform on selected nets in a board design or on an entire schematic.
Interactive Sweeps Dialog Box	Use this dialog box to run sweeps SI simulation on selected nets in a board design, or on an entire schematic.
Interactive Sweeps with Measurements Dialog Box	Use this dialog box to run sweeps SI simulation on the selected nets in a board design or on all the nets in a schematic and display waveforms and measurements.
Layer Mapping Dialog Box	Use the Layer Mapping dialog box to map stackup layers in the source design to stackup layers in the current design. This dialog box opens only when you import a stackup containing more layers than the current design.
Load/Save Waveforms Dialog Box	Use this dialog box to load one or more previously-saved waveform files into the oscilloscope or FastEye Viewer or to save the latest waveforms to a comma-separated values (*.csv*) or an *.lis* file with HyperLynx-specific formatting.

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Topic	Description
Measurements Dialog Box	Use this dialog box to set up the simulation details for the SI and crosstalk measurements that you enable on the Interactive Simulation with Measurements dialog box.
Edit Net Classes Constraints and Stimuli Dialog Box	Use this dialog box to set up constraints for General Batch simulation.
Model Connections Dialog Box	Use this dialog box to enable the software to automatically detect port connections for an S-parameter model, or create, import, or remove a connection file (.connect) that is associated with a S-parameter or SPICE model.
New HyperLynx Full-Wave Project Dialog Box	Use this dialog box to start creating a new HyperLynx Full-Wave Solver project file (*.v3d*) project file based on defaults or an existing project file.
Options for New Terminators Dialog Box	Use this dialog box to specify reference designator and part type information for Quick Terminators that you include in back annotation.
Padstack Editor Dialog Box	Use this dialog box to set or edit pad and anti-pad properties for padstacks.
Padstack Manager Dialog Box	Use this dialog box to manage the padstacks in the schematic.
PDN Model Extractor Wizard	Use this wizard to set up and export a power-distribution network (PDN) to an S-parameter model.
PDN Net Manager Dialog Box	Use this dialog box to view, add, and delete power supply nets.
Pin Group Manager Dialog Box	Use this dialog box to group IC power-supply pins together, making it possible for decoupling analysis to view PDN impedance through the pins in the group in parallel.
Power Network Dialog Box	Use this dialog box to verify that the software recognizes power and signal nets correctly, visualize power net connections in the PDN, assign power integrity models, or begin setup for batch DC drop simulation.
Preferences Dialog Box	Use this dialog box to view and edit properties that affect how the software operates and helps you set up a working environment that best suits your design and the way you work.
QPL-File Editor	Use the QPL-File Editor to create or edit a *.QPL* automapping file, which assigns models and values to components with specific part names.
REF-File Editor	Use the REF-File Editor to create or edit a model assignment for a reference designator in your design.
Reporter Dialog Box	Use this dialog box to display simulation messages and results.

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Topic	Description
Restore Session Edits Dialog Box	The software saves any board changes you make during a session, such as IC output buffer enable/disable settings and stackup edits. These changes are saved to the BoardSim User Data session file (*.BUD*). When the board finishes loading, this dialog box opens, allowing you to choose which session file to load with your board.
Save MultiBoard Session Edits Dialog Box	Use this dialog box to specify how to save changes for a board that is used more than once in a multiple board project.
Script Editor	Use the VBScript Environment or Python Environment window to write and debug scripts.
Select Active Layers Dialog Box	Use this dialog box to select the layer on which you want to place objects in the PDN Editor and to select which layers to display.
Select Directories for IC-Model Files Dialog Box	Use this dialog box to specify one or more directories on the computer or network that contain IC models available for simulation.
Select Directories for Stimulus Files Dialog Box	Use this dialog box to specify one or more directories on your computer or network that contain stimulus files (*.EDS*).
Select IC Model Dialog Box	Use the Select IC Model dialog box to display the contents of IC model libraries, to find a model in a library, and to assign a model to a reference designator in the design.
Select Method of Simulating Vias Dialog Box	The software can automatically calculate via inductance and capacitance or you can manually specify the information. These options enable you to simulate the effects of padstacks in the current PCB layout and to perform “what if” experiments to simulate the effects of changed padstacks.
Select the Instance Dialog Box	Use this dialog box to select an instance of a board design (and its session edits) to load into the software.
Set Coupling Thresholds Dialog Box	Use this dialog box to enable electrical or geometric coupling thresholds, and set coupling threshold options.
Set Directories Dialog Box	Use this dialog box to set the folder location for designs, models, stimulus, reports, and so on.
Set Reference Net Dialog Box	Use this dialog box to specify the net that provides return current paths for the selected power supply pin(s).
Set Spectrum Analyzer Probing (EMC) Dialog Box	Use this dialog box to set a probe for EMC simulation. The software accounts for differential-mode radiation caused by current flowing on a single-ended signal trace and an ideal return path. The software does not report radiation for differential-pair signal traces.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
Setup Anti-Pads and Anti-Segments Dialog Box	Use this dialog box to specify clearances among objects on the same stackup layer. Clearances created by anti-pads in the board file and the PDN Editor are used for accurate power-integrity simulation and PI/SI co-simulation, and for board display.
Simulation Controls Dialog Box	Use this dialog box to specify advanced simulation options for running signal integrity simulations. This includes additional waveform probe locations, simulation engine options, and a way to enable SI/PI co-simulation.
Simulation Results Dialog Box	Use this dialog box to display delay, SI, and crosstalk measurements for the nets simulated by the Interactive Simulation with Measurements Dialog Box or Interactive Sweeps with Measurements Dialog Box.
Specify Device Kit for Current Design Dialog Box	Use this dialog box to choose a specific device kit and specify additional information about the device kit.
Specify DFE Dialog Box	Use this dialog box to specify tap weight values for decision-feedback equalization (DFE) circuitry in the receiver. You can specify tap weights by typing values in the spreadsheet or by reading in a *.TAPS* file containing the tap weights synthesized by a previous run of the wizard.
Specify Pre-Emphasis Dialog Box	Use this dialog box to specify tap weight values for pre-emphasis circuitry in the driver. You can specify tap weights by typing values in the spreadsheet or by reading in a *.TAPS* file containing the tap values synthesized by a previous run of the wizard.
Spectrum Analyzer Dialog Box	Electromagnetic simulation (EMC) allows you to measure radiated emissions from a selected net on your board design. You can compare simulation results to government regulation limits, such as FCC for the United States, to help you decide whether to change the design. You can also use a current probe to measure peak current for a net across a frequency range for schematic and board designs.
SPICE Options Dialog Box	Use this dialog box to specify SPICE simulation parameters, options, and include files.
Stackup Manager Dialog Box	For a design that has multiple stackups, use this dialog box to edit the master stackup or a local stackup.
Statistical Contour Chart Dialog Box	Use this dialog box to display a nested series of eye opening contours and their bit error rate (BER).
Sweep Manager Dialog Box - Setup Tab	Use this tab to define the set of design property values (sweep range) to apply to a design property during sweep simulations.
Sweep Manager Dialog Box - Simulation Cases Tab	Use this tab to display the combination of design property values for each sweep simulation, to optionally stop sweep simulations if a simulation fails, and report failed simulations.

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Topic	Description
Sweeping Dialog Box	Use this dialog box to define sweep ranges. The format of this dialog box depends on whether the design property is swept by numerical values or by named values. For example, use numerical values to sweep dielectric thicknesses and use named lists to sweep IC process corners.
Synthesize DFE Dialog Box	Use this dialog box to specify how many taps exist in the receiver equalization circuitry.
Synthesize Pre-Emphasis Dialog Box	Use this dialog box to specify how many taps exist in the driver pre-emphasis circuitry and how many of them are pre-taps.
Synthesized DFE Weights Dialog Box	Use this dialog box to display optimum decision-feedback equalization (DFE) tap weight values and to save them to a file (*.TAPS*).
Synthesized Pre-Emphasis Weights Dialog Box	Use this dialog box to display optimum pre-emphasis tap weight values and to save them to a file (*.TAPS*).
Target-Z Wizard	Use this wizard to specify the peak transient current transmitted through a pair of power supply nets, the nominal voltage provided by the voltage-regulator module (VRM) and its ripple, and read the output of the target-Z calculation.
Translator Options Dialog Box	For some PCB design systems, key information needed to create BoardSim boards is not stored in a predictable way. Use this dialog box to provide data such as attribute names, to indicate how the information is stored and which information to include during translation.
Via Model Extractor Wizard	Use this wizard to set up and export signal vias to S-parameter models.
Via Properties Dialog Box	Use this dialog box to specify the type of electrical model and via properties for a signal via symbol in a schematic.
Via Visualizer Dialog Box	Use this dialog box to display the electrical and geometric properties of a signal via or a pair of coupled signal vias, and to export a SPICE netlist representing via electrical properties. You can also use this dialog box to display the geometric properties of vias connected to a power-supply net.
View Options Dialog Box	Use this dialog box and right-click menus to control how the board viewer displays objects.
Viewing Filter Dialog Box	Use this dialog box to control the visibility of individual stackup layers and highlighted nets in the board viewer.
xPCB/xDX View	Use this dialog box to view layout designs stored in *.CCE* format in xPCB/xDX View or the BoardSim board viewer.

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Topic	Description
Zooming and Examining a FastEye Diagram	You can use any of the controls in the following table to examine FastEye diagrams.

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3D-Area Creation Options

Scope: BoardSim

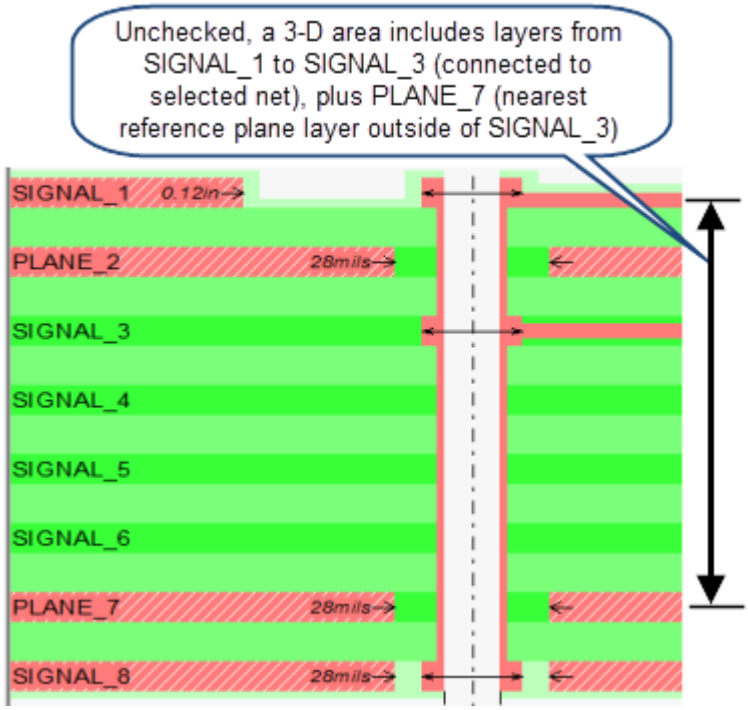
To access: From the 3D-Area Manager dialog box, click **Options**

Use this dialog box to define options for creating new 3-D areas for selected nets, and for finding board areas that match existing 3-D areas.

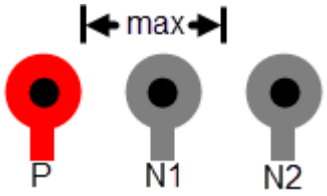
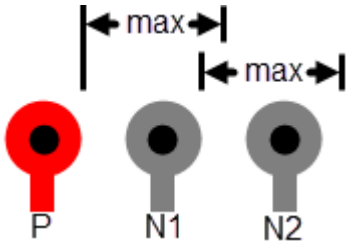
Fields

Field	Description
Extra objects to include in 3D areas section	
SMD pads one layer above/ below plane	Checked, includes an SMD pad that is immediately above or below a reference plane layer. This option can create areas with two layers: One layer with the SMD pad and another layer for the reference plane.
Single-ended nets	Checked, includes single-ended nets. Unchecked, creates areas only for differential pairs.

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Field	Description
Layers outside (via connection span + nearest references)	Unchecked, includes stackup layers that connect a via or pin to traces on a selected net, plus the nearest reference plane layers outside of the signal layers connected to a selected net. The nearest reference plane layer can be several layers away from a connected signal layer. A 3-D area for the figure below includes all layers, except for SIGNAL_8. Checked, includes all stackup layers for a via or pin, even layers that do not connect to a selected net. In this case, a 3-D area for the figure below would additionally include layer SIGNAL_8. 
Distance thresholds section	
Stitching-via search radius	Specifies the maximum distance between a via or pin on a selected net to a stitching via to include in an area. The software measures distance from pad edge to pad edge.
Max distance from area boundary to objects	Specifies the maximum distance from an area boundary to nearby objects to include in an area. The distance is measured from an area boundary to an edge of a neighboring object, such as a via pad or trace. The software adjusts (that is, moves out) the area boundary to include a nearby object.

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Field	Description
Coupling distance between victim & aggressor pins/pads	Specifies the maximum horizontal distance between a pin or pad on a primary net to a pin or pad on a neighboring net that you want to include in a 3-D area. The software measures distance from pad edge to pad edge. For an example measurement, see Figure 8-3 on page 470. For information about "primary net", see "3-D Areas That Include Multiple Nets" on page 467.
Limit 3D area to primary net and coupled nets	Checked, a 3-D area includes only neighboring nets that are within the maximum coupling distance you specify (with the "Coupling distance between victim & aggressor pins/pads" option) from a primary net. In the figure below, the 3-D area includes N1 and excludes N2.  Unchecked, a 3-D area includes: • Neighboring nets within the maximum coupling distance from a primary net. Let us call these neighboring nets an "initial set of neighboring nets". • Neighboring nets within the maximum coupling distance from an *initial set of neighboring nets*. The software repeats this inclusion until it finds no additional nets. Unchecking this option is useful when you want to create a 3-D area for all pins in a connector. In the figure below, the 3-D area includes N1 and N2. 
Max solve frequency	Specifies the upper solving frequency that HyperLynx Full-Wave Solver uses to create an S-parameter model that represents a 3-D area.
Max mismatches for area matching section	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Port position Signal-via position Stitching-via position	Specifies the maximum position mismatch that can exist between an object in an existing 3-D area and an object in a potential new 3-D area.
Stitching-via count	Specifies the maximum stitching via count mismatch that can exist between an existing 3-D area and a potential matching 3-D area.
Drill diameter	Specifies the maximum drill diameter mismatch that can exist between an existing 3-D area and a potential matching 3-D area.
Match only if total area within	Specifies the maximum metal area mismatch, on a layer by layer basis, that can exist between an existing 3-D area and a potential matching 3-D area.

Related Topics

Modeling a Board Area With an S-Parameter Model

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

3D-Area Manager Dialog Box

Scope: BoardSim

To access: **Analyze 3D EM > Manage Areas for 3D EM Simulation**

Use this dialog box to create or edit 3-D areas, and run HyperLynx DRC to find other matching 3-D areas in your design. The software can represent a 3-D area with an S-parameter model that you create with HyperLynx Full-Wave Solver.

Fields

Field	Description
Selected nets	Displays selected nets. Select Nets — Opens Select Nets dialog box, so you can change net selection.
Filter nets and area by boards (Available only for MultiBoard projects)	Checked, the: • Selected nets list contains nets only for selected board(s). • 3D Areas table contains 3-D areas only for selected board(s).
3D areas	Lists 3-D areas that you defined and any matching 3-D areas found by HyperLynx DRC. Table columns: • Left (unlabeled) — Select to highlight a 3-D area in the board viewer. If you enable “Zoom to selection”, the board viewer will zoom to a highlighted 3-D area. • On — Checked, represent a 3-D area with an S-parameter model. Unchecked, represent a 3-D area with board design information. • Name — 3-D area name. 3-D areas automatically defined by HyperLynx DRC use this naming format: *3D_<victim_net name>_x_and_y_coordinates_for_3-D_area_center>_<primary_net_quantity>_PNT*. • Model — File name of an S-parameter model assigned to the 3-D area. Models automatically defined by HyperLynx DRC use this naming format: *3D_PV_<signal_pin_and_via_quantity>_P_<port_quantity>_N_<signal_net_quantity>_<x_and_y_coordinates_for_3-D_area_center>.sNp*. Names displayed with red text and ending with “(N/A)” indicate that you have not solved an area to create a model.
Show only areas on selected nets	Checked, the 3D areas table contains areas only for selected nets.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
View Models	Opens the Touchstone Viewer and loads S-parameter models for areas selected in the table.
Zoom to selection	Checked, the board viewer zooms and pans to show a 3-D area that you select with the left (unlabeled) column in the 3D areas table.
HLAS Options Folder	Specifies an optional folder that contains HyperLynx Advanced Solver options files that override default settings. See "HyperLynx Advanced Solvers Option Files for 3-D Areas" on page 551.
Manually edit areas section	
New	Opens the Add or Edit 3D Area dialog box, where you can define a new 3-D area and assign an S-parameter model to it.
Edit	With one 3-D area selected in the table — Opens the Add or Edit 3D Area dialog box, where you can edit a 3-D area. With more than one 3-D area selected in the table — Opens the Edit 3D Areas dialog box, where you can edit solver options.
Delete	Deletes 3-D area(s) you selected in the table and deletes the *found_3D_area_patterns.txt* file.
Import	Enables you to import the text file (...*MentorGraphics*\<*release*>*SDD_HOME**hyperlynx64**HypFiles**found_3D_area_patterns.txt*) that contains the found set of 3-D areas generated after HyperLynx DRC runs the 3D Area Pattern Match rule.
Automatically create areas section	
Create Areas for Selected Nets	Runs HyperLynx DRC (in the background) to find 3D areas, and adds new 3-D areas and model names to the table. 3-D area name format is *3D_<victim_net name>_x_and_y_coordinates_for_3-D_area_center>_<primary_net_quantity>_PNT*. Model name format is *3D_PV_<signal_pin_and_via_quantity>_P_<port_quantity>_N_<signal_net_quantity>_<x_and_y_coordinates_for_3-D_area_center>.sNp*.
Find Matches for Highlighted Areas	Runs HyperLynx DRC to find matching 3-D areas for areas you selected in the table. You can use the left (unlabeled) table column to select one or more 3-D areas to match. The search scope depends on whether you have selected nets: • No nets selected — Search all nets for matching 3-D areas. • Nets selected — Search only selected nets for matching 3-D areas.
Options	Opens the 3D-Area Creation Options dialog box, where you can specify area matching options.

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Field	Description
Solve Highlighted Areas	Runs HyperLynx Full-Wave Solver (in the background) on the 3-D area(s) you selected in the table. Click **Stop** to abort a solve.
Solve on Multiple Machines	Opens the Distributed Solve dialog box and loads the HyperLynx Advanced Solvers project file (PHYS) for each 3-D area you selected. You can use this dialog box to solve a 3-D area with a job distribution cluster. You can also use "local mode", which enables you to solve on a remote Linux computer or multiple simulator instances on one computer. See "Job Distribution" in the *HyperLynx Advanced Solvers User Guide*.

Related Topics

Modeling a Board Area With an S-Parameter Model

Exporting Part of a Board Design for Analysis in HyperLynx Full-Wave Solver

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Add Connector Model Dialog Box

To access:

- In BoardSim, open the MultiBoard Project Wizard (**File > New MultiBoard Project**), click **Next** to advance until connector model options appear, select **Advanced** as the Connector model, then click the **Assign** button.

Use this dialog box to assign a passive SPICE and S-Parameter (Touchstone) model to a connector.

Fields

Field	Description
Model	Specifies a type of model to assign.
Files list	Lists available models. S-parameter (Touchstone) files display an *SNP* extension, such as *S4P*. SPICE files display an *SP* extension.
Device	Lists the device models available for a SPICE model.
Library Path	Opens the Select Directories for IC-Model Files dialog box, to identify folders that contain models you want to assign.
Left Side Ports Right Side Ports	Enables you to assign model ports to component pins, power-supply nets, or nothing (not connected). • Port — Displays the name of a port. • Fltr — Type text in this field to filter the list or ports. • Arrow buttons (>> <<) — Move a port from one side to the other
Auto-Place Ports	Distributes all the ports on the board depending on the directions you select from the dropdown list.
View Model File	(Available for Touchstone models.) Opens the HyperLynx Touchstone and Fitted-Poles Viewer, which enables you to view S-parameter model data.
Edit Model File	(Available for SPICE models.) Opens the HyperLynx File Editor, which enables you to edit the script content of the model file.
Edit Parameters	Opens the Eldo/SPICE Model Parameters dialog box, where you can specify parameter-value pairs.
Model Connections	Opens the Model connections dialog box.

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Add or Edit 3D Area Dialog Box

Scope: BoardSim

To access: **Analyze 3D EM > Manage Areas for 3D EM Simulation**, then do one of the following:

- To add an area, click **New**.
- To edit an area, select an area from the list and click **Edit**.

Use this dialog box to define an area and topology on which to run a 3-D electromagnetic (EM) simulation to create a S-parameter model.

Fields

Field	Description
Name	Defines the name of the 3-D area. The software uses this name to create the corresponding S-parameter file name. **Tip:** Create separate 3-D areas for different structures on the board, or for variations of one area (such as sets of stackup layers or area fills connected to reference nets).
Board (Available only for MultiBoard projects)	Defines the board in a MultiBoard project that contains the 3-D area you want to export.
Apply to all same boards (Available only for MultiBoard projects that use multiple instances of a board)	Checked, uses the current dialog box settings for all copies (instances) of that board in your MultiBoard project. Default value is checked. Unchecking this option is not recommended unless you are familiar with how it works.
Area of Interest	Defines the area to export. Select a shape and fill out the associated fields to define that shape: • Rectangle — Defines the area of the rectangle. Enter coordinates for the sides or click and drag a rectangle over a section on the board viewer. Check “Whole Board” to export the entire board. • Polygon — Defines the area of the polygon. Enter coordinates for the sides or click and drag each line of the polygon in the board viewer. To close the polygon, click **Close** or drag a line to the starting point of the first line. To remove a line from the polygon, click to the left of the table entry to highlight the entire row, then click **Delete**. • Snap To The Grid — Checked, snaps a dragged line or corner to the coordinate resolution you enter. Does not apply to coordinates you enter.

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Field	Description
Included Data Area — Enables you to filter the board data to determine what to export. **Tip:** To reduce 3-D EM simulation run time, consider exporting only the minimum amount of data required to simulate the topology of interest.	
Check Primary Nets	Displays nets included in the 3-D area. If an included net is part of an *electrical net*, this field displays all members of the electrical net. Checked, a net is a crosstalk victim. Unchecked, a net is a crosstalk aggressor. To change net inclusion, click **Select Nets** to open the "Select Net by Name" dialog box.
Supply Nets	Displays power-supply nets enclosed by the selected area, on any stackup layer.
Layers	Displays stackup layers. Check a layer to include data from that layer in the export.
Ports Area — Enables you to specify the ports and port numbering for the S-parameter model created by the software to represent the 3-D area.	
Choose Ports	Displays the nets, pins, and passive components available as ports for the S-parameter model. Check an object to include it as a port. **Caution:** If the 3-D area includes a passive component that you want to model, ensure that you select the single port (formed across both pins) for that component in the Choose Ports list. The software represents an unchecked series passive component port as an open circuit.
Port Map	Displays the S-parameter model port index for each net you selected in the Choose Ports list. Click in the # column to change the port index.
Simulation Area — Enables you to specify the location of the S-parameter model and simulation options.	
Model File	Defines the location of the S-parameter model. Enter or browse to the location. The default file name uses the form *<name>.sNp*, where N is the number of ports that you enable in the Choose Ports list.
Minimum Frequency Maximum Frequency	Displays the frequency range for the S-parameter model created by 3-D EM simulation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Use Absorbing Boundaries	Checked, the software models the area boundary edges as an absorbing material (PCB) rather than a reflective material (air) and eliminate artificial resonances from the model. The software slightly expands the exported area on all sides to provide room for the absorbing boundary. You can compare results with this option enabled/disabled to evaluate how well-isolated the area is from the rest of your design.
Use External Reference	Checked, for areas under a BGA, the software adds a plane layer to the 3-D model that represents the reference layer of the BGA die to create a more accurate representation of the design. Select this option when a board area includes only a partial reference layer or inconsistent reference throughout the area under the BGA. This ensures that the software creates ports for BGA pins in a consistent location (at the external reference).
View	Opens the Touchstone Viewer and displays the S-parameter model assigned to the 3-D area.
CCE File	Defines the location of a CCE file that represents the 3-D area. Click **Export** to create a CCE file.
Export	Saves a CCE file that represents the 3-D area.
Simulate Stop	• Simulate — In the background, HyperLynx 3D Solver creates an S-parameter model representing the 3-D area and assigns it to the area. • Stop — Stops an in-progress simulation.
Advanced Solver	Opens HyperLynx 3D Solver and loads a Simulation that represents the 3-D area. You can review and change 3-D area properties, and then mesh and solve to create an S-parameter model. • Keep Existing Project — If an existing project has the same name as the 3-D area, HyperLynx 3D Solver opens the saved project and does not create a new project that uses information from this dialog box. • Create New Project — If an existing project has the same name as the 3-D area, HyperLynx 3D Solver creates a new project and overwrites the existing project. **Note:** This option is available when you have acquired the 3D Area Model Export and Full-Wave Solver licenses.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
3D Explorer Variability	• Open 3D Explorer — Supports post-layout design variations. Use this option to evaluate the effects of PCB manufacturing tolerances. • Create Pre-Layout Template — Supports pre-layout design variations (more flexible). Use this option to optimize your design, such as adding or moving stitching vias near a signal via. These options open HyperLynx 3D Explorer and load a template to represent the 3-D area. 3D Explorer can help you set up many "what-if" design variations, run a solver on each variation, create an S-parameter model for each variation, and consolidate results into a report. You can then choose an S-parameter model for one of the design variations to assign to the 3-D area. See "Exploring Design Variations" in the *HyperLynx Advanced Solvers User Guide*.

Related Topics

Modeling a Board Area With an S-Parameter Model

Exporting Part of a Board Design for Analysis in HyperLynx Full-Wave Solver

Add Signal Via Dialog Box

Scope: LineSim

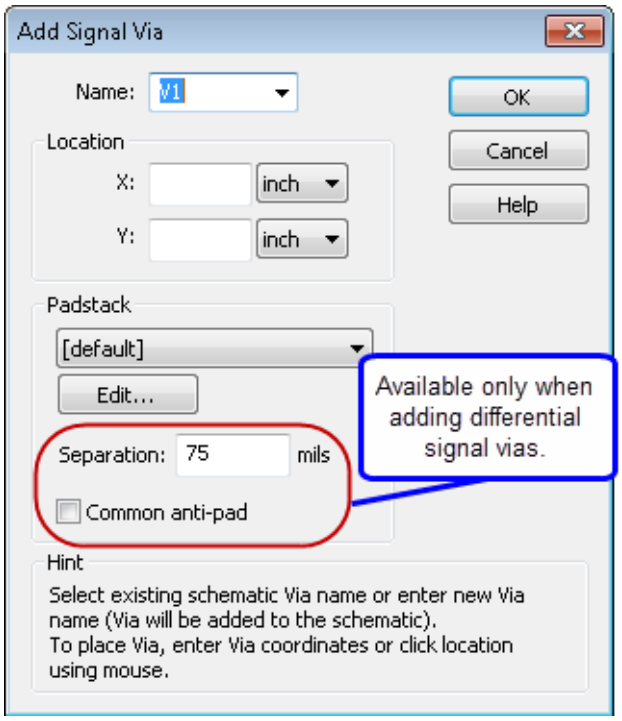
To access: In the PDN Editor, select Add Signal Via or Add Differential Signal Via .

Use this dialog box to add single-ended and differential vias in the PDN.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

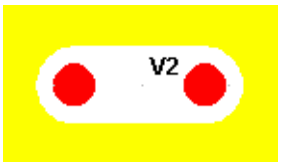
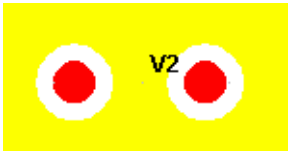
Signal vias are not shared between the PDN Editor and LineSim schematic editor unless you add them to the PDN Editor first. If you are setting up for power-integrity simulation and want the signal via in the schematic to interact with the PDN, add the via using the PDN Editor.



Fields

Field	Description
Name	Defines the name of the via.
Location	Defines the location of the via. Enter coordinate values or click a location in the PDN Editor to populate the X, Y fields. **Note:** For a differential via, the X, Y coordinates define the mid-point between the two via barrels.
Padstack	Defines the padstack for the via. Click **Edit** to open the Padstack Editor dialog box, which enables you to edit parameters for the selected padstack.
Separation	For differential vias, defines the separation between the via barrel centerlines.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Common anti-pad	Checked, uses a single anti-pad that encloses both barrels in a differential via.  Unchecked, uses individual anti-pads for each barrel in a differential via. 

Related Topics

Creating a PDN Design

Add/Edit Decoupling Capacitor(s) Dialog Box

Scope: LineSim

To access: In the PDN Editor, do one of the following:

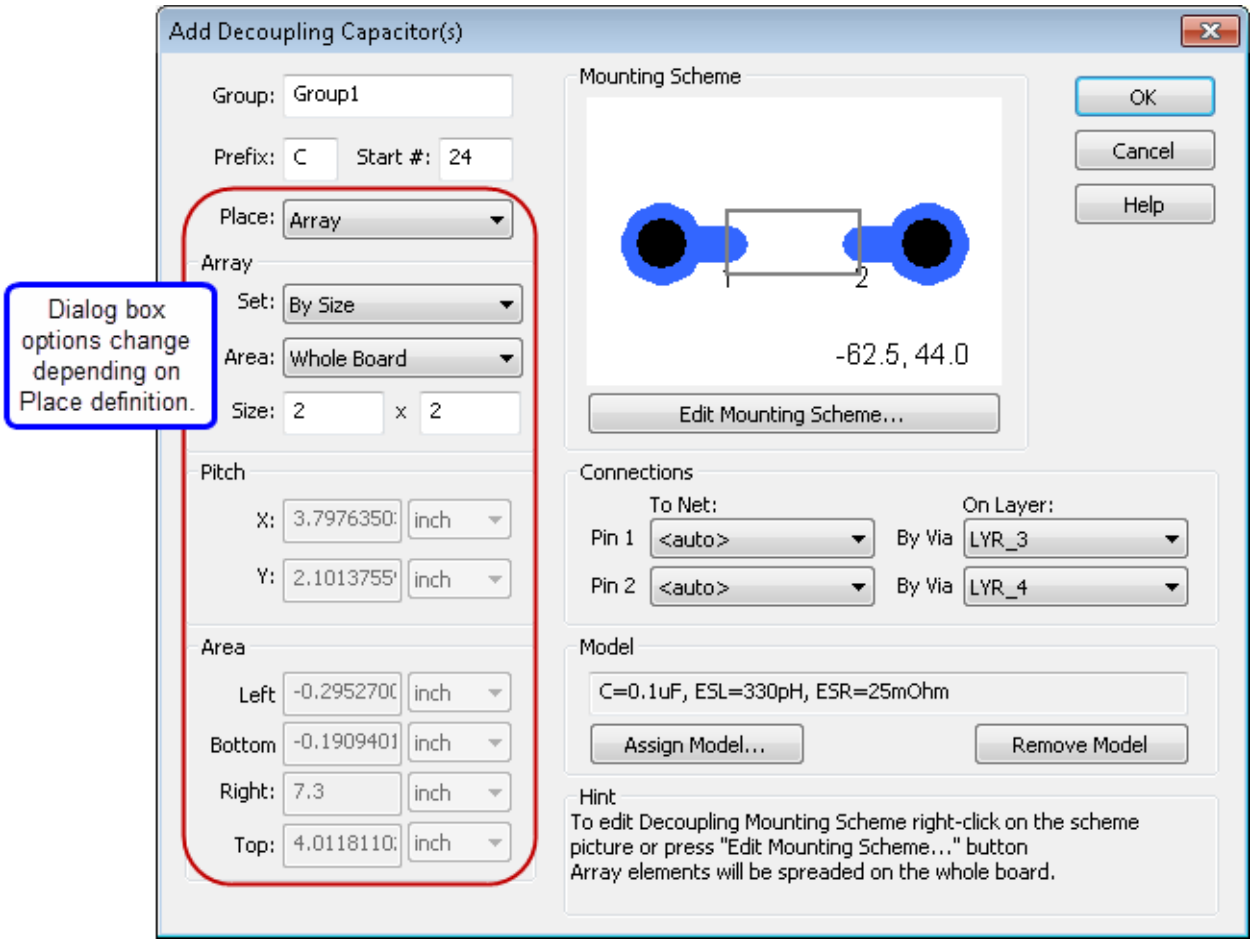
- To add a decoupling capacitor, click
- To edit a decoupling capacitor, double-click a capacitor.

Use this dialog box to add or edit decoupling capacitors to the PDN.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

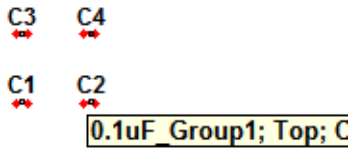
To include a power supply net in simulation that is connected by a series passive component to a power supply net you have probed, check “Allow AC supply models for non-capacitor components” in the Preferences dialog box. See Preferences Dialog Box - Power Integrity Tab. If you check this option, the wizard displays the Supply Component Models page instead of the Check Capacitor Models page.



Fields

Field	Description
Place	Defines the placement method for the capacitor(s). The dialog box options change depending on whether you select **Single** or **Array** from the dropdown list.
Name	Defines the name of the single capacitor. For example, C1.
Location	Defines the location of the single capacitor. Enter coordinate values or click a location in the PDN Editor to populate the X, Y fields.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Group	Defines the name of the capacitor array. **Note:** Each capacitor array acts as a single object in the PDN Editor. Edits apply to all the members of the array. 
Prefix	Defines the reference designator prefix for all the capacitors in the array.
Start #	Defines the number from which to begin numbering the capacitors in the array.
Array	• Set — Defines the array By Size (number of columns and rows) or By Pitch (distance between capacitors). • Area — Defines the area in which to distribute the array, across the Whole Board or within a selected Rectangular Area. • Size — Defines the number of columns and rows in the array.
Pitch	Defines the distances between the capacitors on the X and Y axes of the array.
Area	Defines the boundary of the rectangular area in which to place the array. Enter coordinate values, or select a region in the PDN Editor.
Mounting Scheme	Enables you to manipulate the mounting scheme. • Open the popup in the view window to add/edit Vias, Pins, and Traces, delete objects or segments, load a pre-existing scheme, and save the current scheme. • Click **Edit Mounting Scheme** to enlarge the view window.
Connections	Defines the via connections in the mounting scheme. Default is Via 1 on the left and Via 2 on the right. • To Net — Defines the power-supply net for each via. Select a net from the dropdown list. Select **<auto>** when the IC pin connects to a stackup layer with one power-supply net. **Note:** Use the PDN Net Manager Dialog Boxto add nets. • On Layer — Defines the stackup layer(s) for each via. Select a layer from the dropdown list. Select **<multiple>** to define more than one layer.
Model	Displays the defined model. Click **Assign Model** to open the Assign / Edit Capacitor Model Dialog Box, which enables you to define the model. Click **Remove Model** to remove the model.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Creating a PDN Design

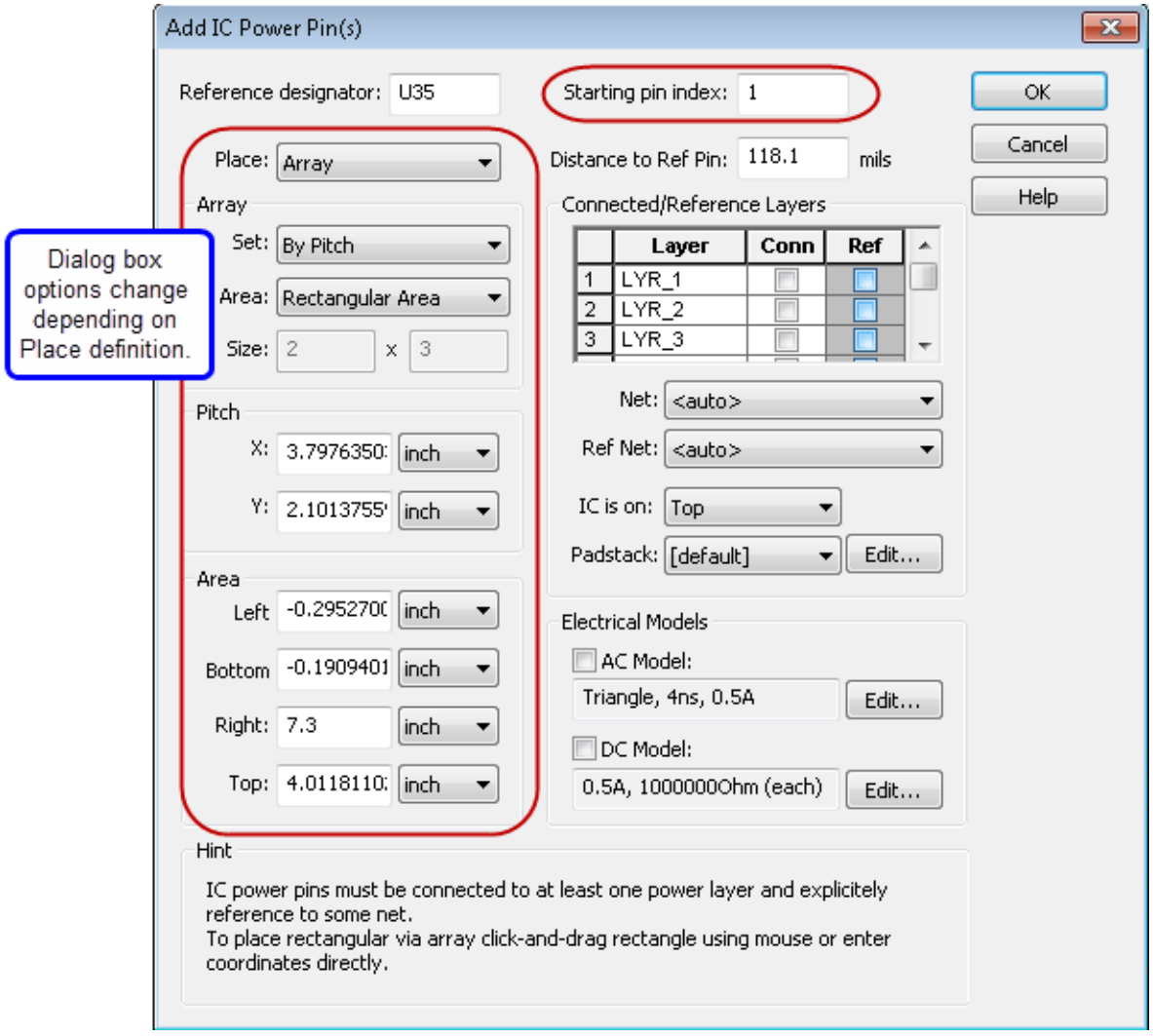
Add/Edit IC Power Pin(s) Dialog Box

Scope: LineSim

To access: In the PDN Editor, do one of the following:

- To add an IC power pin, click 
- To edit an IC power pin, double-click a power pin.

Use this dialog box to add or edit IC power-supply pins in a power-distribution network (PDN).



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Reference designator	Defines the reference designator for a single pin or an array. **Note:** Each IC pin array acts as a single object in the PDN Editor.
Place	Defines the placement method for the IC power pin(s). The dialog box options change depending on whether you select **Single** or **Array** from the dropdown list.
Pin name	Defines the pin number for a single pin.
Starting pin index	Defines the starting pin number for the array.
Distance to Ref Pin	Defines the separation distance between the IC power pin (or the starting IC pin in the array) and the reference pin. Accept the default or enter a new number. **Note:** Set the default in the Default separation between IC power and reference pins option on the Preferences dialog box - Power Integrity tab.
Location	Defines the location of the IC power pin. Enter coordinate values or click a location in the PDN Editor to populate the X, Y fields.
Array	• Set — Defines the array either By Size (number of columns and rows) or By Pitch (distance between pins). • Area — Defines the area in which to distribute the array, across the Whole Board or within a selected Rectangular Area. • Size — Defines the number of columns and rows in the array.
Pitch	Defines the distances between the IC pins on the X and Y axes of the array.
Area	Defines the boundary of the rectangular area in which to place the array. Enter coordinate values, or select a region in the PDN Editor.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Connected/Reference Layers	• Layer — Defines the stackup layer name. • Conn — Checked, defines a stackup layer that carries current for the IC pin. Unchecked, does not carry current. • Ref — Checked, identifies a stackup layer that carries return current for the IC pin. Unchecked, does not carry return current. **Restriction:** This column is read only when you enable the "Automatically assign reference layers" option in Preferences Dialog Box - Power Integrity Tab. • Net — Defines the power-supply net that carries current for the IC pin. Select a net from the dropdown list. Select **<auto>** when the IC pin connects to a stackup layer with one power-supply net. • Ref Net — Defines the power-supply net that carries return current for the IC pin. Select a net from the dropdown list. Select **<auto>** when the IC pin connects to a stackup layer with one power-supply net. • IC is on — Defines the side of the board on which the VRM pin is located. • Padstack — Defines the padstack. Click **Edit** to open the Padstack Editor Dialog Box, which enables you to modify the selected padstack. **Note:** Use the Padstack Manager Dialog Box to add nets.
Electrical Models	Enables you to select and define the power integrity model(s) to assign to the IC power-supply pin. • AC Model — Checked, assigns the defined AC model. Click **Edit** to open the Edit AC Power Pin Model Dialog Box, and define the electrical characteristics and stimulus waveform of the model. AC models typically represent I/O buffer switching and IC core-logic power on/off transitions. • DC Model — Checked, assigns the defined DC model. Click **Edit** to open the Edit DC Power Pin Model Dialog Box, and define the electrical characteristics of the current sink model. DC models represent static loads, such as IC power-supply pins connected only to non-switching circuitry.

Related Topics

Creating a PDN Design

Adding an Eye Mask to a FastEye Diagram

The FastEye Viewer enables you to overlay a FastEye diagram with a mask that displays the "keep out" regions that the eye must avoid.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

1. In the Show area, check “Eye mask”.
2. You can select an eye mask from a library of popular communication protocols or create your own eye mask and add it to the library.
3. If the eye mask position is not centered within the bit interval, you can do any of the following:
   - In the Cursors area, click **Adjust Mask**, then drag the eye mask to the new position. The new timing offset values are automatically written to the **Eye Mask** tab on the Configure Eye Diagram dialog box.
   - Type the exact offset into the eye mask boxes on the **Eye Mask** tab on the Configure Eye Diagram dialog box.

Related Topics

FastEye Viewer

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Advanced AMI Options Dialog Box

To access: **Simulate SERDES > Run SERDES Batch Simulation > Advanced Eye Diagram > Advanced AMI Options**

Use this dialog box to set advanced AMI-related options, such as the number of samples per bit interval, number of bits per call to AMI DLL, number of starting bits to ignore, or advanced troubleshooting options.

Fields

Field	Description
Samples per symbol interval	Specifies the number of samples per symbol interval to adjust the display resolution of eye diagram contours in the FastEyeViewer and HyperLynx Eye Density/BER Viewer. Large numbers will slow analysis considerably.
Number of symbols per GetWave call to AMI DLL	Specifies the number of stimulus symbols to send as a block to an AMI *dll* or *.so* file. You might change this value when advised by the model creator to optimize interaction with certain AMI *dll* or *.so* files.
Let AMI models determine number of symbols to ignore	Checked, the software uses values specified in the Tx and Rx AMI models.
Ignore first ___ thousand symbols simulated	Specifies the number of initial symbols to exclude from an eye diagram. Specify a sufficient number of bits for the driver and receiver circuits to stabilize and reach normal operating conditions. The software only uses this value if it is larger than values specified in the Tx and Rx AMI models. See Factors that Affect the Number of Warmup Bits.
Advanced troubleshooting options	
Log all AMI interface function inputs and outputs	Generates the basic files that the SERDES analysis engine uses, including a configuration file (*.cnfg*), and primary and any crosstalk step response waveform files (*.lis*). Also generates impulse responses before and after calls to AMI_Init() functions and any time-domain waveforms before and after calls to AMI_GetWave() functions. These files are useful for advanced users that are working with a Siemens representative to troubleshoot AMI models.
Ignore receiver AMI model clock ticks	Checked, the simulator creates its own clock ticks. Enable this option when you think that the Rx AMI model generates invalid clock ticks.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Advanced Channel Characterization Dialog Box

To access: **SERDES > Run SERDES Batch Wizard > Advanced Eye Diagram > Advanced**

Use this dialog box to set advanced channel characterization options, such as the characterization type, channel response, linearity check, and reporting options.

Fields

Field	Description
Characterization type Area	
Step	Derives the channel impulse response from the response at the receiver to a step waveform.
Step/Pulse response	Derives the channel impulse response from the response at the receiver to a step waveform. If you want to check channel linearity, select this option and enable “Check channel linearity”. The software runs a step and pulse input separately, and compares the two channel responses to determine linearity.
PRBS	Derives the channel impulse response from a PRBS waveform. See Bit Sequence for Automatic Channel Characterization.
Simulator Options	Opens the Simulation Controls Dialog Box, which you use to specify the simulation engine, simulation resolution, SPICE simulation options, or waveform generation options.
Channel response Area	
Default (recommended)	De-select this option to investigate non-linearity or other analysis-related errors reported by the wizard.
ISI effects are gone after	Specify the number of bit intervals needed for transient activity to settle. The wizard reports an error if it detects unsettled activity at the end of the channel-characterization simulation. See inter-symbol interference (ISI). The default value is conservative, and provides ample time for interference (due to the topology and reflection characteristics of the channel) to die out. **Restriction:** You can type values from 50 to 1000, in bit intervals.
Number of warmup bits before the Tx/ channel are stable	Specify a sufficient number of bits for the driver and receiver circuits to stabilize and reach normal operating conditions. See Factors that Affect the Number of Warmup Bits. This option adds bits to the beginning of the overall bit sequence used to create the channel-response extraction waveforms. **Restriction:** You can specify a value from 0 to 255, in bit intervals.
Linearity check Area	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Check channel linearity (Available for Step/Pulse Response characterization type only)	Checked, compares the pulse and step waveforms and reports non-linear channel-response. See Checking Channels for Linear and Time-Invariant Behavior. Disable this option for any of the following reasons: • You are absolutely certain the channel is linear. • You previously ran the wizard with the same input waveform files, and linearity checking passed. • You are creating only a worst-case bit sequence and are not creating a FastEye diagram.
Non-linearity limit (Available for Step/Pulse Response characterization type only)	Specifies the channel non-linearity threshold. The wizard reports an error if the difference of the energy between the actual and calculated pulse responses exceeds the Non-linearity limit value. Channel analysis checks the linearity of the channel by comparing the energy in the actual pulse response to the energy in the calculated pulse response, where the calculated pulse response is the difference between the actual step response and the actual step response negated and delayed. See Checking Channels for Linear and Time-Invariant Behavior.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Advanced Jitter Settings Dialog Box

To access: From the Edit Stimulus or Configure Eye Diagram dialog box, click **Advanced**

Use this dialog box to specify jitter settings, save jitter settings to a file, or load jitter settings from a file.

Fields

Field	Description
Add Gaussian jitter	Checked, adds jitter with a normal distribution that has no sigma limit. You can specify jitter in terms of standard deviation or peak-to-peak. "Frequency" is the rate at which the jitter offset varies. See "Gaussian Jitter" on page 556 and "Units for Gaussian and Uniform Jitter" on page 562.
Add uniform jitter	Checked, adds jitter with an even distribution "Mean" divides a jitter range into two parts of equal area. See "Uniform Jitter" on page 560 and "Units for Gaussian and Uniform Jitter" on page 562.
Add sinusoidal jitter	Checked, adds sinusoidal jitter. **Note:** You can usually set "Initial phase" to zero degrees. You might specify a non-zero initial phase value for "short" simulations that are not long enough to contain many periods of slowly-changing jitter. Sinusoidal jitter usually shifts slowly relative to the bit rate. "Frequency" is the rate at which a jitter offset varies. See "Sinusoidal Deterministic Jitter" on page 559.
Add Tx duty cycle distortion	Checked, adds duty cycle distortion jitter. See "Duty Cycle Distortion Jitter" on page 555. **Restriction:** The Digital Oscilloscope dialog box does not apply duty cycle distortion jitter.
Jitter template Area You can save jitter settings to a jitter template file (JIT) or load jitter settings from a file.	

AMI File Assignment Dialog Box

To access: **Simulate SERDES > Run IBIS-AMI Channel Analysis**, select **Configure AMI Models** page, and click **Assign AMI Files**

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Use this dialog box to assign *.AMI* and *.DLL (Windows)/.so (Linux)* files to the channel driver and receiver.

If you assign IBIS models containing [Algorithmic Model] keywords to the channel driver and receiver ICs, this dialog box automatically displays the location of the *.AMI* and *.DLL/.so* files. You can override the assignments made by [Algorithmic Model] keywords. The *.FEW* file stores your changes, not the IBIS model.

Note

.DLL files are executable files. Make sure the IBIS model specifies *.DLL*/*.so* files for all the computer platforms you run simulation on. Store *.DLL*/*.so* files in the same folder as the IBIS model or in another folder displayed in the Model-library file path(s) list in the Set Directories dialog box.

Fields

Field	Description
AMI Tx files Area	
Tx AMI .ami path	Location of the external ASCII file that contains parameters for the driver.
Tx AMI .dll or .so path	Location of the external compiled file that contains signal-processing functions for the driver.
AMI Rx files Area	
Rx AMI .ami path	Location of the external ASCII file that contains parameters for the driver.
Rx AMI .dll or .so path	Location of the external compiled file that contains signal-processing functions for the receiver.

Usage Notes

The *.DLL*/*.so* and *.AMI* paths and Browse buttons are unavailable when an IBIS model specifies the *.DLL*/*.so* or *.AMI*. To change the *.DLL*/*.so* or *.AMI* path, edit the IBIS model.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

AMI Model Testing Dialog Box

To access: **Simulate SERDES > Run SERDES Batch Wizard > Advanced Eye Diagram > Setup**

Use this dialog box to set AMI model troubleshooting options.

Note

This dialog box is available when only one differential pair is selected for simulation on the Analysis Options page of the SERDES Batch Wizard.

Fields

Field	Description
Display BER plots in AMI Sweeps Viewer	Checked, the software accumulates subsequent simulation runs in a a single instance of the HyperLynx Eye Density/BER Viewer, so you can easily compare simulation results when troubleshooting.
Log all AMI interface function inputs and outputs	Checked, generates the files that the SERDES analysis engine uses, including a configuration file (.cnfg), and primary and any crosstalk step response waveform files (.lis). Also generates impulse responses before and after calls to AMI_Init() functions and any time-domain waveforms before and after calls to AMI_GetWave() functions. These files are useful for advanced users who work with a Siemens representative to troubleshoot AMI models.
Ignore receiver AMI model clock ticks	Checked, the simulator creates its own clock ticks. Enable this option when the Rx AMI model generates invalid clock ticks.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Assign / Edit Capacitor Model Dialog Box

Scope: BoardSim

To access: **Models > Edit Decoupling-Capacitor Models**. From the Assign Decoupling-Capacitor Models dialog box, double-click a capacitor group.

Scope: LineSim

To access: From the PDN Editor, double-click a decoupling capacitor symbol. From the Edit Decoupling Capacitor(s) dialog box, click **Assign Model**.

Use this dialog box to assign a model to a decoupling capacitor.

Fields

Field	Description
Capacitor model includes mounting inductance (no additional inductance will be calculated/added during analysis)	Check when the capacitor model includes the effects of the via and its connectivity to the capacitor package. This situation happens when the vendor did not de-embed the capacitor model from how the capacitor package was mounted in the test fixture.
Library Models	Load or save capacitor models from or to libraries. **Note:** You can load and save capacitor models to directories that are listed in the Model-library file path area of the Set Directories dialog box (**Setup > Options > Directories**). If you need to load or save models from or to another directory, add the path to the Model-library file path area.
Model type = Simple C-L-R	
ESR	Specifies the equivalent series resistance of the capacitor. This is a frequency-dependent value. Choose a value somewhere near the frequency the mounted capacitor will resonate. The frequency range is typically 1 - 100 MHz.
Auto-calculate	Select to have the software automatically calculate ESL, based on the size of capacitor package body and mounting via length.

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Field	Description
Capacitor Size	Specifies the capacitor size as one of the following: • <Auto-estimate>—The software automatically calculates the package body dimensions. • <Custom>—You enter the package body dimensions. • <predefined_body_size>—You select a standard package body size. Width is the width of rectangular SMD pads. For Auto-estimate, if the SMD pad width appears to be unrealistic, the software makes the width to be one-half the capacitor body length. Length is the distance between the capacitor body pins (BoardSim) or capacitor body ports (LineSim). In LineSim, use the Decoupling Mounting Scheme Editor Dialog Box to view capacitor body port distances. Height is the distance between conductors inside the capacitor body and the PCB surface. For Auto-estimate and <predefined_body_size>, the value is always 39 mils or 1 mm. BoardSim automatically determines whether the capacitor is located on the top or bottom of the board. In LineSim, you use the Decoupling Mounting Scheme Editor dialog box to specify whether the capacitor is located on the top or bottom of the board.
Specify value	Specifies the equivalent series inductance of the capacitor. This is a frequency-dependent value. Choose a value somewhere near the frequency the mounted capacitor will resonate. The frequency range is typically 1 - 100 MHz. ESL is usually much smaller than the mounting inductance.
Resonance	Displays the resonant frequency of an unmounted capacitor. At this frequency and higher, the impedance of the capacitor increases. Figure 10-1 shows an example resonant frequency plotted for a Z-parameter model.
Model type = SPICE	
SPICE Files	Specifies the model file. To specify folders that contain models, see Set Directories Dialog Box.
Sub-circuits	Specifies the sub-circuit representing the capacitor behavior. The number of sub-circuit ports and capacitor pins must be the same.
Show only compatible subcircuits	Check to hide sub-circuits with a number of ports that is different than the number of capacitor pins.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Spreadsheet	Specifies how model notes map to pin names. The software displays the read-only spreadsheet only when the number of sub-circuit ports and capacitor pins is the same.
Model type = Touchstone	
Files	Specifies the model file. To specify folders that contain models, see Set Directories Dialog Box.
Show only compatible models	Check to hide models with a number of ports that is different than the number of capacitor pins.
Two-port model type	Specifies how the capacitor was connected when measured during characterization. Figure 10-2 and Figure 10-3 show the measurement setup for series and shunt models. Check Autodetect to automatically identify series and shunt models. The software relies on the conductance between ports 1 and 2 to choose between series and shunt types. Series capacitors have small conductance, especially at low frequencies. Shunt capacitors have large inductance. **Note:** This area is hidden unless a two-port or one-port model is selected in the Files list.

Usage Notes

Figure 10-1. Capacitor Resonant Frequency Example

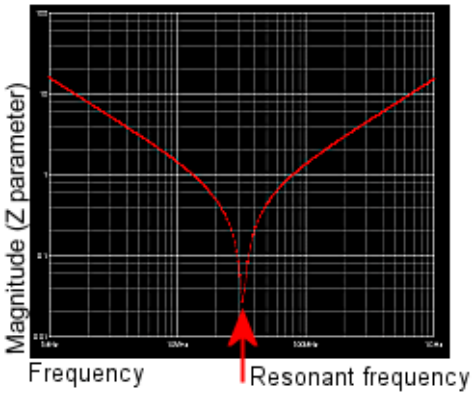
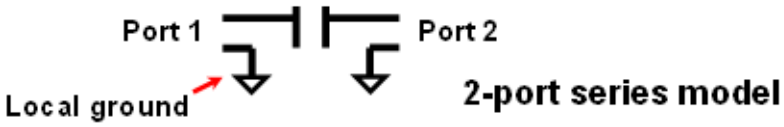
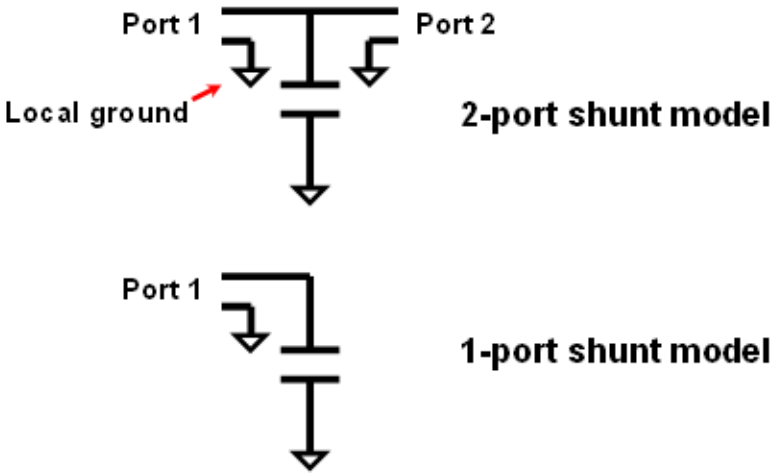


Figure 10-2. Measurement Setup for Series Model



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-3. Measurement Setup for Shunt Model



Related Topics

Assign Decoupling-Capacitor Models Dialog Box

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Assign Decoupling-Capacitor Groups Dialog Box

Scope: BoardSim

To access: **Models > Edit Decoupling-Capacitor Groups**

Use this dialog box to edit decoupling-capacitor groups.

Fields

Field	Description
Decoupling capacitors	Displays capacitors not in a group.
Capacitor groups	Displays capacitor groups. The Ref Des column uses the QPL_prefix to identify model assignments you have made with a *.QPL* file. The Value column contains capacitance values from the *.REF* file, *.QPL* file, or design file (from highest to lowest priority). The software uses these values only to create the initial membership of decoupling-capacitor groups. Power-integrity simulation never uses these values. The Max Dimen column contains the maximum component dimension, which comes from the design file.
Auto-Grouping	Opens the Auto-Grouping Dialog Box.

Related Topics

Decoupling Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Assign Decoupling-Capacitor Models Dialog Box

Scope: BoardSim

To access: **Models > Edit Decoupling-Capacitor Models**

Use this dialog box to assign a model to a decoupling capacitor or a series passive component that connects two power supply nets.

Note

Assign a model to a series passive component that connects power supply nets if you want to include those nets in simulation. To assign a model to a passive component, in the Preferences dialog box, check "Allow AC supply models for non-capacitor components". See "Preferences Dialog Box - Power Integrity Tab" on page 995.

When this option is checked, the title of this dialog box (Assign Decoupling-Capacitor Models) becomes "Assign Power Integrity AC Models for Supply Components".

Fields

Field	Description
Show these nets only	Check to filter the spreadsheet. To display decoupling capacitors and groups connected to a specific pair of power-supply nets, select a power-supply net in each list. To display decoupling capacitors and groups connected to one specific power-supply net and any other power-supply net, select a power-supply net in one list, and then select **<All Supply Nets>** in the other list.
Group/Ref Des	Displays capacitor groups. The Group/Ref Des column uses the QPL_ prefix to identify model assignments you have made with a .QPL file. The Value column contains capacitance values from the *.REF* file, *.QPL* file, or design file (from highest to lowest priority). The software uses these values only to create the initial membership of decoupling-capacitor groups, and does not use them for power-integrity simulation.
Assign Model	Opens the Assign / Edit Capacitor Model Dialog Box.
Edit Groups	Opens the Assign Decoupling-Capacitor Groups Dialog Box.

Related Topics

Decoupling Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Assign IC Component Model Dialog Box

Scope: LineSim

To access: Double-click an IC component symbol.

Use this dialog box to assign an IBIS model to an IC component symbol, hide pins that you do not plan to include in simulation, and to edit pin location.

Objects

Object	Description
Libraries Devices	Specifies an IBIS model assignment when you click **OK**.
Library Path	Opens the Select Directories for IC-Model Files dialog box, where you can add a folder that contains a model you want to assign.
Edit Model File	Opens the Visual IBIS Editor, where you can view or edit the contents of an assigned IBIS model.
First spreadsheet column	Moves a pin up or down a side of a component. See Hiding or Moving an IC Component Pin.
Visible	Uncheck to hide an IBIS model pin that you do not plan to include in simulation.
Side	Specifies whether a pin is located on the left or right side of a component. For information about moving a block of pins, see Hiding or Moving an IC Component Pin.
Auto-Place Ports	Moves all pins to a pattern that you specify. **Note:** The software uses spreadsheet row numbers when moving pins. For example, the "Odd to Left, Even to Right" pattern moves pins on odd-numbered spreadsheet rows to the left side of a component.
Configure	Opens the Configure IC Component Symbol dialog box, where you can edit the pin location for many pins at once.

Related Topics

Creating a Schematic Design

Assign Models Dialog Box

To access:

- For board designs, select a net, then select **Models > Assign Models/Values by Net**.
- For schematic designs, double-click an IC buffer or ferrite bead symbol.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Use this dialog box to assign a model to a specific IC pin or ferrite bead. For board designs, this dialog box has additional tabs enabling you to edit passive component values, assign Quick Terminators, and to display series bus switch connectivity.

The tab or item you select on the dialog box determines the fields available.

Fields

Field	Description
Apply to all similar boards (Available only for MultiBoard projects that use multiple instances of a board)	Indicates whether to use the current dialog box settings for all copies (instances) of that board in your MultiBoard project. Default value is “selected.” Unchecking this option is not recommended unless you are familiar with how it works.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Pins	Displays the pins for the selected net and associated nets. Double-click the reference designator/pin to open one of the following dialog boxes, depending on the type of component you choose: • Select IC Model dialog box, which enables you to find and assign an IC component model to the selected pin. • Select Package dialog box, which enables you to view and select a package connectivity description. Status icons display next to pin names. • indicates that a model is assigned. • indicates that a model is not assigned. The component is modeled as an electrical open during simulation. • indicates that a driver model is assigned. • indicates that a receiver model is assigned. • R or Q indicates that the model is currently assigned from a *.REF* or *.QPL* file. • MDL indicates that the pin/net was brought into the simulation with the coupling coming from the package model rather than electrical/geometric thresholds. • identifies a connector pin for a MultiBoard design or a pin that has a model assigned from an *.EBD/EMD* model. • identifies a pin on an aggressor net when crosstalk simulation is enabled. • identifies a pin on the net that is connected to a series bus switch, as identified by an IBIS [Series Pin Mapping] keyword. The Pins list highlights all series bus switch pins connected in series with the selected pin. • identifies a pin on the net that is connected to a differential resistor Quick Terminator.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Buffer Settings (part 1 of 2)	The available buffer state options are determined by information in the model ([Model] Model-type): • Input—Indicates a receiving or input state. In this state, an I/O or bidirectional model driver is turned off, and the pin is receiving. For input-only pins, this is the only available state. • Output Hi-Z—Indicates a passive, high-impedance state in which a driver or output model is turned off. This selection is available exclusively for output-only models that have tristate capability; the high-impedance state of a bidirectional model is selected with state Input. Data sheets sometimes refer to an IC pin high-impedance state, meaning that the pin is I/O capable, but the driving circuitry is disabled and the pin is in a high-impedance receiving state. For these types of pins, select Input as the buffer state. • Output—Indicates a driving or output state. In the oscilloscope or in EZwave, the model follows the Driver Waveform setting (it falls on a falling edge and rises on a rising edge). If the stimulus is set to waveform type Oscillator, the model starts with a rising edge. • Output Inverted—Indicates an inverted driving or output state (rises on a falling edge and falls on a rising edge). If the stimulus is set to waveform type Oscillator, the model starts with a falling edge. This setting is most useful for differential-driver pairs, where one pin model must be inverted relative to the other.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Buffer Settings (part 2 of 2)	• Stuck High—Indicates a driving or output state, in which the model stays high and never switches. For interactive SI simulation, the model stays high regardless of the Driver Waveform setting. This setting is useful for wired-OR or wired-AND buses for which you want to investigate the effect of one driver switching while other drivers remain static. It is also important for crosstalk simulations, in which the driver on the victim net is simulated in a static state. • Stuck Low—Indicates a driving or output state, in which the model stays low and never switches. For interactive SI simulation, the model stays low regardless of the Driver Waveform setting. This setting is useful for wired-OR or wired-AND buses for which you want to investigate the effect of one driver switching while other drivers remain static. It is also important for crosstalk simulations, in which the driver on the victim net is simulated in a static state.
Vcc pin Vss pin	For IBIS models, the software determines the connected power supply net by referencing the assigned model ("use model's internal value"). To override information in an IBIS model, select "When assigning a model to an IC pin, use a power supply net connected to the IC" in the Preferences dialog box (**Setup > Options > General** tab), and select the power supply nets that are connected to the Vcc and Vss pins.
Select	Opens the Select IC Model dialog box, which enables you to find and assign an IC component model to the selected pin. Click **Remove** to remove the model from the pin. Connectivity Picture: A graphical representation of how the components in a network package are connected. If a package is too long to fit in the display area, the picture is truncated. • Internal components (resistors or capacitors) are displayed as little boxes • Package pins are displayed in blue • Connections are displayed in the following colors: • Black for connections between independent pins and component ends • Maroon for connections between common pin #1 (for example, power supply pin #1) and component ends • Green for connections between common pin #2 and component ends

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Edit Model File	Opens the model file in the HyperLynx Visual IBIS Editor.
Connectivity area	Displays the internal pin connections in a component package.
Impedance/Frequency Graph (Available only for ferrite bead models)	Displays the behavior described in the assigned model file, if values for R series, L series and C series keywords are included in the model file.
Assign model's pin name	Checked, uses the pin name from the [Pin] keyword in the model you have assigned to an IC component. **Restriction:** The software applies this option only when you first assign a model. After a model is assigned, the software ignores this option.

Related Topics

Assigning a Model to an IC Pin

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Assign Power Integrity Models Dialog Box

Use this dialog box to assign power-integrity models to power-supply pins and values to supply-net resistors, inductors and other components.

Scope: BoardSim

To access: **Models > Assign Power Integrity Models**

Scope: Xpedition Layout

To access: **Analysis Control dialog box > HyperLynx > Assign PI Models**

Topic	Description
Assign Power Integrity Models Dialog Box - IC Tab	Use this tab to assign power-integrity models to IC power-supply pins.
Assign Power Integrity Models Dialog Box - Other Supply-Net Components Tab	Use this tab to interactively assign values to non-resistor/ inductor components, such as high-current power FETs, that connect the power-supply net that you plan to simulate to another power-supply net.
Assign Power Integrity Models Dialog Box - Supply-Net Inductors Tab	Use this tab to interactively assign values to inductors that connect the power-supply net to another power-supply net or to a voltage-regulator-module (VRM).
Assign Power Integrity Models Dialog Box - Supply-Net Resistors Tab	Use this tab to interactively assign values to resistors that connect the power supply net that you plan to simulate to another power supply net or to a voltage-regulator-module (VRM).

Assign Power Integrity Models Dialog Box - IC Tab

Scope: BoardSim

To access: From the Assign Power Integrity Models Dialog Box, select the **IC** tab.

Use this tab to assign power-integrity models to IC power-supply pins.

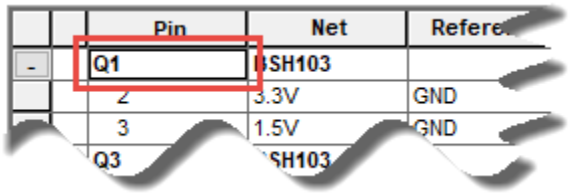
The types of models you assign depend on the type of power-integrity simulation you plan to run.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Filters	Enables you to filter the contents of the table for the selected Component Types. Enter a string in the Reference Designator and/or Power-Supply net field and click **Apply**. The fields support the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character. To clear the filter, delete the contents of the field and click **Apply**. **Note:** The Power-Supply Net field is available if accessed from the DC Drop Analysis dialog box.
Include attached nets	(Available only if you access this dialog box from the DC Drop Analysis dialog box.) Checked, adds to the table the pins from power-supply nets associated to the selected net by a series component.
<table>	Displays the power-supply pins available for assignment. Click the box to the left of the pin to select the row. Use **Shift-click** or **Ctrl-click** to select multiple rows.
AC Model	• **Assign** — Opens the Edit AC Power Pin Model dialog box, which enables you to specify the electrical characteristics and stimulus waveform of the source model assigned to the selected IC power-supply pin(s). • **Remove** — Removes AC models from the selected IC power-supply pin(s).
DC Sink Model	• **Assign** — Opens the Edit DC Power Pin Model dialog box, which enables you to specify the electrical characteristics of the current sink model assigned to the selected IC power-supply pin(s). • **Remove** — Removes DC models from the selected IC power-supply pin(s).
VRM Model	• **Assign** — Opens the Assign VRM Model dialog box, which enables you to specify the electrical characteristics of the voltage-regulator model (VRM) assigned to the selected IC power-supply pin(s). • **VRM Manager** — Opens the VRM Manager dialog box, which enables you to view all VRM models, and assign, edit, or delete models as needed.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Reference Net	• **Assign** — Opens the Set Reference Net dialog box, which enables you to identify the net and stackup layer(s) used to provide return-current paths for the selected power-supply pin(s). • **Remove** — Removes the reference net from the selected IC power-supply pin(s).
Save Model	Opens the Save DC Models dialog box, where you can save a DC Sink model for reuse in another simulation. **Tip:** To save a component model, click in the pin or net column of the component (not the entire row):  and click **Save Model**. See Usage Notes.

Usage Notes

The Save DC Models dialog box saves the model you select in an HLPIMODEL file to the design directory. If an IBIS model assignment exists, the model_name defaults to the IBIS file name. Otherwise, the model_name defaults to the Part Name. You can change the model_name as needed, and assign the model as follows:

- Assign the model by REF file, which creates an REF file containing the selected model assignment. An “R” appears in the column to the left of the component pins after you click **OK**.
- Assign Model by QPL file, which enables you to specify the QPL file to use when making the model assignment. A “Q” appears in the column to the left of the component pins after you click **OK**.
- Save the model and keep current assignment, which saves the model file without making a model assignment.

Related Topics

Assigning VRM Source, DC Sink, and AC Models

Running DC Drop Simulation From Layout Software

Power Integrity Models

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Assign Power Integrity Models Dialog Box - Other Supply-Net Components Tab

Scope: BoardSim

To access: Open the Assign Power Integrity Models Dialog Box, select the **Other Supply-Net Components** tab.

Use this tab to interactively assign values to non-resistor/inductor components, such as high-current power FETs, that connect the power-supply net that you plan to simulate to another power-supply net.

These series components “associate” power-supply nets in the same way that series components associate signal nets, which means the associated power-supply nets are included in the power-integrity simulation.

Note

Only DC drop simulation supports series components that associate one power-supply net to another power-supply net.

The software uses resistance values for ICs and connectors for DC power-integrity analyses. The inductance of “other” supply-net components is assumed to be 0 nanohenry for AC power-integrity simulations.

Fields

Field	Description
Filters	Enables you to filter the contents of the tables. Enter a string in the Reference Designator and/or Power-Supply net field and click **Apply**. The fields support the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character. To clear the filter, delete the contents of the field and click **Apply**. **Note:** The Power-Supply Net field is available if accessed from the DC Drop Analysis dialog box.
Include attached nets	(Available only if you access this dialog box from the DC Drop Analysis dialog box.) Checked, adds the pins from power-supply nets associated to the selected net by a series component.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Select IC	Lists the ICs available. Select an IC to display the pins for that component. If this list does not contain an IC you expected to see, the software may not have correctly identified: • A power-supply net for the IC. See Verifying That Power Supply and Signal Nets are Recognized Correctly. • The reference designator prefix for the IC. See Verifying That Component Types are Recognized Correctly.
Select First Pin	Lists the pins available for selection. Click the box to the left of the Pin to select the first pin of the component. Click **Connect** to connect the first and second pins.
Select Second Pin	Lists the pins available for selection. Click the box to the left of the Pin to select the second pin of the component. Click **Connect** to connect the first and second pins.
Connections	Lists the connected pins and their current resistance values. To change the resistance value of a connection do the following: 1. Select a row on the table. 2. Enter or edit the value in the Resistance, Ohm field. 3. Click **Assign**.
Resistance, Ohm	Defines the resistance value to assign to selected pin connections.

Related Topics

Assigning VRM Source, DC Sink, and AC Models

Running DC Drop Simulation From Layout Software

Power Integrity Models

Assign Power Integrity Models Dialog Box - Supply-Net Inductors Tab

To access: Open the Assign Power Integrity Models Dialog Box, select the **Supply-Net Inductors** tab.

Use this tab to interactively assign values to inductors that connect the power-supply net to another power-supply net or to a voltage-regulator-module (VRM).

For DC drop simulation, these series components “associate” one power-supply net to another power-supply net in the same way that series components associate signal nets, which means the associated power-supply nets are included in the power-integrity simulation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

Only DC drop simulation supports series components that associate one power-supply net to another power-supply net.

Fields

Field	Description
Filters	Enables you to filter the contents of the tables. Enter a string in the Reference Designator and/or Power-Supply net field and click **Apply**. The fields support the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character. To clear the filter, delete the contents of the field and click **Apply**. **Note:** The Power-Supply Net field is available if accessed from the DC Drop Analysis dialog box.
Include attached nets	(Available only if you access this dialog box from the DC Drop Analysis dialog box.) Checked, adds the pins from power-supply nets associated to the selected net by a series component.
Select Inductor	Specifies the inductors that connect one power-supply net to another. You can assign (or remove) Inductance and Resistance values. If an inductor displays a checkmark and an “R” or a “Q” next to it, that means the values were read in from a *.REF* or *.QPL* file, respectively. To change assignment, select an inductor and click one of the following: • **Assign** — Assigns the values defined in the Inductance and Resistance fields and displays a checkmark next to the inductor. • **Remove** — Removes the values. If this list does not contain an inductor you expected to see, the software may not have correctly identified: • A power-supply net for the inductor. See Verifying That Power Supply and Signal Nets are Recognized Correctly. • The reference designator prefix for the inductor. See Verifying That Component Types are Recognized Correctly.
Inductance	Defines the inductance of selected inductor.
Resistance	Defines the resistance of selected inductor. Use resistance values for DC power-integrity simulations.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Assigning VRM Source, DC Sink, and AC Models

Running DC Drop Simulation From Layout Software

Power Integrity Models

Assign Power Integrity Models Dialog Box - Supply-Net Resistors Tab

To access: Open the Assign Power Integrity Models Dialog Box, select the **Supply-Net Resistors** tab.

Use this tab to interactively assign values to resistors that connect the power supply net that you plan to simulate to another power supply net or to a voltage-regulator-module (VRM).

For DC drop simulation, these series components "associate" one power supply net to another power supply net in the same way that series components associate signal nets, which means the associated power supply nets are included in the power-integrity simulation.

Note

Only DC drop simulation supports series components that associate one power supply net to another power supply net.

Fields

Field	Description
Filters	Enables you to filter the contents of the tables. Enter a string in the Reference Designator and/or Power-Supply net field and click **Apply**. The fields support the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character. To clear the filter, delete the contents of the field and click **Apply**. **Note:** The Power-Supply Net field is available if accessed from the DC Drop Analysis dialog box.
Include attached nets	(Available only if you access this dialog box from the DC Drop Analysis dialog box.) Checked, adds the pins from power supply nets associated to the selected net by a series component.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Select Resistor	Lists the resistors to which you can assign (or remove) Resistance values. If a resistor displays a checkmark and an "R" or a "Q" next to it, that means the values were read in from a *.REF* or *.QPL* file, respectively. To change assignment, select a resistor and click one of the following: • **Assign** — Assigns the values defined in the and Resistance field and displays a checkmark next to the inductor. • **Remove** — Removes the values. If this list does not contain a resistor you expected to see, the software may not have correctly identified: • A power supply net for the resistor. See Verifying That Power Supply and Signal Nets are Recognized Correctly. • The reference designator prefix for the resistor. See Verifying That Component Types are Recognized Correctly.
Package	Defines the network package in which the resistor is housed and depicts the package in the Connectivity section.
Resistance	Defines the value for the selected resistor.

Related Topics

Assigning VRM Source, DC Sink, and AC Models

Running DC Drop Simulation From Layout Software

Power Integrity Models

Assign S-Parameter/SPICE Model Dialog Box

To access:

- In BoardSim, open the MultiBoard Project Wizard (**File > New MultiBoard Project**), select **Advanced** as the Connector model, then click the **Assign** button.
- In LineSim, click **Add S-Parameter/SPICE Model to schematic** (), place symbol on the Free-Form Schematic Editor, then double-click the symbol.

Use this dialog box to assign passive SPICE and S-Parameter (Touchstone) models to a package/connector symbol.

Fields

Field	Description
Model Type	Specifies a type of model to assign.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Existing Files	Lists available models. S-parameter (Touchstone) files display an *SNP* extension, such as *S4P*. SPICE files display an *SP* extension.
Create in 3D Explorer	Opens HyperLynx 3D Explorer, to create an S-parameter model. For help, see “Exploring Design Variations” in the *HyperLynx Advanced Solvers User Guide*.
Devices	Lists the device models available for a SPICE model.
Library Path	Opens the Select Directories for IC-Model Files dialog box, to identify folders that contain models you want to assign.
Ports	Enables you to assign model ports to component pins, power-supply nets, or nothing (not connected). • Name — Displays the name of a port. • Connection — Assigns a port to a connector reference designator, a power-supply net, or nothing. NC means “not connected”. • Pin — For a port that has been assigned to a connector reference designator, assigns a port to a connector pin. Available in BoardSim and a multiple-board project: • Board — Assigns a board or board instance, such as B00, to a port. Available in LineSim: • Visible — Enables you to select whether or not to make the port visible in the schematic. • Side — Defines the side of the board on which the port exists. To change multiple port sides at once, highlight all the ports (rows) you want to change, then open the popup on the Side location you want and select **Apply to Selection**. • Circuit Connection — Displays the location of the circuit connection.
Auto-Place Ports	(Available in LineSim.) Distributes all the ports on the board depending on the directions you select from the dropdown list.
View Model File	(Available for Touchstone models.) Opens the HyperLynx Touchstone and Fitted-Poles Viewer, which enables you to view S-parameter model data.
Edit Model File	(Available for SPICE models.) Opens the HyperLynx File Editor, which enables you to edit the script content of the model file.
Edit Parameters	Opens the Eldo/SPICE Model Parameters dialog box, where you can specify parameter-value pairs.

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Field	Description
Model Connections	Opens the Model Connections dialog box, where you can enable the software to automatically detect port connections for an S-parameter model, or create, import, or remove a connection file (.connect) that is associated with a S-parameter or SPICE model.
Load from PJH	(Available when you assign an advanced connector between boards in a multiple-board project.) Enables you to load model data to the dialog box from an existing *PJH* file.

Related Topics

Setting Up a Multiple Board Design

Creating a Schematic Design

Assign Stimulus Dialog Box

To access, do one of the following:

- From the Digital Oscilloscope Dialog Box or Interactive Simulation Dialog Box (and more dialog boxes), select **Per-net/pin** and click **Assign**.
- Choose the **Setup > Stimulus** menu item.
- In a board design, select a net and right-click an IC pin or trace segment on selected net and choose the **Edit Stimulus** menu item.

  When you right-click an IC pin or trace segment for an unselected net, the Assign Stimulus dialog box displays information for the selected net.
- From a schematic, right-click an IC symbol and choose the **Edit Stimulus** menu item.

Use this dialog box to assign stimulus to specific pins or nets in the design. For a board design, you can assign stimulus to individual pins or to a net (and all its pins). For a schematic, you can assign stimulus to individual pins. You can also import a DDRx stimulus (*.txt*) file.

You can sort the spreadsheet by clicking a column header.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Options

Option	Description
Show inherited values	Checked, shows inherited values in green text. A value assigned to a net class is inherited by all nets in the class unless you override the inherited value with a specific value. The “All” net class includes all nets and cannot be changed or deleted. For details about how to edit inherited and specific stimulus settings, see “Running General Batch Simulation” on page 175.
Net Class/Net/Pin column	Shows the name of the Net Class, Net, and Pin. When a multiple-board project is loaded, the name of the board is added to the net or component name. For example, U1_B02.14, where _B02 identifies the board. For a schematic, this column always displays pin names.
Stimulus column	Assigns the selected stimulus to the net or pin. <default> represents a rising edge. You cannot edit the default stimulus. This column displays stimulus names for stimulus files located in the stimulus file folder (the design folder by default) that you specify in the Set Directories Dialog Box.
Initial Delays column	Specifies the offset of the stimulus, in ns. The dash (-) means no delay. When you enter 0 ns, it is automatically replaced with a dash. Dashes make it easier to see non-zero delay values.
Filters	Filters the spreadsheet to display nets or pins whose names contain the filter string. Use the asterisk (*) wildcard to match any number of characters. Use the question mark (?) wildcard to match any one character. Alphabetic characters are case insensitive. To display all spreadsheet rows, enter * and click **Apply**. For example, to display pin names for U7, click in the Net/Pin field, enter u7*, and then click **Apply**.
Import DDRx Stimulus	Enables you to import a DDRx stimulus text file from a simulation. The DDRx Wizard automatically creates this file from information on the DDRx Batch-Mode Wizard - Stimulus and Crosstalk Page.
Edit Stimulus	Opens the Edit Stimulus Dialog Box to enable you to specify a custom stimulus.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Assigning a Stimulus

Add/Edit VRM Dialog Box

To access (BoardSim): Assign Power Integrity Models dialog box, **IC** tab, VRM Model section, **VRM Manager** button, VRM Manager dialog box, **New** button

To access (LineSim): PDN Editor window, VRM Manager toolbar icon, VRM Manager dialog box, **New** button

Use this dialog box to specify the electrical characteristics of the voltage-regulator model (VRM) assigned to the IC power supply pin.

A VRM is a form of DC-to-DC converter and is used as a voltage source in simulation.

Restriction

If you plan to run AC power-integrity simulation, you must assign a reference net when assigning a VRM model. See Assigning VRM Source, DC Sink, and AC Models.

Fields

Note

Dialog box fields change depending on the model type that you select.

Field	Description
Name	Defines a name for the VRM. The name displays in the VRM Manager dialog box.
Model	Defines the type of VRM model to assign to the IC pin. • Simple — Enables basic fields. Use this option to model a VRM that includes a buck switching regulator. • Advanced — Enables more specific fields. Use this option to model a VRM that includes a linear regulator.
Voltage	Defines the “on” voltage of the VRM.
Number of Phases	Defines the number of phases. Choose 1 if the VRM is not a multi-phase VRM.
Sense Line	Specifies the pin where the VRM detects voltage to adjust the VRM output, if your VRM has a sense line.
Max Voltage Boost	Specifies the maximum additional voltage increase that the VRM output can generate to compensate for IR drop, if your VRM is capable of this.
Phases area	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
I %	Specifies the percentage of total current that flows through the phase.
Power Net	Specifies the net that the VRM connects to.
Power Pin(s)	Specifies the power pin(s) at the output of the VRM.
Ref Net	Specifies the reference net that the VRM connects to.
Ref Pin(s)	Specifies the reference pin(s) near the output of the VRM.
Resistance	(Simple model only) Defines the “on” resistance of the VRM. This value represents the total series resistance of the equivalent VRM circuit, not the value of a component within the circuit. If you do not know the VRM resistance, enter **0** for the resistance value.
Inductance	(Simple model only) Defines the “on” inductance of the VRM. This value represents the total series inductance of the equivalent VRM circuit, not the value of a component within the circuit. For example, the inductance may come from the inductance of pins or cables that connect the VRM to the PCB. If you do not know the VRM inductance, enter **0** for the inductance value. **Note:** Only AC power-integrity simulation uses this value.
RO	(Advanced model only) Defines the “on” resistance of the VRM. It can represent the following: • The I/V behavior of the VRM. • The value of the resistor inside the VRM that is between the VRM sense point and where the VRM connects to the PCB. If you do not know the VRM resistance, enter **0** milliohms.
L out	(Advanced model only) Defines the “on” inductance of the VRM. For example, the inductance may come from the inductance of pins or cables that connect the VRM to the PCB. If you do not know the VRM inductance, enter **0** nanohenries. This value is used only for AC power-integrity simulation.
R flat	(Advanced model only) Defines the equivalent series resistance (ESR) of the capacitor associated with the VRM.
L slew	(Advanced model only) Defines the V(*dt*/*di*) value so that current ramps up in the linear VRM in the same amount of time that it ramps up in the PCB. **Note:** Only AC power-integrity simulation uses this value.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Power Integrity Models

Assign Power Integrity Models Dialog Box - IC Tab

Auto-Create Groups Options Dialog Box

Scope: BoardSim

To access: From the Pin Group Manager dialog box, click **Create Options**.

Use this dialog box to choose the types of IBIS model information the software can use to automatically create power supply pin groups.

Fields

Field	Description
Use [Pin Mapping] section in IBIS model	Check to use information from a [Pin Mapping] keyword. To use a [Pin Mapping] keyword with data that does not match the board design, uncheck any of the following options: • Disable empty busses — A bus named in a [Pin Mapping] keyword connects to no power supply pins in the design. • Bus Pins are to be listed in [Pin] section — A [Pin Mapping] keyword names a pin that is not defined by a [Pin] keyword. • Bus Pins cannot belong to signal board nets — A pin in a [Pin] keyword connects to a non-power supply net in the design.
Use [Pin] section in IBIS model	Check to use information from a [Pin] keyword. To not use a [Pin] keyword with a POWER or GND pin that connects to a non-power supply net in the design, check Supply IBIS Pins cannot belong to signal board nets. To not create an empty power supply pin group, because a [Pin] keyword names a POWER or GND pin that does not connect to a power supply net in the design, check Don’t create empty groups.
All board Pins are to be listed in [Pin] section	Check to stop using an IBIS model if a [Pin] keyword does not list all pins on the component.

Related Topics

Pin Group Manager Dialog Box

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Auto-Grouping Dialog Box

Scope: BoardSim

To access: **Models > Edit Decoupling-Capacitor Groups**, click **Auto-Grouping** button

Use this dialog box to automatically create decoupling-capacitor groups.

Fields

Field	Description
Selected capacitors only	Creates a new capacitor group that contains capacitors you have selected in the "Decoupling capacitors" spreadsheet from the Assign Decoupling-Capacitor Groups Dialog Box.
All available capacitors (preserve existing groups)	Adds all unassigned capacitors in the "Decoupling capacitors" spreadsheet from the Assign Decoupling-Capacitor Groups Dialog Box to one or more new capacitor groups. The software automatically assigns decoupling capacitors with the same capacitance and maximum pin-to-pin dimensions to the same group. The software does not change the contents of existing groups.
Remove existing groups and perform complete regrouping	Deletes all capacitor groups (including groups you have manually created) and automatically assigns all capacitors to new groups. The software assigns decoupling capacitors with the same capacitance and maximum pin-to-pin dimensions to the same group.

Related Topics

Assign Decoupling-Capacitor Groups Dialog Box

Batch DC Drop Simulation Dialog Box

To access: **Simulate PI > Run DC Drop Batch Simulation**

Use this dialog box to simulate DC drop for multiple power supply nets at a time.

Fields

Field	Description
Board	Displays information for all boards or a selected board. For information about board IDs, such as B00, see "MultiBoard Project Board IDs" on page 566. (Available for a multiple-board design.)
Session File	Displays the path to the simulation settings file (*.dcs*).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Select Power/Ground Nets to Analyze	Lists the power and ground nets in your design that are available for simulation. Check a box to include a net in the simulation. Edit the constraint thresholds, using the following units: • Max Voltage Drop — % • Max Current Density — mA/mil^2 (English) or A/mm^2 (Metric) • Max Via Current — mA **Note:** To override this global maximum via current constraint and specify padstack-specific constraints, click **Via Constraints**. See the description on another row in this table. The simulation reports highlights results that exceed constraint values. To filter the list, enter a string in the Filter field and click **Apply**. The field supports the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character. To clear the filter, delete the contents of the Filter field and click **Apply**.
Assign Models	Opens the Assign Power Integrity Models dialog box, which enables you to assign models to IC power supply pins.
Via Constraints	Opens the **Plating/Constraints** tab of the Backdrilling/ Plating/Constraints Settings dialog box, which enables you to specify padstack-specific maximum current constraints and plating properties. See “Accounting for Per-Padstack Plating and Constraints” on page 137.
Include Reference Nets	Checked, includes reference nets in the simulation. Enable to include the reference nets for the selected net in simulation. Ensure that you specified a reference net for each VRM and IC pin when you assigned PI models. **Restriction:** The software automatically enables this option when you run thermal/DC drop co-simulation.
Create PowerScope Data	Checked, creates graphical simulation results that you can load into the HyperLynx PI PowerScope dialog box, and are used by the optional HTML report.

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Field	Description
Create Reports	Checked, creates simulation results in a format of your choice. **Note:** If you want the HTML report to contain screen captures and links that display the location of measurements in your board design, check Create PowerScope Data.
Write power-map files for Simcenter Flotherm/XT/ FLOEFD	Checked, saves power dissipation results for the simulated nets to a *Thermal<net_name>.txt* or *Thermal_<net_name>_<board_ID>.txt* file that you can import into any of these Siemens EDA tools: • Simcenter Flotherm • Simcenter Flotherm XT • Simcenter FLOEFD • HyperLynx ThermalSim
DC Drop Resistances DC + Resistances	Specifies whether to report DC drop, resistance between pairs of VRM and DC load pins, or both. **Note:** When an IC sink pin connects to a net with a multiple-phase VRM with two or more pins, the software measures resistance between an IC sink pin and each VRM pin. The software temporarily disconnects VRM and DC sink models for all non-measured pins before performing a measurement.

Related Topics

Running Batch DC Drop Simulation or Thermal Co-Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

General Batch Wizard

Use this wizard to set up and run general batch SI and EMC simulations.

To access: **Simulate SI > Run General Batch Wizard** ()

Topic	Description
General Batch Wizard - Quick Analysis Terminator Wizard Page	Use this page to specify the Terminator Wizard information types to include in the Summary report file. You can omit unwanted information to reduce clutter in the report.
General Batch Wizard - Quick-Analysis Interconnect Statistics Page	Use this page to select the interconnection statistics to include in the Summary report file. You can omit unwanted information to reduce clutter in the report.
General Batch Wizard - EMC Page	Use this page to select nets for radiated emissions simulation and specify electromagnetic compatibility (EMC) simulation options.

General Batch Wizard - Quick Analysis Terminator Wizard Page

To access: **Simulate SI > Run General Batch Simulation**, check Run Quick analysis and crosstalk estimation on selected nets on the **Overview** page, and check Suggest termination changes and optimal values on the **Quick Analysis** page.

Use this page to specify the Terminator Wizard information types to include in the Summary report file. You can omit unwanted information to reduce clutter in the report.

Fields

Field	Description
Include this terminator-suggestion information	• Total IC input capacitance — Checked, reports the IC input capacitance. • Effective trace impedance — Checked, reports the nominal characteristic impedance of a trace whose value has been adjusted to account for the presence of lumped IC capacitance loading.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Report these Terminator Wizard warnings	• Termination length violations — Checked, reports nets with traces that are too long for either the driver slew time or the connection between the driver IC pin and the series resistor. • Do not report length violations if any resistors found on net — Checked, reports length violations for nets that connect to a resistor of less than 350 ohms.
Coupling Settings	Opens the Coupling Settings dialog box. **Note:** Settings you specify in the Coupling Settings dialog box do not affect crosstalk estimates. They do affect results for options you enable on the Terminator Wizard and Interconnect Statistics pages.

Related Topics

Running General Batch Simulation

General Batch Wizard - Quick-Analysis Interconnect Statistics Page

To access: **Simulate SI > Run General Batch Simulation**, check Run Quick analysis and crosstalk estimation on selected nets on the **Overview** page, and check the Show interconnect statistics option on the **Quick Analysis** page.

Use this page to select the interconnection statistics to include in the Summary report file. You can omit unwanted information to reduce clutter in the report.

Fields

Field	Description
Min and max trace impedance	Checked, writes minimum metal Z0 and maximum metal Z0 data to the Interconnect Statistics section of the report.
Total trace length	Checked, writes total metal length data to the report.
Total trace capacitance (w/o ICs)	Checked, writes total metal capacitance data to the report, and enables you to select whether or not to include data for named physical nets (when the electrical net consists of more than one physical net).
Total trace inductance	Checked, writes total metal inductance data to the report, and enables you to select whether or not to include data for named physical nets.
Total trace resistance	Checked, writes total metal resistance data to the report, and enables you to select whether or not to include data for named physical nets.
Average trace impedance	Checked, writes average metal Z0 data to the report.

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Field	Description
Total trace delay (copper only; w/o ICs)	Checked, writes total metal delay data, which includes the summed propagation delay for all trace segments, in the report. **Tip:** Run detailed simulation to get the driver-to-receiver delay that includes net topology, possible reflections and ringing, receiver thresholds, and so on.
Coupling Settings	Opens the Coupling Settings dialog box. **Note:** Settings you specify in the Coupling Settings dialog box do not affect crosstalk estimates. They do affect results for options you enable on the Terminator Wizard and Interconnect Statistics pages.

Related Topics

Running General Batch Simulation

General Batch Wizard - EMC Page

To access: **Simulate SI > Run General Batch Simulation**, check Run EMC on selected nets on the **Overview** page.

Use this page to select nets for radiated emissions simulation and specify electromagnetic compatibility (EMC) simulation options.

Fields

Field	Description
Simulations options section	
IC-model corners	Selected, defines the IBIS IC model corner to use during simulation. Fast-strong usually generates the largest currents and radiation. For more information on min/typ/max data in an IBIS model, see IC Operating Settings.
Include radiation from	• Printed-circuit traces — Checked, models trace radiation. • Component packages — Checked, models the component package radiation. ICs can radiate as much or more than PCB traces. • Multipath from earth ground — Checked, models the floor as grounded and non-absorbing. For more information on multipath behavior, see Multipath Correction.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Distance from antenna to PCB	Defines the distance between the antenna and the PCB. Default is 3 meters. **Tip:** If you plan to perform real EMC laboratory measurements on the board, use the same distance you will use later in the laboratory.
Regulatory constraints	Specifies the limits to use for the simulation and class of products. • FCC — Checked, uses the limits of the United States. • CISPR — Checked, uses the limits of Europe. • VCCI — Checked, uses the limits of Japan. • User-defined — Checked, uses the limits you define. • Class A — Checked, uses the limits for industrial products. • Class B — Checked, uses the limits for consumer products.

Related Topics

Running General Batch Simulation

Bathtub Chart Dialog Box

To access: Run SERDES Wizard IBIS AMI simulation and view HTML results, or FastEye Channel Analyzer - View Analysis Results Page, select **Bathtub curves**, and run analysis

Use this dialog box to display and document bathtub curves. Bathtub curves help identify valid data sampling locations by reporting the bit error rate (BER) as a function of the sampling location across the unit interval (UI, same as bit interval) at several voltage offsets.

The Y axis represents the BER and the X axis represents the unit interval.

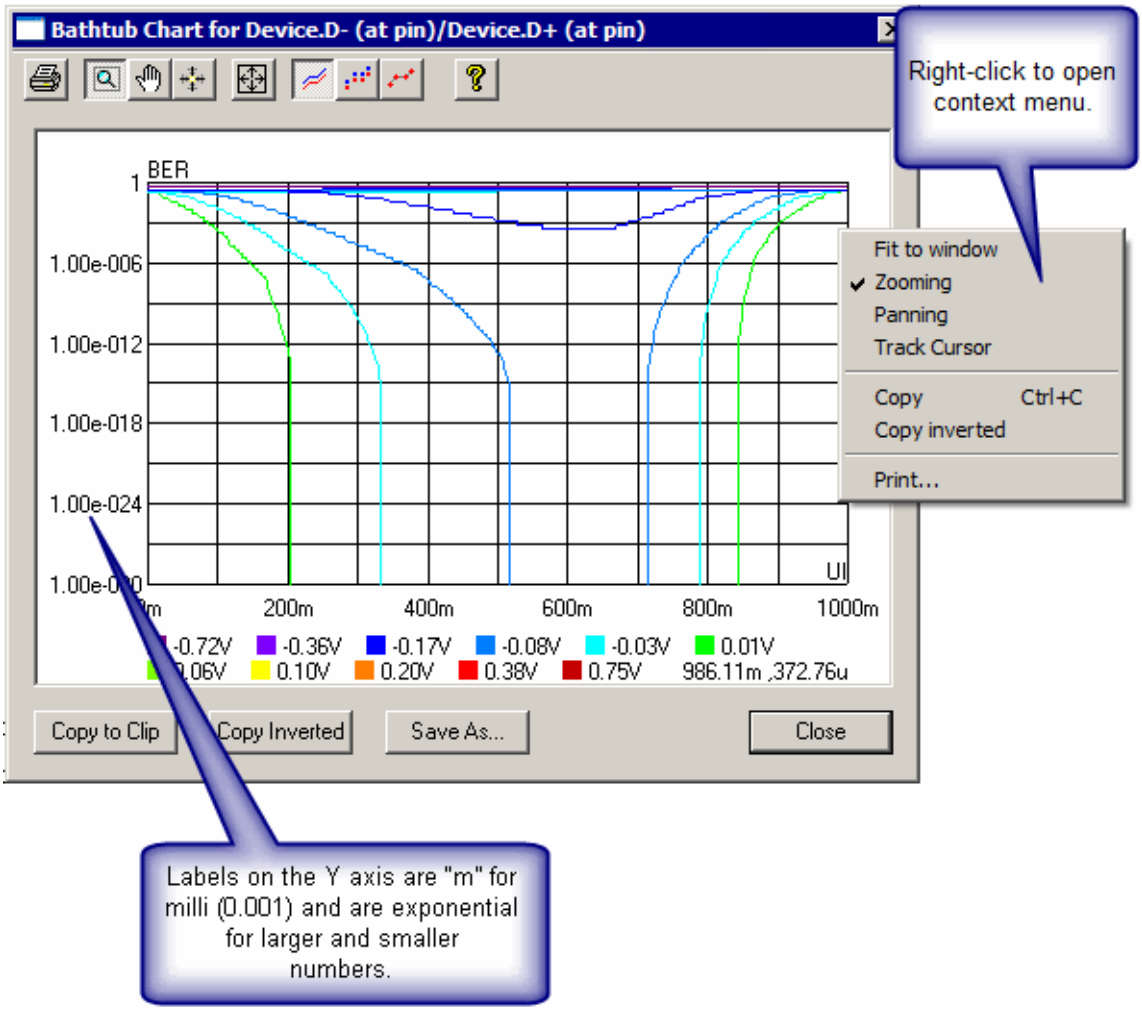
Bathtub curves indicate the quality of sampling locations across the UI by providing the probability of failure at each sampling location. If a point on the bathtub curve is located at 0.5 on the Y axis, an equal probability of bit transmission success and failure exists at that sampling location. By contrast, eye diagrams leave it to you to judge the probability of failure at sampling locations.

BER is directly related to the signal-to-noise ratio. Channel behaviors that can increase BER include reflections, jitter (random, deterministic), crosstalk, loss, and so on.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

The software displays a bathtub chart based on the middle eye when you specify a PAM-4 stimulus and display all three (unconsolidated) eyes.



Fields

Field	Description
Print (right-click) 	Print the graph with a white background.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Zooming (right-click) 	Zoom in by doing the following: 1. Position the mouse pointer over one corner of the zoom box you want to create, and then drag to define the other corner of the zoom box. 2. Release the mouse button to magnify the contents of the zoom box to fill the graph.
Panning (right-click) 	Pan by dragging the graph across the dialog box.
Track Cursor (right-click) 	Attach measurement crosshairs to a waveform by selecting the waveform to measure. As you move the mouse horizontally, the measurement crosshairs tracks the selected curve.
Fit to window (right-click) 	Fit the entire curve to window.
	Display only lines between curve vertices (no vertice dots).
	Display only curve vertices (no lines).
	Display both lines and vertice dots.
	Open Help for the dialog box.
Copy (right-click)	Copy graph to the clipboard and use a white background. This option uses less printer ink or toner if you print it out. You can paste the image from the Windows clipboard into another program such as Microsoft WordPad.
Copy inverted (right-click)	Copy graph to the clipboard and use a black background. You can paste the image from the Windows clipboard into another program such as Microsoft WordPad.
Save As	Save the numerical bathtub data to a file. You can open the file with a spreadsheet application, such as Microsoft Excel.

Unit Interval Origin

The origin of the UI axis is based on the time of the maximum voltage for the initial transition of the waveform. As a result, 0.5 on the UI axis may not be located at the “center” of the bathtub curve. Asymmetric bathtub curves can also move the center of the bathtub away from 0.5 on the UI axis.

When multiple bathtub curves are displayed, note that their UI origins are set by independent maximum voltages and do not necessarily align to each other or to the start of simulation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Bathtub Curve Sampling Voltage Offsets

The Bathtub Chart dialog box displays BER curves for several sampling voltage offsets, which can represent sampling threshold level variations in the system. A legend near the bottom of the X axis maps the color of the BER curve to the sampling voltage offset.

The nominal sampling voltage (or voltage origin—0 V) is the median value of the high and low voltage levels, which is the crossover voltage for differential signaling and linear models. The curve associated with the nominal sampling voltage maps to the left-most legend color square.

Change Trace Widths Dialog Box

Scope: BoardSim

To access: **Edit > Trace Widths**

Use this dialog box to change the width of a trace on nets or stackup layers you specify. This capability helps you perform “what if” experiments to improve signal quality.

Fields

Field	Description
Choosing Which Segments to Change	The Select Trace Segments To Change area in the Change Trace Widths dialog box gives you considerable flexibility in choosing which trace segments to alter. Each of the sub-selections below can be set independently of the other two selections:
Choosing Which Nets	In the Traces On These NETS area, you can choose to make width changes on only a selected net, or on all of the board's nets. If you are choosing a particular net, you can change the order in which the nets are listed in the Selected Net combo box by selecting the desired radio button in the Sort Nets By area.
Choosing Which Layers	In the AND On These LAYERS area, you can choose whether to change widths on only a selected stackup layer, or on all layers. The Selected Layer combo box lists all of the layers in the current board stackup.
Choosing a Range of Widths	In the AND With WIDTHS In This RANGE area, you can choose to limit the changes to only trace segments in a selected range of original widths, or to all segments regardless of width. If you choose to enter a range, you enter the minimum and maximum widths in the range.

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Field	Description
Conditions are ANDed	Notice that the three selection criteria (nets, layers, and width range) are ANDed together. To eliminate one of the criteria, click the All radio button in that criterion's area. For example, to eliminate the width range, click the All Widths radio button.
Restoring Original Widths	To restore the original trace widths used in your PCB layout, you must re-load the board file. When you re-load, the width changes made in the previous session are discarded and the original widths from the board file are restored.

Usage Notes

Possible Bad Effects from Width Changes

Changing the widths of a board's traces usually does not affect the electrical validity of the traces, but this cannot always be guaranteed. Generally, narrowing traces (i.e., making them narrower) is usually safe; widening traces (if the board is densely routed) may cause electrical problems.

For example, if you widen a trace too much such that it touches another trace, BoardSim may connect the two traces together (because you've “shorted” the widened trace to the other trace).

In a given signal-integrity simulation, BoardSim only “looks” at the net you've chosen for simulation, plus any associated nets. Therefore, it's not as dangerous to widen traces as it might seem. Problems only arise if the widened trace touches another segment on the same net or a segment on an associated net.

In rare cases, narrowing a trace may cause electrical problems. This could occur, for example, if a trace connects to a pad marginally, at the edge of the trace only. Narrowing the trace could cause the trace-to-pad connection to be opened.

Related Topics

Solving Problems Found in Simulation

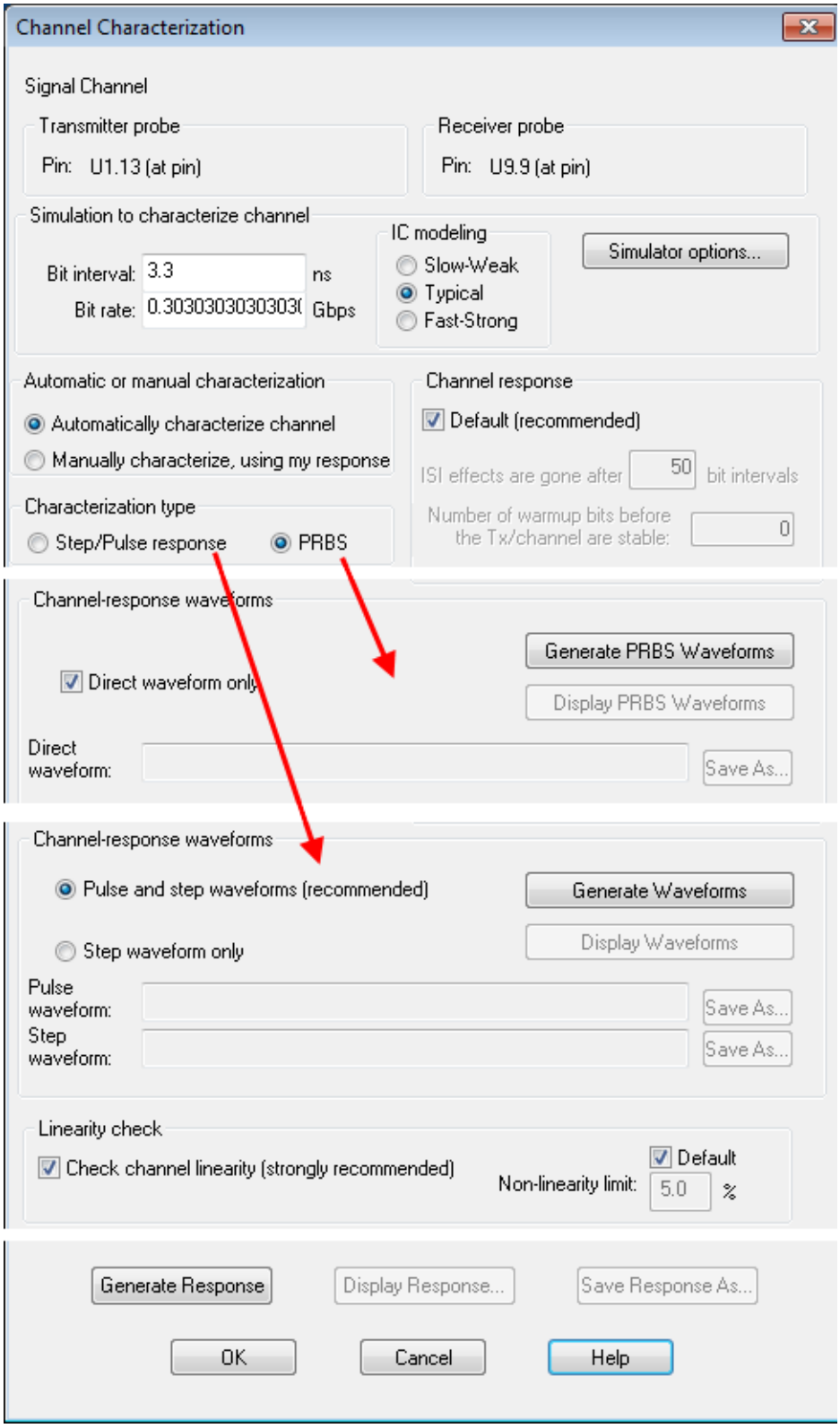
Channel Characterization Dialog Box

To access: Select **SI Simulation > Run FastEye Channel Analysis**, select the **Set Up Channel Characterization** page, select **New/View** or select a crosstalk spreadsheet row and select **Characterize Selected**

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Use this dialog box to set up simulation properties for a new channel characterization. Optionally, you can view channel-response waveforms automatically created by channel analysis or manually created by this dialog box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Signal Channel Crosstalk Channel	Signal Channel—Indicates a signal channel without crosstalk modeling or a victim channel with crosstalk modeling. Crosstalk Channel—Indicates an aggressor channel. Note that many options are unavailable for aggressor channels.
Transmitter probe Area Receiver Probe Area	
Pin	Name of the channel pin to probe. You can run channel analysis for one single-ended channel or one differential channel at a time. If an expected differential probe does not appear, manually create a differential probe with the oscilloscope.
Simulation to characterize channel Area	
Bit interval Bit rate	When choosing between Bit interval and Bit rate, use the option that provides the best accuracy. For example, to test the channel at 333 Mb/s, you can specify a bit rate of 0.333 Gb/s instead of a bit interval of 3.00300300300 ns. Editing the Bit interval value updates the Bit rate value, and vice versa. The values may have been previously set by any of the following sources, sorted in descending priority: The fitted-poles (*.PLS*) file is used to characterize the channel by the software in IBIS AMI simulation. The *.PLS* file contains a comment that specifies the bit interval. The Bit interval and Bit rate values used for standard eye diagrams and specified in the Configure Eye Diagram Dialog Box - Stimulus Tab. The simulation time step for characterization simulation is based on the bit interval.
IC modeling area	IC-model operating conditions: • Slow-Weak • Typical • Fast-Strong
Simulator options	Opens the Simulation Controls Dialog Box.
Automatic or manual characterization Area	
Automatically characterize channel	Automatically create channel-response waveforms and extract a fitted-poles channel characterization file from them.
Manually characterize, using my responses	Use a previously-saved channel characterization file. See External Characterization Files.
Characterization type Area	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Step/Pulse response	Drive step response and pulse response waveforms and analyze the channel response.
PRBS	Drive a PRBS (pseudorandom binary sequence) waveform and analyze the channel response. The PRBS waveform usually provides enough time for the initial conditions for the channel (such as with DC blocking capacitors) and transmitter/receiver ICs to settle out. See Bit Sequence for Automatic Channel Characterization.
Channel response Area	
Default (recommended)	You typically de-select this option to investigate non-linearity or other analysis-related errors reported by the wizard.
ISI effects are gone after	Specify the number of bit intervals needed for transient activity to settle. The wizard reports an error if it detects unsettled activity at the end of the channel-characterization simulation. See inter-symbol interference (ISI). The default value is conservative and provides ample time for interference, due to the topology and reflection characteristics of the channel, to die out. **Restriction:** You can type values from 50 to 1000, in bit intervals.
Number of warmup bits before the Tx/channel are stable	Specify a sufficient number of bits for the driver and receiver circuits to stabilize and reach normal operating conditions. See Factors that Affect the Number of Warmup Bits. This option adds bits to the beginning of the overall bit sequence used to create the channel-response extraction waveforms. **Restriction:** You can type values from 0 to 255, in bit intervals.
Channel-response waveforms Area - The contents of this area depends on the option you enable in the Characterization type area.	
Direct waveform only	Select to specify only one waveform, instead of a pair of direct and inverted waveforms.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Pulse and step waveforms	Generate both step response and pulse response waveforms. This option is recommended for FastEye channel analysis because it enables the wizard to check the channel for linear response behavior and to identify the DC values for logic zero and one, which are required for accurate results. This option is not recommended for IBIS-AMI channel analysis because the simulation engine requires linear response behaviors and accepts only one channel-response waveform. Linear response behavior means that there is no difference in the channel response to a rising versus falling edge.
Step waveform only	Generate only the step waveform. This option is recommended for IBIS-AMI channel analysis because the simulation engine requires linear response behavior. Linear response behavior means that there is no difference in the channel response to a rising versus falling edge.
Pulse waveform Step waveform Save As	After you optionally Generate Waveforms, you can save them to disk. **Restriction:** These options are available when you select Step/ Pulse response.
Direct waveform Inverted waveform Save As	After you optionally Generate Waveforms, you can save them to disk. **Restriction:** These options are available when you select PRBS.
Generate Waveforms	Optionally generate the waveforms while this dialog box is open. This enables you to display the waveforms and save them to disk.
Display PRBS Waveforms Display Waveforms	After you optionally Generate Waveforms, you can display them. These waveforms are extracted from the fitted-poles file and their length is based on the unit interval. You may want to display channel-response waveforms to investigate unexpected channel analysis results or to learn details about the channel characterization. For example, a good step response waveform eventually settles to the opposite voltage and a good pulse response waveform has a single peak and the leading edge starts its transition at about the same time as the step response.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Browse	Load a channel-response waveform file that you either saved by selecting *Save As* in this dialog box or created with third-party software and then selected Manually characterize, using my responses. Waveform files that you create with third-party software must meet the requirements listed in Requirements for External Channel-Response Waveforms. By contrast, waveform files created by this dialog box automatically meet the requirements.
Linearity check Area	
Check channel linearity (strongly recommended)	Compare the pulse and step waveforms and report non-linear channel-response. See Checking Channels for Linear and Time-Invariant Behavior. You can disable this option for any of the following reasons: • You are absolutely certain the channel is linear. • You previously ran the wizard with the same input waveform files, and they passed linearity checking. • You are creating only a worst-case bit sequence and are not creating a FastEye diagram.
Default	Enable to apply the default Non-linearity limit, which is the maximum value that has been empirically tested to provide FastEye channel analysis results that are very close to time domain SPICE simulations. Disable to investigate non-linearity or other analysis related errors reported by the wizard. The wizard reports an error if the difference of the energy between the actual and calculated pulse responses exceed the Non-linearity limit value.
Non-linearity limit	The channel non-linearity threshold. The wizard reports an error if the channel non-linearity exceeds this value. Channel analysis checks the linearity of the channel by comparing the energy in the actual pulse response to the energy in the calculated pulse response, where the calculated pulse response is the difference between the actual step response and the actual step response negated and delayed. See Checking Channels for Linear and Time-Invariant Behavior.

Factors that Affect the Number of Warmup Bits

Provide enough simulation time for the driver and receiver circuits to stabilize and reach normal operating conditions. Some drivers and receivers contain circuitry that requires several bits to stabilize and produce linear or representative signal transitions.

For example, bigger drivers and receivers may need several bits to set internal flip flops to a known state.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The number of warmup bits depends on any of the following:

- Your knowledge of the driver/receiver model and channel interaction.
- Documentation that comes with a HyperLynx device kit.
- Documentation that comes from the driver/receiver model vendor.
- Troubleshooting efforts—If the extracted step/pulse-response waveforms look incorrect and you suspect that channel stabilization could be a factor, try adding more bits.

Requirements for External Channel-Response Waveforms

To run FastEye/IBIS-AMI channel analysis with step- and pulse-response or PRBS waveforms that you create using third-party software, verify they meet the requirements contained in this section.

After manually editing the waveform files, you can display them in the Digital Oscilloscope Dialog Box or in EZwave to verify their contents.

General requirements:

- Waveform files can contain waveforms for more than one signal or differential pair.
- For differential channels, the simulation waveforms must represent the differential behavior between the pins in the differential pair. In other words, do not specify waveforms for the individual pins in a differential pair.
- If you provide a PRBS waveform:
  - The simulation stimulus should consist of a PRBS bit pattern with a minimum bit order of 6. This produces a waveform consisting of 63 bits ().
  - The stimulus must have two or more repetitions.
  - The stimulus bit order must correspond to the shortest PRBS sequence exceeding ISI length.
  - The PRBS taps and seed value of the stimulus pattern must match those which the FastEye or AMI Wizard would use in an automatically generated PRBS stimulus.
- If you provide separate PRBS direct and inverted waveform files, they must have the same number of simulation cycles.

  For example, you could create the direct waveforms with the initial stimulus state of “Low” and create the inverted waveforms with the initial stimulus state of “High” while inverting the logic state of each bit in the stimulus.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Provide simulation waveforms that meets or exceeds the minimum extraction time (also known as “correlation time”) of either of the following:

- SI x bit interval. You specify the bit interval and ISI values in the Channel Characterization Dialog Box.
- 15 x maximum driver delay.

Provide enough time for the channel to settle down.

The simulation used to create the waveforms must allow enough time for the channel to dissipate its responses to the stimulus. The suggested simulation run time for each pulse-response and step-response waveform is the greater of the following:

- Bit interval * ISI_propagation_length_(in_bits), where ISI is inter-symbol interference

  Physically, this value is how long (in bits) the response to a single transition lasts.
- Longest delay * 15

For example waveforms, see Analog Channel Characterization Step/Pulse Waveform Descriptions.

Provide enough simulation time for the driver and receiver circuits to stabilize and reach normal operating conditions.

Some drivers and receivers contain circuitry that requires several simulation cycles to stabilize and produce linear or representative signal transitions. For example, bigger drivers and receivers may need several simulation cycles to set internal flip flops to a known state.

Provide enough simulation time for series capacitors to charge.

Series decoupling capacitors in the channel may not always start with the correct DC initial condition, and the channel may drift for some time before reaching normal DC values.

Provide a quiet time (a period of stimulus inactivity) between the group of simulation cycles used to stabilize the driver and receiver circuits and the group of simulation cycles used to provide step- and pulse-responses to characterize the channel behavior.

The quiet time should be long enough for reflections to end. Knowing how long to wait for the channel to settle may require experimentation because it depends on both the channel length and the influence of discontinuities and terminations.

Step- and pulse-response waveforms start at the same logic state and identical voltage.

While Figure 10-5 and Figure 10-6 show waveforms starting at logic zero, the wizard accepts externally-generated waveforms starting at logic one. If the step-response waveform starts at logic one, the pulse-response waveform must also start at logic one. Similarly, if the step-

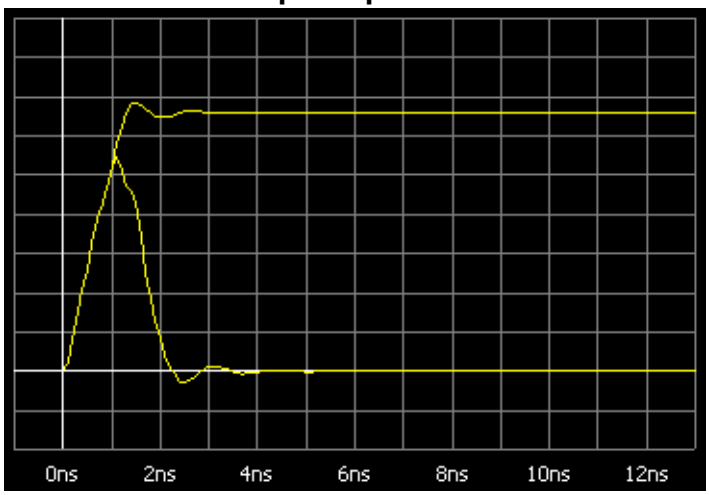
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

response waveform starts at logic zero, the pulse-response waveform must also start at logic zero.

Waveforms automatically created by the wizard start at logic zero.

The starting voltages for the step- and pulse-response waveforms must be identical. If not, FastEye/IBIS-AMI channel analysis reports non-linearity and uses only the step-response waveform. See Figure 10-4.

Figure 10-4. Pulse and Step Response Waveforms - Zoomed In



Pulse-response waveforms start and end within a voltage tolerance.

The starting and ending voltages must be within 1% of the peak value of the pulse-response waveform.

Align the initial transition times for the step- and pulse-response waveforms.

The initial rising (or falling) transition must begin at the same time in the step- and pulse-response waveforms.

Remove un-needed portions of the waveforms.

The goal is to provide the step-response behavior in one waveform file and the pulse-response behavior in another waveform file.

Remove waveform activity preceding and following the step-response or pulse-response behavior. Remember that the waveforms must provide enough time for the channel to settle down.

Some complex SPICE models need to run for several simulation cycles for the driver circuitry to reach normal switching behaviors. Remove this “circuitry start up” portion of the waveform preceding the step-response or pulse-response behavior.

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If you remove waveform activity preceding the step-response or pulse-response behavior, make sure the start up portion of the final waveforms are identical, that is they originate from the same voltage and transition time.

For differential channels, the simulation waveforms must represent the differential behavior between the pins in the differential pair.

In other words, do not specify waveforms for the individual pins in a differential pair.

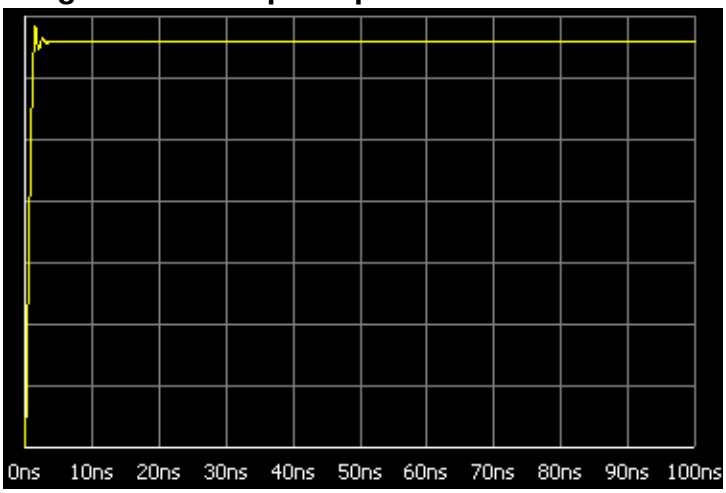
Analog Channel Characterization Step/Pulse Waveform Descriptions

The Channel Characterization Dialog Box can use the following types of step/pulse waveforms:

Single edge step— Step-response curves show where reflections occur and how long energy is stored in the channel. From a bit sequence perspective, a step-response waveform resembles the 011111111111 bit sequence, with many trailing ones. See Figure 10-5.

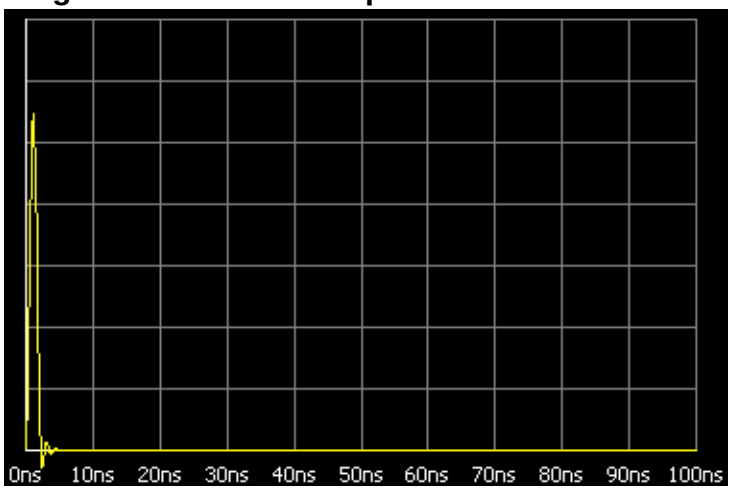
Single pulse— Pulse-response curves show information about ISI and serve as a building block for the FastEye/IBIS-AMI analysis contour. From a bit sequence perspective, a pulse-response waveform resembles the 010000000000 bit sequence, with many trailing zeroes. See Figure 10-6.

Figure 10-5. Step Response Waveform - Full



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Figure 10-6. Pulse Response Waveform - Full



Supported Analog Channel Characterization Waveform File Formats

The wizard accepts waveform files in the following formats:

- .LIS—Created by SPICE simulation.
- .LIS (HyperLynx)—Created by the Digital Oscilloscope Dialog Box.
- .CHI—Created by ADMS.
- .CSV—Created by ICX, ICX Pro, and the Digital Oscilloscope Dialog Box.

Related Topics

FastEye Channel Analyzer - Set Up Channel Characterizations Page

Configure Eye Diagram Dialog Box - Eye Mask Tab

To access:

- From the Digital Oscilloscope Dialog Box, select **Eye Diagram > Global > Configure**
- From the Digital Oscilloscope Dialog Box, select **Eye Diagram > Per-Net/Pin > Assign**
- From the HyperLynx Eye Density/BER Viewer, select **Edit > Configure Eye Mask**
- From the FastEye Viewer, select **Configure**

Use this tab to edit eye mask properties for eye diagram analysis. You can load existing eye masks from a library or save new eye masks into a library.

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The *BSW.mask* file is located in the same folder as the HyperLynx application file *bsw.exe* (Windows) or *bsw* (Linux). For example, *C:\MentorGraphics\<release>\SDD_HOME\hyperlynx\bsw.exe*. On Windows, you can learn the folder name by right-clicking the **HyperLynx Simulation Software** Start menu item, and then selecting **Properties**. The Target field contains the folder name.

Options

Option	Description
Mask Library Area	
Mask Name	Displays either of the following: • Names of eye masks stored in the *User.mask* and *BSW.mask* files located in the HyperLynx installation directory. • <unnamed>, which enables you to create a new eye mask without affecting the values for an existing eye mask.
Save 	Saves the currently-displayed eye mask to the library file named *User.mask*, which is located in the HyperLynx installation directory.
Save as 	Saves the currently-displayed eye mask to a new mask name in the library file named *User.mask*, which is located in the HyperLynx installation directory.
Delete 	Removes the currently-displayed mask from the library file named *User.mask*, which is located in the HyperLynx installation directory. **Restriction:** You cannot remove eye masks saved in the *BSW.mask* file.
Scale Area	
Time, Voltage	Scales the eye mask values to verify timing or voltage margins.
Offset Area	
<offset time and voltage>	Repositions the eye mask in the Digital Oscilloscope Dialog Box. These values change when you enable Adjust Mask and drag the eye mask to a new position on the oscilloscope screen.
BER Threshold Area	

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Option	Description
<BER value>	Displays the pass/fail threshold for the Pass/fail eye mask spreadsheet column in the HyperLynx Eye Density/BER Viewer. Enter values in decimal or scientific notation, such as 1e-12. **Restriction:** This area is available only when you open this dialog box from the HyperLynx Eye Density/BER Viewer.
Mask Display Area	
<mask time and voltage>	Enter values to either create a new mask or override the values for a mask in the library. **Restriction:** The eye mask fields are available only when the values in the Scale Area are 100%.

Configure Eye Diagram Dialog Box - Stimulus Tab

To access:

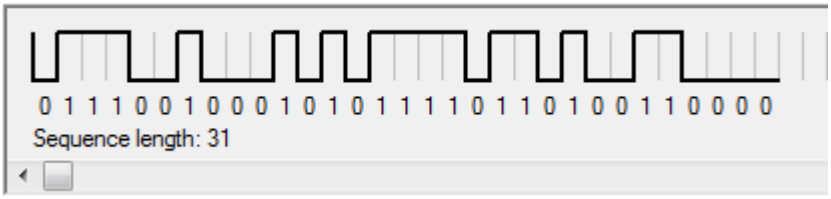
- From the Digital Oscilloscope Dialog Box, choose **Eye Diagram > Global > Configure**
- From the Interactive Simulation Dialog Box, choose **Eye Diagram > Global > Configure**
- From the Interactive Sweeps Dialog Box, choose **Eye Diagram > Global > Configure**

Use this tab to define complex driver or transmitter stimulus.

Fields

Field	Description
Stimulus Name	Displays the path of a stimulus file (EDS) you have opened. An EDS file contains wave shape and timing information.
	Loads a stimulus file.
	Saves stimulus to a file. Using meaningful stimulus names, such as PRBS_128 or 250MHz_clock, can help you to recognize the stimulus contents when assigning stimulus to specific pins or nets.
	Saves stimulus to a new file or location.
	Permanently deletes a stimulus file.

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Field	Description
Bit pattern Area	
Sequence	Specifies the type of stimulus.
Bit order	Specifies the number of bits in the sequence. The number of bits is 2**<bit_order> - 1. (Available for PRBS stimulus.)
Initial state	Specifies a high or low initial driver value.
Display and edit area	 You can use this area to create custom bit patterns when you select <Custom> Sequence. See “Creating a Custom Stimulus” on page 492.
Stimulus Area	
Bit interval	Specifies the length of the unit interval, in ns. Editing this value also updates the Bit Rate value. When choosing between the Bit interval and Bit Rate properties, use the one that provides the best accuracy. For example, to test the channel at 333 Mb/s, you can specify a bit rate of 0.333 Gb/s instead of a bit interval of 3.003003003 ns.
Bit Rate	Specifies the number of bits transmitted through a channel, in gigabits per second. Editing this value also updates the Bit Interval value.
Sequence reps	Specifies the number of times to repeat the stimulus. If you enable jitter, the software applies unique jitter to each sequence repetition.
Period	Specifies the period in ns. Editing this value also updates the Frequency value. (Available for Toggling stimulus.)
Frequency	Specifies the frequency in megahertz. Editing this value also updates the Period value. (Available for Toggling stimulus.)
Duty cycle	Specifies the time, in percentage of the period, that a stimulus is high. (Available for Toggling stimulus.)

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Display Area	
Skip first	Specifies the number of bits to exclude from the eye diagram. You can use this option when the driver and receiver circuits need a number of warmup bits to stabilize and reach normal operating conditions.
Show	Specifies the number of eyes to display in the Digital Oscilloscope dialog box.
Jitter Area For advice about choosing a jitter distribution, see “Jitter Distribution Types” on page 553. When using more than one jitter distribution type, you can isolate the contribution of each type by enabling one type at a time and running separate simulations. Do not specify the jitter produced by the following effects, unless you have a specific reason to do so: • PCB layout effects, such as impedance mismatches and signal dispersion. • Data-dependent effects, such as ISI, duty-cycle distortion, pseudo-random bit sequence periodicity.	
Gaussian (s)	Checked, adds jitter with a normal distribution that has no sigma limit. A histogram consisting of a large number of Gaussian distributed jitter values resembles a bell curve. Specifies the jitter magnitude and units. See “Gaussian Jitter” on page 556.
Uniform (pk-pk)	Checks, adds jitter with an even distribution. A histogram consisting of a large number of uniform-distributed jitter values resembles a rectangle. Uniform produces a worst-case distribution more quickly than Gaussian. Specifies the jitter magnitude and units. See “Uniform Jitter” on page 560.
Sine (1/2 pk-pk) Frequency	Checked, adds sinusoidal jitter. Specifies the jitter magnitude, jitter frequency (the rate at which the jitter offset varies), and units. See “Sinusoidal Deterministic Jitter” on page 559.
DCD (pk-pk)	Checked, adds duty cycle distortion jitter. Specifies the jitter magnitude and units. See “Duty Cycle Distortion Jitter” on page 555. **Note:** The Digital Oscilloscope dialog box does not apply duty cycle distortion jitter.

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Field	Description
Advanced	Opens the Advanced Jitter Settings Dialog Box, where you can specify additional jitter settings, save jitter settings to a file, or load jitter settings from a file.

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Configure IC Component Symbol

Scope: LineSim

To access: Open the Assign IC Component Model Dialog Box and click **Configure**.

Use this dialog box to hide pins that you do not plan to include in simulation, and to edit pin location.

Fields

Field	Description
IBIS pin list	Specifies hidden pins.
Symbol pins in order of appearance	Specifies visible pins and their location on an IC component symbol.
Up Down	For pins selected in the right spreadsheet, moves pins up or down.

Related Topics

Hiding or Moving an IC Component Pin

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Confirm Connections Dialog Box

Scope: LineSim

To access: **Edit > Connect Symbols**

Use this dialog box to reverse connections between two IC component symbols.

Note

The Signal column is blank for a pin on an S-parameter or SPICE model component.

Fields

Field	Description
Reverse order	Check to reverse the order of wires connected to a symbol.
Connect	Applies connections defined by the spreadsheet.

Related Topics

Hiding or Moving an IC Component Pin

Connect Nets with Manhattan Routing Dialog Box

Scope: BoardSim

To access: **Edit > Connect Nets with Manhattan Routing**

Use this dialog box to perform “what if” trace routing and component placement experiments on your board design by editing the length of a routed, unrouted, or partially-routed signal net.

Manhattan routing allows you to simulate a board design that contains unrouted nets, or routed nets that you want to reroute. The software models these nets as transmission lines, and calculates a routed length by using a Manhattan algorithm, where net paths follow the x or y direction only (no diagonal paths). The software uses an uncoupled transmission line to represent net routing from one component pin to another, and it does not account for coupling to neighboring traces, loss, and impedance discontinuities (such as a via or different trace widths). As a result of this net modeling, SI simulation results are approximate.

You can either manually specify the net length or have the software calculate it for you (by accounting for the distance between component pins and a routing factor that you specify). When a net has three or more component pins, the software proportionally distributes the overall net length among the transmission lines that represent the net. For information about automatic net length calculation, see “Automatic Net Length Calculation” on page 755.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

Even though this dialog box can change the virtual routing of a power supply net, you should change only the virtual routing for signal nets.

Fields

Field	Description
All unrouted nets	You can choose to route all unrouted nets, or you can route specific nets from the list of unrouted nets.
Selected nets only	
Selected nets and associated nets	
Design file	For MultiBoard projects, specifies the single board upon which you want to operate.
Specify Manhattan multiplier	Enables you to specify a multiplier value. For more information, see "Automatic Net Length Calculation" on page 755.
Specify length	Enables you to specify the length of traces created by Manhattan routing.

Usage Notes

- Manhattan routing is created only for one selected board ID for a multiple-board project (Manhattan routing stops at the board's external connector).
- This dialog box opens automatically when all of the following are true:
  - The board contains any number of completely unrouted nets.
  - Manhattan routing information for the completely unrouted nets has not been saved to the *.BUD* file, or restored from the *.BUD* file using the Restore Session Edits dialog box.
  - You click **Yes** when the software asks whether you want to create Manhattan routing at board-load time.
- To unroute routed nets, select:

  Edit > Unroute Routed Nets

Automatic Net Length Calculation

The software automatically calculates the Manhattan routing's net length for all re-routed nets using this equation:

simulation net length = Calculated Manhattan Length x Manhattan multiplier

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- Calculated Manhattan Length—refers to the sum of all pin-to-pin segment lengths for the net, where the software calculates the Manhattan length for each segment.

  Manhattan length refers to the shortest connection than can be made between two points on a board while using only the board's X-Y routing tracks (that is, no diagonal traces). The Manhattan length is the sum of the segment's X length and Y length, where X length = abs(X1 – X2) and Y length = abs(Y1 – Y2). For nets with several pins or pads, the Manhattan length is calculated for each segment, which is then summed into the Calculated Manhattan length.

- Manhattan multiplier—a factor used to compensate for non-ideal routing (that is, routing that avoids collisions with other board elements) or to compensate for 45-degree routing segments. You may increase or decrease the Manhattan multiplier to take these issues into account.

Related Topics

Solving Problems Found in Simulation

Coupling Settings Dialog Box

Scope: BoardSim

To access: **Setup > Coupling Settings**

Use this dialog box to specify trace, package, and area fill coupling settings.

Fields

Field	Description
Trace to trace coupling section	
Include trace to trace coupling	Checked, includes coupling among traces. **Note:** When you uncheck this option and interactively select multiple nets, simulation still accounts for trace-to-trace coupling among selected nets that meet coupling thresholds.
Include self-coupling	Checked, includes coupling between trace segments on the same net whose routing forms meanders, trombones, or accordions with tight spaces. **Note:** When you enable geometric coupling thresholds in the Set Coupling Thresholds Dialog Box and set Horizontal Neighbor Limit to 0, the software does not model self coupling (even when you check this option). **Note:** When you enable electrical coupling thresholds in the Set Coupling Thresholds Dialog Box, the software models self coupling (even when you uncheck this option).

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Field	Description
Include non-parallel coupling	Checked, includes coupling between non-parallel traces, up to 35 degrees. **Note:** Coupling between area fills and non-parallel traces is always modeled for all angles.
S-parameter/Package coupling section	
Include S-parameter coupling	Checked, enables package coupling, connector coupling, and 3D-EM solved area and power-aware-SI model coupling.
Use same threshold for all	Checked, enables you to set a single threshold that applies to packages, connectors, 3D-EM solved areas, and power aware SI models.
Frequency range	Specifies the frequency range within which coupling thresholds apply. **Note:** The frequency range threshold does not apply to RLC matrix or SPICE package models.
Include package coupling	Checked, includes coupling defined by IBIS package models. • For S-parameter package models, the software uses the dB threshold value. • For RLC matrix package models (defined within the [Define Package Model] keyword), the software determines the coupling ratio for each pin and compares the value 20*log(C_mutual/C_self) to the dB threshold value. See “Coupling Ratio for RLC Matrix Package Coupling” on page 489. • For SPICE package models, the software considers *all* pins in the SPICE circuit as coupled if you specify a threshold value below -6 dB (such as the default -20 dB). If you specify a value between 0 and -6 dB, the software considers *all* pins to be uncoupled.
Include connector coupling	Checked, accounts for coupling within assigned S-parameter models that describe connections between boards in a multiple-board design (Advanced connector models). See “Setting Up a Multiple Board Design” on page 57. The software does not use this threshold for SPICE Advanced connector models, and considers all pins in the SPICE circuit as coupled.
Include 3D-EM solved area and Power Aware SI coupling	Checked, accounts for coupling within 3D-EM solved areas (S-parameter models) and power aware SI S-parameter models.
Trace to area fill coupling section	

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Field	Description
Include trace to area fill coupling	Checked, models coupling between a selected signal net and both of the following: • Area fills and hatched planes. Enter fill values as defined on the dialog box. Note: If you enter a search distance for an area fill that is smaller than the search distance for traces, the simulation instead uses the "Maximum distance from aggressor" value from the Set Coupling Thresholds Dialog Box. Note: The software models area fills (solid and hatched) and power supply nets as ideally grounded conductors. • Routed power supply nets, using settings from the "Use geometric thresholds (advanced)" section of the Set Coupling Thresholds Dialog Box. Unchecked, the software calculates trace impedance assuming that plane layers are completely solid.
Area fills search distance	See Trace to Area Fill Coupling Examples.
Area fill grid	Smaller values result in approximated area fill shapes with more details, but also increase the simulation run time by increasing the number of field solver cross sections and the number of elements in the simulation netlist. **Note:** If your design contains hatched area fills or planes, set the grid to a value that is equal to, or less than, the distance between the hatching shapes. See Area Fill Edge Approximation Examples.

Related Topics

Accounting for Coupling

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Coupling Settings Dialog Box (LineSim)

Scope: LineSim

To access: In LineSim, click **Setup > Coupling Settings**

Enables you to specify which coupled nets are included with those you select in the design area.

Fields

Field	Description
Force select all nets	Checked, this option disables individual net selection. The software considers all nets selected for all stimulation types, regardless of coupling interaction between nets. Checking this box disables all other options in the Coupling Settings dialog box.
Include S-parameter coupling	Checked, the software considers coupling between nets within an assigned or referenced S-parameter model.
Use same threshold for all	When checked, sets a single threshold that applies to nets regardless of package or SI model content.
Include blackbox coupling	Checked, considers coupling relationships between nets within a connected S-parameter model that is assigned as an S-parameter/SPICE model symbol (represented in a schematic as a black box—see “Assign S-Parameter/SPICE Model Dialog Box” on page 719). When the coupling relationship between a selected net and any additional nets exceeds the threshold, the software considers each additional net that is coupled and includes it in the set of nets selected for simulation.
Include package coupling	Checked, the software considers coupling relationships between nets within an S-parameter package model referenced from an IBIS model that is assigned to a buffer symbol (see “Modeling a Package in an IBIS File” on page 642). When the coupling relationship between a selected net and any additional nets exceeds the threshold, the software considers each additional net that is coupled and includes it in the set of nets selected for simulation.
Include t-line coupling	Checked, the software considers any nets connected by transmission line symbol coupling regions to be coupled. See “Viewing Electrical Field Lines in LineSim for Coupling Regions” on page 369. When you select a net for simulation, nets connected to the selected net by a coupling region are also included in the set of selected nets.

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Field	Description
Frequency range	Specifies the frequency range within which coupling thresholds apply.

CTLE Settings Dialog Box

To access: **SI Simulation > Run FastEye Channel Analysis**, select the **Add Pre-Emphasis/DFE/CTLE** page, select **Specify taps/parameters**, select **Add CTLE**, then click **Specify parameters**.

FastEye Analysis allows you to simulate the effect of a CTLE high-pass peaking filter at the receiver for your design.

FastEye analysis can determine the most effective CTLE parameters for your design when you select Synthesize optimal values on the **Add Pre-Emphasis/DFE/CTLE** page. If you want to specify specific values from manufacturers data, select Specify taps/parameters and type the values in CTLE Settings dialog box.

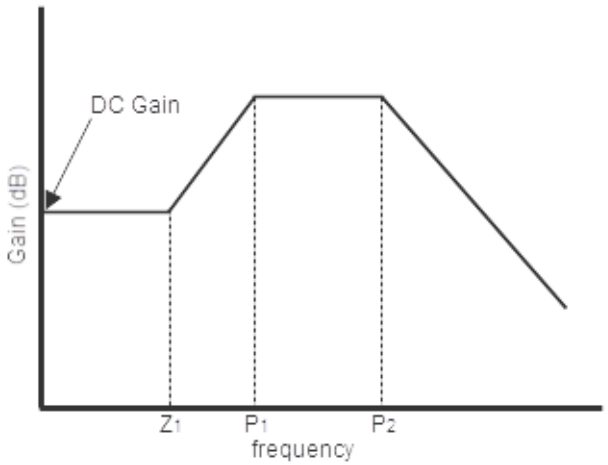
This dialog box also appears to provide the optimal CTLE values when simulation completes, if you check Synthesized filter settings on the View Analysis Results page.

Fields

Field	Description
DC Gain	See DC Gain in the Bode plot below.
Zero frequency	See Z_1 in the Bode plot below.
Pole 1 frequency	See P_1 in the Bode plot below.
Pole 2 frequency	See P_2 in the Bode plot below.

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The following Bode plot shows how the parameters listed in the CTLE Parameters dialog box form the filter:



Related Topics

FastEye Channel Analyzer - Add Pre-Emphasis/DFE/CTLE Page

DC Drop Analysis Dialog Box

To access: **Simulate PI > Run DC Drop Simulation (PowerScope)**

Use this dialog box to interactively simulate DC drop for power-supply nets.

Fields

Field	Description
Board (Available for a multiple-board design.)	Displays information for all boards or a selected board. For information about board IDs, such as B00, see “MultiBoard Project Board IDs” on page 566.
Power/Ground Net to Analyze	Displays the power-supply nets in the design. Select a net to display its copper areas, pads, traces, and power-integrity models. Enable “Include Reference net(s)” to include reference nets in the simulation.
Include Reference net(s)	Enable to include the reference nets for the selected net in simulation. Ensure that you have specified a reference net for each VRM and IC pin when you assigned PI models.

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Field	Description
Connected (Available for a board design)	Displays nets that are connected to the selected net, and are included in simulation. **Note:** For a multiple-board design, if you select a net that uses an Advanced interconnect model to connect to a net on another board, the software excludes the net on the other board from display and simulation.
Assigned Models	Displays DC sink or VRM model assignments for components that are connected to the selected net or any nets associated with the selected net. The label "<ref>" identifies models assigned to components on reference nets. • Assign — Opens the Assign Power Integrity Models dialog box, which enables you to assign models to the IC power-supply pins. (Available for a board design.) • Show all Pins — Checked, displays pins without model assignments. "<none>" means the pin has no DC sink or VRM model. "(disconnected)" applies only to IC pins and means the pin does not connect to the power-supply net through a via or trace segments. • Zoom to Selection — Checked, zooms the display pane view to a selected reference designator and pin.
Display Areas	Displays stackup layers that contain metal areas and trace segments connected to the selected power-supply net. Click a stackup layer to highlight its geometries in the display pane.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Constraints	To specify constraints, type values in the following fields: • Max Voltage Drop — Defines the maximum acceptable decrease in percent from the operating voltage for the power-supply net. • Max Current Density — Defines the maximum acceptable current that can flow across a power-supply net structure, such as an area fill or trace. • Max Via Current — Defines the maximum acceptable current for vias connected to power-supply nets. This is a global constraint that you can override with a padstack-specific constraint. To specify padstack-specific constraints or plating properties, click **Via Constraints**, then specify constraints on the **Plating/Constraints** tab of the Backdrilling/Plating/Constraints Settings dialog box. See "Running DC Drop Simulation" on page 198 and "Accounting for Per-Padstack Plating and Constraints" on page 137. The Reporter Dialog Box highlights simulation results that exceed constraint values.
Reports	Checked, saves simulation results. Type a report name and choose report options. The software saves report files and HTML folders to the design folder.
Write power-map files for Simcenter Flotherm/XT/ FLOEFD	Checked, saves power dissipation results for the simulated nets to a *Thermal<net_name>.txt* or *Thermal_<net_name>_<board_ID>.txt* file that you can import into any of these Siemens EDA tools: • Simcenter Flotherm • Simcenter Flotherm XT • Simcenter FLOEFD • HyperLynx ThermalSim
DC Drop Resistances DC + Resistances	Specifies whether to report DC drop, resistance between pairs of IC sink and VRM pins, or both. **Note:** When an IC sink pin connects to a net with a multiple-phase VRM with two or more VRM pins, the software measures resistance between an IC sink pin and each VRM pin. The software temporarily disconnects VRM and DC sink models for all non-measured pins before performing a measurement.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Pre-Process Geometry	Pre-processes the design and displays anti-pads and anti-segments in the display pane. Pre-processing does the following: • Checks net connectivity through copper pours, traces, pads, and vias. • Prepares copper areas (in memory only) for simulation by removing overlapping metal and by creating antipads where needed, using the default padstack values.
Show PowerScope	Opens the HyperLynx PI PowerScope dialog box, which enables you to view graphical DC drop simulation results.
Simulate	Starts the simulation process. When simulation completes, the HyperLynx PI PowerScope dialog box and Reporter dialog box open to display graphical and textual simulation results.

Related Topics

Running DC Drop Simulation

DDR2 Slew Rate Derating Dialog Box

To access: From the Measurement toolbar in the Digital Oscilloscope Dialog Box, select **Derate DDR2**

Reports derated slew rates for DDR2 data, address, and control pins, and to report non-derated slew rates for DDR differential clock and strobe pins. As part of filling out a timing budget spreadsheet for the DDR2 signal, you can use the reported values to look up the following derating values in datasheet tables: delta tDS, delta tDH, delta tIS, and delta tIH.

Derated slew rate measurements take into account the nominal slew rate of the waveform and the slew rate of a tangent line for the waveform. The tangent line is plotted from the VREF DC level to the input threshold voltage as specified in the JEDEC DDR2 SDRAM Specification JESD79-2B.

When this dialog box is open, the Digital Oscilloscope Dialog Box temporarily displays only the latest waveform for the pin selected in the Waveform list. The oscilloscope hides the waveforms for other pins, as well as any previous results, until you close this dialog box.

Any time multiple transitions exist in the waveform, such as when using oscillator stimulus or running an eye diagram analysis, the oscilloscope reports the minimum and maximum slew rates taken across all transitions.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Options

Option	Description
Input Area	
Waveform	Specifies the waveform to measure.
Thresholds Area	
Speed grade	Specifies the speed grade for the design.
Standard levels for Setup Standard levels for Hold Standard differential levels Custom levels for Setup Custom levels for Hold Custom differential levels	Specifies the type of measurement. For Standard differential levels, measure the slew rate and perform no derating.
Vref (dc) Vinh (ac) Vinl (ac)	Specifies reference and threshold voltages when you select a custom levels option.
Result Area—Displays the slew rate.	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDR5 Wizard

Use this wizard to set up the parameters required to simulate a single-ended net (such as DQ) for a DDRx interface that supports equalization. If you have run DDRx batch simulation and want to investigate results for a single-ended net with equalization, you can simulate that net with the DDR5 Wizard. The wizard can account for equalization behaviors represented by IBIS-AMI models, sweep AMI parameters, and report results as eye diagrams.

Topic	Description
DDR5 Wizard Page - Output Options	Use this page to specify report, waveform storage, and various results options.
DDR5 Wizard Page - Stimulus	Use this page to assign stimulus to pins and to set advanced AMI-related options.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDR5 Wizard Page - Output Options

To access: Choose the **Simulate SI > Run DDR5 Interactive Simulation** menu item, then select the **Output Options** page.

Use this page to specify report, waveform storage, and various results options.

Fields

Field	Description
Sub-directory for results	Specifies the folder that stores results. To avoid overwriting previous results, check “Add time tags to the folder name”. The root folder location depends on options that you set in the “Reports and log files directory” section of the Set Directories Dialog Box.
Show waveforms	Checked, opens EZwave and shows waveforms used by the simulator. If you enable the “Advanced DDRx AMI” simulation flow, you can see step response waveforms for victim and aggressor nets.
Save waveforms	Checked, saves EZwave waveform database files (*.wdb*) to the results sub-folder.
Restore DC offset in eye diagram	Checked, sets the vertical center of an eye diagram to DC level ((V_dc_high - V_dc_low)/2) from an RX waveform, instead of 0 V.
Clear previous eyes	Checked, in the Eye Density Viewer, delete eye diagrams created by previous simulations.
Create AMI model debug info	Checked, adds *Debugfolder* to the results folder.
Open Eye Density Viewer	Displays an eye diagram from a previous simulation.

Related Topics

Running Simulation on a DDRx Single-Ended Net With IBIS-AMI Models

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDR5 Wizard Page - Stimulus

To access: Choose the **Simulate SI > Run DDR5 Interactive Simulation** menu item, then select the **Stimulus** page.

Use this page to assign stimulus to pins and to set advanced AMI-related options.

Fields

Field	Description
Assign Stimulus	Opens the Assign Stimulus dialog box, where you can manually edit stimulus or import stimulus previously created for this design by the DDRx Batch-Mode Wizard. **Restriction:** If you enable "AMI Statistical" on the Channel page, the software ignores a bit pattern you specify, but applies a stimulus bit rate that you specify.
Number of bits to simulate	Specifies the number of bits that follow the number of ignored bits. An eye diagram displays this number of bits because it excludes ignore bits.
Samples per bit interval	Specifies the number of samples per bit interval to adjust the display resolution of eye diagram contours in the HyperLynx Eye Density/BER Viewer. Large numbers will slow analysis considerably.
Number of bits per call to AMI DLL	Specifies the number of stimulus bits to send as a block to an AMI *.dll* or *.so* file. You might change this value when advised by the model vendor to optimize interaction with certain AMI *dll* or *.so* files.
Let AMI model determine number of bits to ignore	Enabled, the software reads an *.ami* file to find the number of initial bits to exclude from an eye diagram.
Ignore first <number> bits simulated	Specifies the number of initial bits to exclude from an eye diagram. Specify a sufficient number of bits for the driver and receiver circuits to stabilize and reach normal operating conditions. The software only uses this value if it is smaller than values specified in the Tx and Rx AMI models. See "Factors that Affect the Number of Warmup Bits" on page 742.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
ISI effects are gone after <number> unit intervals	Specifies the number of unit intervals (bits) needed for transient activity to settle. The wizard reports an error if it detects unsettled activity at the end of the channel-characterization simulation. See inter-symbol interference (ISI). The default value is conservative, and provides ample time for interference (due to topology and reflection characteristics of the channel) to die out. **Restriction:** You can type values from 50 to 1000, in unit intervals.

Related Topics

Running Simulation on a DDRx Single-Ended Net With IBIS-AMI Models

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDRx Batch-Mode Wizard

Use this wizard to set up the parameters required to run a batch simulation on a DDRx interface.

The set of wizard pages you see depends on options you enable throughout the wizard. For example, if you disable the Address and Command timing option on the Nets to Simulate page, the wizard does not display the Address and Command Nets page.

The wizard stores page settings to a setup file (*.ddr*).

The color of a non-highlighted page name indicates the following:

- White — You have visited the page, even if you did not edit any values.
- Red — The page is accessible and you have not yet visited the page.

Topic	Description
DDRx Batch-Mode Wizard - Address and Command Nets Page	Use this page to specify address and command nets in the memory interface.
DDRx Batch-Mode Wizard - Clock Nets Page	Use this page to specify the clock nets in the memory interface.
DDRx Batch-Mode Wizard - Control Nets Page	Use this page to specify control nets in the memory interface.
DDRx Batch-Mode Wizard - Controller Page	Use this page to specify the reference designator of the memory controller IC.
DDRx Batch-Mode Wizard - Data Nets Page	Use this page to specify the data and data mask nets in the memory interface.
DDRx Batch-Mode Wizard - Data Strobes Page or Write Data Strobes Page	This page is accessible only when you have selected a memory controller on the **Controller** page and a data timing cycle on the **Nets to Simulate** page.
DDRx Batch-Mode Wizard - Read Data Strobes Page	Use this page to specify read data strobe nets in a memory interface.
DDRx Batch-Mode Wizard - Disable Nets Page	Use this page to exclude nets from “what if” simulations on specific nets with problems, or when running an initial screening simulation on a subset of nets for a byte lane.
DDRx Batch-Mode Wizard - DRAM Signals Page	Use this page to review or assign memory interface nets connected to each DRAM and to report the DRAM organization.
DDRx Batch-Mode Wizard - DRAMs Page	Use this page to associate pin reference designators with slot and rank locations on DRAM ICs.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
DDRx Batch-Mode Wizard - Equalization Page	This page displays equalization settings provided by IBIS-AMI models assigned to controller and DRAM pins. You can also use this page to override default IBIS-AMI model settings and save your changes to AMI settings files (*.ami*).
DDRx Batch-Mode Wizard - IBIS Models Page	Use this page to review memory controller, DRAM, PLL and register model assignments.
DDRx Batch-Mode Wizard - IBIS Model Selectors Page	Use this page to specify IBIS [Model Selector] keyword values for DDRx signals that do not use on-die termination (ODT).
DDRx Batch-Mode Wizard - Initialization Page	Use this page to manage DDRx setup files and to specify the type of DDRx interface in your design.
DDRx Batch-Mode Wizard - Introduction Page	This page summarizes the capabilities and usage of the DDRx batch simulation wizard.
DDRx Batch-Mode Wizard - Leveling&Calibration Page	This page specifies leveling and calibration delays for write/ read operations.
DDRx Batch-Mode Wizard - Nets to Simulate Page	Use this page to select the types of nets in the DDRx interface to simulate and the type of timing measurement voltage thresholds to use.
DDRx Batch-Mode Wizard - ODT Models Page	Use this page to specify on-die termination (ODT) models for data, data strobe, and data mask nets.
DDRx Batch-Mode Wizard - ODT Behavior Page	Use this page to set or verify the ODT enable/disable settings for the memory controller and DRAMs for all the possible read/ write operations.
DDRx Batch-Mode Wizard - PLLs and Registers Page	Use this page to specify the slots and reference designators of the PLLs and registers used by RDIMM ICs.
DDRx Batch-Mode Wizard - Quality Checks Page	Use this page to specify custom limits for signal integrity measurements, or accept JEDEC default values.
DDRx Batch-Mode Wizard - Report Options Page	Use this page to specify the simulation setup audit, report type and formatting, and waveform storage options.
DDRx Batch-Mode Wizard - Round Trip Time Page	Use this page to set up round trip time measurements by entering values into the table or importing the values using an RTT_Limits file.
DDRx Batch-Mode Wizard - Simulate Page	Use this page to save wizard settings, save a log file, or open the DDRx Batch-Mode - Run Simulation Dialog Box dialog box, which you use to launch simulation.
DDRx Batch-Mode Wizard - Simulation Options Page	Use this page to specify detailed simulation options.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
DDRx Batch-Mode Wizard - Stimulus and Crosstalk Page	Use this page to specify unique per-bit stimulus and crosstalk options for your DDRx interface simulation. You can also include the effects of simultaneous switching noise (SSN) in your simulation.
DDRx Batch-Mode Wizard - Sweep Manager Page	Use this page to define the set of design property values (sweep range) to apply to a design property during sweep simulations.
DDRx Batch-Mode Wizard - Timing Models Page	Use this page to specify the memory interface data rate and the device speed grade and timing models for the memory controller and DRAM devices. You can also use this page to display, edit, and verify timing models.
DDRx Batch-Mode Wizard - Vref Training Page	Use this page to specify how the software defines the reference voltage (Vref) during read measurements.

DDRx Batch-Mode Wizard - Address and Command Nets Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Addr/Comm Nets** page. This page is accessible only when you have selected a memory controller on the **Controller** page and have enabled Address and Command timing on the **Nets to Simulate** page.

Use this page to specify address and command nets in the memory interface.

Fields

Field	Description
Controller Nets	Displays the list of available controller nets. • Click the box to the left of the net name to select the net you want to include as an address and command net. • Right-click a net name to view its associated nets. • Double right-click the net name to view the net topology. Right-click a net name to display its associated nets. When an automatically-mapped net has an arbitrary name, such as $123, and one of its associated nets has a meaningful name with respect to the DDRx interface, displaying associated nets provides a way to help you confirm the correctness of the automatic net assignment.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Filter	Defines the filter string. Enter a string in the field and click **Apply**. The field supports the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character. To clear the filter, delete the contents of the Filter field and click **Apply**.
	Adds the selected controller net to the Clock Nets table.Click Ctrl+  to replace the nets in the Clock Nets table with the selected net(s) from the Controller Nets table.
Address and Command Nets	Displays the list of Clock nets assignments. • Click the box to the left of the net name to select the net. • Right-click on the net name to view the associated nets. Click **Remove Selected** to move the selected nets to the Controller Nets table.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - Clock Nets Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Clock Nets** page. This page is accessible only when you have selected a memory controller on the **Controller** page and have enabled Address and Command timing on the **Nets to Simulate** page.

Use this page to specify the clock nets in the memory interface.

Fields

Field	Description
Controller Nets	Displays the list of available controller nets. • Click the box to the left of the net name to select the net you want to include as a clock net. • Right-click a net name to view its associated nets. When an automatically-mapped net has an arbitrary name, such as $123, and one of its associated nets has a meaningful name with respect to the DDRx interface, displaying associated nets provides a way to verify an automatic net assignment.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Filter	Defines the filter string. Enter a string in the field and click **Apply**. The field supports the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character. To clear the filter, delete the contents of the Filter field and click **Apply**.
	Adds the selected controller net to the Clock Nets table. Click Ctrl+  to replace the nets in the Clock Nets table with the selected net(s) from the Controller Nets table.
Clock Nets	Displays the list of Clock nets assignments. • Click the box to the left of the net name to select the net. • Right-click on the net name to view the associated nets. Click **Remove Selected** to move the selected nets to the Controller Nets table.

DDRx Batch-Mode Wizard - Control Nets Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Addr/Comm Nets** page. This page is accessible only when you have selected a memory controller on the **Controller** page and have enabled Address and Command timing on the **Nets to Simulate** page.

Use this page to specify control nets in the memory interface.

Restrictions

- For LPDDR<any> memory interfaces, do not use the string “CKE” in an IBIS signal name unless the signal is a CKE control net. When a control net connects to a DRAM pin assigned to an IBIS signal with “CKE” anywhere in its name, the software identifies it as the CKE control net.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Controller Nets	Displays the list of available controller nets. • Click the box to the left of the net name to select the net you want to include as a clock net. • Right-click a net name to view its associated nets. When an automatically-mapped net has an arbitrary name, such as $123, and one of its associated nets has a meaningful name with respect to the DDRx interface, displaying associated nets provides a way to verify an automatic net assignment.
Filter	Defines the filter string. Enter a string in the field and click **Apply**. The field supports the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character. To clear the filter, delete the contents of the Filter field and click **Apply**.
	Adds the selected controller net to the Control Nets table. Click Ctrl+  to replace the nets in the Control Nets table with the selected net(s) from the Controller Nets table.
Control Nets	Displays the list of Control nets assignments. Click the box to the left of the net name to select the net. Click **Remove Selected** to move the selected nets to the Controller Nets table.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - Controller Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Controller** page.

Use this page to specify the reference designator of the memory controller IC.

Caution

Do not select an EBD/EMD component as a controller, DRAM, clock buffer, or RDIMM register.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field

Field	Description
Board	Defines the MultiBoard project. Select the board that contains the memory controller.
Memory Controller Reference Designator Selector	Displays the list of available parts in the board. Click the box to the left of the reference designator to select the part to use for the memory controller. • Ref Des, Part Name — Identifies the device. Right-click on a specific table cell in either column to view the connected net topologies for that part. • Model File — The name of the model file. Right-click on a specific model cell to view the path to the model library file. Double-click the cell to open the model in the Visual IBIS Editor. • Model Component — The [Component] name in the IBIS model.
	Assigns or overwrites the current selection in the table as the memory controller.
Filter By	Defines the type of object by which to filter.
Filter	Defines the filter string. Enter a string in the field and click **Apply**. The field supports the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character. To clear the filter, delete the contents of the Filter field and click **Apply**.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - Data Nets Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Data Nets** page. This page is accessible only when you have selected a memory controller on the **Controller** page, selected a data timing cycle on the **Nets to Simulate** page, and identified data strobes on the **Data Strobes** page.

Use this page to specify the data and data mask nets in the memory interface.

When measuring timing, DDRx batch simulation reports the delay between the strobe and the data/data mask nets. Data mask nets are only simulated for write operations and are grouped separately from data nets.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Controller Nets	Displays the list of available controller nets. • Click the box to the left of the net name to select the net you want to include as a strobe net. For differential signaling, select only the positive member of the differential pair. • Right-click on the net name to view the associated nets. • Double right-click the net name to view the net topology.
Filter	Defines the filter string. Enter a string in the field and click **Apply**. The field supports the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character. To clear the filter, delete the contents of the Filter field and click **Apply**.
	Adds the selected controller net to the Data /Mask Nets to Strobe Assignments table without removing existing nets.
Data/Mask Nets to Strobe Assignments	Displays the list of Data and Mask nets assignments. • Click the box to the left of the strobe name to select the net. • Right-click on the net name to view the associated nets. • Double right-click the net name to view the net topology Click **Remove Selected** to move the selected nets to the Controller Nets table.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - Data Strobes Page or Write Data Strobes Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Data Strobes** or **Write Data Strobes** page.

This page is accessible only when you have selected a memory controller on the **Controller** page and a data timing cycle on the **Nets to Simulate** page.

Use this page to specify data strobe nets, or write data strobe nets, depending on the type of memory interface.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
My data strobes are	(Available only for DDR2 interfaces) Enables you to select the type of nets to use as data strobes
Controller Nets	Displays the list of available controller nets. • Click the box to the left of the net name to select the net you want to include as a strobe net. For differential signaling, select only the positive member of the differential pair. • Right-click a net name to view its associated nets. When an automatically-mapped net has an arbitrary name, such as $123, and one of its associated nets has a meaningful name with respect to the DDRx interface, displaying associated nets provides a way to verify an automatic net assignment.
Filter	Defines the filter string. Enter a string in the field and click **Apply**. The field supports the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character. To clear the filter, delete the contents of the Filter field and click **Apply**.
	Adds the selected controller net to the Data Strobe Nets table. Click Ctrl+  to replace the nets in the Data Strobe Nets table with the selected net(s) from the Controller Nets table.
Data Strobe Nets	Displays the list of strobe nets. • Click the box to the left of the net name to select the net. • Right-click on the net name to view the associated nets. • Double right-click the net name to view the net topology. Click **Remove Selected** to move the selected nets to the Controller Nets table.

Related Topics

Running a DDRx Memory Interface Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDRx Batch-Mode Wizard - Read Data Strobes Page

To access: Choose the **Simulate SI > Run DDRx Batch Simulation** menu item, then select the **Read Data Strobes** page.

This page is accessible only when you have selected a memory controller on the **Controller** page and a data timing cycle on the **Nets to Simulate** page.

Use this page to specify read data strobe nets in a memory interface.

Fields

Field	Description
Controller Nets	Displays a list of available controller nets. Click the box to the left of the net name to select the net you want to include as a strobe net. For differential signaling, select only the positive net within the differential pair. Right-click on a net name to see its *associated net*s. When an automatically-mapped net has an arbitrary name, such as $123, and one of its associated nets has a meaningful name with respect to the DDRx interface, displaying associated nets provides a way to verify an automatic net assignment.
Filter	Specifies a filter string. Type a string in the field and click **Apply**. The field supports the following wildcard characters: • * — Matches any number of characters. • ? — Matches any one character. To clear the filter, delete the contents of the Filter field and click **Apply**.
	Adds a selected controller net to the Read Strobe Nets to Write Strobe Assignments table. Click Ctrl-  to replace nets in the Read Strobe Nets to Write Strobe Assignments table with selected net(s) from the Controller Nets table.
Read Strobe Nets to Write Strobe Assignments	Displays a list of read strobe nets. Usage: • Right-click a net name to view its associated nets. • Double right-click a net name to see its net topology. • Click the box to the left of a net name to select it, then click **Remove Selected** to move a selected net to the Controller Nets table.

Related Topics

Running a DDRx Memory Interface Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDRx Batch-Mode Wizard - Disable Nets Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Disable Nets** page.

Use this page to exclude nets from "what if" simulations on specific nets with problems, or when running an initial screening simulation on a subset of nets for a byte lane.

The set of nets in the table depends on options you enable on the Nets to Simulate page. For example, if you disable "Clock-to-strobe skew timing" on the Nets to Simulate page, the table on this page does not display strobe nets.

Fields

Field	Description
Simulate?	Checked, simulates the net(s).
Net Type	Displays the type of net and groups them by color.
Net Name	Displays the name of the net. • Right-click a net name to view its associated nets. When an automatically-mapped net has an arbitrary name, such as $123, and one of its associated nets has a meaningful name with respect to the DDRx interface, displaying associated nets provides a way to verify an automatic net assignment. • Double right-click a net name to view its net topology.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - DRAM Signals Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **DRAM Signals** page. This page is accessible only when you have selected a memory controller and at least one DRAM IC on the **Controller** and **DRAM** pages, respectively.

Use this page to review or assign memory interface nets connected to each DRAM and to report the DRAM organization.

Assign the same reference designator to all pins that belong to the same DRAM or controller component. This enables the DDRx Wizard to identify the IC symbols that collectively belong to the controller and each DRAM, based on the reference designator assignment.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Perform Automatic Net Mapping	Identifies a DRAM for each controller net.
Undo Automatic Net Mapping	Restores the previous set of connections.
View Controller Net Topology	Opens a dialog box that displays the topology of controller net, to help you see how a net connects to component pins and other nets.
<table>	Displays the connections between the controller and the DRAM ICs. • Right-click a reference designator to open a dialog box that can help you see how a net connects to component pins and other nets. • Right-click a net to view its associated nets. When an automatically-mapped net has an arbitrary name, such as $123, and one of its associated nets has a meaningful name with respect to the DDRx interface, displaying associated nets provides a way to verify an automatic net assignment. • Double right-click on a specific controller net to view its connected net topology. **Note:** If the table is blank, verify that you have opened a setup file on the **Initialization** page, assigned IBIS models to the controller and DRAM ICs, and that your design contains circuits that permit signal-path tracing.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - DRAMs Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **DRAMS** page.

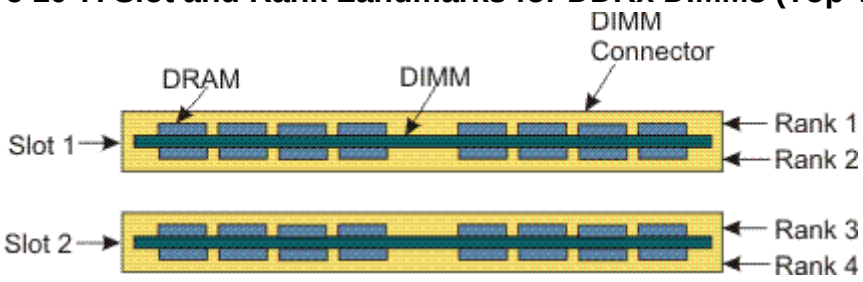
Use this page to associate pin reference designators with slot and rank locations on DRAM ICs.

Use slot and rank numbers to indicate DRAM locations within the DDRx interface, as shown in Figure 10-7 and Figure 3-1. DDRx simulation supports the following DRAM configurations:

- The memory controller and DRAM components are located on the same board.
- The memory controller and DIMMs are located on different boards in a multiple-board design.
- Slot and Rank Landmarks for DDRx DIMMs (Top View)

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-7. Slot and Rank Landmarks for DDRx DIMMs (Top View)



If the memory controller and DRAM ICs are located on the same board, as opposed to a multiple-board project where DIMMs plug into a board with the memory controller, you must know which DRAM instances work together during a memory operation so you can assign them to slots and ranks. A rank is a group of DRAMs that is controlled by a single, unique, chip select signal. The number of chip select signals on the memory controller determines the number of memory ranks. Ranks contain 64 non-ECC bits or 72 ECC bits.

Tip

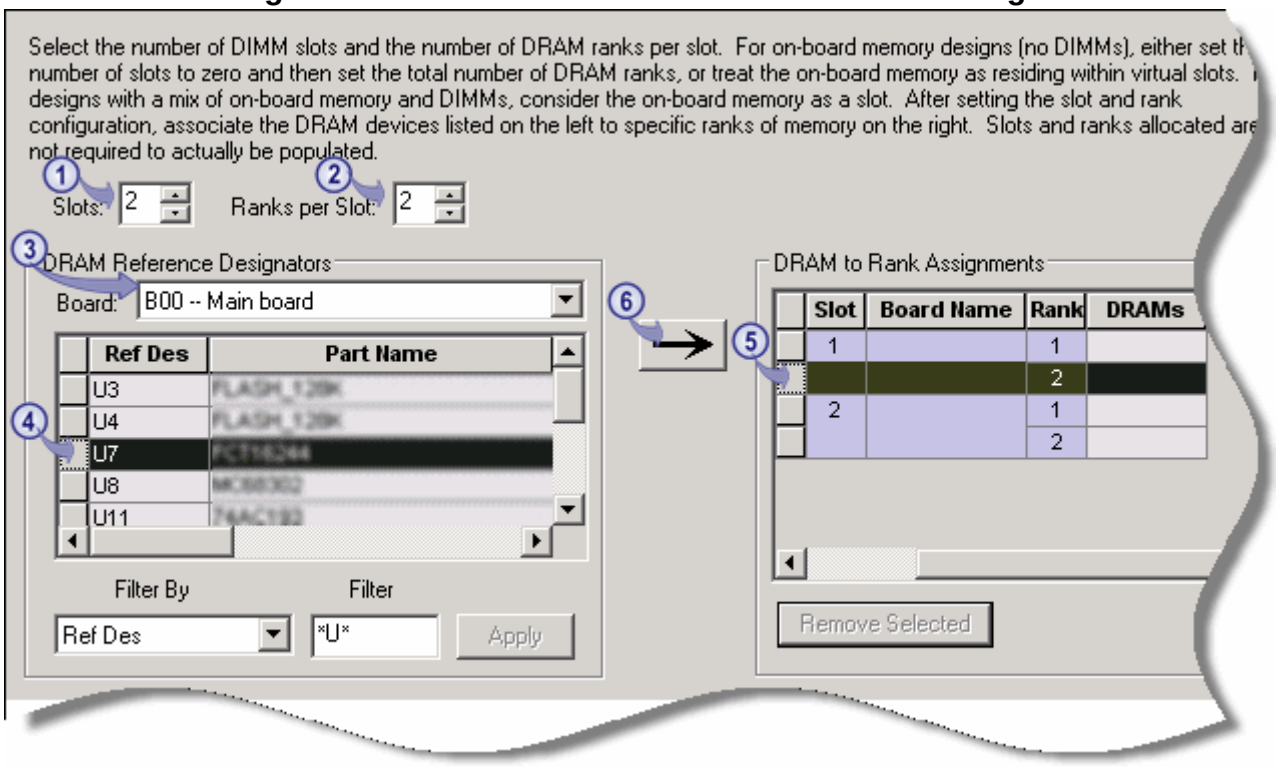
To display the DDRx-related nets for a reference designator or part name, right-click a reference designator or part name in the spreadsheet.

Note

The DDRx Wizard supports the use of stacked-die DRAMs if you have EBD/EMD models that define them and point to IBIS models. The DDRx Wizard does not support the use of EBD/EMD models that point to other EBD/EMD models. See Specifying Locations for Stacked-Die DRAMs.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-8. DDRx Batch-Mode Wizard - DRAMs Page



Fields

Field	Description
Slots 	Defines the number of slots. If the design includes stacked-die DRAM, see the detailed procedure in Specifying Locations for Stacked-Die DRAMs. **Tip:** If your design does not contain DIMMs, set the number of slots to 1. **Note:** If your design includes unused slots, you can specify the slots as unused and leave the unused slots empty.
Ranks per Slot 	Defines the number of ranks for the number of dies that are stacked. For example, a stacked quad-die DRAM has 4 ranks. **Note:** You can specify more ranks per slot than are actually used and leave the unused ranks empty.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
DRAM Reference Designators 3	Displays the list of DRAM models on the designated board. For stacked-die DRAMs, see Specifying Locations for Stacked-Die DRAMs. Click the box to the left of the reference designator to select the DRAMs you want to map. • Ref Des, Part Name — Identifies the DRAM. Right-click on a specific table cell in either column to view the connected net topologies for that DRAM. • Model File — Displays the name of the model file. Right-click on a specific model cell to view the path to the model library file. Double-click the cell to open the model in the Visual IBIS Editor. • Model Component — The [Component] name in the IBIS model. **Note:** When you specify the DRAM reference designators, include the associated ECC DRAM in the selection.
→	Assigns the selected DRAM to the selected rank. Click Ctrl+→ to replace an existing assignment with a new assignment.
Filter By	Defines the type of object by which to filter.
Filter	Defines the filter string. Enter a string in the field and click **Apply**. The field supports the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character. To clear the filter, delete the contents of the Filter field and click **Apply**.
DRAM to Rank Assignments 4 5	Displays the list of slots and ranks you specified. To make an assignment: 1. Click the row header(s) in the spreadsheet to select the reference designator for the DRAM(s) you want to map. 2. Select the row header for the slot or rank map destination of the reference designator. 3. Click → to perform the mapping. To replace an existing assignment with a new assignment, select the assignment, select a DRAM and press Ctrl+ → Click **Remove Selected** to remove the selected assignments. **Tip:** You can also select multiple DRAMS by clicking and dragging the mouse in the RefDes column. Similarly, you can select a particular rank by clicking in a cell in the DRAMs column.
Remove Selected	Removes the selected assignments.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Running a DDRx Memory Interface Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDRx Batch-Mode Wizard - Equalization Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Equalization** page.

This page displays equalization settings provided by IBIS-AMI models assigned to controller and DRAM pins. You can also use this page to override default IBIS-AMI model settings and save your changes to AMI settings files (*.ami*).

Fields

Field	Description
Use Individual Settings For	Specifies the memory interface elements to display in the "Settings tree". For example, to display all controller pins in a byte lane for a write operation: 1. Select "For Lane". 2. Select "Controller Pins". 3. Select "Write cycles". 4. Select the name of a byte lane.
Settings tree (left side of the page)	Displays AMI settings. Select a setting to display its data.
Settings data (right side of the page)	Displays data for a selected AMI setting. Use the Value field to override a default settings value, unless the model vendor has made it read only. **Note:** When you edit values, the software saves your changes to AMI settings files (*.ami*) located in the *\<design>\AMI_SETTINGS* folder. The Description field displays setting documentation provided by the IBIS-AMI model. **Note:** The "Ignore initial DQ bits" and "Ignore initial ACC bits" options on the Simulation Options page override a value specified by an AMI Ignore_bits parameter displayed on this page.
Save As Load	Saves or loads an AMI settings file (*.ami*). You can use this capability to re-use AMI settings for a different design.
Reset Pin Reset All	Restores default settings from IBIS-AMI models and deletes corresponding AMI settings files (*.ami*) located in the *\<design>\AMI_SETTINGS* folder.
Use setting for all Controller pins Use setting for all DRAM pins	Copies a selected setting value to all pins for a controller component or all DRAM components.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - IBIS Models Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **IBIS Models** page. This page is accessible only when you have selected a memory controller and DRAM ICs on the **Controller** and **DRAM** pages, respectively.

Use this page to review memory controller, DRAM, PLL and register model assignments.

If an IBIS model is missing or incorrect, edit the *.REF* or *.QPL* automapping file to assign IBIS or EBD/EMD models to reference designators. Note that for a schematic, you can also interactively select IBIS models, but not EBD/EMD models. The dialog explains how to edit automapping files.

Fields

Field	Description
Device	Lists the available components.
Type	Displays the type of each component listed.
IBIS File	Displays the component IBIS models. • Black text indicates currently assigned models. Right-click the IBIS File cell to display the file pathname. Double-click the cell to open the model file in the Visual IBIS Editor. • Red text indicates errors in locating previously-assigned models, or unassigned models. To correct the location errors, close the wizard and add the folder that contains the models to the Model-library file path field on the Set Directories dialog box. To assign new models, click the **Assign Component Models** button.
IBIS Component	Displays the components associated with the assigned IBIS models.
Assign Component Models	Opens the REF-File Editor dialog box, which enables you to assign IBIS models.

Related Topics

Assigning a Model or Value to an Entire Component Using a .REF File

Assigning a Model or Value to an Entire Component Using a .QPL File

Running a DDRx Memory Interface Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDRx Batch-Mode Wizard - IBIS Model Selectors Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **IBIS Model Selectors** page.

Use this page to specify IBIS [Model Selector] keyword values for DDRx signals that do not use on-die termination (ODT).

Fields

Field	Description
List Items By	• IBIS Component — Selected, displays all devices with the same IBIS component in the same row. This enables you to quickly assign the same [Model Selector] keyword value to all devices using the same IBIS component. • Device Reference Designator — Selected, displays each device in its own row. This enables you to assign unique [Model Selector] keyword values to individual devices.
Non-ODT IBIS Model Selectors	Lists the IBIS components and their properties. Click a Model to select the [Model Selector] keyword value.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - Initialization Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Initialization** page.

Use this page to manage DDRx setup files and to specify the type of DDRx interface in your design.

A DDRx setup file (*.ddr*) contains configuration settings that the wizard requires to run DDRx batch simulation. The wizard saves a setup file before closing, which you can import during a future wizard session. See “DDRx Setup File” on page 511.

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Fields

Field	Description
Setup File	Displays the setup file name associated with the loaded design. You can create a new file by proceeding through the wizard and saving the configurations when prompted on the Simulate page. Or you can specify an existing setup file in the Edit, Verify or Import fields. **Note:** Although you can import a setup file from any location, the wizard always saves the setup file to the results folder.
Edit	Opens an existing setup file in the HyperLynx File Editor, enabling you to make changes as necessary. **Note:** Add custom comments only to the "Notes" section of the setup file. Comments created in other sections are deleted when you save the file.
Verify	Validates the syntax of an existing setup file.
Import	Imports configuration settings from a setup file into the wizard and displays its contents in the wizard pages. **Restriction:** Although you can import a setup file from any location, the wizard always saves the setup file to the <design> folder. See Design Folder and HyperLynx Files.
Reset	Clears data from the wizard to create a new setup file.
This DDR interface is	Specifies the DDRx interface type and whether to buffer the address, control, and clock signals in the interface. For some DDRx interfaces, you additionally specify whether your memory has unbuffered or registered signals. A registered DIMM (RDIMM) contains registers and a phase-lock-loop (PLL) to buffer the following types of signals in the DDRx interface: Address, control, and clock. **Note:** The DDRx Wizard supports the simulation of systems with registered DIMMs (RDIMMs) only when the register and PLL are separate components, or a single component on the RDIMM.
DDRx Data Rate	Specifies the data rate in MT/s (megatransfers per second). Select a value from the dropdown list, or create a custom value by selecting <custom> and entering a new value. When you change the data rate, the data rate value on the Timing Models page also changes.
WCK:CK Ratio (LPDDR5 interface only)	Specifies the write and read strobe frequency to command/address clock frequency ratio.

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Field	Description
Supply Voltage is (DDR3 interface only)	Specifies the VCC. DDR3U corresponds to 1.25 V and DDR3L corresponds to 1.35 V.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - Introduction Page

To access: **Simulate SI > Run DDRx Batch Simulation**, select the **Introduction** page.

This page summarizes the capabilities and usage of the DDRx batch simulation wizard.

To prevent this page from displaying, uncheck “Always show this page when starting the Wizard.” You can always display this page by clicking Introduction in the contents pane.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - Leveling&Calibration Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Leveling&Calibration** page.

This page is accessible when your memory interface has leveling and calibration capabilities for read/write operations and you enable “Data timing” or “Clock-to-strobe skew timing” on the **Nets to Simulate** page.

This page specifies leveling and calibration delays for write/read operations.

The skew values in the table on this page specify how the absolute initial delay value differs from the default initial delay value. For example, when the table specifies all zeros, the absolute initial delay is the same as the default value (as when there is no leveling). For more information, see “Write/Read Leveling for DDRx” on page 517.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Automatically run simulation 2 times	Checked, runs two simulation passes. The first pass runs with no delays and creates a delays file that the software applies for the second pass. The second pass also creates a delays file, and the wizard saves both files in the results directory. Unchecked, runs one simulation pass and creates a delays file in the design directory. Both options create a delays file with values that compensate for the delay between the clock and DQS. For information about these two options, see “Creating and Applying a DDRx Delays File” on page 490.
My Controller can calibrate skews between:	Specifies the delays calculated and applied during simulation: • CLK and each lane — Checked, calibrates write delays for entire lanes, including strobe, data, and mask nets. This allows the strobe to be aligned with the clock when arriving at the memory. • DQS and each DQ signal of that lane for WRITE transactions — Checked, calibrates write delays for data and mask nets with respect to their strobe. • DQS and each DQ signal of that lane for READ transactions: • Each DQ can be individually delayed to optimize it with respect to the DQS — Selected, each data net is delayed to optimize timing with respect to its strobe. • Only the DQS signal can be adjusted to optimize it to all the DQ signals — Selected, the strobe is delayed with respect to all the data nets in order to optimize timing across the worst-case data nets. • CLK and each Address/Command signal — Checked, adjusts each address bit to make the clock centered around the address bits at the DRAM. For more information, see “Types of Calibration” on page 792.
In table, set skews for IC model corner:	Enables you to view and specify delays in the table for one type of IC model corner at a time. Check “Same skews for all corners” to apply the Typ corners to the Fast-strong and Slow-weak corners.
Use values from ...	• Use values from the table on this page — Selected, applies only the skew correction values you specify in the table to the listed nets. • Use values from the DDRx Delays external file — Selected, applies leveling information contained in the specified file. Click the **Select** button to specify the file. After applying the file, the wizard updates values in the table.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Write/Read Leveling Skews table	Defines the write/read leveling delay values for strobe, data, and data mask signals. Populate the table with the skew value types you specify in the "My controller can calibrate skews between" field. The table includes the data strobe nets that you specified in the Data Strobes page. See "DDRx Batch-Mode Wizard - Data Strobes Page or Write Data Strobes Page" on page 777. To set all delays in the table to 0, click the **Clear All** button.

Types of Calibration

The type of calibration you select on this page depends on the capabilities of the controller in your design. Options for calibration are as follows:

- DQS-CLK write leveling. An adjustment to ensure that DQS lines up optimally with CLK at the DRAM. Note that the all signals associated with the same lane as the DQS will be delayed by the same amount with this type of calibration.
- DQS-DQ during write. Also known as per-bit data write calibration/leveling/deskew. The controller delays each DQ bit with respect to the strobe to optimally center the strobe with respect to the data at the DRAM.
- DQS-DQ during read.

  The controller delays each received DQ signal during a read in order to optimally sample the signals using the strobe.

  The controller delays the strobe to center the strobe between the two extreme data bit delays within that lane.

  For controllers that can calibrate each data bit with respect to the strobe, any static delays within the driving DRAM are calibrated away and are therefore not applicable. In such cases, if the selected technology provides dynamic output uncertainty of the DRAM, the wizard uses that value as the output uncertainty. If Jedec provides no value for dynamic uncertainty of the DQ pins for the selected technology, the wizard assumes a nominal value of 1% of UI.

  With LPDDR4, for example, tQW incorporates the dynamic output uncertainty of each pin and this cannot be calibrated away by the controller. The wizard uses this value to calculate the dynamic uncertainty for controllers with the ability to calibrate each DQ bit during reads. For systems with LPDDR4 controllers that do not have the ability to calibrate each DQ bit with respect to the DQS, the wizard uses the tDQSQ and tQH values (which incorporate static uncertainties across pins) instead to calculate the output uncertainty.

  Continuing with the DDR4 example, there is no dynamic output uncertainty stated, since tDQSQ and tQH incorporate uncertainty across pins that can be calibrated away by the

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controller (if it has this ability). For controllers that have the ability to calibrate each pin during a read, the wizard assumes that the dynamic output uncertainty is 1% of the UI.

- CLK-ADDR. Some controllers allow the adjustment of each address bit to make the clock centered around the address bits at the DRAM.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - Nets to Simulate Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Nets to Simulate** page.

Use this page to select the types of nets in the DDRx interface to simulate and the type of timing measurement voltage thresholds to use.

Note

The availability of many options on this page depend on which DDRx interface your design contains. This topic describes all possible options, even if they do not apply to your DDRx interface.

Fields

Field	Description
Data timing (relative to strobes)	Checked, enables you to select a simulation cycle.
Clock-to-strobe skew timing	Checked, captures the measurements required to create a write leveling delay file and minimize simulation run time. See Creating and Applying a DDRx Delays File.
Address and Command timing (relative to clocks)	Checked, enables you to select when to have the memory controller issue a memory command. • 1T timing — Selected, on the rising edge of every clock. • 2T timing — Selected, on the rising edge of every second clock. **Tip:** Select 2T timing when the address bus capacitance is too high, such as when most of the slots/ranks are populated with DRAMs.
Timing measurement threshold	Specifies the type of timing measurement voltage threshold to use during simulation. • Use all four receiver AC/DC thresholds • Use only the Vtt threshold — For differential signals, the Vtt threshold is zero volts; for signal-ended signals, the Vtt threshold is $(V_{IH(ac)}+V_{IL(ac)})/2$ volts.

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Field	Description
Advanced Options	
Check round trip times (available only if clock-to-strobe skew timing is enabled on this page)	Checked, performs round trip time measurements. Set the measurements on the DDRx Batch-Mode Wizard - Round Trip Time Page.
Compensate signal launch skews to account for variations in time-to-Vmeasure	Checked, applies an additional initial delay to all signals. Compensated initial delay = initial delay + (tCK/2 - time-to-Vmeas), where time-to-Vmeas is measured by one simulation run using a test load for the driver. When used with round trip time calculations, compensated flight times are included in the round trip time calculations. When not enabled, no compensation is applied **Caution:** This option can change all main measurement results including setup/hold/skew because it changes initial delays for all the signals.
Dynamic termination (3 ODT states available)	Checked, enables you to select ODT states for the controller and each DRAM on the ODT Behavior page.
Data Bus Inversion	Checked, for a given set of eight bits, if more than four bits are low for a DDRx interface (or high for an LPDDRx interface), the entire byte is inverted. A ninth bit, called the DBI bit, is the same as the Data Mask bit and is set low for DDRx and high for LPDDRx, indicating that the byte is flipped. Because the ninth bit is shared with data mask, either data mask or the DBI functionality can be used, and not both. For a given set of nine bits, at least 5 bits are guaranteed to be high for each DDRx interface transaction and low for each LPDDRx transaction.
Controller or DRAM supports equalization	Checked, makes the Equalization page available, where you can review and edit IBIS-AMI parameter values.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - ODT Models Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **ODT Models** page.

Use this page to specify on-die termination (ODT) models for data, data strobe, and data mask nets.

DDRx batch simulation runs with one ODT model at a time. You can change an ODT model and simulate again.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Device	Displays the devices available. Click the box to the left of the device name to select the device.
IBIS Component	Displays the IBIS Component for each device.
Type	Displays the part type.
DQS, DQS#, DQ, DM	Displays the disabled and enabled on-die termination models for specific nets. • To specify a model file for a specific device, click the associated ODT Enabled cell and select a model. (Cells are available when the IBIS model assigned to the device contains an ODT model assignment in the [Model Selector] keyword.) • Red text indicates that the IBIS model for the controller or DRAM component does not contain the [Model Selector] or [Model] keyword for the pins.
Apply These Settings to Similar DRAMs	Copies ODT settings from the selected DRAM to all the other DRAMs of the same IBIS component type.
ODT settings group <#>	Sets an ODT configuration for use in a sweep simulation. (LineSim only). To set a configuration, select an ODT Setting and select the models to include in the setting. Repeat until you have all of your configurations set. Use the DDRx Batch-Mode Wizard - Sweep Manager Page to select the ODT Settings to include in your sweep simulation.

Related Topics

Running a DDRx Memory Interface Simulation

On-Die Termination - ODT

DDRx Batch-Mode Wizard - ODT Behavior Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **ODT Behavior** page.

Use this page to set or verify the ODT enable/disable settings for the memory controller and DRAMs for all the possible read/write operations.

The “DRAM Configuration” label displays the number of slots and ranks in the memory interface that you specified on the DRAMs page. The values are of the form <slot_1_config_value>/<slot_2_config_value> where slot_#_config_value can be any of the following:

2R—DRAMs populate both ranks (2R) of the DIMM.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

1R—DRAMs populate one rank (R) of the DIMM.

<empty>—The slot is empty.

For example, “DRAM Configuration: 2R/2R” means that DIMMs populate both slots and that DRAMs populate both ranks of the DIMMs.

Tip

“Rank” represents a specific location in the memory interface. DRAM components in the same rank function together during read and write operations. Ranks comprise of 64 (non-ECC) or 72 (ECC) bits.

Note that the IBIS model and routing impedance together determine the overall impedance for enabled/disabled ODT settings.

During simulation, ODT is enabled based on a write operation to a slot, rather than the specific rank within a slot. For example, consider a 2R/2R system, in which DIMMs populate both slots and DRAMs populate both ranks of each DIMM. When slot 1 is written to (either side 1 or side 2), ODT is enabled on the front side of slot 2. This means that one simulation can produce waveforms for the receiver pins on both ranks on the DIMM in slot 1. Therefore, there is no need to simulate write operations separately for slot 1/rank 1 and slot 1/rank 2 because the ODT setting is the same for both operations.

For DDR4/LPDDR4 interfaces, three states are available. Each option applies a different IBIS model during simulation.

- Driving state - select ODT Disabled

  Use when the device is driving, for example, during a read cycle from a DRAM, or a write cycle from the controller.

- Receiving state - select ODT1 Enabled

  Use for the DRAM during a write cycle and for the controller during a read cycle.

- Passive state - select ODT2 Enabled

  Use this for all situations where the DRAM is not in a read or write cycle. This option is available for DRAMs only.

Fields

Field	Description
ODT for Write Operations	Enables you to override default ODT enable/disable behaviors on write operations for a DRAM.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
ODT for Read Operations	Enables you to override default ODT enable/disable behaviors on read operations for a DRAM.
Use Defaults	Restores default behavior.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - PLLs and Registers Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **PLLs and Registers** page. This page is accessible only when you select **Registered** for the DDR interface on the **Initialization** page.

Use this page to specify the slots and reference designators of the PLLs and registers used by RDIMM ICs.

Restrictions

The DDRx Wizard supports the simulation of systems with registered DIMMs (RDIMMs) only when the register and PLL are separate components, or a single component on the RDIMM.

Note

The DDRx Wizard does not support hybrid RDIMM interfaces that do not have a PLL or a Register.

If the memory controller and RDIMMs are located on the same board, as opposed to a multiple board project where RDIMMs plug into a board with the memory controller, you must know which RDIMM instances work together during a memory operation before you can assign them to slots and ranks. Use slot and rank numbers to indicate DRAM locations within the DDRx interface, as shown in Figure 10-7. A rank is a group of RDIMMs that are controlled by single, unique, chip select signal. The number of chip select signals on the memory controller determines the supported number of memory ranks. Ranks comprise of 64 (non-ECC) or 72 (ECC) bits.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Effective PLL Clock Input to DRAM/ Register Clock Input Delay	Defines the typical interconnect delay between the output pin of the PLL component and the clock input pin for the DRAM or register component. Enter the typical delay value and the plus/minus tolerance value.
PLL/Register Reference Designators	Displays the list of PLL models on the designated board. Click the box to the left of the reference designator to select the part you want to map. • Ref Des, Part Name — Identifies the part. Right-click on a specific table cell in either column to view the connected net topologies for that part. • Model File — Displays the name of the model file. Right-click on a specific model cell to view the path to the model library file. Double-click the cell to open the model in the Visual IBIS Editor. • Model Component — Displays the model file for the component.
Filter By	Defines the type of object by which to filter.
Filter	Defines the filter string. Enter a string in the field and click **Apply**. The field supports the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character. To clear the filter, delete the contents of the Filter field and click **Apply**.
	Assigns the selected PLL and/or Register to the selected slot. Click Ctrl+  to replace an existing assignment with a new assignment.
RLL/Register to Slot Assignments	Displays the list of slots available. Click the box to the left of the slot to select the slot you want to map the selected PLL or register. Click **Remove Selected** to remove the selected assignments.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - Quality Checks Page

To access: **Simulate SI > Run DDRx Batch Simulation**, select the **Quality Checks** page.

Use this page to specify custom limits for signal integrity measurements, or accept JEDEC default values.

All results are reported in the *DDR_report_Jedec_measurements_<corner_case>.xls* file.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Use JEDEC default values for all measurements	Checked, uses the default values depending on the DDR technology selected on the DDRx Batch-Mode Initialization Page. A negative value disables a measurement. Unchecked, enables you to edit measurement limits in the spreadsheet.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Differential signals: (Availability depends on DDRx interface)	Defines the minimum or maximum values for differential AC, DC, and crossing levels: • Vrx/2 (CK, DQS): Differential peak level. Checks whether the differential voltage exceeds the specified value at least once in every transition. Specify the limit values in the wizard as two comma-separated numbers: the first number is the minimum Vrx requirement for the two single-ended signals comprising the clock, and the second number is the minimum Vinse requirement for the strobe signals. • Min VID(AC): Differential AC level. Checks whether the differential voltage exceeds the specified value at least once in every transition. The default value depends on the AC levels set. AC levels come from the IBIS file for each particular net, or from manual selection on the DDRx Batch-Mode Wizard - Simulation Options Page. • Min VID(DC), Min VIHdiff/VILdiff(DC): Differential DC level. Checks if the differential voltage exceeds and remains beyond this value in every transition. • Max VIX: Differential crossing level. Checks if the positive and negative signals cross each other within VDD/2 +/- VIX. • Min VSEH/VSEL: Checks if the single-ended positive and negative signals of the differential pair exceed this value in every half cycle. The rising edge must be more positive than VDD/2 + VSEH min, and the falling edge must be less than VDD/2 – VSEL max. • Vinse peak-to-peak: Similar to VID(AC) for other technologies. Checks that the differential signal exceeds a minimum level on every transaction. The wizard accepts two comma-separated values: the first value corresponds to the minimum value required for the differential clock, and the second value corresponds to the minimum value required for the differential strobe. • MinVinse (CK, DQS): This is the minimum deviation that both the P and the N components of the differential signal need to deviate from the appropriate Vref voltage (Vref_CA for the Clock or Vref_DQ for the Strobe). This value is similar to the VSEH/VSEL values defined in other technologies but is measured with respect to the Vref rather than with respect to ground. Specify the limit values in the wizard as two comma-separated numbers: the first number is the minimum Vinse requirement for the two single-ended signals comprising the clock, and the second number is the minimum Vinse requirement for the strobe signals.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Check tVAC/ tDVAC (Available for LPDDR2, DDR3 and LPDDR3 interfaces only.)	Checked, verifies that the signal exceeds the appropriate AC threshold for a minimum amount of time. Limits are built-in to the wizard (not editable) and depend on the slew rate and AC threshold level. The wizard defines AC threshold values, including AC150 and AC175. This measurement includes only nets with a specified AC threshold. AC thresholds for each particular net are set on the "DDRx Batch-Mode Wizard - Simulation Options Page" on page 806. Unchecked, disables the measurement.
Monotonicity threshold:	Specifies the maximum amplitude of non-monotonicities to ignore, only when signals are between the DC thresholds. Use this option to filter out glitches that are so small they do not have any effect inside the receiver IC package. See "Usage Notes" on page 801.

Usage Notes

For single-ended nets, the monotonicity threshold check is performed while the waveform is between VIL(DC) and VIH(DC). Simulation uses the following receiver threshold information from the IBIS model (highest priority first):

- VIL(DC)

  [Receiver Thresholds] keyword, Vth - Vinl_dc subparameter

  [Model Spec] or [Model] keyword, Vinl subparameter

- VIH(DC)

  [Receiver Thresholds] keyword, Vth + Vinh_dc subparameter

  [Model Spec] or [Model] keyword, Vinh subparameter

For differential signals, the monotonicity threshold check is performed while the differential waveforms are within the voltage range defined by the Vdiff(DC). Simulation uses the following receiver information from the IBIS model (highest priority first):

- Vdiff(DC)

  [Receiver Thresholds] keyword, Vdiff_dc subparameter

  [Diff Pin] keyword, Vdiff subparameter

Related Topics

Running a DDRx Memory Interface Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDRx Batch-Mode Wizard - Report Options Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Report Options** page.

Use this page to specify the simulation setup audit, report type and formatting, and waveform storage options.

Fields

Field	Description
For simulation, run	Specifies the type of simulation to run. If your selection includes an audit, the simulation run creates a *DDR_audit-<DDRx_setup_file_name>-.xls* file, that contains information about simulation setup problems for all pins in the DDRx interface. **Note:** DDRx memory interface simulations can run for hours when you enable many nets for simulation. To use the time efficiently, use “audit” to enable batch simulation to report common setup problems prior to running simulations.
As a result of the skew/ setup/hold time measurements generate	Specifies one or more measurements reports to generate. • all cases report — Checked, contains all nets and all measurements, for each cycle in the simulation. • worst cases report — Checked, contains all nets and their worst-case measurements. • violation cases report — Checked, contains only nets that violate timing-model requirements.
Reporting options	Specifies the types of reports to create • HTML report — Selected, creates a DDR Simulation Report containing worst-case measurement results. Click a measurement value in the report to open a waveform in EZwave. The software can additionally create a report in PDF format and include EZwave images. This option also creates CSV spreadsheets. • XLS spreadsheets — Selected, writes report files in Microsoft Excel format. **Note:** If your report files include all cases (such as DDR_report_data_allcases_Slow) or you are working in Linux, the software always writes results to a *.csv* file. • CSV spreadsheets — Selected, writes reports to *.csv* file. This format is recommended for reports with more than 5000 rows. The software writes reports to the *<results>\DDR_report* folder.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Save all <DDRx> simulation waveforms to disk	Checked: • In HTML reports and measurement table cells, adds links to open waveforms with EZwave. • Saves simulation waveforms to *.csv* files, which you can use to investigate results by displaying in the oscilloscope or EZwave. • Save all simulated points in waveforms — Checked, saves all simulation waveform data points, which can produce very large *.csv* files. Unchecked, saves one in ten simulation waveform data points.
Create AMI model debug info (If available)	Checked, saves debug files to *DebugFolder* in the results sub-folder.
Output directory	Specifies the name of the results folder. • add time tags to the folder name Checked, adds a time stamp to the end of the output folder name. The root folder location depends on options you set in the "Reports and log files directory" section of the "Set Directories Dialog Box" on page 1016.

Usage Notes

DDRx simulation can create very large *_allcases_ results* files, especially when you apply a PRBS stimulus with a bit order of 8 or more on the DDRx Batch-Mode Wizard - Stimulus and Crosstalk Page. Because Microsoft Excel has problems loading *.xls* files that contain a large number of rows, DDRx simulation automatically switches format for _allcases_ result files to *.csv* when a report file exceeds 5000 rows.

You can use the Windows environment variable named HL_DDR_ALLCASES__MAX_XLS_ROWS to change the threshold from 5000 to another number. This environment variable affects only the format of *_allcases_ results* files.

To bypass the format switch and force DDRx simulation to create very large *.xls* files, set HL_DDR_ALLCASES__MAX_XLS_ROWS to 0.

Related Topics

DDRx Batch Simulation Results

Running a DDRx Memory Interface Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDRx Batch-Mode Wizard - Round Trip Time Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Round Trip Time** page. This page is accessible only when you have enabled the Clock-to-strobe skew timing and Check round trip times fields on the **Nets to Simulate** page.

Use this page to set up round trip time measurements by entering values into the table or importing the values using an RTT_Limits file.

Restriction

Round trip time is measured and reported twice for each DQS net: once with initial edge rising and again with the falling (reading operation). Therefore, the initial delay for the DQS net is compensated by time-to-Vmeas(rise) and by time-to-Vmeas(fall).

Results are reported in the *DDR_report_round_trip_time_*.xls* file where * depends on the IC model corners set in "DDRx Batch-Mode Wizard - Simulation Options Page" on page 806.

Round Trip time uses compensated flight times when "Compensate signal launch skews" field is enabled on the "DDRx Batch-Mode Wizard - Nets to Simulate Page" on page 793. Otherwise, the simulation uses zero for time-to-Vmeas.

Fields

Field	Description
Import Limits	Loads minimum and maximum round trip time limits into the table. The default file name is *RTT_Limits.txt*.
Reset Limits	Clears the values in the table.
Data Strobes	Lists the nets selected in the DDRx Batch-Mode Wizard - Data Strobes Page or Write Data Strobes Page.
RTT-1 Min/Max	Defines the minimum and maximum limits of the Round Trip Time 1. RTT-1 is calculated as CLK flight time plus tDQSCK (DQS from CLK delay uncertainty) plus DQS flight time (read). RTT-1 is specified at the controller where DQS is gated with DQSMASK.
RTT-2 Min/Max	Defines the minimum and maximum limits of the Round Trip Time 2. RTT-2 consists of RTT-1 and includes a 90 degree shift to DQS (read). RTT-2 is specified where read data switches from the DQS domain to the internal CLK domain.

Related Topics

Running a DDRx Memory Interface Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDRx Batch-Mode Wizard - Simulate Page

To access: **Simulate SI > Run DDRx Batch Simulation**, select the **Simulate** page.

Use this page to save wizard settings, save a log file, or open the DDRx Batch-Mode - Run Simulation Dialog Box dialog box, which you use to launch simulation.

All of the information required to write a complete setup file is set before you reach this page. Exiting the wizard and saving any changes automatically writes the setup file to disk.

Fields

Field	Description
Setup Issues That May Affect Batch Simulations area	Displays any issues that exist with your setup. If an audit error occurs, do not start batch simulation. Review the issues and correct them before proceeding with simulation.
Save Log	Saves any issues to a logfile.
Run Batch Simulation	Saves the wizard setup settings in a new or existing *.DDR* file upon exiting the wizard, then opens the DDRx Batch-Mode - Run Simulation dialog box. The Run Simulation Dialog Box loads a DDRx setup file and allows you to launch simulation. This dialog box includes: • Setup - The path to the DDRx setup file. • Browse - Loads a DDRx setup file. • Run - Starts the simulation. • Cancel - If a simulation is not running, exits wizard without running simulation or saving the setup file. If simulation is running, stops the batch simulation. However, the current portion of the batch simulation runs to completion, and the results are saved before simulation stops completely.
Save Setup Data	Saves the setup data, exits the wizard and does not run simulation.

Related Topics

Running a DDRx Memory Interface Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DDRx Batch-Mode Wizard - Simulation Options Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Simulation Options** page.

Use this page to specify detailed simulation options.

Fields

Field	Description
Select IC model corners	Specifies the IBIS model IC operating conditions to use during simulation.
AMI simulation flow (If available)	Specifies a simulation flow. For information, see “DDRx AMI Simulation Flows” on page 149.
When simulating, vary voltage reference values with IC corners	Checked, applies the IC voltage to passive termination components. Unchecked, simulation uses the voltage assigned to the power supply net that is connected to the voltage references.
Measure and report clock jitter	Checked, does the following without changing any other measurements: • Measures the minimum high and low pulses during a multi-cycle waveform (tJIT or tCH/tCL) • Calculates the cumulative errors across several cycles (tERR) • Reports results in *DDR_report_clock_jitter_*.xls* and *DDR_report_clock_jitter_err_*.xls*, where * is Typ, Fast, or Slow. For measurement details, see “Usage Notes” below.
Maximum run-time per net	Specifies the maximum number of minutes to simulate a specific net. **Tip:** Make this value long enough to handle all operations and corner cases. If set too short, some simulations may be excluded.
Ignore initial DQ bits Ignore initial ACC bits (address, command, and control signals) (If available)	Specifies the number of bits needed for a signal to stabilize. The software starts to measure a signal or plot an eye diagram, after this number of bits. These options override a value specified by an AMI Ignore_bits parameter displayed on the Equalization page.
AMI samples per bit	Specifies the number of samples per bit interval to adjust the display resolution of eye diagram contours in the HyperLynx Eye Density/ BER Viewer. Large numbers will slow analysis considerably.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
AC threshold selection (If available)	Specifies the AC threshold values. • AC thresholds from IBIS models — Uses values in the assigned IBIS models. If there is no Receiver Threshold section defined, the wizard uses Vinh and Vinl values from the main [Model] section. • Manual selection of AC thresholds — Enables you to specify an AC threshold for DQ/DM and ACC nets.
Measurement location	Specifies probe locations for the controller and DRAMs: • IBIS Default — Makes all measurements at the location specified in the IBIS file (Timing_location keyword). • At Die — Makes all measurements at the die. • At Pin — Makes all measurements at the pin.

Usage Notes

Measurements for tJIT (lck), tERR(lck) are available when crosstalk is off. To make measurements for both values tJIT and tJIT(lck) or tERR and tERR(lck), run two simulations: with crosstalk and without crosstalk.

Clock Jitter Measurement	Description
tJIT (or tCH/tCL)	Measures the minimum high and low pulses during all multi-cycle waveforms. PCB Jitter High/Low is the difference between tCK/2 and the minimum high/low pulses. Total jitter is PCB Jitter High/Low plus Chip jitter. Chip jitter is taken from timing files and includes absolute and average values. Positive Chip jitter is added to the low pulse jitter and negative chip jitter is added to high pulse jitter.
tERR	Calculates the cumulative error across several cycles. Each cycle consists of two pulses: a high and a low. For each cycle, tERR(n) = (tCK(1) + tCK(2) … + tCK(n)) - tCK * n. tERR is measured for 50 cycles of a waveform. If the clock waveform is shorter (PRBS7 or smaller), fewer cycles are used. Limits for tERR come from the JEDEC spec and are read-only.

Related Topics

Running a DDRx Memory Interface Simulation

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DDRx Batch-Mode Wizard - Stimulus and Crosstalk Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Stimulus and Crosstalk** page.

Use this page to specify unique per-bit stimulus and crosstalk options for your DDRx interface simulation. You can also include the effects of simultaneous switching noise (SSN) in your simulation.

The Wizard automatically saves stimulus from this page for future simulations. See "Usage Notes" on page 809.

Fields

Field	Description
Number of DQ bits Number of ACC bits (address, command, control signals)	Specifies the number of stimulus bits. An eye diagram displays this number of bits. The overall stimulus length is (number ignore bits) plus (number of stimulus bits). An eye diagram excludes ignore bits. **Note:** Although a large number of bits can produce very long simulation run times, they also produce more statistically meaningful simulation results, especially when exploring the effects of inter-symbol interference (ISI). **Restriction:** For address signals only, the software applies a maximum of 32 bits.
Bit Pattern Spreadsheet	
Net Name	List of nets in your interface that are included in the simulation.
Random?	Checked, the Bit Pattern column populates with a random bit pattern. Unchecked, allows you to define a custom stimulus.
Bit Pattern	Specifies the bit pattern used during simulation for the specified net. • If Random is checked, you cannot edit this field. • If Random is unchecked, you can enter your own binary numbers.
Reshuffle	Regenerates the bit pattern for all signals that have Random checked.
Apply Pattern to All Nets	Applies the selected bit pattern to all nets in the spreadsheet, regardless of whether Random is checked or not. Select the net by clicking the box to the left of the signal name.
Hide Disabled Nets	Checked, hides rows for nets that you disabled on the Disable Nets page.

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Field	Description
Use PRBS as Random	Checked, for nets with an X in the Random column, the software applies PRBS stimulus instead of random stimulus. The number of bits you specify determines the PRBS stimulus. For example, if you specify a stimulus of 256 bits, the software applies PRBS8.
Include crosstalk effects in simulations	Checked, includes the effects of crosstalk among various signals in the memory interface. Aggressor nets for signals that are not in the memory interface are simulated and driven with a PRBS stimulus, using the same bit order you specify for non-strobe and non-clock nets.
Coupling Thresholds	Opens the Set Coupling Thresholds Dialog Box, where you can define the neighboring nets to include in simulation.
Include PI effects in SI simulations (Power Aware)	Checked, uses power-aware portions of the IBIS models to simulate Simultaneous Switching Noise (SSN). This option is required to automatically add aggressor SSN pins. See “Accounting for Non-Ideal Power Supplies in SI Simulation” on page 113. **Restriction:** The software ignores this option when your DDRx interface supports equalization and you have not enabled the Advanced DDRx AMI simulation flow option on the Simulation Options page. **Note:** This option can significantly increase the simulation run time.
Save	Save the current settings to a DDR file so you can reuse the stimulus and other settings.

Usage Notes

The DDRx stimulus can help you to investigate poor margins or other batch DDRx results by assigning the stimulus to the interface signals and simulating with the Oscilloscope or EZwave. The wizard automatically saves the DDRx stimulus from this page to a text file in the *StimulusForNets* subdirectory of the results directory. The text file references EDS files in the same subdirectory. To reuse the stimulus, import the text file using the Assign Stimulus Dialog Box.

For additional information about creating custom EDS stimulus files, see “Creating a Custom Stimulus” on page 492.

Simulation automatically applies toggling stimulus to clock and strobe nets, using half the data rate you specified on the Initialization page. For example, if the data rate is 800 MT/s (megatransfers per second), the period of the toggling stimulus is 2.5 ns (800 MT/s / 2 = 400 Mbps, assume Mbps = MHz, 1/400 MHz = 2.5 ns).

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Stimulus delays are derived from signal relationships defined in the timing model.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - Sweep Manager Page

Scope: LineSim

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Sweep Manager** page.

Use this page to define the set of design property values (sweep range) to apply to a design property during sweep simulations.

Fields

Field	Description
[Design property tree]	Displays the design properties that you can sweep. Double-click a design property to open the Sweeping Dialog Box and specify the sweep range. Check a design property to enable sweep ranges. Uncheck to disable (not delete) a sweep range.
Add Range Edit Range	Opens the Sweeping Dialog Box, which enables you to add a new sweep range to the selected design property, or edit an existing range.
Remove Range	Deletes an existing sweep range. **Caution:** Deleting the sweep range for a reference design property also deletes the sweep ranges for its dependent properties.
Copy Range Paste Range	Copies the sweep range from one selected design property to another.
Paste Range as a Lock	Synchronizes and locks the sweep values copied from one design property to the other.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - Timing Models Page

To access: **Simulate SI > Run DDRx Batch Simulation**, and select the **Timing Models** page.

Use this page to specify the memory interface data rate and the device speed grade and timing models for the memory controller and DRAM devices. You can also use this page to display, edit, and verify timing models.

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The easiest way to create a new *timing model* is to copy and edit one of the timing models (*.v*) that are included with the HyperLynx installation. You can also use the HyperLynx Timing Model Wizard to create a timing model.

All timing models must define the same parameters included in the timing models that ship with HyperLynx. All DRAMs must have the same timing requirements and use the same timing model. Differences are flagged in red.

Timing models are located in the Libs folder. For example: *\MentorGraphics\<release>\SDD_HOME\hyperlynx\Libs*

Fields

Field	Description
DDRx Data Rate (MT/s)	Specifies the default data rate for the DDRx interface in MT/s (megatransfers per second). From the dropdown list, select a standard rate or <custom> to enter a new value. All speed grades update with the new data rate value. See Map Custom Data Rates to Standard JEDEC Derating Tables. Changing the data rate also changes the value on the Controller page. See DDRx Batch-Mode Wizard - Controller Page.
Tight timing / Relaxed timing (If available)	Displays either tight or relaxed timing values for setup and hold if specified in the model (parameters tDS_tight, tDS_relaxed, tDH_tight, and tDH_relaxed).
Device	Displays the devices available. Click the box to the left of the device name to select the device.
Type	Displays the part type.
Part Type	Displays the part number. Double-click the number to display a graphic and parameters of the timing model.
Speed	Defines the speed grade for the device. Select a speed from the dropdown list. • Angle brackets (<) signify the default speed. • Red text indicates unverified speed. The speed grade is normally greater than or equal to the data rate. The speed column automatically updates when you change the DDRx interface type on the Initialization page, based on the Data Rate selected.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Model File	Defines the timing model for the device. From the dropdown list, select a model or **<Browse>** to search for a timing model file. Right-click to display the file location. • Angle brackets (<) signify the default timing model. • Red text indicates unverified timing model.
Apply Settings to Similar DRAMs	Copies the data rate and timing model values from the selected DRAM to all other DRAMs in the design. **Tip:** DRAMs in a design typically use the same settings. After you set up one DRAM, apply its settings to all the DRAMs.
View	Displays a graphic and parameters of the selected timing model. Click the square cell to the left of the Device name to select the row and click **View**.
Edit	Opens the HyperLynx Timing Model Editor, which enable you to edit the timing parameters. See Creating Controller and DRAM Timing Models.
Verify	Verifies the timing model syntax and loads parameter values into the wizard for the selected device. Red (unverified) text turns black when verified.
Verify All	Verifies the timing model syntax and loads parameter values into the wizard for all devices. Red (unverified) text turns black when verified.
TM Wizard	Opens the HyperLynx Timing Model Wizard, which enables you to create a simple timing model for a DDRx memory controller. See Creating Controller and DRAM Timing Models.

Related Topics

Running a DDRx Memory Interface Simulation

DDRx Batch-Mode Wizard - Vref Training Page

To access: **Simulate SI > Run DDRx Batch Simulation**, select the **Vref Training** page

Use this page to specify how the software defines the reference voltage (Vref) during read measurements.

Note

For DQ signals in some memory interfaces, Vref depends on the details of your design. For example, the voltage center of a resulting eye diagram depends on the driver strength and the signal load.

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Note

Write measurements at the DRAM are not affected by Vref training because DRAMs use JEDEC guidelines to determine Vref.

Fields

Field	Description
Select way of grouping signals to train Vref for reads:	• Separate Vref for each pin • Single Vref per lane Vref value is calculated using the min and max Vcent value for each lane. For multi-rank, Vref value is calculated using one min and one max value per DRAM group. • Single Vref for all the signals Vref value is calculated using the min and max Vcent values across all signals. Therefore all signals have the same Vref.
Enable separate Vref per rank	Checked, changes how Vref is calculated during a read operation, depending on which signal grouping option you selected. Single Vref per lane – Each rank (specified in the Operation column) in each lane has a different Vref value. For example, if you are simulating an interface with two lanes, each with two ranks, four Vref values are calculated. Single Vref for all signals – Vref value is calculated per rank by taking the average of the largest and smallest Vcent values in the “Voltage at widest eye opening Vcent” column for each given rank in the spreadsheet results.
Use file for custom Vref values	Checked (when you initially enable this option), opens a dialog box where you can specify the location of an existing file or a new file. To define the contents of a new file, see Usage Notes.

Usage Notes

To create a new Vref file:

1. Check “Use file for custom Vref values”, then specify the location of a new file.
2. Click **Edit**.

   An editor opens the file and automatically adds example syntax.
3. Make edits as required, and save the file.

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Related Topics

Running a DDRx Memory Interface Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Decoupling Wizard

Use this wizard to run Lumped Analysis, Quick Analysis, or Distributed Analysis decoupling simulation.

To access: **Simulate PI > Analyze Decoupling**

Topic	Description
Decoupling Wizard - Check Capacitor Models Page	Use this page to verify and edit decoupling capacitor model assignments.
Decoupling Wizard - Check VRM Models Page	Use this page to show or assign VRM models. You can also disable a VRM model so that simulation does not account for its loading.
Decoupling Wizard - Choose a Type of Analysis Page	Use this page to choose the type of decoupling simulation to run.
Decoupling Wizard - Choose Easy / Custom Page	Use this page to choose between default and custom decoupling analysis options.
Decoupling Wizard - Control Frequency Sweep Page	Use this page to edit frequency range and sampling options, both of which affect simulation run time and the resolution of the Z-parameter model.
Decoupling Wizard - Customize Settings Page (Standard Simulation)	Use this page to enable detailed simulation options. Simulating the design with different sets of enabled and disabled options can help you determine how individual types of design properties contribute to decoupling performance.
Decoupling Wizard - Customize Settings Page (Advanced Simulation)	Use this page to enable detailed simulation options. Simulating the design with different sets of enabled and disabled options can help you determine how individual types of design properties contribute to decoupling performance.
Decoupling Wizard - Select IC Pin Group Probes Page	Use this page to select power supply pin groups to include as ports in the Z-parameter model created by decoupling simulation. Decoupling simulation measures PDN impedance through group pins in parallel. The more pin groups you select, the longer simulation takes and the larger the model file becomes.
Decoupling Wizard - Select IC Pin-Pair Probes Page	Use this page to select pin pairs on the power-supply nets to include as ports in the Z-parameter model created by decoupling simulation. The more pin pairs you select, the longer simulation takes and the larger the model file becomes.
Decoupling Wizard - Select IC Power Pins Page	Use this page to select pins on the pair of power-supply nets to include as ports in the Z-parameter model created by decoupling simulation. The more pins you select, the longer simulation takes and the larger the model file becomes.

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Topic	Description
Decoupling Wizard - Select Nets for Analysis Page	Use this page to select power-supply nets that you want to probe.
Decoupling Wizard - Set the Target Impedance Page	Use this page to specify the target impedance of the pair of power-supply nets. The value you specify is displayed as a green reference line in the Touchstone Viewer when you display the output Z-parameter file.
Decoupling Wizard - Start Analysis Page	Use this page to start a new simulation or load the settings for a saved analysis.
Decoupling Wizard - Supply Component Models Page	Use this page to verify and edit model assignments for decoupling capacitors and for series components that connect two power-supply nets.

Decoupling Wizard - Check Capacitor Models Page

To access: **Simulate PI** > **Analyze Decoupling** or **Advanced Decoupling Analysis**, select the Check Capacitor Models page

Use this page to verify and edit decoupling capacitor model assignments.

Fields

Field	Description
Show these nets only	Specifies whether to show capacitors for a specific pair of power supply nets. In LineSim, the bottom portion of the lists display the “built in” power supply nets for signal-integrity simulation, such as Vcc, VpullDn, VpullUp, and VSS. The exporting features do not use these nets.
Value	Specifies capacitance values from a *.QPL* file, *.REF* file, or design file (from highest to lowest priority), if a value exists. **Caution:** Decoupling simulation does not use these values. The software uses them to create an initial membership of decoupling-capacitor groups.
Model Info	Specifies C-L-R values or capacitor models used by power-integrity simulation.
Enabled	Checked, the software uses assigned models during decoupling simulation.
Assign Model Remove Models	Edits or removes models for selected capacitors or capacitor groups.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Edit Groups (Available in BoardSim.)	Manages capacitor groups.

Usage Notes

Double-click a row to edit a model assignment.

Right-click a column heading to sort spreadsheet rows.

Related Topics

Decoupling Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Decoupling Wizard - Check VRM Models Page

To access: **Simulate PI > Analyze Decoupling** or **Advanced Decoupling Simulation**, select the Check VRM Models page

Use this page to show or assign VRM models. You can also disable a VRM model so that simulation does not account for its loading.

Note

This page is unavailable if you select “Quick Analysis” on the “Choose a Type of Analysis” page.

Fields

Field	Description
Show VRM(s) for selected nets only	Checked, the spreadsheet shows VRM models that you have assigned to nets you have selected on the “Select Nets for Analysis” page. Unchecked, the spreadsheet shows VRM models for all nets.
Right-most spreadsheet column	Shows the VRM model status as one of the following: • Checked check box — Complete model assignment. Simulation accounts for passive loading of the VRM. • Unchecked check box — Complete model assignment. Simulation does not account for passive loading of the VRM. • Empty blue square — Incomplete model assignment, such as a missing reference net.
VRM Manager	Opens the Assign Power Integrity Models dialog box, which you can use to add or edit VRM model assignments, or to open the VRM Manager.

Related Topics

Decoupling Simulation

Decoupling Wizard - Choose a Type of Analysis Page

To access: Simulate PI > Analyze Decoupling or Advanced Decoupling Analysis, select the Choose a Type of Analysis page

Use this page to choose the type of decoupling simulation to run.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Advanced Decoupling Wizard	
Decoupling Analysis	Evaluate the ability of a PDN to provide low-impedance paths for specific power supply pin pairs or pin group pairs. Advanced distributed PDN decoupling simulation accounts for the location of each decoupling capacitor, inter-plane capacitance and inductance, the board outline, and the mounting parasitics and loss for decoupling capacitors.
PDN Decoupling Analyzer	The PDN Decoupling Optimizer enables you to analyze the decoupling capacitor locations in a board or schematic (PDN Editor) design to determine the optimum decoupling capacitor for each location. The optimized design includes the minimum number of capacitors, the minimum number of different capacitor types, and has the lowest cost.
Decoupling Wizard	
Quick Analysis	Quickly create a spreadsheet containing detailed information about all decoupling capacitors in the design, including total mounting inductance and mounted resonant frequency. This information enables you to identify ineffective decoupling capacitors.
Lumped Analysis	Model the pair of power supply nets as a series of directly-connected elements and create a Z-parameter model. See Circuit Topology for Lumped Decoupling Simulation.
Distributed Analysis	Model the pair of power supply nets as one or more transmission planes and create a Z-parameter model. See Circuit Topology for Lumped Decoupling Simulation. This method provides the most accurate Z-parameter model, but consumes more memory and run time than lumped analysis.
Quick Analysis	Quickly create a spreadsheet containing detailed information about all decoupling capacitors in the design, including total mounting inductance and mounted resonant frequency. This information enables you to identify ineffective decoupling capacitors.

Related Topics

Decoupling Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Decoupling Wizard - Choose Easy / Custom Page

To access: **Simulate PI > Analyze Decoupling** or **Advanced Decoupling Simulation**, select the Choose Easy / Custom page

Use this page to choose between default and custom decoupling analysis options.

Note

This page is unavailable if you select “Quick Analysis” on the Decoupling Wizard - Choose a Type of Analysis Page.

Fields

Field	Description
Easy	Popular settings are automatically enabled on some of the following wizard pages. Some settings become read only.
Custom	You can edit all settings on the following wizard pages.

Related Topics

Decoupling Simulation

Decoupling Wizard - Control Frequency Sweep Page

To access: **Simulate PI > Analyze Decoupling** or **Advanced Decoupling Analysis**, select the Control Frequency Sweep page

Use this page to edit frequency range and sampling options, both of which affect simulation run time and the resolution of the Z-parameter model.

Note

This page is unavailable if you select “Quick Analysis” on the Decoupling Wizard - Choose a Type of Analysis Page.

Fields

Field	Description
Min frequency	The minimum simulation frequency, in MHz.

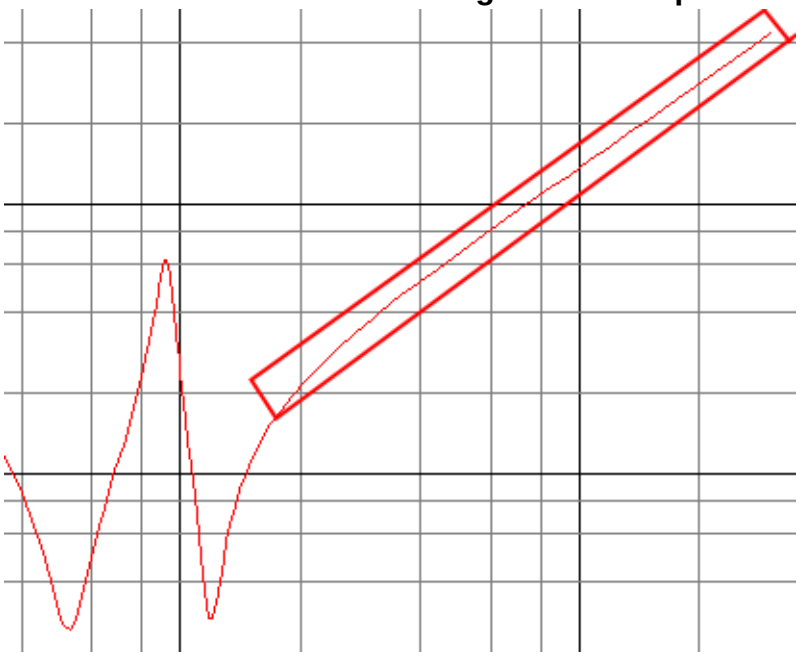
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Max frequency	The maximum simulation frequency, in MHz. Many ICs have in-package decoupling that provide the main decoupling effects above a certain frequency, such as 150 MHz. This means decoupling capacitors and buried capacitance located in the PCB contribute little or no decoupling above this design-dependent frequency.
Adaptive sampling	Varies the sampling step size depending on model characteristics. The adaptive scale is better than logarithmic and linear because it increases the sampling rate near frequencies with resonances. **Note:** This option is available only for standard decoupling simulation.
Logarithmic sampling	Sampling points are distributed at logarithmic intervals across the frequency range. The intervals between sampling points are smaller at lower frequencies and larger for higher frequencies. With logarithmic sampling, every next frequency point is equal to the previous value times a factor K > 1. This produces a constant increase ratio, but the absolute distance between sampling points grows.
Linear sampling	Sampling points are distributed at equal intervals across the frequency range.
Accuracy at resonances	For lumped simulation, selecting High may yield reasonably fast simulation run times. For distributed simulation, you should take the complexity of the design into account. If the design has large numbers of power-supply nets, hundreds of decoupling capacitors, and hundreds or thousands of stitching vias, selecting Low provides preliminary results with decreased simulation run time. After evaluating the preliminary results, you can identify which frequency ranges interest you the most and try running simulation with higher accuracy on each range of interest. **Note:** This option is available when you enable Adaptive sampling.
Minimum number of samples in flat, non-resonant regions	The number of samples you specify applies to flat, non-resonant, regions of an impedance profile. See the enclosed curve region in Figure 10-9. **Note:** This option is available when you enable Adaptive sampling.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Number of samples	The number of samples you specify applies to the entire frequency range. **Note:** This option is available when you enable Adaptive sampling.

Figure 10-9. Flat and Non-Resonant Region of an Impedance Profile



Related Topics

Decoupling Simulation

Decoupling Wizard - Customize Settings Page (Standard Simulation)

To access: **Simulate PI > Analyze Decoupling**, select the Customize Settings page

Use this page to enable detailed simulation options. Simulating the design with different sets of enabled and disabled options can help you determine how individual types of design properties contribute to decoupling performance.

Note

If you selected Easy on the Decoupling Wizard - Choose Easy / Custom Page, options on this page are read only.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Include all power / ground nets in the analysis	Examines transmission planes formed by all power-supply nets that can influence the decoupling behavior of the already-selected pair of power-supply nets. This can happen when transmission planes formed by non-selected power-supply nets receive energy from the analysis. For example, transmission planes formed by other pairs of power-supply nets (possibly including one of the currently-selected nets) can be activated by energy traveling through stitching vias that run vertically throughout the stackup. Checking this option can increase analysis run time because the simulation model and calculations are bigger and more complex. Uncheck to analyze only the pair of power-supply nets specified on the Decoupling Wizard - Select Nets for Analysis Page. You might disable this option if you are convinced that no other pairs of power-supply nets contribute to the decoupling behavior of the selected power-supply nets.
Include capacitor mounting inductance	To determine the contribution of capacitor mounting inductance to the overall decoupling performance, you can run analysis with this option enabled, run it again with this option disabled, and then compare the results.
Remove series inductance unique to each power pin, to see plane decoupling more clearly	Check to evaluate the PDN impedance without the effects of power-supply pin series mounting inductance by excluding the series inductance that is unique to each power-supply pin. For example, you might do this to evaluate the effects of decoupling-capacitor placement or values on PDN impedance. Note: Inductance from IC power-supply pin mounting, such as fanout traces and the portions of mounting vias that lie above the first reference layer, can obscure the impedance of the overall transmission-plane decoupling. Including such inductance in analysis may lead to overly pessimistic (that is, high-impedance) results, because ICs actually use many power pins in parallel, thereby reducing the effect of the series inductance. Uncheck to evaluate the effects of power-supply pin series mounting inductance on PDN impedance.
Include inter-plane capacitance	To determine the contribution of inter-plane capacitance to the overall decoupling performance, you can run analysis with this option enabled, run it again with this option disabled, and then compare the results.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Include poorly connected capacitors	Check to include all decoupling capacitors in the simulation. Simulation normally excludes capacitors with poor or non-ideal mounting properties that make the capacitor useful only at low frequencies. Examples of poor mounting include long connection traces and vias located outside the power-ground plane pairs being decoupled. These types of capacitor mounting usually have large inductances, which cause them to be mostly ineffective at high frequencies, and are difficult to accurately model for simulation. In some cases, only a rough approximation of the inductance can be made for capacitors with this type of mounting.
Enable stitching-via optimization	Check to find stitching vias that are located close together and merge their individual models into an equivalent model. This process is repeated across the transmission plane. Reducing the number of stitching-via models speeds up simulation and reduces memory consumption. In the Tolerance area, select Medium unless you have a specific reason not to. For details about the stitching-via optimization process and the effect of the Tolerance setting, see Stitching-Via Optimization.

Related Topics

Decoupling Simulation

Decoupling Wizard - Customize Settings Page (Advanced Simulation)

To access: **Simulate PI > Advanced Decoupling Analysis**, select the **Customize Settings** page

Use this page to enable detailed simulation options. Simulating the design with different sets of enabled and disabled options can help you determine how individual types of design properties contribute to decoupling performance.

Note

If you selected Easy on the Decoupling Wizard - Choose Easy / Custom Page, options on this page are read only.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Defeature metal areas	Checked, merges small details on complex metal shapes. This option reduces simulation run time, usually with no loss of accuracy. The Resolution option represents the minimum dimension of a feature to include in simulation.
Crop layout data	Checked, excludes metal outside of a perimeter that encloses all probed pin pairs and group pairs. This option significantly reduces simulation run time, but can potentially reduce accuracy.
Show HyperLynx Hybrid Solver GUI	Checked, loads the design and project into HyperLynx Hybrid Solver. If you want to investigate ways to change the design to improve decoupling performance, you can manually edit the project in HyperLynx Hybrid Solver and run simulation from there.

Related Topics

Decoupling Simulation

Decoupling Wizard - Select IC Pin Group Probes Page

To access: **Simulate PI > Analyze Decoupling**, select the Select IC Pin Group Probes page

Use this page to select power supply pin groups to include as ports in the Z-parameter model created by decoupling simulation. Decoupling simulation measures PDN impedance through group pins in parallel. The more pin groups you select, the longer simulation takes and the larger the model file becomes.

Note

This page is available when you enable “Automatically assign reference layers” on the **Power Integrity** tab of the Preferences dialog box and enable “Distributed Simulation” on the Decoupling Wizard - Choose a Type of Analysis Page.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Filter spreadsheet row	Filters the spreadsheet when you: • Select an object or enter a search string in the first column. The search string can include one trailing asterisk wildcard character. • Uncheck the third column to display pin groups that you have not included for simulation.
Spreadsheet check box	Checked, includes the pin group in decoupling analysis. The first column displays the Z-parameter model port number for the checked pin group. Note: You cannot check a pin group that contains a component pin that you have probed on the "Select IC Power Pins" page. Point to a blue check box to see the name of a probed component pin(s). To create an additional pin group, click **Manage Supply Pin Groups**.
Group by Reference Designators	Checked, collapses spreadsheet rows into groups of pins with the same reference designator.
Manage Supply Pin Groups	Opens the Pin Group Manager dialog box, where you can define power supply pin groups.

Related Topics

Decoupling Simulation

Decoupling Wizard - Select IC Pin-Pair Probes Page

To access: **Simulate PI > Advanced Decoupling Analysis**, select the Select IC Pin-Pair Probes page

Use this page to select pin pairs on the power-supply nets to include as ports in the Z-parameter model created by decoupling simulation. The more pin pairs you select, the longer simulation takes and the larger the model file becomes.

Fields

Field	Description
Edit Pin-Pair Probes	Click to add or delete IC power-supply pin pairs displayed in the spreadsheet.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Filter spreadsheet row	Filters the spreadsheet when you: • Select an object or enter a search string in the first column. The search string can include one trailing asterisk wildcard character. • Uncheck the third column to display pin pairs that you have not included for simulation.
Spreadsheet check box	Checked, includes a component pin pair in decoupling simulation. **Note:** You cannot check a pin pair that belongs to a pin group that you have probed on the Select Group Pair Probes page. Point to a blue check box to see the name of a probed pin group.
Group by reference designators	Collapse spreadsheet rows into groups of pins with the same reference designator.

Related Topics

Decoupling Simulation

Decoupling Wizard - Select IC Power Pins Page

To access: **Simulate PI** > **Analyze Decoupling**, select the Select IC Power Pins page

Use this page to select pins on the pair of power-supply nets to include as ports in the Z-parameter model created by decoupling simulation. The more pins you select, the longer simulation takes and the larger the model file becomes.

Fields

Field	Description
Spreadsheet check box	Checked, includes a component pin in decoupling simulation. Note: You cannot check a component pin that belongs to a pin group that you have probed on the “Select IC Pin Group Probes” page. Point to a blue check box to see the name of a probed pin group.
Group by Reference Designators	Collapse spreadsheet rows into groups of pins with the same reference designator.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Add IC Power Pin	(Available in BoardSim). Click to add missing IC power-supply pins to the spreadsheet. You assign reference nets to power-supply pins, to make them available as Z-parameter model ports. If the spreadsheet does not display the added port, the transmission plane does not enclose it with sufficient overlap.

Related Topics

Decoupling Simulation

Decoupling Wizard - Select Nets for Analysis Page

To access: **Simulate PI > Analyze Decoupling** or **Advanced Decoupling Analysis**, select the Select Nets for Analysis page

Use this page to select power-supply nets that you want to probe.

Note

For advanced distributed decoupling simulation, all power-supply nets that you make available on this page are included in simulation if you uncheck the “Ignore metal of all nets that are not probed” option on the Customize Settings page.

Fields

Field	Description
Allow probes at pins of any supply net	Checked, all power-supply nets are available for probing.
Available nets Nets to probe	Double-click an Available net to make it available for probing. Included nets appear in the Nets to probe area.

Usage Notes

In LineSim, the Available nets area displays power-supply nets defined in the PDN Editor. The PDN Editor initially contains a power-supply net for each stackup layer that is assigned the “plane” usage type. If you short together power-supply nets, the Available Nets area displays the name of only one of the power-supply nets. You can short power-supply nets with stitching vias and by specifying two or more connected or reference layers for a IC power-supply pin, decoupling via pin, or VRM pin.

Related Topics

Decoupling Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Creating a PDN Design

Decoupling Wizard - Set the Target Impedance Page

To access: **Simulate PI > Analyze Decoupling** or **Advanced Decoupling Analysis**, select the Set the Target Impedance page

Use this page to specify the target impedance of the pair of power-supply nets. The value you specify is displayed as a green reference line in the Touchstone Viewer when you display the output Z-parameter file.

Fields

Field	Description
Target Z	Specifies the target impedance, in milliOhms.
Calculator	If you do not know the target impedance, but you know peak transient current, nominal VCC, and power-supply ripple, click **Calculator** to open the Target-Z Wizard.

Related Topics

Information Needed to Calculate Target PDN Impedance

Decoupling Simulation

Decoupling Wizard - Start Analysis Page

To access: **Simulate PI > Analyze Decoupling** or **Advanced Decoupling Analysis** > select the Start Analysis page

Use this page to start a new simulation or load the settings for a saved analysis.

Fields

Field	Description
New	Does not use any previous wizard settings.
Use last configuration Note: This option is unavailable after you close the design.	Uses wizard settings currently in memory.
Load saved configuration	Uses wizard settings from a *.DAO* file you have previously saved.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Save settings to file	Checked, saves wizard settings to a *.DAO* file. The default file location is the <design> folder. See Design Folder and HyperLynx Files.

Related Topics

Decoupling Simulation

Decoupling Wizard - Supply Component Models Page

To access: **Simulate PI > Advanced Decoupling Analysis** > select the **Supply Component Models** page

Use this page to verify and edit model assignments for decoupling capacitors and for series components that connect two power-supply nets.

Note

You must check the "Allow AC supply models for non-capacitor components" option in the Preferences dialog box to display this page in the wizard. See Preferences Dialog Box - Power Integrity Tab.

Fields

Field	Description
Show these nets only	Specifies whether to show capacitors for a specific pair of power-supply nets. In LineSim, the bottom portion of the lists display the "built in" power-supply nets for signal-integrity simulation, such as Vcc, VpullDn, VpullUp, and VSS. The exporting features do not use these nets.
Value	Specifies component values from a *.QPL* file, *.REF* file, or design file (from highest to lowest priority), if a value exists. **Caution:** Decoupling simulation does not use these values. The software uses them to create an initial membership of decoupling-capacitor groups.
Model Info	Specifies C-L-R values or passive component models used by power-integrity simulation.
Enabled	Checked, the software uses assigned models during decoupling simulation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Assign Model Remove Models	Edits or removes models for selected components or capacitor groups.
Edit Groups (Available in BoardSim.)	Manages capacitor groups.

Usage Notes

Double-click a row to edit a model assignment.

Right-click a column heading to sort spreadsheet rows.

Related Topics

Decoupling Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Define Constraint Template Dialog Box

Use this tab to specify length and delay constraints for the net and its pin pairs, constraints for differential pairs, FromTos that appear in the FromTos section of the exported template file, and view pin sets and modify the type property for pin sets.

To access: Choose **Export > Constraint Template**. Click **Edit Template**.

Topic	Description
Define Constraint Template Dialog Box - Length/Delay Tab	Use this tab to specify length and delay constraints for the net and its pin pairs. Add only the constraints you need; it is okay to have empty spreadsheet cells.
Define Constraint Template Dialog Box - Diff Pair Tab	Use this tab to specify constraints for differential pairs. Add only the constraints you need; it is okay to have empty spreadsheet cells.
Define Constraint Template Dialog Box - Net Scheduling Tab	Use this tab to specify FromTos that appear in the FromTos section of the exported template file. FromTos represent the pin-to-pin, or pin-to-branch-point, routing sequence of a net or differential pair.
Define Constraint Template Dialog Box - Pin Sets Tab	Use this tab to view pin sets and modify the type property for pin sets. If a schematic does not have any pin sets, the Pin Sets tab is empty. To have a pin set, the schematic must contain a “virtual pin”, where three or more transmission lines directly connect to each other.

Define Constraint Template Dialog Box - Length/Delay Tab

To access: Open the Export Constraint Template Dialog Box, select the **Length/Delay Pair** tab

Use this tab to specify length and delay constraints for the net and its pin pairs. Add only the constraints you need; it is okay to have empty spreadsheet cells.

Fields

Field	Description
Net-level constraint section	Specifies net constraints. Click **Expand To Advanced Mode** >> to see a complete list of constraints. For constraint descriptions, see Table 10-1.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field		Description
Type		• Length specifies physical length. • Delay specifies electrical length, such as signal propagation delay or time of flight (TOF).
Constrained pin pairs section		Specifies pin pair constraints that appear in the PinPairs section of the exported template file. Pin pairs are not the same as FromTos. Pin pairs define electrical pairings that define electrical relationships among component pins. FromTos define physical pairings used to instruct the router to implement traces between component pins. **Restrictions:** • The net must have at least one driver and receiver to add pin pairs to the Constrained Pin Pairs spreadsheet. • For differential nets, virtual pins are not exported to the template file.
	Add All	Adds pin pairs for all IC components, passive components, and virtual pins.
	Add Drvs -> Rcvrs	Adds pin pairs for all combinations of IC components (including virtual pins), but not for passive components.
Expand To Advanced Mode >>		Click to see all constraints.
Switch to Basic Mode <<		Click to see only the Min and Max constraints.

Table 10-1. Constraint Descriptions

Constraint	Description
Min	The minimum acceptable physical routing length or signal propagation delay between component pins.
Max	The maximum acceptable physical routing length or signal propagation delay between component pins.
Match	A text string that identifies the net as belonging to a group of nets with similar length or time of flight delay routing requirements. Constraint Manager supports match names consisting of a single alphabetical or numerical character, or a string of characters starting with an alphabetical character followed by any combination of alphabetical, numerical, and underscore _ characters.
Tol	A tolerance range for net routing delay requirements for nets belonging to the same match name group.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 10-1. Constraint Descriptions (cont.)

Constraint	Description
Formula	A formula, following the rules supported by Constraint Manager, that can be used to create pin pair delay relationships. You can set up delay relationships among similar design objects that would benefit from such associations. For example, you can set one pin pair delay to equal the delay of another (=), specify that the delay of one net must be greater than or less than the delay of another (> or <), add or subtract the delay of one pin pair to or from the delay of another. You can also include constants and variables to define net and pin pair delay with even more detail. LineSim does not check formulas for validity.

Related Topics

Exporting a Constraint Template From LineSim

Define Constraint Template Dialog Box - Diff Pair Tab

To access: Open the Export Constraint Template Dialog Box, select the **Diff Pair** tab

Use this tab to specify constraints for differential pairs. Add only the constraints you need; it is okay to have empty spreadsheet cells.

Fields

Field	Description
Pair Tolerance section	
Constraint Type	• Length specifies physical length. • Delay specifies electrical length, such as signal propagation delay or time of flight (TOF).
Max Tolerance	Specifies the net length tolerance. Net lengths/delays are measured from pin to pin. Tolerance is measured by subtracting one net length/delay from the other, and taking the absolute value of the difference.
Routing convergence section	

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Field	Description
Max Distance	Specifies the maximum distance between the pins and the convergence point. The convergence point is where the traces begin the controlled routing gap, which creates a uniform mutual impedance as the traces are routed.
Max difference/ tolerance	Specifies the maximum difference of the routing lengths between the pins and the convergence point.
Separation Distance section	
Max distance	Specifies the maximum distance between the differential traces when router temporarily exceeds the controlled routing gap to avoid a via, pin, or other obstacle.
Differential Z section	
Target	Specifies the target impedance. **Note:** The physical spacing for differential pairs is specified per-layer (not per-net) in the Constraint Manager "Trace and Via Properties" page. Physical spacing rules are used whenever the differential Z target can not be met.
Tolerance +/-	Specifies the target impedance tolerance.
Net class	Specifies the net class name.

Related Topics

Exporting a Constraint Template From LineSim

Define Constraint Template Dialog Box - Net Scheduling Tab

To access: Open the Export Constraint Template Dialog Box, select the **Net Scheduling** tab

Use this tab to specify FromTos that appear in the FromTos section of the exported template file. FromTos represent the pin-to-pin, or pin-to-branch-point, routing sequence of a net or differential pair.

FromTos are not the same as pin pairs. FromTos define physical pairings that instruct the router to implement traces between component pins. Pin pairs define electrical pairings that define electrical relationships among component pins.

A branch point exists where three or more transmission lines directly connect to each other. The LineSim schematic automatically creates "virtual pins" to serve as branch points. You cannot assign virtual pins to From or To pins.

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Fields

Field	Description
Use Schematic Topology	Automatically adds FromTos for all IC components, passive components, and virtual pins.

Related Topics

Exporting a Constraint Template From LineSim

Define Constraint Template Dialog Box - Pin Sets Tab

To access: Open the Export Constraint Template Dialog Box, select the **Pin Sets** tab

Use this tab to view pin sets and modify the type property for pin sets. If a schematic does not have any pin sets, the Pin Sets tab is empty. To have a pin set, the schematic must contain a "virtual pin", where three or more transmission lines directly connect to each other.

Fields

Field	Description
Name	Specifies the virtual pin name. To change the name of the pin set, Right-click a virtual pin and select **Edit Virtual Pin Name**.
Type	• Balanced — The distance between the virtual pin and all pins in the pin set must be equal. • Unbalanced — The distance between the virtual pin and all pins in the pin set may be different.
Pins	Contains the list of pins that comprise the pin set, which are the pins at the other end of the transmission lines. This can contain device pins and other virtual pins.

Related Topics

Exporting a Constraint Template From LineSim

Design Changes Dialog Box

To access: **Export > Reports > Design Change Summary**

Use the Design Changes dialog box to generate a concise report of all the component changes you have made on your board to improve signal quality or lower radiated emissions. A layout

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designer or service bureau uses this record of changes you want made to your board in its next revision. You could also use the list yourself to drive changes in schematics for the board.

Fields

Field	Description
Stackup	Changes such as thickness adjustments that affect impedance.
Changed components	Changes such as modified terminating component values that improve signal quality.
New components	Changes such as new terminators (Quick Terminators) to improve signal quality.

Usage Notes

The HyperLynx File Editor opens and displays the report. The report file is named *<board_file_name>.txt* and is located in the folder that contains the board file.

Related Topics

Creating a Design Change Report

Quick Terminators

Differential Pair Net Names Dialog Box

Scope: BoardSim

To access: From the HyperLynx shell (with no board or schematic design loaded), select **Setup > Differential Pairs**

Use this dialog box to specify differential pair rules for all board designs and save them to *BSW.INI*.

Create a pairing rule by entering characters and wildcards:

- Alphabetic characters are not case sensitive.
- Use the * wildcard to represent any number of characters.
- Use the ? wildcard to represent any single character.
- Use the *? wildcards together to represent one or more characters.
- Use a \ character to find net names that include *, ?, or \ characters. For example, type “net*” to find a net named “net*”.

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Related Topics

Verifying That Differential Pairs are Recognized Correctly

Differential Pairs Dialog Box

Scope: BoardSim

To access: Load a board design, then choose the **Setup > Differential Pairs** menu item

Use this dialog box to create net name pairing rules that the software can use to automatically recognize differential pairs in your board design. You can also use this dialog box to manually specify differential net pairs and differential pin pairs.

Note

The software saves differential pair rules to the *bsw.ini* file, so they are available to all board designs. The software saves manual pairings to the BUD file for a specific board design, unless you check “Rebuild Differential pairs when loading”.

Fields

Field	Description
Differential pair net names: pos <-> neg	Specifies pairing rules. Enter characters and wildcards, then click **Add**. Character and wildcard usage: • Alphabetic characters are not case sensitive. • Use the ? wildcard to represent any single character. • Use the *? wildcards together to represent one or more characters. • Use a \ character to find net names that include *, ?, or \ characters. For example, type “net*” to find a net named “net*”.
Setup diff twin pins	Opens the “Setup diff twin pins” dialog box, where you can manually define a differential pin pair. Text in this dialog box is case-sensitive. **Note:** The software automatically defines differential pin pairs, and pairs pins with the same padstack, close proximity, and other criteria. You typically manually define a differential pin pair to override an incorrect automatic definition.

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Field	Description
Differential pairs	Displays all the differential pairs on the board. Resize the dialog box to see long net names. **Note:** The list does not display differential pairs identified by differential IBIS model assignments or by parallel terminators.
Rebuild All	Clears the current list of differential pairs from the "Differential pairs" list and re-identifies the differential pairs on the board. When you check "Rebuild differential pairs when loading", add or remove pairing rules, you can click Rebuild All to see an updated list of differential pairs.
Rebuild differential pairs when loading	Checked, ignores differential pairing rules for the current board design. Pairing rules are not applied when you load the board again. This setting is stored in the BUD file.

Related Topics

Verifying That Differential Pairs are Recognized Correctly

Digital Oscilloscope Dialog Box

To access: From the toolbar, click 

Note: On the Welcome Screen, enable Oscilloscope or Both waveform viewers to make this dialog box available (**View > Welcome Screen**).

Use this dialog box to interactively simulate signal integrity and display the results as waveforms or eye diagrams. You can simulate the selected net and its associated nets in your board design, or simulate all the nets in the schematic that have an enabled driver.

You can measure simulation results automatically or manually, and you can save simulation results to a file for further analysis in another program or to share with others.

Fields

Field	Description
Comment (for Clipboard and printer)	Enter a description or comment to include when you print, copy, or save waveforms.
Operation Area	
Standard	Displays simulation results as waveforms.
Eye Diagram	Displays simulation results as an eye diagram.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Start Simulation Start Sweeps	Starts simulation or sweeps. When the Sweep Manager Dialog Box - Setup Tab is open, this button changes to Start Sweeps.
Main screen	Displays simulation results in the large screen near the upper-left corner of the dialog box. Hover over a waveform to display the design pin name or sweep condition in a ToolTip.
Stimulus Area	
Global	Assigns a single driver waveform, either Edge or Oscillator, to all driver pins. For a board design, the software applies global stimulus to all driver pins on the selected net and its associated nets. For a schematic design, the software applies global stimulus to all driver pins in the schematic. You can set the default driver waveform type, and default oscillator frequency and duty cycle, that appears when you first open the oscilloscope. See Preferences Dialog Box - Oscilloscope Tab. The oscilloscope saves oscillator frequency and duty cycle data on a per-pin basis when you set these values interactively.
Per-Net/Pin	Assigns different driver waveforms to nets or pins. You define multiple driver waveforms and manually assign them to specific nets (board design) or pins (schematic design). See Assigning a Stimulus.
Assign	Opens the Assign Stimulus Dialog Box to assign a stimulus to specific pins or nets. **Restriction:** This option is available only when Per-Net/Pin is selected.
Edge	Assigns a stimulus to all drivers in the design. Use an edge stimulus to isolate transmission-line effects and study how a transition settles out while eliminating the confusing effects of additional transitions. **Restriction:** These options are available only when Global is selected.

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Field	Description
Oscillator	Assigns an oscillator stimulus to all drivers in the design, using the following properties: • MHz—Frequency of the stimulus. • Duty—Percentage of the period that waveform is high. Use an oscillator stimulus to study the standing-wave effects of repetitive stimulus. **Restriction:** These options are available only when Global is selected.
Eye diagram Area	
Configure	Opens the Configure Eye Diagram Dialog Box - Stimulus Tab to define global stimulus for all enabled drivers on the selected net or schematic. **Restriction:** This option is available only when Eye Diagram and Global are selected.
Eye Mask	Opens the Configure Eye Diagram Dialog Box - Eye Mask Tab. **Restriction:** This option is available only when Eye Diagram and Per-Net/Pin are selected.
Simulator Area	
Settings in this area also update settings in the Simulation Controls Dialog Box. When HyperLynx closes, options in this area revert to Auto and SI/PI co-simulation reverts to disabled.	
Auto HyperSim ADMS HSPICE	Selects one of the following simulation engine options: • Auto—The software automatically selects the simulator. See Automatic SI Simulator Selection. • HyperSim—The software uses only the HyperSim simulator. • ADMS—The software uses only the ADMS simulator. Use this option when you assign unencrypted SPICE models or ADMS-encrypted models to the design. • HSPICE—The software uses only the HSPICE simulator. Use this option when you assign HSPICE-encrypted models to the design. You install and license HSPICE separately from HyperLynx.

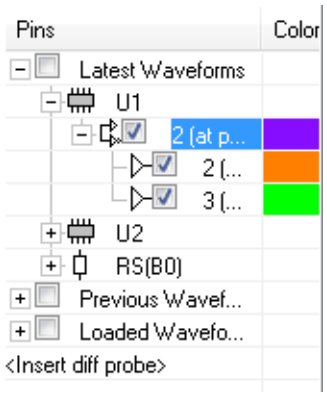
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
SI/PI Co-Sim	Enables highly-accurate signal-via models, which interact with transmission-plane structures in the design. See Accounting for Noise Between Single-Ended Signal Via and Power Planes in SI Simulation (Co-Simulation). Updating this option also updates the value of the SI/PI Co-Simulation option in the Simulation Controls Dialog Box. **Restrictions:** • This option requires the Co-Simulation license. • This option is unavailable when a MultiBoard project is loaded. • This option is unavailable if you enable the ADMS or HSPICE simulator.
Time Resolution	Enables automatic or manual simulation time step size. For information about the software calculates the time step, see Automatic Time Step Calculation for Time-Domain SI Simulation. When you enable the manual option, specify the time step in ps. The value must be a positive and real number from 0.001 to 1000 ps. **Restriction:** This option is unavailable when you enable SI/PI Co-Sim. The power-integrity simulation engine automatically selects the time step and signal-integrity simulation engine.
SPICE Options	Opens the SPICE Options Dialog Box. When you have assigned SPICE models and must specify parameters or include files, use this dialog box to provide them.
IC modeling Area	
Restriction: This area is unavailable when the Sweep Manager Dialog Box - Setup Tab is open.	
Slow-Weak Typical Fast-Strong	Selects the IBIS IC model corner to use during simulation. For information about the combination of min/typ/max data in an IBIS IC model that is used for each IC model corner, see IC Operating Settings. SPICE models use power-supply net voltage settings and do not use this setting. For information about manually editing power-supply voltages, see “Verifying That Power Supply and Signal Nets are Recognized Correctly” on page 62.
Show Area	

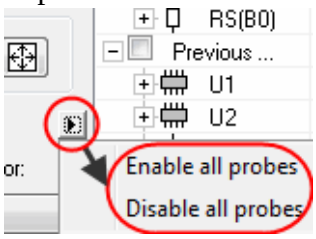
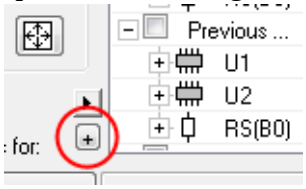
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Overview pane	Displays waveforms for the full simulation, even when you zoom in on waveforms in the Main screen. This capability helps you relate the waveforms displayed in the main screen to the entire waveform. For example, if you were to lose your place in the simulation while zooming in on a specific portion of a waveform in the main screen, the overview pane shows you where this portion is located compared to the entire waveform. For more information, see Overview Pane. **Restriction:** This option is unavailable when Eye Diagram is enabled.
Readout text	Displays the horizontal scale, vertical scale, horizontal delay, and vertical offset value near the top of the Main Screen. Disable the settings readout to reduce clutter.
Eye mask	Displays the eye mask in the Main screen. **Restriction:** This option is unavailable when Standard is enabled.
Probes	Use the “Located” option to probe components with IBIS models at one of the following locations: • Always at the pin • Always at the die • Per each IC model’s setting—When the model contains the [Timing_location] keyword, the oscilloscope assigns the probe to the model-specified location. Otherwise, the oscilloscope assigns the probe to the pin location. For components with SPICE/S-parameter (Touchstone) models, the oscilloscope assigns probes to the external ports in the model. This setting is linked to the Probe locations setting in the FastEye Channel Analyzer - Set Up Channel Characterizations Page. Updating the setting on those pages updates the setting in the oscilloscope, and vice versa.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Pins  	Use the Pins tree to do any of the following: **Note:** The pins tree displays only the available pins based on which nets/components are selected in the viewer. • Enable and disable probes. See Effects of Enabling and Disabling Probes in the Oscilloscope. You can enable or disable all probes at once. See Enable all probes/Disable all probes. The oscilloscope provides a larger version of the Pins tree. See Open probes dial box option. • Edit the waveform color by double-clicking the color square. • Display waveforms for the latest and previous simulations. • Display waveforms that you previously saved to a file and loaded into the oscilloscope. • Manually add differential probes by double-clicking **<Insert diff probe>**. If the reference designator or pin number is truncated, move the mouse pointer over it to display the name in a ToolTip. For differential pin pairs, the Pins tree displays the non-inverted probe to the left of the inverted probe.
Loaded results	Displays waveforms from a file that you manually loaded with Save/Load. This option enables you to compare the saved waveform to the current or previous waveform. **Restriction:** This option is unavailable when Eye Diagram is enabled.
Previous results	Displays waveforms from the previous simulation. The results of each new simulation supersede the previous simulation results. **Restriction:** This option is unavailable when Eye Diagram is enabled.
Latest results	Displays waveforms from the latest simulation. **Restriction:** This option is unavailable when Eye Diagram is enabled.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Visibility	Select one of the following: • **Voltage**—Display voltage waveforms. • **Current**—Display current waveforms. **Restrictions:** • This option is available only for the HyperSim simulator. • Enable Buffer currents prior to simulation to record current waveforms. • The oscilloscope does not display current for nets with a series MOSFET component.
Zoom	Zooms in on the waveforms inside a box that you define. To zoom, position the mouse pointer over one corner of the zoom box you want to create, drag to define the other corner of the zoom box, and then release the mouse button.
Fit to window	Zooms out to fully display the waveforms.
Enable all probes Disable all probes 	Enables or disables all probes at once. See Effects of Enabling and Disabling Probes in the Oscilloscope.
Open Probes dialog box 	Opens the Probes dialog box, which displays the contents of the Pins tree in a much larger form factor. See Effects of Enabling and Disabling Probes in the Oscilloscope.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Thresholds for	Displays IBIS IC model threshold voltages (such as Vinh, Vmeas, and Vdiff) as dashed lines in the Main screen. Displays numeric threshold voltages when you select a specific pin. Main Select any of the following: • All IC Pins—Display all thresholds in the Main screen. • <no pin selected>—Displays no thresholds. • <reference designator.pin>, such as U2.1—Displays the thresholds for the selected pin both in the Main screen and the numeric list located to the right of this option. When the IBIS model provides more than one threshold for the pin, use the numeric list to select the threshold to display.
Cursors Area	
<cursor numeric display>	Displays the cursor location and the location of measurement markers. Also, displays the delta and slope between two measurement markers. Click once in the Main screen to add the first measurement marker. Click again to add the second measurement marker. Click the third time to remove both of the measurement markers.
Track Waveform	Enable to attach a measurement marker to the waveform that you select, and then automatically track the waveform voltage or current as you move the mouse horizontally. Use this capability to make precise waveform measurements. **Restriction:** This option is unavailable when you enable Eye Diagram.
Adjust Mask	Enable to horizontally drag the eye mask to the center of the eye diagram. **Restriction:** This option is available when you enable both Eye Diagram and Eye mask.
Measurements Area	
Use options in this area to automatically measure waveforms.	
Entire	Perform measurements across the entire simulation time.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Region	Perform measurements within a time region that you define by dragging the mouse in the Main screen with the cursor. Use this capability to measure multiple-period waveforms, especially when “warm up” bits are needed to bring the channel, transmitter, or receiver to the normal operating condition. Similarly, when the circuit exhibits a behavior that can distort the measurement, such as a capacitor charging during design power-up, specifying a waveform region enables you to specify a valid measurement window. **Restriction:** This option is unavailable for eye diagrams.
Waveform	Select the waveform to measure. You can measure waveforms from the latest simulation, previous simulation, and waveforms that you manually loaded.
Measurement toolbar	Select an automatic measurement to perform. See Taking Measurements From an Oscilloscope Waveform or Eye Diagram. Some measurements have options that you can specify by selecting the arrow to the right of the main button.  Measurements results are displayed below the measurement toolbar. **Restriction:** Automatic measurements are unavailable for current waveforms and are available for voltage waveforms.
Vertical Area	
Position	Shifts the waveforms and the 0.0 V ground marker in the main screen up or down relative to the grid. By contrast, the vertical scroll bar moves the grids, waveforms, and green ground marker up and down together. This option creates a voltage offset by adding or subtracting voltage to or from the simulation data. **Restriction:** The value range is plus/minus five divisions with a precision of 1/10 division.
Scale	Specifies the vertical scale. Do either of the following: • Select an arrow below the knob. • Drag the knob.
Horizontal Area	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Delay	Moves the specified simulation time to the left edge of the main screen when simulation completes. For example, if you set this value to 25 ns, the oscilloscope aligns the waveforms corresponding to the simulation time of 25 ns to the left edge of the main screen. For Standard operation, this option affects the waveform display only for the next simulation. It has no effect on the latest waveform. For Eye Diagram operation, use this option to center the eye in the oscilloscope. **Restriction:** The value range is 0 ns to 100 ns, with a precision of 1 ps. See Effects on Simulation of Horizontal Scale and Delay Settings.
Scale	Specifies the horizontal scale. Do either of the following: • Select an arrow below the knob. • Drag the knob. See Effects on Simulation of Horizontal Scale and Delay Settings.
Save/Load	Opens the Load/Save Waveforms Dialog Box.
Copy to Clip	Copies the waveforms to the Windows Clipboard in order to paste them into other Windows applications. The image includes information such as the name of the board or schematic, and a time and date stamp.
Erase	Erases loaded, previous, and latest waveforms.
Print	Prints the waveforms.

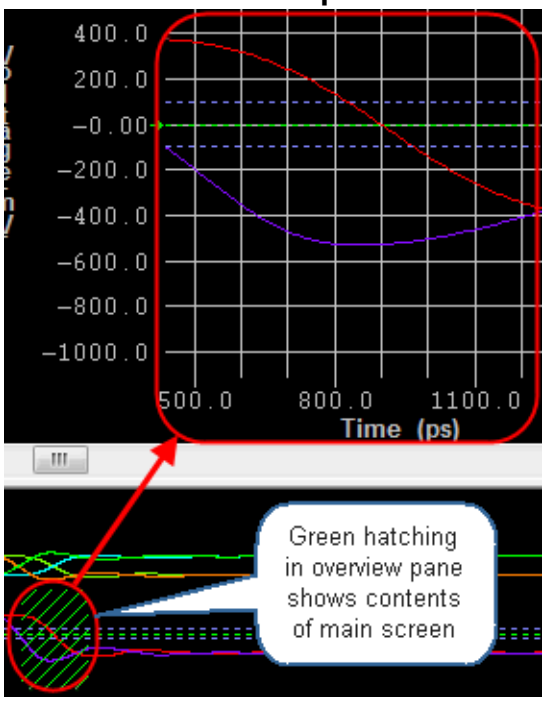
Overview Pane

The Overview pane option opens a pane that is located below the main screen. You can change the relative size of the main screen and overview pane by dragging the window splitter bar that separates them.

The following figure shows that a green hatched pattern in the overview pane identifies the portion of the overall waveforms that are displayed in the main screen.

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Figure 10-10. Oscilloscope Overview Pane



Effects of Enabling and Disabling Probes in the Oscilloscope

The effect of enabling and disabling probes depends on the mode of operation you choose:

- Standard oscilloscope operation—Enable probes to show waveforms and disable probes to hide waveforms. All probes are simulated, whether or not you enable them prior to simulation.
- Eye diagram operation
  - Prior to simulation—Enable probes to simulate them. Disable probes to conserve computer memory.
  - After simulation—Enable probes to show waveforms and disable probes to hide waveforms.

    You cannot use dimmed probe check boxes to show or hide waveforms because simulation data for those probes does not exist. However, you can enable them for the next simulation.

Automatic Time Step Calculation for Time-Domain SI Simulation

Time-domain SI simulation, except IBIS-AMI and FastEye channel analyses, uses information from IC models assigned to the net to calculate the time step:

- IBIS models—Time step = Driver ramp time / 40. The [Ramp] keyword and dV/dt_r and dV/dt_f sub-parameters provide ramp times. When the rise/fall ramp times are not

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symmetrical, simulation uses the shorter time. That is, ramp time is the minimum of dt_r and dt_f.

- SPICE models—Time step = Approximate output switching time / 40. The “Approx. Output Switching Time” option on the IC tab of the Assign Models Dialog Box specifies this value.
- When there is more than one driver on the net, simulation uses the fastest driver.
- When there is an error obtaining information from driver models, the timestep is 10 ps.

FastEye channel analyses use the bit interval to calculate the time step:

- Time step = bit interval / 300. For FastEye channel analysis, the FastEye Channel Analyzer - Define Stimulus Page defines the bit interval.

Effects on Simulation of Horizontal Scale and Delay Settings

- Simulation end or stop time

  The horizontal scale and horizontal delay settings together determine the simulation end time. The oscilloscope instructs the simulator to generate data for ten horizontal time divisions plus the horizontal delay that you specify.

  For example, when the horizontal scale is set to 1 ns/div and the horizontal delay is set to 0 ns, the simulator generates data for 10 ns of simulation time. That is, 1 ns/div times 10 divisions, plus horizontal delay of 0 ns, is 10 ns. If you increase the horizontal scale to 5 ns/div and increase the horizontal delay to 3 ns, the simulator generates data for 53ns of simulation time.

- Memory consumption for simulation

  When you start simulation, the oscilloscope calculates the number of data points the simulator generates. Horizontal scale and horizontal delay settings affect the number of data points. If the computer has insufficient memory to record the simulation results, the oscilloscope displays a warning and does not start simulation.

Related Topics

Running Signal Integrity Simulation with the Oscilloscope Waveform Viewer

Oscilloscope Probes

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Display 3D Board Area Dialog Box

Scope: BoardSim. This dialog box does not support multiple-board projects.

To access: **View > Display in 3D > Area**

Use this dialog box to rotate and view a board design from multiple angles to more easily see net topology and power-distribution network (PDN) implementation. 3-D viewing can be especially useful to see how a signal via passes through metal areas and how a decoupling capacitor connects to a PDN.

Note

Many options in this dialog box are intentionally unavailable, and you cannot enable them. This topic describes the available options.

Fields

Field	Description
Area of Interest	Specifies an outline that encloses an area in the board viewer that you want to view in 3-D. Select a shape and fill out the associated options to define that shape: • Rectangle — Specifies the area of a rectangle. Drag a rectangle over a section in the board viewer. • Polygon — Specifies the area of the polygon. Click and drag each line of the polygon in the board viewer. To close a polygon, click **Close** or drag a line to the starting point of the first line. To remove a line from a polygon, click to the left of the table entry to highlight the entire row, then click **Delete**. See on-screen Help text for additional outline editing capabilities. • Snap To the Grid — Checked, snaps a dragged line or corner to the coordinate resolution you enter. Does not apply to coordinates that you manually type into the Points table.
Included Data	Within an “Area of Interest” outline that you have specified, specifies nets and stackup layers to include when viewing a board area.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Use External Reference **Note:** This option has no effect when you click Export, to view a board area in the 3D PCB Viewer.	Checked, for areas under a BGA, the software adds a plane layer to the 3-D model that represents the reference layer of the BGA die, to create a more accurate representation of the design. Enable this option when a board area includes only a partial reference layer or inconsistent reference throughout the area under the BGA. This ensures that the software creates ports for BGA pins at a consistent location (at the external reference).
Export	Opens a board area in the 3D PCB Viewer.
Advanced Solver	Opens HyperLynx Full-Wave Solver and loads a simulation that represents a board area. You can review and change 3-D area properties, and then mesh and solve to create an S-parameter model. **Note:** Solving a design requires a Full-Wave Solver license.
3D Explorer Variability	Opens HyperLynx 3D Explorer and loads a template to represent a board area. 3D Explorer can help you set up many "what-if" design variations, run a solve on each variation, create an S-parameter model for each variation, and consolidate results into a report. **Note:** Running 3D Explorer and solving a design requires nbgeomexplorer and Full-Wave Solver licenses.

Edit AC Power Pin Model Dialog Box

To access: Open the Assign Power Integrity Models Dialog Box, select the **IC** tab, click **Assign** in the AC Model section.

Use this dialog box to specify the electrical characteristics and stimulus waveform of the current source model assigned to the IC power-supply pin. AC models typically represent I/O buffer switching and IC core-logic power on/off transitions.

Fields

Field	Description
Capacitance	Defines the capacitance of the current source.
Resistance	Defines the resistance of the current source.
Signal Type	Defines the type of signal to assign.

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Field	Description
Amplitude	Defines the maximum current. For double triangle signals, you can specify an amplitude value for each pulse.
Initial Delay	Defines the time from time zero when the current waveform is applied.
Initial Phase	(Available only when Signal Type is Pulse and Shape is Sinusoidal.) Defines the timing offset of the sinusoidal wave.
Rise Time	Defines the time from the start of the rising edge to the end of the rising edge.
Fall Time	Defines the time from the start of the falling edge to the end of the falling edge.
Shape	Defines the waveform for the pulse signal type. • Triangle — Represents large-scale IC structures that switch on and off at the same time. • Double Triangle — Represents large-scale IC structures that switch on and off at two different times (that is, out of phase) and with different amplitudes. • Trapezoid — Represents large-scale IC structures that switch on and off at the same time and with a delay added between the rising and falling edges. The high and low levels are DC loads. • Sinusoidal — Represents how a specific frequency is filtered by decoupling capacitors. • Gaussian — Represents large-scale IC structures, such as I/O buses or large blocks of core logic, that switch on and off at the same time. However, higher frequencies are “rounded off” due to the filtering effect of the package that connects the PCB to the silicon.
Delay 1-2	(Available only when Signal Type is Pulse and Shape is Double Triangle.) Defines the time from the end of the first pulse to the start of the second pulse.
Pulse Time	(Available only when Signal Type is Pulse and Shape is Trapezoid.) Defines the width of the pulse, from the start of the rising edge to the end of the falling edge.

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Field	Description
Max Freq	(Available only when Signal Type is Pulse and Shape is Gaussian.) Defines the maximum frequency of the signal spectrum. To calculate the signal spectrum you use third-party software to apply the Fourier transformation to a function describing the pulse shape.
Period	Checked, repeats the stimulus for the specified time. The plane noise simulation time has precedence over the period length in the AC model. For example, if the AC model contains a repeating current waveform that extends beyond the simulation time, the current waveform is truncated. Use the stop value in the HyperLynx PI PowerScope Dialog Box to set the simulation time.

Related Topics

Assigning VRM Source, DC Sink, and AC Models

Edit DC Power Pin Model Dialog Box

To access: Open the Assign Power Integrity Models Dialog Box, select the **IC** tab, click **Assign** in the DC Sink Model section.

Use this dialog box to specify the electrical characteristics of the current sink model assigned to the IC power-supply pin.

DC models represent static loads, such as IC power-supply pins connected only to non-switching circuitry.

Fields

Field	Description
Apply Current to	Specifies how to distribute the current to each DC model. • Each Sink — Each DC model receives the value specified. • Whole Group — Each DC model receives the value specified divided by the number of selected power-supply pins. For example, (5 A)/(10) = 0.5 A per DC model.
Current	Defines the current source of the DC model.
Resistance	Defines the resistance.

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Related Topics

Assigning VRM Source, DC Sink, and AC Models

Edit Power-Supply Nets Dialog Box

To access: **Setup > Power Supplies**

Use the Edit Power-Supply Nets dialog box to edit power supply net properties.

Spreadsheets

Spreadsheet	Description
Apply to all instances of board	Checked, applies power-supply net settings to all boards with that net. Unchecked, applies power-supply net settings to the board specified by the “Design file” list. (Available when you have a multiple-board design open.)
Select supply nets	Select check boxes to identify each power-supply net. The filter is case-insensitive and supports wildcards (* for any number of characters, ? for any one character). To select or deselect multiple spreadsheet rows, do either of the following: 1. Select a block of rows—Click the first row, press and hold Shift, click the last row, continue to press Shift, and then click the check box in any of the selected rows. 2. Select non-adjacent rows—Click a row, press and hold Ctrl, click other rows, continue to press Ctrl, and then click the check box in any of the selected rows. When you first open the board, BoardSim automatically identifies power-supply nets and selects their check boxes. See “How BoardSim Recognizes Power Supply Nets” on page 548.
Power Network	Opens the Power Network dialog box. Enables you to visualize power net connections in the PDN, and find any obscurely-named nets that connect to the nets that you specified as power-supply nets. Also enables you to assign power integrity models or begin setup for batch DC drop simulation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Spreadsheet	Description
Edit supply voltages	Enter the voltage for each power-supply net. Voltages enclosed by angle brackets < > indicate an automatically assigned value. The filter is case-insensitive and supports wildcards (* for any number of characters, ? for any one character). You can add/remove power-supply nets from this spreadsheet by selecting/clearing check boxes in the Select Supply Nets spreadsheet.
Assign supply nets to plane layers	Select the power-supply net for each plane layer. This option has the effect of flooding the stackup layer with metal and assigning it to the power-supply net you specify. Make this assignment only when the stackup layer really should be flooded with metal. This spreadsheet will be empty for many designs. Usually it is necessary to make this assignment if some portion of the power-supply net does not have an explicit metal area defined in the board file. That is, an entire stackup layer is somehow assumed to be filled with metal and assigned to a power-supply net. The spreadsheet will also be empty when the stackup contains no plane layers. The Supply Net list contains stackup layers that function as plane layers (AC grounds). Use the "Usage" column in the stackup editor spreadsheet to identify plane layers. See "Defining the Basic Stackup" on page 1233.

Usage Notes

Settings in this dialog box are written to the BoardSim session data (*.bud*) file, which is located in the same folder as the HyperLynx executable file (*bsw.exe* - Windows, *bsw* - Linux). For example, *C:\MentorGraphics\<release>\SDD_HOME\hyperlynx*.

Related Topics

Verifying That Power Supply and Signal Nets are Recognized Correctly

Edit Stimulus Dialog Box

To access:

- From the Assign Stimulus Dialog Box, click **Edit Stimulus**.

Use this dialog box to create or edit a stimulus.

Stimulus is saved in an ASCII-formatted *.eds* file. You can use a stimulus file located on the computer or network, such as in a stimulus library for a design project.

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Fields

Fields	Description
Stimulus Name	Select to display the list of *.eds* files contained in the folders specified in the Stimulus File Path(s) area of the Set Directories Dialog Box. Displays the path of the currently-loaded stimulus file. *.eds* files contain wave shape and timing information.
Open 	Opens an existing stimulus file. If you open a stimulus file that is located in a non-default folder, that folder is automatically added to the Stimulus File Path(s) field of the Set Directories Dialog Box.
Save 	Saves the currently-loaded stimulus to a file. Using meaningful stimulus names, such as PRBS_128 or 250MHz_clock, can help you to recognize the stimulus contents when assigning stimulus to specific pins or nets.
Save as 	Saves the currently-loaded stimulus to a new file or location.
Delete 	Permanently deletes the currently-loaded stimulus file from the computer or network location.
Bit pattern Area	
Sequence	Select a type of stimulus. The availability of some options in this dialog box depends on the sequence you select.
Bit	Select the number of bits in the sequence. The number of bits is 2**<bit_order> - 1. **Restriction:** his option is available only for the PRBS stimulus.
Direction	Select **Falling** or **Rising**. **Restriction:** This option is available for only the Edge stimulus.
Initial	Select **High** or **Low**. **Restriction:** This option is unavailable for the USB 2.0 compliance stimulus.
Display and edit area	This area is read-only when you load an existing stimulus file, and is editable when you select the <Custom> Sequence.
Stimulus Area — The options available in this area depend on the Sequence you select.	

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Fields	Description
Bit interval	Length of the unit interval, in ns. Editing this value also updates the Bit Rate value. When choosing between the Bit interval and Bit Rate properties, use the one that provides the best accuracy. For example, to test the channel at 333 Mb/s, you can specify a bit rate of 0.333 Gb/s instead of a bit interval of 3.003003003 ns.
Bit Rate	Number of bits transmitted through the channel, in gigabits per second. Editing this value also updates the Bit interval value.
Sequence reps	Number of times to repeat the stimulus. If you enable jitter, the software applies unique jitter to each sequence repetition.
Period	Period in ns. Editing this value also updates the Frequency value.
Frequency	Frequency in megahertz. Editing this value also updates the Period value.
Duty cycle	Time, in percentage of the period, an oscillating stimulus is high.
Jitter Area	
<include jitter type>	Enable one or more of the following types of jitter: • Gaussian — A normal distribution with no sigma limit. A histogram consisting of a large number of Gaussian-distributed jitter values resembles a bell curve. • Uniform — An even distribution. A histogram consisting of a large number of uniform-distributed jitter values resembles a rectangle. Uniform produces a worst-case distribution more quickly than Gaussian. • Sine — Sinusoidal deterministic jitter represents a non-Gaussian distribution with a finite peak-to-peak amplitude. • DCD (pk-pk) — Duty cycle distortion (DCD) deterministic jitter represents a non-Gaussian and bimodal distribution of signals. **Note:** When using more than one jitter distribution type, you can isolate the contribution of each type by enabling one type at a time and running separate simulations. See “Jitter Distribution Types” on page 553 and “Jitter Applications” on page 561.
<jitter magnitude>	Width of the distribution. See “Units for Gaussian and Uniform Jitter” on page 562.

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Fields	Description
Frequency	For sine jitter, frequency is the rate at which the jitter offset varies. See “Units for Gaussian and Uniform Jitter” on page 562.
Advanced	Opens the Advanced Jitter Settings Dialog Box, to provide additional jitter options.

Related Topics

Assigning a Stimulus

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Edit Transmission Line Dialog Box

Use this dialog box to edit transmission line properties and define coupling regions.s

Scope: LineSim

To access: Right-click a coupled transmission line and choose **Edit Type and Values**.

Topic	Description
Edit Transmission Line Dialog Box - Add/Move to Coupling Region Tab	Use this tab to assign a transmission line to a new or existing coupling region. You can select multiple uncoupled transmission line symbols at the same time and add them to a coupling region.
Edit Transmission Line Dialog Box - Cables Tab	Use this tab to assign properties to a cable transmission line.
Edit Transmission Line Dialog Box - Connectors Tab	Use this tab to assign properties for connectors.
Edit Transmission Line Dialog Box - Edit Coupling Regions Tab	Use this tab to edit geometric properties for a coupling region cross section. On this tab, transmission lines represent PCB trace segments.
Edit Transmission Line Dialog Box - Loss Tab	Use this tab to view the resistance or attenuation frequency range for the transmission line.
Edit Transmission Line Dialog Box - Transmission-Line Type Tab	Use this tab to specify properties for transmission lines used to model PCB trace segments, connectors, cables, wires, and so on.
Edit Transmission Line Dialog Box - Values Tab	Use this tab to define transmission line properties.

Edit Transmission Line Dialog Box - Add/Move to Coupling Region Tab

Scope: LineSim

To access: Right-click a coupled transmission line and choose **Edit Type and Values** to open the Edit Transmission Line dialog box. Select the **Add** or **Move to Coupling Region** tab.

Use this tab to assign a transmission line to a new or existing coupling region. You can select multiple uncoupled transmission line symbols at the same time and add them to a coupling region.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
coupling region	Specifies how two or more conductors, usually PCB traces, are coupled together electromagnetically. Such a region exists when the conductors lie near each other for some length. Coupling regions are inherently two dimensional in that they have a fixed cross section over their entire length.

Related Topics

Creating a Schematic Design

Edit Transmission Line Dialog Box - Cables Tab

Scope: LineSim

To access: Double-click a transmission line to open the Edit Transmission Line dialog box, select the **Transmission-line Type** tab, then choose **Cable** transmission-line type.

Use this tab to assign properties to a cable transmission line.

Fields

Field	Description
Electrical properties	Displays the electrical property values for the industry-standard cable types you select. **Tip:** If a model is not available for the cable used in your design, create a simple transmission line model to represent the cable.
Cable length	Defines the length of the transmission-line.

Related Topics

Creating a Schematic Design

Edit Transmission Line Dialog Box - Connectors Tab

Scope: LineSim

To access: Double-click a transmission line to open the Edit Transmission Line dialog box, select the **Transmission-line Type** tab, then choose **Connectors** transmission-line type.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Use this tab to assign properties for connectors.

Fields

Field	Description
Connector Information	Displays the property and other pertinent file information for the connector you select in the Connectors section.
Connectors	Lists the single line model library files (SLM) each of which represents a single connector.
Pin models	Assigns the pin model to use for the connector.
File Viewer	Displays the source file information in the selected format. The *.SLM* source files contain information on grounding schemes, which help determine impedance and delay for other pins.

Related Topics

Creating a Schematic Design

Edit Transmission Line Dialog Box - Edit Coupling Regions Tab

Scope: LineSim

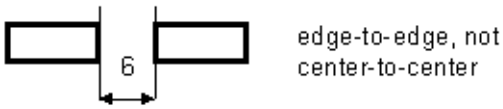
To access: Double-click a coupled transmission line to open the Edit Transmission Line dialog box and select the **Edit Coupling Regions** tab.

Use this tab to edit geometric properties for a coupling region cross section. On this tab, transmission lines represent PCB trace segments.

Fields

Field	Description
Coupling regions	Displays the contents of coupling regions in both tree view and graphical forms. You can select the transmission line in either view to edit. Use arrow buttons in the “Move trace” section to move the selected transmission line left or right of other traces in the same layer, or up or down to another layer. Neighboring transmission lines shift position to accommodate the moved separations. Check “Auto zoom” to view a trace or traces at close range. Uncheck to display the reference conductor, when the coupling region contains one.

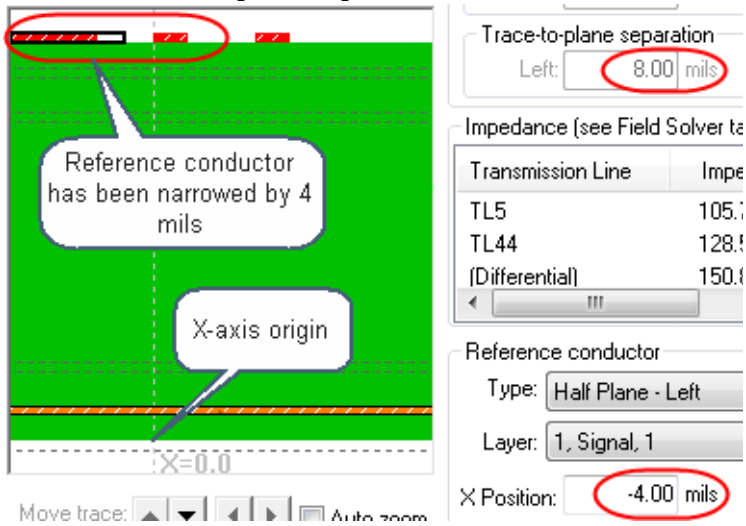
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Coupling region section	
Name	Defines the name of the coupling region.
Length	Defines the length of the coupling region. The length applies to all transmission lines, and their associated traces, in the coupling region.
Stackup (For schematics with multiple stackups)	Defines the name of the stackup.
Edit Stackup	Opens the Stackup Editor, which enables you to verify or edit the stackup for your coupling region.
Transmission line section	
X position	Defines the distance to move all the traces on a layer left or right. Use this when you want to shift the traces on one layer relative to the traces on another layer, such as when you have differential routing with broadside coupling.
Trace width	Defines the width of the selected trace.
Layer	Defines the layer on which to move the selected transmission line. The current stackup determines the list of layers from which to choose.
Trace-to-trace separation	Defines the distance from the edge of the selected trace to the edge of other traces, using separate left and right trace-to-trace separation values.  If the transmission line is located on a stackup plane layer rather than a signal layer, and is bordered on either side by plane copper rather than other traces, LineSim uses the trace-to-plane separation parameters instead. **Note:** You can change the default trace-to-trace separation on the **LineSim** tab of the Preferences dialog box.
Trace-to-plane separation	Defines the distance from the edge of the selected transmission line to the edge of the plane copper. You can specify unique values for left and right trace-to-plane separations. **Note:** You can change the default trace-to-trace separation on the **LineSim** tab of the Preferences dialog box.
Impedance (see Field Solver tab, View button) section	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Table	Displays a summary of the field solver calculations, which includes diagonal impedance values and, for two-trace coupling regions, the differential impedance. Check “Auto calc” to automatically calculate the impedance of the transmission line any time you change a parameter that affects the electrical data. Uncheck “Auto calc” and click **Calculate** to manually update the impedance. This can speed up GUI operation when your coupling is large and complex. **Note:** The field solver actually calculates more data than is visible in this dialog box, including recommended terminations, capacitance/inductance/characteristic-impedance matrices, propagation speeds, and so on. See the “Numerical results” section of the **Field Solver** tab.
Reference conductor section	
Type	Defines a reference conductor: • Solid Plane — Adds solid metal on both sides of a trace or traces, using Trace-to-plane separation values. • Half Plane - Left — Adds solid metal to the left of X position, using Trace-to-plane separation values. • Half Plane - Right — Adds solid metal on the right of X position, using Trace-to-plane separation values. • Ground Trace — Adds a grounded trace, using X position and Width values. • Hatched Plane — Adds hatched metal everywhere on a stackup layer. **Restriction:** You cannot assign Hatched Plane to a layer with traces or another Hatched Plane reference conductor. • Spacer (remove stackup’s plane) — Ignores the plane layer defined in the stackup for this cross section when no other type of reference conductor exists on the layer.
Layer	Defines the stackup layer that contains a reference conductor.

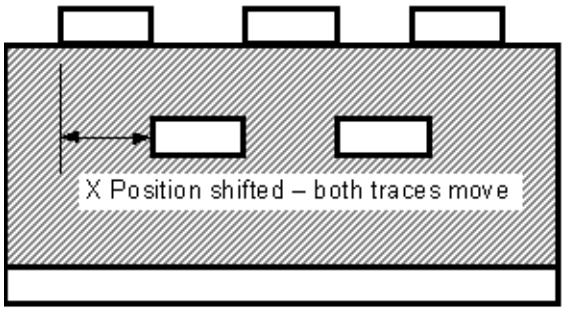
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
X Position (For Half Plane and Ground Trace)	Specifies the location for a reference conductor. The software narrows the reference conductor width as needed to maintain the separation. For example, the figure below shows that the “Half Plane - Left” reference conductor has been narrowed to maintain the “Trace-to-plane separation” of 8 mils.  **Note:** If the reference conductor is too close to the transmission line, the Trace-to-plane separation value has higher priority than the X position value.
Width (For Ground Trace)	Defines the width of a Ground Trace reference conductor.
Add Ref. Conductor	Adds a Solid Plane reference conductor to the selected coupling region.
Delete All Ref. Conductors (For a coupling region with a reference conductor)	Removes all reference conductors from the coupling region.
Delete (For a coupling region with a reference conductor)	Remove a single, selected reference conductor.
Configure Hatched Plane (For a coupling region with a Hatched Plane type of reference conductor)	Opens the Configure Hatched Plane Reference Conductor dialog box, where you can specify properties for a hatched reference plane.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Usage Notes

The figure below shows where the traces on the lower signal layer have been shifted to the right, so that they are aligned in a staggered way relative to the traces on the upper signal layer.



Related Topics

Creating a Schematic Design

Edit Transmission Line Dialog Box - Loss Tab

Scope: LineSim

To access: Select **Enable Lossy Simulation** .Double-click a transmission line to open the Edit Transmission Line dialog box, select the **Loss** tab.

Use this tab to view the resistance or attenuation frequency range for the transmission line.

Fields

Field	Description
Attenuation	Displays signal attenuation information for the transmission line. The blue curve represents the combined resistive and dielectric attenuation.
Surf. Roughness	Select to include the effects of conductor surface roughness in loss calculations.
Per unit length	Unchecked, displays the resistance or attenuation for the full transmission line length. Checked, displays the resistance or attenuation on a per-unit basis. This information can be helpful if the net/trace must meet a per-unit resistance or attenuation design constraint.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Propagation mode (Available only for coupled transmission lines)	Specifies the propagation mode applied by the field solver. If the transmission line couples to one other transmission line, you can select **Differential(+-)** or **Common(++)**. + and - are the voltage polarity of the stimulus applied to coupled transmission lines. For example Differential(+-) indicates that the field solver stimulates coupled transmission lines with opposite polarity signals. If the transmission line couples to two or more other transmission lines, you can select **<mode number><stimulus list>**. The stimulus list values can be +, -, 0 (no signal). For example, if the transmission line couples to two other transmission lines, the Propagation mode list contains 1(+-+), 2(+++), and 3(-+-).
Dielectric loss dominates at (Available when you check both Resistive and Dielectric)	Displays the frequency at which the dielectric attenuation curve crosses the resistive attenuation curve.

Related Topics

Creating a Schematic Design

Edit Transmission Line Dialog Box - Transmission-Line Type Tab

Scope: LineSim

To access: Double-click a transmission line to open the Edit Transmission Line dialog box and select the **Transmission-Line Type** tab.

Use this tab to specify properties for transmission lines used to model PCB trace segments, connectors, cables, wires, and so on.

Fields

Field	Description
Transmission-line type section — Specifies the type of model to use for the transmission line.	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Uncoupled	Defines the model type for a single transmission line. The type you select opens another tab that enables you set the property values for that transmission line: • Simple—Use when you know the characteristic impedance and propagation delay. • Stackup—Use when you know the trace properties and want to link the transmission line to a global stackup. • Microstrip—Use when you know the trace and stackup properties, and the conductor is on an outer-layer trace with a solid AC ground plane to only one side. • Microstrip Over Hatch—Use when you know the trace, stackup, and hatch properties, and the conductor is on an outer-layer trace with a hatched AC ground plane to only one side. • Buried Microstrip—Use when you know the trace and stackup properties, and the conductor is on an inner-layer trace with a solid AC ground plane to only one side. • Stripline—Use when you know the trace and stackup properties, and the conductor is on an inner-layer trace with a solid AC ground plane to both sides. • Stripline Between Hatch—Use when you know the trace, stackup, and hatch properties, and the conductor is on an inner-layer trace with a hatched AC ground plane to both sides. • Wire Over Ground—Use when you know the wire and wire-to-AC ground geometries. The wire must have a circular cross section. • Cable—Use when you implement the transmission line as an industry standard cable. If a model is not available for the cable in your design, create a simple transmission line model to represent the cable. • Connector—Use when you implement the transmission line as an industry standard connector and a first-order model is sufficient.
Coupled	Defines the Stackup model for a transmission line that electrically couples to one or more other transmission lines: • Stackup—Use when you know the trace properties, want to link the transmission line to a global stackup, and know *coupling region* properties. • Dot to Left, Dot to Right—Defines the coupling direction and location of the coupling dot on a transmission line symbol. LineSim considers the dotted ends of the transmission lines in a coupling region to be electromagnetically coupled. See “Coupling Dots” on page 485.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Transmission-line properties (For a Simple transmission line)	Enables you to define electrical properties: • Z0—Defines the characteristic impedance. *Z0 = sqrt(L/C)* • Delay—Defines the propagation delay. *delay = length * sqrt(LC)* • R—Defines the resistance. • Comment—Defines the label to include on the transmission line.
Transmission line to paste	Copies the properties from one transmission line to another.

Related Topics

Creating a Schematic Design

Edit Transmission Line Dialog Box - Values Tab

Scope: LineSim

To access: Double-click a transmission line to open the Edit Transmission Line dialog box, select the **Transmission-line Type** tab, then choose one of the following uncoupled transmission-line types:

- Stackup
- Microstrip
- Microstrip Over Hatch
- Buried Microstrip
- Stripline
- Stripline Between Hatch
- Wire Over Ground

Use this tab to define transmission line properties.

Note

The contents of this tab depend on the type of transmission line that you select on the **Transmission-Line Type** tab. Some fields may be unavailable for the transmission line type you have selected.

Fields

Field	Description
Advanced	Opens the Advanced Impedance Options dialog box, which enables you to define the bulk resistivity and temperature coefficient values.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Conductor thickness - T	Defines the amount of base copper used to make a trace.
Dielectric constant - Er	Defines the dielectric constant of an insulating material.
Dielectric height - H	Defines the height or thickness of the dielectric between a trace or wire and an AC-ground or reference plane.
Dielectric height - H1 Dielectric height - H2	
Hatch Angle - HA	Defines the angle between the centerlines of a signal trace and hatch traces.
Hatch Offset - HO	Defines the distance from the centerline of a signal trace to the intersection of centerlines of hatch traces.
Hatch Pitch - HP	Defines the distance between hatch traces.
Hatch Width - HW	Defines the cross section width of a hatch trace.
Layer	Defines the stackup layer for a conductor.
Length - L	Defines the length of a trace, cable, or wire used to calculate line delay and total L, C, and R.
Loss tangent - Lt	Defines the loss tangent for an insulating material.
Plating thickness - P	Defines the amount of copper plating used over an outer-layer trace.
Radius - R	Defines the radius of a circular wire.
Solder Mask diel. const. - SEr	Defines the dielectric constant of a solder mask material.
Solder Mask height- SH	Defines the height or thickness of a dielectric between a trace and an AC-ground or reference plane.
Width - W	Defines the cross section width of a trace. This value cannot be determined from copper weight because it depends on the results of an etching process.

Related Topics

Creating a Schematic Design

Export Constraint Template Dialog Box

Scope: LineSim

To access, do one of the following:

- **Export > Constraint Template**
- Right-click a driver IC symbol or pin, and choose **Create Constraint Template**

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Use this dialog box to generate a constraint template file based on a selected net, or modify an existing template file.

Constraint templates are a reusable set of constraints that you can apply to similar nets in different designs. Constraint templates contain electrical constraints (such as maximum transmission-line lengths), physical constraints, FromTos (net scheduling), IC model assignments, and so on.

Fields

Field	Description
Edit Template	Click to set up or modify the template before generation using the “Define Constraint Template Dialog Box” on page 832. If you do not edit the template, the software generates a default template.
Template Name	The name of the template that displays in the Constraint Template Editor and Constraint Manager.
Template File	The pathname of the generated template file.
Open generated template in the Constraint Template Editor	Checked, automatically opens the generated template file after clicking **OK**.
Update Constraint Manager with generated template	Checked, automatically updates the Constraint Manager after clicking **OK**. This option is available only when Constraint Manager is currently running and it exported the original schematic. See *Constraint Manager User's Manual*.

Related Topics

Exporting a Constraint Template From LineSim

Export Nets to S-Parameters in Batch Mode Dialog Box

Context: BoardSim

To access: **Select Export > Nets to S-parameter model (Batch Mode)**

Use this dialog box to create S-parameter models from multiple single nets or differentially paired nets. The software creates an S-parameter model for each net or differential pair of nets that you select in the dialog box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Fields	Description
Spreadsheet columns	• On—Checked, exports the single net or pair of differential nets. • Net—Lists the name of a single net or one of the nets in a differential pair. • Associated Nets—Lists the nets that are connected or coupled to each other, based on the coupling threshold setting. • Ports—Lists pins for all associated nets. • File—Lists the name of the S-parameter file that is created.
Filter	Enter a filter string and click **Apply**. The filter is case-insensitive and supports wildcards (* for any number of characters, ? for any one character).
Edit Ports (Available when you select a net from the On column)	Opens the Edit Ports dialog box, to assign port numbers or NC (to reduce the size of the exported model by excluding a pin that you do not need to evaluate) to component pins.
Diff Pairs	Opens the Differential Pairs Dialog Box. If you do not see a differential pair that you want to export, use this dialog box to pair the member nets.
Minimum and Maximum Frequency	Specifies the default sampling simulation frequency range that the software uses when creating the S-parameter model. You can calculate an approximate maximum frequency limit from the signal rising or falling edge time. For example: Maximum frequency = 0.35 / (rise time or fall time)

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields	Description
Sweeping Type	• Adaptive— The number of sampling points varies depending on model characteristics. The number of sampling points is higher near resonant frequencies and other high-rate-of-change events. Adaptive creates accurate model files that are smaller than model files created using the Linear or Logarithmic sweeping type. • Linear— Sampling points are distributed at equal intervals across the frequency range. • Logarithmic— Sampling points are distributed at logarithmic intervals across the frequency range. The intervals between sampling points are smaller at lower frequencies and larger for higher frequencies. With logarithmic sampling, every next frequency point is equal to the previous value times a factor K > 1. This produces a constant increase ratio, but the absolute distance between sampling points grows.
Number of points (Available when Sweeping Type is Linear)	Specifies the number of sampling points. Decrease the number of points to reduce the time it takes to create models. Increase the number of points to increase model accuracy.
Points per decade (Available when Sweeping Type is Logarithmic)	Specifies the number of sampling points per decade. Decrease the number of points to reduce the time it takes to create models. Increase the number of points to increase model accuracy.
Tolerance (Available when Sweeping Type is Adaptive)	Adjusts the tolerance level. Set the tolerance level to High to reduce the time it takes to create models. Set the tolerance level to Low to increase model accuracy.
Type	Specify a pre-defined or custom impedance to which the model parameters are normalized.
Value (Available when Type is Custom)	Specifies the impedance to which the model parameters are normalized.
Save Session	If you want to use dialog box settings again, click **Save Session** to export settings to a batch S-parameter export session file (*.BSE*).
Load Session	If you want to load previously-saved dialog box settings, click **Load Session** to open a batch S-parameter export session file (*.BSE*).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Export to LineSim Free-Form Schematic Dialog Box

To access: **Export > Net To > Free-Form Schematic** (menu item)

Use this dialog box to create a LineSim schematic for selected signal and power supply nets.

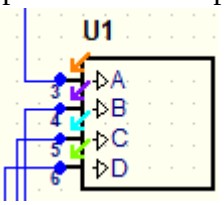
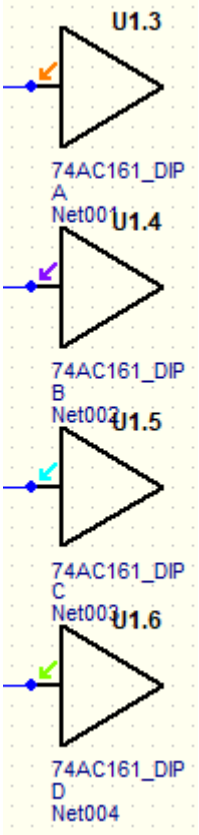
When you have a multiple-board project open and select a net that connects to a net on another board, the exported schematic includes the net and the board-to-board connector.

Power supply nets that you export to a schematic can include power-distribution network elements.

Fields

Field	Description
Output File	Specifies the location of the exported schematic.
Export to Free-Form Schematic Editor	Checked, exports information for selected signal nets.
Expand into EBD/EMD	Checked, includes the electrical contents of EBD/EMD models assigned to pins on the net. (Available when an EBD/EMD model is assigned.)
Export coupled segments	Checked, exports segments and area fills that couple to selected signal nets.
Export 3D Areas	Checked, exports 3D areas. Examples of 3D areas include: • A signal via, its feeding traces, nearby stitching vias, and plane layers or other reference metal areas. • A signal trace that is routed over a gap in the plane layer. For information about creating 3D areas, see “Exporting Part of a Board Design for Analysis in HyperLynx Full-Wave Solver” on page 417.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Use IC component symbol if more than <number> pins	Checked, when selected nets connect to more than the specified number of pins on the same IC component, exports an IC component symbol to represent that component. For example:  Unchecked, exports an IC symbol for each IC pin on selected nets. For example: 
Export to PDN Editor	Checked, exports information for the selected power supply nets. (Available when you have not loaded a multiple-board project.)

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Export Vias as	Displays vias in the exported schematic as one of the following: • Schematic Symbols — Includes physical via properties, such as padstack geometries, to model vias with single or differential via components. Select this option for the best simulation correlation between BoardSim and LineSim. • Electrical Models — Uses via simulation options (set in the Select Method of Simulation Vias dialog box) to model vias as sets of L, C, and transmission-line components. If you enable this option, the software may use several transmission lines and capacitors to represent the via. • Do not export — Excludes vias from the exported schematic. Select this option to isolate the effects of vias on the selected nets by creating two versions of the schematic, one with and one without via models, simulating both schematics, then comparing results. (Available when you uncheck Export to PDN Editor, which always exports vias as schematic symbols.)
Open exported file in LineSim	Checked, opens the exported schematic in LineSim. (Available with a LineSim license.)

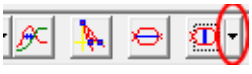
Related Topics

Exporting a Net from BoardSim to LineSim

Eye Height Sampling Dialog Box

To access. Open the Digital Oscilloscope Dialog Box and select **Eye Diagram**. From the measurement toolbar below the waveform display and next to the Eye Height button, select the menu and click **Options**.

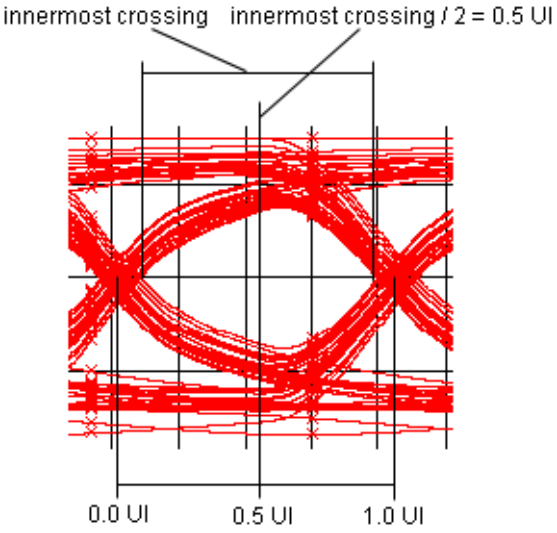
Use this dialog box to specify the time within the unit interval (UI) you want to measure the height of the eye diagram. If you know when the receiver circuitry actually samples the logic state of the waveform, enter that time.



The eye height sampling location is also used to find high and low level voltages used by automatic measurements for eye diagrams.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
	Specifies an offset, in unit interval (UI), from the origin of the UI. Unit interval is the same as bit interval. The software calculates the UI origin by finding the midpoint of the innermost horizontal crossings, identifying that time as 0.5 UI, and then subtracting UI / 2 to find 0.0 UI. 

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FastEye Channel Analyzer

Use the FastEye Channel Analyzer to simulate a SERDES channel to investigate how channel topology, Rx/Tx equalization and pre-emphasis parameters, jitter, and crosstalk, affect channel performance when you do not have IBIS-AMI models that describe transmitters or receivers in your SERDES design.

To access: **SI Simulation > Run FastEye Channel Analysis**

Topic	Description
FastEye Channel Analyzer - Add Jitter Page	Use this page to add jitter to the input stimulus. FastEye channel analysis supports duty cycle distortion, Gaussian, sine, and uniform jitter distributions.
FastEye Channel Analyzer - Add Pre-Emphasis/DFE/ CTLE Page	Use this page to specify transmitter pre-emphasis and receiver equalization.
FastEye Channel Analyzer - Choose Fitting/Convolution Page	Use this page to select the analysis method used by FastEye channel analysis engine.
FastEye Channel Analyzer - Choose New/Saved Analysis Page	Use this page to start an all-new FastEye channel analysis or to load settings saved from a previous FastEye channel analysis.
FastEye Channel Analyzer - Define Statistical Stimulus Page	Use this page to specify the stimulus applied to the channel during FastEye channel analysis, when using the statistical simulation engine.
FastEye Channel Analyzer - Define Stimulus Page	Use this page to specify the stimulus to apply to the channel during FastEye channel analysis, when using the time domain simulation engine.
FastEye Channel Analyzer - Choose Analysis Type Page	Use this page to enable the FastEye or statistical simulation engine, or to only create a worst-case stimulus file.
FastEye Channel Analyzer - Introduction Page	Use this page to familiarize yourself with FastEye Channel Analyzer wizard capabilities, inputs, and types of results.
FastEye Channel Analyzer - Set Up Channel Characterizations Page	Use this page to set up simulation to create a new analog channel characterization, create optional crosstalk files, or load existing characterization or crosstalk files.
FastEye Channel Analyzer - Set Up Crosstalk Analysis Page	Use this page to set up crosstalk options.

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Topic	Description
FastEye Channel Analyzer - View Analysis Results Page	Use this page to choose the types of analysis results to create and to specify the number of samples per bit interval. You can also use this page to view simulation results that are still in memory or load simulation results that you have manually saved to disk.

FastEye Channel Analyzer - Add Jitter Page

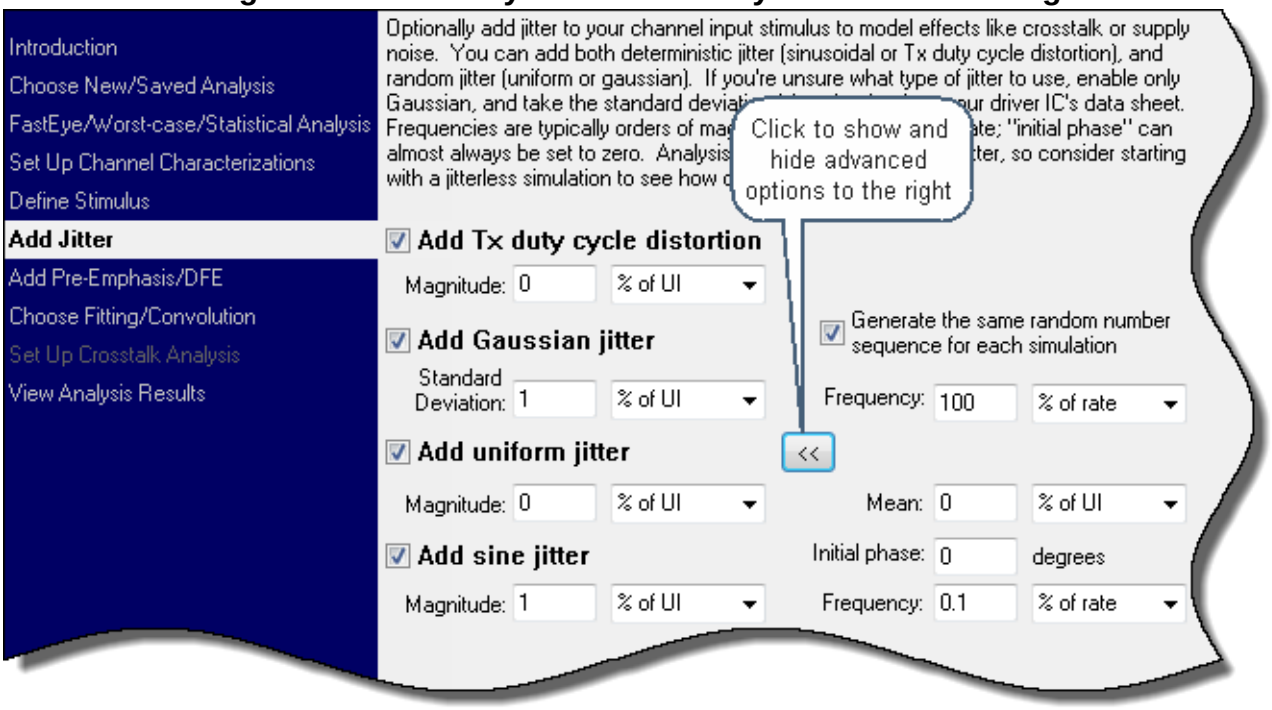
To access: **Simulate SERDES > Run FastEye Channel Analysis**, select the Add Jitter page

Use this page to add jitter to the input stimulus. FastEye channel analysis supports duty cycle distortion, Gaussian, sine, and uniform jitter distributions.

Note

Jitter increases FastEye channel analysis run time. When planning to run very long bit sequences or many stimulus repetitions, you might first run analysis without jitter to see how closed the eye already is.

Figure 10-11. FastEye Channel Analyzer - Add Jitter Page



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Options

Option	Description
Add Tx duty cycle distortion	Apply deterministic Tx duty cycle distortion jitter to the stimulus. See “Duty Cycle Distortion Jitter” on page 555.
• Magnitude—value	The magnitude represents one half the overall width of the distribution.
• Magnitude—units	Specify jitter units as an absolute (such as nanoseconds) or relative value (such as a percentage of the unit interval set for the simulation).
Generate the same random number sequence for each simulation **Restriction:** This option is available only for Gaussian and uniform jitter. Sine and duty cycle distortion jitter are always repeatable.	Enable this option to make the analysis results repeatable. This option may be helpful if you make termination or topology changes and want to use exactly the same jitter to compare results, or if you want to correlate your results with another person.
Add Gaussian jitter	Apply random jitter to the stimulus. See “Gaussian Jitter” on page 556.
• Standard Deviation—magnitude	Specify jitter width (or magnitude) at one standard deviation (that is, one sigma). Increasing the value of sigma increases (on average) the deviation of the timing of waveform transitions away from the ideal switching time. You specify the width of one sigma and FastEye channel analysis derives the width of other sigmas from it. The sigmas are equally spaced from one another. If you want the maximum jitter value (on average) to exceed three sigma, the bit sequence must contain at least 370 bits. This value is based on reference information about the Gaussian distribution and its relationship to the confidence interval. See Table 8-15 on page 558.
• Standard Deviation—units	Specify jitter units as an absolute (for example, in nanoseconds) or relative value (for example, a fraction of the unit interval set for the simulation). See “Units for Gaussian and Uniform Jitter” on page 562.
• Frequency—value and units	Jitter frequency is the rate at which the jitter offset varies from bit to bit. Median frequency is the frequency that divides the jitter spread into two parts of equal area. **Note:** This is an advanced setting. Use the default advanced setting values, unless you have a reason to change them.
Add uniform jitter	Apply uniform jitter to the stimulus, See Uniform Jitter.

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Option	Description
• Magnitude—value	The magnitude represents one half the overall width of the distribution. See Figure 8-38 and Figure 8-39.
• Magnitude—units	Specify jitter units as an absolute (for example, in nanoseconds) or relative value (for example, a fraction of the unit interval set for the simulation). See “Units for Gaussian and Uniform Jitter” on page 562.
• Mean—value and units	The mean represents the center of the possible jitter value range. A non-zero mean value offsets the center of the distribution away from the ideal switching time. **Note:** This is an advanced setting. Use the default values, unless you have a reason to change them.
Add sine jitter	Apply deterministic sinusoidal jitter to the stimulus. See “Sinusoidal Deterministic Jitter” on page 559.
• Magnitude—value and units	The magnitude represents one half the overall width of the distribution. See Figure 8-37.
• Initial phase	Initial phase of the sinusoidal jitter in degrees. You can usually set this value to zero degrees. You might specify a non-zero initial phase value for “short” simulations that are not long enough to contain many periods of slowly-changing jitter. Sinusoidal jitter usually shifts slowly relative to the bit rate. See Figure 8-35 and Figure 8-36. **Note:** This is an advanced setting. Use the default values, unless you have a reason to change them.
• Frequency—value and units	Jitter frequency is the rate at which the jitter offset varies. **Note:** This is an advanced setting. Use the default values, unless you have a reason to change them.

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

FastEye Channel Analyzer - Add Pre-Emphasis/ DFE/CTLE Page

To access: **Simulate SERDES > Run FastEye Channel Analysis**, select the Add Pre-Emphasis/DFE/CTLE page

Use this page to specify transmitter pre-emphasis and receiver equalization.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

You can:

- Add pre-emphasis to the driver.
- Add decision-feedback equalization (DFE) or continuous time linear equalization (CTLE) to the receiver.
- Identify optimum values for pre-emphasis and DFE taps to create the best-possible eye opening.

In many SERDES channels, the IC driver contains circuitry that performs pre-emphasis and the IC receiver contains circuitry that performs equalization (usually DFE or CTLE). These digital-signal processing (DSP) technologies modify transmitted and received waveforms to compensate for channel distortions, such as high-frequency losses.

If the channel driver/receiver implements pre-emphasis/DFE or CTLE, enabling the FastEye channel analysis options on this wizard page can help avoid pessimism by applying the same eye-opening methods during analysis that are used in system operation.

Caution

If you provide external characterization waveforms that include the effects of pre-emphasis or CTLE, leave these filters disabled so that their effects are not duplicated.

Options

Option	Description
Synthesize optimal values	Automatically identify optimal tap weights. Use this option to see tap weight or parameter values that open the eye the most. These values may be used to program the IC driver/receiver values. To view the optimal values as results when analysis completes, check Synthesized filter settings on the View Analysis Results page.
Specify taps/parameters	Manually specify taps and tap weights by enabling this option and selecting **Taps/weights or Specify parameters**. Use this option to see how effective the current settings are.
Tx Area	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Add pre-emphasis	Add pre-emphasis to the step response and pulse response waveforms. Enable this option if the IC model supports pre-emphasis behavior, but pre-emphasis is disabled when producing the waveforms. This condition is true whether the wizard automatically creates the waveforms or you provide external waveforms. Disable this option if any of the following conditions are true: • The IC model does not support pre-emphasis behavior. • The IC model supports pre-emphasis behavior and it was enabled when producing the waveforms. Because the waveforms already contain pre-emphasis, disabling this option avoids adding it again. This condition is true whether the wizard automatically creates the waveforms or you provide external waveforms. • The IC model does support pre-emphasis and you want to learn how much pre-emphasis opens the eye by temporarily disabling it during FastEye channel analysis.
Taps/weights	If you enabled Specify taps/parameters, opens the Specify Pre-Emphasis Dialog Box. If you enabled Synthesize optimal values, opens the Synthesize Pre-Emphasis Dialog Box. To determine optimal tap weights, see Identifying Optimum Tap Weights.
Rx Area	
Add DFE	Add decision-feedback equalization to the receiver. Disable this option if the receiver IC does not support DFE. If the receiver IC does support DFE, you can learn how much DFE opens the eye by temporarily disabling it during FastEye channel analysis. **Restriction:** You cannot add DFE if you selected Only generate worst-case sequence on the FastEye Channel Analyzer - Choose Analysis Type page. **Note:** DFE is not available when you convert a stimulus to PAM-4 encoding on the Define Stimulus or Define Statistical Stimulus page.
Taps/weights	If you enabled Specify taps/weights, opens the Specify DFE Dialog Box. If you enabled Synthesize optimal values, opens the Synthesize DFE Dialog Box. To determine optimal tap weights, see Identifying Optimum Tap Weights.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Add CTLE	Add continuous time-linear equalization to the receiver. Disable this option if the receiver IC does not support CTLE. If the receiver IC does support CTLE, you can learn how much CTLE opens the eye by temporarily disabling it during FastEye channel analysis.
Specify parameters	If you enabled Specify taps/parameters, opens the CTLE Settings Dialog Box. If you enable Synthesize optimal values, the software determines the optimal CTLE parameter values, and this option is unavailable.

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

FastEye Channel Analyzer - View Analysis Results Page

FastEye Channel Analyzer - Choose Analysis Type Page

FastEye Channel Analyzer - Choose Fitting/ Convolution Page

To access: **Simulate SERDES > Run FastEye Channel Analysis**, select the Choose Fitting/ Convolution page

Use this page to select the analysis method used by FastEye channel analysis engine.

Options

Option	Description
Complex-pole fitting	Table 10-2 on page 885 compares the strengths of each analysis method. Additional advantages exist for complex-pole fitting. See "Model Channel Frequency Response with Complex-Pole Models" on page 564.
Convolution	

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Table 10-2. Comparing the Strengths of Complex-Pole Fitting and Convolution

	Wide Spectrum	**Narrow Spectrum or Few Resonance Peaks**
Long Response	Both may be slow For example, a long cable with low loss and strong reflections.	Complex-pole fitting is faster For example, channels with just a few narrow resonances may have very long response times (take a long time for transient residuals to die out).
Short Response	Convolution is faster For example, channels producing short and sharp-edged pulse response, which can be characterized by low loss, but no reflections and high-Q resonances.	Both may be fast

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

FastEye Channel Analyzer - Choose New/Saved Analysis Page

To access: **Simulate SERDES > Run FastEye Channel Analysis**, select the Choose New/Saved Analysis page

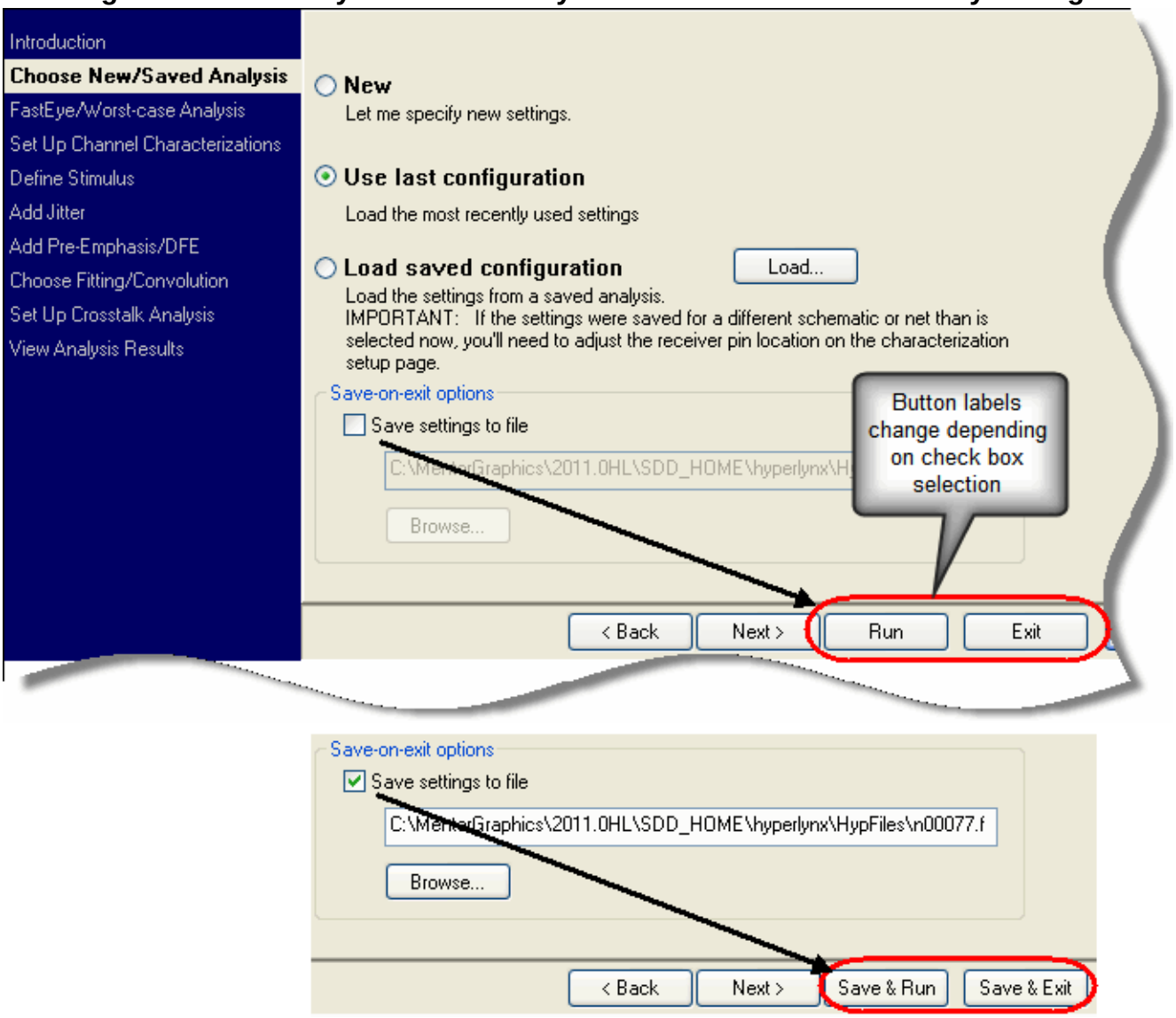
Use this page to start an all-new FastEye channel analysis or to load settings saved from a previous FastEye channel analysis.

The FastEye Channel Analyzer wizard saves its settings to the *.FEW* (FastEye wizard) file, which is located in the design folder unless you specify another location. See Design Folder and HyperLynx Files.

- The SERDES / FastEye / AMI license is required to run FastEye channel analysis.
- The BoardSim PI Only or LineSim PI Only license cannot be checked out when you run IBIS-AMI channel analysis.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-12. FastEye Channel Analyzer - Choose New/Saved Analysis Page



Options

Option	Description
New	Select to start a new analysis using default wizard settings.
Use last configuration	Select to load settings from memory. Wizard settings are stored in memory until you close HyperLynx. Loading settings from memory may be useful when analyzing similar structures, such as different instances of a channel.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Load saved configuration	Select and select **Load** to browse to a wizard settings file (*.FEW*). Loading previously saved settings may useful when analyzing similar structures, such as different instances of a channel, or to reproduce previous analysis results. After loading the saved settings, you can edit and save them to another file.
Load	Select to browse to a wizard settings file (*.FEW*).
Save-on-exit options Area	
Save settings to file	Select and select **Browse** to save wizard settings to a new or existing file. You can also type the file location in the box. Selecting this option causes the button labels **Save & Run** and **Save & Exit** to display near the bottom of the wizard page. Deselecting this option causes the button labels **Run** and **Exit** to display.
Browse	Select to browse to a wizard settings file (*.FEW*).

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

FastEye Channel Analyzer - Define Statistical Stimulus Page

To access: **Simulate SERDES > Run FastEye Channel Analysis**, select Perform statistical analysis on the FastEye Channel Analyzer - Choose Analysis Type page, and then select the Define Statistical Stimulus page

Use this page to specify the stimulus applied to the channel during FastEye channel analysis, when using the statistical simulation engine.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Options

Option	Description
Bit interval Bit rate	Specify the value for the bit interval or rate. When choosing between the Bit interval and Bit rate options, use the one that provides the best accuracy. For example, to test the channel at 333 Mbps, you can specify a bit rate of 0.333 Gbps instead of a bit interval of 3.00300300300 ns. Editing the Bit interval value updates the Bit rate value, and vice versa. The existing values may come from any of the following sources, sorted in descending priority: 1. The Bit interval and Bit rate values in the Channel Characterization Dialog Box. 2. The fitted-poles (*.PLS*) file used to characterize the channel and loaded on the FastEye Channel Analyzer - Set Up Channel Characterizations Page. The *.PLS* file contains a comment that specifies the bit interval. 3. The Bit interval and Bit rate values used for standard eye diagrams, which are specified in the Stimulus tab of the Configure Eye Diagram dialog box.
Bit pattern Area	
Type	Select any of the following types of bit patterns: • PRBS—Pseudorandom binary sequence. **Note:** When you enable PRBS, statistical simulation applies a completely unconstrained bit pattern, where ones and zeros can be applied in any sequence and are not correlated with other bits. By contrast, a PRBS bit pattern with a bit order of seven cannot have eight or more ones or zeroes in a row. • 8B/10B, 64B/66B, 128B/130B—Randomly-generated characters that obey the signaling protocol. Statistical simulation takes the selected bit pattern type into account, although it does not literally apply the pattern bit by bit. **Note:** 8b10b stimulus types are not available when you check “Convert to PAM-4”.
Signaling Type Area	
Convert to PAM-4	Checked, the software converts an NRZ stimulus to PAM-4 encoding by using a 2-bit Gray code. 8b10b stimulus types are not available when you check “Convert to PAM-4”. Statistical analysis assumes that all PAM-4 symbols are uncorrelated and have equal probability.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

FastEye Channel Analyzer - Choose Analysis Type Page

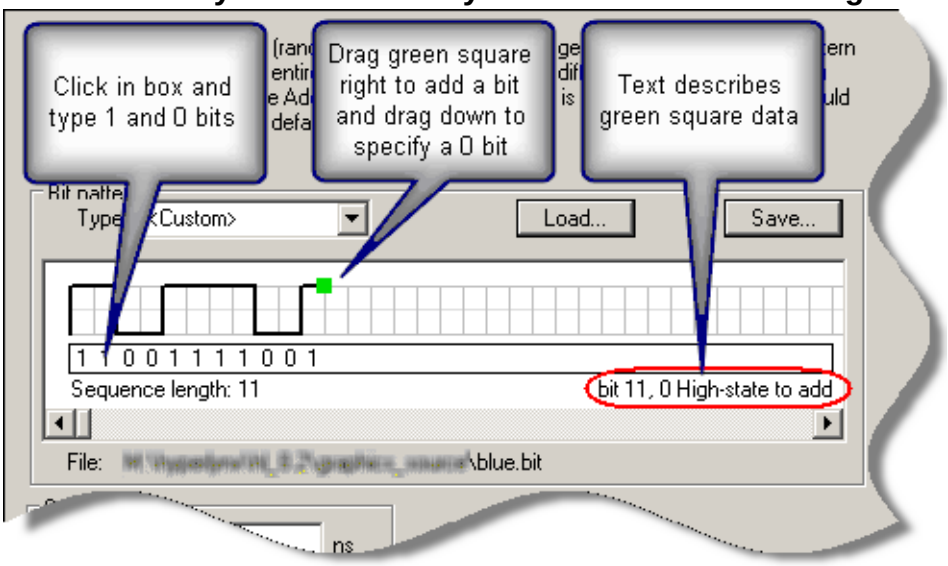
Configure Eye Diagram Dialog Box - Stimulus Tab

FastEye Channel Analyzer - Define Stimulus Page

To access: **Simulate SERDES > Run FastEye Channel Analysis**, select Perform FastEye analysis and optionally generate worst-case sequence on the FastEye Channel Analyzer - Choose Analysis Type page and select the Define Stimulus page

Use this page to specify the stimulus to apply to the channel during FastEye channel analysis, when using the time domain simulation engine.

Figure 10-13. FastEye Channel Analyzer - Define Stimulus Page - Custom



Options

Option	Description
Bit pattern Area	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Type	Select any of the following types of bit patterns: • Worst-case PRBS—The sequence of pseudorandom bits that close the eye the most. • Worst-case 8B/10B—The sequence of characters that close the eye the most. • PRBS—Pseudorandom binary sequence. • 8B/10B, 64B/66B, 128B/130B—Randomly-generated characters that obey the signaling protocol. • Custom—Bit sequences that you define or load from a bit stimulus (.BIT) file. **Note:** 8b10b stimulus types are not available when you check "Convert to PAM-4".
# of 10 bit Characters # of 66, or 130 bit Sequences	Specify how many 10-bit characters or 66/130 bit sequences the bit sequence will contain. **Restriction:** This option is available only for the 8B/10B, 64B/66B, and 128B/130B stimulus types.
Bit Order	Select the bit order to determine the number of bits in the sequence. The number of bits is $2^{bit\ order}$ - 1. For example, if the bit order is 6, the number of bits is 63 (that is, 2^6 - 1). **Restriction:** This option is available only for the PRBS stimulus type.
Checks per UI	Select the number of points per UI for which the worst-case bit sequence is determined. See Checks Per UI. This value cannot exceed the Samples per bit interval value in the FastEye Channel Analyzer - View Analysis Results Page. This check makes sure the overall analysis can show the details found when determining the worst-case bit stimulus. **Restriction:** This option is available only for the Worst-case PRBS and Worst-case 8B/10B stimulus types.
Load	Load an existing bit stimulus file or create a new bit pattern. The file location is displayed near the bottom of the Bit pattern area. **Restriction:** This option is available only for the Custom stimulus type.
Save	Save the bit pattern to a *.BIT* file. The FastEye Channel Analyzer wizard saves the bit pattern file to the design folder unless you specify another location. See Design Folder and HyperLynx Files. **Restriction:** This option is available only for the Custom stimulus type.
Signaling Type Area	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Convert to PAM-4	Checked, the software converts an NRZ stimulus to PAM-4 encoding by using a 2-bit Gray code. 8b10b stimulus types are not available when you check “Convert to PAM-4”.
Stimulus length Area	
Bit interval	Specify the value for the bit interval or rate.
Bit rate	When choosing between the Bit interval and Bit rate options, use the one that provides the best accuracy. For example, to test the channel at 333 Mbps, you can specify a bit rate of 0.333 Gbps instead of a bit interval of 3.00300300300 ns. Editing the Bit Interval value updates the Bit Rate value, and vice versa. The existing values may come from any of the following sources, sorted in descending priority: 1. The Bit interval and Bit rate values in the Channel Characterization Dialog Box. 2. The fitted-poles (*.PLS*) file used to characterize the channel and loaded on the FastEye Channel Analyzer - Set Up Channel Characterizations Page. The *.PLS* file contains a comment that specifies the bit interval. 3. The Bit interval and Bit rate values used for standard eye diagrams and specified in the Stimulus tab of the Configure Eye Diagram dialog box
Pattern repetitions	Specify the number of times to run the pattern during simulation.

Worst-Case Bit Patterns - FastEye

The FastEye Channel Analyzer can create the worst-case bit sequence that closes the eye the most. However, if the buffer is non-linear, the worst-case bit pattern may not close the eye the most in absolute terms.

If the driver or receiver models are not sufficiently linear to produce accurate results in FastEye channel analysis, you can still use the worst-case bit pattern to run standard-eye diagram simulation in the Digital Oscilloscope Dialog Box or Interactive Simulation Dialog Box.

You can also save the worst-case bit pattern to use as stimulus for third-party simulators and analysis software. You may need to reformat the file depending on the requirements of the third-party software.

The FastEye Channel Analyzer can create the following types of worst-case bit patterns:

Pseudo-random bit sequence (PRBS)—Sequence of bits

8B/10B—Sequence of characters that complies with the encoding protocol

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

You can calculate the length of a worst-case bit pattern using this formula:

<checks_per_UI> x ISI x 2

Where:

<checks_per_UI> — the number of locations in the bit interval at which a donor worst-case bit sequence is determined. The FastEye Channel Analyzer calculates the final sequence from the several donor sequences. See “Checks Per UI” on page 892.

ISI — the inter-symbol interference history length

2 — indicates that the overall sequence consists of the worst-case sequence and an inverted version of it

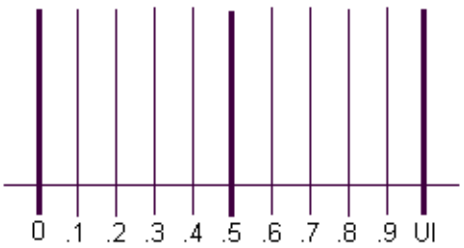
You specify the value of these parameters in the FastEye Channel Analyzer - Define Stimulus Page.

Checks Per UI

If you select worst-case PRBS or worst-case 8B/10B bit patterns, the FastEye Channel Analyzer wizard creates multiple temporary worst-case bit sequences, one for each equally-spaced sampling location in the UI. From this set of temporary bit sequences, the wizard calculates a final worst-case bit sequence that it uses during analysis or saves to a file.

For example, if you specify eleven sampling locations, the wizard creates eleven temporary worst-case bit sequences, one for each equally-spaced sampling location. See Checks Per UI - 11 Sampling Locations. It then creates a cumulative worst-case bit sequence from the eleven temporary bit sequences.

Checks Per UI - 11 Sampling Locations



Use Checks per UI on the FastEye Channel Analyzer - Define Stimulus Page to specify the number of sampling locations. The default value provides a reasonable balance of accuracy and run time. You might decrease the value to reduce analysis run time when repeating the stimulus many times, such as when testing for a bit-error rate (BER) of 1e-12.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

The FastEye Channel Analyzer wizard does not apply jitter when determining worst-case bit sequences.

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

FastEye Channel Analyzer - Choose Analysis Type Page

Digital Oscilloscope Dialog Box

Interactive Simulation Dialog Box

FastEye Channel Analyzer - Choose Analysis Type Page

To access: **Simulate SERDES > Run FastEye Channel Analysis**, select the **Choose Analysis Type** page

Use this page to enable the FastEye or statistical simulation engine, or to only create a worst-case stimulus file.

When you know the channel contains a non-linear element (such as the driver) and suspect that FastEye channel analysis results will be invalid, you may want to create only the worst-case stimulus and use it to simulate the channel in the time domain.

Statistical analysis does not create worst-case bit patterns or an eye diagram. It does create BER plots, bathtub curves, and so on.

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

FastEye Channel Analyzer - Introduction Page

To access: **Simulate SERDES > Run FastEye Channel Analysis**, select the **Introduction** page

Use this page to familiarize yourself with FastEye Channel Analyzer wizard capabilities, inputs, and types of results.

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

FastEye Channel Analyzer - Set Up Channel Characterizations Page

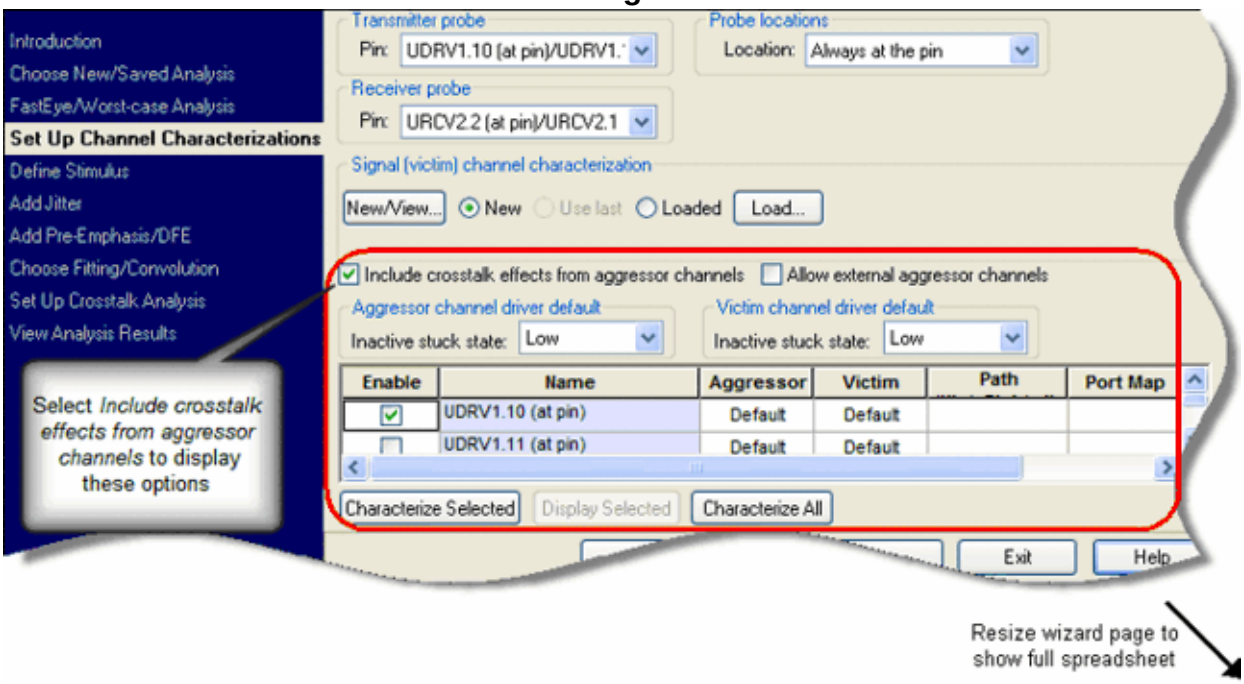
To access: **Simulate SERDES > Run FastEye Channel Analysis**, select the **Set Up Channel Characterizations** page

Use this page to set up simulation to create a new analog channel characterization, create optional crosstalk files, or load existing characterization or crosstalk files.

Note

Wizard settings include the channel-characterization file and probe location for a specific LineSim schematic or BoardSim selected net. If you load settings for a different schematic or selected net, be sure to update the Loaded and probe-related options on this page.

Figure 10-14. FastEye Channel Analyzer - Set Up Channel Characterizations Page



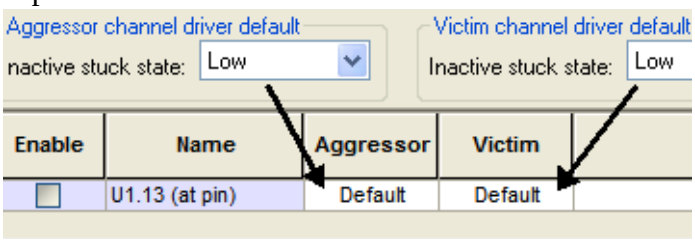
Options

Option	Description
Transmitter probe, Receiver probe Areas	
Pin	Differential or single-ended channel pin to probe. If an expected differential probe does not appear, manually create a differential probe with the oscilloscope (Pins > <Insert diff probe>). You can run FastEye channel analysis for one single-ended or differential channel at a time.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Probe locations Area	
Location	Probe location.
Signal (victim) channel characterization Area	
New/View	Open the Channel Characterization Dialog Box to do either of the following: • Set up simulation properties for a new channel characterization. • View channel characterization waveforms manually created from the Channel Characterization Dialog Box or automatically created when you run analysis to completion.
New	Create a new channel characterization when you run channel analysis. Selecting **New** deletes from memory a previous channel characterization.
Use last	Use a previous channel characterization that is still in memory. You might use this option when using the same channel topology and probe locations with different analysis settings. This option is unavailable if you do any of the following: • Change the net topology, such as editing stackup layer properties or the probe location • Select another net (BoardSim only) • Select **New** • Close HyperLynx Enabling this option makes unavailable nearly all the other options on this page.
Loaded Load	Browse to an existing fitted-poles (*.PLS*) or S-parameter file (*.S2P*, *.S4P*) that contains results from a previous channel characterization. Enabling this option makes unavailable nearly all the other options on this page.
Include crosstalk effects from aggressor channels	Channel characterization includes the crosstalk effects from nearby aggressor channels/nets. If you do not see any aggressors in the spreadsheet, some likely causes are: • None of the aggressors have enabled transmitter/driver pins • In BoardSim, none of the aggressors exceed the coupling threshold • You have not loaded any optional external crosstalk files **Restriction:** The Crosstalk license is required to run crosstalk simulation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Allow external aggressor channels	Optionally, account for crosstalk effects on the victim channel receiver from a channel or net that you have characterized outside of the wizard. Click **Channel +** to add an external crosstalk file.
Aggressor channel driver default Area Victim channel driver default Area	
Inactive stuck state	The default value to use in the spreadsheet. See Aggressor and Victim options. 
Spreadsheet columns	
Note: The spreadsheet contains a row for each aggressor channel/net in the design with the transmitter/driver set to output. Aggressors are characterized one at a time. BoardSim identifies an aggressor channel/net when it exceeds the coupling threshold you set. LineSim identifies an aggressor channel/net when it is part of the same coupling region as the victim channel.	
Enable	Select to either use the channel as an aggressor to or to receive actions from the **Characterize Selected** or **Display Selected** buttons. This option links to the same option on the FastEye Channel Analyzer - Set Up Crosstalk Analysis Page.
Name	Reference designator, pin name, and probe location for an aggressor transmitter/driver IC pin. The Location option determines the probe location.
Aggressor	The transmitter/driver state to apply to the aggressor when it does not switch during victim characterization. When characterizing crosstalk at the victim Rx, aggressors switch one a time. For example, if the design contains aggressors A and B, crosstalk characterization runs twice: • Aggressor A is set to the selected inactive state when aggressor B switches. • Aggressor B is set to the selected inactive state when aggressor A switches. If the value is Default, use the Inactive stuck state option value.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Victim	The transmitter/driver state to apply to the victim during characterization when an aggressor switches. If the value is Default, use the Inactive stuck state option value.
Path	Location of the file representing the crosstalk received at the victim receiver pin and caused by a transmitter/driver switching high or low to create a step response. Select the cell to browse to the file. Running analysis to completion automatically characterizes aggressor channels internal to the design. For external aggressor channels, you can specify SPICE (*.LIS*), fitted-poles (*.PLS*), or Touchstone (.S2P, .S4P) files. To delete the path to an external aggressor channel file, right-click the cell and select **Yes**. For external file requirements, see "External Characterization Files" on page 539. Point to the cell with the mouse to display a ToolTip containing the full file path.
Port Map	Select a port mapping for an S-parameter file. L and R represent ports on the left and right sides of the simulation symbol. For example, L13R24 means that ports 1 and 3 are on the left side and ports 2 and 4 are on the right side. See S-Parameter Port Numbering. **Restriction:** This cell does not display port-mapping options when you load a fitted-poles model or .LIS file in the Path cell because they have known port mapping.
Characterize Selected	Optionally, characterize a single aggressor crosstalk channel. Aggressor channel characterization is done automatically when you run analysis, but you may want to do this manually to investigate the crosstalk contribution of a specific channel. **Restriction:** This button is unavailable unless you enable a spreadsheet row.
Display Selected	Optionally, after you manually characterize an aggressor crosstalk channel, you can display it. **Restriction:** This button is unavailable unless you enable a spreadsheet row.
Characterize All	Optionally, characterize all the aggressor crosstalk channels. Aggressor channel characterization is done automatically when you run analysis, but you may want to do this manually to investigate the crosstalk contribution of one or more specific channels.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
+ Channel	Add a new spreadsheet row to add crosstalk effects on the victim channel receiver from a channel or net that you have characterized outside of the wizard. You might use this capability when running channel analysis from a set of S-parameter files that represent channel behavior measured on PCB hardware. **Restriction:** This option is unavailable if "Allow external aggressor channel" is disabled.
- Channel	Remove the selected spreadsheet row for a channel or net that you have characterized outside of the wizard. **Restriction:** This option is unavailable if "Allow external aggressor channel" is disabled.

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

FastEye Channel Analyzer - Set Up Crosstalk Analysis Page

To access: **Simulate SERDES > Run FastEye Channel Analysis**, select the **Set Up Crosstalk Analysis** page

Use this page to set up crosstalk options.

Restrictions

- This page is unavailable unless you enable "Include crosstalk effects from aggressor channels" on the FastEye Channel Analyzer - Set Up Channel Characterizations Page.
- The Crosstalk license is required to run crosstalk simulation.

Options

Option	Description
Crosstalk timing Area	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Synchronous	Victim and aggressor channels are phase locked. There is a constant phase among transitions for all aggressor and victim channels. This option can take into account the effects of many causes of crosstalk, such as the mutual delays between rising/falling edges in different channels. Even when you select this option, note that the channel interconnect can spread out the arrival of aggressor crosstalk on the victim net.
Asynchronous	Victim and aggressor channels are not phase locked. There is an arbitrary phase of transitions among the aggressor and victim channels. This option can take into account statistically independent sources of crosstalk. Reports only the average crosstalk effect obtained on the final eye diagram along the unit interval.
Analysis type Area	
Time domain	Select when the aggressor transmitter/output switching is non-linear.
Statistical	Select when the aggressor transmitter/output switching is linear.
Default stimulus Area—The Default value in the Stimulus column in the spreadsheet applies values defined in this area.	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Type Parameter	• Random—Random binary sequence. Each bit value is chosen randomly from either 0 or 1, with an equal probability and without any dependence on the value of the preceding bits. The pattern is not periodic and can contain all possible bit combinations. **Restriction:** The Parameter option is unavailable for this bit pattern type. • 8B10B, 64B66B, 128B130B—Non-periodic word pattern that complies with encoding rules. For each state, while considering imparity and running length, the algorithm defines the set of allowed candidates for the next word. From this set and with equal probability, it randomly selects the next word. The algorithm determines the new state and repeats the process. **Restriction:** The Parameter option is unavailable for this bit pattern type. • PRBS—Periodic pseudo-random bit pattern, that is the minimum length sequence (M-sequence) or PRBS of the bit order N (where N = 3 to 31). A linear feedback shift register of the length N generates the bit pattern. One period of PRBSN is 2^N-1. It contains all possible combinations in N bits, except for “N zeros”. The Parameter option is the bit order that determines the number of bits in the sequence. The number of bits is $2^{bit\ order}$ - 1. For example, if the bit order is 6, the number of bits is 63 (that is, 2^6 - 1). Bit Sequence File—Custom bit sequences loaded from a bit stimulus (*.BIT*) file. See Creating a Custom Stimulus. The Parameter option shows the location of the loaded file.
Signaling Type Area	
Convert to PAM-4	Checked, converts an NRZ stimulus to PAM-4 encoding by using a 2-bit Gray code. 8b10b stimulus types are not available when you check “Convert to PAM-4”. The software applies the converted stimulus to both aggressor and victim nets.
Spreadsheet columns	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Notes: • The spreadsheet contains a row for each aggressor channel/net in the design with the transmitter/driver set to output. Aggressors are characterized one at a time. • BoardSim identifies an aggressor channel/net when it exceeds the coupling threshold you set. • LineSim identifies an aggressor channel/net when it is part of the same coupling region as the victim channel. • If spreadsheet cells are not visible, you can either make the dialog box larger or drag the scroll bars.	
Enable	Select to use the channel as an aggressor. This option links to the same option on the FastEye Channel Analyzer - Set Up Channel Characterizations Page.
Name	Reference designator, pin name, and probe location for an aggressor transmitter/driver IC pin. The Location option on the FastEye Channel Analyzer - Set Up Channel Characterizations Page determines the probe location.
Stimulus Parameter	Default applies the stimulus specified in the Default stimulus area. There is no Parameter for the Default stimulus type. You can specify per-channel stimulus. For information about the stimulus types and parameters, see “Type Parameter” in this table. Note that Bit_File in the spreadsheet is the same stimulus type as Bit Sequence File in the “Default stimulus” area.

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

FastEye Channel Analyzer - View Analysis Results Page

To access: **Simulate SERDES > Run FastEye Channel Analysis**, select the **View Analysis Results** page

Use this page to choose the types of analysis results to create and to specify the number of samples per bit interval. You can also use this page to view simulation results that are still in memory or load simulation results that you have manually saved to disk.

To save displayed results to disk, use the save features available from the dialog box that displays the results. For example, save the BER plot from the HyperLynx Eye Density/BER Viewer.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Options

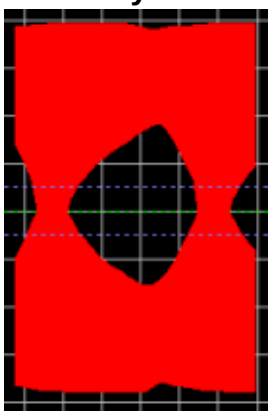
Option	Description
Eye diagram	Display analysis results as an eye diagram. To save eye diagrams, use the FastEye Viewer (detailed waveforms only) or the Windows clipboard (detailed waveforms and contours). **Restriction:** This option is unavailable when you enable Perform statistical analysis in the FastEye Channel Analyzer - Choose Analysis Type page.
All traces	Display detailed eye-diagram waveforms. The wizard automatically overrides this option and displays eye diagram contours if you run extremely long bit sequences, when it becomes impractical to store and display waveform traces for individual bit intervals.
Contours only	Display only filled-in inner and outer perimeters of the eye diagram. For an example, see Figure 10-15 on page 904.
BER plots	Display eye density plots or bit error rate plots in the HyperLynx Eye Density/BER Viewer. BER plots help identify valid data sampling locations by reporting BER as a function of the sampling location across the unit interval (UI, same as bit interval) and voltage. The color of the contour indicates its BER.
Bathtub curves	Display the Bathtub Chart Dialog Box. Use this dialog box to display and document bathtub curves. Bathtub curves help identify valid data sampling locations by reporting the bit error rate (BER) as a function of the sampling location across the unit interval (UI, same as bit interval) at several voltage offsets.
Statistical contours	Display the Statistical Contour Chart Dialog Box. Use this dialog box to display a nested series of eye opening contours and their bit error rate (BER).
Synthesized filter settings	Display the Synthesized DFE Weights Dialog Box, Synthesize Pre-Emphasis Dialog Box, and CTLE Settings Dialog Box when analysis finishes. These dialog boxes allow you to see the synthesized values and save them to files. **Restriction:** This option is available when you enable Synthesize optimal values on the FastEye Channel Analyzer - Add Pre-Emphasis/DFE/CTLE Page.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Save worst-case stimulus to file	Optionally save the bit values that close the eye the most to a file. Use the FastEye Channel Analyzer - Define Stimulus Page to specify whether the sequence is unconstrained or constrained to support 8B/10B protocol requirements. You can use the sequence to simulate the channel in the time domain. **Note:** This option is unavailable when you enable Perform statistical analysis in the FastEye Channel Analyzer - Choose Analysis Type page.
<file_name>	The default file name is of form <*design*>-*worst-case.bit*, where <design> is the file name of the LineSim schematic or BoardSim board. The default file location is the <design> folder. See "Design Folder and HyperLynx Files" on page 520.
Consolidate PAM4 eyes into a single eye	Checked, shows eye diagram simulation results (for a PAM4-encoded stimulus) as a single overlaid eye. Unchecked, shows three eyes (one for each voltage level) on a single diagram.
Samples per bit interval	Specify the number of samples per bit interval to adjust the display resolution of eye diagram contours in the FastEye Viewer and HyperLynx Eye Density/BER Viewer. Type a number from 16 to 32. Large numbers will slow analysis considerably. **Restriction:** This option is unavailable when you enable Perform statistical analysis in the FastEye Channel Analyzer - Choose Analysis Type page.
View	Optionally, re-open an analysis results window. Analysis windows open automatically when analysis completes. If you close an analysis window, you can re-display the results until you close the wizard. There is a View button for each type of analysis output. The bottom two buttons are unavailable until you run analysis to completion.
Load	Optionally, open previously-saved bathtub charts (*.*BTD*) and statistical contour charts (*.*SCD*). The default file location is the <design> folder. See "Design Folder and HyperLynx Files" on page 520.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-15. Example FastEye Channel Analysis Contour



Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

FastEye Viewer

FastEye Channel Analyzer - Choose Analysis Type Page

FastEye Viewer

Use the FastEye Viewer to display and measure FastEye diagrams created by the FastEye Channel Analyzer.

The FastEye Viewer is a special version of the Digital Oscilloscope Dialog Box that contains only the features needed to display and measure FastEye diagrams.

Related Topics

Analyzing a SERDES Channel Using the FastEye Channel Analyzer Wizard

Field Solver Dialog Box

To access:

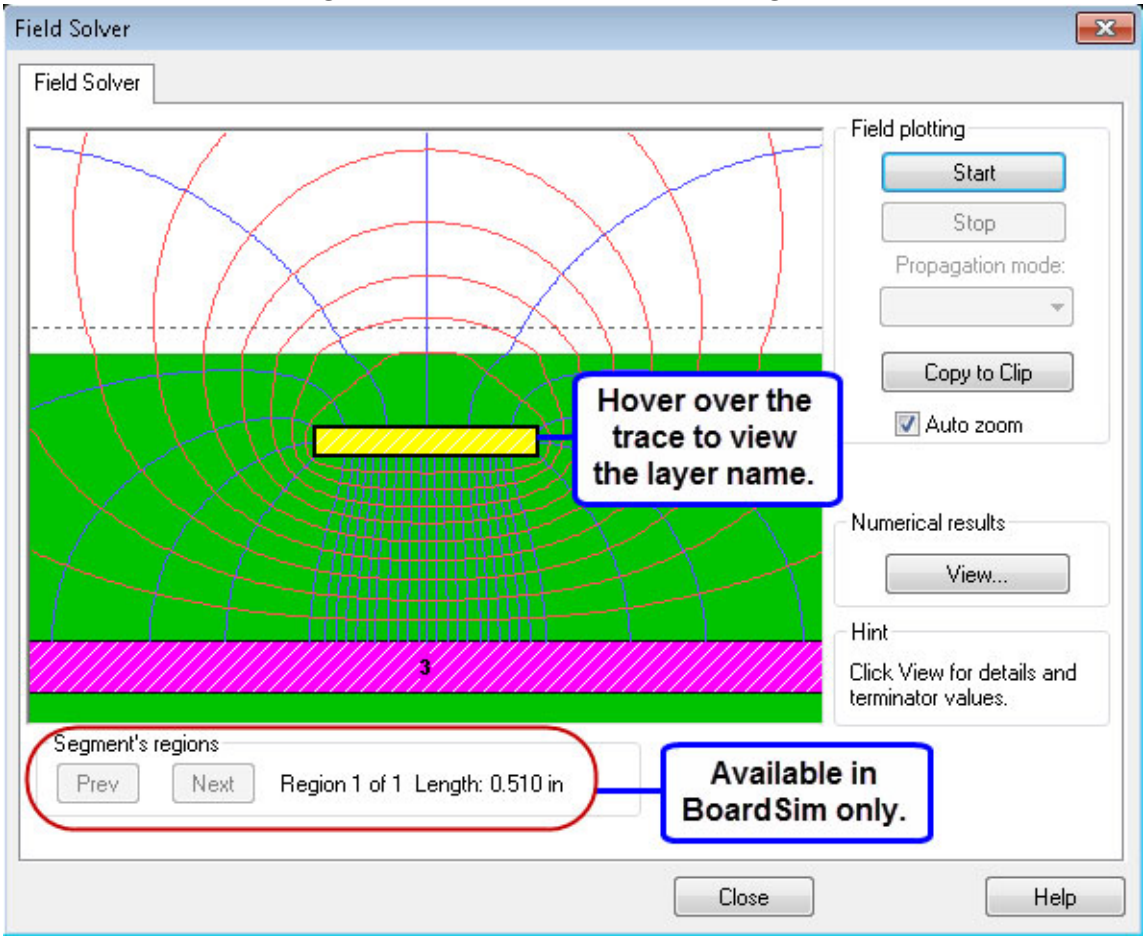
- In LineSim, double-click a coupled transmission line to open the Edit Transmission Line dialog box. Select the **Field Solver** tab.
- In BoardSim, right-click a trace segment, and select **View Field-Solver Output**.

Use this dialog box to run the field solver and view electrical field lines.

When a signal travels along a conductor, it transfers energy in the form of a wave that consists of time-changing electric and magnetic fields. The field solver predicts these fields, given a specific cross section containing conductive traces and various dielectrics.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-16. Field Solver Dialog Box



Fields

Field	Description
Start	Calculates and displays the electric field lines and electric equipotentials.
Stop	Interrupts or stops the field-line plotting.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Propagation mode	Enables you to set the propagation mode depending on the number of other transmission lines coupled to the selected transmission line. If one transmission line — the dropdown list contains **Differential(+-)** and **Common(++)**, where "+" and "-" are the voltage polarity of the stimulus applied to the coupled transmission lines. For example, Differential(+-) indicates the field solver stimulates the coupled transmission lines with opposite polarity signals. If two or more transmission lines — the dropdown list contains **#(<polarity list>)**, where # is the mode number, and "+", "-", and "0" are the signal voltage polarity of the stimulus. For example if there are three coupled transmission lines, the Propagation Mode list may contain 1(+-+), 2(+++), and 3(-+-).
Copy to Clip	Copies the field solver graphic to a clipboard.
Auto Zoom	Checked, zooms to the trace or traces. Unchecked, displays the reference conductor contained in a coupling region.
Numerical results	Displays a detailed report of the field solver calculations when you click **View**.
Segment's regions	(Available in BoardSim only.) If two or more coupling regions exist on the selected net, moves you to previous (Prev) or next (Next) coupling region.
Stackup	(Available in BoardSim only, for board designs that have multiple stackups.) Displays the name of the stackup definition for the trace.

Related Topics

How Field Solver Results Display

Find Component Dialog Box

To access: **View > Find Component**

Use this dialog box to locate a specific component or pin on a board or power distribution network (PDN) layout. After you specify the component or pin, the view changes to show and highlight the component or pin. To remove the highlighting, click in the board viewer.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Option	Description
Filter	Specifies a text string used to filter the contents of the Select Component list. You can specify a combination of reference designator characters and wildcard characters and click **Apply**. For example, to display all IC components with the U reference designator prefix, type u*. Filtering is case insensitive and supports the * wildcard (match any number of characters) and ? wildcard (match any one character).

Free-Form Schematic Editor

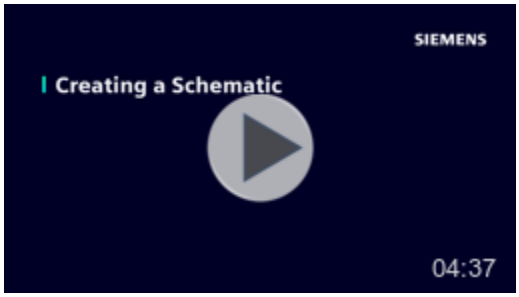
To access: Select **File > Open Schematic** or **File > New Free-Form Schematic**

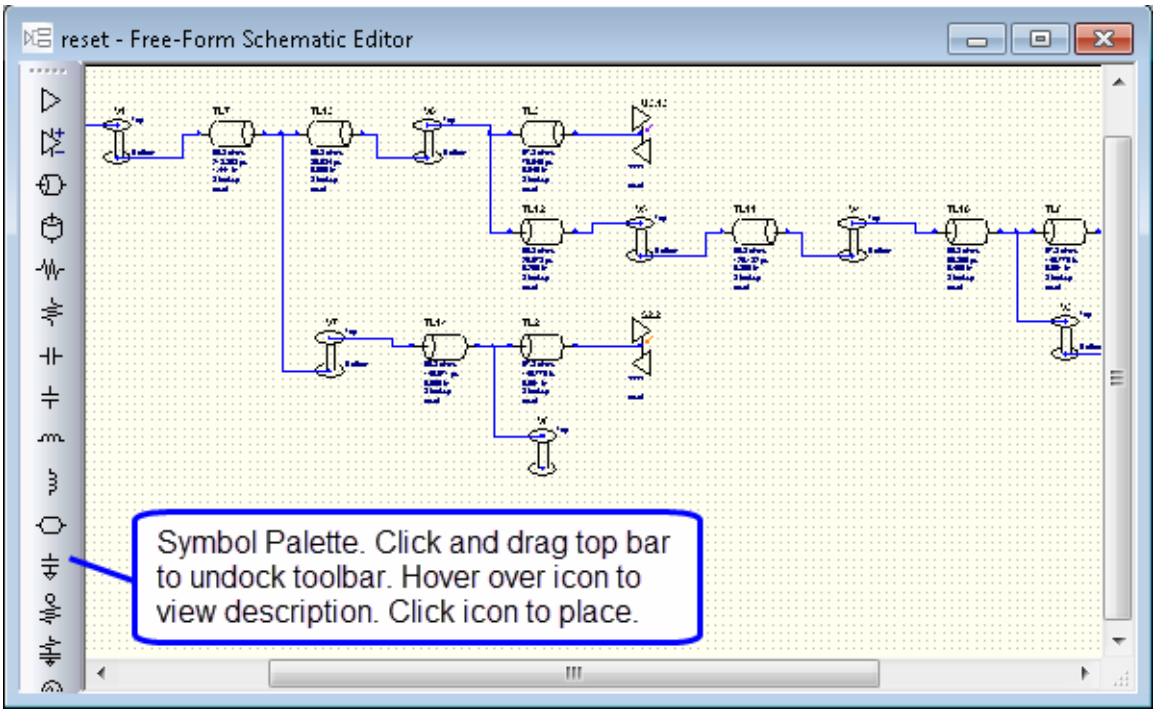
Use this interface to draw or modify a signal-integrity schematic.

You can place symbols and draw wires to capture circuit topology. The editor reads and writes design data files with an *FFS* file extension.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The following video contains additional information:





Manipulating the Free-Form Schematic Editor

If you want to...	Do the following...
Place a symbol	1. Click an icon on the Symbol Palette to select a symbol. 2. Click the workspace to place the symbol. To place the symbol at the nearest X/Y coordinate, select **View > Snap to Grid** from the toolbar
Move a symbol	1. Select a symbol on the workspace. The object turns red and an open square indicates it is selected. 2. Drag the object to a new location and release the mouse.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to...	Do the following...
Connect the symbols	Use on of the following methods to add a net: • Click and drag a line from the pin of one symbol to the pin of another symbol. The netline turns green and adjusts automatically when connected. • Click and drag a selected symbol so a pin touches the pin of another symbol. When the other pin turns red, drag the selected symbol away. The software automatically draws the net.
Connect the symbols by abutment	1. Click and drag a selected symbol so a pin touches the pin of another symbol. 2. When the other pin turns red, drag the selected symbol away. The net appears automatically.
Rearrange the schematic automatically	Choose one of the following to make the schematic more compact or visually appealing: • **Edit > Auto Place > Auto Arrange**. • **Edit > Auto Place > Force Left-To-Right** (favors a horizontal layout).

Related Topics

Creating a Schematic Design

Generate Back-Annotation File/Data Dialog Box

Scope: BoardSim

To access: **Export > ECO Back-Annotation File**

Use this dialog box to save certain types of changes you made to your board design in a *.ECO* file so that you can pass those changes back to your board layout program or schematic editor. For example, if you changed values for several termination components, you can use this dialog box to automatically pass the new values back to your layout program.

Fields

Field	Description
Target program	Specifies the layout program or schematic editor that you want to receive the back-annotation data (contained in a *.ECO* file). You can optionally send this data directly to the program if it is currently running.
Information to back annotate	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Changed values for existing passive components	Checked, includes values that you interactively changed. Back annotation does not include values you specify in *.REF* or *.QPL* files.
New terminators	Checked, includes values of Quick Terminators that you added. When you check this option, the dialog box displays a **Next** button, which opens the Options for New Terminators Dialog Box.
IC-model assignments	Checked, includes IC models assigned by the *.REF* file. Back annotation does not include IC models assigned by *.QPL* files or by interactive selections. Also, back annotation data does not include ferrite bead models.
Power-supply nets and voltages	Checked, includes nets that you added or subtracted from the power supply nets list, as well as changed voltages.

Highlight Net Dialog Box

To access:

- In BoardSim or the LineSim PDN Editor, select **View > Highlight Net**
- From the Viewing Filter Dialog Box, click **Highlight Net**

Use this dialog box to locate nets in the board viewer or PDN Editor. Nets are only highlighted and not selected for simulation.

Options

Option	Description
List of nets	The length and names of signal and power-supply nets in the design. When you select a net in this list, the board viewer dims all the other nets.
Filter	Type the filter value and select **Apply**. Supported wildcard characters include the asterisk (*) (substitute any number of characters) and question mark (?) (substitute one character).
Design file	If a MultiBoard project is open, select the board ID.
Sort nets by	Select an option to sort the list of nets.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Highlight using	Select the color(s) to use for net highlighting. Select one of the following: • Layer colors—Display each net segment with the color associated with the stackup layer on which the net segment is located. If the net is routed on different stackup layers, it will likely be displayed in more than one color. • User color and <color_shape> —Use one color for the entire net, even if it is routed on different stackup layers. Select the color shape to select another color. • In the LineSim PDN Editor, click the color rectangle to select another highlight color.
Include associated nets (BoardSim only)	Highlight associated nets. Most of the time, you probably want to highlight associated nets. If you highlight a net and then unroute it, the associated nets will still be displayed even after unrouting is complete. Clear the check box to avoid highlighting associated nets for an unrouted net.
Highlight	If you select a net in the list, selecting **Highlight** applies the properties in the Highlight using area to it.
Remove Highlight	If you select a net in the list, selecting **Remove Highlight** unhighlights it.
Remove All	Select **Remove All** to unhighlight all nets.

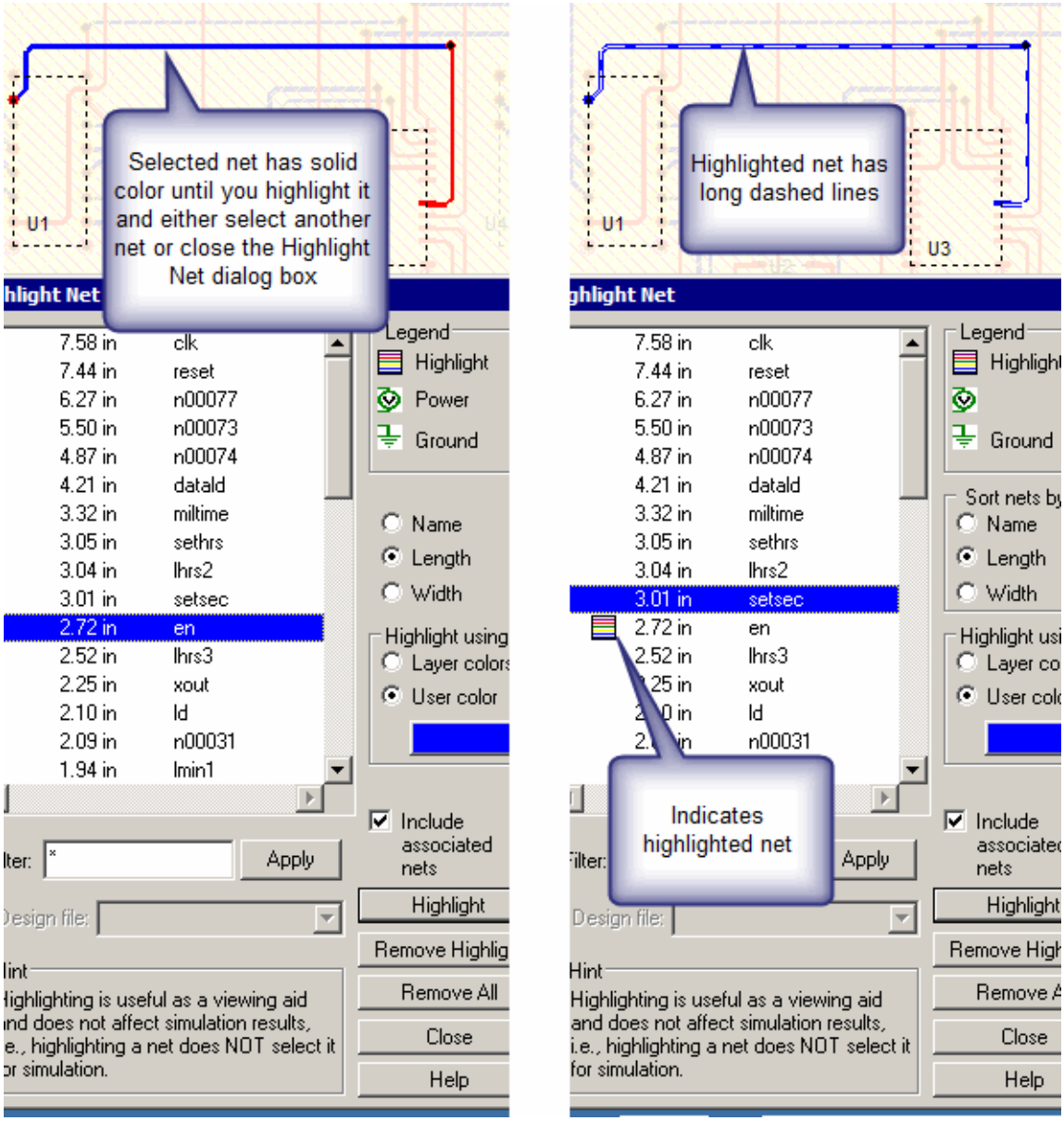
Appearance of Highlighted Nets

The board viewer displays highlighted nets as long dashed lines in colors that you choose. By contrast, selected nets appear as solid lines in colors defined in the stackup.

See Table 10-17.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-17. Appearance of Highlighted Nets



BoardSim Crosstalk displays crosstalk aggressor nets as short dashed lines in colors defined in the stackup. You can distinguish crosstalk aggressor nets from highlighted nets by the length of the dashed lines:

- Highlighted nets have long dashes.
- Crosstalk aggressor nets have short dashes.

If a net is both highlighted for viewing and selected for analysis, the board viewer displays the net as a selected net.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If a net is both highlighted for viewing and is an aggressor net, the board viewer displays the net as an aggressor net.

Related Topics

Viewing a Board in 2-D

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

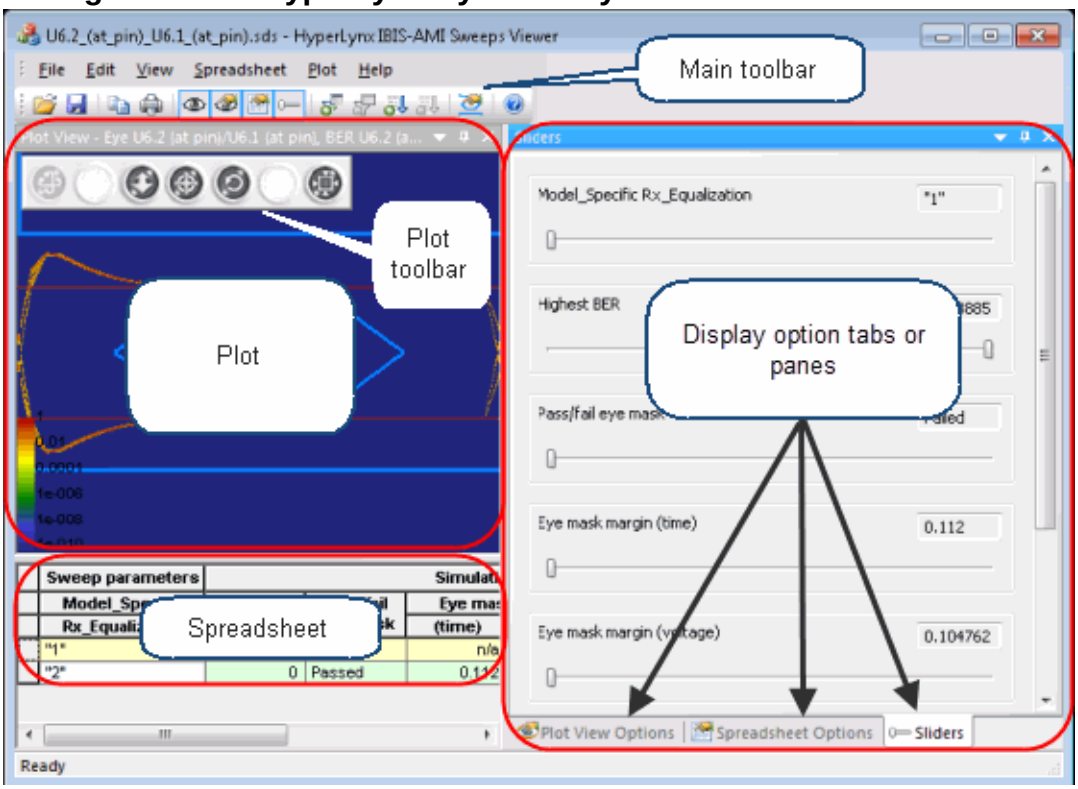
HyperLynx Eye Density/BER Viewer

Use this dialog box to display eye density and BER simulation results.

To access:

- Automatically opens after running IBIS-AMI sweep simulation.
- **Simulate SERDES > Open Eye Density/BER Viewer** (menu item)

Figure 10-18. HyperLynx Eye Density/BER Viewer GUI Overview



You can move, hide, and detach GUI objects, using the methods described in “Pane Organization in the HyperLynx Eye Density/BER Viewer” on page 915. For information on simulating a SERDES Channel, see “Analyzing Multiple SERDES Channels Using the SERDES Wizard - IBIS AMI” on page 225.

Topic	Description
Pane Organization in the HyperLynx Eye Density/BER Viewer	You can hide, automatically hide, and detach the panes in the HyperLynx Eye Density/BER Viewer.
HyperLynx Eye Density/BER Viewer - Menus	The Eye Density/BER Viewer contains several menus to help you view your simulation results.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
HyperLynx Eye Density/ BER Viewer - Main Toolbar	The following table describes the main toolbar buttons.
HyperLynx Eye Density/ BER Viewer - Plot View Pane	Use this pane to display IBIS-AMI simulation results.
HyperLynx Eye Density/ BER Viewer - Spreadsheet	Use the spreadsheet to display the combination of model parameter values, numerical simulation results, and pass/fail results.
HyperLynx Eye Density/ BER Viewer - Plot View Options Pane	Use this pane to set display options for the Plot View Pane.
HyperLynx Eye Density/ BER Viewer - Spreadsheet Options Pane	Use this pane to edit display options for the spreadsheet.
HyperLynx Eye Density/ BER Viewer - Sliders Pane	Use a slider on this pane to quickly scan through a large number of simulations to see results for a specific slider category. The slider automatically selects the corresponding spreadsheet row and displays simulation results.

Pane Organization in the HyperLynx Eye Density/ BER Viewer

You can hide, automatically hide, and detach the panes in the HyperLynx Eye Density/BER Viewer.

If you want to...	Do the following...
Manually Hide or Show a Pane	From the pane title bar, either: • Select > **Hide**. • **Right-click > Hide**.
Automatically Hide or Show Panes	• Select vertical pushpin to enable auto hide. • Select horizontal pushpin to disable auto hide. • Select > **Auto Hide** to toggle auto hide. **Right-click > Auto Hide** to toggle auto hide.
Detach a Pane	Double-click the title bar. **Right-click > Floating**.

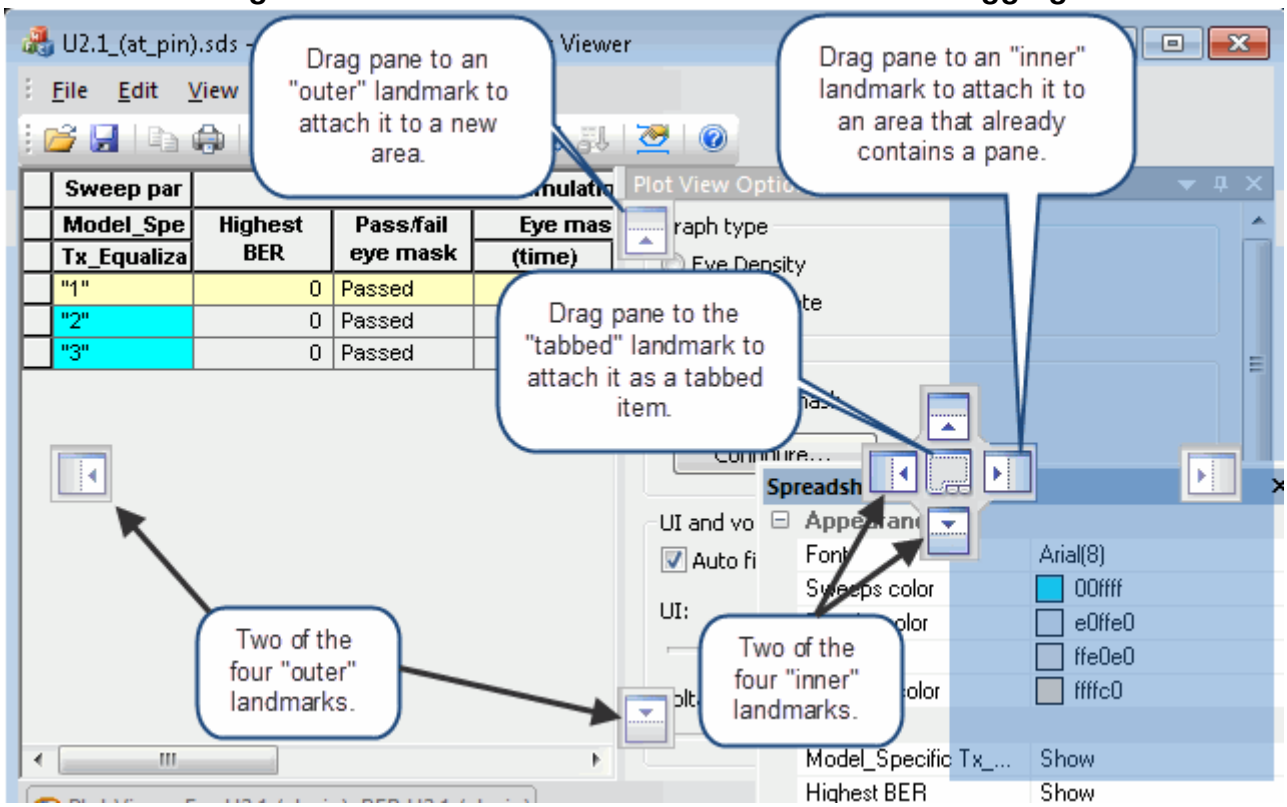
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to...	Do the following...
Attach a Pane	From the pane title bar, do any of the following: Double-click the title bar. **Right-click > Floating**. To attach the pane as a tab, drag the pane to an attached pane, and then drag it to a tabbed landmark. To attach the pane to the overall viewer dialog box, drag it to an outer landmark. To attach the pane to another pane, drag it to an attached pane, and then drag it to an inner landmark. See the following figures:

Landmark Locations

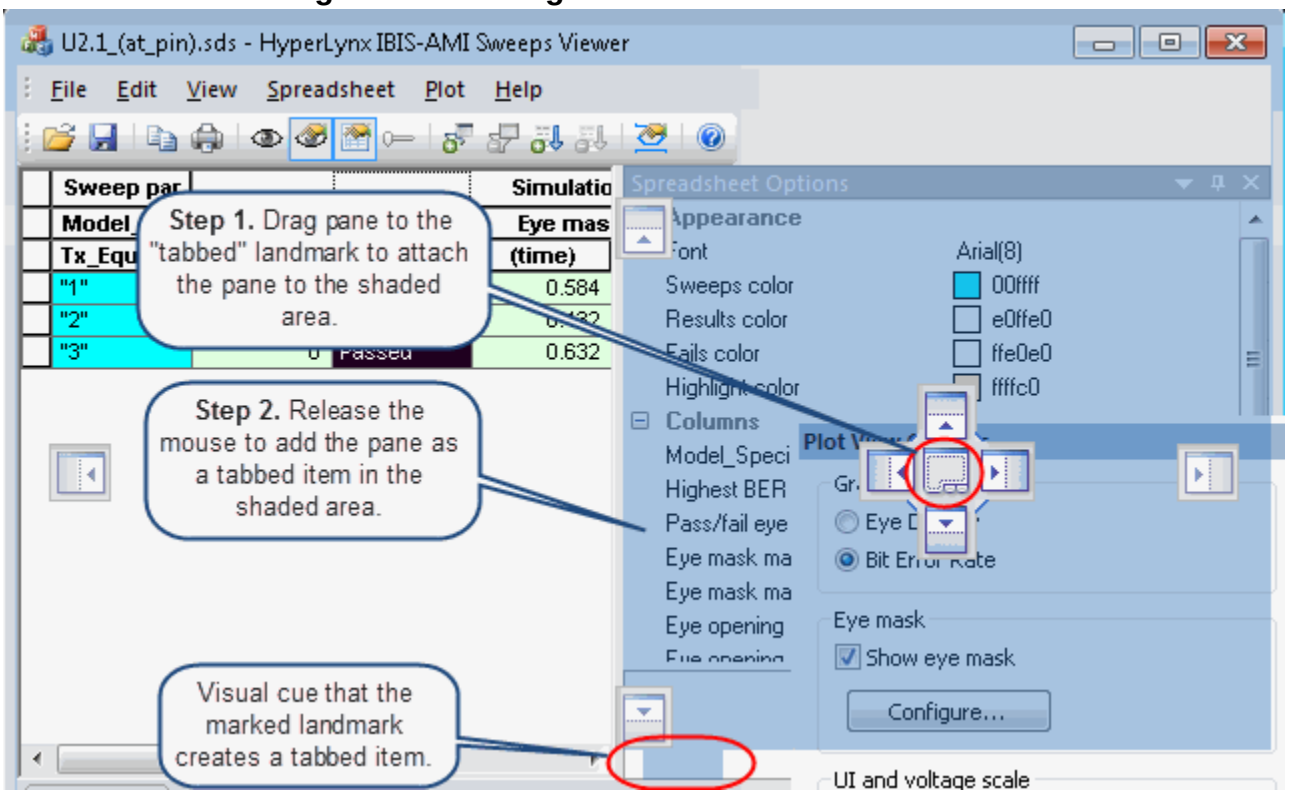
The following figures show the landmark locations:

Figure 10-19. Attachment Landmarks for Pane Dragging



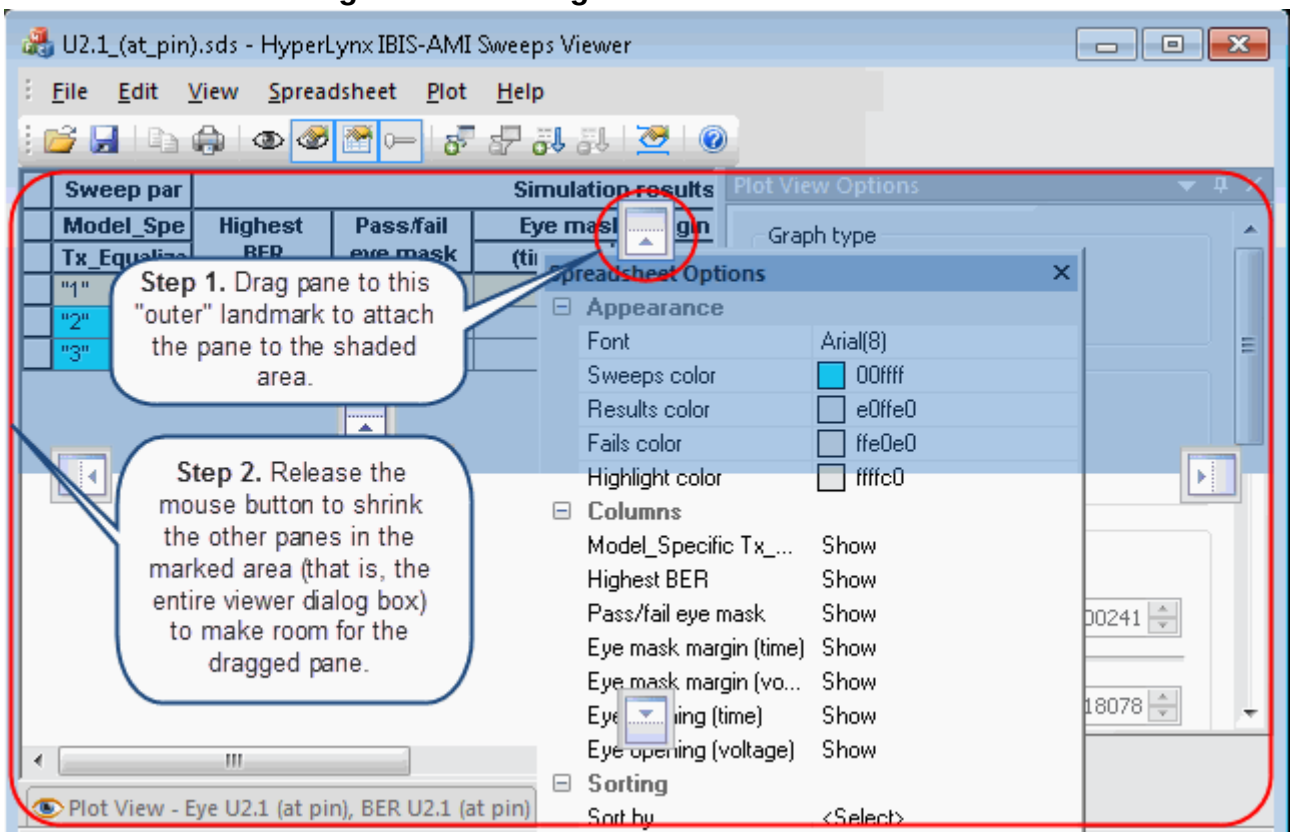
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-20. Drag Pane to Make a Tabbed Item



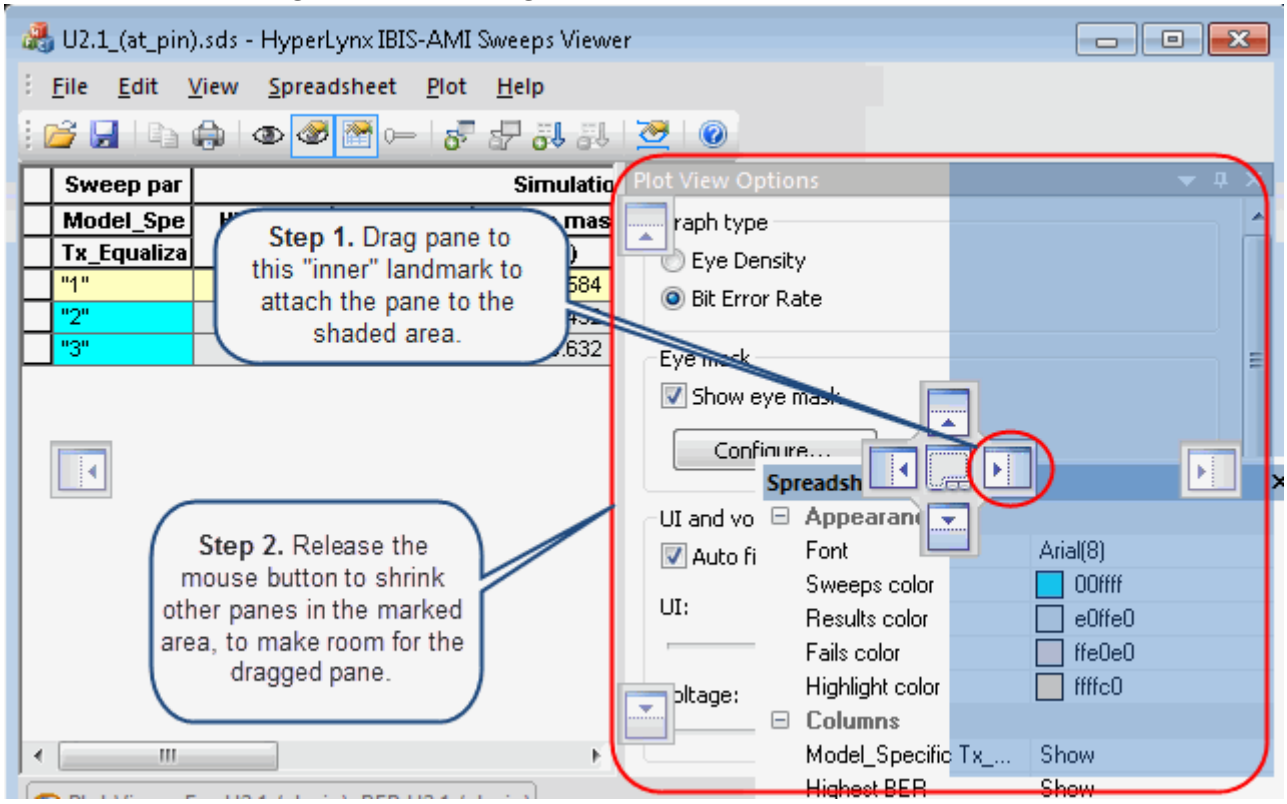
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-21. Drag Pane to Make a New Area



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-22. Drag Pane to Attach to Another Pane



HyperLynx Eye Density/BER Viewer - Menus

The Eye Density/BER Viewer contains several menus to help you view your simulation results.

File Menu

Item	Description
Open	Opens a simulation data storage (*.sds*) file previously written by running a IBIS-AMI sweep simulation or FastEye channel analysis. For information about the location and contents of *.sds* files, see the Results section of "Analyzing Multiple SERDES Channels Using the SERDES Wizard - IBIS AMI" on page 225.
Save	Saves the currently-loaded simulation data storage (*.sds*) file. For example, if you edited the eye mask in such a way that made an eye change from a fail to a pass, the updated spreadsheet pass value is saved to the *.sds* file.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Item	Description
Save As	Saves currently-loaded simulation data storage (*.sds*) file to a new file. If you save the file to a new folder, the BER and eye density plots (*.tps*) files associated with the *.sds* file are also copied to the new folder.
Print	Prints the spreadsheet.
Print Preview	Opens print preview.
Print Setup	Opens the Print Setup dialog box.
<numbered_list>	Displays a numbered list of previously-opened *.sds* files. Select a file to open it.
Exit	Closes the viewer.

Edit Menu

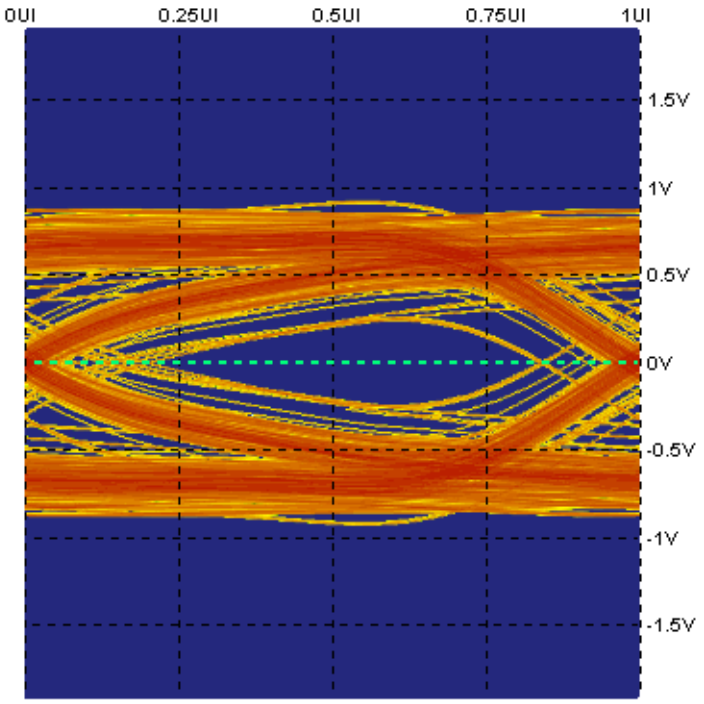
Item	Description
Copy	Copies selected spreadsheet cells, so you can paste the text to another application, such as Microsoft Notepad.
Configure Eye Mask	Opens the Configure Eye Mask dialog box.

Spreadsheet Menu

Item	Description
Add Filter	Opens the Add Filter dialog box, where you specify the filter conditions for hiding spreadsheet rows.
Remove All Filters	Permanently deletes all filters and displays all the spreadsheet rows.
Add Sorting	Opens the Add Sorting dialog box, where you specify the conditions for sorting spreadsheet rows.
Remove All Sorting	Permanently deletes all sorting.
Columns > <column_name>	Select to display the named spreadsheet column. This option also controls which sliders the Sliders pane displays.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Plot Menu

Item	Description
Eye Density	Formats the results as an eye density plot. For an example plot, see Figure 10-23 on page 923.
Bit Error Rate	Formats the results as a bit error rate plot. For an example plot, see Figure 10-24 on page 924.
Show Eye Mask	Overlays an eye mask over the plot.
Show Grid	Overlays the plot with a grid, where UI is on the X axis and voltage is on the Y axis. 
Auto fit to window	Scales the simulation results to fit in the window.
Auto Range	Adjusts the zoom scale and data range to fit the results to the screen.
Show Navigation Bar	Displays a toolbar in the plot display area. This toolbar has buttons to help you change the display of a plot.
Copy To Clipboard	Copies a plot to the clipboard.

Related Topics

Pane Organization in the HyperLynx Eye Density/BER Viewer

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HyperLynx Eye Density/BER Viewer - Main Toolbar

The following table describes the main toolbar buttons.

Toolbar Contents

Button	Description
Open an existing document 	Opens a simulation data storage (*.SDS*) file previously written by IBIS-AMI sweep simulation. For information about the location and contents of *.SDS* files, see the Results section of "Analyzing Multiple SERDES Channels Using the SERDES Wizard - IBIS AMI" on page 225.
Save the existing document 	Saves the currently-loaded simulation data storage (*.SDS*) file. For example, if you edited the eye mask in such a way that made an eye change from a fail to a pass, the updated spreadsheet pass value is saved to the *.SDS* file.
Copy the selection and put it on the clipboard 	Copies selected spreadsheet cells, so you can paste the text to another application, such as Microsoft Notepad.
Print the active document 	Prints the spreadsheet.
Show or hide Plot View pane	Select to display the HyperLynx Eye Density/BER Viewer - Plot View Pane.
Show or hide Plot Options pane 	Select to display the HyperLynx Eye Density/BER Viewer - Plot View Options Pane.
Show or hide Spreadsheet Options pane 	Select to display the HyperLynx Eye Density/BER Viewer - Spreadsheet Options Pane.
Show or hide Sliders pane 	Select to display the HyperLynx Eye Density/BER Viewer - Sliders Pane.
Add new data filter 	Opens the Add Filter dialog box, where you specify the filter conditions for hiding spreadsheet rows.
Remove all current filters 	Permanently deletes all filters and displays all the spreadsheet rows.
Add data sorting 	Opens the Add Sorting dialog box, where you specify the conditions for sorting spreadsheet rows.
Remove all current sorting 	Permanently deletes all sorting and displays spreadsheet rows in the original order.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Button	Description
Edit eye mask parameters	Opens the Configure Eye Diagram dialog box to edit eye mask properties for eye diagram analysis. You can load existing eye masks from a library or save new eye masks into a library.
Copy to clip	Copy plot to the clipboard.
Display help	Opens Help for this dialog box.

Related Topics

Pane Organization in the HyperLynx Eye Density/BER Viewer

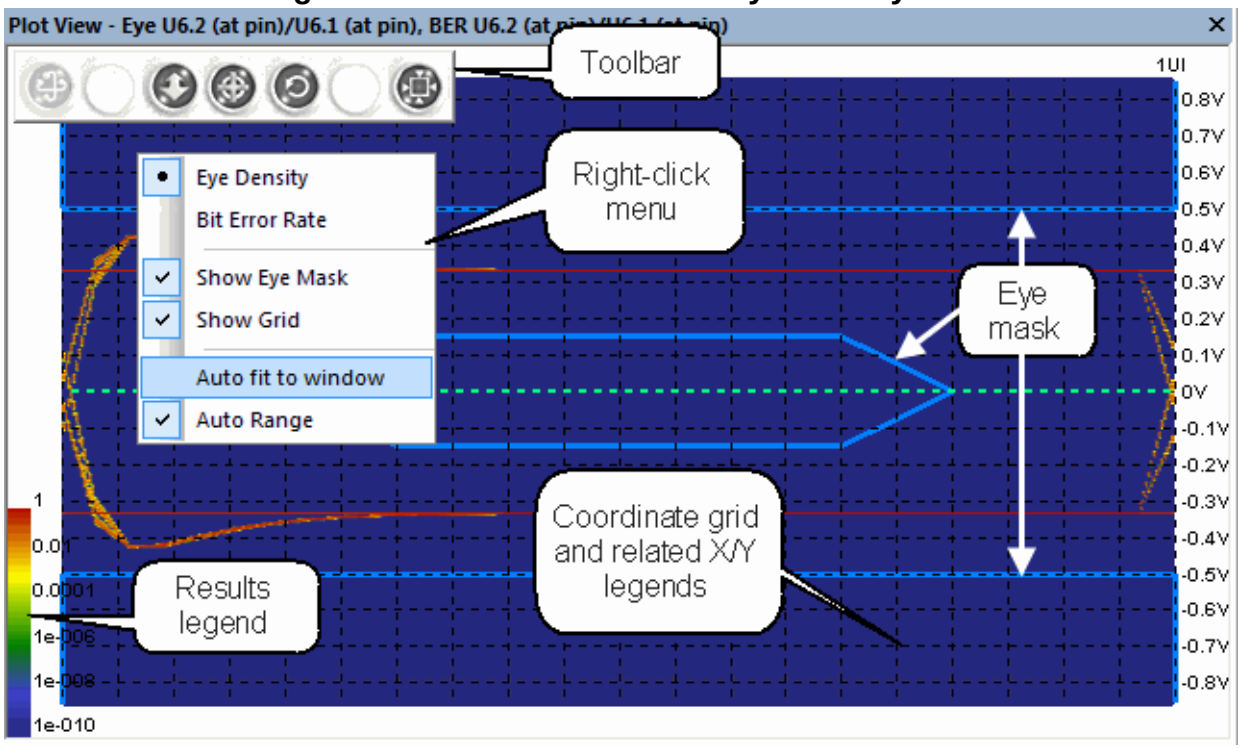
HyperLynx Eye Density/BER Viewer - Plot View Pane

To access: From the HyperLynx Eye Density/BER Viewer, select **View > Plot View**.

Use this pane to display IBIS-AMI simulation results.

You can format the results as an eye density plot or a bit error rate (BER) plot. Figure 10-23 and Figure 10-24 show the 2-D view. A 3-D view is also available.

Figure 10-23. Plot View Pane - Eye Density Plot



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-24. Plot View Pane - BER Plot

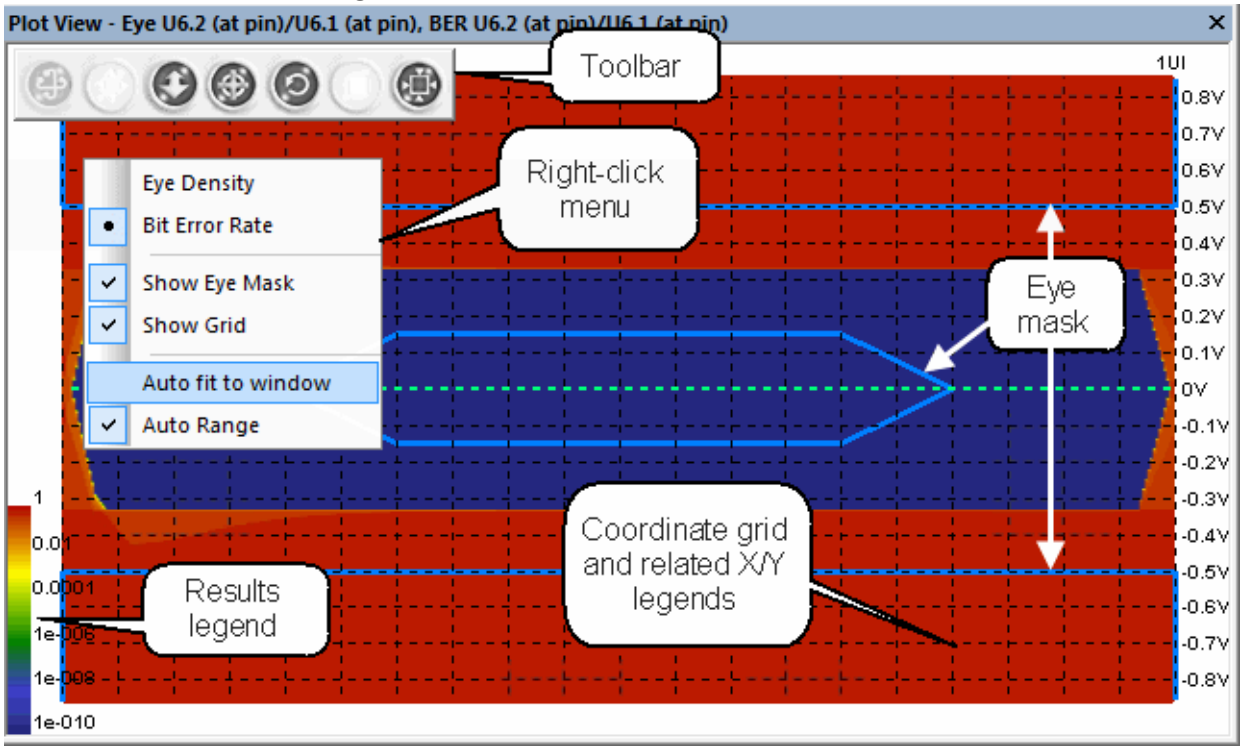
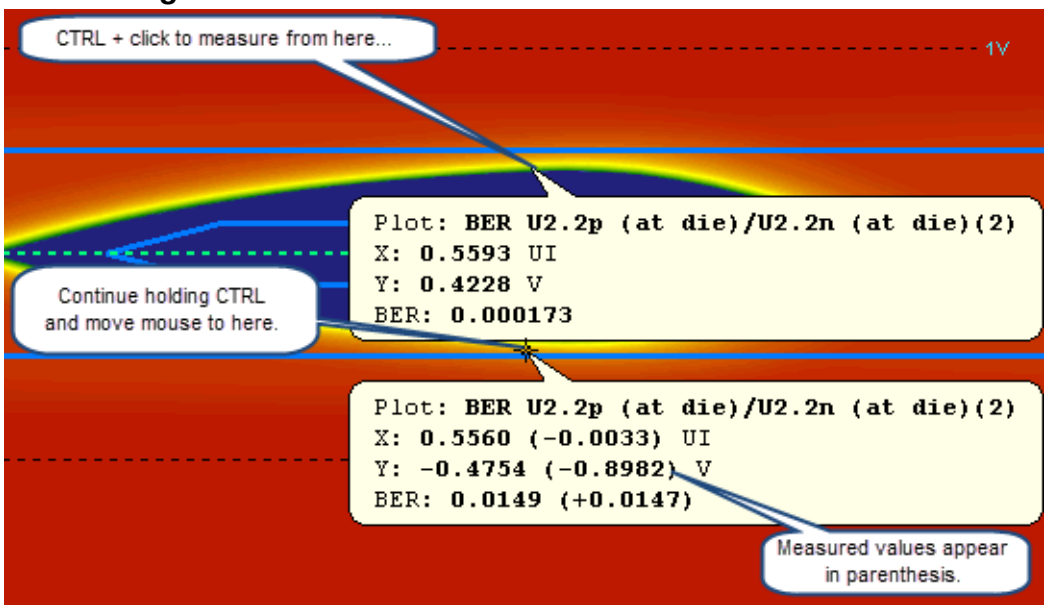


Figure 10-25. Plot View Pane - Manual Measurement



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Options

Table 10-3. Plot View Pane Contents

Option	Description
Right-click menu	The right-click menu duplicates some of the options located in the HyperLynx Eye Density/BER Viewer - Plot View Options Pane.
Eye mask	Optionally overlays the plot with an eye mask that provides a visual way to determine whether the simulation results intersect a “kaput” area. To load a pre-defined eye mask or edit the current eye mask, select either: • **Edit eye mask parameters** button on the main toolbar. • **Configure** on the Plot View Options pane.
Results legend	Maps the color of the results to a BER.
Coordinate grid and related X/Y legends	Optionally overlays the plot with a grid, where UI is on the X axis and voltage is on the Y axis. 1 unit interval = 1 bit interval. To show or hide the grid, select **Show grid**.

Toolbar - Plot View

Use the toolbar located in the Plot View Pane to enable turn, zoom, and other view-enhancement features. The Inspect button displays numerical results when you point to the simulation results shape.

Table 10-4. HyperLynx Eye Density/BER Viewer - Toolbar - Plot View Contents

Button	Description
Turn	Drag in the plot to rotate simulation results. The rotation axis is in the center of the maximum X/Y extents of the simulated geometries. **Restriction:** This mode is unavailable when top view is enabled.
Pan	Drag in the plot to move simulation results without zooming or rotating. **Keyboard shortcut:** Press **Shift** to enable.
Zoom	Drag up to zoom out. Drag down to zoom in. Use the mouse wheel to zoom in and out. **Keyboard shortcut:** Press **Alt** to enable.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 10-4. HyperLynx Eye Density/BER Viewer - Toolbar - Plot View Contents

Button	Description
Inspect 	Point to a shape to display X/Y coordinates and numerical simulation results values in a ToolTip. **Keyboard shortcut:** Press **Ctrl** to enable.
Default View 	Restores initial rotation and zoom.
3-D / 2-D, Top View Only 	Toggles between the 3-D and 2-D views. The 2-D view displays results as seen from above.
Fit to View 	Resizes the results so they fit on the screen.

Related Topics

Pane Organization in the HyperLynx Eye Density/BER Viewer

HyperLynx Eye Density/BER Viewer - Spreadsheet

To access: Open the HyperLynx Eye Density/BER Viewer

Use the spreadsheet to display the combination of model parameter values, numerical simulation results, and pass/fail results.

Note

The software bases measurements on the middle eye when you specify a PAM-4 stimulus and display all three (unconsolidated) eyes.

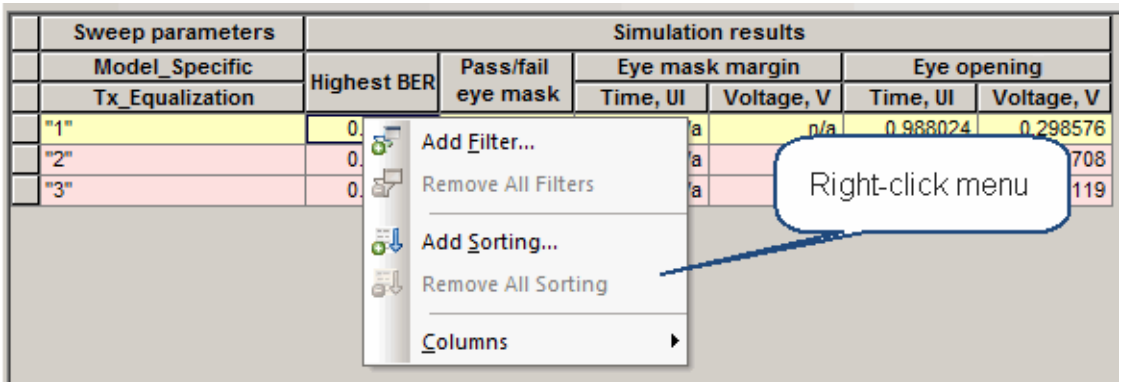
You can sort spreadsheet rows by Highest BER, Eye mask margin (time), and so on using the Right-click menu.

Select anywhere in a spreadsheet row to display the corresponding set of:

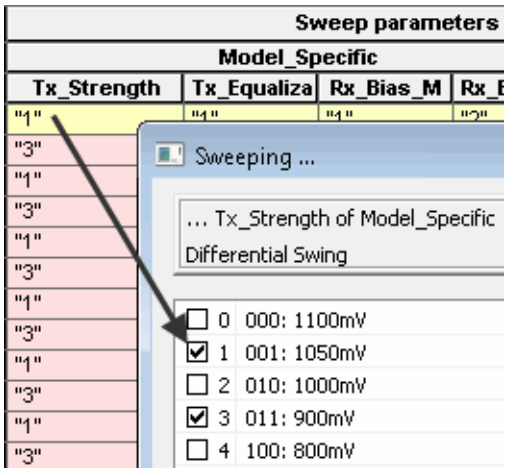
- Simulation results in the Plot View pane
- Values in the Sliders pane

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-26. HyperLynx Eye Density/BER Viewer Spreadsheet



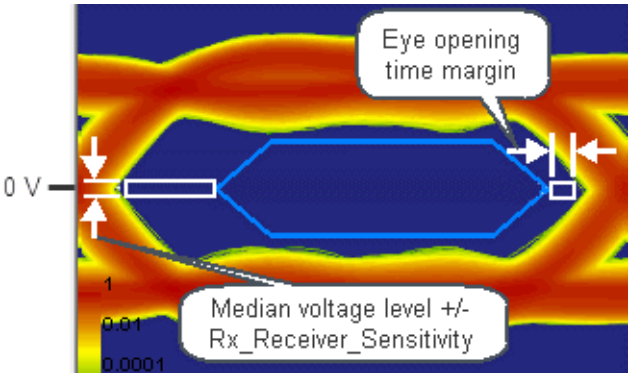
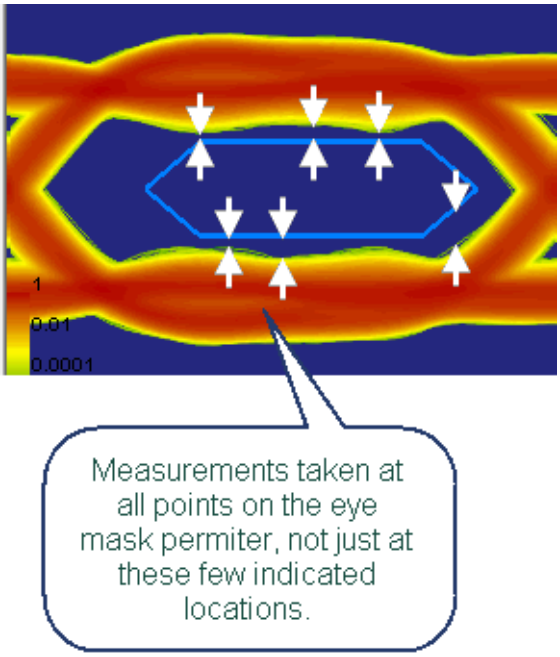
Spreadsheet Columns

Column Name	Description
Simulation date (Not available when you run sweeps with the IBIS-AMI Channel Analyzer.)	Date and time for a simulation.
Probe name (Not available when you run sweeps with the IBIS-AMI Channel Analyzer.)	Receiver probe that you specified on the “Set Up Channel Characterizations” page of the IBIS-AMI Channel Analyzer or FastEye Channel Analyzer.
Sweep parameters Columns	
<swept_parameters> (Available when you run sweeps with the SERDES Wizard.)	IBIS-AMI model parameters that you selected to sweep in the SERDES Wizard Parameter Sweep page. The spreadsheet displays some sweep ranges as a list of quoted integers, such as “0”, “2”. The integers map to spreadsheet rows in the Sweeping Dialog Box. 

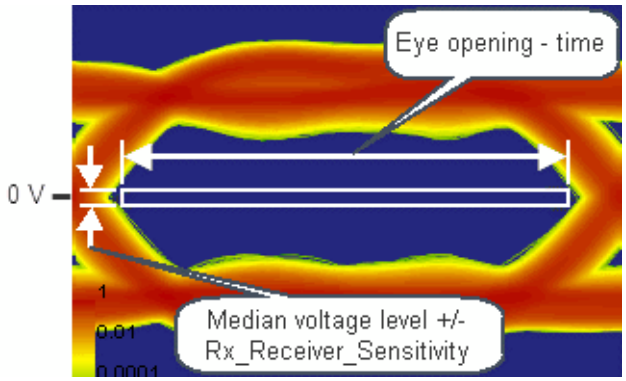
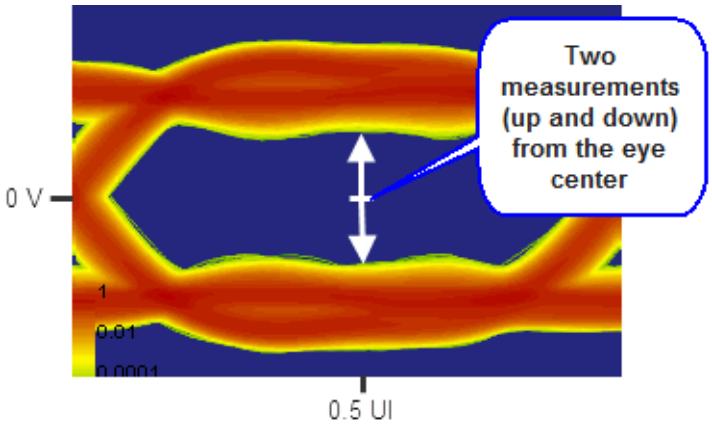
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Description
Simulation results Columns	
The viewer automatically calculates all the data in these columns. This means: • The *.sds* file does not store the spreadsheet results. The viewer recalculates them when you open the *.sds* file. • The number and composition of the spreadsheet columns does not depend on the initial set of swept parameters. • These values depend on the eye mask parameters. If you change the eye mask or BER threshold value, viewer recalculates all the data in these columns.	
Highest BER	Highest bit error rate value measured along the perimeter of the diamond-shaped eye mask polygon. **Tips:** • You may need to widen the spreadsheet column to see the full measurement value. • The 0 value represents 1e-10 or lower. If you place the eye mask over the BER plot, you can see where the eye mask touches the BER plot. This cell reports the highest BER measurement touched by the eye mask. For example measurements, see “Highest BER Examples” on page 931. If the highest BER is less than the BER Threshold value, the “Pass/fail eye mask” value is Pass. Use the Eye Mask tab on the Configure Eye Diagram dialog box to set the BER threshold.
Pass/fail eye mask	Pass if Highest BER is less than the BER Threshold value. Use the Eye Mask tab on the Configure Eye Diagram dialog box to set the BER threshold. The spreadsheet always displays the pass/fail result, even if the eye mask is not displayed.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Description
Eye mask margin - Time, UI	Minimum distance, in time (in UI), between the eye plot and eye mask, as measured at the median voltage level (0 V) +/- Rx_Receiver_Sensitivity. The reserved parameter Rx_Receiver_Sensitivity is optional. When it is not specified, the value defaults to 0.  The value is “n/a” when there is no gap between the eye mask and the eye plot.
Eye mask margin - Voltage, V	Minimum vertical distance, in voltage, between the eye plot and eye mask. The voltage margin is measured at all points on the eye mask perimeter, not just the few places indicated in the figure below.  The value is n/a if there is no gap between the eye mask and the eye plot.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Description
Eye opening - Time, UI	The distance in time between the right and left sides of the inner boundary of the eye diagram, as measured at the median voltage level (0 V) +/- Rx_Receiver_Sensitivity. The reserved parameter Rx_Receiver_Sensitivity is optional. When it is not specified, the value defaults to 0.  Eye opening measurements do not include a guardband. By contrast, test bench equipment may apply a guardband, such as three sigma.
Eye opening - Voltage, V	The minimum distance in voltage from the median voltage level (0 V) to the top and bottom sides of the inner boundary of the eye diagram, as measured at the eye center (0.5 UI). The AMI specification indicates the eye center is formed by Rx clocks and reflects the actual sampling point after CDR (clock and data recovery).  Eye opening measurements do not include a guardband. By contrast, test bench equipment may apply a guardband, such as three sigma.
Right-click menu	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Column Name	Description
	Duplicates many of the options available from the HyperLynx Eye Density/BER Viewer - Spreadsheet Options Pane. Right-click anywhere in the spreadsheet to open this menu.

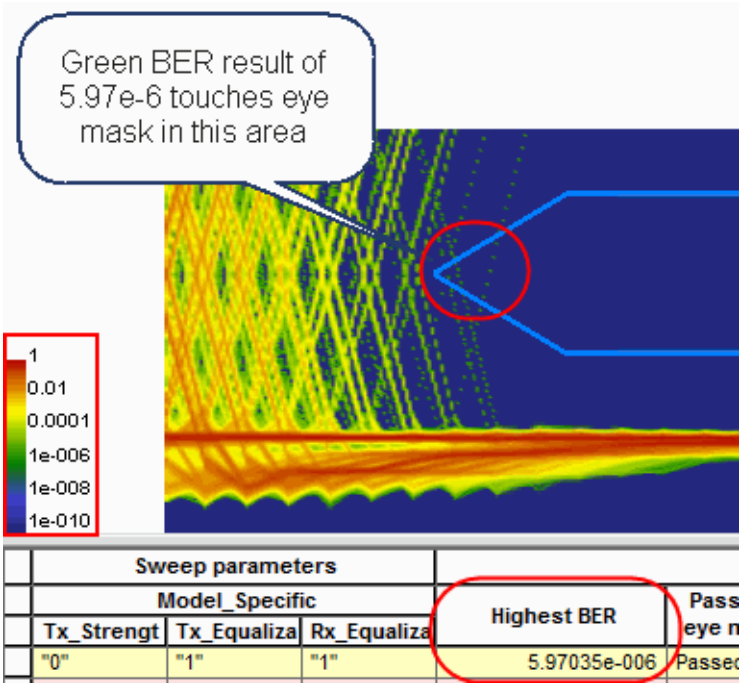
Highest BER Examples

Figure 10-27 and Figure 10-28 show example measurements for the Highest BER spreadsheet cell.

Note

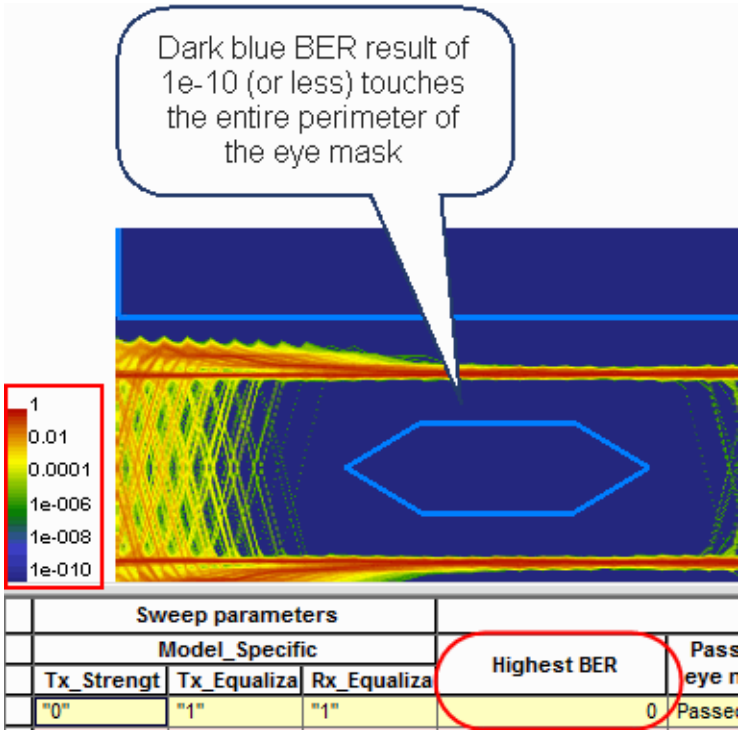
For simplicity, these figures do not show the effects of the Rx_Receiver_Sensitivity reserved parameter.

Figure 10-27. Highest BER is 5.97e-6



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-28. Highest BER is Very Low



Related Topics

Pane Organization in the HyperLynx Eye Density/BER Viewer

HyperLynx Eye Density/BER Viewer - Plot View Options Pane

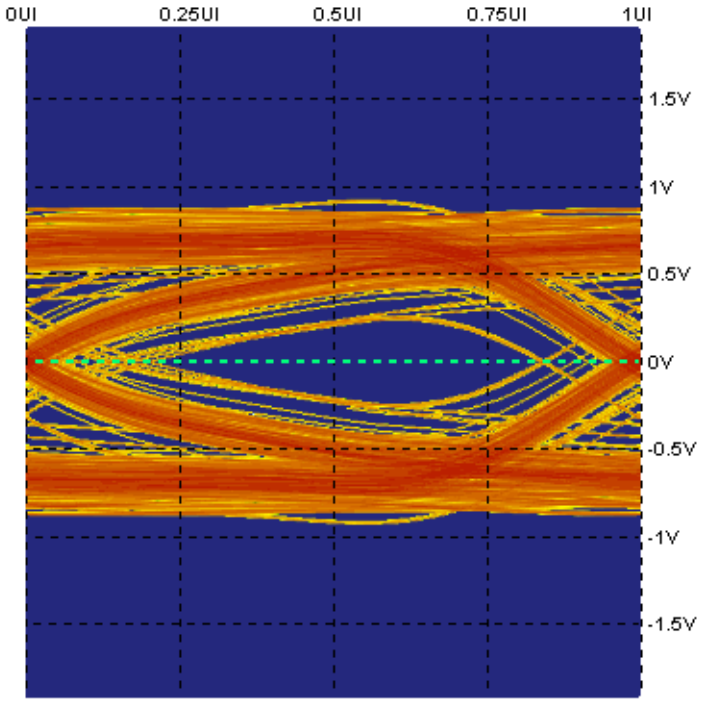
To access: From the HyperLynx Eye Density/BER Viewer, select **View > Plot Options**.

Use this pane to set display options for the Plot View Pane.

Options

Option	Description
Graph type Area	
Eye Density	Select to format the simulation results as an eye density plot. For an example plot, see Figure 10-23 on page 923.
Bit Error Rate	Select to format the simulation results as a BER plot. For an example plot, see Figure 10-24 on page 924.
Eye mask Area	
Show eye mask	Select to overlay an eye mask over the plot.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Configure	Opens the Configure Eye Diagram dialog box to edit eye mask properties for eye diagram analysis. You can load existing eye masks from a library or save new eye masks into a library.
UI and voltage scale Area	
Auto UI and voltage	Scales the simulation results to fit in the window.
UI	X-axis scale. 1 unit interval = 1 bit interval.
Voltage	Y-axis scale.
Range Area	
Auto range	Adjusts the zoom scale and data range to fit the results to the screen.
Zoom	Zoom scale.
Move	Defines the lowest BER value to display. Values below this will not display.
Coordinate grid Area	
Show grid	Select to overlay the plot with a grid, where UI is on the X axis and voltage is on the Y axis. 
UI	Drag the slider to select the time interval, in UI, of vertical grid lines.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Voltage	Drag the slider to select the voltage interval, in volts, of horizontal grid lines.

Related Topics

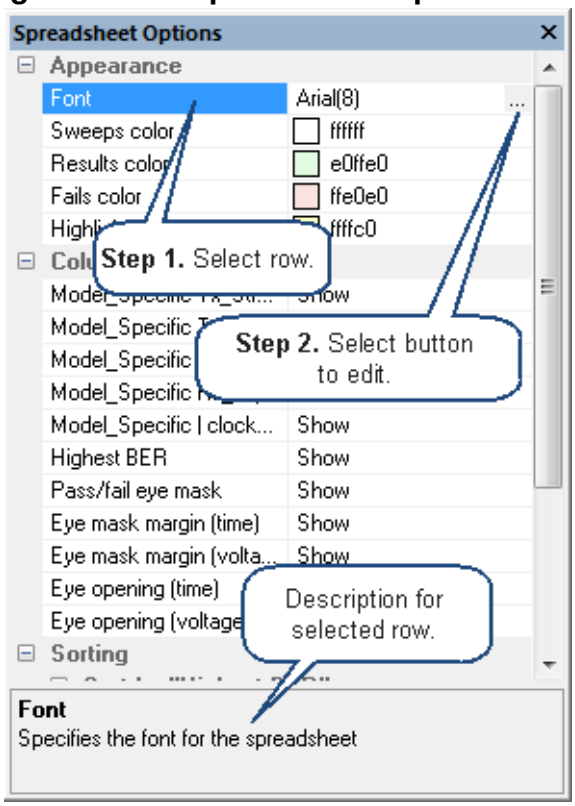
Pane Organization in the HyperLynx Eye Density/BER Viewer

HyperLynx Eye Density/BER Viewer - Spreadsheet Options Pane

To access: From the HyperLynx Eye Density/BER Viewer, select **View > Spreadsheet Options**.

Use this pane to edit display options for the spreadsheet.

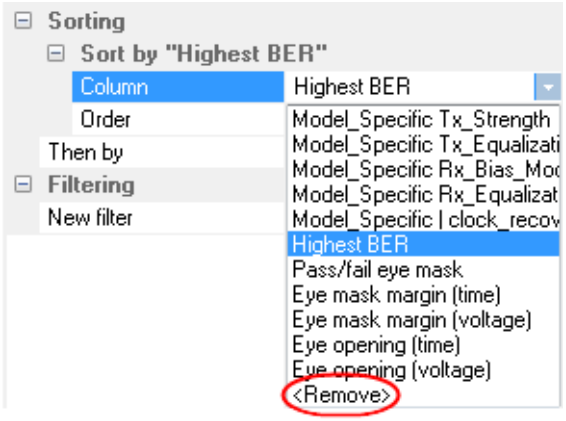
Figure 10-29. Spreadsheet Options Pane



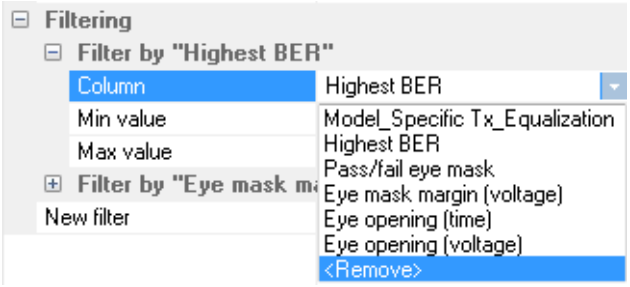
Options

Option	Description
Appearance Area	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Font	Specifies the appearance of characters in rows containing simulation results rows.
Sweeps color	Specifies the background for the left-most group of columns for unselected rows that pass simulation. For information about pass/fail criteria, see “Pass/fail eye mask” in HyperLynx Eye Density/BER Viewer - Spreadsheet.
Results color	Specifies the background color for rows that pass simulation. For information about pass/fail criteria, see “Pass/fail eye mask” in HyperLynx Eye Density/BER Viewer - Spreadsheet.
Fails color	Specifies the background color for rows that fail simulation. For information about pass/fail criteria, see “Pass/fail eye mask” in HyperLynx Eye Density/BER Viewer - Spreadsheet.
Highlight color	Specifies the background color of the selected spreadsheet row.
Columns Area	
<various>	Specifies whether to show or hide the column, and the sliders in the Sliders pane. To only show a few columns, next to All columns, select **Hide all**, and then select **Show** next to the columns that you want to see. Columns under the Sweep parameters heading display the set of IBIS-AMI model parameters that you selected to sweep in the SERDES Wizard Parameter Sweep page.
Sorting Area	
Sort by	Specifies one or more criteria to sort spreadsheet rows. To remove existing sorting criteria, select **<Remove>** at the bottom of the Column list. 
Filtering Area	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
New filter	Specifies one or more criteria to remove spreadsheet rows. To remove an existing filter, select <**Remove**> at the bottom of the Column list. 

Related Topics

Pane Organization in the HyperLynx Eye Density/BER Viewer

HyperLynx Eye Density/BER Viewer - Sliders Pane

To access: From the HyperLynx Eye Density/BER Viewer, select **View > Sliders**.

Use a slider on this pane to quickly scan through a large number of simulations to see results for a specific slider category. The slider automatically selects the corresponding spreadsheet row and displays simulation results.

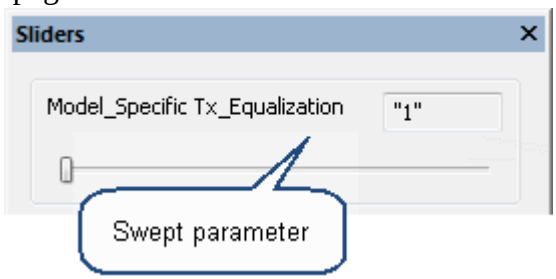
Note

The software bases measurements on the middle eye when you specify a PAM-4 stimulus and display all three (unconsolidated) eyes.

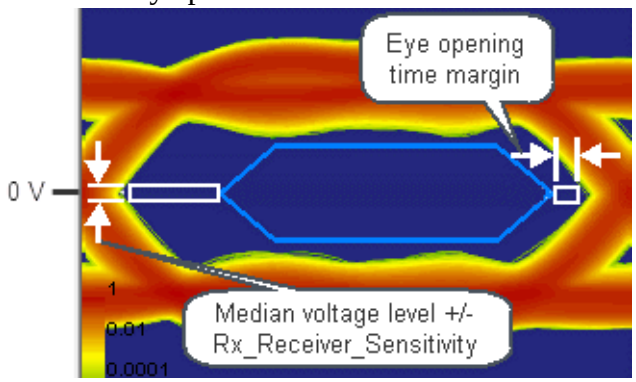
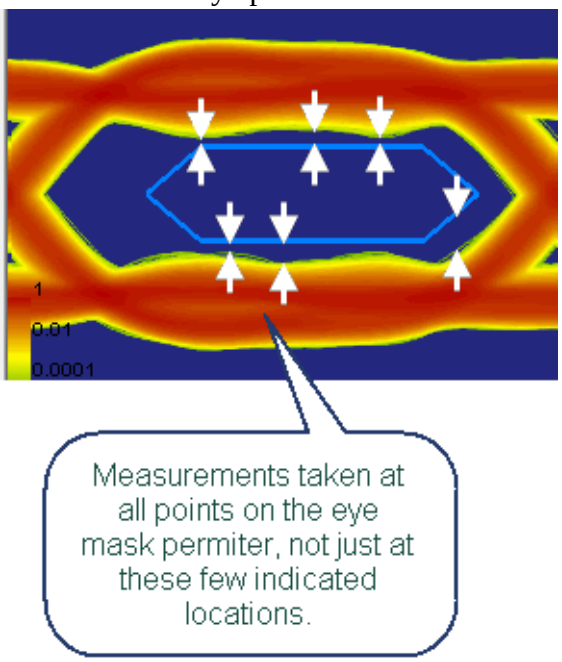
Options

Option	Description
Simulation date (Not available when you run sweeps with the IBIS-AMI Channel Analyzer.)	Date and time for a simulation.
Probe name (Not available when you run sweeps with the IBIS-AMI Channel Analyzer.)	Receiver probe that you specified on the “Set Up Channel Characterizations” page of the IBIS-AMI Channel Analyzer or FastEye Channel Analyzer.

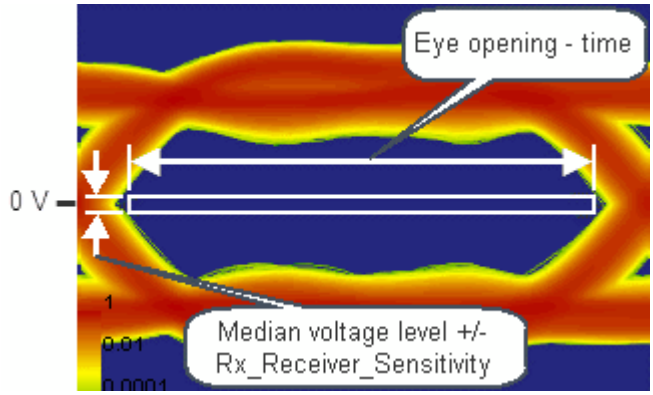
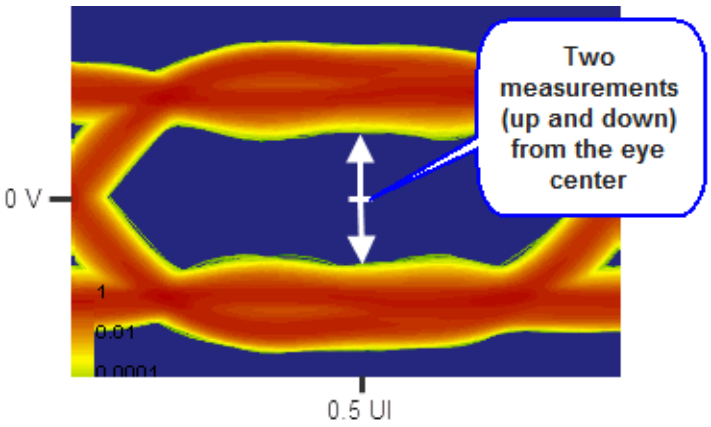
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
<swept_parameters> (Available when you run sweeps with the IBIS-AMI Channel Analyzer.)	Displays the set of IBIS-AMI model parameters that you selected to sweep in the SERDES Wizard Parameter Sweep page.  The Sliders pane displays some sweep ranges as a list of quoted integers, such as “0”, “2”. The integers map to spreadsheet rows in the Sweeping Dialog Box.
Highest BER	Highest bit error rate value measured along the perimeter of the diamond-shaped eye mask polygon. **Tip:** You may need to widen the spreadsheet column to see the full measurement value. • The 0 value represents 1e-10 or lower. If you place the eye mask over the BER plot, you can see where the eye mask touches the BER plot. This cell reports the highest BER measurement touched by the eye mask. For example measurements, see “Highest BER Examples” on page 931. If the highest BER is less than the BER Threshold value, the Pass/fail eye mask value is Pass. Use the Eye Mask tab on the Configure Eye Diagram dialog box to set the BER threshold.
Pass/fail eye mask	Pass if Highest BER is less than the BER Threshold value. Use the Eye Mask tab on the Configure Eye Diagram dialog box to set the BER threshold. The spreadsheet always displays the pass/fail result, even if the eye mask is not displayed.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Eye mask margin (time)	Minimum distance, in time, between the eye plot and eye mask, as measured at the median voltage level (0 V) +/- Rx_Receiver_Sensitivity. The reserved parameter Rx_Receiver_Sensitivity is optional. When it is not specified, the value defaults to 0. The value is n/a if there is no gap between the eye mask and the eye plot. 
Eye mask margin (voltage)	Minimum vertical distance, in voltage, between the eye plot and eye mask. The voltage margin is measured at all points on the eye mask perimeter, not just the few places indicated in the figure below. The value is n/a if there is no gap between the eye mask and the eye plot. 

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Eye opening (time)	The distance in time between the right and left sides of the inner boundary of the eye diagram, as measured at the median voltage level (0 V) +/- Rx_Receiver_Sensitivity. The reserved parameter Rx_Receiver_Sensitivity is optional. When it is not specified, the value defaults to 0.  Eye opening measurements do not include a guardband. By contrast, test bench equipment may apply a guardband, such as three sigma.
Eye opening (voltage)	The minimum distance in voltage from the median voltage level (0 V) to the top and bottom sides of the inner boundary of the eye diagram, as measured at the eye center (0.5 UI). The AMI specification indicates the eye center is formed by Rx clocks and reflects the actual sampling point after CDR (clock and data recovery).  Eye opening measurements do not include a guardband. By contrast, test bench equipment may apply a guardband, such as three sigma.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Pane Organization in the HyperLynx Eye Density/BER Viewer

HyperLynx Eye Density/BER Viewer

HyperLynx PI PowerScope Dialog Box

To access: This dialog box opens automatically when a power-integrity simulation started from HyperLynx SI/PI completes. To manually open the HyperLynx PI PowerScope:

- In BoardSim, **Simulate PIRun DC Drop Simulation (PowerScope)** (menu item), **Show PowerScope** (button).
- In LineSim, **Simulate PIShow Previous DC Drop Results (PowerScope)** (button).
- In Analysis Control (in Xpedition Layout), **Hazards View > Use PowerScope** (button).

Use this dialog box to display graphical simulation results for planes and metal areas in the design.

When you run plane noise and decoupling capacitor simulation, this interface displays the voltage difference between the two metal regions that form the transmission plane.

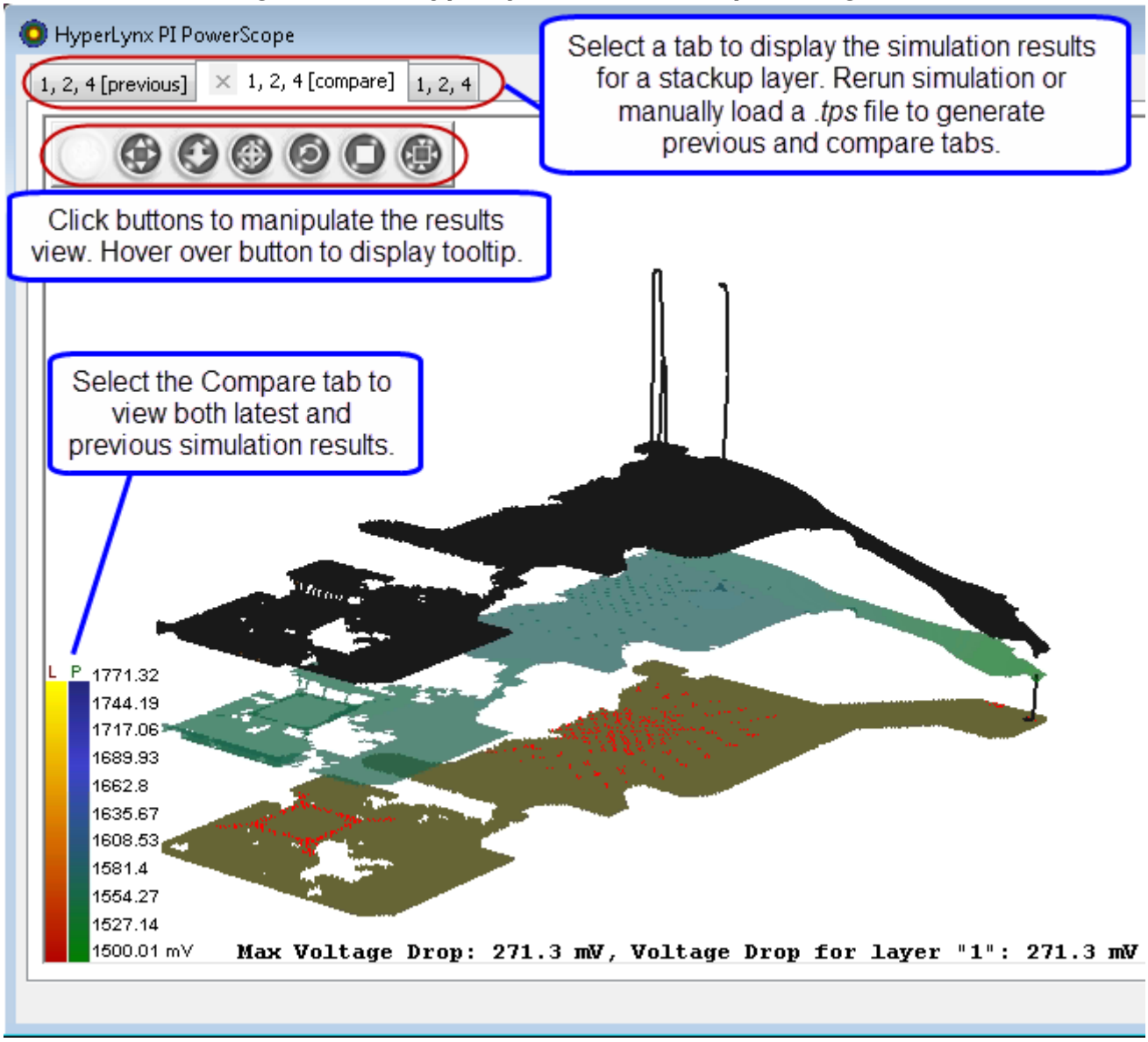
For DC drop, each tab displays the voltage drop only on stackup layers for a specific power-supply net.

Note

The results display contains approximate plane geometries, but simulation uses precise geometries. This enables fast rotating, zoom, and panning of the results image.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-30. HyperLynx PI PowerScope Dialog Box



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Fields	Description
Plane Noise Simulation (Available when plane-noise simulation opens this dialog box.)	Enables you to control the plane noise simulation run. Click **Pause** | **Stop** to temporarily or permanently halt the simulation, **Resume | Restart Simulation** to resume a paused simulation or launch a subsequent simulation starting at time zero. • Time — Displays the current simulation time when the simulation is running. • Stop • In BoardSim, displays the simulation time from the Plane Noise Analysis dialog box. • In LineSim, defines the simulation time for plane-noise simulation. The value has precedence over the period length (for pulse signal types) in the AC model. For example, if the AC model contains a repeating current waveform that extends beyond the simulation time, the current waveform is truncated.
Positioning Options	Enables you to set the position and scale of the results display. Drag the slider to quickly edit. If you type in a value, press **Enter**. • Span — Defines the vertical scale. • Origin — Defines the vertical offset. • Auto span & origin — Checked, automatically changes the span and origin values to display the full extent of the simulation results. • Thresholds — Checked, enables you to add and edit horizontal reference planes to help you see simulation values that exceed a limit.
Visual Options — Contains the Show, Model View, and Graph Type sections, which enable you to define and enhance the visual image of the results display.	
Show	Enables you to show (when checked) the results in solid form, wire form, or both. Check **Hide invisible** to hide regions of the reference threshold planes located behind the simulation results surface.
Model View	Enables you to include (when checked) the following with the display results: • Grid — The simulation grid. **Note:** When you run DC drop simulation (FEM), the simulation grid is finer than the visible grid, especially at the end of simulation. • TPlane model — The transmission-plane model as a red (top)/green (bottom) rectangle and model ports (pins and vias) as small circles. Circles indicate where sink and source models attach to the transmission-plane model. • Meshed model — The squared copper outline and model ports.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields	Description
Graph type	Enables you to select how to display (graph) the simulation results. The field names vary depending on the type of simulation you run: • DC Drop Voltage or Noise voltage — Selected, displays as three-dimensional with the height determined by voltages across the plane geometries. • DC Current Vectors or Surface and capacitor currents — Selected, displays as two-dimensional vectors: the length and color indicate current density, and the angle indicates current flow direction. Check Meshed model to see approximate vector origins. • DC Current Density or Surface current — Selected, displays as three-dimensional with the height determined by current flow across a plane. • DC Via Current — Selected, displays as three-dimensional with the height determined by current flow within a via. • Show maximums — Checked, displays maximum amplitude result (Z axis) for every X/Y location on the three-dimensional surface over the entire simulation. **Note:** This option is not available for DC Drop when you load a TPS file. Generally, TPS files are located in the <design> folder but you can move them elsewhere.
Layer Options	Lists the stackup layers. Check to include geometries on the stackup layer. Uncheck to hide geometries for that layer. • <REF> — (DC Drop simulation only) Indicates that the stackup layer is a reference net for another stackup layer being simulated. The algorithm relies on reference nets assigned to VRM models on IC power-supply pins. See “Assign Power Integrity Models Dialog Box - IC Tab” on page 712.
Save Load	Save the plot to a *.tps* file. You can load a *.tps* file to compare it to an already-loaded plot.

Related Topics

DC Drop Simulation

Simulating PDN Decoupling - Distributed

Running Plane Noise Simulation

IBIS AMI Parameter Editor

To access: **Simulate SERDES > Run IBIS-AMI Channel Analysis**, select the Configure AMI Models page, and select **Configure Tx AMI** or **Configure Rx AMI**

Use this editor to display and change AMI parameter values.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you make changes, the editor saves your changes into a new *.AMI* file named *<original_file_name>_settings.ami* or to a file or location you specify. This behavior preserves the contents of the original *.AMI* file.

Optionally, you can sweep IBIS-AMI parameters by specifying sweep ranges on the SERDES Wizard Parameter Sweep page. Sweep simulations temporarily override values you set here.

The IBIS specification, starting in version 5.0, defines AMI parameters and usage. Changes to parameter values may be based on IC datasheets, design kit documentation, your own knowledge, and so on.

You can use this editor to add and remove jitter parameters. Use a text editor to add or remove other types of parameters.

Options

Option	Description
IBIS AMI file	Displays the path of the *<original_file_name>_settings.AMI* file that contains edited parameter values. This box is blank unless you either load an existing *<original_file_name>_settings.AMI* file or save your edits to a new file.
Parameter tree area	Displays the parameters in the *.AMI* file. Select a parameter name in the tree to display its contents.
Parameter value area	Displays the contents of the selected parameter. Some parameters in the Reserved_Parameters branch are read-only because the model vendor supplies information about the model *.DLL* or *.so* file that you cannot change. For example, you cannot edit the GetWave_Exists parameter. Single-value parameters are also read-only. For example, if the *.AMI* file contains a parameter defined as (Format Value 0.0), the parameter value is read-only. The idea is to use the one value that the model developer declared to be valid. Select **View/Edit Table** to open the Ami Table Editor to read or change parameter table contents. This button is available only when you select a parameter in the tree whose contents are in the table format.
Reset	Discard the current parameters and load the contents of the original *.AMI* file.
Save As	Save the current parameters to a new *.AMI* file.
Save	Saves your changes into a new *.AMI* file named *<original_file_name>_settings.ami*. This behavior preserves the contents of the original *.AMI* file.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Load	Load parameters from an *.AMI* file.
Exit	Close the editor.

Import Constraints from Constraint Manager Dialog Box

To access: Open the Edit Net Classes Constraints and Stimuli Dialog Box, click **Import / Export Net Classes and Constraints**, then click **Import from Constraint Manager**.

Use this dialog box to select the Constraint Manager project that contains the constraints or net groupings (net classes) that you want to import.

Options

Option	Description
Constraint Manager project	Specifies the location of the project file (*.prj*).
Schematic Layout	Specifies whether to import schematic or layout data from the design database (iCDB).
Design	Specifies the design name, using names from the project file (*.prj*).

Related Topics

Importing and Exporting Constraints, Net Classes, or Stimulus

License Status Dialog Box

To access: **Setup > Options > License Status**

Use this dialog box to see the status of currently acquired licenses.

Fields

Field	Description
Pre Layout Post Layout Pre and Post Layout	Displays license status. Use the HyperLynx Licensing dialog box (which automatically appears when you open the software) to acquire or release licenses.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Cross Licensing	Displays the currently selected non-HyperLynx product cross license.
Translators	Opens the Installed Translators dialog box, which displays the status of translator licenses.
PCLS_OK	Opens the pcls_ok utility to help you troubleshoot licensing problems. pcls_ok can report key licensing information in the environment or registry, and can test license acquisition. (Available on computers running Windows.)
Show license dialog on startup Warn of expiring licenses	Use the HyperLynx Licensing dialog box (which automatically appears when you open the software) to edit these options.

Related Topics

Setting Up the Software

Interactive Simulation Dialog Box

To access: Click 

Note: On the Welcome Screen, enable **EZwave** or **Both** waveform viewers to make this dialog box available.

Use this dialog box to run interactive SI simulation on selected nets in a board design or an entire schematic. When simulation completes, EZwave displays waveforms.

Options

Option	Description
Operation Area	
Standard	Displays simulation results as waveforms.
Eye diagram	Displays simulation results as an eye diagram.
Stimulus Area	
Global	Assigns oscillator or edge stimulus to all driver pins on all nets. The software applies global stimulus to all driver pins on the selected net and its associated nets on a board and to all driver pins on a schematic. When you enable this option, the options Edge/Oscilloscope and Rising Edge/Falling edge become available.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Per-Net/Pin	Displays the Assign button so that you can manually assign specific stimulus to specific driver nets or pins. Per-net/pin stimulus enables you to simulate timing relationships among nets/pins, such as the following: • Crosstalk investigations with different waveforms on aggressor nets, to help examine the pattern dependency of crosstalk • Source-synchronous signaling, such as DDRx and similar technologies, where one IC transmits both the clock and data signals (as opposed to a master system clock, which is transmitted by a different IC) with typically slightly different timing.
Assign	Opens the Assign Stimulus Dialog Box to assign stimulus to specific pins or nets. **Restriction:** This option is available only when you select Per-Net/Pin.
Edge	Select the signal edge to display.
Oscillator	Enter values into the MHz and Duty fields. The duty cycle value defines the percentage of the period that the driver is high.
Configure	Opens the Configure Eye Diagram Dialog Box - Stimulus Tab, which you use to define global stimulus for all enabled drivers on the selected net in a board design or schematic. (Available when Eye Diagram and Global are selected.)
Start Simulation	Runs simulation.
Simulation Controls	Opens the Simulation Controls Dialog Box, which you use to set up additional simulation options.
IC modeling Area	
Slow-Weak Typical Fast-Strong	Selects the IBIS IC model corner to use during simulation. For information about the combination of min/typ/max data in an IBIS IC model that is used for each IC model corner, see IC Operating Settings. SPICE models use power supply net voltage settings and do not use this setting. For information about manually editing power supply voltages, see Edit Power-Supply Nets Dialog Box. (Unavailable when you enable a sweep range for IC operating parameters in the Sweep Manager Dialog Box - Setup Tab.)
Simulation duration Area	
	Auto — Simulation runs the amount of time needed to complete the stimulus sequence you specify in the Stimulus area. Manual — Simulation runs the amount of time you enter.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Eye diagrams Area	
Restriction: This area is available when you enable Eye diagram operation.	
Process only receiver waveforms	Displays waveforms only for receivers. You can manually display driver waveforms in EZwave. See Displaying Waveform Results in EZwave.
Plot eye waveforms	Displays waveforms in EZwave. When running a long simulation, you can hide waveforms to save time and memory. The software always stores EZwave waveforms in an in-memory database. See Displaying Waveform Results in EZwave.
Put into calc database	Writes eye diagram waveforms into an EZwave database named “calc”.
How to plot	Plots waveforms in one of the following ways: • Overlaid waveforms — Plots waveforms in a single row. • Stacked waveforms — Plots each waveform in a new row. • In separate windows — Plots each waveform in a different window.
Keep initial offset	Initial offset represents the delay (from time zero) the software uses as a starting point for building the eye diagram. Unchecked, the software finds the offset for the top displayed waveform and applies it to all displayed waveforms. Checked, the software finds and applies waveform-specific offsets. Unchecked, compares eye diagrams for different pins on a net. The default is unchecked.
Skip first	Defines the number of bits at the start of simulation to exclude from the eye diagram. You may want to skip these bits when the channel, transmitter, or receiver uses them to reach normal operating conditions.
Show 3 eyes	Formats the eye diagram to display two full eyes.
Clear previous results	Deletes waveforms for previous simulations from EZwave.

Related Topics

Running Signal Integrity Simulation with the EZwave Waveform Viewer

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Interactive Simulation with Measurements Dialog Box

To access: Click 

Note: On the Welcome Screen, enable **EZwave** or **Both** waveform viewers to make this dialog box available.

Use this dialog box to run interactive SI simulation and select the types of measurements to perform on selected nets in a board design or on an entire schematic.

When simulation completes, EZwave displays waveforms and the Simulation Results Dialog Box displays measurements.

Options

Options	Description
Stimulus Area	
Global	Assigns oscillator stimulus. For a board design, the software applies global stimulus to all driver pins on the selected net and its associated nets. For a schematic design, the software applies global stimulus to all driver pins in the schematic.
Per-Net/Pin	Displays the Assign button, so that you can manually assign specific stimulus to specific driver nets or pins. Per-net/pin stimulus enables you to simulate timing relationships among nets/pins, to investigate issues such as: • Crosstalk with different waveforms on aggressor nets, to help examine the pattern dependency of crosstalk • Source-synchronous signaling, such as DDRx and similar technologies, where one IC transmits both the clock and data signals (as opposed to a master system clock, that is transmitted by a different IC) with typically slightly different timing
Initial state Area	
Low High MHz Duty	Specify the initial logic state, frequency, and duty cycle for the oscillator stimulus. The duty cycle value defines the percentage of the period that the driver is high.
Assign	Opens the Assign Stimulus Dialog Box to assign stimulus to specific pins or nets. (Available when you select Per-Net/Pin.)

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Options	Description
Start Simulation	Runs simulation.
Simulation Controls	Opens the Simulation Controls Dialog Box, which you optionally use to set up additional simulation options.
IC modeling Area	
Slow-Weak Typical Fast-Strong	Selects the IBIS IC model corner to use during simulation. If you enable Crosstalk, consider Fast-Strong to obtain the maximum crosstalk results. For information about the combination of min/typ/max data in an IBIS IC model that is used for each IC model corner, see IC Operating Settings. SPICE models use power supply net voltage settings and do not use this setting. For information about manually editing power supply voltages, see Edit Power-Supply Nets Dialog Box. (Unavailable when you enable a sweep range for IC operating parameters in the Sweep Manager Dialog Box - Setup Tab.)
Analysis type Area	
Delay and signal integrity	Runs delay and SI analysis and writes the measurements to the Interconnect Delays and Signal Integrity tabs in the Simulation Results Dialog Box.
Crosstalk	Runs crosstalk analysis and writes the measurements to the Crosstalk tab in the Simulation Results Dialog Box.
Measurements	Opens the Measurements dialog box, which you use to set up the analysis details for SI and crosstalk measurements.
Nets and constraints Area - Boards Only	
Selected net	Displays the name of the selected net. See Selecting Nets for SI Simulation. When a multiple-board design is loaded, the software appends the board name (such as B00) to the net name.
Set Constraints	Opens the "Edit Net Classes Constraints and Stimuli Dialog Box" on page 963, so that you can edit constraints for the selected net. **Restriction:** You cannot select additional nets in the spreadsheet.
Run analysis for Area - Schematics Only	
Entire schematic	Simulates all the nets in the schematic.
Net	Displays the name of the net you selected for simulation. To select a different net for simulation, select a schematic symbol that is connected to the net.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Options	Description
Store analysis waveforms	Stores waveforms in a temporary database and automatically displays them in EZwave. You can save the database to disk from EZwave. Disable this option to save runtime when running long simulations.
Clear previous results	Deletes the waveforms and measurements for previous simulations from EZwave and the Simulation Results Dialog Box.

Related Topics

Running Signal Integrity Simulation with the EZwave Waveform Viewer

Interactive Sweeps Dialog Box

To access: Click  , from the Sweep Manager Dialog Box - Setup Tab, click **Run Sweeps**

Note: On the Welcome Screen, enable **EZwave** or **Both** waveform viewers to make this dialog box available.

Use this dialog box to run sweeps SI simulation on selected nets in a board design, or on an entire schematic.

Options

Option	Description
Operation Area	
Standard	Formats simulation results as waveforms.
Eye diagram	Formats simulation results as an eye diagram.
Stimulus Area	
Global	Assigns edge or oscillator stimulus. For a board design, the software applies global stimulus to all driver pins on the selected net and its associated nets. For a schematic design, the software applies global stimulus to all driver pins in the schematic. When you enable this option, the options Edge/Oscilloscope and Rising edge/Falling edge become available.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Per-Net/Pin	Displays the Assign button so that you can manually assign specific stimulus to specific driver nets or pins. Per-net/pin stimulus enables you to simulate timing relationships among nets/pins, to investigate issues such as: • Crosstalk with different waveforms on aggressor nets, to help examine the pattern dependency of crosstalk • Source-synchronous signaling, such as DDRx and similar technologies, where one IC transmits both the clock and data signals (as opposed to a master system clock, that is transmitted by a different IC) with typically slightly different timing
Assign	Opens the Assign Stimulus Dialog Box to assign stimulus to specific pins or nets. (Unavailable when you select Global.)
Edge	Select an edge type.
Oscillator	Enter values into the MHz and Duty fields. The duty cycle value defines the percentage of the period that the driver is high.
Configure	Opens the Configure Eye Diagram Dialog Box - Stimulus Tab, which you use to define global stimulus for all enabled drivers on the selected net (board design) or schematic. (Available when Eye diagram and Global are selected.)
Start Sweeps	Runs sweeps simulation.
Simulation Controls	Opens the Simulation Controls Dialog Box, which you use to set up additional simulation options.
IC modeling Area	
Slow-Weak Typical Fast-Strong	Selects the IBIS IC model corner to use during simulation. For information about the combination of min/typ/max data in an IBIS IC model that is used for each IC model corner, see IC Operating Settings. SPICE models use power-supply net voltage settings and do not use this setting. For information about manually editing power-supply voltages, see Edit Power-Supply Nets Dialog Box.
Simulation duration Area	
<value_field>	Enter the simulation run time. When you enable Per-net/pin and apply a long stimulus sequence, such as for an eye diagram, manually calculate the simulation run time to apply the full stimulus sequence.
Eye diagrams Area	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Restriction: This area is available when you enable the Eye diagram option.	
Process only receiver waveforms	Displays waveforms only for receivers. You can manually display driver waveforms in EZwave. See Displaying Waveform Results in EZwave.
Plot eye waveforms	Displays waveforms in EZwave. When running a long simulation, you can hide waveforms to save time and memory. HyperLynx always stores EZwave waveforms in an in-memory database. See Displaying Waveform Results in EZwave.
Put into calc database	Writes eye diagram waveforms into an EZwave database named "calc".
How to plot	Plots waveforms in one of the following ways: • Overlaid — Plots waveforms in a single row. • Stacked waveforms — Plots each waveform in a new row. • In separate windows — Plots each waveform in different window.
Keep initial offset	Uncheck to compare eye diagrams for different pins on a net. The default value is unchecked. Initial offset represents the delay (from time zero) the software uses as a starting point for building the eye diagram. Unchecked, the software finds the offset for the top displayed waveform and applies it to all displayed waveforms. Checked, the software finds and applies waveform-specific offsets.
Skip first	Defines the number of bits at the start of simulation to exclude from the eye diagram. Skip these bits when the channel, transmitter, or receiver uses them to reach normal operating conditions.
Show 3 eyes	Formats the eye diagram to display two full eyes.
Clear previous results	Deletes the waveforms for previous simulations from EZwave.

Related Topics

Running Signal Integrity Simulation with the EZwave Waveform Viewer

Interactive Sweeps with Measurements Dialog Box

To access: Click  , from the Sweep Manager Dialog Box - Setup Tab, click **Run Sweeps**

Note: On the Welcome Screen, enable **EZwave** or **Both** waveform viewers to make this dialog box available.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Use this dialog box to run sweeps SI simulation on the selected nets in a board design or on all the nets in a schematic and display waveforms and measurements.

When simulation completes, the Simulation Results Dialog Box displays measurements and EZwave displays waveforms.

Options

Option	Description
Stimulus Area	
Global	Assigns oscillator stimulus. For a board design, the software applies global stimulus to all driver pins on selected nets and associated nets. For a schematic design, the software applies global stimulus to all driver pins in the schematic.
Per-net/pin	Displays the Assign button so that you can manually assign specific stimulus to specific driver nets (board design) or pins (schematic). Per-net/pin stimulus enables you to simulate timing relationships among nets/pins, to investigate issues such as: • Crosstalk with different waveforms on aggressor nets, to help examine the pattern dependency of crosstalk • Source-synchronous signaling, such as DDRx and similar technologies, where one IC transmits both the clock and data signals (as opposed to a master system clock that is transmitted by a different IC) with typically slightly different timing
Assign	Opens the Assign Stimulus Dialog Box to assign stimulus to specific pins or nets. (Available when you select Per-net/pin.)
Initial state Area	
Low High MHz Duty	Specify the initial logic state, frequency, and duty cycle for the oscillator stimulus. The duty cycle value defines the percentage of the period that the driver is high. (Available when you select Global.)
Start Simulation	Runs sweeps simulation.
Simulation Controls	Opens the Simulation Controls Dialog Box, which you optionally use to set up additional simulation options.
IC modeling Area	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Slow-Weak Typical Fast-Strong	Selects the IBIS IC model corner to use during simulation. If you enable Crosstalk, consider enabling Fast-Strong to obtain the maximum crosstalk results. For information about the combination of min/typ/max data in an IBIS IC model that is used for each IC model corner, see IC Operating Settings. SPICE models use power supply net voltage settings and do not use this setting. For information about manually editing power supply voltages, see Edit Power-Supply Nets Dialog Box. (Unavailable when you enable a sweep range for IC operating parameters in the Sweep Manager Dialog Box - Setup Tab.)
Analysis type Area	
Delay and signal integrity	Runs delay and SI analysis and writes the measurements to the Interconnect Delays and Signal Integrity tabs in the Simulation Results Dialog Box.
Crosstalk	Runs crosstalk analysis and writes the measurements to the Crosstalk tab in the Simulation Results Dialog Box.
Measurements	Opens the Measurements dialog box, which you use to set up the analysis details for SI and crosstalk measurements.
Nets and constraints Area - Board Only	
Selected net	Displays the name of the selected net. See Selecting Nets for SI Simulation.
Set Constraints	Opens the "Edit Net Classes Constraints and Stimuli Dialog Box" on page 963.
Run analysis for Area - Schematics Only	
Entire schematic	Simulates all the nets in the schematic.
Net	Displays the net to simulate. To select a different net for simulation, keep this dialog box open and select a schematic symbol connected to another net.
Store analysis waveforms	Stores waveforms in a temporary database and automatically displays them in EZwave. You can save the database to disk from EZwave. Disable this option to save runtime or memory resources when running many sweeps.
Clear previous results	Deletes the waveforms and measurements for previous simulations from EZwave and the Simulation Results Dialog Box.

Related Topics

Running Signal Integrity Simulation with the EZwave Waveform Viewer

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Layer Mapping Dialog Box

To access: Opens automatically when you import a stackup containing more layers than the currently loaded design.

Use the Layer Mapping dialog box to map stackup layers in the source design to stackup layers in the current design. This dialog box opens only when you import a stackup containing more layers than the current design.

Fields

Field	Description
Used spreadsheet	Column identifies stackup layers that are used by nets in the board or schematic.
Destination Layer	To map stackup layers between the source and current designs: 1. Click the Destination Layer cell to specify any of the following mappings: <layer_name>—Replace the current Destination Layer with the imported Source Layer. Not Imported—Do not use the Source Layer. New Layer—Insert the Source Layer. New layer names are of form new_layer_<number>. 2. Click **OK**.

Related Topics

Exporting and Importing a Stackup

Load/Save Waveforms Dialog Box

To access: From the Digital Oscilloscope Dialog Box or FastEye Viewer, select **Save/Load**

Use this dialog box to load one or more previously-saved waveform files into the oscilloscope or FastEye Viewer or to save the latest waveforms to a comma-separated values (*.csv*) or an *.lis* file with HyperLynx-specific formatting.

Waveform files contain both voltage versus time and, for the HyperSim simulator only, current versus time data.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Options

Option	Description
Load	Loads an existing waveform file. The Digital Oscilloscope Dialog Box can read HyperLynx-generated *.lis* and EZwave *.csv* files.
Save As	Saves the waveform, using the options that you set below. **Restriction:** Eye diagrams cannot be saved.
CSV (comma-separated values)	Saves the waveform to the CSV format. See Contents of Waveform Files in CSV Format. **Restriction:** The maximum number of data points is 2^20 (1,048,576). Microsoft Excel 2003 and earlier supports 64K rows, so if you select the Microsoft Excel format, the maximum number of data points is 2^16 (65536) minus the number of rows used by header information. Alternatively, you can make a copy of the CSV file and use a text editor delete the header rows to read in Excel the maximum number of data rows.
Microsoft Excel (w/ extended header)	Use this option if you plan to analyze the waveform data using Microsoft Excel or another external application that can tolerate the extra rows at the top of the file that document the probe names. Windows regional settings determine the decimal and column delimiter characters.
EZwave	Use this option if you plan to display the waveforms in Siemens EDA EZwave. The top row contains information about waveform names. To support file formatting required by the Siemens EDAviewers, decimal delimiting is made with a period and column delimiting is made with a comma regardless of the Windows regional settings.
Generic (w/ simple header)	Use this option if you plan to analyze the waveforms in a mathematics application, such as Matlab. The top row contains information about waveform names. Windows regional settings determine decimal and column delimiter characters.
... using 2^	Enable to force the number of waveform data points to be a power of two. External applications implementing FFT algorithms can directly read files containing data in quantity of powers of two.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
... with delay	Preserves the horizontal delay that you added in the Digital Oscilloscope Dialog Box.
HyperLynx .LIS	Saves the waveform to the HyperLynx *.lis* format.

Measurements Dialog Box

To access: From the Interactive Simulation with Measurements Dialog Box, select **Measurements.**

Use this dialog box to set up the simulation details for the SI and crosstalk measurements that you enable on the Interactive Simulation with Measurements dialog box.

Fields

Field	Description
Measurements interval Area	
Start cycle	Specifies the simulation cycle at the beginning of the measurement region. The value must be an integer from 1 to 300. Specify a sufficient number of cycles for the driver and receiver circuits to stabilize and reach normal operating conditions.
End cycle	Specifies the simulation cycle at the end of the measurement region. That value must be an integer from 2 to 301, and must be larger than the Start cycle value.
Edge	Specifies the type of waveform transition to use for the measurements: Both, Falling, or Rising.
Simulation run time Area	
Auto for required number of cycles	Automatically determines the simulation run time based on the following: • Sufficient time to simulate all the measurement cycles specified in the Measurements interval Area. • Sufficient time for net to settle to a reasonably steady state, taking into account the total flight time.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Fixed	Runs simulation for the time value that you enter. The value must be a positive real number from 0.001 to 6500 ns. Note: There is no interaction between this option and the simulation run time specified in the Simulation duration Area in the Interactive Simulation Dialog Box.
Crosstalk analysis settings Area	
Analysis type Area — Chooses the algorithm used to enable drivers on aggressor and victim nets, and switch drivers on aggressor nets during crosstalk simulation.	
Total crosstalk only - drivers automatically selected	Switches all aggressor nets during crosstalk simulation. For algorithm details, see Crosstalk Simulation Algorithm for Total Contributions.
Total plus individual contributions - drivers automatically selected	Runs both of the following types of simulation: • Switches all aggressor nets in crosstalk simulation. For algorithm details, see Crosstalk Simulation Algorithm for Total Contributions. • Switches individual aggressor nets in crosstalk simulation. For algorithm details, see Crosstalk Simulation Algorithm for Individual Contributions. For an example of how this option works on a victim net and its two aggressor nets, see Example Results for Total Plus Individual Contributions Crosstalk Algorithm.
Victim logic states Area	
High Low Tri-state (if legal)	Specifies the constant state of the victim net (high, low, high-impedance) that enables crosstalk simulation (when the driver supports that state).
Stimulus for neighboring nets Area	
Passive	The behavior depends on which of the following options you enable in the General batch simulation or the Interactive Simulation with Measurements Dialog Box: • Delay and signal integrity—Sets all drivers on neighboring aggressor nets to the high-impedance state. • Crosstalk—Sets drivers on neighboring aggressor nets to the state defined by the Independent option.
Switch with	Drives neighboring aggressor nets with the same stimulus as the selected net.

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Field	Description
Switch against	Drives neighboring aggressor nets with the opposite stimulus as the selected net.
Independent	The stimulus applied to neighboring aggressor nets depends on the type of stimulus you enable in the Interactive Simulation with Measurements Dialog Box: • When you enable Global stimulus, drives neighboring aggressor nets with the same stimulus as the selected victim net. With global stimulus, this option and the "Switch with" option behave the same. • When you enable "Per-net/pin stimulus," drives neighboring aggressor nets with the stimulus assigned in the Assign Stimulus Dialog Box.

Crosstalk Simulation Algorithm for Total Contributions

When you enable "Total crosstalk only - drivers automatically selected," crosstalk simulation uses the algorithm below to enable drivers on aggressor nets and simulate crosstalk on victim nets. All aggressor nets switch for each simulation in step 3. The software performs the following:

1. For all neighboring aggressor nets:
   a. Selects and enables a driver based on switching speed. For details, see "Crosstalk Simulation Algorithm for Individual Contributions" below.
   b. Assigns stimulus to the driver based on options that you enable in the Stimulus Area of the Interactive Simulation with Measurements Dialog Box.
2. Sets the selected victim net to the first logic state that you enable in the "Victim logic states" area of the Measurements Dialog Box.
3. Simulates the circuit.
4. If needed, repeats steps 2—3.

Crosstalk Simulation Algorithm for Individual Contributions

When you enable "Total plus individual contributions - drivers automatically selected," crosstalk simulation uses the algorithm below to enable drivers on aggressor nets and simulate crosstalk on victim nets. One aggressor net switches for each simulation in step 4. The software performs the following:

1. Runs crosstalk simulation using the algorithm described in "Crosstalk Simulation Algorithm for Total Contributions" above.

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2. Sets the driver on the selected victim net to the first logic state that you enable in the "Victim logic states" area of the Measurements Dialog Box.

   When a selected victim net has more than one driver, simulation automatically enables the driver with the fastest switching time, which is most likely to create the largest reflections. If all the drivers have the same switching time, simulation enables a random driver.

3. For the first neighboring aggressor net:

   a. Selects and enables a driver.

      When a neighboring aggressor net has more than one driver, simulation automatically enables the driver with the fastest switching time, which is most likely to create the most crosstalk. If all the drivers have the same switching time, simulation enables a random driver.

   b. Assigns stimulus to the driver based on options that you enable on the General batch simulation Crosstalk page or Interactive Simulation with Measurements Dialog Box.

   c. For the other neighboring aggressor nets, disables all drivers that can be set to high-impedance state.

4. Simulates the circuit.

5. If more than one neighboring aggressor net exists, repeats steps 3—4.

6. If more than one victim net logic state is enabled, repeats steps 2—5.

Example Results for Total Plus Individual Contributions Crosstalk Algorithm

When you enable "Total plus individual contributions - drivers automatically selected" and there is one victim net (clk) and two aggressor nets (rhrs1 and rsec1), batch simulation runs the following three simulations:

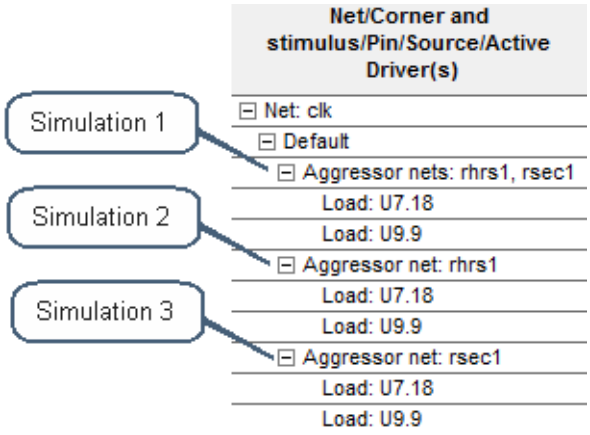
1. rhrs1 and rsec1 both switch. This is the total crosstalk measurement.

2. rhrs1 switches and rsec1 is high impedance. This is the first individual contribution measurement.

3. rsec1 switches and rhrs1 is high impedance. This is the second individual contribution measurement.

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The Simulation Results Dialog Box reports the results:



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Edit Net Classes Constraints and Stimuli Dialog Box

To access: **Setup > Net Classes and Constraints**

Use this dialog box to set up constraints for General Batch simulation.

Objects

Object	Description
Constraints Tab	
Show inherited values	Checked, shows inherited values in green text. A value assigned to a net class is inherited by all nets in the class unless you override the inherited value with a specific value. The “All” net class includes all nets and cannot be changed or deleted.
Push net selection to	• **Spreadsheet** – Selects the nets in the spreadsheet table to match the selected set of nets in the board viewer. • **Board Viewer** – Selects the nets in the board viewer to match the selected set of nets in the spreadsheet table.
Spreadsheet table of constraint values or stimulus settings	See “Running General Batch Simulation” on page 175 for details on how to edit inherited and specific constraint values or stimulus settings.
Only nets selected in board viewer	Filters the list of nets in the spreadsheet table to only show the nets that are selected in the board viewer.
Set Up Net Classes	Opens the Edit Net Classes dialog box to enable you to create and edit groups of nets (net classes) to select and assign settings to many nets at once. See “Defining Net Classes” on page 69.
Import / Export Constraints	Opens the Import / Export Net Grouping and Constraints dialog box to enable you to import net classes and constraints from a CSV file, a CSE constraints file, or directly from Constraint Manger. You can also export to a CSV file.
Stimulus Tab	
Load Model Info	Gets data from assigned models and displays it in the Models and Edge Rate columns of the spreadsheet table.
Edit Stimulus	Opens the Edit Stimulus dialog box to enable you to specify a custom stimulus.
Import DDRx Stimulus	Enables you to import a list of stimulus assignments from a DDR simulation as a DDRx stimuli list text file.

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Model Connections Dialog Box

To access:

- In BoardSim, open the MultiBoard Project Wizard (**File > New MultiBoard Project**), select **Advanced** as the Connector model, click the **Assign** , then click **Model Connections**.
- In LineSim, click **Add S-Parameter/SPICE Model to schematic** (), place symbol on the Free-Form Schematic Editor, double-click the symbol, then click **Model Connections**.

Use this dialog box to enable the software to automatically detect port connections for an S-parameter model, or create, import, or remove a connection file (.connect) that is associated with a S-parameter or SPICE model.

Fields

Field	Description
Library	Specifies the path to the model library.
Connections	Specifies the path to the connections file.
Status	Displays any connection information errors in the header of an S-parameter file. To recheck the status, click **Reload**.
Preview Connections	Lists connections.
Fltr	Type text to filter the list of connections.
Import Connections	Click to select a .connect file, or an S-parameter model file that contains connection information in the header.
Auto-detect Connections	Enables the software to automatically detect connections between ports in (S-parameter model only)
Save Connections	Click to save a .connect file.
Remove Connections	Click to delete the .connect file that is associated with the model file. **Note:** You must manually remove connection information in the header of an S-parameter file.

New HyperLynx Full-Wave Project Dialog Box

Context: LineSim

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

To access: From the Via Properties Dialog Box - Via Properties Tab, select **HyperLynx Full-Wave Solver** from the 3D EM Modeling list, click **New**.

Use this dialog box to start creating a new HyperLynx Full-Wave Solver project file (*.v3d*) project file based on defaults or an existing project file.

The project file contains:

- Geometric information — Such as padstack names and differential via separation
- Simulation set up information — Such as the number of ports and simulation frequency range

Fields

Field	Description
Project File	Specifies the location of the project file.
Use Template Area	
Default	Creates a new project file, based on hard-coded defaults.
File	Specifies an existing project file whose data you want to include in the new project file.

Options for New Terminators Dialog Box

Scope: BoardSim

To access: Generate Back-Annotation File/Data Dialog Box, check “New terminators” and click **Next**.

Use this dialog box to specify reference designator and part type information for Quick Terminators that you include in back annotation.

When you include new terminators (Quick Terminators) in your back-annotation data, this dialog box automatically becomes part of the back-annotation process. The information in this dialog box is necessary because your layout program or schematic editor requires that the back annotation data contain the reference designator and part type for each new terminator component.

This dialog box can assign only a single part type each to all new resistor and capacitor components. When your design requires multiple component types, this limit causes at least one of the components to have the wrong part type. In this case, you manually edit the part type attribute in your layout program or schematic editor.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

How to Avoid Redundant Terminating Components

Quick Terminator components are stored in the *.BUD* file and are present in the back annotation data. After back-annotating Quick Terminator components to your layout program or schematic editor, the next board file you create contains real terminating components in place of the virtual ones created by Quick Terminator. Now you have potentially redundant terminating components in your design; the real ones in the board file and the virtual ones in the *.BUD* file.

To prevent redundant terminating components when reading in the new board file, uncheck Quick Terminators in the Restore Session Edits dialog box. BoardSim then ignores the new terminating components described in the *.BUD* file.

Padstack Editor Dialog Box

Scope: LineSim

To access: **Setup > Padstacks** to open the Padstack Manager dialog box, then click **New** or select a padstack and click **Edit**.

Use this dialog box to set or edit pad and anti-pad properties for padstacks.

Fields

Field	Description
Layer span	Defines the ends of the via barrel. <top> and <bottom> represent the outer layers of the board, no matter the actual layer names. To model a surface mount device pad, specify the same surface layer in both the From and To fields, and check the SMD box.
Drill	Defines the diameter of the via barrel. • Finished Diameter — Specifies the inside diameter of the via barrel after plating. • Actual Diameter — Specifies the outside diameter of the via barrel, which is (Finished Diameter) + (2 * plating thickness). The “Via Barrel Plating” thickness option on the **Default Padstack** tab of the Preferences dialog box specifies the plating thickness.
Board	Lists a board in a multiple-board project. This read-only option appears only for a padstack in a schematic created by exporting a net in a BoardSim MultiBoard project to LineSim.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
<Layer table>	Lists the layers, and the pad and anti-pad properties on each layer. Click **Add** to add a layer to the table. To remove a layer, select a layer row and click **Delete**. • Pad Shape — <None> means either that the layer has no pad, or that the layer has a pad with the same diameter as the drill size. • Anti-Pad Shape — <Auto> assigns the same shape as the pad shape and uses the Clearance value from the **Default Padstack** tab of the Preferences dialog box. • Clearance — When you select Clearance for the Anti-Pad Shape, denotes the minimum clearance between the pad and the anti-pad. This value overrides the clearance value set on the **Default Padstack** tab of the Preferences dialog box.
Edit Stackup (Not available when you open this dialog box from the Via Properties dialog box.	Opens the Stackup Editor, which enables you define the board stackup.
Draw proportionally	Checked, displays the layers showing proportional thicknesses.
Fit to window	Checked, displays the entire via(s).
Use layer colors	Checked, displays the layers using colors defined in the Stackup Editor.

Related Topics

Editing a Padstack Definition

Padstack Manager Dialog Box

Scope: LineSim

To access: **Setup > Padstacks**

Use this dialog box to manage the padstacks in the schematic.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Padstack Name	Lists the padstacks in the schematic. • To rename a padstack, select a name and type in a new name. • To add a new padstack, click **New** to open the Padstack Editor, which enables you to define the properties of the padstack. • To edit a padstack, select a padstack from the list and click **Edit** to open the Padstack Editor. • To copy an existing padstack, select a padstack from the list and click **Copy**. Rename the padstack as desired. • To remove a padstack, select a padstack from the list and click **Delete**.
Type	Lists the via type.
Layer Span	Lists the top and bottom stackup layers of the via barrel.
Drill Size	Lists the finished diameter of the via barrel.
Pad Size	Displays the pad shape and either the pad size for the <default> layer or the largest pad size of all the layers in the padstack. Pad shapes are identified as follows: • D — Round pads. • # x # — Square or rectangular pads. For example, 70 x 80. • # o # — Oval and oblong pads. For example, 70 o 80.

Related Topics

Editing a Padstack Definition

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

PDN Model Extractor Wizard

Use this wizard to set up and export a power-distribution network (PDN) to an S-parameter model.

To access:

- From LineSim, **Export > Model > PDN & Channel Model**

Wizard Page	Description
PDN Model Extractor Wizard - Check Capacitor Models Page	Use this page to review and edit decoupling capacitor model assignments.
PDN Model Extractor Wizard - Choose Easy / Custom Page	Use this page to choose between default and custom export options.
PDN Model Extractor Wizard - Control Frequency Sweep Page	Use this page to edit simulation frequency range and sampling options, both of which affect simulation run time and the resolution of the exported S-parameter model.
PDN Model Extractor Wizard - Customize Settings Page	Use this page to enable detailed PDN model extraction options.
PDN Model Extractor Wizard - Normalization Impedance Page	Use this page to specify the normalization impedance for the exported S-parameter model.
PDN Model Extractor Wizard - Select Signal Vias Page	Use this page to select signal vias to include as ports in the exported S-parameter model.
PDN Model Extractor Wizard - Start Analysis Page	Use this page to start a new export session or load the settings for a saved export.

PDN Model Extractor Wizard - Check Capacitor Models Page

To access:

- From LineSim, **Export > Model > PDN & Channel Model**

Use this page to review and edit decoupling capacitor model assignments.

For a description of options on this page, see Assign / Edit Capacitor Model Dialog Box.

Related Topics

Exporting a PDN to an S-Parameter Model

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

PDN Model Extractor Wizard - Choose Easy / Custom Page

To access:

- From LineSim, **Export > Model > PDN & Channel Model**

Use this page to choose between default and custom export options.

Fields

Field	Description
Easy	Enables popular options on other wizard pages. Some enabled options become read-only.
Custom	Makes all options on all wizard pages available for you to edit.

Related Topics

Exporting a PDN to an S-Parameter Model

PDN Model Extractor Wizard - Control Frequency Sweep Page

To access:

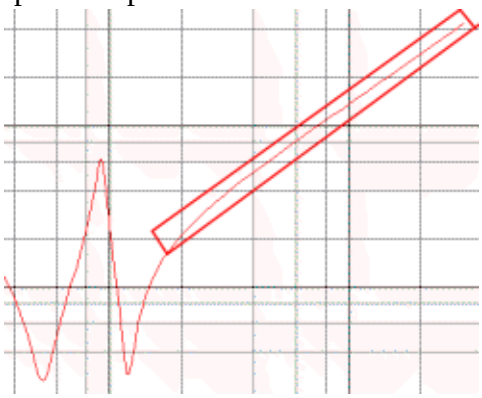
- From LineSim, **Export > Model > PDN & Channel Model**

Use this page to edit simulation frequency range and sampling options, both of which affect simulation run time and the resolution of the exported S-parameter model.

Fields

Field	Description
Min frequency Max frequency	Specifies the frequency range of the simulation and exported S-parameter model. Many ICs have in-package decoupling that provide the main decoupling effects above a certain frequency, such as 300 to 350 MHz. This means decoupling capacitors and buried capacitance located in the PCB contribute little or no decoupling above this design-dependent frequency.
Adaptive sampling	Varies the sampling step size depending on model characteristics. The adaptive scale is better than logarithmic and linear because it increases the sampling rate near frequencies with resonances.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Logarithmic sampling	Distributes sampling points across the frequency range at logarithmic intervals. The intervals between sampling points are smaller at lower frequencies and larger for higher frequencies. With logarithmic sampling, every next frequency point is equal to the previous value times a factor K > 1.
Linear sampling	Distributes sampling points across the frequency range at equal intervals.
Accuracy at resonances	For adaptive sampling, specifies the relative accuracy at resonant frequencies. For lumped analysis, enabling the High option may still yield reasonably fast simulation run times. For distributed analysis, you should take the complexity of the design into account. If the design has a large number of power supply nets, hundreds of decoupling capacitors, and hundreds or thousands of stitching vias, enabling the Low option provides preliminary results with decreased analysis run time. After evaluating the preliminary results, you can identify which frequency ranges interest you the most and try running analysis with higher accuracy on each range of interest.
Minimum number of samples in flat, non-resonant regions	For adaptive sampling, specifies the number of samples the software applies to flat, non-resonant regions of an impedance profile. The figure below shows a flat and non-resonant region of an example impedance profile. 
Number of samples	For logarithmic and linear sampling, specifies the number of samples the software applies to the entire frequency range.

Related Topics

Exporting a PDN to an S-Parameter Model

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

PDN Model Extractor Wizard - Customize Settings Page

To access:

- From LineSim, **Export > Model > PDN & Channel Model**

Use this page to enable detailed PDN model extraction options.

Extracting models with different sets of enabled and disabled options can help you determine how individual types of design properties contribute to PDN model impedance.

Fields

Field	Description
Include capacitor mounting inductance	Specifies whether to account for decoupling capacitor mounting inductance.
Enable stitching-via optimization	Specifies whether to reduce simulation run time and memory consumption by automatically finding stitching vias that are located close together, and then merging their individual models into an equivalent model. See Stitching-Via Optimization.

Related Topics

Exporting a PDN to an S-Parameter Model

PDN Model Extractor Wizard - Normalization Impedance Page

To access:

- From LineSim, **Export > Model > PDN & Channel Model**

Use this page to specify the normalization impedance for the exported S-parameter model.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Normalization impedance	Specifies the normalization impedance for the exported S-parameter model. Models with different normalization impedances can make them harder to understand visually when they are displayed in the Touchstone Viewer. For example, if you export two S-parameter models for the same differential via pair, but specify 25 ohms normalization impedance for one model and 50 ohms for the other, the S-parameter data may look different. For an example, see Figure 10-36. Note: Specifying the exact normalization impedance is not a simulation problem because simulators produce identical results whether the S-parameter model has a 50-ohm impedance or another impedance.

Related Topics

Exporting a PDN to an S-Parameter Model

PDN Model Extractor Wizard - Select Signal Vias Page

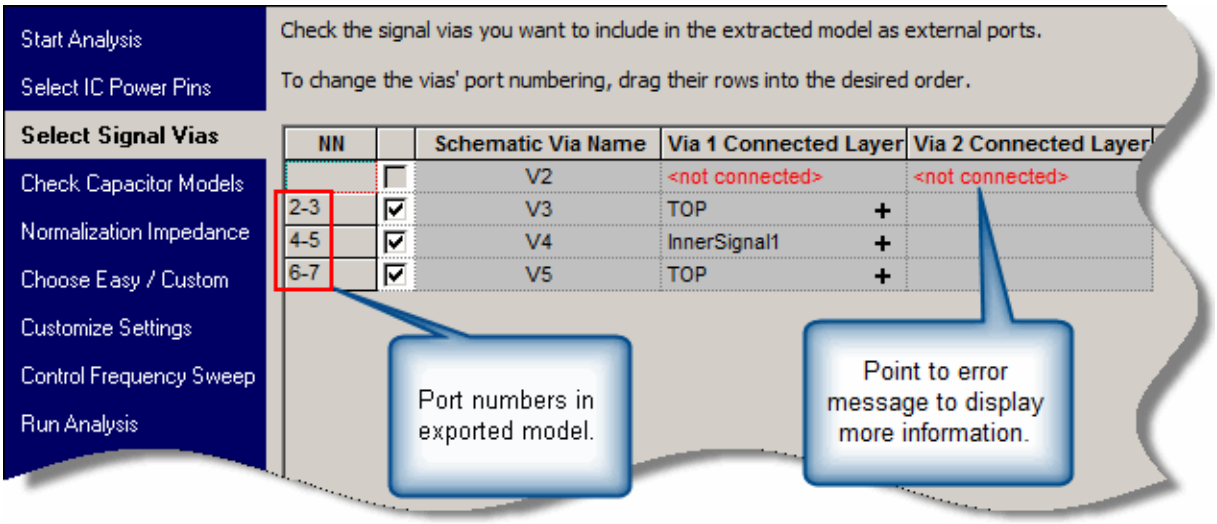
To access:

- From LineSim, **Export > Model > PDN & Channel Model**

Use this page to select signal vias to include as ports in the exported S-parameter model.

For schematic designs, an individual member of a differential pair is modeled as a single-ended via, unless the differential via symbol in the PDN Editor connects symmetrically to transmission-line symbols.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.



Fields

Field	Description
NN	Displays port numbers for the exported model. Port numbering on this page resumes port numbering started on the “Select IC Power Pins” page. To renumber ports, drag one or more rows to the correct location. As you drag, a red horizontal line appears. When you release the mouse button, the dragged rows move to the spreadsheet row below the red line.
Select Net	Enables controls on the toolbar and board viewer to select a signal net (and dim other nets), to help you find the signal via to export. (Available in BoardSim.)

Related Topics

Exporting a PDN to an S-Parameter Model

Selecting Nets for SI Simulation

PDN Model Extractor Wizard - Start Analysis Page

To access:

- From LineSim, **Export > Model > PDN & Channel Model**

Use this page to start a new export session or load the settings for a saved export.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Use last configuration	Selected, the wizard uses settings from the current software session. (Available when you have opened and closed the wizard in the current LineSim session.)
Load saved configuration	Selected, enables you to open a wizard settings file (*.DAO*).
Save settings to file	Checked, saves settings to a wizard settings file (*.DAO*). The default file location is the <design> folder. See Design Folder and HyperLynx Files.

Related Topics

Exporting a PDN to an S-Parameter Model

PDN Net Manager Dialog Box

Scope: LineSim

To access: From the PDN Editor toolbar, select **Net Manager** 

Use this dialog box to view, add, and delete power supply nets.

Fields

Field	Description
Net	Lists the available net names, which are either exported from the software or created by you.
Voltage	Lists the voltage associated with the net name as exported from the software. Note: The values listed in this column are for reference only. The voltage values displayed in the PDN Net Manager are not used for simulation purposes.
<new>	Adds a new spreadsheet row. Select **<new>**. A row is added to the table with a placeholder net name and voltage value. Edit the net name and voltage value, and press **OK**.

Related Topics

Creating a Schematic Design

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Pin Group Manager Dialog Box

Scope: BoardSim

To access: **Models > Assign Power Integrity Pin Groups**

Use this dialog box to group IC power-supply pins together, making it possible for decoupling analysis to view PDN impedance through the pins in the group in parallel.

Note

Each component can have two or more pin groups. For example, one pin group could consist of VCC pins (primary net) and GND pins (reference net) and another pin group for the same component could consist of GND pins (primary net) and VCC pins (reference net).

Fields

Field	Description
Filter row	Displays objects that you specify by either: • Selecting an object name from a list. • Entering text that exactly matches the object name. Text is case sensitive.
Create Groups	Automatically creates pin groups. Does not overwrite an existing pin group.
Create Options	Opens the Auto-Create Groups Options box.
Right-click in first column	Displays options for a pin group or component, including: • Show Component — In BoardSim only, zooms and pans to the component in the board viewer. • Edit Group — Opens the Edit Pin Group dialog box, where you can specify a set of power-supply pins in the group, a model, and a reference net. **Note:** Advanced distributed decoupling simulation supports only the shorted model. • Auto-create Groups — Automatically creates groups by using information from IBIS models (if assigned) and power-supply nets. • IBIS Group — Automatically creates pin groups by using information only from the IBIS model assigned to the component. • Delete Group — Deletes a pin group. **Note:** To recreate a deleted pin group with the Create Groups button, you must first delete all pin groups for the component.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Nets columns	Displays the primary and reference nets for the pin group. You can select a reference net from the list. The reference net column displays N/A when there is more than one reference net for the primary net. The software does not simulate a pin group with N/A in the reference net column.

Related Topics

Creating Power Supply Pin Groups

Auto-Create Groups Options Dialog Box

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Power Network Dialog Box

To access: From the Edit Power Supply Nets dialog box, click **Power Network**.

Use this dialog box to verify that the software recognizes power and signal nets correctly, visualize power net connections in the PDN, assign power integrity models, or begin setup for batch DC drop simulation.

Fields

Field	Description
Power- supply nets area	
Select	Select All, None, or Matched. Matched select the nets found by a search.
Search	Supports the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character.
Table of nets and voltages	Checked, identifies the net as a power-supply net.
Remove from power-supplies	Removes the checked nets from the set of recognized power supply nets. The software considers removed nets to be signal nets.
Connected components area	
Select	Select All, None, or Matched. Matched select the nets found by a search.
Search	Supports the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character.
Table of components	Checked, identifies the net connected to the component as a power-supply net.
Connected nets area	
Select	Select All, None, or Matched. Matched select the nets found by a search.
Search	Supports the following wildcard characters: • * — Matches any number of characters • ? — Matches any one character.
Table of connected nets	Checked, identifies the net as a power-supply net.
Add to power-supplies	Adds the checked nets to the set of recognized power supply nets.
PI Models	Opens the Assign Power Integrity Models dialog box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Power area	Shows a visual representation of the PDN, including power supply nets and connected components. Click a net or component to assign a PI model.
Run Batch DC Drop	Opens the Batch DC Drop Simulation dialog box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Preferences Dialog Box

Use this dialog box to view and edit properties that affect how the software operates and helps you set up a working environment that best suits your design and the way you work.

Properties specified in this dialog box apply to all designs.

Tab	Description
Preferences Dialog Box - Advanced Tab	Use this dialog box to specify advanced simulation options that affect which algorithms the tool uses during simulation.
Preferences Dialog Box - BoardSim Tab	Use this dialog box to specify BoardSim-specific preferences for vias, net handling, crosstalk, and default trace separation.
Preferences Dialog Box - Simulators Tab	Use this dialog box to specify how the software works with ADMS, HSPICE, and HyperLynx Advanced Solver simulators.
Preferences Dialog Box - Default Padstack Tab	Use this dialog box to set the initial properties for the <default> layer of new padstacks created in the Padstack Manager dialog box in the schematic editor.
Preferences Dialog Box - Default Stackup Tab	Use this dialog box to set the properties of new layers created in the Stackup Editor and the stackup properties for a new schematic.
Preferences Dialog Box - General Tab	Use this dialog box to define general signal-integrity simulation settings and board temperature settings.
Preferences Dialog Box - LineSim Tab	Use this dialog box to specify LineSim-specific preferences.
Preferences Dialog Box - Oscilloscope Tab	Use this dialog box to specify the default properties used by the Digital Oscilloscope dialog box.
Preferences Dialog Box - Power Integrity Tab	Use this dialog box to specify options for power-integrity simulations.

Preferences Dialog Box - Advanced Tab

To access: **Setup > Options > General** tab

Use this dialog box to specify advanced simulation options that affect which algorithms the tool uses during simulation.

Caution

Siemens EDA recommends against changing these options except under special circumstances. Contact Siemens for technical support before changing the default settings.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
BoardSim	
Inform users of zero-length segments	Checked, reports a warning when trying to simulate a net that contains a zero-length routing segment in a board design (which can result from a PCB-layout tool that does not clean up its database). Since zero-length segments rarely cause a problem, it is almost always advisable to leave this option unchecked.
Treat test points as IC pins	Checked, enables you to assign IC models or Quick Terminators to test points on your board. Test points are not available until you check this option and reload the board. Unchecked, the software filters out test points at board-load time. EZwave always probes test points, whether you check this option or not. By contrast, you cannot attach oscilloscope probes to test points unless you enable this option and assign an IC model to them.
HYP-file CURVE records	
Linearize curves with radii <	Specifies the minimum radius of a curve that is preserved as a curve (not converted to a line) when the software loads the board. This option works on both valid and invalid curves. Replacements take place in memory and do change the *.hyp* file. Specify 0 to disable this option.
Don't load boards with invalid curves	Checked, the software does not load the board if it contains one or more invalid curves. See "CURVE Subrecords with Invalid Coordinates" on page 495.
Convert invalid curves to lines	Checked, converts invalid curves to lines. Replacements take place in memory and do not change the *.hyp* file. See "CURVE Subrecords with Invalid Coordinates" on page 495.
Load boards with invalid curves	Checked, keeps invalid curves and loads the board design. See "CURVE Subrecords with Invalid Coordinates" on page 495.
For EMC, ignore traces shorter than	Specifies the minimum trace segment length to include in EMC simulation. If you set this option to a length that is longer than most of the trace segments in a trace that contains many tight and repeated turns, the omission of the short trace segments can cause the predicted radiation to be too low for that trace.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
For Crosstalk, ignore coupling regions shorter than	Specifies the minimum length of a coupling region to include in crosstalk simulation when you enable the electrical coupling threshold. This results in a performance improvement with no accuracy penalty (unless the value is increased to something too large) because short coupling regions contribute virtually no crosstalk but can consume significant CPU power during simulation.
Differential Via Search Padstack Size Factor	Specifies the maximum distance between two vias below which the software simulates the vias differentially and enables the Via Visualizer to display both vias. The distance you specify is a multiple of pad diameter between the two via centerlines.
General	
Combine line segments where possible	Checked, the software combines adjacent trace segments that have the same impedance into a single, longer segment. This optimization produces a faster transient simulation.
High accuracy field solver	Checked, forces the field solver to run with higher-than-normal spatial resolution. The default, non-high-accuracy mode, provides excellent accuracy for nearly any problem on which the simulation would normally run. However, in rare cases (such as extremely narrow trace separations), you may prefer the higher-accuracy mode. Note that high-accuracy mode runs substantially slower than normal mode, so enable only when absolutely necessary.
Use field-solver cache	Checked, the field solver uses a cache to prevent recalculating cross sections that it has already encountered and analyzed. Such repeated cross sections occur frequently during crosstalk analysis on a given design. Unchecked, disables the cache, which can seriously degrade field solver performance. Since there is rarely any reason to disable the cache, it is advisable to not uncheck this option.
Assume distant ground if there are no plane layers	Checked, when plane layers do not exist in the stackup, the field solver assumes that a distant plane layer exists during its calculations. If you use double-sided or flexible boards with no plane layers, checking this option eliminates the need to manually create a “fake” plane layer in the stackup to enable simulation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Do not delete generated simulation netlists	Checked, retains the SPICE netlist and simulation run files that the software creates when: • Exporting an S-parameter model for a schematic or net. • Running ADMS or HSPICE simulations. The Simulation Messages dialog box usually reports the line numbers of the SPICE netlist or simulation run file in simulation errors or warnings. Unchecked, the software deletes these files when the export process completes. The software writes these files to the same folder as the schematic or board file. See “Design Folder and HyperLynx Files” on page 520.
Enable AMI DLL string editing	Checked, enables AMI DLL string editing by displaying **Edit Tx AMI DLL String** and **Edit Rx AMI DLL String** buttons on the SERDES Wizard Endpoint 1 and Endpoint 2 pages.
Settle Series Capacitors Before Every Simulation	Checked, assigns a settled voltage to a series capacitor on a signal net. The settled voltage represents a voltage that would result by driving a net with both of the following conditions: • a transmitter/driver with symmetrical pullup/pulldown circuits • a long sequence of balanced stimulus, such as an oscillating signal with a 50% duty cycle Enabling this option means that simulation can obtain a settled voltage for a DC blocking capacitor on a SERDES channel without having you run a very long time-domain simulation or assign a special ADMS or SPICE model to the differential pair. The software runs two extra DC solutions prior to running transient simulation. The first DC solution is based on normal stimulus and the second DC solution is based on an inverted stimulus. The settled voltage is the average capacitor voltage for the two DC solutions. **Note:** Advanced Eye Diagram analysis (in the SERDES Wizard) automatically assigns a settled voltage to series capacitors, even when this option is unchecked.
Limit non-HyperSim features to	Checked, specifies the maximum number of threads that HyperLynx features other than HyperSim can use.
IBIS Models Processing	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Strip V-t and I-t non-switching time, keeping all V-t and I-t tables in a model time correlated	Checked, removes the initial non-switching time from the V-t and I-t waveform tables in IBIS models, while keeping all V-t and I-t tables in the models time-correlated. However, you should leave this option unchecked under almost all circumstances because the software is able to process fast clock pulses correctly even when initial delay is not removed. For details about the main steps in the algorithm, see “Initial Delay Removal Algorithm” on page 1283. This option edits the tables only in memory and does not modify the model file. To remove non-switching time from waveform tables in the IBIS model file, use the Visual IBIS Editor. See “Removing Initial Delays from IBIS Models” on page 1282.
Use lumped representation of IBIS package parasitics	Checked, the software models IBIS package parasitics as lumped elements. Simulation run time increases when you check this option. Check this option only if you have a specific reason to model with lumped elements, such as results from a special correlation study or the package routing is not implemented as transmission lines (such as for wire-bonded packages). Unchecked, the software models IBIS package parasitics as either equivalent transmission lines or coupled transmission lines. If you have a flip-chip package that is constructed like a miniature PCB, uncheck this option. See “Modeling Package Parasitics” on page 641.

Related Topics

Setting Up the Software

Preferences Dialog Box - BoardSim Tab

To access: **Setup > Options > General**, select the **BoardSim** tab

Use this dialog box to specify BoardSim-specific preferences for vias, net handling, crosstalk, and default trace separation.

Fields

Field	Description
Vias	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Synthesize missing pads	Checked, under certain conditions, the software automatically creates pads of the specified diameter when they are missing from the board file.
Pad diameter	Specifies the diameter of pads automatically created when the synthesize missing pads option is checked.
Synthesize missing drill holes	Checked, if the *.HYP* file contains padstacks with no drill holes, the software automatically synthesizes the missing drill holes.
Drill-hole diameter	Specifies the diameter for missing drill holes.
Net Handling	
Assume net is a power supply if...	Specifies the number of capacitors and vias for automatically identifying power supply nets. If this identification is ever wrong, and you do not want to change the threshold, you can remove the misidentified net from the power-supplies list using the power supply net editor. See Edit Power-Supply Nets Dialog Box.
Remove redundant metal from a board's nets as the board is loaded	Checked, cleans all signal nets, eliminating redundant metal and combining overlapping structures when possible into fewer large structures at board-loading time. Mentor Graphics generally recommends that you not check this option so that your designs load faster. The software calculates net lengths when you open a board. If you uncheck this option, the net lengths displayed in the Select Net by Name dialog box include the effects of redundant metal.
Crosstalk options	
Maximum number of aggressor nets	Specifies the maximum number of aggressor nets per crosstalk simulation.
Default trace separations	
Trace to trace	Specifies the default test trace separation value here. This value is used if the *.HYP* file and *.BUD* file do not contain test trace separation for stackup layers. **Note:** Setting the value here also changes the Trace to trace value on the **LineSim** tab.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Trace to plane	Specifies the default trace-to-metal-area clearance, when the *.HYP* file contains anti-pads, but does not specify a clearance. This field is one of several sources of trace-to-metal-area clearance values. If you set the clearance value in the Setup Anti-Pads and Anti-Segments Dialog Box, this value is no longer used because it has lower priority. **Note:** Setting the value here also changes the Trace to plane value on the **LineSim** tab.
3D Areas	
Replace 3D Area with corresponding S-parameter model during SI analysis	Checked, the software models any defined 3D areas with S-parameter models generated by 3D EM simulation (**Export > 3D Area**).
Crop Area Fills	
Don't crop area fills with board outline (Allegro boards with complex board outline) Need to reload board	Specifies whether to avoid cropping or deleting area fills that have a board outline property. Reload your board to see the effects of changing this option value.

Related Topics

Setting Up the Software

Preferences Dialog Box - Simulators Tab

To access: **Setup > Options > General**, select the **Simulators** tab

Use this dialog box to specify how the software works with ADMS, HSPICE, and HyperLynx Advanced Solver simulators.

Fields

Field	Description
ADMS options	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Use native HyperLynx version	Selected, uses the limited version of ADMS that is automatically installed with the software. This simulator supports SPICE models containing text encrypted for the Eldo simulator. This simulator does not support the following: • Transistors or diodes located in the plain text (non-encrypted) portion of an Eldo model. • Waveform probes in user-supplied SPICE models. • User-supplied VHDL-AMS models.
Use Full ADMS	Selected, uses the full version of ADMS that supports unencrypted SPICE models and SPICE models containing text that is encrypted for the Eldo simulator. You must install the full ADMS license to use this option.
ADMS model syntax type	• Native Eldo— Select this option when your design uses SPICE models with Eldo-specific syntax. • HSPICE compatible (including Eldo encrypted)— Select this option when your design uses SPICE models with HSPICE-specific syntax. The models can contain text encrypted for Eldo. Note that you cannot simulate nets with a mixture of native Eldo syntax models and HSPICE-specific syntax models.
HSPICE options	
MultiCPU	Checked, enables HSPICE to use more than one CPU. Select any number of CPUs up to the capacity of your licenses. This option is available only for HSPICE. Older versions of HSPICE that support one CPU simply ignore this option.
Found at	Specifies the location of the HSPICE simulator software that you have enabled. The path is based on the following environment variables set by SPICE simulation software installation: • HSPICE (Windows)—the path is *%installdir%\bin\hspice.exe* • HSPICE (Linux)—the path is *$installdir/bin/hspice*
SPICE-deck file extensions	
SPICE-deck file extensions	Specifies the list of SPICE model filename extensions that the software recognizes. The Select IC Model dialog box uses this list when you assign models.
3D HyperLynx Advanced Solvers	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
External version is installed in	Specifies the location of the HyperLynx Advanced Solvers software that you install and license separately from HyperLynx SI/PI. For example: *C:\MentorGraphics\HL_AdvancedSolvers_6.1.0\SDD_HOME\Nimbic*
Preferred solver	Specifies the Full-Wave solver version that the software uses for 3D EM simulation.

Related Topics

Setting Up the Software

Preferences Dialog Box - Default Padstack Tab

To access: **Setup > Options > General, select the Default Padstack** tab

Use this dialog box to set the initial properties for the <default> layer of new padstacks created in the Padstack Manager dialog box in the schematic editor.

Fields

Field	Description	Default Value
Pad		
Shape	Selects the pad shape: • Round • Rectangular • Oval • Oblong — An oblong pad is rectangular with rounded corners.	Round
Width	Specifies the horizontal width of pad. For round pads, this is the diameter.	24 mils
Height	Specifies the vertical height of the pad. This option is disabled for round pads.	24 mils
Drill		
Finished hole size	Specifies the diameter of the drill hole after plating.	13 mils
Differential Pairs		
Separation	Specifies the distance between the centers of the via barrels.	75 mils
<Auto> Anti-Pads		

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description	Default Value
Clearance	Specifies the distance between the pad and automatically-created anti-pad. This value is applied when you select <Auto> in the Anti-pad Shape column of the spreadsheet in the Padstack Editor Dialog Box.	10 mils
Via Barrel Plating		
Enable per-padstack via plating	Checked, the software supports padstack-specific plating values. For information about specifying per-padstack plating and via current constraints, see "Accounting for Per-Padstack Plating and Constraints" on page 137. Unchecked, the software applies global plating values (specified by other options in this section of the dialog box) to all padstacks.	Unchecked
Padstack Plating Settings (Available when you have enabled the "Enable per-padstack via plating" option)	Opens the **Plating/Constraints** tab on the Backdrilling/ Plating/Constraints Settings dialog box, where you can specify design-specific and padstack-specific plating and constraint values. See "Accounting for Per-Padstack Plating and Constraints" on page 137.	
Thickness	Specifies the thickness of the metal plating that forms the wall of a via barrel. This option is unavailable when you check the "Vias are conductively filled" option.	1.25 mils
Vias are conductively filled	Checked, the via barrel is completely filled with metal. This option can affect DC drop simulation. See "Running DC Drop Simulation" on page 198.	Unchecked
Metal	Specifies the type of metal for plating via barrels. Select "Custom" to specify a metal type with a unique resistivity.	Copper
Resistivity	Specifies the resistivity in Ohm-meters of the plating metal for via barrels. This option is read-only unless you select "Custom" in the Metal list.	

Related Topics

Setting Up the Software

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Preferences Dialog Box - Default Stackup Tab

To access: **Setup > Options > General, select the Default Stackup** tab

Use this dialog box to set the properties of new layers created in the Stackup Editor and the stackup properties for a new schematic.

Fields

Field	Description
Test Trace Area	
Width	Default trace width. On a board design, you can manually edit trace width. See Editing Trace Widths.

Related Topics

Setting Up the Software

Preferences Dialog Box - General Tab

To access: **Setup > Options > General**

Use this dialog box to define general signal-integrity simulation settings and board temperature settings.

Fields

Field	Description
IC-Model Voltage References	
When assigning a model to an IC pin, always use the model's internal values	Selected, connects the IC pin to a virtual net that is set to the internal voltage specified in the IC model. Select this option unless you have a compelling reason to not use it. Selecting this option ensures the power supply voltage does not fall outside the voltage range specified for buffer and clamp data in the IC model.

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Field	Description
When assigning a model to an IC pin, use a power supply net connected to the IC	Selected, connects the IC pin to an external power supply net that the software automatically identifies. If the software cannot identify the power supply net, it connects the IC pin to a "virtual" net that is set to the internal voltage specified in the IC model. **Note:** The IC-model power supply voltage options on the General tab do not apply to SPICE models. SPICE models always connect to an external power supply net whose voltage is specified in the Edit Power-Supply Nets dialog box.
When simulating, vary voltage reference values with IC corners	Checked, allows the software to simulate using the power supply voltages specified in an IBIS model for various IC corners. Unchecked, simulation uses the power supply voltage specified in an IBIS model for only the Typical corner. If a power supply voltage value is not available, the software uses the voltage value specified for the connected net in one of the following dialog boxes: "Set Power-Supply Voltages and Nets" (schematic design) or Edit Power-Supply Nets Dialog Box (board design). This option does not apply to the following: • ICs with assigned SPICE models. • ICs with assigned IBIS models that you specified to use external power supply nets. • Batch simulation behavior. See IC Operating Settings.
Analysis options Area	
Board temperature	Specifies the temperature that the field solver uses to calculate transmission line resistance.
SPICE temperature	Specifies the temperature value in any SPICE netlist created by the software. The software uses this value to write a .TEMP statement to the SPICE netlist. This value does not have any effect on transmission line resistance calculations made by the field solver.
Use in SPICE simulations	Checked, writes the SPICE temperature value to any SPICE netlist read by the software. For example, if the SPICE temperature is 27 C and you select the ADMS simulator in the Digital Oscilloscope Dialog Box, the SPICE netlist contains the following statement: `.temp 27.000000`
Default Driver Characteristic	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Rise/fall time	Specifies in nanoseconds the time in which the default driver IC switches high and low (0%-100%). If the rise and fall times differ, enter the faster of the two. Note that for board designs, this field and the Rise/Fall Time field on the Default IC Model Setting dialog box (**Setup > Coupling Thresholds, Change Default IC Model** button) are linked. Editing either box automatically updates the value in the other box. The software uses the default rise/fall time for several types of analysis. The Terminator Wizard uses this information when evaluating nets for signal-integrity violations when a driver model is not assigned to the nets. The Via Visualizer uses this information to calculate the knee frequency (Fknee) and other electrical properties of a via.

Related Topics

Setting Up the Software

Edit Power-Supply Nets Dialog Box

Preferences Dialog Box - LineSim Tab

To access: **Setup > Options > General, select the LineSim** tab

Use this dialog box to specify LineSim-specific preferences.

Fields

Field	Description
Default trace separations	
Trace to trace	Specifies the default trace-to-trace a separation to use when you add a transmission line to a coupling region. **Note:** Setting the value here also changes the Trace to Trace value on the **BoardSim** tab.
Trace to plane	Specifies the default trace-to-plane separation to use when you add a transmission line to a coupling region. **Note:** Setting the value here also changes the Trace to plane value on the **BoardSim** tab.

Related Topics

Setting Up the Software

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Preferences Dialog Box - Oscilloscope Tab

To access: **Setup** > **Options** > **General** > **Oscilloscope** tab

Use this dialog box to specify the default properties used by the Digital Oscilloscope dialog box.

Fields

Field	Description
Operation — Controls the way waveforms display in the oscilloscope.	
Standard	Selected, displays waveforms over the full simulation time.
Eye Diagram	Selected, displays waveforms over the bit interval by cutting up the waveform into bit-interval lengths and overlaying them. Use an eye diagram to analyze a SERDES channel.
Driver Waveform — Selects the type of waveform stimulus the driver applies.	
Edge / Osc	• Edge — Select to set up an edge stimulus. • Osc — If the oscilloscope operation is set to Standard, the stimulus is a repetitive clock waveform. If the oscilloscope operation is set to Eye Diagram, the stimulus is toggling.
Rising Edge / Falling Edge	Applies only when you select Edge stimulus.
Stimulus — Specifies stimulus options.	
General	Selected, applies a global stimulus, where you define a single driver waveform and the oscilloscope automatically assigns it to all driver pins. For a board design, the software applies global stimulus to all driver pins on the selected net and its associated nets. For a schematic design, the software applies global stimulus to all driver pins in the schematic.
Per net/pin	Selected, applies per-net/pin stimulus, where you define multiple driver waveforms and manually assign them to specific driver nets (board design) or pins (schematic design). Per-net/pin stimulus enables you to simulate timing relationships among nets/pins, such as the following: • Crosstalk investigations with different waveforms on aggressor nets, to help examine the pattern dependency of crosstalk • Source-synchronous signaling, such as DDRx and similar technologies, where one IC transmits both the clock and data signals with typically slightly different timing. Note that a FastEye diagram does not use per-net/pin stimulus.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Oscillation Parameters — Default oscillator properties. The oscilloscope saves oscillator frequency and duty cycle data on a per-pin basis when you set their values interactively.	
Duty Cycle	Specifies the percentage of the period that the driver is high.
IC Modeling - Controls the default option for IC modeling during simulation in the oscilloscope.	
Slow-Weak	See IC Operating Settings.
Typical	
Fast-Strong	
Show - Selects objects to display in the Oscilloscope window.	
Overview pane	Checked, displays the Overview pane.
Readout text	Checked, displays the following on the main oscilloscope screen: horizontal scale, vertical scale, horizontal delay, and vertical offset values. Uncheck this option to reduce clutter on the main oscilloscope screen.
Eye mask	Checked, displays the eye mask in the display for the oscilloscope.
Mark data points	Checked, affects the size of the data points drawn on the main screen. Typically, the data points are not visible until you zoom way in.
Vertical - Default display position and scale.	
Position	Specifies how far to shift waveforms and the 0.0 V position marker in the main screen up or down relative to the grid. The vertical position creates a voltage offset by adding or subtracting voltage to or from the simulation data. When changing the vertical position, the grids remain stationary while the waveforms and ground marker move up and down.
Scale	Specifies vertical voltage scale in volts per division.
Horizontal — Default display delay and scale. These values affect how the simulation runs.	
Delay	Specifies the simulation time displayed at the left edge of the main screen when simulation completes. For example, if you set this value to 25 ns, the oscilloscope aligns the waveforms corresponding to the simulation time of 25 ns to the left edge of the main screen. For Standard operation, this option affects the waveform display only for the next simulation. It has no effect on the latest waveform. For Eye diagram operation, use this option to center the eye in the oscilloscope. Note that the value range is 0 ns to 100 ns, with a precision of 1 ps. See Effects on Simulation of Horizontal Scale and Delay Settings.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Scale	Specifies the horizontal scale in seconds per division. This setting and the horizontal delay settings together determine the simulation end/stop time. The oscilloscope instructs the simulator to generate data for ten horizontal time divisions plus the horizontal delay that you specify. When you click the Start Simulation or Start Sweeps button, the oscilloscope calculates the number of data points the simulator generates based partly on your horizontal scale and horizontal delay settings. The horizontal scale setting typically does not affect the simulation timestep because it is just one of several factors used by the software to calculate the simulation timestep. However, a very small horizontal scale value can decrease the simulation timestep.
In Standard operation, record data only for probes with check marks	Checked, records simulation results for only the probes you enable prior to starting simulation. You might enable this option to speed up simulation or avoid memory errors caused by storing simulation results for all probes. In standard operation, the oscilloscope automatically assigns probes to all pins in a schematic design or all pins on the selected and associated nets in a board design. If you disable this option, the oscilloscope stores waveform data for all probed pins (even for probes that were disabled prior to starting simulation). which enables you to view their waveform data without re-simulating.
Printing (Also applies to copy to clipboard operations.)	Specifies the location of the legend, in relation to the plot. The auto-detect option puts the legend below the plot when the legend width is more than half of the width of the plot or the legend is too wide to fit beside a plot.

Related Topics

Setting Up the Software

Digital Oscilloscope Dialog Box

Preferences Dialog Box - Power Integrity Tab

To access: **Setup > Options > General**, select the **Power Integrity** tab

Use this dialog box to specify options for power-integrity simulations.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
High-accuracy mode	Checked, makes the simulation grid twice as dense as standard for all types of power-integrity analysis, including time domain (FDTD), frequency domain and DC Drop simulation.
DC drop options	
Separate nets if resistor exceeds	Specifies the minimum value of a resistor connecting two power supply nets that causes the software to simulate the nets separately. Background: In a PCB design, power supply nets can be connected to each other by resistors. These resistors are either small, assuming that both nets actually form a single power supply circuit or the resistors are huge, to prevent DC current from flowing between nets with different supply voltages. Nets connected by a resistor with a value greater than the specified value are simulated separately. Nets with resistors smaller than this value are considered electrically connected and are simulated together.
Frequency- and time-domain analysis options (not DC drop)	
Minimum void size	Specifies the minimum void size that is taken into account during simulation. Background: Generally, the presence of small voids does not significantly affect wave propagation in the planar waveguides, but taking them into account has a large impact on memory and performance. The default value of 120 mils (3 mm) ensures that most small voids (antipads) are ignored. However, if voids are numerous and densely distributed, you may want to decrease the value to, say, 10 mil and see how that affects simulation.
Minimum metal-area size	Specifies the smallest metal shape taken into account during simulation. The size of the metal area is calculated by multiplying this value times itself.

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Field	Description
Default separation between IC power and reference pins	Specifies the default distance between an IC power pin and the ground pin that provides its return current. This value is used when distributed decoupling analysis and signal-via bypass analysis cannot find a pin on the reference net near the pin to which you assigned an AC current sink model. The return current pin must be on the same component and connect to a transmission plane that interacts with the pin with the AC current sink model. Distributed decoupling analysis and PDN S-parameter model exporting use this value to calculate the inductance of the differential portion of the power supply pin mounting vias. In a schematic design, you can override this value. See Add/Edit IC Power Pin(s) Dialog Box.
Automatically assign reference layers	Checked, ensures good simulation correlation when exporting power supply nets from your board design to a schematic (PDN Editor). Unchecked, allows detailed control over which reference layers to include in the PI simulation circuit. This makes the related check boxes located in the Set Reference Net Dialog Box and Add/Edit IC Power Pin(s) Dialog Box available for checking or unchecking. **Note:** Advanced distributed decoupling simulation does not use this option.

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Field	Description
Pin-to-area connection search distance (BoardSim)	For standard distributed decoupling simulation, specifies a distance between a decoupling capacitor pin and the metal area that forms part of the transmission plane, which is used as follows: • If the distance is less than the specified value, the software calculates the impedance of this connection and adds it to the electrical circuit. • If the distance is more than the specified value, the software assumes a "virtual" connection of the pin, at its present location, to the metal area and the related transmission plane. In this case, virtual means that the electrical circuit contains the decoupling capacitor capacitance and not its mounting inductance. This is also true if there is no physical connection between the decoupling capacitor pin and the metal area. This behavior supports the early stages of PCB layout, where the decoupling capacitors are placed, but detailed routing connections have not yet been made. • If the specified value is 0.0, the software calculates the impedance of this connection, no matter how long the distance is. It only uses a virtual connection to model decoupling capacitors that have no connection to the metal area. When the software creates the electrical circuit for AC power-integrity analysis, you can use this option to influence the portion of the circuit that represents decoupling capacitor mounting. Capacitors with virtual connections have approximate mounting inductance and are available to both lumped and distributed decoupling analysis.
Allow AC supply models for non-capacitor components	Checked, displays the Supply Component Models page in the advanced distributed decoupling simulation wizard. You can use this page to assign an RLC, SPICE, or Touchstone model to a series resistor or inductor that connects pins on two power supply nets. **Note:** Use the Assign Power Integrity Models dialog box to assign simple (one value) models to two-pin series components.
Advanced - Plane-noise and co-simulation grid (FDTD)	

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Field	Description
Define Grid	Selects the grid definition for the FDTD simulator. The grid size affects simulator spatial resolution and performance (more resolution = less performance). • Auto — Sets the grid size to a value that the simulator to a default value that is good for most designs. Choosing Auto is strongly recommended. • By Cell size — Sets the grid based on the X/Y size of the cells. • By Dimension — Sets the grid based on the number of cells used to cover the board in the X and Y dimensions. Note that even though you enter the X and Y dimensions separately, the simulator adjusts them to make the cells approximately square because simulation may have problems with geometrically unbalanced cells.

Related Topics

Setting Up the Software

Decoupling Wizard - Supply Component Models Page

QPL-File Editor

To access: **Models > Assign Models/Values by Part Name**

Use the QPL-File Editor to create or edit a *.QPL* automapping file, which assigns models and values to components with specific part names.

Requirement: SPICE and Touchstone models must be assigned interactively to a pin.

Note

If you create a new *.QPL* file, make sure to add the path of new *.QPL* file to the Qualified-Parts-List File(s) field in the Set Directories Dialog Box.

When you open the QPL File Editor, one of the following occurs, depending on the contents of the Qualified-Parts-List File(s) (QPL) field in the Set Directories dialog box (**Setup > Options > Directories**):

- If the field contains no files, the editor starts with a new file. After you add rows to the file and click **OK**, the editor prompts you to specify the name of the file.
- If the field contains only one file, the editor opens it.
- If the field contains multiple files, the Select QPL-file dialog box opens, which enables you to select a file to edit or create a new file.

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To edit a different file or a new file, select **File > Open** or **File > New**.

Fields

Field	Description
Part Type Info area	Use this section to specify part type information.
Part type	Select the component type for model assignment.
Part name	Type the part name.
Description	Type a brief description of the component or a general comment. Comments can contain only printable characters, except for commas, which are field delimiters. Comments can be up to 80 characters long.
Model/value to insert area	Specify a model or value for a part. The contents of this area depends on the Part Type list item you select in the Part Type Info area. Select **By Value** to provide a value, or **By Model** to assign a model.
Part type = IC	To assign an IC model: 1. Select a library. 2. Select a component in the library. 3. Click **Assign Model**. **Note:** To search for a model, click **Find Model**.
Part type = Resistor, Capacitor, or Inductor	If you select **By Value**: • To assign a value to a discrete (two pin) passive component, click **Discrete**, type a value, then click **Assign Model**. • To assign a value to a networked resistor or capacitor, click **Package / network**, select the package from the list, type values, then click **Assign Model**. If you select **By Model**: 1. To assign a model, in the Model/Value To Insert area, click **Select Model**. 2. The Select IC models dialog box opens. 3. Select a library, select a component in the library, and click **OK**. 4. In the Model/Value To Insert area, click **Assign Model**. See “The Connectivity Picture” description in this table. If you want to model package parasitics, include parasitic R, L, and C values in the IBIS model that you assign to the component.

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Field	Description
Package List	The information in the Packages list includes: • Package name • Package shape (SIP or DIP) • Total number of pins on the package The package names in BSW.PAK are detailed, so the shape and number of pins are usually obvious just from reading the name. The packages in the list are populated when BoardSim loads your board from the file *BSW.PAK*. Any additional package definitions you create and save in the file *USER.PAK* display at the end of the list.
The Connectivity Picture	The connectivity picture attempts to show you graphically how the components in a network package are connected. The following points apply to the connectivity picture: • Internal components (resistors or capacitors) are displayed only as little boxes • Package pins are displayed in blue • Connections are displayed in the following colors: • Black for connections between independent pins and component ends • Maroon for connections between common pin #1 (i.e., power supply pin #1) and component ends • Green for connections between common pin #2 and component ends • As much of a package as will fit in the Connectivity area is displayed; if a package is too long, its picture is truncated.

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Field	Description
Part type = Decoupling Capacitor	1. Enter a **Part name** in the Part type info area. 2. Select the type of model to assign: • Simple C-L-R — Enter values for capacitance, ESR (equivalent series resistance), and ESL (equivalent series inductance). If you know the exact amount of ESL, select **ESL by value** and enter the value. If you do not know the exact value for ESL, select **ESL by capacitor size** and select your packaging from the menu or select **<Custom>** and enter the Width and Length of your packaging. HyperLynx computes the ESL based on these dimensions. • Library — Select a library and a model. Use the Assign / Edit Capacitor Model Dialog Box to create decoupling-capacitor libraries. • SPICE — Select a library and a device, and then assign capacitor pin names (numbers) to model node names. • Touchstone — Select a library (but not a device) and assign capacitor pin names (numbers) to model port index numbers. 3. Select **Includes mounting inductance** only if the decoupling capacitor model includes the effects of the via and its connectivity to the capacitor package. This situation happens when the vendor does not separate the capacitor model from how the capacitor package is mounted in the test fixture. 4. Click **Assign Model**. **Note:** If you assign a model by entering the C-L-R values, the values do not display in the Value fields of the QPL-file model assignment section; they are saved to the QPL file in the order you enter them: Capacitance, ESR, ESL. To view the values, select a Part Name and look in the Model/value to insert section of the QPL file Editor or open the QPL file in a text editor.
QPL-file model assignment area	Displays the contents of the *.QPL* file. • To sort spreadsheet rows, click a column header. • To edit an assignment, double-click a row in the lower spreadsheet. Press Ctrl+Z to undo an edit. **Note:** Finding text in the spreadsheet (**Search > Find**) is case sensitive.

Related Topics

Assigning a Model or Value to an Entire Component Using a .QPL File

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

REF-File Editor

To access: **Models > Assign Models/Values by Reference Designator (.REF file)**

Use the REF-File Editor to create or edit a model assignment for a reference designator in your design.

Fields

Field	Description
Design's parts list	Select the reference designator to receive the model or value assignment. The left-most column indicates whether an automapping file has assigned a model or value to the reference designator. A checkmark followed by an R or Q indicates that the model (ICs) or value (passive component) assignment came from a *.REF* or *.QPL* file.
Filter	Use any of the following methods to filter the spreadsheet: • Click a column header to sort the rows • Enter a string in the filter box and click **Apply**. The filter box supports the * (substitute any number of characters) and ? (substitute one character) wildcards. • If needed, check or uncheck the options below the filter box.
Model/value to insert	Specify a model or value for a reference designator. The contents of this area depends on the component type you select in the spreadsheet in the Design's Parts List area.
REF-file model assignment	*.REF* file contents.
Remove	Removes the selected line from the REF-file model assignment spreadsheet. Use Ctrl-Z to undo any remove operations.

The part name is displayed only to help you identify which device or particular reference designator refers to. The software does not use the Part Name data in a design file when mapping reference designators on your board to models in a *.REF* file.

Related Topics

Assigning a Model or Value to an Entire Component Using a .REF File

Reporter Dialog Box

To access: Automatically opens to display simulation results or messages. To manually open the Reporter dialog box select **View > Simulation Reports > [Open | Open Most Recent | <simulation or export type>]**.

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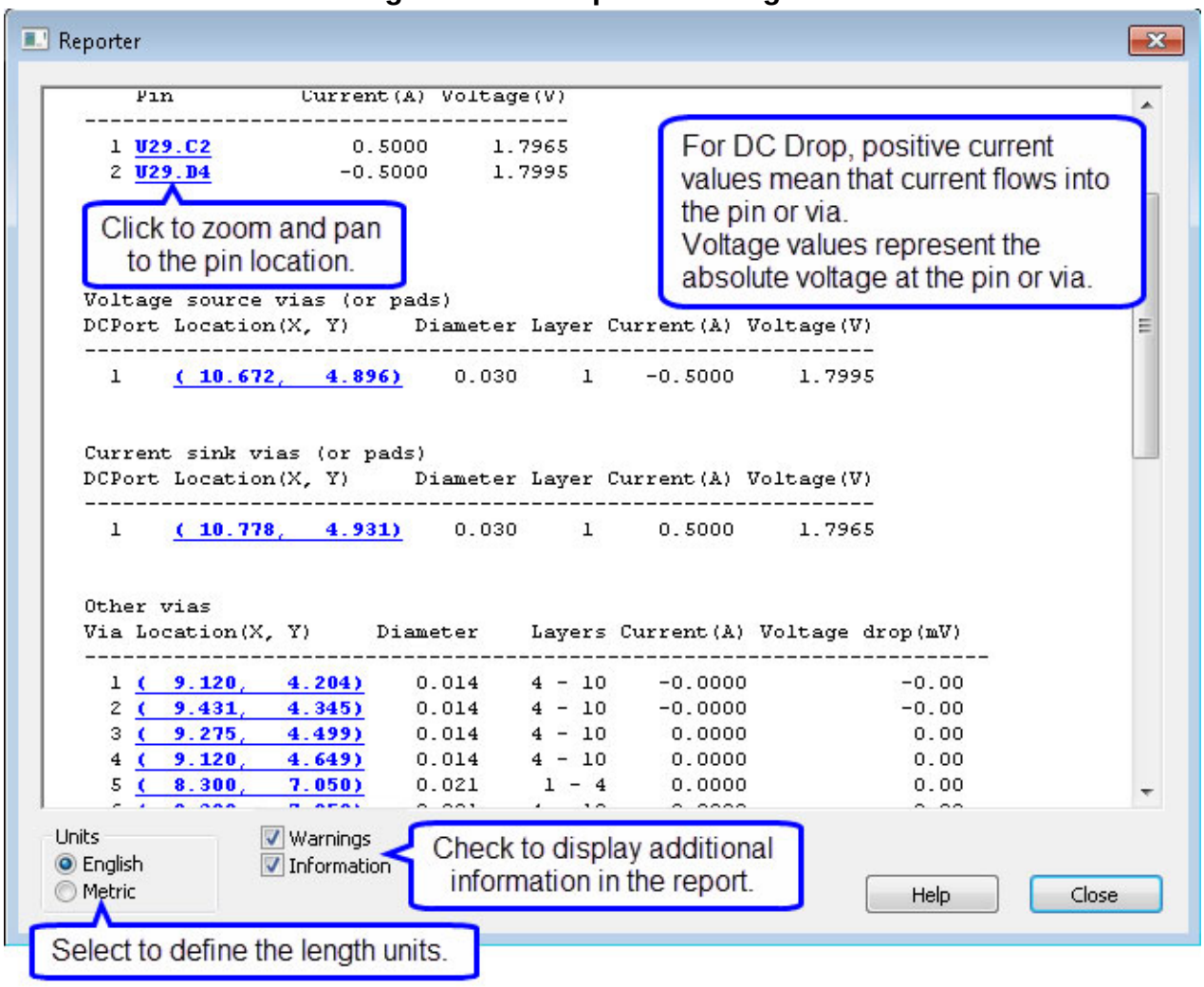
Use this dialog box to display simulation messages and results.

The contents and formatting in the window differ depending on the type of simulation you run, and can display as either a result (*.TXT*) file or a log (*.LOG*) file.

Note

This dialog box automatically looks for report and log files in the <design> folder for the currently-loaded design. You can override this behavior and browse for files located in other folders. See Design Folder and HyperLynx Files

Figure 10-31. Reporter Dialog Box



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Restore Session Edits Dialog Box

To access: Open a BoardSim board that you changed during a previous session.

The software saves any board changes you make during a session, such as IC output buffer enable/disable settings and stackup edits. These changes are saved to the BoardSim User Data session file (*.BUD*). When the board finishes loading, this dialog box opens, allowing you to choose which session file to load with your board.

Description

For a board design, the software saves the last two versions of the session file. The *.BUD* file contains the most recent set of changes. The *.BBD* file (backup *.BUD* file) contains the second most recent set of changes. These files are stored in the same folder as the board file.

Objects

Object	Description
Stackup	Restore stackup definitions, such as geometric or electrical property edits. If you edit and save stackup definitions for a board design that was exported from Xpedition Layout, uncheck this option to restore the original exported stackup data.
Component data	Restore interactive component settings, such as per-pin IC model assignments and passive component value edits.
Manhattan routes	Restore Manhattan routing. **Restrictions**: • The Manhattan Routing check box is unavailable when you clear the Stackup check box. Because you can interactively add a stackup layer and then use that new layer for Manhattan routing, you can only restore Manhattan routing edits when you also restore stackup edits. • Unrouting changes are not saved in the *.BUD* file. For example, if you unroute a net without re-routing it, the unrouting changes do not appear when you reload your board.
Quick Terminators	Restore Quick Terminators.
Stackup Areas	Restore stackup areas for a board design with multiple stackups. If you edit and save stackup areas for a board design that was exported from Xpedition Layout, uncheck this option to restore the original exported stackup area data.
Previous	Load session edits from the most recent session file (*.BUD*).

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Object	Description
Backup to previous	Load session edits from the second most recent session file (*.BBD*). You might want to load the *.BBD* file instead of the *.BUD* file if, for example, you accidentally saved unwanted data from your last session.
Do not show this dialog again for this design	Be sure to set your options carefully before checking this option. Once checked, you cannot change any of the restore options for the loaded board.

Related Topics

Opening a Design

Save MultiBoard Session Edits Dialog Box

To access: Load a multiple board design that uses more than one instance of a board design file > Make unique changes to an instance > **File** > **Save BoardSim Session File**

Use this dialog box to specify how to save changes for a board that is used more than once in a multiple board project.

Fields

Field	Description
How to save	Specifies how to save changes for board design instances: • **Common file for all instances**—Save changes for the selected instance to one *.BUD* file, which applies to all instances of that board. • **Separate file for each instance**—Save changes for each instance to its own *.BUD* file. For example, use this option when you want to save IC driver/receiver settings on a per-instance basis.
Selected instance	Specifies the board design instance with the changes you want to apply all instances. Changes are lost for non-selected instances. (Available when you select **Common file for all instances**.)

Related Topics

Setting Up a Multiple Board Design

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Script Editor

To access: Choose the **Developer** > **VBScript Editor** or **Python Editor** menu item.

Use the VBScript Environment or Python Environment window to write and debug scripts.

Note

Acquire the Scripting license to edit and run scripts.

Fields

Field	Description
Toolbar icons — Hover over an icon to display its description and keyboard shortcut. Many icons are self-explanatory and this table describes a subset of icons.	
Debug script 	Runs a script in debug mode, which is designed to collect and display data to help you locate problems.
Step Into 	If the next statement calls a function, the debugger moves the cursor to the first line inside the function. Otherwise, it runs the next statement.
Step Over 	If the next statement calls a function, the debugger completes the function, and then takes you to the statement that follows the statement that called the function. Otherwise, it runs the next statement.
Step Out 	If the debugger has entered a function, it completes the function. It then takes you to the statement that follows the statement that called the function. Otherwise, it runs the next statement.
Step to Cursor 	Runs a script up to the location of the cursor. In this context, the cursor behaves as a breakpoint.
Tabs **Restriction:** For a computer running Linux, only the **Object Browser** tab is available.	
Watch	Displays values at a breakpoint for watched data when you run a script in debug mode. To watch a variable, type the variable name in the Expression column and then press Enter.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Breakpoints	Displays breakpoints. A conditional breakpoint pauses a script when a condition is true. To add a conditional breakpoint, type a variable name and condition in the Condition column, then press Enter. When you specify a condition, use syntax supported by the script language. For example, if you have a Python script and you want to pause script execution when a variable named TraceWidth equals 8, enter TraceWidth==8.
Call Stack	Displays the sequence of methods and modules called at a breakpoint when you run a script in debug mode.
Object Browser	Displays objects and their methods and properties. You can right-click an item to copy its name or value so that you can paste it into the editor.

Related Topics

HyperLynx SI PI Automation

Scripting Tips and Tricks

Select Active Layers Dialog Box

Scope: LineSim

To access: Click **Select Active Layer(s)**  .

Use this dialog box to select the layer on which you want to place objects in the PDN Editor and to select which layers to display.

Some symbols can exist only on specific layers. For example, when adding copper voids, you must indicate on which plane layers the copper void exists. To accomplish this, specify the active layers before you add the symbol to the PDN or when you select it for editing. For example, to locate a new void area on the stackup layer named TOP, you set the active layer to TOP and then add the void area symbol. The software places copper and void areas on the active layers by default.

Related Topics

Creating a PDN Design

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Select Directories for IC-Model Files Dialog Box

To access: **Setup > Options > Directories**, in the Model-library file path(s) section, click **Edit**

Use this dialog box to specify one or more directories on the computer or network that contain IC models available for simulation.

The software also searches the directories you specify to find connector models (*.SLM* files).

Note

If you add directories or change precedence using this dialog box, you must regenerate the model finder index file by choosing **Models > Generate Model Finder Index**. This allows you to search for IC models, as explained in "Searching for an IC Model in Model Directories" on page 631.

Fields

Field	Description
Directory list	Lists the directories to search for IC model files. Check to include the directory in the search. Uncheck to exclude the directory from the search. The topmost checked file has highest precedence. Use the following buttons to modify the list: • **Add** — Adds a single directory only. • **Add with Subfolders** — Adds a directory and all of its subdirectories. • **Delete** — Removes the selected directory. • **Up**, **Down** — Modifies the precedence of the selected directory.
Add design folder	Checked, adds reference model library files stored in the design folder.
Add design folder subfolders	Checked, adds reference model library files stored in subfolders inside the design folder.
Update subfolders each time design is opened	Checked, refreshes the list of subfolders you previously specified.
Search a maximum of n subfolders	Checked, allows you to specify the maximum number of subfolders per directory to search for IC model files. Note that the software searches subfolders in alpha-numeric order, and there is no way to change search precedence.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Import	Loads model directory information from another project, computer, or design kit. You can: • Browse to the previously-exported model path file. Use this option when transferring library values from one project or computer to another. • Browse to the *BSW.INI* file with a [BSW_LIBRARY] section that contains the model file directories to use. Use this option when transferring library values from one installation of HyperLynx to another. *BSW.INI* is in the *hyperlynx* folder, *C:\MentorGraphics\<latest_release>\SDD_HOME\hyperlynx*.
Export	Saves the model directory settings to a *<exported_model_directory_setting_file>.ini* file. The default export directory comes from the *.hyp* and *.ffs* file path area in the Set Directories dialog box.

Related Topics

Set Directories Dialog Box

Setting Up the Software

Select Directories for Stimulus Files Dialog Box

To access: **Setup > Options > Directories**, from the Stimulus file path(s) area, click **Edit.**

Use this dialog box to specify one or more directories on your computer or network that contain stimulus files (*.EDS*).

Fields

Field	Description
Add	Adds a single directory, but no subdirectories, to the model-search path. The directory is added to the bottom of the list and has the lowest precedence. Clear the check box next to a path to ignore library files in that location.
Add with Subfolders	Adds a directory and all of its subdirectories.
Add design folder	Checked, allows the software to reference any model library files stored in the design folder.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Add design folder subfolders	Checked, allows the software to reference model library files stored in subfolders inside the design folder.
Delete	Deletes the selected directories. If you delete all directories, the default directory is used.
Up Down	Changes the precedence of the selected directory.
Import	Loads stimulus directory information from another project, computer, or design kit. You can do one of the following: • Browse to the previously-exported stimulus path file. This option is useful when transferring stimulus files from one project to another. • Browse to the *BSW.INI* file with a [BSW_LIBRARY] section containing the stimulus file directories to use. This option is useful when transferring library values from one installation to another. *BSW.INI* is located in the *SDD_HOME\hyperlynx* directory.
Export	Saves library paths to a *.INI* file.

Related Topics

Set Directories Dialog Box

Setting Up the Software

Select IC Model Dialog Box

Use the Select IC Model dialog box to display the contents of IC model libraries, to find a model in a library, and to assign a model to a reference designator in the design.

Fields

Field	Description
Selected Device Area	Contains basic information for the selected device, if available in the model, including: • Name of the library, device, signal, and pin • I/O type, such as bidirectional, input-only, output-only, and so on • Switching thresholds for inputs and outputs These values are used in BoardSim to calculate timing delays and are currently unused in LineSim.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Model Selector button	Available when you select an IBIS model with the [Model Selector] keyword.
Notes box	Click to view model information supplied by the author such as author's name and model creation date, the model copyright, and revision history.
Select a Library Device and Signal-Pin Area	Displays the available libraries. Click a library name to filter the list and select a model for the pin.
Find Model	Click to search all libraries for a model.
Select By Area	If available for the selected device, displays either signal or pin information. Click the Pin or Signal option to specify the information type to display.

Related Topics

Assigning a Model to an IC Pin

Select Method of Simulating Vias Dialog Box

Scope: BoardSim

To access: Select **Setup > Via Simulation Method**

The software can automatically calculate via inductance and capacitance or you can manually specify the information. These options enable you to simulate the effects of padstacks in the current PCB layout and to perform "what if" experiments to simulate the effects of changed padstacks.

When you choose to include or exclude via models from simulation, the following capabilities use your choice:

- SI simulation (batch and interactive)
- Terminator Wizard
- Export to SPICE (using the SPICE Writer option)

Via Visualizer results are best suited for the Auto-calculate option. If you use either of the User-supplied options, the Via Visualizer displays approximate via electrical properties based on the values you supply.

Note

The software ignores options in this dialog box when you enable "SI/PI Co-Simulation" in the Digital Oscilloscope Dialog Box or Simulation Controls Dialog Box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Include via L and C Include via C	Check, enables via modeling. You can also enable and disable via modeling with the Enable Via Modeling  button on the toolbar. **Restriction:** The Via Models license is required to enable via modeling. Otherwise, the software can model a via only as a lumped capacitance. To model vias as a lumped capacitance in this case, check the "Include via C" option, and then select **OK**.
Include capacitance of SMD pads	Checked, models SMD pad capacitance. **Restriction:** If multiple vias touch the component-pin pad, the software models only one of them.
Include via stub inductance	Checked, includes the L value from via stubs.
Via-modeling method Area	
Design file	For a MultiBoard project, selects the board to which you apply settings. You must explicitly specify settings for each board in the project.
Auto-calculate	Specifies that the software extracts L/C values from padstack and trace geometry data in the board file. The software generates a via model that takes into account the following geometric properties: • Layers on which connected traces enter and exit the via • Layer positions and sizes of all pads in the padstack • Positions of AC ground planes in the stackup • Size of the antipads separating via barrels from AC ground planes • Fringing capacitance resulting from coupling between the via barrel and AC ground layer In addition, the software takes into account changes in impedance due to differential via pairs. The software does not take into account the reduced impedance due to local decoupling capacitors.
User-supplied global L and C	Allows you to specify one set of L/C values for all padstacks. The L/C you provide represents the L/C for each padstack, and the software converts the L/C into a transmission line with the equivalent Z0 and delay values.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
User-supplied padstack-specific L and C	Allows you to specify L/C values for each padstack. For padstacks that connect two or more signal layers, you can choose to provide L and C yourself or have the software calculate L and C. The spreadsheet displays the calculated L and C that you can override on a per-padstack basis. In the spreadsheet, the calculated L and C are based on the full length of the via barrel. This means the same L/C values are used for all instances of the padstack, regardless of which layers are used for traces entering and exiting the via. The spreadsheet does not contain unused padstacks or padstacks connecting to only one signal layer. You can scan the spreadsheet to identify padstacks with large L or C values. The software does not take into account which layers are used by traces to enter and exit the via. L and C values in the spreadsheet correspond to the full length of the via. To obtain C for the equivalent transmission line representing the padstack, use the following equation: C (padstack transmission line) = C (total) - C (entry pad) - C (exit pad) Where: C (total) represents the value in the C cell of the spreadsheet in this dialog box. C (entry pad) and C (exit pad) represent C of the outside capacitors displayed in the Via Visualizer.

Related Topics

Setting Up the Software

Via Visualizer Dialog Box

Select the Instance Dialog Box

To access: Automatically opens when you open a board design that is used more than one time in a multiple board design, and you have saved changes for each board design instance.

Use this dialog box to select an instance of a board design (and its session edits) to load into the software.

Related Topics

Save MultiBoard Session Edits Dialog Box

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Set Coupling Thresholds Dialog Box

Scope: BoardSim

To access: **Setup > Coupling Thresholds**

Use this dialog box to enable electrical or geometric coupling thresholds, and set coupling threshold options.

Fields

Field	Description
Use electrical thresholds section	
Include nets with coupled voltages greater than	Specifies the minimum coupling voltage to use when locating aggressor nets that couple to the selected victim net. For example, to include an aggressor net that generates more than 150 mV of crosstalk in the selected victim net, enter 150.
Default IC model characteristics	Displays the rise/fall time for the default IC, which is used for any net without a driver IC model.
Change Default IC Model	Opens the Default IC Model Settings dialog box, where you can edit the default IC model. For a description of the "Rise/fall time" option, see "Preferences Dialog Box - General Tab" on page 990.
Use geometric thresholds (advanced) section **Note:** For information about how the software applies options described in this section, see "Horizontal and Vertical Geometric Search Range for Coupled Nets" on page 546.	
Maximum distance from aggressor D	Specifies the maximum horizontal search distance to use when locating aggressor nets for the selected victim net. Measurements start and end at trace centerlines.
Minimum total coupled segment length L	Specifies the minimum coupled trace length to use when locating aggressor nets for the selected victim net. When the selected victim net and aggressor net form multiple *coupling region*s, this option represents the minimum sum of coupling region lengths. Shorter values can increase simulation run time, especially when aggressor nets have many corners or curves.
Horizontal Neighbor Limit	Specifies the maximum number of aggressor nets on either side of each *coupling region* for a selected victim net and its neighboring nets. Nets located directly above or below the selected victim net do not count toward this limit.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Vertical Layer Limit	Specifies the maximum number of metal layers above and below the selected victim net to use when locating aggressor nets. **Note:** The software does not use this option when searching for coupled area fills and routed power-supply traces.
Coupling Settings	Opens the Coupling Settings dialog box.

Related Topics

Accounting for Coupling

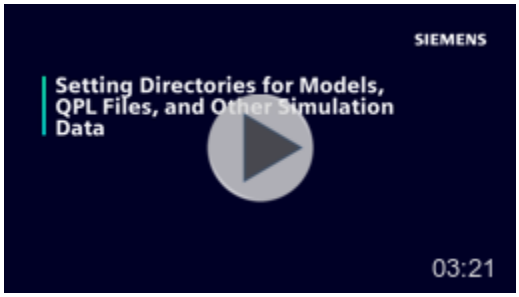
Horizontal and Vertical Geometric Search Range for Coupled Nets

Set Directories Dialog Box

To access: **Models > Edit Model Library Paths or Setup > Options > Directories**

Use this dialog box to set the folder location for designs, models, stimulus, reports, and so on.

The following video contains additional information:



Options

Option	Description
HYP and FFS file path Area	
The design file directory is shared among *.HYP*, *.FFS*, *.CCE* (CADCAM Professional compressed and encrypted), and ODB++ files. The directory setting is a convenience and it specifies the default directory for the Open File dialog box. You can store board and schematic files anywhere you want and in multiple directories and subdirectories.	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Browse	Use to select the directory you want to use as the default. You can also type the directory path in the HYP and FFS file path area. **Restriction:** You must disable Use directory of last-opened file before selecting Browse or editing the path.
Default	Restores the directory to its default value.
Use directory of last-opened file	Select the check box to always use the directory from which you last loaded a design file (rather than a particular “fixed” directory). Clear the check box to modify the directory.
Model-library file path(s) Area	
Edit	Opens the Select Directories for IC-Model Files Dialog Box. Use this dialog box to specify one or more directories on the computer or network that contain IC models available for use. Clear the check box next to a directory or subdirectory path to restrict HyperLynx from accessing library files in that location.
Default	Deletes all existing model directory values and specifies the directory that shipped with HyperLynx.
Add design folder	Allows HyperLynx to reference any model library files stored in the design folder.
Add design folder subfolders	Allows HyperLynx to reference model library files stored in subfolders inside the design folder.
Qualified parts list file(s) (QPL) Area	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Add File	Opens the Choose a File to Add to QPL Path dialog box. The Choose A File To Add To QPL Path dialog box opens. Select a file by double-clicking it, or selecting it and selecting **Open**. The new file path has lowest precedence. You can also add files by typing the *.QPL* file path into the Qualified Part File(s) box. If you specify multiple file paths, use a semicolon (;) to separate them. The software can read multiple *.QPL* files from different directories on the computer or network. For example, you can use the files *\\central_CAD\vendorFiles\master.qpl*, which is maintained by a central CAD group, and *C:\MentorGraphics\<latest_release>\SDD_HOME\hyperlynx\HypFiles\project.qpl*, which supplements or corrects the CAD group's IC mapping for your project. Directory paths can be absolute or relative. The examples above illustrate absolute directory paths. To use the relative path approach, store the *.QPL* file(s) in the directory specified in the Model Library File Path box in this dialog box, then precede the *.QPL* file name with ".\", such as *.\foo.qpl*. Precedence among the *.QPL* pathnames decreases from left to right in the Qualified Part File(s) text box; the left-most *.QPL* pathname has the highest precedence.
Use QPL file(s) to assign models	Enables QPL file automapping.
Stimulus file path(s)	
Edit	Opens the Select Directories for Stimulus Files Dialog Box. Clear the check box next to a directory or subdirectory path to restrict HyperLynx from accessing library files in that location.
Default	Select to set the path to the original value.
Add design folder	Allows HyperLynx to reference any stimulus files stored in the design folder.
Add design folder subfolders	Allows HyperLynx to reference stimulus files stored in subfolders inside the design folder.
Most recently used files Area	
Limit list	Enter the maximum number of files to display in File > Recent Files area. If the box is blank, the default maximum of eight is used.
Reports and log files directory - Specifies the location for report and log files. This capability enables you to use separate directories for designs and outputs from simulation.	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Browse	Select to specify a non-project directory. **Restriction:** The Browse button is unavailable if the "Use Project Directory For Reports And Log Files" check box is selected.
Default	Restores the default directory.
Use project directory for reports and log files	Select the check box to use the same directory that is specified in the .HYP and .FFS file path box. Clear the check box to specify a non-project directory. Then type or browse to the directory.
PCB flow path - Specifies the location of an Xpedition flow release, including the SDD_HOME folder. HyperLynx uses this location when reading constraint data from Constraint Manager.	
From SDD_HOME environment variable	Checked, uses the value of the $SDD_HOME environment variable and displays it in the "PCB flow path" field. If the environment value is not set, the "PCB flow path" field is blank.

Related Topics

Setting Up the Software

Set Reference Net Dialog Box

To access: Open the Assign Power Integrity Models Dialog Box, select the **IC** tab, then click **Assign** in the Reference Net section.

Use this dialog box to specify the net that provides return current paths for the selected power supply pin(s).

Fields

Field	Description
Reference Net list	Defines the net that provides return current paths for the IC power supply pin(s) (selected in the Assign Power Integrity Models dialog box).
Filter	Filters the contents of the Reference Pin list by reference designator. Begin typing a reference designator in the field, and search results automatically appear in the list. The filter is case-sensitive.
Reference Pin List	Defines a pin connected to the reference net that you selected in the Reference Net list.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Set Spectrum Analyzer Probing (EMC) Dialog Box

To access: **Simulate SI > Attach Spectrum Analyzer Probe**

Use this dialog box to set a probe for EMC simulation. The software accounts for differential-mode radiation caused by current flowing on a single-ended signal trace and an ideal return path. The software does not report radiation for differential-pair signal traces.

For schematic designs, you specify a pin at which to set a current probe. For board designs, you specify either a pin at which to set a current probe or a distance at which to set an antenna probe.

Fields

Field	Description
When Probe type is Current...	
Current probe	Specifies a pin when the probe type is current.
Pins	List of available pins to which you can assign a current probe.
When Probe type is Antenna...	
Antenna probe	Specifies a distance between board and antenna when the probe type is antenna.
Antenna and board position	You can let the software automatically determine antenna and board position for maximum radiation, or you can manually specify those positions.
Include radiation from	Specifies the sources of radiation to include in EMC simulation.

Related Topics

Spectrum Analyzer Dialog Box

Setup Anti-Pads and Anti-Segments Dialog Box

Scope: BoardSim

To access: **Setup > Anti-Objects**

This menu item is unavailable for the following conditions:

- BoardSim—You have loaded the design in *.HYP* format and both of the following conditions are true:
  - The *.HYP* file contains no area shapes of type PLANE.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- You have not assigned a power-supply net to any plane layer in the Edit Power-Supply Nets dialog box. See Verifying That Power Supply and Signal Nets are Recognized Correctly.

Note

For MultiBoard projects, the above conditions must be true for all boards.

- BoardSim—You have loaded the design in .CCE or ODB++ format, but have not assigned a power-supply net to any plane layer.
- LineSim—The PDN Editor window is not selected, such as when it is located behind the LineSim schematic window.

Use this dialog box to specify clearances among objects on the same stackup layer. Clearances created by anti-pads in the board file and the PDN Editor are used for accurate power-integrity simulation and PI/SI co-simulation, and for board display.

BoardSim also uses anti-segment information when simulating and displaying boards.

In BoardSim, processing this information for a very large board can noticeably increase simulation and viewing times, depending on the way the *.HYP* file defines plane shapes. You can disable the simulation and display of anti-pads and anti-segments. You might do this temporarily to speed up preliminary investigations and simulations.

Options

Option	Description
Show anti-objects in board viewer and use in analysis (better accuracy, worst performance)	Select to display anti-pads and anti-segments in the board viewer or PDN Editor, and use clearances formed by anti-pads and anti-segments in simulation. In BoardSim, anti-pads are always displayed when the board file contains explicit anti-pad geometry information, even when you clear this check box. In BoardSim, this option links to the Anti-Object option in the View Options Dialog Box, so these two options always have the same value.
Force user-defined clearances	Select to override clearance information contained in the board file or PDN Editor padstack with global values you define. The clearances you specify are applied to all pads and trace segments in the design.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Anti-pad clearance	If you enable Force user-defined clearances, define anti-pad clearances. The initial value comes from the Clearance field in the Preferences Dialog Box - Default Padstack Tab (if you change the value in one tab, it is automatically copied to the other tab). When you save the design settings, the value in this dialog box is saved to the *.PJH* file and overrides the value in the Preferences dialog box.
Anti-segment clearance	If you enable Force user-defined clearances, specify the anti-segment clearance. The initial value comes from the Trace to plane field in Preferences Dialog Box - LineSim Tab or Preferences Dialog Box - BoardSim Tab (if you change the value in one tab, it is automatically copied to the other tab). When you save the design settings, the value in this dialog box is saved to the *.PJH* file and overrides the value in the Preferences dialog box.

Advanced users can override clearances defined in the design. You might do this when running "what if" power-integrity simulations and PI/SI co-simulations to see the effects of clearances between via anti-pads and other metal shapes. The clearance values you provide apply to all pads and trace segments in the BoardSim or PDN Editor design. These clearance values are not used for trace separations in coupling regions in the LineSim schematic editor.

Figure 10-32 shows how the display of an anti-pad in the board viewer changes in response to visibility and clearance value options.

Figure 10-32. Example Anti-Pad Visibility and Clearance Options

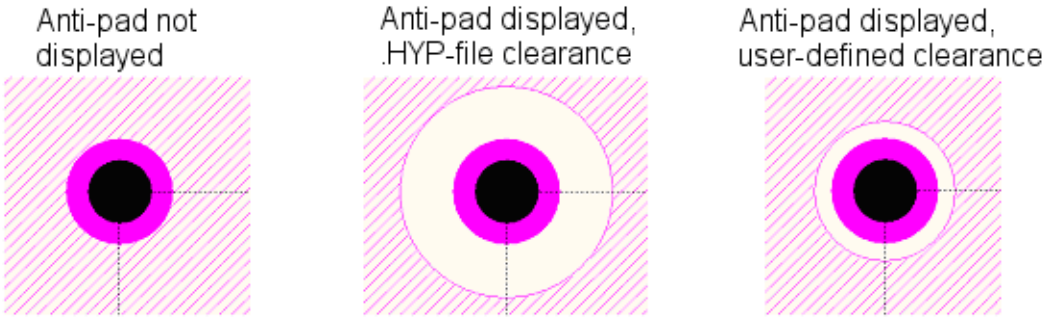
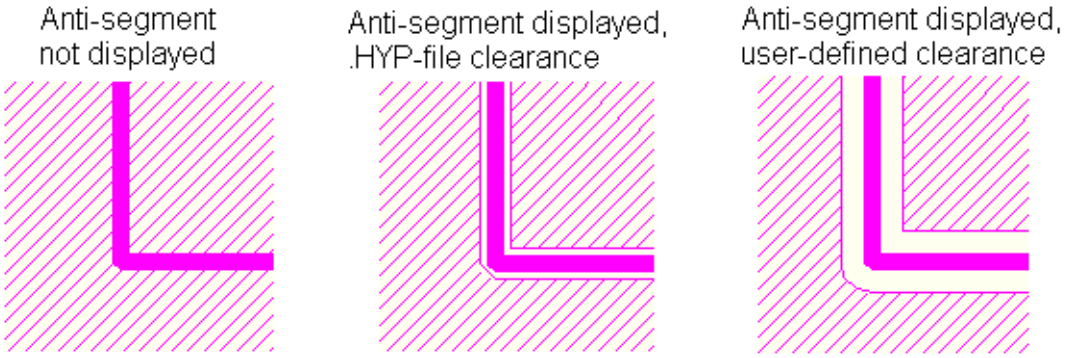


Figure 10-33 shows how the display of anti-segments in the board viewer changes in response to visibility and clearance value options.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-33. Example Anti-Segment Visibility and Clearance Options



Simulation Controls Dialog Box

To access: **Setup > Simulation Controls**

Use this dialog box to specify advanced simulation options for running signal integrity simulations. This includes additional waveform probe locations, simulation engine options, and a way to enable SI/PI co-simulation.

Caution

When you close HyperLynx, options in this dialog box return to their default values.

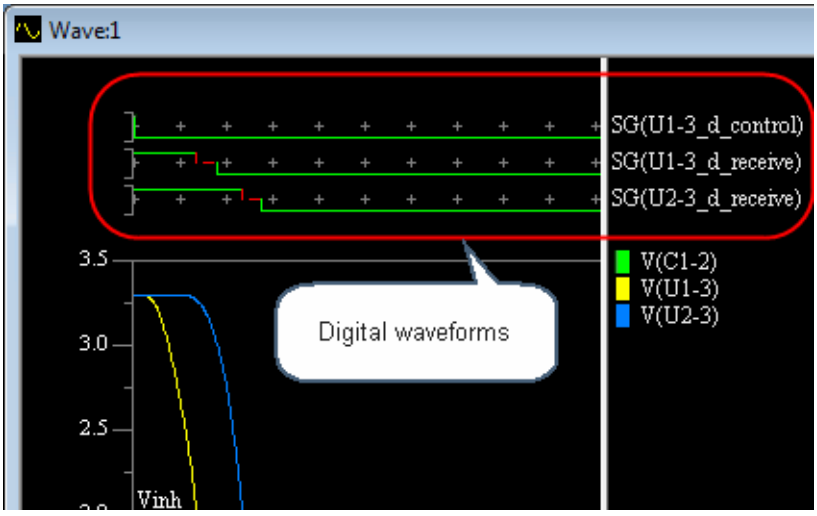
Options

Option	Description
Simulation engine Area	
Note: This setting also updates the Simulator Area setting in the Digital Oscilloscope Dialog Box.	
Auto	Automatically selects the simulator. See Automatic SI Simulator Selection.
Manual	Specifies which SI simulator to use: • HyperSim — Supports IBIS models, S-parameter models and passive SPICE elements. • ADMS — Use the version of ADMS that is enabled in ADMS options on the Preferences Dialog Box - Simulators Tab. • HSPICE — Use for HSPICE-encrypted models You install and license HSPICE separately from HyperLynx. See Supported SI Models and Simulators.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
SI/PI Co-Sim	Enables signal-via models that interact with transmission-plane structures in the design. See Accounting for Noise Between Single-Ended Signal Via and Power Planes in SI Simulation (Co-Simulation). Changing this option also changes the value of the SI/PI Co-Sim option in the Digital Oscilloscope Dialog Box. **Restrictions:** • This option is unavailable when a MultiBoard project is loaded. • This option is unavailable if you set the Manual option to ADMS or HSPICE.
Simulation resolution Area	
Auto	Automatically calculates the simulation time step. For information about the software calculates the time step, see Automatic Time Step Calculation for Time-Domain SI Simulation.
Manual	Specifies the time step for the simulator. The value must be a positive, real, number from 0.001 to 1000 ps. **Restriction:** This option is unavailable when you enable SI/PI Co-Simulation. The power-integrity simulation engine automatically sets the time step for itself and the signal-integrity simulation engine.
Additional waveforms Area	
Restriction: These options do not apply to simulations run by the Digital Oscilloscope Dialog Box.	
Coupled electrical nets	Probes coupled signal nets that exceed the coupling threshold set in the Set Coupling Thresholds Dialog Box.
Buffer currents	Probes buffer current. Positive current waveform values represent current flowing into the buffer.
Vias	Probes the signal via(s) on each stackup layer that connects to the net.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Test waveforms	Probes the standalone driver rise/fall waveforms, where the driver switches into the test load specified by the IBIS driver model [Model] keyword and the following sub-parameters: • Single-ended driver—Rref, Vref, Cref, Cref_rising, Cref_falling, Rref_rising, Rref_falling, Vref_rising, Vref_falling, Vmeas_rising, Vmeas_falling • Differential driver—Rref_diff, Cref_diff, and these optional subparameters: Rref, Vref, Cref **Restriction:** HyperLynx does not generate the test waveform when a SPICE model is assigned to the driver.
Pin and die (overrides IBIS)	Probes both the external pin and die locations for ICs. When this option is disabled and an IBIS model is assigned to the IC receiver, probe at the location(s) specified by the [Component] keyword and its SI_location and Timing_location subparameters. The default value of Pin is used when the subparameter is absent.
Digital waveforms	Probes and displays the logic 1/0 waveform for the following: • Receiver, which is logic1 when the waveform at the receiver is above Vinh and logic 0 when the waveform is below Vinl. **Note:** The receiver thresholds are subject to hysteresis parameters in the [Model Spec] keyword, including the Vinl, Vinh+, Vinh-, Vinl+ subparameters. The receiver thresholds also account for the [Receiver Thresholds] keyword and its subparameters. • Driver stimulus, which is labeled "*d_control" in EZwave.  This option has no effect when the net has only SPICE models because these models do not provide threshold information. Digital waveforms are not plotted for eye diagrams.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
SPICE Options	Opens the SPICE Options Dialog Box. When you have assigned SPICE models, are running full ADMS or HSPICE, and must specify parameters or include files, use this dialog box to provide them.

Simulation Results Dialog Box

To access: From the Interactive Sweeps with Measurements Dialog Box, click **Start Simulation**. This dialog box automatically opens when simulation completes.

Use this dialog box to display delay, SI, and crosstalk measurements for the nets simulated by the Interactive Simulation with Measurements Dialog Box or Interactive Sweeps with Measurements Dialog Box.

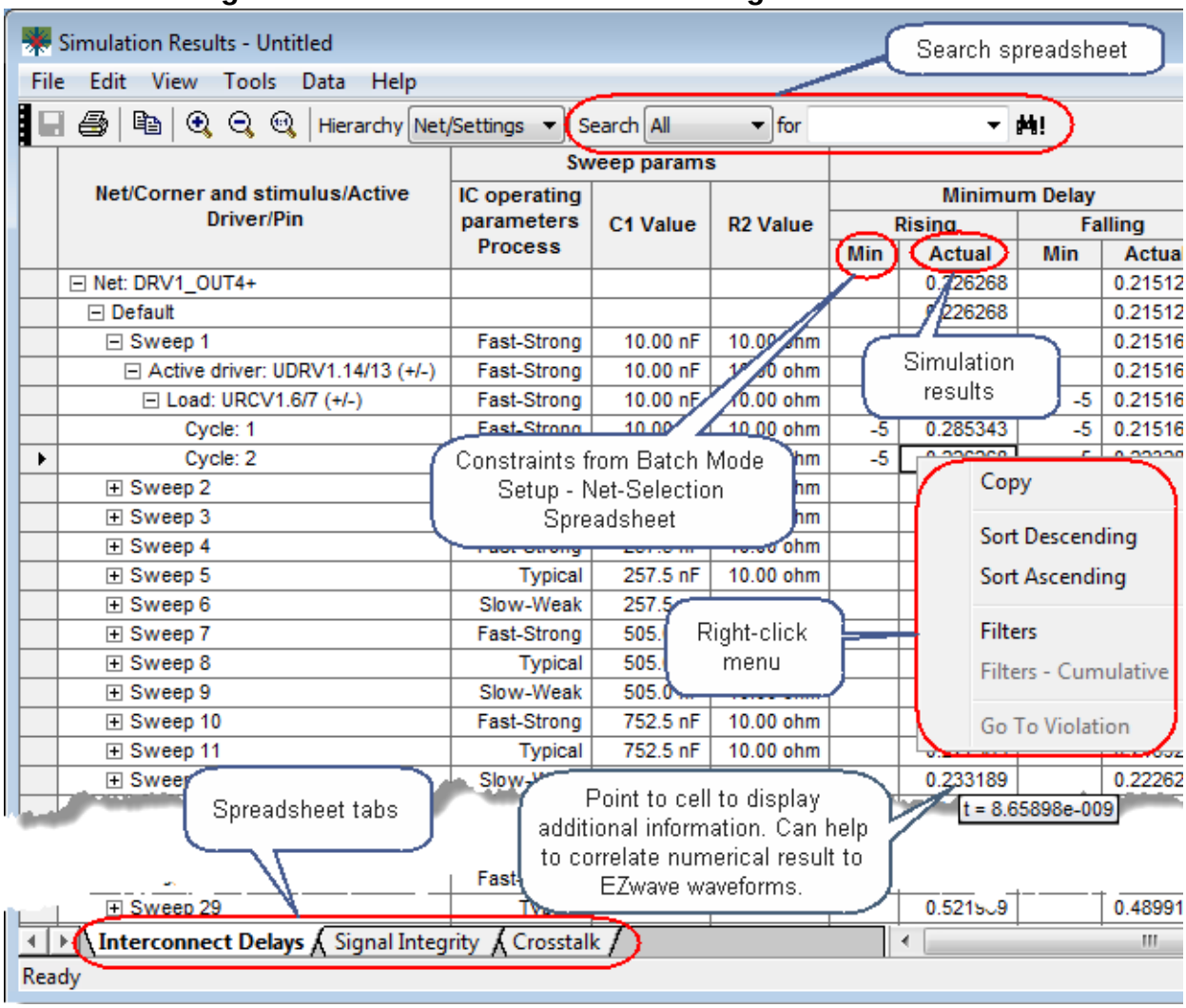
Red cells identify simulation results that exceed the limits that you set in the "Edit Net Classes Constraints and Stimuli Dialog Box" on page 963. Yellow cells identify near-failing simulation results that are within 10% of the limit. A lighter shade of red or yellow indicate that you must expand the spreadsheet tree to display the row that contains the failing or near-failing simulation result.

The asterisk (*) indicates that a critical error prevented the measurement. For example, delay cannot be measured when the receiver input voltage does not reach Vinh or Vinl.

A blank value in a cell indicates the measurement was not applicable, and therefore not taken. For example, when simulation settings are configured to measure only rising-edge delay, falling-edge delays are not reported.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-34. Simulation Results Dialog Box - Overview



This dialog box contains options for the following areas:

- Options - Menus and Toolbar Contents
- Options - Interconnect Delays Tab
- Options - Signal Integrity Tab
- Options - Crosstalk Tab

Note

When you run sweep simulations, the dialog box displays additional columns and rows that are not described here.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Options - Menus and Toolbar Contents

Option	Description
File Menu	
Save Copy As	Saves the proprietary binary *.SRD* file that stores the spreadsheet contents. You cannot use the File menu to re-open the *.SRD* file. To open an *.SRD* file, run the *SimResServer.exe* file located in the hyperlynx or hyperlynx64 folder. You can use this capability to display previous simulation results.
Edit Menu	
Copy	Copies the selected spreadsheet cell or range of cells to the clipboard, so you can paste the values to Microsoft Excel or another application. **Restriction:** Spreadsheet column headings are not copied.
Tools Menu	
Options	Opens the Options dialog box, where you can edit color coding and automatic file save options.
Data Menu	
Filters	Displays a filter row in the spreadsheet. Use filters to hide spreadsheet rows containing measurement data, based on the filter criteria that you enter. Select the filter cell in the spreadsheet column to enter criteria.
Filters - Cumulative	Provides a way to remove spreadsheet rows from the remaining subset of rows that meet the criteria set by Filters. You can set multiple cumulative filters.
Toolbar	
Copy	Copies the selected spreadsheet cell or range of cells to the clipboard, so you can paste the values to Microsoft Excel or another application. **Restriction:** Spreadsheet column headings are not copied.
Zoom	Zooms and restores 100% scale.
Search 	Searches for text in the current spreadsheet tab. Optionally, select the search scope from the list, type a string into the box, and press **<Enter>** or click the icon. Search is case sensitive.

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Options - Interconnect Delays Tab

Min and Max values come from constraints set in the “Edit Net Classes Constraints and Stimuli Dialog Box” on page 963.

Restriction

Max and Min cells are blank for a schematic.

Column Heading	Description
Net/Corner and stimulus/ Active Driver/Pin	Displays the net, driver pin, receiver pin, and cycle for the measurement.
Simulated Delay Time (ns)	
Minimum Delay - Rising	
Min	Displays Min Delay, which is a constraint from the “Edit Net Classes Constraints and Stimuli Dialog Box” on page 963.
Actual	delay = receiver time - driver time Where: • delay is for a rising-edge transition. • receiver time is the first crossing of Vil at the receiver. • driver time is the crossing of Vmeas at the driver test waveform. For measurement details, see Min Delay.
Minimum Delay - Falling	
Min	Displays Min Delay.
Actual	delay = receiver time - driver time Where: • delay is for a falling-edge transition. • receiver time is the first crossing of Vih at the receiver. • driver time is the crossing of Vmeas at the driver test waveform. For measurement details, see Min Delay.
Maximum Delay - Rising	
Max	Displays Max Delay.

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Column Heading	Description
Actual	delay = receiver time - driver time Where: • delay is for a rising-edge transition. • receiver time is the final crossing of Vih at the receiver. • driver time is the crossing of Vmeas at the driver test waveform. For measurement details, see Max Delay.
Maximum Delay - Falling	
Max	Displays Max Delay.
Actual	delay = receiver time - driver time Where: • delay is for a falling-edge transition. • receiver time is the final crossing of Vil at the receiver. • driver time is the crossing of Vmeas at the driver test waveform. For measurement details, see Max Delay.
Worst Case	
Shortest	Minimum of the following measurements: • Minimum Delay - Rising • Minimum Delay - Falling
Longest	Maximum of the following measurements: • Maximum Delay - Rising • Maximum Delay - Falling
Time of Flight (ns)	Displays the delay based only on interconnect properties. This measurement takes into account the length and characteristic impedance of each trace segment between the driver and receiver pins. This measurement ignores the loading effect of driver and receiver models. This value is not used when testing delay constraints.

Options - Signal Integrity Tab

Restriction

Max and Min cells are blank when viewing a schematic.

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Column Heading	Description
Net/Corner and stimulus/ Active Driver/Pin	Displays the net, driver pin, receiver pin, and cycle for the measurement.
Static Overshoot Voltage (V)	
Low	
Max	Displays Static Overshoot Low.
Actual	overshoot = power rail - measurement Where: • overshoot is for a falling-edge transition. • power rail is the low rail voltage. • measurement is the minimum voltage at the receiver. For measurement details, see Static Overshoot Low. Hover over the cell to display the simulation time for the measurement.
High	
Max	Displays Static Overshoot High.
Actual	overshoot = measurement - power rail Where: • overshoot is for a rising-edge transition. • measurement is the maximum voltage at the receiver. • power rail is the high rail voltage. For measurement details, see Static Overshoot High. Hover over the cell to display the simulation time for the measurement.
Dynamic Overshoot	
Low - Voltage (V)	
Max	Displays Dynamic Overshoot Low.

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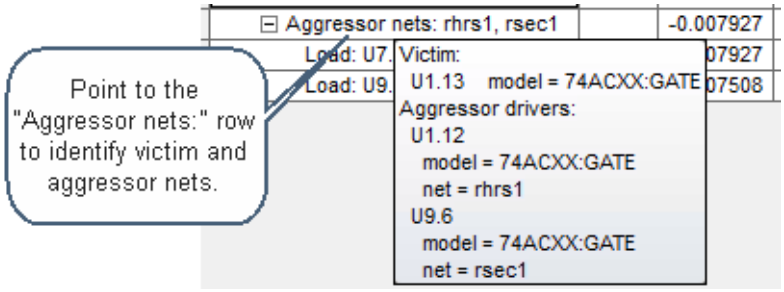
Column Heading	Description
Actual	overshoot = power rail - measurement Where: • overshoot is for a falling-edge transition. • power rail is the low rail voltage. • measurement is the minimum voltage at the receiver. For measurement details, see Dynamic Overshoot Low. Hover over the cell to display the measurement margin. margin = limit - overshoot Where: • margin is for a falling-edge transition. • limit is Dynamic Overshoot Low • overshoot is Actual for the low static overshoot voltage
Low - Time (ns)	
Max	Displays Dynamic Overshoot Time.
Actual	The amount of time the waveform spends below the minimum acceptable static voltage. For measurement information, see Dynamic Overshoot Time. Hover over the cell to display the simulation times for the start and end of the measurement.
High - Voltage (V)	
Max	Displays Dynamic Overshoot High.
Actual	overshoot = measurement - power rail Where: • overshoot is for a rising-edge transition. • measurement is the maximum voltage at the receiver. • power rail is the high rail voltage. For measurement details, see Dynamic Overshoot High. Hover over the cell to display the measurement margin. margin = limit - overshoot Where: • margin is for a rising-edge transition. • limit is Dynamic Overshoot High • overshoot is Actual for the high static overshoot voltage
High - Time (ns)	

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Column Heading	Description
Max	Displays Dynamic Overshoot Time.
Actual	The amount of time the waveform spends above the maximum acceptable static voltage. For measurement information, see Dynamic Overshoot Time. Hover over the cell to display the simulation times for the start and end of the measurement.
Ringback Margin (V)	
Low	
Min	Displays Ringback Margin Low.
Actual	Maximum ringback voltage measured at the receiver. For measurement details, see Ringback Margin Low.
High	
Min	Displays Ringback Margin High.
Actual	Minimum ringback voltage measured at the receiver. For measurement information, see Ringback Margin High.
Non-monotonic	
	“Rising” or “Falling” when a rising or falling transition reverses direction while between receiver thresholds. Figure 8-53 shows a non-monotonicity that is not reported in this column because it does not occur between Vih and Vil. Hover over the cell to display the time of the non-monotonicity with the greatest amplitude.

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Options - Crosstalk Tab

Column Heading	Description
Net/Corner and stimulus/Pin/ Source/Active Driver(s)	Displays the net, driver pin, receiver pin, and cycle for the measurement. Load rows represent input or high-impedance output pins on the victim net. 
Crosstalk (V)	
Logic Low	The driver on the victim net is stuck low.
Logic High	The driver on the victim net is stuck high, when that state is available.
Tristate	The driver on the victim net is high impedance, when that state is available.
Max	Value of the Crosstalk constraint.
Actual	crosstalk = measurement - steady state Where: • crosstalk is for both rising- and falling-edge transitions. • measurement is the maximum deviation of the receiver voltage from its steady state voltage. • steady state is the steady-state DC voltage at the victim receiver. For measurement information, see Crosstalk.

Specify Device Kit for Current Design Dialog Box

To access: Select **Setup** > **Device Kit**

Use this dialog box to choose a specific device kit and specify additional information about the device kit.

Device kits typically contain advanced IC models and design examples that you simulate to learn about the technologies they implement. Device kits are not available in the shipping

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version of the software but are rather shipped and installed separately. See the documentation provided with the device kit for installation and configuration information.

Device kits provide *<device_kit>.INI* files containing model path information and sometimes simulator environment information. The software uses the *<device_kit>.INI* file model path information to automatically edit model library path settings in the Set Directories Dialog Box.

Related Topics

Specifying Device Kits

Specify DFE Dialog Box

To access: FastEye Channel Analyzer - Add Pre-Emphasis/DFE/CTLE Page > check Add DFE, click **Specify taps/weights**

Use this dialog box to specify tap weight values for decision-feedback equalization (DFE) circuitry in the receiver. You can specify tap weights by typing values in the spreadsheet or by reading in a *.TAPS* file containing the tap weights synthesized by a previous run of the wizard.

Fields

Field	Description
Tap	The zero row represents the main tap. The positive-numbered rows represent post-taps. For information about the assumed implementation of the DFE structure, see Pre-Emphasis and DFE Structures.
Weight	Specify values for all taps implemented by the receiver. Note: Specify all tap values for the range. For example, if the receiver implements taps 0, 1, 2 and 3, you cannot omit the values for taps 1 and 2.
Load	Browse to a file (*.TAPS*) containing the tap weights synthesized by a previous run of the wizard.

Specify Pre-Emphasis Dialog Box

To access: FastEye Channel Analyzer - Add Pre-Emphasis/DFE/CTLE Page > check Add driver pre-emphasis, click **Specify taps/weights**

Use this dialog box to specify tap weight values for pre-emphasis circuitry in the driver. You can specify tap weights by typing values in the spreadsheet or by reading in a *.TAPS* file containing the tap values synthesized by a previous run of the wizard.

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Fields

Fields	Description
Tap	Negative-numbered rows represent pre-taps. The zero row represents the main tap. Positive-numbered rows represent post-taps. For information about the assumed implementation of the pre-emphasis structure, see Pre-Emphasis and DFE Structures.
Weight spreadsheet cell	Specify values for all taps implemented by the driver. Note: Specify all tap values for the range. For example, if the driver has the taps -2, -1, 0, 1, 2 and 3, you cannot you cannot omit the values for taps -1, 0, 1, or 2.
Load	Browse to a file (*.TAPS*) containing the tap weights synthesized by a previous run of the wizard.

Spectrum Analyzer Dialog Box

Context: BoardSim and LineSim, for a single board design

To access: **Simulate SI > Run Interactive EMC Simulation**

Electromagnetic simulation (EMC) allows you to measure radiated emissions from a selected net on your board design. You can compare simulation results to government regulation limits, such as FCC for the United States, to help you decide whether to change the design. You can also use a current probe to measure peak current for a net across a frequency range for schematic and board designs.

Use this dialog box to set up and run EMC simulation, and to display results.

Fields

Field	Description
Comment	Specifies text to include when you print, copy, or save waveforms.
Probe	LineSim: Specifies a pin at which to set a current probe. BoardSim: Specifies either a pin at which to set a current probe or a distance at which to set an antenna probe. Click **Set** to open the Set Spectrum Analyzer Probing (EMC) Dialog Box.
Stimulus	Specifies the waveform to apply to the net.

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Field	Description
IC Modeling	Selects the IBIS IC model corner to use during simulation. For information about the combination of min/typ/max corner data in an IBIS IC model, see IC Operating Settings.
Mini oscilloscope display	Shows the time-domain waveform of current at the driver IC.
Display	Specifies results display options for the Spectrum display. Auto Scale does not adjust the horizontal scale. If you select Auto Scale and no radiated-emissions readings are visible, try increasing the horizontal scale.
Spectrum display	Shows the frequency-domain results of the simulation. Note: As you change the settings in the Horizontal area, you may see a hatched region on the right side of the display. The hatching covers the frequency area in which the response has fallen by 3 dB or more, so that data there is not valid.
Horizontal	Central freq — Specifies the frequency value shown at the middle of the X axis of the display. Scale — Specifies the cycles per division on the X axis of the display.
Vertical	Offset — Shifts the 0 dB uV/m line up or down relative to the grid on the display.
Regulations	Specifies one or more sets of governmental EMC regulations to check for compliance against simulation results. Before using this option, you must first set an antenna probe. Click **User** to define your own EMC limits.
Copy to clip	Copies results to the clipboard.
Save as CSV	Saves results in CSV format. When the antenna probe is used, the CSV file contains columns representing frequency versus radiated emission data. When the current probe is used, the CSV file contains columns representing frequency versus current data.
View Points	Saves the spectrum analysis results in text file with content similar to the CSV file but is easier to read.

Limitations

- EMC simulation supports one enabled driver on the selected net.
- EMC simulation using the antenna probe is not available for MultiBoard projects.
- EMC simulation cannot simulate nets with SPICE models.
- LineSim EMC cannot predict component-package radiation.
- LineSim EMC cannot simulate the effect of differently oriented segments.

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- LineSim EMC differential-pair simulations are optimistic because LineSim does not have physical information about the separation between the traces.

SPICE Options Dialog Box

To access: From the Digital Oscilloscope Dialog Box or Simulation Controls Dialog Box, click **SPICE Options**

Use this dialog box to specify SPICE simulation parameters, options, and include files.

Settings in this dialog box apply to the design you have loaded and are saved to the board design session file (*.BUD*) or schematic file (*.FFS*).

Options

Option	Description
SPICE Added Statements Area	
Add Params Edit Params	Add or edit simulation parameter (*.param*) values.
Add Options Edit Options	Add or edit simulation option (*.option*) values.
SPICE Include Files Area	
<value_field>	Displays the paths of previously-added SPICE *.INC* files.
Add File(s)	Select to specify include files with the Select SPICE Include Files dialog box. The Select SPICE Include Files dialog box displays SPICE files from all folders in the model library path. Use the Set Directories Dialog Box to display and edit model library paths. You can use an include file to specify an .IC statement for HSPICE. For more information about using the .IC statement, see the documentation provided with HSPICE.

Related Topics

Exporting a Net to a SPICE Netlist

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Stackup Manager Dialog Box

To access: **Setup > Stackup > Stackup Manager**

For a design that has multiple stackups, use this dialog box to edit the master stackup or a local stackup.

Objects

Object	Description
Edit	Opens the Stackup Editor for the stackup you have selected in the spreadsheet.
New	Opens the Create Stackup dialog box, to create a new local stackup.
Copy	Creates a new stackup, based on the stackup you have selected in the spreadsheet.
Delete	Deletes the stackup you have selected in the spreadsheet.
Common Layers	Opens the Common layers dialog box, which you can use to assign a metal layer from the master stackup to a metal layer for a local stackup. The software electrically connects the layers you have assigned to each other.
Select Areas	Highlights in the board viewer all stackup areas that use the stackup you have selected in the spreadsheet. (BoardSim only.)
Select TLs Vias	Highlights in the schematic viewer all transmission line and via symbols that use the stackup you have selected in the spreadsheet. (LineSim only.)
Preview	Graphically displays all the stackups in the spreadsheet. If you want to change the location of a stackup in the Preview area, click and release its row in the spreadsheet, and drag the cell in the first column up or down.
Show master stackup	Displays the master stackup. Uncheck when your design uses only local stackups and you want to reduce the number of stackups in the Preview area. (Available when you enable Preview.)

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Related Topics

Modeling a Board Design With Multiple Stackups

Statistical Contour Chart Dialog Box

To access: Run SERDES Wizard - IBIS AMI and click View Contour in the Advance Eye tab of HTML results, or FastEye Channel Analyzer - View Analysis Results Page > select **Statistical contours** and run analysis

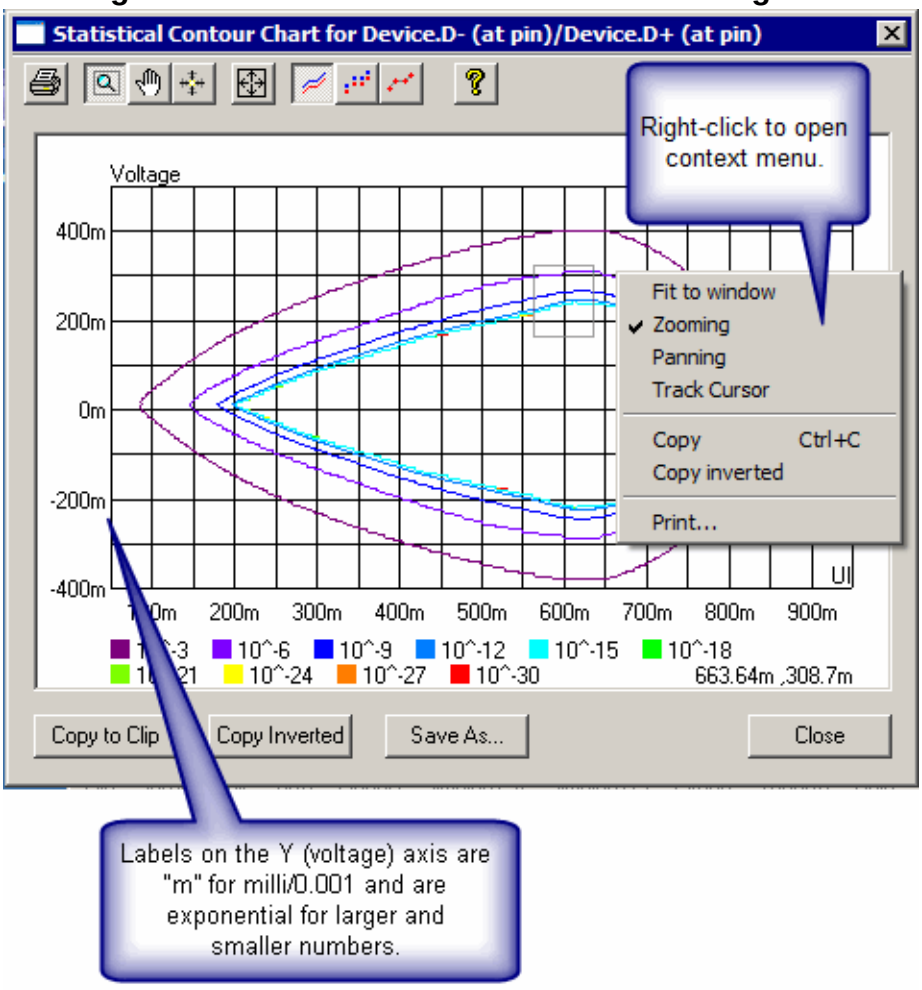
Use this dialog box to display a nested series of eye opening contours and their bit error rate (BER).

The color of the contour indicates its BER. Like bathtub curves, statistical contours indicate the quality of sampling locations across the unit interval (UI, same as bit interval). An advantage of statistical contours over bathtub curves is that the inner-eye contours display both sampling time and voltage information.

The Y axis represents the sampling voltage and the X axis represents the sampling time in terms of the unit interval (UI).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-35. Statistical Contour Chart Dialog Box



Fields

Field	Description
Print (right-click) 	Print the graph with a white background.
Zooming (right-click) 	Zoom in by doing the following: Position the mouse pointer over one corner of the zoom box you want to create, and then drag to define the other corner of the zoom box. Release the mouse button to magnify the contents of the zoom box to fill the graph.
Panning (right-click) 	Pan by dragging the graph across the dialog box.

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Field	Description
Track Cursor (right-click) 	Attach measurement crosshairs to a waveform by selecting the waveform to measure. As you move the mouse horizontally, the measurement crosshairs tracks the selected curve.
Fit to window (right-click) 	Fit the entire curve to window.
	Display only lines between curve vertices (no vertex dots).
	Display only curve vertices (no lines).
	Display both lines and vertex dots.
	Open Help for the dialog box.
Copy (right-click)	Copy graph to the clipboard and use a white background.
Copy inverted (right-click)	Copy graph to the clipboard and use a black background.
Save As	Save the numerical contour data to a file. You can open the file with a spreadsheet application, such as Microsoft Excel.

Sweep Manager Dialog Box - Setup Tab

To access: Choose **Simulate SI > Run Interactive Sweeps** or click  ,  , or 

Use this tab to define the set of design property values (sweep range) to apply to a design property during sweep simulations.

Restriction

When you enable crosstalk for a board design, this dialog box does not identify whether components, such as passive components and ICs, are part of the victim net or an aggressor net. The Assign Models dialog box identifies pins on aggressor nets with a  symbol.

When you load a multiple-board design, the Setup tab displays the board ID (such as B00) for each design property. Sweep values are saved in the schematic *.FFS* file and in the board *.BUD* file.

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Options

Option	Description
Design property tree Stackup layers Power supplies IC modeling Passive components Programmable buffers Transmission lines Coupling regions	Displays the design properties that you can sweep. Expand the tree and double-click a design property to open the Sweeping Dialog Box and specify the sweep range.
Run Sweeps	The dialog box that opens when you click **Run Sweeps**, depends on how you accessed the Sweep Manager. •  opens the Digital Oscilloscope Dialog Box. •  , opens the Interactive Sweeps Dialog Box. •  opens the Interactive Sweeps with Measurements Dialog Box.
Add Range Edit Range	After you select a design property in the "Design property tree", click to open the Sweeping Dialog Box to add a new sweep range or edit an existing sweep range.
Remove Range	Select a design property and click **Remove Range** to delete the existing sweep range. **Caution:** Deleting the sweep range for a reference design property also deletes the sweep ranges for any dependent properties. See the description below for "Paste Range as a Lock". Uncheck a design property to disable, but not delete, sweep ranges for an entire hierarchical branch or individual design properties.
Copy Range	Select a design property and click **Copy Range** to make the existing sweep range available for pasting to another design property of the same type.
Paste Range	Select a design property and click **Paste Range** to paste a previously-copied sweep range to it.
Paste Range as a Lock	Select a design property and click **Paste Range as a Lock** to: • Paste a previously-copied sweep range to it. • Synchronize the sweep values for the copied from (reference) design property and the pasted to (dependent) design property. This ensures that the dependent design property always has the same sweep value as the reference design property.

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Option	Description
Sweep simulations requested	Displays the number of simulations required to perform all of the sweeps.

Related Topics

Parametric Sweeps

Running Signal Integrity Simulation

Sweep Manager Dialog Box - Simulation Cases Tab

To access: For Digital Oscilloscope, click  . For EZwave, click  , or  .

Use this tab to display the combination of design property values for each sweep simulation, to optionally stop sweep simulations if a simulation fails, and report failed simulations.

Options

Option	Description
Design property value spreadsheet	Displays the combination of design property values for each sweep simulation.
Run Sweeps	Opens the Digital Oscilloscope Dialog Box. Near the upper-right corner of the oscilloscope, select **Start Sweeps**.
Stop sweeping if error occurs	Stops all sweep simulations when a simulation error occurs. This capability enables you to investigate the failing simulation when it happens, instead of waiting for the remaining sweep simulations to finish. The left column in the spreadsheet displays a ! character to identify a failing sweep simulation. Point to the ! character to display a ToolTip that contains either the specific error or a recommendation to interactively simulate the specific sweep condition.
Sweep simulations requested	Displays the number of simulations needed to perform all the sweeps.

Related Topics

Tips for Running Simulation with Parametric Sweeps

Running Signal Integrity Simulation

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Sweeping Dialog Box

To access: Double-click a tree item in any of the following:

- Sweep Manager Dialog Box - Simulation Cases Tab
- DDRx Batch-Mode Wizard - Sweep Manager Page
- SERDES Wizard - Parameter Sweep Page

Use this dialog box to define sweep ranges. The format of this dialog box depends on whether the design property is swept by numerical values or by named values. For example, use numerical values to sweep dielectric thicknesses and use named lists to sweep IC process corners.

You can type values in decimal, scientific notation, or decimal scale format. For example, 0.02, 2e-2, or 20m. Scientific notation values are automatically converted to decimal scale format. See Decimal Scaling Suffixes for Sweeps.

You do not need to type units because units are pre-assigned and not editable. If you delete or change unit text, the original units are restored when you re-open the dialog box.

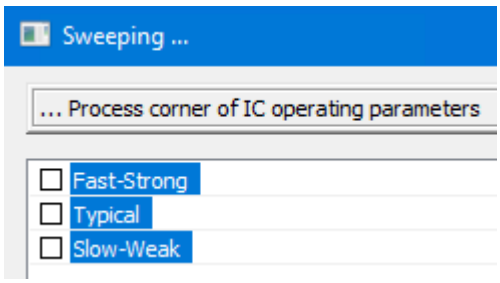
Fields

Field	Description
Information area	Names the swept design property or IBIS-AMI model parameter. For IBIS-AMI sweeps, this area also displays the contents of the Description statement in the *.AMI* model, if it exists.
By initial / final values	Begin sweep simulation at initial sweep range value and end at the final value. Use either “Simulation count” or Increment to determine the number of steps.
Initial Final	Starting and ending values of the sweep range. **Note:** Values in the Initial and Final boxes do not have to be in any particular order. If the value in the Final box is less than the value in the Initial box, the Increment box label automatically changes to Decrement.
Simulation count	Number of simulations. When you specify the number of simulations or step size, the dialog box automatically calculates the other value when you select outside the current box.
Increment	Step size. When you specify the number of simulations or step size, the dialog box automatically calculates the other value when you select outside the current box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
By tolerance	Begin sweep simulation at (Value - Tolerance) and end at (Value + Tolerance). Use either “Simulation count” or Increment to determine the number of steps. If you specify a negative increment value, the sweep range begins at the center value plus the tolerance and ends at the center value minus the tolerance.
Value	Center value of the sweep range.
Simulation count	Number of simulations. When you specify the number of simulations or step size, the dialog box automatically calculates the other value when you select outside the current box.
Tolerance	Maximum deviation from the center value.
Increment	Step size. When you specify the number of simulations or step size, the dialog box automatically calculates the other value when you select outside the current box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
By list	Begin sweep simulation at the first list item and run simulation for each item in the list. When you define the sweep range by “By initial / final values” or “By tolerance”, this area displays the individual design property values. To save simulation time, delete unwanted design property values by enabling “By list” and deleting values. You can select multiple rows (using Shift-Click or Ctrl-Click), then check or uncheck list items for all selected rows. For example, all the items in the figure below are selected and unchecked, and the dialog box checks all items when you click a checkbox for any row.  **Notes:** • Some types of IC models, such as SPICE models, do not respond to IC operating condition values. • For IBIS models, assigning a sweeping model to a specific IC pin is restricted to models located in the current model library file. • Sweeping models for a programmable IC buffer is restricted to models located in the [Model Selector] keyword in the current IBIS model. When sweeping pins for a reference designator that maps to a [Model Selector] keyword, all the pins receive the same model assignment for each simulation. If the [Model Selector] keyword contains a description of each model, the second column displays the description.
Simulation count	The number of simulations that will run, based on your sweep range.

Decimal Scaling Suffixes for Sweeps

Use the following table decimal scaling suffixes to define your sweeps.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table 10-5. Supported Scaling Factor Suffixes for Sweeps

Suffix	Name	Scale
M	mega	1,000,000x
K or k	kilo	1,000x
m	milli	0.001x
u or U	micro	1e-6x
n or N	nano	1e-9x
p or P	pico	1e-12x

Related Topics

Parametric Sweeps

Tips for Running Simulation with Parametric Sweeps

Synthesize DFE Dialog Box

To access: FastEye Channel Analyzer - Add Pre-Emphasis/DFE/CTLE Page > select **Synthesize optimal values**, check Add DFE, and click **Taps/weights**

Use this dialog box to specify how many taps exist in the receiver equalization circuitry.

Fields

Field	Description
Total number of taps	Specify the number of taps contained in the receiver equalization circuitry. For information about the assumed implementation of the DFE structure, see Pre-Emphasis and DFE Structures.

Synthesize Pre-Emphasis Dialog Box

To access: FastEye Channel Analyzer - Add Pre-Emphasis/DFE/CTLE Page > select **Synthesize optimal values**, check Add pre-emphasis, and click **Taps/weights**

Use this dialog box to specify how many taps exist in the driver pre-emphasis circuitry and how many of them are pre-taps.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Total number of taps	Specify the overall number of taps in the driver pre-emphasis circuitry. For information about the assumed implementation of the pre-emphasis structure, see Pre-Emphasis and DFE Structures.
Number of pre-taps section	
Don't care	Have the wizard choose the number of pre-taps.
Number	Manually specify the number of pre-taps.

Synthesized DFE Weights Dialog Box

To access: FastEye Channel Analyzer - Add Pre-Emphasis/DFE/CTLE Page > select **Synthesize optimal values**, check Add DFE > FastEye Channel Analyzer - View Analysis Results Page > select **Synthesized filter settings** > run analysis

Use this dialog box to display optimum decision-feedback equalization (DFE) tap weight values and to save them to a file (*.TAPS*).

Note

If you close this dialog box, you must rerun FastEye channel analysis to re-open it.

Fields

Field	Description
Tap	The zero row represents the main tap. The positive-numbered rows represent post-taps. For information about the assumed implementation of the DFE structure, see Pre-Emphasis and DFE Structures.
Weight	The optimum weight values for the DFE circuitry.
Save As	Save the tap and weight values to a tap weights file (*.TAPS*).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Synthesized Pre-Emphasis Weights Dialog Box

To access: FastEye Channel Analyzer - Add Pre-Emphasis/DFE/CTLE Page > select **Synthesize optimal values**, check Add pre-emphasis > FastEye Channel Analyzer - View Analysis Results Page > check Synthesized filter settings > run analysis

Use this dialog box to display optimum pre-emphasis tap weight values and to save them to a file (.*TAPS*).

Note

If you close this dialog box, you must rerun FastEye channel analysis to re-open it.

Fields

Field	Description
Tap	Negative-numbered rows represent pre-taps. The zero row represents the main tap. Positive-numbered rows represent post-taps. For information about the assumed implementation of the pre-emphasis structure, see Pre-Emphasis and DFE Structures.
Weight	The optimum weight values for the pre-emphasis circuitry.
Save As	Save the tap and weight values to a tap weights file (.*TAPS*).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Target-Z Wizard

Use this wizard to specify the peak transient current transmitted through a pair of power supply nets, the nominal voltage provided by the voltage-regulator module (VRM) and its ripple, and read the output of the target-Z calculation.

To access: **Simulate PI > Analyze Decoupling** > select the Set the Target Impedance page, click **Calculator**

Topic	Description
Target-Z Wizard - Finish Page	Use this page to read the output of the target-Z calculation.
Target-Z Wizard - Specify Peak Transient Current Page	Use this page to specify the peak transient current transmitted through a pair of power supply nets.
Target-Z Wizard - Specify Supply Voltage and Max Ripple Page	Use this page to specify the nominal voltage provided by the voltage-regulator module (VRM) and its ripple.

Target-Z Wizard - Finish Page

To access: **Simulate PI > Analyze Decoupling** > select the Set the Target Impedance page, click **Calculator**

Use this page to read the output of the target-Z calculation.

Fields

Field	Description
Target Z	Displays the output of the target-Z calculation.

Related Topics

Decoupling Wizard - Set the Target Impedance Page

Target-Z Wizard - Specify Peak Transient Current Page

To access: **Simulate PI > Analyze Decoupling** > select the Set the Target Impedance page, click **Calculator**

Use this page to specify the peak transient current transmitted through a pair of power supply nets.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you run distributed decoupling analysis and include the series inductance that is unique to IC power supply pins, consider specifying in this dialog box the amount of peak transient current that you specify one of the AC PI models. You calculate this by taking the overall peak transient current for the IC and dividing it by the number of power pins with AC power pin models. The effect of this calculation is to increase the target impedance. See the "Remove series inductance unique to each power pin, to see plane decoupling more clearly" option on the Decoupling Wizard - Customize Settings Page (Standard Simulation).

Fields

Field	Description
Peak transient current	Peak IC core and I/O transient current. See Information Needed to Calculate Target PDN Impedance.

Related Topics

Information Needed to Calculate Target PDN Impedance

Decoupling Wizard - Set the Target Impedance Page

Target-Z Wizard - Specify Supply Voltage and Max Ripple Page

To access: **Simulate PI > Analyze Decoupling** > select the Set the Target Impedance page, click **Calculator**

Use this page to specify the nominal voltage provided by the voltage-regulator module (VRM) and its ripple.

Fields

Field	Description
Max. percentage ripple	Specify ripple as an offset from the nominal DC voltage. Do not specify ripple as the peak-to-peak range of the nominal DC voltage.

Related Topics

Information Needed to Calculate Target PDN Impedance

Decoupling Simulation

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Translator Options Dialog Box

To access: **File > New Board (Run PCB Translator)**  , select a file to translate and click **Options** from the Translate File Dialog Box.

For some PCB design systems, key information needed to create BoardSim boards is not stored in a predictable way. Use this dialog box to provide data such as attribute names, to indicate how the information is stored and which information to include during translation.

Available options depend on which translator you are using.

Note

Do not set non-standard command-line options unless you are advised to do so.

Fields

Field	Description
Standard options Area	
VALUE Alias	The name of the attribute used to specify part values. The name is case insensitive. **Example**: value Available for the following translators: • Siemens EDA Board Station RE • Specctra DSN Translator • Zuken CR-3000
Library File Property	The name of the attribute used to specify the filename of the IC model library. **Example**: HYP_LIB Available for the following translators: • Cadence Allegro • Siemens EDA Board Station RE
Device Model Property	The name of the attribute used to specify IC model names. **Examples**: COMP_PART_NUMBER, HYP_DEVICE Available for the following translators: • Cadence Allegro • Siemens EDA Board Station RE
Default capacitance units	Select the default unit for capacitance, to use when no units are provided. Available for the following translators: • Siemens EDA Board Station RE

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Include partial plane areas and copper pours	Select to write partial plane areas and copper pour (polygon fills) information to the *.HYP* file. This allows you to see area fills in the board viewer. **Restriction:** Enable this option if you plan to run power-integrity analysis. **Note:** Enabling this option produces larger *.HYP* files and can increase drawing times in the BoardSim board viewer. Available for the following translators: • Cadence Allegro • Siemens EDA Board Station RE • Specctra DSN Translator • Zuken Visula/CADStar for Windows Translator • Zuken CR-3000 • Zuken CR-5000 Translator
Increase details for power integrity	Select to write precise clearances among metal regions, detailed capacitor mounting information, and other physical information about the power-distribution network. This option writes explicit anti-pad values to the *.HYP* file and may cause the Setup Anti-Pads and Anti-Segments Dialog Box to be unavailable. Available for the following translators: 1. Cadence Allegro 2. Siemens EDA Board Station RE 3. Zuken CR-5000 Translator
Device Name Field	Select the name of the attribute to specify IC part names: Symbol or Part_number. The *.CMP* file provides this information. Available for the following translators: • Siemens EDA Board Station RE
Non-Standard Command-Line Options	Leave this field empty. Do not use this option unless you are advised to do so. **Note:** Board Station RE only: If the design contains variants, use the -a <variant_name> option.

Related Topics

Translating a Board Design

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Via Model Extractor Wizard

Use this wizard to set up and export signal vias to S-parameter models.

To access:

- From BoardSim, choose **Export > Signal-Via Model**
- From LineSim, choose **Export > Model > Signal-Via Model**

Wizard Page	Description
Via Model Extractor Wizard - Check Capacitor Models Page	Use this page to review and edit decoupling capacitor model assignments.
Via Model Extractor Wizard - Choose Easy / Custom Page	Use this page to choose between default and custom export options.
Via Model Extractor Wizard - Control Frequency Sweep Page	Use this page to edit frequency range and sampling options, both of which affect simulation run time and the resolution of the exported S-parameter model.
Via Model Extractor Wizard - Customize Settings Page	Use this page to enable detailed signal-via-model extraction options.
Via Model Extractor Wizard - Select Signal Via Page	Use this page to select the via or differential via pair to export.
Via Model Extractor Wizard - Set Model Type Page	Use this page to specify the normalization impedance for the exported S-parameter model. For differential vias, use this page to specify the propagation mode information to include in the exported model.
Via Model Extractor Wizard - Start Analysis Page	Use this page to start a new export session or load the settings for a saved export.

Via Model Extractor Wizard - Check Capacitor Models Page

To access:

- From BoardSim, choose **Export > Signal-Via Model**
- From LineSim, choose **Export > Model > Signal-Via Model**

Use this page to review and edit decoupling capacitor model assignments.

For a description of options on this page, see Assign / Edit Capacitor Model Dialog Box.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Via Model Extractor Wizard - Choose Easy / Custom Page

To access:

- From BoardSim, choose **Export > Signal-Via Model**
- From LineSim, choose **Export > Model > Signal-Via Model**

Use this page to choose between default and custom export options.

Fields

Field	Description
Easy	Enables popular options on other wizard pages. Some enabled options become read-only.
Custom	Makes all options on all wizard pages available for you to edit.

Via Model Extractor Wizard - Control Frequency Sweep Page

To access:

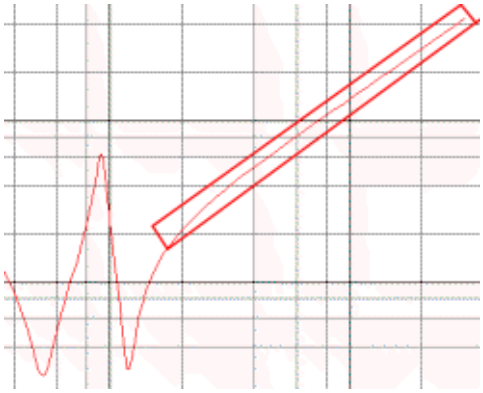
- From BoardSim, choose **Export > Signal-Via Model**
- From LineSim, choose **Export > Model > Signal-Via Model**

Use this page to edit frequency range and sampling options, both of which affect simulation run time and the resolution of the exported S-parameter model.

Fields

Field	Description
Min frequency Max frequency	Specifies the frequency range of the simulation and exported S-parameter model. Many ICs have in-package decoupling that provide the main decoupling effects above a certain frequency, such as 300 to 350 MHz. This means decoupling capacitors and buried capacitance located in the PCB contribute little or no decoupling above this design-dependent frequency.
Adaptive sampling	Varies the sampling step size depending on model characteristics. The adaptive scale is better than logarithmic and linear because it increases the sampling rate near frequencies with resonances.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Logarithmic sampling	Distributes sampling points across the frequency range at logarithmic intervals. The intervals between sampling points are smaller at lower frequencies and larger for higher frequencies. With logarithmic sampling, every next frequency point is equal to the previous value times a factor K > 1.
Linear sampling	Distributes sampling points across the frequency range at equal intervals.
Accuracy at resonances	For adaptive sampling, specifies the relative accuracy at resonant frequencies. For lumped analysis, enabling the High option may still yield reasonably fast simulation run times. For distributed analysis, you should take the complexity of the design into account. If the design has a large number of power supply nets, hundreds of decoupling capacitors, and hundreds or thousands of stitching vias, enabling the Low option provides preliminary results with decreased analysis run time. After evaluating the preliminary results, you can identify which frequency ranges interest you the most and try running analysis with higher accuracy on each range of interest.
Minimum number of samples in flat, non-resonant regions	For adaptive sampling, specifies the number of samples the software applies to flat, non-resonant, regions of an impedance profile. The figure below shows a flat and non-resonant region of an example impedance profile. 
Number of samples	For logarithmic and linear sampling, specifies the number of samples the software applies to the entire frequency range.

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Via Model Extractor Wizard - Customize Settings Page

To access:

- From BoardSim, choose **Export > Signal-Via Model**
- From LineSim, choose **Export > Model > Signal-Via Model**

Use this page to enable detailed signal-via-model extraction options.

Extracting models with different sets of enabled and disabled options can help you determine how individual types of design properties contribute to via model impedance.

Fields

Field	Description
Include capacitor mounting inductance	Specifies whether to account for decoupling capacitor mounting inductance.
Enable stitching-via optimization	Specifies whether to reduce simulation run time and memory consumption by automatically finding stitching vias that are located close together and merging their individual models into an equivalent model. See Stitching-Via Optimization.

Via Model Extractor Wizard - Select Signal Via Page

To access:

- From BoardSim, choose **Export > Signal-Via Model**
- From LineSim, choose **Export > Model > Signal-Via Model**

Use this page to select the via or differential via pair to export.

Restrictions

- You can export a model for one via or differential via pair at a time.
- In BoardSim, both differential vias must connect to the same two stackup layers.
- In LineSim, you can export models only for vias connected to stackup type (coupled or uncoupled) transmission lines.

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Fields

Field	Description
Check box	Checked, includes the via or via pair behavior in the exported S-parameter model. (Available in LineSim.)
Single via Differential via	Specifies whether the via is single ended or part of a differential via pair. (Available in BoardSim.)
Pan to	Click to display and highlight the via or via pair in the center of the board viewer. (Available in BoardSim.)

Via Model Extractor Wizard - Set Model Type Page

To access:

- From BoardSim, choose **Export > Signal-Via Model**
- From LineSim, choose **Export > Model > Signal-Via Model**

Use this page to specify the normalization impedance for the exported S-parameter model. For differential vias, use this page to specify the propagation mode information to include in the exported model.

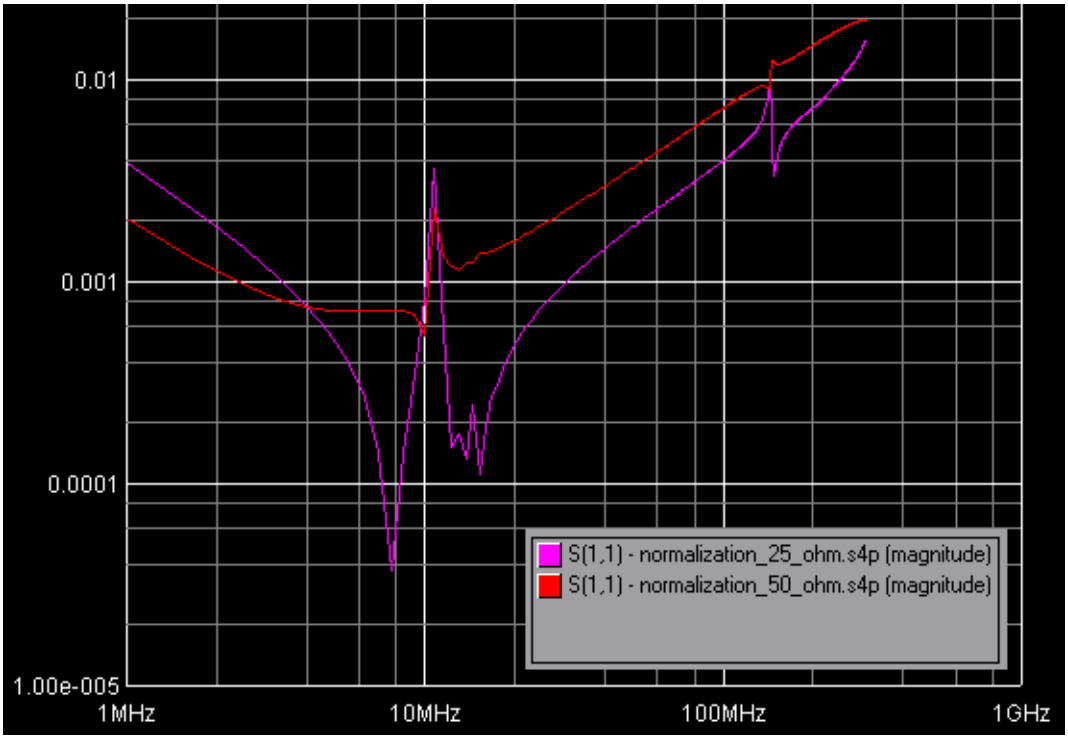
Fields

Field	Description
Normalization impedance	Specifies the normalization impedance for the exported S-parameter model. Models with different normalization impedances can make them harder to understand visually when they are displayed in the Touchstone Viewer. For example, if you export two S-parameter models for the same differential via pair, but specify 25 ohms normalization impedance for one model and 50 ohms for the other, the S-parameter data may look different. For an example, see Figure 10-36. However, specifying the exact normalization impedance is not a simulation problem because simulators produce identical results whether the S-parameter model has a 50-ohm impedance or another impedance.

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Field	Description
4-port standard model 2-port differential-mode model 2-port common-mode model	Specifies the propagation mode information to include in the exported S-parameter model for a differential signal-via pair. See below for information about choosing a propagation mode.

Figure 10-36. Exported S-Parameter Models for Same Via Pair at 25 and 50 Ohms



Propagation Modes for Exported Differential Models

"Standard" exported differential via models contain both differential and common propagation-mode information. Standard differential via models are good for both simulation and viewing. These models have four ports and are easy to add to LineSim schematics.

You can also export via models containing only differential-mode information or only common-mode information, which are good for viewing, but not for simulation. Comparing these models to standard via models can show the contribution of individual differential and common modes to signal loss. Asymmetric geometries or asymmetric differential signal transitions can cause some of the signal energy to convert to common mode.

Via models containing only differential-mode information do not account for the effects of stitching vias. Advanced users can use third-party 3-D simulation software to create via models containing only differential-mode information, and then use the Touchstone Viewer to combine

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them with common-mode models exported from the software. See Combine to Standard Mode Dialog Box.

Via Model Extractor Wizard - Start Analysis Page

To access:

- From BoardSim, choose **Export > Signal-Via Model**
- From LineSim, choose **Export > Model > Signal-Via Model**

Use this page to start a new export session or load the settings for a saved export.

Fields

Field	Description
Use last configuration	Selected, the wizard uses settings from the current software session. (Available when you have opened and closed the wizard in the current BoardSim or LineSim session.)
Load saved configuration	Selected, enables you to open a wizard settings file (*.DAO*).
Save settings to file	Checked, saves settings to a wizard settings file (*.DAO*). The default file location is the <design> folder. See Design Folder and HyperLynx Files.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Via Properties Dialog Box

Use this dialog box to specify the type of electrical model and via properties for a signal via symbol in a schematic.

Scope: LineSim

To access: Double-click a via or differential via pair in the schematic.

Restrictions:

- Set up padstacks for the schematic before assigning them to vias. See "Padstack Manager Dialog Box" on page 967.
- Signal vias are not shared between the PDN Editor and free-form schematic editor unless you add them to the PDN Editor first. If you are setting up for power-integrity simulation and want the signal via in the schematic to interact with the PDN, use the PDN Editor to add the via. See "Creating a PDN Design" on page 77 and "Add Signal Via Dialog Box" on page 685.
- LineSim schematics can represent signal vias, but not decoupling capacitor vias, stitching vias, and vias with thermal spokes.
- You have the Via Models license.
- To export via models to HyperLynx Full-Wave Solver, you have the 3D Via Model Export license.

Topic	Description
Via Properties Dialog Box - Backdrilling Settings Tab	Use this tab to specify backdrilling properties for a signal via symbol in a schematic.
Via Properties Dialog Box - Feeding Traces Tab	Use this tab to specify geometric properties for feeding traces that connect to a via in a schematic.
Via Properties Dialog Box - Global Parameters Tab	Use this tab to specify 3-D electromagnetic simulation parameters for a signal via or differential via pair in a schematic.
Via Properties Dialog Box - Sections Tab	If a via has more than one finished diameter, use this tab to divide the via into sections and assign a padstack to each section.
Via Properties Dialog Box - Stitching Vias Tab	Use this tab to specify stitching via properties for a signal via or differential via pair in a schematic.
Via Properties Dialog Box - Via Properties Tab	Use this tab to specify the type of electrical model and geometric properties for a signal via symbol in a schematic.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Via Properties Dialog Box - Backdrilling Settings Tab

Use this tab to specify backdrilling properties for a signal via symbol in a schematic.

Fields

Field	Description
Backdrilling (Available when you enable backdrilling. From the Backdrilling/Plating/Constraints Settings dialog box (**Setup > Backdrilling/Plating/ Constraints Settings** menu item), ensure the "Backdrilling enabled" button appears. If needed, click the "Backdrilling disabled" button.)	Specifies backdrilling properties. For information about "Setback", see *backdrill setback*.Use this tab to specify the type of electrical model and geometric properties for a signal via symbol in a schematic.
View 3D (Available when you set 3D EM Modeling to HyperLynx Full-Wave Solver.)	Opens HyperLynx Full-Wave Solver to display the geometries used by 3-D electromagnetic simulation. The geometries can include the signal via, signal traces, metal areas, and stitching vias.
Simulate (Available when you set 3D EM Modeling to HyperLynx Full-Wave Solver.)	Runs a simulation in HyperLynx Full-Wave Solver and creates an S-parameter model that represents the 3-D area.
View Model (Available when you set 3D EM Modeling to HyperLynx Full-Wave Solver.)	Displays an S-parameter file previously created by HyperLynx Full-Wave Solver to represent this via. The default file name is *via_FFS_V<model_version>.sNp* where: <model_version> automatically increments to avoid overwriting a previously-generated model file N is the number of model ports

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Sync global parameters (Available when you set 3D EM Modeling to HyperLynx Full-Wave Solver, and when there is a mismatch between via and design settings.)	Replaces via-specific settings with global design settings. You can use the Via Properties dialog box to specify via-specific settings (such as non-functional pads, surface roughness, and trapezoidal trace shapes). Use this option to replace via-specific settings in this dialog box with settings from your design. You use the Setup menu and **Manufacturing** tab of the Stackup Editor to specify values for your design.

Related Topics

Modeling a Via with a 3D EM Model in a Schematic

Creating a Schematic-PDN Design

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Via Properties Dialog Box - Feeding Traces Tab

Use this tab to specify geometric properties for feeding traces that connect to a via in a schematic.

Note

This tab is available when you set the 3D EM Modeling option on the **Via Properties** tab to HyperLynx Full-Wave Solver.

Fields

Field	Description
Connected	Checked, adds a feeding trace to the signal via, using the specified stackup layer.
Length (Available for a single-ended via.) Length 1 (L 1) Length 2 (L 2) (Available for a differential via.)	Specifies the distance between the via centerline and the edge of the geometry box for 3-D EM simulation. **Note:** To visualize a geometry box, vias, and trace segments, click **View 3D**. Length 1 and Length 2 partially specify the distance between the via centerline and the edge of the geometry box used by 3-D EM simulation. Other geometries contributing to this distance include trace-to-trace and via-to-via separation. The coupled transmission lines properties includes the trace-to-trace separation. **Note:** Length 2 is unavailable if you check “by Angle”. The default values provide a good balance between model accuracy and simulation run times for general board stackups and signals up to the 15 to 20 GHz frequency range. The software automatically subtracts feeding trace lengths from lengths of transmission lines connected to a via. This subtraction takes place during simulation and does not affect the length specified in a transmission-line symbol.
Width (W)	Specifies the feeding trace width.
Separation (Available for a differential via.)	Trace-to-trace separation (edge-to-edge) for a differential pair, as defined in the Edit Transmission Line dialog box - **Edit Coupling Regions** tab.
by Angle (Available for a differential via.)	Unchecked, displays the calculated angle between the L1 and L2 trace segments. See “Connected Trace Angle for Differential Pairs” on page 1066 Checked, specifies the angle you type. Length 2 displays the calculated length for L2.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Direction	Specifies whether you attach a feeding trace to the left or right side of a signal via. To attach a feeding trace to the left side of the signal via, set Direction to Input. To attach a feeding trace to the right side of the signal via, set Direction to Output. Displays <multiple> when you select multiple items in the Connected area and there is at least one Input item and one Output item.
View 3D	Opens HyperLynx Full-Wave Solver to display the geometries used by 3-D electromagnetic simulation. The geometries can include the signal via, signal traces, metal areas and stitching vias.
Simulate	Automatically runs simulation in HyperLynx Full-Wave Solver and creates an S-parameter model that represents the 3-D area.
View Model	Display the S-parameter file created by HyperLynx Full-Wave Solver. **Note:** Run simulation to completion before selecting View Model. The default file name format is *via_FFS_V<model_version>.s#p* where: <model_version> automatically increments to avoid overwriting a previously-generated model file # is the number of model ports
Sync global parameters (Available when there is a mismatch between via and design settings.)	Replaces via-specific settings with global design settings. You can specify settings (such as non-functional pads, surface roughness, and trapezoidal trace shapes) for an individual signal via. Use this option to replace settings in this dialog box (that apply to an individual signal via) with settings from your design. You use the Setup menu and **Manufacturing** tab of the Stackup Editor specify values for your design.

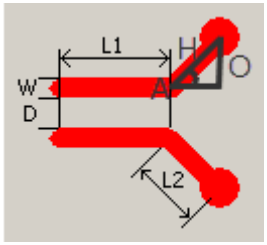
Usage Notes

Connected Trace Angle for Differential Pairs

Figure 10-37 on page 1067 shows the landmarks used in this section.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-37. Connected Trace Angle for Differential Pairs



Example:

(All lengths in mils)

H = L2 = 30.5. L2 is the same for both vias.

Separation (distance between via centerlines) = 75

D = 10

W = 6

Note

The Edit Transmission Line Dialog Box - **Edit Coupling Regions** tab provides the values for D and W.

To calculate the angle:

Create a right triangle as shown by the black lines in Figure 10-37.

H = hypotenuse. O = opposite. A = angle.

Solve for O:

O = (Separation / 2) - (D / 2) - (W / 2)

O = (75 / 2) - (10 / 2) - (6 / 2) = 29.5

Solve for A:

A = arcsin (O / H)

A = arcsin (29.5 / 30.5) = 75.3 degrees

Related Topics

Modeling a Via with a 3D EM Model in a Schematic

Creating a Schematic-PDN Design

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Via Properties Dialog Box - Global Parameters Tab

Use this tab to specify 3-D electromagnetic simulation parameters for a signal via or differential via pair in a schematic.

Note

This tab is available when you set the 3D EM Modeling option on the **Via Properties** tab to HyperLynx Full-Wave Solver.

Fields

Field	Description
Simulation parameters Area	
Model Type	• Normal — Model all objects with actual metal stackup layer thicknesses. This option provides the best accuracy at the expense of a longer run time. • Accelerated — Model traces and vias with actual metal stackup layer thicknesses, but model other objects (such as metal pours and plane layers) with an infinitely thin metal stackup layer thickness. This option provides good accuracy and shorter run time.
Boundary Size	Specifies the distance from the edge of the geometry box used by 3-D electromagnetic simulation to either the centerline of the stitching via or the signal via (if there are no stitching vias). The default value (75 mils or 1.905 mm) provides a good balance between model accuracy and simulation run times for general board stackups and signals up to the 15 to 20 GHz frequency range.
Use Absorbing Boundaries	Checked, eliminates artificial resonances during simulation by modeling the area boundary edges as an absorbing material (such as a PCB dielectric) rather than a reflective material (such as air).
Frequencies Area	
Minimum	Checked, specifies the lowest frequency that simulation runs and writes to the S-parameter model. 10 MHz is good for common cases. If you are not interested in frequency responses below 1 GHz or higher, you can specify a larger value. Unchecked, the software calculates the minimum frequency based on via structure and the maximum frequency, and displays <auto>.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Maximum	Specifies the highest frequency that simulation runs and writes to the S-parameter model. You can calculate the maximum frequency from the signal rising or falling edge time. For example, below is a popular equation: Maximum frequency = 0.35 / (signal rise time or fall time)
Meshing	Specifies the highest frequency the simulation mesh can accurately represent. The maximum simulation frequency cannot exceed this value.
Num Points (Available when you enable Minimum.)	Specifies the number of frequencies written to the generated S-parameter model. The software linearly distributes the frequencies. 100 to 200 points are appropriate unless you want a generated S-parameter model to represent a very narrow frequency range. When you uncheck Minimum, the software calculates the minimum frequency based on via structure and the maximum frequency, and displays <auto>. **Note:** HyperLynx Full-Wave Solver uses an advanced searching algorithm to solve only the critical frequency points. The simulation run times for values of 100 and 200 can be very close because the software solves the same number of critical frequency points in both cases.
Global Via parameters Area	
Remove Non-Functional Pads	Checked, omit via pads on stackup layers that do not connect to feeding traces. **Note:** Simulation does not omit pads on the top and bottom stackup layers because they are required for the plating process during PCB manufacturing.
Default anti-pad clearance	Specifies the distance between a pad and metal for a plane stackup layer or an explicit metal area.
Plating Metal	Specifies the type of metal used to plate via barrels. Select “Custom” to specify a metal type with a unique resistivity.
Resistivity (Available when you set Plating Metal to Custom.)	Specifies the resistivity in Ohm-meters of plating metal.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Enable Surface Roughness	Checked, simulation accounts for surface roughness of feeding traces. Enable this option only if the board uses a dielectric material with extremely low loss. Surface roughness has a negligible effect on simulation results for dielectric materials with higher loss. For example, if a board uses FR-4 dielectric material and the simulation applies high frequency stimulus (at which surface roughness begins to have a measurable effect), surface roughness losses are negligible compared to dielectric losses. Signal loss is related to surface roughness. While increased roughness increases adhesion strength, it can also increase high-frequency signal loss. As the frequency increases, the current density near the conductor surface increases because the skin depth decreases. At some high frequency, the skin depth approaches the surface roughness amplitude, and the signal follows the irregular contours of the conductor surface, which increases the distance the signal travels.
Specify roughness by (Available when you enable Enable Surface Roughness.)	Specifies the method used to measure conductor roughness. See “Roughness” on page 1257.
Model type (Available when you enable Enable Surface Roughness.)	Specifies the conductor roughness model. See “Roughness” on page 1257.
Factor (Available when you enable Enable Surface Roughness.)	Defines a conductor roughness correction factor. See “Roughness” on page 1257.
Global trace parameters Area	
Trace-to-plane separation	Defines the distance from each side of a feeding trace to metal on a plane stackup layer. This option applies only to a trace on a plane layer.
Use Trapezoidal trace shapes	Checked, model feeding traces as trapezoidal shapes, using the average trace width defined by trapezoidal trace shape settings from the **Manufacturing** tab of the Stackup Editor.
View 3D	Opens HyperLynx Full-Wave Solver to display the geometries used by 3-D electromagnetic simulation. The geometries can include the signal via, signal traces, metal areas and stitching vias.
Simulate	Runs simulation in HyperLynx Full-Wave Solver to create an S-parameter model that represents the via.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
View Model	Displays the S-parameter file created by HyperLynx Full-Wave Solver, after simulation runs to completion. The default file name format is *via_FFS_V<model_version>.s#p* where: <model_version> automatically increments to avoid overwriting a previously-generated model file # is the number of model ports
Sync global parameters (Available when there is a mismatch between via and design settings.)	Replaces via-specific settings with global design settings. You can specify via-specific settings (such as non-functional pads, surface roughness, and trapezoidal trace shapes). Use this option to replace via-specific settings in this dialog box with settings from your design. You use the Setup menu and **Manufacturing** tab of the Stackup Editor to specify values for your design.

Related Topics

Modeling a Via with a 3D EM Model in a Schematic

Creating a Schematic-PDN Design

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Via Properties Dialog Box - Sections Tab

If a via has more than one finished diameter, use this tab to divide the via into sections and assign a padstack to each section.

Note

When you define sections for a via, the software represents it with a padstack named AutoPadstk_<number> and displays it on the **Via Properties** tab.

Fields

Field	Description
Layers	Specifies the range of stackup layers for the overall via or for a section of a via. If needed, use the **Add** button to divide a via into sections.
Offset	The software does not use this data.
Drill	Displays the actual drill diameter (not the post-plating finished diameter) of a padstack.
Padstack	Specifies the padstack assigned to the overall via or to a section of a via.
View 3D (Available when you set 3D EM Modeling to HyperLynx Full-Wave Solver.)	Opens HyperLynx Full-Wave Solver to display the geometries used by 3-D electromagnetic simulation. The geometries can include the signal via, signal traces, metal areas, and stitching vias.
Simulate (Available when you set 3D EM Modeling to HyperLynx Full-Wave Solver.)	Runs a simulation in HyperLynx Full-Wave Solver and creates an S-parameter model that represents the 3-D area.
View Model (Available when you set 3D EM Modeling to HyperLynx Full-Wave Solver.)	Displays an S-parameter file previously created by HyperLynx Full-Wave Solver to represent this via. The default file name is *via_FFS_V<model_version>.sNp* where: <model_version> automatically increments to avoid overwriting a previously-generated model file N is the number of model ports

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Sync global parameters (Available when you set 3D EM Modeling to HyperLynx Full-Wave Solver, and when there is a mismatch between via and design settings.)	Replaces via-specific settings with global design settings. You can use the Via Properties dialog box to specify via-specific settings (such as non-functional pads, surface roughness, and trapezoidal trace shapes). Use this option to replace via-specific settings in this dialog box with settings from your design. You use the Setup menu and **Manufacturing** tab of the Stackup Editor to specify values for your design.

Related Topics

Modeling a Via with a 3D EM Model in a Schematic

Creating a Schematic-PDN Design

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Via Properties Dialog Box - Stitching Vias Tab

Use this tab to specify stitching via properties for a signal via or differential via pair in a schematic.

Note

This tab is available when you set the 3D EM Modeling option on the **Via Properties** tab to HyperLynx Full-Wave Solver.

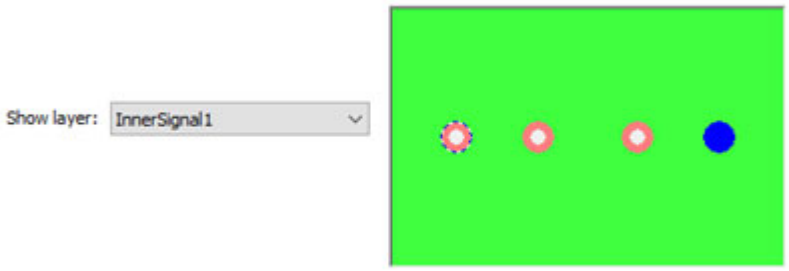
Note

If you have a multiple-board or multiple-stackup design, stitching vias you define on this dialog box use the board or stackup you select on the **Via Properties** tab.

Fields

Field	Description
Offset X	Single-ended via — The X offset from the centerline of a barrel for a signal via to the center line of a barrel for a stitching via. Differential via — The X offset from the midpoint between the differential vias to the center line of a barrel for a stitching via.
Offset Y	Single-ended via — The Y offset from the centerline of a barrel for a signal via to the center line of a barrel for a stitching via. Differential via — The Y offset from the midpoint between the differential vias to the center line of a barrel for a stitching via.
Padstack	Name of the padstack to use. **Note:** Define padstacks for the schematic before assigning them to vias.
Connected Layers	Metal stackup layers connected to stitching vias.
Backdrilling (Available when you enable backdrilling. From the Backdrilling/Plating/ Constraints Settings dialog box (**Setup > Backdrilling/ Plating/Constraints Settings**), ensure the “Backdrilling enabled” button appears. If it does not, click the “Backdrilling disabled” button.)	Specifies backdrilling properties. For information about “Setback”, see backdrill setback.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Add **Multiple Add**	Opens the Add/Edit Stitching Via dialog box, where you can add one or more stitching vias.
Remove	For a stitching via that you select in the table, removes the stitching via from the via symbol.
Edit	For a stitching via that you select in the table, opens the stitching via in the Add/Edit Stitching Via dialog box.
Show layer	Displays signal via and stitching via (if assigned) geometries for the specified stackup layer. 
View 3D	Opens HyperLynx Full-Wave Solver to display the geometries used by 3-D electromagnetic simulation. The geometries can include the signal via, signal traces, metal areas and stitching vias.
Simulate	Automatically runs simulation in HyperLynx Full-Wave Solver and creates an S-parameter model that represents the 3-D area.
View Model	Displays the S-parameter file created by HyperLynx Full-Wave Solver. **Note:** Run simulation to completion before selecting View Model. The default file name form is *via_FFS_V<model_version>.s#p* where: <model_version> automatically increments to avoid overwriting a previously-generated model file # is the number of model ports
Sync global parameters (Available when there is a mismatch between via and design settings.)	Replaces via-specific settings with global design settings. You can specify via-specific settings (such as non-functional pads, surface roughness, and trapezoidal trace shapes). Use this option to replace via-specific settings in this dialog box with settings from your design. You use the Setup menu and **Manufacturing** tab of the Stackup Editor to specify values for your design.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Modeling a Via with a 3D EM Model in a Schematic

Creating a Schematic-PDN Design

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

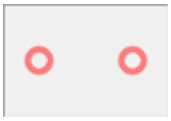
Via Properties Dialog Box - Via Properties Tab

Use this tab to specify the type of electrical model and geometric properties for a signal via symbol in a schematic.

Fields

Field	Description
3D EM Modeling	Specifies the type of electrical via model: • None — Use 2-D decomposition and field solvers. • HyperLynx Full-Wave Solver — Use the built-in version of HyperLynx Full-Wave Solver to extract via geometries and create an S-parameter model to represent them. Adds a “3D” label to the via symbol in the schematic.
Connected Layers (Available when you set 3D EM Modeling to None.)	Specifies the number of stackup layers connected to the via. This value determines how many ports appear on a via symbol in the schematic. **Note:** For an SMD pad, this value determines how many ports appear on a via symbol in the schematic and not the number of connected stackup layers.
3D EM Project File (Available when you set 3D EM Modeling to HyperLynx Full-Wave Solver.)	Specifies the location of a via project file (*.v3d*). The project file contains geometric information (such as feeding trace lengths and stitching via configuration) and simulation set up information (such as number of ports and simulation frequency range). The default file naming convention is: *via_FFS_<via_reference_designator>_<file_version>.v3d* where: <via_reference_designator> is the reference designator located next to the via symbol in the schematic <file_version> is an integer starting from 01 The design folder is the default project file location. See “Design Folder and HyperLynx Files” on page 520. For information about using a 3-D EM via definition in another schematic design, see “Via Project File and S-Parameter File Reuse” on page 1079.
New (Available when you set 3D EM Modeling to HyperLynx Full-Wave Solver.)	Opens the New HyperLynx Full-Wave Project dialog box, to start the process of creating a new via project file (*.v3d*).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Stackup (Available when your schematic represents a design with multiple stackups.)	Specifies the name of the stackup definition to use.
Edit (next to Stackup)	Opens the Stackup Editor to edit stackup properties.
Padstack Via 1 Padstack Via 2 Padstack	Specifies the name of the padstack to use. To create a padstack, use the Padstack Manager Dialog Box. To edit the default padstack, use the Preferences Dialog Box - Default Padstack Tab.
Edit (next to padstack)	Opens the Padstack Editor dialog box to edit padstack properties.
Via Separation (Available for a differential via.)	Specifies the distance between the centerlines of the via barrels.
Via Angle (Available for a differential via.)	Specifies the clockwise rotation of a via pair. Via Angle = 0 degrees:  Via Angle = 45 degrees: 
Common Anti-Pad (Available for a differential via.)	Checked, encloses both vias with a single anti-pad.
Via Visualizer	Opens the Via Visualizer dialog box to display padstack properties.
Distance to Plane (Available for editing when you uncheck Use Padstack.)	The column(s) on the right side of the table display: • <a number> — On a plane layer, represents the distance from a plane to a via. On a non-plane layer, represents the distance from a via to nearby metal (such as a metal area). The number represents the distance between objects on the same layer. • No metal — On a non-plane layer, the via is not close to any metal. **Note:** If you exported a net from BoardSim and disable the Use Padstack option, LineSim uses the exact clearance values from your board design.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Use Padstack	Checked, displays properties for the selected padstack. Unchecked, you can edit the via-to-reference-plane distance. To add reference plane metal to a stackup layer for only this via, replace the “No metal” value with a clearance.
View 3D (Available when you set 3D EM Modeling to HyperLynx Full-Wave Solver.)	Opens HyperLynx Full-Wave Solver to display the geometries used by 3-D electromagnetic simulation. The geometries can include the signal via, signal traces, metal areas, and stitching vias.
Simulate (Available when you set 3D EM Modeling to HyperLynx Full-Wave Solver.)	Runs a simulation in HyperLynx Full-Wave Solver to create an S-parameter model that represents the 3-D area.
View Model (Available when you set 3D EM Modeling to HyperLynx Full-Wave Solver.)	Displays an S-parameter file previously created by HyperLynx Full-Wave Solver to represent this via. The default file name is *via_FFS_V<model_version>.sNp* where: <model_version> automatically increments to avoid overwriting a previously-generated model file N is the number of model ports
Sync global parameters (Available when you set 3D EM Modeling to HyperLynx Full-Wave Solver, and when there is a mismatch between via and design settings.)	Replaces via-specific settings with global design settings. You can use the Via Properties dialog box to specify via-specific settings (such as non-functional pads, surface roughness, and trapezoidal trace shapes). Use this option to replace via-specific settings in this dialog box with settings from your design. You use the Setup menu and **Manufacturing** tab of the Stackup Editor to specify values for your design.

Usage Notes

Via Project File and S-Parameter File Reuse

You can directly reuse the via project file (*.v3d*) and its associated S-parameter file (*.sNp*) in other LineSim schematics. Notes:

- If you locate the pair of files in a shared folder, changing the via project file or S-parameter file affects all the schematics that use them.
- If you collect via project files into a central location on the network, be sure to copy the associated S-parameter files to the same folder. For example, copy both *via_FFS_V1.v3d* and *via_FFS_V1.s4p* to the same folder.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- If there are parameter mismatches between the schematic and via project file, the software warns you and does not attempt to automatically change any parameters. For example, a parameter mismatch occurs when the schematic uses trace width X, but the via project file uses trace width Y.

You can use an existing via project file as a template to quickly create new via design.

Related Topics

Modeling a Via with a 3D EM Model in a Schematic

Creating a Schematic-PDN Design

Via Visualizer Dialog Box

To access:

- In BoardSim, right-click a via and choose **View Via Properties**.
- In LineSim, right-click a via and choose **Properties**. From the Via Properties dialog box, click **View**.

Use this dialog box to display the electrical and geometric properties of a signal via or a pair of coupled signal vias, and to export a SPICE netlist representing via electrical properties. You can also use this dialog box to display the geometric properties of vias connected to a power-supply net.

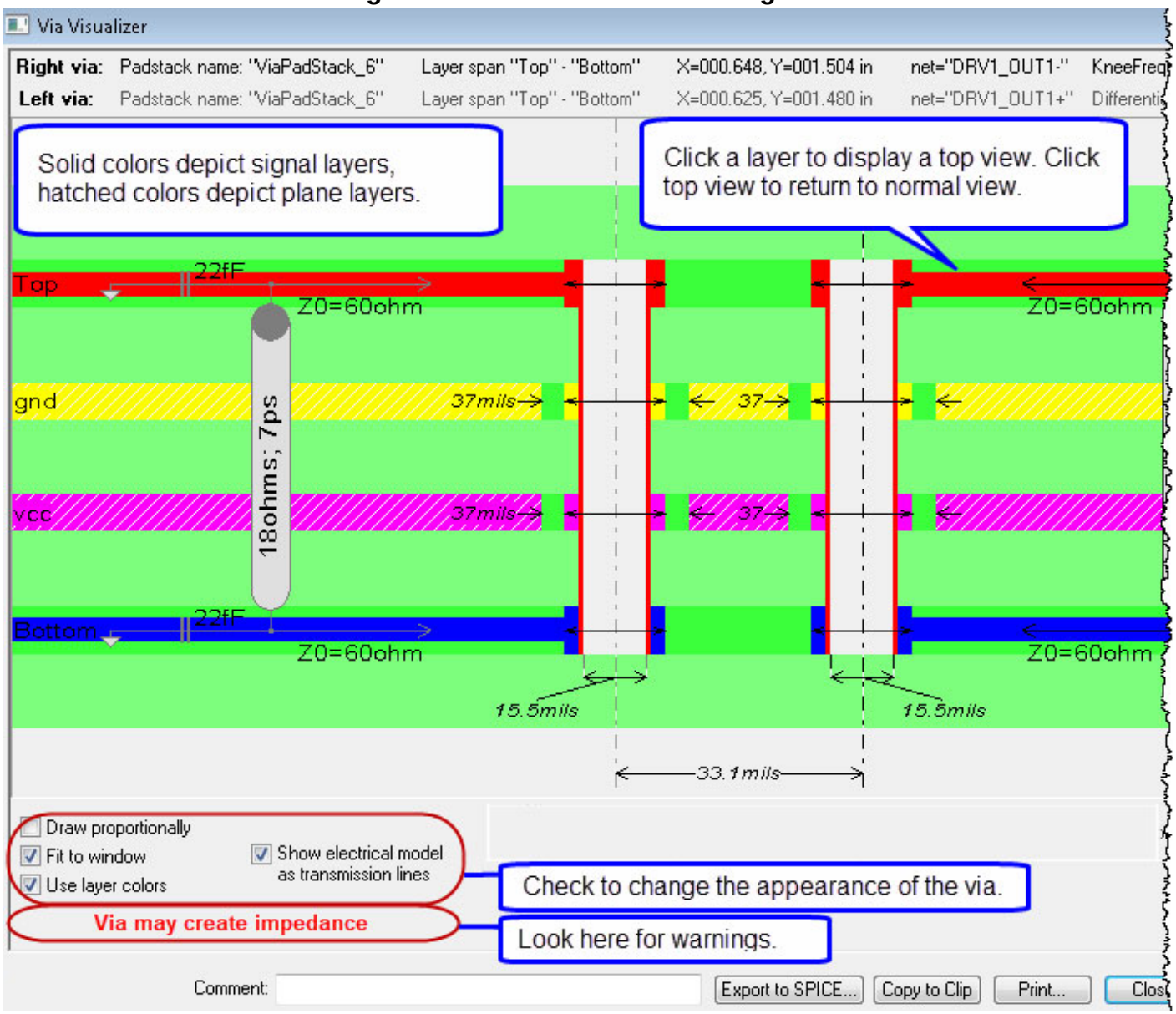
You can use via properties to help judge the effects of a signal via on propagation delay and signal integrity.

Note

In LineSim, if no transmission lines connect to a differential via pair, the Via Visualizer calculates the differential impedance based on a full layer span. For a definition of “layer span”, see via.

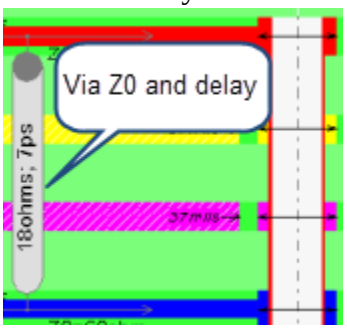
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure 10-38. Via Visualizer Dialog Box



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Via Z0 and delay 	Available for signal vias and calculated at the *knee frequency* displayed in this dialog box. **Note:** Make sure that the knee frequency is higher than the 3 dB frequency (approximately 0.35 / rise time) of the signal you apply during simulation.
Draw proportionally	Checked, displays the layers showing proportional thicknesses. This option may render some labels invisible because of size restrictions. Hover over a feature to view more information.
Fit to window	Checked, displays the entire via(s). This option may render some property labels invisible.
Use layer colors	Checked, displays the layers using colors defined in the stackup editor.
Show electrical model as transmission lines	Checked, displays a signal via modeled as a transmission line rather than as capacitors and inductors.
Comment	Enables you to include a comment with your printout or clipboard figure. **Note:** The comment does not appear in the SPICE list.
Export to SPICE	Exports a signal via electrical model to a SPICE netlist. **Note:** This button is available for a signal via.

Related Topics

Visualizing the Geometric and Electrical Characteristics of a Signal Via

Accounting for Backdrilling

Accounting for Per-Padstack Plating and Constraints

View Options Dialog Box

To access: **View > Options**

Use this dialog box and right-click menus to control how the board viewer displays objects.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

You can perform the following operations from this dialog box or the right-click menu for an object in the board viewer:

- View or hide reference designator labels
- Emphasize the appearance of objects
- Highlight objects on the fly
- View or hide pads and anti-pads
- Remove highlighting for all nets

Options

Option	Description
Show reference designators for	Select the type of components you want to display reference designators for. You can also right-click in the board viewer and select Show Reference Designators and select the items to display or hide. To help you identify components, the board viewer normally displays the reference designator label next to the component outline. On very dense boards, or when a board is viewed at a distance (zoomed far out), the reference designator labels may crowd together and overlap. In this case, you can improve layout legibility by hiding reference designator labels by object type.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Emphasize objects by brightness	The board viewer displays a selected net at full brightness, while displaying other objects at reduced brightness. You can also display at full brightness other types of nets related to the selected net. • None — All objects display the same. • Selected net and highlighted nets — Display selected net and highlighted nets at full brightness and all other objects at reduced brightness. • Associated nets — Displays nets associated to selected net by conductivity and differential IBIS models in full brightness. **Note:** This option is unavailable if the Selected Net and Highlighted Nets check box is cleared. • Coupled nets — Displays nets associated to selected nets by coupling. **Note:** This option is unavailable if the Associated nets check box is cleared or you do not have the BoardSim Crosstalk license. If nets appear to be too complicated or seem to be connected to other nets they should not be, you probably have undetected power supply nets. See Verifying That Power Supply and Signal Nets are Recognized Correctly. Turning off associated-net viewing only removes the extraneous nets from the board viewer, not from the simulation database.
Highlight on the fly	Some objects in the board viewer can temporarily change color, or highlight, when you move the pointer over them. In a crowded area in the layout, this feature can help you accurately position the pointer for right-click operations, such as selecting a net for interactive simulation. You can control on-the-fly highlighting by object type. Select the objects to enable highlighting on the fly.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
Show objects	You can choose to view or hide anti-pads generated by BoardSim. Anti-pads, however, are always displayed when the .HYP file contains explicit anti-pad geometry information. • Pads—Display pads connected to vias or pins. • Anti-pads —Display the BoardSim-generated clearance between an object, such as a pad, trace, or via, and the plane layer on which it resides. This check box is unavailable when the *.HYP* file does not contain copper pour information. This option links to the Show Anti-Objects In Board Viewer option in the Setup Anti-Pads and Anti-Segments Dialog Box, so these two options always have the same value.
Remove Highlights	Select to remove highlighting for all nets.

Related Topics

Viewing a Board in 2-D

Viewing Filter Dialog Box

Scope: BoardSim

To access: Right-click over an empty area in the board viewer and choose **Show Viewing Filter**.

Use this dialog box to control the visibility of individual stackup layers and highlighted nets in the board viewer.

On dense boards with many layers in the PCB stackup, it can be difficult to view objects on inner layers because they be obscured by objects on outer layers. In these situations, you can temporarily hide the outer layers.

For example, in a dense 10-layer stackup, you might have trouble seeing the details on inner layers 5 and 6. To work around this, you could use this dialog box to hide layers 1-4 and 7-10 in the board viewer.

Tip

Drag a dialog box edge to change its height or width.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Stackup Layers pane	Display/hide stackup layers by selecting/clearing the check box. If a MultiBoard project is open, expand the tree for a specific board to edit its layer visibility options. Edit the color of the stackup layer by clicking the color square.
• Show all	Display all stackup layers. If a MultiBoard project is open, applies to all boards.
• Hide all	Hide all stackup layers. If a MultiBoard project is open, applies to all boards.
Dimming and Transparency	
• Disable Dimming	Displays all objects in board viewer at full brightness. Another way to disable dimming is to right-click over an empty area in the board viewer and select Disable Dimming.
• Enable Transparency	Displays board layers with transparency.
• Global Transparency	Sets the transparency for all board layers.
• Areas Transparency	Sets the transparency level for metal plane areas.
Highlighted Nets pane	Display/hide highlighted nets by selecting/clearing the check box. If you hide stackup layers in the Stackup Layers pane, geometries on hidden layers are not displayed for nets you highlight in this pane. If a MultiBoard project is open, expand the tree for a specific board to edit the visibility options for highlighted nets on that board.
• Show all	Display all highlighted nets. If a MultiBoard project is open, applies to all boards.
• Hide all	Hide all highlighted nets. If a MultiBoard project is open, applies to all boards.
• Highlight Net	Opens Highlight Net Dialog Box.
• Disable Dimming	Displays all objects in board viewer at full brightness. Another way to disable dimming is to right-click over an empty area in the board viewer and select Disable Dimming.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Viewing a Board in 2-D

xPCB/xDX View

Scope: BoardSim

To access: **File > Run xPCB/xDX View**

Use this dialog box to view layout designs stored in *.CCE* format in xPCB/xDX View or the BoardSim board viewer.

xPCB/xDX View has the following advantages over the board viewer:

- Displays artwork layers and manufacturing data not supported by the BoardSim *.HYP* file format
- Displays very large layouts in a viewer that is faster than the board viewer

You can create these files by exporting a design from Siemens EDA Xpedition Layout, PADS Professional Layout, or CAMCAD Professional.

For information about zooming and panning, see Zooming and Panning.

BoardSim can directly load *.CCE* files. See Opening a Design.

To enable selecting nets in xPCB/xDX View for simulation in BoardSim, load the *.CCE* file for the design into BoardSim and load the *.CCE* file for the same design into xPCB/xDX View.

Options

Option	Description
File menu	
Open	Opens a *.CCE* file.
Select menu	
Net for BoardSim Simulation	Opens the Select Net for BoardSim Analysis dialog box. The filter box supports wildcard characters. Use the * wildcard to match any number of characters. Use the ? wildcard to match any one character.
	Displays the board X/Y origin and lines representing the X and Y axes.
	Fits the full board in the window. Does not show any objects that might lie outside the boundary of the board.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Option	Description
	Fits all objects in the window. Use this option when the layout contain objects that lie outside the boundary of the board.
	Zoom the view to the selected objects.
	Show the previous view.
	Show the next view.
	Opens the Display Control dialog box from which you can control the visibility of objects in xPCB/xDX View. The dialog box has the following three tabs: • Layers— Toggle the visibility of layout layers. You can change the color of the layers by double-clicking the color column for the layer to change. • Graphics Classes—Toggle the visibility of the layout graphics classes. • Insert Types—Toggle the visibility of the layout insert types.

Zooming and Panning

Table 10-6 shows how to use the mouse to zoom and pan.

xPCB/xDX View and Xpedition Layout have the same zoom and pan behaviors.

Table 10-6. Zoom and Pan Commands for xPCB/xDX View

Operation	Button	Mouse Action
Zoom in	Middle	Click or scroll wheel (roll forward)
Zoom out	Middle	Shift+<click> or scroll wheel (roll backward)
Zoom in area	Middle or Right	Shift+<press and drag> left to right
Zoom out area	Middle or Right	Shift+<press and drag> right to left
Pan	Middle	<press and drag>

Related Topics

Viewing a Board in 2-D

Zooming and Examining a FastEye Diagram

You can use any of the controls in the following table to examine FastEye diagrams.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Controls

Control Type	Procedure
Zoom	Click **Zoom**  , position the mouse pointer over one corner of the zoom box you want to create, drag to define the other corner, and then release the mouse button.
Fit to window	Click **Fit to Window** 
Voltage offset	Type a value in volts in the Vertical Position box or click an arrow next to the box. Use the vertical position option to shift the waveforms, and the 0.0 V ground position marker, in the main screen up or down relative to the grid. By contrast, the vertical scroll bar moves the grids, waveforms, and green ground marker up and down together. The vertical position controls create a voltage offset by adding or subtracting voltage to or from the data. When changing the vertical position, the grids remain stationary while the waveforms and ground marker move up and down. The available vertical position range is plus/minus five divisions with a precision of 1/10 division.
Voltage scale	Type a value in millivolts per division in the Vertical Scale box or click an arrow next to the box.
Time offset	Click an arrow next to the Horizontal Delay box. Use this control to center the eye in the screen. The available horizontal delay range is 0ns to 100.000ns, with a precision of 1ns.
Time scale	Click an arrow next to the Horizontal Scale box.
Scroll	Drag the scroll bar to the right or bottom of the screen to move the grids, eye diagram, and eye mask (if it is enabled) as a group.
Settings readout	Check or uncheck Readout text. The horizontal scale, vertical scale, horizontal delay, and vertical offset values are displayed near the top of the screen. Disable the readout to reduce clutter.

Related Topics

FastEye Viewer

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Appendix A
Learning

Understanding HyperLynx behavior can involve technical concepts that sometimes require additional explanation. Refer to this information as needed.

Topic	Description
Board Design Tutorials	The board design tutorials walk you through how to use BoardSim to simulate a design after PCB placement and routing, and perform a system simulation of a multiple-board design. In the process, you will become familiar with BoardSim simulation types, as well as the graphical user interface.
Schematic Design Tutorials	The schematic design tutorials walk you through opening a design project in LineSim, editing schematics and running several simulations. In the process, you will become familiar with LineSim simulation types, as well as the graphical user interface.
Tutorial Reference Information	There is background information related to the technology covered in the board design and schematic design tutorials.
Preparing a Schematic for DDRx Batch Simulation	Use this procedure to assign schematic symbol properties to enable the DDRx Wizard to identify components and nets in the DDRx interface. You can also add text comments to label nets in the schematic.
HyperLynx SI PI Automation	To help you learn and use ActiveX automation for HyperLynx SI/PI, the installation includes several introduction and reference documents, and example scripts.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Board Design Tutorials

The board design tutorials walk you through how to use BoardSim to simulate a design after PCB placement and routing, and perform a system simulation of a multiple-board design. In the process, you will become familiar with BoardSim simulation types, as well as the graphical user interface.

You can perform signal-integrity simulations for a selected signal net or in batch mode for a large number of signal nets. Similarly, you can perform DC voltage drop simulations for a selected power supply net or in batch mode for a large number of power supply nets.

Some of the designs for these tutorials contain old signaling technology and circuit geometries. However, the objective of the tutorials is to show you how to obtain simulation results to help you validate your design or measure its performance.

Note

You can make changes to the tutorial designs and save them. To restore the original tutorial design or model files, extract them from tutorial_golden_files.zip, which is located in the same folder as the HyperLynx executable file (bsw.exe). Example location:

\MentorGraphics\<release>\SDD_HOME\hyperlynx64\tutorial_golden_files.zip

Tutorial	Description
Advanced Via Modeling	BoardSim has the ability to model vias in very-high-speed signal paths.
Visualizing the Geometric and Electrical Characteristics of a Signal Via	Because accurate via modeling is so important in high-speed designs, you can use the Via Visualizer to examine in detail the characteristics of any via in your board design.
Checking the Signal Quality of a Net Crossing Two Boards	The BoardSim MultiBoard option can evaluate signals when they reach the receiver ICs on the daughter boards of a system consisting of a main board and two smaller plug-in PCBs. Some nets in the system start on the main board and run through connectors onto both of the plug-in boards.
Simulating the clk Net	BoardSim allows you to run simulation with detailed simulation waveforms displaying in an oscilloscope.
DC Voltage Drop Simulation	BoardSim can run a DC voltage drop simulation for one or more power-supply nets.
Analyzing Crosstalk on the Virtex-4 Demo Board	BoardSim can simulate crosstalk on a board design.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Advanced Via Modeling

BoardSim has the ability to model vias in very-high-speed signal paths.

At GHz frequencies, a lossy transmission line can increase receiver delay times when taking the loss into consideration. A second phenomenon is often equally noticeable: the electromagnetic effects of PCB vias. Vias, specifically via inductance and capacitance, can cause unexpected delays and signal distortion, especially as frequencies grow higher. This tutorial looks at how to model vias with BoardSim.

Note

This tutorial describes signal-via modeling that does not take the power-distribution network (PDN) into account. See "Running Signal-Via Bypass Simulation" on page 268 and "Exporting a PDN to an S-Parameter Model" on page 403.

This tutorial also does not describe how to represent signal vias in free-form schematics with S-parameter models created by 3-D electromagnetic simulators. See "Via Properties Dialog Box - Via Properties Tab" on page 1077.

Restrictions and Limitations

- When you launch BoardSim by exporting a board from PADS Professional Layout, PADS HyperLynx DC Drop, or PADS HyperLynx DDR, BoardSim can open only CCE files.

Prerequisites

- A Via Models license. This is required to include inductance in via models and to choose among advanced via-modeling options.

Procedure

1. Open either board design:
   a. Choose the **File > Open Board** menu item, then select *demo.cce* or *demo.hyp*.
   b. When prompted to restore session edits, click **OK**.
2. Choose the **Select > Net by Name for SI Analysis** menu item, then double-click the net named clk.

   This net is poorly routed from a high-speed standpoint because it contains multiple vias.

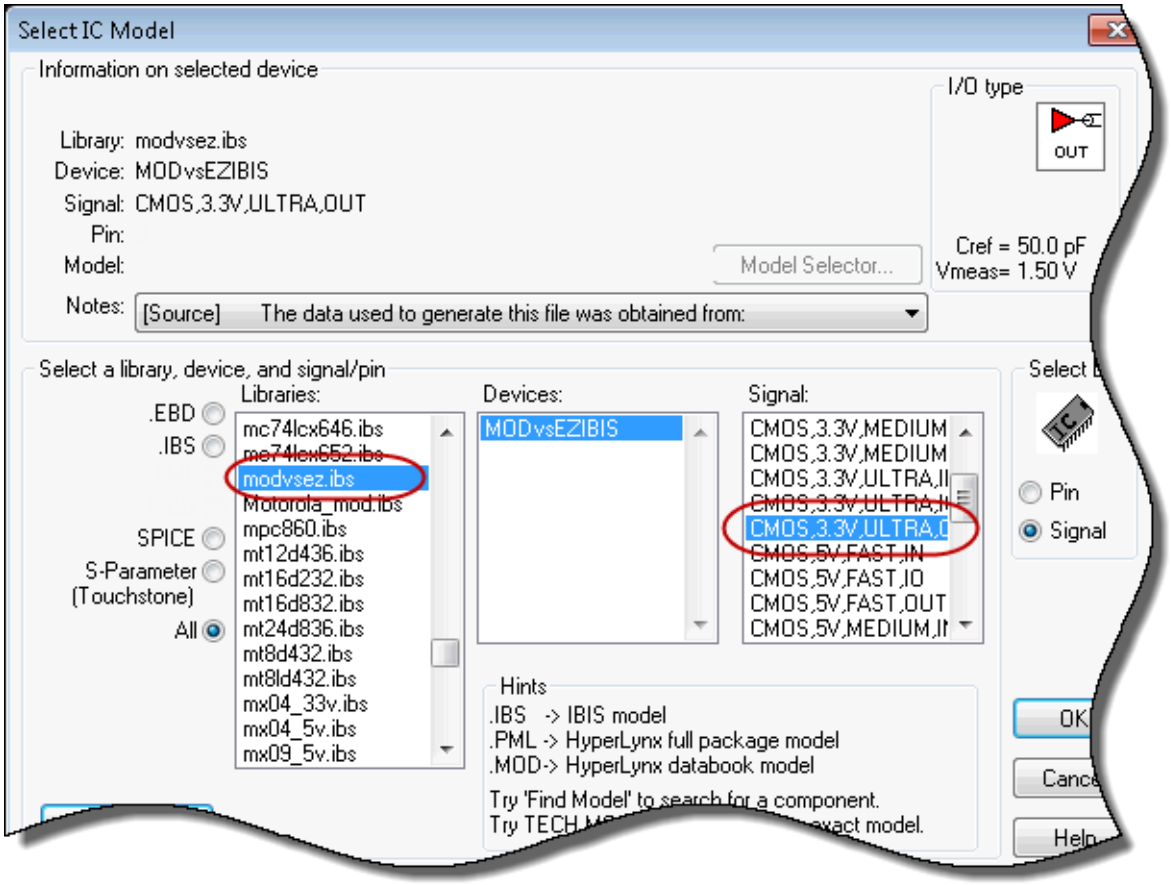
   The effects of vias are easiest to see with fast switching edges because vias look electrically longer to higher-frequency signals and cause more signal distortion.

   a. Choose the **Models > Assign Models/Values By Net** menu item.
   b. In the Pins list, double-click pin U1.13. The Select IC Model dialog box opens.
   c. In the Libraries list, select modvsez.ibs.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

d. In the Signal list, select the pin CMOS,3.3V, ULTRA,OUT.

e. Click **OK**.



f. Click **Close**.

3. Simulate without via modeling.

   Simulating without via modeling provides the capability of isolating the effects of vias on signal integrity. You can run simulation with via modeling enabled, disable via modeling, run simulation again, and compare the waveforms.

   a. Choose the **Setup > Via Simulation Method** menu item. The Select Method of Simulating Vias dialog box opens.

      The Include Via L and C option controls whether to use any via modeling during simulation.

   b. Uncheck “Include Via L and C” to turn off all via modeling. The other via modeling options become unavailable.

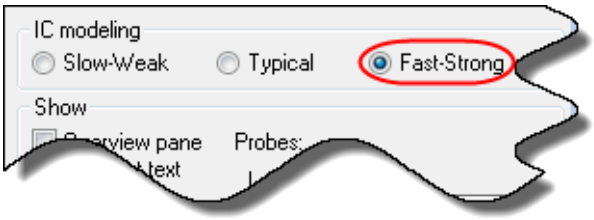
   c. Click **OK**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

d. Choose the **Simulate SI > Run Interactive Simulation** or **Run Interactive Simulation in Oscilloscope** menu item. The Digital Oscilloscope opens.

   In the Stimulus area, select Edge and “Falling edge”.

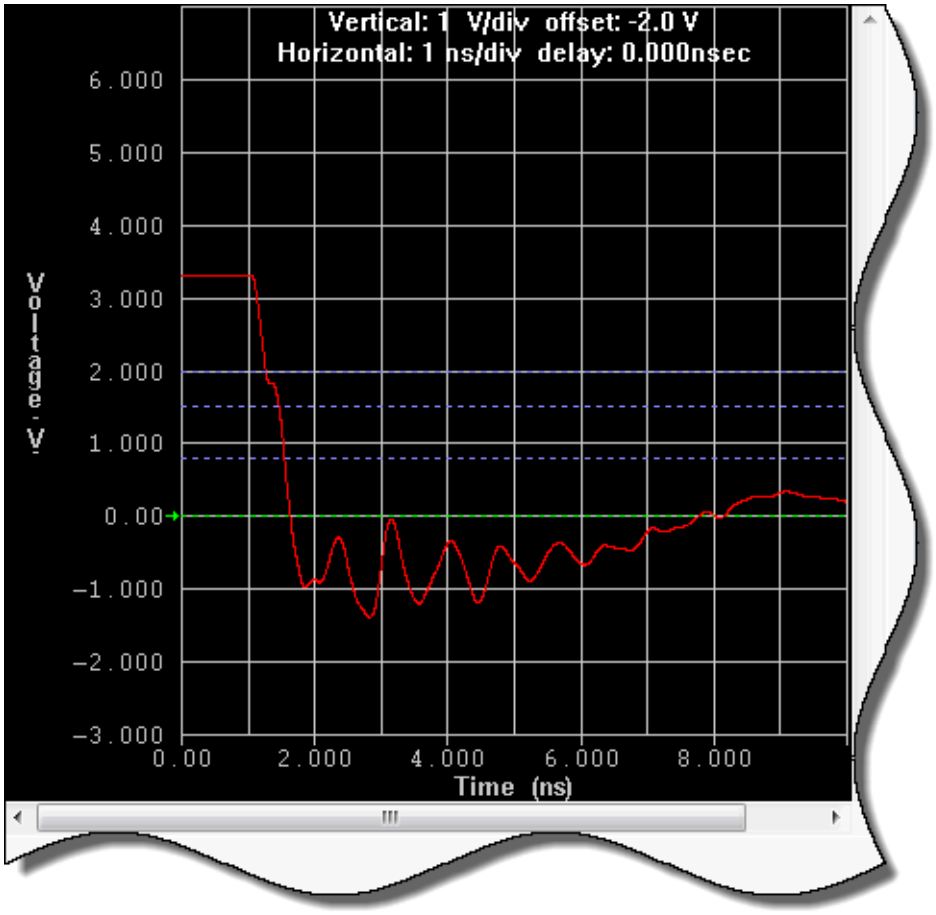
   In the IC modeling area, select Fast-Strong to simulate with the fastest driver edge.



e. Click **Start Simulation**. A waveform appears.

f. In the Show area, under Probes, locate the Pins list.

g. Expand U9 and check pin 9. Uncheck all other probes.



4. Re-simulate with via modeling enabled.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

a. Minimize the oscilloscope.

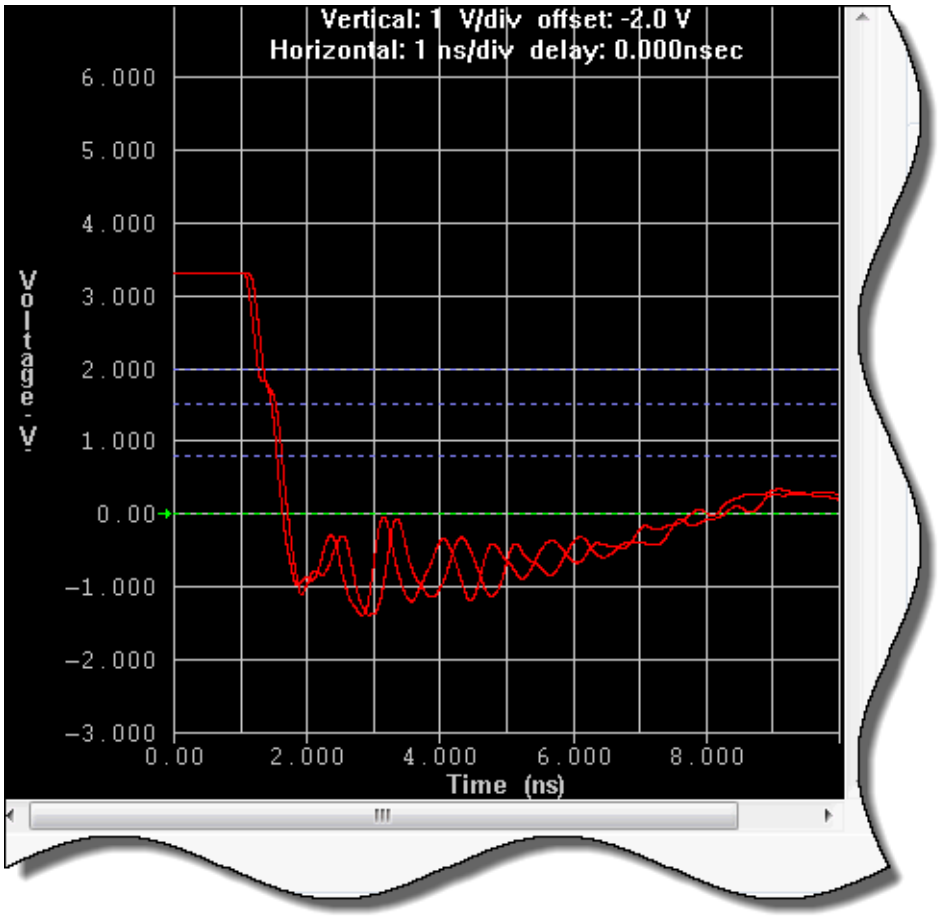
b. Choose the **Setup > Via Simulation Method** menu item. The Select Method of Simulating Vias dialog box opens.

c. Check “Include Via L and C” and select Auto-calculate. This option uses built-in automatic modeling algorithms for vias.

d. Click **OK**.

e. Restore the oscilloscope.

f. Click **Start Simulation**. A new waveform appears.



Comparing the two waveforms, the delay at the receiver ICs displays a clear difference. The delays are pushed out when via modeling is added to the simulation. The effect is similar to increased delay caused by the addition of lossy transmission line simulation. This means that for accurate delay calculations, it is often important to use accurate via modeling. Note that in the BoardSim general batch SI wizard, you can enable both lossy transmission line and via modeling.

5. Set up via modeling. This step briefly discusses the various methods of via modeling supported by BoardSim.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

a. Click **Close** to close the oscilloscope.

b. Choose the **Setup > Via Simulation Method** menu item.

c. Select “User-supplied padstack-specific L and C”. A spreadsheet appears and values are filled in after a short delay.

The selections in the top half of the dialog box offer three types of via modeling:

Table A-1. BoardSim Via Modeling Types

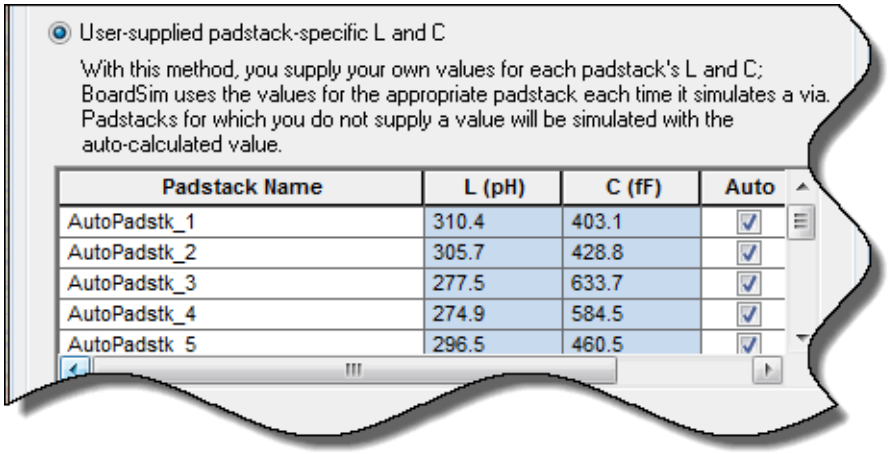
Via Modeling Type	Description
Auto-calculate	The most powerful method runs internal algorithms to automatically model each instance of a via. These algorithms decompose each via into sections and call fast solvers, section-by-section, accounting for detailed effects such as the frequency-dependent inductance of a via as it changes the reference planes of its signal. Also included are the effects of different signal entry and exit layers.
User-supplied global L and C	Allows you to provide a single L and C value to be used for all vias on a board.
User-supplied padstack-specific L and C	A more-advanced type of user-supplied value, managed in a spreadsheet; you can mix auto-calculated values for some vias and specify custom values for certain other padstacks.

Most customers prefer the accuracy of the automated algorithms. However, you can supply your own inductance and capacitance values based on the results of external electromagnetic extractions or lab-measured data. Look briefly at the contents of the padstack spreadsheet. Until you disable some options, each pad stack shows its auto-calculated value. Note the typical values: hundreds of pH and fF, for L and C respectively.

The inductance value is frequency-dependent for padstacks containing signals that change reference planes at least once. Although BoardSim displays the inductance value

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

in the spreadsheet at f=250 MHz, during simulation it uses the knee frequency of the driver-IC switching edge for each net as the calculation frequency.



Visualizing the Geometric and Electrical Characteristics of a Signal Via

Because accurate via modeling is so important in high-speed designs, you can use the Via Visualizer to examine in detail the characteristics of any via in your board design.

Note

While the Via Visualizer can show vias on a signal or power-supply net, this topic describes only signal net behavior.

Accessible directly from the BoardSim board viewer, the Via Visualizer automatically shows you both the geometric and electrical model of a single via, or a pair of coupled differential vias. Note that accurately modeling via coupling is essential for accurate simulation results in SERDES and other high-speed differential paths. This exercise looks briefly at the Via Visualizer for a sample differential via pair.

Restrictions and Limitations

- When you launch BoardSim by exporting a board from PADS Professional Layout, PADS HyperLynx DC Drop, or PADS HyperLynx DDR, BoardSim can open only CCE files.

Prerequisites

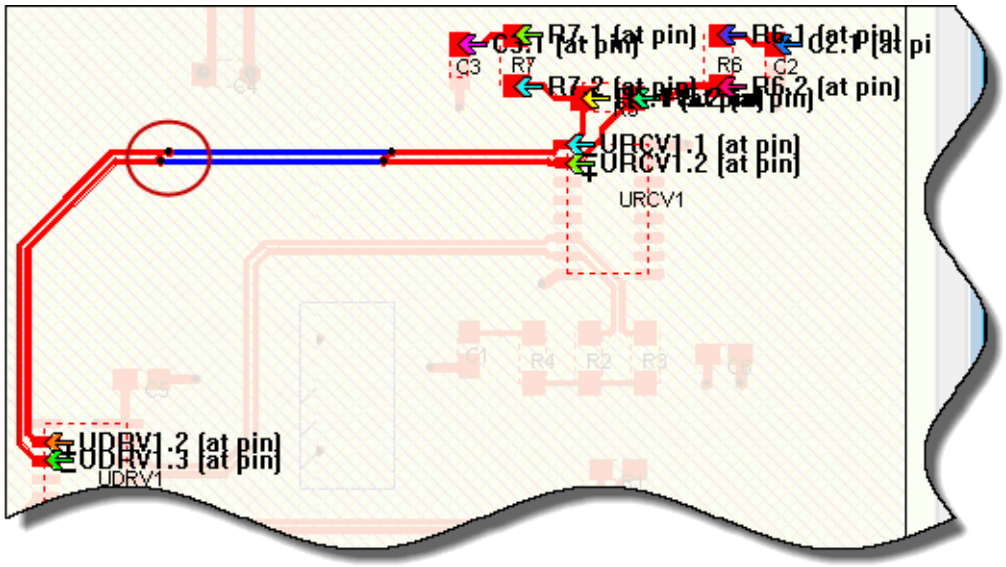
- A Via Models license, which is required to view vias in the board viewer.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

1. Open either board design:
   a. Choose the **File > Open Board** menu item, then select *demodiff.cce* or *demodiff.hyp*.
   b. When prompted to restore session edits, click **OK**.
2. Set up and examine one of the via pairs with the Via Visualizer:
   a. Choose the **Select > Net by Name for SI Analysis** menu item, then double-click the net DRV1_OUT1+.
   b. Verify the **Setup > Enable Trace Coupling** menu item is enabled.

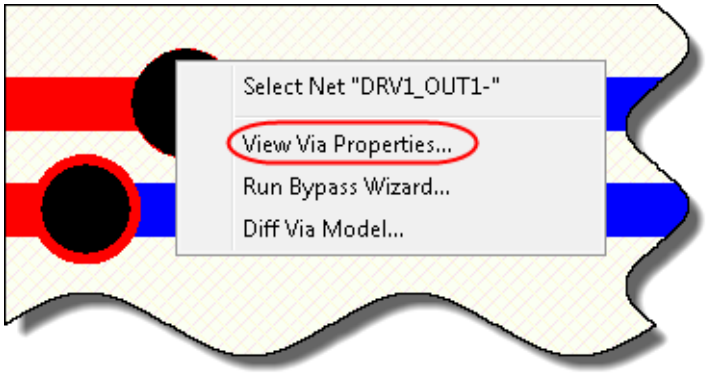
      Note that in the board viewer, other nets are dimmed while net DRV1_OUT1+ and its companion net DRV1_OUT1- appear as a differential pair.



   c. Choose the **View > Zoom Area** menu item, then zoom in on either of the two via pairs for the selected net.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

d. Right-click one of the two vias in the pair and choose the **View Via Properties** menu item.



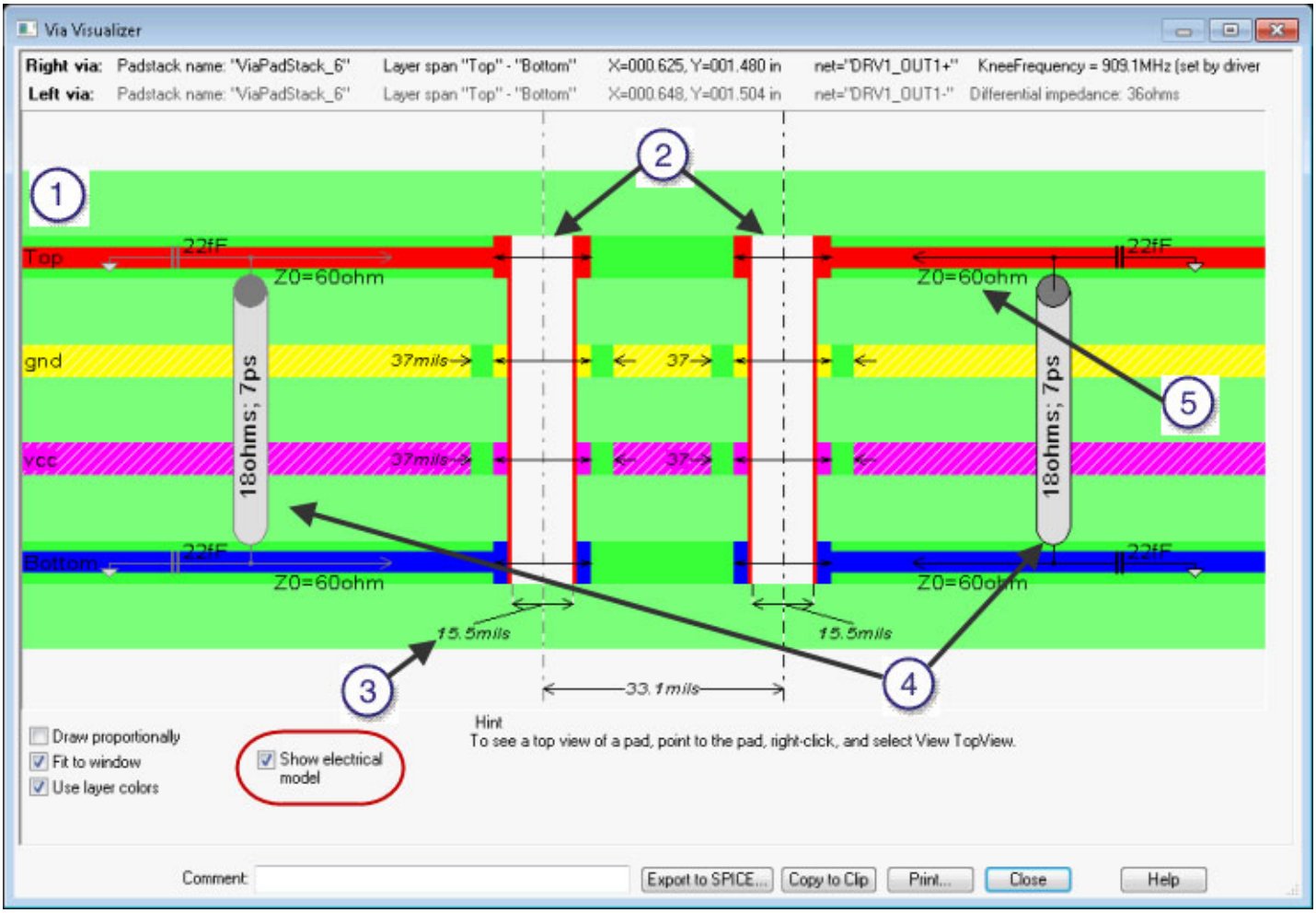
The entire via circle pad turns black when selected. Be careful not to highlight one of the connecting trace segments. The Via Visualizer opens and displays the selected via pair.

e. Resize the window, if necessary, to see the entire graphic in the Visualizer.

After running a fast geometric/electrical check, the Via Visualizer recognizes the selected via as a coupled, partner via, and displays the via as a pair. As the dialog box opens, the Visualizer runs the BoardSim Fast Via Calculator to determine the coupled electrical characteristics of the via pair.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

f. Check “Show electrical model as transmission lines”.



The Via Visualizer displays the following information:

Reference	Description
1	The detailed stackup of the PCB. Signal layers are shown in solid color and plane layers with hatched colors. All metal is displayed in its stackup layer color. Click a layer to display a top view for it. Click the top view to restore the initial side view.
2	The visual geometry of each via in the differential pair, including connected traces, pads and anti-pads, and finished drill hole (after plating).
3	Labeling for all geometric dimensions, including pad shapes/diameters, anti-pad diameters, finished drill hole diameter (after plating), and separation between the two vias.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Reference	Description
4	The electrical model for each via (including the effects of coupling between vias), including the impedance and delay of the via (drawn as a labeled transmission line), 3-D pad capacitance for entry and exit layers (drawn as a lumped capacitor). The software calculates impedance and delay at the *knee frequency*.
5	Connecting traces labeled with their impedance value.

3. BoardSim calculates this information automatically whenever via modeling is enabled. The Via Visualizer makes the modeling information explicit and accessible to you, rather than completely hidden as it is in other signal-integrity tools.
4. If needed, you can adjust backdrilling and plating properties to perform "what if" analysis or fix an incorrect property value. See "Accounting for Backdrilling" on page 135 and "Accounting for Per-Padstack Plating and Constraints" on page 137.
5. Note that you can easily create a SPICE sub-circuit of the entire via structure by clicking **Export to SPICE** located at the bottom of the dialog box.

Related Topics

Via Visualizer Dialog Box

Checking the Signal Quality of a Net Crossing Two Boards

The BoardSim MultiBoard option can evaluate signals when they reach the receiver ICs on the daughter boards of a system consisting of a main board and two smaller plug-in PCBs. Some nets in the system start on the main board and run through connectors onto both of the plug-in boards.

The design is a system consisting of a main board and two identical smaller plug-in PCBs. Some nets in the system start on the main board but run through connectors onto both of the plug-in boards.

Note

You cannot run this tutorial when you launch BoardSim by exporting a board from PADS Professional Layout, PADS HyperLynx DC Drop, or PADS HyperLynx DDR. The tutorial uses a MultiBoard project that contains HYP files, which BoardSim cannot open when launched this way.

Prerequisites

- A MultiBoard license, which is required to load and simulate multiple-board designs.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

1. Load a MultiBoard Project:

   Choose the **File > Open MultiBoard Project** menu item, then double-click *demo_multiboard.pjh*.

   If prompted to restore session edits, click **OK**.

   BoardSim loads each of the boards in the project, similar to how it loads a single board design file. All three boards in this example are visible at the same time in the board viewer.

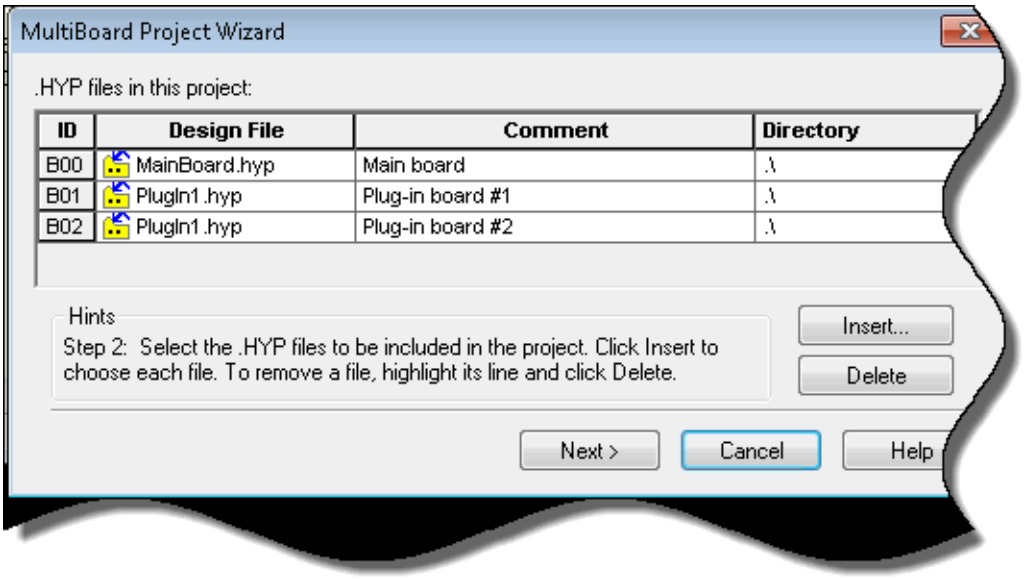
   The PJH file points to the board design files that make up the MultiBoard project.

2. Construct a project using MultiBoard Project Wizard:

   BoardSim makes it easy to connect multiple boards together using the MultiBoard Project Wizard. This example shows how this MultiBoard project was constructed.

   a. Choose the **Edit > MultiBoard Project** menu item. The MultiBoard Project Wizard opens.

      The first page of the wizard lists the boards in the project. Note that a comment now appears beside the HYP file name for each board. These labels also appear in the BoardSim dialog boxes.



      If you were to add a board to the project, just click **Insert** and choose a HYP or CCE file for the board. But this example uses only the three currently-loaded boards.

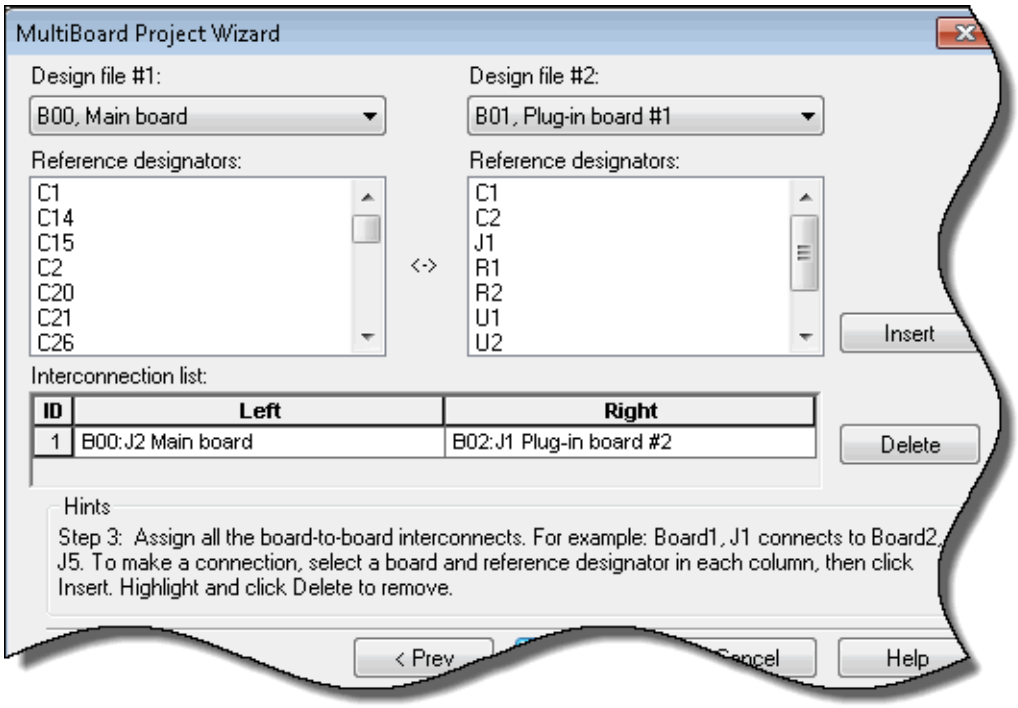
   b. Click **Next**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The second page of the wizard shows the connections between the boards. A single entry in the Interconnection list includes any two connector halves whose pin names match. BoardSim automatically does the pin-by-pin mating.

In the Interconnection list, the main board connector J2 is connected to plug-in board 2, connector J1.

c. You can also remove board connections. Select the top spreadsheet row (ID 1), and click **Delete**.



Note

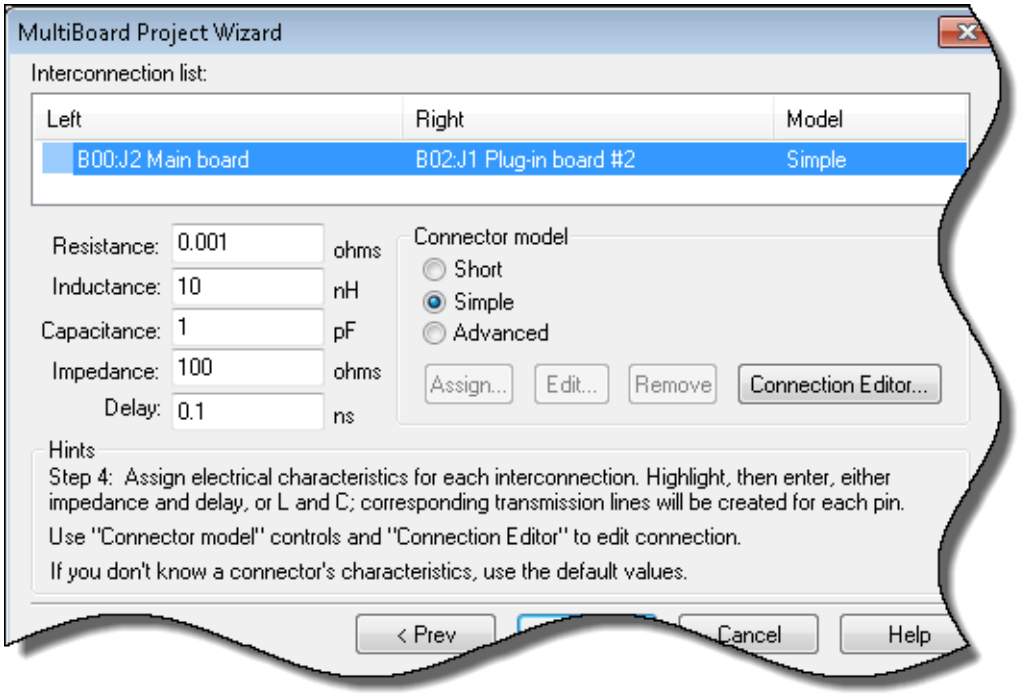
If you have connectors with pin names that do not match, or a connector half that connects to more than one other connector, you can list explicit pin-by-pin connections. See “Setting Up a Multiple Board Design” on page 57 for details.

d. Click **Next** again.

This page shows the electrical characteristics of each board-to-board connector. You can specify the electrical behavior of a connector by providing either a capacitance and inductance, or a delay and impedance. The corresponding transmission lines are created for each pin. For most connectors, use the information from the manufacturer.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

e. Click **Finish** again, click **Yes** to save session edits, then click **OK**.

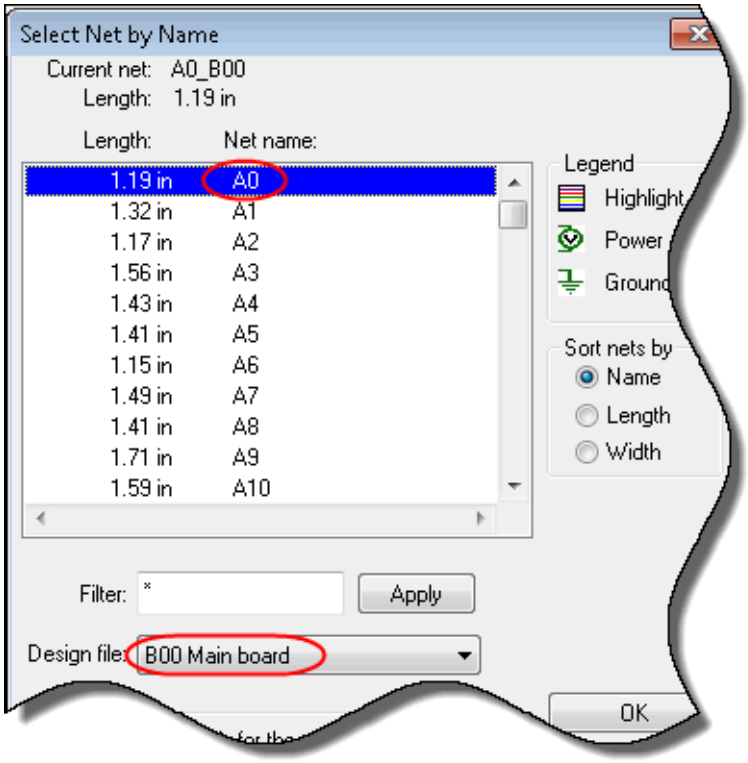


3. Simulate net A0. Net A0 is driven from the main board and has receiver ICs on each of the plug-in boards (but only B02 is connected in this example).

   a. Choose the **Select > Net by Name for SI Analysis** menu item, then double-click net A0. If you cannot see A0 in the list, in the Sort Nets By area, click Name to sort alphabetically.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

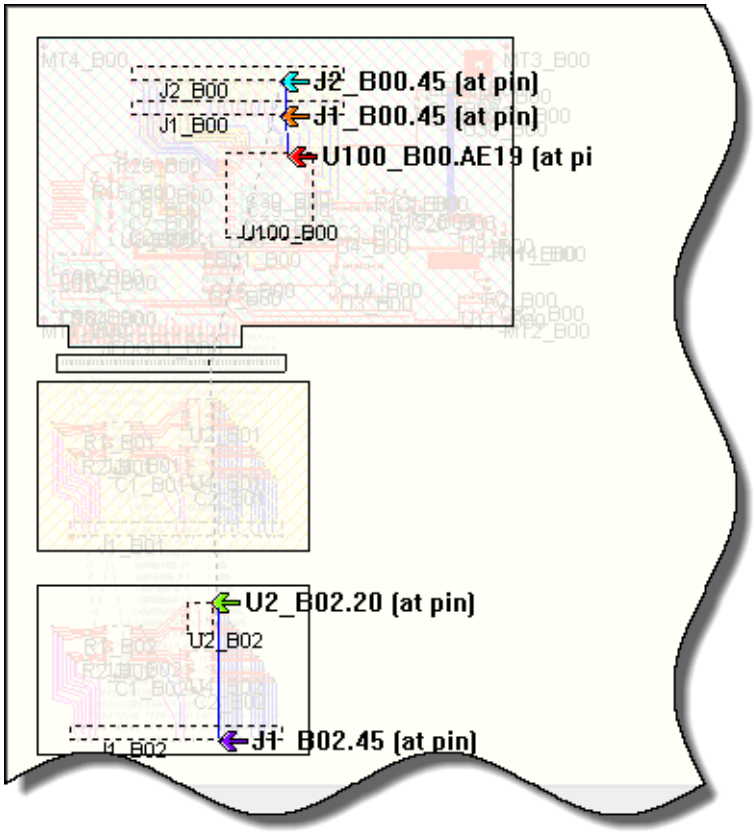
Note that Design File is set to B00 Main board, meaning the net is selected from the main board.



The dialog box closes, and net A0 is highlighted on the main board, along with the nets on the plug-in boards to which it connects (only B02 is connected in this

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

example). Rats nest lines show the connections between boards, through connectors. Notice the arrows on the boards which indicate the location of oscilloscope probes.



4. Set up and run simulation.

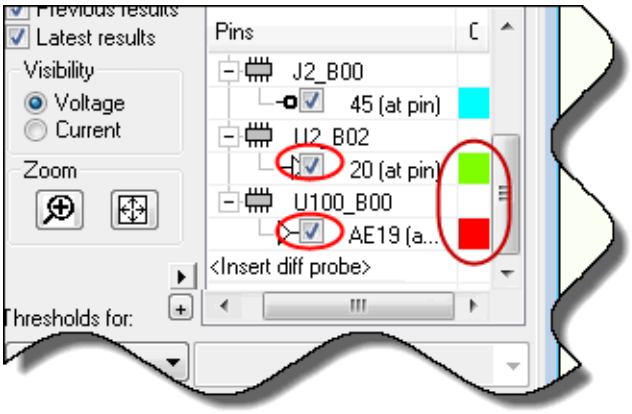
   a. Chose the **Simulate SI > Run Interactive Simulation** or **Run Interactive Simulation in Oscilloscope** menu item. The Digital Oscilloscope opens.

   b. In the Stimulus area, select Oscillator.

   c. In the Pins spreadsheet in the Show area, verify that under U100_B00, pin AE19 is checked, meaning that the probe at that pin is activated. Pin AE19 is the driver pin on the main board, B00.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

d. In the Pins spreadsheet in the Show area, verify that under U2_B02, pin 20 is checked, meaning that the probe at that pin is activated. Pin U2.20 is the receiver pin on the plug-in board #2.



e. Click **Start Simulation**.

The green and red waveforms show the signals at the receiver-IC pins on the plug-in boards. Note that the receivers show some overshoot.

Simulating with just the main board or just one of the plug-in PCBs displays different waveforms. Only by combining the traces from both boards in the simulation do you see the actual, system-level waveforms.

Related Topics

MultiBoard Simulation of Signals Spanning Multiple Boards

Simulating Multiple Boards

Setting Up a Multiple Board Design

Board Design Tutorials

Simulating the clk Net

BoardSim allows you to run simulation with detailed simulation waveforms displaying in an oscilloscope.

Restrictions and Limitations

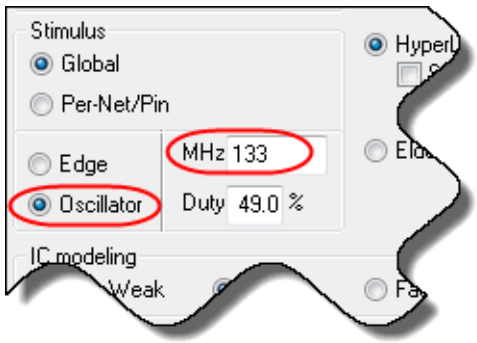
- When you launch BoardSim by exporting a board from PADS Professional Layout, PADS HyperLynx DC Drop, or PADS HyperLynx DDR, BoardSim can open only CCE files.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

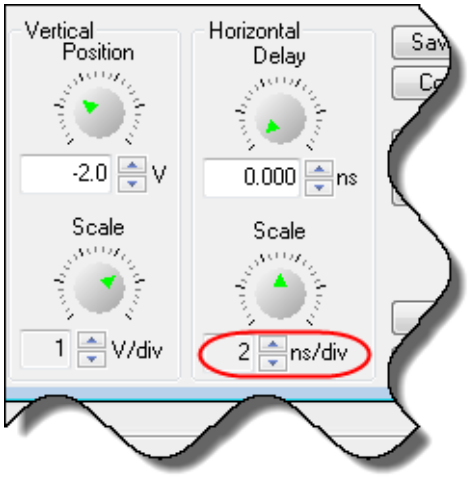
1. Open either board design:
   a. Choose the **File > Open Board** menu item, then select *demo.cce* or *demo.hyp*.
   b. When prompted to restore session edits, click **OK**.
2. Choose the **Select > Net by Name for SI Analysis** menu item, then double-click the net clk.

   Nets display in the board viewer much like in a PCB-layout tool. Each layer has its own color and all aspects of the metal routing including vias and component pads are displayed. The colors on a net correspond to different routing layers. Note also that the routing for the selected net appears in the foreground with all other nets still visible, but dimmed in the background. This allows you to see the context of the selected net.
3. Assign a driver model to U1.13.
   a. Choose the **Models > Assign Models/Values by Net** menu item. The Assign Models dialog box opens.
   b. In the Pins list, double-click pin U1.13. The Select IC Model dialog box opens.
   c. In the Libraries list, select modvsez.ibs.
   d. In the Signal list, select the signal CMOS,3.3V,FAST,OUT.
   e. Click **OK** and **Close**.
4. Simulate the net clk.
   a. Choose the **Simulate SI > Run Interactive Simulation** or **Run Interactive Simulation in Oscilloscope** menu item. The Digital Oscilloscope opens.
   b. In the Stimulus area, select Oscillator.
   c. Verify that the value of MHz is 133.



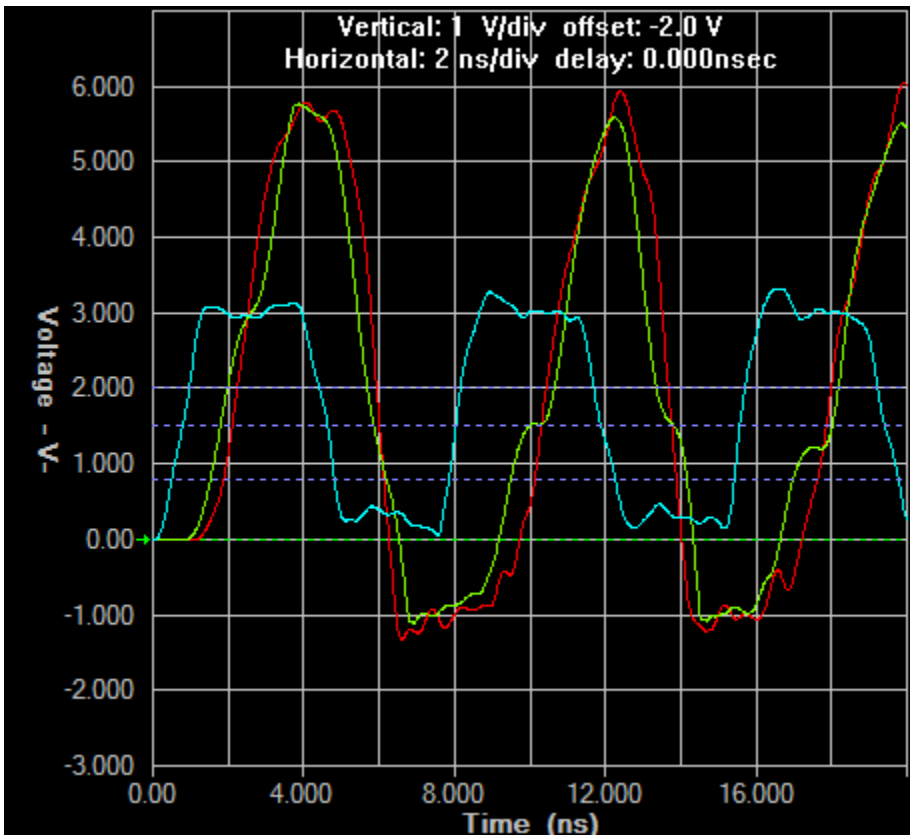
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

d. Change the Horizontal Scale timebase to 2 nsec/div.



e. Click **Start Simulation**.

The waveforms appear in the digital oscilloscope. The waveforms display the voltages at the receivers, U7 and U9, and show significant overshoot. Also, there is considerable high-frequency content in the waveform, which is a potential source of radiated-emissions trouble.



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

5. Improve the signal quality of the net clk by using the Termination Wizard.

   The Terminator Wizard simulates the transmission line and suggests how to fix a net with signal-integrity problems. This step runs the Terminator Wizard interactively on clk to find out how to improve the signal quality of the net.

   An AC terminator was added at the far end of the line (resistor + capacitor to ground) to handle anticipated transmission-line problems. However, the terminator is not functioning correctly.

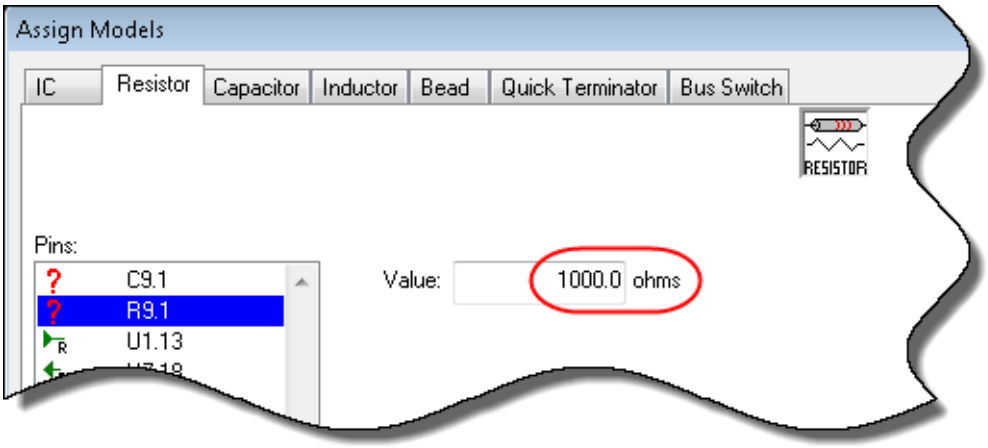
   a. Examine the AC terminator.

      i. Minimize the Digital Oscilloscope.

      ii. Choose the **Models > Assign Models/Values by Net** menu item. The Assign Models dialog box opens.

      iii. In the Pins list, choose R9. The content of the Models area changes to show a resistor.

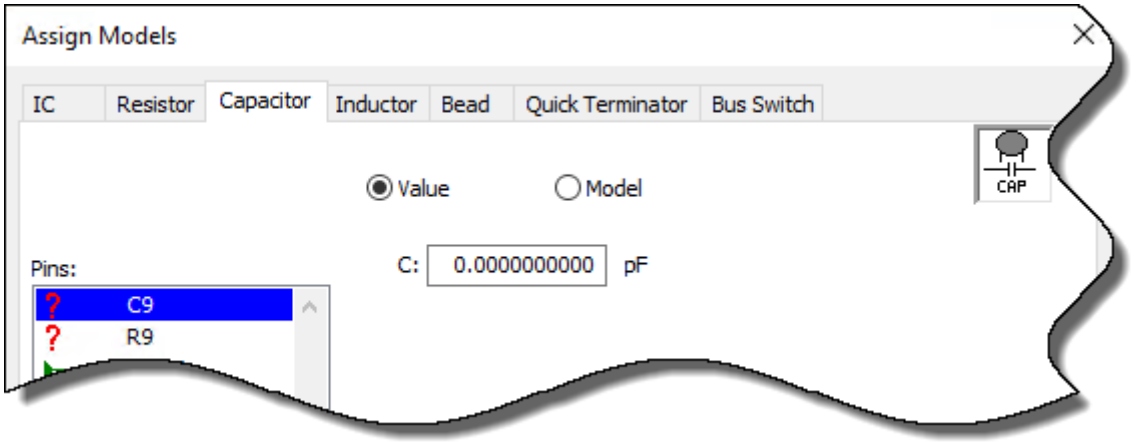
      The resistor value is 1000 ohms, which is too large for proper AC termination. It is possible that the value was just a placeholder.



      iv. In the Pins list, choose C9. The Models area shows a capacitor.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The capacitor value is 0 pF. This is probably too small for a net as long as clk.



b. Run the Terminator Wizard for a recommendation.

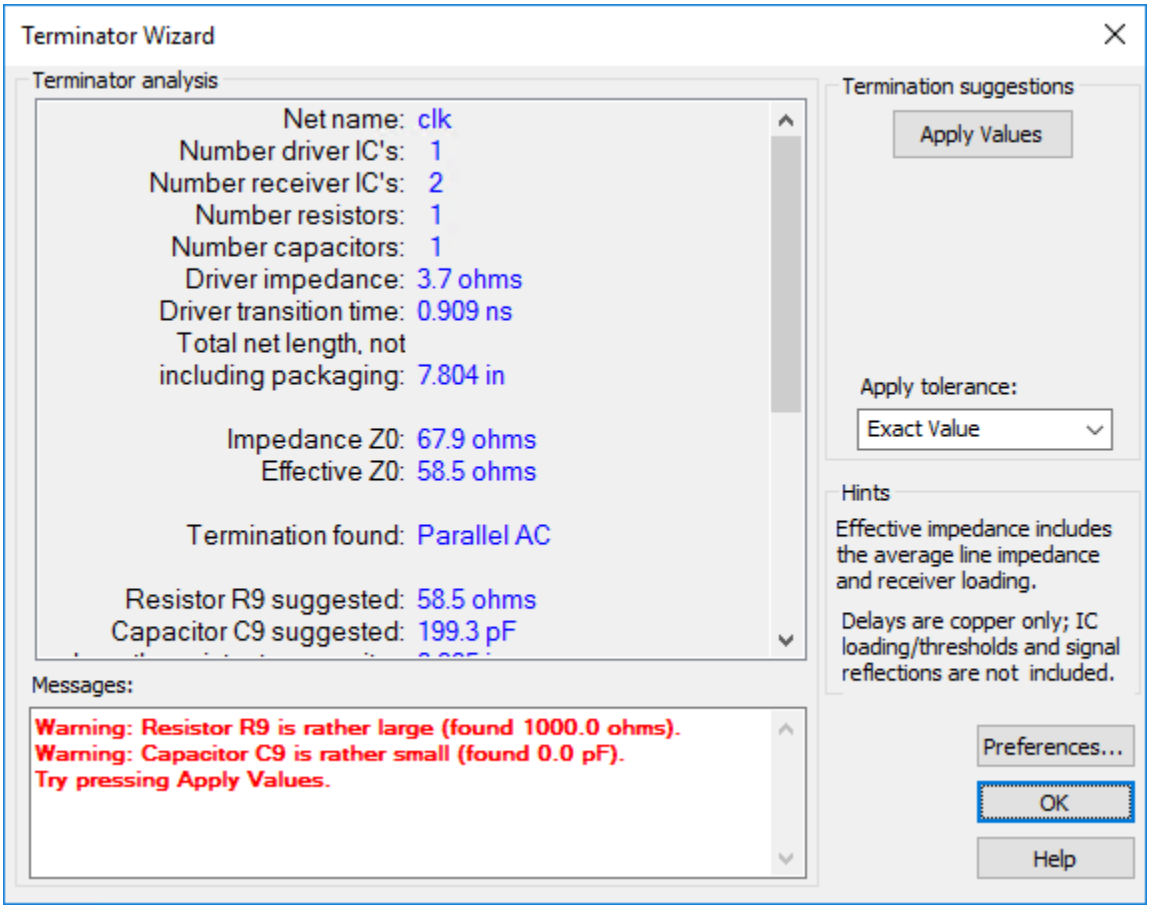
   i. Click **Close**.

   ii. Choose the **Simulate SI > Optimize Termination** menu item. The Terminator Wizard opens.

   The wizard automatically simulates the net, presents a list of trace statistics, and suggests termination values. In this case, the wizard correctly determines that the termination type is parallel AC, and makes suggestions for the optimum values of R and C. In these calculations, BoardSim automatically accounts for such

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

effects as capacitive loading of receiver ICs, total line length, and driver impedance.



iii. In the Termination suggestions area, click **Apply Values**.

The component values recommended by the Terminator Wizard are exported to the resistor and capacitor on net clk.

iv. Click **OK**.

The next step is to re-simulate to see their effect.

6. Re-simulate the clk net.

a. Restore the oscilloscope.

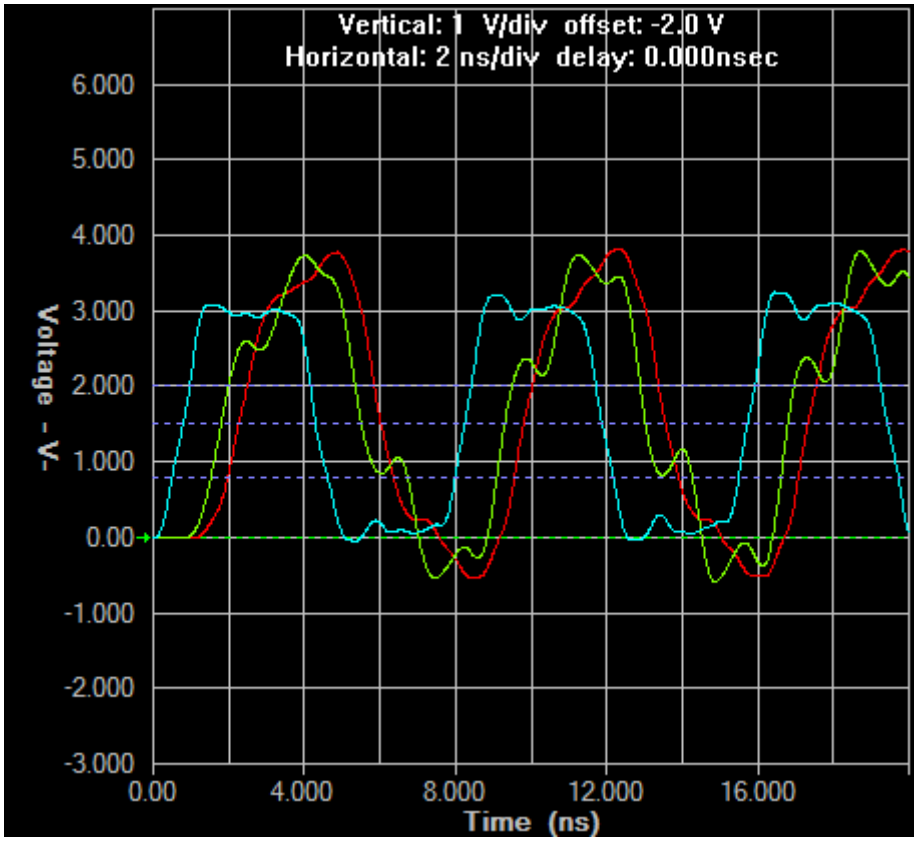
b. Click **Erase** to clear the old waveforms.

c. Click **Start Simulation**.

The overshoot at the receiver ICs is eliminated and the high-frequency content in the waveform is reduced. The signal is much improved from the signal integrity

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

standpoint, and the optimal terminating-component values were found automatically by the Termination Wizard.



One aspect of simulating net clk that this example did not discuss is IC models. Because signal-integrity and crosstalk problems are caused by fast-switching driver ICs, accurately modeling ICs when simulating is crucial. BoardSim ships with many digital IC models and also makes it easy to add new models from IC vendors and add them to the library.

The next step shows how ICs on net clk were matched to some of those models.

7. Look at the reference-designator-to-IC mappings provided with the PCB.

   One way to specify IC models in BoardSim is to map the reference designators (or part names) of the IC to components in the BoardSim model libraries.

   a. Close the oscilloscope.

   b. Choose the **Models > Assign Models/Values by Reference Designator** menu item.The REF-File Editor opens. A REF file maps the reference designators on a specific board to IC models.

      The Design's parts list contains all of the ICs on the board. The Model/value to insert area enables you to choose IC models for each reference designator. These

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

pairings are displayed in the bottom half of the dialog box in the text area and stored in a file named *demo.ref*.

This example uses a simple REF automapping file that maps reference designators U1, U2, U8, and U9 to various IC models. When you, or the batch-mode engine, selects a net with mapped ICs, the models for the IC pins on that net are assigned automatically.

Note

For IBIS models, a REF file only works if the pin names in the IBIS model match the pin names of the device to which it is assigned.

c. Map the reference designator U7 to model type 74AC174_DIP in library 74AC_pml.ibs.

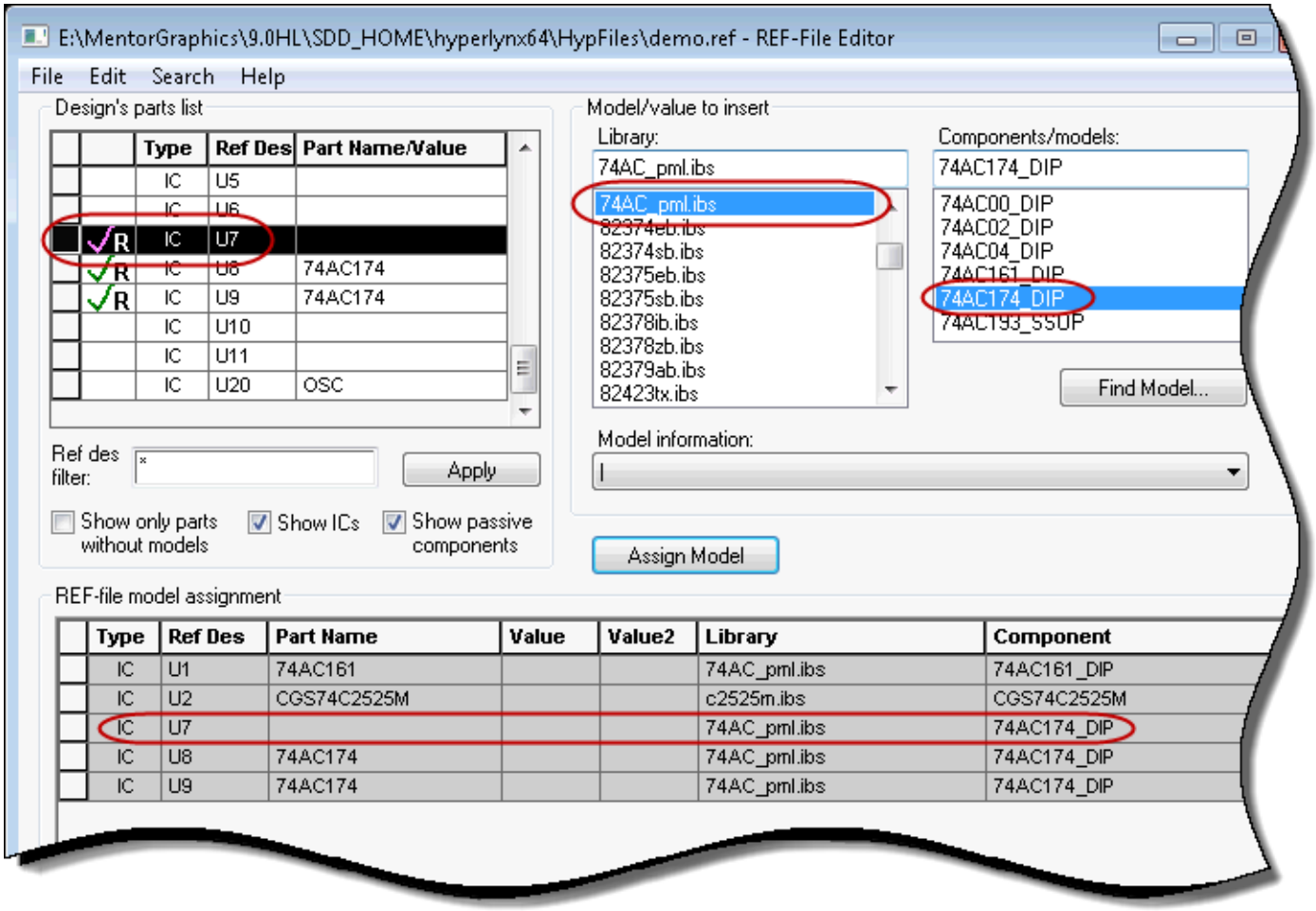
   i. In the Design's parts list, click the spreadsheet row for reference designator U7. You can uncheck "Show passive components" to help find U7.

   ii. In the "Model/value to insert > Library" list, select 74AC_pml.ibs.

   iii. In the Components/models list, select 74AC174_DIP.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

iv. Click **Assign Model**. The new mapping appears in the assignment table.



v. Choose the **File > Save** menu item on the REF-File Editor toolbar.

When simulating any net attached to IC U7, the model 74AC174_DIP is used automatically for all U7 pins.

BoardSim also supports a similar method that maps corporate part names to IC models. This additional method uses an editor and a file called QPL, or qualified part list. These mappings are available for reuse on multiple boards/projects, which is a benefit since corporate names for components rarely change. For more information, see “Assigning Models to Components and Pins” on page 85.

8. Interactively choose a pin model.

Occasionally, you may need to run a quick simulation before you have an model for an IC. Siemens EDA supplies library files named *modvsez.ibs* and *Generic_mod.ibs* that contain technology-oriented models. To use these libraries, you need only know whether an IC is CMOS or bipolar, and approximately how fast it switches (super-fast, fast, or slow).

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

Gigabit-per-second, SERDES-style designs cannot use approximate models due to the required high speeds. These designs often use vendor-supplied SPICE or IBIS-AMI models.

Sometimes it is easier to choose IC models interactively for only the nets of interest rather than mapping reference designators, as when simulating a partially-routed board.

a. Close the REF-File Editor. Discard changes when prompted.

b. Choose the **Models > Assign Models/Values by Net** menu item. The Assign Models dialog box opens.

c. In the Pins list, double-click U1.13. The Select IC Model dialog box opens.

d. Select Generic_mod.ibs and double click the model 74AC11X:LINE-DRV. The Select IC Model dialog box closes and U1, pin 1 is now modeled as a 74AC11X line driver.

e. Click **Close**.

9. Generate a report of board changes.

   As you improve the signal integrity and crosstalk behavior of the nets on your board, a list of component changes accumulates to make to your schematics and/or PCB layout. BoardSim automatically tracks these changes, and conveniently outputs them in a report called the Design Change Summary.

   a. Choose the **Export > Reports > Design Change Summary** menu item. The Design Changes dialog box opens.

   b. Click **Finish**. The report opens.

      Page down in the report to see that the stackup on the board, changed component values, and added components such as Quick Terminators (if any) are all recorded and ready to be handed to those responsible for implementing the changes.

Related Topics

Assigning Models to Components and Pins

DC Voltage Drop Simulation

BoardSim can run a DC voltage drop simulation for one or more power-supply nets.

Prerequisites

The DC Drop license is required to run DC drop simulation.

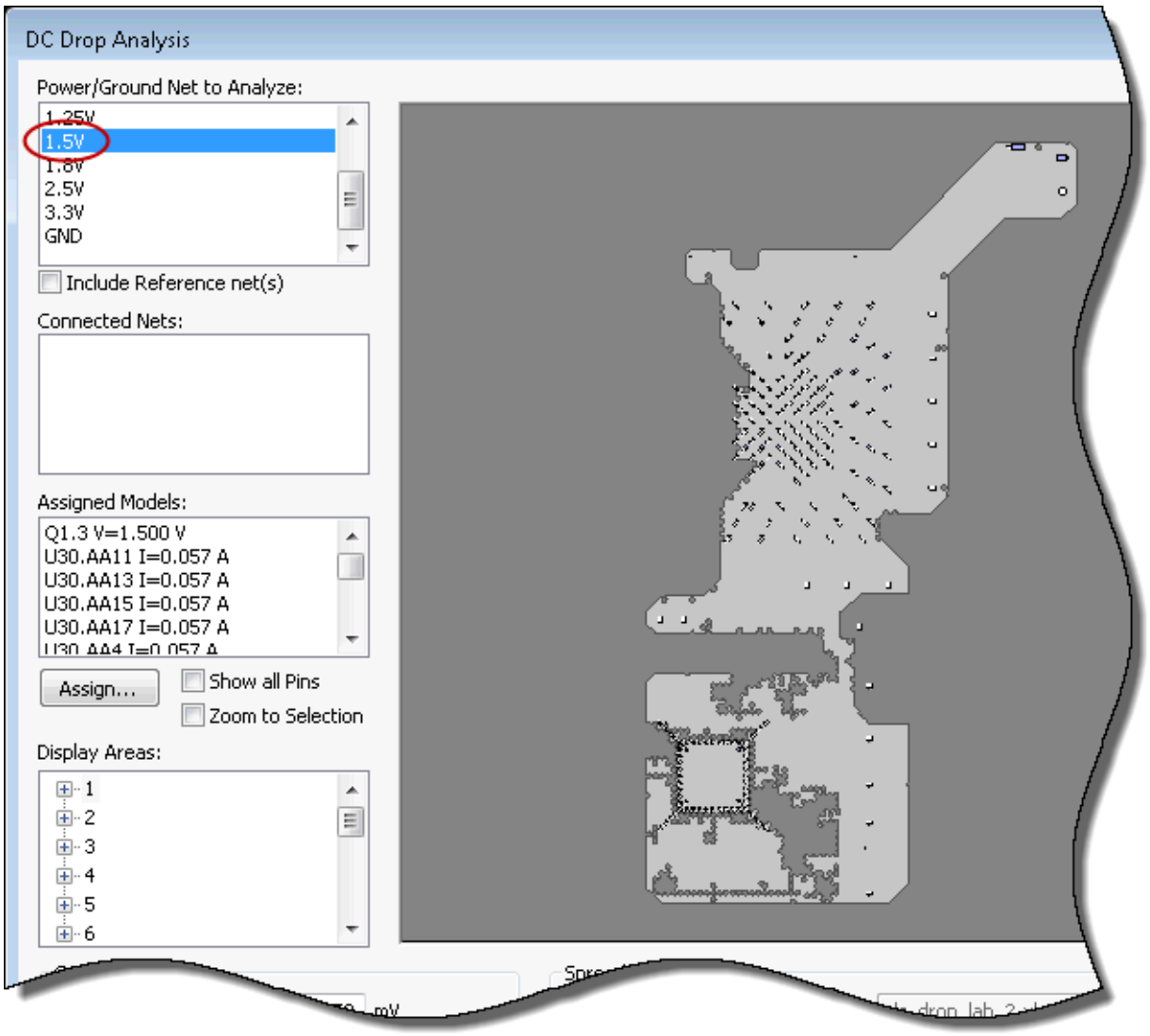
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

1. Open a board design:
   a. Choose the **File > Open Board** menu item, then double-click *dc_drop_lab_2.cce*.
   b. When prompted to restore session edits, click **OK**.
2. Choose the **Simulate PI > Run DC Drop Simulation (PowerScope)** menu item.

   The DC Drop Analysis dialog box opens.
3. Select a power-supply net to simulate.

   From the Power/Ground Net to Analyze list, select the net 1.5V. The viewer displays an outline of the image of the selected net.



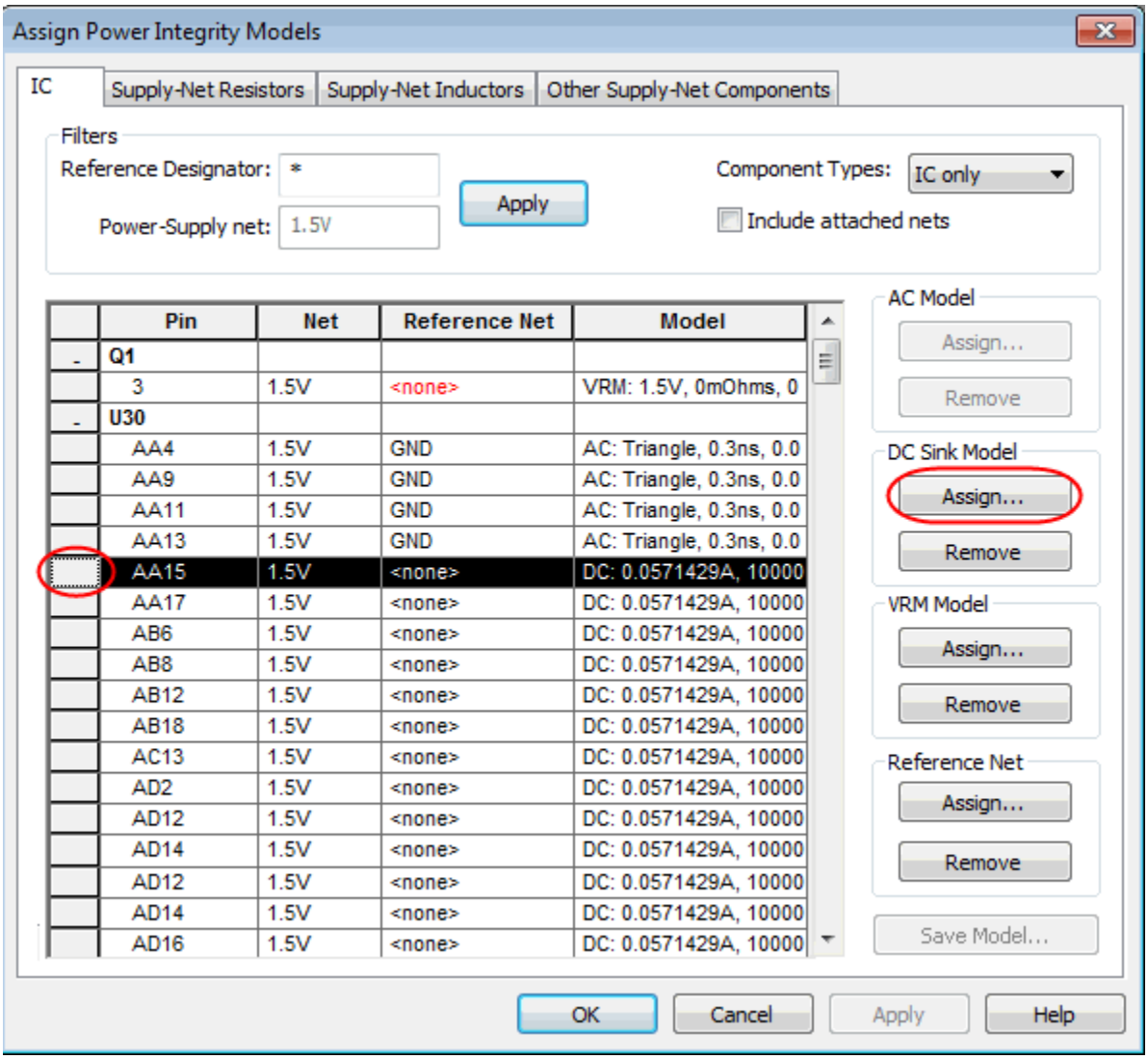
4. Assign models to pins.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The Assigned Models area displays all the models assigned to the different pins. Click **Assign**. The Assign Power Integrity Models dialog box opens.

To run a DC Drop simulation, at least two pins have models assigned to them: one for a DC Model (sink) and the other for a VRM Model (source).

a. Select pin AA15 on component U30. Select the pin by clicking on the gray box next to its number (hold down the Ctrl or Shift key for selecting more than one pin). This highlights the table row and activates the Assign and Remove buttons of the AC, DC, and VRM models. However, assigning AC Models to the pin is irrelevant to this simulation.



b. From the DC Sink Model area, click **Assign**. The Edit DC Supply Pin Model dialog box opens.

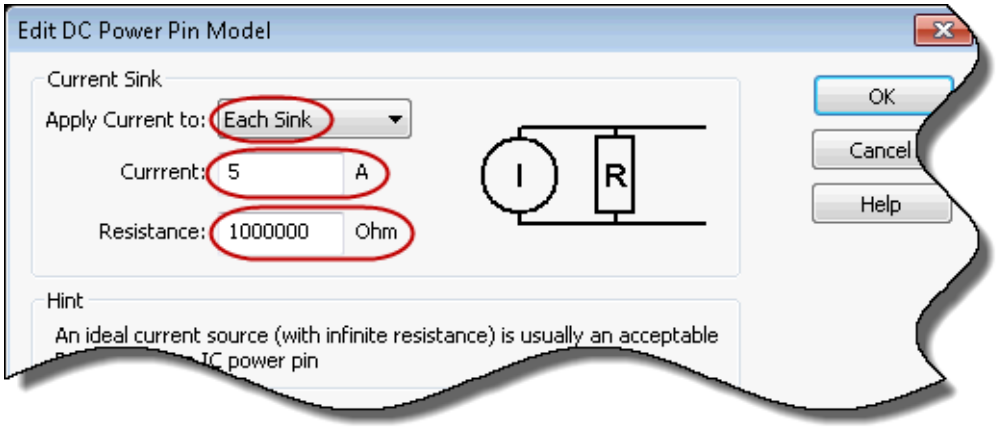
   i. In the “Apply Current to” list, select Each Sink.

   ii. In the Current field, type 5.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

iii. In the Resistance field, type 1000000.

iv. Click **OK**.



c. Repeat the same steps for assigning a VRM Model to a different pin on the power-supply net.

i. Select pin 3 on component Q1.

ii. In the Assign Power Integrity Models dialog box, from the VRM Model area, click **Assign**. The Add/Edit VRM dialog box opens.

iii. In the Model list, select Simple.

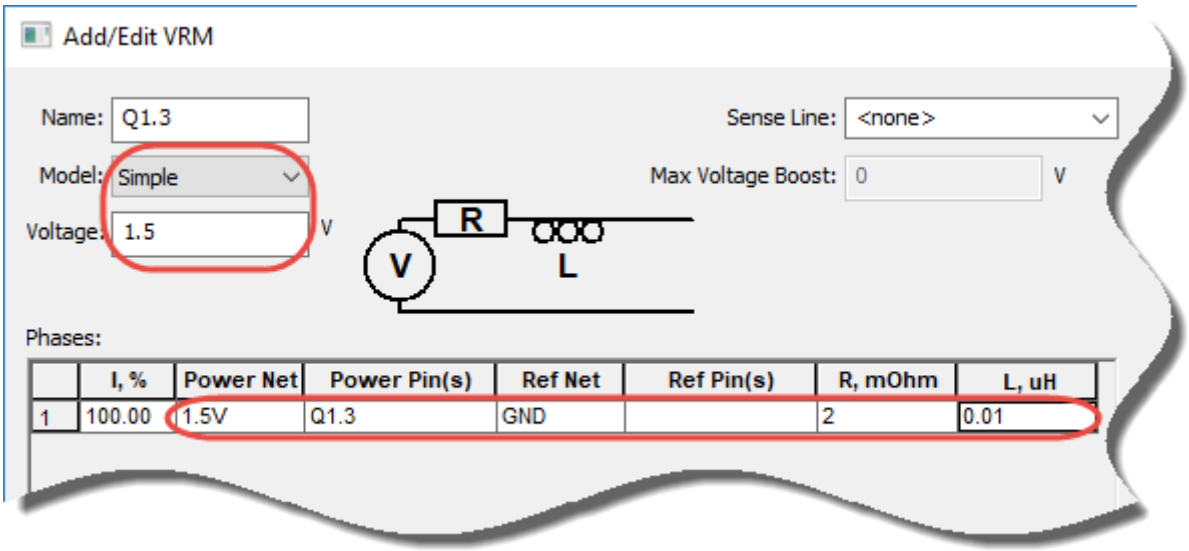
iv. In the Voltage list, type 1.5.

v. In the R table column, type 2.

vi. In the L table column, type 0.01.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

vii. Click **OK**.



After assigning a model to a pin, the model/pin pair displays in the Assigned Models area of the DC Drop Analysis dialog box.

Click **OK** to close the Assign Power Integrity Models dialog box.

5. Check Create Report.
6. Click **Simulate** to run the simulation.

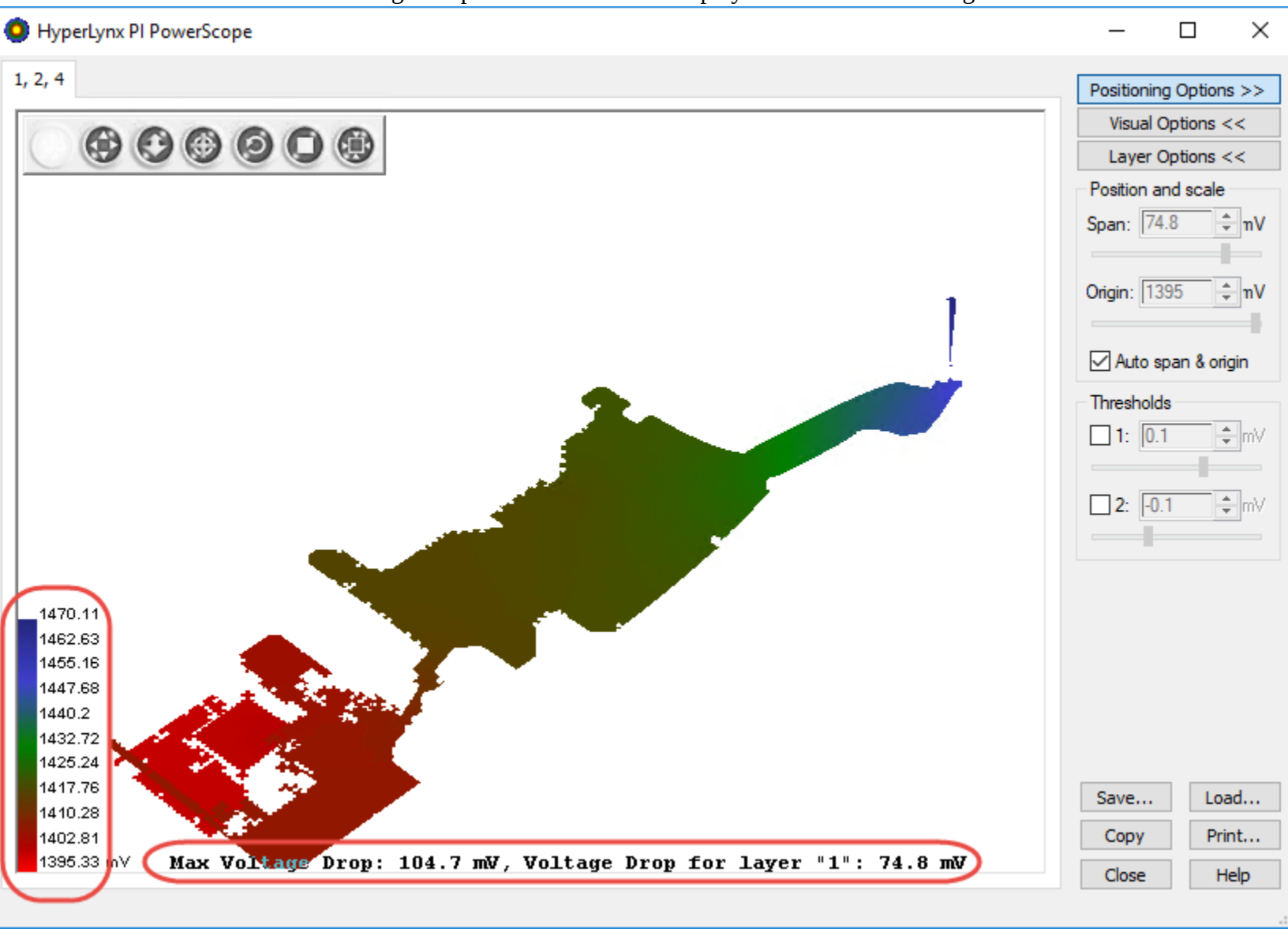
   The Running DC Drop Simulation dialog box appears and displays simulation progress.

   When simulation completes, the software opens:

   - The DC Drop Simulation Report dialog box that displays an easy-to-navigate report that includes screen captures of measurement locations in the board design.
   - The Reporter dialog box that displays a textual report that shows the current and voltage of the pins that you assigned models, and the voltage source and current sink vias.
   - The HyperLynx PI PowerScope dialog box that displays 3-D graphical simulation results and uses a color scale to represent DC drop: dark blue represent the lowest

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DC voltage drop value and red represents the highest DC voltage drop value. The total Voltage Drop numerical value is displayed below the 3-D image.



Use buttons to manipulate the image.

Button	Function
	Turn image
	Shift image
	Zoom in

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

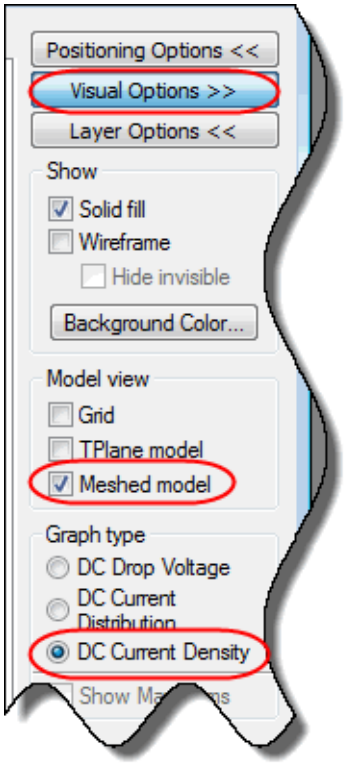
Button	Function
	Inspect image
	Default view
	Top view
	Fit image to window

7. Add a reference plane to the model. The best way to detect a problem in a design is by looking at the DC Current Density graph and displaying a reference plane.

   a. Click **Visual Options**.

   b. In the Model view area, check “Meshed model”.

   c. In the Graph type area, select DC Current Density.



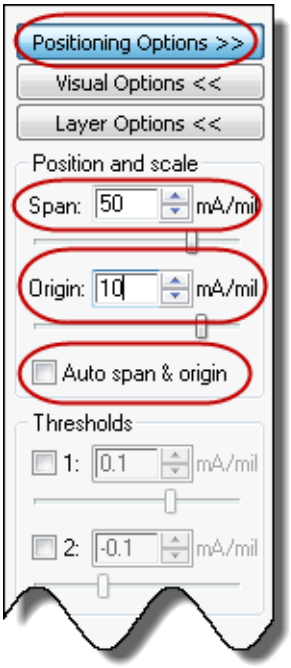
   d. Click **Positioning Options**.

   e. Uncheck “Auto span & origin”. This enables you to modify the current density by modifying the Span value, and visualize how the current density is changing on a board area.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

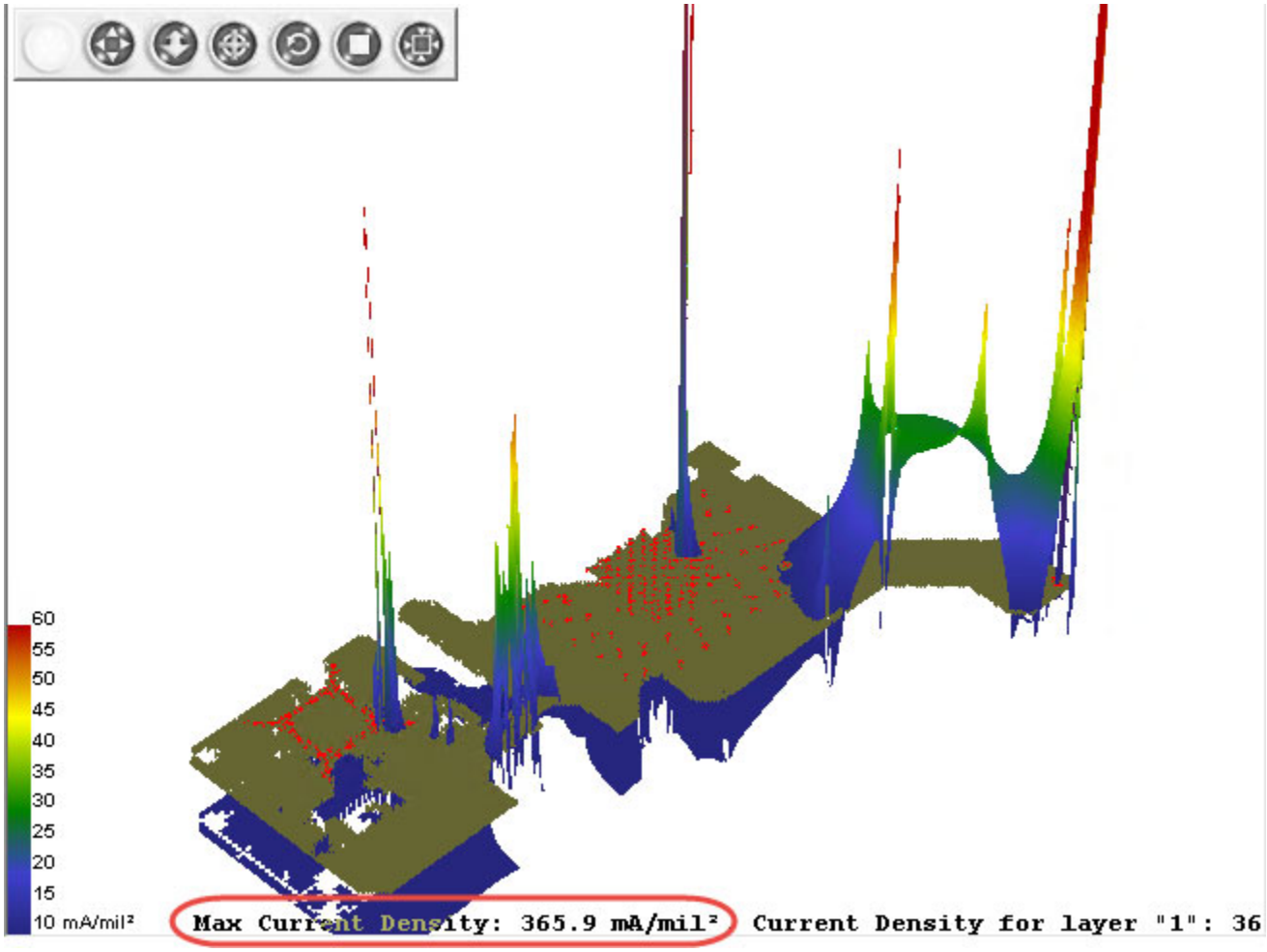
f. In the Span field, type 50 to view the current density across the board.

g. In the Origin field, type 10.



Note the maximum current density is over 365 mA/mil^2.

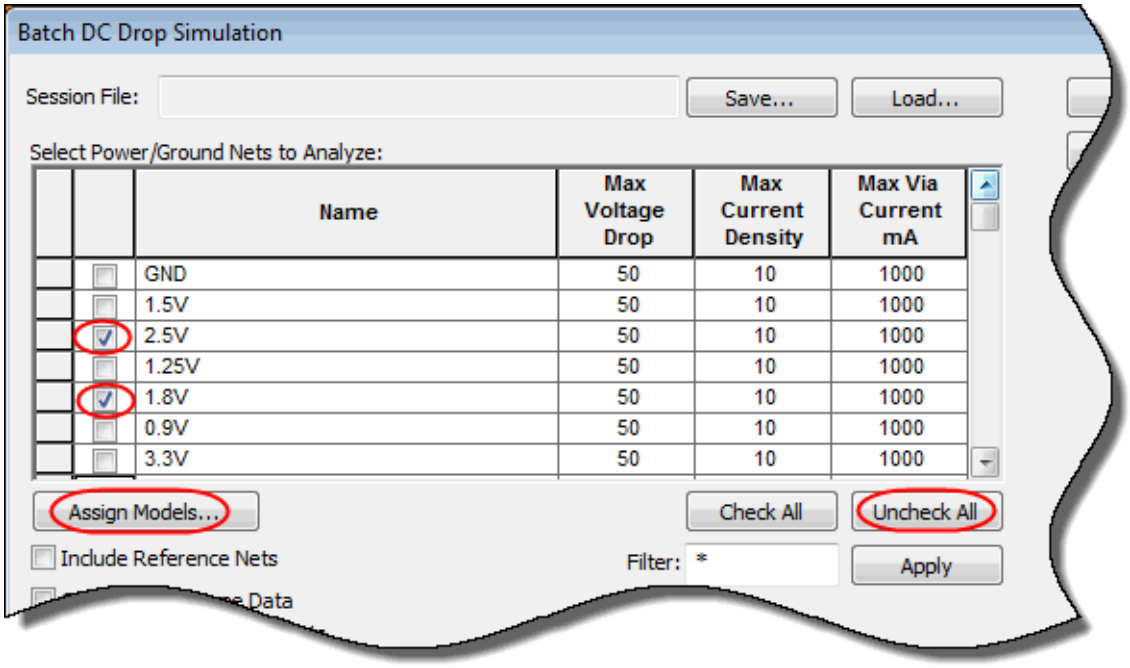
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.



8. Perform a batch DC drop simulation.
   a. Close the various reporting dialog boxes.
   b. Choose the **Simulate PI > Run DC Drop Batch Simulation** menu item. The Batch DC Drop Simulation dialog box opens.
   c. Click **Uncheck All**.

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d. Check the nets 2.5V and 1.8V nets to include them in simulation.



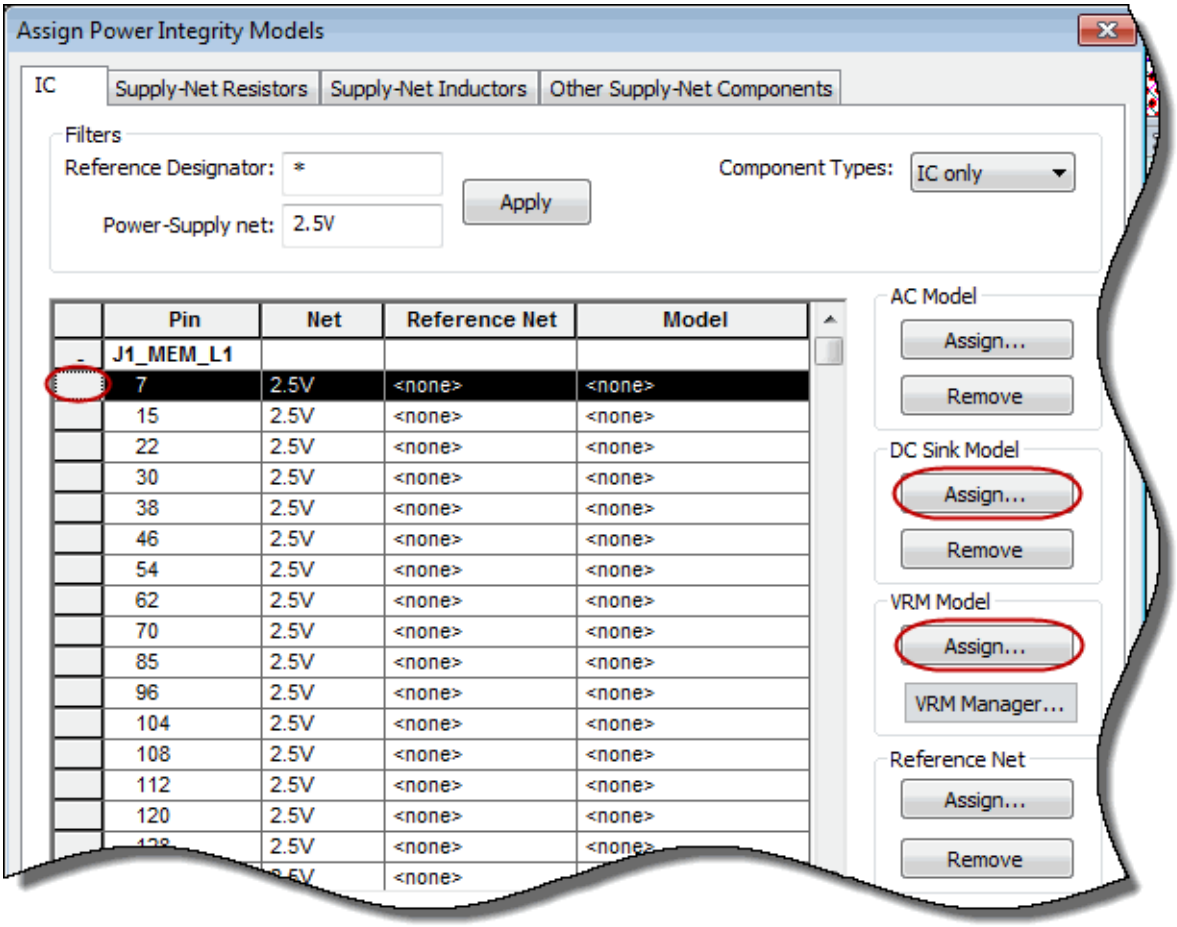
e. Assign models to pins.

A DC drop simulation requires at least two pins with models assigned to them: One for a DC Sink Model (sink) and the other for a VRM Model (source). AC models are ignored for DC drop simulation.

i. Click **Assign Models**. The Assign Power Integrity Models dialog box opens. AC, DC, VRM models or reference nets are assigned to the pins.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

ii. Assign models to the 2.5V rail. In the “Power-Supply net” filter, type 2.5V, then click **Apply**.



f. Select pin 7 of J1_MEM_L1:

i. To assign a model to a pin, select the gray box next to it.

Note that you can select multiple pins by holding down the Shift or Ctrl key.

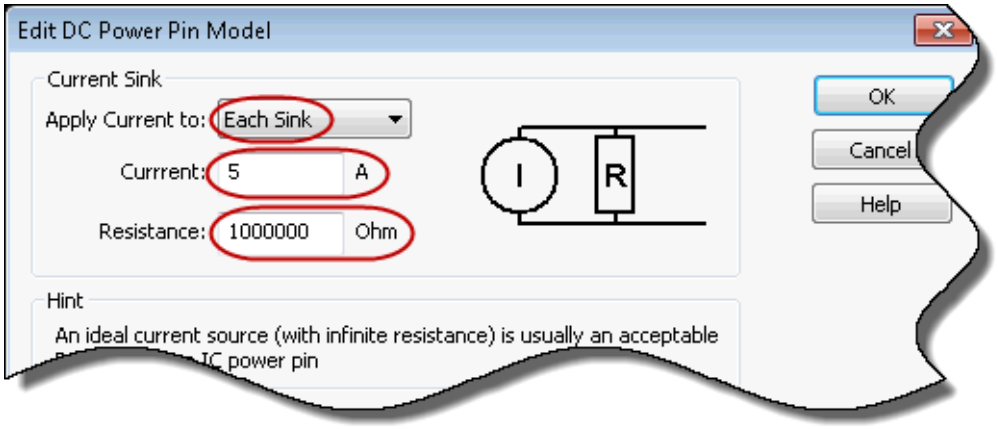
ii. In the DC Sink Model area, click **Assign**. The Edit DC Supply Pin Model dialog box opens.

iii. In the “Apply Current to” list, select Each Sink.

iv. In the Current field, type 5. In the Resistance field, type 1000000.

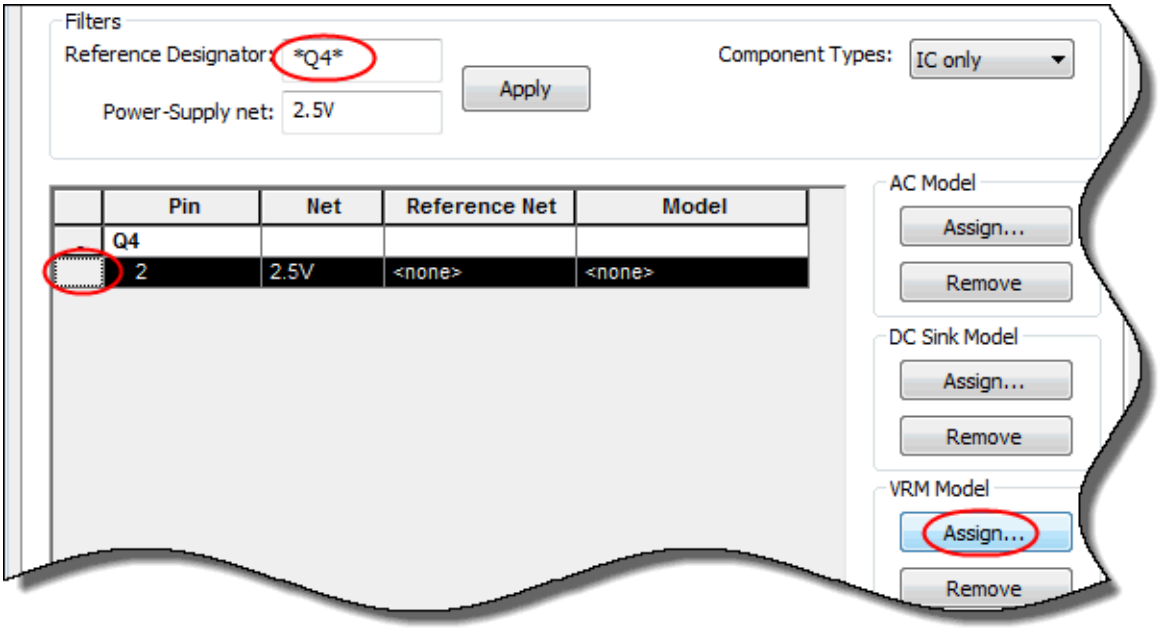
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

v. Click **OK**.



g. Type *Q4* into the Reference Designator filter and click **Apply**.

i. Select pin 2 of Q4.



ii. In the VRM Model area, click **Assign**. The Add/Edit VRM Model dialog box opens.

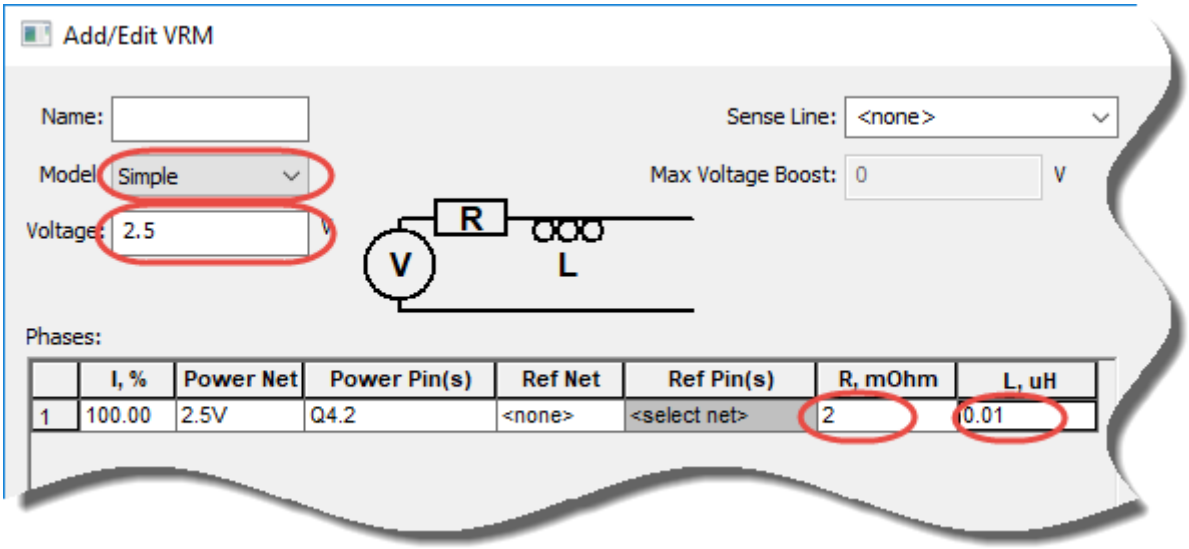
iii. In the Model list, select Simple.

iv. In the Voltage field, type 2.5.

v. In the R column, type 2. In the L column, type 0.01.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

vi. Click **OK**.



h. Assign models to the 1.8V rail.

i. In the Reference Designator field, type *.

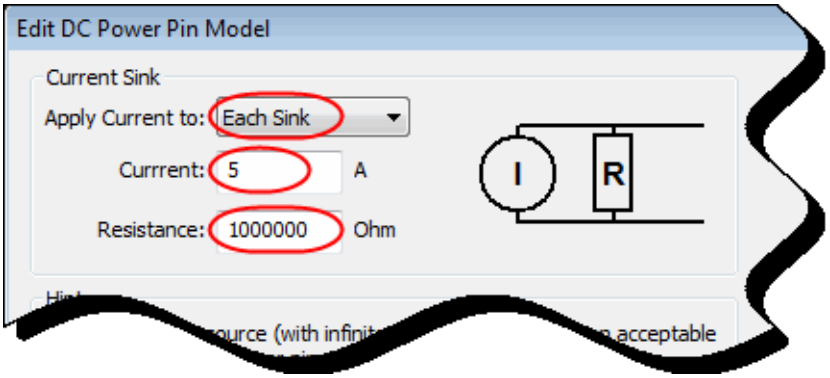
ii. In the “Power-Supply net” field, type 1.8V, then click **Apply**.

iii. Select pin U29.AA8 and, in the DC Sink Model area, click **Assign**. The Edit DC Power Pin Model dialog box opens.

iv. In the “Apply Current to” list, select Each Sink.

v. In the Current field, type 5. In the Resistance field, type 1000000.

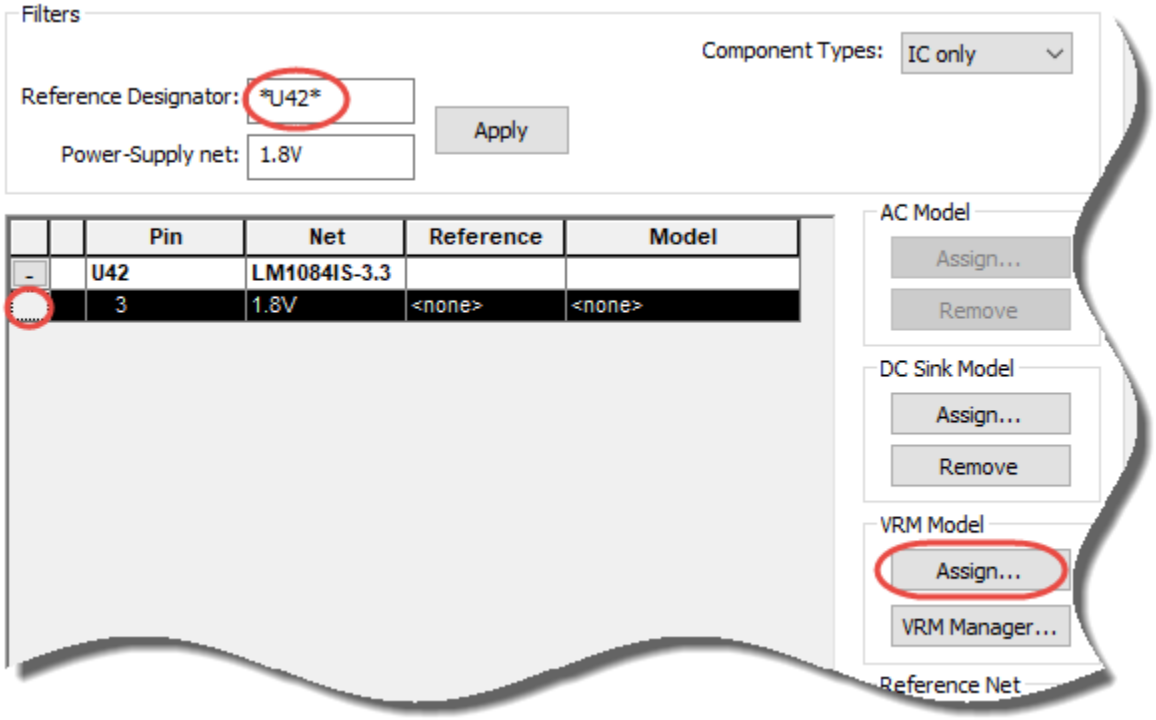
vi. Click **OK**.



vii. Set the Reference Designator to U42* and click **Apply**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

viii. Select pin 3 of U42 and in the VRM Model area, click **Assign**. The Add/Edit VRM dialog box opens.

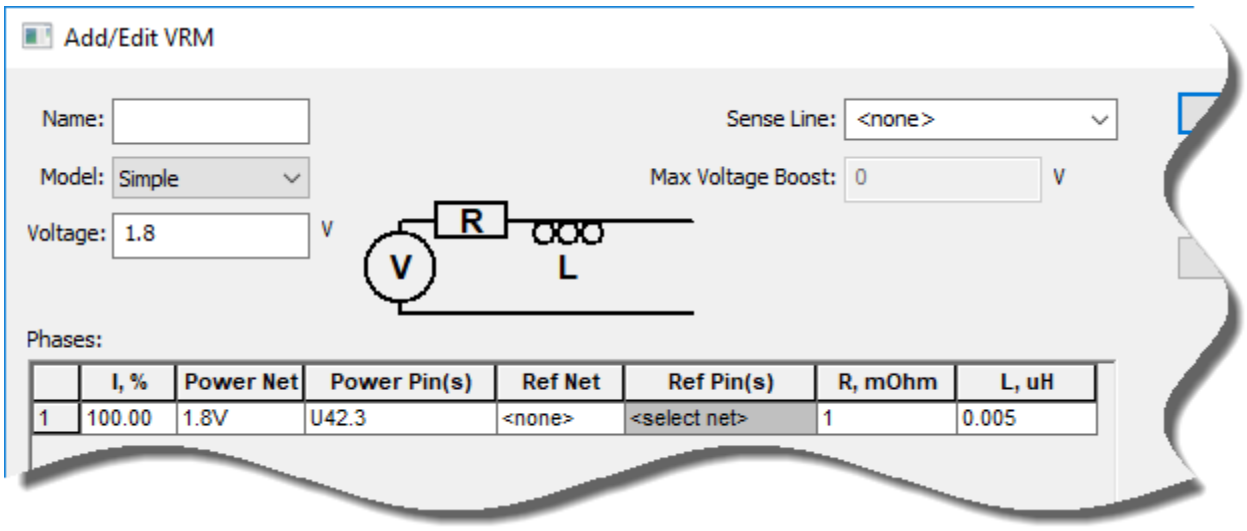


ix. In the Model list, select Simple.

x. In the Voltage field, type 1.8.

xi. In the R column, type 1. In the L column type 0.005.

xii. Click **OK**.



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

i. Click **OK** to close the Assign Power Integrity Models dialog box.

9. Run a DC Drop batch simulation.

   a. From the Batch DC Drop Simulation dialog box, check Create PowerScope Data, check Create Reports, ensure HTML is selected, then click **Run**.

      When simulation completes, the software opens:

      - The DC Drop Simulation Report dialog box that displays an easy-to-navigate report that includes screen captures of measurement locations in the board design.

        You can click a link for a pin or via to open the HyperLynx PI PowerScope and display simulation results for it.

      - The Reporter dialog box that displays a textual report that shows the current and voltage of the pins that you assigned models, and the voltage source and current sink vias.

Related Topics

DC Drop Simulation

Running DC Drop Simulation for Multiple Power Nets That Share a Common Ground Net

Running Batch DC Drop Simulation or Thermal Co-Simulation

Analyzing Crosstalk on the Virtex-4 Demo Board

BoardSim can simulate crosstalk on a board design.

This tutorial has a design goal of guaranteeing that no more than 110 mV of crosstalk can be coupled onto the “victim” net from any nearby “aggressor” nets when using a DDR memory interface on the Virtex-4 demo design. This tutorial predicts crosstalk on a DDR DQS net.

Crosstalk, like other signal-integrity problems, can negatively impact your final design and manifest as false clocking, intermittent data errors, or other difficult-to-find and potentially serious problems. It can also be difficult to know where crosstalk is likely to occur, and eliminating it can be more troublesome than fixing single-trace signal-integrity problems.

A typical net in a modern digital system is in close proximity to many trace segments belonging to other nets - especially on wide, parallel buses such as DDR. This makes the net a potential victim of crosstalk generated by the other nearby aggressor traces.

The most important step to analyzing such a situation is accurately identifying all of the aggressors that contribute significantly to crosstalk on the victim net. When simulating crosstalk on a layout, aggressors are automatically selected using an algorithm that chooses only those neighboring nets with the potential to generate crosstalk above a specified threshold on victim

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nets. This threshold is conveniently described in electrical terms (that is, mV of crosstalk) rather than being geometric, although you have the option of using geometric thresholds, if you prefer.

Note

You cannot run this tutorial when you launch BoardSim by exporting a board from PADS Professional Layout, PADS HyperLynx DC Drop, or PADS HyperLynx DDR. The tutorial uses a multiple-board project that contains HYP files, which BoardSim cannot open when opened this way.

Prerequisites

- A MultiBoard license, which is required to load and simulate multiple-board designs.
- A Crosstalk license, which is required to run crosstalk simulation.

Procedure

1. Load the MultiBoard project:

   Choose the **File > Open MultiBoard Project** menu item, then double-click *virtex4_sdram_multiboard.pjh*.

   If prompted to connect unrouted nets, click **No**.

2. Choose the **Setup > Enable Lossy Simulation** menu item and verify that it is disabled.

3. Choose the **Setup > Enable Trace Coupling** menu item and verify that it is disabled.

4. Choose the **Setup > Enable Area Fill Coupling** menu item and verify that it is disabled.

   Note the DDR interface on this board. One of the most important signals on a DDR interface is a strobe, which acts as the clock for the interface. This step simulates the DDR strobe signal named SDRAM_DQS2.

5. Automatically find aggressor nets.

   An important feature of BoardSim Crosstalk is that it automatically identifies which other nets are coupled strongly enough to the selected victim net to be aggressors. For more information on this powerful capability, see the “Electrical Versus Geometric Thresholds” on page 1204.

   a. Choose the **Select > Net by Name for SI Analysis** menu item.

      Note that the “Design file” list displays B00 Virtex4 Demo, meaning the list of nets are located on the main board.

   b. In the Filter box, type *DQS2 and click **Apply**.

   c. Double-click net SDRAM_DQS2.

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The board viewer highlights SDRAM_DQS2 on the main board and the nets on the DIMM plug-in board to which it connects.

d. Choose the **Setup > Coupling Thresholds** menu item. The Set Coupling Thresholds Dialog Box opens.

e. Select “Use electrical thresholds”.

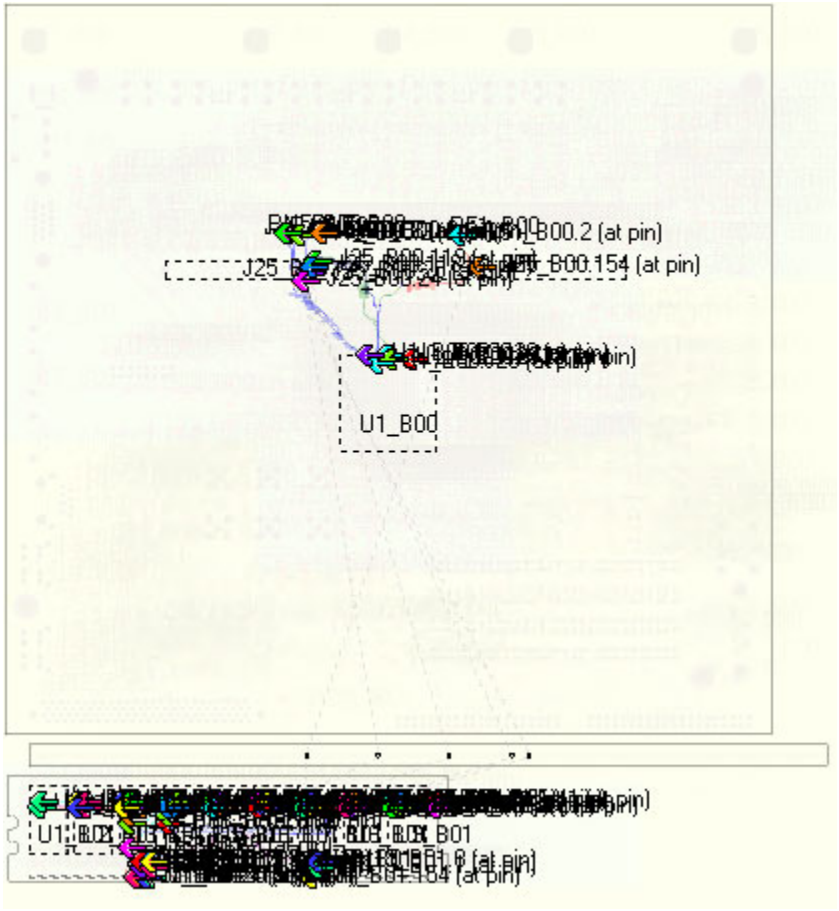
f. In the “Include nets with coupled voltages greater than” field, type 110.

g. Click **OK**.

h. Choose the **Setup > Enable Trace Coupling** menu item and verify that it is enabled. Rats nest lines show the connections between boards, through connectors.

Note

To see rats nets lines, you may have to set the Dimness option to a very low value. This option is available from the Board Settings tab, on the right edge of the board viewer.



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

For this example, BoardSim Crosstalk searches for all aggressor nets that contribute 110 mV or more crosstalk to the selected victim net. Note that this threshold is adjustable.

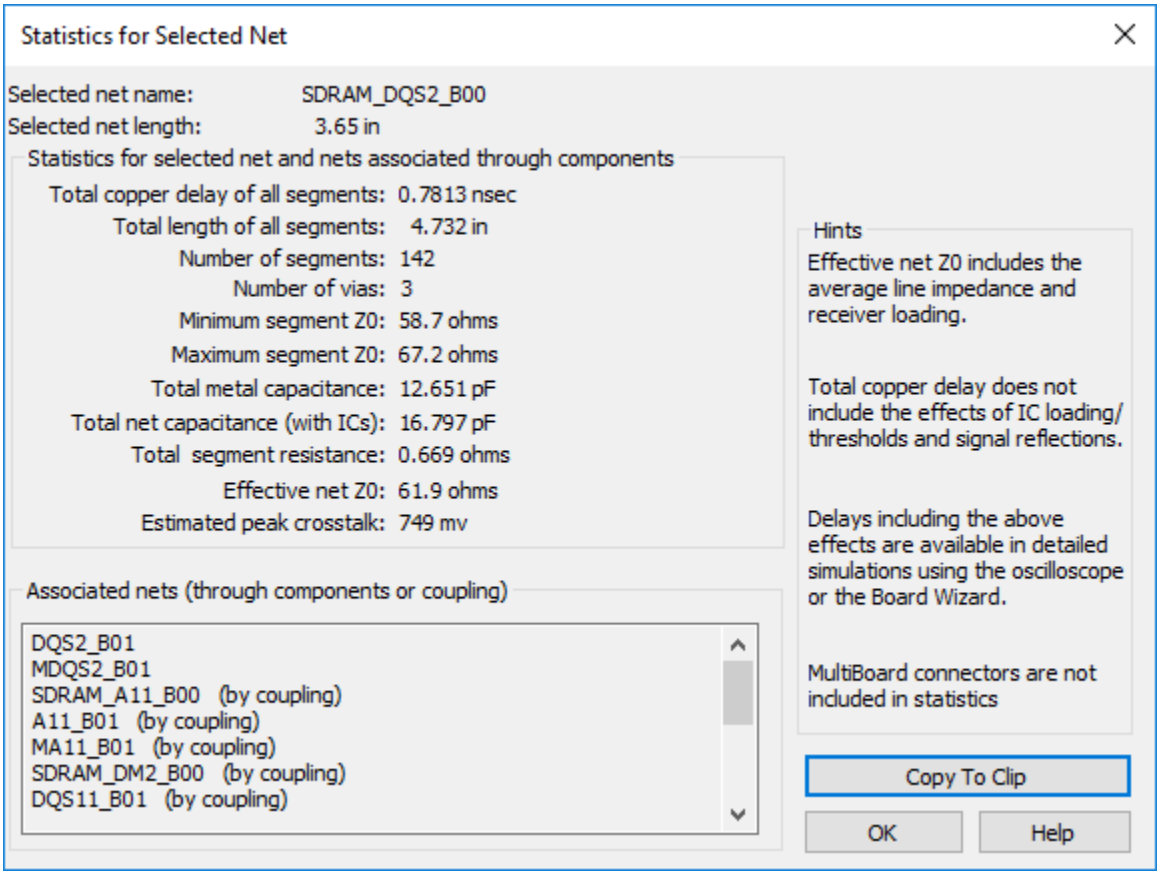
In addition to net SDRAM_DQS2 being visible in the foreground of the board viewer, you can also see several other highlighted nets with dashed lines. The highlighted dashed lines indicate that the surrounding dashed traces are aggressors to SDRAM_DQS2. BoardSim predicts that these aggressor nets have the potential to cause more than 110 mV of crosstalk on the victim net SDRAM_DQS2.

The next step is to look at a report of the crosstalk on the victim net.

i. Choose the **Export > Reports > Net Statistics** menu command. The Statistics for Selected Net dialog box opens.

The Associated Nets area displays the names of the aggressor nets and identifies them with the (by coupling) label.

The other nets listed, DQS2_B01 and MDQS2_B01, are associated to aggressor nets through series resistors or a connection through the DIMM providing a path of conductivity.



j. Click **OK**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

6. Set up IC models for simulation.

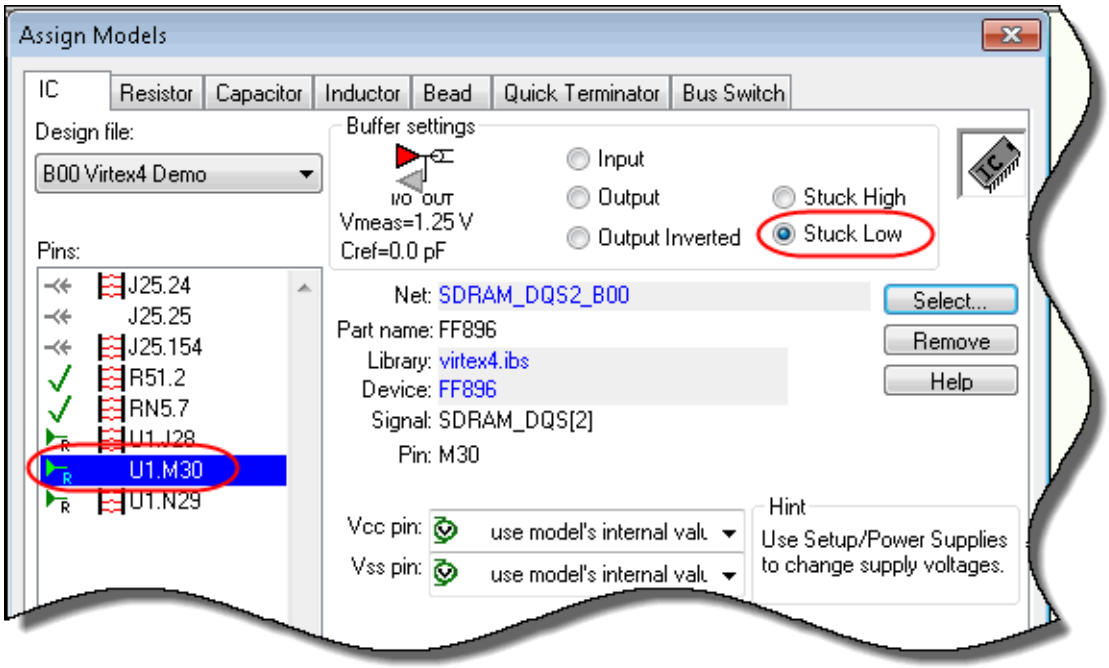
   During crosstalk simulations, BoardSim Crosstalk is capable of simulating any number of victim and aggressor nets, and each victim or aggressor is either actively switching or static, that is, stuck high or low. However, it is much easier to see the crosstalk amplitude and waveform if the driver IC of the victim net is not switching.

   a. Choose the **Models > Assign Models/Values by Net** menu item.

      In the Pins list, note that some pins have a coupled icon just to the left of the reference-designator/pin label. These are the component pins on the aggressor nets. Pins on the selected victim net do not have an icon.

   b. Select U1.M30, the driver IC of the victim net.

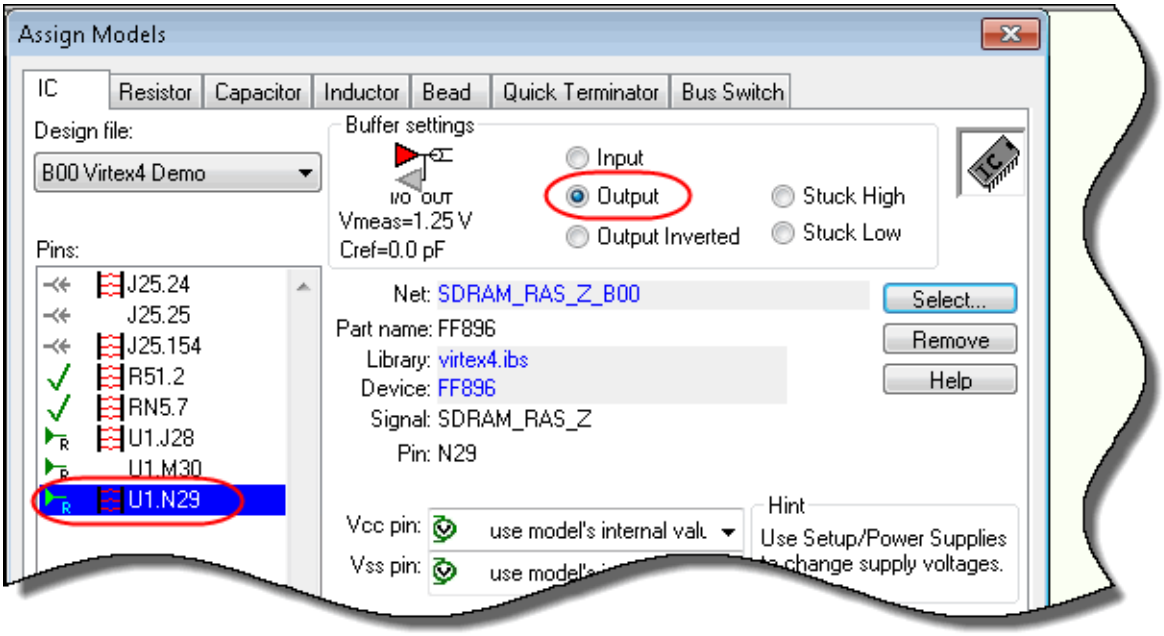
   c. In the Buffer Settings area, select Stuck Low.



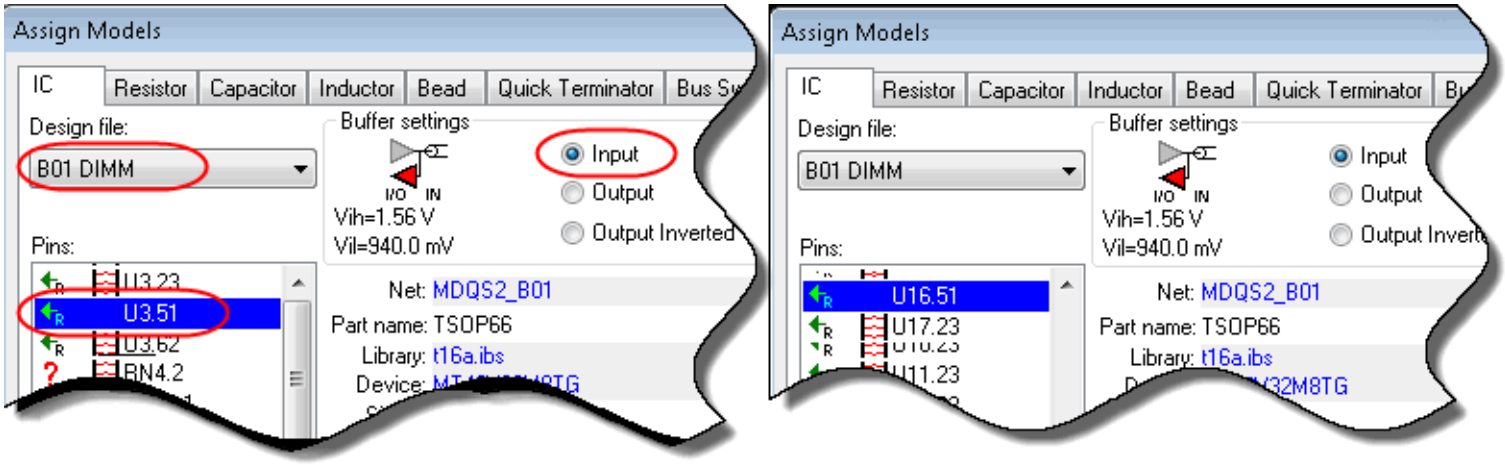
   d. In the Pins list, select U1.N29.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

e. In the Buffer Settings area, select Output.



f. In the Design file list, select B01 DIMM and in the Buffer Settings area, select Input for pins U3.51 and U16.51.



g. Click **Close**.

7. View the crosstalk coupling regions.

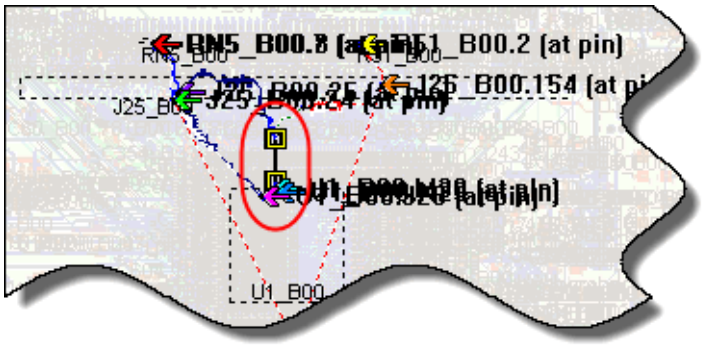
   Before simulating to see how much crosstalk appears on net SDRAM_DQS2, you can view the crosstalk coupling regions, that is, sections along the coupled nets which generate the crosstalk. Viewing the physical and electrical properties of a coupling region can help you understand how each net contributes to the coupling in the region.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

a. Choose the **View > Coupling Regions** menu item.

b. Move the dialog box so you can view the visible nets.

c. In the board viewer, note the set of segments highlighted in black with yellow boxes as endpoint markers.



d. In the Coupling Region dialog box, click **Next**. A different coupling region is highlighted.

The Coupling Region viewer contains the names of the coupled nets, information about how far apart they are in the currently displayed region, and a graphical stackup cross section showing the nets.

e. Click **Impedance** to add an impedance and termination summary to the viewer. You can resize the window vertically to more easily see its contents, or re-size individual panes within the window.

Coupling regions in the viewer are sorted from strongest coupling to weakest.

Note that even this simple net requires several different coupling regions to be accurately simulated. For nets on a dense board, it is common to have a hundred or more regions. BoardSim Crosstalk automatically models all regions.

f. Click **Close** to close the Coupling Region viewer.

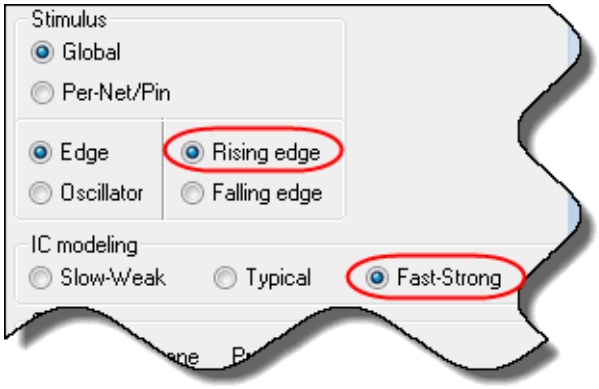
8. Simulate net SDRAM_DQS2 interactively and measure the crosstalk:

a. Choose the **Simulate SI > Run Interactive Simulation** or **Run Interactive Simulation in Oscilloscope** menu item. The Digital Oscilloscope opens.

b. In the Stimulus area, select Rising Edge.

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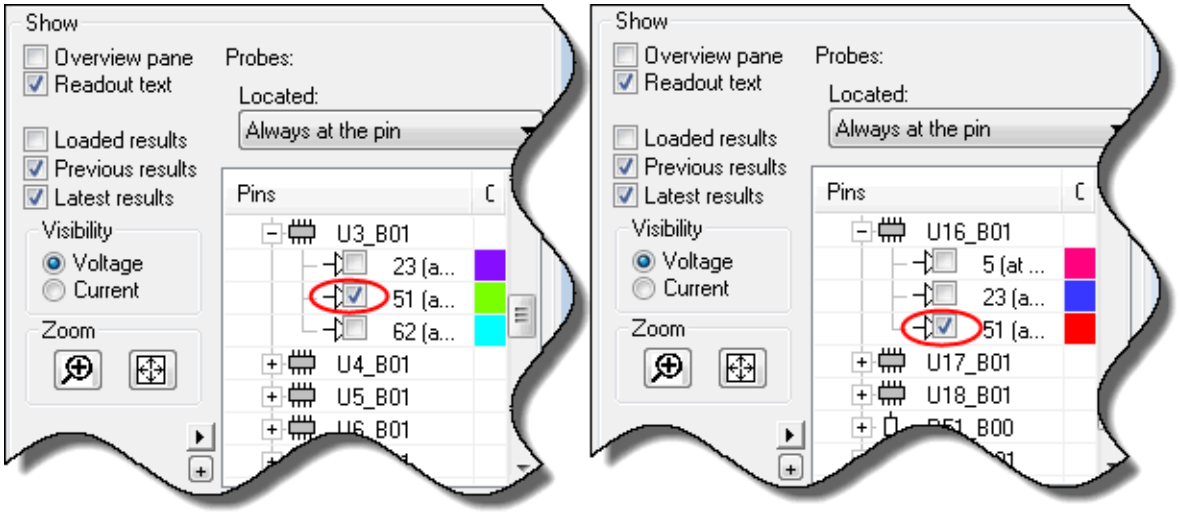
c. In the IC modeling area, select Fast/Strong.



d. In the Show > Probes area, in the Pins list, de-select all probes.

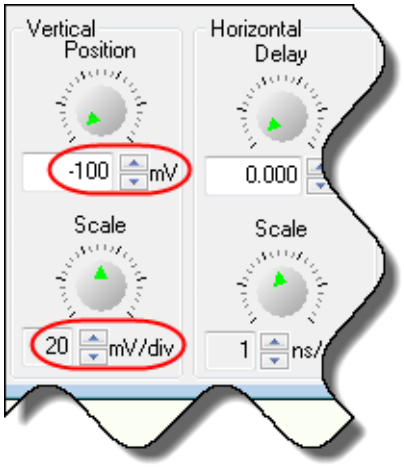
e. In the Show > Probes area, in the Pins list, expand U3_B01, and select pin 51. Pin 51 is the first victim receiver pin on the DIMM.

f. In the Show > Probes area, in the Pins list, expand U16_B01, and verify that the probe at pin 51 is enabled. Pin 51 is the second victim receiver pin on the DIMM.



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g. In the Vertical area, change the vertical position to -100 mV and the vertical scale to 20 mV/div.



h. Click **Start Simulation**. The crosstalk simulation runs.

The waveforms are from the probes at the receiver ICs on the victim nets.

i. Change the Horizontal Scale to 1 ns/div to see the crosstalk clearly.

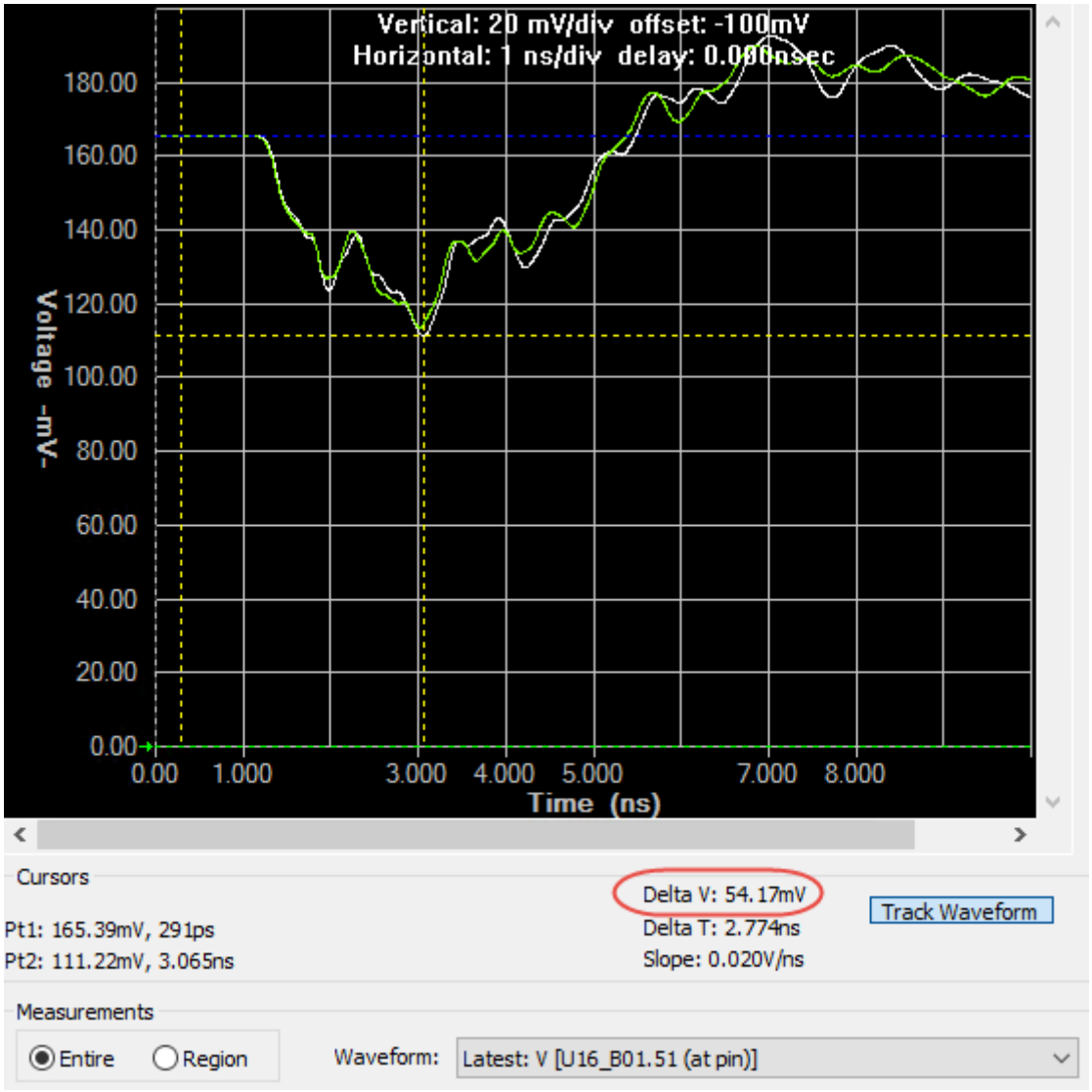
j. Below the waveform display, click **Track Waveform**.

k. In the waveform display, hover over the waveform to locate U16_B01.51 and click to select the waveform.

9. Click to place a probe cursor at the pre-crosstalk, steady state condition on the left, and click to place a second cursor at the largest departure from this value for either waveform.

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The observed Delta V value is about 54 mV as a result of coupling from the aggressor net.



This particular net still falls well within the originally selected threshold of 110 mV. However, if the crosstalk limit is as low as 54 mV, the design has a problem.

BoardSim simulates any mixture of victim and aggressor traces. In fact, the simulator makes no distinction between the two. Generally, the preference is to have the victim nets - the nets on which you want to measure crosstalk- stuck either low or high. However, in this simulation, SDRAM_DQS2 can also switch, making the net both an aggressor to the other nets AND their victim.

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Schematic Design Tutorials

The schematic design tutorials walk you through opening a design project in LineSim, editing schematics and running several simulations. In the process, you will become familiar with LineSim simulation types, as well as the graphical user interface.

Some of the tutorial designs contain very old signaling technology and circuit geometries. However, the objective of the tutorials is to show you how to obtain simulation results to help you design your board or measure its performance.

Note

You can make changes to the tutorial designs and save them. To restore the original tutorial design or model files, extract them from *tutorial_golden_files.zip*, which is located in the same folder as the HyperLynx executable file (*bsw.exe*). Example location:

\MentorGraphics\<release>\SDD_HOME\hyperlynx64\tutorial_golden_files.zip

Tutorial	Description
Simulating a Simple Clock Net	LineSim can help you make important signal-integrity decisions about your clock net before you even begin drawing a logic schematic.
Simulating a Series-Terminated Net with an IBIS Model	Adding a parallel AC terminator to the end of the net is one way to terminate a clock net. This tutorial shows a net that is series terminated and uses an IBIS-format model for the driver IC.
Simulating Using Lossy Transmission Models	In SERDES-based systems, it is common for signals to be greatly attenuated before arriving at receiver ICs.
Modeling a PCB Stackup	LineSim includes an editor to help you create and plan PCB stackups. You can create a stackup and tie any of the transmission lines in a schematic to the stackup.
Achieving a Specific Differential Impedance	Differential signaling takes advantage of the coupling between neighboring traces. When you design a differential pair, you deliberately couple the two traces together strongly, so that any signal induced by external noise on one is also induced on the other, and then rejected by the differential receiver at the ends of the lines.
Setting Up a SPICE Simulation	LineSim includes two built-in SPICE simulators and supports integration with HSPICE. Using SPICE, HyperLynx wraps itself around your choice of SPICE engine for a more user-friendly simulation environment than with just raw SPICE.
Including Touchstone Models in a LineSim Schematic	You can include and set up a typical Touchstone model in a LineSim schematic.

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Tutorial	Description
Planning Minimum Trace Separation on a Bus	You can use LineSim to plan minimum trace separation on a bus.
Signal-Integrity Simulation of a DDR Data Path	HyperLynx LineSim allows you to develop constraints for PCB placement and routing that provide the greatest chance of producing a successful first-prototype board.

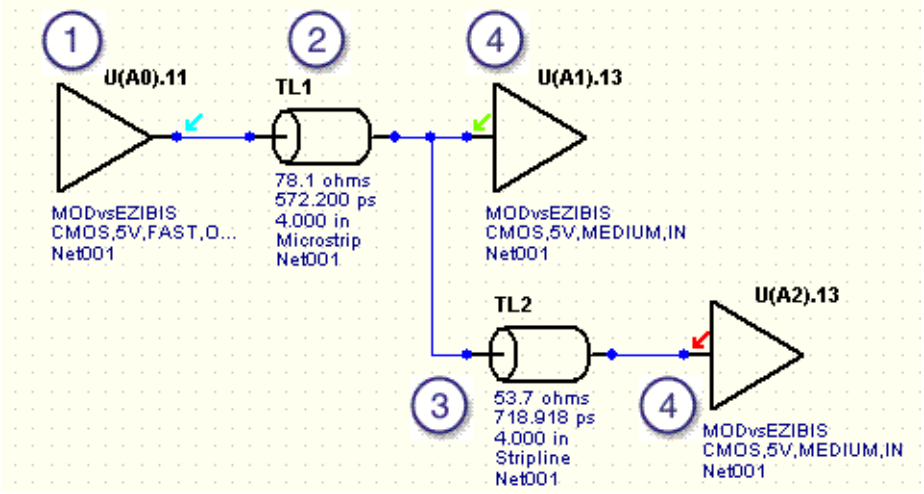
Simulating a Simple Clock Net

LineSim can help you make important signal-integrity decisions about your clock net before you even begin drawing a logic schematic.

The following is a video version of this tutorial:



This schematic is a simple clock net with a driver IC, a PCB trace routed on the outer layer of the board to a receiver IC, and a trace routed on an inner layer to another receiver. The arrows on the schematic indicate the location of oscilloscope probes and the arrow color identifies the corresponding waveform displayed on the oscilloscope.



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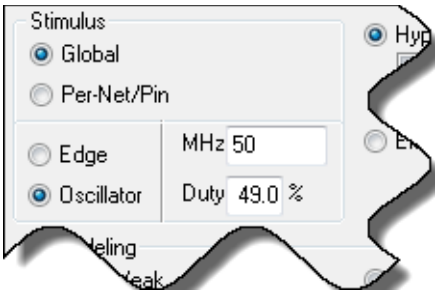
Table A-2. Clock Circuit Description

Reference	Description
1	Driver IC
2	Microstrip transmission-line symbols
3	Stripline transmission-line symbols
4	Receiver ICs

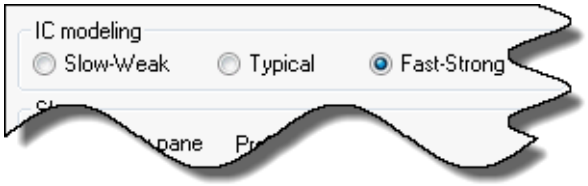
Procedure

1. Open the schematic. Choose the **File > Open Schematic** menu item, then double-click *clock.ffs*.
2. Run an interactive simulation:
   a. Choose the **Simulate SI > Run Interactive Simulation** or **Run Interactive Simulation in Oscilloscope** menu item. The Digital Oscilloscope Dialog Box opens.
   b. In the Stimulus area, select Oscillator.

      In the MHz field, type 50.

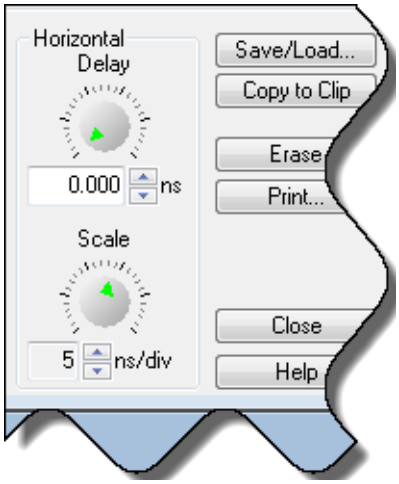


   c. In the “IC modeling” area, select Fast-Strong.



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d. Change the Horizontal Scale timebase to 5 ns/div.

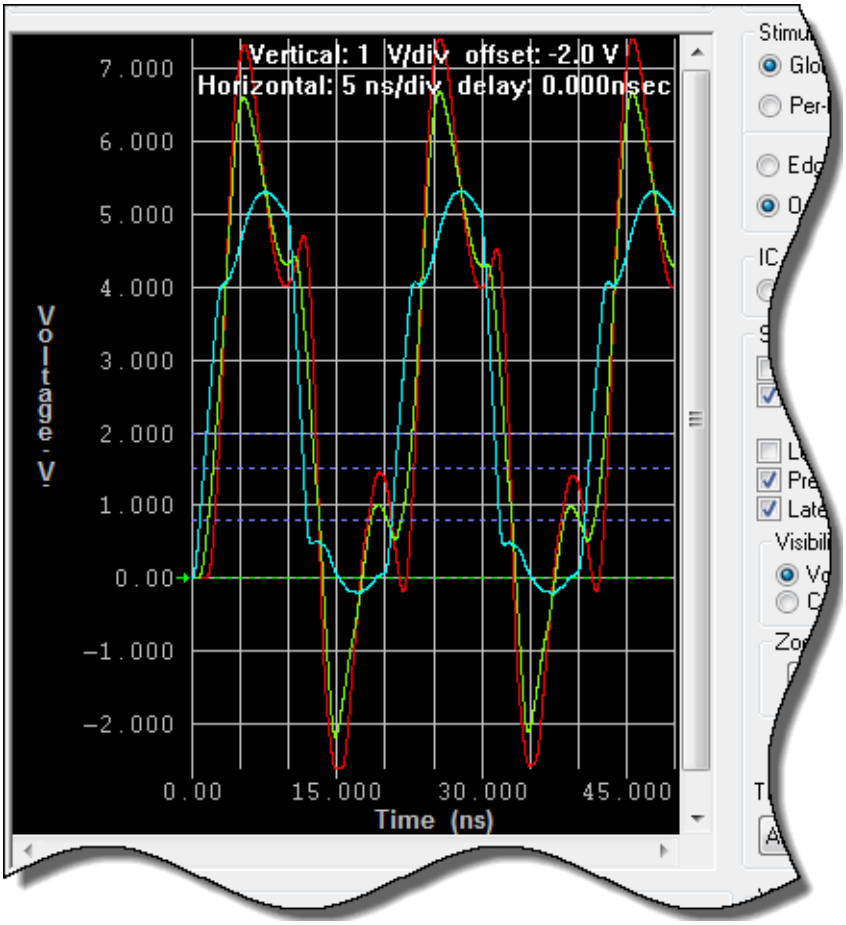


e. Click **Start Simulation**.

The waveforms on the screen are the actual voltages seen if you built the clock net described on the schematic. The voltages at the receiver ICs show a large amount of overshoot / undershoot - so much that the receiver ICs see at least one extra clock edge

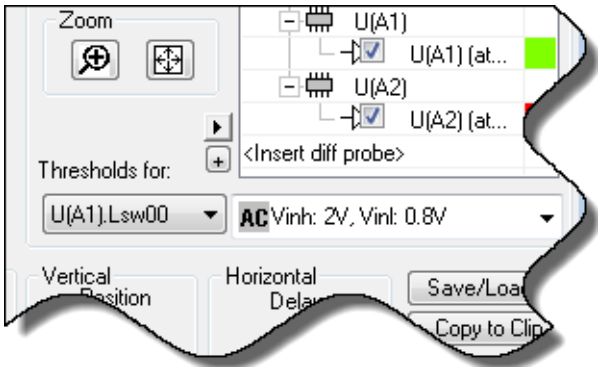
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

per cycle. Plotting the receiver IC thresholds in the oscilloscope displays this problem more clearly.



3. In the “Thresholds for” list, select U(A2).13.

   Two dashed, blue horizontal lines appear in the oscilloscope: the receiver ICs 0.8 V and 2.0 V input thresholds, Vil and Vih. LineSim automatically reads these thresholds from the receiver IC models. The falling-edge waveform crosses the Vil threshold three times, which can cause double-clocking. The resulting board is likely to fail using this clock net.



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4. Minimize the oscilloscope.
5. Fix the clock net.

   LineSim helps you determine how to fix this clock net. Adding a termination, such as adding passive components to match transmission-line impedances, is a good way of fixing many kinds of basic signal-integrity problems. This step demonstrates how to edit a schematic in LineSim by adding a terminator to the end of the net.

   a. Click **Close**.

   b. Click the **Add RC terminator to schematic**  symbol from the component palette and place it near receiver IC U(A2).13. Connect the resistor to the receiver.

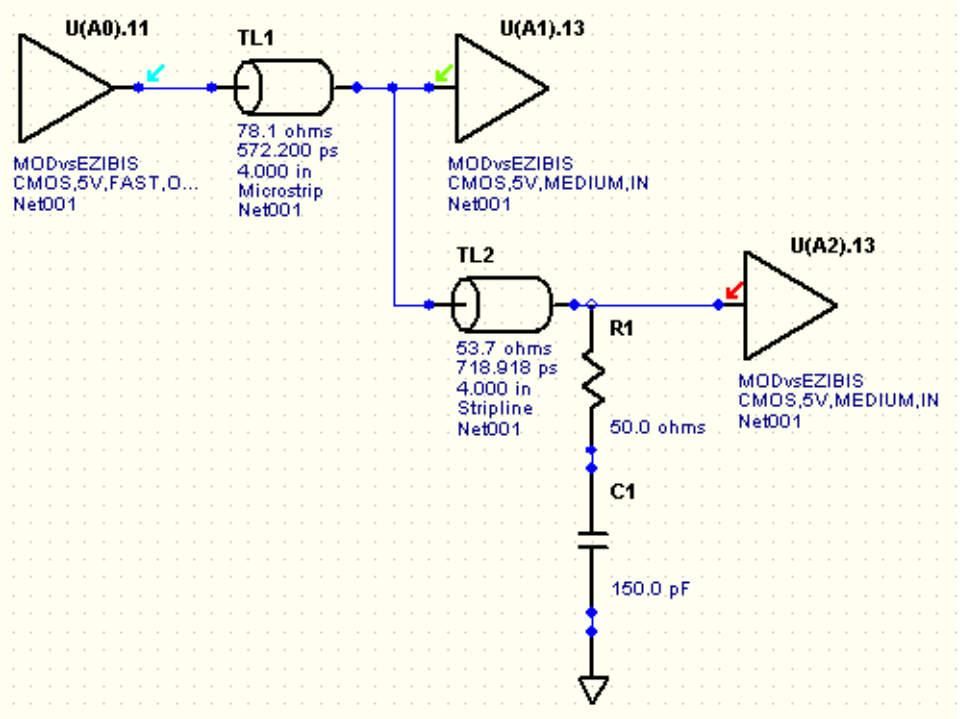
   c. Double-click the resistor. The Assign Models dialog box opens.

   d. Change the resistance to 50 ohms and click **OK**.

   e. Right-click the capacitor, then choose the **Assign Model or Edit Value** menu item. The Assign Models dialog box opens.

   f. Change the capacitance to 150 pF and click **OK**.

   The schematic below shows the component value updates:



   The 50-ohm value for the terminating resistor is a guess based on the fact that a terminator must match the impedance of the transmission line it is terminating. Note that TL2 in the schematic has an impedance of about 50 ohms. The capacitor value is also a guess. Generally, the longer the termination line, the larger the capacitor value. LineSim

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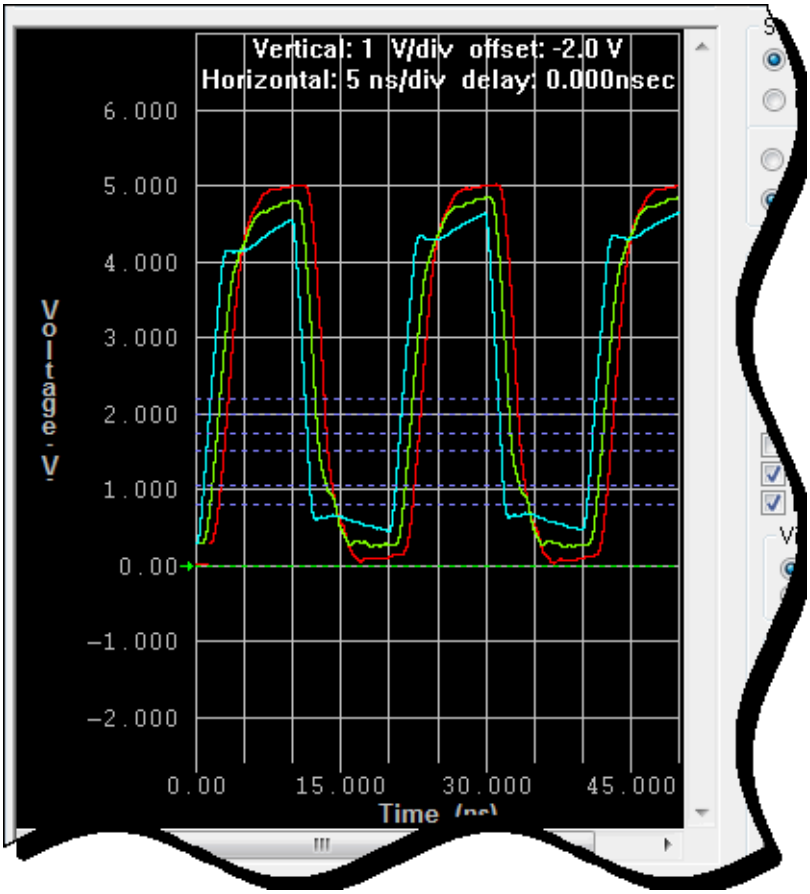
can automatically find the best terminating-component values, to eliminate the need for guessing.

6. Simulate the terminated net. This simulation checks to see if the clock net has an improved waveform.

   a. Restore the oscilloscope, which you previously minimized.

   b. Click **Start Simulation**.

   In this simulation, the receiver IC waveforms look considerably better - almost all of the overshoot is gone. By increasing the value of the capacitor, you can further tune the waveform to eliminate all of the negative overshoot. The ability to perform “what if” analysis is a LineSim strength.



LineSim has an even easier way to determine optimal termination values, by using the Terminator Wizard, as shown in the tutorial named Simulating a Series-Terminated Net with an IBIS Model.

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Simulating a Series-Terminated Net with an IBIS Model

Adding a parallel AC terminator to the end of the net is one way to terminate a clock net. This tutorial shows a net that is series terminated and uses an IBIS-format model for the driver IC.

On this net, an IC modeled with an IBIS model drives a transmission line and receiver IC. The driver is series-terminated with a resistor, whose value is temporarily 0.0 ohms.

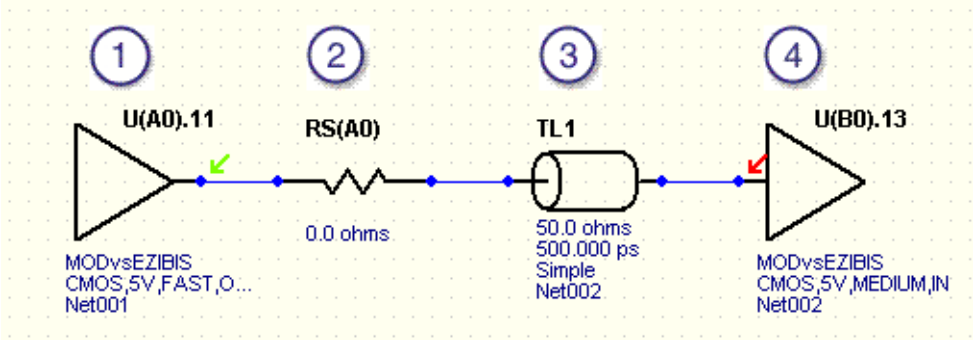


Table A-3. Simulating a Series-Terminated Net with an IBIS Model

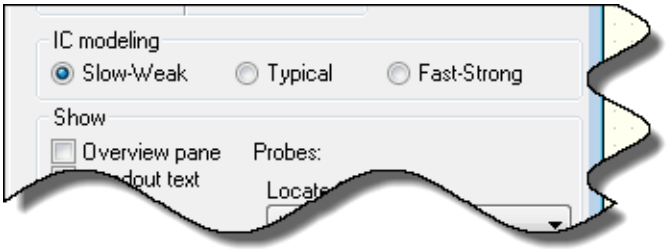
Reference	Description
1	IC driver
2	Series resistor
3	Simple transmission line
4	Receiver IC

Procedure

1. Choose the **File > Open Schematic** menu item, then double-click *ser_ibs.ffs*.
2. Choose the **Simulate SI > Run Interactive Simulation** or **Run Interactive Simulation in Oscilloscope** menu item.
3. In the Stimulus area, select Edge and “Falling edge”.

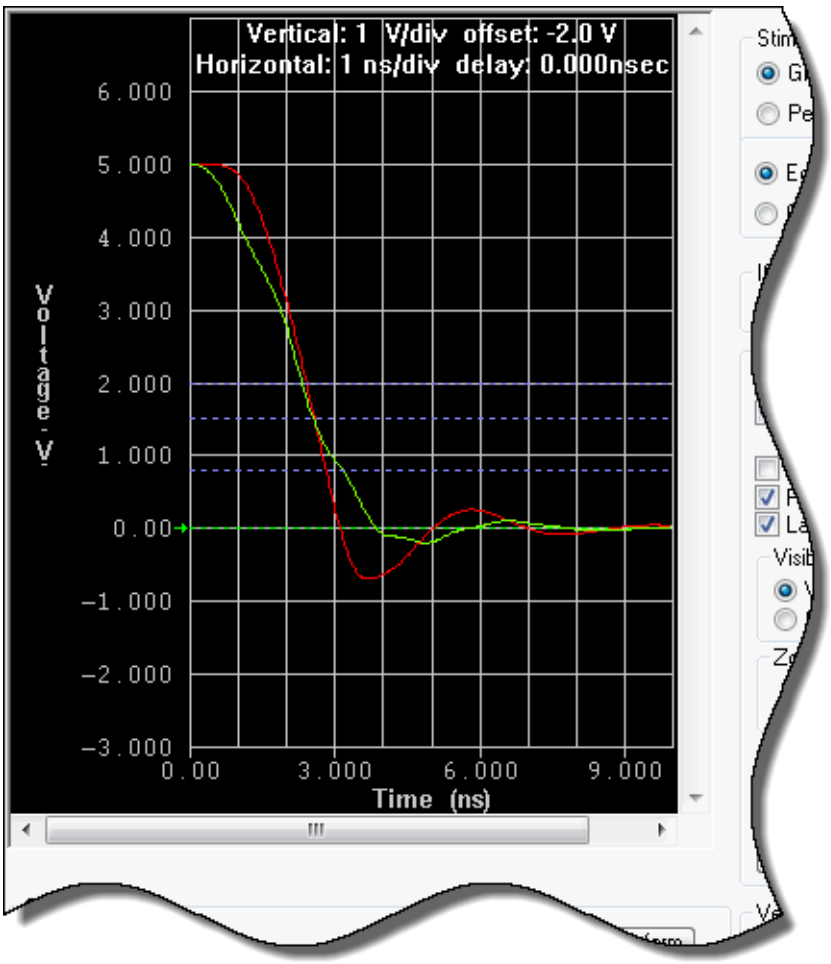
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

4. In the “IC modeling” area, click Slow-Weak.



5. Click **Start Simulation**.

   The simulator runs, showing that the falling-edge signal on this net displays very little ringing.



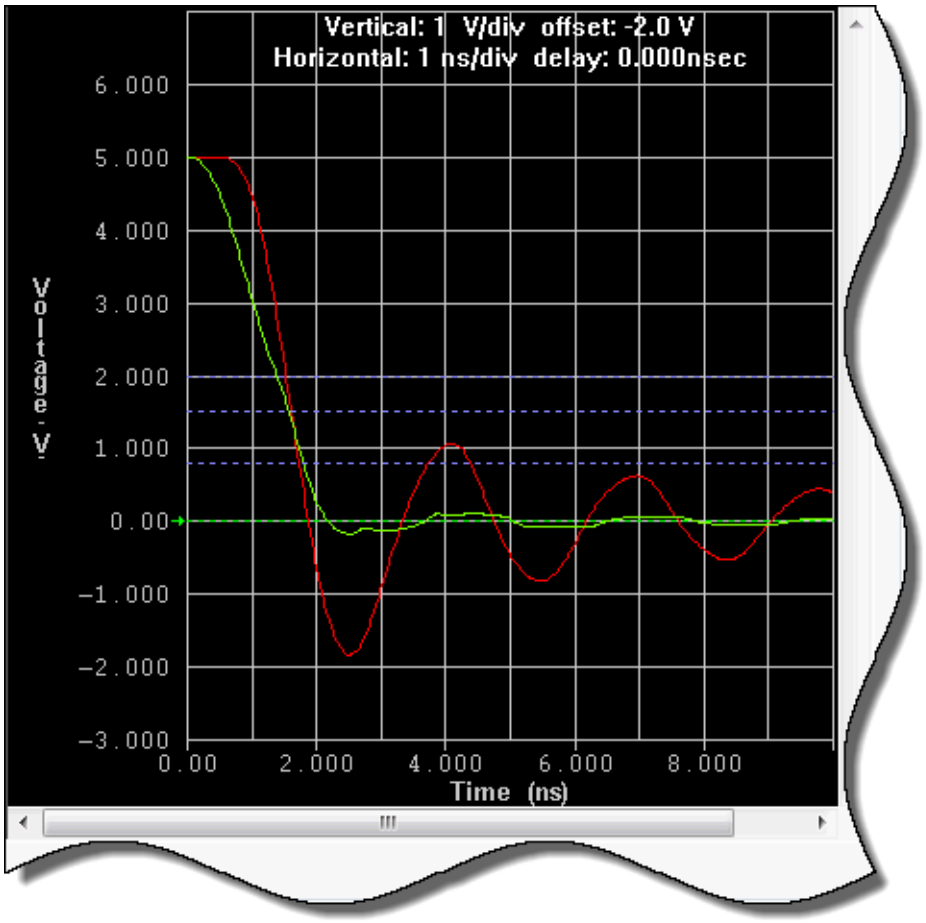
   IBIS models can include minimum, typical and maximum data. Change the model to run with fast/strong parameters.

6. In the “IC modeling” area, select Fast-Strong.

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7. Click **Start Simulation**.

   The simulation runs again, plotting over the previous simulation results. The new waveform now displays considerable ringing. The net needs a termination to protect against the faster versions of the driver IC.



   LineSim uses the Terminator Wizard to find the optimal termination. The Terminator Wizard simulates nets in detail and automatically selects the best termination types and values to use.

8. Minimize the oscilloscope.

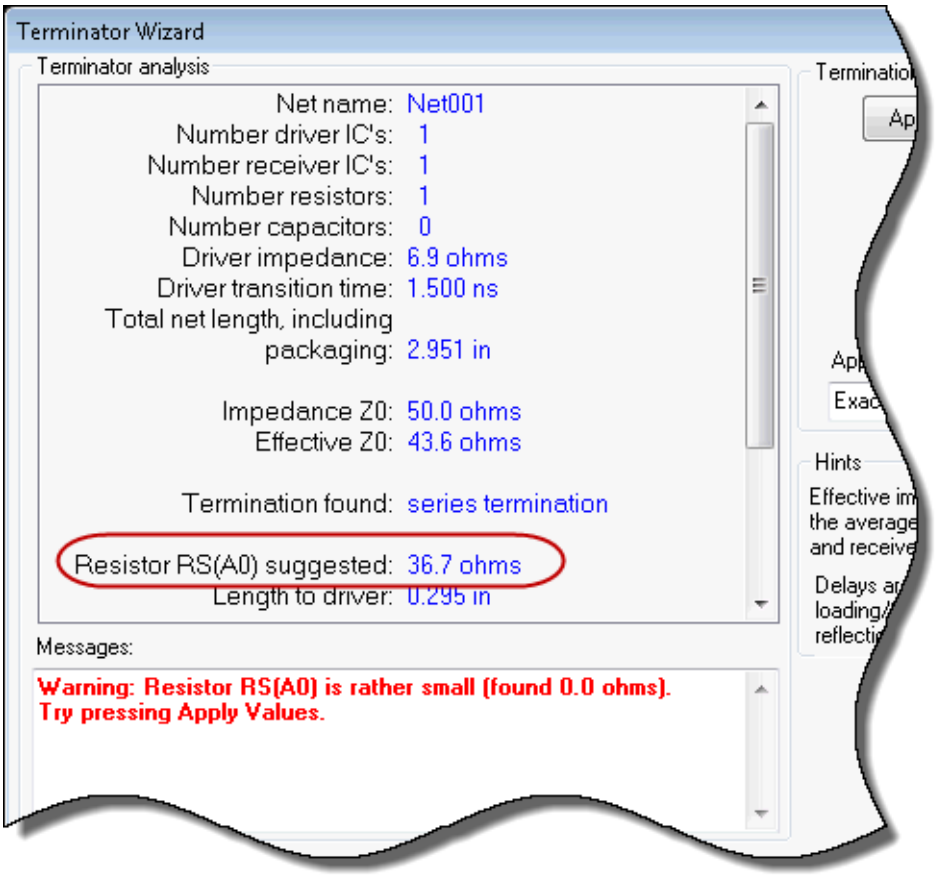
9. Choose the **Simulate SI > Optimize Termination** menu item. The Select Net for Terminator Wizard dialog box opens.

10. In the “Select a device pin” list, select U(A0).11 and click **OK**. The dialog box closes, and the Terminator Wizard opens and performs its analysis.

    When the Terminator Wizard runs, LineSim automatically simulates the selected net, presents a list of trace statistics, and, makes suggestions for termination values at the bottom of the list. In this case, the Wizard correctly determines that the termination type is series, and makes suggestions for the optimum value of R. In these calculations,

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LineSim automatically accounts for such effects as capacitive loading of receiver ICs, total line length, and driver impedance. The wizard recommends a 36.7-ohm terminating resistor.



11. Click **Apply Values** to apply the recommended value to the resistor in the schematic.
12. Click **OK** to close the Terminator Wizard.

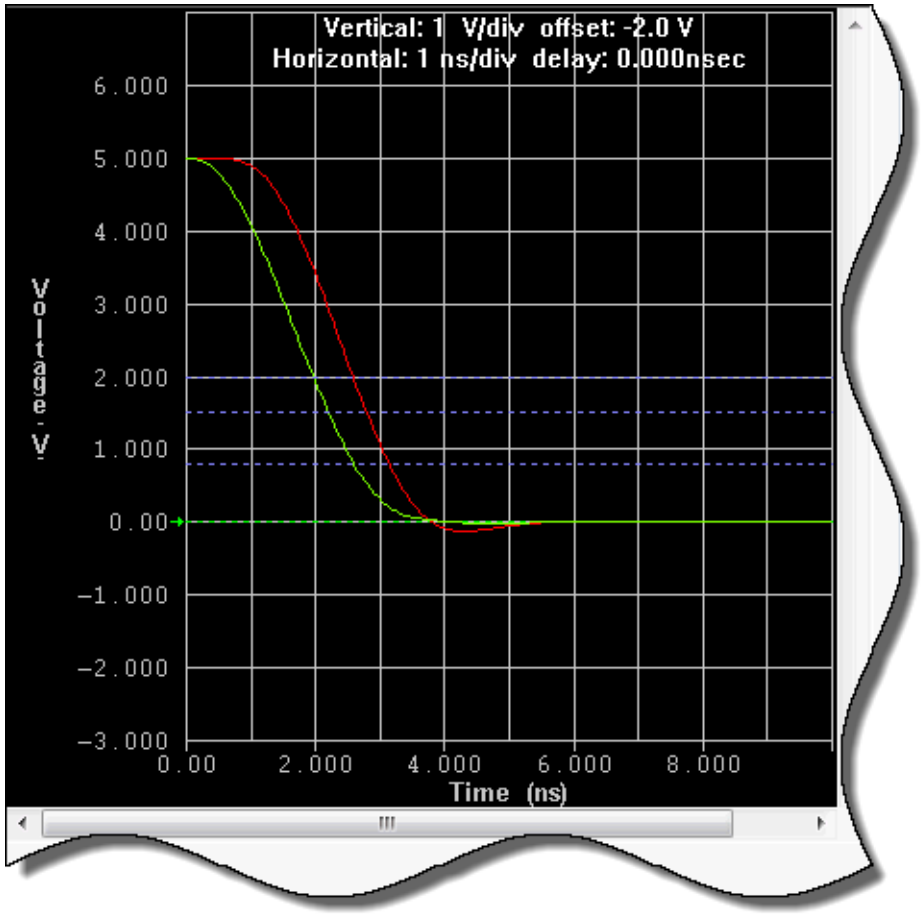
    In the schematic editor, the resistor value changed from 0 ohms to the recommended value of 36.7 ohms. The next step is to simulate to see if the terminator works.
13. Restore the oscilloscope and click **Erase** to clear any waveforms.
14. In the “IC modeling” area, select Typical to run simulations using typical IC data.
15. Click **Start Simulation**.

Results

The waveform improvement is dramatic. At the receiver (red trace), the signal is nearly perfect. By allowing just a small amount of undershoot at the receiver, the Terminator Wizard identified

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the least possible delay to the receiver IC, yet ensured that the low-side clamp diode of the receiver does not turn on.



The Terminator Wizard is a sophisticated tool. In the analysis you just ran, it automatically determined and displayed the following information:

- Switching impedance of the driver IC (average of high-side and low-side values)
- Driver slew time (average of high and low)
- Total net physical length
- Nominal characteristic impedance of the net
- Adjusted, effective impedance of the net, given receiver-IC loading
- The kind of terminator to use
- Topology of the net, so that the Wizard knows what termination style to recommend if no terminator is present
- Driver-to-series-resistor stub length, in case the distance is too long
- The optimal termination value to use, given all of the above.

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Related Topics

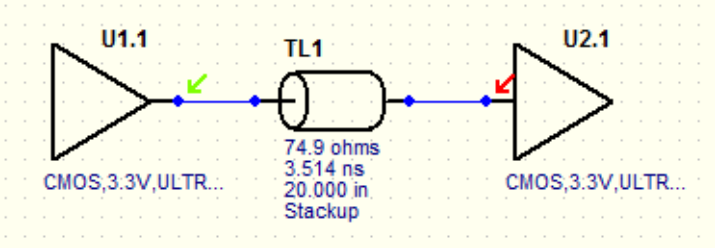
Modeling a Transmission Line

Simulating Using Lossy Transmission Models

In SERDES-based systems, it is common for signals to be greatly attenuated before arriving at receiver ICs.

To account for loss, HyperLynx uses the well-known and trusted W-element algorithm with some added improvements.

This example compares the results of simulations run with and without loss.



The simple schematic contains a driver IC, 20 inches of transmission line buried in FR-4 on an inner layer of the PCB, and a receiver IC. The ICs were modeled using a generic 3.3 V CMOS driver with a nominal switching time of 300 ps. Note that many SERDES-type drivers have faster edges.

Prerequisites

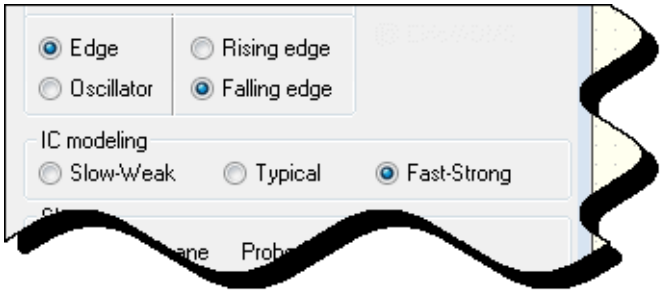
- The Lossy Lines license is required to model lossy transmission lines.
- The transmission line model style must be tied to a PCB stackup or cross section because prediction of loss is based on cross section geometry and materials.

Procedure

1. Open the schematic. Choose the **File > Open Schematic** menu item, then double-click *lossy.ffs*.
2. Simulate the circuit with no loss.
   a. Choose the **Setup > Enable Lossy Simulation** menu item and verify that it is disabled.
   b. Choose the **Simulate SI > Run Interactive Simulation** or **Run Interactive Simulation in Oscilloscope** menu item. The Digital Oscilloscope opens.
   c. In the Stimulus area, select Edge and “Falling edge”.

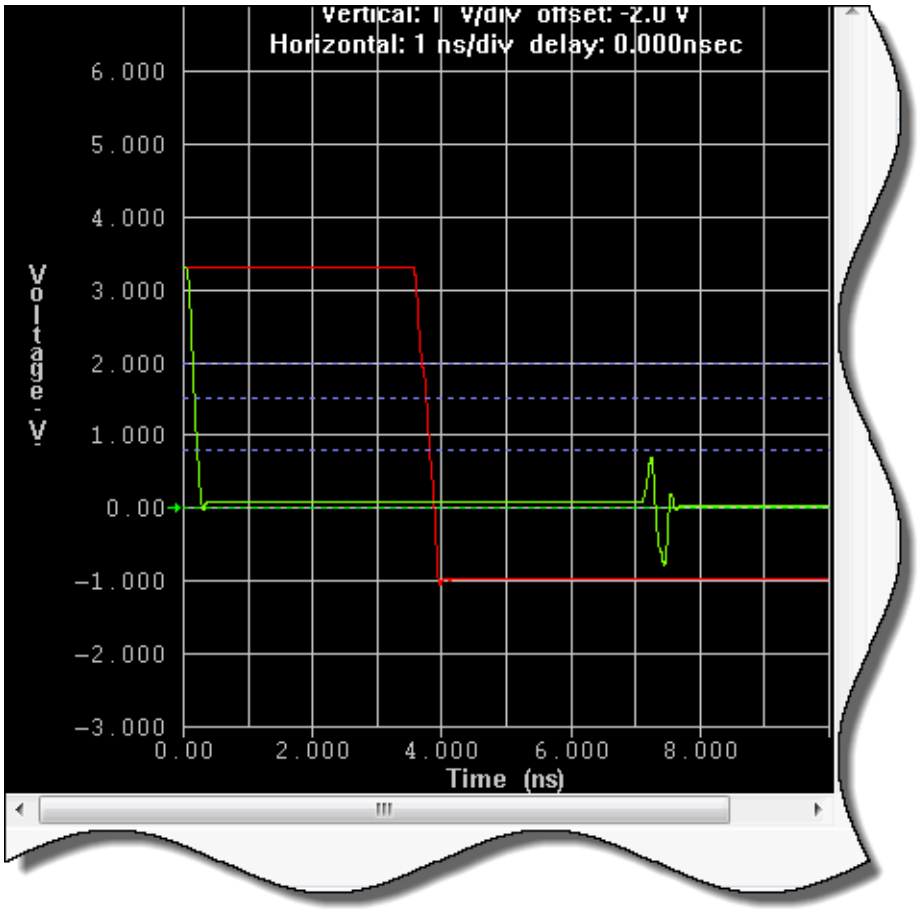
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

d. In the “IC modeling” area, select Fast-Strong. This provides the fastest possible edge time for this driver, resulting in the most loss.



e. Click **Start Simulation**.

In the oscilloscope display, note a sharp driver waveform in green, and after a time delay corresponding to the 20 inches of trace length, a similarly sharp receiver waveform in red.



3. Simulate the circuit with loss enabled to see if there is any visible difference.

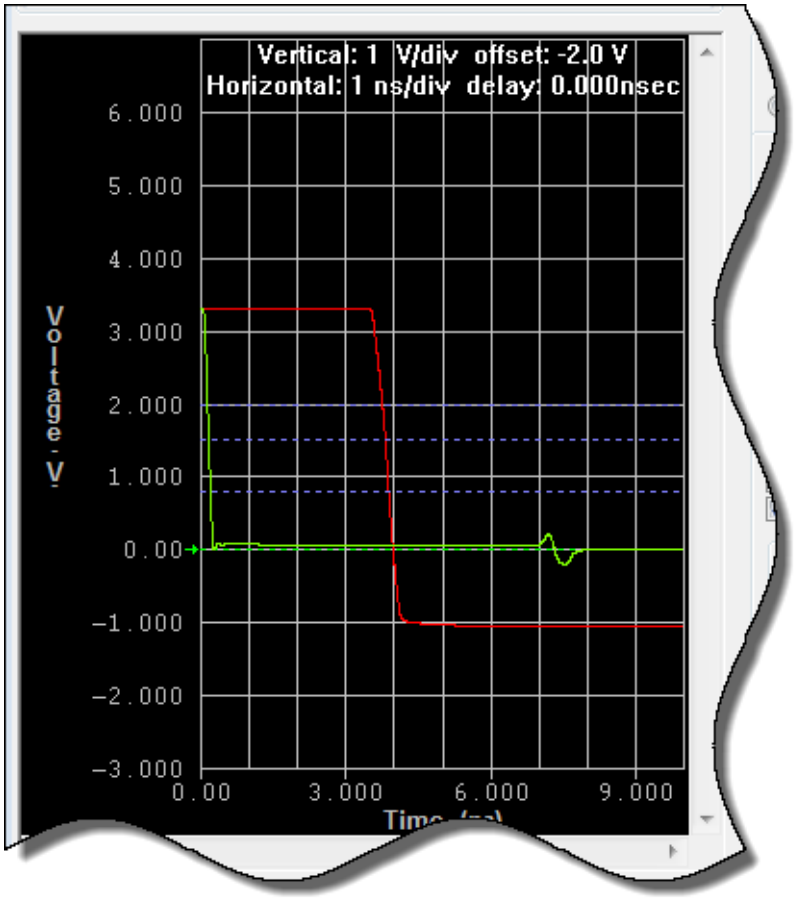
   a. Choose the **Setup > Enable Lossy Simulation** menu item and enable it.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

b. In the oscilloscope, click **Start Simulation**.

The new waveform at the receiver in red does look different from the previous, lossless waveform. This waveform shows more delay and has less amplitude than the lossless waveform. The differences would be even stronger with a faster switching edge, longer trace, or a lossier dielectric material in the PCB.

Notice the disturbance in the green driver waveform, just past 7 ns. This disturbance is due to a reflection from the input capacitance of the receiver IC and is much more attenuated in the lossy simulation than in the lossless simulation.



4. View the loss waveform in the frequency domain.

In GHz-level designs, it is useful to show loss and other effects in the frequency domain. Some specifications, for example, discuss total loss in dB terms at a key frequency. LineSim gives you an easy way to view loss in the frequency domain for any transmission line in a schematic.

Use the following steps to view the loss associated with the 20-inch transmission line in the current schematic.

a. Close the oscilloscope.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

b. In the schematic, double-click the transmission line.

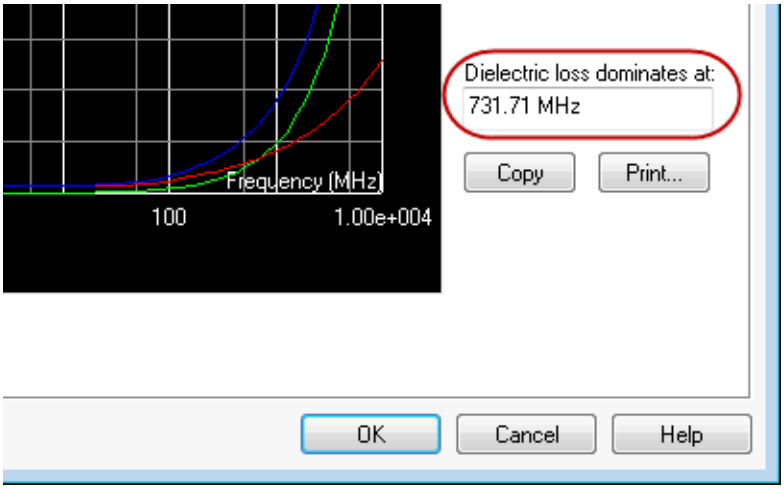
c. In the Edit Transmission Line dialog box, select the **Loss** tab. In the Loss viewer, the graph shows attenuation, or loss, versus frequency and plots three curves:

- Resistive loss (from skin effect) in red
- Dielectric loss in green
- Total loss in blue

5. View the dielectric loss and resistive loss crossover frequency.

Dielectric loss increases with frequency more quickly than resistive loss. Dielectric loss increases linearly with frequency and skin effect only as the square root of frequency. As a result, dielectric loss at some point begins to dominate resistive loss.

The exact crossover frequency is shown in the figure below.



You can also visually see the exact location of the cross-over point:

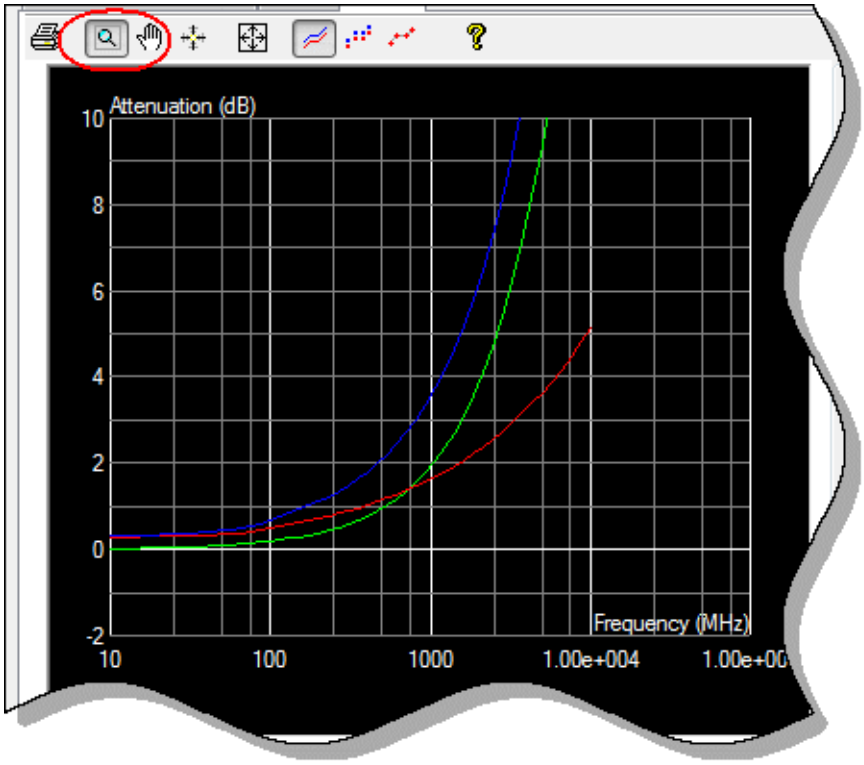
a. Look for the area in the lower right of the loss graph, where the red and green curves intersect.

b. To ensure that zooming is enabled, right-click the graph and choose the **Zooming** menu item.

c. Drag a rectangle that encloses the red/green curve intersection.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The green curve crosses over the red one a little above 700 MHz, meaning that at that frequency, dielectric loss becomes more dominant than skin-effect loss.



Related Topics

LineSim GHz Simulation

Modeling a PCB Stackup

LineSim includes an editor to help you create and plan PCB stackups. You can create a stackup and tie any of the transmission lines in a schematic to the stackup.

For each line, you specify a trace width and select where to place the layer of the stackup the line is located. If you subsequently change the stackup, every line tied to the stackup automatically changes its impedance and delay.

But the stackup editor is more than just a way to manage a stackup during simulation of a transmission-line schematic or a PCB. The stackup editor is a powerful tool for planning

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

stackups, designing controlled impedances, and even documenting stackups for your PCB fabricator.

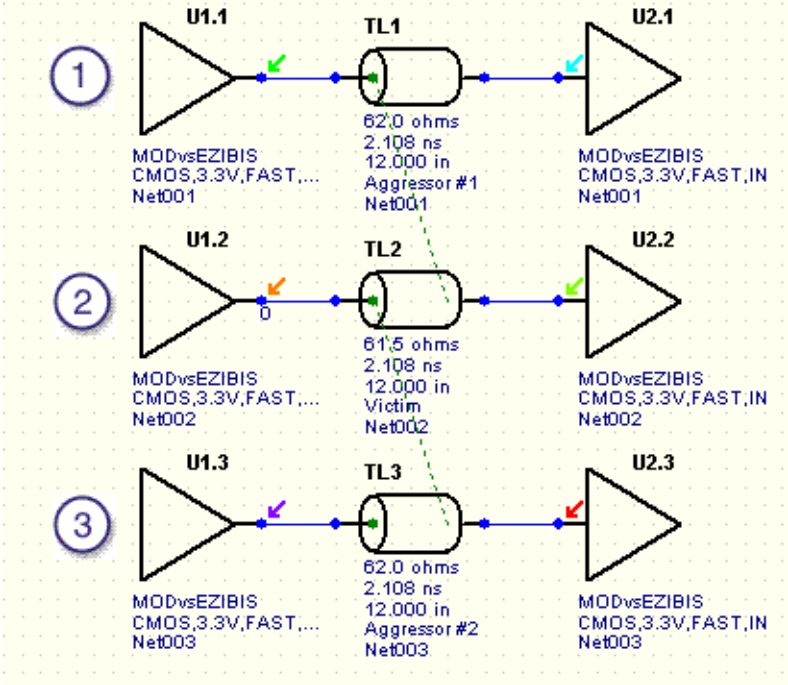


Table A-4. Modeling a PCB Stackup

Reference	Description
1	Aggressor #1
2	Victim
3	Aggressor #2

Prerequisites

- Acquire the Field Solver license.
- Acquire the Crosstalk license to create and simulate coupling regions in LineSim.

Procedure

1. You can view the stackup in any LineSim design. A default stackup is created every time you create a new LineSim schematic. The details of the default stackup are user-definable. To view a stackup, do the following:
2. Choose the **File > Open Schematic** menu item, then double-click *xt_trace_separation.ffs*. A schematic appears in the LineSim editor.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

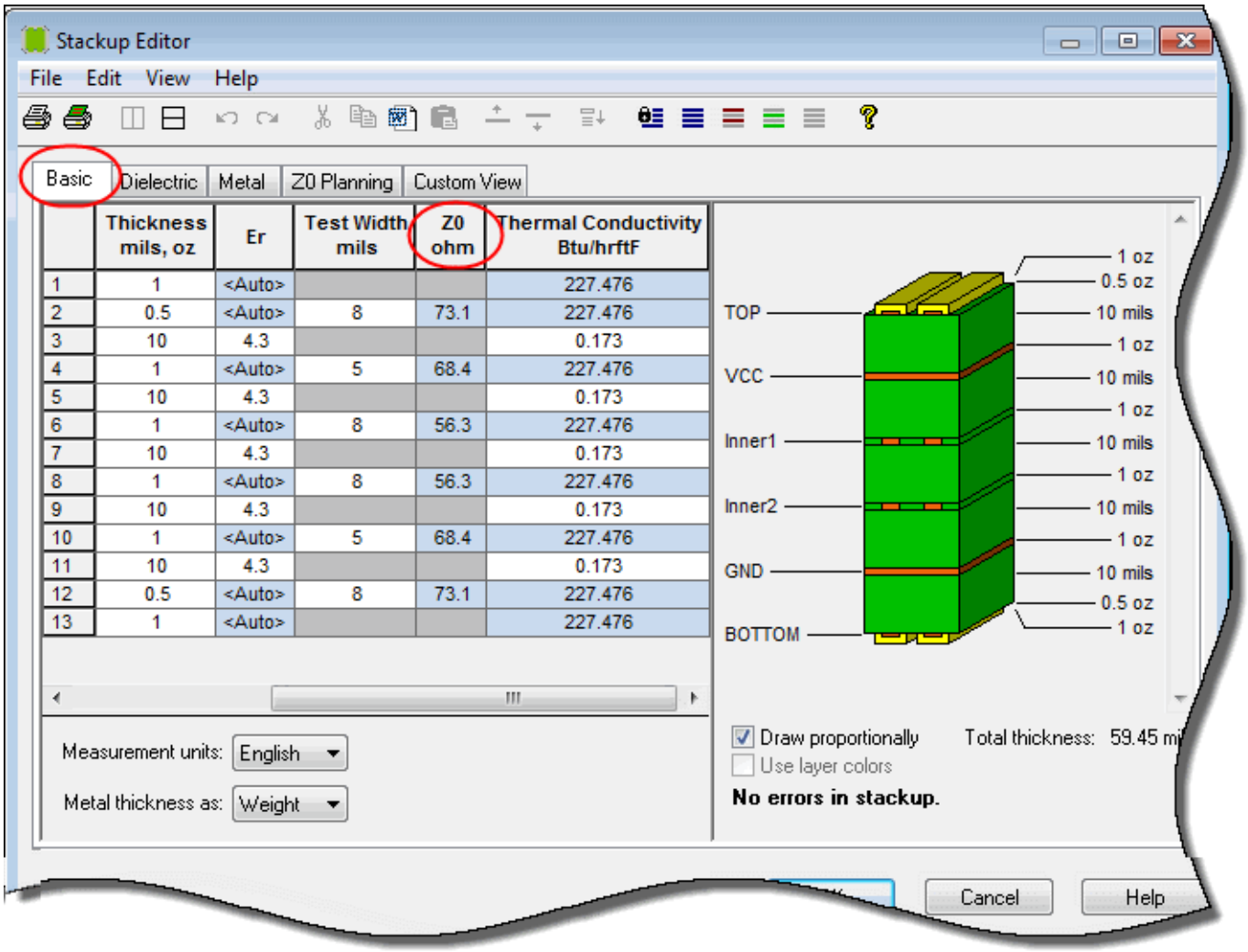
3. Choose the **Setup > Stackup > Edit** menu item. The Stackup Editor dialog box opens.

   The spreadsheet displays the data defining the stackup. Each row represents one layer in the stackup, either metal or dielectric.

   The graphical view on the right visually summarizes key data in the stackup, such as layer names and thicknesses.

4. Verify that the **Basic** tab is selected.

   The impedance of each signal layer in the stackup is displayed in the Z0 column, which is located at the far right of the stackup spreadsheet (you may need to scroll the spreadsheet to see that column). Since impedance can be calculated only for a specific trace width, specify the trace width for each layer in the Test Width column. Any changes to characteristics affecting the impedance such as trace width or dielectric constant immediately changes the impedance value so you can instantly see how changes affect the impedance. See "Impedance Planning" on page 1219 for more information on impedance planning.

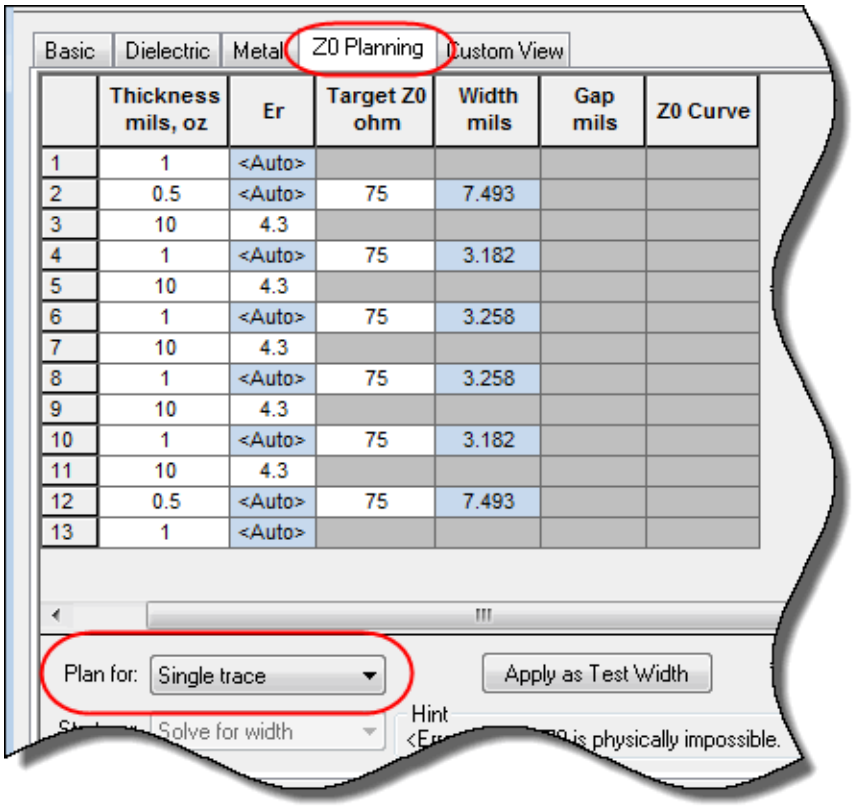


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5. Change the dielectric thickness to see how it affects trace impedance.

   a. Select the **Z0 Planning** tab.

   b. In the "Plan for" list, select "Single trace". This setting specifies that the example deals with the impedances of single traces, rather than differential pairs.

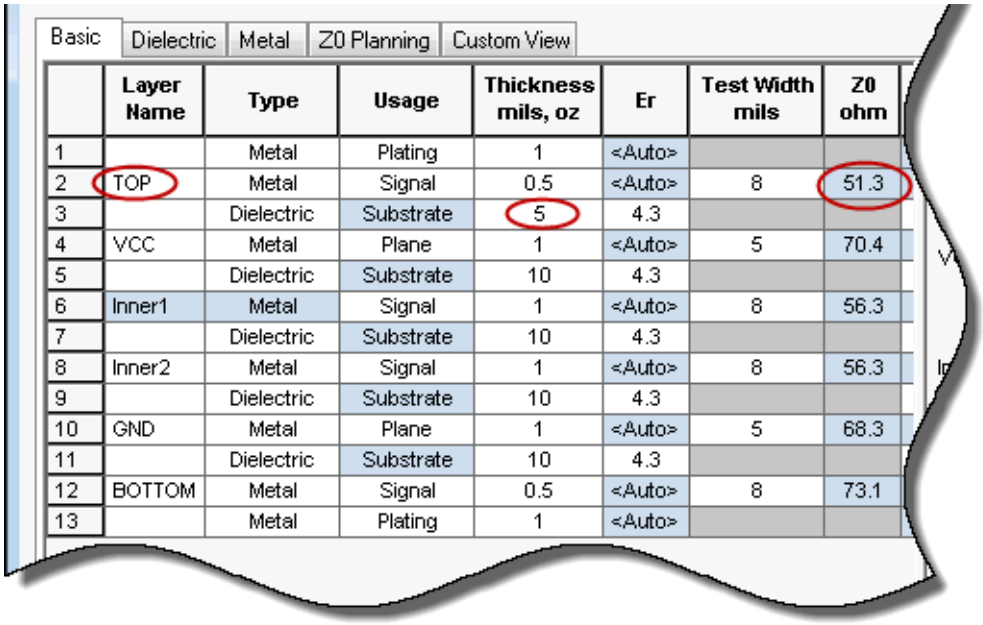


   c. Select the **Basic** tab.

   d. In row 3 of the spreadsheet (the top-most dielectric layer), type 5 into the Thickness column.

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The impedance value for the TOP layer (row 2 of the spreadsheet) changes from 73.1 ohms to 51.3 ohms.



	Layer Name	Type	Usage	Thickness mils, oz	Er	Test Width mils	Z0 ohm
1		Metal	Plating	1	<Auto>		
2	TOP	Metal	Signal	0.5	<Auto>	8	51.3
3		Dielectric	Substrate	5	4.3		
4	VCC	Metal	Plane	1	<Auto>	5	70.4
5		Dielectric	Substrate	10	4.3		
6	Inner1	Metal	Signal	1	<Auto>	8	56.3
7		Dielectric	Substrate	10	4.3		
8	Inner2	Metal	Signal	1	<Auto>	8	56.3
9		Dielectric	Substrate	10	4.3		
10	GND	Metal	Plane	1	<Auto>	5	68.3
11		Dielectric	Substrate	10	4.3		
12	BOTTOM	Metal	Signal	0.5	<Auto>	8	73.1
13		Metal	Plating	1	<Auto>		

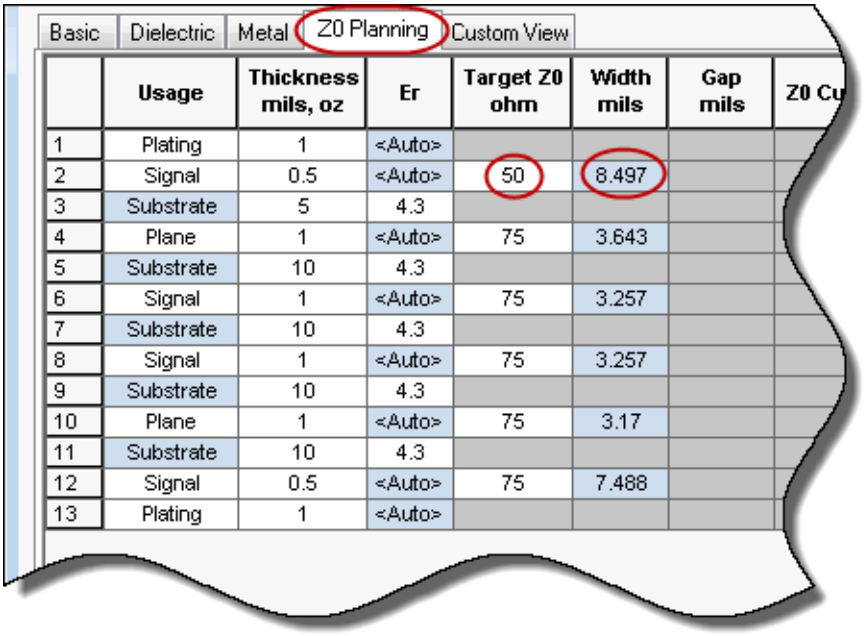
The**Z0 Planning** tab gives you an even faster way of planning impedances. You can use the Target Z0 column to specify a desired impedance for each signal layer. In this example, 75 ohms is achieved on the bottom layer at about 7.5 mils width, and on the other signal layers at about 3.3 mils. You can change the target impedance to calculate a different geometric solution, as shown in the next step.

e. Click the **Z0 Planning** tab.

f. In row 2 of the spreadsheet, type 50 into the Target Z0 column.

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g. The value of the width for layer 2 is now about 8.4 mils, which is the trace width required on this layer to match the new target Z0.



	Usage	Thickness mils, oz	Er	Target Z0 ohm	Width mils	Gap mils	Z0 Cu
1	Plating	1	<Auto>				
2	Signal	0.5	<Auto>	50	8.497		
3	Substrate	5	4.3				
4	Plane	1	<Auto>	75	3.643		
5	Substrate	10	4.3				
6	Signal	1	<Auto>	75	3.257		
7	Substrate	10	4.3				
8	Signal	1	<Auto>	75	3.257		
9	Substrate	10	4.3				
10	Plane	1	<Auto>	75	3.17		
11	Substrate	10	4.3				
12	Signal	0.5	<Auto>	75	7.488		
13	Plating	1	<Auto>				

6. Use impedance planning for differential pairs.

   Differential signaling is widely used at very high data rates, especially in SERDES designs. The Z0 Planning feature can help you find target differential impedances as easily as single-trace impedances. This step shows how to find differential impedances.

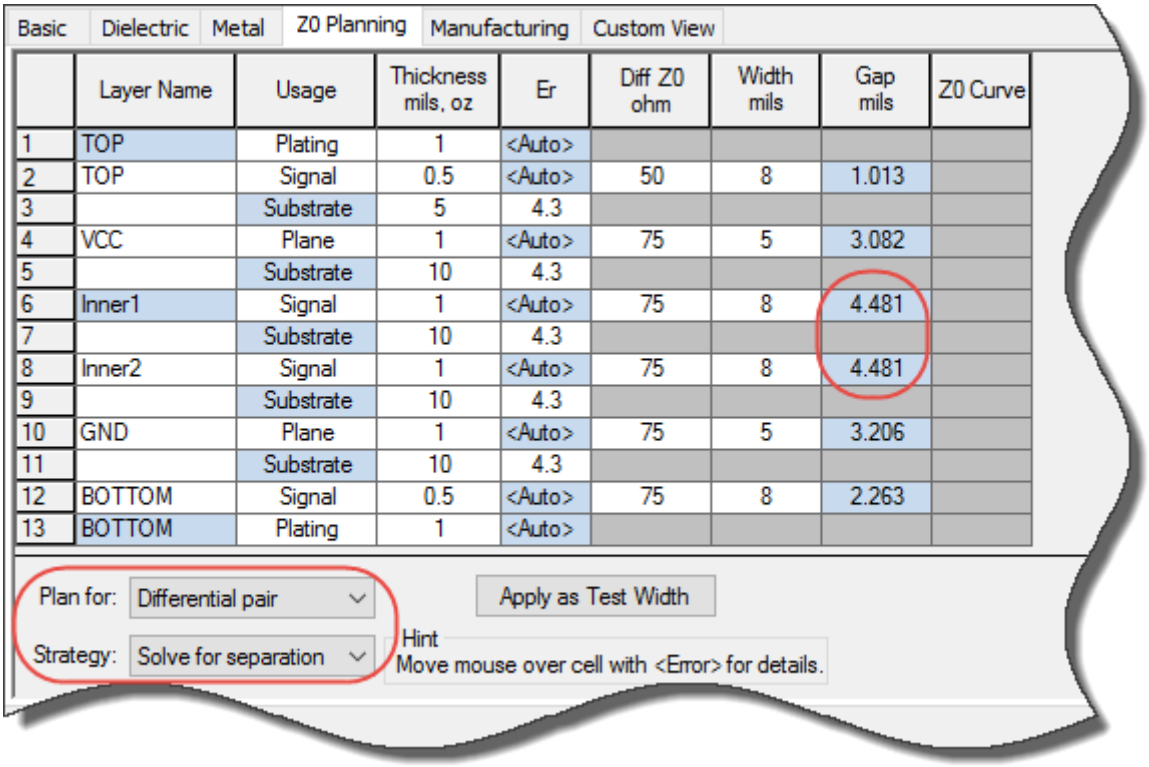
   a. In the “Plan for” list, select “Differential pair”. The title of the Target Z0 column changes to Diff Z0.

   b. In the Strategy list, select “Solve for separation”.

      With these changes, the stackup editor now interprets the target Z0 values as the differential impedance of a pair of traces. Setting Strategy to “Solve for separation”

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enables the Stackup Editor to calculate the required distance between traces (Gap) using the specified trace width and target differential impedance values.



	Layer Name	Usage	Thickness mils, oz	Er	Diff Z0 ohm	Width mils	Gap mils	Z0 Curve
1	TOP	Plating	1	<Auto>				
2	TOP	Signal	0.5	<Auto>	50	8	1.013	
3		Substrate	5	4.3				
4	VCC	Plane	1	<Auto>	75	5	3.082	
5		Substrate	10	4.3				
6	Inner1	Signal	1	<Auto>	75	8	4.481	
7		Substrate	10	4.3				
8	Inner2	Signal	1	<Auto>	75	8	4.481	
9		Substrate	10	4.3				
10	GND	Plane	1	<Auto>	75	5	3.206	
11		Substrate	10	4.3				
12	BOTTOM	Signal	0.5	<Auto>	75	8	2.263	
13	BOTTOM	Plating	1	<Auto>				

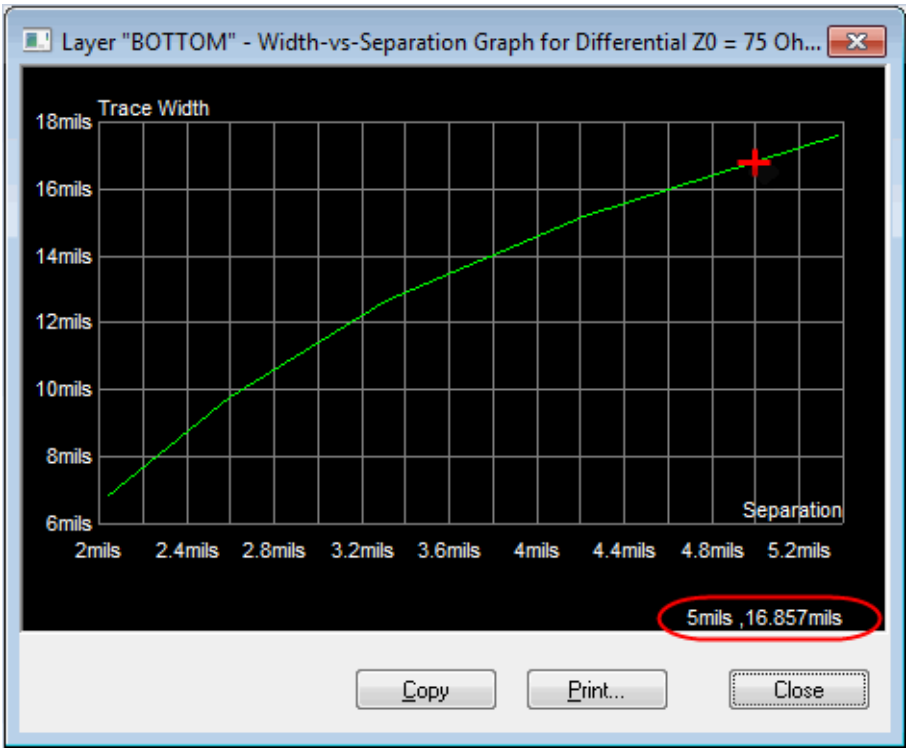
For the inner signal layers, the solved-for separation value is about 4.5 mils.

For differential pairs, you can also specify the separation and solve for trace width or solve for both separation and width simultaneously. The following steps solves for both.

c. In the Strategy list, select “Solve for both”. The Width and Gap columns gray out and the View buttons appear in the Z0 Curve column for each layer.

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d. In row 12, the layer BOTTOM, click **View** in the Z0 Curve column. The field solver runs and after a few seconds, a dialog box appears.



The resulting graph shows a curve of constant 75-ohm differential impedance, the target impedance for this layer, enabling you to choose a range of either trace widths or separations. You can also read the corresponding value. For example, the curve shows that at 5 mils separation, you need a trace about 17 mils wide to achieve 75 ohms differential impedance.

Achieving a Specific Differential Impedance

Differential signaling takes advantage of the coupling between neighboring traces. When you design a differential pair, you deliberately couple the two traces together strongly, so that any signal induced by external noise on one is also induced on the other, and then rejected by the differential receiver at the ends of the lines.

However, differential-pair design involves issues such as determining which geometries to pick to achieve a specific differential impedance. Terminating differential traces can also be challenging.

The LineSim crosstalk option is a powerful tool for differential-signal applications because of the built-in boundary-element field solver. The field solver automatically calculates differential impedances, determines coupling parameters, and suggests termination values.

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IC vendors or bus specifications commonly recommend specific differential-impedance targets for differential signaling. LineSim can help you plan for differential impedances. For example, when you enter a differential trace pair into the LineSim schematic editor, LineSim immediately makes the differential impedance of the pair available to you. See “Edit Transmission Line Dialog Box - Edit Coupling Regions Tab” on page 862.

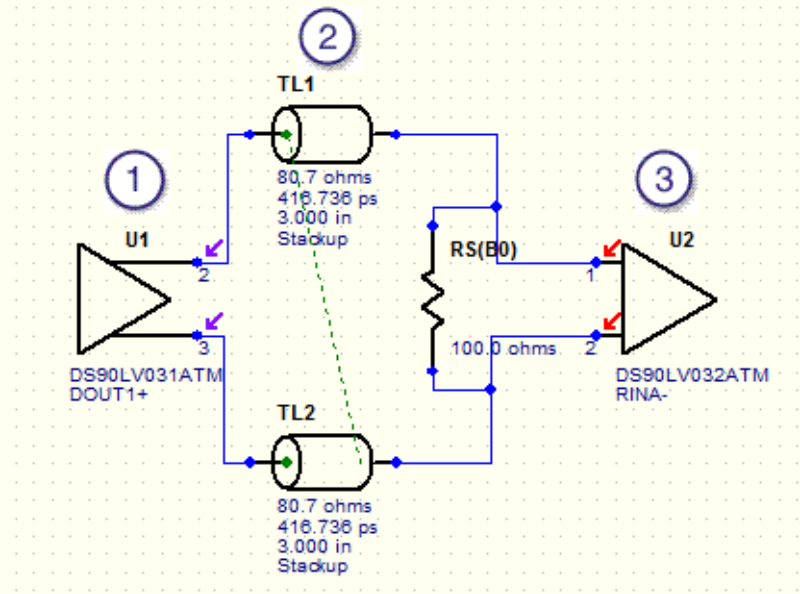


Table A-5. XT_Coupled_Differential Circuit Description

Reference	Description
1	A differential driver IC, using a high-speed, low-swing LVDS driver pair. Total swing voltage is about 400 mV.
2	Two transmission lines coupled together, as shown by the dashed line between them.
3	A differential receiver terminated with a 100-ohm differential resistor.

Prerequisites

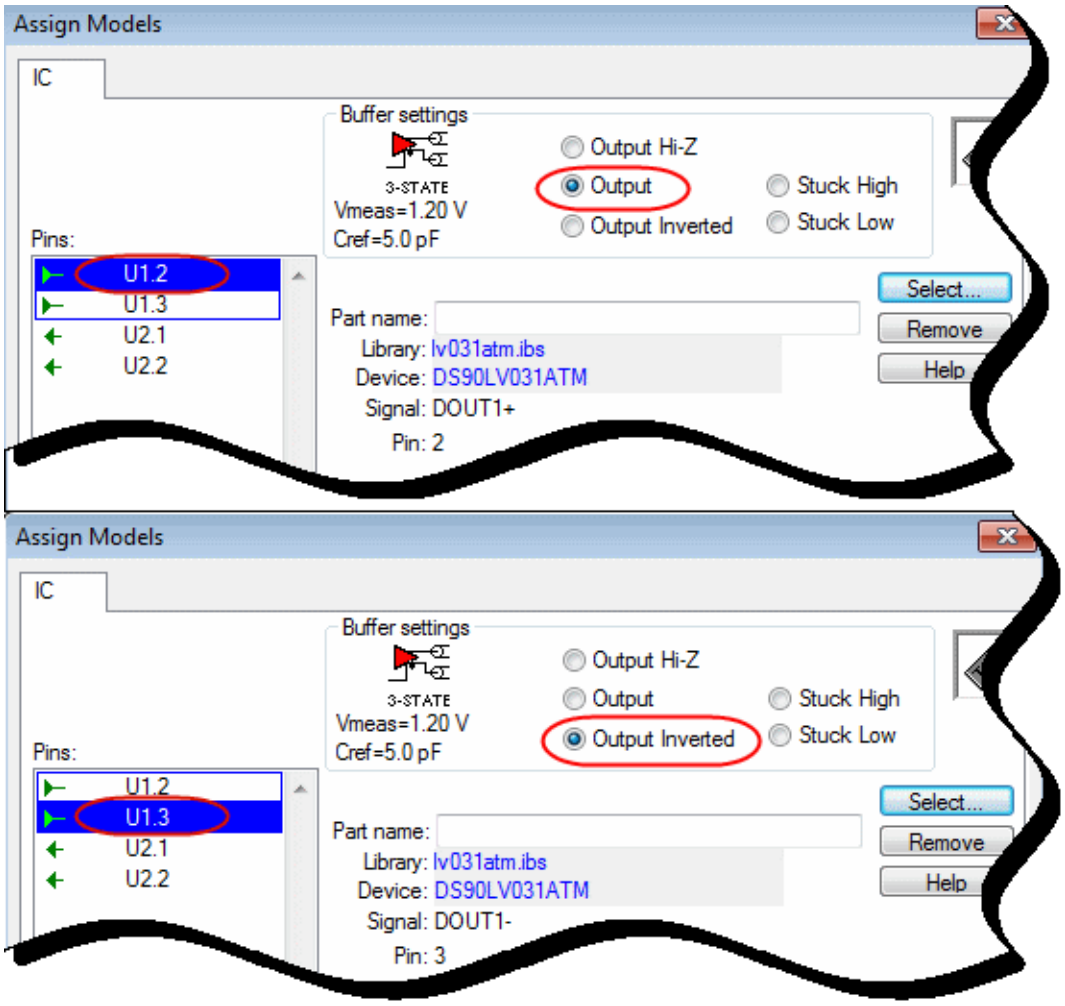
- The Crosstalk license is required to create and simulate coupling regions in LineSim.
- The Advanced Scope license is required to create eye diagrams with the oscilloscope.
- The Lossy Lines license is required to model lossy transmission lines.
- This procedure assumes familiarity with the trace-separation example and the concept of a coupling region. For information about adding coupling regions to transmission lines, see Creating a Schematic Design.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

1. Open the schematic. Choose the **File > Open Schematic** menu item, then double-click *xt_coupled_differential.ffs*.
2. Look at the driver IC setup:
   a. Double-click the driver-IC U1. The Assign Models dialog box opens.
   b. In the Pins list, select IC pin U1.2 and then pin U1.3. In the “Buffer settings” area to the right, note that these two pins have opposite polarity: Output and Output Inverted, respectively. These settings make the two pins switch differentially.

      Because pins U1.2 and U1.3 are a differential pair, if you change the polarity of one pin, the polarity of the other pin changes automatically.



   c. Click **OK** .

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Suppose the design goal, perhaps specified by the driver-IC manufacturer, is to achieve a 100-ohm differential impedance with the trace pair. The differential terminator is already set to this value. LineSim can help you plan for this impedance requirement.

3. Determine differential impedance of coupled traces:

   a. Double-click either of the transmission lines in the schematic. The Edit Transmission Line dialog box opens.

   b. Select the **Edit Coupling Regions** tab.

   c. In the Impedance area, locate the (Differential) row. This shows the differential impedance for the current geometric properties of the coupling-region cross section.

      The differential impedance is currently 123 ohms, considerably higher than the design goal of 100 ohms.

4. Decrease the differential impedance by reducing the trace separation.

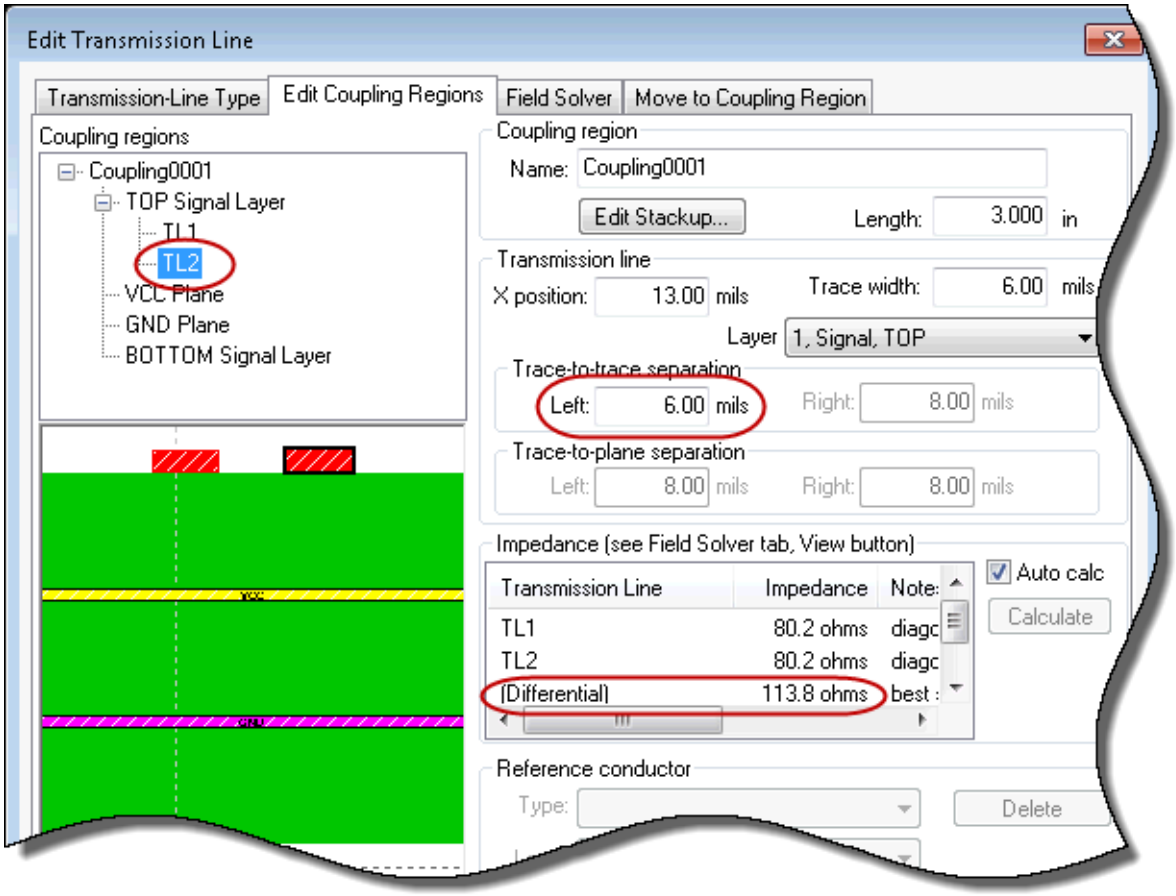
   One way to decrease differential impedance is to strengthen the coupling between the traces. The following steps interactively adjust the differential impedance to achieve the design goal of 100 ohms. You can also solve for differential impedances in the stackup editor. See step 6 in Modeling a PCB Stackup.

   a. In the “Coupling regions” area, select TL2.

   b. In the “Trace-to-trace separation” area, type 6 and press Enter.

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The differential impedance has dropped to 113ohms, which is better than before, but still too high.



5. Other aspects other than trace separation affect differential impedance. This step decreases the stackup dielectric thickness.

   a. Click **Edit Stackup**. The stackup editor opens.

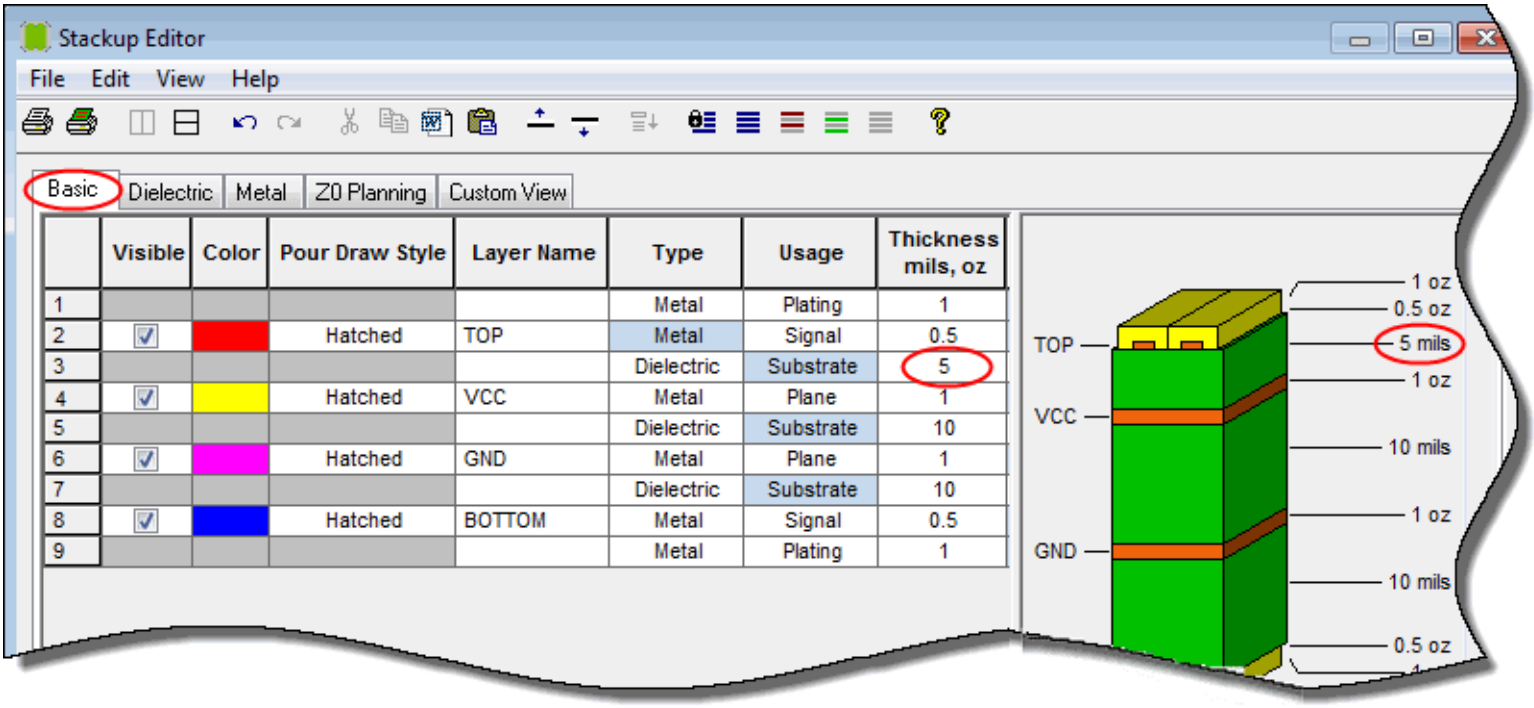
   b. Select the **Basic** tab.

   c. In row 3 of the spreadsheet, click in the Thickness column, type 5, and press Enter.

   d. Verify in the graphical stackup view, located at the far right, that the top-most dielectric layer displays as 5 mils thick.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

e. Click **OK** to close the editor.



f. Back in the Edit Transmission Line dialog box, check the new differential-impedance value in the Impedance area.

The differential impedance is now reduced to about 97 ohms. This is very close to the design goal.

g. Click **OK**.

6. The Impedance area on the Edit Coupling Regions tab gives only a brief summary of the data calculated by the HyperLynx field solver. This step shows you how to view detailed results.

a. Choose the **Setup > Enable Lossy Simulation** menu item, and verify that it is enabled.

b. From the Edit Transmission Line dialog box, select the **Field Solver** tab.

c. In the "Numerical results" area, click **View**. The HyperLynx File Editor displays a report file. Use the scroll bar to view the sections of the report:

- The Impedance and Termination Summary gives a detailed list of possible termination values to use for the differential pair.
- The Physical Input Data records the cross section simulation for future reference.

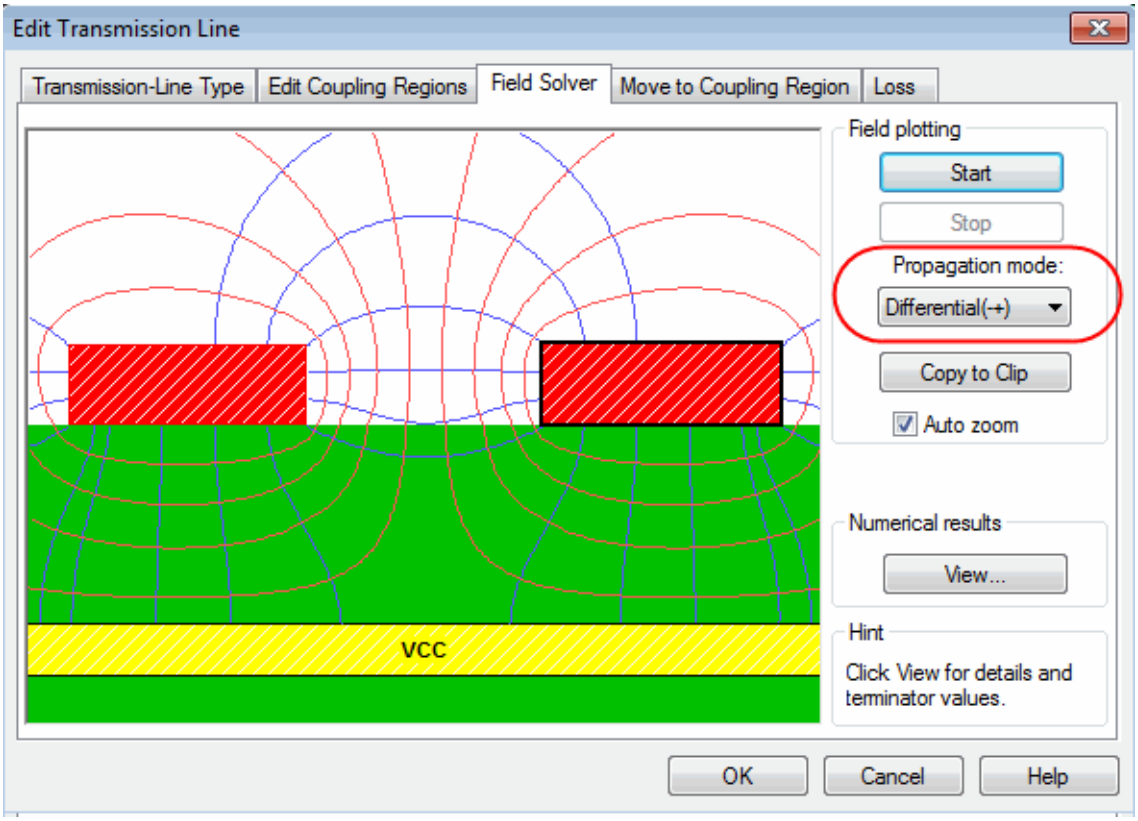
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- The Field-Solver Output Data section gives the detailed electrical characteristics of the cross section, including characteristic-impedance matrix, capacitance matrix, inductance matrix, and propagation speeds.
- The Lossy Data Output section contains trace loss data, including skin-effect, as well as dielectric loss information. This section is unavailable if lossy simulation is not enabled.

7. You can also plot the field lines calculated by the field solver to help give you an intuitive understanding for how a cross section is coupled:

   a. Close the file editor.

   b. In the Edit Transmission Line dialog box, in the “Field plotting” area, in the “Propagation mode”, select Differential(+-)., then click **Start**.



   The field solver calculates and plots the field lines. Electric-field lines are shown in blue, and electric equipotential lines display in red. The plot assumes opposed, differential currents in the two traces.

8. Simulate the differential circuit to see the resulting waveforms:

   a. Click **OK** to close the Edit Transmission Line dialog box.

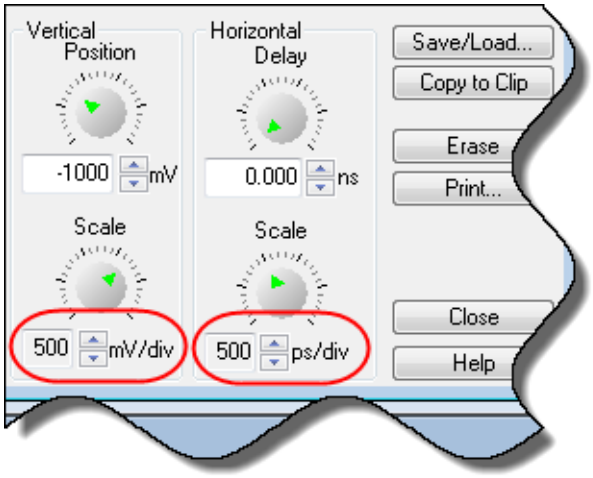
   b. Choose the **Simulate SI > Run Interactive Simulation** or **Run Interactive Simulation in Oscilloscope** menu item. The Digital Oscilloscope opens.

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c. Change the stimulus to “Falling edge”.

d. Change the Horizontal Scale timebase to 500 ps/div.

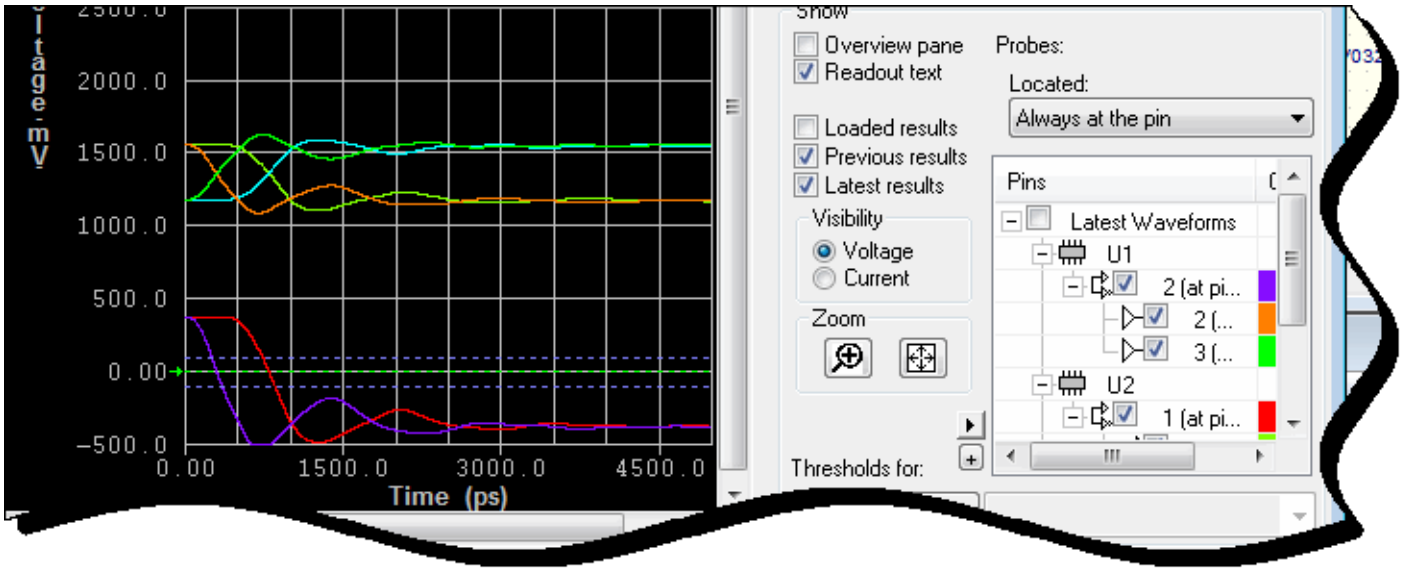
e. Change the Vertical Scale to 500 mV/div.



f. Click **Start Simulation**.

Results

Recall the LVDS drivers have approximately 400 mV of total swing. The digital oscilloscope shows the driver and receiver signals differentially, and also as single-ended waveforms.



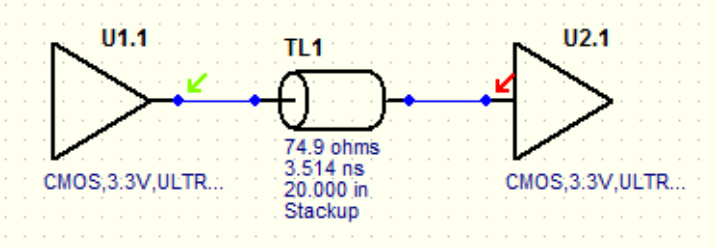
LineSim supports other transmission line types. See “Modeling a Transmission Line” on page 1219 for details.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Setting Up a SPICE Simulation

LineSim includes two built-in SPICE simulators and supports integration with HSPICE. Using SPICE, HyperLynx wraps itself around your choice of SPICE engine for a more user-friendly simulation environment than with just raw SPICE.

This tutorial walks you through how to set up a SPICE model and demonstrates how SPICE results are loaded automatically into the oscilloscope after simulation. This example changes the schematic driver model to a SPICE model.



This tutorial provides two SPICE models. For simplicity, the models are built with linear elements and do not include transistor models from a semiconductor vendor.

Prerequisites

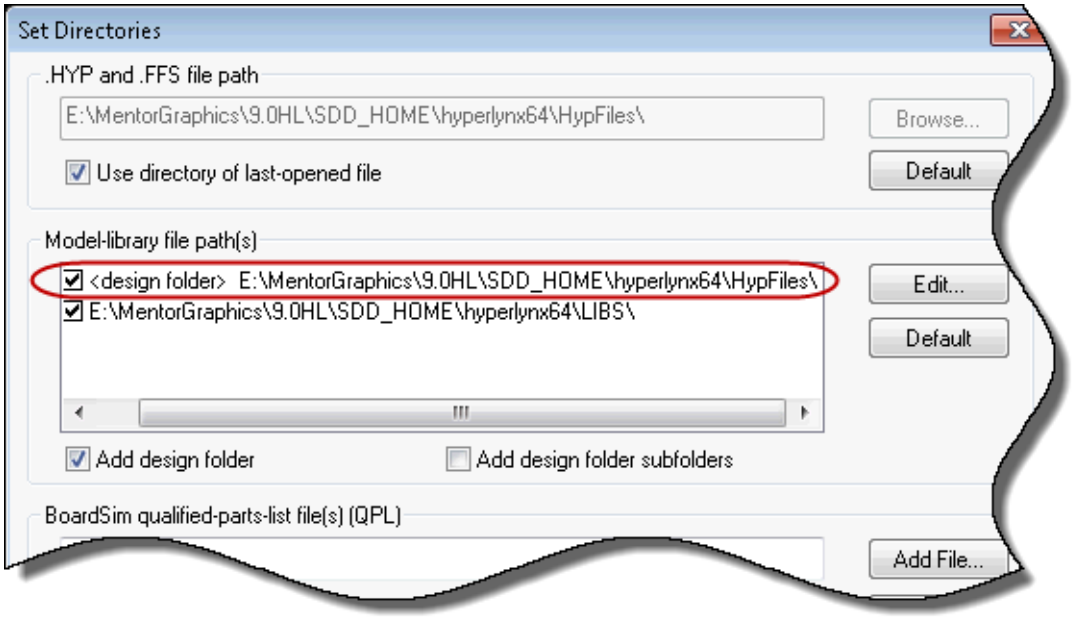
- The SPICE Output license is required to run SPICE simulation.
- You install and license HSPICE separately from HyperLynx. Note that this tutorial uses a built-in SPICE simulator.

Procedure

1. Set the model path:
   a. Choose the **Models > Edit Model Library Paths** menu item. The Set Directories dialog box opens.
   b. Verify that either of the following folders is present:
      - *\MentorGraphics\<release>\SDD_HOME\hyperlynx64\HypFiles*

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- *\MentorGraphics\<release>\SDD_HOME\hyperlynx\HypFiles*



c. If not, in the “Model-library file path(s)” area, click **Edit**. The Select Directories for IC-Model Files dialog box opens.

d. Click **Add** and select one of the folders specified in step b.

e. Click **Select Folder**.

f. Close the Select Directories for IC-Model Files and Set Directories dialog boxes.

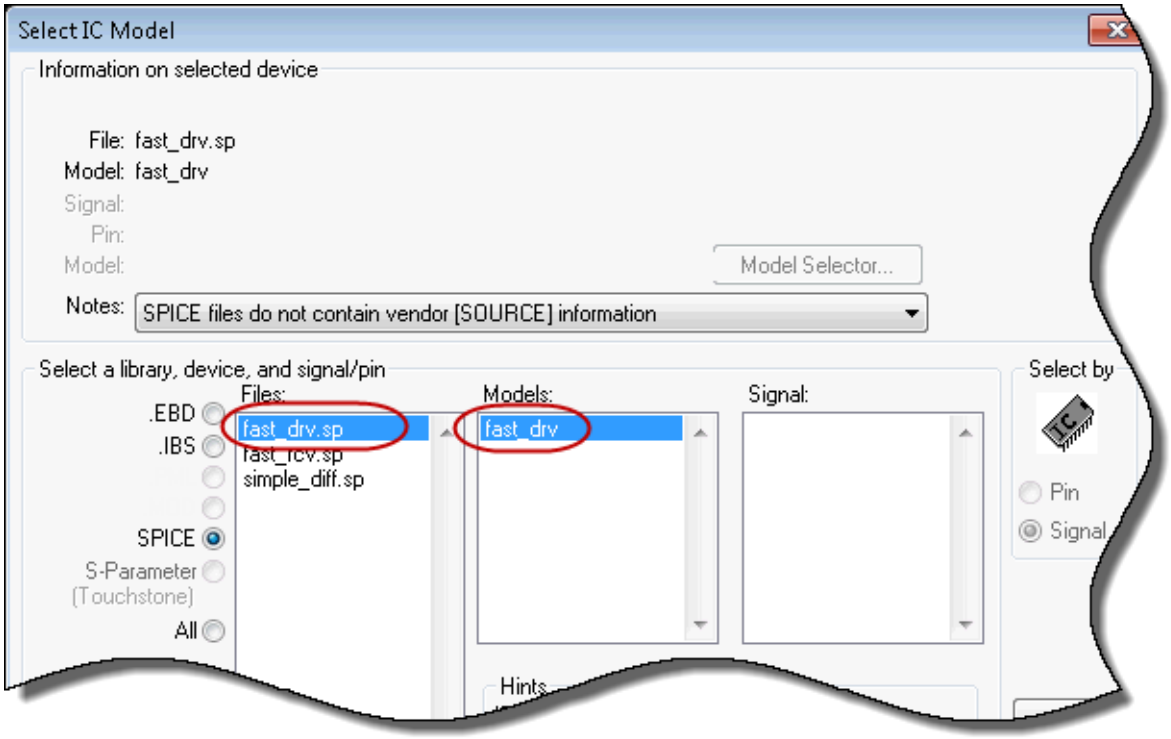
2. Open the schematic. Choose the **File > Open Schematic** menu item, then double-click *lossy.ffs*.

3. In the schematic, double-click the driver-IC symbol at the left end of the transmission line. The Assign Models dialog box opens.

4. In the Pins list, double-click pin U1.1. The Select IC Model dialog box opens.

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5. Select SPICE.



6. In the Files list, select fast_drv.sp.
7. In the Models list, double-click fast_drv.

   In the Assign Models dialog box, a port-mapping spreadsheet appears. The purpose of the spreadsheet is to help you make connections to all of the ports on the SPICE model. The model ports are the external connection points to connect a stimulus waveform, power supplies, and output pins.

   This particular SPICE model contains four ports:

   - Vin - where the model expects to be stimulated
   - Vout - output pin, connected in the schematic
   - Vcc and Gnd - the model power supply pins

   LineSim automatically recognizes the names Vcc and Gnd and connects them to the LineSim built-in power supplies. However, if a SPICE model has obscure supply names, you can connect them manually.

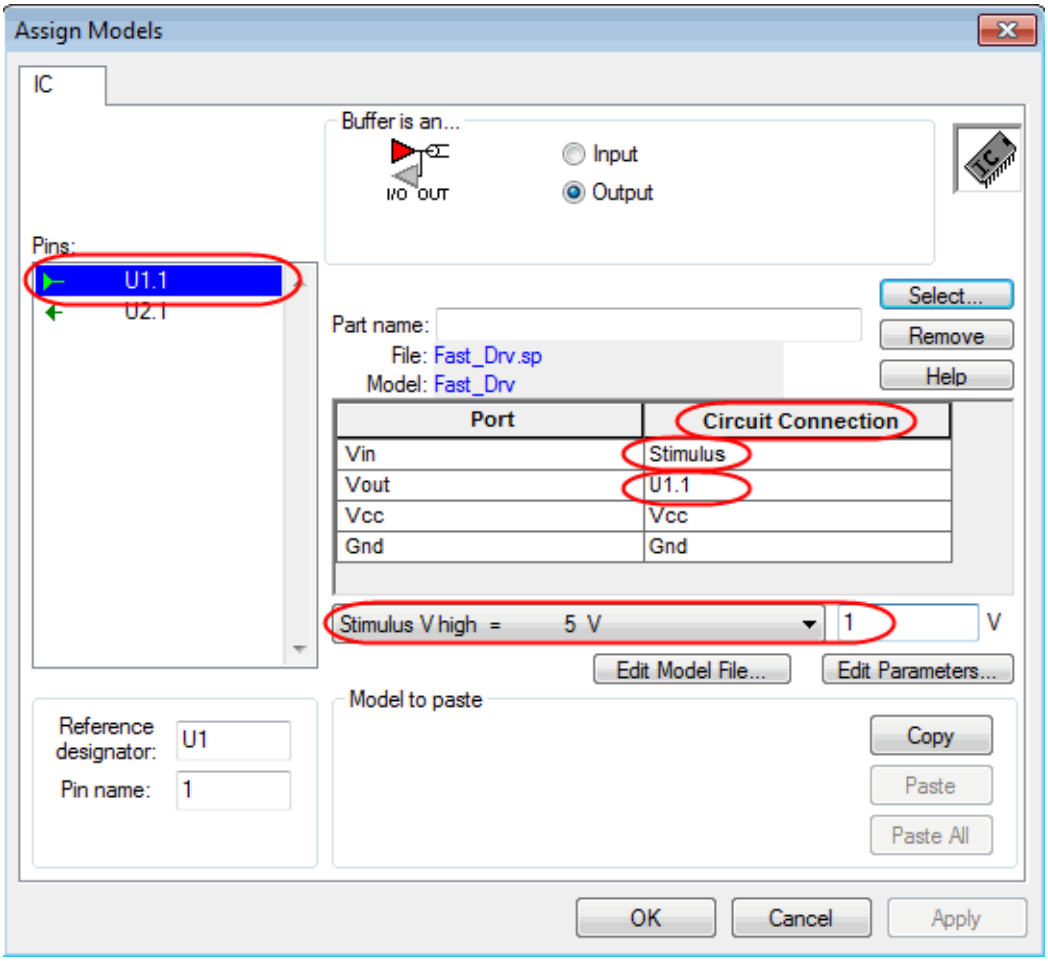
8. Manually connect stimulus and output ports:

   a. In the spreadsheet, click the Circuit Connection cell for the Vin port to open a list of possible connections. The Vin port of the SPICE model expects digital stimulus, so select Stimulus in the list.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

b. Click the Circuit Connection cell for the Vout port to open the port. The Vout port is the pin on the SPICE model to connect in the schematic. Select U1.1, the driver-pin name in the schematic.

c. In the list below the spreadsheet, select Stimulus V high and type 1 into the box to the right of the list.



9. Click **OK**.

The SPICE driver model is now assigned and connected to the schematic symbol. The next step is to assign and connect a SPICE receiver model.

Note

An important difference between SPICE and IBIS IC models is that SPICE models have explicit ports. With a SPICE model, you manually connect power supplies, one or more input stimulus pins, optional control pins, and one or more output pins.

Compare this to using IBIS models: the input/stimulus, power supply, and control pins are implicit and LineSim can connect them automatically for you. This difference results because SPICE models are inherently lower level than IBIS models. Fortunately,

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through its port-mapping spreadsheet, making and managing these extra connections are easy using LineSim and BoardSim.

10. Change the receiver model to a SPICE model:

    a. In the schematic, double-click the driver-IC symbol at the right end of the transmission line. The Assign Models dialog box opens.

    b. In the Pins list, double-click pin U2.1. The Select IC Model dialog box opens.

    c. Select SPICE.

    d. In the Files list, select model fast_rcv.sp.

    e. In the Models list, double-click model Fast_Rcv. The Assign Models dialog box opens.

    f. In the spreadsheet, click the Circuit Connection cell for the Vin port. The Vin port is the pin on the SPICE model to connect in the schematic. Select U2.1, the receiver-pin name in the schematic.

    g. Click **OK**.

    SPICE models are now assigned.

11. Run a SPICE simulation.

    This step shows you how to launch a simulation and how SPICE waveforms are automatically read back into the oscilloscope.

    a. Choose the **Simulate SI > Run Interactive Simulation** or **Run Interactive Simulation in Oscilloscope** menu item. The Digital Oscilloscope opens.

    b. In the Stimulus area, select “Rising edge”.

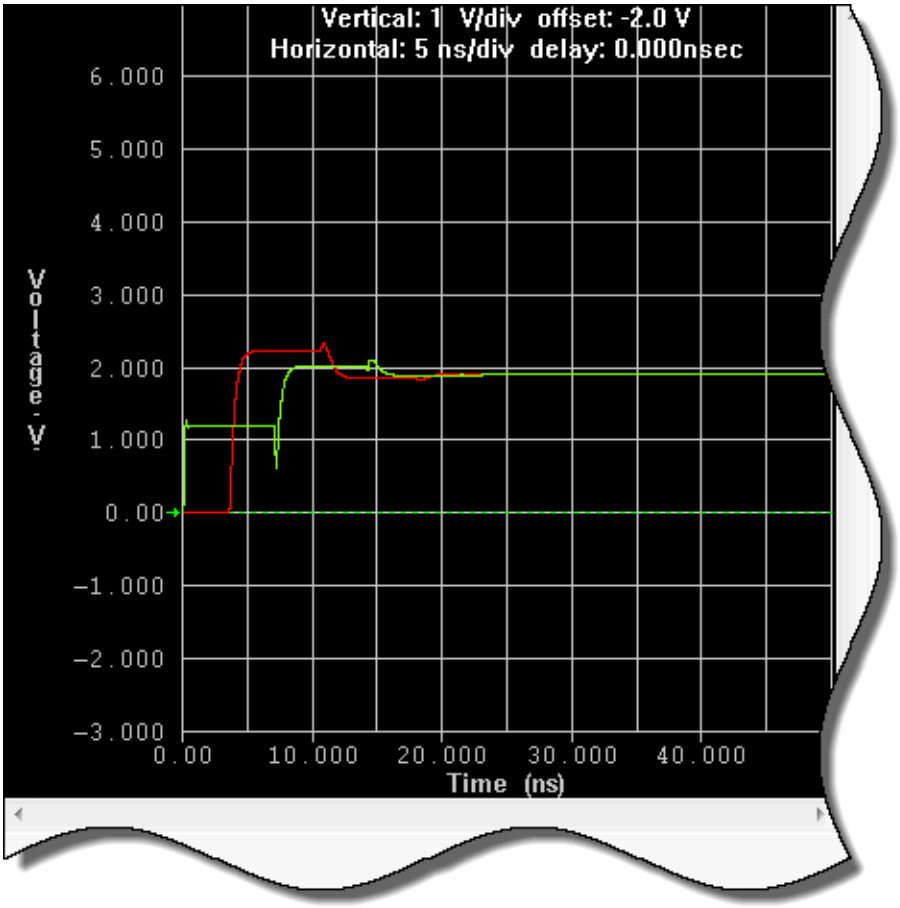
    c. Change the Horizontal Scale to 5 ns/div.

    d. Change the Vertical Scale to 1 V/div.

    e. Click **Start Simulation**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

f. View SPICE results in the oscilloscope.



12. Choose the **File > Close** menu item and do not save your changes. This is because another tutorial uses *lossy.ffs* with its initial IBIS model assignments.

Related Topics

Integrated SPICE Simulations

Including Touchstone Models in a LineSim Schematic

You can include and set up a typical Touchstone model in a LineSim schematic.

This tutorial modifies an existing schematic and demonstrates how to add a example 4-port Touchstone model.

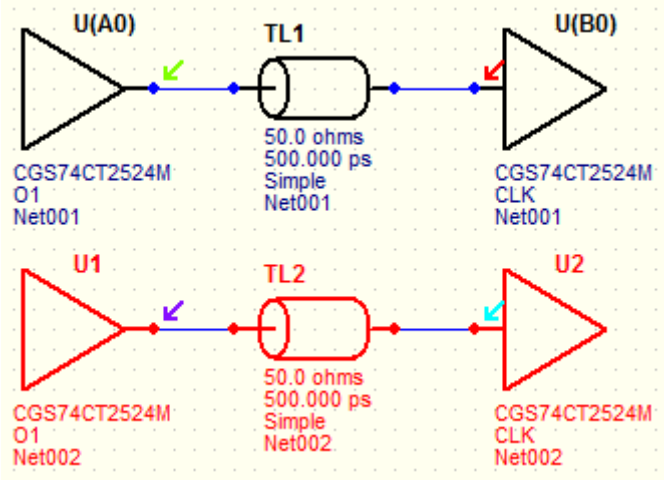
Prerequisites

The GHz license bundle license is required to simulate nets containing Touchstone models.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

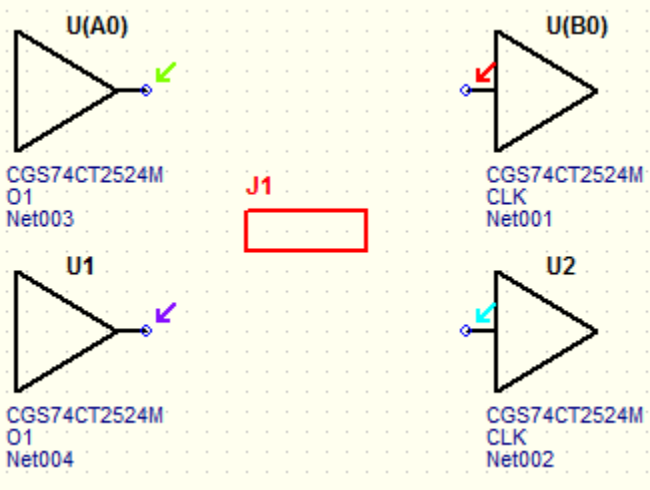
1. Open the schematic. Choose the **File > Open Schematic** menu item, then double-click *ser_touchstone.ffs*.
2. Modify the schematic:
   a. This example uses a 4-port Touchstone connector model, which requires two drivers for the input side and two receivers for the output. The easiest way to create this is to make a copy of the existing driver-tline-receiver circuitry:
      i. Select all of the existing symbols by dragging a box around them with the mouse. Selected symbols are highlight in red.
      ii. Copy and paste the selected symbols. A copy of the symbols appears in the editor, highlighted in red. Note that the new symbols have unique, new reference designators.
      iii. Drag the new symbols as a group to just below the existing symbols so that the drivers, transmission lines and receivers are immediately above/below each other.



   b. Delete the existing transmission lines. Over each transmission line, right-click, then select the **Delete** menu item.
   c. Add the Touchstone interconnect model. On the drawing palette, click **Add S-Parameter/SPICE Model to schematic**  . A red symbol appears. Move the red

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symbol to a position between the four dangling wires in the schematic, and click to place it.



The empty symbol on the schematic can house any passive, non-driving SPICE or Touchstone model. The next step is to assign a 4-port S-parameter model to the empty symbol.

3. Assign a Touchstone model:

   a. Double-click the empty S-parameter/SPICE model symbol. The Assign S-Parameter/SPICE Model dialog box opens.

   b. In the “Model type”, select Touchstone.

   c. In the Existing Files list, select *example.sp4*.

   d. In the Auto-Place Ports list, assign model port numbering by selecting Odd to Left, Even to Right

   e. Click **OK**.

      In the schematic, the empty symbol is replaced by one labeled example.s4p showing two ports to the left and two ports to the right. Assume that this model is arranged with two internal paths: one from port 1 to port 2, and the other from port 3 to port 4. The two interconnects are most likely coupled to each other.

      For more information about Touchstone model port numbering, see S-Parameter Port Numbering.

   f. Wire the IC-buffer symbols to the nearest ports:

      i. Connect driver pin U(A0)to J1 Port1 by moving the pointer to the end of the wire for U(A0) (the pointer shape becomes a plus sign +), dragging a wire to Port1 of the connector, and releasing the mouse button.

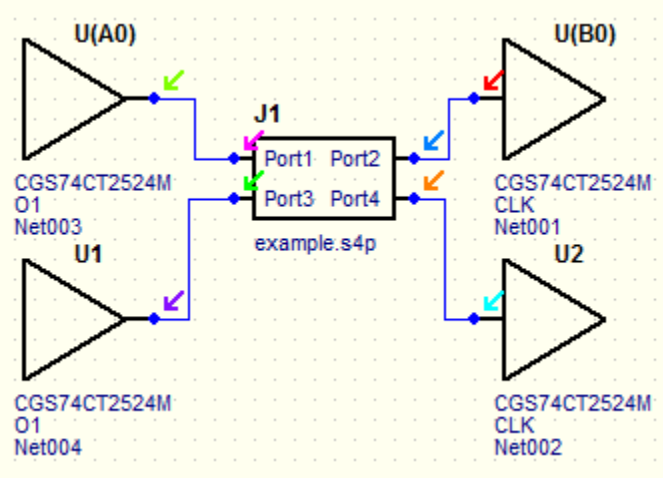
      ii. Repeat to connect driver pin U1 to J1 Port3.

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g. For variety, connect the remaining two wires in a different way. Click receiver symbol U(B0) and drag it until the end of its wire touches the end of the wire for J1 Port2. Release the mouse briefly, then press down again and drag U(B0) away from J1. A wire appears between the two symbols.

h. Repeat to connect U2 to J1 Port4.

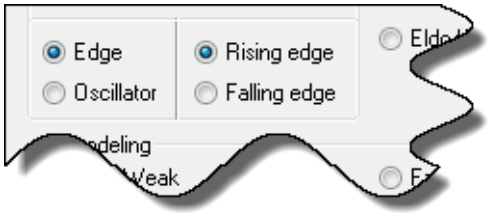
The resulting schematic is shown below:



4. Run simulation:

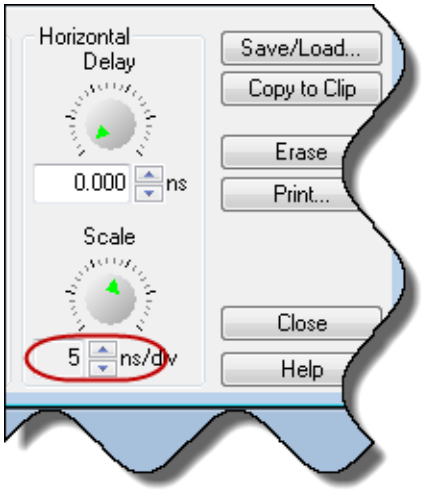
a. Choose the **Simulate SI > Run Interactive Simulation** or **Run Interactive Simulation in Oscilloscope** menu item. The Digital Oscilloscope opens.

b. In the Stimulus area, select “Rising edge”.



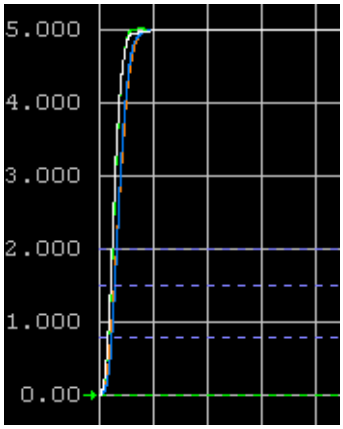
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

c. Change the Horizontal Scale to 5 ns/div.



d. Click **Start Simulation**.

5. View SPICE results in the oscilloscope.



Related Topics

Touchstone (S-Parameter) Modeling

Viewing and Converting Touchstone and Fitted-Poles Models

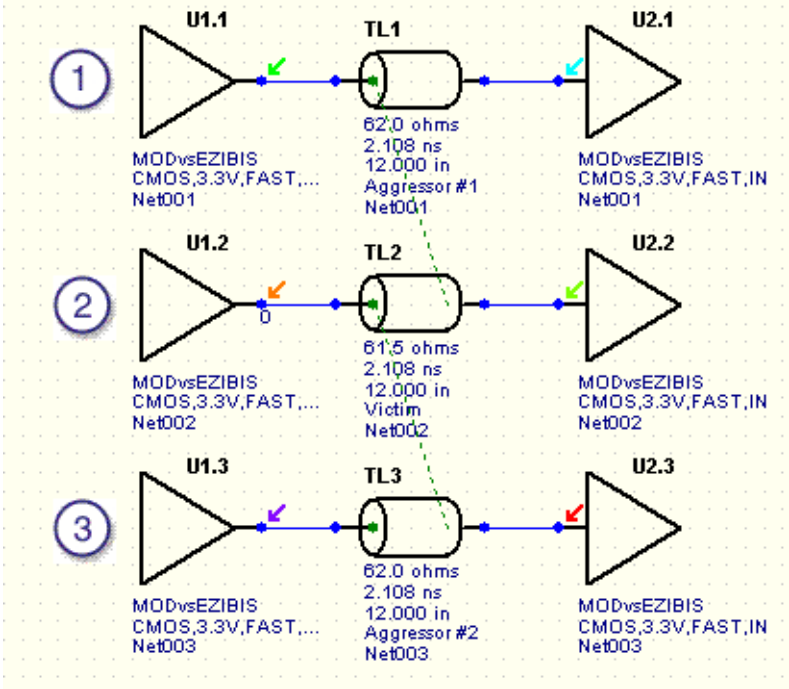
Planning Minimum Trace Separation on a Bus

You can use LineSim to plan minimum trace separation on a bus.

The goal of this tutorial is to design a bus with a constraint of no more than 500 mV of crosstalk between any of the signals on the bus. This tutorial uses a traditional synchronous-style design and demonstrates how the LineSim crosstalk option helps you meet this design goal and develop the proper routing constraints to achieve it.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

A typical parallel-style bus in a digital system contains many parallel traces – 16, 32, 64, and even more signals. However, when simulating to predict crosstalk on such a bus, simulating all of the signals is not necessary. Instead, you can take advantage of the fact that the crosstalk driven into a given victim trace comes predominantly from two other traces: the neighboring traces on either side of the victim trace. Therefore, simulation typically includes a set of three traces, as shown in this example.



In the schematic, the three transmission lines represent the side-by-side traces on the bus described above. The triangular IC-driver symbols at the left end of each line show that all three traces are being driven from the left side. Each line also has a receiver IC at its right end. The ICs are modeled with a generic 3.3-V fast CMOS model.

Prerequisites

The Crosstalk license is required to create coupling regions in LineSim schematics.

Procedure

1. Open the schematic. Choose the **File > Open Schematic** menu item, then double-click *xt_trace_separation.ffs*.

   If the entire schematic does not appear on the screen, choose the **View > Fit to Window** menu item.

2. Disable lossy modeling for this example. Choose the **Setup > Enable Lossy Simulation** menu item and verify that it is disabled.

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3. Set up the driver ICs for simulation:

   a. Double-click any of the left-end driver-IC symbols in the schematic. The Assign Models dialog box opens.

   b. Select pin U1.2, the driver on the middle trace. In the Buffer Settings area, verify that pin U1.2 is set to Stuck Low. This means that it does NOT switch during simulation.

   c. Click **OK** to close the dialog box. Back in the schematic editor, the middle-trace driver has a 0 near its symbol, indicating visually that it is set to Stuck Low.

   The reason the driver ICs have the middle trace stuck low and outer traces switching is so that the middle signal is the victim in the simulation and the outer signals are the aggressors. The goal is to see how much crosstalk develops on the middle trace when its neighboring traces switch. However, the middle trace is not completely undriven. Instead, a driver-IC model is applied in a static state. Modeling driver ICs on victim traces is very important since low-impedance drivers reflect rather than absorb crosstalk energy.

   LineSim can simulate any mixture of victim and aggressor traces. In fact, the simulator makes no distinction between the two. Generally, aggressors are traces that are actively switching and victims are those on which the resulting crosstalk is observed. For example, if the middle trace of this simulation was also switching, it would be both an aggressor to the other traces AND their victim.

4. Define trace coupling. This step shows you how the cross section of the region is defined geometrically.

   a. Double-click on any of the transmission lines in the schematic. The Edit Transmission Line dialog box opens.

   b. Select the **Edit Coupling Regions** tab.

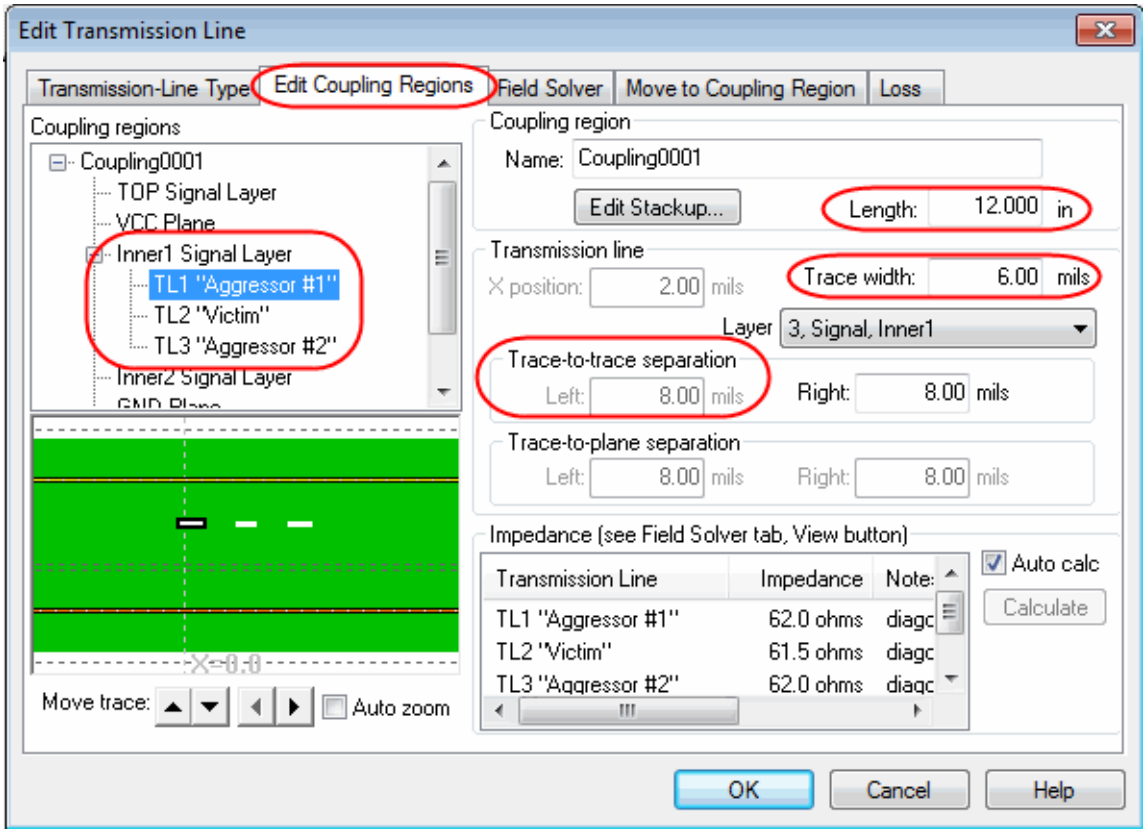
   c. The dialog box allows you to completely define the geometry of the coupling region. The Coupling Regions list shows a tree of the region stackup layers and transmission lines, and a graphical view of the current definition. The various edit boxes on the right let you change geometric parameters for the currently selected trace, or in some cases, globally for the entire region. The Impedance list in the lower right summarizes the resulting electrical characteristics. Additional electrical data is available from the field solver in the topic “Achieving a Specific Differential Impedance” on page 1164.

      This coupling region is currently defined as follows:

      - Traces are together on an inner, stripline layer
      - Traces are 6 mils wide and 8 mils apart (edge-to-edge)

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- The cross section of the region applies over a length of 12 inches



5. Run a simulation with the existing coupling to see how much crosstalk occurs. The design goal is no more than 500 mV of crosstalk voltage.

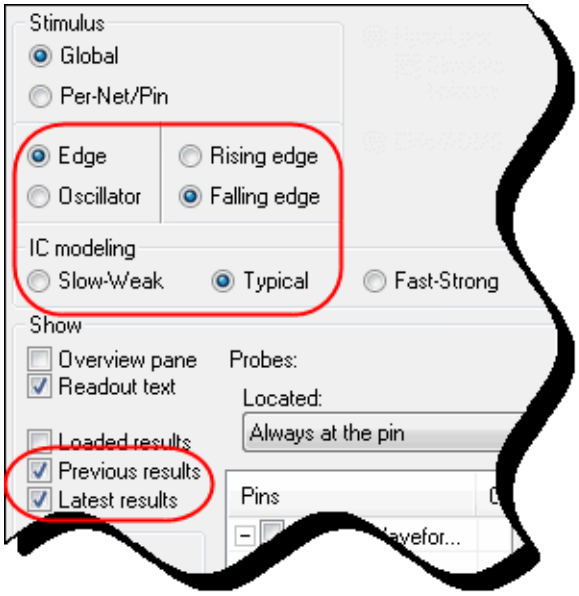
   a. Click **OK** to close the Edit Transmission Line dialog box.

   b. Choose the **Simulate SI > Run Interactive Simulation** or **Run Interactive Simulation in Oscilloscope** menu item. The Digital Oscilloscope opens.

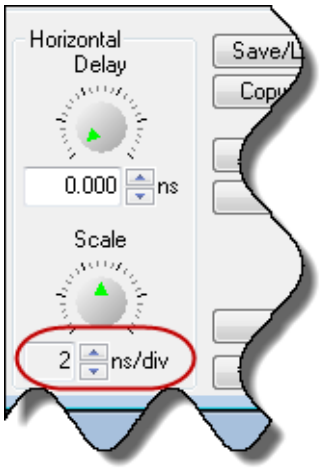
   c. Verify that the Stimulus is set to Edge, Falling Edge, and the IC Modeling is set to Typical.

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d. In the Show area, check “Latest results” and “Previous results”.



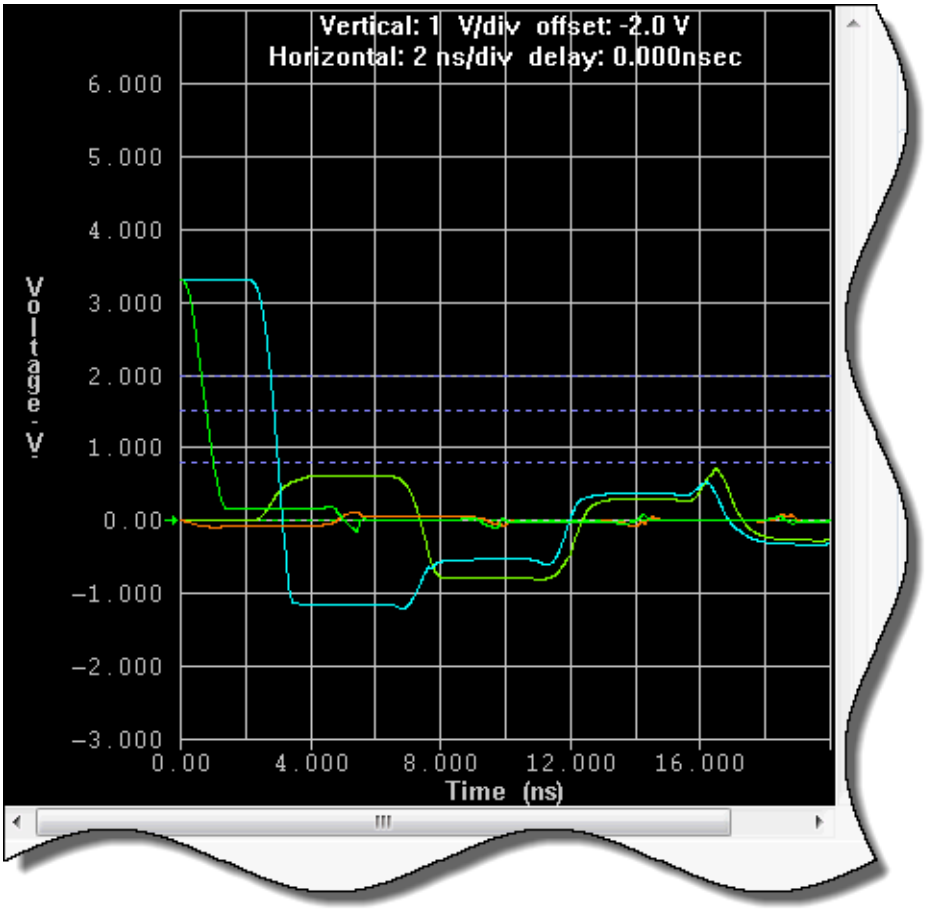
e. Change the Horizontal Scale timebase to 2 ns/div.



f. Change the Horizontal Scale to 2 ns/div.

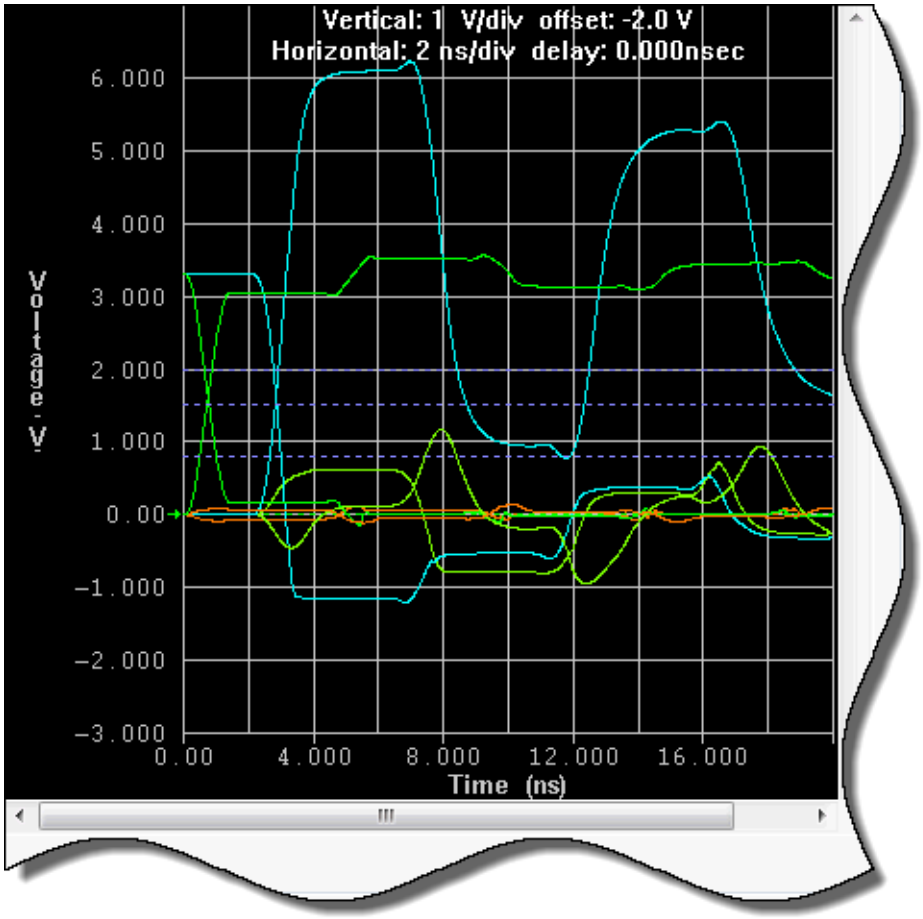
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

g. Click **Start Simulation**. The results display automatically.



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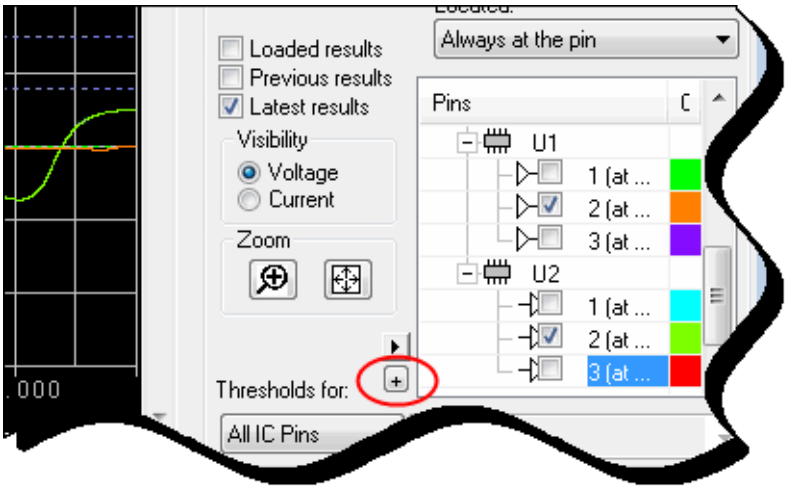
6. When the simulation is complete, change the Stimulus to Rising Edge and click **Start Simulation** to re-simulate.



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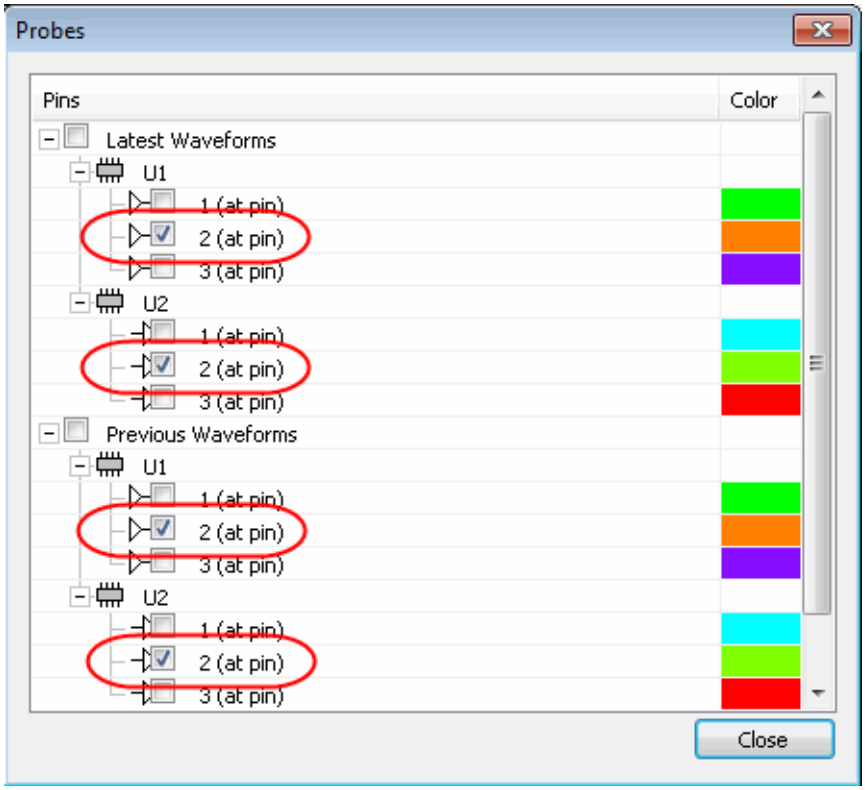
7. To make it easier to see the crosstalk on the victim net, hide the waveforms for pins on the aggressor nets:

   a. Click the plus + button to the bottom-left of the Pins spreadsheet.

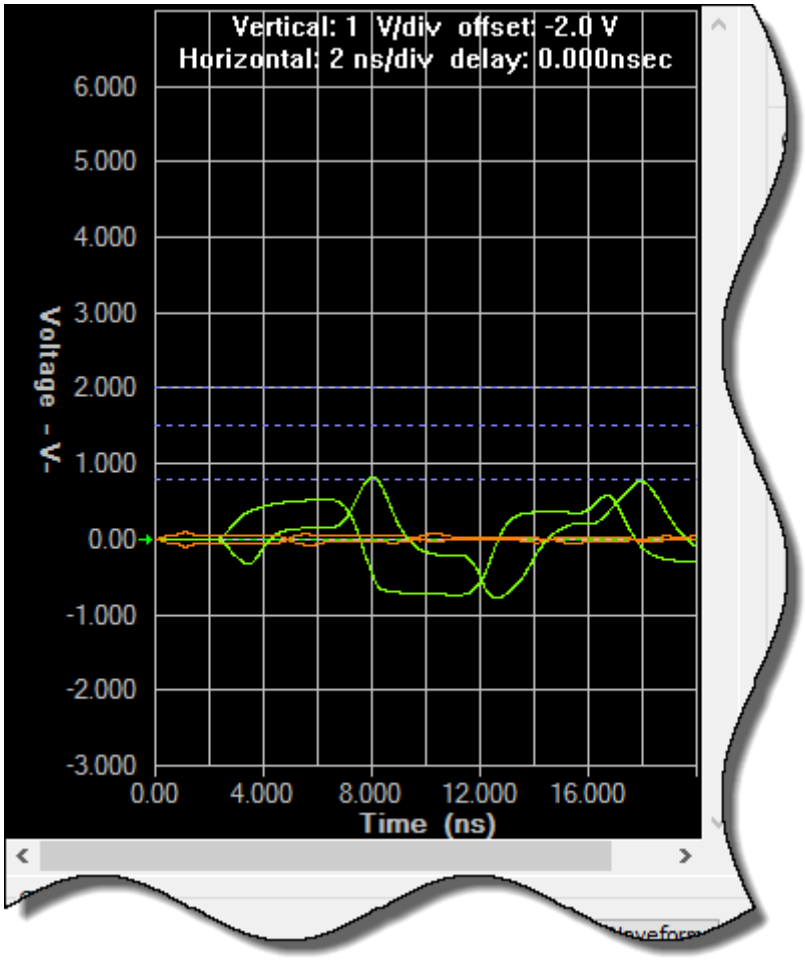


Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

b. Clear all the check boxes, except for U1.2 and U.2.2 for the Latest Waveforms and Previous Waveforms branches, then click **Close**.



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The waveforms show the crosstalk voltages on the middle, victim trace, at the receiver and driver ends, respectively. That the <orange> waveform hardly moves is no surprise, since this end of the line is held low by a low-impedance CMOS driver. But the situation is very different at the <green> receiver end. Nearly 1 V of crosstalk is present when the aggressor signals are driving high, which is well above the design criterion of 500 mV maximum crosstalk.

8. To see which waveforms correspond to which driver edge, toggle Previous results on and off. The waveform that persists is for the rising-edge simulation.

   During simulation, LineSim uses its boundary-element field solver to convert all of the geometric data entered into electromagnetic coupling parameters. This example does not look at the results generated by the field solver. The results are available in the Field Solver tab of the Edit Transmission Line dialog box. See the differential-pair example to view the output of the field solver in details.

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Note

The backward-crosstalk pulse reflecting off the driver IC of the victim line generates the 1 V problem. With a little experience using LineSim, you will be able to comfortably distinguish forward crosstalk from backward crosstalk. Backward crosstalk persists for twice the delay length of the aggressor net that creates it. To see this, compare the length in time of the pulses in the <green> waveform to the transmission-line delay reported in the schematic.

9. One way to decrease the crosstalk is to increase the separation between the traces. This step walks you through editing the coupling region to increase the trace separation.

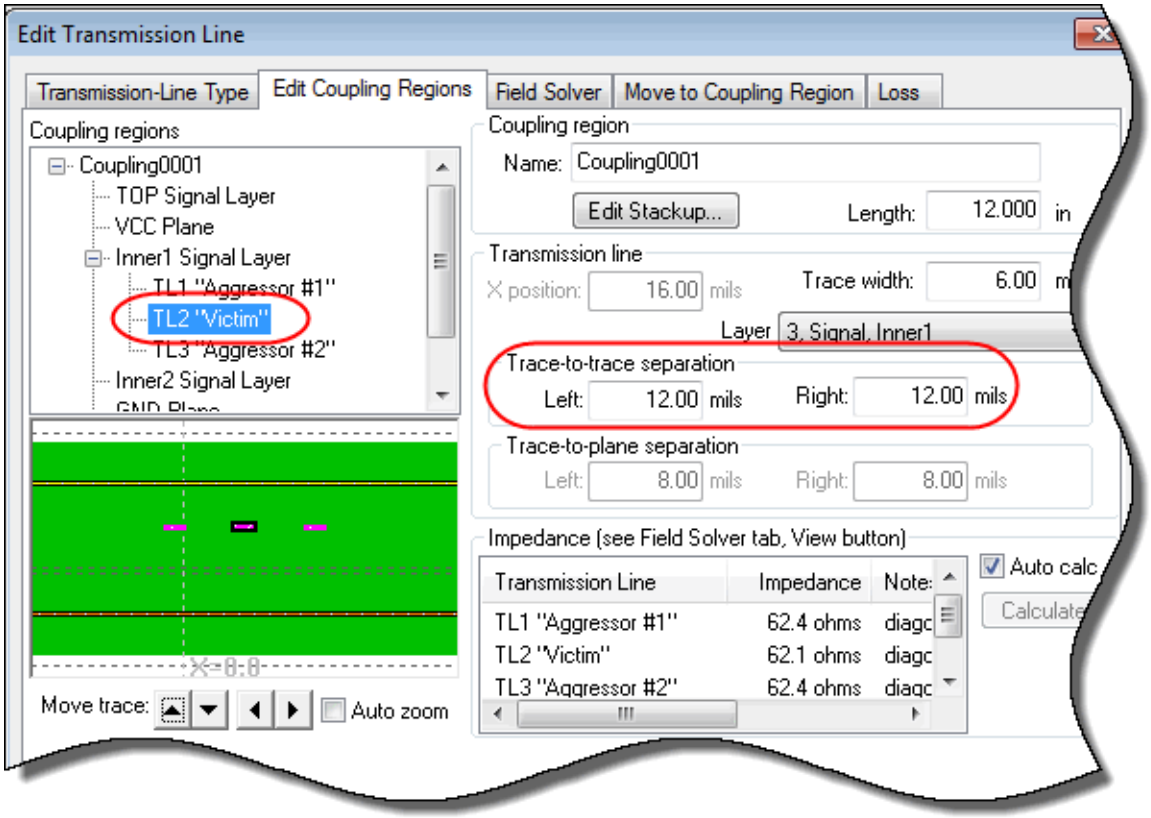
   a. Minimize the oscilloscope.

   b. Double-click one of the transmission lines in the schematic.

   c. In the Edit Transmission Line dialog box, select the **Edit Coupling Regions** tab.

   d. In the Coupling Regions list, select the middle trace by clicking once on transmission line TL2 in the tree list or clicking on the middle trace in the graph.

   e. In the “Trace-to-trace separation” area, increase the separation from the aggressor traces to 12 for both Left and Right. The separations become wider in the graphical viewer.



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f. Click **OK**.

g. Restore the minimized oscilloscope and click **Start Simulation**.

The maximum crosstalk is reduced. However, nearly 700 mV is still well above the acceptable level.

10. Decrease crosstalk by changing the stackup dielectric thickness.

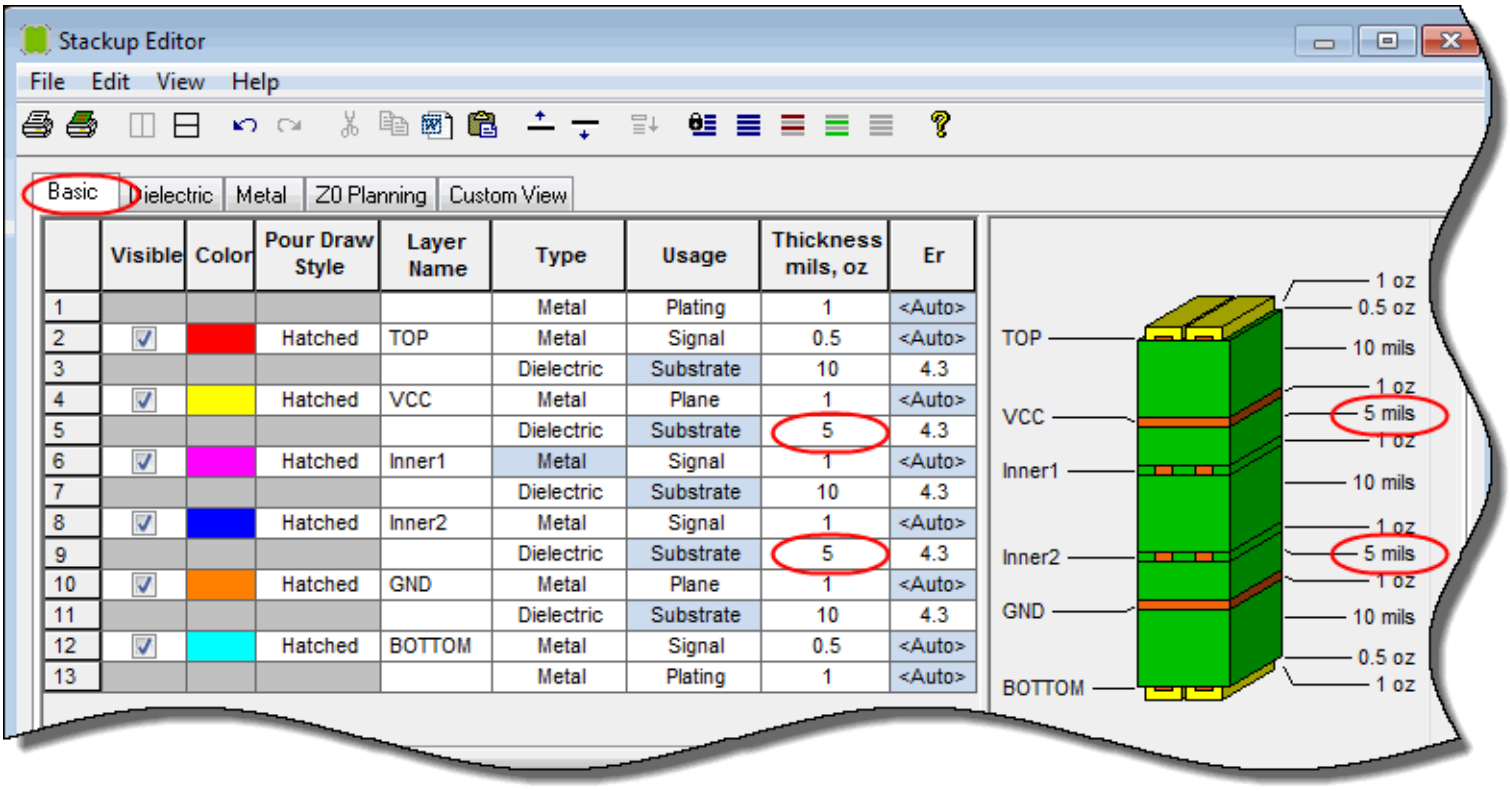
Changing trace separation is not the only way to affect crosstalk. One that is sometimes overlooked is the PCB stackup. A simple stackup change can further decrease the amount of crosstalk on the bus.

a. Choose the **Setup** > **Stackup** > **Edit** menu item. The Stackup Editor opens.

b. Change the value of the Thickness cell for the dielectric between layers VCC and Inner1 (row 5 of the spreadsheet) to 5.

c. Change the dielectric layer between layers Inner2 and GND (row 9) to 5 mils thick.

d. In the graphical stackup viewer, verify that the desired layers display as 5 mils thick.



	Visible	Color	Pour Draw Style	Layer Name	Type	Usage	Thickness mils, oz	Er
1					Metal	Plating	1	<Auto>
2	☑		Hatched	TOP	Metal	Signal	0.5	<Auto>
3					Dielectric	Substrate	10	4.3
4	☑		Hatched	VCC	Metal	Plane	1	<Auto>
5					Dielectric	Substrate	5	4.3
6	☑		Hatched	Inner1	Metal	Signal	1	<Auto>
7					Dielectric	Substrate	10	4.3
8	☑		Hatched	Inner2	Metal	Signal	1	<Auto>
9					Dielectric	Substrate	5	4.3
10	☑		Hatched	GND	Metal	Plane	1	<Auto>
11					Dielectric	Substrate	10	4.3
12	☑		Hatched	BOTTOM	Metal	Signal	0.5	<Auto>
13					Metal	Plating	1	<Auto>

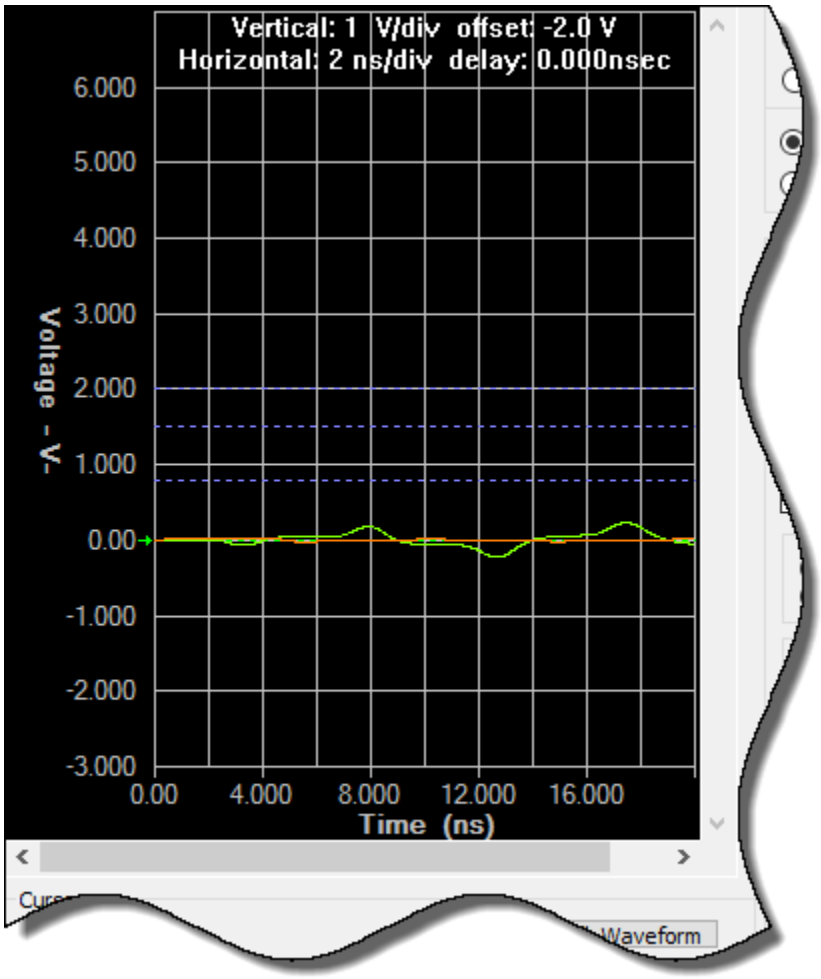
e. Click **OK**.

f. Restore the oscilloscope and click **Start Simulation**.

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Results

The maximum crosstalk at the receiver end of the victim trace is sharply reduced to less than 250 mV. This meets the design goal.



In general, crosstalk is a complex effect that is influenced by many different factors such as:

- Driver-IC technology
- Trace separation
- Trace width
- Line length and line-end termination
- PCB stackup including layer ordering and dielectric thickness/material.

  Note that crosstalk generally requires more-complex termination than single-line reflections.

LineSim lets you rapidly explore many different options to see which combinations most effectively meet your requirements. One of the most powerful uses for LineSim is the

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development of routing guidelines and constraints. For example, in this case, the routing for the bus must be set to a minimum trace separation of 12 mils and two of the dielectrics in the stackup must be 5 mils thick.

Related Topics

Modeling a Transmission Line with Coupling

Trace Coupling

Signal-Integrity Simulation of a DDR Data Path

HyperLynx LineSim allows you to develop constraints for PCB placement and routing that provide the greatest chance of producing a successful first-prototype board.

This example attaches oscilloscope probes and simulates a DDR data path using LineSim to view the signal integrity of the data bus.

The file *ddr_4dimm_data_min.ffs* represents a typical topology for a DDR data path implemented in the LineSim schematic editor. Although the schematic is drawn so that the entire design fits on a single page, it can easily be stretched out to fit on multiple pages. The DDR bus can run at various speeds. This tutorial attempts to make the design work at 266 Mbps (133 MHz). The design incorporates the minimum interconnect lengths allowed by the JEDEC specification with trace widths and stackups designed for the desired impedances.

The design incorporates the elements in Table A-6:

Table A-6. Circuit Description - DDR Data Path

Reference	Description
1	A DDR controller, represented by an IBIS model for a Xilinx Virtex-4 SSTL2 driver.
2	Several transmission lines representing extra package parasitics (recommended by Xilinx in their IBIS model), breakout routing, and routing to the first DIMM module; plus a series termination resistor just after the breakout.
The first of four DIMM modules, consisting of the following elements:	
3	A transmission line representing the DIMM connector.
4	Transmission lines representing routing to a series resistor on the DIMM, and then more routing.
5	A T in the routing, with transmission-line branches going to two SDRAM data inputs, each represented by a Micron Technology MT46V16M8A0 IBIS model.
6	A transmission line representing more routing, between DIMM 1 and DIMM 2 connectors.

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Table A-6. Circuit Description - DDR Data Path (cont.)

Reference	Description
7	Then DIMM2 (a copy of DIMM 1 structure); DIMM2-DIMM3 routing; DIMM3; DIMM3-DIMM4 routing; and DIMM4
8	A parallel pull-up-resistor terminator, to 1.25 V.

Prerequisites

The Lossy Lines license is required to model lossy transmission lines.

Procedure

1. Open the schematic. Choose the **File > Open Schematic** menu item, then double-click *ddr_4dimm_data_min.ffs*.

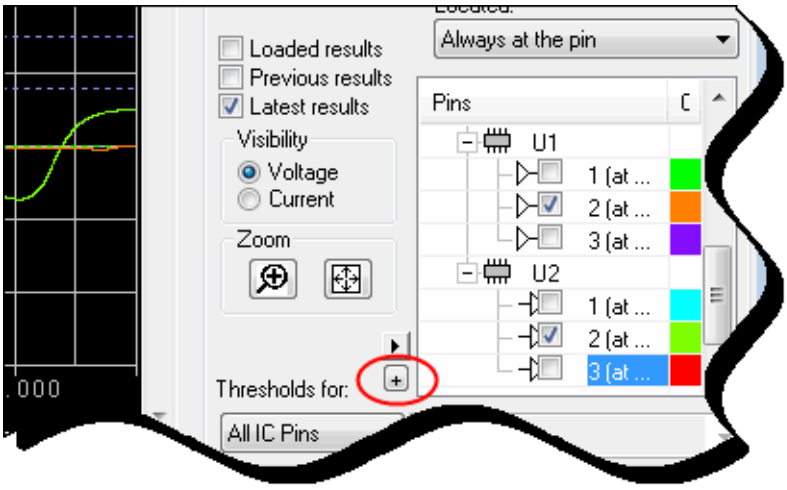
   The schematic displays in the free form editor, which functions as a standard schematic capture tool: you choose symbols from a palette, and wire them together.

2. Enable lossy modeling. From the **Setup > Enable Lossy Simulation** menu item, verify that it is enabled.

3. Verify oscilloscope probes:

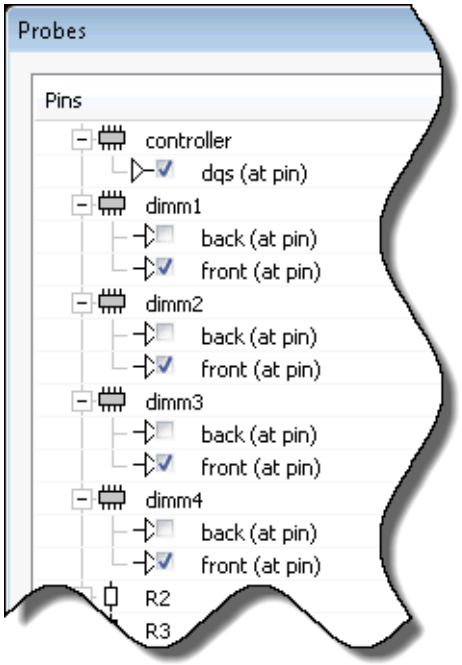
   a. Choose the **Simulate SI > Run Interactive Simulation** or **Run Interactive Simulation in Oscilloscope** menu item.

   b. Click the plus + button to the left of the Pins tree.



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

c. In the Probes dialog box, verify that controller.dqs has a probe attached. Repeat for pins dimm1.front, dimm2.front, dimm3.front, and dimm4.front.



In the schematic, a colored probe displays at each selected pin.

d. In the Stimulus area, click Oscillator. Verify the frequency is 133 MHz.

e. In the IC modeling area, verify that Typical is selected.

f. Change the Horizontal Scale to 2 ns/div.

g. Change the Vertical Scale to 500 mV/div.

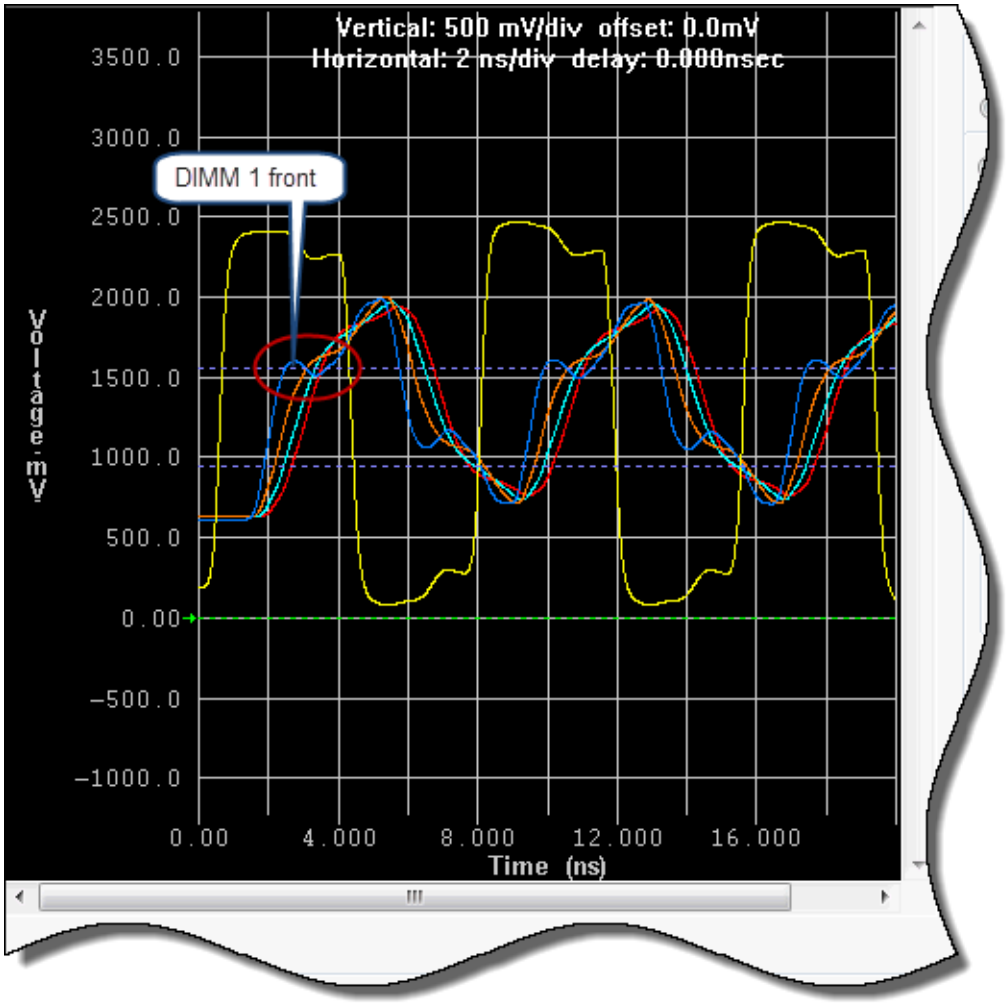
h. In the “Threshold for” list, select pin dimm1.back. The Vil and Vih threshold values of the receivers are plotted in the oscilloscope as dashed blue lines.

i. Click **Start Simulation**.

The resulting waveforms show a problem at the left-most of the DIMMs. A noticeable anti-reflection is causing the DIMM 1 received signal to dip back slightly above the high

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threshold. This means that a double-clock on DIMM 1 is a possibility which can cause a failure in the field.



4. One way to improve the waveforms on the data path is by changing the termination values, especially the series resistor at the driver or pull-up at the end of the bus, which are not on the DIMM modules and therefore under our control. This step changes the pull-up terminator value.

   a. Minimize the oscilloscope.

   b. In the schematic, right-click the pull-up resistor at the far right edge and choose the **Assign Model or Edit Value**. The Assign Models dialog box opens.

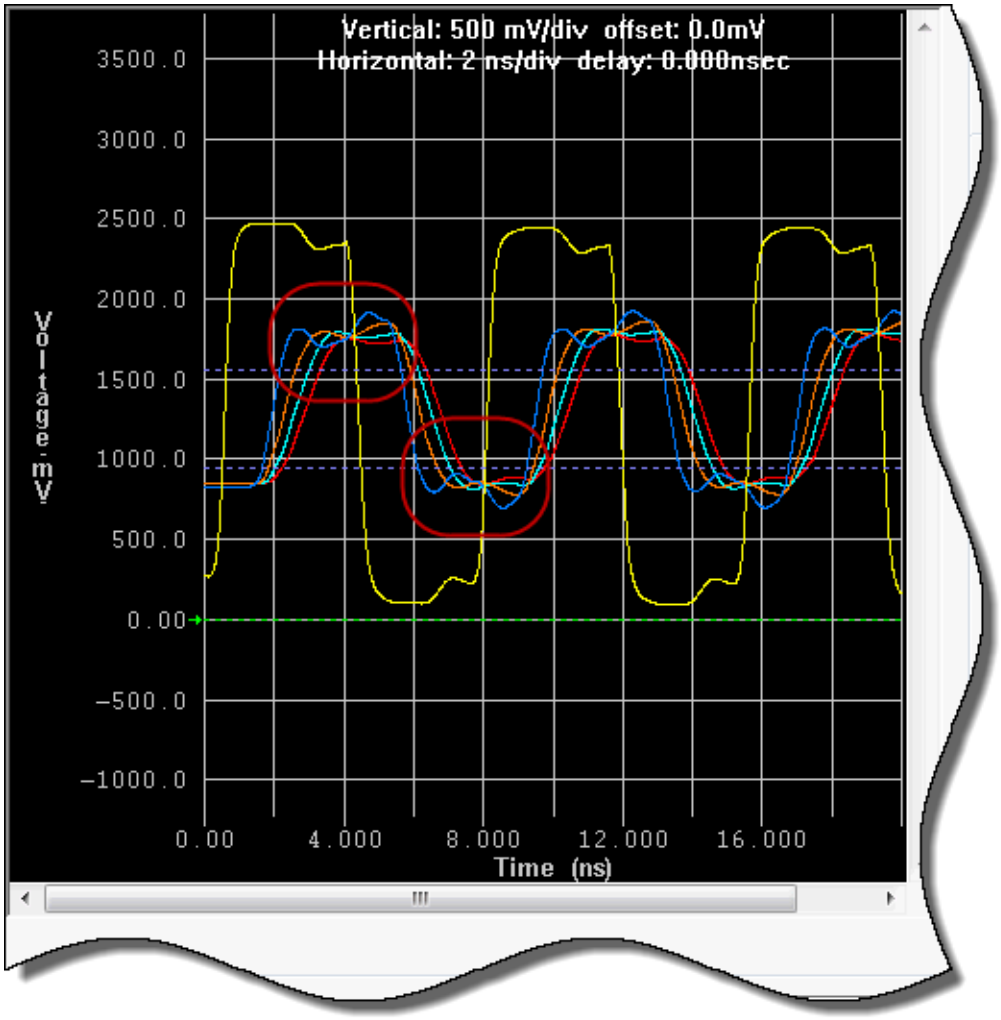
   c. Change the resistance to 22 ohms and click **OK**.

   d. Restore the oscilloscope. Click **Erase** to clear any existing waveforms.

   e. Click **Start Simulation** to re-simulate.

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The rising edge waveform is improved. The ringback at DIMM 1 is well above the Vih threshold. However, the signal is still marginal on the falling edge versus Vil.



To improve the falling edge, the schematic currently uses the minimum possible interconnect lengths per the JEDEC specification. Sometimes, increasing the routing length can help.

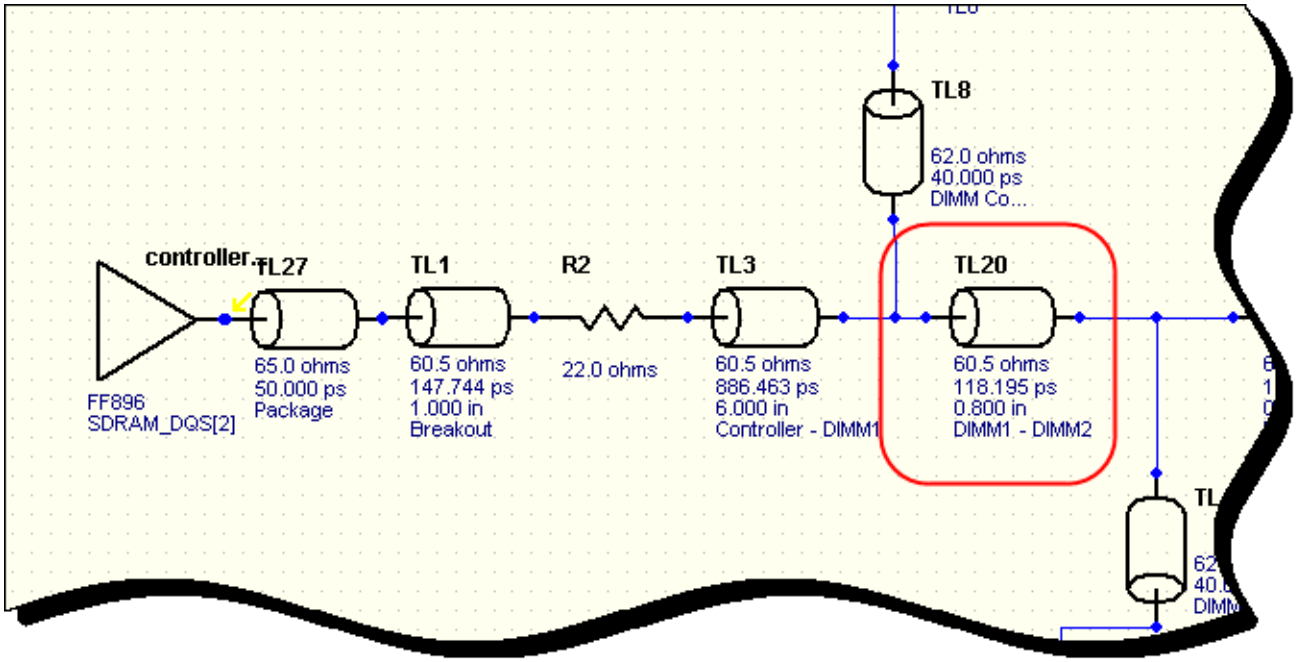
5. Increase the routing length:

   a. Minimize the oscilloscope.

   b. If the component text is too small to read, choose the **View > Zoom Area In** menu item, then draw a box that encloses the driver symbol on the left and the four symbols to the right of the driver.

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c. Locate TL20, labeled DIMM1 – DIMM2.



d. Double-click TL20. The Edit Transmission Line dialog box opens.

e. Click the **Values** tab.

f. Change the Length to 1.2 and click **OK**.

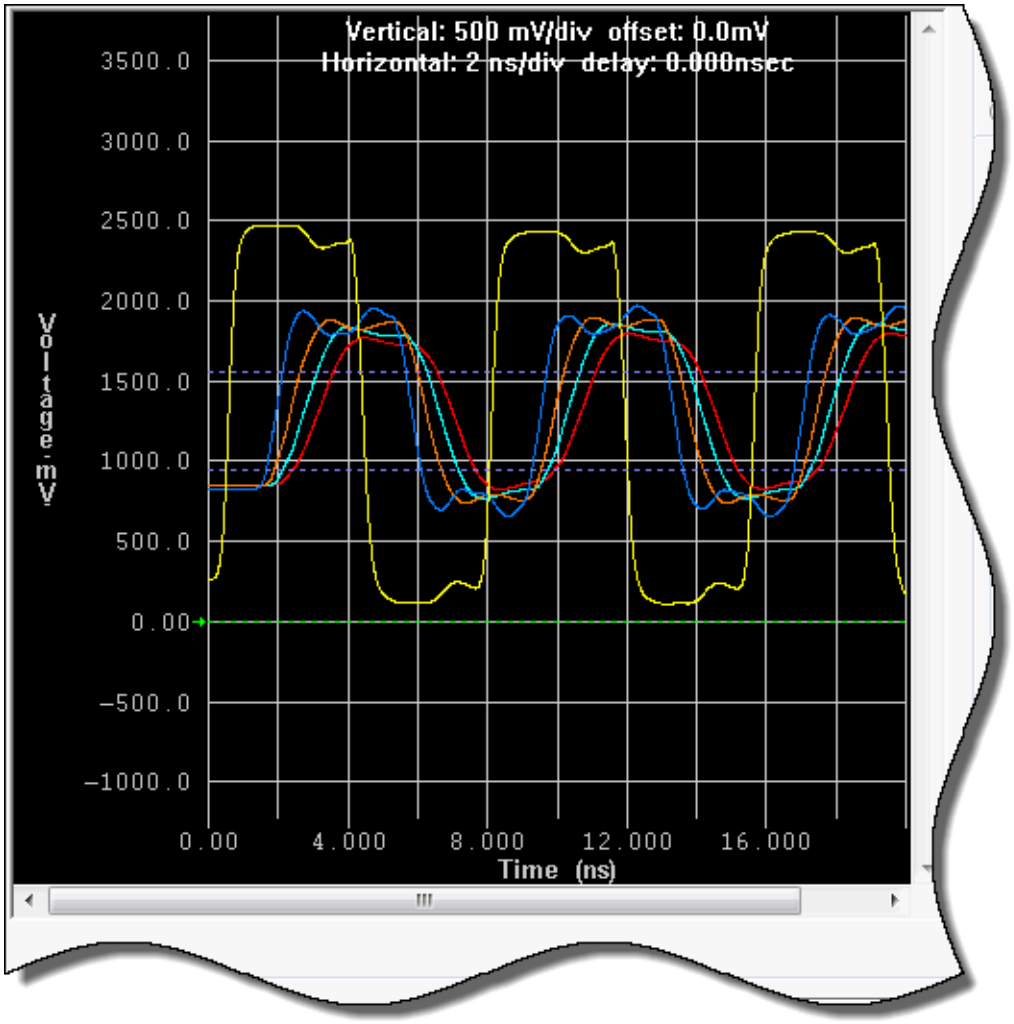
g. Repeat steps d through f for transmission lines TL26 and TL53. They are located to the right of TL20.

h. Restore the oscilloscope, click **Erase** to clear the waveforms.

i. Click **Start Simulation** to re-simulate.

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Success! The signal quality on all receivers on the DDR data bus is now clean enough to work reliably.



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Tutorial Reference Information

There is background information related to the technology covered in the board design and schematic design tutorials.

Topic	Description
MultiBoard Simulation of Signals Spanning Multiple Boards	Many designs involve multiple, interconnected PCBs, such as a motherboard with one or more memory modules plugged in, or a system consisting of several boards joined by connectors and cables. The BoardSim MultiBoard option adds the ability to load multiple boards simultaneously, virtually interconnect them, and simulate them together as a system.
Electrical Versus Geometric Thresholds	Selecting aggressors for crosstalk simulation can include the nearest-neighbor, geometric zone, or electrical estimation. Because crosstalk simulation is CPU-intensive, the simulation time is affected by the number of selected aggressor nets. Therefore, selecting only those nets that are significantly coupled to the victim net maximizes simulation efficiency.
Signal-Integrity Simulation	Signal-integrity simulation is examining the quality of the digital signals on a printed circuit board.
Crosstalk Simulation	Crosstalk simulation is a particular category of signal-integrity simulation that looks specifically at unwanted noise generated between signals.
GHz Simulation	GHz simulation is a general term for the collection of special techniques used to simulate gigabit-per-second (Gbps) SERDES-based designs.
Eye Diagrams Introduction	Very-high-speed SERDES-style designs are usually examined in the time domain using an eye diagram. An eye diagram superimposes large numbers of bit transitions to build a view of a data stream in which jitter and eye opening can be readily viewed.
Multi-Bit Stimulus Introduction	BoardSim makes the generation of eye diagrams fairly easy. Set-up activities, such as defining a stimulus pattern, are much easier in HyperLynx than directly in SPICE. Additionally, when simulations are performed using IBIS models, eye diagrams are created quickly.
BoardSim Crosstalk and Differential-Signal Simulation	The BoardSim Crosstalk option adds the ability to perform crosstalk simulation of a board after layout.
Post-Layout Simulation: BoardSim and Batch Mode	Using the data from your actual routed PCB layout, BoardSim moves the HyperLynx simulation into the post-layout phase of your design cycle.

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Topic	Description
Simulating Multiple Boards	The BoardSim MultiBoard option adds the ability to load multiple boards simultaneously, interconnect them, and simulate them together as a system.
Adding IC Models to Your Existing Libraries	You can add several types of IC models to your existing libraries.
MultiBoard Simulation with EBD/EMD Models	In HyperLynx, the mapping of an EBD/EMD model to a reference designator occurs in the *.REF* or *.QPL* IC Automapping files, just as with any other IBIS model.
Translating a Board into a BoardSim Format	Before using BoardSim, you must translate your PCB layout into a format that BoardSim can read.
Multi-Bit Stimulus	Some designers of SERDES-based designs today use standalone SPICE netlists to create eye diagrams. While possible and sometimes even necessary because a certain IC model is available only in SPICE format, using raw SPICE for eyes is usually cumbersome and time-consuming. SPICE simulations often run very slowly, and setting up for simulation especially generating stimulus patterns, is awkward. LineSim, by contrast, makes the generation of eye diagrams fairly easy.
Crosstalk Simulation - LineSim Tutorial	Crosstalk simulation is a particular category of signal-integrity simulation that looks specifically at unwanted noise generated between signals.
Modeling a Transmission Line with Coupling	LineSim can model transmission lines as part of a coupling region, which is a region containing multiple lines coupled together in the same PCB cross section.
LineSim Crosstalk Simulation	LineSim allows you to quickly construct a schematic and simulate it to see the resulting waveforms. The LineSim crosstalk-simulation option adds line-to-line coupling into your schematics.
Using LineSim for Differential-Signal Simulation	LineSim coupled-line simulation features are valuable in the design of differential signals because the same line-to-line coupling that causes crosstalk on unrelated signals also creates differential impedance and other electrical characteristics important in differential signaling. Differential pairs are common in very-high-speed design and are used widely in gigabit-per-second, SERDES-based designs.

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Topic	Description
Touchstone (S-Parameter) Modeling	Due to high frequency designs, especially in SERDES technology, SPICE models are not the only type of model appearing in signal-integrity simulations. Increasingly, models for passive interconnect structures such as connectors and IC packages are provided in Touchstone format. Touchstone models are used in RF/microwave engineering to accurately characterize ultra-high-speed devices and structures.
Impedance Planning	In high-speed design, it is often necessary to plan stackups and trace widths in such a way that traces have certain desired characteristic impedances. Some standard buses, for example, mandate that trace impedances be within a certain range, such as 60 - 100 ohms.
Modeling a Transmission Line	When you activate a transmission line in a schematic, you can model it in any of several ways.
IC Modeling With HyperLynx	HyperLynx provides a diverse set of modeling options to accommodate your IC-modeling needs.
Why IC Models are Important	A number of factors affect how a trace on a PCB behaves from a signal-integrity standpoint: the geometric properties of the trace itself; how it connects to other traces; how the board layers are stacked up; what materials the board is manufactured from; and so forth. A signal-integrity/ crosstalk/GHz-level simulator must accurately model all of these parameters.
Trace Coupling	The LineSim crosstalk option lets you add coupling information to any LineSim schematic.
LineSim GHz Simulation	SERDES-based design, a technology that emphasizes very-high-speed data streams traveling on narrow, serialized data paths uses gigabit-per-second signaling. Gigabit-per-second signaling requires lossy simulation and advanced via modeling, and sometimes even the use of SPICE-based driver models.
Integrated SPICE Simulations	Vendors of some very-high-speed driver and receiver ICs make models available for their components only in SPICE format because some devices have subtle behavior which is difficult to model accurately in the IBIS behavioral format. In almost all cases, these SPICE models are provided in a proprietary format: either HSPICE or Eldo/ADMS. Additionally, some models are encrypted, which hides all model details from the user.

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MultiBoard Simulation of Signals Spanning Multiple Boards

Many designs involve multiple, interconnected PCBs, such as a motherboard with one or more memory modules plugged in, or a system consisting of several boards joined by connectors and cables. The BoardSim MultiBoard option adds the ability to load multiple boards simultaneously, virtually interconnect them, and simulate them together as a system.

Each board can be in the form of a *.HYP* file, a *.CCE* file, or a type of IBIS board model called *.EBD/EMD*, electrical board/module description. If the system under simulation consists entirely of your own PCBs, you will likely load all of your boards into BoardSim as *.HYP* or *.CCE* files. If some of the boards come from a third party (for example, memory modules), those 3rd-party boards might be provided in EBD/EMD format.

Electrical Versus Geometric Thresholds

Selecting aggressors for crosstalk simulation can include the nearest-neighbor, geometric zone, or electrical estimation. Because crosstalk simulation is CPU-intensive, the simulation time is affected by the number of selected aggressor nets. Therefore, selecting only those nets that are significantly coupled to the victim net maximizes simulation efficiency.

Often, the greatest amount of crosstalk on a given section of a victim net is due to the nearest trace on either side. However, a fast driver can cause a more distant net to be the strongest aggressor. Using a traditional geometric coupling window or zone to identify aggressors ignores faster drivers on distant nets, while nets in closer proximity with slow drivers are included needlessly. This scenario can lead to a significant underestimation of the crosstalk on the victim net.

If you chose a different approach and increased the width of the coupling zone, you might catch further-away aggressor nets, but in many cases you would also include many nets which are not significant aggressors and whose presence would simply slow your simulations.

By default, BoardSim Crosstalk uses electrical thresholds. This approach has several major benefits. First, more distant nets with fast drivers are correctly found by the aggressor-finding algorithm. Second, nearby nets with slower drivers are included only if they contribute crosstalk above the threshold you specify. The result is a minimum but correct set of nets to simulate, which can cut simulation time significantly, and increase accuracy. Finally, electrical thresholds make crosstalk easier to visualize by presenting it as mV of noise rather than in geometric limits.

Signal-Integrity Simulation

Signal-integrity simulation is examining the quality of the digital signals on a printed circuit board.

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As driver ICs switch faster and faster, more and more boards suffer from signal degradations such as overshoot and undershoot, ringing, non-monotonicity, crosstalk, and excessive settling delays. When these become serious enough, the logic on a board can begin to fail.

Signal-integrity concerns are critical for very-high-speed SERDES-based designs using serializer/deserializer technology. Special methods such as eye diagrams are used to judge whether signal quality is sufficient for a data stream to be recovered at the receiver IC. Simulation must include such advanced effects as lossy transmission lines and complex via modeling.

Why do signal-integrity effects occur, and why so much more today than before? The answer lies in the transmission-line behavior of the metal traces on a PCB. When a lower-frequency digital signal (that is, a signal that switches relatively slowly) travels along a board trace, the trace itself is almost invisible from a circuit standpoint. But when a higher-frequency signal (that is, a signal that switches more quickly) travels along the same trace, the trace exhibits circuit characteristics that distort and degrade the signal. The problems get worse at high frequencies; at gigabit speeds, signals are sometimes attenuated by trace loss by more than 50% before arriving at receivers.

The trend behind these problems is the driver IC switching rate. The reason that fewer designs exhibited transmission-line effects in the past is that many of the ICs switched more slowly than the ICs common today. For example, consider a 6-inch, 6-mil-wide trace on the outer layer of a board, 5 mils above a ground plane. If driven with an older logic family with a switching time of 3 ns, there are only a few visible transmission-line effects. But when the same trace is driven by a modern CMOS logic IC (switching time = 750 ps), the signal at the end of the trace overshoots by more than a volt, and rings for more than 20 ns.

A rule of thumb to use is if the switching time of a driver IC on a trace is shorter in nanoseconds than the length of the trace in inches, the signal will suffer from transmission-line effects. This means that a driver IC with a 1 ns switching time will create transmission-line effects on any trace 1 inch or longer.

A number of factors affect how a trace on a PCB behaves from a signal-integrity standpoint:

- The geometric properties of the trace itself
- How it connects to other traces
- How the board layers are stacked up
- The materials used to manufacture the board

A signal-integrity/crosstalk/GHz-level simulator must accurately model all of these parameters.

Another important modeling factor involves the ICs on the trace, especially the driver IC. It is typically the driver IC that causes transmission-line problems, because of fast rise and fall times. Receiver ICs also play a role, especially as a result of their input capacitance and diode-clamping effects.

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For accurate signal-integrity of driver ICs, each of the following must be considered for simulation:

- Switching time (rising and falling edges)
- Switching impedance (rising and falling edges)
- Switching shape (rising and falling edges)
- Clamp diodes (high and low side)
- Output capacitance

For receiver ICs, the following are important: Input capacitance, clamp diodes (high and low side), and input resistance.

Note

Gigabit-per-second, SERDES-style designs, do not use approximate models because the very high speeds and accuracy required for this type of design. In fact, many times vendor-supplied SPICE or IBIS-AMI models are required. See “Setting Up a SPICE Simulation” on page 1172 or “Analyzing Multiple SERDES Channels Using the SERDES Wizard - IBIS AMI” on page 225.

Crosstalk Simulation

Crosstalk simulation is a particular category of signal-integrity simulation that looks specifically at unwanted noise generated between signals.

Crosstalk occurs when two or more nets on a PCB are coupled to each other. Such coupling can arise any time two nets are routed next to each other for any significant length. When a signal is driven on one of the lines, the electric and magnetic fields it generates cause an unexpected signal to also appear on the nearby line.

Crosstalk is a particularly hard phenomenon to anticipate and control without simulation because there is almost no way of intuitively knowing how much crosstalk voltage and current will develop due to a given coupling. Many complex factors combine to create an unwanted crosstalk signal: The length over which the traces are coupled, the distance between the traces, their positions in the PCB stackup, what driver ICs are used on both the aggressor and the victim lines, whether or not the lines are terminated, and so forth. In order to resolve all of these factors and produce an accurate simulation, the HyperLynx software uses a fast, built-in field solver that can calculate the electromagnetic properties that govern the line-to-line coupling.

GHz Simulation

GHz simulation is a general term for the collection of special techniques used to simulate gigabit-per-second (Gbps) SERDES-based designs.

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This type of signaling has appeared in the past few years as a solution to the problem of how to push data rates into the multi-Gbps range, where classic parallel, synchronous bus techniques become nearly impossible to manage. SERDES data channels are serial (hence the need for SERializers and DESerializers), extremely fast, and travel over interconnect without explicit clocks. Sophisticated receiver ICs use techniques such as equalization to recover these signals after they are seriously degraded by propagation across a PCB or down a cable.

GHz-level simulation must account for lossy effects and the electrical complexities of vias. Loss refers to the phenomenon in which PCB-trace resistance and the heating of dielectric materials (like FR-4) cause signals to lose amplitude (that is, attenuate) and suffer shape distortion (disperse). These effects are hardly noticed at the frequencies present in a 2-ns driver edge, but for the frequencies that make up a 200-ps edge, they can be quite severe. Accurately analyzing loss is difficult because lossy effects are frequency-dependent, and digital signals contain a wide range of frequencies. The situation is similar with vias: to a 2-ns signal edge, a via is hardly noticeable, but to a 200-ps edge a via has significant electrical complexity. To accurately simulate GHz-level designs, complex via modeling is needed.

Sub-GHz designs are typically characterized with simple waveforms and delay values, but GHz-level designs require special techniques like eye diagrams and jitter measurement. An eye diagram takes the results of a simulation driven by a long, multi-cycle bit sequence, superimposes each bit period over the top of all others, and presents a waveform that looks something like a human eye. How open the middle of the eye is at the receiver IC is a key factor in judging how likely the receiver is to recover each bit of arriving data. The tendency of the bits in a complex stream to wiggle around each other (in voltage and time) is called jitter. A data channel with too much jitter will have a high bit error rate and be unreliable.

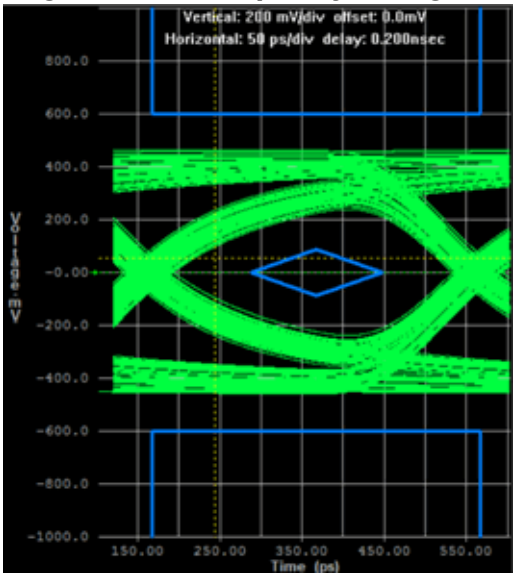
Eye Diagrams Introduction

Very-high-speed SERDES-style designs are usually examined in the time domain using an eye diagram. An eye diagram superimposes large numbers of bit transitions to build a view of a data stream in which jitter and eye opening can be readily viewed.

Figure A-1 illustrates a sample eye diagram.

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Figure A-1. Sample Eye Diagram



Many modern oscilloscopes can run either in traditional, single-edge mode or in eye-diagram mode. Likewise, the oscilloscope can run in either standard or eye-diagram mode.

Generating an eye diagram with a simulation tool is more difficult than generating one in the lab with real hardware. In the lab, it takes only a brief amount of time to capture hundreds of millions of bit cycles from a data stream. But in a software-based simulator, especially if advanced IC modeling is required, it can take several minutes to generate a thousand or even a few hundred cycles. Additionally, test equipment is readily available to generate statistically useful bit sequences in the lab. However, when using software, the user must create the stimulus to drive the generation of an eye diagram.

Note that eye diagrams can only be constructed by driving a sequence of bits down a trace. This means that in order to generate an eye diagram, you must define multi-bit stimulus. Thus, these two features – eye diagrams and multi-bit driving – are tightly linked.

Eyes are judged by the extent of their opening. Each SERDES technology typically specifies a minimum allowed opening, which is translated into an eye mask that visually defines a keep-out region. If a given eye penetrates this region, the eye fails to meet the specification minimum requirements for signal quality. Even if the eye is open, the mask gives a quick visual impression of how much margin is in the eye.

Similar to eye patterns, LineSim includes built-in masks (such as USB 2.0) and allows you to define your own.

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Multi-Bit Stimulus Introduction

BoardSim makes the generation of eye diagrams fairly easy. Set-up activities, such as defining a stimulus pattern, are much easier in HyperLynx than directly in SPICE. Additionally, when simulations are performed using IBIS models, eye diagrams are created quickly.

Some designers of SERDES-based designs today use standalone SPICE netlists to create eye diagrams. While possible, and sometimes even necessary because a certain IC model is available only in SPICE format, using raw SPICE for eye diagrams is usually cumbersome and time-consuming. SPICE simulations often run very slowly, and setting up for simulation, especially, generating stimulus patterns, is awkward.

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BoardSim Crosstalk and Differential-Signal Simulation

The BoardSim Crosstalk option adds the ability to perform crosstalk simulation of a board after layout.

BoardSim offers batch and interactive post-layout signal-integrity simulation. As with other signal-integrity problems, crosstalk can negatively impact your final design and manifest as false clocking, intermittent data errors, or other difficult-to-find and potentially serious problems. It can also be difficult to know where crosstalk is likely to occur, and eliminating it can be even trickier than fixing single-trace signal-integrity problems.

Topic	Description
How BoardSim Crosstalk Simulation Works	BoardSim enables you to simulate in both batch and interactive modes. Batch-mode simulation includes detailed simulation (with timing and crosstalk data saved into a report file), as well as a Quick Analysis feature that can rapidly scan your entire PCB.
BoardSim Crosstalk for Differential-Signal Simulation	The BoardSim coupled-line simulation features are also valuable in the design of differential signals, since the same line-to-line coupling that causes unwanted crosstalk on unrelated signals also generates the differential impedance and other electrical characteristics important in differential signaling.
Automatically Finding Aggressor Nets	An important feature of BoardSim Crosstalk is that it automatically identifies which other nets are coupled strongly enough to the selected victim net to be aggressors.

How BoardSim Crosstalk Simulation Works

BoardSim enables you to simulate in both batch and interactive modes. Batch-mode simulation includes detailed simulation (with timing and crosstalk data saved into a report file), as well as a Quick Analysis feature that can rapidly scan your entire PCB.

An aspect of Quick Analysis is a crosstalk feature that can provide a list – sorted from most to least – of the amount of crosstalk that could potentially appear on each net of your board. This list is particularly powerful because it helps you determine very quickly which nets on your board are likely to have crosstalk trouble, and merit further investigation.

BoardSim Crosstalk also offers a unique way of automatically determining which nets are coupled to any net that is selected for simulation (interactively or in batch mode). Rather than forcing you to specify a geometric zone around each net in which to find aggressor nets, BoardSim Crosstalk allows you simply specify an electrical coupling threshold. For example, you can say, I want to include all nets in simulation that could generate 100 mV or more of

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crosstalk on my victim nets, and BoardSim will automatically find them for you. This is a much easier, less-error-prone, more-powerful way of finding aggressor nets than by crude geometric methods.

BoardSim crosstalk features allow you to:

- Quickly predict which nets are likely to suffer the most crosstalk, and have BoardSim determine automatically which nets are the likely aggressors
- Use electrical rather than geometric thresholds, for more-accurate and faster simulations. Geometric thresholds are available, too, in case you prefer them. See "Electrical Versus Geometric Thresholds" on page 1204.
- Simulate a large number of nets in batch mode, with the numerical results of each net (timing, overshoot, crosstalk) saved into a report file
- Simulate interactively to see in oscilloscope waveforms the exact amplitude of crosstalk on a victim net
- See the effects on crosstalk results of changing parameters like stackup layer, dielectric thickness, driver-IC slew rate, driver impedance, line termination, and so forth
- Confidently design high-speed buses and other PCB structures that meet tight timing and low-crosstalk-noise requirements
- Select termination strategies that greatly reduce or eliminate the crosstalk seen at receiver ICs

BoardSim Crosstalk for Differential-Signal Simulation

The BoardSim coupled-line simulation features are also valuable in the design of differential signals, since the same line-to-line coupling that causes unwanted crosstalk on unrelated signals also generates the differential impedance and other electrical characteristics important in differential signaling.

Specifically, you can use BoardSim to:

- Determine the differential impedance of trace pairs on your routed board, and observe the effects of stackup layer, dielectric thickness, and so forth
- Accurately simulate differential signals, taking into account the coupling between traces and the presence of nearby aggressor and reference (power/ground) traces
- Simulate both differential- and common-mode propagation, or any mix of the two
- Easily design terminations that work for both the differential- and common-mode components of your signals

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Automatically Finding Aggressor Nets

An important feature of BoardSim Crosstalk is that it automatically identifies which other nets are coupled strongly enough to the selected victim net to be aggressors.

Guessing is eliminated and, as with other crosstalk-simulation tools, you do not have to specify a geometric zone which you hope is wide enough to include all of the important aggressor nets. For more information about this capability, refer to “Electrical Versus Geometric Thresholds” on page 1204.

Post-Layout Simulation: BoardSim and Batch Mode

Using the data from your actual routed PCB layout, BoardSim moves the HyperLynx simulation into the post-layout phase of your design cycle.

Typically, when using BoardSim after placement and routing, the simulation is based on the actual routing details of your board. However, you can also simulate a board as soon as placement is complete and before routing. BoardSim creates routing using Manhattan routes. Alternatively, you can use BoardSim when your board is placed and only partially routed.

BoardSim reads the data representing a routed PCB and performs signal-integrity and crosstalk simulation on the actual layout. In BoardSim, signal-integrity and crosstalk results appear either as signal waveforms in an oscilloscope when using interactive mode, or in a multi-net simulation report when using batch mode. Eye diagrams for high-speed serial designs are produced in the BoardSim oscilloscope.

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Simulating Multiple Boards

The BoardSim MultiBoard option adds the ability to load multiple boards simultaneously, interconnect them, and simulate them together as a system.

Each board can be described by a.*HYP*, *.CCE* , or *ODB++* design file, or a type of IBIS board model called EBD/EMD (Electrical Board/Module Description). Some third party boards, for example, memory modules, may be provided in EBD/EMD format.

The EBD format is part of the IBIS specification. IBIS is best-known for modeling IC buffers. However, the EBD/EMD format allows the modeling of interconnect, and is used to represent PCBs, complex IC packages, and so forth. The EMD format can model coupling and also reference s-parameter and SPICE models.

The main difference between a board design file and an EBD/EMD model is that a board design file is a physical representation of the PCB: it contains details such as trace routing and stackup, which can be viewed. EBD/EMD models, on the other hand, are an electrical representation of the PCB: the interconnection is represented as transmission lines, with previously calculated inductance and capacitance, or impedance and delay. An *.EBD/EMD* file cannot be viewed because there is no physical information to display. Also, an *.EBD* file cannot represent coupling. However, either type of file can include the effects of plug-in modules and boards in a multiple-board simulation.

Topic	Description
Other Simulation Types and MultiBoard Designs	The primary differences between simulating with one board versus multiple boards using the BoardSim MultiBoard option include board-to-board connectors and simulating the complete net that spans more than one board. In addition to simulating interactively, you can run a Board Wizard batch simulation on multiple boards, enabling delay calculations for a complete multiple-board system.
Simulating with EBD/EMD Models	Generally, EBD/EMD models are treated as IC models rather than explicitly as *.HYP* boards. The mapping of an EBD/EMD model to a reference designator happens in the *.REF* or *.QPL* automapping files, as with any other IBIS model.

Other Simulation Types and MultiBoard Designs

The primary differences between simulating with one board versus multiple boards using the BoardSim MultiBoard option include board-to-board connectors and simulating the complete net that spans more than one board. In addition to simulating interactively, you can run a Board Wizard batch simulation on multiple boards, enabling delay calculations for a complete multiple-board system.

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BoardSim is hardly any more difficult to use for multiple-board simulation than for single boards. If your connectors use consistent pin names between the mating halves, you can usually set up a multiple-board project in a few minutes.

Simulating with EBD/EMD Models

Generally, EBD/EMD models are treated as IC models rather than explicitly as *.HYP* boards. The mapping of an EBD/EMD model to a reference designator happens in the *.REF* or *.QPL* automapping files, as with any other IBIS model.

Sometimes, you may have a PCB in a multiple-board design that is modeled with an IBIS EBD/EMD file rather than a board design file. This is typically a third party board that is included in your system, for example, memory modules. See “Simulating Multiple Boards” on page 1213 for a general description of EBD/EMD and how it differs technically from board design file formats, such as.*HYP*.

After auto-mapping an EBD/EMD model and beginning simulation, BoardSim automatically creates a board representation of the EBD/EMD model in memory, and its circuit effects are included in simulations. You can probe inside an EBD/EMD model in the same way plug-in boards in the multiple-board system are probed. A board design file offers better ease-of-use than both EBD and EMD because it can be viewed and can model coupling.

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Adding IC Models to Your Existing Libraries

You can add several types of IC models to your existing libraries.

Topic	Description
Adding IBIS Models	HyperLynx includes a Visual IBIS Editor that allows you to easily view, syntax-check, and maintain IBIS models that you receive from vendors and other third parties.
SPICE and Touchstone Models	LineSim GHz and BoardSim GHz offer direct integration with SPICE (your choice of HSPICE or ADMS).

Adding IBIS Models

HyperLynx includes a Visual IBIS Editor that allows you to easily view, syntax-check, and maintain IBIS models that you receive from vendors and other third parties.

See Creating and Editing IBIS Models for details.

SPICE and Touchstone Models

LineSim GHz and BoardSim GHz offer direct integration with SPICE (your choice of HSPICE or ADMS).

SPICE IC models can be attached to component pins directly in HyperLynx, in just the same way as IBIS models are assigned. SPICE or Touchstone passive-interconnect models can be placed into a special symbol in the LineSim free-form schematic editor. Simulation occurs in SPICE under the control of HyperLynx and results are automatically read back and displayed in the Digital Oscilloscope Dialog Box. See “Setting Up a SPICE Simulation” on page 1172 for a demonstration of SPICE integration.

MultiBoard Simulation with EBD/EMD Models

In HyperLynx, the mapping of an EBD/EMD model to a reference designator occurs in the *.REF* or *.QPL* IC Automapping files, just as with any other IBIS model.

When your multiple-board design includes a third-party board (such as a memory module), an IBIS EBD/EMD file that describes the board is often available. Generally, EBD/EMD models are treated as IC models rather than explicitly as a board described by a design file.

See “Assigning Models to Components and Pins” on page 85 for additional information.

When you describe your design by creating a MultiBoard project using an EBD/EMD model and begin simulation, the software automatically creates a board representation of the model in memory, and its circuit effects are automatically included in simulations. You can even probe

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inside an EBD/EMD model. However, you cannot physically view an EBD/EMD file, and EBD files are not able to model coupling. When you have a choice, always use board design files rather than EBD/EMD files for the simulation accuracy obtained from including coupling.

Translating a Board into a BoardSim Format

Before using BoardSim, you must translate your PCB layout into a format that BoardSim can read.

In some PCB-layout tools, a BoardSim translator is built-in and accessed from a menu. For other tools, you can use an external translator supplied with BoardSim. Either way, the end result is a file that BoardSim can read.

For a complete list of translators, see “Opening a Design” on page 55.

Multi-Bit Stimulus

Some designers of SERDES-based designs today use standalone SPICE netlists to create eye diagrams. While possible and sometimes even necessary because a certain IC model is available only in SPICE format, using raw SPICE for eyes is usually cumbersome and time-consuming. SPICE simulations often run very slowly, and setting up for simulation especially generating stimulus patterns, is awkward. LineSim, by contrast, makes the generation of eye diagrams fairly easy.

As the example Multi-Bit Stimulus shows, set up activities such as defining a stimulus pattern are much easier in HyperLynx than in SPICE. Eye diagrams are created quickly when simulations are performed using IBIS models. However, simulations do not need to run simulations in the HyperLynx native simulator. Eye diagrams are created as easily for SPICE simulations as they are for HyperLynx simulations.

Crosstalk Simulation - LineSim Tutorial

Crosstalk simulation is a particular category of signal-integrity simulation that looks specifically at unwanted noise generated between signals.

Crosstalk occurs when two or more nets on a PCB are coupled to each other. Such coupling can arise any time two nets are routed next to each other for any significant length. When a signal is driven on one of the lines, the electric and magnetic fields it generates cause an unexpected signal to also appear on the nearby line.

Crosstalk is a particularly hard phenomenon to anticipate and control unless you are able to simulate it because there is almost no way of intuitively knowing how much crosstalk voltage and current will develop due to a given coupling. Many complex factors combine to create an unwanted crosstalk signal: the length over which the traces are coupled, the distance between the traces, their positions in the PCB stackup, what driver ICs are used on both the aggressor

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and the victim lines, whether or not the lines are terminated, and so forth. In order to resolve all of these factors and produce an accurate simulation, the HyperLynx software uses a fast, built-in field solver that can calculate the electromagnetic properties that govern the line-to-line coupling.

Modeling a Transmission Line with Coupling

LineSim can model transmission lines as part of a coupling region, which is a region containing multiple lines coupled together in the same PCB cross section.

From this kind of line definition, LineSim can perform detailed crosstalk analyses. If the coupling region contains a pair of lines, LineSim can predict differential impedances and perform coupled-pair simulation.

To determine the characteristics of a set of coupled transmission lines, LineSim invokes a fast boundary-element field solver. The field solver can extract the electromagnetic properties of any cross section including matrix impedances, capacitances, inductances, skin- and dielectric-loss parameters, and propagation velocities, and incorporate them into a simulation. You can view the detailed parameters of any cross section in a report file and also plot the field lines for the region.

You can view several regions as part of two schematics supplied with this demo. See “Achieving a Specific Differential Impedance” on page 1164 and Planning Minimum Trace Separation on a Bus for more details on coupled-line simulations.

LineSim Crosstalk Simulation

LineSim allows you to quickly construct a schematic and simulate it to see the resulting waveforms. The LineSim crosstalk-simulation option adds line-to-line coupling into your schematics.

With this capability, you can:

- Accurately predict how much crosstalk occurs when two or more PCB traces are routed near each other
- Efficiently specify maximum parallelism, minimum line separation, and other routing constraints
- See the effects on crosstalk waveforms of trace separation; trace width; dielectric thickness; driver-IC edge rate and impedance; parallel run length; and so forth
- Confidently design high-speed buses that meet tight timing and low-crosstalk-noise requirements
- Learn the difference between forward and backward crosstalk, and develop an intuitive sense of when crosstalk occurs and how to minimize it

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- Implement resistor-termination strategies that can greatly reduce or eliminate end-of-the-line crosstalk

Using LineSim for Differential-Signal Simulation

LineSim coupled-line simulation features are valuable in the design of differential signals because the same line-to-line coupling that causes crosstalk on unrelated signals also creates differential impedance and other electrical characteristics important in differential signaling. Differential pairs are common in very-high-speed design and are used widely in gigabit-per-second, SERDES-based designs.

Specifically, you can use LineSim to:

- Accurately simulate differential signals, taking full account of the coupling between traces
- Explore termination options for differential signals, and determine when a single line-to-line resistor is sufficient or when a full array termination is required

The following sections include examples of how the LineSim crosstalk-simulation option makes preventing crosstalk and designing differential signals easier.

Touchstone (S-Parameter) Modeling

Due to high frequency designs, especially in SERDES technology, SPICE models are not the only type of model appearing in signal-integrity simulations. Increasingly, models for passive interconnect structures such as connectors and IC packages are provided in Touchstone format. Touchstone models are used in RF/microwave engineering to accurately characterize ultra-high-speed devices and structures.

Touchstone models are based in the frequency domain, which makes them fundamentally different from other model types such as IBIS and SPICE. The Touchstone format is describes several types of network parameters: S - scattering, Z - impedance, and Y - admittance parameters. Each type treats the structure being modeled as a multi-port black box, and provides a matrix describing how each port behaves at every frequency of interest.

Because the models are expressed in the frequency domain, they can not be directly simulated in the time domain. Fortunately, most SPICE simulators, including HSPICE and Eldo/ADMS, can use Touchstone models in transient simulation, using one of several possible techniques, the most-common being convolution. A particular strength of the Eldo/ADMS simulator is its ability to fit S-parameter models using complex poles, which has various advantages over simple convolution such as model compression, speed, passivation, and causality enforcement.

In signal-integrity, S-parameter models are by far the most common type of Touchstone model in use. One typical way to create such a model is by measurement using a vector network

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analyzer, which directly outputs S-parameter data. Another possibility is to model the structure using 3-D electromagnetic software which also outputs S parameters.

Impedance Planning

In high-speed design, it is often necessary to plan stackups and trace widths in such a way that traces have certain desired characteristic impedances. Some standard buses, for example, mandate that trace impedances be within a certain range, such as 60 - 100 ohms.

Designers often rely on reference books or closed-form equations to perform such impedance planning. However, the HyperLynx stackup editor provides a faster, more accurate way to plan impedances than manual methods. The Z0 Planning tab, located in the stackup editor, has a back solver that instantly calculates geometric values such as trace width based on desired target impedances. The results are more accurate because rather than relying on equations, which are approximate and suffer from significant error outside certain geometric ranges, the stackup editor runs a field solver in the background to accurately calculate impedance and delay. The field solver is a simulator that takes into account the actual geometries and material properties of the net being simulated.

Modeling a Transmission Line

When you activate a transmission line in a schematic, you can model it in any of several ways.

- Electrically, by directly entering its characteristic impedance and propagation delay
- Geometrically, by tying it individually to a particular kind of PCB cross section geometry
- Geometrically, by tying it to a PCB stackup created for the whole schematic
- As a connector, cable, or wire

When modeling a transmission line geometrically with a cross sectional area, you can select from these cross section types:

- Microstrip
- Buried microstrip
- Stripline

Prerequisites

None.

Procedure

1. In the schematic, double-click the transmission line you want to model. The Edit Transmission Line dialog box opens.

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2. In the Transmission Line Type area, select the appropriate modeling type:

Model Type	Description
Simple Microstrip Buried Microstrip Stripline, Stripline Between Hatch	Directly enter the electrical properties for a line (such as impedance and delay). The Microstrip, Buried Microstrip, and Stripline types let you model a line with a geometric cross section. The corresponding electrical properties are calculated automatically.
Stackup	Ties the transmission line to a layer stackup that applies globally to the entire schematic. See Layer Stackups for more details on creating and editing a stackup. When you change the stackup, all of the impedances and delays of the transmission lines tied to the stackup are automatically updated.
Wire Over Ground Cable	Model a transmission line as an industry-standard cable or as a wire over a ground plane.
Connector	Model the transmission line as any of over 100 connectors. The connector library contains detailed electrical characterizations of a wide array of its comprehensive product line. In most cases, the library contains an equivalent model when using a connector from a different vendor. You can also model the connector as a simple transmission line by directly entering the appropriate impedance and delay.

3. Specify properties for the transmission line. For information about available properties, see “Edit Transmission Line Dialog Box - Transmission-Line Type Tab” on page 867.
4. Click **OK**.

   New electrical values are calculated and exported to the transmission line in the schematic.

   The stackup method is the most versatile and powerful. For example, you can tie every transmission line in a complex schematic to a stackup. Then any experiments you make in the stackup, such as changing dielectric thicknesses or re-ordering layers immediately affect every line in the schematic.

IC Modeling With HyperLynx

HyperLynx provides a diverse set of modeling options to accommodate your IC-modeling needs.

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Table A-7. Supported IC-Modeling Formats

Format	Description
IBIS	The Input/Output Buffer Information Specification (IBIS) is an industry-standard format for high-speed PCB simulation and is supported by a large number of simulation and IC vendors. These vendors are able to create detailed, accurate IBIS models without including proprietary information about how buffers are actually constructed. Nearly all semiconductor vendors now offer IBIS models for download from their Web and FTP sites. Simply place vendor IBIS models in a subdirectory of your choosing, point HyperLynx to the directory, and the models are immediately available for use - even in the middle of a HyperLynx session. Although IBIS models generally come directly from semiconductor vendors, the HyperLynx IBIS Wizard allows you to create an IBIS model for a custom IC or a new device which is not yet modeled by the vendor. The Wizard walks you through questions about the device characteristics and automatically generates a syntactically correct IBIS model, ready for simulation.
IBIS EBD/EMD	IBIS capability in HyperLynx includes support for the IBIS Electrical Board/Module Description sub-format, a method of modeling small boards and modules. Many memory modules, for example, are represented in EBD/EMD files that can be downloaded from the manufacturer. Some module providers also make their data available directly in a board design file format such as *.HYP*, *.CCE*, or ODB++.
IC Automapping	The IC Automapping feature lets you map IC reference designators or part numbers to models using a convenient pop-up list. Each IC pin then automatically attaches to the proper model when you simulate. Reference-designator mapping is useful on a per-board basis. Part-number mapping is leveraged across all designs because it is based on corporate part numbers that rarely change.
SPICE Models	Models for the ICs used in some gigabit-per-second, SERDES-based technologies are available from semiconductor vendors only in the form of SPICE files, often in encrypted format to hide proprietary details. In recognition of this fact, HyperLynx provides direct integration with two important SPICE simulators: HSPICE and Siemens EDA's HSPICE-compatible Eldo/ADMS. See "Integrated SPICE Simulations" on page 1224.

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Table A-7. Supported IC-Modeling Formats (cont.)

Format	Description
Touchstone Models	Models for passive interconnect structures such as connectors and IC packages used in SERDES and other high-frequency designs are increasingly provided in the form of Touchstone models. The Touchstone format comes from the RF/microwave world. I it allows devices to be characterized (in the frequency domain) using so-called network parameters: S (scattering), Z (impedance), or Y (admittance) parameters. Note that Touchstone data is captured in the frequency domain, either from software-based electromagnetic extraction or measurements taken from a vector network analyzer. As such, these models cannot by directly simulated in the time domain. Fortunately, the Eldo/ADMS simulator has a robust technique for accurately fitting frequency-domain models using complex poles and residues, and making them usable in the time domain. You can include Touchstone models directly in HyperLynx simulations. The LineSim schematic editor includes a special symbol designed specifically to house SPICE and Touchstone passive-interconnect models. For an example of setting up a simulation using a Touchstone connector model, see “Including Touchstone Models in a LineSim Schematic” on page 1177.

Why IC Models are Important

A number of factors affect how a trace on a PCB behaves from a signal-integrity standpoint: the geometric properties of the trace itself; how it connects to other traces; how the board layers are stacked up; what materials the board is manufactured from; and so forth. A signal-integrity/ crosstalk/GHz-level simulator must accurately model all of these parameters.

Another important modeling factor involves the ICs on the trace, especially the driver IC. It is typically the driver IC that causes transmission-line problems, because of fast rise and fall times. Receiver ICs also play a role, especially as a result of their input capacitance and diode-clamping effects.

Note: For driver ICs, each of the following must be considered for accurate signal-integrity simulation: switching time (rising and falling edges), switching impedance (rising and falling edges), switching shape (rising and falling edges), clamp diodes (high and low side), and output capacitance.

For receiver ICs, the following are important: input capacitance, clamp diodes (high and low side), and input resistance.

Trace Coupling

The LineSim crosstalk option lets you add coupling information to any LineSim schematic.

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The drawing for this example was created by entering a LineSim schematic with three transmission lines and their driver and receiver ICs. Once the schematic is drawn, information about how the three lines are coupled together is added. You can couple any line in a schematic by right-clicking it and changing its type to coupled stackup. Any number of coupling regions can be defined, and any line can be added into any coupling region.

When a transmission line is coupled, it displays differently in the schematic editor than when uncoupled. The transmission lines in this schematic have rats nest lines between them, indicating that they are coupled together.

Once transmission lines are gathered into a coupling region, the cross section properties and length of the region are defined to match the problem you want to simulate. The definition you make is geometric and LineSim converts this data into electromagnetic parameters.

LineSim GHz Simulation

SERDES-based design, a technology that emphasizes very-high-speed data streams traveling on narrow, serialized data paths uses gigabit-per-second signaling. Gigabit-per-second signaling requires lossy simulation and advanced via modeling, and sometimes even the use of SPICE-based driver models.

LineSim performs pre-layout simulation for GHz-level designs. HyperLynx provides the following GHz-level simulation:

- Lossy transmission-line simulation. See “Simulating Using Lossy Transmission Models” on page 1153.
- Advanced via modeling.
- Eye diagrams. See “Multi-Bit Stimulus” on page 1216.
- Integrated SPICE simulation (HSPICE and Eldo/ADMS. See “Integrated SPICE Simulations” on page 1224.
- Touchstone (S-parameter) support. See “Including Touchstone Models in a LineSim Schematic” on page 1177.

As driver-IC switching times grow shorter, the frequency content of the resulting signal increases. An older-style design with a fundamental frequency of 133 MHz can have significant energy content at several higher harmonics and little content at or above 1 GHz. In contrast, gigabit-per-second designs use very-high-speed serialized bit streams that demand extremely sharp switching edges which have harmonic content well above the 1-GHz level.

As a result, physical effects (collectively called loss) that play a minor role in traditional designs become important in GHz-level designs. Therefore, simulator results must include the effects of signal loss. PCBs can produce two types of losses:

- Skin effect, which is loss due to the resistance in the trace metal

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- Dielectric loss, which is loss due to the nature of the surrounding dielectric layers.

Note

The FR-4 material used in typical PCB manufacturing is particularly prone to loss, compared to other more-expensive types of dielectric.

Both of these effects are difficult to simulate in the time domain because each effect worsens as signal frequency increases. Skin effect refers to the fact that the current in a trace tends to crowd more and more to the edges of the trace cross section as frequency increases. Because there is more crowding at higher frequencies, there is more resistance. Dielectric loss works similarly; the higher the signal frequency, the higher the loss.

These two factors combine to change the shape of a signal launched at a driver IC as it travels down a trace: higher-frequency components of a signal are attenuated more severely than lower, which tends to soften the shape of the signal and drop its amplitude. Shape changes also result from the fact the different frequencies propagate at different speeds. The sum total of these lossy effects changes how a signal appears at the end of a PCB trace. This in turn means that timing and other critical signal-quality factors are significantly altered by loss.

Integrated SPICE Simulations

Vendors of some very-high-speed driver and receiver ICs make models available for their components only in SPICE format because some devices have subtle behavior which is difficult to model accurately in the IBIS behavioral format. In almost all cases, these SPICE models are provided in a proprietary format: either HSPICE or Eldo/ADMS. Additionally, some models are encrypted, which hides all model details from the user.

HSPICE is a well-known industry SPICE simulator. Eldo/ADMS is an alternative SPICE-based simulator from Siemens EDAwith a robust HSPICE-compatible mode. Eldo/ADMS also offers some features not found in HSPICE, such as an industry-leading dielectric model for lossy transmission lines and a unique method of simulating S-parameter models in the time domain. To run signal-integrity simulations with SPICE-based models, HyperLynx provides a seamless interface to both HSPICE and Eldo/ADMS.

Because SPICE models are necessary under some circumstances, designers conclude that they need to drop their traditional signal-integrity tools and proceed using only a SPICE simulator. There are a number of disadvantages to this approach:

- Raw SPICE has a primitive, unfriendly user interface compared to HyperLynx, wasting valuable design time on set up
- SPICE interconnect netlists require manual creation using a text editor. Even for simple “what if” scenarios, this is time-consuming and error prone. Additionally, post-route scenarios are essentially impossible.

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- Field solutions to convert geometric cross sections into electrical parameters also require manual creation. Again, this is error-prone and impossible for complex interconnect scenarios which involve many different cross sections
- Stimulus for an eye diagram require manual generation – a tedious and error-prone task

A better approach is to integrate the SPICE simulation engine into the robust, friendly HyperLynx environment, an approach that offers the best of both worlds: you get the extra accuracy of running SPICE IC models, and all the convenience and productivity of the HyperLynx environment.

Note

In the past, all digital buffers were modeled accurately in the non-proprietary IBIS format. Now, vendors believe some devices are more accurately represented in using SPICE models. However, very advanced driver models reportedly have simplified analog characteristics to enforce strict linearity. Early indications are that IBIS modeling may once again suffice to capture the analog behavior. Also, SERDES-type I/Os have increasingly large amounts of associated digital logic to generate pre-emphasized driver signals or implement receiver equalization circuits, which may have to be modeled in mixed analog/digital languages such as VHDL-AMS or Verilog-AMS. SPICE is unable to handle significant amounts of digital logic.

Related Topics

Setting Up a SPICE Simulation

Preparing a Schematic for DDRx Batch Simulation

Use this procedure to assign schematic symbol properties to enable the DDRx Wizard to identify components and nets in the DDRx interface. You can also add text comments to label nets in the schematic.

Procedure

1. Open your schematic in LineSim.
2. Double-click the IC symbol to edit its reference designator.

   Assign the same reference designator to all pins that belong to the same DRAM or controller component. This enables a DDRx wizard to identify the IC symbols that collectively belong to the controller and each DRAM, based on the reference designator assignment.

   Figure A-2 shows a schematic that contains a few of the data, strobe, and data mask nets in a DDRx interface. After adding the IC symbols to the schematic, reference designator

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U1 was assigned to the controller IC symbols on the left side and reference designator U2 was assigned to the DRAM IC symbols on the right side.

Figure A-2. Reference Designators in Schematic for Controller and DRAM

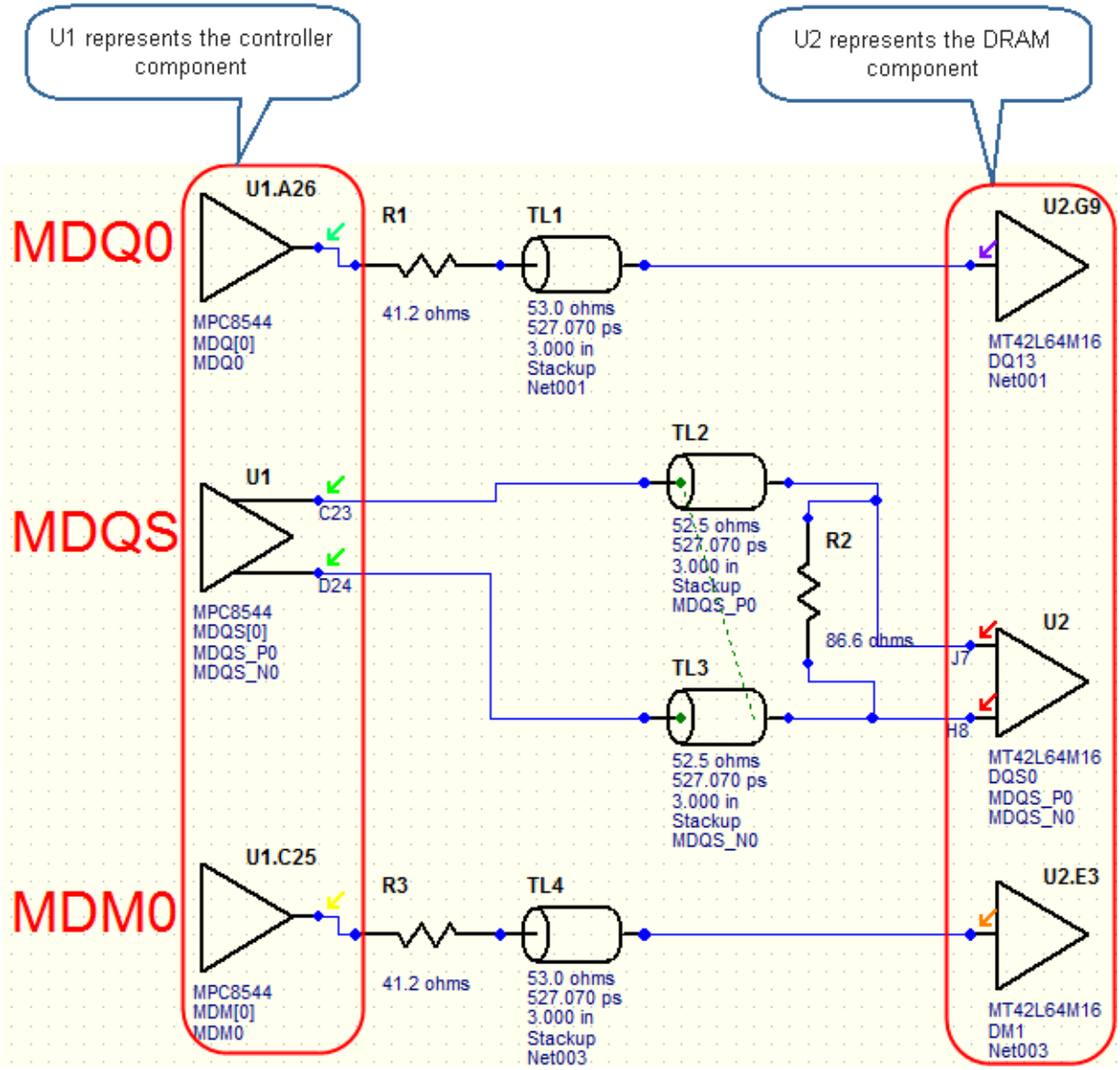


Figure A-3 shows that U1 is assigned as the controller on the Controller page of the DDRx Wizard.

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Figure A-3. Controller Reference Designator Assignment in DDRx Wizard

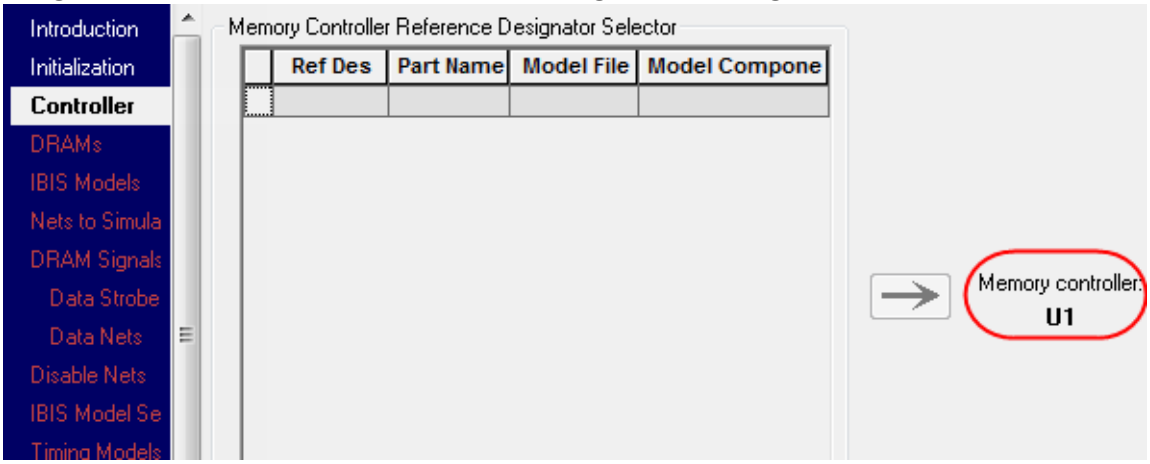
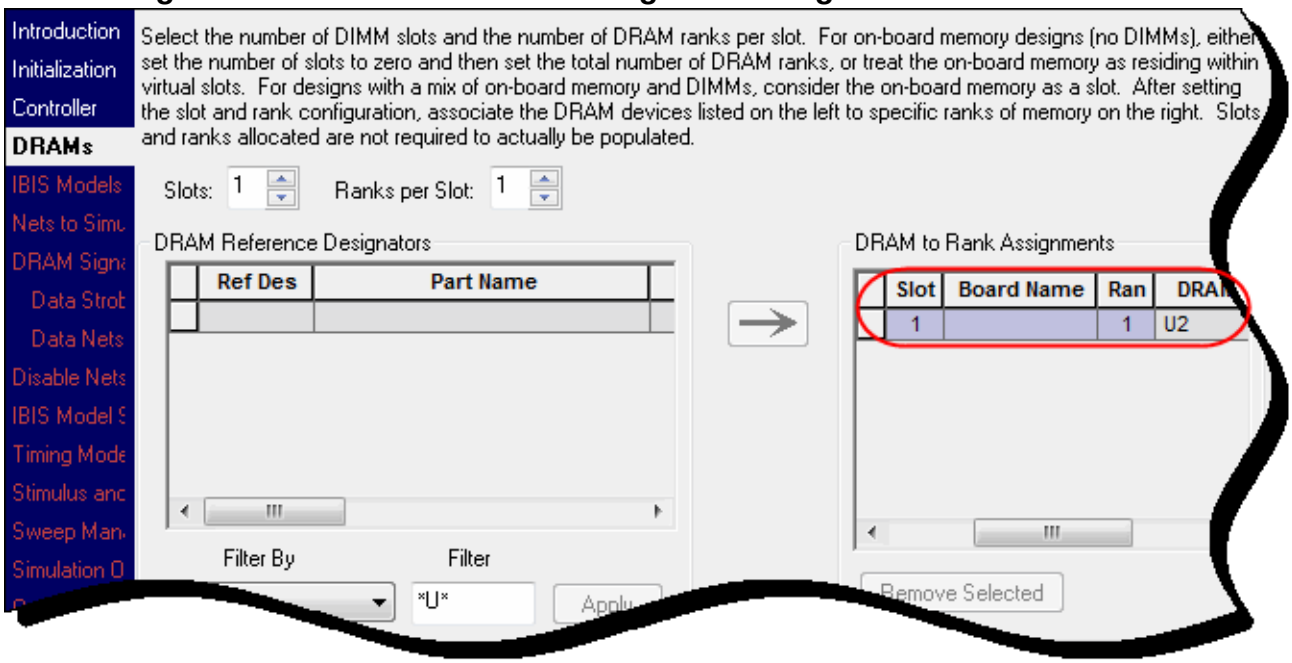


Figure A-4 shows that U2 is assigned as a DRAM on the DRAMs page of the DDRx Wizard.

Figure A-4. DRAM Reference Designator Assignment in DDRx Wizard

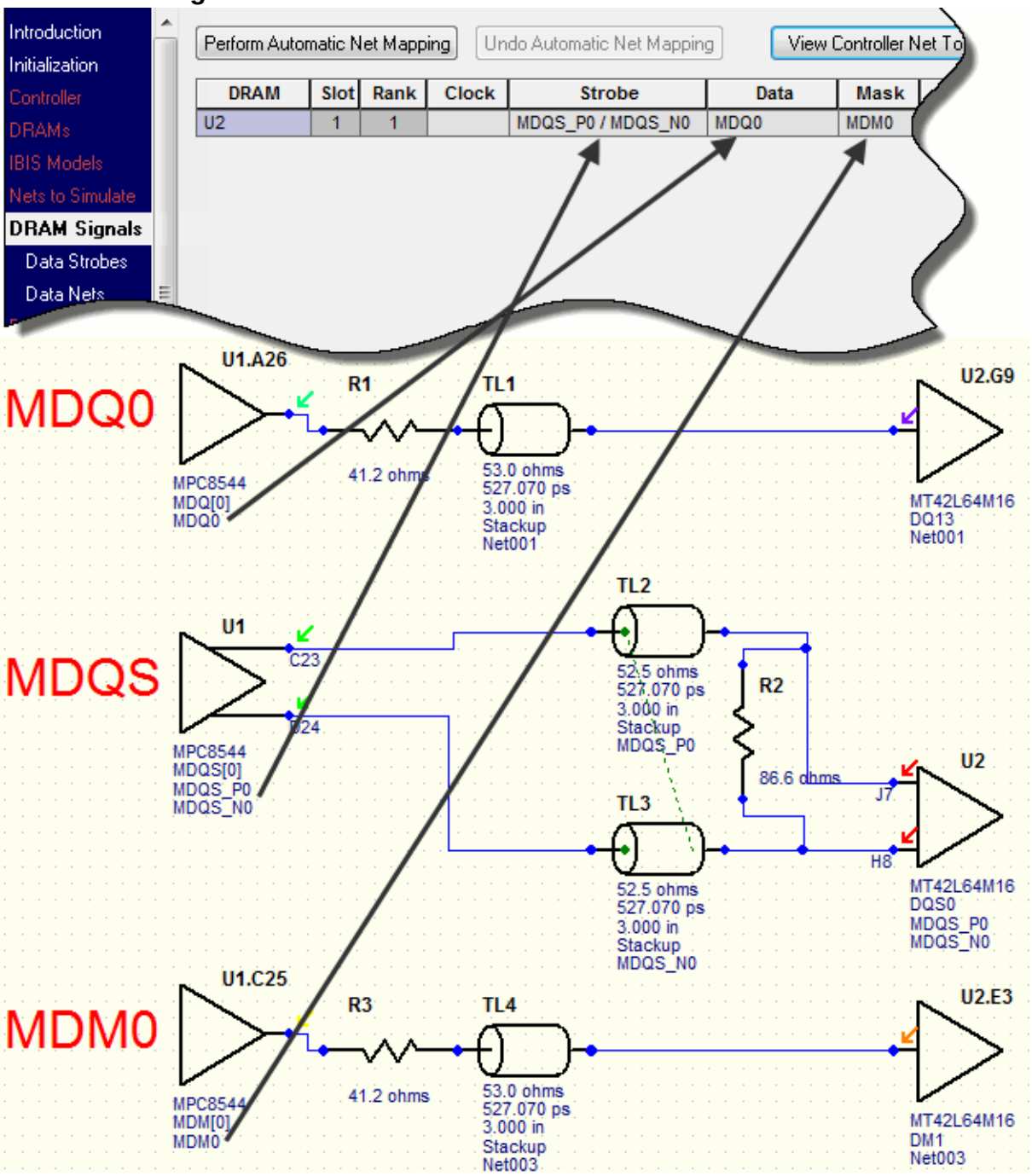


3. Assign the names you want to appear in the DRAM Signals page of the DDRx Wizard.

   Figure A-5 shows how the DRAM Signals page in the DDRx Wizard displays net names from the schematic.

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Figure A-5. Schematic Net Names in the DDRx Wizard



4. Optionally, add text comments that make it easier to identify nets in the DDRx interface that you implemented in the schematic.

   The large red text near the left side in Figure A-5 are examples of text comments. This capability is useful when the schematic contains many nets and the net names that appear below the component symbols are too small to read when you zoom out to display all the nets at once.

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Related Topics

Running a DDRx Memory Interface Simulation

HyperLynx SI PI Automation

To help you learn and use ActiveX automation for HyperLynx SI/PI, the installation includes several introduction and reference documents, and example scripts.

See ...*MentorGraphics*\<*release*>*SDD_HOME**hyperlynx64**HypFiles**HL_Automation_Starter*.

Note

HyperLynx supports scripts written in Visual Basic and Python.

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Appendix B
Layer Stackups

A layer stackup for a PCB design defines the arrangement and materials of the signal and plane layers in a multiple layer design.

You create a stackup by specifying the order of the layers along with their physical and electrical characteristics (such as thickness, plating, dielectric properties, and impedance values). You may need to perform multiple analyses and experiment with different values to achieve the optimal electrical characteristics for each layer in the stackup.

Topic	Description
Stackup Editor	The Stackup Editor provides a spreadsheet and a dynamically updated cross section of the PCB. From some tools, you can define the layer arrangement and specify the physical and electrical characteristics for the layer stackup. From other tools, some or all stackup properties are read only.
Defining the Basic Stackup	You can define the basic layer stackup for your design and display impedance characteristics for test traces with the Stackup Editor.
Exporting a Stackup	You can reuse stackups among designs. Exporting a proven stackup can save time when preparing a design for simulation (such as in HyperLynx SI/PI) or for development (such as in Constraint Manager). You can also create a backup copy of a stackup, which is helpful when performing multiple “what if” experiments.
Defining Trace Width and Separation to Meet Target Impedance	You can determine trace width and differential pair spacing to meet required impedance specifications; view the characteristic impedance and DC resistance of an existing stackup; perform “what if” experiments to identify stackup properties that achieve specific characteristic impedances and DC resistances.
Basic Factors That Affect Trace Impedance	You can define a combination of dielectric layer thickness, trace width, and dielectric constant values for your design that helps you to create signal nets with the required impedance. To improve correlation between calculated and actual trace impedance values, contact your board vendor to obtain detailed information about dielectric constant and trace etch values.

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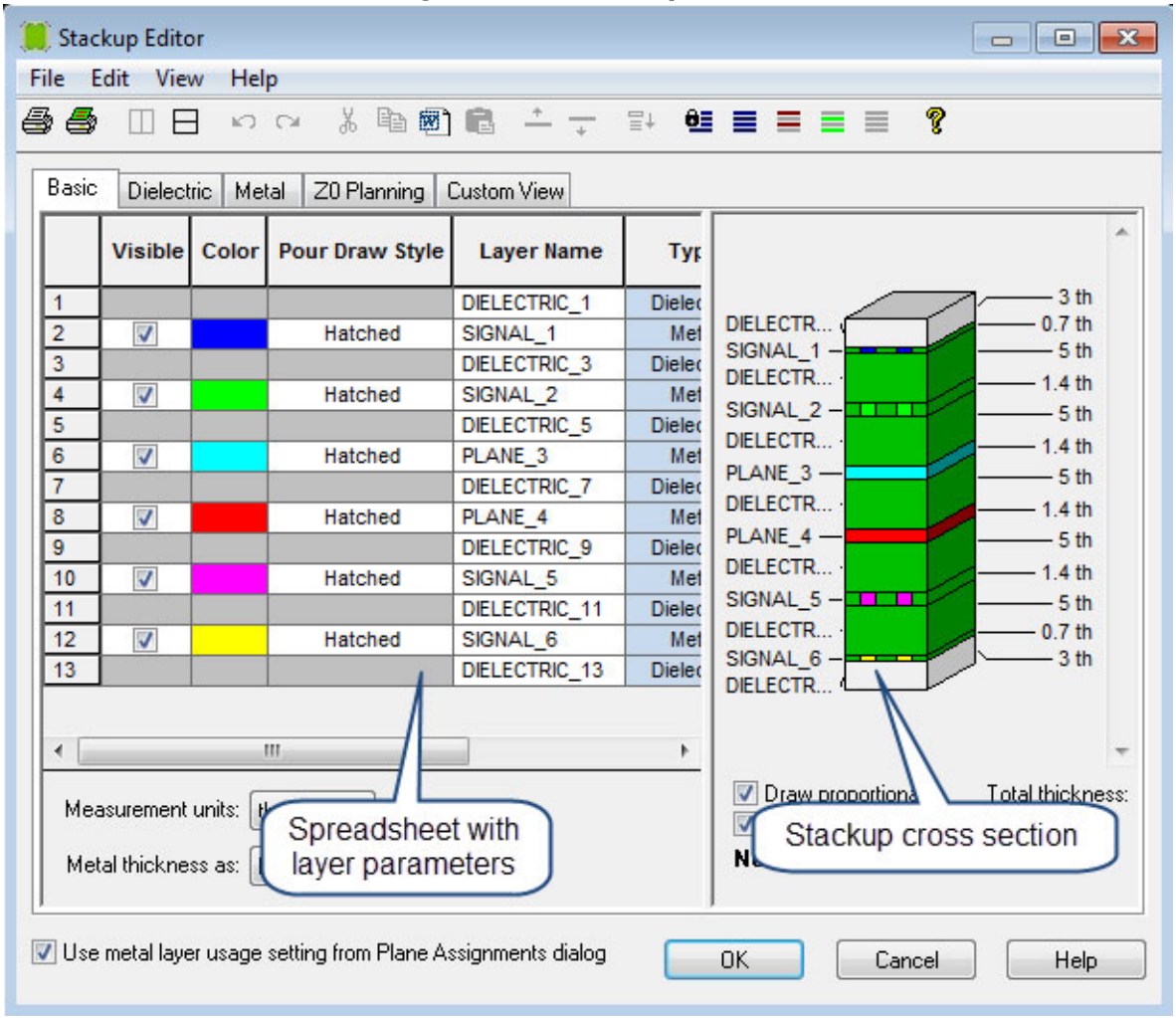
Topic	Description
Setting Up a Custom View	You can facilitate your analysis and calculations by customizing the columns in the Stackup Editor.
Stackup Editor Reference	The following topics describe the Stackup Editor user interface.
Stackup Terminology	The Stackup Editor uses special terms in the spreadsheet. You should understand these terms and their applications as stackup layer parameters.

Stackup Editor

The Stackup Editor provides a spreadsheet and a dynamically updated cross section of the PCB. From some tools, you can define the layer arrangement and specify the physical and electrical characteristics for the layer stackup. From other tools, some or all stackup properties are read only.

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Figure B-1. Stackup Editor



Related Topics

Defining the Basic Stackup

Stackup User Interface

Stackup Terminology

Defining the Basic Stackup

You can define the basic layer stackup for your design and display impedance characteristics for test traces with the Stackup Editor.

Tip

You can open the Stackup Editor for two designs at the same time and use Ctrl-C and Ctrl-V to copy and paste values between them.

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Tip

When you add, move, paste, or delete layers, correct any warnings or errors that appear in the status line of the picture pane to preserve the integrity of the overall stackup or a Board Outline (for layout tools).

Restrictions and Limitations

- The Stackup Editor ignores plane voids and hatching areas in your design. The Stackup Editor calculates impedance based on a solid plane layer and assumes the PDN (power-distribution network) provides low-inductance return current paths for the signal traces.

Prerequisites

- You understand information described in "Basic Factors That Affect Trace Impedance" on page 1239.
- If you plan to model the cross section for traces as a trapezoidal shape, you have acquired the Lossy Lines license.

Procedure

1. Open the Stackup Editor:

If you have a design with one stackup and use...	Do this...
HyperLynx® SI/PI	1. Disable the **Setup > Stackup > Enable Multiple Stackups** menu item (choose the menu item to deactivate its check mark). 2. Choose the **Setup > Stackup > Edit** menu item. 3. If you have opened a multiple-board design, from the Stackup Editor toolbar, select a board from the Board list.

If you have a design with multiple stackups and use...	Do this...
HyperLynx SI/PI	1. Enable the **Setup > Stackup > Enable Multiple Stackups** menu item (choose the menu item to activate its check mark). 2. Choose the **Setup > Stackup > Stackup Manager** menu item. The Stackup Manager dialog box opens. 3. If you have opened a multiple-board design, select a board from the Board list. 4. Select a stackup and click **Edit**.

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2. In any tab, do any of the following:

If you want to...	Do the following...
Add a new layer.	Right-click the spreadsheet row next to the new layer, select the **Insert Above** or **Insert Below** menu item, then select the layer type. **Note:** For HyperLynx SI/PI, if the added layer does not have the correct properties, you can edit the default stackup properties. From BoardSim® or LineSim®, choose the **Setup > Options > General** menu item. From the Preferences dialog box, click the **Default Stackup** tab. **Note:** If you add a signal or plane layer, the software adds a dielectric layer to separate it from other metal layers.
Cut or copy and paste a layer.	1. Select the row numbers you want to cut or copy, then select **Cut** or **Copy** from the popup menu. 2. Right-click in the row below where you want to insert the copied rows, then choose the **Paste** menu item.
Change the order of the layers.	1. Select the row number you want to move. 2. Click and drag the selected row to its new location. Alternatively, in the picture pane, click and drag a layer to a new location in the stackup cross-section. **Note:** If the design in HyperLynx SI/PI has partial vias, the Stackup Editor prevents you from dragging signal layers to locations that break the connectivity among signal layers used by partial vias.
Set the default layer types and parameters. (Available only in Constraint Manager, Xpedition Layout, Xpedition Package Designer, and PADS Professional Layout.)	Check “Use metal layer usage setting from the Plane Assignments dialog”. This option assigns the layer parameters in the Stackup Editor based on the settings you previously defined in the Plane Assignments dialog box.
Edit layer parameters.	Click the cell for the parameter you want to change, then enter a new value (or select a new value from the dropdown list). To copy data to multiple cells in the same column: 1. Select the layers into which you want to paste data. 2. Right-click over the cell containing the data to copy and choose the **Apply to Selection** menu item.

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If you want to...	Do the following...
Change a layer type.	Click the Usage cell for the layer you want to change, then select the new layer type from the dropdown list. **Note:** The software may not allow you to permanently change the Usage type. Changing the Usage type temporarily enables you to experiment with different settings while analyzing the stackup characteristics
Delete a layer.	Select the row numbers you want to delete, then press Delete.

3. Click the **Dielectric** tab and define the dielectric characteristics for each layer.
4. Click the **Metal** tab and define the metal characteristics for each layer.
5. To change the bulk resistivity or temperature coefficient for a layer, click the Metal cell, then select <Custom> from the dropdown list to activate the Bulk R and T coef cells.

   The software uses bulk resistivity when calculating DC resistances for trace segments on the layer. Bulk resistivity is considered to be an advanced parameter that you normally do not need to change.
6. To analyze resistance and attenuation over a frequency range, click **View** in the Loss Curve column for the layer you want to evaluate.

   The Loss-vs-Frequency Graph dialog box graphs the effects of different frequency ranges on resistance and attenuation for the layer.
7. Click the **Manufacturing** tab and define metal surface roughness and trace etch parameters. You can also represent a trace cross section as a trapezoidal shape.

Results

After you have defined the basic layer stackup, you can analyze the stackup and specify more precise dielectric and trace characteristics.

Related Topics

Verifying the Stackup Definition

Exporting and Importing a Stackup

Stackup User Interface

Stackup Terminology

Defining Trace Width and Separation to Meet Target Impedance

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Exporting a Stackup

You can reuse stackups among designs. Exporting a proven stackup can save time when preparing a design for simulation (such as in HyperLynx SI/PI) or for development (such as in Constraint Manager). You can also create a backup copy of a stackup, which is helpful when performing multiple “what if” experiments.

Procedure

1. Choose the **File > Export** menu item.
2. Specify the STK file location and click **Save**.

Defining Trace Width and Separation to Meet Target Impedance

You can determine trace width and differential pair spacing to meet required impedance specifications; view the characteristic impedance and DC resistance of an existing stackup; perform “what if” experiments to identify stackup properties that achieve specific characteristic impedances and DC resistances.

You can change the test trace width on each layer to calculate the characteristic impedance for the following:

- A less commonly used trace width on the net
- An experimental trace width to see how different impedances could be achieved with different widths

The Stackup Editor calculates impedance based on a solid plane layer and assumes the PDN (power-distribution network) provides low-inductance return current paths for the signal traces. The software ignores plane void and hatching areas in your design.

Procedure

1. Choose the **View > Calculate Z0** menu item and verify that it is enabled. This enables automatic characteristic impedance calculation.
2. Do any of the following operations:

If you want to...	Do the following...
Calculate the characteristic impedance for a test trace width.	1. Click the **Metal** tab. 2. Type a new value in the Test Width column for the layer you want to analyze and press Enter. The characteristic impedance value in the Z0 column and the picture pane update.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to...	Do the following...
Calculate single trace widths based on characteristic impedances.	1. Click the **Z0 Planning** tab. 2. Select “Single trace” from the “Plan for” dropdown list. 3. Type a new value (in ohms) in the Target Z0 column for the layer you want to analyze, then press Enter. The required width in the Width column updates.
Calculate trace widths for a differential pair based on characteristic impedances.	1. Click the **Z0 Planning** tab. 2. Select “Differential pair” from the “Plan for” dropdown list. 3. Select “Solve for width” from the Strategy dropdown list. 4. Type a new target characteristic impedance value (in ohms) in the Diff Z0 column for the signal layer. 5. Type a new trace separation in the Gap cell for the signal layer, then press Enter. The required width in the Width column updates.
Calculate trace separations for a differential pair based on characteristic impedances.	1. Click the **Z0 Planning** tab. 2. Select “Differential pair” from the “Plan for” dropdown list. 3. Select “Solve for separation” from the Strategy dropdown list. 4. Type a new target characteristic impedance value (in ohms) in the Diff Z0 column for the signal layer. 5. Type a new value in the Width column for the signal layer, then press Enter. The required separation in the Gap column updates.
Calculate trace widths and separations for a differential pair based on characteristic impedances.	1. Click the **Z0 Planning** tab. 2. Select “Differential pair” from the “Plan for” dropdown list. 3. Select “Solve for both” from the Strategy dropdown list. 4. Type a new target characteristic impedance value (in ohms) in the Diff Z0 column for the signal layer. 5. In the Z0 Curve column, click **View**. 6. Examine the width versus separation curves to determine the appropriate values.

3. If you want to copy values from the Width column on the **Z0 Planning** tab to the Test Width column on the **Basic** tab, click **Apply as Test Width**.

Related Topics

Width-vs-Separation Graph Dialog Box

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

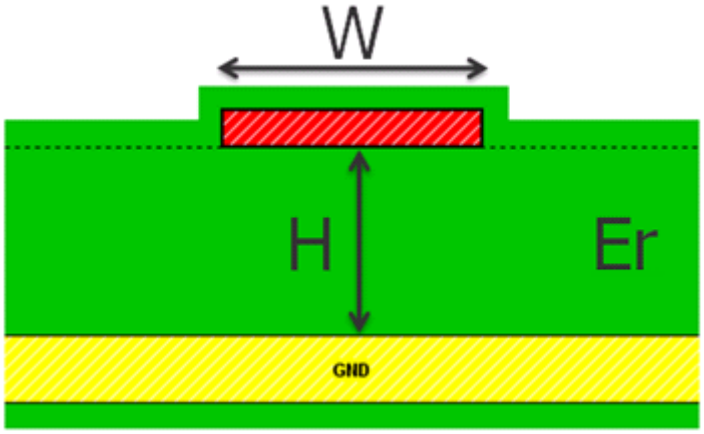
Basic Factors That Affect Trace Impedance

Basic Factors That Affect Trace Impedance

You can define a combination of dielectric layer thickness, trace width, and dielectric constant values for your design that helps you to create signal nets with the required impedance. To improve correlation between calculated and actual trace impedance values, contact your board vendor to obtain detailed information about dielectric constant and trace etch values.

Figure B-2 shows the three main factors that determine trace impedance.

Figure B-2. Trace Impedance Factors



Where:

W is the trace width.

H is the height above a reference plane. This is also known as a dielectric layer thickness.

Er is the dielectric constant.

Trace impedance increases as the height above a reference plane increases, and impedance decreases as trace width and dielectric constant increase.

To determine a combination of values for trace width and the height above a reference plane, one strategy is to begin with the minimum trace width that does not incur additional manufacturing cost. You can then define the height above a reference plane to meet the target impedance.

Note

For differential pairs, trace-to-trace spacing also affects impedance. See Defining Trace Width and Separation to Meet Target Impedance.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

To see the most realistic calculated impedance values, provide accurate information in the Stackup Editor for dielectric constant and trace etch properties. Examples:

- Fiberglass and resin content can vary with thickness, so you may need to specify different dielectric constant values for different layer thicknesses.
- Fiberglass and resin content can vary around traces. When a board manufacturer presses together multiple PCB layers, the resin tends to flow into the cavities around traces and create resin-rich areas with lower dielectric constant values. You can account for this condition by using the **Dielectric** tab to specify dielectric values for metal layers.
- A trace typically has a cross section shape that is more of a trapezoid than a rectangle. The etching process used during board manufacturing produces a narrower trace on the side that contacts the etching chemical the longest. You can specify trapezoidal trace shapes on the **Manufacturing** tab.

Related Topics

Stackup Editor - Dielectric Tab

Stackup Editor - Manufacturing Tab

Setting Up a Custom View

You can facilitate your analysis and calculations by customizing the columns in the Stackup Editor.

You can arrange the spreadsheet columns in any order and hide certain columns so that only the information you want to work with is visible.

Procedure

1. Click the **Custom View** tab, then click **Customize**.

   Tip

   You can also customize the view from any tab by choosing the **View > Customize** menu item.

2. In the Customize Spreadsheet dialog box, do any of the following:

If you want to...	Do the following...
Hide a column.	Uncheck the column you want to hide.
Display a column.	At the bottom of the columns list, check the column you want to display.
Change the order of the columns.	Select an item row and use the up/down arrow to reorder the columns.

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Related Topics

Stackup User Interface

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Stackup Editor Reference

The following topics describe the Stackup Editor user interface.

Stackup User Interface 1243
Stackup Editor - Basic Tab 1244
Stackup Editor - Dielectric Tab 1245
Stackup Editor - Metal Tab. 1247
Stackup Editor - Z0 Planning Tab 1248
Stackup Editor - Manufacturing Tab 1249
Stackup Editor - Custom View Tab 1251
Loss-vs-Frequency Graph Dialog Box 1251
Width-vs-Separation Graph Dialog Box 1252
Stackup Verifier Dialog Box 1252

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

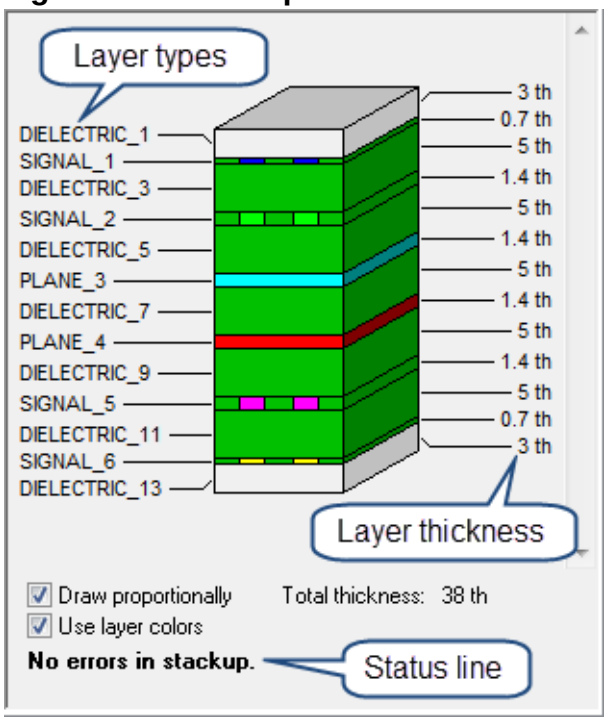
Stackup User Interface

Use the Stackup Editor to define the layer stackup parameters for a design. For board designs with multiple stackup definitions, use the Stackup Editor to define each stackup.

The Stackup Editor consists of a tabbed spreadsheet and a picture pane.

The picture pane is common to all of the spreadsheet tabs. It shows a cross section of the layer stackup with the layer types and the thickness, Er, or Z0 values of each layer.

Figure B-3. Stackup Editor Picture Pane



If there are no errors and the stackup is electrically valid, the status line at the bottom of the picture pane shows: No errors in stackup.

If there are errors in the stackup, the Stackup Editor reports them immediately. The status line shows the errors in red text and reports multiple errors in sequence, one-at-a-time.

Topic	Description
Stackup Editor - Basic Tab	Use this tab to define the basic stackup information, set measurement units and metal thickness, and view the characteristic impedance for a test width.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
Stackup Editor - Dielectric Tab	Use this tab to define the dielectric characteristics of the stackup. Dielectric information includes technology, loss tangent (for lossy transmission line simulation), the dielectric constant measurement frequency, and whether to calculate the dielectric constant and loss tangent for the metal layer from surrounding dielectric layers.
Stackup Editor - Metal Tab	Use this tab to define the metal characteristics of the stackup.
Stackup Editor - Z0 Planning Tab	Use this tab to plan and define the characteristic impedance of the stackup. Impedance planning enables you to calculate the optimal physical data when you supply the target impedance for a single trace or for differential pair traces.
Stackup Editor - Manufacturing Tab	Use this tab to specify metal surface roughness and trace etch parameters. SI simulation and the field solver accounts for surface roughness and trapezoidal trace shapes when specified.
Stackup Editor - Custom View Tab	Use this tab to set up a customized view of the spreadsheet information derived from the other tabs in the Stackup Editor. The fields shown in the **Custom View** tab are configurable. By default, all of the columns are shown. You can choose to display (or not display) any of the columns that are available in the other tabs of the spreadsheet.

Stackup Editor - Basic Tab

To access: Select the **Basic** tab.

Use this tab to define the basic stackup information, set measurement units and metal thickness, and view the characteristic impedance for a test width.

Note

For information about stackup editing restrictions, see “Defining the Basic Stackup” on page 1233.

Fields

Field	Description
Visible	Checked, makes the layer visible in the HyperLynx SI/PI board viewer and PDN Editor. For other tools, this option has no effect.

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Field	Description
Color	Defines the color of the layer in the HyperLynx SI/PI board viewer and PDN Editor. For other tools, defines only the color of the layer in the picture pane.
Pour Draw Style	Defines the drawing style (None, Solid, Hatched, Outline) of metal pours and planes displayed in the HyperLynx SI/PI board viewer and PDN Editor.
Layer Name	Defines the name of the layer.
Type	Defines whether the layer material is a conductor or an insulator.
Usage	Defines the main function of the layer. See Usage.
Thickness	Defines the thickness of the layer.
Er	Defines the dielectric constant (Er) value of the layer.
Test Width	Defines the width of the test trace used to automatically calculate the characteristic impedance.
Z0	Displays the calculated characteristic impedance (Z0) value of the layer.
Thermal Conductivity	Defines the thermal conductivity of the layer. See Thermal Conductivity.
Description	Contains information that you add.
Measurement units	Defines the measurement units. For HyperLynx SI/PI only, this setting also updates the setting on the Units dialog box (Setup > Options > Units).
Metal thickness as	Defines the metal thickness units. For HyperLynx SI/PI only, this setting also updates the setting on the Units dialog box (Setup > Options > Units).

Related Topics

Stackup Terminology

Stackup Editor - Dielectric Tab

To access: Select the **Dielectric** tab.

Use this tab to define the dielectric characteristics of the stackup. Dielectric information includes technology, loss tangent (for lossy transmission line simulation), the dielectric constant measurement frequency, and whether to calculate the dielectric constant and loss tangent for the metal layer from surrounding dielectric layers.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Fields

Field	Description
Layer Name	Defines the name of the layer.
Type	Defines whether the layer material is a conductor or insulator.
Usage	Defines the main function of the layer. See Usage. **Note:** To specify dielectric layer functions for board designs with multiple stackups, check **Enable rigid-flex dielectric support** (HyperLynx SI/PI). **Note:** Solder Mask is also known as conformal coating or SMOBC (solder mask over bare copper).
Technology	Defines whether the material for the dielectric layer is rigid prior to baking (Core), semi-soft prior to baking (Prepreg), or is flexible before and after baking (Flex Core). See Technology. **Note:** This field is blank and read-only when you set Usage to Adhesive, Cover Layer, or Stiffener.
Er	Defines the dielectric constant (Er) value of the layer. Enter the Er value or check “Calculate Er for metal layers from surrounding dielectrics” to automatically calculate the Er value. **Note:** To display the calculated values, check “Calculate Er for metal layers from surrounding dielectrics”, then uncheck the option.
Loss Tangent	Defines the loss tangent value of the layer. See Loss Tangent. Enter the loss tangent value or check “Calculate Er for metal layers from surrounding dielectrics” to automatically calculate the loss tangent value. **Note:** To display the calculated values, check “Calculate Er for metal layers from surrounding dielectrics”, then uncheck the option.
Measurement Frequency GHz	Defines a frequency used to measure dielectric constant (Er) and loss tangent. If your PCB manufacturer provides data for multiple frequencies, choose one of the data points in the range between 1 and 10 GHz, then enter that measurement value and associated frequency into the table.
Thermal Conductivity	Defines the thermal conductivity of the layer. See Thermal Conductivity. The Metric unit is W/m-C—Watts / (meter * degrees Celsius). The English unit is Btu/hrftF—British Thermal Units / (hour * feet * degrees Fahrenheit).

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Field	Description
Enable rigid-flex dielectric support (Available only with HyperLynx SI/PI.)	For the Usage column, adds dielectric layer functions for board designs with multiple stackups. **Note:** This option is checked and read-only when you specify a dielectric layer function in the Usage column that supports a board design with multiple stackups.

Related Topics

Stackup Terminology

Stackup Editor - Metal Tab

To access: Select the **Metal** tab.

Use this tab to define the metal characteristics of the stackup.

Fields

Field	Description
Layer Name	Defines the name of the layer.
Type	Defines whether the layer material is a conductor or insulator.
Usage	Defines the main function of the layer. See Usage.
Attached Dielectric Layer	Defines a dielectric layer that is attached to a metal layer. You can use this information to help specify how you want a PCB manufacturer to build the layer stackup.
Metal	Defines the metal material for the layer.
Thickness	Defines the thickness of the layer.
Er	Defines the dielectric constant (Er) value of the layer. Enter the Er value or check “Calculate Er for metal layers from surrounding dielectrics” to automatically calculate the Er value. **Note:** To display the calculated values, check “Calculate Er for metal layers from surrounding dielectrics”, then uncheck the option.
Bulk R	Displays the calculated bulk resistivity of the layer. See Bulk Resistivity.
T coef	Displays the calculated temperature coefficient of the layer. See Temperature Coefficient.
Test Width	Defines the width of the test trace used to automatically calculate the characteristic impedance.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Z0	Displays the calculated characteristic impedance (Z0) value of the layer.
Loss Curve	When you click **View**, displays the Loss vs Frequency graph.
Thermal Conductivity	Defines the thermal conductivity of the layer. See Thermal Conductivity. The Metric unit is W/m-C—Watts / (meter * degrees Celsius). The English unit is Btu/hrftF—British Thermal Units / (hour * feet * degrees Fahrenheit).

Related Topics

Loss-vs-Frequency Graph Dialog Box

Stackup Terminology

Stackup Editor - Z0 Planning Tab

To access: Select the **Z0 Planning** tab.

Use this tab to plan and define the characteristic impedance of the stackup. Impedance planning enables you to calculate the optimal physical data when you supply the target impedance for a single trace or for differential pair traces.

Fields

Field	Description
Layer Name	Defines the name of the layer.
Usage	Defines the main function of the layer. See Usage.
Thickness	Defines the thickness of the layer.
Er	Defines the dielectric constant (Er) value of the layer.
Target Z0 Diff Z0	Defines the target Z0 value when solving for a single trace. Defines the differential Z0 value when solving for a differential pair.
Width	Displays the calculated trace width for a single trace or for a differential pair when solving for width. Defines the trace width for each member of a differential pair when solving for separation.
Gap	Displays the calculated gap (trace-to-trace clearance) for a differential pair when solving for separation.

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Field	Description
Z0 Curve	When you click **View**, displays the Width vs Separation graph for a differential pair when solving for both separation and trace width.
Apply as Test Width	Copies values from the Width column on this tab to the Test Width column on the **Basic** tab.

Related Topics

Defining Trace Width and Separation to Meet Target Impedance

Width-vs-Separation Graph Dialog Box

Stackup Terminology

Stackup Editor - Manufacturing Tab

To access: Select the **Manufacturing** tab.

Use this tab to specify metal surface roughness and trace etch parameters. SI simulation and the field solver accounts for surface roughness and trapezoidal trace shapes when specified.

Surface roughness parameters quantify how the copper foil surface zigzags vertically away from an averaged smooth surface. SI simulators use roughness values to calculate conductor-related transmission line losses. Loss increases as the depth of the surface roughness approaches the skin depth of the signal current.

Etch factor parameters quantify how etching chemicals produce traces with a trapezoidal cross section. SI simulators and field solvers use etch factor values when calculating trace impedance.

Fields

Field	Description
Layer Name	Defines the name of the layer.
Technology	Defines whether the material for the dielectric layer is rigid prior to baking (Core), semi-soft prior to baking (Prepreg), or is flexible before and after baking (Flex Core). See Technology.
Process	Defines the manufacturing process used to form the copper foil on the layer. See Process.
Treated Side	Defines the side of the layer with intentionally-increased surface roughness.

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Field	Description
Roughness (Top) Roughness (Bottom)	Defines the surface roughness amplitude parameter (Rrms, Ra or Rz) value provided by the PCB fabrication vendor. You can change this value when you choose Custom from the Process column. See Roughness.
Etch Factor	Defines the trapezoidal shape caused by etching (X/T), where: • X is equal to (width of the wide side of the trace) minus (width of the narrow side of the trace). • T is the trace thickness. See Etch Factor.
Narrow Side	Defines the side of the layer that first contacts the etching chemical during PCB manufacture. For traces on inner layers, the narrow side is the prepreg side of the trace. See Etch Factor.
Include surface roughness	Enables HyperLynx SI simulation to model surface roughness. Enable this option only if the board uses a dielectric material with extremely low loss. This is because the effects of surface roughness on simulation results are negligible on dielectric materials with higher loss. For example, if the board uses FR-4 dielectric material and the simulation applies high frequency stimulus (at which surface roughness begins to have a measurable effect), surface roughness losses are negligible compared to dielectric losses. Signal loss is related to surface roughness. While increased roughness increases adhesion strength, it can also increase high-frequency signal loss. As the frequency increases, the current density near the conductor surface increases because the skin depth decreases. At some high frequency, the skin depth approaches the surface roughness amplitude, and the signal follows the irregular contours of the conductor surface, which increases the distance the signal travels.
Use trapezoidal trace shapes	Enables HyperLynx SI simulation and its field solver to model traces as trapezoidal shapes.
specified as	Defines a method used to measure conductor roughness. See “Roughness” on page 1257.
using model type	Defines a conductor roughness model. See “Roughness” on page 1257.
with factor (Available when model type is Modified Hammerstad)	Defines a conductor roughness correction factor. See “Roughness” on page 1257.

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Related Topics

Stackup Terminology

Stackup Editor - Custom View Tab

To access: Select the **Custom View** tab.

Use this tab to set up a customized view of the spreadsheet information derived from the other tabs in the Stackup Editor. The fields shown in the **Custom View** tab are configurable. By default, all of the columns are shown. You can choose to display (or not display) any of the columns that are available in the other tabs of the spreadsheet.

Loss-vs-Frequency Graph Dialog Box

To access: Select the **Metal** tab, then click **View** in the Loss Curve column of any layer row.

Use this dialog box to view the Loss vs Frequency graph.

Note

Unlike the LineSim transmission line editor, the Stackup Editor does not provide conductor length and coupling information to the field solver, so per-unit and propagation mode results are unavailable.

Fields

Field	Description
Show	Specifies the type of graph to display. Click the color box to change the color of the graph. • **Resistance** — Selected, displays the Resistance graph. • **Attenuation** — Selected, displays the Attenuation graph. Check **Resistive** to show resistive effects. Check **Dielectric** to show dielectric effects.
Frequency range	Specifies the minimum and maximum limits for the frequency range.

Tip

Drag the cursor frame around a small region of the graph to zoom in on more detail. Right-click in the graphic for more viewing options.

Related Topics

Stackup Editor - Metal Tab

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Width-vs-Separation Graph Dialog Box

To access: Choose the **Z0 Planning** tab, then click **View** in the Z0 Curve column for any layer row.

Use this dialog box to view and analyze the Width vs Separation graph.

Tip

Drag the cursor frame around a small region of the graph to zoom in on more detail. Right-click in the graphic for more viewing options.

Related Topics

Defining Trace Width and Separation to Meet Target Impedance

Stackup Editor - Z0 Planning Tab

Stackup Verifier Dialog Box

To access:

- Opens automatically when you open a design that has incorrect stackup parameters or an incorrect metal layer type.
- In HyperLynx SI/PI, choose **Setup > Stackup > Check** (unavailable for a board design with multiple stackups).

Use this dialog box to view stackup error and per-layer metal usage information or to assign plane and signal types to metal layers.

Note

The appearance and contents of the Stackup Verifier dialog box depend on whether it displays stackup error information, metal usage information, advanced metal usage settings, or all at the same time.

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Stackup Terminology

The Stackup Editor uses special terms in the spreadsheet. You should understand these terms and their applications as stackup layer parameters.

Topic	Description
Bulk Resistivity	Every signal and plane layer must have a bulk resistivity value for the metal material. The Stackup Editor uses the bulk resistivity value in conjunction with the temperature coefficient value to calculate DC resistances for trace segments on each layer.
Dielectric Constant and Permittivity	Some PCB layout tools do not provide dielectric permittivity (Er) and loss tangent values for metal layers. The Stackup Editor requires an Er value so the field solver can calculate electrical properties such as Z0.
Etch Factor	As traces are etched from top to bottom, the etching chemical remains in contact with the top of the trace longer than the bottom. This makes the top of the trace narrower than the bottom and gives the trace a trapezoidal cross section.
Loss Tangent	If your PCB design has high-speed signals or if it has signals that propagate over very long or very narrow conductors, you can improve PCB modeling by specifying the dielectric material loss tangent parameter. With loss tangent information, a simulator is better able to predict frequency-dependent transmission line losses.
Metal	The Metal parameter indicates the metal material used on the layer. You can select a standard material or a custom material. If you select a standard material, the spreadsheet automatically supplies the bulk resistivity and temperature coefficient values. If you select a custom material, you must enter the bulk resistivity and temperature coefficient values manually.
Process	The Process parameter specifies the manufacturing process used to make the copper foil on the layer.
Roughness	The Roughness parameter specifies the random small-scale “bumpiness” of a metal surface.
Technology	The Technology parameter defines dielectric layer properties related to PCB fabrication.

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Topic	Description
Temperature Coefficient	Every signal and plane layer must have a Temperature Coefficient value for the metal layer material. The Stackup Editor uses the Temperature Coefficient value in conjunction with the Bulk Resistivity value to calculate DC resistances for trace segments on each layer.
Thermal Conductivity	The default value for thermal conductivity for dielectric layers applies to FR-4 material. The default value for metal layers applies to copper. If you change the dielectric technology or metal material, the thermal conductivity value does not update automatically; you must enter the appropriate thermal conductivity value for the dielectric or metal you choose for the layer.
Usage	Metal Usage parameters include Signal, Solid Plane, and Plating (used only for outer metal layers). Split/Mixed and Flooded Signal parameters are available when you open the Stackup Editor by an Xpedition Enterprise product.

Bulk Resistivity

Every signal and plane layer must have a bulk resistivity value for the metal material. The Stackup Editor uses the bulk resistivity value in conjunction with the temperature coefficient value to calculate DC resistances for trace segments on each layer.

The bulk resistivity default value applies to copper.

Bulk resistivity is considered an advanced parameter, one that you normally do not need to change. Signal and plane layers automatically default to the bulk resistivity of the metal selected in the Metal column. To edit the bulk resistivity value, set the Metal parameter to <Custom>.

The DC resistance of a piece of metal conductor varies with temperature. When HyperLynx SI/PI calculates the DC resistance of a trace, it uses the following equation to include the effects of temperature:

$$R = R_b[1 + T_c(T - 20)]$$

where:

Rb is the bulk resistivity of the trace metal at 20 degrees C.

Rb is the bulk resistivity of the trace metal at 20 degrees C.

Tc is the temperature coefficient.

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T is the temperature at which the simulation is being run. If the temperature for the bulk resistivity of the trace metal is not 20 degrees C, subtract that temperature value from T within the parentheses.

Dielectric Constant and Permittivity

Some PCB layout tools do not provide dielectric permittivity (Er) and loss tangent values for metal layers. The Stackup Editor requires an Er value so the field solver can calculate electrical properties such as Z0.

When calculating Er and dielectric constant values for inner metal layers, the Stackup Editor emulates a stackup with alternating core and prepreg dielectric layers. During PCB manufacturing, metal traces are placed on substrate or core dielectric layers and sink slightly into the adjacent prepreg layer (which is slightly spongy prior to baking).

Use the following algorithms to determine the correct Er and loss tangent values for metal layers in the Stackup Editor:

- If the metal layer is an outer layer, then assign Er = 1 and loss tangent = 0 (which are the values for air).
- If the metal layer is adjacent to a dielectric layer with Er = 1, then assign Er = 1 and loss tangent = 0.
- If the metal layer is adjacent to a dielectric layer with the Usage value set to Solder Mask, then copy the Er and loss tangent values from the adjacent dielectric layer.
- For inner metal layers that are not described by the conditions above, use this algorithm:
    - If the metal layer is an “odd” inner layer (counting down from the top and counting only metal layers), then copy the Er and loss tangent values from the dielectric layer above the metal layer.
    - If the metal layer is an “even” inner layer (counting down from the top and counting only metal layers), then copy the Er and loss tangent values from the dielectric layer below the metal layer.

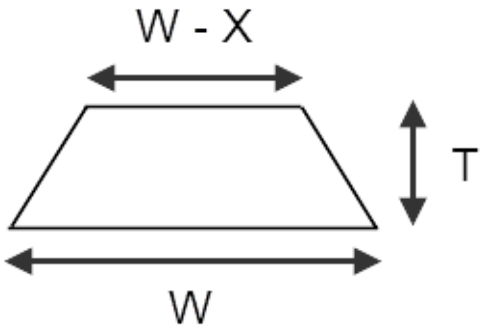
Etch Factor

As traces are etched from top to bottom, the etching chemical remains in contact with the top of the trace longer than the bottom. This makes the top of the trace narrower than the bottom and gives the trace a trapezoidal cross section.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The Etch Factor parameter defines the trapezoidal shape for the HyperLynx SI simulator and field solver.

$$\text{Etch Factor} = \frac{X}{T}$$



Where:

X = (Width of wide side of trace) - (Width of narrow side of trace)

W is the width of the wide side of the trace.

T is the thickness of the trace.

Loss Tangent

If your PCB design has high-speed signals or if it has signals that propagate over very long or very narrow conductors, you can improve PCB modeling by specifying the dielectric material loss tangent parameter. With loss tangent information, a simulator is better able to predict frequency-dependent transmission line losses.

The Loss Tangent column is displayed only on the Dielectric tab. The Er column is displayed on the Metal tab because it can affect the value in the Z0 column. Loss tangent does not affect Z0, therefore it is not displayed on the Metal tab.

Metal

The Metal parameter indicates the metal material used on the layer. You can select a standard material or a custom material. If you select a standard material, the spreadsheet automatically

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

supplies the bulk resistivity and temperature coefficient values. If you select a custom material, you must enter the bulk resistivity and temperature coefficient values manually.

Process

The Process parameter specifies the manufacturing process used to make the copper foil on the layer.

- Rolled — Copper foil is formed by repeatedly squeezing a copper billet through a pair of rollers.
- Electrodeposited — Copper foil is formed by electrodepositing copper onto a rotating steel drum. The side of the foil contacting the drum has the surface roughness of the (usually smooth) drum and the other side of the foil is rough and nodular.
- Custom — If you have reliable roughness values from the PCB vendor, select this option on the Manufacturing tab of the Stackup Editor and type the value into cells in the Roughness columns. Otherwise select one of the other options and use the default values. For the definition of roughness, see Roughness.

Roughness

The Roughness parameter specifies the random small-scale “bumpiness” of a metal surface.

PCB manufacturers intentionally increase the surface roughness of copper foil to improve its adhesion to dielectric resin. The top and bottom sides can have different roughness values.

Roughness Amplitude Parameters

Roughness is commonly measured in terms of amplitude parameters R_{rms}, R_a, or R_z. You can choose any amplitude parameter for displaying or specifying roughness values in the Stackup Editor.

Roughness Models

The Stackup Editor supports a set of industry-standard roughness models to represent conductor loss for simulation.

The models include a built-in power loss factor that represents the ratio of power loss for a rough conductor to an ideal smooth conductor.

$$\text{Power loss factor} = \frac{\text{Power loss}_{\text{rough conductor}}}{\text{Power loss}_{\text{smooth conductor}}}$$

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The power loss factor value varies over frequency from 1 to increasingly large values at high frequencies. This accounts for decreased skin depth as current crowds closer and closer to the surface.

Supported roughness models:

- Cannonball — Similar to Huray or “Snowball”, except that the Cannonball model uses only one ball size.

  The software uses the roughness amplitude that you specify to calculate the ball size and power loss factor.

  Use this model unless you have a specific reason to not use it.

- Modified Hammerstad

  The “Factor” field on the **Manufacturing** tab of the Stackup Editor sets the upper limit of the power loss factor for the simulation model. Use the default power loss factor value of 2 unless you have a specific reason to use a different value.

- Hammerstad — Similar to Modified Hammerstad.

  Note

  HyperLynx SI/PI 9.4.2 and older used this model. Its power loss factor has an upper limit of 2.

Technology

The Technology parameter defines dielectric layer properties related to PCB fabrication.

Note

The Stackup Editor does not use values from the Technology column. The Technology column is provided so you can document the stackup more completely.

- Core — Material is rigid prior to baking and typically serves as the foundation of a PCB during fabrication.
- Flex Core — Material is flexible before and after baking and typically serves as the foundation of a flexible area of a PCB during fabrication.
- Prepreg — Material is semi-soft and must be baked to become rigid. Signal traces can sink a little into prepreg dielectric layers prior to baking.

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Temperature Coefficient

Every signal and plane layer must have a Temperature Coefficient value for the metal layer material. The Stackup Editor uses the Temperature Coefficient value in conjunction with the Bulk Resistivity value to calculate DC resistances for trace segments on each layer.

The Temperature Coefficient default value applies to copper.

Temperature Coefficient is considered an advanced parameter, one that you normally do not need to change. Signal and plane layers automatically default to the temperature coefficient of the metal selected in the Metal column. To edit the Temperature Coefficient value, set the Metal parameter to <Custom>.

Thermal Conductivity

The default value for thermal conductivity for dielectric layers applies to FR-4 material. The default value for metal layers applies to copper. If you change the dielectric technology or metal material, the thermal conductivity value does not update automatically; you must enter the appropriate thermal conductivity value for the dielectric or metal you choose for the layer.

Note

HyperLynx SI and Thermal do not use the thermal conductivity values in the Stackup Editor.

Usage

Metal Usage parameters include Signal, Solid Plane, and Plating (used only for outer metal layers). Split/Mixed and Flooded Signal parameters are available when you open the Stackup Editor by an Xpedition Enterprise product.

Dielectric Usage parameters include Substrate and Solder Mask.

- Substrate — The layer is not flexible and has a flat profile, even when it covers raised features such as signal traces or components. A substrate layer has a variable thickness and a thinner cross section where it covers high spots on the board.
- Solder Mask — The layer is a protective material that prevents damage to traces on the outside of a board. It tends to have a bumpy profile where it covers raised features such as signal traces or components or fills metal voids. A solder mask layer has a relatively uniform thickness over the entire board surface.

Additional dielectric Usage parameters are available for board designs with flexible stackup areas.

- Adhesive — The layer bonds one layer to another. For example, it can bond a metal layer to a Flex Substrate or Cover Layer.

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- Cover Layer — The layer is a thicker form of Solder Mask.
- Flex Substrate — The layer is flexible.
- Stiffener — The layer is rigid and on the outside of a board. For example, if you want to mount a component to a flexible area in the board design, add a Stiffener layer to the area below the component. In Xpedition Layout, you use a Stiffener object to define the location of a Stiffener layer.

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Appendix C
Creating and Editing IBIS Models

You can create, edit, verify, and maintain IBIS (I/O Buffer Information Specification) device models with the HyperLynx® Visual IBIS Editor. The editor includes IBIS syntax checkers, a graphical waveform view, and an IBIS model wizard.

Topic	Description
Verifying IBIS Models	You can use the Visual IBIS Editor to verify an IBIS (IO Buffer Information Specification) model. IBIS models that you create or download from an IC vendor may have defects in them that prevent an IBIS simulator from running or yielding the expected results.
Checking IBIS File Syntax	You can perform syntax and limited data checking on the IBIS model without leaving the editor. If you are creating or editing a model, you can periodically check the syntax of the model to see if you have introduced errors.
Correcting V-T and V-I Table Mismatches Automatically	A common syntax-checking warning that the IBIS committee parser reports is a mismatch between the endpoint DC voltages in a V-T table and the endpoint DC voltages predicted by the intersection of the load line and the V-I tables.
Viewing V-I and V-T Curves	You can display V-I or V-T table data as a curve, which helps you to find errors in the data, such as a mistyped number or bad sign.
Graphically Editing V-I and V-T Curves	You can modify V-I and V-T table data of an IBIS model by adding, removing, or moving data points in the curve data.
Troubleshooting IBIS Models	You can troubleshoot common problems you encounter when you check an IBIS model or when you view the model waveforms and curves.
Creating IBIS Models with the Easy IBIS Wizard	The Easy IBIS Wizard is a model-generation utility that automatically generates an IBIS model from the component characteristics you provide. The wizard generates an IBIS model that is syntactically correct and ready to simulate.
Removing Initial Delays from IBIS Models	You can reduce simulation run time or correct models that contain unnecessary initial delay using the Remove Initial Delays dialog box.
Initial Delay Removal Algorithm	It is important to understand each of the methods for removing initial delay before deciding to modify your IBIS models.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
Guidelines for Documenting and Printing IBIS Models	Document your IBIS models, especially if the IBIS committee parser reports warning messages.
Graphical User Interface	It is helpful to understand the overall interface of the Visual IBIS Editor.
Editor Operations	The editor panes are linked so you can quickly display or edit data in another pane. You can also customize the appearance of the editor.
View IBIS Data Dialog Box	The graphical viewer displays the table data as a set of curves. If data is available, the min, typ, and max curves appear in different colors. The graphical viewer scales itself automatically to best fit the data.

Verifying IBIS Models

You can use the Visual IBIS Editor to verify an IBIS (IO Buffer Information Specification) model. IBIS models that you create or download from an IC vendor may have defects in them that prevent an IBIS simulator from running or yielding the expected results.

The main process to verify an IBIS model includes:

1. Checking IBIS File Syntax
2. Viewing V-I and V-T Curves
3. Troubleshooting IBIS Models
4. Graphically Editing V-I and V-T Curves

Checking IBIS File Syntax

You can perform syntax and limited data checking on the IBIS model without leaving the editor. If you are creating or editing a model, you can periodically check the syntax of the model to see if you have introduced errors.

The editor includes the official EIA-656 (IBIS) validation-checking program (IBIS committee parser) and the HyperLynx parser.

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Procedure

1. Run the syntax check on the IBIS file using one or more of the following methods:

If you want to...	Do the following...
Use the IBIS committee parser	Click or choose the **IBIS > Check Syntax (committee parser)** menu item.
Use the HyperLynx parser	1. Set parser options with the **IBIS > Set HyperLynx Parser Options** menu item. 2. Click or choose the **IBIS > Check Syntax (HyperLynx parser)** menu item.

Note

To toggle warnings in the output window, click . Errors are always displayed.

2. Review errors and warnings in the output window.

   If you used the IBIS committee parser, you can cross probe from any error/warning in the output window that includes a line number to the line in the model file.

If you want to...	Do the following...
Go to a specific error/warning	In the output window, double-click the error/warning.
Go to the next error/warning	Choose the **IBIS > Go to Next Error** menu item.
Go to the previous error/warning	Choose the **IBIS > Go to Previous Error** menu item.

3. Resolve errors and warnings.

   a. Fix all IBIS errors.

      IBIS errors can cause simulators to not run or to create incorrect results.

   b. Review and judge all IBIS warnings, cautions, and notes. Fix if required. Document acceptable warnings in the [Notes] section of the IBIS file.

      IBIS warnings can cause simulators to create incorrect results, but warnings are generally accepted by simulators.

4. Rerun the IBIS parsers until there are no errors and only warnings that are both acceptable and documented.

Results

You are now ready to view the V-I curves and V-T waveforms.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

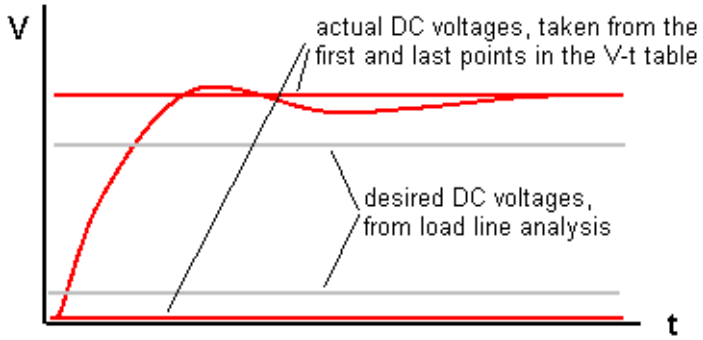
Related Topics

Viewing V-I and V-T Curves

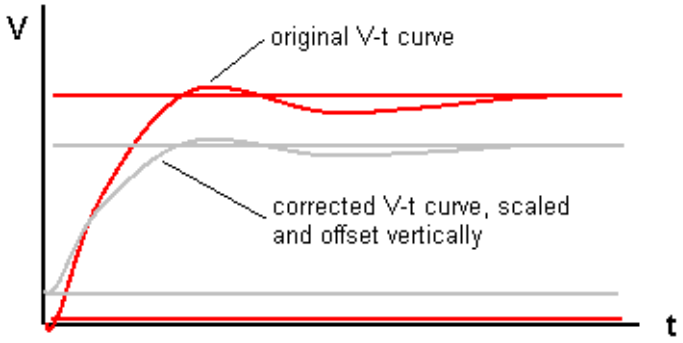
Correcting V-T and V-I Table Mismatches Automatically

A common syntax-checking warning that the IBIS committee parser reports is a mismatch between the endpoint DC voltages in a V-T table and the endpoint DC voltages predicted by the intersection of the load line and the V-I tables.

The following illustrates how the endpoint DC voltages from the V-T table do not match the endpoint DC voltages predicted by the intersection of the load line and V-I tables:



The following illustrates how the editor scales and vertically offsets the original V-T curve to create the corrected V-T curve:



Prerequisites

- Before using the editor to automatically correct mismatches, you should understand why the mismatches occurred. See V-T and V-I Table Data are Mismatched.
  - Look for reactance in the V-T data extraction process.
  - Check the voltage references used in the V-I data sweeps.

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- You have enabled this licensed (visualibis) feature. (Choose the **Edit > Preferences** menu item, then check “Enable licensed features”.)

Procedure

Correct the mismatch with any of the following methods.

If you want to....	Do the following...
Fix a single model	In the tree-view pane, right-click the model and choose the **Correct I-V/V-T Mismatches** menu item.
Fix all models in the IBIS file	Choose the **IBIS > Correct All I-V/V-T Mismatches** menu item.

Viewing V-I and V-T Curves

You can display V-I or V-T table data as a curve, which helps you to find errors in the data, such as a mistyped number or bad sign.

The graphical display shows the min, typ, and max curves, if available, each in a different color. The display scales automatically to best fit the table data.

Procedure

1. In the tree-view window, select a [Model], a pin, or a table keyword, and then select either “View data” from the popup menu or choose the **IBIS > View Data** menu item.
2. In the View IBIS Data dialog box, click a tab and view the curves using any of the following methods:

If you want to...	Do the following...
Zoom to a selected area on a curve	1. Click  or select **Zooming** from the popup menu. 2. Select the area you want to enlarge.
Pan	1. Click  or select **Panning** from the popup menu. 2. Move the curves to the desired position.
Display exact values for points on a curve as you move the cursor	1. Click or select **Track Cursor** from the popup menu. 2. Click the curve. 3. Move the pointer and view data points in the lower-right corner of the window.

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If you want to...	Do the following...
Measure between two points on a curve	1. Click  or select **Measure** from the popup menu. 2. Click the curve. 3. Click the first measurement point. A pair of measurement crosshairs attaches to the measurement point. 4. Click the second measurement point. The Measurement Results dialog box displays the measurement data.
Fit the curve to the window	Click  or select **Fit to window** from the popup menu.
Fit the curve to user-defined extent	1. Click  or choose the **View > Extents** menu item. 2. Enter the values for the bottom-left and upper-right corners of the extent.

Related Topics

View IBIS Data Dialog Box

Graphically Editing V-I and V-T Curves

You can modify V-I and V-T table data of an IBIS model by adding, removing, or moving data points in the curve data.

Note

You must have a license (visualibis) to edit and save data that you have modified in the editor (choose the **Edit > Preferences** menu item and check “Enable licensed features”.

Procedure

1. Display the curve you want to edit. If necessary, zoom in to the portion of the curve you want to change.

   Note

   Display curves as points only ().

2. Click  or select **Editing** from the popup menu.

3. Click a curve.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

4. Select the editing mode and modify the curve:

If you want to...	Do the following...
Move points	1. Select **Move Point** from the popup menu or click  . 2. If required, select move options from the popup menu: • **Rubber Curve** — When you move a point, other points move to maintain a smooth curve. • **Vertical Motion** — You can move the point only vertically. 3. Click points and drag them.
Insert points	1. Select **Insert Point** from the popup menu or click  . 2. Click to insert points.
Move points to specific locations	1. Select **Exact Position** from the popup menu or click  . 2. Click points, then enter values for the new position.
Delete points	1. Select **Delete Point** from the popup menu or click  . 2. Click points to delete them.

5. To edit points on another curve, choose **Select** from the popup menu, then click the curve you want to edit.
6. If needed, repeat Steps 4-5.
7. In the View IBIS Data dialog box, choose the **File > Save** menu item, then choose the **File > Close** menu item.
8. In the editor, save the IBIS file.

Results

If you change IBIS data, you should recheck the IBIS file syntax.

If you make changes that produce endpoint DC voltage mismatches between the V-I and V-T tables, the editor prompts you whether to correct the mismatches automatically. See V-T and V-I Table Data are Mismatched.

Related Topics

Checking IBIS File Syntax

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Troubleshooting IBIS Models

You can troubleshoot common problems you encounter when you check an IBIS model or when you view the model waveforms and curves.

Symptom	Description
Table Data Has the Wrong Sign	Check that the data has the correct sign. Sign errors in [Pullup] data occur frequently.
Table Data Has the Wrong Units	Check that the scaling factor was correctly entered for both the X-axis and the Y-axis.
IBIS Model Exhibits Unexpected Noise	Check the waveforms for non-monotonicity. The test hardware and setup can introduce noise or other artifacts into the data table. These artifacts may not represent how the IO buffer switches in a “clean” environment. Models that contain table data with noise or other artifacts of the test setup can lead to simulation output waveforms that are artificially noisy.
One Curve of Typ-Min-Max is Non-Monotonic	Check curves for non-monotonicity in a single curve when multiple curves exist. Individual data points can have the wrong value or the wrong scaling.
Model Has Typ-Min-Max Data Incorrectly Ordered	Check that the columns of data columns are ordered correctly. IBIS requires the order typical, minimum, then maximum.
V-I Data Does Not Pass Through the Origin	Check that V-I curves pass through the origin (zero volts and zero amperes).
Simulation Tools Report Missing V-I and V-T Tables	Check that required V-I and V-T tables exist in the model. The IBIS model for an output buffer must contain the [Pullup] and [Pulldown] tables, and optionally the [Rising Waveform] and [Falling Waveform] tables.
V-T and V-I Table Data are Mismatched	Check that the DC behavior of an output or bidirectional buffer at the start and end of a transition from high to low or low to high is consistent with the V-I table for the same load.
Paired Curves Do Not Have the Opposite Polarity	Check that the data for paired curves has opposite polarity.
Ramp Table Data Have Zero or Negative Values	Check that [Ramp] data contains positive time values for the dV/dt_r and dV/dt_f subparameters. A zero or a negative value is incorrect and can cause problems during simulation.

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Symptom	Description
Vmeas Voltage Does Not Cross V-I Data	Check that the Vmeas subparameter for the [Model] defines a voltage that crosses the V-I table data. The IBIS committee parser reports a syntax warning when the IO buffer cannot drive past Vmeas using the specified Rref and Vref subparameter values.

Table Data Has the Wrong Sign

Check that the data has the correct sign. Sign errors in [Pullup] data occur frequently.

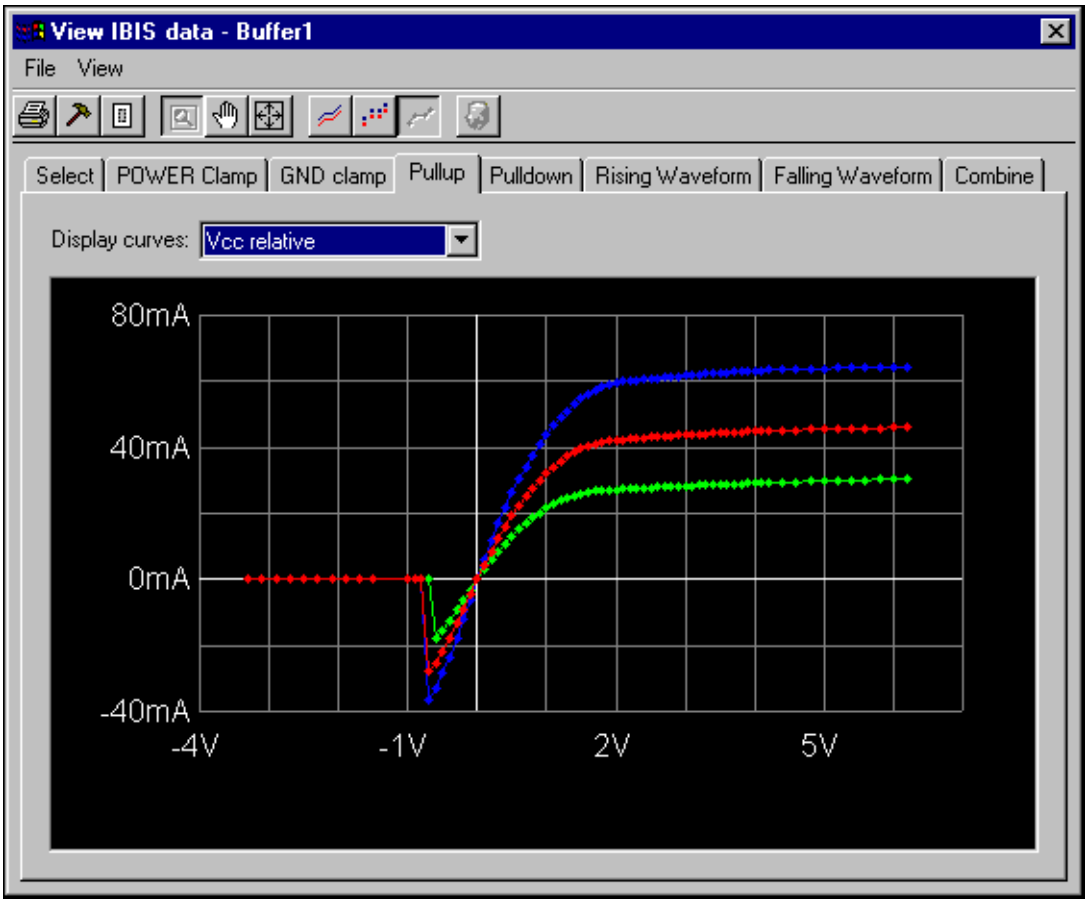
What to look for

- The IBIS committee parser reports an increasing current for [Pullup]. For example:

  ```
  WARNING - Model Buffer1: Pullup has Increasing Current
  ```

- The curve shows a positive current at voltages where a negative current is expected.

  The IBIS specification defines positive current as flowing into the component.



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Description

Recheck how the data for the [Pullup] table was collected. For example, was the negative curve tracer probe connected to OUT to collect [Pullup] data?

The voltage in a [Pullup] table is referenced to VCC. A positive [Pullup] voltage value represents an offset toward ground.

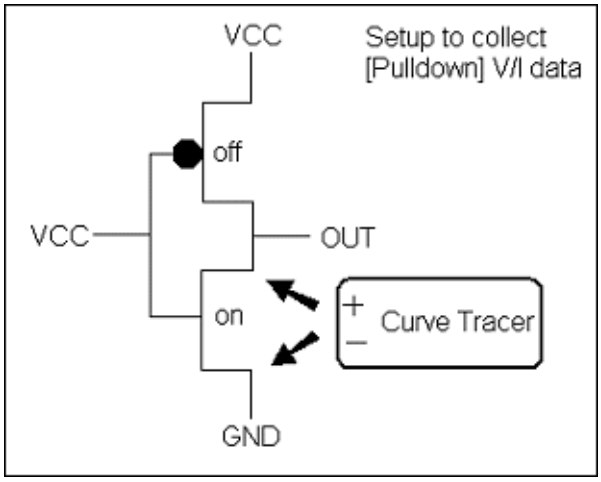
The equation to convert the [Pullup] table voltage to a ground-referenced value is:

```
VCC - [Pullup] table voltage = ground-referenced voltage
```

For example if VCC=3.3V and [Pullup] table voltage=1.3V, the ground-referenced voltage is 2V.

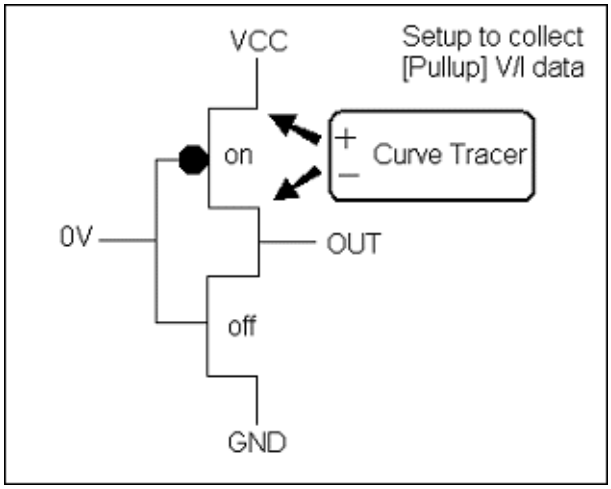
The following figures show how the connections between the curve tracer and the CMOS driver change when collecting [Pullup] data versus [Pulldown] data.

Connect the negative curve tracer probe to GND to collect [Pulldown] data.



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Connect the negative curve tracer probe to OUT to collect [Pullup] data.



The following shows a [Pullup] V-I table fragment and the calculated pin voltage when VCC = 3.3V.

-2V	+1000mA	3.3V - (-2V) = 5.3V
-1V	+100mA	3.3V - (-1V) = 4.3V
0V	0mA	3.3V - 0V = 3.3V
+1V	-20mA	3.3V - 1V = 2.3V
+2V	-35mA	3.3V - 2V = 1.3V

Related Topics

Viewing V-I and V-T Curves

Checking IBIS File Syntax

Table Data Has the Wrong Units

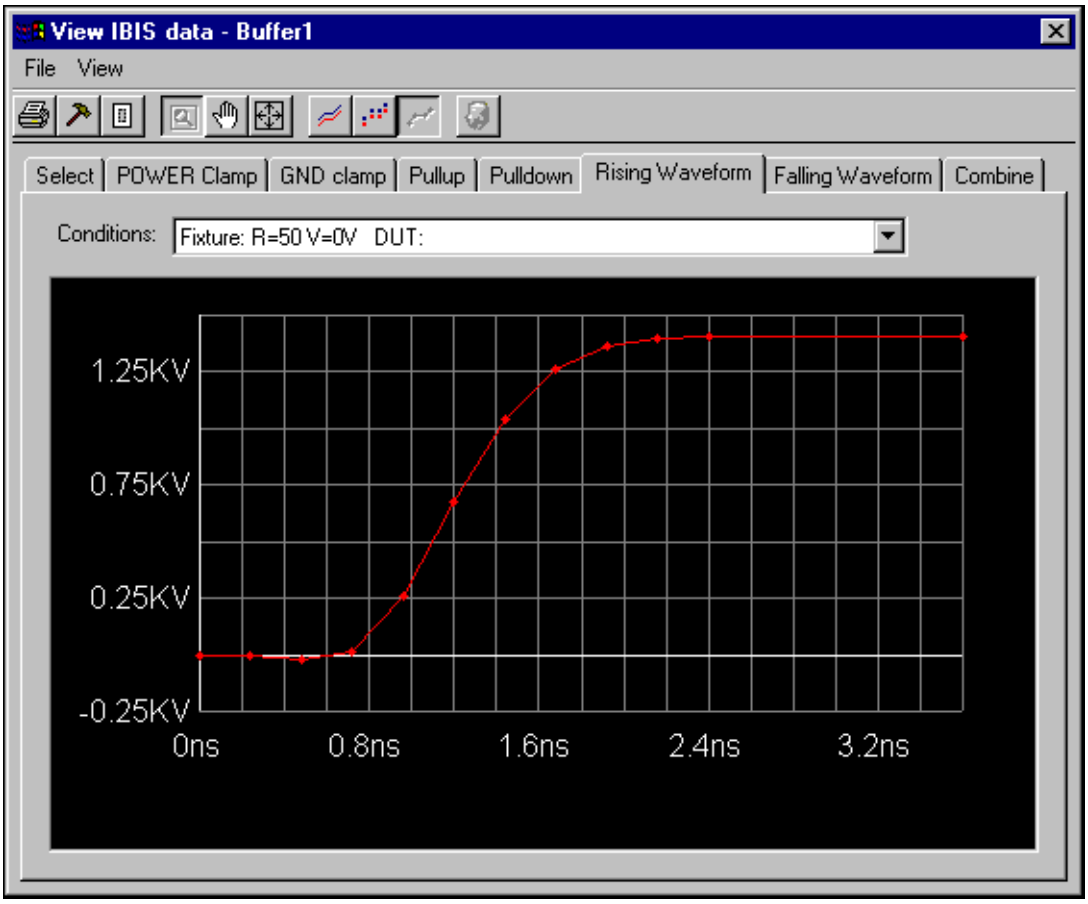
Check that the scaling factor was correctly entered for both the X-axis and the Y-axis.

What to look for

The following example shows [Rising Waveform] data with an amplitude exceeding 1250V. Assuming the model represents a 3.3V product, the amplitude is too large. This error can result

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from a scaling factor being dropped, such as the omission of “mV” in the [Rising Waveform] table.



Related Topics

Viewing V-I and V-T Curves

IBIS Model Exhibits Unexpected Noise

Check the waveforms for non-monotonicity. The test hardware and setup can introduce noise or other artifacts into the data table. These artifacts may not represent how the IO buffer switches in a “clean” environment. Models that contain table data with noise or other artifacts of the test setup can lead to simulation output waveforms that are artificially noisy.

What to look for

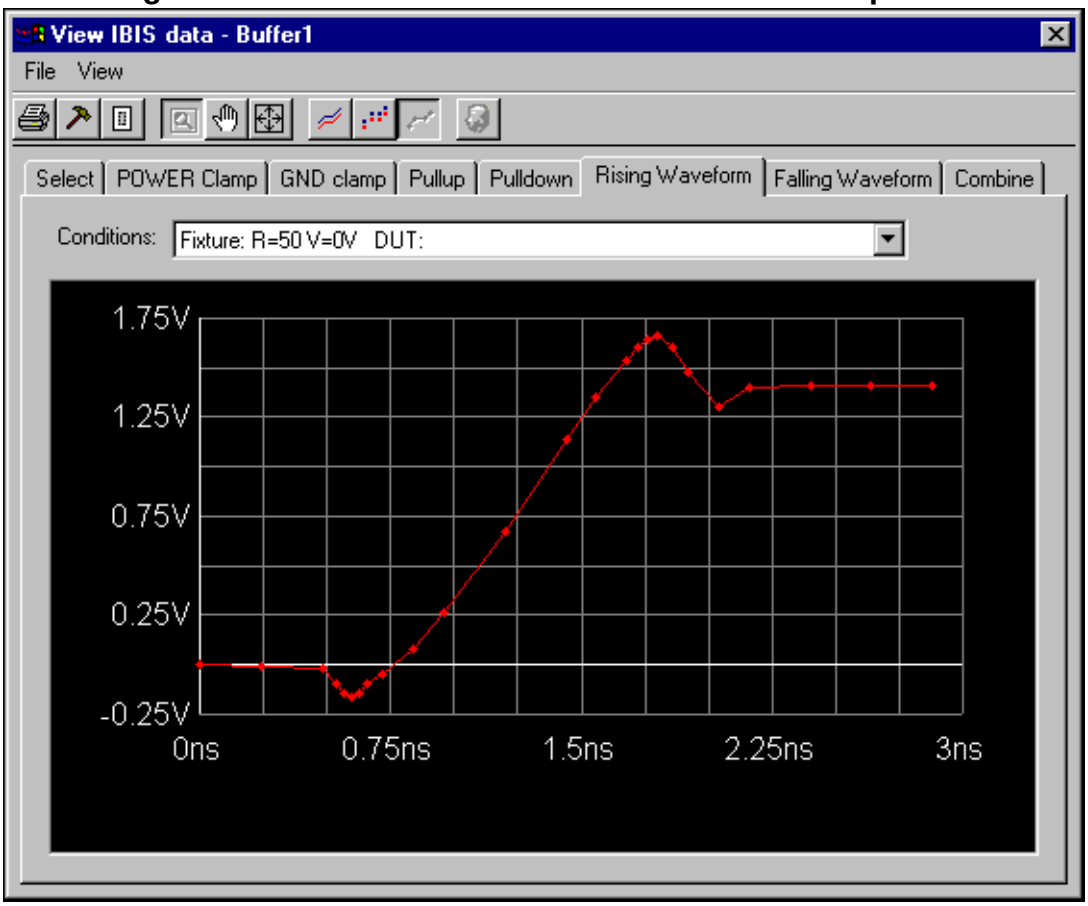
- The IBIS committee parser reports non-monotonic data. For example:

```
WARNING (line 80) - Pulldown Typical data is non-monotonic
WARNING (line 80) - Pulldown Minimum data is non-monotonic
WARNING (line 80) - Pulldown Maximum data is non-monotonic
```


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- The waveform shows non-monotonistic or ringing characteristics. For example, The following shows a non-monotonic dip near 500ps and ringing that starts near 1.75ns.

Figure C-1. IBIS Model Error - Data with Noise Properties



Description

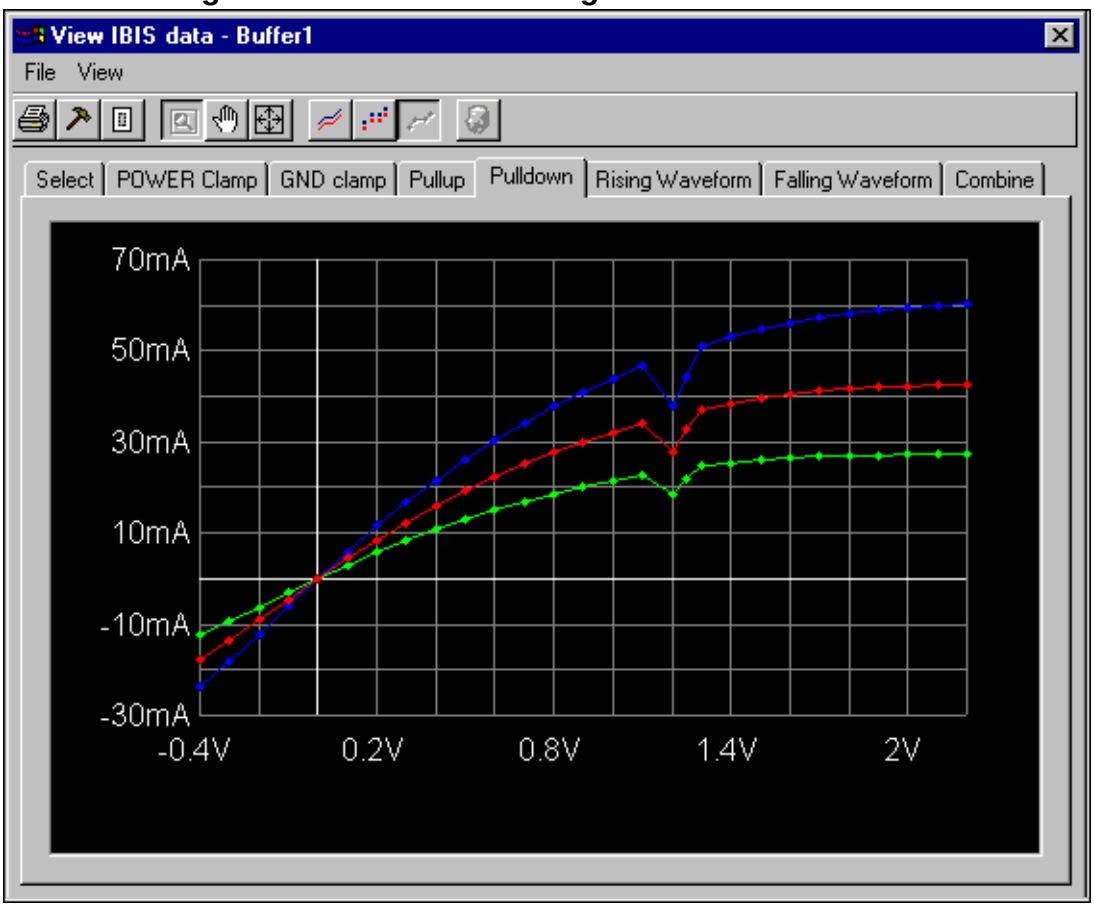
Non-monotonic data requires additional investigation.

Figure C-2 shows a non-monotonic dip near 1.2V. This may be acceptable because most simulators try to clean the data. However, these negative-resistance regions can cause simulation problems and probably do not reflect real device behavior. (Negative-resistance describes a segment of a V-IV-I curve where the current shifts toward 0mA while voltage shifts away from 0V.)

Re-examine Figure C-1. Does it represent non-monotonic data? Yes. If the data sign values were corrected, would it be a problem? No. Figure C-1 is an example of non-monotonic data resulting from incorrect separation of the clamp current and output driver current. The simulator will add these two curves back together so they are not a problem, unlike the non-monotonic dip in Figure C-2.

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Figure C-2. Curves Showing Non-Monotonic Data



Related Topics

Viewing V-I and V-T Curves

Checking IBIS File Syntax

One Curve of Typ-Min-Max is Non-Monotonic

Check curves for non-monotonicity in a single curve when multiple curves exist. Individual data points can have the wrong value or the wrong scaling.

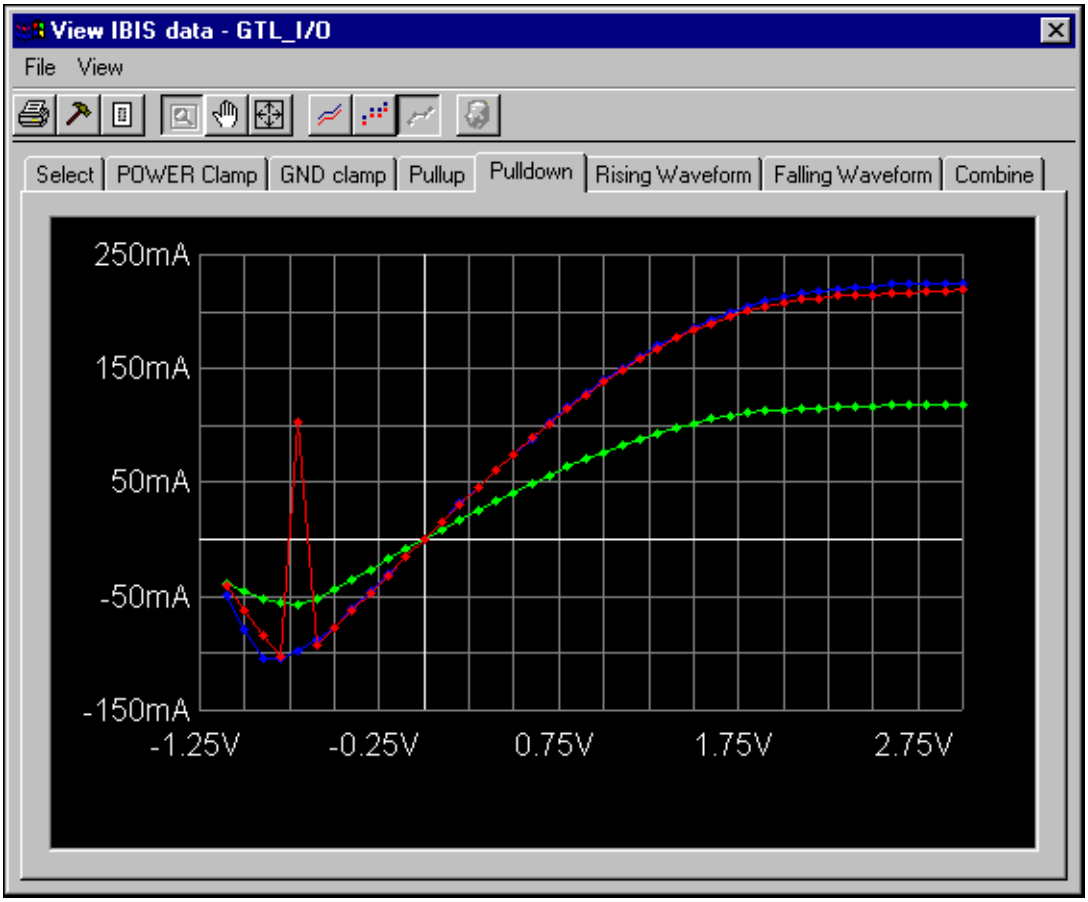
What to look for

- The IBIS committee parser sometimes reports spurious data points as non-monotonic data. For example:

```
WARNING (line 465) - Pulldown Typical data is non-monotonic
```


Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- The curve shows a bad data point on one curve.



Related Topics

Viewing V-I and V-T Curves

Checking IBIS File Syntax

Model Has Typ-Min-Max Data Incorrectly Ordered

Check that the columns of data columns are ordered correctly. IBIS requires the order typical, minimum, then maximum.

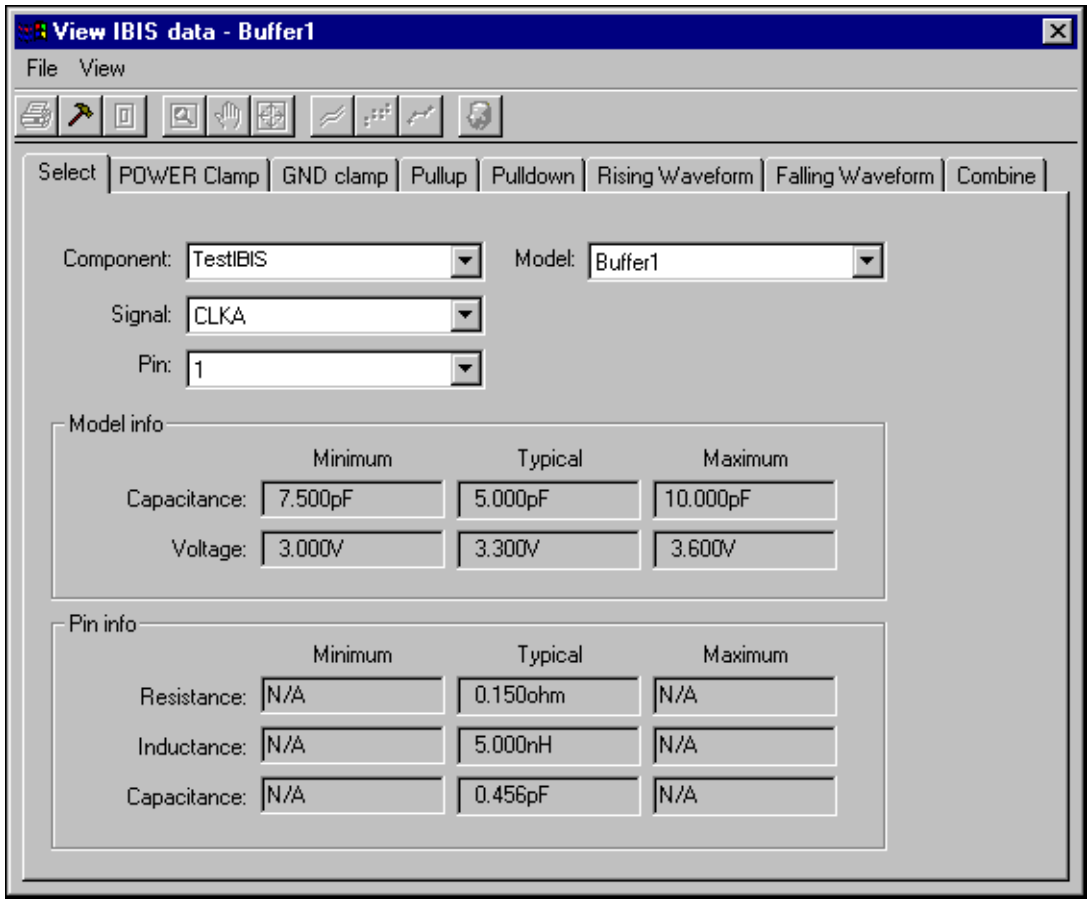
What to look for

- The IBIS committee parser reports that a value is not in the correct order. For example:

```
WARNING (line 56) - Typ value is not in between min and max
```


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- The View Data dialog box, **Select** tab shows that the values in the Model Info area are not in the correct order. The typical capacitance is less than the minimum capacitance, which is incorrect.



Related Topics

Viewing V-I and V-T Curves

Checking IBIS File Syntax

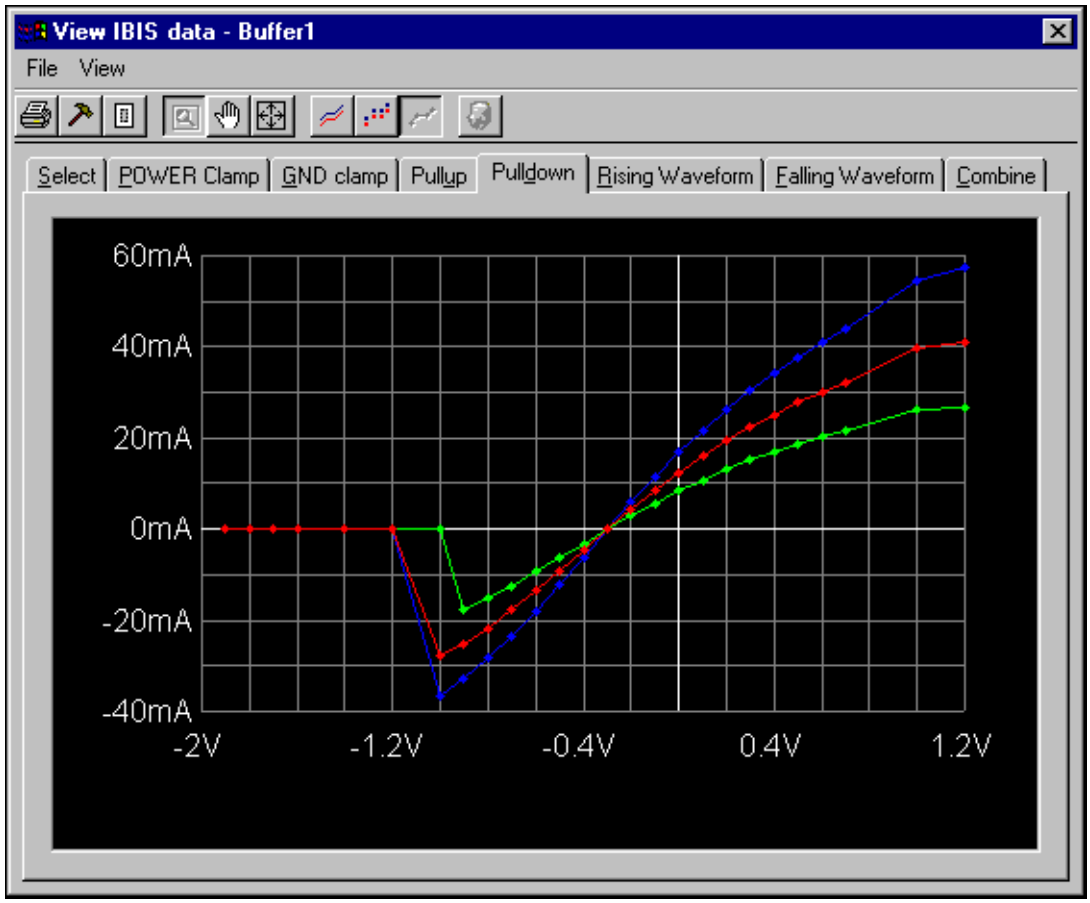
V-I Data Does Not Pass Through the Origin

Check that V-I curves pass through the origin (zero volts and zero amperes).

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What to look for

- The V-I curve has non-zero current at 0V. For example, the following shows V-I curves with zero current at about -0.3V.



Related Topics

Viewing V-I and V-T Curves

Simulation Tools Report Missing V-I and V-T Tables

Check that required V-I and V-T tables exist in the model. The IBIS model for an output buffer must contain the [Pullup] and [Pulldown] tables, and optionally the [Rising Waveform] and [Falling Waveform] tables.

What to look for

- HyperLynx LineSim® or BoardSim® reports that an output or bidirectional model does not contain the required model tables.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

V-T and V-I Table Data are Mismatched

Check that the DC behavior of an output or bidirectional buffer at the start and end of a transition from high to low or low to high is consistent with the V-I table for the same load.

What to look for

- The IBIS committee parser reports the DC endpoints differ from the equivalent load applied to the I-V tables. For example:

```
WARNING-Model Buffer1: The [Rising Waveform]
with [R_fixture]=50 Ohms and [V_fixture]=0V
has TYP column DC endpoints of 0.00V and 2.47v, but
an equivalent load applied to the model's I-V tables yields
different voltages ( 0.00V and 2.58V),
a difference of 0.00% and 4.41%, respectively.
```

Description

If the IBIS committee parser reports this warning you should:

- Look for reactance in the V-T data extraction process.
- Check the voltage references used in the V-I data sweeps.

The IBIS committee parser uses load line analysis to correlate V-T and V-I table data. Each V-T table specifies V_fixture and R_fixture parameters that can be converted to a load line. When the parser calculates the intersections between the load line and V-I curves, it recognizes the intersections to predict the starting and ending DC voltages (or “endpoints”) in the V-T tables. The V-I and V-T data are mismatched when load line analysis predicts a DC voltage that is over 2% different from the V-T table endpoint.

Most simulators tolerate mismatches greater than 2%. Mismatches between 0%-10% are accepted by most simulators. Larger mismatches can cause simulation problems.

IBIS models may have multiple [Rising Waveforms] or [Falling Waveform] V-T tables. For example, the V_fixture value may be 3.3V in one [Falling Waveform] table and 0V in another [Falling Waveform] table. The parser checks all the V-T tables for mismatches with the V-I tables.

For a more detailed description, see “I-V and V-T Matching” in the *IBIS Modeling Cookbook* (prepared by the IBIS Open Forum).

Related Topics

Correcting V-T and V-I Table Mismatches Automatically

Viewing V-I and V-T Curves

Checking IBIS File Syntax

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Paired Curves Do Not Have the Opposite Polarity

Check that the data for paired curves has opposite polarity.

- Pullup and Pulldown
- Rising Waveform and Falling Waveform
- POWER Clamp and GND Clamp

Related Topics

Viewing V-I and V-T Curves

Ramp Table Data Have Zero or Negative Values

Check that [Ramp] data contains positive time values for the dV/dt_r and dV/dt_f subparameters. A zero or a negative value is incorrect and can cause problems during simulation.

The IBIS committee parser does not check for zero or negative dV/dt_r and dV/dt_f subparameter values. However, HyperLynx LineSim and BoardSim detect this problem.

Related Topics

Viewing V-I and V-T Curves

Vmeas Voltage Does Not Cross V-I Data

Check that the Vmeas subparameter for the [Model] defines a voltage that crosses the V-I table data. The IBIS committee parser reports a syntax warning when the IO buffer cannot drive past Vmeas using the specified Rref and Vref subparameter values.

What to look for

- The IBIS committee parser reports a warning, for example:

```
WARNING - Model 'PCI': TYP VI curves cannot drive through Vmeas=1.5V
given load Rref=40 Ohms to Vref=0V
```

Related Topics

Viewing V-I and V-T Curves

Checking IBIS File Syntax

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Creating IBIS Models with the Easy IBIS Wizard

The Easy IBIS Wizard is a model-generation utility that automatically generates an IBIS model from the component characteristics you provide. The wizard generates an IBIS model that is syntactically correct and ready to simulate.

Note

You must have a license (visualibis) to use the wizard. (Choose the **Edit > Preferences** menu item, then check “Enable licensed features”.)

Restrictions and Limitations

- There is no direct support for ECL or pseudo-ECL models. To generate such a model, consider the following options:
  - Start with a standard technology type (such as CMOS), then modify the resulting IBIS file to make it work for your ECL buffer. (See the IBIS specification for details on the changes needed.) The CMOS model will be incorrect in some important respects (such as table reference voltages and model type). However, the model provides you a “skeleton” model to work with.
  - Find an existing ECL IBIS model and modify it with the main editor. (Do not use the Easy IBIS Wizard).

    The IBIS syntax allows most numerical values without limits, but IBIS simulators will sometimes have trouble with very large or small values. The wizard may not warn you about unreasonable values and could generate a model that some simulators may not correctly simulate.
- The wizard creates IBIS v1.1 models; it does not create models that contain features from later versions of the IBIS language, such as V-T tables.

Procedure

1. Choose the **IBIS > Easy IBIS Wizard** menu item. The wizard opens.

   Note

   Click individual fields to display hints in the Hint area.

2. On the “Starting Information” page, enter basic information for the IBIS model or open an existing model.

   For a new model, enter a specific name. Typically it includes manufacturer, device, and package information.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

If you open an existing model not previously created with the wizard, you must manually add the pin model information into the wizard (Step 7). The wizard does not read pin models from an existing IBIS model. Instead, the pin model information that you enter into the wizard is stored elsewhere.

3. On the “General Information” page, enter header information.

   The typical copyright form is “Copyright <year>, <manufacturer name>, All Rights Reserved”.

4. On the “Source Information” page, describe how the data was derived. For example, measured or simulated. These fields are limited to 80 characters.

5. On the “Notes” page, enter contact details and notes. The Notes field is limited to 80 characters.

   Notes should include clarifying detail about the model that is not already captured in the other fields. Typical information includes caveats about data missing from the model, descriptions of when the model is most accurate (and when it is not), and so forth.

6. On the “Device Package” page, enter the package, pin count, and default parasitics.

7. On the “Edit Pins” page, enter pin-specific data. The pin-specific data overrides the default parasitics defined in the previous step. Use the following operations to edit the spreadsheet.

If you want to...	Do the following...
Edit an individual cell	Click the cell and enter/select a data item.
Assign a data item to multiple pins	1. Select multiple pins using the Ctrl or Shift keys. 2. In the Group Edit field, select the data column you want to edit, for example: Models. 3. Select or enter the data value, then click **Apply**.

8. If required, edit, copy, add, or delete pin models:

   a. Click **Edit Models**.

   b. In the “Edit Models” page, select the model and the type of operation to perform.

   c. In the “General Information” page, enter or modify the basic information for the pin model.

   d. In the “Operating Parameters” page, enter or modify the operating voltage, die capacitance, thresholds, and scaling factors.

   e. In the “Clamp Diodes” page, set up or modify the clamps. You can also create custom clamps.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

Check the IC data sheet for the type and strength (strong, typical, weak) of clamp diode to specify. If no information is available, select Silicon Typical.

f. On the “Output High Parameters” page, enter or modify parameters for output, I/O, tristate, and open sink buffers.

g. On the “Output Low Parameters” page, enter or modify parameters for output, I/O, tristate, and open source buffers.

h. On the “Load and Logic” page, enter or modify the output polarity and the load circuit data for output and I/O buffers.

i. Click **Finish** to close the Edit Models dialog box.

9. click **Finish** to save your model and close the Easy IBIS Wizard.

Related Topics

Guidelines for Documenting and Printing IBIS Models

Viewing V-I and V-T Curves

Removing Initial Delays from IBIS Models

You can reduce simulation run time or correct models that contain unnecessary initial delay using the Remove Initial Delays dialog box.

The software can remove the initial non-switching time from V-T and I-T waveform tables in an IBIS model.

Prerequisites

- Understand how the software removes initial delays from your IBIS model. See Initial Delay Removal Algorithm.

Procedure

1. Choose the **IBIS > Remove Initial Delays** menu item. The Remove Initial Delays dialog box opens.
2. To account for the initial non-switching time in I-T tables when the software calculates the initial delays to remove, check “Allow using [Composite Current] section”.

   If you disable this option, the software uses only V-T tables to calculate the initial delays to remove from both V-T and I-T tables.
3. Select an option from the Correlation list:
   - Make All V-t Tables in a Model Time Correlated — Correlates all data in the columns of all V-T tables. Select this option unless you have a reason not to.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

- Process Rising and Falling V-t Independently — Correlates all [Rising Waveform] tables to each other, then correlates all [Falling Waveform] tables to each other.
- Process Corners Independently — Correlates min data in all tables to each other, correlates max data in all tables to each other, and then correlates typ data in all tables to each other.

4. If you want to additionally correlate [Composite Current] tables to the tables in the option you chose above, check "Allow using [Composite Current] section."
5. Specify the percentage of voltage and current (range: 1 to 20) that determines the switching threshold, then click **OK**.

Note

You can also remove initial delay from all IBIS models in a folder or a single IBIS model by running IBISVTC from a command line. Display the complete command line syntax by running IBISVTC with no options.

For example: *SDD_HOME/hyperlynx/IBISVTC*

Initial Delay Removal Algorithm

It is important to understand each of the methods for removing initial delay before deciding to modify your IBIS models.

The software provides the following algorithms to remove initial delay:

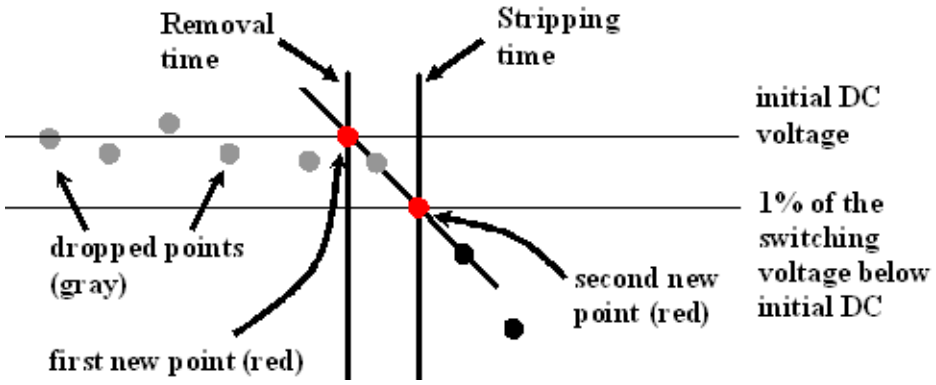
- a single curve (main algorithm)

  Note that you should understand the single curve algorithm before trying to understand the methods that apply to the other curves.
- all curves in all tables
- all rising curves and all falling curves
- all min curves, all typ curves, and all max curves

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Single Curve - Main Algorithm

The following graphic shows a falling-edge waveform with a slightly noisy initial DC non-switching time.



The software determines the initial delay and new start point for an individual curve as follows:

1. Calculates the initial DC voltage (with the first and last V-T table entries). (Top horizontal line in figure.)
2. Calculates the point (right red point in figure) where the curve crosses a 1% noise threshold from the initial DC voltage, and then removes all data points (gray points) before this new point.

   Note that you can modify the 1% noise threshold when you set up the software to remove initial delays.
3. Creates the new start point (left red point) by extrapolating linearly back to the initial DC voltage using the new 1% noise threshold voltage point and the next curve data point.
4. Adjusts the remaining data points to reflect the new time value of new starting point.

Note

The software applies a similar algorithm to determine the initial delay and new starting point for an I-T waveform in a [Composite Current] table. The software calculates the initial current as the first I-T table entry. The 1% noise threshold applies to the current range (maximum I-T table entry - minimum I-T table entry).

All Tables Correlated Method

The IBIS specification recommends that all V-T and I-T waveforms are correlated. If the switching waveform for one table is delayed relative to the waveform of another table, the delay is intentional and describes the true behavior of the device.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The software correlates the data as follows:

1. Calculates the initial switching delay of each curve in the model independently using the main (single curve) algorithm.

   The curve with the smallest switching delay determines the “global” removal time.

2. Deletes all data points before the “global” removal time and adjusts the remaining data points to reflect the new 0.0 time value.

Rising and Falling Table Treated Separately Method

Sometimes a model that is created from measured data has rising- and falling-edge tables that are not correlated. For example, a clocked oscilloscope waveform that contains a rising and then a falling edge may have been split to create the two waveform tables. It is possible that the raw time data from the second half of the waveform was copied directly into a V-T table or I-T table. A model created this way will not perform properly if initial delays are removed using the “All Tables Correlated” method. For this type of model, use the “Rising and Falling Treated Separately” correlation method.

Caution

This method can destroy asymmetric duty cycles and make eye diagrams overly optimistic.

The software correlates the data as follows:

1. Calculates the initial switching delay of each rising V-T curve and I-T curve (if it is present) in the model independently using the main (single curve) algorithm.

   The curve with the smallest switching delay determines the “global” removal time.

2. Deletes all data points before the “global” removal time and adjusts the remaining data points to reflect the new 0.0 time value.

3. Repeats the process for all falling curves in the model.

Corners Treated Separately Method

The software correlates all min curves in all tables, correlates all typ curves in all tables, then correlates all max curves in all tables. This process removes the unique initial delay for all min curves, for all typ curves, and for all max curves.

The software does not correlate delay between the min/typ/max curves when:

- Different test circuits/conditions are used for each of the min/typ/max curves.
- Part of the initial delay results from internal logic that is already accounted for in the model.

The software correlates the data as follows:

1. Calculates the initial switching delay of each min curve in the model independently.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The curve with the smallest switch delay determines the “corner-specific” removal time.

2. Deletes all data points before the “corner-specific” removal time and adjusts the remaining data points to reflect the new 0.0 time value.
3. Repeats the process for all typ curves, then all max curves.

   This creates gaps in the V-T and I-T table data because the data points in each min, typ, and max curve were adjusted independently (NA in graphic).

Time	V(typ)	V(min)	V(max)
0.00000E-11	5.08000E-06	6.24000E-06	3.95000E-06
8.00000E-11	-1.30000E-03	NA	NA
1.40000E-10	NA	1.38000E-02	NA
2.20000E-10	NA	NA	6.95000E-02
3.20000E-10	-1.16200E-02	NA	NA
3.80000E-10	NA	2.43150E-01	NA
4.60000E-10	NA	NA	7.75550E-01
...			

4. Creates data points to fill gaps by extrapolating surrounding values.

Related Topics

Removing Initial Delays from IBIS Models

Guidelines for Documenting and Printing IBIS Models

Document your IBIS models, especially if the IBIS committee parser reports warning messages.

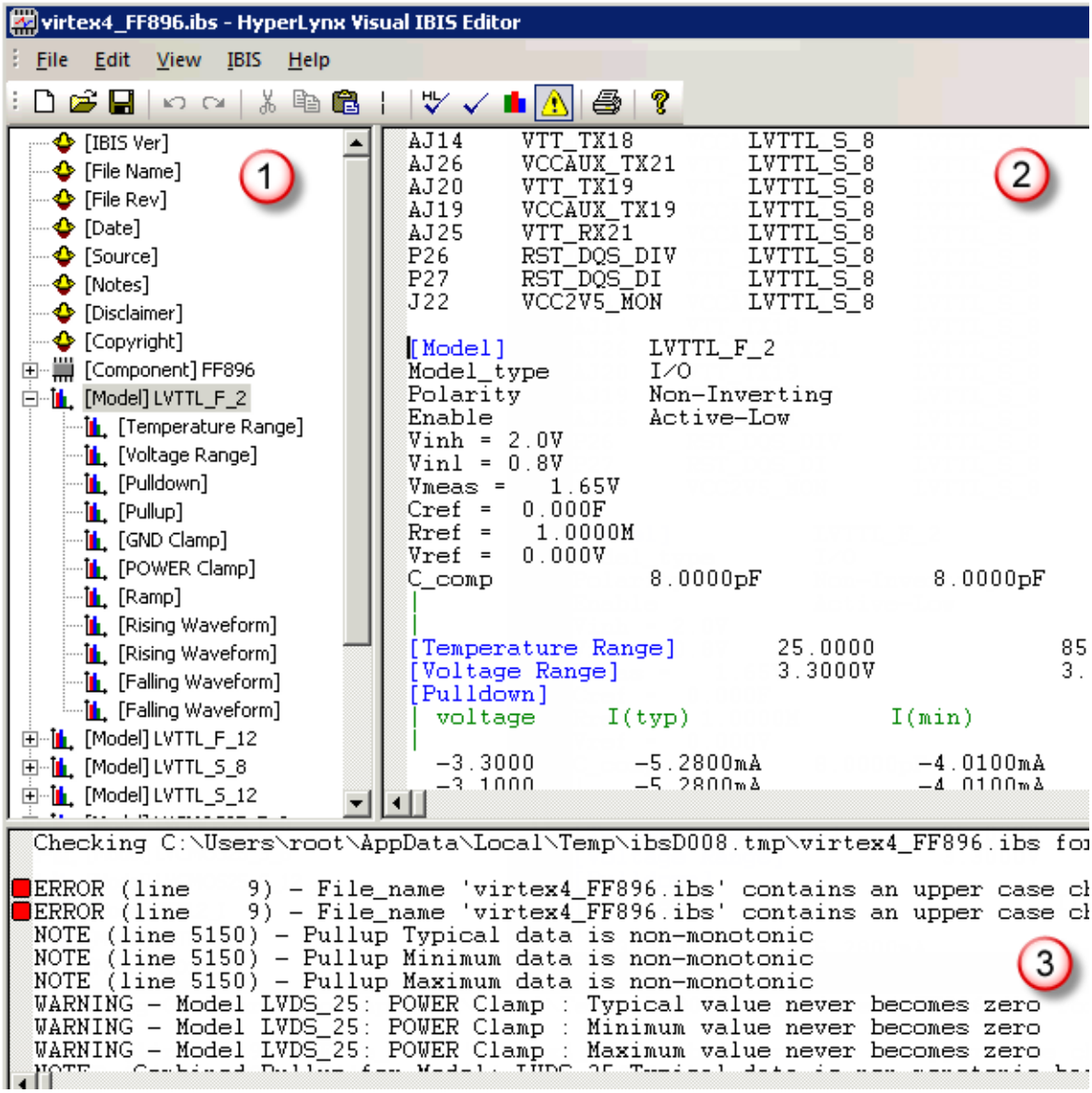
You can document or print IBIS models and waveforms as follows:

If you want to...	Do the following....
Document warnings produced during a syntax check, also test bench measurement information, and so forth	Explain each warning, caution, or other note in the [Notes] section of the IBIS model.
Print waveforms	In the View IBIS Data dialog box  , select the waveform tab you want to print, then click  or choose **File** > **Print**.
Print the IBIS model file	In the main editor, click  or choose **File** > **Print**.

Graphical User Interface

It is helpful to understand the overall interface of the Visual IBIS Editor.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.



The editor panes display the IBIS file content and structure.

Panes

Pane	Description
1 (Tree-view)	Displays the keywords in the IBIS file.
2 (Edit)	Defines the source text in the IBIS file. The edit pane displays keywords, comments, and other text in different colors.
3 (Output)	Displays the results of IBIS syntax checker or other messages.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Editor Operations

Editor Operations

The editor panes are linked so you can quickly display or edit data in another pane. You can also customize the appearance of the editor.

Operations

If you want to...	Do the following...
Display table data graphically	In the tree-view pane, right-click the keyword and choose **View data**.
Display the file contents for a specific keyword	In the tree-view pane, double-click the keyword or click the keyword, then choose **Go to Item** from the popup menu.
Synchronize the tree-view with the source file (edit window)	In the source file (edit pane), right-click on the line and choose **Synch Content** from the popup menu.
Find a specific pin or signal in the IBIS file	In the tree-view pane, select a component, pin, or signal and then choose **Find signal** or **Find pin** from the popup menu.
Jump to a line in the source file (edit window) that contains an IBIS syntax error	In the output pane, double-click a warning or error message (the message must contain a line number). **Tip:** Press **F4** (**IBIS > Go to Next Error**) or **Ctrl+F2** (**Go to Previous Error**) to traverse the errors.
Edit the IBIS file (edit pane)	
Select a rectangular block of text	Press **Alt** while selecting the area of text.
Find text in the IBIS file	1. Press **Ctrl+F** or choose **Edit > Find.** 2. Type the text and select options. **Tip:** Use **F3** to find the next text match.
Replace text in the IBIS file	1. Press **Ctrl+H** or choose **Edit > Replace.** 2. Type the text strings and select options.
Comment/uncomment lines in the IBIS file	In the edit pane, press **Ctrl+M** or click **Toggle comment** ( ).
Add or clear bookmarks	Choose **View > Toggle Bookmark** or **View > Clear bookmarks**.
Convert tabs to spaces in an IBIS file	Choose **Edit > Convert Tabs** and enter the number of spaces.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to...	Do the following...
Set tabs in an IBIS file	Choose **Edit** > **Preferences**, check **Keep tabs**, and enter the Tab size. **Note:** Although the IBIS specification allows tabs, you should avoid them because different tools expand tabs in different ways.
Editor Layout and Appearance	
Hide editor panes	Select or clear items in the View menu.
Set font and colors for the edit or output panes	Choose **Edit** > **Preferences**.

Related Topics

Graphical User Interface

Viewing V-I and V-T Curves

Graphically Editing V-I and V-T Curves

View IBIS Data Dialog Box

To access: In the HyperLynx Visual IBIS Editor, right-click a keyword in the Tree-View Pane and select **View Data**.

The graphical viewer displays the table data as a set of curves. If data is available, the min, typ, and max curves appear in different colors. The graphical viewer scales itself automatically to best fit the data.

Tip

To view data for a different table without leaving the View IBIS Data dialog box, click the **Select/Info** tab and then select a new value in the Component, Signal, Pin, or Model list.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.



Fields

Field/Tab	Description
Toolbar and field items	
1	Editing and Viewing buttons.
2	Editing Point buttons (available in Editing mode only).
3	Line Display and Preferences buttons.
4	Value of the current data point.
5	When in editing mode, the selected curve is highlighted in white.
Tabs	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field/Tab	Description
6 Select/Info	Defines the parameters of the currently selected Component, Model, Signal, and Pin. You can change any of these options to view different table data in the curve or waveform tabs.
POWER Clamp Pullup	**Display curves** — The IBIS specification requires V-I data for [Pullup] and [POWER Clamp] tables to be relative to VCC. However, you can view this data relative to ground by choosing **Ground relative** from the dropdown list.
Rising Waveform Falling Waveform	**Conditions** — Defines V-T tables for different test loads.
Composite Current (Rising/Falling)	Displays the edge current waveforms time-correlated to the Rising Waveform and Falling Waveform data.
Waveform AND Current	Displays the selected combination (sum) of clamp and V-I table data (if available) as a single curve. The curve shows the model behavior for the combined voltage and current ranges. You can view the data relative to ground by choosing **Ground relative** from the dropdown list.
Golden Waveforms	Displays the “golden” waveforms if the IBIS model contains this data. Use the waveforms to compare accuracy.The golden waveforms show actual measured or simulated waveforms with the model in a simple circuit. This provides a method to compare the simulation results with the vendor’s results.
Series Switch	Displays the current passing through a Series_switch model in either an On or Off state.
ISSO_PU ISSO_PD	Displays the current waveform as voltage is swept from -Vcc to Vcc. For ISSO_PU, you can view the data relative to ground by choosing from the dropdown list.

Related Topics

Viewing V-I and V-T Curves

Graphically Editing V-I and V-T Curves

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Appendix D
Viewing and Converting Touchstone and Fitted-Poles Models

Use the Touchstone and Fitted-Poles Viewer to understand the contents and judge the quality of Touchstone and fitted-poles models.

Topic	Description
Viewing and Measuring Model Curves	Seeing model data formatted as a curve can make it easier for you to analyze the model.
Viewing Insertion Loss, Return Loss, and Coupling for an S-Parameter Model	You can view insertion loss, return loss, and coupling S-parameters for a Touchstone model. When observing a model that has many ports or unknown port connections, you can enable the software to automatically show insertion loss, return loss, and coupling S-parameters for all corresponding port combinations.
Evaluating SERDES Channel Compliance From an S-Parameter Model in the Touchstone Viewer	If you have a model of a SERDES channel that uses a supported protocol, the software can quickly calculate figures of merit (FOM) without using IBIS AMI transmitter and receiver models. This method provides a quick way to evaluate performance that is particularly useful when your design includes many channels and you do not have IBIS AMI models.
Reporting Connectivity Among Ports	You can identify ports that connect directly to each other, such as ports for connector pins that short together.
Checking and Fixing Passivity and Causality Errors	You can check Touchstone and fitted-poles models for passivity and causality errors. You can fix passivity, causality, and symmetry errors by running the software to update the model or convert it to another form.
Checking S-Parameter Model Quality	You can check S-parameter model quality by displaying model data in a series of graphs and examining the graphs for characteristics that indicate whether the model is good or bad.
Cascade Multiple S-Parameter Models in Series	S-parameter models are increasingly used in SERDES and other high-speed simulations to characterize portions of the interconnect between drivers and receivers. If multiple elements in a signal path are described with S parameters, the circuit topology may contain a “chain” or “cascade” of S-parameter models connected in series.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Topic	Description
Convert and Fix Touchstone and Fitted-Poles Models	You can convert a Touchstone model to another type (such as S-parameter to Z-parameter), reduce the number of ports (to reduce the model file size), and so on.
Supporting Information for the Touchstone Viewer	Understanding Touchstone viewer behavior can involve technical concepts that sometimes require additional explanation.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Viewing and Measuring Model Curves

Seeing model data formatted as a curve can make it easier for you to analyze the model.

Topic	Description
Viewing Touchstone and Fitted-Poles Model Curves	You can display model data as a curve to help you see attenuation, phase shift, and so on. If you have two or more models that represent variations of a design or different model extraction options, you can overlay their curves to compare them.
Zooming and Other Curve Viewing Operations	You can better see the details of a curve by using zoom and other viewing operations.
Measuring Between Two Points on a Curve	You can attach markers to a curve to accurately measure the delta between two points.
Adding Targets or Thresholds	You can plot a threshold (target) curve to help you see model parameters that exceed a limit. The target consists of one or more line segments, where you specify the location of each vertex.

Viewing Touchstone and Fitted-Poles Model Curves

You can display model data as a curve to help you see attenuation, phase shift, and so on. If you have two or more models that represent variations of a design or different model extraction options, you can overlay their curves to compare them.

Your own experience or education, such as an electrical engineer, provides the best guidance on how to choose and interpret a curve.

Note

The viewer can display Touchstone and PLS files simultaneously because it synthesizes the behavior of the poles at any frequency.

Restrictions and Limitations

- The Touchstone and Fitted-Poles Viewer does not support Touchstone models containing H- and G-parameters.

Prerequisites

- You have acquired the Touchstone Viewer license.

Procedure

1. Choose the **File** > **Open Touchstone** or **Open Fitted Poles** menu item, select one or more files, and then click **Open**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

2. In the Files window, display a curve for one or more models:

If you want to display...	Do this...
A curve for an individual model.	Either: • Check a model name. • Right-click a model name and select the **Show** menu item.
A curve for each model.	Right-click a model name and choose the **Select All** menu item.

3. In the Parameters spreadsheet, display a curve for a port pair by doing one of the following:

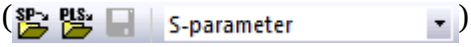
   - Check an individual port pair.
   - Right-click a port pair and select an option.

   If you display more than one model with the same number of ports, your selection applies to all models. This behavior is helpful when you compare models.

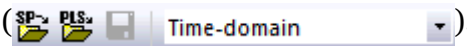
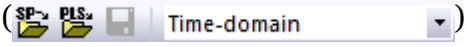
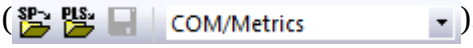
   If the Display window cannot fully display the spreadsheet and you do not want to scroll, do one of the following:

   - Undock the Display window and make it larger.
   - Click the plus sign button toward the upper-left corner of the spreadsheet to open a dialog box with a larger spreadsheet.

4. Select a display type:

If you want to display...	Do this...
<standard curve>	1. Select S-parameter (). 2. From the Display list (below the Parameters spreadsheet), select a display type.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

If you want to display...	Do this...
Time-domain response — Stimulate the selected port(s) and plot the responses in the time domain. A time-domain response waveform can provide a more intuitive view of model data, such as the time delay and signal quality between ports on a net.	1. Select Time-domain (). 2. Specify options in the Time-Domain window. See the time-domain table in HyperLynx Touchstone and Fitted-Poles Viewer Dialog Box. **Note:** To avoid an overly-complex waveform with a series of stair steps that can be difficult to evaluate, set the Stop Time value to twice the propagation time through the connector or topology represented by the Touchstone model. For an example measurement, see "Measure Delay Between Touchstone File Ports" on page 1353.
TDR impedance — Reproduce the behavior of a time-domain reflectometer by displaying impedance over time for a selected port pair. **Restriction:** TDR impedance is available only for standard mode Touchstone models.	1. Select Time-Domain (). 2. From the Display list (below the Parameters spreadsheet), select TDR Impedance. 3. Specify options in the Time-Domain window. See the time-domain table in HyperLynx Touchstone and Fitted-Poles Viewer Dialog Box. **Note:** Event times for TDR impedance plots have a 2X factor built in. See "Event Times in TDR Impedance Plots" on page 1352.
Protocol-specific metric — Provides a way to check the quality of 802.x (backplane Ethernet) interfaces. **Restriction:** The Protocol-specific metrics format is available when a standard model with 4 ports, 8 ports, and other multiples of 4 ports is loaded.	1. Select COM/Metrics (). 2. Specify options in the Protocol-specific Metrics window. See "Protocol-Specific Metrics Window Contents" on page 1377. For more information, see "Evaluating SERDES Channel Compliance From an S-Parameter Model in the Touchstone Viewer" on page 1303.

5. If you have loaded a fitted-poles model, specify the frequency range of the curves in the Full-fit Range area.

   Because a fitted-poles model does not contain frequency range information, you can display its curves across a practically unlimited frequency range.

Results

You can load multiple models into the viewer, to compare models produced by different model-extraction options or representing variations of a design.

Related Topics

Measuring Between Two Points on a Curve

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Zooming and Other Curve Viewing Operations

Adding Targets or Thresholds

HyperLynx Touchstone and Fitted-Poles Viewer Dialog Box

Zooming and Other Curve Viewing Operations

You can better see the details of a curve by using zoom and other viewing operations.

Procedure

Adjust your view of a curve:

If you want to...	Do this...
Zoom in	1. Right-click the chart and choose the **Zoom** () menu item. 2. Drag the zoom rectangle to enclose the area you want to enlarge.
Fit curve to window	Right-click the chart and choose the **Fit to Window** () menu item.
Pan	1. Right-click the chart and choose the **Pan** () menu item. 2. Drag a curve to a new position.
See logarithmic axis scales	On the toolbar, click **Log scale X axis**  or **Log scale Y axis**  . **Note:** Logarithmic scales are unavailable if one or more curves contains negative values.
See the port(s) for a curve	Hover over a curve to see its port numbers enclosed by parentheses.
See curves as lines or vertices	On the toolbar, click **View lines between vertices**  or **View vertices** .

Related Topics

Measuring Between Two Points on a Curve

Adding Targets or Thresholds

Measuring Between Two Points on a Curve

You can attach markers to a curve to accurately measure the delta between two points.

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Procedure

1. Right-click the chart and choose the **Track/Measure** () menu item.
2. Click a curve and drag the marker to the first measurement point.

   A second marker appears.
3. Drag the marker to the second measurement point, click, and see the delta.
4. To end tracking mode or track another curve, right-click the chart and choose the**Track/ Measure** menu item.

Related Topics

Zooming and Other Curve Viewing Operations

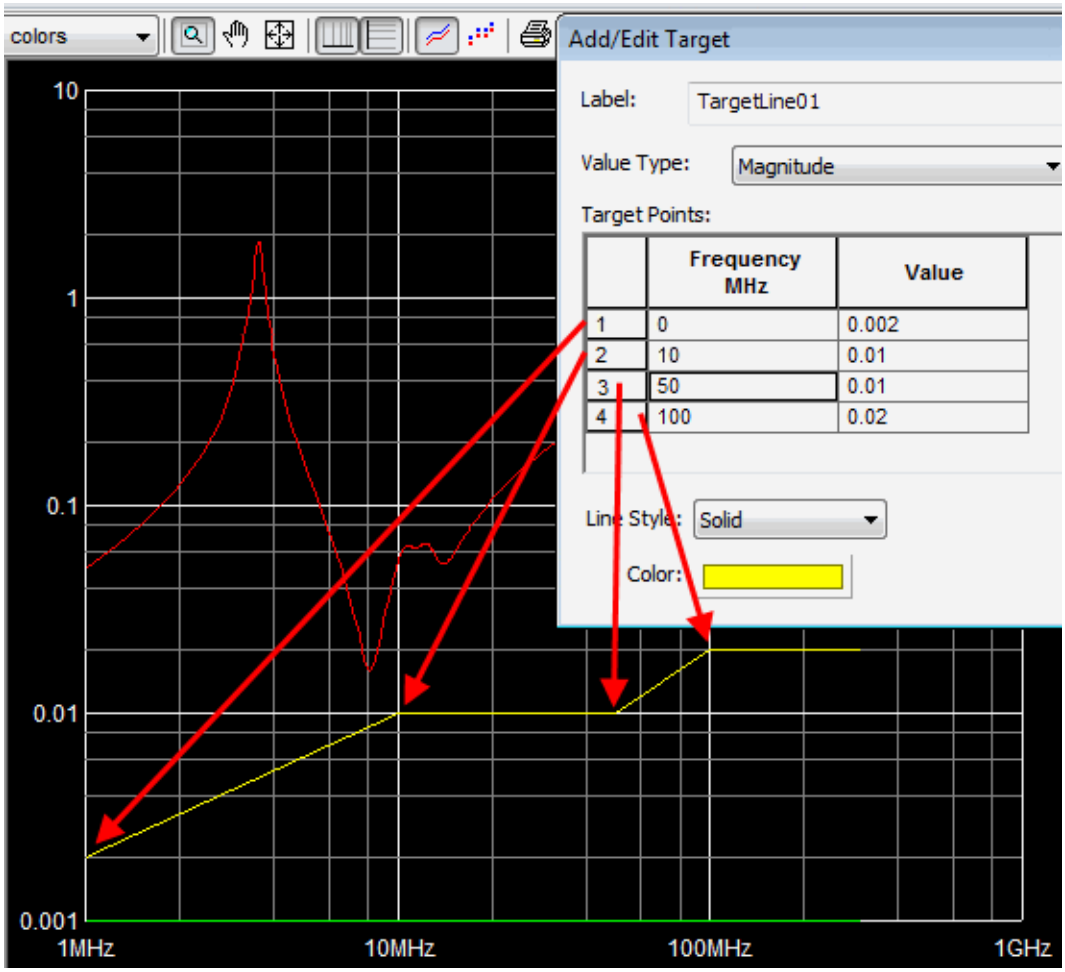
Adding Targets or Thresholds

Adding Targets or Thresholds

You can plot a threshold (target) curve to help you see model parameters that exceed a limit. The target consists of one or more line segments, where you specify the location of each vertex.

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Figure D-1. Example Target



Procedure

1. Choose the **View > Targets** menu item. The Manage Targets dialog box opens.
2. Click **Add** to create a new target. The Add/Edit Target dialog box opens.
3. Optionally, edit the value in the Label field.
4. In the Value Type list, select the units for the Y-axis value that you specify in the next step.
5. Type frequency and Y-axis values for a point, and then press Enter.

   The Value spreadsheet cell represents the target data, using the units that you set in step 4.

   The 0 MHz frequency is read only.

   A new spreadsheet row appears, ready to accept values for additional points.

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Note

The equation for Magnitude in DB display mode is Y(dB) = 20 * log(Y). For example, to locate the target at -20 dB in the Magnitude in DB display mode, specify Magnitude in step 4 and enter 0.1 in the Value spreadsheet cell.

6. To add a target data point, select the spreadsheet row (by clicking the cell in the left-most column), and choose the **Add Point** menu item.

   The new target data point appears below the selected spreadsheet row.

7. To delete a target data point, select the spreadsheet row to delete (by clicking the cell in the left-most column), and choosing the **Delete Point** menu item.

8. Optionally, edit Line Style and Color values.

9. Close the dialog boxes.

Results

Targets that you manually create persist when you open different models. By contrast, when you run decoupling analysis and create a Z-parameter model, the target Z value for the model is displayed as an <auto> target that is discarded when you close the model.

When you load a Touchstone file and change the Display value, the existing target value automatically converts to the new Display type. For example, when you create a target value for “Magnitude” and change the Display type to “Magnitude in dB”, the target value changes to the equivalent value in dB.

Viewing Insertion Loss, Return Loss, and Coupling for an S-Parameter Model

You can view insertion loss, return loss, and coupling S-parameters for a Touchstone model. When observing a model that has many ports or unknown port connections, you can enable the software to automatically show insertion loss, return loss, and coupling S-parameters for all corresponding port combinations.

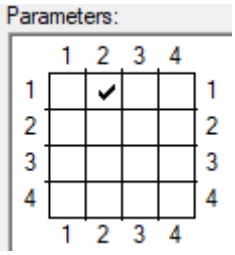
Procedure

1. From the Touchstone and Fitted-Poles Viewer, choose the **File > Open Touchstone** menu item, then open an S-parameter model.

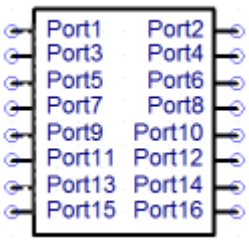
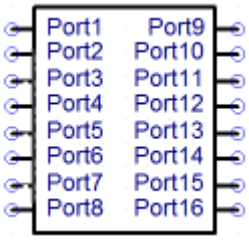
2. In the grid in the Parameters area, check the box corresponding to the appropriate parameter numbers. The matrix X axis represents response ports, and the Y axis

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represents stimulus ports. For example, check the box as shown in the following image to show insertion loss (S21).



3. To show insertion loss or return loss for an S-parameter model with many ports, do one of the following:

If you want to...	Do this...
Show return loss for all ports	Right-click the grid, then choose the **Return Loss** menu item.
Show insertion loss for port pairs with a specific numbering sequence	Right-click the grid, choose the **Show Insertion Loss** menu item, then choose one of the following menu items: • **1-2** — Use when an S-parameter model has an "odd to left, even to right" port numbering sequence. The following image provides an example of this numbering sequence:  • **1-N/2** — Use when an S-parameter model has a "first half to left, second half to right" port numbering sequence. The following image provides an example of this numbering sequence: 
Show insertion loss for connected port pairs	Right-click the grid, then choose the **Show Insertion Loss > Connected** menu item. **Note:** The software considers ports to be connected when the magnitude of a port-to-port insertion loss ratio at its lowest frequency point is greater than 0.33 (about -10 dB).

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If you want to...	Do this...
Show coupling for port combinations that meet specific coupling thresholds	1. Right-click the grid and chose the **Show Coupling** menu item. 2. In the Coupling Parameters dialog box, edit the coupling threshold and frequency range if needed. 3. Edit the port-to-port insertion loss ratio value in the "Ignore DC connections stronger than" area if needed. Decrease this value if the software includes uncoupled nets (connected ports) in coupling plots. 4. Click **OK**.

Evaluating SERDES Channel Compliance From an S-Parameter Model in the Touchstone Viewer

If you have a model of a SERDES channel that uses a supported protocol, the software can quickly calculate figures of merit (FOM) without using IBIS AMI transmitter and receiver models. This method provides a quick way to evaluate performance that is particularly useful when your design includes many channels and you do not have IBIS AMI models.

You can analyze multiple SERDES channels at a time, by opening multiple S-parameter models in the Touchstone Viewer and selecting them for analysis.

Restrictions and Limitations

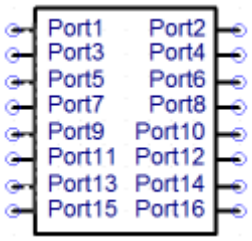
- The Touchstone and Fitted-Poles Viewer does not support Touchstone models containing H- and G-parameters.

Prerequisites

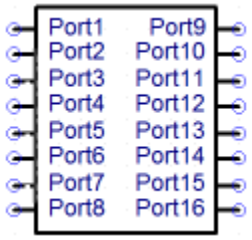
- You have the Complex Pole Fitter license.
- You have an S-parameter model that:
  - Represents the interconnect for a SERDES channel and neighboring SERDES channels with significant coupling to it.
  - Has four ports, or a multiple of four ports, where each set of four ports represents a differential pair.

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- Has one of the following supported port numbering patterns:
  - All odd ports are on the left side of the channel, where a signal path connects an N port on the left to the N + 1 port on the other side. For example:



  - The first half of the ports are on the left side of the channel, where a signal path connects an N port on the left to the N + (total number of ports / 2) port on the other side. For example:

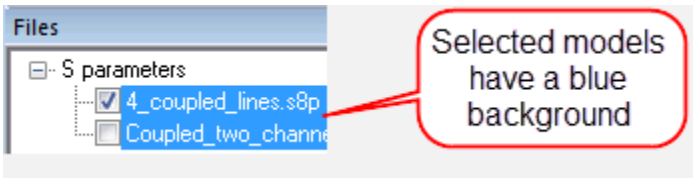


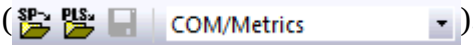
- Excludes package behavior.
- You have a high-quality S-parameter model. See Checking S-Parameter Model Quality.

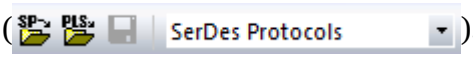
Procedure

1. Open one or more S-parameter models in the Touchstone and Fitted-Poles Viewer with the **File > Open Touchstone** menu item.
2. If you have opened multiple S-parameter models, in the Files window, select Ctrl-click each model you want to analyze.

   A check box does not select a model for analysis; you must click or Ctrl-click a model to select it. Two models are selected in the following figure.



3. For an Ethernet interconnect, select COM/Metrics (

). For other SERDES protocols, select SerDes Protocols (

).

   The Protocol-specific Metrics window opens.

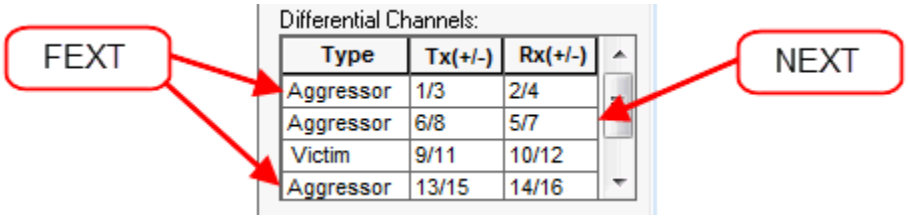
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4. Identify the port-numbering pattern:

If your S-parameter model uses this port-numbering pattern...	Do this...
All odd ports are on the left side of the channel.	From the Port Connectivity list, select "1 - 2".
The first half of the ports are on the left side of the channel.	From the Port Connectivity list, select "1 - (1 + (number of ports /2))".

5. Identify port numbers for victim and optional aggressor differential pairs.

   The port numbering you select in the Tx and Rx columns determine NEXT and FEXT aggressor channel conditions for analysis, as shown in the figure below.



6. From the Analyze list, select one of the SERDES protocols, or select "Channel Operating Margin" for Ethernet.
7. Select the operation mode of your channel.
8. If your S-parameter model includes transmitter and receiver packages, check "Don't add packages", otherwise leave this option unchecked to enable the software to represent the packages with a worst-case (minimally compliant) model.
9. Specify a base output folder name when you have enabled "SerDes Protocols". The software creates a sub-folder with the same name as the S-parameter model file and writes results to it.

   Note
   The software erases all files in an output sub-folder if you rerun analysis.

10. Click **Run Analysis**. The software displays results in an HTML report and writes the contents of the COM Analysis Results window to *R_PassFailAnalysisReport.txt*.

Results

Review analysis results in HTML format or in *R_PassFailAnalysisReport.txt*. If the results are poor (a FOM value fails) various other files in the output folder can help you identify contributing factors. For information about the contents of the output files, see COM Analysis Results.

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You can communicate the optimal equalization settings to the Rx/Tx programmer. The programmer may see a specification that corresponds to these parameters.

Related Topics

Viewing and Converting Touchstone and Fitted-Poles Models

HyperLynx Touchstone and Fitted-Poles Viewer Dialog Box

Reporting Connectivity Among Ports

You can identify ports that connect directly to each other, such as ports for connector pins that short together.

Since Touchstone models typically contain no port names or other helpful information about how the model maps to the physical structure being modeled, it can be difficult to understand even simple model properties, such as “which ports are directly connected to each other?”

The Touchstone and Fitted-Poles Viewer can identify ports on a Touchstone model (but not a fitted-poles model) that connect to each other conductively at DC. This information can be useful when you are connecting the model to a circuit in HyperLynx® LineSim®, but cannot access the documentation indicating which ports connect directly to each other.

Restrictions and Limitations

- The software does not identify connected ports for fitted-poles models.

Procedure

1. In the Files area, right-click over the model name and choose the **View Connectivity** menu item. The Connectivity dialog box opens.
2. Click a port to view its connected ports and connection strengths as one of the following:
   - Strong — Ports probably have strong connectivity (magnitude of port-to-port signal at DC is close to 1)
   - Medium — Ports probably have medium connectivity
   - Weak — Ports probably do not connect or have minimal effect on each other

Related Topics

HyperLynx Touchstone and Fitted-Poles Viewer Dialog Box

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Checking and Fixing Passivity and Causality Errors

You can check Touchstone and fitted-poles models for passivity and causality errors. You can fix passivity, causality, and symmetry errors by running the software to update the model or convert it to another form.

Restriction

Passivity checking is valid only for S-parameter models.

Topic	Description
Passivity and Causality Errors	Passivity errors indicate that energy is created between ports on the model. For example, a backplane connector is a passive component and its model should have passive behavior. The passivity check looks across all the ports and reports an error if the sum of energy coming out of the ports exceeds the energy going into one of the ports.
Automatically Reporting Passivity and Causality Errors	The Touchstone and Fitted-Poles Viewer automatically checks passivity and causality when you load the model. The Files area reports errors by displaying the file name in red characters, followed by the error name(s).
Manually Reporting Passivity and Causality Errors	You can check the model for passivity and causality errors after the model is loaded. You might want to do this if you disabled automatic checking when loading the files.
Fixing Passivity, Symmetry, and Causality Errors	To fix passivity and symmetry errors in a model, use the Convert > Correct Errors > Enforce Passivity | Enforce Symmetry menu.

Passivity and Causality Errors

Passivity errors indicate that energy is created between ports on the model. For example, a backplane connector is a passive component and its model should have passive behavior. The passivity check looks across all the ports and reports an error if the sum of energy coming out of the ports exceeds the energy going into one of the ports.

Errors

An error can result from any number of model-generation problems, including the following:

- Fixture and calibration errors in test bench or measurement equipment, such as a VNA (vectored network analyzer)

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- Limited fitting accuracy caused by insufficient resolution or non-causality of the original sampled data
- Software used to extract the model may not have wide-band capabilities

Restrictions

- Passivity checking is available only for Touchstone models containing S-parameters.
- Causality checking is unavailable for fitted-poles models. The fitted-poles model-creation process enforces causality, producing models that are inherently causal.

Causality errors indicate a propagation speed faster than physics allow or a reversal in the phase trajectory. Models with causality errors can produce simulator instability or incorrect delays. The model may be inherently non-causal, in which case it should be re-created; or it may be causal, but appear non-causal because it is not sampled with enough frequency points in some regions, or contains numerical or measurement noise.

Model data can be slightly non-causal due to unavoidable measurement and simulation errors. For example there may be insufficient frequency resolution in the sample, which typically occurs at low frequencies because of using equidistant frequency points.

Even if no causality errors are reported, the model may still not be absolutely causal because the Touchstone and Fitted-Poles Viewer does not perform an exhaustive causality check of the sampled Touchstone data.

Tip

The fitted-poles model-creation process enforces causality, so fitted-poles models are inherently causal.

To investigate errors, you can start by displaying model data in passivity plot curves (for passivity errors) and trajectory plot curves (for causality errors).

Automatically Reporting Passivity and Causality Errors

The Touchstone and Fitted-Poles Viewer automatically checks passivity and causality when you load the model. The Files area reports errors by displaying the file name in red characters, followed by the error name(s).

The amount of time needed to evaluate a model for passivity and causality errors is short for most models. However if your models take too long to process, you can disable the automatic checking.

Use this procedure to disable automatic passivity and causality checking.

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Procedure

1. Choose the **Edit > Options** menu item. The Options dialog box opens.
2. On the **Files Loading** tab, uncheck any of the options and click **OK**.

Manually Reporting Passivity and Causality Errors

You can check the model for passivity and causality errors after the model is loaded. You might want to do this if you disabled automatic checking when loading the files.

Procedure

1. In the Files area, click the model name so that it is selected. A check mark does not select a model for checking.
2. Do either of the following:
   - On the toolbar, click **Perform Passivity Check**  or **Perform Causality Check**  .
   - In the Files area, right-click over a model name, then choose the **Check > Passivity** or **Causality** menu item.

Fixing Passivity, Symmetry, and Causality Errors

To fix passivity and symmetry errors in a model, use the Convert > Correct Errors > Enforce Passivity | Enforce Symmetry menu.

Converting a Touchstone model into a fitted-poles model fixes causality errors. To do this, use the **Convert > To Fitted Poles** menu to convert the model to a fitted-poles model.

The Enforce Passivity option corrects poles/residues in such a way as to make the approximation strictly passive.

Caution

Do not enforce passivity for active devices, such as amplifiers or active filters.

Note

Use the Enforce Symmetry option on a reciprocal network only. This option is unavailable when a fitted-poles model is loaded.

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Checking S-Parameter Model Quality

You can check S-parameter model quality by displaying model data in a series of graphs and examining the graphs for characteristics that indicate whether the model is good or bad.

Topic	Description
Examples of a High-Quality S-Parameter Model	A high-quality S-parameter model has all of the characteristics listed in this section.
Examples of Low-Quality S-Parameter Models	A low-quality S-parameter model has any of the characteristics listed in this section.

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Examples of a High-Quality S-Parameter Model

A high-quality S-parameter model has all of the characteristics listed in this section.

Sufficiently Wide Frequency Range 1311
Proper Asymptotic Behavior at Zero and Infinite Frequency 1312
Sufficient Resolution 1312
Proper Even and Odd Behavior 1314
Causal Trajectory Plot 1315
Passive Behavior 1317

Sufficiently Wide Frequency Range

The model data range is broad and includes the frequencies of all important resonances.

If the model covers a wide frequency band, which is a requirement for digital signaling, it should consist of data sampled at either of the following scales:

- Adaptive — Varies the sampling step size depending on model characteristics

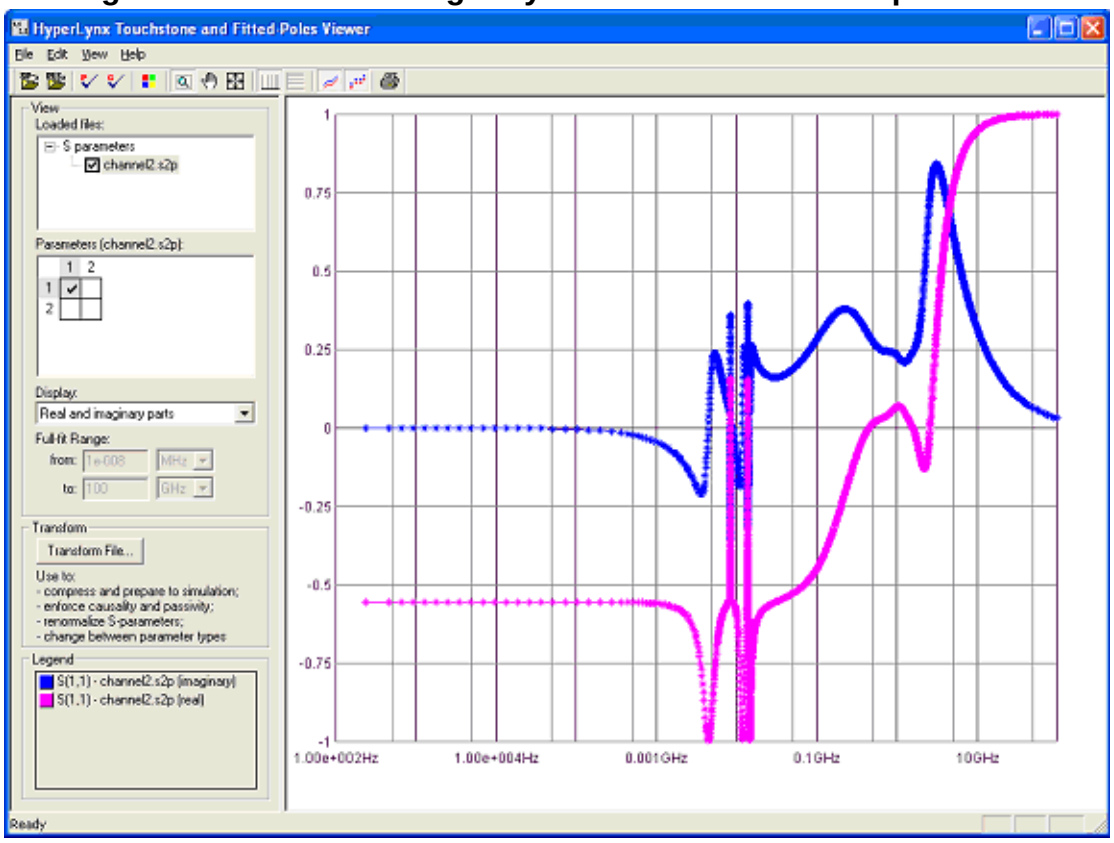
  The adaptive scale is better than logarithmic because it increases the sampling rate near frequencies with resonances.

- Logarithmic

Figure D-2 shows an S-parameter model with a frequency range of approximately 300 Hz to 100 GHz, with the real (pink) and imaginary (blue) portions of the S11 dependence. Logarithmic sampling was used to provide sufficient resolution at low frequencies without producing an overwhelming amount of data at high frequencies.

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Figure D-2. Real and Imaginary Portions of the S11 Dependence



Proper Asymptotic Behavior at Zero and Infinite Frequency

All dependencies should trend asymptotically at low and high frequencies to purely real, which means the imaginary portion should trend to zero.

As previously noted, S-parameter data consists of a complex dependence (per port and between ports) at each frequency. This dependence can be expressed as $H(j\omega) = H_r(\omega) + jH_i(\omega)$, with the real and imaginary parts displayed separately.

Figure D-2 shows how the imaginary (blue) portion of S11 tends to zero at the lowest and highest frequencies.

Sufficient Resolution

The sharpest resonances are represented with sufficient sampling resolution.

If you zoom in with the Touchstone and Fitted-Poles Viewer on sharp peaks or valleys in a dependence, many frequency points exist at each maximum/minimum. For resonances with

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high-Q factors, this is possible only with adaptive sampling, which adds extra points in the region of resonance.

Figure D-3 and Figure D-4 on page 1314 show that adaptive sampling provided sufficient resolution for each resonance.

Figure D-3. Two Resonances with Sufficient Resolution-Moderate Zoom

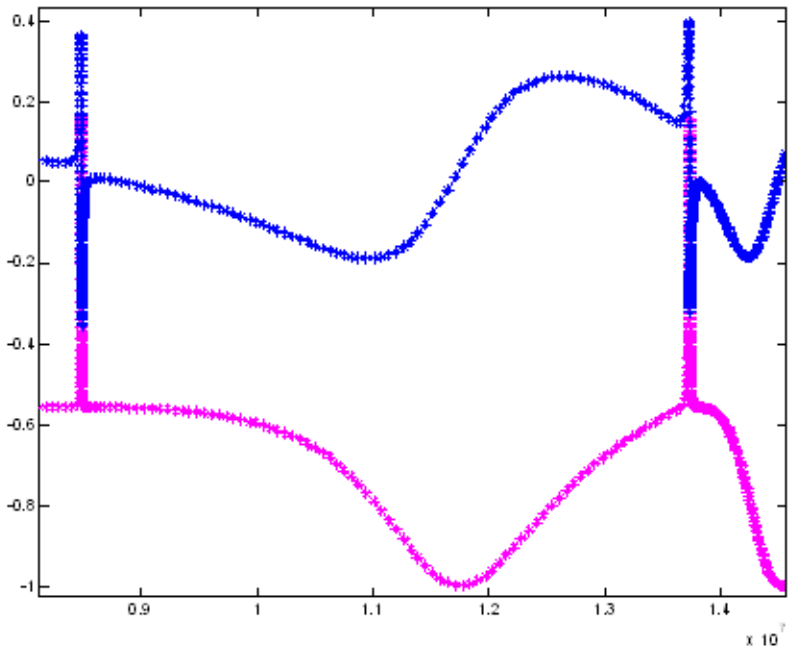
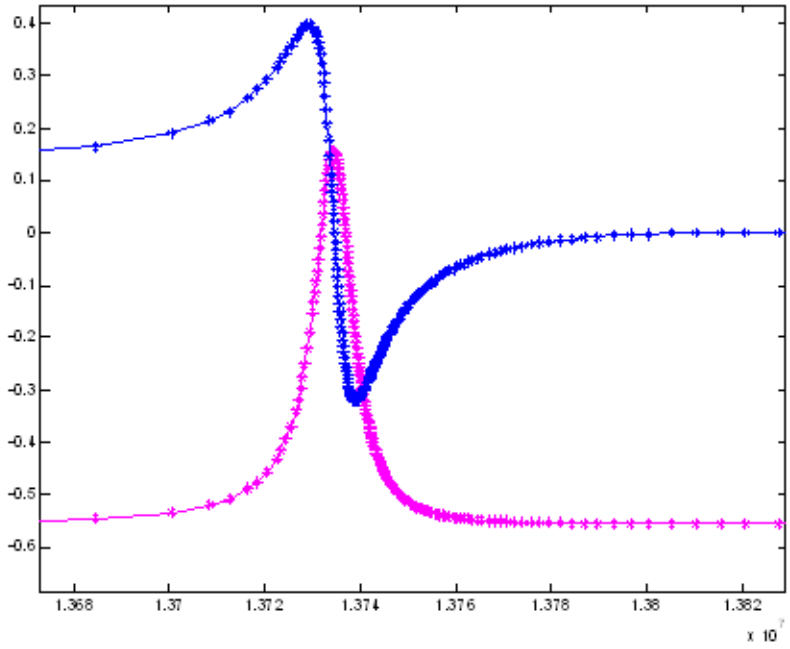


Figure D-4 shows that resonances have large numbers of frequency points, even at an extreme zoom.

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Figure D-4. Two Resonances with Sufficient Resolution-Extreme Zoom



Proper Even and Odd Behavior

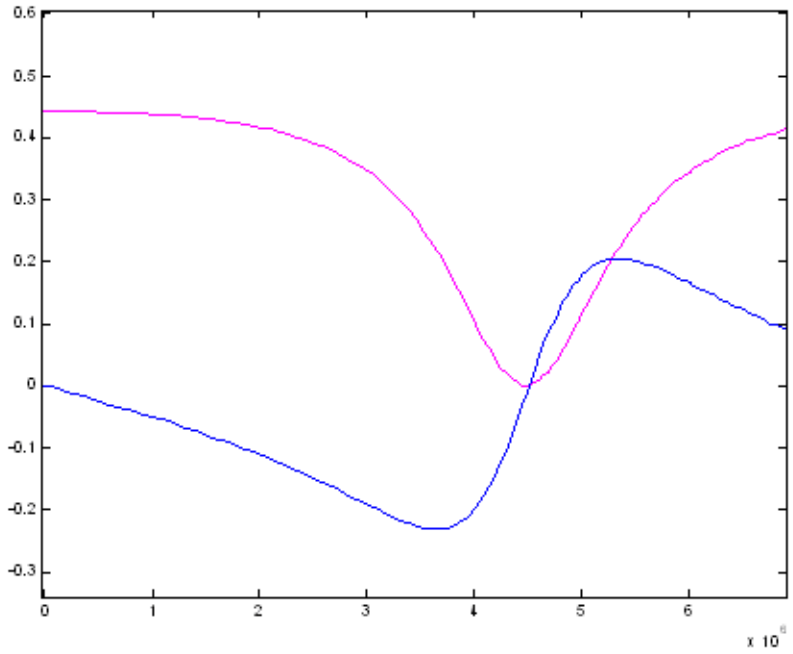
The real part of a dependency is an even function of frequency [f(x) = f(-x)], and the imaginary part is an odd function [f(x) = -f(-x)].

This means that the slope of the real part at DC must be zero so that there is a smooth extension into negative frequencies. The value itself may be non-zero. Conversely, the imaginary part can have a non-zero slope, but must have zero value. The dependence must have enough points defined near zero frequency to clearly meet these requirements.

Figure D-5 shows a model dependency from about 70 MHz down to DC. As required, the real portion (pink) approaches DC with zero slope, and the imaginary portion (blue) approaches DC with zero value.

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Figure D-5. Proper Behavior of Real and Imaginary Parts of the Dependence



Technical Background on System Realness

Although you normally do not build the dependence for negative frequencies, you can. The model time response, including impulse response, is a real function and the impulse response and frequency-domain response are mutual Fourier transforms. If the inverse Fourier transform of a complex frequency response is always a real function indicating system realness, then this complex frequency response must have the following special properties:

- The real part of the dependence must be an even function of frequency.
- The imaginary part must be an odd function.

If these conditions are satisfied, then the imaginary part of the inverse Fourier integral from minus infinity to plus infinity becomes identically zero. Since these functions must also be continuous, it immediately follows that all the following are true:

- Odd number (first, third, and so on) derivatives of the real part $H_r(\omega)$ are zero.
- Even number (zero, second,…) derivatives of the imaginary part $H_i(\omega)$ are also zero.

Causal Trajectory Plot

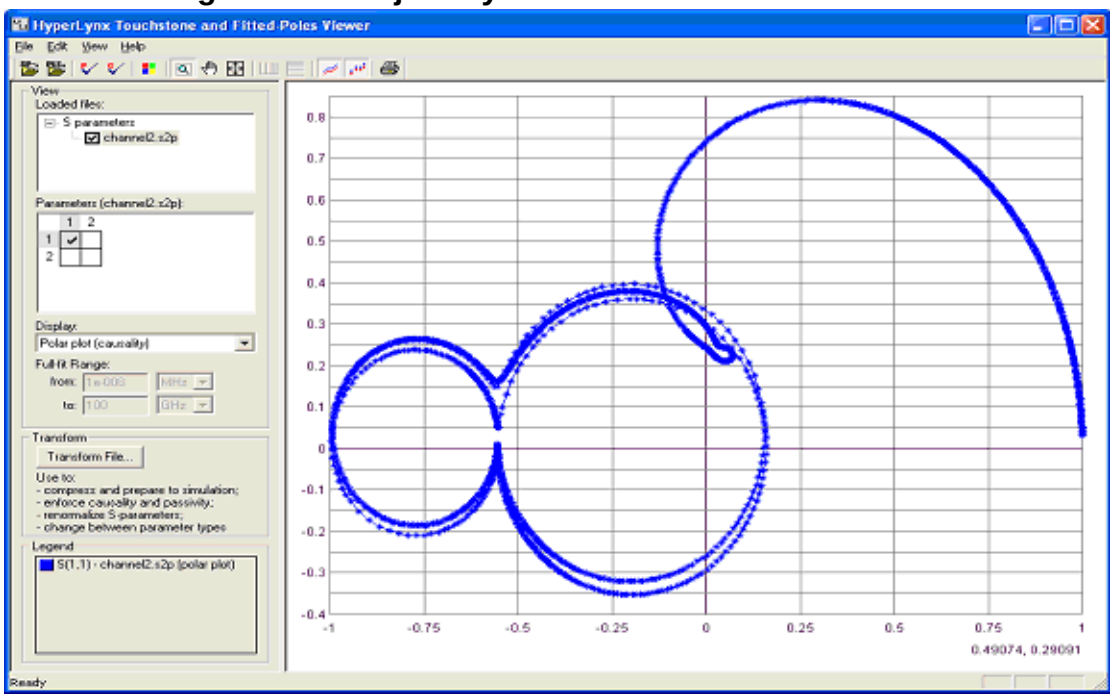
A trajectory plot (that is, a plot of imaginary versus real parts) of any dependency shows a continuous and properly directed trajectory.

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Specifically, the trajectory exhibits continual clockwise rotation, with resonances creating circles in the path. This is a requirement if the model is causal. The trajectory starts and ends on the real axis because the imaginary portion must tend to zero at DC and high frequencies, as required and described in “Proper Even and Odd Behavior” on page 1314.

Figure D-6 shows the trajectory plot corresponding to Figure D-2 on page 1312. The starting point of the trajectory is (-0.554, 0), and the end point is (1, 0). The path is always moving clockwise.

Figure D-6. Trajectory Plot with Clockwise Rotation



Technical Background on Causality

In the trajectory plot of a model dependency, each high-Q resonance creates an almost ideal circle. By contrast, the overlap of several moderate resonances creates a more complicated picture.

For example, both of the following examples have clockwise trajectories:

- An isolated pair of complex conjugate poles creates a complete circle.
- An isolated real pole creates a half-circle.

Taken as a whole, the trajectory could be thought of as a composition of simple clockwise rotations each with its own distinct center, radius, and speed. Sometimes the overlap of such rotations in some small isolated areas gives an impression of reverse or counterclockwise movement. A well-known example is a cycloid, which is a composition of correct trajectories.

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Why must a trajectory plot consist only of such proper, clockwise rotations? This property is closely related to model causality, where cause must precede effect in the model behavior. A realistic system can impose only positive delay. By contrast, a negative delay means the response precedes the input, which cannot be physical. Locally, the delay can be defined through "group delay," which is the derivative of the phase by frequency, negated. Clockwise rotation produces positive group delay, while counterclockwise rotation makes negative delay. Each cyclic component of the trajectory plot corresponds to a primitive time-domain impulse response of the following form:

$$h(t < 0) = 0$$

$$h(t \geq 0) = e^{-\alpha t}(A_1 \cos \omega t + A_2 \sin \omega t)$$

Each response, separately and any combination of such responses, thus obeys the following requirement:

$$h(t < 0) = 0$$

Thus, the sum of causal dependencies is a causal dependence. Less known is that the product of causal dependencies is also causal. This is related to a property of convolution where you may cascade several causal models and the signal propagating through all of them is properly delayed. In the time domain, cascading means convolving impulse responses of the models, and in the frequency domain it corresponds to finding their product, which must also remain causal. The inverse is also true. If you multiply causal and non-causal frequency dependencies, the product may well be non-causal.

Passive Behavior

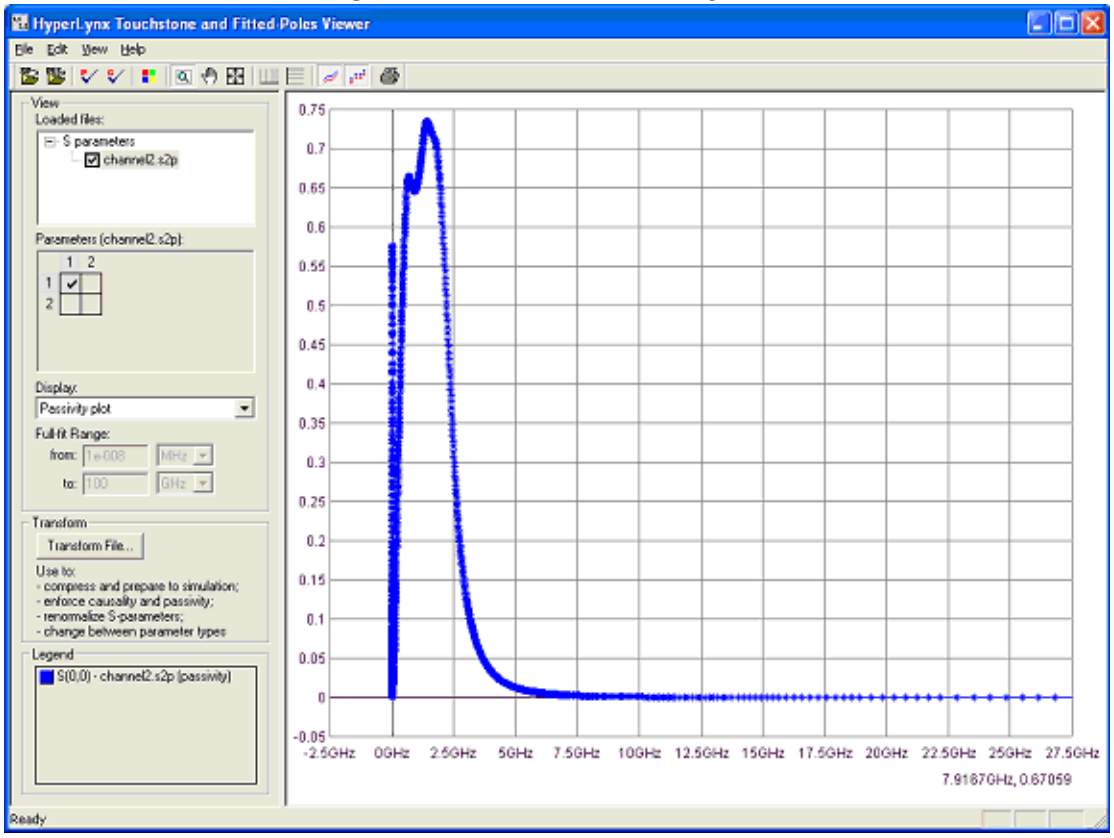
For an interconnect model, the passivity plot must be greater than zero at all frequencies.

In signal-integrity simulations, S-parameter models almost always represent interconnect structures, which can dissipate energy but cannot produce energy (unlike an active device, such as an amplifier).

Figure D-7 shows the passivity plot corresponding to Figure D-2 on page 1312. The value is greater than zero at all frequencies. Note also that, unlike the previous plots shown, the passivity graph is a property of the entire model and not a property of one of the model dependencies/ parameters, such as S11 or S21.

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Figure D-7. Good Passivity Plot



Comment on Passivity

Completely passive models can behave in many ways.

- Absorb/dissipate active power. For example, a model of a resistor can transform input energy into heat or radiate it.
- Reflect active power back into the surrounding circuit.
- Store input energy in an electric field, that is, an ideal capacitor, and then return it back into the surrounding circuit.
- Store input energy in a magnetic field, that is, an ideal inductor, and then return it back into the surrounding circuit.

Under no conditions can a passive model produce return energy exceeding the amount of energy it has received.

By contrast, non-passive models can produce energy. Sometimes non-passive models can cause simulation instability in the form of voltages and currents that increase without limit. For example, connected external resistances (losses) may be able to dissipate the surplus of energy created by a non-passive model and thus prevent the total energy in the simulation from growing.

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In another design using the same model, the surrounding circuitry may exhibit less loss and allow the generated power to accumulate over time and lead eventually to instability. Further, in an “in-between” case, the surplus of the energy may not be enough to cause obvious instability, but be sufficient to make simulation results appear correct but actually be wrong.

Caution

Because of this unpredictability, non-passive S-parameter models are extremely dangerous and should never be used in simulations.

For S parameters, the passivity function for the set of parameters A is defined as

$$P_s(f) = \| I - A(f)A^*(f) \|$$

where

'*' is the Hermitian transpose.

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Examples of Low-Quality S-Parameter Models

A low-quality S-parameter model has any of the characteristics listed in this section.

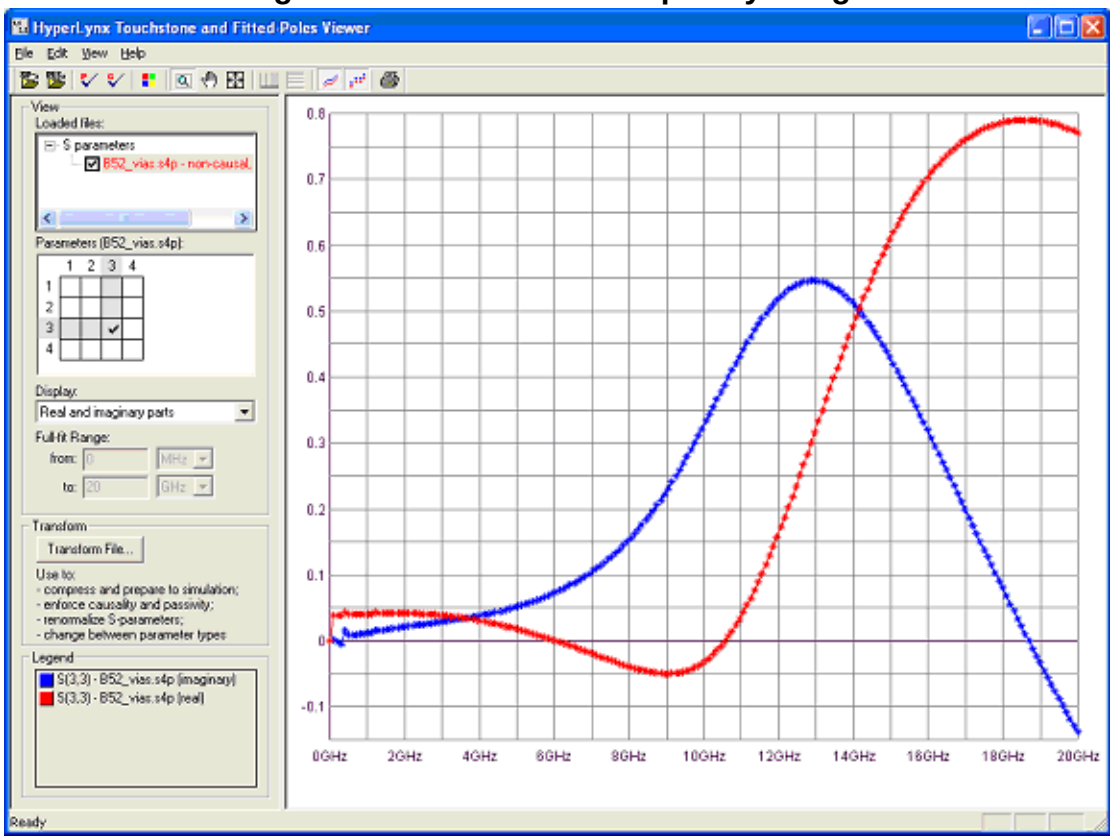
Insufficient Data Range 1320
Insufficient Resolution 1324
Non-Ideal Asymptotic Behavior at DC 1328
Inherent Non-Causality 1331
Artificially Created and Modified Points 1332

Insufficient Data Range

The S-parameter model does not contain data over a sufficiently wide frequency range.

The figure below shows a model defined from nearly DC to 20 GHz. If you are planning to simulate a signal with a fundamental frequency of about 1.5 GHz and are convinced that little energy exists in harmonics above 20 GHz, this model may look sufficient.

Figure D-8. Insufficient Frequency Range



However, there is a problem with this model because its dependency above 20 GHz is ambiguous. In a higher-quality version of the model, the dependency would continue upwards in frequency until it was completely settled. Compare Figure D-8 with Figure D-2 on

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page 1312. Recall the earlier requirement that in a good model, the imaginary part of the dependency tend to zero at high frequency and DC. Figure D-8 shows a non-zero value at high frequencies.

Another problem with the model behavior illustrated by Figure D-8 is a more mathematical consideration. Ideally, you would create it from a perfect, infinite causal dependence by multiplying the causal dependence by a rectangular window function (w(f) = 1 if f < 20 GHz; w(f) = 0 otherwise). This rectangular function is non-causal because it would have a straight-line trajectory plot. But as described in Technical Background on Causality, the product of causal and non-causal models is non-causal and this is always true except for trivial cases of models that are identically zero. Therefore, the truncated model in Figure D-8 is clearly non-causal if taken as is.

If this model were simulated in a convolution-based simulator, some behavior above 20 GHz would be imposed, but the exact behavior is unknown. For example, by applying inverse Fourier transformation to the non-causal function in Figure D-8, such a simulator would get an impulse response that starts somewhere in negative time. Programmatically, this negative portion of the response can be removed and not used in convolution. However, such a mechanistic removal of the negative-time portion of the response would affect the entire model in an unpredictable way. Transforming this truncated response back into the frequency domain would result in a dependence quite different from the original.

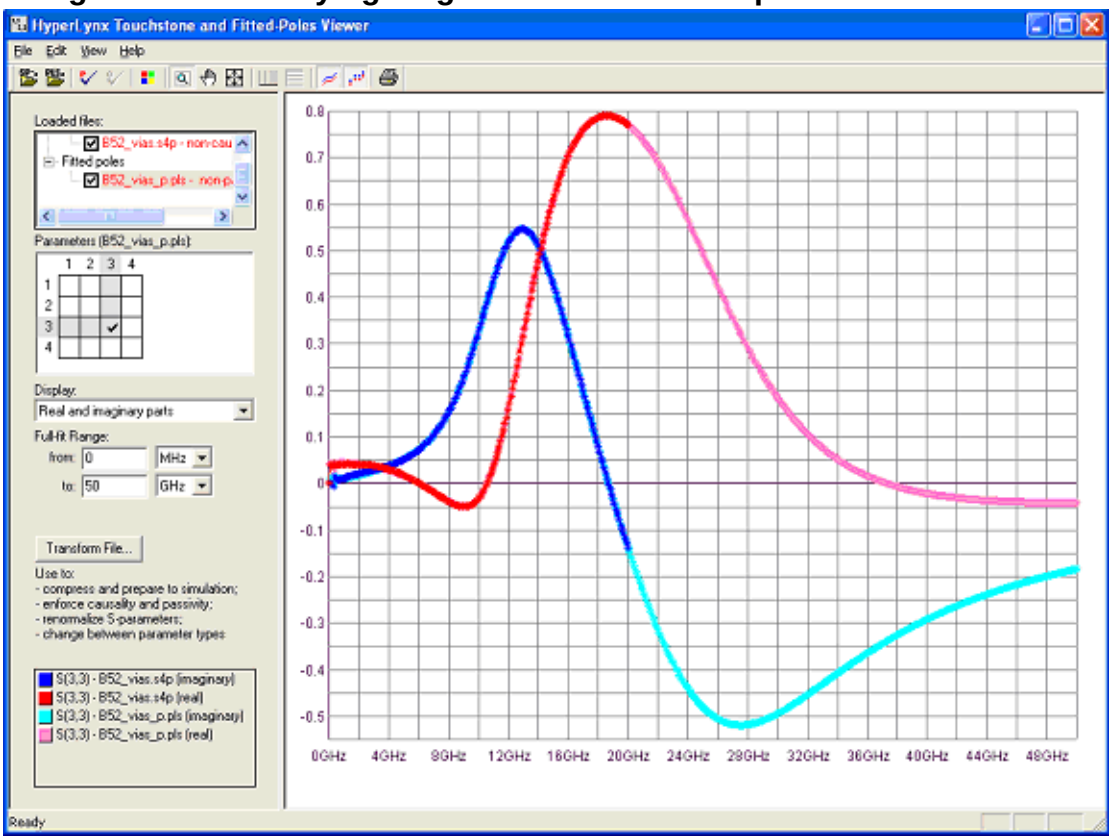
By contrast, using Siemens EDA complex pole fitting (CPF) based approach, you would first fit the model to a set of complex poles and then, before simulating, you could compare the fitted model to the original to see exactly what behaviors are being assumed outside the range of the original model data. For information about CPF, see “Simulating S-Parameter Models in the Time Domain” on page 1350.

The fitted dependence must be a mathematically continuous function. It cannot be forced to zero above 20 GHz. Nor should it be, since it is clear from Figure D-8 that some behavior occurs above that limit. But there is no data in the model to help define the behavior above 20 GHz, and the CPF fitter is forced to guess.

Figure D-9 shows the post-fit model, where the original (red, blue) and post-fit (pink, cyan) dependencies overlay.

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Figure D-9. Overlaying Original and Post-Fit Dependencies of Previous Model



The Touchstone and Fitted-Poles Viewer supports simultaneously loading and displaying the original S-parameter data and the fitted representation, so you can compare pre- and post-fit data. The fitted representation is analytical, meaning that you can extend it to any frequency, to understand its behavior outside the range of the original model. It is also guaranteed to be causal, due to the natural by-product of the fitting process.

The fitted model may look good, but recall that the behavior above 20 GHz is only defined by the data given below 20 GHz. The behavior above 20 GHz is not unique, and the extension process is uncontrollable. In particular, it is occasionally possible to have areas of non-passivity in the extended frequencies above the original 20 GHz limit.

The dependence shown in Figure D-9 is just a single matrix component in a 4x4 S-parameter model. Each of the other components in the model has its own uncontrollable continuations above 20 GHz. Since passivity is a property of the entire model, the combined effect from all such components at high frequencies may be severe. On average, the effect becomes stronger with increasing matrix size. Therefore, for larger matrices, the author of the model must define the high-frequency region more carefully, and non-passive continuations become increasingly unacceptable.

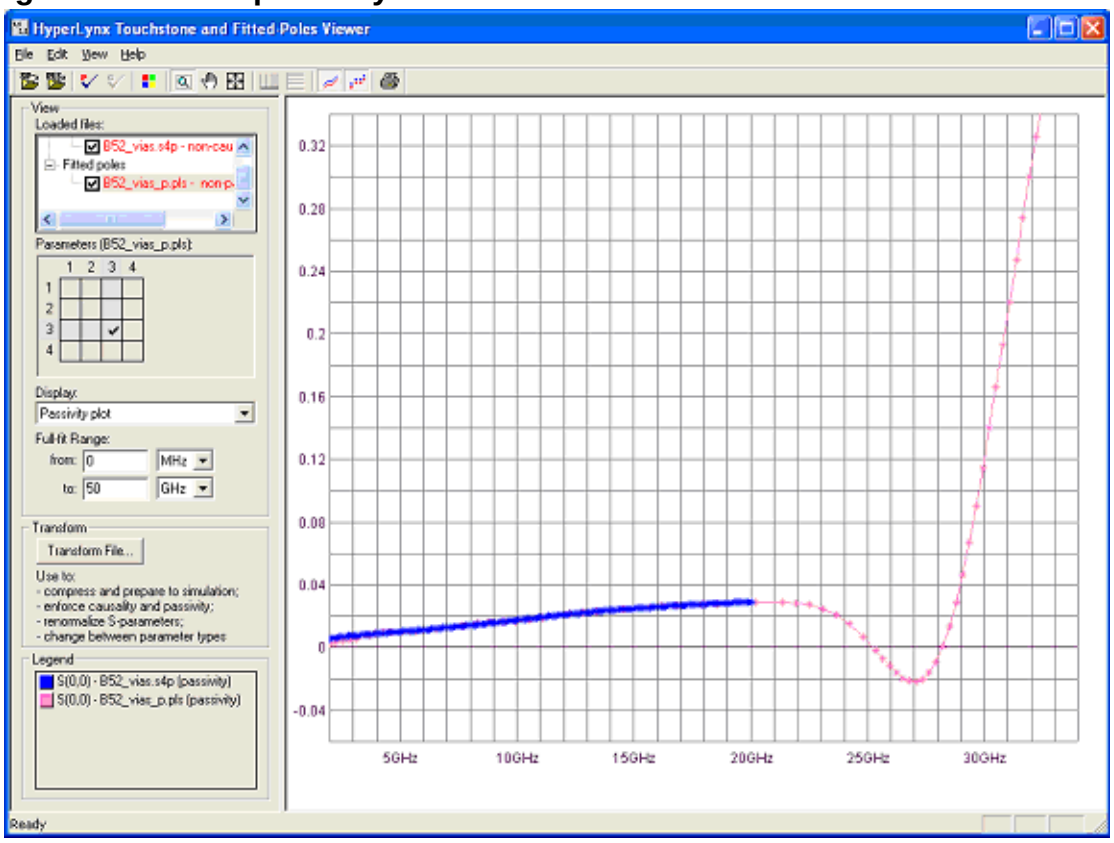
A convolution-based simulator would also make assumptions about the model behavior above 20 GHz, including some that might cause model non-passivity. However, you cannot determine

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these assumptions because the simulator lacks an intermediate fitting step that provides viewable results.

Figure D-10 shows the passivity function of the fitted model that indicates an area of non-passivity between 25 GHz and 28 GHz, which is above the original data range.

Figure D-10. Non-passivity Between 25 GHz and 28 GHz for Previous Model

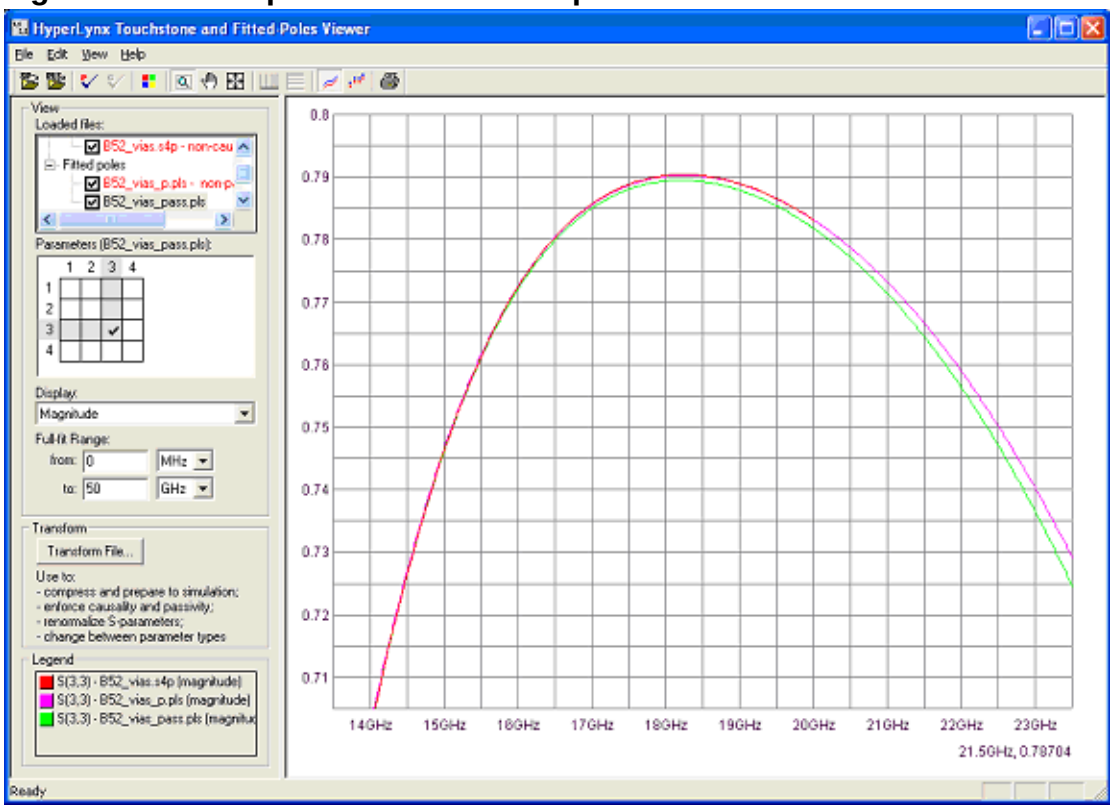


You can use Siemens EDA technology to enforce passivity on the fitted model by slightly modifying the fitted poles residue, which helps avoid any possibility of model instability. However, this post-fit passivation can reduce fit accuracy somewhat.

Figure D-11 shows the differences between the original (red), fitted (magenta) and fitted/passivated (green) dependencies.

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Figure D-11. Comparison of Model Dependencies-Small Differences Exist



All of this trouble, though manageable, could have been avoided if the author of the original model had supplied data to a higher frequency, preferably to a frequency at which the model behavior is settled.

Sometimes, it may be difficult to expand an insufficient frequency range enough to reach the desired asymptotic behavior when any of the following conditions are true:

- The asymptotic region is too far away in frequency from the range of interest.
- The model does not behave properly.
- Measurements cannot be accurately taken at high-enough frequencies.

If so, the model must at least provide data well above the range of interest, even if the dependence does not settle, so that during passivity enforcement the interesting portion of the model will not be significantly affected.

Insufficient Resolution

There are not enough sample points to resolve the model behavior within the supplied data range, especially at resonances.

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This condition prevents CPF fitting from constructing an accurate analytical representation of the model behavior. Poor resolution leaves much ambiguity in a model and there is no way to uniquely define the behavior between given points.

Sometimes, resolution problems manifest themselves as model non-causality. You can check this by viewing the trajectory plot for the model, with the Touchstone and Fitted-Poles Viewer, where non-causality can be seen as irregular behavior and a chaotic-looking trajectory. Such non-smooth data can be the result of either of the following conditions:

- Not having enough sample points, which can be corrected by using a finer frequency grid
- Measurement or simulation noise, which cannot easily be corrected

Figure D-12 shows an example of model data with insufficient resolution. The dependency is apparently oscillatory in the frequency domain, but you need at least 8-10 points per period to validly represent it.

Figure D-12. Insufficient Frequency Resolution

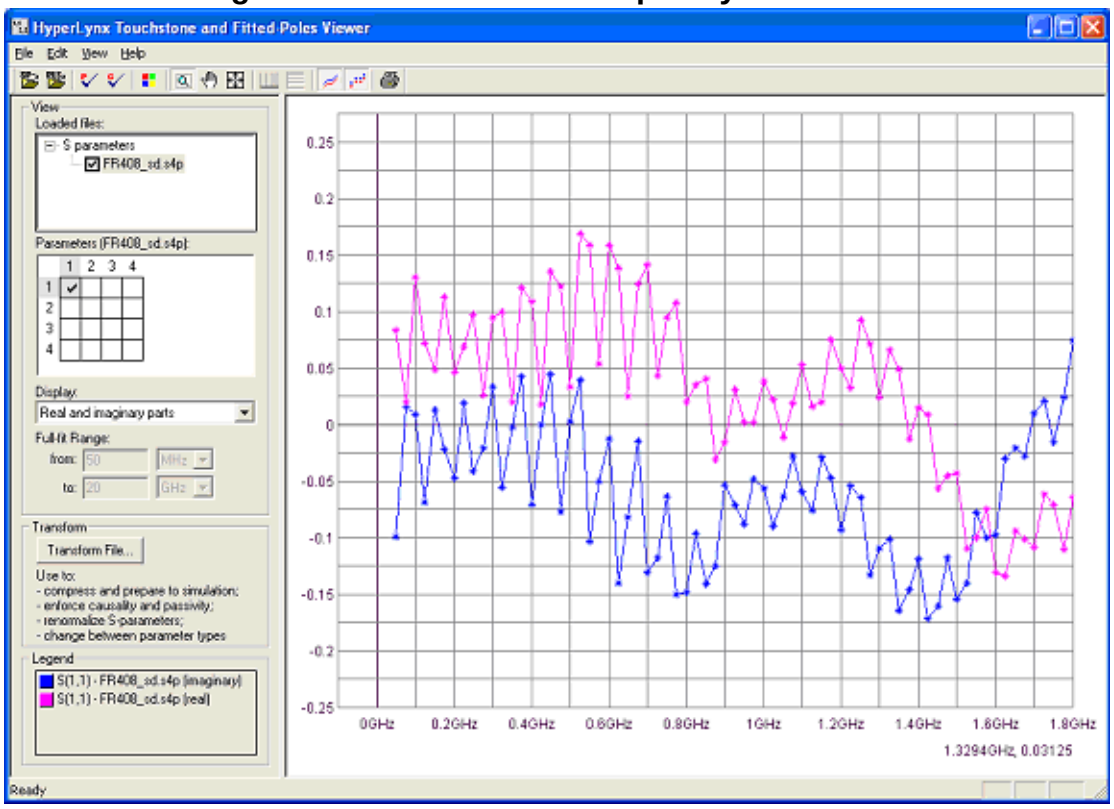
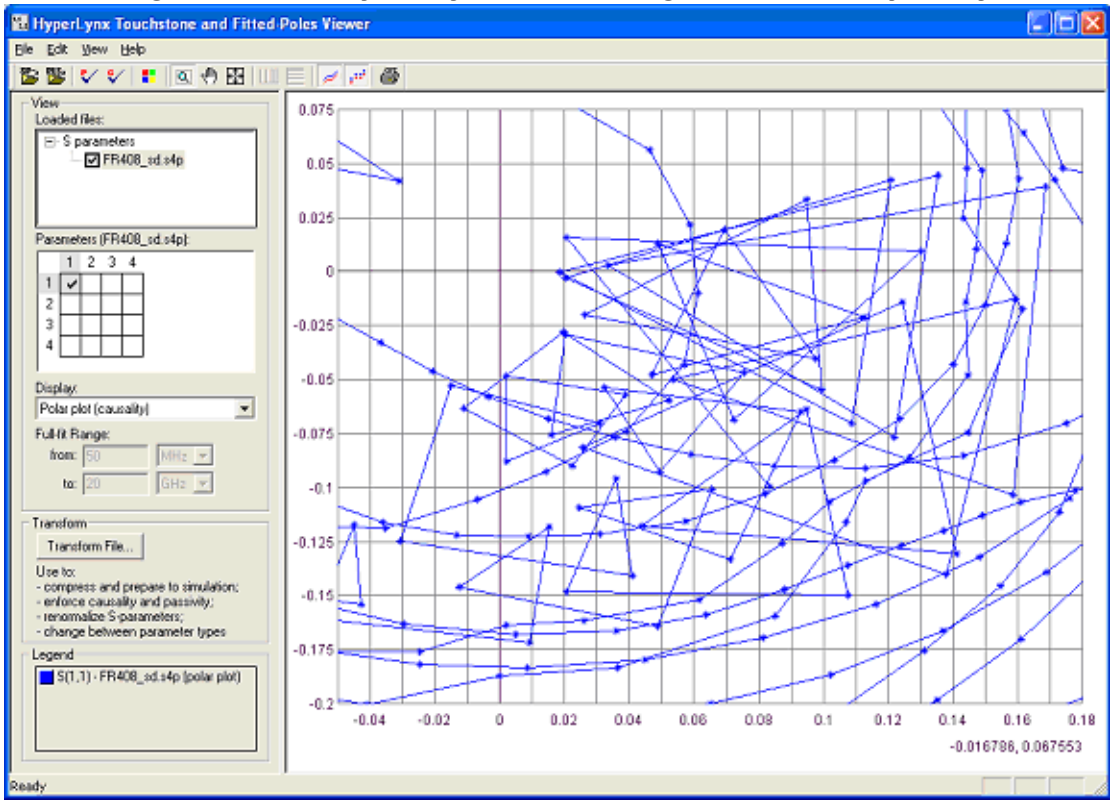


Figure D-13 shows the trajectory plot for the same model has a chaotic trajectory curve.

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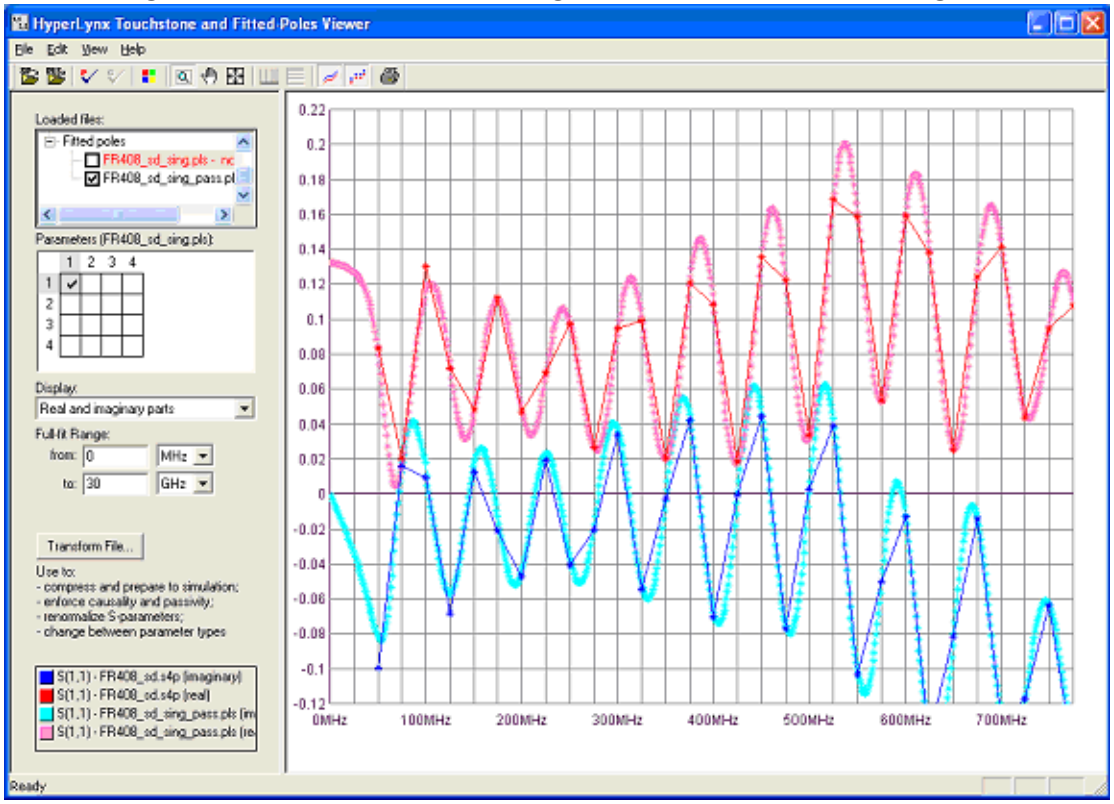
Figure D-13. Trajectory Plot Showing a Chaotic Trajectory



Suppose that, in spite of the overwhelming visual evidence in Figure D-12 on page 1325 and Figure D-13 on page 1326 that the model suffers from serious resolution problems, you attempt to fit it anyway in preparation for simulation. Figure D-14 shows the results.

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Figure D-14. Results of Fitting Model from Previous Figure



The fitted results may actually seem acceptable because the fitter has found a representation that passes through the original data points, is causal, and so on. But notice also how many assumptions have been made about the dependency. For example, the value of each maximum and minimum have no corresponding data points.

The CPF fitter has a high-resolution option and you might be tempted to try it on a model with insufficient original resolution. However, this is the wrong approach. Figure D-15 shows, on an admittedly artificial case of extreme under-resolution, the high-resolution fit causes catastrophic accuracy/passivity degradation.

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Figure D-15. Extreme Case of Under-Resolved Original Data

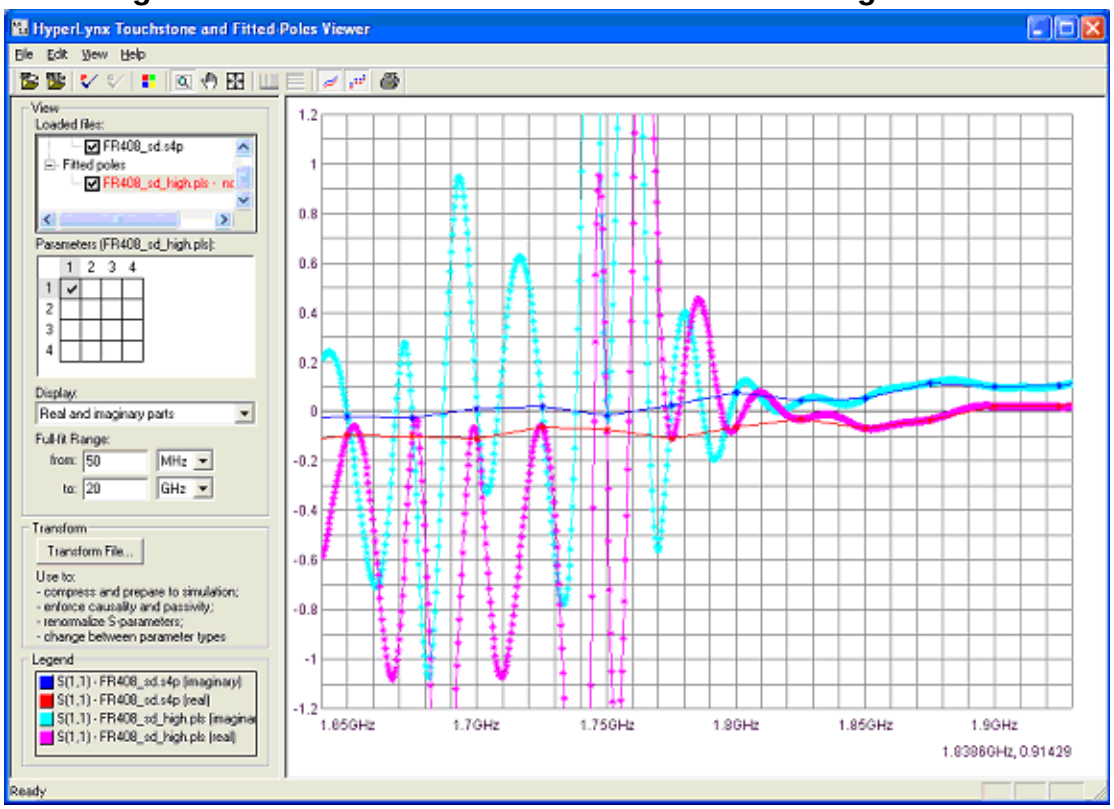


Figure D-15 also shows that the fitter generates ambiguous results because it has a larger-than-reasonable degree of freedom (that is, more poles are allowed), in combination with under-sampled input data. Note that the fitter approximation at the original points (blue/red) in the model is quite good and is sometimes even better than would occur with ordinary precision fitting. However, the fitting is essentially uncontrollable between the original points. Therefore, if a model suffers from insufficient frequency resolution, do not run high-precision fitting.

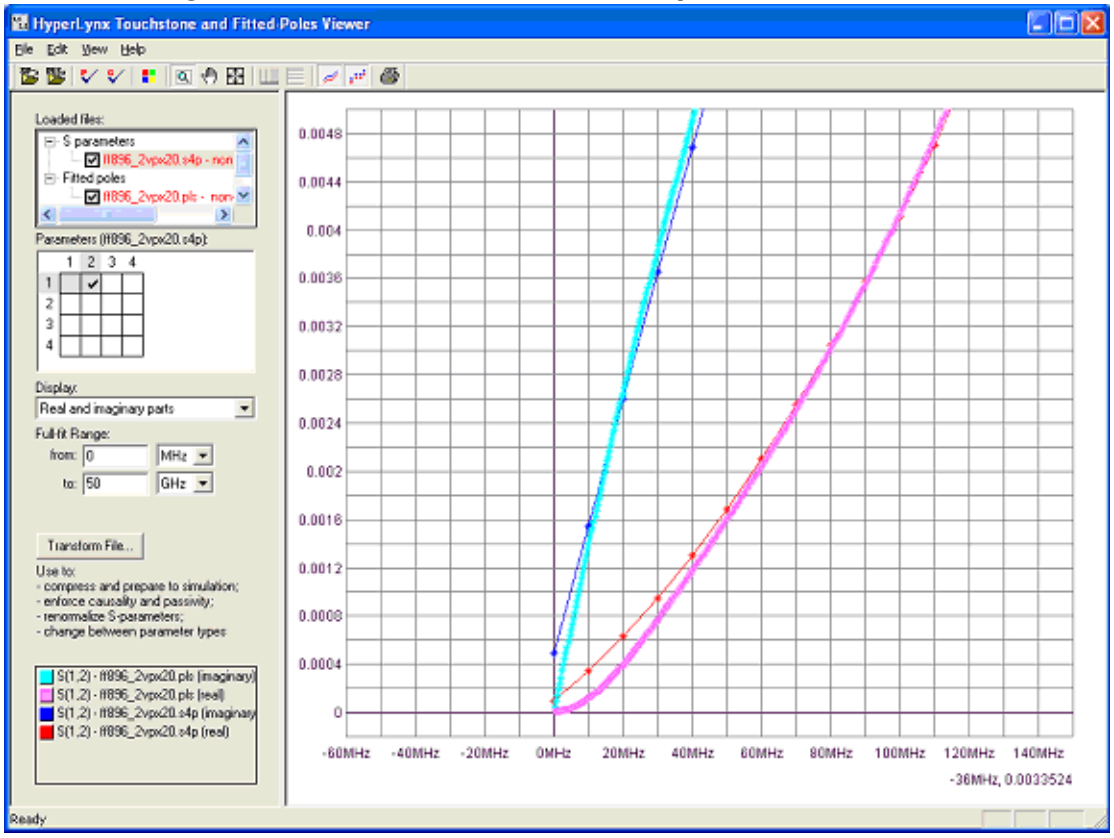
Non-Ideal Asymptotic Behavior at DC

The real part of a model dependence must be an even function having a slope of zero at DC and the imaginary part must be an odd function with a value of zero at DC. If the original data in a model does not obey these requirements, then the fitted dependence may deviate from the original as the fitter attempts to impose the correct behavior.

Figure D-16 and Figure D-17 show examples with the fitted data (magenta, cyan) deviating from the original (red, blue) near DC. This effect may seem minor, but even a small deviation may be important because it defines the model behavior at/near DC and may strongly affect the results in a time-domain simulation. For example, the effect may produce a wrong switching-voltage range or incorrect results over the time extent of a very long eye diagram. A good S-parameter model should use some type of variable sweep, logarithmic at least, to provide finer resolution near DC.

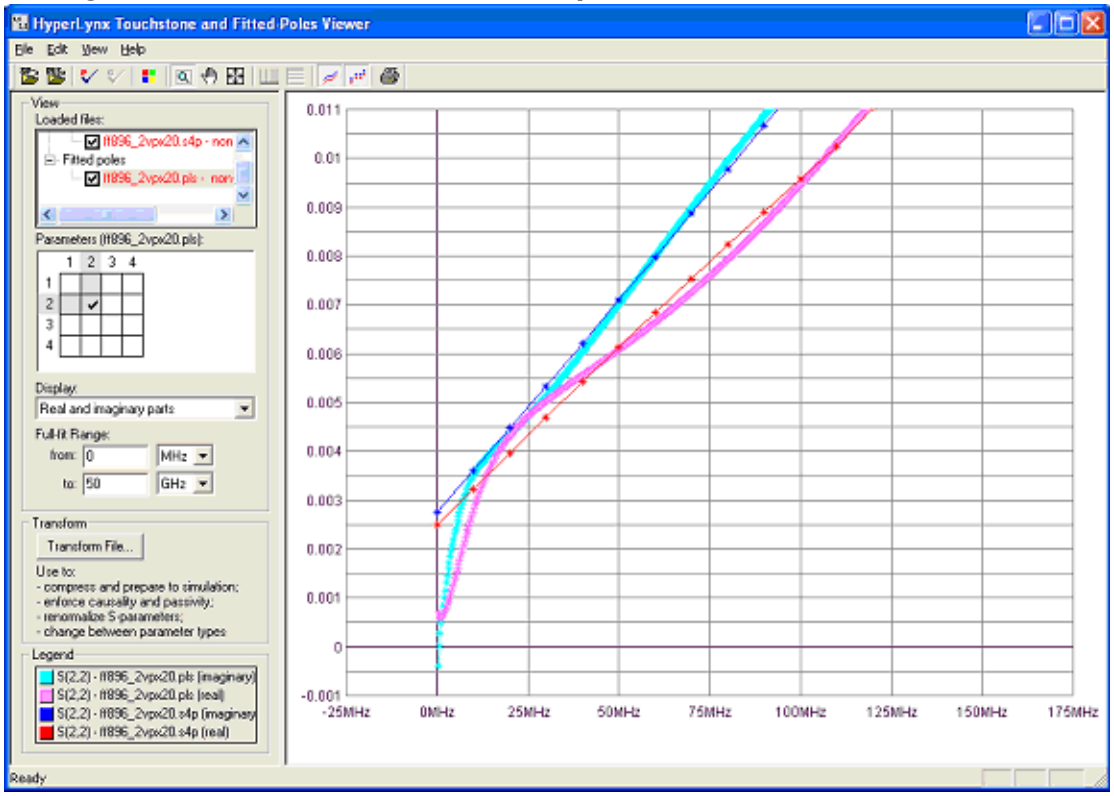
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Figure D-16. Deviation of Fitted Dependence Near DC



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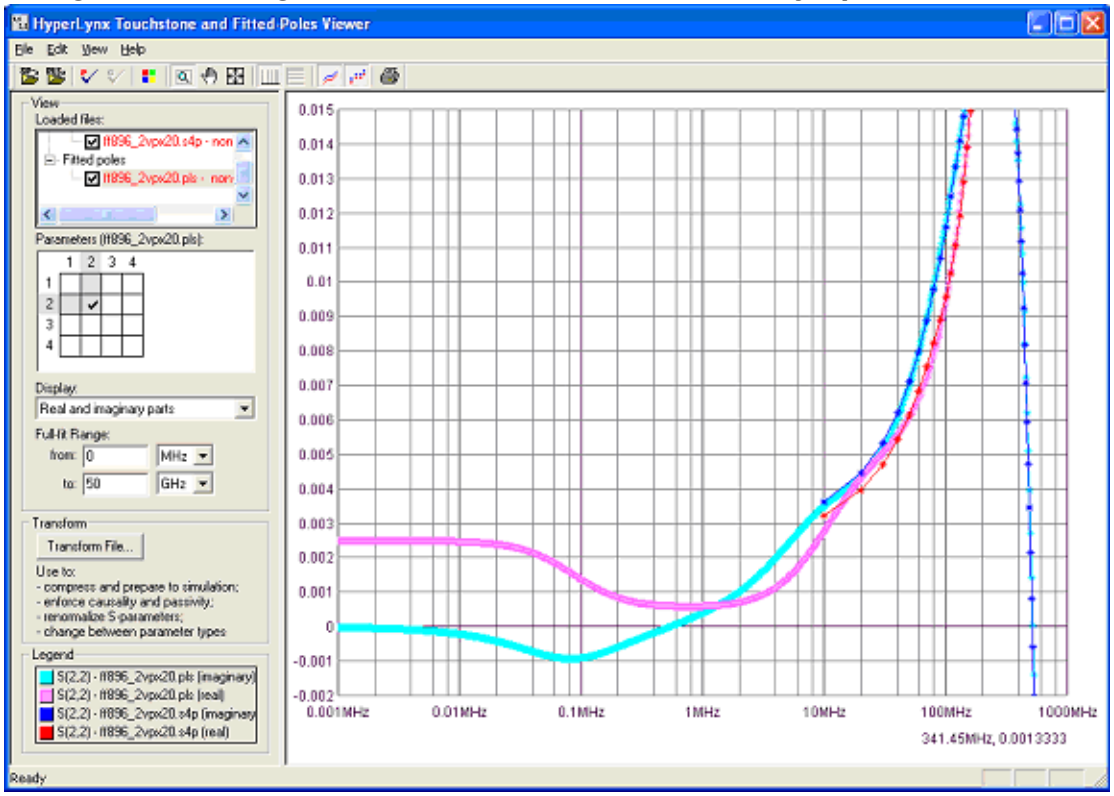
Figure D-17. Deviation of Fitted Dependence Near DC-More Severe



In Figure D-17, it may appear that the fitted data itself does not meet the low-frequency asymptotic requirements. However Figure D-18 displays the same data in a logarithmic frequency scale, which makes the asymptotes easier to see and shows that the data has the correct behavior.

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Figure D-18. Logarithmic Scale Shows Correct Asymptotic Behavior



Inherent Non-Causality

A model can be inherently non-causal, even when it has sufficient resolution.

In this case, the trajectory plot exhibits trajectory errors with regions of counterclockwise rotations, rather than clockwise rotation. Note that the trajectory errors may not be chaotic or unsmooth, as Figure D-12 and Figure D-13 show.

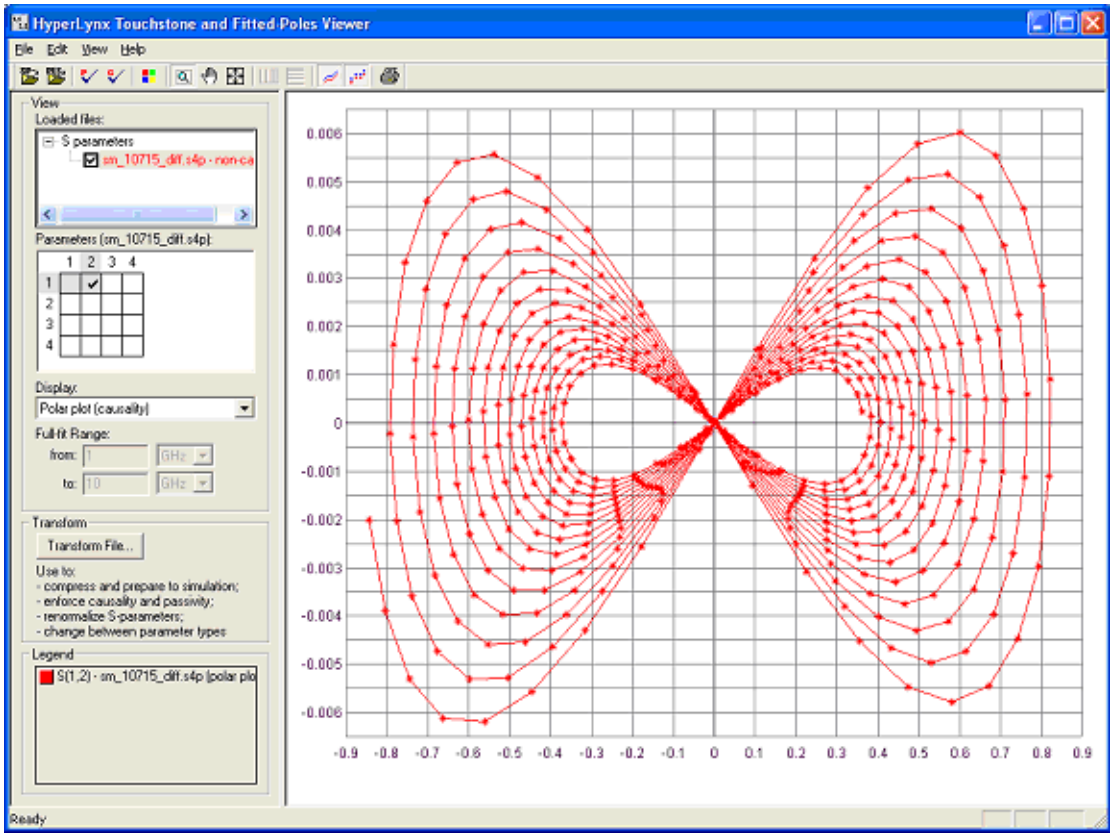
Note

The figures in "Insufficient Resolution" on page 1324 show that insufficient frequency resolution can generate non-causality. For example, if you take valid and causal data, resample it with a larger, coarser step size, the result is non-causal data.

Figure D-19 shows an extreme case where the wrong sign was used for all imaginary components. The result is a trajectory where much of the path is counterclockwise. This is a useless model and simulations run with it produce meaningless results.

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Figure D-19. Inherent Non-Causality with Substantial Counterclockwise Rotation



Artificially Created and Modified Points

The model contains data points that were manually entered.

Unfortunately, attempts to manually correct bad models can unexpectedly introduce the types of problems previously described, such as insufficient resolution, non-causality, or non-realness. Further, it is possible for a model to exhibit all three of these problems.

One common cause of trouble is correcting or adding missing points near DC, to force proper behavior or to balance differential/common-mode components. For example, in an attempt to restore passivity, model consumers sometimes apply a brute-force method of simply scaling the values at frequencies where passivity violations occur. As a rule, these attempts do not improve the model quality, but instead introduce other problems like those detailed in the previous topics. Unfortunately, it is almost impossible to guess the missing points or properly correct existing ones. Therefore, it is almost always safer to use a model as is, rather than try to manually correct it. If correction seems necessary, the model should be sent back to the supplier for re-generation.

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Cascade Multiple S-Parameter Models in Series

S-parameter models are increasingly used in SERDES and other high-speed simulations to characterize portions of the interconnect between drivers and receivers. If multiple elements in a signal path are described with S parameters, the circuit topology may contain a “chain” or “cascade” of S-parameter models connected in series.

For example, Figure D-20 represents a SERDES channel using the following sequence of S-parameter models for an entire signal path:

1. Driver (transmitter) package
2. Breakout (including differential via pair)
3. Transmission lines
4. Connector
5. Transmission lines
6. Breakout (including differential via pair)
7. Receiver package

Figure D-20. SERDES Channel Represented by a Cascaded Series of S-Parameter Models

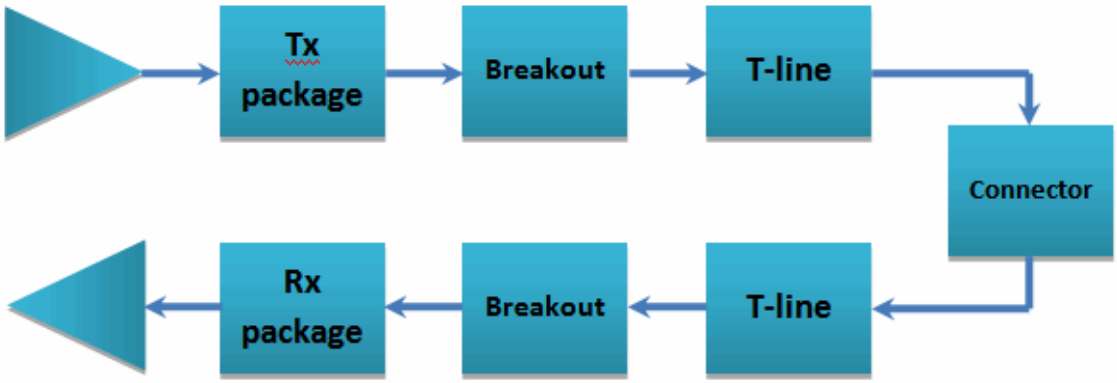
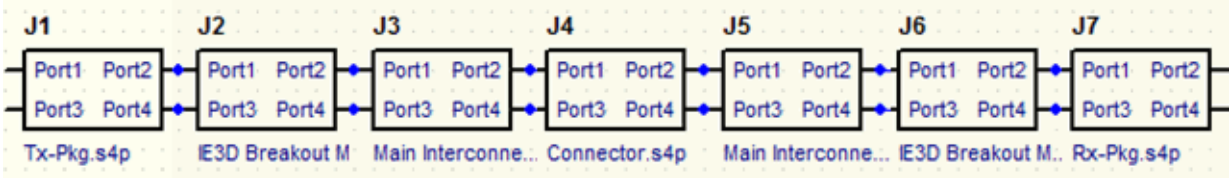


Figure D-21 shows the interconnect portion of the circuit in Figure D-20 as it appears in LineSim.

Figure D-21. Interconnect Portion of SERDES Channel in a LineSim Schematic



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Since the channel is differential (that is, it contains a + and – side), each block is a 4-port S-parameter model.

HyperLynx (and other circuit simulators, such as SPICE) can simulate a chain of S-parameter models, but special care is required in the underlying algorithms to ensure accurate results. The following sections explain why accurate cascading of S-parameter models is algorithmically complex, and what features Touchstone and Fitted-Poles Viewer 2.0 (and newer) provides to address the problem.

Algorithmic Complexity of S-Parameter Cascading **1334**
High-Accuracy Cascading. **1337**
Applying Cascading to Simulation of Certain IBIS-AMI Models **1341**

Algorithmic Complexity of S-Parameter Cascading

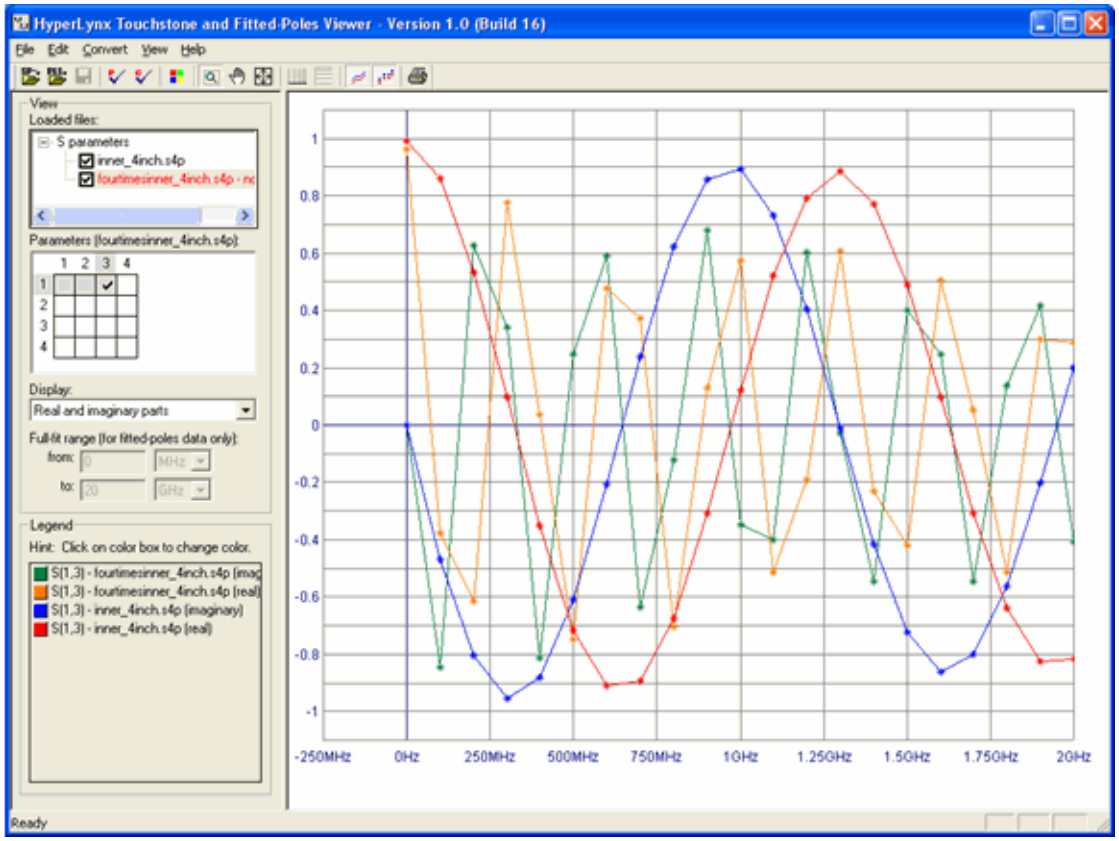
S-parameter models contain data sampled at a discrete set of frequencies.

In a chain of S-parameter models such as shown in Figure D-21, rarely would each model use the same frequencies. Thus, producing a single, cascaded model requires that each sub-model be represented continuously, so that its precise value is known at every frequency in the final cascaded model.

Less obvious is that even if the constituent S-parameter models were sampled at the same frequencies, a cascaded model using those frequencies could easily be undersampled (and thus, simulate inaccurately). For example, Figure D-22 shows the real and imaginary parts of an S-parameter model representing 4 inches of a differential pair in red and blue; the sampling is quite satisfactory. But if four copies of the model are cascaded (with no change in sampling resolution) to represent a 16-inch pair, the orange and green plots result—the sampling is clearly undersampled.

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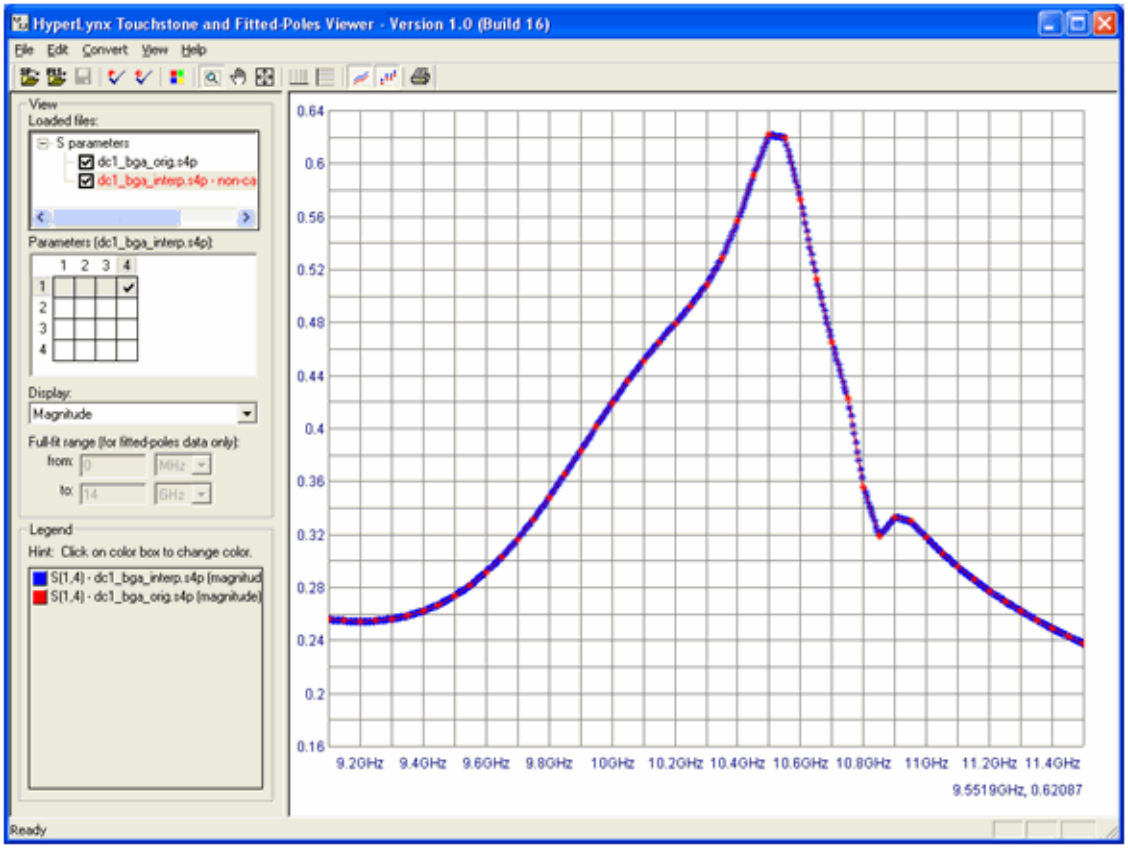
Figure D-22. Real and Imaginary Parts for Non-Cascaded and Undersampled Cascaded S-Parameter Models



Generating new frequency points in an S-parameter model requires interpolation, which sounds relatively easy, but is not. Commercial simulators use a variety of interpolation algorithms, many of which are imperfect. Worse, the user typically has no visibility into these algorithms, and their effects can sometimes be wrong in subtle ways. For example, one widely used SPICE simulator, if instructed to perform an AC-sweep of an S-parameter model to produce a version with finer frequency resolution, produces the magnitudes in Figure D-23, where the original points from simulation are in red and the denser-sampled output in blue.

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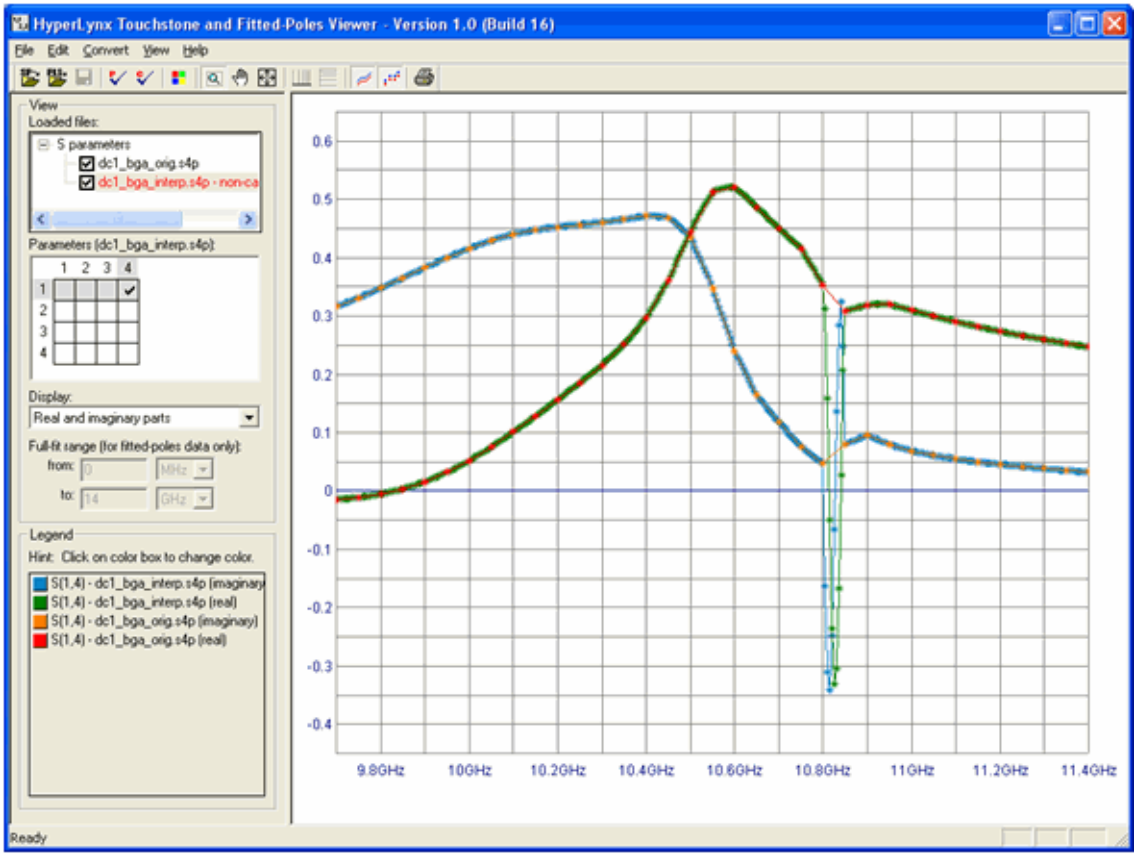
Figure D-23. SPICE-Simulator AC-Sweep Interpolation of an Original S-Parameter Model - Magnitude Plot



The results look quite good, until in Figure D-24, the plot is switched to real/imaginary parts, where the behavior of the new model between 10.80 and 10.85 GHz is clearly wrong.

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Figure D-24. SPICE-Simulator AC-Sweep Interpolation of an Original S-Parameter Model - Real and Imaginary Plot



Another challenge for cascading algorithms is the possibility that the channel may contain series DC-blocking (or AC coupling) capacitors, or a series resonant circuit (such as an L-C structure in an IC package model). Series capacitors or resonant structures make the channel non-transparent at some frequencies. In one commonly used algorithm, the input S-parameter models are each converted to T parameters. The cascaded T-parameter model is found by simple matrix multiplication of the T-parameter sub-models. The resulting T-parameter model is converted to the final cascaded S-parameter model with a standard transformation.

Unfortunately, the transformation from S to T and from T back to S parameters involves inversion of sub-matrices, such as S21. With non-transparent channels, S21 is singular (or close) at certain frequencies, causing numerical instability in the transformations, which often leads to poor results.

High-Accuracy Cascading

You can combine a series of 4-port S-parameter models into a single model.

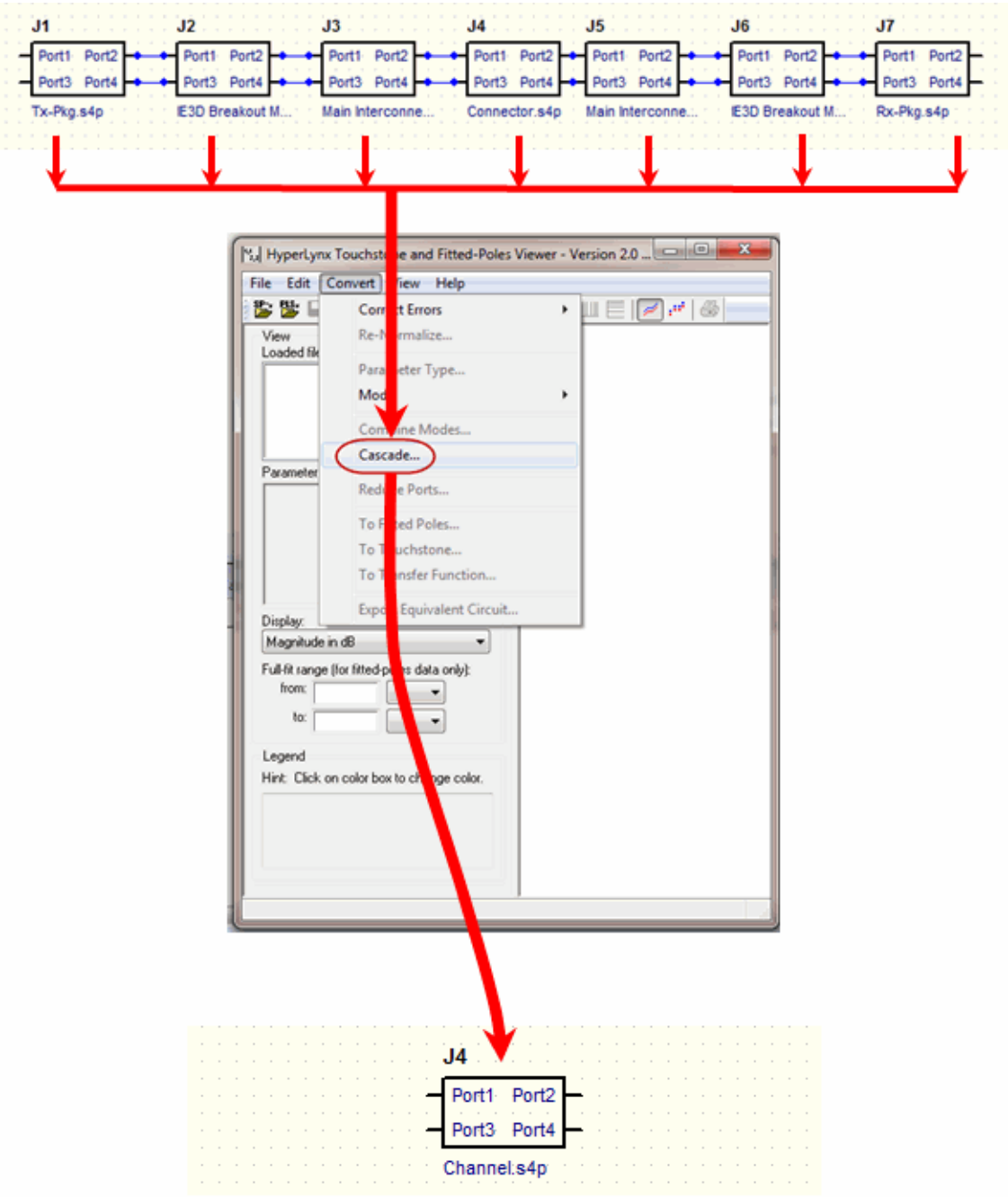
The Touchstone and Fitted-Poles Viewer implements a sophisticated algorithm for cascading 4-port S-parameter models. It automatically selects the proper sampling resolution for the cascaded output model, and uses a proprietary interpolation method that avoids any non-

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physical numerical artifacts. The model quality is equally good for symmetric or asymmetric, passive or non-passive, and transparent or non-transparent cases.

To achieve the highest possible simulation accuracy, you are encouraged to replace chains of 4-port S-parameter models with a single, cascaded model produced in the Touchstone and Fitted-Poles Viewer, as shown in Figure D-25.

Figure D-25. S-Parameter Cascading Workflow



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Procedure

1. Choose the **Models > Edit Touchstone Models** menu item or click  . The Touchstone and Fitted-Poles Viewer opens.

   This step applies to opening the Touchstone and Fitted-Poles Viewer from HyperLynx. This documentation does not provide instructions to open the Touchstone and Fitted-Poles Viewer from other Siemens EDA products.

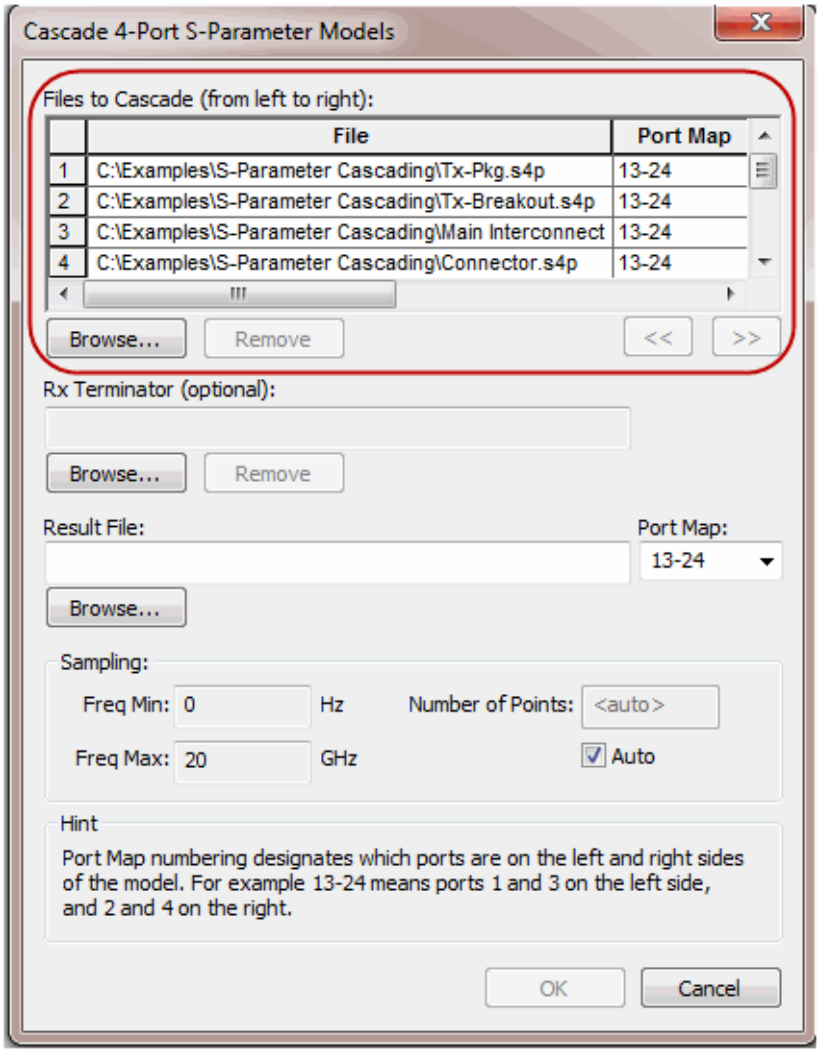
2. Choose the **Convert > Cascade** menu item. The Cascade 4-Port S-Parameters dialog box opens.

3. Below the “Files to Cascade” spreadsheet, click **Browse** to select 4-port S-parameter models that you want to cascade, in left-to-right, driver-to-receiver “signal-flow” order in the schematic.

   Referring to Figure D-25, when you finish this step, the spreadsheet looks something like Figure D-26.

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Figure D-26. Files to Cascade Spreadsheet - Example Contents



4. Verify in the Port Map column that the port ordering for each model is correctly specified. The default value of “13-24” means that the model has ports 1 and 3 on the left side, and 2 and 4 on the right (fairly standard for differential-channel models). If needed, click a Port Map cell and select a different ordering.

5. In the Result File box, specify the location of the output cascaded model, and select the port ordering from the Port Map list.

6. Click **OK**. The Touchstone and Fitted-Poles Viewer produces the new cascaded model and automatically opens it.

7. The minimum and maximum frequencies in the cascaded model are determined automatically from the input models: the minimum frequency is the highest of the starting frequencies in the input models, and maximum frequency is the lowest of the input ending frequencies. The number of points in the cascaded model is chosen automatically by default, but you can uncheck the Auto check box and type a number into the Number of Points field.

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8. (Optional) You can include receiver-end termination in the cascaded model by selecting a 2-port S-parameter model in the Rx Terminator box.
9. Click **OK**.

Results

When the cascading operation completes, the Touchstone and Fitted-Poles Viewer shows the output model. You can use other features in the Touchstone and Fitted-Poles Viewer to inspect the model to confirm that it is a high-quality model. See "Checking S-Parameter Model Quality" on page 1310.

Related Topics

HyperLynx Touchstone and Fitted-Poles Viewer Dialog Box

Applying Cascading to Simulation of Certain IBIS-AMI Models

The cascading feature is also useful when simulating with certain IBIS-AMI models (of advanced SERDES driver/Tx and receiver/Rx buffers). Although the AMI standard requires the analog portions of such models to be provided as IBIS models, some AMI models instead use S-parameter characterizations.

Note

This is a questionable approach, since S-parameter models are linear, and the buffer silicon may not be. Nevertheless, some AMI models are made this way.

Procedure

1. Gather the 4-port S-parameter models needed to represent the interconnect portion of the channel. If necessary, create S-parameter models for certain sections by drawing their circuits in a LineSim schematic and choosing the **Export > S-Parameter Model** menu item.
2. Locate the S-parameter models that represent the Tx and Rx analog stages of the AMI model. The silicon vendor whose AMI you are simulating can help; you can also look in the ASCII.AMI file accompanying the model, for entries such as:

```
(Model_Specific
Tstonefile (Usage Info) (Type String)
(List "Xs.s4p" "S.s4p" "T.s4p" "F.s4p" "XF.s4p"))
```

3. Open the Touchstone and Fitted-Poles Viewer.
4. Chain the S-parameter models (including the AMI portions) in the proper order. See "High-Accuracy Cascading" on page 1337.

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When adding models to the Files to Cascade table in the Cascade 4-Port S-Parameter Models dialog box, start with the Tx analog-stage S-parameter model, add interconnect models, and then end with the Rx model.

5. Convert the cascaded model into a transfer function by choosing the **Convert > To Transfer Function** menu item, then doing the following in the Convert to Transfer Function dialog box:

   a. Set Port Map to match the order in the cascaded channel model.

   b. When working with AMI models, uncheck Default Resistance and Conductance and set all values to 0.

   c. Specify an output file location, then click **OK**.

   The result is a 1-port S-parameter file.

   Note

   When you update the Default Resistance and Conductance values, there is no series termination at the input end and no line-to-ground or line-to-line termination at the output end. These are the proper values when the cascaded channel has AMI S-parameter models attached at the input and output ends. However, if the transfer function conversion operation is applied to a “pure” interconnect channel (with no attached AMI models), then it is sensible to apply 50-ohm termination at both ends. To do so, disable the check box, and set Z1 = Z2 = 50 ohms, Y1 = Y2 = 0.02 1/ohms [Siemens], and Y12 = 0.0 1/ohms.

6. Convert the output file from step 4 to fitted-poles form, by choosing the **Convert > To Fitted Poles** menu item. If the model is complex (has a wide frequency range and contains many details, such as resonances, within its range), then it may be necessary to increase the “Maximum complexity order” option value from its default to a value of 1000. The result is a 1-port PLS file.

7. In LineSim, run the AMI wizard to perform the simulation (choose the **Simulate SI > Run IBIS-AMI Channel Analysis** menu item). Because the channel (including attached Tx and Rx analog stages) was characterized in the preceding steps, on the Set up Channel Characterization page of the FastEye Channel Analyzer wizard, click **Load**. This reads the file created in step 5.

8. Proceed as normal in the rest of the wizard.

Convert and Fix Touchstone and Fitted-Poles Models

You can convert a Touchstone model to another type (such as S-parameter to Z-parameter), reduce the number of ports (to reduce the model file size), and so on.

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Use options in the Convert menu to convert or repair Touchstone and fitted-poles models.

You can switch between standard and mixed-mode S-parameter models on-the-fly without performing a conversion.

Restriction

You must have the Complex Pole Fitter license.

Table D-1. Convert Menu Contents

Option	Description
Correct Errors	• **Enforce Passivity**—Make a Touchstone model passive. • **Enforce Symmetry**—Make a Touchstone model symmetric. See "Checking and Fixing Passivity and Causality Errors" on page 1307.
Re-Normalize	Opens the Re-Normalize dialog box. Select to re-normalize the characteristic impedance of a Touchstone model.
Parameter Type	Opens the Convert Parameter Type Dialog Box. Select to convert a Touchstone model of one parameter type (S, Y, or Z) to another (S, Y, or Z).
Mode	• **General**—Opens the Convert Mode Dialog Box with the "Convert to" option fully available. • **Standard to Mixed**—Opens the Convert Mode dialog box with the "Convert to" option set to Mixed Mode and read only. • **Standard to Differential**—Opens the Convert Mode dialog box with the "Convert to" option set to Differential Mode and read only. **Restriction:** This option is only available for Touchstone models with 4 ports, 8 ports, or other multiples of 4 ports.
Combine Modes	Opens the Combine to Standard Mode Dialog Box. Select to combine common and differential mode models. This option is only available for a differential mode Touchstone or fitted-poles model.
Cascade	Opens the Cascade Four-Port S-Parameter Models Dialog Box. Select to cascade (with high accuracy) multiple 4-port S-parameter models into a single S-parameter model.
Reduce Ports	Opens the Reduce Number of Ports Dialog Box. Select to remove unused ports from a Touchstone model. This option reduces the file size.

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Table D-1. Convert Menu Contents (cont.)

Option	Description
To Fitted Poles	Opens the Convert to Fitted Poles Dialog Box. Select to convert a Touchstone model to a fitted-poles model.
To Touchstone	Opens the Convert to Touchstone Dialog Box. Select to convert fitted-poles models to Touchstone models.
To Transfer Function	Opens the Convert to Transfer Function Dialog Box. Select to convert a 4-port Touchstone model to a transfer function, in the form of a 1-port Touchstone model. **Restriction:** This option is only available for Touchstone models with 4 ports, 8 ports, or other multiples of 4 ports.
Export Equivalent Circuit	Opens the Save As dialog box. There are no exporting options. Select to convert an S-parameter fitted-poles model to an equivalent SPICE circuit (INC). You cannot use this to convert a Touchstone model to an equivalent SPICE circuit. Validate the exported model by running an ADMS simulation with the original PLS model and then the equivalent circuit model and comparing the results. You cannot load the equivalent circuit file into the Touchstone and Fitted-Poles Viewer.

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Supporting Information for the Touchstone Viewer

Understanding Touchstone viewer behavior can involve technical concepts that sometimes require additional explanation.

Topic	Description
Touchstone and Fitted-Poles Models	Touchstone and fitted-poles models use n-port network parameter data to represent passive interconnect networks and active devices. However understanding the quality or contents of these models is difficult without help from analytic software, such as the Touchstone and Fitted-Poles Viewer.
Simulating S-Parameter Models in the Time Domain	S-parameter models are frequency-domain characterizations because they describe the behaviors of a set of ports at a number of different frequencies. Because of these frequency dependencies, the simulator uses special techniques to solve S-parameter models in a time-domain simulation.
Electrical Circuits Used for TDR Impedance Plots	The stimulus/probe location and port termination used by the Touchstone Viewer for TDR impedance plots can help you interpret the results, especially for the Y-axis units and amplitude.
Event Times in TDR Impedance Plots	Event times in the plot have a factor of two built in because the signal source and receiving probe are both located at the same port, creating a round trip. This behavior is similar to physical TDR equipment, where the incident voltage wave propagates to the location of the impedance discontinuity and back to the probe.
Electrical Circuits Used for Time-Domain Responses	The stimulus type and location, probe location, and port termination used by the Touchstone Viewer for time-domain responses can help you interpret the results, especially for the Y axis units and amplitude.
Stimulus Options for Time-Domain Responses	This topic describes the stimulus options for time-domain responses.
Removing the Effect of Measurement Fixtures From an S-Parameter Model (De-embedding)	You can remove the effects of measurement fixtures, such as test equipment connectors, from a measured S-parameter model to create an S-parameter model that represents only the PCB structures that you want to analyze (the network).
Edit the Appearance of Curves and Legends	You can change the appearance of the contents of the chart to make them easier to view or improve printing legibility.
COM Analysis Results	Common operating margin (COM) analysis creates a report file and various other files. If a report file shows poor results, the other files can help you identify problems to fix in your design.

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Topic	Description
Touchstone Viewer Dialog Boxes	Understanding the meaning of dialog box options can sometimes require additional explanation.

Touchstone and Fitted-Poles Models

Touchstone and fitted-poles models use n-port network parameter data to represent passive interconnect networks and active devices. However understanding the quality or contents of these models is difficult without help from analytic software, such as the Touchstone and Fitted-Poles Viewer.

Touchstone models are in a format originally developed by Agilent Technologies and then adopted by the EIA/IBIS Open Forum. Because S-parameter models are in the Touchstone format, they appear incomprehensible when viewed as text. What you see is a long, continuous listing of frequencies and associated complex numbers, such as magnitude/angle and real/imaginary pairs, that describe the relationships among the model ports.

You typically obtain Touchstone models from a component vendor or generate your own models with a test bench and a vectored network analyzer (VNA).

Touchstone models containing S-, Y-, or Z-parameter data are often used to represent equivalent circuits for backplane connectors and IC packages. Part of this popularity resulted because VNAs make it relatively easy to collect n-port network parameter data for a circuit and create a Touchstone model for it.

Fitted-poles models are in a proprietary format and represent Siemens EDA's preferred way to simulate Touchstone S-parameter models. A fitted-poles model contains a set of complex poles/residues representing frequency behavior in a semi-analytical way. You typically obtain fitted-poles models by running an ADMS simulation, which automatically converts an S-parameter model into a fitted-poles model to decrease simulation run time and model file parsing time. You can manually obtain fitted-poles model using the Touchstone and Fitted-Poles Viewer, which allows you to convert a Touchstone model into a fitted-poles model.

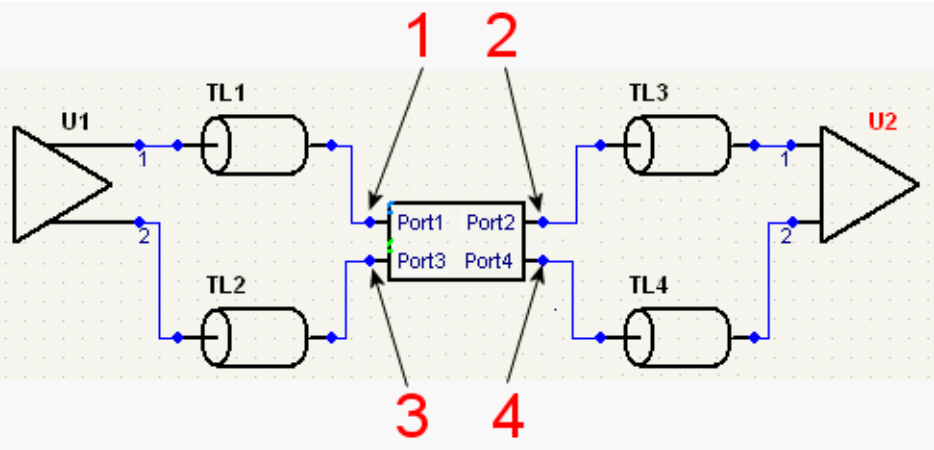
S-Parameter Port Numbering

On single-ended and differential nets, the ports of S-parameter models can have any ordering, if the ordering of the ports in the S-parameter model follows the same sequence that the HyperLynx LineSim schematic uses. The simulators and the Touchstone and Fitted-Poles Viewer automatically use the sequence specified in the schematic.

Touchstone models do not contain information to tell you how to associate the data sets they contain with signal names on the circuit element or component which they describe. Figure D-27 shows the recommended ordering for the ports of an S-parameter model.

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Figure D-27. Recommended S-Parameter Port Numbering



You can extend the recommended red numbering scheme to S-parameter models with more than four ports.

The electrical significance of each S-parameter plot depends on the way you order the S-parameter information. For example, the single mode S(2,1) S-parameter could be the transmission loss (recommended), or the reflection coefficient (not recommended) depending on how the information in the S-parameter file is ordered.

The descriptions for individual S-parameter plots in this document rely on the use of the recommended port numbering scheme in Figure D-27. If you use another scheme, you must replace the S-parameter coefficients in the description in Table D-2 with those that match your scheme.

Table D-2 provides descriptions for the individual standard mode S-parameters corresponding to an incident signal at port one.

The port numbering convention for S-parameters is S(<receive_port><stimulus_port>). The responding port is always listed first. For example, S(2,1) represents the signal received at port 2 for the signal transmitted at port 1. This document uses a comma to separate the port numbers.

Note

The Touchstone and Fitted Poles Viewer display control grid always displays the standard S-parameter descriptions.

Table D-2. Receive Port and Stimulus Port Combinations

Receive Port and Stimulus Port Combination	Means
S(1,1)	Reflection coefficient; also known as return loss S(1,1) = (Sine wave output from port 1)/(Sine wave into port 1)

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Table D-2. Receive Port and Stimulus Port Combinations (cont.)

Receive Port and Stimulus Port Combination	Means
S(2,1)	Transmission loss; also known as insertion loss S(2,1) = (Sine wave output from port 2)/(Sine wave into port 1)
S(3,1)	Near-end or backward crosstalk S(3,1) = (Sine wave output from port 3)/(Sine wave into port 1)
S(4,1)	Far-end, or forward, crosstalk S(4,1) = (Sine wave output from port 4)/(Sine wave into port 1)

Mixed Mode S-Parameters

To understand the performance of elements of differential nets, you should use mixed mode S-parameters instead of standard mode S-parameters.

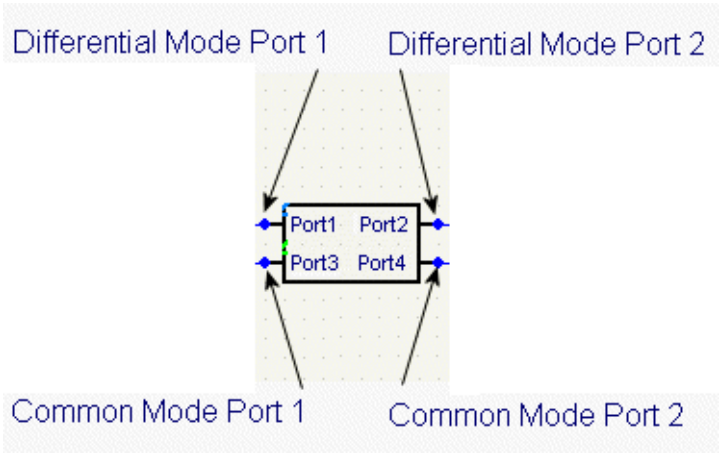
You can convert standard mode S-parameters to mixed-mode S-parameters using the Convert Mode dialog box.

Tip

It is important that the S-parameter port numbers in the Convert Mode dialog box matches the numbering in the schematic and that you use the preferred port numbering sequence.

If you use the recommended S-parameter port ordering, see Figure D-27, the propagation of the common mode signal and differential mode signal components of a signal are shown in Figure D-28.

Figure D-28. S-Parameter Port Numbering for Mixed Mode S-Parameter Models



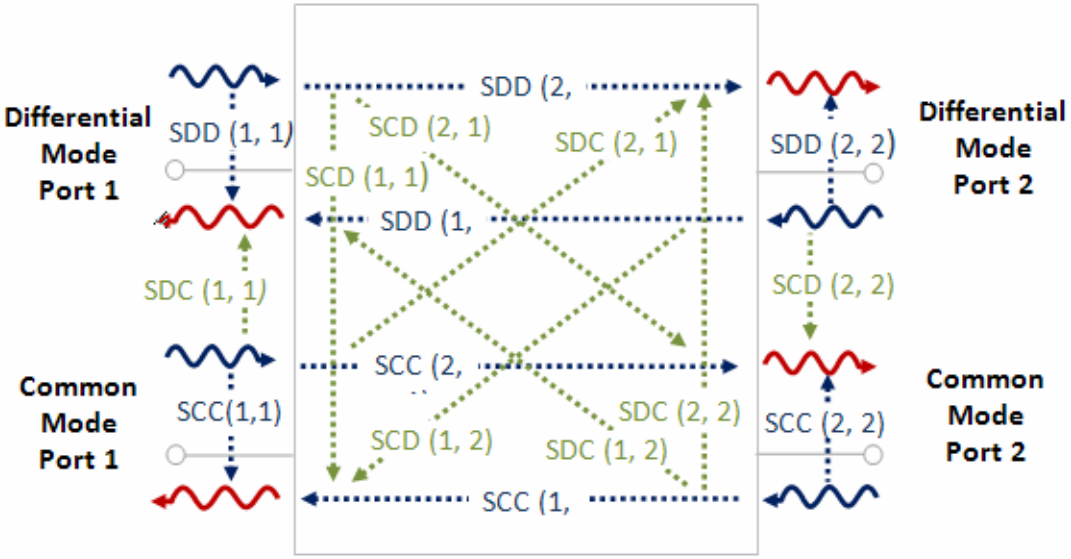
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

The mixed mode S-parameters describe the propagation of the differential signal from port 1 to port 2. In particular they describe the propagation and reflection:

- Differential component of the signal, SDD(n,n)
- Common component of the signal, SCC(n,n)
- Component of the signal converted from differential mode to common mode, SCD(n,n)
- Component of the signal converted from common mode to differential mode, SDC(n,n)

Figure D-29 shows all of the mixed mode parameters for a four-port S-parameter element.

Figure D-29. Mixed-Mode S-Parameters for a 4 Port Model



The meaning of the individual S-parameter coefficients are similar to those of their standard S-parameter equivalents. For a signal originating at port 1:

- Differential Mode
  - SDD(1, 1) — Differential mode reflection also known as return loss
  - SDD(2, 1) — Differential mode transmission loss also known as insertion loss
- Common Mode
  - SCC(1, 1) — Common mode reflection also known as return loss
  - SCC(2, 1) — Common mode transmission loss also known as insertion loss
- Differential to Common Mode Conversion
  - SCD(1, 1) — Common mode reflection due to the forward differential mode signal

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

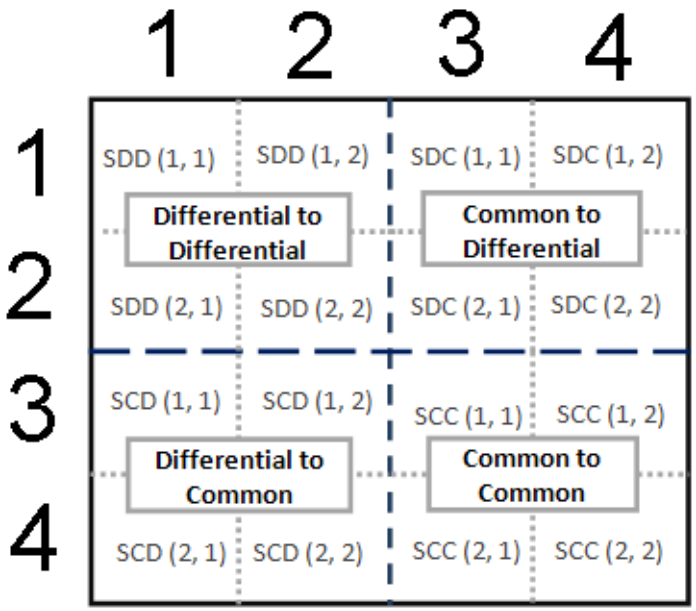
  - SCD(2,1) — Common mode transmission due to the forward differential mode signal

- Common to Differential Mode Conversion
  - SDC(1, 1) — Differential mode reflection due to the forward common mode signal
  - SDC(2, 1) — Differential mode transmission due to the forward common mode signal

Looking at Figure D-30, you can obtain the S-parameter propagation for signals originating at port 2 by replacing 1 for 2 and 2 for 1 in all the descriptions above. For example, SDD(2,2) is the differential mode reflection or return loss for signals originating at port 2.

The Touchstone and fitted poles viewer does not display the mixed-mode S-parameter coefficients in its display control grid. For a 4 port S-parameter model, the grid will have 16 entries. You can use Figure D-30 to select the correct grid square for a specific mixed-mode S-parameter curve:

Figure D-30. Mixed Mode S-Parameter Matrix



Simulating S-Parameter Models in the Time Domain

S-parameter models are frequency-domain characterizations because they describe the behaviors of a set of ports at a number of different frequencies. Because of these frequency dependencies, the simulator uses special techniques to solve S-parameter models in a time-domain simulation.

Most simulators solve S-parameter behavior in time using a relatively straightforward method involving inverse Fourier transformation and direct convolution. It is beyond the scope of this

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topic to describe convolution, but a description can be found in nearly any undergraduate text on communications theory.

By contrast, Siemens EDA simulators use complex pole fitting (CPF) to simulate S parameters in the time domain. Prior to simulation, the CPF method fits an S-parameter model to a set of complex poles, which accurately represents the frequency dependencies of the model. These poles can then be simulated directly in time. Fitting is required only once and the resulting poles can be reused.

Compared to convolution, CPF has these important advantages:

- **Performance —** CPF is typically 5-7 times faster than convolution.
- **Accuracy —** To increase speed, convolution-based simulators are often forced to truncate the time-domain response they use to characterize an S-parameter model. This can cause accuracy problems, especially at lower frequencies. CPF avoids truncation.
- **Model compression —** S-parameter models are often very large, for example, over 500 MB, which decreases overall simulation speed and makes them difficult to transfer from person to person. By contrast, after CPF fitting, a model is greatly compressed 10-100 times. The fitted model is completely self-contained, allows all types of analyses (DC, AC, and transient), and serves as a complete replacement for the original sampled data.
- **Causality —** A common problem with S-parameter models is the lack of causality, which is described in more detail in “Inherent Non-Causality” on page 1331. CPF automatically fixes causality problems during the fitting process.
- **Passivity —** The same truncation used in convolution to increase speed may also introduce non-passivity into a model. CPF avoids truncation.

The explicit fitting step performed by CPF makes the input model quality problems explicit and visible. Convolution-based simulators make similar transformations, but only internally. Thus you have no visibility into what changes were needed and no control over how they occur. This is an advantage of CPF because, after fitting, you can inspect data on which the simulator actually runs and compare it to the original data.

Electrical Circuits Used for TDR Impedance Plots

The stimulus/probe location and port termination used by the Touchstone Viewer for TDR impedance plots can help you interpret the results, especially for the Y-axis units and amplitude.

Even though this dialog box obtains the time-domain response indirectly, you can think of the stimulus as a unit step, which is 0 for all t < 0 and 1 for t >= 0. The unit step stimulus produces a waveform that resembles a set of stairs.

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Note

Notes for Figure D-31, Figure D-32, and Figure D-33: For termination information for inactive ports, see the description for "Other ports" in "HyperLynx Touchstone and Fitted-Poles Viewer Dialog Box" on page 1373.

Figure D-31. Electrical Circuit Used for TDR Impedance Plots - Single-Ended

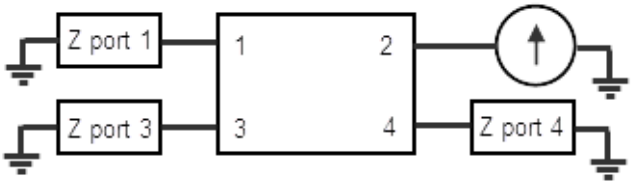


Figure D-32. Electrical Circuit Used for TDR Impedance Plots - Mixed Mode Plot Type and Differential Mode

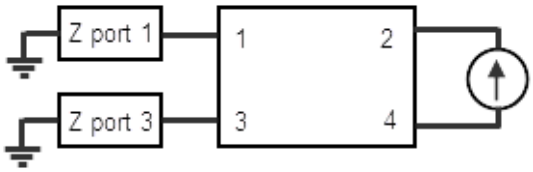
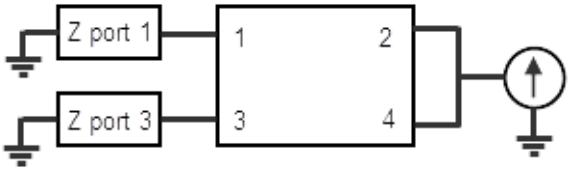


Figure D-33. Electrical Circuit Used for TDR Impedance Plots - Mixed Mode Plot Type and Common Mode



Event Times in TDR Impedance Plots

Event times in the plot have a factor of two built in because the signal source and receiving probe are both located at the same port, creating a round trip. This behavior is similar to physical TDR equipment, where the incident voltage wave propagates to the location of the impedance discontinuity and back to the probe.

Figure D-34 shows how to use the Touchstone Viewer to measure the delay through the connector or other topology represented by the Touchstone file. Use the delay to identify the stair step interval in Figure D-35. The 4-port Touchstone file in the figure uses standard port numbering, as described in S-Parameter Port Numbering. Also, the geometry represented by the Touchstone file includes a differential pair with two nearby stitching vias.

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Figure D-34. Measure Delay Between Touchstone File Ports

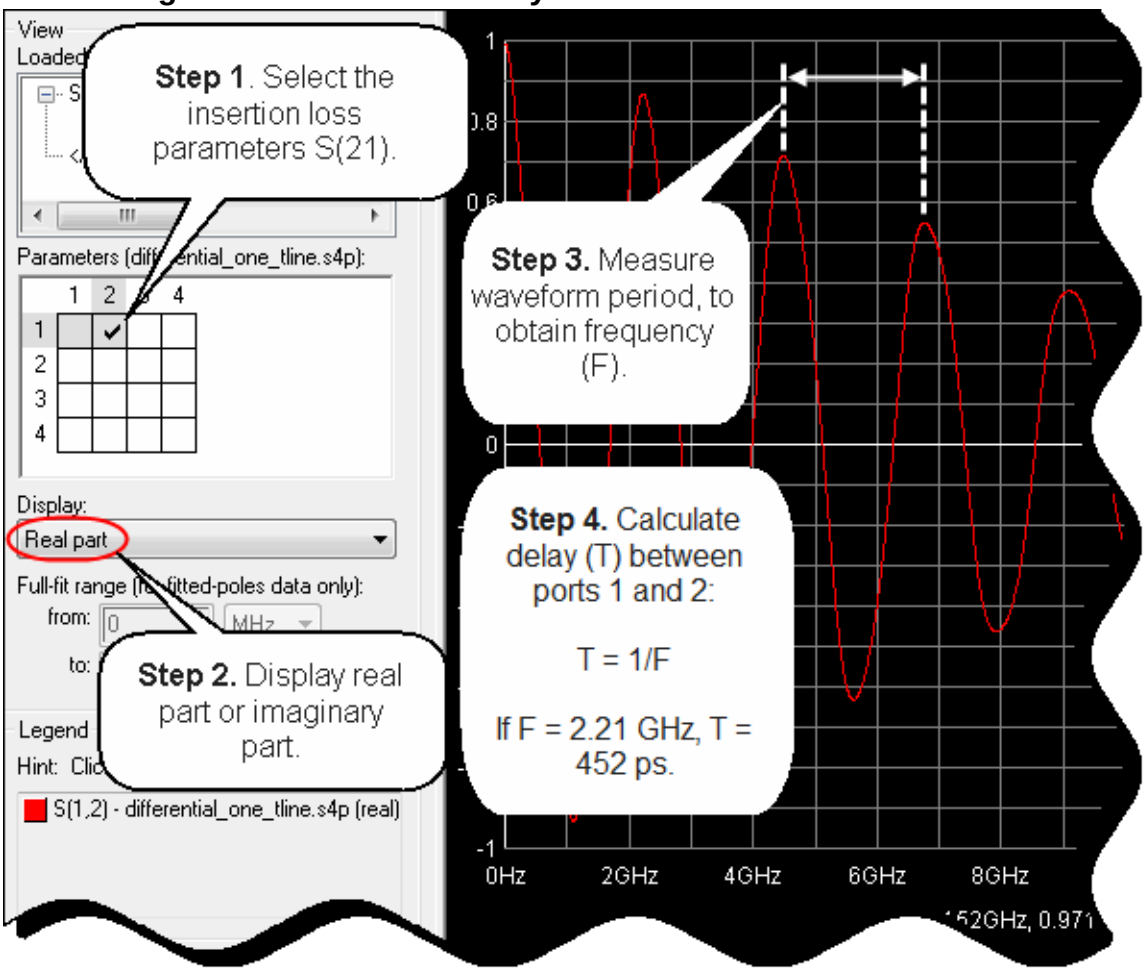


Figure D-35 shows a single-ended plot to illustrate how the reported time for impedance changes is twice the insertion delay.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure D-35. Measure Stair Step Intervals - Single-Ended

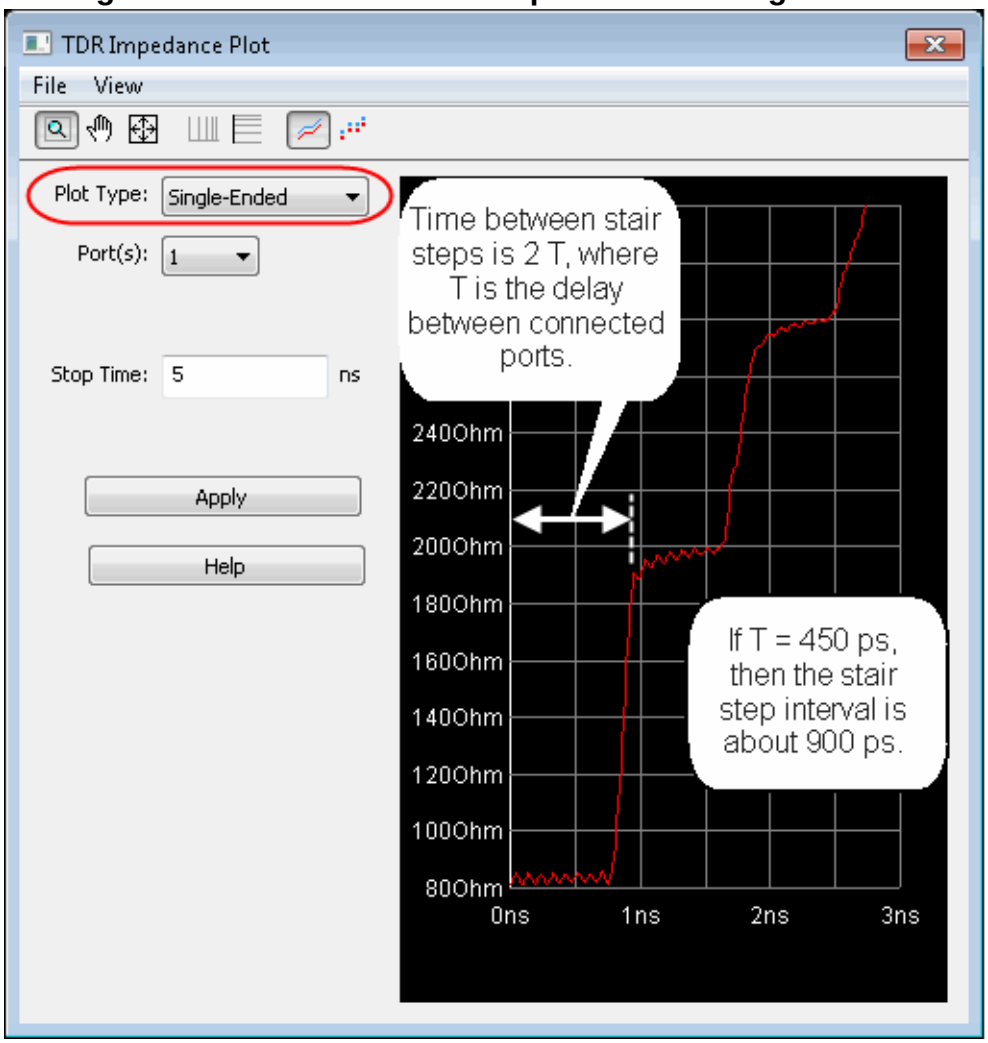


Figure D-36 shows a mixed mode plot to illustrate how the reported time for impedance changes are two times the insertion delay from port to port.

Note

For mixed mode plots, there is no general rule about the stair step interval being twice the delay between connected ports. For example, consider a Touchstone model extracted from a coupled two conductor transmission line. With weak coupling, all delays could be close to each other (for single ended, mixed mode differential, and mixed mode common cases). With strong coupling, the delays may become different, because even and odd mode propagation velocities on coupled microstrip traces are not equal.

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Figure D-36. Measure Stair Step Intervals - Differential

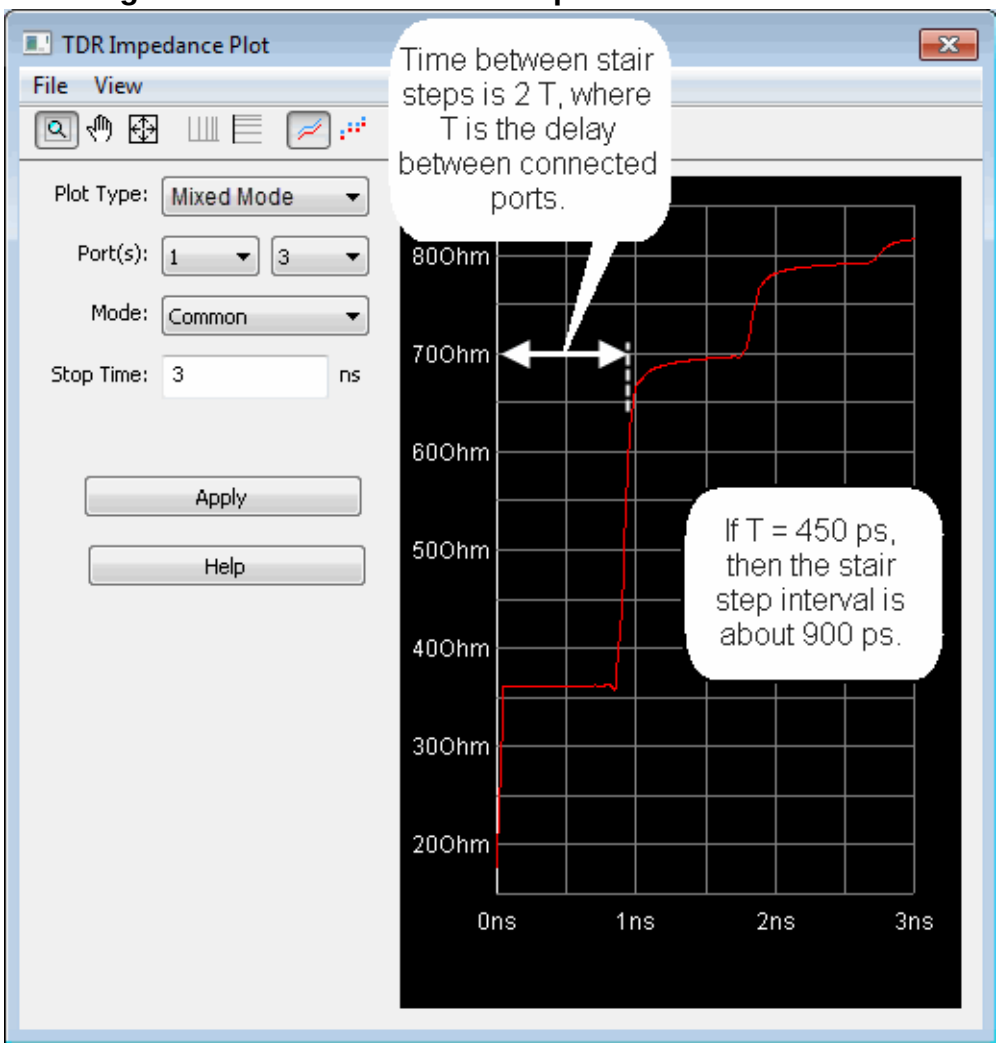
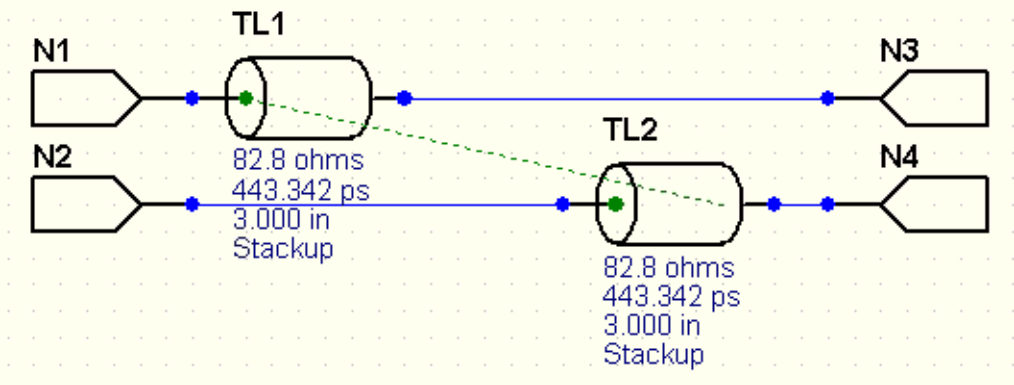


Figure D-34, Figure D-35, and Figure D-36 show deliberately simple plots. The Touchstone model was exported from LineSim, using the schematic in Figure D-37.

Figure D-37. TDR Impedance Plots - Schematic for Extracted Touchstone Model



Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Electrical Circuits Used for Time-Domain Responses

The stimulus type and location, probe location, and port termination used by the Touchstone Viewer for time-domain responses can help you interpret the results, especially for the Y axis units and amplitude.

Note

Reference to ground is not drawn for model ports in the figures below, to reduce clutter.

Electrical Circuit Used for Time-Domain Responses - S-Parameters

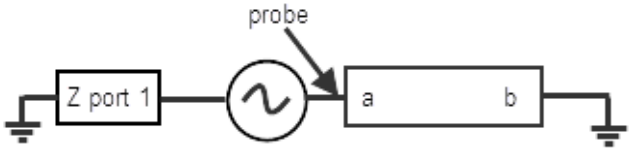
Figure D-38. S(a,b) shows the ratio of the reflected wave at port a to the magnitude of the incident wave, when the incident wave is applied to port b, and the incident waves at all other ports (if any) are assumed to be zero. The incident wave is created by a wave source whose internal impedance equals the normalizing impedance for port b.

Note

In this context, the source is an incident wave and the response is a reflected wave.

All ports, except for b, are terminated to ground by a normalizing impedance of the value specified in the Touchstone model. The Touchstone model can either specify the same normalizing impedance for all ports or specify a unique normalizing impedance for each port.

Figure D-38. Electrical Circuit Used for Time-Domain Responses - S(a,b)



The incident and reflected waves are formed by port voltage and current in the following ways:

Reflected wave $$Wa = \frac{(Va - (Za \times Ia))}{\sqrt{Za}}$$

where:

Va is the voltage at port a

Za is the normalizing impedance for port a

Ia is the current at port a

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Incident wave $Wb = \frac{(Vb + (Zb \times Ib))}{\sqrt{Zb}}$

where:

Vb is the voltage at port b

Zb is the normalizing impedance for port b

Ib is the current at port b, where current enters the port

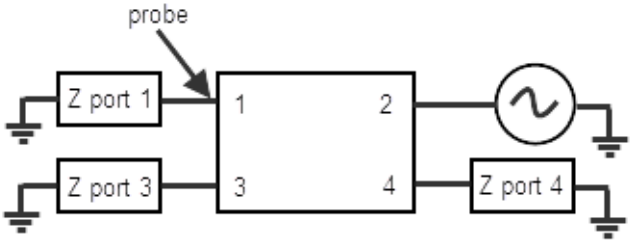
The response is a dimensionless ratio: $\frac{Wa}{Wb}$

When all normalizing impedances are identical in the Touchstone file, $\sqrt{Z}$ can be omitted because the ratio stays the same. But when normalizing impedances are not identical, the equations must include them.

For information about the available types of incident waves, see "Stimulus Options for Time-Domain Responses" on page 1359.

Figure D-39 shows an S(1,2) example for a four-port model.

Figure D-39. Electrical Circuit Used for Time-Domain Responses - S(1,2)



Electrical Circuit Used for Time-Domain Responses - Z-Parameters

Figure D-40. Electrical Circuit Used for Time-Domain Responses - Z(a,b)

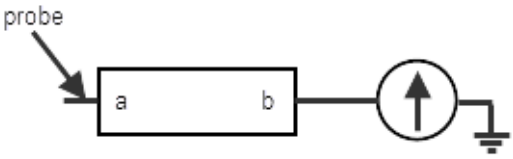


Figure D-40 Z(a, b) shows the voltage at port a when the stimulus from an ideal current source is applied to port b. All ports, except for b, are open.

where:

The ideal current source at port b has infinite internal impedance.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Current enters port b.

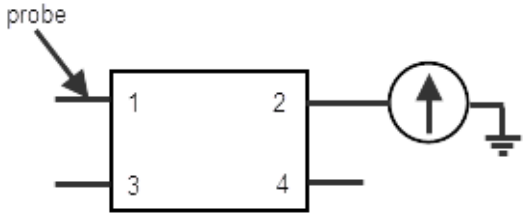
The response at port a has the dimension of impedance.

For Z(a,a), the current is applied and voltage is measured at the same port.

For information about the available types of incident waves, see "Stimulus Options for Time-Domain Responses" on page 1359.

Figure D-41 shows a Z(1,2) example for a four-port model.

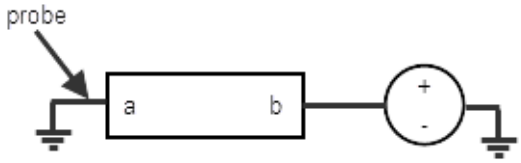
Figure D-41. Electrical Circuit Used for Time-Domain Responses - Z(1,2)



Electrical Circuit Used for Time-Domain Responses - Y-Parameters

See Figure D-42.

Figure D-42. Electrical Circuit Used for Time-Domain Responses - Y(a,b)



Y(a, b) shows the current entering port a when the stimulus from an ideal voltage source is applied to port b. All ports, except for b, are grounded by ideal conductors.

where:

The ideal voltage source at port b has zero internal impedance.

Voltage enters port b.

The response at port a has the dimension of conductance, assuming unit magnitude of step or pulse.

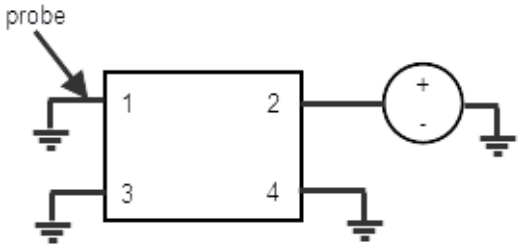
For Y(a,a), the voltage is applied and current is measured at the same port.

For information about the available types of incident waves, see "Stimulus Options for Time-Domain Responses" on page 1359.

Figure D-43 shows a Y(1,2) example for a four-port model.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure D-43. Electrical Circuit Used for Time-Domain Reponses - Y(1,2)



Stimulus Options for Time-Domain Responses

This topic describes the stimulus options for time-domain responses.

Note

t0 is time zero.

Dirac Impulse

Can be thought of as a derivative of an ideal unit step or as a rectangular pulse of height 1/H and width H when H approaches 0. To keep the area under the curve equal to one, the amplitude is practically infinite. The Dirac impulse can also be thought of as infinite at t0 and 0 everywhere else.

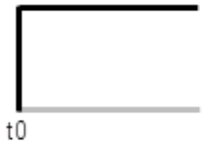
Figure D-44. Stimulus - Dirac Impulse



Unit Step

— 0 for all t < 0 and 1 for t >= 0.

Figure D-45. Stimulus - Unit Step

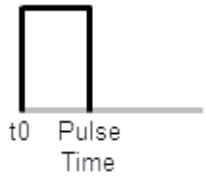


Rectangular

1 between t0 and the start of the falling transition (pulse time), and 0 otherwise.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure D-46. Stimulus - Rectangular



Trapezoidal

The sum of two separate ramped rising and falling transitions: x(t)= a(t) +b(t)

where:

x(t) is the summed wave that is applied to the Touchstone model port. Note that the probed response may be slightly distorted because it is not x(t), but the response on the port connected to input x(t).

a(t) is the rising transition that goes up linearly from 0 to 1 between t0 and t = rise time, and then stays a constant 1.

b(t) is the falling transition that stays 0 until t = pulse time (the start of the falling transition), goes down linearly from 0 to -1 for fall time, and then stays a constant -1.

The stimulus becomes 0 at t > max(Rise Time, Pulse Time+Fall Time).

Figure D-47 shows a single pulse, where Rise Time is less than Pulse Time.

Figure D-47. Stimulus - Trapezoidal Pulse, Rise Time is Less Than Pulse Time

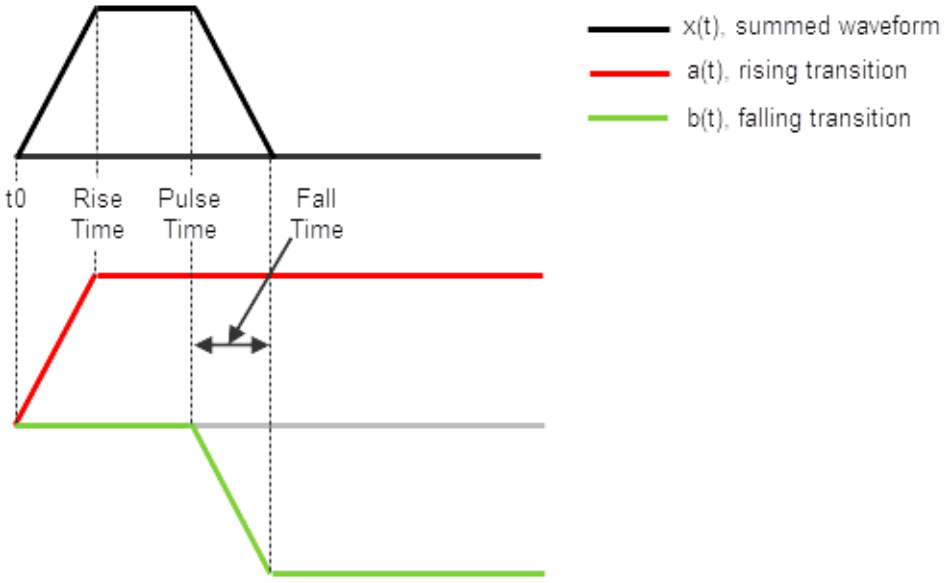
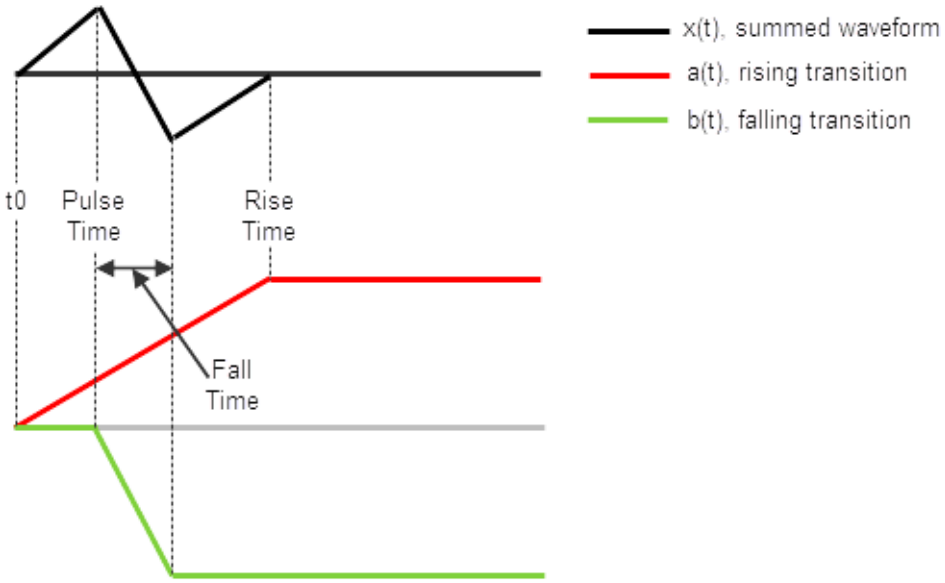


Figure D-48 shows a double-pulse, where Rise Time is greater than Pulse Time.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure D-48. Stimulus - Trapezoidal Pulse, Rise Time is Greater Than Pulse Time



Removing the Effect of Measurement Fixtures From an S-Parameter Model (De-embedding)

You can remove the effects of measurement fixtures, such as test equipment connectors, from a measured S-parameter model to create an S-parameter model that represents only the PCB structures that you want to analyze (the network).

Prerequisites

- You have opened the measured S-parameter model that represents the network, including the measurement fixture.
- You have S-parameter files that represent the measurement fixture on both sides of the network.
- All three S-parameter files have a 1 -> N+1 port numbering pattern and the same number of ports, with ports mapped to the correct signals in your design.

Procedure

1. Choose the **Convert > De-embed** menu item.
2. Next to Left S-Parameter, click **Browse** and select the file that represents the measurement fixture on the left side of the network.
3. Next to Right S-Parameter, click **Browse** and select the file that represents the measurement fixture on the right side of the network.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

4. Next to De-embedded Result, type the name and location of the resulting de-embedded S-parameter file.
5. Click **OK**.

   The de-embedded file opens in the viewer.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Edit the Appearance of Curves and Legends

You can change the appearance of the contents of the chart to make them easier to view or improve printing legibility.

Editing Curve Colors for the Current Session **1363**
Editing Default Curve Colors **1363**
Editing Parameter Curve Color Modes **1364**
Editing Chart Appearance Properties **1364**

Editing Curve Colors for the Current Session

You can edit the color of curves for the current session.

Procedure

1. In the Legend window, click the color square for the curve.
2. Select the new color and click **OK**.

   The color is for the current session only and the default color is applied when you next open the model file.

Editing Default Curve Colors

You can edit the set of colors used to display parameter curves.

Restrictions and Limitations

- This setting applies only when you set the parameter curve color mode to “Color schemes”. See “Editing Parameter Curve Color Modes” on page 1364.

Procedure

1. On the toolbar, click **Edit plot colors**  . The Edit Colors dialog box opens.
2. Select a waveform type from the “Data type” list.
3. Select a color scheme from the “Color scheme” list.
4. Click **Edit**, edit the color scheme properties, then click **OK**.

   The color sequence distinguishes parameters of the same type when multiple models are open at the same time.
5. To manage or add your own color schemes, click **Organize**, manage the color schemes, and then click **OK**.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

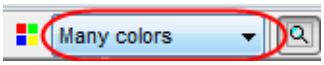
Results

The color scheme is applied to any currently displayed parameters, even if you have edited their color by performing the procedure described in "Editing Curve Colors for the Current Session" on page 1363.

Editing Parameter Curve Color Modes

You can choose the set of colors used to display parameter curves.

Procedure

Select an item from the "Select color mode" 

list:

- Many colors — Assign a unique color to the curves for each pair of ports, from the large set of available colors.
- One color per file — Assign the same color to the curves for all pairs of ports.

  This option can make comparing files easier.
- Color schemes — Assign a unique color to the curves for each pair of ports, from the small set of available colors. The Display setting determines the available set of colors.

Editing Chart Appearance Properties

You can change the legend font, chart colors (except for curves), and vertex size.

Procedure

1. Choose the **Edit > Options** menu item. The Options dialog box opens.
2. On the **Chart Control** tab, edit properties as needed.

   Vertex size is in pixels.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

COM Analysis Results

Common operating margin (COM) analysis creates a report file and various other files. If a report file shows poor results, the other files can help you identify problems to fix in your design.

COM Report File 1365
COM Miscellaneous Output Files 1367
Example PMF and CDF Plots. 1369

COM Report File

Analysis reports the figure of merit (FOM) and channel operating margin (COM) values and various supporting information. For parameter and results definitions, see the appropriate IEEE specification (802.3bj, 802.3bm).

Report Item	Description
Differential connectors	Identifies a channel: • VICTIM — A “through” channel. • FEXT — An aggressor channel whose transmitter is located “far” from the receiver for the victim channel. • NEXT — An aggressor channel whose transmitter is located “near” the receiver for the victim channel.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Report Item	Description
FOM optimization results	FOM — A simplified measure that is mostly proportional to COM. The software computes FOM by varying all modifiable equalization parameters, and then reports COM for the combination of parameters that maximizes FOM.
	A_s (A_s) — Signal magnitude (in volts). This is the value of the bit response at its best sampling location, which is the point near the peak where DFE is able to force the first post-cursor to zero.
	Noise contributors (in volts): • Sigma_ISI — Contribution from ISI (pre- and post-cursors of the bit response itself), considered to be a square root of the "variance" of noise. • Sigma_J — Similar to Sigma_ISI but the effect comes from input jitter. • Sigma_N — Noise at the receiver equalizer. • Sigma_TX — Noise from the transmitter. • Sigma_XT — Noise from crosstalk.
	Best FFE — Optimal pre-tap (C-1), main tap (C0), and post-tap (C1) settings for a feed forward equalizer, used to calculate FOM.
	Best CTLE — Optimal settings for a continuous time linear equalizer, used to calculate FOM. The units for DCGain is dB and the units for other settings is hertz. See an IEEE specification for settings definitions. **Note:** The reported "Best CTLE Zero freq" is not the CTLE zero frequency, but instead is Zerofreq / 10^(DC_gain/20), which equals the frequency of the first pole.
COM results	• A_s — Signal magnitude (in volts). This is the value of the bit response at its best sampling location, which is the point near the peak where DFE is able to force the first post-cursor to zero. • DER_0 — Detector error ratio, as specified for the operation mode used by analysis. • A_{ni} — Noise magnitude. Analysis accounts for the same contributors (ISI, input jitter, noise at the receiver and transmitter, and crosstalk), but found more accurately, without assuming a Gaussian distribution. • COM — Channel operating margin (in dB), which is $20*\log10(A_s/A_{ni})$.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Related Topics

Evaluating SERDES Channel Compliance From an S-Parameter Model in the Touchstone Viewer

COM Miscellaneous Output Files

You can view various output files to identify factors (such as excessive crosstalk or ISI) that may contribute to poor COM analysis results.

You can use EZwave® to open *.LIS* files.

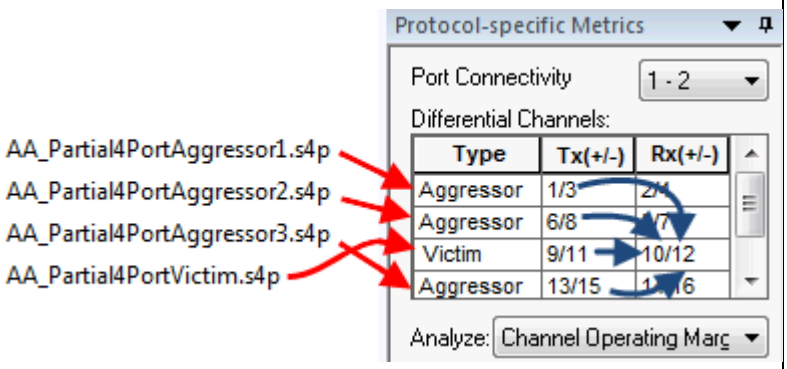
You can use Microsoft Excel or Matlab to open *.TXT* files. It can be helpful to display these files with a logarithmic scale.

Note

Length in the file names below represents Long or Shrt.

File	Description
AA_Partial4PortVictim.s4p AA_Partial4PortAggressor#.s4p	S-parameter models that represent how a signal propagates from the transmitter on each channel to the receiver on the victim net. For a victim channel, the S-parameter model represents how a signal propagates from a transmitter to a receiver on the same pair of differential nets. This model represents a “through” channel. For an aggressor channel, the S-parameter model represents how a signal propagates from a transmitter on an aggressor net to the receiver on the victim net. This model represents a “crosstalk” channel. In the figure below, blue arrows show pairs of input and output ports for an example set of **Partial4Port*s4p* files.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

File	Description
	
AB_Differential2PortVictim.s2p AB_Differential2PortAggressor#.s2p	Similar to AA_Partial4Port... files, but the S-parameter models contain only differential mode information.
B_ShortestPackage.s2p B_LongestPackage.s2p	S-parameter models that represent package behavior.
B_ShortestTline.s2p B_LongestTline.s2p	S-parameter models that represent the behavior of transmission-lines inside the package.
C_Pkg*Length*_DiffModelWithPackagesVictim.s2p C_Pkg*Length*_DiffModelWithPackagesAggressor#.s2p	S-parameter models that represent the differential portion of a channel with packages attached to both ends.
D_Pkg*Length*_VoltTransferNoRxNoiseFilterVictim.s1p D_Pkg*Length*_VoltTransferNoRxNoiseFilterAggressor#.s1p	S-parameter models that represent the scalar transfer function of the victim channel or aggressor channel(s), with packages attached and termination applied.
D_Pkg*Length*_VoltTransferWithRxNoiseFilterVictim.s1p D_Pkg*Length*_VoltTransferWithRxNoiseFilterAggressor#.s1p	Similar to D_Pkg*Length*_VoltTransferNoRx... files, but adds noise from the receiver filter. This behavior is equivalent to lowering the transfer coefficient at high frequencies.
E_Pkg*Length*_NonEqPulseResponseVictim.lis E_Pkg*Length*_NonEqPulseResponseAggressor#.lis	List files with non-equalized bit or pulse responses found from scalar transfer functions by running IFFT.
F_Pkg*Length*_EqualizedPulseResponseVictim.lis F_Pkg*Length*_EqualizedPulseResponseAggressor#.lis	List files with pulse responses equalized by FFE and CTLE, using parameters that produce the best FOM.
G_Pkg*Length*_PMF_WithISI.txt	Text files that describe the probability mass function (PMF) of noise, including ISI.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

File	Description
G_Pkg*Length*_PMF_WithISI_DD.txt	Similar to G_Pkg*Length*_PMF_WithISI.txt, but adds Dual-Dirac jitter.
G_Pkg*Length*_PMF_WithISI_DD_XTALK.txt	Similar to G_Pkg*Length*_PMF_WithISI_DD.txt, but adds crosstalk.
G_Pkg*Length*_PMF_WithISI_DD_XTALK_NOISE.txt	Similar to G_Pkg*Length*_PMF_WithISI_DD_XTALK.txt, but adds Gaussian noise from Tx, Rx and random jitter sources.
H_Pkg*Length*_CDF_WithISI.txt	List files that describe the cumulative distribution function (CDF), which is found by integrating the contents of G_Pkg*Length*_PMF_WithISI.txt.
H_Pkg*Length*_CDF_WithISI_DD.txt	List files that describe the cumulative distribution function (CDF), which is found by integrating the contents of G_Pkg*Length*_PMF_WithISI_DD.txt.
H_Pkg*Length*_CDF_WithISI_DD_XTALK.txt	List files that describe the cumulative distribution function (CDF), which is found by integrating the contents of G_Pkg*Length*_PMF_WithISI_DD_XTALK.txt.
H_Pkg*Length*_CDF_WithISI_DD_XTALK_NOISE.txt	List files that describe the cumulative distribution function (CDF), which is found by integrating the contents of G_Pkg*Length*_PMF_WithISI_DD_XTALK_NOISE.txt. **Note:** The software uses this CDF data to find the magnitude of noise (A_{ni}), at the probability level equal to DER_0.

Related Topics

Evaluating SERDES Channel Compliance From an S-Parameter Model in the Touchstone Viewer

Example PMF and CDF Plots

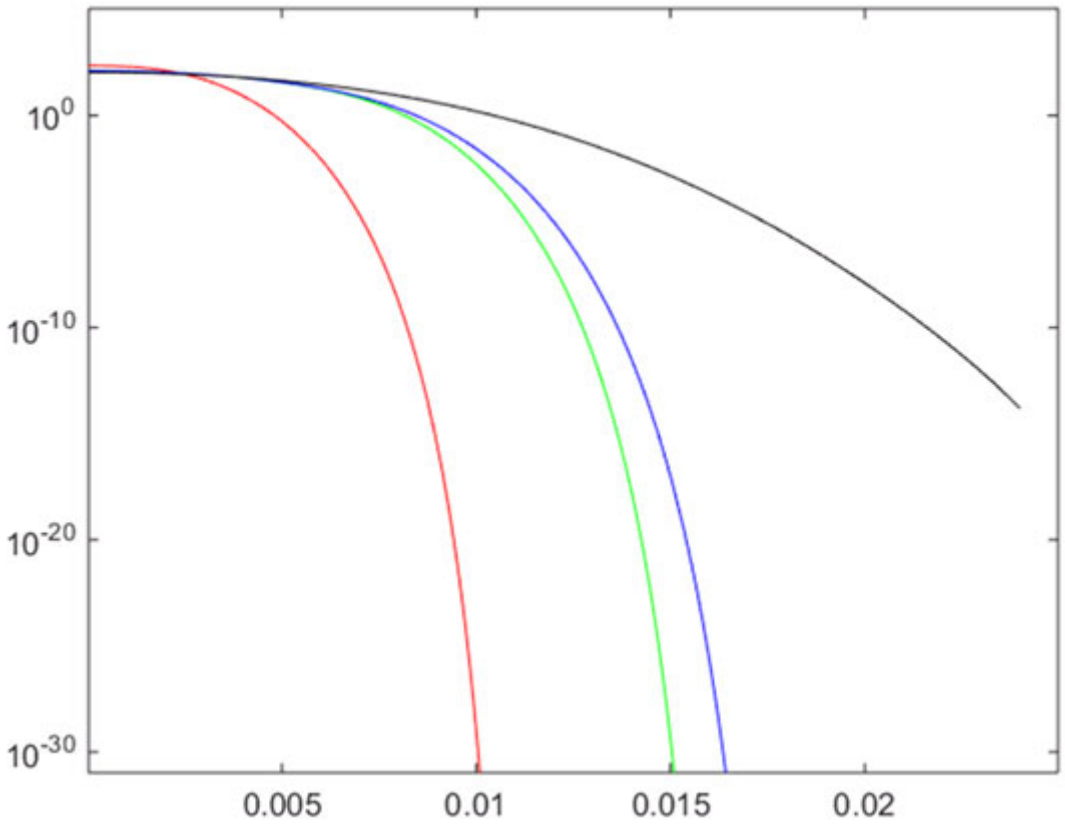
You can use probability mass function (PMF) and cumulative distribution function (CDF) plots to see the contribution of various types of noise components.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

In the example plots below, the largest noise contributors are ISI (red), Dual-Dirac noise (horizontal distance between red and green), and Tx/Rx noise with random jitter (horizontal distance between blue and black). The crosstalk contribution is not large (horizontal distance between green and blue).

The example PMF plot below contains these curves:

- Red — Data from G_Pkg*Length*_PMF_WithISI.txt.
- Green — Data from G_Pkg*Length*_PMF_WithISI_DD.txt.
- Blue — Data from G_Pkg*Length*_PMF_WithISI_DD_XTALK.txt.
- Black — Data from G_Pkg*Length*_PMF_WithISI_DD_XTALK_NOISE.txt.



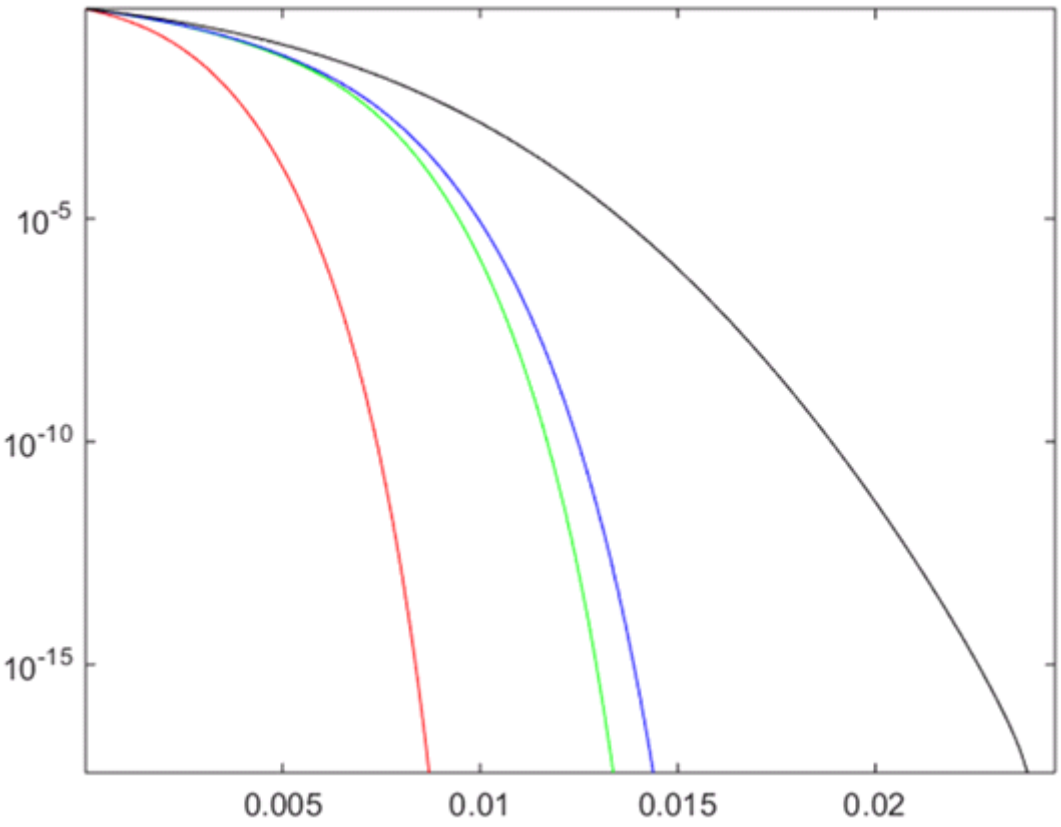
The example CDF plot below contains these curves:

- Red — Data from H_Pkg*Length*_CDF_WithISI.txt.
- Green — Data from H_Pkg*Length*_CDF_WithISI_DD.txt.
- Blue — Data from - H_Pkg*Length*_CDF_WithISI_DD_XTALK.txt.
- Black — Data from H_Pkg*Length*_CDF_WithISI_DD_XTALK_NOISE.txt.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Note

You can find the noise magnitude (A_{ni}) by taking the horizontal offset at a given DER_0 probability (typically at 1e-5 if error correction is specified, or 1e-12 if not) of the above plot.



Related Topics

Evaluating SERDES Channel Compliance From an S-Parameter Model in the Touchstone Viewer

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Touchstone Viewer Dialog Boxes

Understanding the meaning of dialog box options can sometimes require additional explanation.

HyperLynx Touchstone and Fitted-Poles Viewer Dialog Box **1373**
Cascade Four-Port S-Parameter Models Dialog Box **1377**
Combine to Standard Mode Dialog Box **1379**
Convert Mode Dialog Box **1379**
Convert Parameter Type Dialog Box **1380**
Convert to Fitted Poles Dialog Box **1381**
Convert to Touchstone Dialog Box **1382**
Convert to Transfer Function Dialog Box **1382**
Reduce Number of Ports Dialog Box **1383**

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

HyperLynx Touchstone and Fitted-Poles Viewer Dialog Box

To access: **Models > Edit Touchstone Models**

Use this dialog box to graphically display the contents of Touchstone and fitted-poles models.

Fields

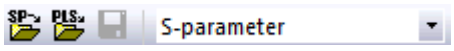
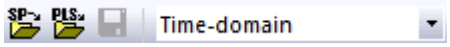
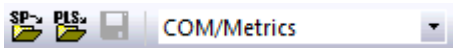
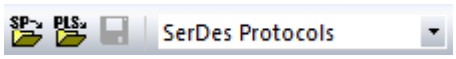
Field	Description
	Displays curves in the frequency domain.
	Displays curves in the time domain.
	Analyzes curves and provides information: • Channel operating margin (COM). The software supports the checks defined by the IEEE 802.3bj and IEEE 802.3bm standards, Annex 93A. • Protocol-specific metrics. The software supports the checks defined by the IEEE802.3-2012 standard, section five, Annex 69B.
	Analyzes curves and provides protocol-specific metrics.

Table D-3. Files Window Contents

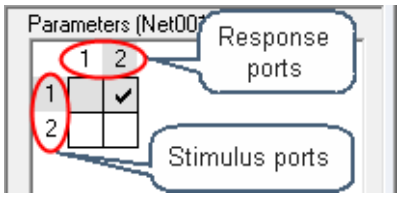
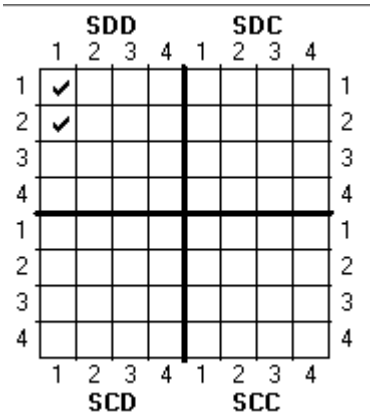
Objects	Description
S parameters Y parameters Z parameters Fitted poles	Checked, the software displays a curve for the model. Use the Parameters spreadsheet to display a curve for a specific model port pair. You can Shift-click or Ctrl-click to select ranges of items, then right-click a selection to hide or show in the Display window.

Table D-4. Display Window Contents

Objects	Description
Display Window **Note:** The Display window is available for the S-parameter and Time-domain modes.	

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table D-4. Display Window Contents (cont.)

Objects	Description
Parameters (For Time-domain mode, displays only when you choose **Time-Domain Response**.)	Checked, in Standard mode, the software displays a curve for the port pair.  Checked, in Mixed mode, the software displays the curve for the differential port pair. See Mixed Mode S-Parameters in S-Parameter Port Numbering for an explanation of the mixed mode parameter labels. 

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table D-4. Display Window Contents (cont.)

Objects	Description
Display	When mode is S-parameter, displays model data in a format you choose. • <standard curve format> • Magnitude in DB • Magnitude • Angle • Magnitude and angle • Real part • Imaginary part • Real and imaginary parts • Trajectory plot (causality) • Passivity plot Magnitude represents attenuation and angle represents phase shift. When mode is Time-domain, displays model data in a format you choose. • **Time-Domain Response** — Stimulate the selected port(s) and plot the responses in the time domain. • **TDR Impedance** — Reproduce the behavior of a time-domain reflectometer by displaying impedance over time for selected port(s). **Restriction:** This format is available only for standard mode Touchstone models.
Full-fit Range (fitted-poles only)	For a fitted-poles model, you can specify the frequency range of a curve. Because a fitted-poles model does not contain frequency range information, you can display its curves across a practically unlimited frequency range. For a Touchstone model, this field is read-only and displays the frequency range of the model data.

Table D-5. Time-Domain Window Contents

Objects	Description
Time-Domain Window Display: **Time-Domain Response**	
Impulse Shape	Shape of the stimulus. See “Stimulus Options for Time-Domain Responses” on page 1359.

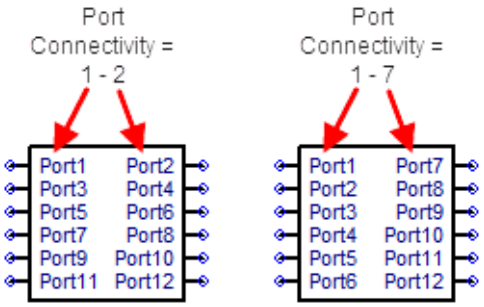
Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table D-5. Time-Domain Window Contents (cont.)

Objects	Description
Stop Time	End time for a falling transition. The falling transition for the stimulus decreases linearly from 0 to -1 starting at the Pulse Time and ending at (Pulse Time + Fall Time).
Display: **TDR Impedance**	
Plot Type	Type of stimulus and plot: • **Single Ended**—Apply stimulus to a single port. Single-ended plots can identify impedance changes seen by a signal propagating through the physical structure represented by the Touchstone model. • **Mixed Mode**—Apply stimulus to a pair of ports. Mixed mode plots can identify differential or common-mode impedance changes in the time domain.
Port(s)	Port(s) to observe. For mixed mode plots, select a pair of ports that drive the same end of a coupled pair of transmission lines. For a four-port model with standard port numbering, this means ports 1-3 or 2-4. The stimulus and probe are assigned to the same port, resulting in a round-trip response.
Other ports (Available only when there are additional ports.)	Type of circuit connection for inactive ports: • **Terminated**—Connect inactive ports to ground through a termination impedance that is equal to the normalization impedance selected for that port. The values can be different among ports. • **Grounded**—Short inactive ports to ground. • **Non-connected**—Leave inactive ports disconnected.
Pulse Time	Start time for a falling transition. This is an absolute time, starting from t0 (time zero).
Rise Time	End time for a rising transition. The rising transition increases linearly from 0 volt to 1 volt between t0 and this value.
Stop Time	Overall length of time to display in the chart.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Table D-6. Protocol-Specific Metrics Window Contents

Objects	Description
Port Connectivity	Specify the port numbering pattern for a model: • **1 - 2** — Odd ports are on the left side and even ports are on the right side. • **1 - ((total number of ports / 2) + 1)** — The first half of ports are on the left side and the second half of ports are on the right side. For example, the figure below shows the supported numbering patterns for a 12 port model. 
Differential Channels	Specify whether the differential channel is an aggressor or victim. Specify the signal direction by identifying the connection of the differential channel ports to the transmitter and receiver.
Analyze	Specify a type of analysis: • Protocol-specific metrics, as defined by the IEEE802.3-2012 standard, section five, Annex 69B. • Channel operating margin (COM), as defined by the IEEE 802.3bj and IEEE 802.3bm standards, Annex 93A.
Operation Mode	For Protocol-specific Metrics, the IEEE802.3-2012 standard, section five, Annex 69B defines these options. For Channel Operating Margin, the IEEE 802.3bj and IEEE 802.3bm standards, Annex 93A, define these options.
Metrics	
Templates	

Cascade Four-Port S-Parameter Models Dialog Box

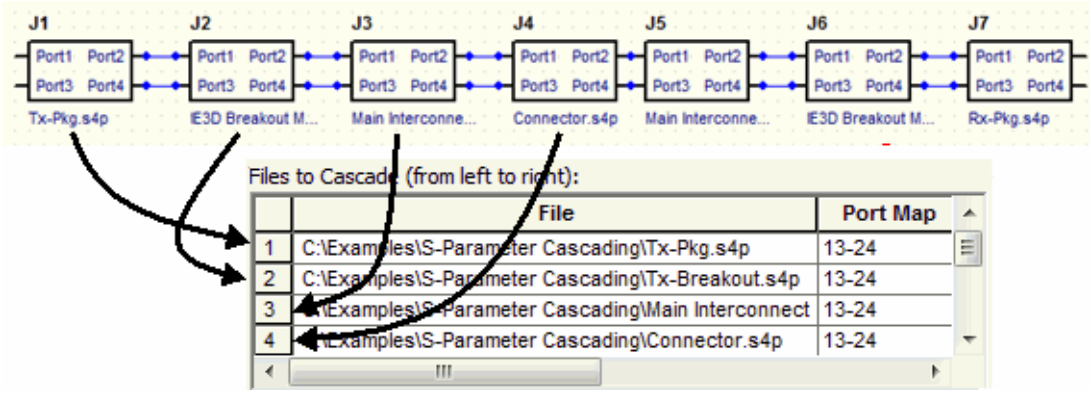
To access: **Convert > Cascade**

Use this dialog box to cascade (with high accuracy) multiple 4-port S-parameter models into a single S-parameter model. The cascading feature is useful if you model a SERDES channel as a serial chain of S-parameter models representing portions of the interconnect between drivers and receivers.

See High-Accuracy Cascading.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure D-49. Order of Models in Spreadsheet



Fields

Field	Description
Files to Cascade (from left to right) Area	
File	4-port S-parameter model that represents an element, such as a connector or interconnect, in the SERDES channel. Assign models to spreadsheet rows in the transmitter-to-receiver order of the SERDES channel, where the top row contains the model for the element closest to the transmitter and the bottom row contains the model for the element closest to the receiver. See Figure D-49. S-parameter models should be in “standard” format. However, if you use mixed mode S-parameters, then every model you cascade must be mixed mode and have the same normalizing impedance and port map.
Port Map	Order of the ports. The default value of *13-24* indicates that the model has ports 1 and 3 on the left side, and 2 and 4 on the right. See “S-Parameter Port Numbering” on page 1346.
Browse	Select a 4-port S-parameter model and add it to the bottom of the spreadsheet.
Remove	Select a row header and click **Remove**.
<<	Moves the selected row up. Select a row header and click <<.
>>	Moves the selected row down. Select a row header and click >>.
Rx Terminator (optional) Area	
	Optional 2-port S-parameter model that represents termination at the receiver. The model should be in the “standard” format with port 1 connecting to the positive and port 2 connecting to the negative side of the receiver termination.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Result File Area	
	Location and file name of the cascaded S-parameter model.
Port Map	Order of the ports. The default value of *13-24* indicates that the model has ports 1 and 3 on the left side, and 2 and 4 on the right. See "S-Parameter Port Numbering" on page 1346.
Sampling Area	
Freq Min	Highest starting frequency in the set of models in the spreadsheet.
Freq Max	Lowest ending frequency in the set of models in the spreadsheet.
Number of Points	Calculated automatically when you enable Auto.
Auto	Disable to manually specify Number of Points.

Related Topics

Cascade Multiple S-Parameter Models in Series

Convert and Fix Touchstone and Fitted-Poles Models

Combine to Standard Mode Dialog Box

To access: **Convert > Combine Modes**

Use this dialog box to combine differential and common mode models into a standard mode model.

Related Topics

Convert and Fix Touchstone and Fitted-Poles Models

Convert Mode Dialog Box

To access:

- **Convert > Mode**
- **Convert standard mode to mixed mode**

- **Convert standard mode to differential mode** 

Use this dialog box to convert a Touchstone model to another mode.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Restrictions and Limitations

- This dialog box is available when you load a 4-port, 8-port, or another multiple of four ports Touchstone model.
- This dialog box is unavailable when a fitted-poles model is loaded.

Fields

Field	Description
Convert to	Standard mode—Convert a mixed mode model to a standard-mode model. Mixed Mode—Convert a standard mode model to a mixed mode model. Differential Mode—Convert a standard mode model to a differential mode model. The resulting model has two ports that contain the same information as the differential ports (SDD#,#) of a mixed mode model. This option is read only when you use the mm or dm toolbar button to open this dialog box.
Differential Pair	Identify the input and output ports (including their positive or negative behavior) of a differential pair. If the model has eight or more ports, the software applies the port numbering pattern that you assign for the first four ports to each remaining group of four ports. For example, if you open an 8-port model and assign ports 1 and 3 as inputs and ports 2 and 4 as outputs, the software assigns ports 5 and 7 as inputs and ports 6 and 8 as outputs.

Related Topics

Touchstone and Fitted-Poles Models

Convert and Fix Touchstone and Fitted-Poles Models

Convert Parameter Type Dialog Box

To access: **Convert > Parameter Type**

Use this dialog box to change the parameter type of your Touchstone model.

Restriction

This menu option is unavailable when a fitted-poles model is loaded.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Options

Option	Description
Convert to	Select a different parameter type (S, Y, or Z) to convert your model to.

Related Topics

Convert and Fix Touchstone and Fitted-Poles Models

Convert to Fitted Poles Dialog Box

To access: **Convert > To Fitted Poles**

Use this dialog box to convert a Touchstone model into a fitted-poles model.

You can specify the maximum fitting order, how to perform the fix, the fitting precision, and more.

Fields

Field	Description
Used points limit	Sets a limit to the number of points in the original data that the software uses to perform the fit. Ideally, you would use all the points given in the Touchstone file to produce an analytical approximation. However, if too many points are given, the fitting procedure becomes increasingly slow. On average, 5000 points can be fitted in a reasonable time, however this time also depends on the number of ports and the number of poles (order of complexity). You may want to limit the number of points first to see if the fit produces what you expect and then, if it works, transform the data with a larger number of points for the final results.
Fitting precision	High—Increase fitting accuracy by allowing more poles than in normal mode. Selecting High slows the conversion and generates a larger model. Normal—Enable when performance is more important than high accuracy. This produces a smaller fitted-poles file.
Allows extrapolation to Zero frequency	Yes—Restores a missing point at zero frequency (DC) by extrapolating the curve from the low-frequency points given in the Touchstone file. If the DC point exists in the Touchstone file, selecting Yes has no effect. No—The DC point exists in the Touchstone file.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Field	Description
Maximum complexity order	If the default value is insufficient, set the order (val/2) to the maximum order of complexity for fitting in CPF. For very complicated (sharp, irregular) dependencies it is sometimes reasonable to reduce the order of complexity, especially if you have reasons not to trust the data at higher frequencies.
Maximum Q-factor for poles	The default value is usually sufficient. Increasing the Q-factor can increase the accuracy of the fit.
Perform delay extraction	Yes—Perform delay extraction. This reduces the complexity of the fitted model. No—Create a fitted model that does not impose limitations on the transient solution step.

Related Topics

Convert and Fix Touchstone and Fitted-Poles Models

Convert to Touchstone Dialog Box

To access: **Convert > To Touchstone**

Use this dialog box to convert a fitted-poles model to a Touchstone model.

Fields

Field	Description
Sampling Algorithm	The adaptive algorithm distributes sampling points across the frequency range in a variable way, by increasing the sampling rate near frequencies with resonances. You cannot choose a different algorithm.
Resolution Factor	Specifies the maximum value of the norm of the matrix increment over two neighboring points ||S(i+1)-S(i)||.

Related Topics

Convert and Fix Touchstone and Fitted-Poles Models

Convert to Transfer Function Dialog Box

To access: **Convert > To Transfer Function**

Use this dialog box to convert a 4-port Touchstone model to a transfer function, in the form of a 1-port Touchstone model.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Restriction

This option is available only if a 4-port Touchstone model is loaded.

Fields

Field	Description
Port Map	Order of the ports. The default value of *13-24* indicates that the model has ports 1 and 3 on the left side, and 2 and 4 on the right. See S-Parameter Port Numbering.
Default Resistance and Conductance	Select when analyzing an interconnect channel with the following properties: • The channel has 50-ohm termination at both ends • No attached AMI S-parameter models Disable this option and enter values when the input ports have no series termination and the output ports have no line-to-ground or line-to-line termination. Disable this option and enter values when you analyze a cascaded SERDES channel with AMI S-parameter models attached at the input and output ends. In this case, the input ports have no series termination and the output ports have no line-to-ground or line-to-line termination.

Related Topics

Cascade Multiple S-Parameter Models in Series

Convert and Fix Touchstone and Fitted-Poles Models

Reduce Number of Ports Dialog Box

To access: **Convert > Reduce Ports**

Use this dialog box to remove ports from a Touchstone model that are not being used in simulation, to decrease the size of the model.

Restriction

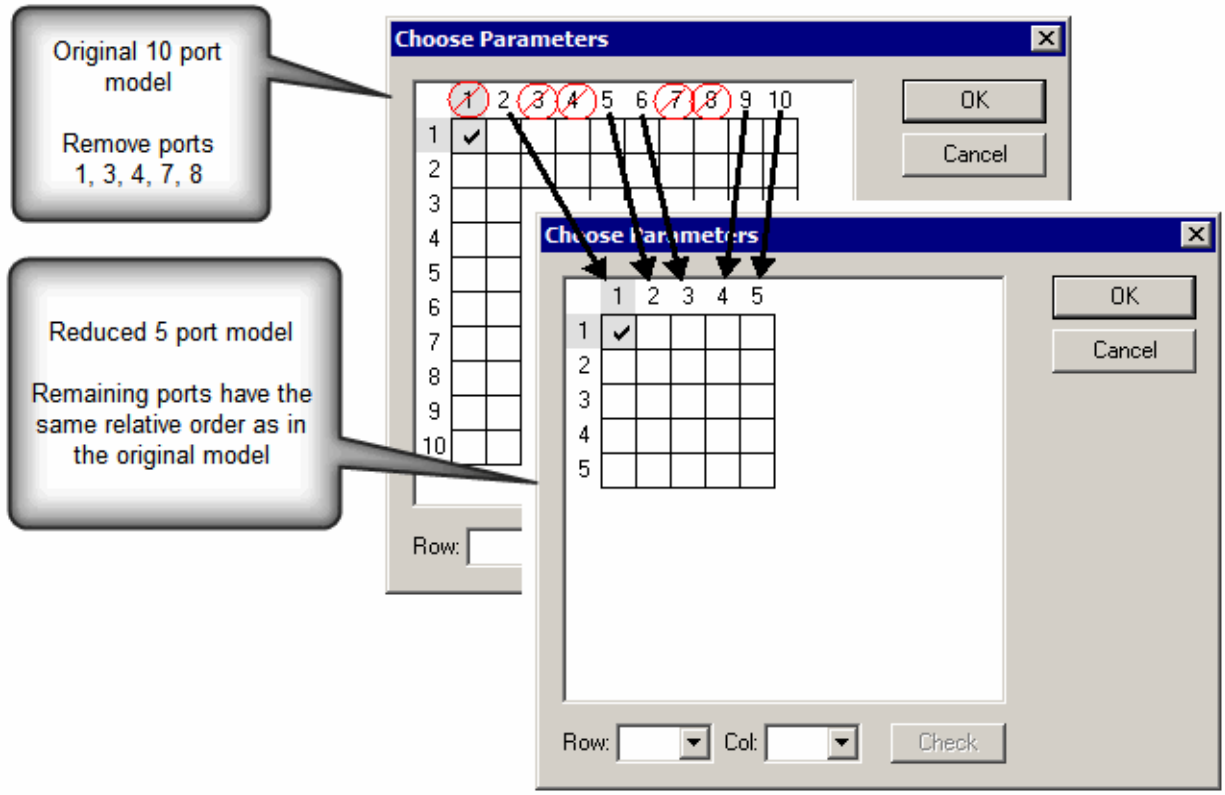
This menu option is unavailable when a fitted-poles model is loaded.

To remove a port, set the State value to Grounded, Terminated, or Non-connected. If you do not want to remove a port, set the State value to Retained.

If you remove ports, the relative order of the remaining ports is preserved. For example, if the original model has 10 ports and you remove ports 1, 3, 4, 7, and 8 (by terminating, grounding or disconnecting them), the result is a 5-port model whose port sequence corresponds to the following original numbers: [2, 5, 6, 9, 10].

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

Figure D-50. Mapping Ports Between Original Model and Reduced-Port Model



Related Topics

Convert and Fix Touchstone and Fitted-Poles Models

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Glossary

aggressor net
A net transmitting signals and causing unwanted voltage noise (crosstalk) on nearby (victim) nets.

anti-pad
An isolation shape providing clearance between a pad and surrounding metal, such as an AC ground plane or area fill. An anti-pad defines the shape and size of the clearance area (copper void) that should be created while pouring around a pin or via if it intersects a copper pour polygon.

anti-segment
An isolation shape providing clearance between a trace segment and surrounding metal, such as an AC ground plane or area fill. An anti-segment defines the shape and size of the clearance area (copper void) that should be created while pouring around a trace if it intersects a copper pour polygon.

antipad
See anti-pad.

antisegment
See anti-segment.

associated net
A net electrically connected to one or more other nets by conduction or coupling that exceeds a crosstalk threshold.

See also: electrical net

attenuation
A reduction of the amplitude of a signal due to losses in the net carrying the signal.

automapping
A method that assigns an IC model or passive component value to all the eligible pins on a PCB component with a specific reference designator or part type.

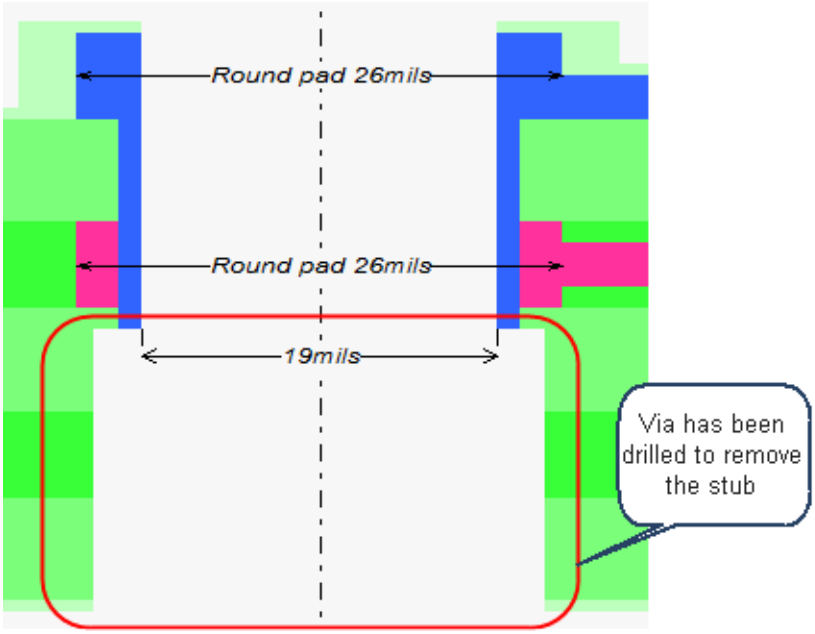
.REF automapping files assign a model or value to pins on a component with a specific reference designator.

.QPL (qualified parts list) automapping files assign a model or value to pins on all components with a specific part name, regardless of its reference designator.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

backdrilling

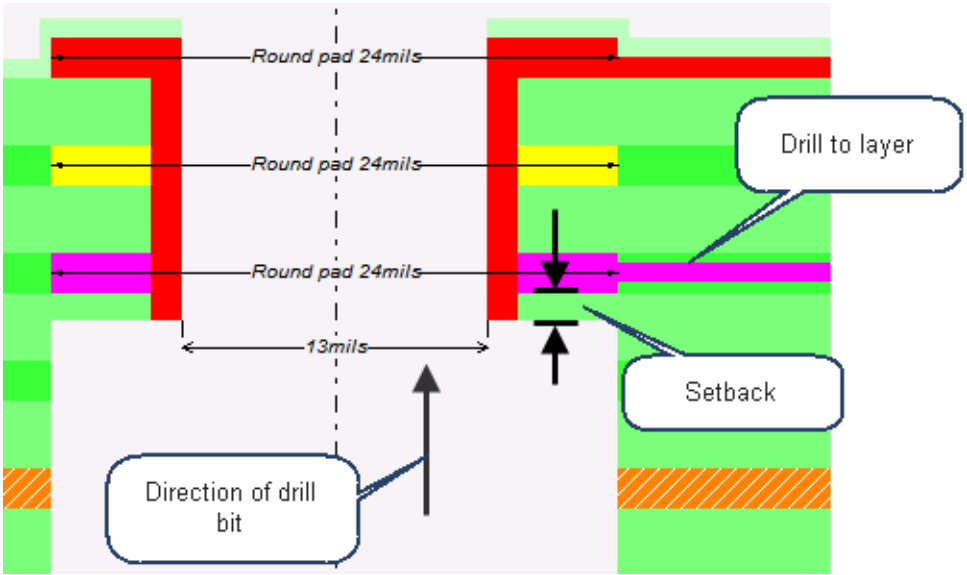
Removing a via stub during PCB manufacturing by drilling from one side of the board.



See also: backdrill setback

backdrill setback

Length of the remaining material measured from the end of the drill bit to the “drill to layer”.



See also: backdrilling

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backward crosstalk

The coupling (crosstalk) on a victim net that flows in the direction opposite of the signal transmitted by a nearby aggressor net. Backward crosstalk flows toward the IC pin on the victim net located closest to the switching driver on the aggressor net. The waveform usually has no resemblance to the coupled signal.

Also known as “near-end crosstalk”.

See also: aggressor net, victim net

barrel

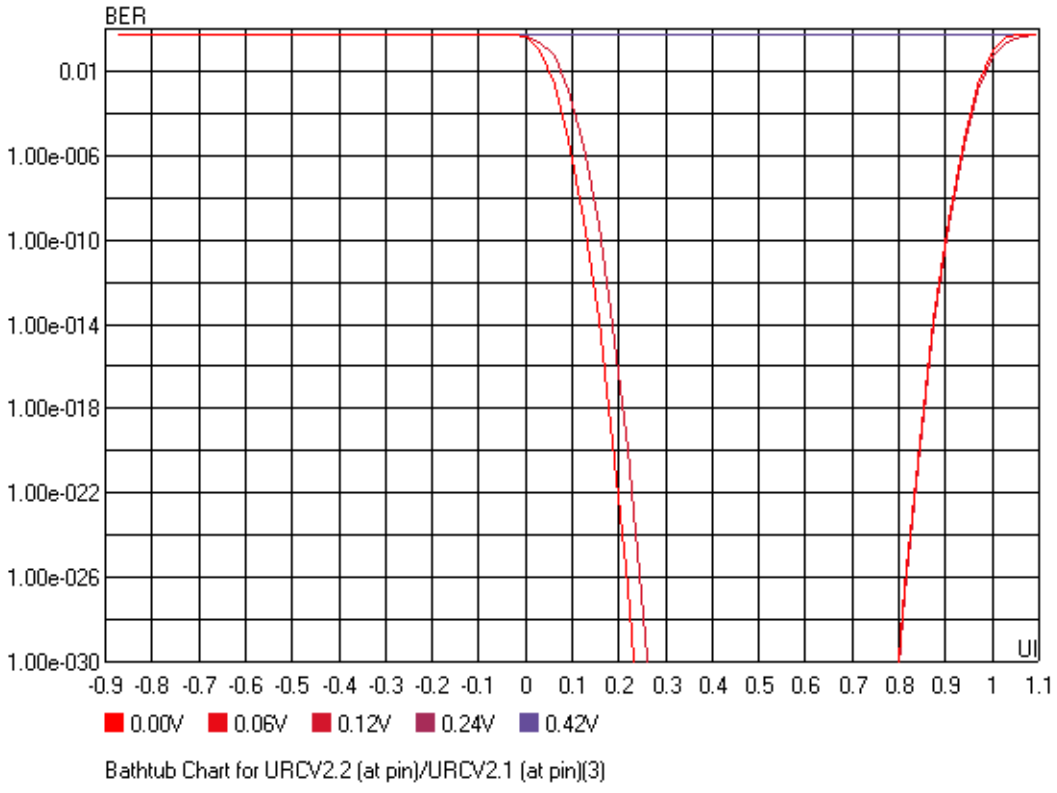
In a via, the round metal tube (plated hole) that penetrates PCB layers.

See also: via

bathtub curve

A graph showing the probability of bit errors at a receiver for various sampling locations across the bit interval. Bathtub curves indicate the quality of sampling locations by plotting the probability of a bit-transmission failure at each sampling location. “Bathtub curves” are so-named because their overall appearance resembles the cross section of a bathtub.

Figure E-1. Bathtub Curve



BEM

See boundary-element method (BEM).

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bit error rate (BER)

The probability of a bit error at a receiver at a specific sampling location in the bit interval. Bit error rates can be calculated by analysis or test equipment. When measured with test equipment, the bit error rate is (number_of_incorrectly_received_bits / number_of_total_sent_bits).

See also: bathtub curve

bit error ratio (BER)

See bit error rate (BER).

bit interval

The duration of an individual bit transmitted in a data stream. Also known as unit interval (UI).

blind via

A via connecting a trace on a surface PCB layer to a trace on an inner PCB layer. Blind vias do not penetrate through the entire board to both surface layers.

See also: via

BoardSim

The HyperLynx board design analysis environment. HyperLynx displays board designs in the board viewer.

boundary-element method (BEM)

Type of field solver used to perform power-integrity analysis in the frequency domain.

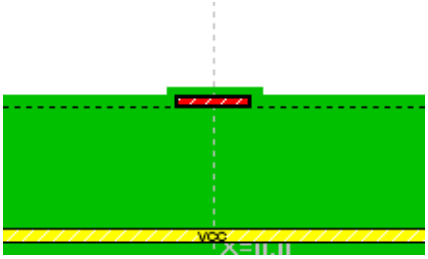
BUD file

A file containing interactive design changes for a BoardSim board, such as stackup edits and interactive IC model assignments.

buried microstrip

A trace routed on an inner layer of the PCB, with a dielectric layer and air on one side and a dielectric plus a plane layer on the other side.

Figure E-2. Buried Microstrip



See also: microstrip, stripline

buried capacitance

A pair of plane layers in a PCB separated only by a dielectric layer, so that no signal layer is between them. For high-speed designs where one plane layer is ground and the other is power, the dielectric layer can be made sufficiently thin to provide a bypass path for return currents.

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buried via

A via completely contained within the inner layers of a board and that does not penetrate either surface of the PCB.

See also: via

bypass capacitor

A capacitor connected to a transmission plane that is used to provide a low-impedance path for return currents for a digital signal to move between transmission-plane layers.

A bypass capacitor can also act as a decoupling capacitor.

See also: transmission plane, power-distribution network (PDN), decoupling capacitor

causality error

Data in a Touchstone model that indicate a propagation speed faster than physics allow, or a reversal in the phase trajectory.

channel - data channel

A set of drivers, receivers, and physical interconnections of a net configured to transmit data according to a communications interface standard, such as PCI Express.

characteristic impedance - Z0

Resistance to current flow caused by the resistive, inductive and capacitive effects of a transmission line. Impedance is affected by layer stackup dimensions and materials, and trace dimensions and clearances.

Characteristic impedance is a property unique to the distributed nature of transmission lines. Because transmission lines consist of a continuous mixture of capacitance and inductance, they “look” instantaneously like a resistance to a transmitted signal.

common mode

A current or voltage transmitted at the same time by both members of a differential pair.

constraints

A set of rules that define how PCB component pins are to be connected, such as maximum propagation delay, net scheduling, and so on.

continuous time linear equalization (CTLE)

Circuitry of an IC receiver that restores content of a signal that is lost when the signal is transmitted through the channel. CTLE is typically implemented as a “peaking” filter.

copper pour

A shape created by poured metal on a PCB layer. Sometimes also called “pour outline” or “plane area.” The pouring operation implemented by the PCB design system automatically creates clearance areas around pins, traces, and vias on other nets.

See also: anti-pad, plane area, pour outline

copper void

A polygon-shaped area inside a copper pour that is kept free of metal when pouring occurs.

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A copper void is sometimes also called a “pour cutout.”

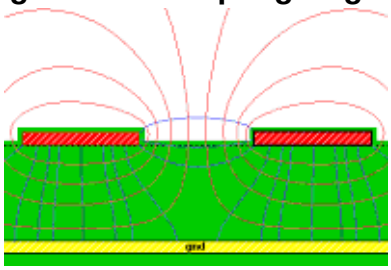
coupled transmission lines

Two or more conductors sufficiently close together to exchange energy with each other when transmitting signals. For differential pairs, coupling provides a way to control signal impedance and to reject interference from nearby aggressor nets.

coupling region

A two-dimensional cross section of the PCB that contains two or more coupled conductors, and assumed to have constant geometry over some specified length.

Figure E-3. Coupling Region



crosstalk

The unwanted coupling of voltages and currents among neighboring nets that are transmitting signals.

See also: backward crosstalk, forward crosstalk

cutout

See copper void and pour cutout.

decision-feedback equalization (DFE)

The circuitry of an IC receiver that restores the high-frequency content of a signal that is lost when the signal is transmitted through the channel.

decoupling capacitor

A capacitor connected to a transmission plane that is used to quickly store and release energy for local power-distribution network (PDN) delivery and to lower transmission-plane impedance.

A decoupling capacitor can also act as a bypass capacitor.

See also: transmission plane, power-distribution network (PDN), bypass capacitor

design rule check (DRC)

An analysis of the geometric and electrical properties of a PCB design that reports the location of structures and layout configurations that can produce operational failures or degrade signal integrity, power integrity, electromagnetic compatibility (EMC), and so on. Design rule checking takes into account routing, component placement, copper pours and voids, stackup, and electrical properties.

A design rule check can use both the layout data and connectivity data of the PCB design to locate design rule violations. Design rule checks can also include specific design rules specified in a rule file.

Note - Viewing PDF files within a web browser causes some links not to function. Use HTML for full navigation.

DFE
See decision-feedback equalization (DFE).

DIMM
Dual in-line memory module

dielectric
A material that does not conduct electricity and is used in PCB designs to insulate conductors and encapsulate components.

dielectric constant
Ratio of the charge stored by air to the charge stored by a specific material. A dielectric constant indicates the ability of a material to store a charge.

differential impedance
For a pair of symmetric-coupled traces, the differential impedance is the trace-to-trace resistance that will properly terminate a pair of signals driven in differential mode.

differential net
A special kind of electrical net, used by a differential pair, formed by the paired combination of two other electrical nets.

See also: electrical net

differential pair
Two conductors deliberately routed parallel to each other and with a constant trace-to-trace gap to provide uniform impedance and noise immunity from neighboring aggressor nets.

DRAM
Dynamic random-access memory

driver
An IC pin transmitting a signal on the net.

edge rate
The speed at which a device transitions from one logic state to the other, specified as volts/time. Rise and fall time depend on the edge rate, plus other factors affecting signal integrity.

effective series inductance
See equivalent series inductance (ESL).

effective series resistance
See equivalent series resistance (ESR).

electrical board description (EBD)
An extension of the IBIS specification that describes the electrical behavior of a complicated interconnection. An EBD model contains physical information, such as trace length and stackup, but not coupling.

See also: IBIS

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electrical net

A set of nets on the PCB that are connected only by passive components, such as resistors and capacitors.

See also: differential net, associated net

electromagnetic compatibility (EMC)

Determining whether the radiated emissions of a design fall below government limits at every frequency. Excessive radiated emissions can cause electromagnetic interference in nearby devices, possibly affecting their operation. Government limits include U.S. FCC, European CISPR, and Japanese VCCI.

electromagnetic interference (EMI)

See electromagnetic compatibility (EMC).

equivalent series inductance (ESL)

In a capacitor simulation model, ESL includes the effects of the inductance of the terminal leads.

ESL and the capacitor capacitance value determine the self-resonance frequency of the capacitor. F = 1/(2*pi*sqrt(LC)).

equivalent series resistance (ESR)

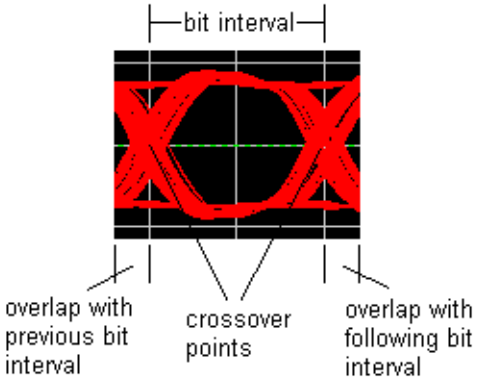
In a capacitor simulation model, ESR includes the effects of the resistance of the dielectric and plate materials, electrolytic solution, and terminal leads.

eye diagram

Cutting up a waveform into bit interval lengths and overlaying the waveform fragments to see how much the timing variation of the waveform crossover points erodes the range of valid data sampling locations. The waveform crossover points cluster around the bit interval boundaries, and the overall visual effect vaguely resembles a human eye.

Signal distortion related to ISI is typically caused by high-frequency signal attenuation and by residual transient responses, such as crosstalk and reflections, to previous signal transitions on the interconnect.

Figure E-4. Eye Diagram



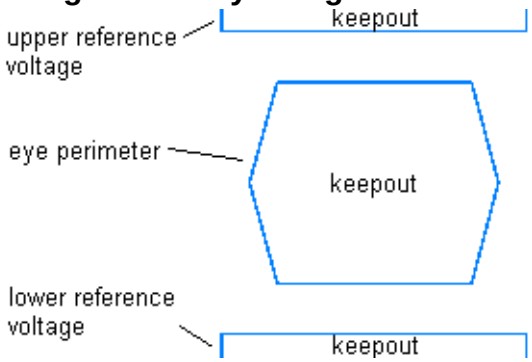
See also: inter-symbol interference (ISI), FastEye diagram

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eye diagram mask

The “keep out” areas that waveforms in an eye diagram should avoid. Waveforms crossing into keep out areas may violate signal-protocol requirements or other performance requirements.

Figure E-5. Eye Diagram Mask



fall time

The time it takes for a signal to switch from the logic one state to the logic zero state.

FastEye diagram

An eye diagram produced by analyzing the response of a net to a step stimulus and a pulse stimulus. By contrast, “standard” eye diagrams are specially-formatted waveforms created by time-domain simulations.

See also: eye diagram

FFS file

Geometric and electrical PCB design information in a format that can be read by the HyperLynx LineSim schematic editor. FFS files contain nets, active component (IC) names, passive component values, stackup, pad stacks, and so on.

field solver

A simulation program that reports the electrical characteristics of a system of conductors and dielectrics, using one or more of the basic equations of electromagnetic theory, such as “Maxwell's equations.” Field solvers can report the capacitances, inductances, propagation velocities, and characteristic impedances of a coupling region cross section.

See also: coupling region

finite-difference time-domain (FDTD)

The type of simulator to perform electromagnetic field analysis using finite-difference modeling in time domain. One of its applications is plane-noise analysis.

flight time

The amount of time that elapses between the time when the signal on an output buffer driving a design load crosses the threshold voltage and the time where the signal crosses the threshold voltage at the receiver. You can evaluate flight time at V_{il}, V_{ref}, and V_{ih}, and use the maximum and minimum values to calculate timing margins.

See also: flight-time compensation

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flight-time compensation

Mathematically adjusting the calculated interconnect delay to account for the driver switching into the actual PCB interconnect load, as opposed to the driver switching into an arbitrary test fixture load specified by the IC model or component datasheet.

See also: flight time

fknee

See knee frequency.

forward crosstalk

The coupling (crosstalk) on a victim net that flows in the same direction of the signal transmitted by a nearby aggressor net, as seen at the end of the victim net farthest from the signal source of the aggressor. Forward crosstalk flows toward the IC pin on the victim net located away from the switching driver on the aggressor net.

Also known as “far-end crosstalk.”

See also: aggressor net and victim net

FDTD

See finite-difference time-domain (FDTD).

glitch

A non-monotonic area of a waveform caused by ringing, reflections, or other signal integrity factors. A glitch passing through the threshold voltage of a receiver can cause system operational failure.

See also: non-monotonic

HYP file

Geometric and electrical PCB design information in a format that can be read by HyperLynx BoardSim. .HYP files contain nets, active component (IC) names, passive component values, stackup, pad stacks, and so on.

IBIS

IBIS stands for “I/O Buffer Information Specification” and is an industry-standard IC model format that describes IC behavior without revealing how the IC is implemented.

IBIS-AMI

IBIS-AMI is an industry standard that uses algorithmic code to model the complex and non-linear transformations of signal waveforms inside transmitters and receivers. Shared executable library files (.DLL) implement the algorithmic code and protect intellectual property (IP). Typical AMI .DLL files contain proprietary algorithms for transmitter pre-emphasis, receiver equalization and DFE, and receiver clock and data recovery. The algorithmic modeling interface (AMI) standard first appeared in I/O Buffer Information Specification (IBIS) version 5.0.

IC automapping

See automapping.

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imparity

The DC balance of an 8B/10B code word. One of the goals when using 8B/10B coded patterns is to support DC balance. While even the random pattern produces an equal number of 1 and 0 bits on average, the idea of imparity in 8B/10B coded patterns is to enforce DC balance in the short term. 8B/10B word can have any of the following imparity characteristics:

- Balanced—Word has an equal number of 1 and 0 bits.
- Positive imparity—Word has more 1 bits than 0 bits.
- Negative imparity—Word has more 0 bits than 1 bits.

The degree of imparity in each word is minimal. For example, a six bit word cannot have more than four 1s or 0s. Continuing the six bit word example, the word has positive imparity when it has four 1 bits and two 0 bits (that is, imparity = +2).

For a sequence of 8B/10B words, the 8B/10B word generator supports minimal running imparity using the following algorithm:

1. Set running imparity to -1.
2. If the running imparity is negative (for example, -1), the next word can only be neutral (that is, 0) or with positive imparity (for example, +2). A neutral word does not change running imparity. A positive word changes the sign of the running imparity (for example -1 + 2 = +1).
3. If the running imparity is positive (for example, +1), the next word can only be neutral (that is, 0) or with negative imparity (for example, -2). A neutral word does not change running imparity. A negative word changes the sign of the running imparity (for example, +1 - 2 = -1).
4. Repeat steps 2-3.

impulse response

A time derivative of the step response. The behavior does not come directly from simulation. Instead the process of finding the impulse response includes simulating the step response and taking the derivative numerically. This term applies to IBIS-AMI and FastEye channel analysis.

See also: pulse response, step response

insertion loss

A reduction of the amplitude of a signal due to adding a device to the net carrying the signal. In terms of S-parameters, insertion loss results when the absolute voltage of the incident signal is more than the absolute voltage of the transmitted voltage.

interconnect delay

See flight time.

inter-symbol interference (ISI)

A form of signal distortion in very fast digital signaling, where the effects of a previous bit linger or persist on a net while more bits are sent. In slower digital signaling, where the time length of a bit was much longer than the delay on the net, it was common that whenever a bit was sent, that its signal on the receiver pin would look identical to the previous and next time it was sent. But when digital signaling speed increases to the point where the length of a bit is shorter than the delay of its net, multiple bits can co-exist on the net and interfere with each other.

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The result is that the rising or falling edge for a signal is no longer guaranteed to look the same every time it arrives at a receiver pin. Specifically, its arrival delay might be a little longer or shorter, and its voltage amplitude might be a little larger or smaller.

ISI can spread the arrival time of individual data bits in a net so much that the receiver cannot reliably distinguish among them. Signal distortion that is related to ISI is typically caused by high-frequency signal attenuation and by residual transient responses, such as crosstalk and reflections, to previous signal transitions on the interconnect. In an eye diagram, ISI-related signal distortion can appear as jitter, voltage overshoot or undershoot, and so on.

See also: eye diagram, FastEye diagram

JEDEC

Joint Electron Device Engineering Council, which is an international standards organization that governs many electrical-engineering specifications.

jitter

The distribution of signal transition times away from the ideal time.

knee frequency

The frequency at which the band width of the net begins to significantly attenuate the energy content of a switching signal.

LineSim

The HyperLynx schematic design analysis environment. HyperLynx displays schematic designs in the Free-Form Schematic Editor and the PDN Editor.

lossy

The attenuation of the energy in a switching signal due to the skin effect (resistance caused by currents crowding the surface of the conductor) and dielectric loss (resistance caused by heating of the dielectric material).

Manhattan routing

Routing traces only on the X-Y routing tracks on the PCB. All corners are 90 degree angles.

memory controller

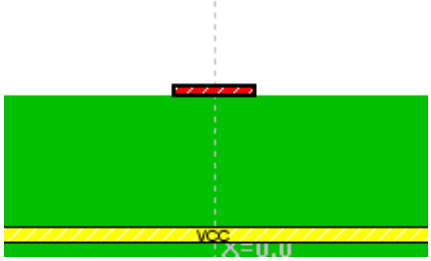
The memory Controller is the component on the main board that interfaces between the DRAMs and the central-processing-unit (CPU). The memory controller can also be of different technology types: FPGA, microcontroller or chipsets. In some of the latest CPUs, the memory controller circuitry has been integrated into the core. Memory controllers come in different packages with variations in pin-counts.

microstrip

A trace routed on the PCB surface. It has air on one side and a dielectric plus a plane layer on the other side.

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Figure E-6. Microstrip



See also: buried microstrip, stripline

microvia

A miniature via that typically goes through only a few surface layers of the board. Microvias do not change reference planes, do not have pads, and pass through only very thin stackup layers. Because microvias do not have pads, they can help you route traces to packages with closely-spaced pins. Microvias are typically drilled by a laser.

mixed plane layer

See partial plane layer.

mounting

Traces and vias that connect a component to a board design. For example, mounting connects a decoupling capacitor pin to a power supply net.

non-monotonic

Data that reverse a trend of an ever-increasing or ever-decreasing numeric sequence.

Figure E-7. Non-monotonic



overshoot

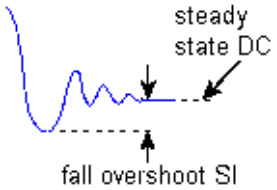
The portion of a signal transition that extends beyond the steady-state voltage (overshoot-signal integrity) or the power rail voltage (overshoot).

Figure E-8. Rising Overshoot - Signal Integrity



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Figure E-9. Falling Overshoot - Signal Integrity



pad

A shape that connects a component pin or a via to a metal layer in the PCB. Pads of through-pin components have plated through-holes in them. Pads of surface mount components have no drilled holes in them.

See also: pad clearance

pad clearance

The minimum gap or space between the pad and other conductive objects on the same metal layer.

padstack

A round metal tube (plated hole) and set of pads penetrating some or all PCB layers, that electrically connects pads on different board layers.

A via is a specific instance of a padstack.

See also: via

PAK file

A HyperLynx package model format describing the electrical connections in resistor and capacitor network packages.

partial plane layer

A stackup layer containing both signals and AC ground regions.

passivity error

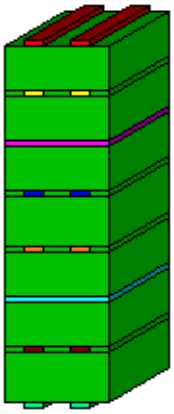
A passivity error exists if the sum of energy coming out of the ports exceeds the energy going into one of the ports. Touchstone models for passive components, such as a connectors, should be passive. Any number of model-generation problems can produce passivity errors.

PCB stackup

A set of metal and dielectric layers stacked over one another, like a deck of cards, to form a printed circuit board (PCB).

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Figure E-10. PCB Stackup



power-delivery network

See power-distribution network (PDN).

power-distribution network (PDN)

A network of PCB elements providing power to ICs and passive components, and providing return current paths for digital signals and IC power supply pins.

PDNs consist of metal areas (like copper fills or plane layers), IC packages, vias, bypass/decoupling capacitors (and their mounting), trace segments, buried capacitances, and so on.

power-distribution system (PDS)

See power-distribution network (PDN).

PJH file

HyperLynx project file. For individual LineSim schematics or BoardSim boards, the PJH file contains various simulation settings and preferences. For BoardSim MultiBoard projects, the PJH file contains the list of boards in the project, electrical properties for board-to-board interconnections, and various simulation settings.

plane area

See copper pour.

plane layer

A solid or patterned metal layer in the PCB stackup that is tied to a DC voltage, such as VCC or ground. Plane layers provide return current paths and electromagnetic shielding.

pour cutout

See copper void.

pour outline

See copper pour.

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power integrity

Maintaining a constant power source voltage with little or no voltage drop or electromagnetic interference between points on a printed circuit board or within a circuit. Power integrity can become a greater problem for designs operating at high switching speeds.

See also: voltage drop

pre-emphasis

The circuitry of an IC driver that increases the high-frequency content of the transmitted signal, to help overcome the high-frequency losses imposed by the channel on the signal.

prepreg

A PCB stackup layer that is spongy or lacks stiffness until it is cured. A prepreg layer is typically adjacent to a rigid layer.

pulse response

A response of a channel on an isolated bit, where the bit causes a rising and falling transition, such as in a 00000000001000000000 bit sequence. This behavior is also known as a bit response. This term applies to IBIS-AMI and FastEye channel analysis.

See also: step response, impulse response

QPL (qualified parts list) file

See automapping.

quick terminator

A simulation-only termination component. Quick terminators are used during “what if” experiments in HyperLynx BoardSim, to identify termination configurations that improve the signal integrity of the net.

rank

A group of DDRx DRAMs that are tied to a single, unique, chip select signal. The number of chip select signals on the memory controller determines the supported number of memory ranks. A rank is usually made up of 64 bits in a DDRx interface. The number of DRAMs that makes up a rank depends on the width of the DRAMs used. For example, if you use x16 DRAMs, 4 DRAMs make up a rank. Note that if you have two sided DIMMs, this does not mean you have two ranks per DIMM.

receiver

An IC pin that observes signals transmitted by a driver on the net.

REF file

See automapping.

reference designator

A unique name identifying a specific instance of a physical component in the design. Reference designators typically consist of an alphabetic prefix corresponding to the function of the component, and a sequential numeric suffix identifying the instance of the component. In

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software, reference designators are often stored as the value of the REF property associated with a specific component or package.

Common conventions for reference designator alphabetic prefixes include U to represent IC components, R to represent resistors, and C to represent capacitors. Thus, a component with a U4 reference designator represents a specific IC in the design. Similarly, R17 represents a specific resistor, and C35 represents a specific capacitor. Reference designators are sometimes (but not always) assigned on a grid system according their physical location on the printed circuit board.

reference plane

A plane layer in the PCB that is tied to a DC voltage, and through which return current flows for digital signals and IC current sinks.

A signal might switch reference planes when it passes through a via or passes over a gap or slot in the current reference plane, if another plane layer provides a return current path with a lower impedance.

reflection

The portion of energy in a high-speed signal that is sent back toward the driver as the signal meets an impedance change in the transmission line. Reflections can cause ringing and overshoot.

return current

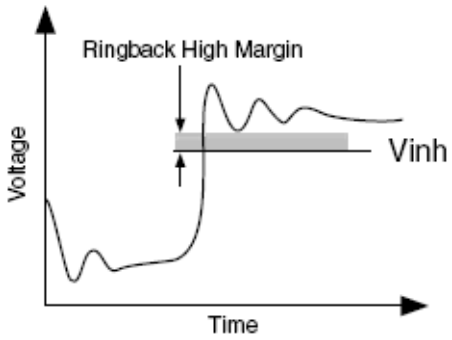
Return current flows from the load to the source through structures located in the power-distribution network (PDN).

ringback

How much the waveform returns toward the timing threshold voltage, after initially passing through it. Excessive ringback can cause unwanted switching at the receiver, because the waveform passes through the timing threshold more than once.

The figure below shows how far the rising waveform “falls back” after first passing through the receiver logic high timing threshold.

Figure E-11. Ringback High Example



ripple

DC voltage is often generated by using a power supply whose output is a stepped down, rectified, and filtered AC source voltage. Any remaining super-imposed alternating voltage on top of the DC voltage is called the ripple voltage.

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rise time

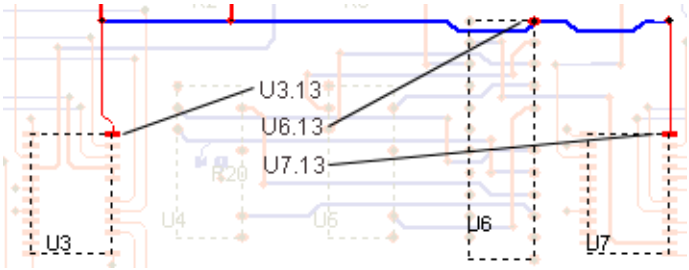
The time it takes for a signal to transition from the logic low state to the logic high state.

round robin

A driver-enabling algorithm that produces a series of simulations, where each simulation represents a specific driver on the net taking a turn driving the net. Only one driver is enabled for each simulation. Nets with multiple bidirectional, three-state, open-drain, or open-collector IC pins are simulated multiple times, once for each driver driving the net.

In the following figure, round robin runs the simulations listed in the table, one simulation for each driver taking its turn to drive the net.

Figure E-12. Round Robin



Simulation	U3.13 Driver	U6.13 Driver	U7.13 Driver
1	enabled	disabled	disabled
2	disabled	enabled	disabled
3	disabled	disabled	enabled

round trip time

The time required for the clock flight time (write) plus the DQS flight time (read) plus the uncertainty. The measurement is taken for the rising and falling edges, and before and after a 90-degree phase shifting of each DQS signal. Round Trip Time measurements do not change any other measurements.

segment

The portion of a trace between two vertices on the same layer.

shorted loop inductance

Reported in HTML results from advanced decoupling analysis, shorted loop inductance is the loop inductance for a decoupling capacitor, calculated as if the capacitor pins were shorted together. The value does not include the inductance of the capacitor body.

signal integrity

The quality of a signal at a receiver pin. Signal integrity can be judged by measuring delay/timing, ringing, overshoot, multiple threshold transitions, and so on.

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signal layer

A stackup layer used to route signal traces instead of serving as a ground or fixed voltage function.

See also: mixed plane layer

SLM model

HyperLynx single-transmission line model used to model uncoupled connectors. SLM stands for “single (transmission-)line model.”

solder mask

A screened or laminated dielectric coating on the surface of a PCB. The coating prevents solder from adhering to selected areas and forming bridges (unwanted conductive paths) between traces and pads during soldering. Solder mask is also known as “conformal coating” and “SMOBC” (solder mask over bare copper).

source synchronous

A method that adds clock information to the data stream. This method avoids using a global clock signal, which can introduce skew and jitter problems. Communication interface protocols, such as DDRx, determine how to add/remove clock information to/from the data stream.

split plane layer

A plane layer with isolated areas tied to different DC voltages.

See also: plane layer

step response

A response of a channel on an isolated transition, such as a logic 0 to a logic 1. This behavior is also known as an edge response. This term applies to IBIS-AMI and FastEye channel analysis.

See also: step response, impulse response

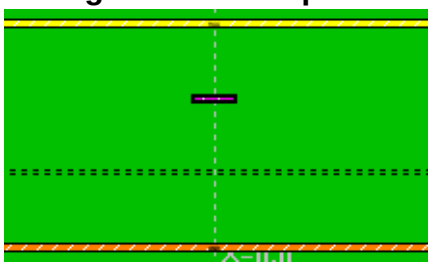
stitching via

A stitching (or shorting or caging) via is one which shorts together metal on two plane layers. In a sense, a stitching via is a perfect capacitor; it provides an extremely low-impedance, high-bandwidth connection between planes.

stripline

A trace routed on an inner layer of the PCB, with plane layers on both sides.

Figure E-13. Stripline



See also: buried microstrip and microstrip

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surface mount device (SMD)

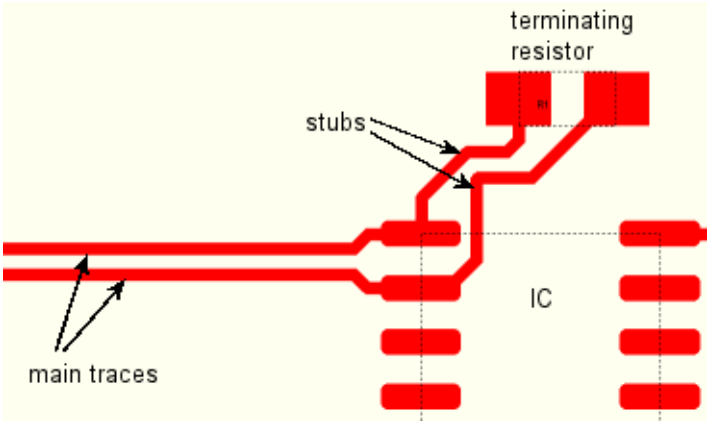
A component whose pins attach to the surface of a circuit board rather than running through the circuit board. A surface mount component does not require component holes on the circuit board. Surface mount components also allow you to place two components at the same x,y location on the circuit board; that is, one surface mount component on the top surface and one surface mount component on the bottom surface.

See also: through-hole device

stub signal trace

A branch off the main path of the signal trace, typically leading to a secondary load such as a terminator.

Figure E-14. Stub Signal Trace



stub via

A portion of the via barrel that is not used to transmit the signal. Stubs are formed when the via barrel extends beyond the signal layers used to transmit the signal.

See also: via

terminator

One or more components added to the net to improve signal integrity. Terminators work by absorbing or redistributing reflected energy caused by impedance mismatches in the circuit.

test point

A pad or via added to the board whose purpose is to apply or sense a signal for testing by an automated test process or for manual contact.

through-hole device

A component whose pins run through the circuit board (and usually come out the opposite side) rather than staying only on the surface.

See also: surface mount device (SMD)

time of flight

See flight time.

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timing model

A model that contains the maximum or minimum setup and hold times for each type of receiver pin (such as data and address) relative to the associated strobe/clock, maximum skew between certain pin pairs, signal launch delay of one pin relative to another, and so on.

topology

The geometric or logical layout of traces, signal layers, IC pins, vias, and so on used to implement a net in a PCB.

Touchstone model

A model using n-port network parameter data to represent passive interconnect networks and active devices. Touchstone models containing S-, Y-, or Z-parameter data are often used to represent equivalent circuits for backplane connectors and IC packages. Part of this popularity resulted because vectored network analyzers (VNAs) make it relatively easy to collect n-port network parameter data for a circuit and create a Touchstone model for it.

The Touchstone model format was originally developed by Agilent Corporation and has been adopted by the EIA/IBIS Open Forum.

transmission line

Any form of conductor that carries a signal from a source to a load. The transmission time is usually long compared to the speed or rise time of the signal, so that coupling, impedance, and terminators are important to preserving signal integrity.

A model of a well-behaved signal-transmission path, commonly formed by a series of routed PCB trace segments which have a well-defined return-current path in near proximity.

transmission plane

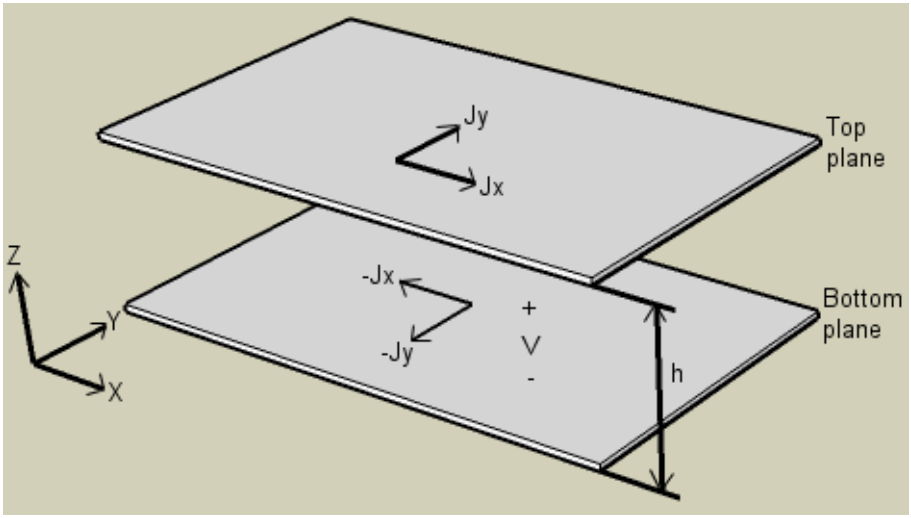
A cavity formed by two metal planes or metal regions on a PCB that stores and propagates energy to IC power supply pins. The metal regions are not mechanically connected. If the metal regions have different X/Y geometries, the transmission plane exists where the metal regions overlap each other.

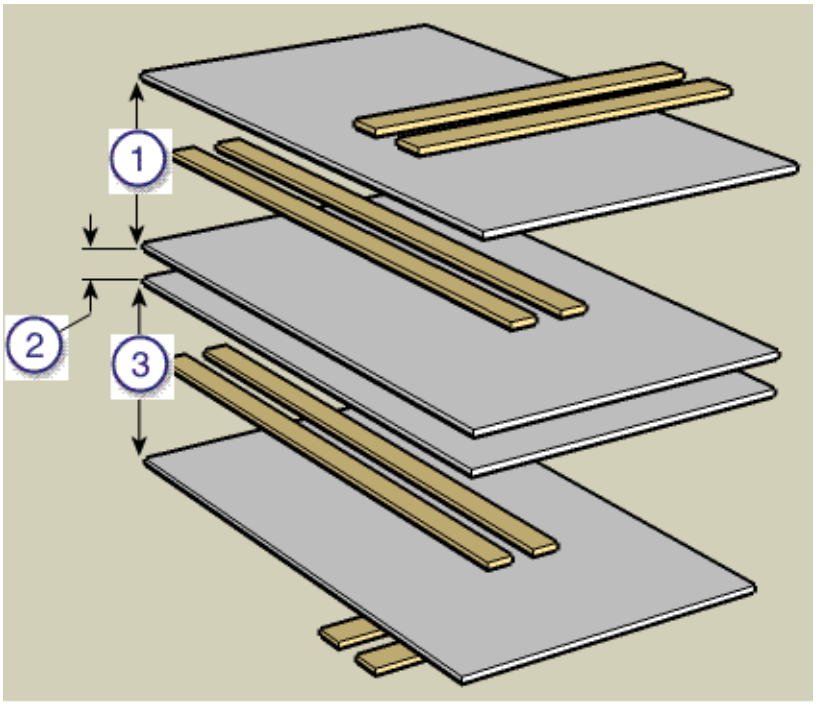
Figure E-15. Transmission Plane - Model

The figure below shows three transmission planes formed by four plane layers. For visual clarity, the figure does not contain geometries that create additional transmission planes, such as splits on plane layers and copper pours on signal layers.

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Figure E-16. Transmission Plane - Three T-Planes in an Eight-Layer Stackup





See “Transmission Planes Overview” on page 522.

tube

See barrel.

unit interval (UI)

See bit interval.

unrouted net

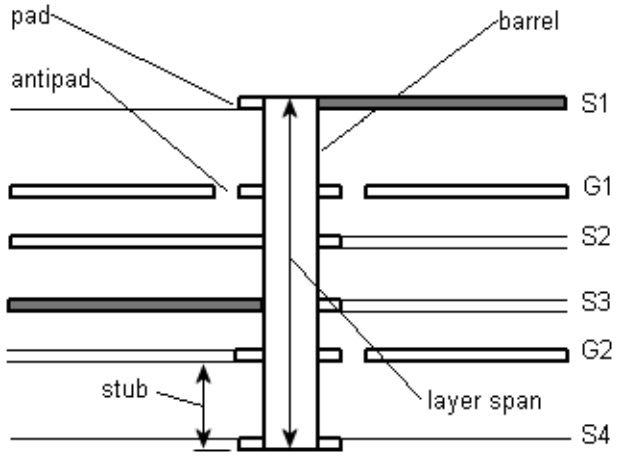
A net whose pin-to-pin connections are defined, but whose pin-to-pin routing is not fully defined.

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via

For signal integrity and traditional usage, a via is an instance of a padstack that connects traces on different metal layers on a circuit board, connects traces to a component pin, or shorts together AC ground planes. A via enables a net to connect to another layer of the circuit board.

For power integrity, a via is any object that can transmit current vertically through a transmission plane. Examples of power-integrity vias include signal vias, stitching vias, mounting pins of decoupling and bypassing capacitors, and IC power supply pins.



See also: padstack, transmission plane

victim net

A net receiving unwanted noise (crosstalk voltage) from nearby (aggressor) coupled nets that are transmitting signals.

voltage drop

The decrease in voltage due to Ohm's law operating on the current and resistance through the power network. Voltage drop occurs through the package pins, bond wires and pads, and on the metal layers of the PCB.

voltage-regulator-module (VRM) model

A switching power supply providing power to ICs and other PCB components. A VRM is a type of DC-to-DC converter.

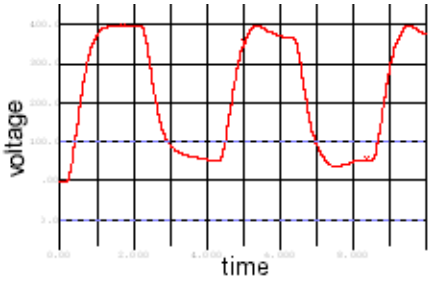
Depending on the electrical requirements, VRMs can range from large plug-in boards to much smaller, integrated, components.

waveform

A graph showing the voltage of a circuit pin at various points in time.

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Figure E-17. Waveform



Z_0

See characteristic impedance - Z0.

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Third-Party Information

Details on open source and third-party software that may be included with this product are available in the *<your_software_installation_location>/legal* directory.

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